

Symbol	Meaning	Section
d	Nominal fastener diameter, mm.	10.10.3.3
d	Diameter, mm.	10.10.7
d	Roller diameter, mm.	10.10.7
d	Nominal fastener diameter, mm.	10.20.7
d	Overall beam depth, mm.	10.20.15
d_b	Beam depth, mm.	10.10.10.6
d_b	Nominal diameter (body or shank diameter), mm.	10.17.4
d_c	Column depth, mm.	10.10.10.6
d_c	Overall column depth, mm.	10.20.9
d_z	Overall panel zone depth between continuity plates, mm.	10.20.9
e	Eccentricity in a truss connection, positive being away from the branches, mm.	10.11.2.1
e	EBF link length, mm.	10.20.15
f_a	Required axial stress at point of consideration of LRFD or ASD load combinations, MPa.	10.8.2
$f_{b(w,z)}$	Required flexural stress at the point of consideration (major axis, minor axis) using LRFD or ASD load combinations, MPa.	10.8.2
f'_{cm}	Specified minimum compressive strength of concrete at elevated temperatures, MPa.	10.18.2
f_o	Stress due to D + R (the nominal dead load + the nominal load due to rainwater or snow exclusive of the ponding contribution, MPa.	10.16.2
f_v	Required shear strength per unit area, MPa.	10.10.3.7
g	Transverse center-to-center spacing (gage) between fastener gage lines, mm.	10.2.3.13
g	Gap between toes of branch members in a gapped K-connection, neglecting welds, mm.	10.11.2.1
h	Clear distance between flanges less the fillet or corner radius for rolled shapes; for built-up sections, the distance between adjacent lines of fasteners or the clear distance between flanges when welds are used; for tees, the overall depth; for rectangular HSS, the clear distance between the flanges less the inside corner radius on each side, mm.	10.2.4.2, Table 6.10.8

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
h	Distance between centroids of individual components perpendicular to the member axis of buckling,mm.	10.5.6.1
h	Distance between horizontal boundary element centerlines, mm.	10.20.17
h_c	Twice the distance from the centroid to the following: the inside face of the compression flange less the fillet or corner radius, for rolled shapes; the nearest line of fasteners at the compression flange or the inside faces of the compression flange when welds are used, for built-up sections, mm.	10.2.4.2
h_o	Distance between flange centroids, mm.	10.6.2.2
h_o	Distance between flange centroids, mm	10.20.9
h_p	Twice the distance from plastic neutral axis to the nearest line of fasteners at the compression flange or inside face of compression flange when welds are used, mm	10.2.4.2
h_{sc}	Hole factor	10.10.3.8
j	Factor defined by Eq. 6.10.141 for minimum moment of inertia for a transverse stiffener	10.7.2.2
k	Distance from outer face of flange to the web toe of fillet, mm.	10.10.10.2
k	Outside corner radius of HSS, which is permitted to be taken as $1.5t$ if unknown, mm.	10.11.1.3
k_c	Coefficient for slender unstiffened elements, mm.	Table 6.10.1
k_s	Slip-critical combined tension and shear coefficient	10.10.3.9
k_v	Web plate buckling coefficient	10.7.2.1
l	Largest laterally unbraced length along either flange at the point of load, mm.	10.10.10.4
l	Length of bearing,mm.	10.10.7
l	Length of connection in the direction of loading,mm.	Table 6.10.2
l	Unbraced length between stitches of built-up bracing members,mm.	10.20.13
l	Unbraced length of compression or bracing member, mm.	10.20.13

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
n	Number of nodal braced points within the span	10.19.2
n	Threads per mm.	10.17.4
p	Ratio of element i deformation to its deformation at maximum stress	10.10.2.4
p	Projected length of the overlapping branch on the chord	10.11.2.2
q	Overlap length measured along the connecting face of the chord beneath the two branches	10.11.2.2
r	Governing radius of gyration, mm.	10.5.2
r	Governing radius of gyration, mm.	10.20.13
r_{crit}	Distance from instantaneous center of rotation to weld element with minimum $\frac{\Delta u}{r_i}$ ratio, mm.	10.10.2.4
r_i	Minimum radius of gyration of individual component in a built-up member, mm.	10.5.6.1
r_{ib}	Radius of gyration of individual component relative to its centroidal axis parallel to member axis of buckling, mm.	10.5.6.1
$\bar{r}_o$	Polar radius of gyration about the shear center, mm.	10.5.4
r_t	Radius of gyration of the flange components in flexural compression plus one-third of the web area in compression due to application of major axis bending moment alone	10.6.4.2
r_{ts}	Effective radius of gyration used in the determination of L_r for the lateral-torsional buckling limit state for major axis bending of doubly symmetric compact I-shaped members and channels	10.6.2.2
r_x	Radius of gyration about geometric axis parallel to connected leg, mm.	10.5.5
r_y	Radius of gyration about y-axis, mm.	10.5.4
r_y	Radius of gyration about y-axis, mm.	10.20.9
r_z	Radius of gyration for the minor principal axis, mm.	10.5.5

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
s	Longitudinal center-to-center spacing (pitch) of any two consecutive holes, mm.	10.2.3.13
t	Thickness of element, mm.	10.2.4.2
t	Thickness of element, mm.	Table 6.10.8
t	Wall thickness, mm.	10.5.7.2
t	Angle leg thickness, mm.	10.6.10.2
t	Width of rectangular bar parallel to axis of bending, mm.	10.6.11.2
t	Thickness of connected material, mm.	10.10.3.10
t	Thickness of plate, mm.	10.4.5.1
t	Design wall thickness for HSS equal to 0.93 times the nominal wall thickness for ERW HSS and equal to the nominal wall thickness for SAW HSS, mm.	10.2.3.12
t	Total thickness of fillers, mm.	10.10.5
t	Thickness of connected part, mm.	10.20.7
t	Thickness of column web or doubler plate, mm.	10.20.9
t	Design wall thickness of HSS main member, mm.	10.11.3.1
t	Design wall thickness of HSS main member, mm.	10.11.2.1
t	Design wall thickness of HSS member, mm.	10.11.1.1
t_b	Design wall thickness of HSS branch member, mm.	10.11.2.1
t_b	Design wall thickness of HSS branch member, mm.	10.11.3.1
t_{bf}	Thickness of beam flange, mm.	10.20.9
t_{bj}	Thickness of the overlapped branch, mm.	10.11.2.3
t_{cf}	Thickness of the column flange, mm.	10.10.10.6

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
t_{cf}	Thickness of column flange, mm.	10.20.9
t_f	Thickness of the loaded flange, mm.	10.10.10.1
t_f	Thickness of flange, mm.	10.20.17
t_{fc}	Compression flange thickness, mm.	10.6.4.2
t_p	Thickness of plate, mm.	10.11.1.1
t_p	Thickness of the attached transverse plate, mm.	10.11.2.3
t_p	Thickness of tension loaded plate, mm.	10.17.3
t_p	Thickness of panel zone including doubler plates, mm.	10.20.9
t_s	Web stiffener thickness, mm.	10.19.2
t_w	Web thickness, mm.	Table 6.10.1
t_w	Thickness of web, mm.	Table 6.10.8
t_w	Thickness of element, mm.	10.5.7.1
t_w	Column web thickness, mm.	10.10.10.6
t_w	Beam web thickness, mm.	10.19.3
w	Width of cover plate, mm.	10.6.13.3
w	Weld leg size, mm.	10.10.2.2
w	Plate width, mm.	Table 6.10.2
w	Leg size of the reinforcing or contouring fillet, if any, in the direction of the thickness of the tension-loaded plate, mm.	10.17.3
w_z	Width of panel zone between column flanges, mm.	10.20.9
x	Parameter used for determining the approximate fundamental period	Appendix P.2
x_o, y_o	Coordinates of the shear center with respect to the centroid, mm.	10.5.4

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
$\bar{x}$	Connection eccentricity,mm.	Table 6.10.2
y	Subscript relating symbol to weak axis	
z	Subscript relating symbol to minor principal axis bending	
Z_b	Minimum plastic section modulus at the reduced beam section, mm ³	10.20.9
α	Factor used in Eq. 6.10.2.2	10.3.2.1
α	Separation ratio for built-up compression members $=h/(2r_{ib})$	10.5.6.1
α	Angle of diagonal members with the horizontal	10.20.12
α	Angle of web yielding in radians, as measured relative to the vertical	10.20.17
β	Reduction factor given by Eq. 6.10.159	10.10.2.2
β	The width ratio; the ratio of branch diameter to chord diameter $= D_b/D$ for round HSS; the ratio of overall branch width to chord width $= B_b/B$ for rectangular HSS	10.11.2.1, 10.11.3.1
β	Compression strength adjustment factor	10.20.16
β_T	Brace stiffness requirement excluding web distortion, N-mm/radian.	10.19.2
β_{br}	Required brace stiffness	10.19.2
β_{eff}	Effective width ratio; the sum of the perimeters of the two branch members in a K-connection divided by eight times the chord width	10.11.2.1
β_{eop}	Effective outside punching parameter	10.11.2.3
β_{sec}	Web distortional stiffness, including the effect of web transverse stiffeners, if any, N-mm/radian.	10.19.2
β_w	Section property for unequal leg angles, positive for short legs in compression and negative for long legs in compression	10.6.10.2
Δ	First-order interstory drift due to the design loads,mm.	10.3.2.2

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
Δ	Design story drift	10.20.15
Δ_b	Deformation quantity used to control loading of test specimen (total brace end rotation for the sub-assembly test specimen; total brace axial deformation for the brace test specimen)	Appendix R.2
Δ_{bm}	Value of deformation quantity, Δ_b , corresponding to the design story drift	Appendix R.6
Δ_{by}	Value of deformation quantity, Δ_b , at first significant yield of test specimen	Appendix R.6
Δ_h	First-order interstory drift due to lateral forces, mm.	10.3.2.1
Δ_i	Deformation of weld elements at intermediate stress levels, linearly proportioned to the critical deformation based on distance from instantaneous center of rotation, r_i , mm.	10.10.2.4
Δ_m	Deformation of weld element at maximum stress, mm.	10.10.2.4
Δ_u	Deformation of weld element at ultimate stress (fracture), usually in element furthest from instantaneous center of rotation, mm.	10.10.2.4
δ	Deformation quantity used to control loading of test specimen	Appendix Q.6
δ_y	Value of deformation quantity δ at first significant yield of test specimen	Appendix Q.6
ρ'	Ratio of required axial force P_u to required shear strength V_u of a link	10.20.15
γ	The chord slenderness ratio; the ratio of one-half the diameter to the wall thickness $= D/(2t)$ for round HSS; the ratio of one-half the width to wall thickness $= B/(2t)$ for rectangular HSS	10.11.2.1, 10.11.3.1
ξ	The gap ratio; the ratio of the gap between the branches of a gapped K-connection to the width of the chord $= g/B$ for rectangular HSS	10.11.2.1

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
η	The load length parameter, applicable only to rectangular HSS; the ratio of the length of contact of the branch with the chord in the plane of the connection to the chord width $= N/B$, where $N = H_b/\sin \theta$	10.11.2.1, 10.11.3.1
λ	Slenderness parameter	10.6.3
λ_p	Limiting slenderness parameter for compact element	10.2.4
λ_{pf}	Limiting slenderness parameter for compact flange	10.6.3
λ_{pw}	Limiting slenderness parameter for compact web	10.6.4
λ_r	Limiting slenderness parameter for noncompact element	10.2.4
λ_{rf}	Limiting slenderness parameter for noncompact flange	10.6.3
λ_{rw}	Limiting slenderness parameter for noncompact web	10.6.4
λ_p, λ_{ps}	Limiting slenderness parameter for compact element	10.20.8
μ	Mean slip coefficient for class A or B surfaces, as Applicable, or as established by tests	10.10.3.8
ϕ	Resistance factor	10.2.3.3
ϕ	Resistance factor	10.20.6
ϕ_b	Resistance factor for flexure	10.6.1
ϕ_b	Resistance factor for flexure	10.20.8
ϕ_c	Resistance factor for compression	10.5.1
ϕ_c	Resistance factor for compression	10.20.8
ϕ_{sf}	Resistance factor for shear on the failure path	10.4.5.1
ϕ_T	Resistance factor for torsion	10.8.3.1
ϕ_t	Resistance factor for tension	10.4.2
ϕ_v	Resistance factor for shear	10.7.1

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
ϕ_v	Resistance factor for shear strength of panel zone of beam-to-column connections	10.20.9
ϕ_v	Resistance factor for shear	10.20.15
ϕ_{total}	Link rotation angle	Appendix Q.2
Ω	Safety factor	10.2.3.4
Ω	Safety factor	10.20.6
Ω_b	Safety factor for flexure	10.6.1
Ω_b	Safety factor for flexure = 1.67	10.20.8
Ω_c	Safety factor for compression	10.5.1
Ω_c	Safety factor for compression = 1.67	10.20.8
Ω_{sf}	Safety factor for shear on the failure path	10.4.5.1
Ω_T	Safety factor for torsion	10.8.3.1
Ω_t	Safety factor for tension	10.4.2
Ω_v	Safety factor for shear	10.7.1
Ω_v	Safety factor for shear strength of panel zone of beam-to-column connections	10.20.9
Ω_o	Horizontal seismic overstrength factor	10.20.4
θ	Angle of loading measured from the weld longitudinal axis, degrees	10.10.2.4
θ	Acute angle between the branch and chord, degrees	10.11.2.1
θ	Acute angle between the branch and chord, degrees	10.11.3.1
θ	Interstory drift angle, radians	Appendix Q.3
ω	Strain hardening adjustment factor	10.20.16
ϵ_{cu}	Strain corresponding to compressive strength of concrete, f'_c	10.18.2
τ_b	Parameter for reduced flexural stiffness using the direct analysis method	10.14.3

<u>Symbol</u>	<u>Meaning</u>	<u>Section</u>
ΣM^*_{pc}	Moment at beam and column centerline determined by projecting the sum of the nominal column plastic moment strength, reduced by the axial stress P_{uc}/A_g , from the top and bottom of the beam moment connection	10.20.9
ΣM^*_{pb}	Moment at the intersection of the beam and column centerlines determined by projecting the beam maximum developed moments from the column face. Maximum developed moments shall be determined from test results	10.20.9

10.1.2.2 Definitions

ACTIVE FIRE PROTECTION	Building materials and systems that are activated by a fire to mitigate adverse effects or to notify people to take some action to mitigate adverse effects.
ADJUSTED BRACE STRENGTH	Strength of a brace in a buckling-restrained braced frame at deformations corresponding to 2.0 times the design story drift.
ALLOWABLE STRENGTH*	Nominal strength divided by the safety factor, R_n/Ω .
ALLOWABLE STRESS	Allowable strength divided by the appropriate section property, such as section modulus or cross-section area.
AMPLIFICATION FACTOR	Multiplier of the results of first-order analysis to reflect second-order effects.
AMPLIFIED SEISMIC LOAD	Horizontal component of earthquake load E multiplied by Ω_o , where E and the horizontal component of E are specified in the Code.
APPLICABLE BUILDING CODE	Building Code under which the structure is designed.
ASD(ALLOWABLE STRENGTH DESIGN)	Method of proportioning structural components such that the allowable strength equals or exceeds the required strength of the component under the action of the ASD load combinations.
ASD LOAD COMBINATION	Load combination in the Code intended for allowable strength design (allowable stress design).

AUTHORITY HAVING JURISDICTION	Organization, political subdivision, office or individual charged with the responsibility of administering and enforcing the provisions of the Code.
AVAILABLE STRENGTH*	Design strength or allowable strength, as appropriate.
AVAILABLE STRESS*	Design stress or allowable stress, as appropriate.
AVERAGE RIB WIDTH	Average width of the rib of a corrugation in a formed steel deck.
AUTHORITY HAVING JURISDICTION (AHJ)	Organization, political subdivision, office or individual charged with the responsibility of administering and enforcing the provisions of this Code.
BATTEN PLATE	Plate rigidly connected to two parallel components of a built-up column or beam designed to transmit shear between the components.
BEAM	Structural member that has the primary function of resisting bending moments. Beam-column. Structural member that resists both axial force and bending moment. Bearing. In a bolted connection, limit state of shear forces transmitted by the bolt to the connection elements.
BEARING (LOCAL COMPRESSIVE YIELDING)	Limit state of local compressive yielding due to the action of a member bearing against another member or surface.
BEARING-TYPE CONNECTION	Bolted connection where shear forces are transmitted by the bolt bearing against the connection elements.
BLOCK SHEAR RUPTURE	In a connection, limit state of tension fracture along one path and shear yielding or shear fracture along another path.
BRACED FRAME	An essentially vertical truss system that provides resistance to lateral forces and provides stability for the structural system.
BRANCH FACE	Wall of HSS branch member.
BRANCH MEMBER	For HSS connections, member that terminates at a chord member or main member.
BUCKLING	Limit state of sudden change in the geometry of a structure or any of its elements under a critical loading condition.

BUCKLING STRENGTH	Nominal strength for buckling or instability limit states.
BUCKLING-RESTRAINED BRACED FRAME (BRBF)	Diagonally braced frame satisfying the requirements of Section 16 in which all members of the bracing system are subjected primarily to axial forces and in which the limit state of compression buckling of braces is precluded at forces and deformations corresponding to 2.0 times the design story drift.
BUCKLING-RESTRAINING SYSTEM	System of restraints that limits buckling of the steel core in BRBF. This system includes the casing on the steel core and structural elements adjoining its connections. The buckling-restraining system is intended to permit the transverse expansion and longitudinal contraction of the steel core for deformations corresponding to 2.0 times the design story drift.
BUILT-UP MEMBER, CROSS-SECTION, SECTION, SHAPE	Member, cross-section, section or shape fabricated from structural steel elements that are welded or bolted together.
CAMBER	Curvature fabricated into a beam or truss so as to compensate for deflection induced by loads.
CASING	Element that resists forces transverse to the axis of the brace there by restraining buckling of the core. The casing requires a means of delivering this force to the remainder of the buckling-restraining system. The casing resists little or no force in the axis of the brace.
CHARPY V-NOTCH IMPACT TEST	Standard dynamic test measuring notch toughness of a specimen.
CHORD MEMBER	For HSS, primary member that extends through a truss connection.
CLADDING	Exterior covering of structure.
COLD-FORMED STEEL STRUCTURAL MEMBER	Shape manufactured by press-braking blanks sheared from sheets, cut lengths of coils or plates, or by roll forming cold-or hot-rolled coils or sheets; both forming operations being performed at ambient room temperature, that is, without manifest addition of heat such as would be required for hot forming.
COLUMN	Structural member that has the primary function of resisting axial force.

COLUMN BASE	Assemblage of plates, connectors, bolts, and rods at the base of a column used to transmit forces between the steel superstructure and the foundation.
COMBINED SYSTEM	Structure comprised of two or more lateral load-resisting systems of different type.
COMPACT SECTION	Section capable of developing a fully plastic stress distribution and possessing a rotation capacity of approximately three before the onset of local buckling.
COMPARTMENTATION	The enclosure of a building space with elements that have a specific fire endurance.
COMPLETE-JOINT-PENETRATION GROOVE WELD (CJP)	Groove weld in which weld metal extends through the joint thickness, except as permitted for HSS connections.
COMPOSITE	Condition in which steel and concrete elements and members work as a unit in the distribution of internal forces.
CONCRETE CRUSHING	Limit state of compressive failure in concrete having reached the ultimate strain.
CONCRETE HAUNCH	Section of solid concrete that results from stopping the deck on each side of the girder in a composite floor system constructed using a formed steel deck.
CONCRETE-ENCASED BEAM	Beam totally encased in concrete cast integrally with the slab.
CONNECTION	Combination of structural elements and joints used to transmit forces between two or more members.
CONVECTIVE HEAT TRANSFER	The transfer of thermal energy from a point of higher temperature to a point of lower temperature through the motion of an intervening medium.
CONTINUITY PLATES	Column stiffeners at the top and bottom of the panel zone; also known as transverse stiffeners.
CONTRACTOR	Fabricator or erector, as applicable.
COPE	Cutout made in a structural member to remove a flange and conform to the shape of an intersecting member.
COVER PLATE	Plate welded or bolted to the flange of a member to increase cross-sectional area, section modulus or moment of inertia.

CROSS CONNECTION	HSS connection in which forces in branch members or connecting elements transverse to the main member are primarily equilibrated by forces in other branch members or connecting elements on the opposite side of the main member.
DEMANDCRITICAL WELD	Weld so designated by these Provisions.
DESIGNEARTHQUAKE	The earthquake represented by the design response spectrum as specified in the Code.
DESIGNSTORYDRIFT	Amplified story drift (drift under the design earthquake, including the effects of inelastic action), determined as specified in the Code.
DESIGN-BASIS FIRE	A set of conditions that define the development of a fire and the spread of combustion products throughout a building or portion thereof.
DESIGN LOAD*	Applied load determined in accordance with either LRFD load combinations or ASD load combinations, whichever is applicable.
DESIGN STRENGTH*	Resistance factor multiplied by the nominal strength, ϕR_n .
DESIGN STRESS RANGE	Magnitude of change in stress due to the repeated application and removal of service live loads. For locations subject to stress reversal it is the algebraic difference of the peak stresses.
DESIGN STRESS*	Design strength divided by the appropriate section property, such as section modulus or cross section area.
DESIGN WALL THICKNESS	HSS wall thickness assumed in the determination of section properties.
DIAGONAL BRACING	Inclined structural member carrying primarily axial force in a braced frame.
DIAGONAL STIFFENER	Web stiffener at column panel zone oriented diagonally to the flanges, on one or both sides of the web.
DIAPHRAGM PLATE	Plate possessing in-plane shear stiffness and strength, used to transfer forces to the supporting elements.
DIAPHRAGM	Roof, floor or other membrane or bracing system that transfers in-plane forces to the lateral force resisting system.

DIRECT ANALYSIS METHOD	Design method for stability that captures the effects of residual stresses and initial out-of-plumbness of frames by reducing stiffness and applying notional loads in a second-order analysis.
DIRECT BOND INTERACTION	Mechanism by which force is transferred between steel and concrete in a composite section by bond stress.
DISTORTIONAL FAILURE	Limit state of an HSS truss connection based on distortion of a rectangular HSS chord member into a rhomboidal shape.
DISTORTIONAL STIFFNESS	Out-of-plane flexural stiffness of web.
DOUBLE CURVATURE	Deformed shape of a beam with one or more inflection points within the span.
DOUBLE-CONCENTRATED FORCES	Two equal and opposite forces that form a couple on the same side of the loaded member.
DOUBLER	Plate added to, and parallel with, a beam or column web to increase resistance to concentrated forces.
DRIFT	Lateral deflection of structure.
DUAL SYSTEM	Structural system with the following features (1) an essentially complete space frame that provides support for gravity loads; (2) resistance to lateral load provided by moment frames (SMF, IMF or OMF) that are capable of resisting at least 25 percent of the base shear, and concrete or steel shearwalls, or steel braced frames (EBF, S CBF or OCBF); and (3) each system designed to resist the total lateral load in proportion to its relative rigidity.
DUCTILE LIMIT STATE	Ductile limit states include member and connection yielding, bearing deformation at bolt holes, as well as buckling of members that conform to the width-thickness limitations of Table 6.10.18. Fracture of a member or of a connection, or buckling of a connection element, is not a ductile limit state.
ECCENTRICALLY BRACED FRAME (EBF)	Diagonally braced frame meeting the requirements of Section 15 that has at least one end of each bracing member connected to a beam a short distance from another beam-to-brace connection or a beam-to-column connection.
EFFECTIVE LENGTH FACTOR, K	Ratio between the effective length and the unbraced length of the member.

EFFECTIVE LENGTH	Length of an otherwise identical column with the same strength when analyzed with pinned end conditions.
EFFECTIVE NET AREA	Net area modified to account for the effect of shear lag.
EFFECTIVE SECTION MODULUS	Section modulus reduced to account for buckling of slender compression elements.
EFFECTIVE WIDTH	Reduced width of a plate or slab with an assumed uniform stress distribution which produces the same effect on the behavior of a structural member as the actual plate or slab width with its nonuniform stress distribution.
ELASTIC ANALYSIS	Structural analysis based on the assumption that the structure returns to its original geometry on removal of the load.
ELEVATED TEMPERATURES	Heating conditions experienced by building elements or structures as a result of fire, which are in excess of the anticipated ambient conditions.
ENCASED COMPOSITE COLUMN	Composite column consisting of a structural concrete column and one or more embedded steel shapes.
END PANEL	Web panel with an adjacent panel on one side only.
ENGINEER OF RECORD	Engineer having authority or license from government approved Authority to sign and seal engineering and contract documents.
END RETURN	Length of fillet weld that continues around a corner in the same plane. Engineer of record. Licensed professional responsible for sealing the contract documents. Expansion rocker. Support with curved surface on which a member bears that can tilt to accommodate expansion.
EXPANSION ROLLER	Round steel bar on which a member bears that can roll to accommodate expansion.
EXEMPTED COLUMN	Column not meeting the requirements of Eq. 6.10.300 for SMF.
EXPECTED TENSILE STRENGTH *	Tensile strength of a member, equal to the specified minimum tensile strength, F_u , multiplied by R_t .

EXPECTED YIELD STRENGTH	Yield strength in tension of a member, equal to the expected yield stress multiplied by A_g .
EXPECTED YIELD STRESS	Yield stress of the material, equal to the specified minimum yield stress, F_y , multiplied by R_y .
EYEBAR	Pin-connected tension member of uniform thickness, with forged or thermally cut head of greater width than the body, proportioned to provide approximately equal strength in the head and body.
FACTORED LOAD	Product of a load factor and the nominal load. Fastener. Generic term for bolts, rivets, or other connecting devices.
FATIGUE	Limit state of crack initiation and growth resulting from repeated application of live loads.
FAYING SURFACE	Contact surface of connection elements transmitting a shear force.
FILLED COMPOSITE COLUMN	Composite column consisting of a shell of HSS or pipe filled with structural concrete.
FILLER METAL	Metal or alloy to be added in making a welded joint. Filler. Plate used to build up the thickness of one component. Fillet weld reinforcement. Fillet welds added to groove welds.
FILLET WELD	Weld of generally triangular cross section made between intersecting surfaces of elements.
FIRE	Destructive burning, as manifested by any or all of the following: light, flame, heat, or smoke.
FIRE BARRIER	Element of construction formed of fire-resisting materials and tested in accordance with ASTM Standard E119, or other approved standard fire resistance test, to demonstrate compliance with the Building Code.
FIRE ENDURANCE	A measure of the elapsed time during which a material or assembly continues to exhibit fire resistance.
FIRE RESISTANCE	That property of assemblies that prevents or retards the passage of excessive heat, hot gases or flames under conditions of use and enables them to continue to perform a stipulated function.

FIRE RESISTANCE RATING	The period of time a building element, component or assembly maintains the ability to contain a fire, continues to perform a given structural function, or both, as determined by test or methods based on tests.
FIRST-ORDER ANALYSIS	Structural analysis in which equilibrium conditions are formulated on the undeformed structure; second-order effects are neglected.
FITTED BEARING STIFFENER	Stiffener used at a support or concentrated load that fits tightly against one or both flanges of a beam so as to transmit load through bearing.
FLASHOVER	The rapid transition to a state of total surface involvement in a fire of combustible materials within an enclosure.
FLARE BEVEL GROOVE WELD	Weld in a groove formed by a member with a curved surface in contact with a planar member.
FLARE V-GROOVE WELD	Weld in a groove formed by two members with curved surfaces.
FLAT WIDTH	Nominal width of rectangular HSS minus twice the outside corner radius. In absence of knowledge of the corner radius, the flat width may be taken as the total section width minus three times the thickness.
FLEXURAL BUCKLING	Buckling mode in which a compression member deflects laterally without twist or change in cross-sectional shape.
FLEXURAL-TORSIONAL BUCKLING	Buckling mode in which a compression member bends and twists simultaneously without change in cross-sectional shape.
FORCE	Resultant of distribution of stress over a prescribed area.
FORMED SECTION	See cold-formed steel structural member.
FORMED STEEL DECK	In composite construction, steel cold formed into a decking profile used as a permanent concrete form.
FULLY RESTRAINED MOMENT CONNECTION	Connection capable of transferring moment with negligible rotation between connected members.
GAGE	Transverse center-to-center spacing of fasteners.
GAP CONNECTION	HSS truss connection with a gap or space on the chord face between intersecting branch members.

GENERAL COLLAPSE	Limit state of chord plastification of opposing sides of a round HSS chord member at a cross-connection.
GEOMETRIC AXIS	Axis parallel to web, flange or angle leg.
GIRDER FILLER	Narrow piece of sheet steel used as a fill between edge of a deck sheet and flange of a girder in a composite floor system constructed using a formed steel deck.
GIRDER	See Beam.
GIRT	Horizontal structural member that supports wall panels and is primarily subjected to bending under horizontal loads, such as wind load.
GOUGE	Relatively smooth surface groove or cavity resulting from plastic deformation or removal of material.
GRAVITY AXIS	Axis through the center of gravity of a member along its length.
GRAVITY FRAME	Portion of the framing system not included in the lateral load resisting system.
GRAVITY LOAD	Load, such as that produced by dead and live loads, acting in the downward direction.
GRIP (OF BOLT)	Thickness of material through which a bolt passes.
GROOVE WELD	Weld in a groove between connection elements. See also AWS D1.1.
GUSSET PLATE	Plate element connecting truss members or a strut or brace to a beam or column.
HEAT FLUX	Radiant energy per unit surface area.
HEAT RELEASE RATE	The rate at which thermal energy is generated by a burning material.
HORIZONTAL SHEAR	Force at the interface between steel and concrete surfaces in a composite beam.
HSS	Square, rectangular or round hollow structural steel section produced in accordance with a pipe or tubing product specification.
INELASTIC ANALYSIS	Structural analysis that takes into account inelastic material behavior, including plastic analysis.

IN-PLANE INSTABILITY	Limit state of a beam-column bent about its major axis while lateral buckling or lateral-torsional buckling is prevented by lateral bracing.
INSTABILITY	Limit state reached in the loading of a structural component, frame or structure in which a slight disturbance in the loads or geometry produces large displacements.
INTERMEDIATE MOMENT FRAME (IMF)	Moment frame system that meets the requirements of Sec 10.20.10.
INTERSTORY DRIFT ANGLE	Interstory displacement divided by story height, radians.
INVERTED-V-BRACED FRAME	See V-braced frame.
JOINT ECCENTRICITY	For HSS truss connection, perpendicular distance from chord member center of gravity to intersection of branch member work points.
JOINT	Area where two or more ends, surfaces, or edges are attached. Categorized by type of fastener or weld used and method of force transfer.
K-AREA	The <i>k-area</i> is the region of the web that extends from the tangent point of the web and the flange-web fillet (AISC “ <i>k</i> ” dimension) a distance of 38 mm in to the web beyond the “ <i>k</i> ” dimension.
K-BRACED FRAME	A bracing configuration in which braces connect to a column at a location with no diaphragm or other out-of-plane support.
K-CONNECTION	HSS connection in which forces in branch members or connecting elements transverse to the main member are primarily equilibrated by forces in other branch members or connecting elements on the same side of the main member.
KSI	Kip per square inch, a US customary unit of stress.
LOWEST ANTICIPATED SERVICE TEMPERATURE (LAST)	The lowest 1-hour average temperature with a 100-year mean recurrence interval.
LRFD (LOAD AND RESISTANCE FACTOR DESIGN)	Method of proportioning structural components such that the design strength equals or exceeds the required strength of the component under the action of the LRFD load combinations.

LRFD LOAD COMBINATION	Load combination in the Code intended for strength design (load and resistance factor design).
LACING	Plate, angle or other steel shape, in a lattice configuration, that connects two steel shapes together.
LAP JOINT	Joint between two overlapping connection elements in parallel planes.
LATERAL BRACING	Diagonal bracing, shear walls or equivalent means for providing in-plane lateral stability.
LATERAL BRACING MEMBER	Member that is designed to inhibit lateral buckling or lateral- torsional buckling of primary framing members.
LATERAL LOAD RESISTING SYSTEM	Structural system designed to resist lateral loads and provide stability for the structure as a whole.
LATERAL LOAD	Load that produced by wind or earthquake effects, acting in a lateral direction.
LATERAL-TORSIONAL BUCKLING	Buckling mode of a flexural member involving deflection normal to the plane of bending occurring simultaneously with twist about shear center of the cross-section.
LEANING COLUMN	Column designed to carry gravity loads only, with connections that are not intended to provide resistance to lateral loads.
LENGTH EFFECTS	Consideration of the reduction in strength of a member based on its unbraced length.
LIMIT STATE	Condition in which a structure or component becomes unfit for service and is judged either to be no longer useful for its intended function (serviceability limit state) or to have reached its ultimate load-carrying capacity (strength limit state).
LINK	In EBF, the segment of a beam that is located between the ends of two diagonal braces or between the end of a diagonal brace and a column. The length of the link is defined as the clear distance between the end soft wodiagonal braces or between the diagonal brace and the column face.
LINK INTERMEDIATE WEB STIFFENERS	Vertical web stiffeners placed within the link in EBF.

LINK ROTATION ANGLE	Inelastic angle between the link and the beam outside of the link when the total story drift is equal to the design story drift.
LINK SHEAR DESIGN STRENGTH	Lesser of the available shear strength of the link developed from the moment or shear strength of the link.
LOAD	Force or other action that results from the weight of building materials, occupants and their possessions, environmental effects, differential movement, or restrained dimensional changes.
LOAD EFFECT	Forces, stresses and deformations produced in a structural component by the applied loads.
LOAD FACTOR	Factor that accounts for deviations of the nominal load from the actual load, for uncertainties in the analysis that transforms the load into a load effect and for the probability that more than one extreme load will occur simultaneously.
LOCAL BENDING**	Limit state of large deformation of a flange under a concentrated tensile force.
LOCAL BUCKLING**	Limit state of buckling of a compression element within a cross section.
LOCAL CRIPPLING**	Limit state of local failure of web plate in the immediate vicinity of a concentrated load or reaction.
LOCAL YIELDING**	Yielding that occurs in a local area of an element.
LRFD (LOAD AND RESISTANCE FACTOR DESIGN)	Method of proportioning structural components such that the design strength equals or exceeds the required strength of the component under the action of the LRFD load combinations.
LRFD LOAD COMBINATION	Load combination in the Code intended for strength design (load and resistance factor design).
MAIN MEMBER	For HSS connections, chord member, column or other HSS member to which branch members or other connecting elements are attached.
MEASURED FLEXURAL RESISTANCE	Bending moment measured in a beam at the face of the column, for a beam-to-column test specimen tested in accordance with Appendix S.
MECHANISM	Structural system that includes a sufficient number of real hinges, plastic hinges or both, so as to be able to articulate in one or more rigid body modes.

MILL SCALE	Oxide surface coating on steel formed by the hot rolling process.
MILLED SURFACE	Surface that has been machined flat by a mechanically guided tool to a flat, smooth condition.
MOMENT CONNECTION	Connection that transmits bending moment between connected members.
MOMENT FRAME	Framing system that provides resistance to lateral loads and provides stability to the structural system, primarily by shear and flexure of the framing members and their connections.
NET AREA	Gross area reduced to account for removed material.
NODAL BRACE	Brace that prevents lateral movement or twist independently of other braces at adjacent brace points (see relative brace).
NOMINAL DIMENSION	Designated or theoretical dimension, as in the tables of section properties.
NOMINAL LOAD	Magnitude of the load specified by the Code.
NOMINAL RIB HEIGHT	Height of formed steel deck measured from the underside of the lowest point to the top of the highest point.
NOMINAL STRENGTH*	Strength of a structure or component (without the resistance factor or safety factor applied) to resist load effects, as determined in accordance with this Specification.
NONCOMPACT SECTION	Section that can develop the yield stress in its compression elements before local buckling occurs, but cannot develop a rotation capacity of three.
NONDESTRUCTIVE TESTING	Inspection procedure wherein no material is destroyed and integrity of the material or component is not affected.
NOTCH TOUGHNESS	Energy absorbed at a specified temperature as measured in Charpy V-Notch test.
NOTIONAL LOAD	Virtual load applied in a structural analysis to account for destabilizing effects that are not otherwise accounted for in the design provisions.
OUT-OF-PLANE BUCKLING	Limit state of a beam-column bent about its major axis while lateral buckling or lateral-torsional buckling is not prevented by lateral bracing.

ORDINARY CONCENTRICALLY BRACED FRAME (OCBF)	Diagonally braced frame meeting the requirements of Section 14 in which all members of the bracing system are subjected primarily to axial forces.
ORDINARY MOMENT FRAME (OMF)	Moment frame system that meets the requirements of Sec 10.20.11.
OVERLAP CONNECTION	HSS truss connection in which intersecting branch members overlap.
OVER STRENGTH FACTOR, Ω_o	Factor specified by the Code in order to determine the amplified seismic load, where required by these Provisions.
PANEL ZONE	Web area of beam-to-column connection delineated by extension of beam and column flanges through connection, transmitting moment through a shear panel.
PARTIAL-JOINT- PENETRATION GROOVE WELD (PJP)	Groove weld in which the penetration is intentionally less than the complete thickness of the connected element.
PARTIALLY RESTRAINED MOMENT CONNECTION	Connection capable of transferring moment with rotation between connected members that is not negligible.
PASSIVE FIRE PROTECTION	Building materials and systems whose ability to resist the effects of fire does not rely on any outside activating condition or mechanism.
PERFORMANCE-BASED DESIGN	An engineering approach to structural design that is based on agreed-upon performance goals and objectives, engineering analysis and quantitative assessment of alternatives against those design goals and objectives using accepted engineering tools, methodologies and performance criteria.
PERCENT ELONGATION	Measure of ductility, determined in a tensile test as the maximum elongation of the gage length divided by the original gage length.
PERMANENT LOAD	Load in which variations over time are rare or of small magnitude. All other loads are variable loads.
PIPE	See HSS.
PITCH	Longitudinal center-to-center spacing of fasteners. Center-to-center spacing of bolt threads along axis of bolt.

PLASTIC ANALYSIS	Structural analysis based on the assumption of rigid-plastic behavior, in other words, that equilibrium is satisfied throughout the structure and the stress is at or below the yield stress.
PLASTIC HINGE	Yielded zone that forms in a structural member when the plastic moment is attained. The member is assumed to rotate further as if hinged, except that such rotation is restrained by the plastic moment.
PLASTIC MOMENT	Theoretical resisting moment developed within a fully yielded cross section.
PLASTIC STRESS DISTRIBUTION METHOD	Method for determining the stresses in a composite member assuming that the steel section and the concrete in the cross section are fully plastic.
PLASTIFICATION	In an HSS connection, limit state based on an out-of-plane flexural yield line mechanism in the chord at a branch member connection.
PLATE GIRDER	Built-up beam.
PLUG WELD	Weld made in a circular hole in one element of a joint fusing to another element.
PONDING	Retention of water due solely to the deflection of flat roof framing.
POST-BUCKLING STRENGTH	Load or force that can be carried by an element, member, or frame after initial buckling has occurred.
PREQUALIFIED CONNECTION	Connection that complies with the requirements of Appendix N
PRESCRIPTIVE DESIGN	A design method that documents compliance with general criteria established in a Code.
PRETENSIONED JOINT	Joint with high-strength bolts tightened to the specified minimum pretension.
PROPERLY DEVELOPED	Reinforcing bars detailed to yield in a ductile manner before crushing of the concrete occurs. Bars meeting the provisions of Chapter 6 (of Part 6) insofar as development length, spacing and cover shall be deemed to be properly developed.
PROTECTED ZONE	Area of members in which limitations apply to fabrication and attachments.
PROTOTYPE	The connection or brace design that is to be used in the building (SMF, IMF, EBF, and BRBF).

PRYING ACTION	Amplification of the tension force in a bolt caused by leverage between the point of applied load, the bolt and the reaction of the connected elements.
PUNCHING LOAD	Component of branch member force perpendicular to a chord.
PURLIN	Horizontal structural member that supports roof deck and is primarily subjected to bending under vertical loads such as snow, wind or dead loads.
P- δ EFFECT	Effect of loads acting on the deflected shape of a member between joints or nodes.
P- Δ EFFECT	Effect of loads acting on the displaced location of joints or nodes in a structure. In tiered building structures, this is the effect of loads acting on the laterally displaced location of floors and roofs.
QUALITY ASSURANCE	System of shop and field activities and controls implemented by the owner or his/her designated representative to provide confidence to the owner and the building authority that quality requirements are implemented.
QUALITY ASSURANCE PLAN	Written description of qualifications, procedures, quality inspections, resources, and records to be used to provide assurance that the structure complies with the engineer's quality requirements, specifications and contract documents.
QUALITY CONTROL	System of shop and field controls implemented by the fabricator and erector to ensure that contract and company fabrication and erection requirements are met.
RATIONAL ENGINEERING ANALYSIS	Analysis based on theory that is appropriate for the situation, relevant test data if available, and sound engineering judgment.
REDUCED BEAM SECTION	Reduction in cross section over a discrete length that promotes a zone of inelasticity in the member.
REENTRANT	In a cope or weld access hole, a cut at an abrupt change in direction in which the exposed surface is concave.
RELATIVE BRACE	Brace that controls relative movement of two adjacent brace points along length of a beam or column or relative lateral displacement of two stories in a frame.

REQUIRED STRENGTH*	Forces, stresses and deformations acting on the structural component, determined by either structural analysis, for the LRFD or ASD load combinations, as appropriate, or as specified in this Chapter.
RESISTANCE FACTOR, F	Factor that accounts for unavoidable deviations of the nominal strength from the actual strength and for the manner and consequences of failure.
RESTRAINED CONSTRUCTION	Floor and roof assemblies and individual beams in buildings where the surrounding or supporting structure is capable of resisting substantial thermal expansion throughout the range of anticipated elevated temperatures.
REVERSE CURVATURE	See double curvature.
ROOT OF JOINT	Portion of a joint to be welded where the members are closest to each other.
ROTATION CAPACITY	Incremental angular rotation that a given shape can accept prior to excessive load shedding, defined as the ratio of the inelastic rotation attained to the idealized elastic rotation at first yield.
RUPTURE STRENGTH	In a connection, strength limited by tension or shear rupture.
SAFETY FACTOR, Ω	Factor that accounts for deviations of the actual strength from the nominal strength, deviations of the actual load from the nominal load, uncertainties in the analysis that transforms the load into a load effect, and for manner and consequences of failure.
SECOND-ORDER ANALYSIS	Structural analysis in which equilibrium conditions are formulated on the deformed structure; second-order effects (both $P-\Delta$ and $P-\delta$, unless specified otherwise) are included.
SECOND-ORDER EFFECT	Effect of loads acting on the deformed configuration of a structure; includes $P-\Delta$ effect and $P-\delta$ effect.
SEISMIC DESIGN CATEGORY	Classification assigned to a building by the Code based upon its seismic use group and the design spectral response acceleration coefficients.
SEISMIC LOAD RESISTING SYSTEM (SLRS)	Assembly of structural elements in the building that resists seismic loads, including struts, collectors, chords, diaphragms and trusses.

SEISMIC RESPONSE MODIFICATION COEFFICIENT	Factor that reduces seismic load effects to strength level as specified by the Code.
SEISMIC USE GROUP	Classification assigned to a structure based on its use as specified by the Code.
SERVICE LOAD COMBINATION	Load combination under which serviceability limit states are evaluated.
SERVICE LOAD	Load under which serviceability limit states are evaluated.
SERVICEABILITY LIMIT STATE	Limiting condition affecting the ability of a structure to preserve its appearance, maintainability, durability or the comfort of its occupants or function of machinery, under normal usage.
SHEAR BUCKLING	Buckling mode in which a plate element, such as the web of a beam, deforms under pure shear applied in the plane of the plate.
SHEAR CONNECTOR	Headed stud, channel, plate or other shape welded to a steel member and embedded in concrete of a composite member to transmit shear forces at the interface between the two materials.
SHEAR CONNECTOR STRENGTH	Limit state of reaching the strength of a shear connector, as governed by the connector bearing against the concrete in the slab or by the tensile strength of the connector.
SHEAR RUPTURE	Limit state of rupture (fracture) due to shear.
SHEAR WALL	Wall that provides resistance to lateral loads in the plane of the wall and provides stability for the structural system.
SHEAR YIELDING	Yielding that occurs due to shear.
SHEAR YIELDING (PUNCHING)	In an HSS connection, limit state based on out-of-plane shear strength of the chord wall to which branch members are attached.
SHEET STEEL	In a composite floor system, steel used for closure plates or miscellaneous trimming in a formed steel deck.
SHIM	Thin layer of material used to fill a space between faying or bearing surfaces.
SIDESWAY BUCKLING	Limit state of lateral buckling of the tension flange opposite the location of a concentrated compression force.

SIDEWALL CRIPPLING	Limit state of web crippling of the sidewalls of a chord member at a HSS truss connection.
SIDEWALL CRUSHING	Limit state based on bearing strength of chord member sidewall in HSS truss connection.
SIMPLE CONNECTION	Connection that transmits negligible bending moment between connected members.
SINGLE-CONCENTRATED FORCE	Tensile or compressive force applied normal to the flange of a member.
SINGLE CURVATURE	Deformed shape of a beam with no inflection point within the span.
SLENDER-ELEMENT SECTION	Cross section possessing plate components of sufficient slenderness such that local buckling in the elastic range will occur.
SLIP	In a bolted connection, limit state of relative motion of connected parts prior to the attainment of the available strength of the connection.
SLIP-CRITICAL CONNECTION	Bolted connection designed to resist movement by friction on the faying surface of the connection under the clamping forces of the bolts.
SLOT WELD	Weld made in an elongated hole fusing an element to another element.
SNUG-TIGHTENED JOINT	Joint with the connected plies in firm contact as specified in Sec 10.10.
SPECIAL CONCENTRICALLY BRACED FRAME (SCBF)	Diagonally braced frame meeting the requirements of Sec 10.20.13 in which all members of the bracing system are subjected primarily to axial forces.
SPECIAL MOMENT FRAME (SMF)	Moment frame system that meets the requirements of Sec 10.20.9.
SPECIAL PLATE SHEAR WALL (SPSW)	Plate shear wall system that meets the requirements of Sec 10.20.17.
SPECIAL TRUSS MOMENT FRAME (STMF)	Truss moment frame system that meets the requirements of Sec 10.20.12.
SPECIFIED MINIMUM TENSILE STRENGTH	Lower limit of tensile strength specified for a material as defined by ASTM.
SPECIFIED MINIMUM YIELD STRESS	Lower limit of yield stress specified for a material as defined by ASTM.
SPLICE	Connection between two structural elements joined at their ends to form a single, longer element.

STABILITY	Condition reached in the loading of a structural component, frame or structure in which a slight disturbance in the loads or geometry does not produce large displacements.
STIFFENED ELEMENT	Flat compression element with adjoining out-of-plane elements along both edges parallel to the direction of loading.
STIFFENER	Structural element, usually an angle or plate, attached to a member to distribute load, transfer shear or prevent buckling.
STIFFNESS	Resistance to deformation of a member or structure, measured by the ratio of the applied force (or moment) to the corresponding displacement (or rotation).
STRAIN COMPATIBILITY METHOD	Method for determining the stresses in a composite member considering the stress-strain relationships of each material and its location with respect to the neutral axis of the cross section.
STRENGTH LIMIT STATE	Limiting condition affecting the safety of the structure, in which the ultimate load-carrying capacity is reached.
STRESS	Force per unit area caused by axial force, moment, shear or torsion.
STRESS CONCENTRATION	Localized stress considerably higher than average (even in uniformly loaded cross sections of uniform thickness) due to abrupt changes in geometry or localized loading.
STRONG AXIS	Major principal centroidal axis of a cross section.
STRUCTURAL ANALYSIS	Determination of load effects on members and connections based on principles of structural mechanics.
STRUCTURAL COMPONENT	Member, connector, connecting element or assemblage.
STRUCTURAL STEEL	Steel elements as defined in Sec 10.1.3.
STRUCTURAL SYSTEM	An assemblage of load-carrying components that are joined together to provide interaction or interdependence.
STATIC YIELD STRENGTH	Strength of a structural member or connection determined on the basis of testing conducted under slow monotonic loading until failure.

STEEL CORE. AXIAL-FORCE-RESISTING ELEMENT OF BRACES IN BRBF	The steel core contains a yielding segment and connections to transfer its axial force to adjoining elements; it may also contain projections beyond the casing and transition segments between the projections and yielding segment.
T-CONNECTION	HSS connection in which the branch member or connecting element is perpendicular to the main member and in which forces transverse to the main member are primarily equilibrated by shear in the main member.
TENSILE RUPTURE	Limit state of rupture (fracture) due to tension.
TENSILE STRENGTH (OF MATERIAL)	Maximum tensile stress that a material is capable of sustaining as defined by ASTM.
TENSILE STRENGTH (OF MEMBER)	Maximum tension force that a member is capable of sustaining.
TENSILE YIELDING	Yielding that occurs due to tension.
TENSION AND SHEAR RUPTURE	In a bolt, limit state of rupture (fracture) due to simultaneous tension and shear force.
TENSION FIELD ACTION	Behavior of a panel under shear in which diagonal tensile forces develop in the web and compressive forces develop in the transverse stiffeners in a manner similar to a Pratt truss.
TESTED CONNECTION	Connection that complies with the requirements of Appendix O.
THERMALLY CUT	Cut with gas, plasma or laser.
TIE PLATE	Plate element used to join two parallel components of a built-up column, girder or strut rigidly connected to the parallel components and designed to transmit shear between them.
TOE OF FILLET	Junction of a fillet weld face and base metal. Tangent point of a rolled section fillet.
TORSIONAL BRACING	Bracing resisting twist of a beam or column.
TORSIONAL BUCKLING	Buckling mode in which a compression member twists about its shear center axis.
TORSIONAL YIELDING	Yielding that occurs due to torsion.
TRANSVERSE REINFORCEMENT	Steel reinforcement in the form of closed ties or welded wire fabric providing confinement for the concrete surrounding the steel shape core in an encased concrete composite column.

TRANSVERSE STIFFENER	Web stiffener oriented perpendicular to the flanges, attached to the web.
TUBING	See HSS.
TURN-OF-NUT METHOD	Procedure whereby the specified pretension in high-strength bolts is controlled by rotating the fastener component a predetermined amount after the bolt has been snug tightened.
UNBRACED LENGTH	Distance between braced points of a member, measured between the centers of gravity of the bracing members.
UNEVEN LOAD DISTRIBUTION	In an HSS connection, condition in which the load is not distributed through the cross section of connected elements in a manner that can be readily determined.
UNFRAMED END	The end of a member not restrained against rotation by stiffeners or connection elements.
UNRESTRAINED CONSTRUCTION	Floor and roof assemblies and individual beams in buildings that are assumed to be free to rotate and expand throughout the range of anticipated elevated temperatures.
UNSTIFFENED ELEMENT	Flat compression element with an adjoining out-of-plane element along one edge parallel to the direction of loading.
V-BRACED FRAME	Concentrically braced frame (SCBF, OCBF or BRBF) in which a pair of diagonal braces located either above or below a beam is connected to a single point within the clear beam span. Where the diagonal braces are below the beam, the system is also referred to as an inverted-V-braced frame.
VARIABLE LOAD	Load not classified as permanent load.
VERTICAL BRACING SYSTEM	System of shear walls, braced frames or both, extending through one or more floors of a building.
WEAK AXIS	Minor principal centroidal axis of a cross section.
WEATHERING STEEL	High-strength, low-alloy steel that, with suitable precautions, can be used in normal atmospheric exposures (not marine) without protective paint coating.
WEB BUCKLING	Limit state of lateral instability of a web.

WEB COMPRESSION BUCKLING	Limit state of out-of-plane compression buckling of the web due to a concentrated compression force.
WEB SIDESWAY BUCKLING	Limit state of lateral buckling of the tension flange opposite the location of a concentrated compression force.
WELD METAL	Portion of a fusion weld that has been completely melted during welding. Weld metal has elements of filler metal and base metal melted in the weld thermal cycle.
WELD ROOT	See root of joint.
X-BRACED FRAME	Concentrically braced frame (OCBF or SCBF) in which a pair of diagonal braces crosses near the mid-length of the braces.
Y-BRACED FRAME	Eccentrically braced frame (EBF) in which the stem of the Y is Link the EBF system.
Y-CONNECTION	HSS connection in which the branch member or connecting element is not perpendicular to the main member and in which forces transverse to the main member are primarily equilibrated by shear in the main member.
YIELD MOMENT	In a member subjected to bending, the moment at which the extreme outer fiber first attains the yield stress.
YIELD POINT	First stress in a material at which an increase in strain occurs without an increase in stress as defined by ASTM.
YIELD STRENGTH	Stress at which a material exhibits a specified limiting deviation from the proportionality of stress to strain as defined by ASTM.
YIELD STRESS	Generic term to denote either yield point or yield strength, as appropriate for the material.
YIELDING	Limit state of inelastic deformation that occurs after the yield stress is reached.
YIELDING (PLASTIC MOMENT)	Yielding throughout the cross section of a member as the bending moment reaches the plastic moment.
YIELDING (YIELD MOMENT)	Yielding at the extreme fiber on the cross section of a member when the bending moment reaches the yield moment.

Notes:

- (1) Terms designated with * are usually qualified by the type of load effect, for example, nominal tensile strength, available compressive strength, design flexural strength.
- (2) Terms designated with ** are usually qualified by the type of component, for example, web local buckling, flange local bending.

10.1.2.3 Referenced specifications, codes and standards

The following specifications, codes and standards are referenced in this Specification.

ACI International (ACI)

ACI 318	Building Code Requirements for Structural Concrete and Commentary
ACI 318M	Metric Building Code Requirements for Structural Concrete and Commentary

American Institute of Steel Construction, Inc. (AISC)

AISC 303	Code of Standard Practice for Steel Buildings and Bridges
ANSI/AISC 341	Seismic Provisions for Structural Steel Buildings
ANSI/AISC N690	Specification for the Design, Fabrication and Erection of Steel Safety-Related Structures for Nuclear Facilities, including Supplement No. 2
ANSI/AISC N690L	Load and Resistance Factor Design Specification for Steel Safety-Related Structures for Nuclear Facilities

American Society of Civil Engineers (ASCE)

SEI/ASCE 7	Minimum Design Loads for Buildings and Other Structures
ASCE/SFPE 29	Standard Calculation Methods for Structural Fire Protection

American Society of Mechanical Engineers (ASME)

ASME B18.2.6	Fasteners for Use in Structural Applications
ASME B46.1	Surface Texture, Surface Roughness, Waviness, and Lay

American Iron and Steel Institute (AISI)

North American Specification for the Design of Cold Formed Steel Structural Members (AISI/COS/NASPEC 2001).

Code of Standard Practice for Cold-Formed Steel Structural Framing

ASTM International (ASTM)

ASTM A6/A6M-04a	Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling
ASTM A36/A36M	Standard Specification for Carbon Structural Steel
ASTM A53/A53M	Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A193/A193M	Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service
ASTM A194/A194M	Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High Pressure or High-Temperature Service, or Both
ASTM A216/A216M	Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High Temperature Service
ASTM A242/A242M	Standard Specification for High-Strength Low-Alloy Structural Steel
ASTM A283/A283M	Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates
ASTM A307	Standard Specification for Carbon Steel Bolts and Studs, 60,000 psi Tensile Strength
ASTM A325	Standard Specification for Structural Bolts, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength
ASTM A325M	Standard Specification for High-Strength Bolts for Structural Steel Joints (Metric)
ASTM A354	Standard Specification for Quenched and Tempered Alloy Steel Bolts, Studs, and Other Externally Threaded Fasteners
ASTM A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A449	Standard Specification for Quenched and Tempered Steel Bolts and Studs
ASTM A490	Standard Specification for Heat-Treated Steel Structural Bolts, 150 ksi Minimum Tensile Strength
ASTM A490M	Standard Specification for High-Strength Steel Bolts, Classes 10.9 and 10.9.3, for Structural Steel Joints (Metric)
ASTM A500	Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes
ASTM A501	Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing
ASTM A502	Standard Specification for Steel Structural Rivets

ASTM A514/A514M	Standard Specification for High-Yield Strength, Quenched and Tempered Alloy Steel Plate, Suitable for Welding
ASTM A529/A529M	Standard Specification for High-Strength Carbon-Manganese Steel of Structural Quality
ASTM A563	Standard Specification for Carbon and Alloy Steel Nuts
ASTM A563M	Standard Specification for Carbon and Alloy Steel Nuts [Metric]
ASTM A568/A568M	Standard Specification for Steel, Sheet, Carbon, and High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, General Requirements for
ASTM A572/A572M	Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel.
ASTM A588/A588M	Standard Specification for High-Strength Low-Alloy Structural Steel with 345 MPa Minimum Yield Point to 100 mm Thick
ASTM A606	Standard Specification for Steel, Sheet and Strip, High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, with Improved Atmospheric Corrosion Resistance
ASTM A618/A618M	Standard Specification for Hot-Formed Welded and Seamless High-Strength Low-Alloy Structural Tubing
ASTM A673/A673M	Standard Specification for Sampling Procedure for Impact Testing of Structural Steel
ASTM A668/A668M	Standard Specification for Steel Forgings, Carbon and Alloy, for General Industrial Use
ASTM A709/A709M	Standard Specification for Carbon and High-Strength Low-Alloy Structural Steel Shapes, Plates, and Bars and Quenched-and-Tempered Alloy Structural Steel Plates for Bridges
ASTM A751	Standard Test Methods, Practices, and Terminology for Chemical Analysis of Steel Products
ASTM A847	Standard Specification for Cold-Formed Welded and Seamless High-Strength, Low-Alloy Structural Tubing with Improved Atmospheric Corrosion Resistance
ASTM A852/A852M	Standard Specification for Quenched and Tempered Low-Alloy Structural Steel Plate with 485 MPa Minimum Yield Strength to 100 mm Thick
ASTM A913/A913M	Standard Specification for High-Strength Low-Alloy Steel Shapes of Structural Quality, Produced by Quenching and Self-Tempering Process (QST)
ASTM A992/A992M	Standard Specification for Structural Steel Shapes

ASTM A1011/A1011M	Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy and High-Strength Low-Alloy with Improved Formability
ASTM C33	Standard Specification for Concrete Aggregates
ASTM C330	Standard Specification for Lightweight Aggregates for Structural Concrete
ASTM E119	Standard Test Methods for Fire Tests of Building Construction and Materials
ASTM E709	Standard Guide for Magnetic Particle Examination
ASTM F436	Standard Specification for Hardened Steel Washers
ASTM F959	Standard Specification for Compressible-Washer-Type Direct Tension Indicators for Use with Structural Fasteners
ASTM F1554	Standard Specification for Anchor Bolts, Steel, 36, 55, and 105 ksi Yield Strength
ASTM F1852	Standard Specification for “Twist-Off” Type Tension Control Structural Bolt/Nut/Washer Assemblies, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength

American Welding Society (AWS)

AWS D1.1/D1.1M	Structural Welding Code–Steel
AWS A5.1	Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding
AWS A5.5	Specification for Low-Alloy Steel Electrodes for Shielded Metal Arc Welding
AWS A5.17/A5.17M	Specification for Carbon Steel Electrodes and Fluxes for Submerged Arc Welding
AWS A5.18	Specification for Carbon Steel Electrodes and Rods for Gas Shielded Arc Welding
AWS A5.20	Specification for Carbon Steel Electrodes for Flux Cored Arc Welding
AWS A5.23/A5.23M	Specification for Low-Alloy Steel Electrodes and Fluxes for Submerged Arc Welding
AWS A5.25/A5.25M	Specification for Carbon and Low-Alloy Steel Electrodes and Fluxes for Electroslag Welding
AWS A5.26/A5.26M	Specification for Carbon and Low-Alloy Steel Electrodes for Electrode Gas Welding
AWS A5.28	Specification for Low-Alloy Steel Electrodes and Rods for Gas Shielded Arc Welding
AWS A5.29	Specification for Low-Alloy Steel Electrodes for Flux Cored Arc Welding

Research Council on Structural Connections (RCSC)

Specification for Structural Joints Using ASTM A325 or A490 Bolts, 2004

Bangladesh Standards and Testing Institute (BSTI)

BDS 1031	Mild Steel (MS) Pipe and Galvanized Iron (GI) Pipe
BDS ISO 6935Parts 1 and 2	Steel for Reinforcement of Concrete.
BDS 1122	GP Sheet with Corrugation
BDS EN 197-1Cement Part 1	Composition. Specifications and Conformity Criteria for Common Cement
BDS 208	Common Building Clay Bricks
BDS ISO 13006	Ceramic Tiles
BDS 1825	Aluminium and Aluminium Alloys - Extruded Rod/Bar Tube and Profiles - Tolerances on Dimensions and Form.

10.1.3 Material**10.1.3.1 Structural steel materials**

- (a) Regular Structural Steel: Material test reports from an acceptable testing laboratory shall constitute sufficient evidence of conformity with one of the above listed ASTM standards. For hot-rolled structural shapes, plates, and bars, such tests shall be made in accordance with ASTM A6/A6M; for sheets, such tests shall be made in accordance with ASTM A568/A568M; for tubing and pipe, such tests shall be made in accordance with the requirements of the applicable ASTM standards listed above for those product forms. If requested, the fabricator shall provide an affidavit stating that the structural steel furnished meets the requirements of the grade specified.

Structural steel material conforming to one of the following specifications is approved for use under this Specification:

(i) Hot-rolled structural shapes

ASTM A36/A36M, ASTM A529/A529M, ASTM A572/A572M, ASTM A588/A588M, ASTM A709/A709M, ASTM A913/A913M, ASTM A992/ A992M

(ii) Structural tubing

ASTM A500, ASTM A501, ASTM A618, ASTM A847, BDS 1031

(iii) Pipe

ASTM A53/A53M, Gr. B, BDS 1031

(iv) Plates

ASTM A36/A36M, ASTM A242/A242M, ASTM A283/A283M, ASTM A514/A514M, ASTM A529/A529M, ASTM A572/A572M, ASTM A588/A588M, ASTM A709/A709M, ASTM A852/A852M, ASTM A1011/A1011M, BDS 1122

(v) Bars

ASTM A36/A36M, ASTM A529/A529M, ASTM A572/A572M, ASTM A709/A709M, BDS ISO 6935-1, BDS ISO 6935-2

(vi) Sheets

ASTM A606, A1011/A1011M SS, HSLAS, AND HSLAS-F, BDS 1122

- (b) Unidentified Steel: Unidentified steel free of injurious defects is permitted to be used for unimportant members or details, where the precise physical properties and weldability of the steel would not affect the strength of the structure.
- (c) Rolled Heavy Shapes: ASTM A6/A6M hot-rolled shapes with a flange thickness exceeding 50 mm, used as members subject to primary (computed) tensile forces due to tension or flexure and spliced using complete-joint-penetration groove welds that fuse through the thickness of the member, shall be specified as follows. The contract documents shall require that such shapes be supplied with Charpy V-Notch (CVN) impact test results in accordance with ASTM A6/A6M, Supplementary Requirement S30, Charpy V-Notch Impact Test for Structural Shapes – Alternate Core Location. The impact test shall meet a minimum average value of 27 Joules absorbed energy at +21⁰ C.

The above requirements do not apply if the splices and connections are made by bolting. The above requirements do not apply to hot-rolled shapes with a flange thickness exceeding 50 mm that have shapes with flange or web elements less than 50 mm thick welded with complete-joint-penetration groove welds to the face of the shapes with thicker elements.

- (d) Built-Up Heavy Shapes: Built-up cross-sections consisting of plates with a thickness exceeding 50 mm, used as members subject to primary (computed) tensile forces due to tension or flexure and spliced or connected to other members using complete-joint-penetration groove welds that fuse through the thickness of the plates, shall be specified as follows. The contract documents shall require that the steel be supplied with Charpy V-Notch impact test results in accordance with ASTM A6/A6M, Supplementary Requirement S5, Charpy V-Notch Impact Test. The impact test shall be conducted in accordance with ASTM A673/A673M, Frequency P, and shall meet a minimum average value of 27 Joules absorbed energy at +21⁰ C.

The above requirements also apply to built-up cross-sections consisting of plates exceeding 50 mm that are welded with complete-joint-penetration groove welds to the face of other sections.

- (e) Cold Form Sections: Specifications for cold form shapes regarding their use as structural members is not covered in Sec 10. For such type of structural steel, AISI standard (AISI/COS/NASPEC 2001) or equivalent may be followed.

10.1.3.2 Steel castings and forgings

Cast steel shall conform to ASTM A216/A216M, Gr. WCB with Supplementary Requirement S11. Steel forgings shall conform to ASTM A668/A668M. Test reports produced in accordance with the above reference standards shall constitute sufficient evidence of conformity with such standards.

10.1.3.3 Bolts, washers and nuts

Bolt, washer, and nut material conforming to one of the following ASTM specifications is approved for use under this Specification:

- (1) Bolts: ASTM A307, ASTM A325, ASTM A325M, ASTM A449, ASTM A490, ASTM A490M, ASTM F1852
- (2) Nuts: ASTM A194/A194M, ASTM A563, ASTM A563M
- (3) Washers: ASTM F436, ASTM F436M
- (4) Compressible-Washer-Type Direct Tension Indicators: ASTM F959, ASTM F959M

10.1.3.4 Anchor rods and threaded rods

Anchor rod and threaded rod material conforming to one of the following ASTM specifications is approved for use under this Specification:

ASTM A36/A36M, ASTM A193/A193M, ASTM A354, ASTM A449, ASTM A572/A572M, ASTM A588/A588M, ASTM F1554

A449 material is acceptable for high-strength anchor rods and threaded rods of any diameter.

Threads on anchor rods and threaded rods shall conform to the Unified Standard Series of ASME B18.2.6 and shall have Class 2A tolerances.

10.1.3.5 Filler metal and flux for welding

Filler metals and fluxes shall conform to one of the following specifications of the American Welding Society: AWS A5.1, AWS A5.5, AWS A5.17/A5.17M, AWS A5.18, AWS A5.20, AWS A5.23/A5.23M, AWS A5.25/A5.25M, AWS A5.26/A5.26M, AWS A5.28, AWS A5.29, AWS A5.32/A5.32M

10.1.3.6 Stud shear connectors

Steel stud shear connectors shall conform to the requirements of Structural Welding Code—Steel, AWS D1.1.

10.1.4 Structural Design Drawings and Specifications

The design drawings and specifications shall meet the requirements specified in this specification (Sections 10.1 to 10.20) and shall be prepared and presented in an internationally approved standard in accordance with the provisions of Sec 10.13, except for deviations specifically identified in the design drawings and/or specifications and approved by an appropriate authority.

10.2 General Design Requirements

The general requirements for the analysis and design of steel buildings and structures that are applicable to all Sections of Chapter 10 Part 6 are given in this Section.

10.2.1 General Provisions

The design of members and connections shall be consistent with the intended behavior of the framing system and the assumptions made in the structural analysis. Unless restricted by the Code, lateral load resistance and stability may be provided by any combination of members and connections.

10.2.2 Loads and Load Combinations

The loads and load combinations shall be as stipulated in Chapter 2 Part 6 of this Code. For design purposes, the nominal loads shall be taken as the loads stipulated in the Chapter 2

10.2.3 Design Basis

Designs shall be made according to the provisions for Load and Resistance Factor Design (LRFD) or to the provisions for Allowable Strength Design (ASD).

10.2.3.1 Required strength

The required strength of structural members and connections shall be determined by structural analysis for the appropriate load combinations as stipulated in Chapter 2 of Part 6. Design by elastic, inelastic or plastic analysis is permitted. Provisions for inelastic and plastic analysis are as stipulated in Sec 10.15, Inelastic Analysis and Design. The provisions for moment redistribution in continuous beams in Sec 10.15.3 are permitted for elastic analysis only.

10.2.3.2 Limit states

Design shall be based on the principle that no applicable strength or serviceability limit state shall be exceeded when the structure is subjected to all appropriate load combinations.

10.2.3.3 Design for strength using load and resistance factor design (LRFD)

Design according to the provisions for Load and Resistance Factor Design (LRFD) satisfies the requirements of this Specification when the design strength of each structural component equals or exceeds the required strength determined on the basis of the LRFD load combinations as specified in Chapter 2 of Part 6. All provisions of this Specification, except for those in Sec 10.2.3.4, shall apply.

Design shall be performed in accordance with Eq.6.10.1:

$$R_u \leq \phi R_n \quad (6.10.1)$$

Where,

R_u = required strength (LRFD)

R_n = nominal strength, specified in Sections 10.2 to 10.20

ϕ = resistance factor, specified in Sections 10.2 to 10.20

ϕR_n = design strength

10.2.3.4 Design for strength using allowable strength design (ASD)

Design according to the provisions for Allowable Strength Design (ASD) satisfies the requirements of this Specification when the allowable strength of each structural component equals or exceeds the required strength determined on the basis of the ASD load combinations as specified in Chapter 2 of Part 6. All provisions of this Specification, except those of Sec 10.2.3.3, shall apply.

Design shall be performed in accordance with Eq.6.10.2:

$$R_a \leq R_n / \Omega \quad (6.10.2)$$

Where,

R_a = required strength (ASD)

R_n = nominal strength, specified in Sections 10.2 to 10.20

Ω = safety factor, specified in Sections 10.2 to 10.20

R_n/Ω = allowable strength

10.2.3.5 Design for stability

Stability of the structure and its elements shall be determined in accordance with Sec 10.3.

10.2.3.6 Design for connection

Connection elements shall be designed in accordance with the provisions of Sections 10.10 and 10.11. The forces and deformations used in design shall be consistent with the intended performance of the connection and the assumptions used in the structural analysis.

10.2.3.6.1 Simple connection

A simple connection transmits a negligible moment across the connection. In the analysis of the structure, simple connections may be assumed to allow unrestrained relative rotation between the framing elements being connected. A simple connection shall have sufficient rotation capacity to accommodate the required rotation determined by the analysis of the structure. Inelastic rotation of the connection is permitted.

10.2.3.6.2 Moment connection

A moment connection transmits moment across the connection. Two types of moment connections, FR and PR, are permitted, as specified below.

- (a) Fully-Restrained (FR) Moment Connections: A fully-restrained (FR) moment connection transfers moment with a negligible relative rotation between the connected members. In the analysis of the structure, the connection may be assumed to allow no relative rotation. An FR connection shall have sufficient strength and stiffness to maintain the angle between the connected members at the strength limit states.

- (b) Partially-Restrained (PR) Moment Connections: Partially-restrained (PR) moment connections transfer moments, but the relative rotation between connected members is not negligible. In the analysis of the structure, the force-deformation response characteristics of the connection shall be included. The response characteristics of a PR connection shall be documented in the technical literature or established by analytical or experimental means. The component elements of a PR connection shall have sufficient strength, stiffness, and deformation capacity at the strength limit states.

10.2.3.7 Design for serviceability

The overall structure and the individual members, connections and connectors shall be checked for serviceability. Performance requirements for serviceability design are given in Sec 10.12.

10.2.3.8 Design for ponding

The roof system shall be investigated through structural analysis to assure adequate strength and stability under ponding conditions, unless the roof surface is provided with a slope of 20 mm per meter or greater toward points of free drainage or an adequate system of drainage is provided to prevent the accumulation of water. Methods of checking ponding are given in Sec 10.16.

10.2.3.9 Design for fatigue

Fatigue shall be considered in accordance with Sec 10.17, Design for Fatigue, for members and their connections subject to repeated loading. Fatigue need not be considered for seismic effects or for the effects of wind loading on normal building lateral load resisting systems and building enclosure components.

10.2.3.10 Design for fire conditions

Two methods of design for fire conditions are provided in Sec 10.18, Structural Design for Fire Conditions: Qualification Testing and Engineering Analysis. Compliance with the fire protection requirements in Part 4 of this Code shall be required in addition to the requirements of Sec 10.18.

10.2.3.11 Design for corrosion effects

Where corrosion may impair the strength or serviceability of a structure, structural components shall be designed to tolerate corrosion or shall be protected against corrosion.

10.2.3.12 Design wall thickness for HSS

The design wall thickness, t , shall be used in calculations involving the wall thickness of hollow structural sections (HSS). The design wall thickness, t , shall be taken equal to 0.93 times the nominal wall thickness for electric-resistance-welded (ERW) HSS and equal to the nominal thickness for submerged-arc-welded (SAW) HSS.

10.2.3.13 Gross and net area determination

10.2.3.13.1 Gross area

The gross area, A_g , of a member is the total cross-sectional area.

10.2.3.13.2 Net area

The net area, A_n of a member is the sum of the products of the thickness and the net width of each element computed as follows:

In computing net area for tension and shear, the width of a bolt hole shall be taken as 2 mm greater than the nominal dimension of the hole.

For a chain of holes extending across a part in any diagonal or zigzag line, the net width of the part shall be obtained by deducting from the gross width the sum of the diameters or slot dimensions as provided in Sec 10.10.3.2, of all holes in the chain, and adding, for each gage space in the chain, the quantity $s^2/(4g)$.

Where,

s = longitudinal center-to-center spacing (pitch) of any two consecutive holes, mm.

g = transverse center-to-center spacing (gage) between fastener gage lines, mm.

For angles, the gage for holes in opposite adjacent legs shall be the sum of the gages from the back of the angles less the thickness.

For slotted HSS welded to a gusset plate, the net area, A_n , is the gross area the product of the thickness and the total width of material that is removed to form the slot.

In determining net area across plug or slot welds, the weld metal shall not be considered as adding to net area.

10.2.4 Classification of Sections for Local Buckling

Sections are classified as compact, noncompact, or slender-element sections. For a section to qualify as compact its flanges must be continuously connected to the web or webs and the width-thickness ratios of its compression elements must not exceed the limiting width-thickness ratios λ_p from Table 6.10.1. If the width- thickness ratio of one or more compression elements exceeds λ_p , but does not exceed λ_r from Table 6.10.1, the section is noncompact. If the width-thickness ratio of any element exceeds λ_r , the section is referred to as a slender-element section.

10.2.4.1 Unstiffened elements

For unstiffened elements supported along only one edge parallel to the direction of the compression force, the width shall be taken as follows:

- For flanges of I-shaped members and tees, the width b is one-half the full-flange width, b_f .
- For legs of angles and flanges of channels and zees, the width b is the full nominal dimension.

- (c) For plates, the width b is the distance from the free edge to the first row of fasteners or line of welds.
- (d) For stems of tees, d is taken as the full nominal depth of the section.

10.2.4.2 Stiffened elements

For stiffened elements supported along two edges parallel to the direction of the compression force, the width shall be taken as follows:

- (a) For webs of rolled or formed sections, h is the clear distance between flanges less the fillet or corner radius at each flange; h_c is twice the distance from the centroid to the inside face of the compression flange less the fillet or corner radius.
- (b) For webs of built-up sections, h is the distance between adjacent lines of fasteners or the clear distance between flanges when welds are used, and h_c is twice the distance from the centroid to the nearest line of fasteners at the compression flange or the inside face of the compression flange when welds are used; h_p is twice the distance from the plastic neutral axis to the nearest line of fasteners at the compression flange or the inside face of the compression flange when welds are used.
- (c) For flange or diaphragm plates in built-up sections, the width b is the distance between adjacent lines of fasteners or lines of welds.
- (d) For flanges of rectangular hollow structural sections (HSS), the width b is the clear distance between webs less the inside corner radius on each side. For webs of rectangular HSS, h is the clear distance between the flanges less the inside corner radius on each side. If the corner radius is not known, b and h shall be taken as the corresponding outside dimension minus three times the thickness. The thickness, t , shall be taken as the design wall thickness, per Sec 10.2.3.12.
- (e) For tapered flanges of rolled sections, the thickness is the nominal value halfway between the free edge and the corresponding face of the web.

10.2.5 Fabrication, Erection and Quality

Shop drawings, fabrication, shop painting, erection, and quality control shall meet the requirements stipulated in Sec 10.13, Fabrication, Erection, and Quality Control.

10.3 Stability Analysis and Design

This Section addresses general requirements for the stability analysis and design of members and frames of steel buildings and structures.

10.3.1 Stability Design Requirements

Stability shall be provided for the structure as a whole and for each of its elements. Any method that considers the influence of second-order effects (including P- Δ and P- δ effects),

flexural, shear and axial deformations, geometric imperfections, and member stiffness reduction due to residual stresses on the stability of the structure and its elements is permitted. The methods prescribed in this Section and Sec 10.14: Direct Analysis Method, satisfy these requirements. All component and connection deformations that contribute to the lateral displacements shall be considered in the stability analysis. In structures designed by elastic analysis, individual member stability and stability of the structure as a whole are provided jointly by:

- Calculation of the required strengths for members, connections and other elements using one of the methods specified in Sec 10.3.2.2, and
- Satisfaction of the member and connection design requirements in this specification based upon those required strengths.

In structures designed by inelastic analysis, the provisions of Sec 10.15 shall be satisfied.

10.3.1.1 Member stability design requirements

Individual member stability is provided by satisfying the provisions of Sections 10.5 to 10.11. Where elements are designed to function as braces to define the unbraced length of columns and beams, the bracing system shall have sufficient stiffness and strength to control member movement at the braced points. Methods of satisfying this requirement are provided in Sec 10.19.

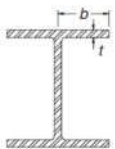
10.3.1.2 System stability design requirements

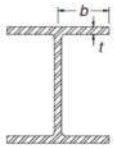
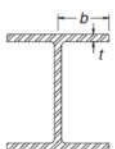
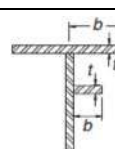
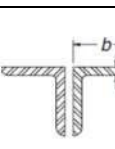
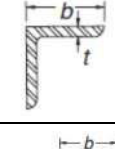
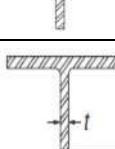
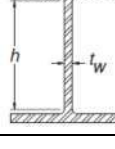
Lateral stability shall be provided by moment frames, braced frames, shear walls, and/or other equivalent lateral load resisting systems. The overturning effects of drift and the destabilizing influence of gravity loads shall be considered. Force transfer and load sharing between elements of the framing systems shall be considered. Braced-frame and shear-wall systems, moment frames, gravity framing systems, and combined systems shall satisfy the following specific requirements:

10.3.1.2.1 Braced-frame and shear-wall systems

In structures where lateral stability is provided solely by diagonal bracing, shear walls, or equivalent means, the effective length factor, K , for compression members shall be taken as 1.0, unless structural analysis indicates a smaller value appropriate. In braced-frame systems, it is permitted to design columns, beams, and diagonal members as a vertically cantilevered, simply connected truss.

Table 6.10.1: Limiting Width-Thickness Ratios for Compression Elements

	Case	Description of Element	Width Thickness Ratio	Limiting Width Thickness Ratio		Example
				λ_p (Compact)	λ_r (Noncompact)	
Unstiffened Element	1	Flexure in flanges of rolled I-shaped sections and channels	b/t	$0.38 \sqrt{E/F_y}$	$1.0 \sqrt{E/F_y}$	

	Case	Description of Element	Width Thick- ness Ratio	Limiting Width Thickness Ratio		Example
				λ_p (Compact)	λ_r (Noncompact)	
Unstiffened Elements	2	Flexure in flanges of doubly and singly symmetric I-shaped built-up sections	b/t	$0.38 \sqrt{E/F_y}$	$0.95 \sqrt{k_c E/F_y}^{[a]}$], [b]	
	3	Uniform compression in flanges of rolled I-shaped sections, plates projecting from rolled I-shaped sections; Outstanding legs of pairs of angles in continuous contact and flanges of channels	b/t	NA	$0.56 \sqrt{E/F_y}$	
	4	Uniform compression in flanges of built-up I-shaped sections and plates or angle legs projecting from built-up I-shaped section	b/t	NA	$0.64 \sqrt{k_c E/F_y}^{[a]}$]	
	5	Uniform compression in legs of single angles, legs of double angles with separators, and all other unstiffened elements	b/t	NA	$0.45 \sqrt{E/F_y}$	
	6	Flexure in legs of single angles	b/t	$0.54 \sqrt{E/F_y}$	$0.91 \sqrt{E/F_y}$	
	7	Flexure in flanges of tees	b/t	$0.38 \sqrt{E/F_y}$	$1.0 \sqrt{E/F_y}$	
Stiffened Elements	8	Uniform compression in stems of tees	d/t	NA	$0.75 \sqrt{E/F_y}$	
	9	Flexure in webs of doubly symmetric I-shaped sections	h/t_w	$3.76 \sqrt{E/F_y}$	$5.70 \sqrt{E/F_y}$	

10.3.1.2.2 Moment frame systems

In frames where lateral stability is provided by the flexural stiffness of connected beams and columns, the effective length factor K or elastic critical buckling stress, F_e , for columns and beam-columns shall be determined as specified in Sec 10.3.2.

10.3.1.2.3 Gravity framing systems

Columns in gravity framing systems shall be designed based on their actual length ($K = 1.0$) unless analysis shows that a smaller value may be used. The lateral stability of gravity framing systems shall be provided by moment frames, braced frames, shear walls, and/or other equivalent lateral load resisting systems. $P-\Delta$ effects due to load on the gravity columns shall be transferred to the lateral load resisting systems and shall be considered in the calculation of the required strengths of the lateral load resisting systems.

10.3.1.2.4 Combined systems

The analysis and design of members, connections and other elements in combined systems of moment frames, braced frames, and/or shearwalls and gravity frames shall meet the requirements of the respective systems.

10.3.2 Calculation of Required Strengths

Except as permitted in Sec 10.3.2.2.2, required strengths shall be determined using a second-order analysis as specified in Sec 10.3.2.1. Design by either second-order or first-order analysis shall meet the requirements specified in Sec 10.3.2.2.

10.3.2.1 Methods of second-order analysis

Second-order analysis shall conform to the requirements in this Section.

10.3.2.1.1 General second-order elastic analysis

Any second-order elastic analysis method that considers both $P-\Delta$ and $P-\delta$ effects may be used. The Amplified First-Order Elastic Analysis Method defined in Sec 10.3.2.1.2 is an accepted method for second-order elastic analysis of braced, moment, and combined framing systems.

10.3.2.1.2 Second-order analysis by amplified first-order elastic analysis

The following is an approximate second-order analysis procedure for calculating the required flexural and axial strengths in members of lateral load resisting systems. The required second-order flexural strength, M_r , and axial strength, P_r , shall be determined as follows:

$$M_r = B_1 M_{nt} + B_2 M_{lt} \quad (6.10.3a)$$

$$P_r = P_{nt} + B_2 P_{lt} \quad (6.10.3b)$$

Where,

$$B_1 = \frac{C_m}{1 - \alpha P_r / P_{e1}} \geq 1 \quad (6.10.4)$$

For members subjected to axial compression, B_1 may be calculated based on first-order estimate $P_r = P_{nt} + P_{lt}$.

For members in which $B_1 \leq 1.05$, it is conservative to amplify the sum of the non-sway and sway moments (as obtained, for instance, by a first-order elastic analysis) by the B_2 amplifier, in other words, $M_r = B_2 (M_{nt} + M_{lt})$.

$$B_2 = \frac{1}{1 - \frac{\alpha \sum P_{nt}}{\sum P_c^2}} \geq 1 \quad (6.10.5)$$

And, $\alpha = 1.00$ (LRFD) $\alpha = 1.60$ (ASD)

M_r = required second-order flexural strength using LRFD or ASD load combinations, N-mm

M_{nt} = first-order moment using LRFD or ASD load combinations, assuming there is no lateral translation of the frame, N-mm

M_{lt} = first-order moment using LRFD or ASD load combinations caused by lateral translation of the frame only, N-mm

P_r = required second-order axial strength using LRFD or ASD load combinations, N

P_{nt} = first-order axial force using LRFD or ASD load combinations, assuming there is no lateral translation of the frame, N

$\sum P_{nt}$ = total vertical load supported by the story using LRFD or ASD load combinations, including gravity column loads, N

P_{lt} = first-order axial force using LRFD or ASD load combinations caused by lateral translation of the frame only, N

C_m = a coefficient assuming no lateral translation of the frame whose value shall be taken as follows:

For beam-columns not subject to transverse loading between supports in the plane of bending,

$$C_m = 0.6 - 0.4(M_1/M_2) \quad (6.10.6)$$

Where, M_1 and M_2 , calculated from a first-order analysis, are the smaller and larger moments, respectively, at the ends of that portion of the member unbraced in the plane of bending under consideration. M_1 / M_2 is positive when the member is bent in reverse curvature, negative when bent in single curvature.

For beam-columns subjected to transverse loading between supports, the value of C_m shall be determined either by analysis or conservatively taken as 1.0 for all cases.

P_{e1} = elastic critical buckling resistance of the member in the plane of bending, calculated based on the assumption of zero sidesway, N

$$P_{e1} = \frac{\pi^2 EI}{(K_1 L)^2} \quad (6.10.7)$$

$\sum P_{e2}$ = elastic critical buckling resistance for the story determined by sideway buckling analysis, N

For moment frames, where sidesway buckling effective length factors K_2 are determined for the columns, it is permitted to calculate the elastic story sidesway buckling resistance as

$$\sum P_{e2} = \sum \frac{\pi^2 EI}{(K_2 L)^2} \quad (6.10.8a)$$

For all types of lateral load resisting systems, it is permitted to use

$$\sum P_{e2} = R_M \frac{\sum HL}{\Delta_H} \quad (6.10.8b)$$

Where,

E = modulus of elasticity of steel = 200 000 MPa

R_m = 1.0 for braced-frame systems;

= 0.85 for moment-frame and combined systems, unless a larger value is justified by analysis

I = moment of inertia in the plane of bending, mm⁴

L = story height, mm

K_1 = effective length factor in the plane of bending, calculated based on the assumption of no lateral translation, set equal to 1.0 unless analysis indicates that a smaller value may be used

K_2 = effective length factor in the plane of bending, calculated based on a sideway buckling analysis

Δ_H = first-order interstory drift due to lateral forces, mm. Where Δ_H varies over the plan area of the structure, Δ_H shall be the average drift weighted in proportion to vertical load or, alternatively, the maximum drift.

$\sum H$ = story shear produced by the lateral forces used to compute Δ_H , N

10.3.2.2 Design requirements

These requirements apply to all types of braced, moment, and combined framing systems. Where the ratio of second-order drift to first-order drift is equal to or less than 1.5, the required strengths of members, connections and other elements shall be determined by one of the methods specified in Sections 10.3.2.2.1 or 10.3.2.2.2, or by the Direct Analysis Method of Sec 10.14. Where the ratio of second-order drift to first-order drift is greater than 1.5, the required strengths shall be determined by the Direct Analysis Method of Sec 10.14.

For the methods specified in Sections 10.3.2.2.1 or 10.3.2.2.2:

Analyses shall be conducted according to the design and loading requirements specified in either Section 10.2.3.3 (LRFD) or Section 10.2.3.4 (ASD).

The structure shall be analyzed using the nominal geometry and the nominal elastic stiffness for all elements.

10.3.2.2.1 Design by second-order analysis

Where required strengths are determined by a second-order analysis:

- (a) The provisions of Sec 10.3.2.1 shall be satisfied.
- (b) For design by ASD, analyses shall be carried out under 1.6 times the *ASD* load combinations and the results shall be divided by 1.6 to obtain the required strengths.
- (c) All gravity-only load combinations shall include a minimum lateral load applied at each level of the structure of $0.002Y_i$, where Y_i is the design gravity load applied at level i . This minimum lateral load shall be considered independently in two orthogonal directions.
- (d) Where the ratio of second-order drift to first-order drift is less than or equal to 1.1, members are permitted to be designed using $K = 1.0$. Otherwise, columns and beam-columns in moment frames shall be designed using a K factor or column buckling stress, F_e , determined from a sideway buckling analysis of the structure. Stiffness reduction adjustment due to column inelasticity is permitted in the determination of the K factor. For braced frames, K for compression members shall be taken as 1.0, unless structural analysis indicates a smaller value may be used.

10.3.2.2.2 Design by first-order analysis

Required strengths are permitted to be determined by a first-order analysis, with all members designed using $K = 1.0$, provided that

- (a) The required compressive strengths of all members whose flexural stiffnesses are considered to contribute to the lateral stability of the structure satisfy the following limitation:

$$\alpha P_r \leq 0.5 P_y \quad (6.10.9)$$

Where,

$$\alpha = 1.0 (\text{LRFD}) \quad \alpha = 1.6 (\text{ASD})$$

P_r = required axial compressive strength under LRFD or ASD load combinations

P_y = member yield strength ($=AF_y$), N.

- (b) All load combinations include an additional lateral load, N_i , applied in combination with other loads at each level of the structure, where

$$N_i = 2.1 \left(\frac{\Delta}{L} \right) Y_i \geq 0.0042 Y_i \quad (6.10.10)$$

Y_i = gravity load from the LRFD load combination or 1.6 times the ASD load combination applied at level i , N

Δ/L = the maximum ratio of Δ to L for all stories in the structure

Δ = first-order interstory drift due to the design loads, mm. Where Δ varies over the plan area of the structure, Δ shall be average drift weighted in proportion to vertical load or, alternatively, maximum drift.

L = story height, mm

This additional lateral load shall be considered independently in two orthogonal directions.

- (c) The non-sway amplification of beam-column moments is considered by applying the B1 amplifier of Sec10.3.2.1 to the total member moments.

10.4 Design of Members for Tension

This Section applies to steel members subject to axial tension caused by static forces acting through the centroidal axis.

10.4.1 Slenderness Limitations

The maximum slenderness (KL/r) limit for design of structural members (except cables and hanger rods) in tension shall be 300 unless it is justified by a comprehensive dynamic analysis (including 2nd order effects if applicable) that a higher slenderness ratio is satisfactory.

Here,

L = laterally unbraced length of the member, mm

r = governing radius of gyration, mm

K = the effective length factor determined in accordance with Sec 10.3.2.

10.4.2 Tensile Strength

The design tensile strength, $\phi_t P_n$, and the allowable tensile strength, P_n/Ω_t , of tension members, shall be the lower value obtained according to the limit states of tensile yielding in the gross section and tensile rupture in the net section.

(a) For tensile yielding in the gross section:

$$P_n = F_y A_g \quad (6.10.11)$$

$$\phi_t = 0.90 \text{ (LRFD)} \quad \Omega_t = 1.67 \text{ (ASD)}$$

(b) For tensile rupture in the net section:

$$P_n = F_u A_e \quad (6.10.12)$$

$$\phi_t = 0.75 \text{ (LRFD)} \quad \Omega_t = 2.00 \text{ (ASD)}$$

Where,

A_e = effective net area, mm²

A_g = gross area of member, mm²

F_y = specified minimum yield stress of the type of steel being used, MPa

F_u = specified minimum tensile strength of the type of steel being used, MPa

When members without holes are fully connected by welds, the effective net area used in Eq.6.10.12 shall be as defined in Sec10.4.3. When holes are present in a member with welded end connections, or at the welded connection in the case of plug or slot welds, the effective net area through the holes shall be used in Eq.6.10.12.

10.4.3 Area Determination

10.4.3.1 Gross area

The gross area, A_g , of a member is the total cross-sectional area.

10.4.3.2 Net area

The net area, A_n , of a member is the sum of the products of the thickness and the net width of each element computed as follows:

In computing net area for tension and shear, the width of a bolt hole shall be taken as 2 mm greater than the nominal dimension of the hole.

For a chain of holes extending across a part in any diagonal or zigzag line, the net width of the part shall be obtained by deducting from the gross width the sum of the diameters or slot dimensions as provided in Sec10.10.3.2, of all holes in the chain, and adding, for each gage space in the chain, the quantity $s^2/(4g)$

Where,

s = longitudinal center-to-center spacing (pitch) of any two consecutive holes, mm.

g = transverse center-to-center spacing (gage) between fastener gage lines, mm.

For angles, the gage for holes in opposite adjacent legs shall be the sum of the gages from the back of the angles less the thickness.

For slotted HSS welded to a gusset plate, the net area, A_n , is the gross area minus the product of the thickness and the total width of material that is removed to form the slot.

In determining net area across plug or slot welds, the weld metal shall not be considered as adding to net area.

10.4.3.3 Effective net area

The effective area of tension members shall be determined as follows:

$$A_e = A_n U \quad (6.10.13)$$

Where, U is the shear lag factor as obtained from Table 6.10.2.

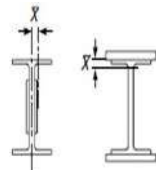
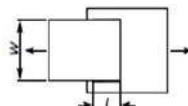
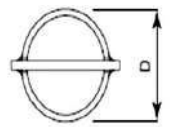
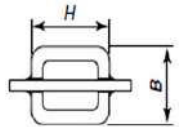
Members such as single angles, double angles and WT sections shall have connections proportioned such that U is equal to or greater than 0.60. Alternatively, a lesser value of U is permitted if these tension members are designed for the effect of eccentricity in accordance with Sections 10.8.1.2 or 10.8.2.

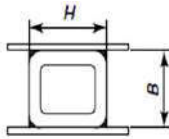
10.4.4 Built-Up Members

For limitations on the longitudinal spacing of connectors between elements in continuous contact consisting of a plate and a shape or two plates, Sec10.10.3.5.

Either perforated cover plates or tie plates without lacing are permitted to be used on the open sides of built-up tension members. Tie plates shall have a length not less than two-thirds the distance between the lines of welds or fasteners connecting them to the components of the member. The thickness of such tie plates shall not be less than one-fiftieth of the distance between these lines. The longitudinal spacing of intermittent welds or fasteners at tie plates shall not exceed 150 mm.

Table 6.10.2: Shear Lag Factors for Connections to Tension Members

Case	Description of Element		Shear Lag Factor, U	Example
1	All tension members where the tension load is transmitted directly to each of cross-sectional elements by fasteners or welds. (except as in Cases 3, 4, 5 and 6)		$U = 1.0$	-
2	All tension members, except plates and HSS, where the tension load is transmitted to some but not all of the cross-sectional elements by fasteners or longitudinal welds (Alternatively, for W, M, S and HP, Case 7 may be used.)		$U = 1 - \bar{X}/I$	
3	All tension members where the tension load is transmitted by transverse welds to some but not all of the cross-sectional elements.		$U = 1.0$ and, A_n = area of the directly connected elements	-
4	Plates where the tension load is transmitted by longitudinal welds only.		$I \geq 2w \dots U = 1.0$ $2w > I \geq 1.5w \dots U = 0.87$ $1.5w > I \geq w \dots U = 0.75$	
5	Round HSS with a single concentric gusset plate		$I \geq 1.3D \dots U = 1.0$ $D \leq I < 1.3D \dots U = 1 - \bar{X} = D/\pi$	
6	Rectangular HSS	with a single concentric gusset plate	$I \geq H \dots U = 1 - \bar{X}/L$ $\bar{X} = \frac{B^2 + 2BH}{4(B + H)}$	

Case	Description of Element		Shear Lag Factor, U	Example
		With two side gusset plates	$I \geq H \dots U = 1 - \bar{X}/L$ $\bar{X} = \frac{B^2}{4(B + H)}$	
7	W, M, S or HP Shapes or Tees cut from these shapes. (If U is calculated per Case 2, the larger value is permitted to be used)	With flange connected with 3 or more fasteners per line in direction of loading	$b_f \geq 2/3d \dots U = 0.90$ $b_f \leq 2/3d \dots U = 0.85$	-
		With web connected with 4 or more fasteners per line in the direction of loading	$U = 0.70$	-
8	Single angles (If U is calculated per Case 2, the larger value is permitted to be used)	With 4 or more fasteners per line in direction of loading	$U = 0.80$	-
		With 2 or 3 fasteners per line in the direction of loading	$U = 0.60$	-

I = Length of connection, mm;
 w = plate width, mm;
 $\bar{X}$ = connection eccentricity, mm;
 B = overall width of rectangular HSS member, measured 90° to the plane of the connection, mm;
 H = overall height of rectangular HSS member, measured in the plane of the connection, mm

10.4.5 Pin-Connected Members

10.4.5.1 Tensile strength

The design tensile strength, ϕP_n and the allowable tensile strength, P_n/Ω_t , of pin-connected members, shall be the lower value obtained according to the limit states of tensile rupture, shear rupture, bearing, and yielding.

- (a) For tensile rupture on the net effective area:

$$P_{\eta} = 2tb_{eff}F_u \quad (6.10.14)$$

$$\phi_t = 0.75 \text{ (LRFD)} \quad \Omega_t = 2.00 \text{ (ASD)}$$

- (b) For shear rupture on the effective area:

$$P_{\eta} = 0.6F_u A_{sf} \quad (6.10.15)$$

$$\phi_{sf} = 0.75 \text{ (LRFD)} \quad \Omega_{sf} = 2.00 \text{ (ASD)}$$

Where,

$$A_{sf} = 2t(a + d/2), \text{ mm}^2$$

a = shortest distance from edge of the pin hole to the edge of the member measured parallel to the direction of the force, mm

$b_{eff} = 2t + 16$, mm but not more than the actual distance from the edge of the hole to the edge of the part measured in the direction normal to the applied force

d = pin diameter, mm

t = thickness of plate, mm

- (c) For bearing on the projected area of the pin, see Sec 10.10.7.

- (d) For yielding on the gross-section, use Eq.6.10.11.

10.4.5.2 Dimensional requirements 1.33 b_{eff}

The pin hole shall be located midway between the edges of the member in the direction normal to the applied force. When the pin is expected to provide for relative movement between connected parts while under full load, the diameter of the pin hole shall not be more than 1 mm greater than the diameter of the pin.

The width of the plate at the pin hole shall not be less than $(2b_{eff} + d)$ and the minimum extension, a , beyond the bearing end of the pin hole, parallel to the axis of the member, shall not be less than $1.33b_{eff}$

The corners beyond the pin hole are permitted to be cut at 45° to the axis of the member, provided the *net area* beyond the pin hole, on a plane perpendicular to the cut, is not less than that required beyond the pin hole parallel to the axis of the member.

10.4.6 Eyebars

10.4.6.1 Tensile strength

The available tensile strength of eyebars shall be determined in accordance with Sec 10.4.2, with A_g taken as the cross-sectional area of the body.

For calculation purposes, the width of the body of the eyebars shall not exceed eight times its thickness.

10.4.6.2 Dimensional requirements

Eyebars shall be of uniform thickness, without reinforcement at the pin holes, and have circular heads with the periphery concentric with the pin hole.

The radius of transition between the circular head and the eyebar body shall not be less than the head diameter.

The pin diameter shall not be less than seven-eighths times the eyebar body width, and the pin hole diameter shall not be more than 1 mm greater than the pin diameter.

For steels having F_y greater than 485 MPa, the hole diameter shall not exceed five times the plate thickness and the width of the eyebar body shall be reduced accordingly.

A thickness of less than 13 mm is permissible only if external nuts are provided to tighten pin plates and filler plates into snug contact. The width from hole edge to plate edge perpendicular to the direction of applied load shall be greater than two-thirds and, for the purpose of calculation, not more than three-fourths times the eyebar body width.

10.5 Design of Members for Compression

This Section addresses members subject to axial compression through the centroidal axis.

10.5.1 General Provisions

The design compressive strength, $\phi_c P_n$, and the allowable compressive strength, P_n/Ω_c , are determined as follows:

The nominal compressive strength, P_n shall be the lowest value obtained according to the limit states of flexural buckling, torsional buckling and flexural-torsional buckling.

For doubly symmetric and singly symmetric members the limit state of flexural buckling is applicable.

For singly symmetric and unsymmetric members, and certain doubly symmetric members, such as cruciform or built-up columns, the limit states of torsional or flexural-torsional buckling are also applicable.

$$\phi_c = 0.90 \text{ (LRFD)} \quad \Omega_c = 1.67 \text{ (ASD)}$$

10.5.2 Slenderness Limitations and Effective Length

The effective length factor, K for calculation of column slenderness, KL/r , shall be determined in accordance with Sec 10.3.

Where,

L = laterally unbraced length of the member, mm

r = governing radius of gyration, mm

K = the effective length factor determined in accordance with Sec 10.3.2.

The maximum limit of slenderness, KL/r , for compression members shall be 150 unless a comprehensive analysis including second order effects (including dynamic effects if any) shows that a higher value is justified.

10.5.3 Compressive Strength for Flexural Buckling of Members without Slender Elements

This Section applies to compression members with compact and non-compact sections, as defined in Sec 10.2.4, for uniformly compressed elements.

The nominal compressive strength, P_n , shall be determined based on the limit state of flexural buckling.

$$P_n = F_{cr} A_g \quad (6.10.16)$$

The flexural buckling stress, F_{cr} , is determined as follows:

$$(a) \text{ When } \frac{KL}{r} \leq 4.71 \sqrt{\frac{E}{F_y}} \quad (\text{or } F_e \geq 0.44 F_y)$$

$$F_{cr} = \left[0.658^{\frac{F_y}{F_e}} \right] F_y \quad (6.10.17)$$

$$(b) \text{ When } \frac{KL}{r} > 4.71 \sqrt{\frac{E}{F_y}} \quad (\text{or } F_e < 0.44 F_y)$$

$$F_{cr} = 0.877 F_e \quad (6.10.18)$$

Where,

F_e = elastic critical buckling stress determined according to Eq.6.10.19, Sec 10.5.4, or the provisions of Sec 10.3.2, as applicable,

$$F_e = \frac{\pi^2 E}{\left(\frac{KL}{r} \right)^2} \quad (6.10.19)$$

10.5.4 Compressive Strength for Torsional and Flexural-Torsional Buckling of Members without Slender Elements

This section applies to singly symmetric and unsymmetric members, and certain doubly symmetric members, such as cruciform or built-up columns with compact and non compact sections, as defined in Sec 10.2.4 for uniformly compressed elements. These provisions are not required for single angles, which are covered in Sec 10.5.5.

The nominal compressive strength, P_n , shall be determined based on the limit states of flexural-torsional and torsional buckling, as follows:

$$P_n = F_{cr} A_g \quad (6.10.20)$$

For double-angle and tee-shaped compression members:

$$F_{cr} = \left(\frac{F_{cry} + F_{crz}}{2H} \right) \left[1 - \sqrt{1 - \frac{4F_{cry}F_{crz}H}{(F_{cry} + F_{crz})^2}} \right] \quad (6.10.21)$$

Where,

F_{cry} is taken as F_{cr} from Eq.6.10.17 or 6.10.18, for flexural buckling about the y-axis of symmetry and $\frac{KL}{r} = \frac{KL}{r_y}$, and

$$F_{crz} = \frac{GJ}{A_g \bar{r}_0^2} \quad (6.10.22)$$

For all other cases, F_{cr} shall be determined according to Eq.6.10.17 or 6.10.18, using the torsional or flexural-torsional elastic buckling stress, F_e determined as follows:

For doubly symmetric members:

$$F_e = \left[\frac{\pi^2 E C_w}{(K_z L)^2} + GJ \right] \frac{1}{I_x + I_y} \quad (6.10.23)$$

For singly symmetric members where y is the axis of symmetry:

$$F_e = \left(\frac{F_{ey} + F_{ez}}{2H} \right) \left[1 - \sqrt{1 - \frac{4F_{ey}F_{ez}H}{(F_{ey} + F_{ez})^2}} \right] \quad (6.10.24)$$

For unsymmetric members, F_e is the lowest root of the cubic equation:

$$(F_e - F_{ex})(F_e - F_{ey})(F_e - F_{ez}) - F_e^2(F_e - F_{ey})\left(\frac{x_0}{\bar{r}_0}\right)^2 - F_e^2(F_e - F_{ex})\left(\frac{y_0}{\bar{r}_0}\right)^2 = 0 \quad (6.10.25)$$

Where,

A_g = gross area of member, mm²

C_w = warping constant, mm⁶

$$\bar{r}_0^2 = x_0^2 + y_0^2 + \frac{I_x + I_y}{A_g} \quad (6.10.26)$$

$$H = 1 - \frac{x_0^2 + y_0^2}{\bar{r}_0^2} \quad (6.10.27)$$

$$F_{ex} = \frac{\pi^2 E}{\left(\frac{K_x L}{r_x}\right)^2} \quad (6.10.28)$$

$$F_{ey} = \frac{\pi^2 E}{\left(\frac{K_y L}{r_y}\right)^2} \quad (6.10.29)$$

$$F_{ez} = \left(\frac{\pi^2 E C_w}{(K_z L)^2} + GJ \right) \frac{1}{A_g \bar{r}_0^2} \quad (6.10.30)$$

G = shear modulus of elasticity of steel = 77200 MPa

I_x, I_y = moment of inertia about the principal axes, mm⁴

J = torsional constant, mm⁴

K_z = effective length factor for torsional buckling

x_0, y_0 = coordinates of shear center with respect to the centroid, mm

$\bar{r}_0$ = polar radius of gyration about the shear center, mm

r_y = radius of gyration about y-axis, mm

10.5.5 Single Angle Compression Members

The nominal compressive strength, P_n , of single angle members shall be determined in accordance with Sec 10.5.3 or Sec 10.5.7, as appropriate, for axially loaded members, as well as those subject to the slenderness modification of Sec 10.5.5(a) or 10.5.5(b), provided the members meet the criteria imposed.

The effects of eccentricity on single angle members are permitted to be neglected when the members are evaluated as axially loaded compression members using one of the effective slenderness ratios specified below, provided that: (1) members are loaded at the ends in compression through the same one leg; (2) members are attached by welding or by minimum two-bolt connections; and (3) there are no intermediate transverse loads.

- (a) For equal-leg angles or unequal-leg angles connected through the longer leg that are individual members or are web members of planar trusses with adjacent web members attached to the same side of the gusset plate or chord:

(i) When $0 \leq \frac{L}{r_x} \leq 80$

$$\frac{KL}{r} = 72 + \frac{0.75L}{r_x} \quad (6.10.31)$$

(ii) When $\frac{L}{r_x} > 80$

$$\frac{KL}{r} = 32 + \frac{1.25L}{r_x} \leq 200 \quad (6.10.32)$$

For unequal-leg angles with leg length ratios less than 1.7 and connected through the shorter leg, KL/r from Eq.6.10.31 and Eq.6.10.32 shall be increased by adding $4[(b_l/b_s)^2 - 1]$, but KL/r of the members shall not be less than $0.95L/r_z$.

- (b) For equal-leg angles or unequal-leg angles connected through the longer leg that are web members of box or space trusses with adjacent web members attached to the same side of the gusset plate or chord:

(i) When $0 \leq \frac{L}{r_x} \leq 75$

$$\frac{KL}{r} = 60 + \frac{0.8L}{r_x} \quad (6.10.33a)$$

(ii) When $\frac{L}{r_x} > 75$

$$\frac{KL}{r} = 45 + \frac{L}{r_x} \leq 200 \quad (6.10.33b)$$

For unequal-leg angles with leg length ratios less than 1.7 and connected through the shorter leg, KL/r from Eq.6.10.33a and 6.10.33b shall be increased by adding $6[(b_l/b_s)^2 - 1]$, but KL/r of the members shall not be less than $0.82L/r_z$.

Where,

L = length of member between work points at truss chord centerlines, mm

b_l = longer leg of angle, mm

b_s = shorter leg of angle, mm

r_x = radius of gyration about geometric axis parallel to connected leg, mm

r_z = radius of gyration for the minor principal axis, mm

- (c) Single angle members with different end conditions from those described in Sections 10.5.5(a) or (b), with leg length ratios greater than 1.7, or with transverse loading shall be evaluated for combined axial load and flexure using the provisions of Sec 10.8. End connection to different legs on each end or to both legs, the use of single bolts or the attachment of adjacent web members to opposite sides of the gusset plate or chord shall constitute different end conditions requiring the use of Sec 10.8 provisions.

10.5.6 Built-up Members

10.5.6.1 Compressive Strength

- (a) The nominal compressive strength of built-up members composed of two or more shapes that are interconnected by bolts or welds shall be determined in accordance with Sections 10.5.3, 10.5.4, or 10.5.7 subject to the following modification. In lieu of more accurate analysis, if the buckling mode involves relative deformations that produce shear forces in the connectors between individual shapes, KL/r is replaced by $(KL/r)_m$ determined as follows:

- (i) For intermediate connectors that are snug-tight bolted:

$$\left(\frac{KL}{r}\right)_m = \sqrt{\left(\frac{KL}{r}\right)_0^2 + \left(\frac{a}{r_i}\right)^2} \quad (6.10.34)$$

- (ii) For intermediate connectors that are welded or pretensioned bolted:

$$\left(\frac{KL}{r}\right)_m = \sqrt{\left(\frac{KL}{r}\right)_0^2 + 0.82 \frac{\alpha^2}{(1+\alpha^2)} \left(\frac{a}{r_{ib}}\right)^2} \quad (6.10.35)$$

Where,

$\left(\frac{KL}{r}\right)_m$ = modified column slenderness of built-up member

$\left(\frac{KL}{r}\right)_0$ = column slenderness of built-up member acting as a unit in the buckling direction being considered

a = distance between connectors, mm

r_i = minimum radius of gyration of individual component, mm

r_{ib} = radius of gyration of individual component relative to its centroidal axis parallel to member axis of buckling, mm

α = separation ratio = $h/2r_{ib}$

H = distance between centroids of individual components perpendicular to the member axis of buckling, mm

- (b) The nominal compressive strength of built-up members composed of two or more shapes or plates with at least one open side interconnected by perforated cover plates or lacing with tie plates shall be determined in accordance with Sections 10.5.3, 10.5.4, or 10.5.7 subject to modification given in Sec 10.5.6.1 (a).

10.5.6.2 Dimensional requirements

Individual components of compression members composed of two or more shapes shall be connected to one another at intervals, a , such that the effective slenderness ratio k_a/r_i of each of the component shapes, between the fasteners, does not exceed three-fourths times the governing slenderness ratio of the built-up member. The least radius of gyration, r_i , shall be used in computing the slenderness ratio of each component part. The end connection shall be welded or pre-tensioned bolted with Class A or B faying surfaces.

At the ends of built-up compression members bearing on base plates or milled surfaces, all components in contact with one another shall be connected by a weld having a length not less than the maximum width of the member or by bolts spaced longitudinally not more than four diameters apart for a distance equal to $1\frac{1}{2}$ times the maximum width of the member.

Along the length of built-up compression members between the end connections required above, longitudinal spacing for intermittent welds or bolts shall be adequate to provide for the transfer of the required forces. For limitations on the longitudinal spacing of fasteners between elements in continuous contact consisting of a plate and a shape or two plates, see Sec 10.10.3.5. Where a component of a built-up compression member consists of an outside plate, the maximum spacing shall not exceed the thickness of the thinner outside plate times $0.75\sqrt{E/F_y}$, nor 305 mm, when intermittent welds are provided along the edges of the components or when fasteners are provided on all gage lines at each section. When fasteners are staggered, the maximum spacing on each gage line shall not exceed the thickness of thinner outside plate times $1.12\sqrt{E/F_y}$ nor 460 mm.

Open sides of compression members built up from plates or shapes shall be provided with continuous cover plates perforated with a succession of access holes. The unsupported width of such plates at access holes, as defined in Sec 10.2.4, is assumed to contribute to the available strength provided the following requirements are met:

- (1) The width-thickness ratio shall conform to the limitations of Sec 10.2.4.
- (2) The ratio of length (in direction of stress) to width of hole shall not exceed two.
- (3) The clear distance between holes in the direction of stress shall be not less than the transverse distance between nearest lines of connecting fasteners or welds.
- (4) The periphery of the holes at all points shall have a minimum radius of 38 mm.

As an alternative to perforated cover plates, lacing with tie plates is permitted at each end and at intermediate points if the lacing is interrupted. Tie plates shall be as near the ends as practicable. In members providing available strength, the end tie plates shall have a

length of not less than the distance between the lines of fasteners or welds connecting them to the components of the member. Intermediate tie plates shall have a length not less than one-half of this distance. The thickness of tie plates shall be not less than one-fiftieth of the distance between lines of welds or fasteners connecting them to the segments of the members. In welded construction, the welding on each line connecting a tie plate shall total not less than one-third the length of the plate. In bolted construction, the spacing in the direction of stress in tie plates shall be not more than six diameters and the tie plates shall be connected to each segment by at least three fasteners.

Lacing, including flat bars, angles, channels, or other shapes employed as lacing, shall be so spaced that the L/r ratio of the flange included between their connections shall not exceed three-fourths times the governing slenderness ratio for the member as a whole. Lacing shall be proportioned to provide a shearing strength normal to the axis of the member equal to 2 percent of the available compressive strength of the member. The L/r ratio for lacing bars arranged in single systems shall not exceed 140. For double lacing this ratio shall not exceed 200. Double lacing bars shall be joined at the intersections. For lacing bars in compression, L is permitted to be taken as the unsupported length of the lacing bar between welds or fasteners connecting it to the components of the built-up member for single lacing, and 70 percent of that distance for double lacing.

For additional spacing requirements, see Sec 10.10.3.5.

10.5.7 Members with Slender Elements

This Section applies to compression members with slender sections, as defined in Sec 10.2.4 for uniformly compressed elements.

The nominal compressive strength, P_n , shall be determined based on the limit states of flexural, torsional and flexural-torsional buckling.

$$P_n = F_{cr} A_g \quad (6.10.36)$$

$$\text{When } \frac{KL}{r} \leq 4.71 \sqrt{\frac{E}{QF_y}} \quad (\text{or } F_e \geq 0.44 QF_y)$$

$$F_{cr} = Q \left[0.658^{\frac{QF_y}{F_e}} \right] F_y \quad (6.10.37)$$

$$\text{When } \frac{KL}{r} > 4.71 \sqrt{\frac{E}{QF_y}} \quad (\text{or } F_e < 0.44 QF_y)$$

$$F_{cr} = 0.877 F_e \quad (6.10.38)$$

Where,

F_e = elastic critical buckling stress, calculated using Eq.6.10.19 and 6.10.23 for doubly symmetric members, Eq.6.10.19 and 6.10.24 for singly symmetric members, and Eq.6.10.25 for unsymmetric members, except for single angles where F_e is calculated using Eq.6.10.19.

$Q = 1.0$ for members with compact and noncompact sections, as defined in Sec 10.2.4, for uniformly compressed elements.

$= Q_s Q_a$ for members with slender-element sections, as defined in Sec 10.2.4, for uniformly compressed elements.

10.5.7.1 Slender unstiffened elements, Q_s

The reduction factor Q_s for slender unstiffened elements is defined as follows:

- (a) For flanges, angles, and plates projecting from rolled columns or other compression members:

$$\text{When } \frac{b}{t} \leq 0.56 \sqrt{\frac{E}{F_y}}$$

$$Q_s = 1.0 \quad (6.10.39)$$

$$\text{When } 0.56 \sqrt{E/F_y} < b/t < 1.03 \sqrt{E/F_y}$$

$$Q_s = 1.415 - 0.74 \left(\frac{b}{t} \right) \sqrt{\frac{F_y}{E}} \quad (6.10.40)$$

$$\text{When } b/t \geq 1.03 \sqrt{E/F_y}$$

$$Q_s = \frac{0.69E}{F_y \left(\frac{b}{t} \right)^2} \quad (6.10.41)$$

- (b) For flanges, angles, and plates projecting from built-up columns or other compression members:

$$\text{When } \frac{b}{t} \leq 0.64 \sqrt{\frac{Ek_c}{F_y}}$$

$$Q_s = 1.0 \quad (6.10.42)$$

When $0.64\sqrt{\frac{Ek_c}{F_y}} < b/t \leq 1.17\sqrt{\frac{Ek_c}{F_y}}$

$$Q_s = 1.415 - 0.65\left(\frac{b}{t}\right)\sqrt{\frac{F_y}{Ek_c}} \quad (6.10.43)$$

When $\frac{b}{t} > 1.17\sqrt{\frac{Ek_c}{F_y}}$

$$Q_s = \frac{0.90Ek_c}{F_y\left(\frac{b}{t}\right)^2} \quad (6.10.44)$$

Where, $k_c = \frac{4}{\sqrt{h/t_w}}$, and shall not be taken less than 0.35 nor greater than 0.76 for calculation purposes.

(c) For single angles

When $\frac{b}{t} \leq 0.45\sqrt{\frac{E}{F_y}}$

$$Q_s = 1.0 \quad (6.10.45)$$

When $0.45\sqrt{E/F_y} < b/t \leq 0.91\sqrt{E/F_y}$

$$Q_s = 1.34 - 0.76\left(\frac{b}{t}\right)\sqrt{\frac{F_y}{E}} \quad (6.10.46)$$

When $b/t > 0.91\sqrt{E/F_y}$

$$Q_s = \frac{0.53E}{F_y\left(\frac{b}{t}\right)^2} \quad (6.10.47)$$

Where, b = full width of longest angle leg, mm

(d) For stems of tees

When $\frac{d}{t} \leq 0.75\sqrt{\frac{E}{F_y}}$

$$Q_s = 1.0 \quad (6.10.48)$$

When $0.75 \sqrt{\frac{E}{F_y}} < d/t \leq 1.03 \sqrt{\frac{E}{F_y}}$

$$Q_s = 1.908 - 1.22 \left(\frac{d}{t} \right) \sqrt{\frac{F_y}{E}} \quad (6.10.49)$$

When $d/t > 1.03 \sqrt{\frac{E}{F_y}}$

$$Q_s = \frac{0.69E}{F_y \left(\frac{d}{t} \right)^2} \quad (6.10.50)$$

Where,

b = width of unstiffened compression element, as defined in Sec 10.2.4, mm

d = the full nominal depth of tee, mm

t = thickness of element, mm

10.5.7.2 Slender unstiffened elements, Q_s

The reduction factor, Q_a , for slender stiffened elements is defined as follows:

$$Q_a = \frac{A_{eff}}{A} \quad (6.10.51)$$

A = total cross-sectional area of member, mm²

A_{eff} = summation of effective areas of the cross section based on the reduced effective width, b_e , mm².

The reduced effective width, b_e , is determined as follows:

- (a) For uniformly compressed slender elements, with $\frac{b}{t} \geq 1.49 \sqrt{\frac{E}{f}}$, except flanges of square and rectangular sections of uniform thickness:

$$b_e = 1.92t \sqrt{\frac{E}{f}} \left[1 - \frac{0.34}{(b/t)} \sqrt{\frac{E}{f}} \right] \leq b \quad (6.10.52)$$

Where, f is taken as F_{cr} with F_{cr} calculated based on $Q = 1.0$.

- (b) For flanges of square and rectangular slender-element sections of uniform thickness

$$\text{with } \frac{b}{t} \geq 1.40 \sqrt{\frac{E}{f}}$$

$$b_e = 1.92t \sqrt{\frac{E}{f}} \left[1 - \frac{0.38}{(b/t)} \sqrt{\frac{E}{f}} \right] \leq b \quad (6.10.53)$$

Where $f = P_n/A_{eff}$

- (c) For axially-loaded circular sections:

$$\text{When } 0.11 \frac{E}{F_y} < \frac{D}{t} < 0.45 \frac{E}{F_y}$$

$$Q = Q_a = \frac{0.038E}{F_y(D/t)} + \frac{2}{3} \quad (6.10.54)$$

Where,

D = outside diameter, mm

t = wall thickness, mm

10.6 Design of Members for Flexure

This Section applies to members subject to simple bending about one principal axis. For simple bending, the member is loaded in a plane parallel to a principal axis that passes through the shear center or is restrained against twisting at load points and supports. The general provisions are provided in Sec 10.6.1. Various Section properties of members are provided in Table 6.10.3.

10.6.1 General Provisions

The design flexural strength, $\phi_b M_n$, and the allowable flexural strength, M_n/Ω_b , shall be determined as follows:

- (a) For all provisions in this Sec 10.6

$$\phi_b = 0.90 \text{ (LRFD)} \quad \Omega_b = 1.67 \text{ (ASD)}$$

And, the nominal flexural strength, M_n , shall be determined according to Sections 10.6.2 to 10.6.12.

- (b) The provisions in this Chapter are based on the assumption that points of support for beams and girders are restrained against rotation about their longitudinal axis.

The following terms are common to the Equations in this Chapter except where noted:

C_b = lateral-torsional buckling modification factor for non-uniform moment diagrams when both ends of the unsupported segment are braced

$$C_b = \frac{12.5M_{max}}{2.5M_{max}+3M_A+4M_B+3M_C} R_m \leq 3.0 \quad (6.10.55)$$

Where,

M_{max} = absolute value of maximum moment in the unbraced segment, N-mm

M_A = absolute value of moment at quarter point of the unbraced segment, N-mm

M_B = absolute value of moment at centerline of the unbraced segment, N-mm

M_C = absolute value of moment at three-quarter point of the unbraced segment, N-mm

R_m = cross-section monosymmetry parameter

= 1.0, doubly symmetric members

= 1.0, singly symmetric members subjected to single curvature bending

= $0.5 + 2 \left(\frac{I_{yc}}{I_y} \right)^2$, singly symmetric members subjected to reverse curvature bending

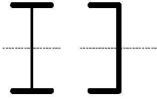
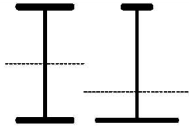
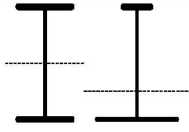
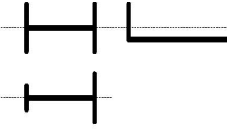
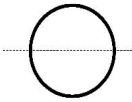
I_y = moment of inertia about the principal y-axis, mm⁴

I_{yc} = moment of inertia about y-axis referred to the compression flange, or if reverse curvature bending, referred to the smaller flange, mm⁴

In singly symmetric members subjected to reverse curvature bending, the lateral- torsional buckling strength shall be checked for both flanges. The available flexural strength shall be greater than or equal to the maximum required moment causing compression within the flange under consideration.

C_b is permitted to be conservatively taken as 1.0 for all cases. For cantilevers or overhangs where the free end is unbraced, $C_b = 1.0$.

Table 6.10.3: Section Types and Selection Table for the Application of Sub-sections of Sec 10.6

Sub-Section in this provision	Cross Section	Flange Slenderness	Web Slenderness	Limit States
10.6.2		C	C	Y, LTB
10.6.3		NC, S	C	LTB, FLB
10.6.4		C, NC, S	C, NC	Y, LTB, FLB, TFY
10.6.5		C, NC, S	S	Y, LTB, FLB, TFY
10.6.6		C, NC, S	N/A	Y, FLB
10.6.7		C, NC, S	C, NC	Y, FLB, WLB
10.6.8		N/A	N/A	Y, LB

Sub-Section in this provision	Cross Section	Flange Slenderness	Web Slenderness	Limit States
10.6.9		C, NC, S	N/A	Y, LTB, FLB
10.6.10		N/A	N/A	Y, LTB, LLB
10.6.11		N/A	N/A	Y, LTB
10.6.12	Unsymmetrical shapes	N/A	N/A	All limit states
Y = yielding, LTB = lateral-torsional buckling, FLB = flange local buckling, WLB = web local buckling, TFY = tension flange yielding, LLB = leg local buckling, LB = local buckling, C = compact, NC = noncompact, S = slender				

10.6.2 Doubly Symmetric Compact I-Shaped Members and Channels Bent about their Major Axis

This Section applies to doubly symmetric I-shaped members and channels bent about their major axis, having compact webs and compact flanges as defined in Sec10.2.4.

The nominal flexural strength, M_n , shall be the lower value obtained according to the limit states of yielding (plastic moment) and lateral-torsional buckling.

10.6.2.1 Yielding

$$M_n = M_p = F_y Z_x \quad (6.10.56)$$

Where,

F_y = specified minimum yield stress of the type of steel being used, MPa

Z_x = plastic section modulus about the x-axis, mm³

10.6.2.2 Lateral –torsional buckling

(a) When $L_b \leq L_p$, the limit state of lateral-torsional buckling does not apply.(b) When $L_p < L_b \leq L_r$

$$M_n = C_b \left[M_p - (M_p - 0.7F_y S_x) \left(\frac{L_b - L_p}{L_r - L_p} \right) \right] \leq M_p \quad (6.10.57)$$

(c) When $L_b > L_r$

$$M_n = F_{cr} S_x \leq M_p \quad (6.10.58)$$

Where,

L_b = length between points that are either braced against lateral displacement of compression flange or braced against twist of the cross section, mm

$$F_{cr} = \frac{C_b \pi^2 E}{\left(\frac{L_b}{r_{ts}} \right)^2} \sqrt{1 + 0.078 \frac{Jc}{S_x h_o} \left(\frac{L_b}{r_{ts}} \right)^2} \quad (6.10.59)$$

And where,

 E = modulus of elasticity of steel = 200000 MPa J = torsional constant, mm⁴ S_x = elastic section modulus taken about the x-axis, mm³The limiting lengths L_p and L_r are determined as follows:

$$L_p = 1.76 r_y \sqrt{\frac{E}{F_y}} \quad (6.10.60)$$

$$L_r = 1.95 r_{ts} \frac{E}{0.7 F_y} \sqrt{\frac{Jc}{S_x h_o}} \sqrt{1 + \sqrt{1 + 6.76 \left(\frac{0.7 F_y S_x h_o}{E Jc} \right)^2}} \quad (6.10.61)$$

Where,

$$r_{ts}^2 = \frac{\sqrt{I_y C_w}}{S_x} \quad (6.10.62)$$

And,

$$\text{For a doubly symmetric I-shape: } c = 1 \quad (6.10.63a)$$

$$\text{For a channel: } c = \frac{h_o}{2} \sqrt{\frac{I_y}{C_w}} \quad (6.10.63b)$$

Where, h_o = distance between the flange centroids, mm

10.6.3 Doubly Symmetric I-Shaped Members with Compact Webs and Noncompact or Slender Flanges Bent about their Major Axis

This Section applies to doubly symmetric I-shaped members bent about their major axis having compact webs and noncompact or slender flanges as defined in Sec10.2.4.

The nominal flexural strength, M_n , shall be the lower value obtained according to the limit states of lateral-torsional buckling and compression flange local buckling.

10.6.3.1 Lateral –torsional buckling

For lateral-torsional buckling, the provisions of Sec 10.6.2.2 shall apply.

10.6.3.2 Compression flange local buckling

(a) For sections with noncompact flanges

$$M_n = \left[M_p - (M_p - 0.7F_y S_x) \left(\frac{\lambda - \lambda_{pf}}{\lambda_{rf} - \lambda_{pf}} \right) \right] \quad (6.10.64)$$

(b) For sections with slender flanges

$$M_n = \frac{0.9Ek_c S_x}{\lambda^2} \quad (6.10.65)$$

Where,

$$\lambda = \frac{b_f}{2t_f}$$

$\lambda_{pf} = \lambda_p$ is the limiting slenderness for a compact flange, Table 6.10.1

$\lambda_{rf} = \lambda_r$ is the limiting slenderness for a noncompact flange, Table 6.10.1

$k_c = \frac{4}{\sqrt{h/t_w}}$ and shall not be taken less than 0.35 nor greater than 0.76 for calculation purposes

10.6.4 Other I-Shaped Members with Compact or Noncompact Webs Bent about their Major Axis

This Section applies to: (a) doubly symmetric I-shaped members bent about their major axis with noncompact webs; and (b) singly symmetric I-shaped members with webs attached to the mid-width of the flanges, bent about their major axis, with compact or noncompact webs, as defined in Section 10.2.4.

The nominal flexural strength, M_n , shall be the lowest value obtained according to the limit states of compression flange yielding, lateral-torsional buckling, compression flange local buckling and tension flange yielding.

10.6.4.1 Compression flange yielding

$$M_n = R_{pc}M_{yc} = R_{pc}F_yS_{xc} \quad (6.10.66)$$

10.6.4.2 Lateral-torsional buckling

(a) When $L_b \leq L_p$, the limit state of lateral-torsional buckling does not apply.

(b) When $L_p < L_b \leq L_r$

$$M_n = C_b \left[R_{pc}M_{yc} - (R_{pc}M_{yc} - F_yS_{xc}) \left(\frac{L_b - L_p}{L_r - L_p} \right) \right] \leq R_{pc}M_{yc} \quad (6.10.67)$$

(c) When $L_b > L_r$

$$M_n = F_{cr}S_{xc} \leq R_{pc}M_{yc} \quad (6.10.68)$$

Where

$$M_{yc} = F_yS_{xc} \quad (6.10.69)$$

$$F_{cr} = \frac{C_b \pi^2 E}{\left(\frac{L_b}{r_t} \right)^2} \sqrt{1 + 0.078 \frac{J}{S_{xc} h_0} \left(\frac{L_b}{r_t} \right)^2} \quad (6.10.70)$$

For, $\frac{I_{yc}}{I_y} \leq 0.23$, J shall be taken as zero.

The stress, F_L , is determined as follows:

(i) For $\frac{S_{xt}}{S_{xc}} \geq 0.7$

$$F_L = 0.7F_y \quad (6.10.71a)$$

(ii) For $\frac{S_{xt}}{S_{xc}} < 0.7$

$$F_L = F_y \frac{S_{xt}}{S_{xc}} \geq 0.5F_y \quad (6.10.71b)$$

The limiting laterally unbraced length for the limit state of yielding, L_p is,

$$L_p = 1.1r_t \sqrt{\frac{E}{F_y}} \quad (6.10.72)$$

The limiting unbraced length for the limit state of inelastic lateral-torsional buckling, L_r , is

$$L_r = 1.95r_t \frac{E}{F_L} \sqrt{\frac{J}{S_{xc}h_0}} \sqrt{1 + \sqrt{1 + 6.76 \left(\frac{F_L S_{xc}h_0}{E J} \right)^2}} \quad (6.10.73)$$

The web plastification factor, R_{pc} , is determined as follows:

$$(i) \text{ For } \frac{h_c}{t_w} \leq \lambda_{pw}$$

$$R_{pc} = \frac{M_p}{M_{yc}} \quad (6.10.74a)$$

$$(ii) \text{ For } \frac{h_c}{t_w} > \lambda_{pw}$$

$$R_{pc} = \left[\frac{M_p}{M_{yc}} - \left(\frac{M_p}{M_{yc}} - 1 \right) \left(\frac{\lambda - \lambda_{pw}}{\lambda_{rw} - \lambda_{pw}} \right) \right] \leq \frac{M_p}{M_{yc}} \quad (6.10.74b)$$

Where

$$M_p = Z_x F_y \leq 1.6 S_{xc} F_y$$

S_{xc}, S_{xt} = elastic section modulus referred to tension and compression flanges, respectively, mm³

$$\lambda = \frac{h_c}{t_w}$$

$\lambda_{pw} = \lambda_p$ limiting slenderness for a compact web, Table 6.10.1

$\lambda_{rw} = \lambda_r$ limiting slenderness for a noncompact web, Table 6.10.1

The effective radius of gyration for lateral-torsional buckling, r_t , is determined as follows:

- (i) For I-shapes with a rectangular compression flange:

$$r_t = \frac{b_{fc}}{\sqrt{12 \left(\frac{h_0}{d} + \frac{1}{6} a_w \frac{h^2}{h_0 d} \right)}} \quad (6.10.75)$$

Where,

$$a_w = \frac{h_c t_w}{b_{fc} t_{fc}} \quad (6.10.76)$$

b_{fc} = compression flange width, mm

t_{fc} = compression flange thickness, mm

- (ii) For I-shapes with channel caps or cover plates attached to the compression flange:

r_t = radius of gyration of the flange components in flexural compression plus one-third of the web area in compression due to application of major axis bending moment alone, mm

a_w = the ratio of two times the web area in compression due to application of major axis bending moment alone to the area of the compression flange components.

10.6.4.3 Compression flange local buckling

- (a) For sections with compact flanges, the limit state of local buckling does not apply.

- (b) For sections with noncompact flanges

$$M_n = \left[R_{pc} M_{yc} - (R_{pc} M_{yc} - F_L S_{xc}) \left(\frac{\lambda - \lambda_{pf}}{\lambda_{rf} - \lambda_{pf}} \right) \right] \quad (6.10.77)$$

- (c) For sections with slender flanges

$$M_n = \frac{0.9 E k_c S_{xc}}{\lambda^2} \quad (6.10.78)$$

Where,

F_L is defined in Eq. 6.10.71a and Eq. 6.10.72b

R_{pc} = is the web plastification factor, determined by Eq. 6.10.74

$k_c = \frac{4}{\sqrt{h/t_w}}$ and shall not be taken less than 0.35 nor greater than 0.76 for calculation purposes

$$\lambda = \frac{b_{fc}}{2t_{fc}}$$

$\lambda_{pf} = \lambda_p$ limiting slenderness for a compact flange, Table 6.10.1

$\lambda_{rf} = \lambda_r$ limiting slenderness for a noncompact flange, Table 6.10.1

10.6.4.4 Tension flange yielding

(a) When $S_{xt} \geq S_{xc}$, the limit state of tension flange yielding does not apply.

(b) When $S_{xt} < S_{xc}$

$$M_n = R_{pt} M_{yt} \quad (6.10.79)$$

Where,

$$M_{yt} = F_y S_{xt}$$

The web plastification factor corresponding to the tension flange yielding limit state, R_{pt} , is determined as follows:

(i) For $\frac{h_c}{t_w} \leq \lambda_{pw}$

$$R_{pt} = \frac{M_p}{M_{yt}} \quad (6.10.80a)$$

(ii) For $\frac{h_c}{t_w} > \lambda_{pw}$

$$R_{pt} = \left[\frac{M_p}{M_{yt}} - \left(\frac{M_p}{M_{yt}} - 1 \right) \left(\frac{\lambda - \lambda_{pw}}{\lambda_{rw} - \lambda_{pw}} \right) \right] \leq \frac{M_p}{M_{yt}} \quad (6.10.80b)$$

Where,

$$\lambda = \frac{h_c}{t_w}$$

$\lambda_{pw} = \lambda_p$, the limiting slenderness for a compact web, defined in Table 6.10.1

$\lambda_{rw} = \lambda_r$, the limiting slenderness for a noncompact web, defined in Table 6.10.1

10.6.5 Doubly Symmetric and Singly Symmetric I-Shaped Members with Slender Webs Bent about Major Axis

This Section applies to doubly symmetric and singly symmetric I-shaped members with slender webs attached to the mid-width of the flanges, bent about their major axis, as defined in Sec10.2.4.

The nominal flexural strength M_n , shall be the lowest value obtained according to the limit states of compression flange yielding, lateral-torsional buckling, compression flange local buckling and tension flange yielding.

10.6.5.1 Compression flange yielding

$$M_n = R_{pg} F_y S_{xc} \quad (6.10.81)$$

10.6.5.2 Lateral-torsional buckling

$$M_n = R_{pg} F_{cr} S_{xc} \quad (6.10.82)$$

(a) When $L_b \leq L_p$, the limit state of lateral-torsional buckling does not apply.

(b) When $L_p < L_b \leq L_r$

$$F_{cr} = C_b \left[F_y - (0.3 F_y) \left(\frac{L_b - L_p}{L_r - L_p} \right) \right] \leq F_y \quad (6.10.83)$$

(c) When $L_b > L_r$

$$F_{cr} = \frac{C_b \pi^2 E}{\left(\frac{L_b}{r_t} \right)^2} \leq F_y \quad (6.10.84)$$

Where,

L_p is defined by Eq.6.10.72

$$L_r = \pi r_t \sqrt{\frac{E}{0.7 F_y}} \quad (6.10.85)$$

R_{pg} is the bending strength reduction factor:

$$R_{pg} = 1 - \frac{a_w}{1200 + 300 a_w} \left(\frac{h_c}{t_w} - 5.7 \sqrt{\frac{E}{F_y}} \right) \leq 1.0 \quad (6.10.86)$$

a_w is defined by Eq.6.10.76 but shall not exceed 10 and

r_t is the effective radius of gyration for lateral buckling as defined in Sec10.6.4.

10.6.5.3 Compression flange local buckling

$$M_n = R_{pg} F_{cr} S_{xc} \quad (6.10.87)$$

- (a) For sections with compact flanges, the limit state of compression flange local buckling does not apply.
- (b) For sections with noncompact flanges

$$F_{cr} = \left[F_y - (0.3 F_y) \left(\frac{\lambda - \lambda_{pf}}{\lambda_{rf} - \lambda_{pf}} \right) \right] \quad (6.10.88)$$

- (c) For sections with slender flanges

$$F_{cr} = \frac{0.9 E k_c}{\left(\frac{b_f}{2 t_f} \right)^2} \quad (6.10.89)$$

Where,

$k_c = \frac{4}{\sqrt{h/t_w}}$, and shall not be taken less than 0.35 nor greater than 0.76 for calculation purposes

$$\lambda = \frac{b_{fc}}{2 t_{fc}}$$

$\lambda_{pf} = \lambda_p$, the limiting slenderness for a compact flange, Table 6.10.1

$\lambda_{rf} = \lambda_r$, the limiting slenderness for a noncompact flange, Table 6.10.1

10.6.5.4 Tension flange yielding

- (a) When $S_{xt} \geq S_{xc}$, the limit state of tension flange yielding does not apply.
- (b) When $S_{xt} < S_{xc}$

$$M_n = F_y S_{xt} \quad (6.10.90)$$

10.6.6 I-Shaped Members and Channels Bent about Their Minor Axis

This Section applies to I-shaped members and channels bent about their minor axis.

The nominal flexural strength, M_n , shall be the lower value obtained according to the limit states of yielding (plastic moment) and flange local buckling.

10.6.6.1 Yielding

$$M_n = M_p = F_y Z_y \leq 1.6 F_y S_y \quad (6.10.91)$$

10.6.6.2 Flange local buckling

- (a) For sections with compact flanges the limit state of yielding shall apply.
 (b) For sections with noncompact flanges

$$M_n = \left[M_p - (M_p - 0.7 F_y S_y) \left(\frac{\lambda - \lambda_{pf}}{\lambda_{rf} - \lambda_{pf}} \right) \right] \quad (6.10.92)$$

- (c) For sections with slender flanges

$$M_n = F_{cr} S_y \quad (6.10.93)$$

Where,

$$F_{cr} = \frac{0.69E}{\left(\frac{b_f}{2t_f} \right)^2} \quad (6.10.94)$$

$$\lambda = \frac{b}{t}$$

$\lambda_{pf} = \lambda_p$, the limiting slenderness for a compact flange, Table 6.10.1

$\lambda_{rf} = \lambda_r$, the limiting slenderness for a noncompact flange, Table 6.10.1

S_y for a channel shall be taken as the minimum section modulus

10.6.7 Square and Rectangular HSS and Box-Shaped Members

This section applies to square and rectangular HSS, and doubly symmetric box-shaped members bent about either axis, having compact or noncompact webs and compact, noncompact or slender flanges as defined in Sec10.2.4.

The nominal flexural strength, M_n , shall be the lowest value obtained according to the limit states of yielding (plastic moment), flange local buckling and web local buckling under pure flexure.

10.6.7.1 Yielding

$$M_n = M_p = F_y Z \quad (6.10.95)$$

Where,

Z = plastic section modulus about the axis of bending, mm^3

10.6.7.2 Flange local buckling

(a) For compact sections, the limit state of flange local buckling does not apply.

(b) For sections with noncompact flanges

$$M_n = M_p - (M_p - F_y S) \left(3.57 \frac{b}{t} \sqrt{\frac{F_y}{E}} - 4.0 \right) \leq M_p \quad (6.10.96)$$

(b) For sections with slender flanges

$$M_n = F_y S_{eff} \quad (6.10.97)$$

Where, S_{eff} is the effective section modulus determined with the effective width of the compression flange taken as:

$$b_e = 1.92t \sqrt{\frac{E}{F_y}} \left[1 - \frac{0.38}{b/t} \sqrt{\frac{E}{F_y}} \right] \leq b \quad (6.10.98)$$

10.6.7.3 Web local buckling

(a) For compact sections, the limit state of web local buckling does not apply.

(b) For sections with noncompact webs

$$M_n = M_p - (M_p - F_y S_x) \left(0.305 \frac{h}{t_w} \sqrt{\frac{F_y}{E}} - 0.738 \right) \leq M_p \quad (6.10.99)$$

10.6.8 Round HSS

This Section applies to round *HSS* having D/t ratios of less than $\frac{0.45E}{F_y}$

The nominal flexural strength, M_n , shall be the lower value obtained according to the limit states of yielding (plastic moment) and local buckling.

10.6.8.1 Yielding

$$M_n = M_p = F_y Z \quad (6.10.100)$$

10.6.8.2 Local buckling

- (a) For compact sections, the limit state of flange local buckling does not apply.
 (b) For noncompact sections

$$M_n = \left(\frac{0.021E}{\frac{D}{t}} + F_y \right) S \quad (6.10.101)$$

- (b) For sections with slender walls

$$M_n = F_{cr} S \quad (6.10.102)$$

Where,

$$F_{cr} = \frac{0.33E}{\frac{D}{t}} \quad (6.10.103)$$

S = elastic section modulus, mm³

10.6.9 Tees and Double Angles Loaded in the Plane of Symmetry

This Section applies to tees and double angles loaded in the plane of symmetry. The nominal flexural strength, M_n , shall be the lowest value obtained according to the limit states of yielding (plastic moment), lateral-torsional buckling and flange local buckling.

10.6.9.1 Yielding

$$M_n = M_p \quad (6.10.104)$$

Where,

$$M_p = F_y Z_x \leq 1.6 M_y \text{ for stems in tension} \quad (6.10.105)$$

$$\leq M_y \text{ for stems in compression} \quad (6.10.106)$$

10.6.9.2 Lateral-torsional buckling

$$M_n = M_{cr} = \frac{\pi \sqrt{EI_y GJ}}{L_b} [B + \sqrt{1 + B^2}] \quad (6.10.107)$$

Where,

$$B = \pm 2.3 \left(\frac{d}{L_b} \right) \sqrt{\frac{I_y}{J}} \quad (6.10.108)$$

The plus sign for B applies when the stem is in tension and the minus sign applies when the stem is in compression. If the tip of the stem is in compression anywhere along the unbraced length, the negative value of B shall be used.

10.6.9.3 Flange local buckling of tees

$$M_n = F_{cr} S_{xc} \quad (6.10.109)$$

S_{xc} is the elastic section modulus referred to the compression flange.

F_{cr} is determined as follows:

- (a) For compact sections, the limit state of flange local buckling does not apply.
- (b) For noncompact sections

$$F_{cr} = F_y \left(1.19 - 0.50 \left(\frac{b_f}{2t_f} \right) \sqrt{\frac{F_y}{E}} \right) \quad (6.10.110)$$

- (c) For slender sections

$$F_{cr} = \frac{0.69E}{\left(\frac{b_f}{2t_f} \right)^2} \quad (6.10.111)$$

10.6.10 Single Angle

This Section applies to single angles with and without continuous lateral restraint along their length.

Single angles with continuous lateral-torsional restraint along the length shall be permitted to be designed on the basis of geometric axis (x, y) bending. Single angles without continuous lateral-torsional restraint along the length shall be designed using the provisions for principal axis bending except where the provision for bending about a geometric axis is permitted.

The nominal flexural strength, M_n , shall be the lowest value obtained according to the limit states of yielding (plastic moment), lateral-torsional buckling and leg local buckling.

10.6.10.1 Yielding

$$M_n = 1.5M_y \quad (6.10.112)$$

Where, M_y = yield moment about the axis of bending, N-mm

10.6.10.2 Lateral-torsional buckling

For single angles without continuous lateral-torsional restraint along the length

(a) When $M_e \leq M_y$

$$M_n = \left(0.92 - \frac{0.17M_e}{M_y}\right) M_e \quad (6.10.113)$$

(b) When $M_e > M_y$

$$M_n = \left(1.92 - 1.17\sqrt{\frac{M_y}{M_e}}\right) M_y \leq 1.5M_y \quad (6.10.114)$$

Where, M_e , the elastic lateral-torsional buckling moment, is determined as follows:

(i) For bending about one of the geometric axes of an equal-leg angle with no lateral-torsional restraint

(a) With maximum compression at the toe

$$M_e = \frac{0.66Eb^4tC_b}{L^2} \left(\sqrt{1 + 0.78\left(\frac{Lt}{b^2}\right)^2} - 1 \right) \quad (6.10.115a)$$

(b) With maximum tension at the toe

$$M_e = \frac{0.66Eb^4tC_b}{L^2} \left(\sqrt{1 + 0.78\left(\frac{Lt}{b^2}\right)^2} + 1 \right) \quad (6.10.115b)$$

M_y shall be taken as 0.80 times the yield moment calculated using the geometric section modulus.

(ii) For bending about one of the geometric axes of an equal-leg angle with lateral-torsional restraint at the point of maximum moment only

M_e shall be taken as 1.25 times M_e computed using Eq.6.10.115a or Eq.6.10.115b.

M_y shall be taken as the yield moment calculated using the geometric section modulus.

(iii) For bending about the major principal axis of equal-leg angles:

$$M_e = \frac{0.46Eb^2t^2C_b}{L} \quad (6.10.116)$$

(iv) For bending about the major principal axis of unequal-leg angles:

$$M_e = \frac{4.9EI_z C_b}{L^2} \left(\sqrt{\beta_w^2 + 0.052 \left(\frac{Lt}{r_z} \right)^2} + \beta_w \right) \quad (6.10.117)$$

Where,

C_b is computed using Eq.6.10.55 with a maximum value of 1.5.

L = laterally unbraced length of a member, mm

I_z = minor principal axis moment of inertia, mm⁴

r_z = radius of gyration for the minor principal axis, mm

t = angle leg thickness, mm

β_w = a section property for unequal leg angles, positive for short legs in compression and negative for long legs in compression. If the long leg is in compression anywhere along the unbraced length of the member, the negative value of β_w shall be used.

10.6.10.3 Leg local buckling

The limit state of leg local buckling applies when the toe of the leg is in compression.

(a) For compact sections, the limit state of leg local buckling does not apply.

(b) For sections with noncompact legs:

$$M_n = F_y S_c \left(2.43 - 1.72 \left(\frac{b}{t} \right) \sqrt{\frac{F_y}{E}} \right) \quad (6.10.118)$$

(c) For sections with slender legs

$$M_n = F_{cr} S_c \quad (6.10.119)$$

Where,

$$F_{cr} = \frac{0.71E}{\left(\frac{b}{t} \right)^2} \quad (6.10.120)$$

b = outside width of leg in compression, mm

S_c = elastic section modulus to the toe in compression relative to the axis of bending, mm³. For bending about one of the geometric axes of an equal-leg angle with no lateral-torsional restraint, S_c shall be 0.80 of the geometric axis section modulus.

10.6.11 Rectangular Bars and Rounds

This Section applies to rectangular bars bent about either-geometric axis and rounds.

The nominal flexural strength, M_n , shall be the lower value obtained according to the limit states of yielding (plastic moment) and lateral-torsional buckling, as required.

10.6.11.1 Yielding

For rectangular bars with $\frac{L_b d}{t^2} \leq \frac{0.08E}{F_y}$ bent about their major axis, rectangular bars bent about their minor axis and rounds:

$$M_n = M_p = F_y Z \leq 1.6 M_y \quad (6.10.121)$$

10.6.11.2 Lateral-torsional buckling

(a) For rectangular bars with $\frac{0.08E}{F_y} < \frac{L_b d}{t^2} \leq \frac{1.9E}{F_y}$ bent about their major axis:

$$M_n = C_b \left[1.52 - 0.274 \left(\frac{L_b d}{t^2} \right) \frac{F_y}{E} \right] M_y \leq M_p \quad (6.10.122)$$

(b) For rectangular bars with $\frac{L_b d}{t^2} > \frac{1.9E}{F_y}$ bent about their major axis:

$$M_n = F_{cr} S_x \leq M_p \quad (6.10.123)$$

Where,

$$F_{cr} = \frac{1.9EC_b}{\frac{L_b d}{t^2}} \quad (6.10.124)$$

t = width of rectangular bar parallel to axis of bending, mm

d = depth of rectangular bar, mm

L_b = length between points that are either braced against lateral displacement of the compression region or braced against twist of the cross section, mm

(c) For rounds and rectangular bars bent about their minor axis, the limit state of lateral-torsional buckling need not be considered.

10.6.12 Unsymmetrical Shapes

This Section applies to all unsymmetrical shapes, except single angles.

The nominal flexural strength, M_n , shall be the lowest value obtained according to the limit states of yielding (yield moment), lateral-torsional buckling and local buckling where

$$M_n = F_n S \quad (6.10.125)$$

Where, S = lowest elastic section modulus relative to the axis of bending, mm^3

10.6.12.1 Yielding

$$F_n = F_y \quad (6.10.126)$$

10.6.12.2 Lateral-torsional buckling

$$F_n = F_{cr} \leq F_y \quad (6.10.127)$$

Where, F_{cr} = buckling *stress* for the section as determined by analysis, MPa

10.6.12.3 Local buckling

$$F_n = F_{cr} \leq F_y \quad (6.10.128)$$

Where, F_{cr} = buckling *stress* for the section as determined by analysis, MPa

10.6.13 Proportions of Beams and Girders

10.6.13.1 Hole reductions

This Section applies to rolled or built-up shapes, and cover-plated *beams* with holes, proportioned on the basis of flexural strength of the gross section.

In addition to the limit states specified in other sections of this Chapter, the nominal flexural strength, M_n , shall be limited according to the limit state of tensile rupture of the tension flange.

For, $F_u A_{fn} \geq Y_t F_y A_{fg}$, the limit state of tensile rupture does not apply.

For, $F_u A_{fn} < Y_t F_y A_{fg}$, the nominal flexural strength, M_n , at the location of the holes in the tension flange shall not be taken greater than:

$$M_n = \frac{F_u A_{fn}}{A_{fg}} S_x \quad (6.10.129)$$

Where,

A_{fg} = gross tension flange area, calculated in accordance with the provisions of Sec 10.4.3.1, mm²

A_{fn} = net tension flange area, calculated in accordance with the provisions of Sec 10.4.3.2, mm²

$Y_t = 1.0$ for $F_y/F_u \leq 0.8$
 $= 1.1$ otherwise

10.6.13.2 Proportioning limits for i-shaped members

Singly symmetric I-shaped members shall satisfy the following limit:

$$0.1 \leq \frac{I_{yc}}{I_y} \leq 0.9 \quad (6.10.130)$$

I-shaped members with slender webs shall also satisfy the following limits:

(a) For $\frac{a}{h} \leq 1.5$

$$\left(\frac{h}{t_w}\right)_{max} = 11.7 \sqrt{\frac{E}{F_y}} \quad (6.10.131)$$

(b) For $\frac{a}{h} > 1.5$

$$\left(\frac{h}{t_w}\right)_{max} = \frac{0.42E}{F_y} \quad (6.10.132)$$

Where, a = clear distance between transverse stiffeners, mm

In unstiffened girders $\frac{h}{t_w}$ shall not exceed 260. The ratio of the web area to the compression flange area shall not exceed 10.

10.6.13.3 Cover plates

Flanges of welded beams or girders may be varied in thickness or width by splicing a series of plates or by the use of cover plates.

The total cross-sectional area of cover plates of bolted girders shall not exceed 70 percent of the total flange area.

High-strength bolts or welds connecting flange to web, or cover plate to flange, shall be proportioned to resist the total horizontal shear resulting from the bending forces on the girder. The longitudinal distribution of these bolts or intermittent welds shall be in proportion to the intensity of the shear.

However, the longitudinal spacing shall not exceed the maximum permitted for compression or tension members in Sec 10.5.6 or 10.4.4, respectively. Bolts or welds connecting flange to web shall also be proportioned to transmit to web any loads applied directly to the flange, unless provision is made to transmit such loads by direct bearing.

Partial-length cover plates shall be extended beyond the theoretical cutoff point and the extended portion shall be attached to the beam or girder by high-strength bolts in a slip-critical connection or fillet welds. The attachment shall be adequate, at the applicable strength given in Sections 10.10.2.2, 10.10.3.8, or 10.2.3.9 to develop the cover plate's portion of the flexural strength in the beam or girder at the theoretical cutoff point.

For welded cover plates, the welds connecting the cover plate termination to the beam or girder shall have continuous welds along both edges of the cover plate in the length a' , defined below, and shall be adequate to develop the cover plate's portion of the strength of the beam or girder at the distance a' from the end of the cover plate.

- (a) When there is a continuous weld equal to or larger than three-fourths of the plate thickness across the end of the plate

$$a' = w \quad (6.10.133)$$

Where, w = width of cover plate, mm.

- (b) When there is a continuous weld smaller than three-fourths of the plate thickness across the end of the plate

$$a' = 1.5w \quad (6.10.134)$$

- (c) When there is no weld across the end of the plate

$$a' = 2w \quad (6.10.135)$$

10.6.13.4 Built-up beams

Where two or more beams or channels are used side-by-side to form a flexural member, they shall be connected together in compliance with Sec 10.5.6.2. When concentrated loads are carried from one beam to another, or distributed between the beams, diaphragms having sufficient stiffness to distribute the load shall be welded or bolted between the beams.

10.7 Design Of Members For Shear

This Section addresses webs of singly or doubly symmetric members subject to shear in the plane of the web, single angles and HSS sections, and shear in the weak direction of singly or doubly symmetric shapes.

10.7.1 General Provisions

Two methods of calculating shear strength are presented below. The method presented in Sec 10.7.2 does not utilize the post buckling strength of the member (tension field action). The method presented in Sec 10.7.3 utilizes tension field action.

The design shear strength, $\phi_v V_n$ and the allowable shear strength, V_n/Ω_v , shall be determined as follows.

For all provisions in this Section except Sec 10.7.2.1(a):

$$\phi_v = 0.90 \text{ (LRFD)} \quad \Omega_v = 1.67 \text{ (ASD)}$$

10.7.2 Members with Unstiffened or Stiffened Webs

10.7.2.1 Nominal shear strength

This Section applies to webs of singly or doubly symmetric members and channels subject to shear in the plane of the web.

The nominal shear strength, V_n , of unstiffened or stiffened webs, according to the limit states of shear yielding and shear buckling, is

$$V_n = 0.6F_y A_w C_w \quad (6.10.136)$$

(a) For webs of rolled I-shaped members with $h/t_w \leq 2.24\sqrt{E/F_y}$

$$\phi_v = 1.00 \text{ (LRFD)} \quad \Omega_v = 1.50 \text{ (ASD)}$$

And,

$$C_v = 1.0 \quad (6.10.137)$$

(b) For webs of all other doubly symmetric shapes and singly symmetric shapes and channels, except round HSS, the web shear coefficient, C_v , is determined as follows:

(i) For $h/t_w \leq 1.10\sqrt{k_v E/F_y}$

$$C_v = 1.0 \quad (6.10.138)$$

(ii) For $1.10\sqrt{k_v E/F_y} < h/t_w \leq 1.37\sqrt{k_v E/F_y}$

$$C_v = \frac{1.10\sqrt{k_v E/F_y}}{h/t_w} \quad (6.10.139)$$

(iii) For $h/t_w > 1.37\sqrt{k_v E/F_y}$

$$C_v = \frac{1.51Ek_v}{(h/t_w)^2 F_y} \quad (6.10.140)$$

Where

A_w = the overall depth times the web thickness, dt_w , mm²

The web plate buckling coefficient, k_v , is determined as follows:

(i) For unstiffened webs with $h/t_w < 260$, $k_v = 5$ except for the stem of tee shapes where, $k_v = 1.2$.

(ii) For stiffened webs,

$$k_v = 5 + \frac{5}{(a/h)^2}$$

$$= 5 \text{ when } a/h > 3.0 \text{ or } a/h > \left[\frac{260}{(h/t_w)} \right]^2$$

Where,

a = clear distance between transverse stiffeners, mm

h = for rolled shapes, the clear distance between flanges less the fillet or corner radii, mm

= for built-up welded sections, the clear distance between flanges, mm

= for built-up bolted sections, the distance between fastener lines, mm

= for tees, the overall depth, mm.

10.7.2.2 Transverse stiffeners

Transverse stiffeners are not required where $h/t_w \leq 2.46\sqrt{E/F_y}$, or where the required shear strength is less than or equal to the available shear strength provided in accordance with Sec 10.7.2.1 for $k_v = 5$.

Transverse stiffeners used to develop the available web shear strength, as provided in Sec 10.7.2.1, shall have a moment of inertia about an axis in the web center for stiffener pairs or about the face in contact with the web plate for single stiffeners, which shall not be less than $at_w^3 j$, where

$$j = \frac{2.5}{(a/h)^2} - 2 \geq 0.5 \quad (6.10.141)$$

Transverse stiffeners are permitted to be stopped short of the tension flange, provided bearing is not needed to transmit a concentrated load or reaction. The weld by which transverse stiffeners are attached to the web shall be terminated not less than four times nor more than six times the web thickness from the near toe to the web-to-flange weld. When single stiffeners are used, they shall be attached to the compression flange, if it consists of a rectangular plate, to resist any uplift tendency due to torsion in the flange. When lateral bracing is attached to a stiffener, or a pair of stiffeners, these, in turn, shall be connected to the compression flange to transmit 1 percent of the total flange force, unless the flange is composed only of angles.

Bolts connecting stiffeners to the girder web shall be spaced not more than 305 mm on center. If intermittent fillet welds are used, the clear distance between welds shall not be more than 16 times the web thickness nor more than 250 mm.

10.7.3 Tension Field Action

10.7.3.1 Limits on the use of tension field action

Consideration of tension field action is permitted for flanged members when the web plate is supported on all four sides by flanges or stiffeners. Consideration of tension field action is not permitted for:

End panels in all members with transverse stiffeners;

Members when a/h exceeds 3.0 or $[260/(h/t_w)]^2$;

$$2 A_w / (A_{fc} + A_{ft}) > 2.5; \text{ or}$$

$$h/b_{fc} \text{ or } h/b_{ft} > 6.0$$

Where,

A_{fc} = area of compression flange, mm²

A_{ft} = area of tension flange, mm²

b_{fc} = width of compression flange, mm

b_{ft} = width of tension flange, mm

In these cases, the nominal shear strength, V_n , shall be determined according to the provisions of Sec 10.7.2.

10.7.3.2 Nominal shear strength with tension field action

When tension field action is permitted according to Sec 10.7.3.1, the nominal shear strength, V_n , with tension field action, according to the limit state of tension field yielding, shall be

For, $h/t_w \leq 1.10\sqrt{k_v E/F_y}$

$$V_n = 0.6F_y A_w \quad (6.10.142)$$

For, $h/t_w > 1.10\sqrt{k_v E/F_y}$

$$V_n = 0.6F_y A_w \left(C_v + \frac{1-C_v}{1.15\sqrt{1+(a/h)^2}} \right) \quad (6.10.143)$$

Where, k_v and C_v are as defined in Sec 10.7.2.1.

10.7.3.3 Transverse stiffeners

Transverse stiffeners subject to tension field action shall meet the requirements of Sec 10.7.2.2 and the following limitations:

$$(b/t)_{st} \leq 0.56 \sqrt{\frac{E}{F_{yst}}}$$

$$A_{st} > \frac{F_y}{F_{yst}} \left[0.15D_s h t_w (1 - C_v) \frac{V_r}{V_c} - 18t_w^2 \right] \geq 0 \quad (6.10.144)$$

Where,

$(b/t)_{st}$ = the width-thickness ratio of the stiffener

F_{yst} = specified minimum yield stress of the stiffener material, MPa

C_v = coefficient defined in Sec 10.7.2.1

D_s = 1.0 for stiffeners in pairs
 = 1.8 for single angle stiffeners
 = 2.4 for single plate stiffeners

V_r = required shear strength at the location of the stiffener, N

V_c = available shear strength; $\phi_v V_n$ (LRFD) or V_n/Ω_v (ASD) with V_n as defined in Sec 10.7.3.2, N

10.7.4 Single Angles

The nominal shear strength, V_n , of a single angle leg shall be determined using Eq. 6.10.136 with $C_v = 1.0$, $A_w = b_t$ where b = width of the leg resisting the shear force, mm and $k_v = 1.2$.

10.7.5 Rectangular HSS and Box Members

The nominal shear strength, V_n , of rectangular HSS and box members shall be determined using the provisions of Sec 10.7.2.1 with $A_w = 2ht$ where h for the width resisting the shear force shall be taken as the clear distance between the flanges less the inside corner radius on each side and $t_w = t$ and $k_y = 5$. If the corner radius is not known, h shall be taken as the corresponding outside dimension minus three times the thickness.

10.7.6 Round HSS

The nominal shear strength, V_n of round HSS, according to the limit states of shear yielding and shear buckling, is

$$V_n = F_{cr} A_g / 2 \quad (6.10.145)$$

Where,

F_{cr} shall be the larger of

$$F_{cr} = \frac{1.60E}{\sqrt{\frac{L_v}{D} \left(\frac{D}{t}\right)^{\frac{5}{4}}}} \quad (6.10.146a)$$

And,

$$F_{cr} = \frac{0.78E}{\left(\frac{D}{t}\right)^{\frac{3}{2}}} \quad (6.10.146b)$$

But shall not exceed $0.6 F_y$

A_g = gross area of section based on design wall thickness, mm^2

D = outside diameter, mm

L_v = the distance from maximum to zero shear force, mm

T = design wall thickness, equal to 0.93 times the nominal wall thickness for ERW HSS and equal to the nominal thickness for SAW HSS, mm

10.7.7 Weak Axis Shear in Singly and Doubly Symmetric Shapes

For singly and doubly symmetric shapes loaded in the weak axis without torsion, the nominal shear strength, V_n , for each shear resisting element shall be determined using Eq. 6.10.136 and Sec 10.7.2.1b with $A_w = b_f t_f$ and $k_v = 1.2$.

10.7.8 Beams and Girders with Web Openings

The effect of all web openings on the nominal shear strength of steel and composite beams shall be determined. Adequate reinforcement shall be provided when the required strength exceeds the available strength of the member at the opening.

10.8 Design of Members for Combined Forces and Torsion

This Section addresses members subject to axial *force* and flexure about one or both axes, with or without torsion, and to members subject to torsion only.

10.8.1 Doubly and Singly Symmetric Members Subject to Flexure and Axial Force**10.8.1.1 Doubly and singly symmetric members in flexure and compression**

The interaction of flexure and compression in doubly symmetric members and singly symmetric members for which $0.1 \leq \left(\frac{I_{yc}}{I_y}\right) \leq 0.9$, that are constrained to bend about a geometric axis (x and/or y) shall be limited by Equations 6.10.147a and 6.10.147b, where I_{yc} is the moment of inertia about the y-axis referred to the compression flange, mm⁴.

For $\frac{P_r}{P_c} \geq 0.2$

$$\frac{P_r}{P_c} + \frac{8}{9} \left(\frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}} \right) \leq 1.0 \quad (6.10.147a)$$

For $\frac{P_r}{P_c} < 0.2$

$$\frac{P_r}{2P_c} + \left(\frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}} \right) \leq 1.0 \quad (6.10.147b)$$

Where,

P_r = required axial compressive strength, N

P_c = available axial compressive strength, N

M_r = required flexural strength, N-mm

M_c = available flexural strength, N-mm

x = subscript relating symbol to strong axis bending

y = subscript relating symbol to weak axis bending

For design according to Sec 10.2.3.3 (LRFD)

P_r = required axial compressive strength using LRFD load combinations, N

$P_c = \phi_c P_n$ = design axial compressive strength, determined in accordance with Sec 10.5, N

M_r = required flexural strength using LRFD load combinations, N-mm

$M_c = \phi_b M_n$ = design flexural strength determined in accordance with Sec 10.6, N-mm

ϕ_c = resistance factor for compression = 0.90

ϕ_b = resistance factor for flexure = 0.90

For design according to Sec 10.2.3.4 (ASD)

P_r = required axial compressive strength using ASD load combinations, N

$P_c = P_n / \Omega_c$ = allowable axial compressive strength, determined in accordance with Sec 10.5, N

M_r = required flexural strength using ASD load combinations, N-mm

$M_c = M_n / \Omega_b$ = allowable flexural strength determined in accordance with Sec 10.6, N-mm

Ω_c = safety factor for compression = 1.67

Ω_b = safety factor for flexure = 1.67

10.8.1.2 Doubly and singly symmetric members in flexure and tension

The interaction of flexure and tension in doubly symmetric members and singly symmetric members constrained to bend about a geometric axis (x and/or y) shall be limited by Eq. 6.10.147a and 6.10.147b,

Where,

For design according to Sec 10.2.3.3 (LRFD)

P_r = required tensile strength using LRFD load combinations, N

$P_c = \phi_t P_n$ = design tensile strength, determined in accordance with Sec 10.4.2, N

M_r = required flexural strength using LRFD load combinations, N-mm

$M_c = \phi_b M_n$ = design flexural strength determined in accordance with Sec 10.6, N-mm

ϕ_t = resistance factor for tension (see Sec 10.4.2)

ϕ_b = resistance factor for flexure = 0.90

For doubly symmetric members, C_b in Sec 10.6 may be increased by $\sqrt{1 + \frac{P_u}{P_{ey}}}$ for axial tension that acts concurrently with flexure,

Where,

$$P_{ey} = \frac{\pi^2 EI_y}{L_b^2}$$

For design according to Sec 10.2.3.4 (ASD)

P_r = required tensile strength using ASD load combinations, N

$P_c = P_n / \Omega_t$ = allowable tensile strength, determined in accordance with Sec 10.4.2, N

M_r = required flexural strength using ASD load combinations, N-mm

$M_c = M_n / \Omega_b$ = allowable flexural strength determined in accordance with Sec 10.6, N-mm

Ω_t = safety factor for tension (see Sec 10.4.2)

Ω_b = safety factor for flexure = 1.67

For doubly symmetric members, C_b in Sec 10.6 may be increased by $\sqrt{1 + \frac{1.5P_a}{P_{ey}}}$ for axial tension that acts concurrently with flexure.

Where,

$$P_{ey} = \frac{\pi^2 EI_y}{L_b^2}$$

A more detailed analysis of the interaction of flexure and tension is permitted in lieu of Equations 6.10.147a and 6.10.147b.

10.8.1.3 Doubly symmetric members in single axis flexure and compression

For doubly symmetric members in flexure and compression with moments primarily in one plane, it is permissible to consider the two independent limit states, in-plane instability and

out-of-plane buckling or flexural-torsional buckling, separately in lieu of the combined approach provided in Sec 10.8.1.1.

- (a) For the limit state of in-plane instability, Eq.6.10.147 shall be used with P_c , M_r , and M_c determined in the plane of bending.
- (b) For the limit state of out-of-plane buckling

$$\frac{P_r}{P_{co}} + \left(\frac{M_r}{M_{cx}} \right)^2 \leq 1.0 \quad (6.10.148)$$

Where,

P_{co} = available compressive strength out of the plane of bending, N

M_{cx} = available flexural-torsional strength for strong axis flexure determined from Sec 10.6, N-mm

If bending occurs only about the weak axis, the moment ratio in Eq.6.10.148 shall be neglected.

For members with significant biaxial moments, $\left(\frac{M_r}{M_c} \right) \geq 0.05$ in both directions, the provisions of Sec 10.8.1.1 shall be followed.

10.8.2 Unsymmetric and Other Members Subject to Flexure and Axial Force

This Section addresses the interaction of flexure and axial *stress* for shapes not covered in Sec 10.8.1. It is permitted to use the provisions of this Section for any shape in lieu of the provisions of Sec 10.8.1.

$$\left| \frac{f_a}{F_a} + \frac{f_{bw}}{F_{bw}} + \frac{f_{bz}}{F_{bz}} \right| \leq 1.0 \quad (6.10.149)$$

Where,

f_a = required axial stress at the point of consideration, MPa

F_a = available axial stress at the point of consideration, MPa

f_{bw}, f_{bz} = required flexural stress at the point of consideration, MPa

F_{bw}, F_{bz} = available flexural stress at the point of consideration, MPa

w = subscript relating symbol to major principal axis bending

z = subscript relating symbol to minor principal axis bending

For design according to Sec 10.2.3.3 (LRFD)

f_a = required axial stress using LRFD load combinations, MPa

$F_a = \phi_c F_{cr}$ = design axial stress, determined in accordance with Sec 10.5 for compression or Sec 10.4.2 for tension, MPa

f_{bw}, f_{bz} = required flexural stress at the specific location in the cross section using LRFD load combinations, MPa

$F_{bw}, F_{bz} = \frac{\phi_b M_n}{S}$ = design flexural stress determined in accordance with Sec 10.6, MPa. Use the section modulus for the specific location in the cross section and consider the sign of the stress.

ϕ_c = resistance factor for compression = 0.90

ϕ_t = resistance factor for tension (Sec 10.4.2)

ϕ_b = resistance factor for flexure = 0.90

For design according to Sec 10.2.3.4 (ASD)

f_a = required axial stress using ASD load combinations, MPa

$F_a = \frac{F_{cr}}{\Omega_c}$ = allowable axial stress determined in accordance with Sec 10.5 for compression, or Sec 10.4.2 for tension, MPa

f_{bw}, f_{bz} = required flexural stress at the specific location in cross section using ASD load combinations, MPa

$F_{bw}, F_{bz} = \frac{M_n}{\Omega_b S}$ = allowable flexural stress determined in accordance with Sec 10.6, MPa. Use the section modulus for the specific location in the cross section and consider the sign of the stress.

Ω_c = safety factor for compression = 1.67

Ω_t = safety factor for tension (Sec 10.4.2)

Ω_b = safety factor for flexure = 1.67

Eq.6.10.149 shall be evaluated using the principal bending axes by considering the sense of the flexural stresses at the critical points of the cross section. The flexural terms are either added to or subtracted from the axial term as appropriate. When the axial *force* is compression, second order effects shall be included according to the provisions of Sec 10.3.

A more detailed analysis of the interaction of flexure and tension is permitted in lieu of Eq.6.10.149.

10.8.3 Members under Torsion and Combined Torsion, Flexure, Shear and/or Axial Force

10.8.3.1 Torsional strength of round and rectangular HSS

The design torsional strength, $\phi_T T_n$ and the allowable torsional strength, T_n/Ω_T , for round and rectangular HSS shall be determined as follows:

$$\phi_T = 0.90 \text{ (LRFD)} \quad \Omega_T = 1.67 \text{ (ASD)}$$

The nominal torsional strength, T_n , according to the limit states of torsional yielding and torsional buckling is:

$$T_n = F_{cr} C \quad (6.10.150)$$

Where,

C is the HSS torsional constant

F_{cr} shall be determined as follows:

For round HSS, F_{cr} shall be the larger of

$$F_{cr} = \frac{1.23E}{\sqrt{\frac{L}{D} \left(\frac{D}{t}\right)^{\frac{5}{4}}}} \quad (6.10.151a)$$

And,

$$F_{cr} = \frac{0.60E}{\left(\frac{D}{t}\right)^{\frac{3}{2}}} \quad (6.10.151b)$$

But, shall not exceed $0.6F_y$,

Where,

L = length of the member, mm

D = outside diameter, mm

For rectangular *HSS*

For $h/t \leq 2.45\sqrt{E/F_y}$

$$F_{cr} = 0.6F_y \quad (6.10.152)$$

For $2.45\sqrt{E/F_y} < h/t \leq 3.07\sqrt{E/F_y}$

$$F_{cr} = 0.6F_y \left(2.45\sqrt{E/F_y} \right) / (h/t) \quad (6.10.153)$$

For $3.07\sqrt{E/F_y} < h/t \leq 260$

$$F_{cr} = 0.458\pi^2 E / (h/t)^2 \quad (6.10.154)$$

10.8.3.2 HSS subject to combined torsion, shear, flexure and axial force

When the required torsional strength, T_r , is less than or equal to 20 percent of the available torsional strength, T_c , the interaction of torsion, shear, flexure and/or axial force for *HSS* shall be determined by Sec 10.8.1 and the torsional effects shall be neglected. When T_r exceeds 20 percent of T_c , the interaction of torsion, shear, flexure and/or axial force shall be limited by

$$\left(\frac{P_r}{P_c} + \frac{M_r}{M_c} \right) + \left(\frac{V_r}{V_c} + \frac{T_r}{T_c} \right)^2 \leq 1.0 \quad (6.10.155)$$

Where,

For design according to Sec 10.2.3.3 (LRFD)

P_r = required axial strength using LRFD load combinations, N

$P_c = \phi P_n$, design tensile or compressive strength in accordance with Sec 10.4 or 10.5, N

M_r = required flexural strength using LRFD load combinations, N-mm

$M_c = \phi_b M_n$, design flexural strength in accordance with Sec 10.6, N-mm

V_r =required shear strength using LRFD load combinations, N

$V_c = \phi_v V_n$, design shear strength in accordance with Sec 10.7, N

T_r =required torsional strength using LRFD load combinations, N-mm

$T_c = \phi_T T_n$, design torsional strength in accordance with Sec 10.8.3.1, N-mm

For design according to Sec 10.2.3.4 (ASD)

P_r = required axial strength using ASD load combinations, N

$P_c = P_n / \Omega$ allowable tensile or compressive strength in accordance with Sec 10.4 or 10.5, N

M_r =required flexural strength using ASD load combinations determined as per Sec10.2.5, N-mm

$M_c = M_n / \Omega_b$, allowable flexural strength in accordance with Sec 10.6, N-mm

V_r = required shear strength using ASD load combinations, N

$V_c = V_n / \Omega_v$, allowable shear strength in accordance with Sec 10.7, N

T_r = required torsional strength using ASD load combinations, N-mm

$T_c = T_n / \Omega_T$, allowable torsional strength in accordance with Sec 10.8.3.1, N-mm

10.8.3.3 Strength of non-HSS members under torsion and combined stress

The design torsional strength, $\phi_T F_n$, and the allowable torsional strength, F_n / Ω_T , for non-HSS members shall be the lowest value obtained according to the limit states of yielding under normal stress, shear yielding under shear stress, or buckling, determined as follows:

$$\phi_T = 0.90 \text{ (LRFD)} \quad \Omega_T = 1.67 \text{ (ASD)}$$

- (a) For the limit state of yielding under normal stress

$$F_n = F_y \quad (6.10.156)$$

- (b) For the limit state of shear yielding under shear stress

$$F_n = 0.6F_y \quad (6.10.157)$$

- (c) For the limit state of buckling

$$F_n = F_{cr} \quad (6.10.158)$$

Where,

F_{cr} = buckling stress for the section as determined by analysis, MPa.

Some constrained local yielding is permitted adjacent to areas that remain elastic.

10.9 Evaluation of Existing Structures

This Section applies to the evaluation of the strength and stiffness under static vertical (gravity) loads of existing structures by structural analysis, by load tests, or by a combination of structural analysis and load tests when specified by the engineer of record or in the contract documents. For such evaluation, the steel grades are not limited to those listed in Sec 10.1.3.1. This Section does not address load testing for the effects of seismic loads or moving loads (vibrations).

10.9.1 General Provisions

These provisions shall be applicable when the evaluation of an existing steel structure is specified for (a) verification of a specific set of design loadings or (b) determination of the available strength of a load resisting member or system. The evaluation shall be performed by structural analysis (Sec 10.9.3), by load tests (Sec 10.9.4), or by a combination of structural analysis and load tests, as specified in the contract documents. Where load tests are used, the engineer of record shall first analyze the structure, prepare a testing plan, and develop a written procedure to prevent excessive permanent deformation or catastrophic collapse during testing.

10.9.2 Material Properties

Determination of Required Tests: The engineer of record shall determine the specific tests that are required from Sections 10.9.2.2 to 10.9.2.6 and specify the locations where they are required. Where available, the use of applicable project records shall be permitted to reduce or eliminate the need for testing.

10.9.2.1 Tensile properties

Tensile properties of members shall be considered in evaluation by structural analysis (Sec 10.9.3) or load tests (Sec 10.9.4). Such properties shall include the yield stress, tensile strength and percent elongation. Where available, certified mill test reports or certified reports of tests made by the fabricator or a testing laboratory in accordance with ASTM A6/A6M or A568/A568M, as applicable, shall be permitted for this purpose. Otherwise, tensile tests shall be conducted in accordance with ASTM A370 from samples cut from components of the structure.

10.9.2.2 Chemical composition

Where welding is anticipated for repair or modification of existing structures, the chemical composition of the steel shall be determined for use in preparing a welding procedure specification (WPS). Where available, results from certified mill test reports or certified reports of tests made by the fabricator or a testing laboratory in accordance with ASTM procedures shall be permitted for this purpose. Otherwise, analyses shall be conducted in accordance with ASTM A751 from the samples used to determine tensile properties, or from samples taken from the same locations.

10.9.2.3 Base metal notch toughness

Where welded tension splices in heavy shapes and plates as defined in Sec 10.1.3.1(d) are critical to the performance of the structure, the Charpy V-Notch toughness shall be determined in accordance with the provisions of Sec 10.1.3.1(d). If the notch toughness so determined does not meet the provisions of Sec 10.1.3.1(d), the engineer of record shall determine if remedial actions are required.

10.9.2.4 Weld metal

Where structural performance is dependent on existing welded connections, representative samples of weld metal shall be obtained. Chemical analysis and mechanical tests shall be made to characterize the weld metal. A determination shall be made of the magnitude and consequences of imperfections. If the requirements of AWS D1.1 are not met, the engineer of record shall determine if remedial actions are required.

10.9.2.5 Bolts and rivets

Representative samples of bolts shall be inspected to determine markings and classifications. Where bolts cannot be properly identified visually, representative samples shall be removed and tested to determine tensile strength in accordance with ASTM F606 or ASTM F606M and the bolt classified accordingly. Alternatively, the assumption that the bolts are ASTM A307 shall be permitted. Rivets shall be assumed to be ASTM A502, Grade 1, unless a higher grade is established through documentation or testing.

10.9.3 Evaluation by Structural Analysis

10.9.3.1 Dimensional data

All dimensions used in the evaluation, such as spans, column heights, member spacings, bracing locations, cross-section dimensions, thicknesses and connection details, shall be determined from a field survey. Alternatively, when available, it shall be permitted to determine such dimensions from applicable project design or shop drawings with field verification of critical values.

10.9.3.2 Strength evaluation

Forces (load effects) in members and connections shall be determined by structural analysis applicable to the type of structure evaluated. The load effects shall be determined for the loads and factored load combinations stipulated in Sec 10.2.2.

The available strength of members and connections shall be determined from applicable provisions of Sections 10.2 to 10.19 of this Specification.

10.9.3.3 Serviceability evaluation

Where required, the deformations at service loads shall be calculated and reported.

10.9.4 Evaluation by Load Tests

10.9.4.1 Determination of load rating by testing

To determine the load rating of an existing floor or roof structure by testing, a test load shall be applied incrementally in accordance with the engineer of record's plan. The structure shall be visually inspected for signs of distress or imminent failure at each load level. Appropriate measures shall be taken if these or any other unusual conditions are encountered.

The tested strength of the structure shall be taken as the maximum applied test load plus the in-situ dead load. The live load rating of a floor structure shall be determined by setting the tested strength equal to $1.2 D + 1.6L$, where D is the nominal dead load and L is the nominal live load rating for the structure. The nominal live load rating of the floor structure shall not exceed that which can be calculated using applicable provisions of the specification. For roof structures, L_r , S , or R as defined in the Symbols, shall be substituted for L . More severe load combinations shall be used if required by specifications of Chapter 2 Part 6.

Periodic unloading shall be considered once the service load level is attained and after the onset of inelastic structural behavior is identified to document the amount of permanent set

and the magnitude of the inelastic deformations. Deformations of the structure, such as member deflections, shall be monitored at critical locations during the test, referenced to the initial position before loading. It shall be demonstrated, while maintaining maximum test load for one hour that the deformation of the structure does not increase by more than 10 percent above that at the beginning of the holding period. It is permissible to repeat the sequence if necessary to demonstrate compliance.

Deformations of the structure shall also be recorded 24 hours after the test loading is removed to determine the amount of permanent set. Because the amount of acceptable permanent deformation depends on the specific structure, no limit is specified for permanent deformation at maximum loading. Where it is not feasible to load test the entire structure, a segment or zone of not less than one complete bay, representative of the most critical conditions, shall be selected.

10.9.4.2 Serviceability evaluation

When load tests are prescribed, the structure shall be loaded incrementally to the service load level. Deformations shall be monitored for a period of one hour. The structure shall then be unloaded and the deformation recorded.

10.9.5 Evaluation Report

After the evaluation of an existing structure has been completed, the engineer of record shall prepare a report documenting the evaluation. The report shall indicate whether the evaluation was performed by structural analysis, by load testing or by a combination of structural analysis and load testing. Furthermore, when testing is performed, the report shall include the loads and load combination used and the load-deformation and time-deformation relationships observed. All relevant information obtained from design drawings, mill test reports and auxiliary material testing shall also be reported. Finally, the report shall indicate whether the structure, including all members and connections, is adequate to withstand the load effects.

10.10 Connections

This Section addresses connecting elements, connectors, and the affected elements of the connected members not subject to fatigue loads.

10.10.1 General Provisions

10.10.1.1 Design basis

The design strength, ϕR_n and the allowable strength $\frac{R_n}{\Omega}$, of connections shall be determined in accordance with the provisions of this Section and the provisions of Sec 10.2.

The required strength of the connections shall be determined by structural analysis for the specified design loads, consistent with the type of construction specified, or shall be a proportion of the required strength of the connected members when so specified herein.

Where the gravity axes of intersecting axially loaded members do not intersect at one point, the effects of eccentricity shall be considered.

10.10.1.2 Simple connection

Simple connections of beams, girders, or trusses shall be designed as flexible and are permitted to be proportioned for the reaction shears only, except as otherwise indicated in the design documents. Flexible beam connections shall accommodate end rotations of simple beams. Some inelastic, but self-limiting deformation in the connection is permitted to accommodate the end rotation of a simple beam.

10.10.1.3 Moment connection

End connections of restrained beams, girders, and trusses shall be designed for the combined effect of forces resulting from moment and shear induced by the rigidity of the connections. Response criteria for moment connections are provided in Sec 10.2.6.3.2.

10.10.1.4 Compression members with bearing joints

- (a) When columns bear on bearing plates or are finished to bear at splices, there shall be sufficient connectors to hold all parts securely in place.
- (b) When compression members other than columns are finished to bear, the splice material and its connectors shall be arranged to hold all parts in line and shall be proportioned for either (i) or (ii) below. It is permissible to use the less severe of the two conditions:
 - (i) An axial tensile force of 50 percent of the required compressive strength of the member; or
 - (ii) The moment and shear resulting from a transverse load equal to 2 percent of the required compressive strength of the member. The transverse load shall be applied at the location of the splice exclusive of other loads that act on the member. The member shall be taken as pinned for the determination of the shears and moments at the splice.

10.10.1.5 Splices in heavy sections

When tensile forces due to applied tension or flexure are to be transmitted through splices in heavy sections, as defined in Sections 10.1.3.1(c) and 10.1.3.1(d), by complete-joint-penetration groove (CJP) welds, material notch-toughness requirements as given in

Sections 10.1.3.1(c) and 10.1.3.1(d), weld access hole details as given in Sec 10.10.1.6 and thermal cut surface preparation and inspection requirements as given in Sec 10.1.3.2.2 shall apply. The foregoing provision is not applicable to splices of elements of built-up shapes that are welded prior to assembling the shape.

10.10.1.6 Beam copes and weld access holes

All weld access holes required to facilitate welding operations shall have a length from the toe of the weld preparation not less than $1\frac{1}{2}$ times the thickness of the material in which the hole is made. The height of the access hole shall be $1\frac{1}{2}$ times the thickness of the material with the access hole, t_w , but not less than 25 mm nor does it need to exceed 50 mm. The access hole shall be detailed to provide room for weld backing as needed.

For sections that are rolled or welded prior to cutting, the edge of the web shall be sloped or curved from the surface of the flange to the reentrant surface of the access hole. In hot-rolled shapes, and built-up shapes with CJP groove welds that join the web-to-flange, all beam copes and weld access holes shall be free of notches and sharp reentrant corners. No arc of the weld access hole shall have a radius less than 10 mm.

In built-up shapes with fillet or partial-joint-penetration groove welds that join the web-to-flange, all beam copes and weld access holes shall be free of notches and sharp reentrant corners. The access hole shall be permitted to terminate perpendicular to the flange, providing the weld is terminated at least a distance equal to the weld size away from the access hole.

For heavy sections as defined in Sections 10.1.3.1(c) and 10.1.3.1(d), the thermally cut surfaces of beam copes and weld access holes shall be ground to bright metal and inspected by either magnetic particle or dye penetrant methods prior to deposition of splice welds. If the curved transition portion of weld access holes and beam copes are formed by predrilled or sawed holes, that portion of the access hole or cope need not be ground. Weld access holes and beam copes in other shapes need not be ground nor inspected by dye penetrant or magnetic particle methods.

10.10.1.7 Placement of welds and bolts

Groups of welds or bolts at the ends of any member which transmit axial force into that member shall be sized so that the center of gravity of the group coincides with the center of gravity of the member, unless provision is made for the eccentricity. The foregoing provision is not applicable to end connections of statically loaded single angle, double angle, and similar members.

10.10.1.8 Bolts in combination with welds

Bolts shall not be considered as sharing the load in combination with welds, except that shear connections with any grade of bolts permitted by Sec 10.1.3.3 installed in standard holes or short slots transverse to the direction of the load are permitted to be considered to share the load with longitudinally loaded fillet welds. In such connections the available strength of the bolts shall not be taken as greater than 50 percent of the available strength of bearing-type bolts in the connection.

In making welded alterations to structures, existing rivets and high strength bolts tightened to the requirements for slip-critical connections are permitted to be utilized for carrying loads present at the time of alteration and the welding need only provide the additional required strength.

10.10.1.9 High-strength bolts in combination with rivets

In both new work and alterations, in connections designed as slip-critical connections in accordance with the provisions of Sec 10.10.3, high-strength bolts are permitted to be considered as sharing the load with existing rivets.

10.10.1.10 Limitations on bolted and welded connections

Pretensioned joints, slip-critical joints or welds shall be used for the following connections:

Column splices in all multi-story structures over 38 m in height

Connections of all beams and girders to columns and any other beams and girders on which the bracing of columns is dependent in structures over 38 m in height

In all structures carrying cranes of over 50 kN capacity: roof truss splices and connections of trusses to columns, column splices, column bracing, knee braces, and crane supports

Connections for the support of machinery and other live loads that produce impact or reversal of load

Snug-tightened joints or joints with ASTM A307 bolts shall be permitted except where otherwise specified.

10.10.2 Welds

All provisions of AWS D1.1 apply under this specification, with the exception that the provisions of the listed Sections apply under this specification in lieu of the cited AWS provisions as follows:

Sec 10.10.1.6 in lieu of AWS D1.1 Section 5.17.1

Sec 10.10.2.2.1 in lieu of AWS D1.1 Section 2.3.2

Table 6.10.5 in lieu of AWS D1.1 Table 2.1

Table 6.10.8 in lieu of AWS D1.1 Table 2.3

Table 6.10.14 in lieu of AWS D1.1 Table 2.4

Sec 10.2.3.9 and Sec 10.17 in lieu of AWS D1.1 Section 2, Part C

Sec 10.13.2.2 in lieu of AWS D1.1 Sections 5.15.4.3 and 5.15.4.4

10.10.2.1 Groove welds**10.10.2.1.1 Effective area**

The effective area of groove welds shall be considered as the length of the weld times the effective throat thickness.

The effective throat thickness of a complete-joint-penetration (CJP) groove weld shall be the thickness of the thinner part joined. The effective throat thickness of a partial-joint-penetration (PJP) groove weld shall be as shown in Table 6.10.4.

The effective weld size for flare groove welds, when filled flush to the surface of a round bar, a 90° bend in a formed section, or rectangular HSS shall be as shown in Table 6.10.5, unless other effective throats are demonstrated by tests. The effective size of flare groove welds filled less than flush shall be as shown in Table 6.10.5, less the greatest perpendicular dimension measured from a line flush to the base metal surface to the weld surface.

Larger effective throat thicknesses than those in Table 6.10.5 are permitted, provided the fabricator can establish by qualification the consistent production of such larger effective throat thicknesses. Qualification shall consist of sectioning the weld normal to its axis, at mid-length and terminal ends. Such sectioning shall be made on a number of combinations of material sizes representative of the range to be used in the fabrication.

10.10.2.1.2 Limitations

The minimum effective throat thickness of a partial-joint-penetration groove weld shall not be less than the size required to transmit calculated forces nor the size shown in Table 6.10.6. Minimum weld size is determined by the thinner of the two parts joined.

10.10.2.2 Fillet welds

10.10.2.2.1 Effective area

The effective area of a fillet weld shall be the effective length multiplied by the effective throat. The effective throat of a fillet weld shall be the shortest distance from the root to the face of the diagrammatic weld. An increase in effective throat is permitted if consistent penetration beyond the root of the diagrammatic weld is demonstrated by tests using the production process and procedure variables.

For fillet welds in holes and slots, the effective length shall be the length of the centerline of the weld along the center of the plane through the throat. In the case of overlapping fillets, the effective area shall not exceed the nominal cross-sectional area of the hole or slot, in the plane of the faying surface.

10.10.2.2.2 Limitations

The minimum size of fillet welds shall be not less than the size required to transmit calculated forces nor the size as shown in Table 6.10.7. These provisions do not apply to fillet weld reinforcements of partial- or complete-joint-penetration groove welds.

Table 6.10.4: Effective Throat of Partial-Joint-Penetration Groove Welds

Welding Process	Welding Position F (flat), H (horiz.), V (vert.), OH (overhead)	Groove Type (AWS D1.1, Figure 3.3)	Effective Throat
Shielded Metal Arc (SMAW)	All	J or U Groove 60° V	Depth of Groove
Gas Metal Arc (GMAW)	All		
Flux Cored Arc (FCAW)			
Submerged Arc (SAW)	F	J or U Groove 60° Bevel or V	Depth of Groove
Gas Metal Arc (GMAW)	F, H	45° Bevel	
Flux Cored Arc (FCAW)			

Welding Process	Welding Position F (flat), H (horiz.), V (vert.), OH (overhead)	Groove Type (AWS D1.1, Figure 3.3)	Effective Throat
Shielded Metal Arc (SMAW)	All	45° Bevel	Depth of Groove Minus 3 mm
Gas Metal Arc (GMAW) Flux Cored Arc (FCAW)	V, OH	45° Bevel	Depth of Groove Minus 3 mm

Table 6.10.5: Effective Weld Sizes of Flare Groove Welds

Welding Process	Flare Bevel Groove ^[a]	Flare V-Groove
GMAW and FCAW-G	$\frac{5}{8}R$	$\frac{3}{4}R$
SMAW and FCAW-S	$\frac{5}{16}R$	$\frac{5}{8}R$
SAW	$\frac{5}{16}R$	$\frac{1}{2}R$

^[a] For Flare Bevel Groove with $R < 10$ mm use only reinforcing fillet weld on filled flush joint. General Note: R = radius of joint surface (can be assumed to be $2t$ for HSS), mm

Table 6.10.6: Minimum Effective Throat Thickness of Partial-Joint-Penetration Groove Welds

Material Thickness of Thinner Part Joined (mm)	Minimum Effective Throat Thickness, ^[a] (mm)
To 6 inclusive	3
Over 6 to 13	5
Over 13 to 19	6
Over 19 to 38	8
Over 38 to 57	10
Over 57 to 150	13
Over 150	16

^[a] See Table 6.10.4.

Table 6.10.7: Minimum Size of Fillet Welds

Material Thickness of Thinner Part Joined (mm)	Minimum Size of Fillet Weld,^[a] (mm)
To 6 inclusive	3
Over 6 to 13	5
Over 13 to 19	6
Over 19	8

[a] Leg dimension of fillet welds. Single pass welds must be used.

Note: See Sec 10.10.2.2.2 for maximum size of fillet welds.

The maximum size of fillet welds of connected parts shall be:

Along edges of material less than 6 mm thick, not greater than the thickness of the material.

Along edges of material 6 mm or more in thickness, not greater than the thickness of the material minus 2 mm, unless the weld is especially designated on the drawings to be built out to obtain full-throat thickness. In the as-welded condition, the distance between the edge of the base metal and the toe of the weld is permitted to be less than 2 mm provided the weld size is clearly verifiable.

The minimum effective length of fillet welds designed on the basis of strength shall be not less than four times the nominal size, or else the size of the weld shall be considered not to exceed $1/4^{\text{th}}$ of its effective length. If longitudinal fillet welds are used alone in end connections of flat-bar tension members, the length of each fillet weld shall be not less than the perpendicular distance between them. For the effect of longitudinal fillet weld length in end connections upon the effective area of the connected member, see Sec 10.4.3.3.

For end-loaded fillet welds with a length up to 100 times the leg dimension, it is permitted to take the effective length equal to the actual length. When the length of the end-loaded fillet weld exceeds 100 times the weld size, the effective length shall be determined by multiplying the actual length by the reduction factor, β ,

$$\beta = 1.2 - 0.002(L/w) \leq 1.0 \quad (6.10.159)$$

Where,

L = actual length of end-loaded weld, mm

w = weld leg size, mm

When the length of the weld exceeds 300 times the leg size, the value of β shall be taken as 0.60.

Intermittent fillet welds are permitted to be used to transfer calculated stress across a joint or faying surfaces when the required strength is less than that developed by a continuous fillet weld of the smallest permitted size, and to join components of built-up members. The effective length of any segment of intermittent fillet welding shall be not less than four times the weld size, with a minimum of 38 mm.

In lap joints, the minimum amount of lap shall be five times the thickness of the thinner part joined, but not less than 25 mm. Lap joints joining plates or bars subjected to axial stress that utilize transverse fillet welds only shall be fillet welded along the end of both lapped parts, except where the deflection of the lapped parts is sufficiently restrained to prevent opening of the joint under maximum loading.

Fillet weld terminations are permitted to be stopped short or extend to the ends or sides of parts or be boxed except as limited by the following:

For lap joints in which one connected part extends beyond an edge of another connected part that is subject to calculated tensile stress, fillet welds shall terminate not less than the size of the weld from that edge.

For connections where flexibility of the outstanding elements is required, when end returns are used, the length of the return shall not exceed four times the nominal size of the weld nor half the width of the part.

Fillet welds joining transverse stiffeners to plate girder webs 19 mm thick or less shall end not less than four times nor more than six times the thickness of the web from the web toe of the web-to-flange welds, except where the ends of stiffeners are welded to the flange.

Fillet welds that occur on opposite sides of a common plane shall be interrupted at the corner common to both welds.

Fillet welds in holes or slots are permitted to be used to transmit shear in lap joints or to prevent the buckling or separation of lapped parts and to join components of built-up members. Such fillet welds may overlap, subject to the provisions of Sec 10.10.2. Fillet welds in holes or slots are not to be considered plug or *slot welds*.

10.10.2.3 Plug and slot welds

10.10.2.3.1 Effective area

The effective shearing area of plug and slot welds shall be considered as the nominal cross-sectional area of the hole or slot in the plane of the faying surface.

10.10.2.3.2 Limitations

Plug or slot welds are permitted to be used to transmit shear in lap joints or to prevent buckling of lapped parts and to join component parts of built-up members.

The diameter of the holes for a plug weld shall not be less than the thickness of the part containing it plus 8 mm, rounded to the next larger odd even mm, nor greater than the minimum diameter plus 3 mm or $2\frac{1}{4}$ times the thickness of the weld.

The minimum center-to-center spacing of plug welds shall be four times the diameter of the hole.

The length of slot for a slot weld shall not exceed 10 times the thickness of the weld. The width of the slot shall be not less than the thickness of the part containing it plus 8 mm rounded to the next larger odd even mm, nor shall it be larger than $2\frac{1}{4}$ times the thickness of the weld. The ends of the slot shall be semicircular or shall have the corners rounded to a radius of not less than the thickness of the part containing it, except those ends which extend to the edge of the part.

The minimum spacing of lines of slot welds in a direction transverse to their length shall be four times the width of the slot. The minimum center-to-center spacing in a longitudinal direction on any line shall be two times the length of the slot.

The thickness of plug or slot welds in material 16 mm or less in thickness shall be equal to the thickness of the material. In material over 16 mm thick, the thickness of the weld shall be at least one-half the thickness of the material but not less than 16 mm.

10.10.2.4 Strength

The design strength, ϕR_n and the allowable strength, R_n/Ω , of welds shall be the lower value of the base material and the weld metal strength determined according to the limit states of tensile rupture, shear rupture or yielding as follows:

Table 6.10.8: Available Strength of Welded Joints, N

Load Type and Direction Relative to Weld Axis	Pertinent Metal	ϕ and Ω	Nominal Strength (F_{BM} or F_W), N	Effective Area (A_{BM} or A_W), mm ²	Required Filler Metal Strength Level ^{[a][b]}
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COMPLETE-JOINT-PENETRATION GROOVE WELDS

Tension Normal to weld axis	Strength of the joint is controlled by the base metal			Matching filler metal shall be used. For T and corner joints with backing left in place, notch tough filler metal is required. Sec 10.10.2.6.	
Compression Normal to weld axis	Strength of the joint is controlled by the base metal			Filler metal with a strength level equal to or one strength level less than matching filler metal is permitted.	
Tension or Compression Parallel to weld axis	Tension or compression in parts joined parallel to a weld need not be considered in design of welds joining the parts.			Filler metal with a strength level equal to or less than matching filler metal is permitted.	
Shear	Strength of the joint is controlled by the base metal			Matching filler metal shall be used. ^[c]	

PARTIAL-JOINT-PENETRATION GROOVE WELDS INCLUDING FLARE VEE GROOVE AND FLARE BEVEL GROOVE WELDS

Tension Normal to weld axis	Base	$\phi = 0.90$ $\Omega = 1.67$	F_y	See 10.10.4	Filler metal with a strength level equal to or less than matching filler metal is permitted.
	Weld	$\phi = 0.80$ $\Omega = 1.88$	$0.60 F_{EXX}$	See 10.10.2.1.1	

Load Type and Direction Relative to Weld Axis	Pertinent Metal	ϕ and Ω	Nominal Strength (F_{BM} or F_W), N	Effective Area (A_{BM} or A_W), mm ²	Required Filler Metal Strength Level ^{[a][b]}
Compression Column to Base Plate and column splices designed per 10.10.1.4(a)	Compressive stress need not be considered in design of welds joining the parts.				Filler metal with a strength level equal to or less than matching filler metal is permitted.
Compression Connections of members designed to bear other than columns as described in 10.10.1.4(b)	Base	$\phi = 0.90$ $\Omega = 1.67$	F_y	See 10.10.4	
	Weld	$\phi = 0.80$ $\Omega = 1.88$	$0.60 F_{EXX}$	See 10.10.2.1.1	
Compression Connections not finished-to-bear	Base	$\phi = 0.90$ $\Omega = 1.67$	F_y	See 10.10.4	
	Weld	$\phi = 0.80$ $\Omega = 1.88$	$0.90 F_{EXX}$	See 10.10.2.1.1	
Tension or Compression Parallel to weld axis	Tension or compression in parts joined parallel to a weld need not be considered in design of welds joining the parts.				
Shear	Base	Governed by 10.10.4			
	Weld	$\phi = 0.75$ $\Omega = 2.00$	$0.60 F_{EXX}$	See 10.10.2.1.1	

Load Type and Direction Relative to Weld Axis	Pertinent Metal	ϕ and Ω	Nominal Strength (F_{BM} or F_W), N	Effective Area (A_{BM} or A_W), mm ²	Required Filler Metal Strength Level ^{[a][b]}
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FILLET WELDS INCLUDING FILLETS IN HOLES AND SLOTS AND SKEWED T-JOINTS

Shear	Base	Governed by 10.10.4			Filler metal with a strength level equal to or less than matching filler metal is permitted.
	Weld	$\phi = 0.75$ $\Omega = 2.00$	$0.60F_{EXX}^{[d]}$	See 10.10.2.2.1	
Tension or Compression Parallel to weld axis	Tension or compression in parts joined parallel to a weld need not be considered in design of welds joining the parts.				

PLUG AND SLOT WELDS

Shear Parallel to faying surface on the effective area	Base	Governed by 10.10.4			Filler metal with a strength level equal to or less than matching filler metal is permitted.
	Weld	$\phi = 0.75$ $\Omega = 2.00$	$0.60F_{EXX}^{[d]}$	10.10.2.3.1	

^[a] For matching weld metal see AWS D1.1, Section 3.3.

^[b] Filler metal with a strength level one strength level greater than matching is permitted.

^[c] Filler metals with a strength level less than matching may be used for groove welds between the webs and flanges of built-up sections transferring shear loads, or in applications where high restraint is a concern. In these applications, the weld joint shall be detailed and the weld shall be designed using the thickness of the material as the effective throat, $\phi = 0.80, \Omega = 1.88$ and $0.60 F_{EXX}$ as the nominal strength.

^[d] Alternatively, the provisions of Sec 10.10.2.4 (a) are permitted provided the deformation compatibility of the various weld elements is considered. Alternatively, Sections 10.10.2.4 (b) and (c) are special applications of Sec 10.10.2.4 (a) that provide for deformation compatibility.

For the base metal

$$R_n = F_{BM} A_{BM} \quad (6.10.160)$$

For the weld metal

$$R_n = F_w A_w \quad (6.10.161)$$

Where,

F_{BM} = nominal strength of the base metal per unit area, MPa

F_w = nominal strength of the weld metal per unit area, MPa

A_{BM} = cross-sectional area of the base metal, mm²

A_w = effective area of the weld, mm²

The values of ϕ , Ω , F_{BM} , and F_w and limitations thereon are given in Table 6.10.8.

Alternatively, for fillet welds loaded in-plane the design strength, ϕR_n and the allowable strength, R_n/Ω of welds is permitted to be determined as follows:

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

(a) For a linear weld group loaded in-plane through the center of gravity

$$R_n = F_w A_w \quad (6.10.162)$$

Where,

$$F_w = 0.60 F_{EXX} (1.0 + 0.50 \sin^{1.5} \theta) \quad (6.10.163)$$

And,

F_{EXX} = electrode classification number, MPa

θ = angle of loading measured from the weld longitudinal axis, degrees

A_w = effective area of the weld, mm²

- (b) For weld elements within a weld group that are loaded in-plane and analyzed using an instantaneous center of rotation method, the components of the nominal strength, R_{nx} and R_{ny} , are permitted to be determined as follows:

$$R_{nx} = \sum F_{wix} A_{wi} \quad R_{ny} = \sum F_{wiy} A_{wi} \quad (6.10.164)$$

Where,

A_{wi} = effective area of weld throat of any i^{th} weld element, mm^2

$$F_{wi} = 0.60 F_{EXX} (1.0 + 0.50 \sin^{1.5} \theta) f(p) \quad (6.10.165)$$

$$f(p) = [p(1.9 - 0.9p)]^{0.3} \quad (6.10.166)$$

F_{wi} = nominal stress in any i^{th} weld element, MPa

F_{wix} = x component of stress, F_{wi}

F_{wiy} = y component of stress, F_{wi}

$\rho = \frac{\Delta_i}{\Delta_m}$, ratio of element i deformation to its deformation at maximum stress

w = weld leg size, mm

r_{crit} = distance from instantaneous center of rotation to weld element with minimum Δ_u/r_i ratio, mm

Δ_I = deformation of weld elements at intermediate stress levels, linearly proportioned to the critical deformation based on distance from the instantaneous center of rotation, $r_i(\text{mm}) = \frac{r_i \Delta_u}{r_{crit}}$.

$\Delta_m = 0.209(\theta + 2)^{-0.32} w$ deformation of weld element at maximum stress, mm

$\Delta_u = 1.087(\theta + 6)^{-0.65} w \leq 0.17w$, deformation of weld element at ultimate stress (fracture), usually in element furthest from instantaneous center of rotation, mm

- (c) For fillet weld groups concentrically loaded and consisting of elements that are oriented both longitudinally and transversely to the direction of applied load, the combined strength, R_n , of the fillet weld group shall be determined as the greater of

$$R_n = R_{wl} + R_{wt} \quad (6.10.167a)$$

Or,

$$R_n = 0.85R_{wl} + 1.5R_{wt} \quad (6.10.167b)$$

Where,

R_{wl} = the total nominal strength of longitudinally loaded fillet welds, as determined in accordance with Table 6.10.8, N

R_{wt} = the total nominal strength of transversely loaded fillet welds, as determined in accordance with Table 6.10.8 without the alternate in Sec 10.10.2.4(a), N

10.10.2.5 Combination of welds

If two or more of the general types of welds (groove, fillet, plug, slot) are combined in a single *joint*, the strength of each shall be separately computed with reference to the axis of the group in order to determine the strength of the combination.

10.10.2.6 Filler metal requirements

The choice of electrode for use with complete-joint-penetration groove welds subject to tension normal to the effective area shall comply with the requirements for matching filler metals given in AWS D1.1.

Filler metal with a specified Charpy V-Notch (CVN) toughness of 27.12 N-m (27 Joule) at 4°C shall be used in the following joints:

Complete-joint-penetration groove welded T and corner joints with steel backing left in place, subject to tension normal to the effective area, unless the joints are designed using the nominal strength and resistance factor or safety factor as applicable for a PJP weld.

Complete-joint-penetration groove welded splices subject to tension normal to the effective area in heavy sections as defined in Sections 10.1.3.1(c) and 10.1.3.1(d).

10.10.2.7 Mixed weld metal

When Charpy V-Notch toughness is specified, the process consumables for all weld metal, tack welds, root pass and subsequent passes deposited in a joint shall be compatible to ensure notch-tough composite weld metal.

10.10.3 Bolts and Threaded Parts

10.10.3.1 High-strength bolts

Use of high-strength bolts shall conform to the provisions of the Specification for Structural Joints Using ASTM A325 or A490 Bolts, hereafter referred to as the RCSC Specification, as approved by the Research Council on Structural Connections, except as otherwise provided in this Specification.

When assembled, all joint surfaces, including those adjacent to the washers, shall be free of scale, except tight mill scale. All ASTM A325/A325M and A490/A490M bolts shall be tightened to a bolt tension not less than given in Table 6.10.9, except as noted below. Except as permitted below, installation shall be assured by any of the following methods: turn-of-nut method, a direct tension indicator, calibrated wrench or alternative design bolt.

- (a) Bolts are permitted to be installed to only the snug-tight condition when used in bearing-type connections.
- (b) Tension or combined shear and tension applications, for ASTM A325 or A325M bolts only, where loosening or fatigue due to vibration or load fluctuations are not design considerations.

Table 6.10.9: Minimum Bolt Pretension, kN*

Bolt Size, mm	A325M Bolts	A490M Bolts
M16	91	114
M20	142	179
M22	176	221
M24	205	257
M27	267	334
M30	326	408
M36	475	595

* Equal to 0.70 times the minimum tensile strength of bolts, rounded off to nearest kN, as specified in ASTM specifications for A325M and A490M bolts with UNC threads.

The snug-tight condition is defined as the tightness attained by either a few impacts of an impact wrench or the full effort of a worker with an ordinary spud wrench that brings the connected plies into firm contact. Bolts to be tightened only to the snug-tight condition shall be clearly identified on the design and erection drawings.

When ASTM A490 or A490M bolts over 25 mm in diameter are used in slotted or oversized holes in external plies, a single hardened washer conforming to ASTM F436, except with 8 mm minimum thickness, shall be used in lieu of the standard washer.

In slip-critical connections in which the direction of loading is toward an edge of a connected part, adequate available bearing strength shall be provided based upon the applicable requirements of Sec 10.10.3.10.

When bolt requirements cannot be provided by ASTM A325/A325M, F1852, or A490/A490M bolts because of requirements for lengths exceeding 12 diameters or diameters exceeding 38 mm, bolts or threaded rods conforming to ASTM A354 Grade BC, A354 Grade BD, or A449 are permitted to be used in accordance with the provisions for threaded rods in Table 6.10.10.

When ASTM A354 Grade BC, A354 Grade BD, or A449 bolts and threaded rods are used in slip-critical connections, the bolt geometry including the head and nut(s) shall be equal to or (if larger in diameter) proportional to that provided by ASTM A325/A325M, or ASTM A490/A490M bolts. Installation shall comply with all applicable requirements of the RCSC Specification with modifications as required for the increased diameter and/or length to provide the design pretension.

10.10.3.2 Size and use of holes

The maximum sizes of holes for bolts are given in Table 6.10.11, except that larger holes, required for tolerance on location of anchor rods in concrete foundations, are permitted in column base details.

Standard holes or short-slotted holes transverse to the direction of the load shall be provided in accordance with the provisions of this specification, unless over-sized holes, short-slotted holes parallel to the load or long-slotted holes are approved by the engineer of record. Finger shims up to 6 mm are permitted in slip-critical connections designed on the basis of standard holes without reducing the nominal shear strength of the fastener to that specified for slotted holes.

Oversized holes are permitted in any or all plies of slip-critical connections, but they shall not be used in bearing-type connections. Hardened washers shall be installed over oversized holes in an outer ply.

Short-slotted holes are permitted in any or all plies of slip-critical or bearing-type connections. The slots are permitted without regard to direction of loading in slip-critical connections, but the length shall be normal to the direction of the load in bearing-type connections. Washers shall be installed over short-slotted holes in an outer ply; when high-strength bolts are used, such washers shall be hardened.

Long-slotted holes are permitted in only one of the connected parts of either a slip-critical or bearing-type connection at an individual faying surface. Long-slotted holes are permitted without regard to direction of loading in slip-critical connections, but shall be normal to the direction of load in bearing-type connections. Where long-slotted holes are

used in an outer ply, plate washers, or a continuous bar with standard holes, having a size sufficient to completely cover the slot after installation, shall be provided. In high-strength bolted connections, such plate washers or continuous bars shall be not less than 8 mm thick and shall be of structural grade material, but need not be hardened. If hardened washers are required for use of high-strength bolts, the hardened washers shall be placed over the outer surface of the plate washer or bar.

Table 6.10.10: Nominal Stress of Fasteners and Threaded Parts, MPa

Description of Fasteners	Nominal Tensile Stress, F_{nt} , MPa	Nominal Shear Stress in Bearing-Type Connections, F_{nv} , MPa
A307 bolts	310 [a][b]	165 [b] [c] [f]
A325/A325M bolts, when threads are not excluded from shear planes	620 [e]	330 [f]
A325/A325M bolts, when threads are excluded from shear planes	620 [e]	414 [f]
A490/A490M bolts, when threads are not excluded from shear planes	780 [e]	414 [f]
A490/A490M bolts, when threads are excluded from shear planes	780 [e]	520 [f]
Threaded parts meeting the requirements of Sec 10.1.3.4, when threads are not excluded from shear planes	$0.75 F_u$ [a][d]	$0.40 F_u$
Threaded parts meeting the requirements of Sec 10.1.3.4, when threads are excluded from shear planes	$0.75 F_u$ [a][d]	$0.50 F_u$

[a] Subject to the requirements of Sec 10.17.

[b] For A307 bolts the tabulated values shall be reduced by 1 percent for each 2 mm over 5 diameters of length in the grip.

[c] Threads permitted in shear planes.

[d] The nominal tensile strength of the threaded portion of an upset rod, based upon the cross-sectional area at its major thread diameter, AD, which shall be larger than the nominal body area of the rod before upsetting times F_y .

[e] For A325 or A325M and A490 or A490M bolts subject to tensile fatigue loading, see Sec 10.2.3.9

[f] When bearing-type connections used to splice tension members have a fastener pattern whose length, measured parallel to the line of force, exceeds 1270 mm, tabulated values shall be reduced by 20 percent.

Table 6.10.11: Nominal Hole Dimensions, mm

Bolt Diameter	Hole Dimensions			
	Standard (Diameter)	Oversize (Diameter)	Short-Slot (Width × Length)	Long-Slot (Width × Length)
M16	18	20	18 × 22	18 × 40
M20	22	24	22 × 26	22 × 50
M22	24	28	24 × 30	24 × 55
M24	27 [a]	30	27 × 32	27 × 60
M27	30	35	30 × 37	30 × 67
M30	33	38	33 × 40	33 × 75
≥M36	$d + 3$	$d + 8$	$(d + 3) \times (d + 10)$	$(d + 3) \times 2.5d$

[a] Clearance provided allows the use of a 25 mm. bolt if desirable.

10.10.3.3 Minimum spacing

The distance between centers of standard, oversized, or slotted holes, shall not be less than $2\frac{2}{3}$ times the nominal diameter, d of the fastener; a distance of $3d$ is preferred.

10.10.3.4 Minimum edge distance

The distance from the center of a standard hole to an edge of a connected part in any direction shall not be less than either the applicable value from Table 6.10.12, or as required in Sec 10.10.3.10. The distance from the center of an oversized or slotted hole to an edge of a connected part shall be not less than that required for a standard hole to an edge of a connected part plus the applicable increment C_2 from Table 6.10.13.

Table 6.10.12: Minimum Edge Distance,^[a] mm, from Center of Standard Hole^[b] to Edge of Connected Part

Bolt Diameter (mm)	At Sheared Edges	At Rolled Edges of
		Plates, Shapes or Bars, or Thermally Cut Edges ^[c]
16	28	22
20	34	26
22	38 ^[d]	28
24	42 ^[d]	30
27	48	34

Bolt Diameter (mm)	At Sheared Edges	At Rolled Edges of Plates, Shapes or Bars, or Thermally Cut Edges ^[c]
30	52	38
36	64	46
Over 36	1.75d	1.25d

Notes: Table 6.10.12

^[a] Lesser edge distances are permitted to be used provided provisions of Sec 10.10.3.10, as appropriate, are satisfied.

^[b] For oversized or slotted holes, see Table 6.10.13.

^[c] All edge distances in this column are permitted to be reduced 3 mm when the hole is at a point where required strength does not exceed 25 percent of the maximum strength in the element.

^[d] These are permitted to be 32 mm at the ends of beam connection angles and shear end plates.

Table 6.10.13: Values of Edge Distance Increment C₂, mm

Nominal Diameter of Fastener(mm)	Oversized Holes	Slotted Holes		
		Long Axis Perpendicular to Edge		Long Axis Parallel to Edge
		Short Slots	Long Slots ^[a]	
≤ 22	2	3	0.75d	0
24	3	3		
≥ 27	3	5		

^[a] When length of slot is less than maximum allowable (see Table 6.10.11). C₂ is permitted to be reduced by one-half the difference between the maximum and actual slot lengths.

10.10.3.5 Maximum spacing and edge distance

The maximum distance from the center of any bolt or rivet to the nearest edge of parts in contact shall be 12 times the thickness of the connected part under consideration, but shall not exceed 150 mm. The longitudinal spacing of fasteners between elements in continuous contact consisting of a plate and a shape or two plates shall be as follows:

- For painted members or unpainted members not subject to corrosion, the spacing shall not exceed 24 times the thickness of the thinner plate or 305 mm.
- For unpainted members of weathering steel subject to atmospheric corrosion, the spacing shall not exceed 14 times the thickness of the thinner plate or 180 mm.

10.10.3.6 Tension and shear strength of bolts and threaded parts

The design tension or shear strength, ϕR_n and the allowable tension or shear strength, R_n/Ω , of a snug-tightened or pretensioned high-strength bolt or threaded part shall be determined according to the limit states of tensile rupture and shear rupture as follows:

$$R_n = F_n A_b \quad (6.10.168)$$

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

Where,

F_n = nominal tensile stress F_{nt} , or shear stress, F_{nv} from Table 6.10.10, MPa

A_b = nominal unthreaded body area of bolt or threaded part (for upset rods, see footnote d, Table 6.10.10, mm²)

The required tensile strength shall include any tension resulting from prying action produced by deformation of the connected parts.

10.10.3.7 Combined tension and shear in bearing-type connection

The available tensile strength of a bolt subjected to combined tension and shear shall be determined according to the limit states of tension and shear rupture as follows:

$$R_n = F'_{nt} A_b \quad (6.10.169)$$

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

Where,

F'_{nt} = nominal tensile stress modified to include the effects of shearing stress, MPa

$$F'_{nt} = 1.3F_{nt} - \frac{F_{nt}}{\phi F_{nv}} f_v \leq F_{nt} \text{ (LRFD)} \quad (6.10.170a)$$

$$F'_{nt} = 1.3F_{nt} - \frac{\Omega F_{nt}}{\phi F_{nv}} f_v \leq F_{nt} \text{ (ASD)} \quad (6.10.170b)$$

F_{nt} = nominal tensile stress from Table 6.10.10, MPa

F_{nv} = nominal shear stress from Table 6.10.10, MPa

f_v = the required shear stress, MPa

The available shear stress of the fastener shall equal or exceed the required shear strength per unit area, f_v .

10.10.3.8 High-strength bolts in slip-critical connections

High-strength bolts in slip-critical connections are permitted to be designed to prevent slip either as a serviceability limit state or at the required strength limit state. The connection must also be checked for shear strength in accordance with Sections 10.10.3.6 and 10.10.3.7 and bearing strength in accordance with Sections 10.10.3.1 and 10.10.3.10.

Slip-critical connections shall be designed as follows, unless otherwise designated by the engineer of record. Connections with standard holes or slots transverse to the direction of the load shall be designed for slip as a serviceability limit state. Connections with oversized holes or slots parallel to the direction of the load shall be designed to prevent slip at the required strength level.

The design slip resistance, ϕR_n and the allowable slip resistance, $\frac{R_n}{\Omega}$ shall be determined for the limit state of slip as follows:

$$R_n = \mu D_u h_{sc} T_b N_s \quad (6.10.171)$$

For connections in which prevention of slip is a serviceability limit state

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

For connections designed to prevent slip at the required strength level

$$\phi = 0.85 \text{ (LRFD)} \quad \Omega = 1.76 \text{ (ASD)}$$

Where,

μ = mean slip coefficient for Class A or B surfaces, as applicable, or as established by tests

= 0.35 for Class A surfaces (unpainted clean mill scale steel surfaces or surfaces with Class A coatings on blast-cleaned steel and hot-dipped galvanized and roughened surfaces)

= 0.50 for Class B surfaces (unpainted blast-cleaned steel surfaces or surfaces with Class B coatings on blast-cleaned steel)

$D_u = 1.13$; a multiplier that reflects the ratio of the mean installed bolt pretension to the specified minimum bolt pretension. The use of other values may be approved by the engineer of record.

h_{sc} = hole factor determined as follows:

- (a) For standard size holes $h_{sc} = 1.00$
- (b) For oversized and short-slotted holes $h_{sc} = 0.85$
- (c) For long-slotted holes $h_{sc} = 0.70$

N_s = number of slip planes

T_b = minimum fastener tension given in Table 6.10.9, kN

10.10.3.9 Combined tension and shear in slip-critical connections

When a slip-critical connection is subjected to an applied tension that reduces the net clamping force, the available slip resistance per bolt, from Sec 10.10.3.8, shall be multiplied by the factor, k_s , as follows:

$$k_s = 1 - \frac{T_u}{D_u T_b N_b} \quad (\text{LRFD}) \quad (6.10.172a)$$

$$k_s = 1 - \frac{1.5 T_a}{D_u T_b N_b} \quad (\text{ASD}) \quad (6.10.172b)$$

Where,

N_b = number of bolts carrying the applied tension

T_a = tension force due to ASD load combinations, kN

T_b = minimum fastener tension given in Table 6.10.9, kN

T_u = tension force due to LRFD load combinations, kN

10.10.3.10 Bearing strength at bolt holes

The available bearing strength, ϕR_n and R_n/Ω , at bolt holes shall be determined for the limit state of bearing as follows:

$$\phi = 0.75 \quad (\text{LRFD}) \quad \Omega = 2.00 \quad (\text{ASD})$$

For a bolt in a connection with standard, oversized, and short-slotted holes, independent of the direction of loading, or a long-slotted hole with the slot parallel to the direction of the bearing force:

- (a) When deformation at the bolt hole at service load is a design consideration

$$R_n = 1.2L_c t F_u \leq 2.4dt F_u \quad (6.10.173a)$$

- (b) When deformation at the bolt hole at service load is not a design consideration

$$R_n = 1.5L_c t F_u \leq 3.0dt F_u \quad (6.10.173b)$$

For a bolt in a connection with long-slotted holes with the slot perpendicular to the direction of force:

$$R_n = 1.0L_c t F_u \leq 2.0dt F_u \quad (6.10.173c)$$

For connections made using bolts that pass completely through an unstiffened box member or *HSS* see Sec 10.10.7 and Eq. 6.10.180,

Where,

d = nominal bolt diameter, mm

F_u = specified minimum tensile strength of the connected material, MPa

L_c = clear distance, in the direction of the force, between the edge of the hole and the edge of the adjacent hole or edge of the material, mm

t = thickness of connected material, mm

For connections, the bearing resistance shall be taken as the sum of the bearing resistances of the individual bolts.

Bearing strength shall be checked for both bearing-type and slip-critical connections. The use of oversized holes and short- and long-slotted holes parallel to the line of force is restricted to slip-critical connections per Sec 10.10.3.2.

10.10.3.11 Special fasteners

The nominal strength of special fasteners other than the bolts presented in Table 6.10.10 shall be verified by tests.

10.10.3.12 Tension fasteners

When bolts or other fasteners in tension are attached to an unstiffened box or *HSS* wall, the strength of the wall shall be determined by rational analysis.

10.10.4 Affected Elements of Members and Connecting Elements

This Section applies to elements of members at connections and connecting elements, such as plates, gussets, angles, and brackets.

10.10.4.1 Strength of elements in tension

The design strength, ϕR_n and the allowable strength, R_n/Ω , of affected and connecting elements loaded in tension shall be the lower value obtained according to the limit states of tensile yielding and tensile rupture.

For tensile yielding of connecting elements:

$$R_n = F_y A_g \quad (6.10.174)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

For tensile rupture of connecting elements:

$$R_n = F_u A_e \quad (6.10.175)$$

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

Where,

A_e = effective net area as defined in Sec 10.4.3.3, mm²; for bolted splice plates, $A_e = A_n \leq 0.85 A_g$

10.10.4.2 Strength of elements in shear

The available shear yield strength of affected and connecting elements in shear shall be the lower value obtained according to the limit states of shear yielding and shear rupture:

For shear yielding of the element:

$$R_n = 0.60 F_y A_g \quad (6.10.176)$$

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

For shear rupture of the element:

$$R_n = 0.60F_u A_{nv} \quad (6.10.177)$$

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

Where,

$$A_{nv} = \text{net area subject to shear, mm}^2$$

10.10.4.3 Block shear strength

The available strength for the limit state of block shear rupture along a shear failure path or path(s) and a perpendicular tension failure path shall be taken as

$$R_n = 0.60F_u A_{nv} + U_{bs}F_u A_{nt} \leq 0.6F_y A_{gv} + U_{bs}F_u A_{nt} \quad (6.10.178)$$

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

Where,

$$A_{gv} = \text{gross area subject to shear, mm}^2$$

$$A_{nt} = \text{net area subject to tension, mm}^2$$

$$A_{nv} = \text{net area subject to shear, mm}^2$$

Where, the tension *stress* is uniform, $U_{bs} = 1$; where the tension stress is non-uniform, $U_{bs} = 0.5$.

10.10.4.4 Strength of elements in compression

The available strength of connecting elements in compression for the limit states of yielding and buckling shall be determined as follows.

For $KL/r \leq 25$

$$P_n = F_y A_g \quad (6.10.179)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

For $KL/r > 25$ the provisions of Sec 10.5 apply.

10.10.5 Fillers

In welded construction, any filler 6 mm or more in thickness shall extend beyond the edges of the splice plate and shall be welded to the part on which it is fitted with sufficient weld to transmit the splice plate load, applied at the surface of the filler. The welds joining the splice plate to the filler shall be sufficient to transmit the splice plate load and shall be long enough to avoid overloading the filler along the toe of the weld. Any filler less than 6 mm thick shall have its edges made flush with the edges of the splice plate and the weld size shall be the sum of the size necessary to carry the splice plus the thickness of the filler plate.

When a bolt that carries load passes through fillers that are equal to or less than 6 mm thick, the shear strength shall be used without reduction. When a bolt that carries load passes through fillers that are greater than 6 mm thick, one of the following requirements shall apply:

For fillers that are equal to or less than 19 mm thick, the shear strength of the bolts shall be multiplied by the factor $[1 - 0.0154(t - 6)]$, where t is the total thickness of the fillers up to 19 mm;

The fillers shall be extended beyond the joint and the filler extension shall be secured with enough bolts to uniformly distribute the total force in the connected element over the combined cross section of the connected element and the fillers;

The size of the joint shall be increased to accommodate a number of bolts that is equivalent to the total number required in (2) above; or

The joint shall be designed to prevent slip at required strength levels in accordance with Sec 10.10.3.8.

10.10.6 Splices

Groove-welded splices in plate girders and beams shall develop the nominal strength of the smaller spliced section. Other types of splices in cross sections of plate girders and beams shall develop the strength required by the forces at the point of the splice.

10.10.7 Bearing Strength

The design bearing strength, ϕR_n and the allowable bearing strength, R_n/Ω , of surfaces in contact shall be determined for the limit state of bearing (local compressive yielding) as follows:

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

The nominal bearing strength, R_n , is defined as follows for the various types of bearing:

For milled surfaces, pins in reamed, drilled, or bored holes, and ends of fitted bearing stiffeners:

$$R_n = 1.8F_y A_{pb} \quad (6.10.180)$$

Where,

F_y = specified minimum yield stress, MPa

A_{pb} = projected bearing area, mm²

For expansion rollers and rockers:

If $d \leq 635$ mm

$$R_n = 1.2(F_y - 90)ld/20 \quad (6.10.181)$$

If $d > 635$ mm

$$R_n = 30.2(F_y - 90)l\sqrt{d}/20 \quad (6.10.182)$$

Where,

d = diameter, mm

l = length of bearing, mm

10.10.8 Column Bases and Bearing on Concrete

Proper provision shall be made to transfer the column loads and moments to the footings and foundations.

In the absence of the Code regulations, the design bearing strength, $\phi_c P_n$ and the allowable bearing strength, $\frac{P_n}{\Omega_c}$ for the limit state of concrete crushing are permitted to be taken as follows:

$$\phi_c = 0.60 \text{ (LRFD)} \quad \Omega_c = 2.50 \text{ (ASD)}$$

The nominal bearing strength, P_p , is determined as follows:

On the full area of a concrete support:

$$P_p = 0.8f'_c A_1 \quad (6.10.183)$$

On less than the full area of a concrete support:

$$P_p = 0.8f'_c A_1 \sqrt{A_2/A_1} \leq 1.7f'_c A_1 \quad (6.10.184)$$

Where,

A_1 = area of steel concentrically bearing on a concrete support, mm²

A_2 = maximum area of the portion of the supporting surface that is geometrically similar to and concentric with the loaded area, mm²

10.10.9 Anchor Rods and Embedments

Anchor rods shall be designed to provide the required resistance to loads on the completed structure at the base of columns including the net tensile components of any bending moment that may result from load combinations stipulated in Sec 10.2.2. The anchor rods shall be designed in accordance with the requirements for threaded parts in Table 6.10.10.

Larger oversized and slotted holes are permitted in base plates when adequate bearing is provided for the nut by using structural or plate washers to bridge the hole.

When horizontal forces are present at column bases, these forces should, where possible, be resisted by bearing against concrete elements or by shear friction between the column base plate and the foundation. When anchor rods are designed to resist horizontal force the base plate hole size, the anchor rod setting tolerance, and the horizontal movement of the column shall be considered in the design.

10.10.10 Flanges and Webs with Concentrated Forces

This Section applies to single- and double-concentrated forces applied normal to the flange(s) of wide flange sections and similar built-up shapes. A single- concentrated force can be either tensile or compressive. Double-concentrated forces are one tensile and one compressive and form a couple on the same side of the loaded member.

When the required strength exceeds the available strength as determined for the limit states listed in this Section, stiffeners and/or doublers shall be provided and shall be sized for the difference between the required strength and the available strength for the applicable limit state. Stiffeners shall also meet the design requirements in Sec 10.10.10.8. Doublers shall also meet the design requirement in Sec 10.10.10.9.

Stiffeners are required at unframed ends of beams in accordance with the requirements of Sec 10.10.10.7.

10.10.10.1 Flange local bending

This Section applies to tensile single-concentrated forces and the tensile component of double-concentrated forces.

The design strength, ϕR_n and the allowable strength, R_n/Ω , for the limit state of flange local bending shall be determined as follows:

$$R_n = 6.25t_f^2 F_{yf} \quad (6.10.185)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

Where,

F_{yf} = specified minimum yield stress of the flange, MPa

t_f = thickness of the loaded flange, mm

If the length of loading across the member flange is less than $0.15b_f$, where b_f is the member flange width, Eq. 6.10.185 need not be checked.

The value of R_n shall be reduced by 50 percent, when the concentrated force to be resisted is applied at a distance from the member end that is less than $10t_f$.

When required, a pair of transverse stiffeners shall be provided.

10.10.10.2 Web local yielding

This Section applies to single-concentrated forces and both components of double-concentrated forces.

The available strength for the limit state of web local yielding shall be determined as follows:

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

The nominal strength, R_n shall be determined as follows:

When the concentrated force to be resisted is applied at a distance from the member end that is greater than the depth of the member d

$$R_n = (5k + N)F_{yw}t_w \quad (6.10.186)$$

When the concentrated force to be resisted is applied at a distance from the member end that is less than or equal to the depth of the member d ,

$$R_n = (2.5k + N)F_{yw}t_w \quad (6.10.187)$$

Where,

k = distance from outer face of the flange to the web toe of the fillet, mm

F_{yw} = specified minimum yield stress of the web, MPa

N = length of bearing (not less than k for end beam reactions), mm

t_w = web thickness, mm

When required, a pair of transverse stiffeners or a doubler plate shall be provided.

10.10.10.3 Web Crippling

This Section applies to compressive single-concentrated forces or the compressive component of double-concentrated forces.

The available strength for the limit state of web local crippling shall be determined as follows:

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

The nominal strength, R_n shall be determined as follows:

When the concentrated compressive force to be resisted is applied at a distance from the member end that is greater than or equal to $d/2$:

$$R_n = 0.80t_w^2 \left[1 + 3 \left(\frac{N}{d} \right) \left(\frac{t_w}{t_f} \right)^{1.5} \right] \sqrt{\frac{EF_{yw}t_f}{t_w}} \quad (6.10.188)$$

When the concentrated compressive force to be resisted is applied at a distance from the member end that is less than $d/2$:

For $N/d \leq 0.2$

$$R_n = 0.40t_w^2 \left[1 + 3 \left(\frac{N}{d} \right) \left(\frac{t_w}{t_f} \right)^{1.5} \right] \sqrt{\frac{EF_{yw}t_f}{t_w}} \quad (6.10.189a)$$

For $N/d > 0.2$

$$R_n = 0.40t_w^2 \left[1 + \left(\frac{4N}{d} - 0.2 \right) \left(\frac{t_w}{t_f} \right)^{1.5} \right] \sqrt{\frac{EF_{yw}t_f}{t_w}} \quad (6.10.189b)$$

Where,

d = overall depth of the member, mm

t_f = flange thickness, mm

When required, a transverse stiffener or pair of transverse stiffeners, or a doubler plate extending at least one-half the depth of the web shall be provided.

10.10.10.4 Web sidesway buckling

This Section applies only to compressive single-concentrated forces applied to members where relative lateral movement between the loaded compression flange and the tension flange is not restrained at the point of application of the concentrated force.

The available strength of the web shall be determined as follows:

$$\phi = 0.85 \text{ (LRFD)} \quad \Omega = 1.76 \text{ (ASD)}$$

The nominal strength, R_n for the limit state of web sidesway buckling shall be determined as follows:

If the compression flange is restrained against rotation:

For $(h/t_w)/(l/b_f) \leq 2.3$

$$R_n = \frac{C_r t_w^3 t_f}{h^2} \left[1 + 0.4 \left(\frac{h/t_w}{l/b_f} \right)^3 \right] \quad (6.10.190)$$

For $(h/t_w)/(l/b_f) > 2.3$, the limit state of web sidesway buckling does not apply.

When the required strength of the web exceeds the available strength, local lateral bracing shall be provided at the tension flange or either a pair of transverse stiffeners or a doubler plate shall be provided.

If the compression flange is not restrained against rotation

For $(h/t_w)/(l/b_f) \leq 1.7$

$$R_n = \frac{C_r t_w^3 t_f}{h^2} \left[0.4 \left(\frac{h/t_w}{l/b_f} \right)^3 \right] \quad (6.10.191)$$

For $(h/t_w)/(l/b_f) > 1.7$ the limit state of web sidesway buckling does not apply.

When the required strength of the web exceeds the available strength, local lateral bracing shall be provided at both flanges at the point of application of the concentrated forces.

In Equations 6.10.190 and 6.10.191, the following definitions apply:

b_f = flange width, mm

C_r = 6.62×10^6 MPa when $M_u < M_y$ (LRFD) or $1.5 M_a < M_y$ (ASD) at the location of the force

= 3.31×10^6 MPa when $M_u \geq M_y$ (LRFD) or $1.5 M_a \geq M_y$ (ASD) at the location of the force

h = clear distance between flanges less fillet or corner radius for rolled shapes;
distance between adjacent lines of fasteners or clear distance between flanges
when welds are used for built-up shapes, mm

l = largest laterally unbraced length along either flange at the point of load, mm

t_f = flange thickness, mm

t_w = web thickness, mm

10.10.10.5 Web compression buckling

This Section applies to a pair of compressive single-concentrated forces or the compressive components in a pair of double-concentrated forces, applied at both flanges of a member at the same location.

The available strength for the limit state of web local buckling shall be determined as follows:

$$R_n = \frac{24t_w^3 \sqrt{EF_{yw}}}{h} \quad (6.10.192)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

When the pair of concentrated compressive forces to be resisted is applied at a distance from the member end that is less than $d/2$, R_n shall be reduced by 50 percent.

When required, a single transverse stiffener, a pair of transverse stiffeners, or a doubler plate extending the full depth of the web shall be provided.

10.10.10.6 Web panel zone shear

This Section applies to double-concentrated forces applied to one or both flanges of a member at the same location. The available strength of web panel zone for limit state of shear yielding shall be determined as follows:

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

The nominal strength, R_n , shall be determined as follows:

When the effect of panel-zone deformation on frame stability is not considered in the analysis:

For $P_r \leq 0.4P_c$

$$R_n = 0.60F_y d_c t_w \quad (6.10.193)$$

For $P_r > 0.4P_c$

$$R_n = 0.60F_y d_c t_w \left(1.4 - \frac{P_r}{P_c}\right) \quad (6.10.194)$$

When frame stability, including plastic panel-zone deformation, is considered in the analysis:

For $P_r \leq 0.75P_c$

$$R_n = 0.60F_y d_c t_w \left(1 + \frac{3b_{cf} t_{cf}^2}{d_b d_c t_w} \right) \quad (6.10.195)$$

For $P_r > 0.75P_c$

$$R_n = 0.60F_y d_c t_w \left(1 + \frac{3b_{cf} t_{cf}^2}{d_b d_c t_w} \right) \left(1.9 - \frac{1.2P_r}{P_c} \right) \quad (6.10.196)$$

In Equations 6.10.193 to 6.10.196, the following definitions apply:

A = column cross-sectional area, mm²

b_{cf} = width of column flange, mm

d_b = beam depth, mm

d_c = column depth, mm

F_y = specified minimum yield stress of the column web, MPa

P_c = P_y , N (LRFD)

P_c = 0.6 P_y , N (ASD)

P_r = required strength, N

P_y = $F_y A$, axial yield strength of the column, N

t_{cf} = thickness of the column flange, mm

t_w = column web thickness, mm

When required, doubler plate(s) or a pair of diagonal stiffeners shall be provided within the boundaries of the rigid connection whose webs lie in a common plane. See Sec 10.10.10.9 for doubler plate design requirements.

10.10.10.7 Unframed ends of beams and girders

At unframed ends of beams and girders not otherwise restrained against rotation about their longitudinal axes, a pair of transverse stiffeners, extending the full depth of the web, shall be provided.

10.10.10.8 Additional stiffeners requirements for concentrated forces

Stiffeners required to resist tensile concentrated forces shall be designed in accordance with the requirements of Sec 10.4 and welded to the loaded flange and the web. The welds to the flange shall be sized for the difference between the required strength and available limit state strength. The stiffener to web welds shall be sized to transfer to the web the algebraic difference in tensile force at the ends of the stiffener.

Stiffeners required to resist compressive concentrated forces shall be designed in accordance with the requirements in Sections 10.5.6.2 and 10.10.4.4 and shall either bear on or be welded to the loaded flange and welded to the web. The welds to the flange shall be sized for the difference between the required strength and the applicable limit state strength. The weld to the web shall be sized to transfer to the web the algebraic difference in compression force at the ends of the stiffener. For fitted bearing stiffeners, see Sec 10.10.7.

Transverse full depth bearing stiffeners for compressive forces applied to a beam or plate girder flange(s) shall be designed as axially compressed members (columns) as per the requirements of Sections 10.5.6.2 and 10.10.4.4.

The member properties shall be determined using an effective length of $0.75h$ and a cross section composed of two stiffeners and a strip of the web having a width of $25t_w$ at interior stiffeners and $12t_w$ at the ends of members. The weld connecting full depth bearing stiffeners to the web shall be sized to transmit the difference in compressive force at each of the stiffeners to the web.

Transverse and diagonal stiffeners shall comply with the following additional criteria:

The width of each stiffener plus one-half the thickness of the column web shall not be less than one-third of the width of the flange or moment connection plate delivering the concentrated force.

- (a) Thickness of a stiffener shall not be less than one-half the thickness of the flange or moment connection plate delivering the concentrated load, and greater than or equal to the width divided by 15.

- (b) Transverse stiffeners shall extend a minimum of one-half the depth of the member except as required in Sec. 10.10.10.5 and 10.10.10.7.

10.10.10.9 Additional doubler plate requirements for concentrated forces

Doubler plates required for compression strength shall be designed according to the requirements of Sec 10.5.

Doubler plates required for tensile strength shall be designed in accordance with the requirements of Sec 10.4.

Doubler plates required for shear strength (Sec 10.10.10.6) shall be designed following the provisions of Sec 10.7.

In addition, doubler plates shall comply with the following criteria:

- (a) The thickness and extent of the doubler plate shall provide the additional material necessary to equal or exceed the strength requirements.
- (b) The doubler plate shall be welded to develop the proportion of total force transmitted to doubler plate.

10.11 Design of Hss and Box Member Connections

This Section covers member strength design considerations pertaining to connections to HSS members and box sections of uniform wall thickness. See also Section 10.10 for additional requirements for bolting to HSS.

10.11.1 Concentrated Forces on HSS

10.11.1.1 Notation and definitions

The notation and their definitions related to this Section are given in Sec 10.1.2.

10.11.1.2 Limits of applicability

The criteria herein are applicable only when the connection configuration is within the following limits of applicability.

For HSS:

$$\text{Strength: } F_y \leq 360 \text{ MPa}$$

$$\text{Ductility: } \frac{F_y}{F_u} \leq 0.8$$

Other limits apply for specific criteria.

10.11.1.3 Concentrated force distributed transversely

10.11.1.3.1 Criterion for round HSS

When a concentrated force is distributed transversely to the axis of the HSS the design strength, ϕR_n and the allowable strength, R_n/Ω , for the limit state of local yielding shall be determined as follows:

$$R_n = F_y t^2 [5.5/(1 - 0.81 B_p/D)] Q_f \quad (6.10.197)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

Where, the Equations for Q_f is given in Sec 10.11.2.2 (Eq. 6.10.209).

Additional limits of applicability are:

$$0.2 < B_p/D \leq 1.0.$$

$$\text{For T-connections: } D/t \leq 50$$

$$\text{For cross-connections: } D/t \leq 40$$

10.11.1.3.2 Criterion for rectangular HSS

When a concentrated force is distributed transversely to the axis of the HSS the design strength, ϕR_n and the allowable strength, R_n/Ω , shall be the lowest value according to the limit states of local yielding due to uneven load distribution, shear yielding (punching) and sidewall strength.

Additional limits of applicability are:

$$0.25 < B_p/B \leq 1.0$$

$$\text{For loaded HSS wall } B/t \leq 35$$

For the limit state of local yielding due to uneven load distribution in the loaded plate,

$$R_n = \left[\frac{10}{\frac{F_y t}{\left(\frac{B}{t}\right)}} \right] B_P \leq F_{yP} t_P B_P \quad (6.10.198)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

For the limit state of shear yielding (punching),

$$R_n = 0.6 F_y t [2t_P + 2B_{ep}] \quad (6.10.199)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

Where, $B_{ep} = 10B_P / (B/t) \leq B_P$

This limit state need not be checked when $B_P > (B - 2t)$, nor when $B_P < 0.85B$.

For the limit state of sidewall under tension loading, the available strength shall be taken as the strength for sidewall local yielding. For the limit state of sidewall under compression loading, available strength shall be taken as the lowest value obtained according to the limit states of sidewall local yielding, sidewall local crippling and sidewall local buckling.

This limit state need not be checked unless the chord member and branch member (connecting element) have the same width ($\beta = 1.0$).

For the limit state of sidewall local yielding,

$$R_n = 2F_y t [5k + N] \quad (6.10.200)$$

$$\phi = 1.0 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

Where, k = outside corner radius of the HSS, which is permitted to be taken as $1.5t$ if unknown, mm.

For the limit state of sidewall local crippling, in T-connections,

$$R_n = 1.6t^2 \left[1 + \frac{3N}{(H-3t)} \right] (EF_y)^{0.5} Q_f \quad (6.10.201)$$

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.0 \text{ (ASD)}$$

Where, the Equations for Q_f is given in Sec 10.11.2.3 (Eq. 6.10.218).

For the limit state of sidewall local buckling in cross-connections,

$$R_n = \left[\frac{48t^3}{(H-3t)} \right] (EF_y)^{0.5} Q_f \quad (6.10.202)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

Where, the Equations for Q_f is given in Sec 10.11.2.3 (Eq. 6.10.218).

The nonuniformity of load transfer along the line of weld, due to the flexibility of the HSS wall in a transverse plate-to-HSS connection, shall be considered in proportioning such welds. This requirement can be satisfied by limiting the total effective weld length, L_e , of groove and fillet welds to rectangular HSS as follows:

$$L_e = 2 \left[\frac{10}{\left(\frac{B}{t}\right)} \right] \left[\frac{(F_t t)}{(F_{yp} t_p)} \right] B_p \leq 2B_p \quad (6.10.203)$$

Where, L_e = total effective weld length for welds on both sides of the transverse plate, mm.

In lieu of Eq. 6.10.203, this requirement may be satisfied by other rational approaches.

10.11.1.4 Concentrated Force distributed longitudinally at the center of the HSS diameter or width and acting perpendicular to the HSS Axis

When a concentrated force is distributed longitudinally along the axis of the HSS at the center of the HSS diameter or width, and also acts perpendicular to the axis direction of the HSS (or has a component perpendicular to the axis direction of the HSS), the design strength, ϕR_n and the allowable strength, R_n/Ω , perpendicular to the HSS axis shall be determined for the limit state of chord plastification as follows.

10.11.1.4.1 Criterion for round HSS

An additional limit of applicability is:

$D/t \leq 50$ for T-connections and $D/t \leq 40$ for cross-connections

$$R_n = 5.5F_y t^2 \left(1 + \frac{0.25N}{D} \right) Q_f \quad (6.10.204)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

Where, the Equation for Q_f is given in Sec 10.11.2.2 (Eq. 6.10.209).

10.11.1.4.2 Criterion for rectangular HSS

An additional limit of applicability is:

B/t for the loaded HSS wall ≤ 40

$$R_n = \left[\frac{F_y t^2}{(1 - t_p/B)} \right] \left[\frac{2N}{B} + 4 \left(1 - \frac{t_p}{B} \right)^{0.5} Q_f \right] \quad (6.10.205)$$

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

Where, $Q_f = (1 - U^2)^{0.5}$

The Equation for U is given in Sec 10.11.2.3 (Eq. 6.10.220).

10.11.1.5 Concentrated force distributed longitudinally at center of HSS width and acting parallel to HSS axis

When a concentrated force is distributed longitudinally along the axis of a rectangular HSS and also acts parallel but eccentric to the axis direction of the member, the connection shall be verified as follows:

$$F_{yp} t_p \leq F_u t \quad (6.10.206)$$

10.11.1.6 Concentrated axial force on the end of a rectangular HSS with a cap plate

When a concentrated force acts on the end of a capped HSS and the force is in the direction of the HSS axis, the design strength, ϕR_n and the allowable strength, R_n/Ω , shall be determined for the limit states of wall local yielding (due to tensile or compressive forces) and wall local crippling (due to compressive forces only), with consideration for shear lag, as follows.

If $(5t_p + N) \geq B$, the available strength of HSS is computed by summing the contributions of all four HSS walls.

If $(5t_p + N) < B$, the available strength of the HSS is computed by summing the contributions of the two walls into which the load is distributed.

For the limit state of wall local yielding, for one wall,

$$R_n = F_y t [5t_p + N] \leq B F_y t \quad (6.10.207)$$

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

For the limit state of wall local crippling, for one wall,

$$R_n = 0.8t^2 \left[1 + \left(\frac{6N}{B} \right) \left(\frac{t}{t_p} \right)^{1.5} \right] \left[\frac{EF_y t_p}{t} \right]^{0.5} \quad (6.10.208)$$

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

10.11.2 HSS-To-HSS Truss Connections

HSS-to-HSS truss connections are defined as connections that consist of one or more branch members that are directly welded to a continuous chord that passes through the connection and shall be classified as follows:

When the punching load ($P_r \sin \theta$) in a branch member is equilibrated by beam shear in the chord member, the connection shall be classified as a T-connection when the branch is perpendicular to the chord and a Y-connection otherwise.

When the punching load ($P_r \sin \theta$) in a branch member is essentially equilibrated (within 20 percent) by loads in other branch member(s) on the same side of the connection, the connection shall be classified as a K-connection. The relevant gap is between the primary branch members whose loads equilibrate. An N-connection can be considered as a type of K-connection.

When the punching load ($P_r \sin \theta$) is transmitted through the chord member and is equilibrated by branch member(s) on the opposite side, the connection shall be classified as a cross-connection.

When a connection has more than two primary branch members or branch members in more than one plane, the connection shall be classified as a general or multiplanar connection.

When branch members transmit part of their load as K-connections and part of their load as T-, Y-, or cross-connections, the nominal strength shall be determined by interpolation on the proportion of each in total.

For the purposes of this Specification, the centerlines of branch members and chord members shall lie in a common plane. Rectangular HSS connections are further limited to have all members oriented with walls parallel to the plane. For trusses that are made with HSS that are connected by welding branch members to chord members, eccentricities within the limits of applicability are permitted without consideration of the resulting moments for the design of the connection.

10.11.2.1 Notation and Definitions

The notation and their definitions related to this Section are given in Sec 10.1.2.

10.11.2.2 Criteria for round HSS

The interaction of stress due to chord member forces and local branch connection forces shall be incorporated through the chord-stress interaction parameter Q_f .

When the chord is in tension,

$$Q_f = 1$$

When the chord is in compression,

$$Q_f = 1.0 - 0.3U(1 + U) \quad (6.10.209)$$

Where, U is the utilization ratio given by

$$U = \left| \frac{P_r}{(A_g F_c)} + \frac{M_r}{(S F_c)} \right| \quad (6.10.210)$$

And,

P_r = required axial strength in chord, N; for K-connections, P_r is to be determined on the side of the joint that has the lower compression stress (lower U)

M_r = required flexural strength in chord, N-mm.

A_g = chord gross area, mm²

F_c = available stress, MPa.

S = chord elastic section modulus, mm³

For design according to Section 10.2.3.3 (LRFD):

$P_r = P_u$ = required axial strength in chord, using LRFD load combinations, N

$M_r = M_u$ = required flexural strength in chord, using LRFD load combinations, N-mm.

$F_c = F_y$, MPa.

For design according to Section 10.2.3.4 (ASD):

$P_r = P_a$ = required axial strength in chord, using ASD load combinations, N

$M_r = M_a$ = required flexural strength in chord, using ASD load combinations, N-mm.

$f_c = 0.6f_y$, MPa.

10.11.2.2.1 Limits of applicability

The criteria herein are applicable only when the connection configuration is within the following limits of applicability:

- (a) Joint eccentricity: $-0.55D \leq e \leq 0.25D$, where D is the chord diameter and e is positive away from the branches
- (b) Branch angle: $\theta \geq 30^\circ$
- (c) Chord wall slenderness: ratio of diameter to wall thickness less than or equal to 50 for T-, Y- and K-connections; less than or equal to 40 for cross-connections
- (d) Tension branch wall slenderness: ratio of diameter to wall thickness less than or equal to 50
- (e) Compression branch wall slenderness: ratio of diameter to wall thickness $\leq 0.05E/F_y$.
- (f) Width ratio: $0.2 \leq \frac{D_b}{D} \leq 1.0$ in general, and $0.4 \leq \frac{D_b}{D} \leq 1.0$ for gapped K-connections
- (g) If a gap connection: g greater than or equal to the sum of the branch wall thicknesses
- (h) If an overlap connection: $25\% \leq O_v \leq 100\%$, where $O_v = \left(\frac{q}{p}\right) \times 100\%$. P is the projected length of the overlapping branch on the chord; q is the overlap length measured along the connecting face of the chord beneath the two branches. For overlap connections, the larger (or if equal diameter, the thicker) branch is a “thru member” connected directly to the chord.

- (i) Branch thickness ratio for overlap connections: thickness of overlapping branch to be less than or equal to the thickness of the overlapped branch.
- (j) Strength: $F_y \leq 360$ MPa. for chord and branches
- (k) Ductility: $F_y/F_u \leq 0.8$

10.11.2.2.2 Branches with axial loads in T-, Y- and cross-connections

For T- and Y- connections, the design strength of the branch ϕP_n or the allowable strength of the branch, P_n/Ω , shall be the lower value obtained according to the limit states of chord plastification and shear yielding (punching).

For the limit state of chord plastification in T- and Y-connections,

$$P_n \sin \theta = F_y t^2 [3.1 + 15.6\beta^2] \gamma^{0.2} Q_f \quad (6.10.211)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

For the limit state of shear yielding (punching),

$$P_n = 0.6 F_y t \pi D_b \left[\frac{(1 + \sin \theta)}{2 \sin^2 \theta} \right] \quad (6.10.212)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

This limit state need not be checked when $\beta > (1 - 1/\gamma)$.

For the limit state of chord plastification in cross-connections,

$$P_n \sin \theta = F_y t^2 \left[\frac{5.7}{(1 - 0.81\beta)} \right] Q_f \quad (6.10.213)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

10.11.2.2.3 Branches with axial loads in K-connections

For K-connections, the design strength of the branch, ϕP_n and the allowable strength of the branch, P_n/Ω , shall be the lower value obtained according to the limit states of chord plastification for gapped and overlapped connections and shear yielding (punching) for gapped connections only.

For the limit state of chord plastification,

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

For the compression branch:

$$P_n \sin \theta = F_y t^2 \left[2.0 + \frac{11.33 D_b}{D} \right] Q_g Q_f \quad (6.10.214)$$

Where, D_b refers to the compression branch only, and

$$Q_g = \gamma^{0.2} \left[1 + \frac{0.024 \gamma^{1.2}}{e^{\left(\frac{0.5g}{t} - 1.33\right)} + 1} \right] \quad (6.10.215)$$

In gapped connections, g (measured along the crown of the chord neglecting weld dimensions) is positive. In overlapped connections, g is negative and equals q .

For the tension branch,

$$P_n \sin \theta = (P_n \sin \theta)_{\text{compression branch}} \quad (6.10.216)$$

For the limit state of shear yielding (punching) in gapped K-connections,

$$P_n = 0.6 F_y t \pi D_b \left[\frac{(1 + \sin \theta)}{2 \sin^2 \theta} \right] \quad (6.10.217)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

10.11.2.3 Criteria for rectangular HSS

The interaction of stress due to chord member forces and local branch connection forces shall be incorporated through the chord-stress interaction parameter Q_f .

When the chord is in tension,

$$Q_f = 1$$

When the chord is in compression in T -, Y -, and cross-connections,

$$Q_f = 1.3 - 0.4U/\beta \leq 1 \quad (6.10.218)$$

When the chord is in compression in gapped K -connections,

$$Q_f = 1.3 - 0.4U/\beta_{eff} \leq 1 \quad (6.10.219)$$

Where, U is the utilization ratio given by

$$U = \left| \frac{P_r}{(A_g F_c)} + \frac{M_r}{(S F_c)} \right| \quad (6.10.220)$$

And,

P_r = required axial strength in chord, N. For gapped K-connections, P_r is to be determined on the side of the joint that has the higher compression stress (higher U).

M_r = required flexural strength in chord, N-mm.

A_g = chord gross area, mm²

F_c = available stress, MPa.

S = chord elastic section modulus, mm³

For design according to Section 10.2.3.3 (LRFD):

$P_r = P_u$ = required axial strength in chord, using *LRFD load combinations*, N

$M_r = M_u$ = required flexural strength in chord, using LRFD load combinations, N-mm.

$F_c = F_u$, MPa.

For design according to Section 10.2.3.4 (ASD):

$P_r = P_a$ = required axial strength in chord, using *ASD load combinations*, N.

$M_r = M_a$ = required flexural strength in chord, using ASD load combinations, N-mm.

$F_c = 0.6F_y$, MPa.

10.11.2.3.1 Limits of applicability

The criteria herein are applicable only when the connection configuration is within the following limits:

- (a) Joint eccentricity: $-0.55H \leq e \leq 0.25H$, where H is the chord depth and e is positive away from the branches
- (b) Branch angle: $\theta \geq 30^\circ$

- (c) Chord wall slenderness: ratio of overall wall width to thickness less than or equal to 35 for gapped K-connections and T-, Y- and cross-connections; less than or equal to 30 for overlapped K-connections
- (d) Tension branch wall slenderness: ratio of overall wall width to thickness less than or equal to 35
- (e) Compression branch wall slenderness: ratio of overall wall width to thickness less than or equal to $1.25(E/F_{yb})^{0.5}$ and also less than 35 for gapped K-connections and T-, Y- and cross-connections; less than or equal to $1.1(E/F_{yb})^{0.5}$ for overlapped K-connections
- (f) Width ratio: ratio of overall wall width of branch to overall wall width of chord greater than or equal to 0.25 for T-, Y-, cross- and overlapped K-connections; greater than or equal to 0.35 for gapped K-connections
- (g) Aspect ratio: $0.5 \leq \text{ratio of depth to width} \leq 2.0$
- (h) Overlap: $25\% \leq O_v \leq 100\%$, where $O_v = \left(\frac{q}{p}\right) \times 100\%$. p is the projected length of the overlapping branch on the chord; q is the overlap length measured along the connecting face of the chord beneath the two branches. For overlap connections, the larger (or if equal width, the thicker) branch is a “thru member” connected directly to the chord
- (i) Branch width ratio for overlap connections: ratio of overall wall width of overlapping branch to overall wall width of overlapped branch greater than or equal to 0.75
- (j) Branch thickness ratio for overlap connections: thickness of overlapping branch to be less than or equal to the thickness of the overlapped branch
- (k) Strength: $F_y \leq 360 \text{ MPa}$. for chord and branches
- (l) Ductility: $F_y/F_u \leq 0.8$
- (m) Other limits apply for specific criteria

10.11.2.3.2 Branches with axial loads in T-,Y- and cross-connections

For T-, Y- and cross-connections, the design strength of the branch, ϕP_n or the allowable strength of the branch, P_n/Ω , shall be the lowest value obtained according to the limit states of chord wall plastification, shear yielding (punching), sidewall strength and local yielding due to uneven load distribution. In addition to the limits of applicability in Section 10.11.2.3a, β shall not be less than 0.25.

For the limit state of chord wall plastification,

$$P_n \sin \theta = F_y t^2 \left[\frac{2\eta}{(1-\beta)} + \frac{4}{(1-\beta)^{0.5}} \right] Q_f \quad (6.10.221)$$

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

This limit state need not be checked when $\beta > 0.85$.

For the limit state of shear yielding (punching),

$$P_n \sin \theta = 0.6 F_y t B [2\eta + 2\beta_{eop}] \quad (6.10.222)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

In Eq. 6.10.222, the effective outside punching parameter $\beta_{eop} = 5\beta/\gamma$ shall not exceed β .

This limit state need not be checked when $\beta > (1 - 1/\gamma)$, nor when $\beta < 0.85$ and $B/t \geq 10$.

For the limit state of sidewall strength, the available strength for branches in tension shall be taken as the available strength for sidewall local yielding. For the limit state of sidewall strength, the available strength for branches in compression shall be taken as the lower of the strengths for sidewall local yielding and sidewall local crippling. For cross-connections with a branch angle less than 90° , an additional check for chord sidewall shear failure must be made in accordance with Section 10.7.5.

This limit state need not be checked unless the chord member and branch member have same width ($\beta = 1.0$)

For the limit state of local yielding,

$$P_n \sin \theta = 2F_y t [5k + N] \quad (6.10.223)$$

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

Where,

k = outside corner radius of the HSS, which is permitted to be taken as $1.5t$ if unknown, mm.

N = bearing length of the load, parallel to the axis of the HSS main member, $H_b/\sin \theta$, mm.

For the limit state of sidewall local crippling, in T- and Y-connections,

$$P_n \sin \theta = 1.6t^2 \left[1 + \frac{3N}{(H-3t)} \right] (EF_y)^{0.5} Q_f \quad (6.10.224)$$

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

For the limit state of sidewall local crippling in cross-connections,

$$P_n \sin \theta = \left[\frac{48t^3}{(H-3t)} \right] (EF_y)^{0.5} Q_f \quad (6.10.225)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

For the limit state of local yielding due to uneven load distribution,

$$P_n = F_{yb} t_b [2H_b + 2b_{eoi} - 4t_b] \quad (6.10.226)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

Where,

$$b_{eoi} = \left[\frac{10}{\left(\frac{B}{t}\right)} \right] [F_y t / (F_{yb} t_b)] B_b \leq B_b \quad (6.10.227)$$

This limit state need not be checked when $\beta < 0.85$.

10.11.2.3.3 Branches with axial loads in gapped K-connections

For gapped K-connections, the design strength of the branch, ϕP_n or the allowable strength of the branch, P_n/Ω , shall be the lowest value obtained according to the limit states of chord wall plastification, shear yielding (punching), shear yielding and local yielding due to uneven load distribution. In addition to the limits of applicability in Sec 10.11.2.3.1, the following limits shall apply:

$$(a) \quad B_b/B \geq 0.1 + \gamma/50$$

$$(b) \quad \beta_{eff} \geq 0.35$$

$$(c) \quad \zeta \leq 0.5(1 - \beta_{eff})$$

$$(d) \quad \text{Gap: } g \text{ greater than or equal to the sum of the branch wall thicknesses}$$

$$(e) \quad \text{The smaller } B_b > 0.63 \text{ times the larger } B_b$$

For the limit state of chord wall plastification,

$$P_n \sin \theta = F_y t^2 [9.8 \beta_{eff} \gamma^{0.5}] Q_f \quad (6.10.228)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

For the limit state of shear yielding (punching),

$$P_n \sin \theta = 0.6 F_y t B [2\eta + \beta + \beta_{eop}] \quad (6.10.229)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

In the above equation, the effective outside punching parameter $\beta_{eop} = 5\beta/\gamma$ shall not exceed β .

This limit state need only be checked if $B_b < (B - 2t)$ or the branch is not square.

For the limit state of shear yielding of the chord in the gap, available strength shall be checked in accordance with Sec 10.7. This limit state need only be checked if the chord is not square.

For the limit state of local yielding due to uneven load distribution,

$$P_n = F_{yb} t_b [2H_b + B_b + b_{eoi} - 4t_b] \quad (6.10.230)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

Where,

$$b_{eoi} = \left[\frac{10}{\left(\frac{B}{t}\right)} \right] [F_y t / (F_{yb} t_b)] B_b \leq B_b \quad (6.10.231)$$

This limit state need only be checked if the branch is not square or $B/t < 15$.

10.11.2.3.4 Branches with axial loads in overlapped K-connections

For overlapped K-connections, the design strength of the branch, ϕP_n or the allowable strength of the branch, P_n/Ω shall be determined from the limit state of local yielding due to uneven load distribution,

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

For the overlapping branch and for overlap $25\% \leq O_v \leq 50\%$, measured with respect to overlapping branch,

$$P_n = F_{ybi} t_{bi} \left[\left(\frac{O_v}{50} \right) (2H_{bi} - 4t_{bi}) + b_{eoi} + b_{eov} \right] \quad (6.10.232)$$

For the overlapping branch, and for overlap $50\% \leq O_v < 80\%$ measured with respect to overlapping branch,

$$P_n = F_{ybi} t_{bi} [2H_{bi} - 4t_{bi} + b_{eoi} + b_{eov}] \quad (6.10.233)$$

For the overlapping branch and for overlap $80\% \leq O_v \leq 100\%$ measured with respect to overlapping branch,

$$P_n = F_{ybi} t_{bi} [2H_{bi} - 4t_{bi} + B_{bi} + b_{eov}] \quad (6.10.234)$$

Where, b_{eoi} is the effective width of the branch face welded to the chord,

$$b_{eoi} = \left[\frac{10}{\left(\frac{B}{t} \right)} \right] [(F_y t) / (F_{ybi} t_{bi})] B_{bi} \leq B_{bi} \quad (6.10.235)$$

b_{eov} is the effective width of the branch face welded to the overlapped brace,

$$b_{eov} = \left[\frac{10}{\left(\frac{B_{bj}}{t_{bj}} \right)} \right] [(F_{ybj} t_{bj}) / (F_{ybi} t_{bi})] B_{bi} \leq B_{bi} \quad (6.10.236)$$

B_{bi} = overall branch width of the overlapping branch, mm.

B_{bj} = overall branch width of the overlapped branch, mm.

F_{ybi} = specified minimum yield stress of the overlapping branch material, MPa.

F_{ybj} = specified minimum yield stress of the overlapped branch material, MPa.

H_{bi} = overall depth of the overlapping branch, mm.

T_{bi} = thickness of the overlapping branch, mm.

t_{bj} = thickness of the overlapped branch, mm.

For the overlapped branch, P_n shall not exceed P_n of the overlapping branch, calculated using Eq. 6.10.232, Eq. 6.10.233 or Eq. 6.10.234, as applicable, multiplied by the factor $(A_{bj}F_{ybj}/A_{bi}F_{ybi})$.

Where,

A_{bi} = cross-sectional area of the overlapping branch

A_{bj} = cross-sectional area of the overlapped branch

10.11.2.3.5 Welds to branches

The nonuniformity of load transfer along the line of weld, due to differences in relative flexibility of HSS walls in HSS-to-HSS connections, shall be considered in proportioning such welds. This can be considered by limiting the total effective weld length, L_e , of groove and fillet welds to rectangular HSS as follows:

In T-, Y- and cross-connections,

For $\theta \geq 50^\circ$

$$L_e = \frac{2(H_b - 1.2t_b)}{\sin \theta} + (B_b - 1.2t_b) \quad (6.10.237)$$

For $\theta \geq 60^\circ$

$$L_e = \frac{2(H_b - 1.2t_b)}{\sin \theta} \quad (6.10.238)$$

Linear interpolation shall be used to determine L_e for values of θ between 50° and 60° .

In gapped K-connections, around each branch,

For $\theta \geq 50^\circ$

$$L_e = \frac{2(H_b - 1.2t_b)}{\sin \theta} + 2(B_b - 1.2t_b) \quad (6.10.239)$$

For $\theta \geq 60^\circ$

$$L_e = \frac{2(H_b - 1.2t_b)}{\sin \theta} + (B_b - 1.2t_b) \quad (6.10.240)$$

Linear interpolation shall be used to determine L_e for values of θ between 50° and 60° .

In lieu of the above criteria in Equations 6.10.237 to 6.10.240, other rational criteria are permitted.

10.11.3 HSS-To-HSS Moment Connections

HSS-to-HSS moment connections are defined as connections that consist of one or two branch members that are directly welded to a continuous chord that passes through the connection, with the branch or branches loaded by bending moments. A connection shall be classified

As a T-connection when there is one branch and it is perpendicular to the chord and as a Y-connection when there is one branch but not perpendicular to the chord.

As a cross-connection when there is a branch on each (opposite) side of the chord.

For the purposes of this Specification, the centerlines of the branch member(s) and the chord member shall lie in a common plane.

10.11.3.1 Notation and definitions

The notation and their definitions related to this Section are given in Sec 10.1.2.

10.11.3.2 Criteria for round HSS

The interaction of stress due to chord member forces and local branch connection forces shall be incorporated through the chord-stress interaction parameter Q_f .

When the chord is in tension,

$$Q_f = 1$$

When the chord is in compression,

$$Q_f = 1.0 - 0.3U(1 + U) \quad (6.10.241)$$

Where, U is the utilization ratio given by

$$U = \left| \frac{P_r}{A_g F_c} + \frac{M_r}{S F_c} \right| \quad (6.10.242)$$

And,

P_r = required axial strength in chord, N.

M_r = required flexural strength in chord, N-mm.

A_g = chord gross area, mm²

F_c = available stress, MPa.

S = chord elastic section modulus, mm³

For design according to Section 10.2.3.3 (LRFD):

$P_r = P_u$ = required axial strength in chord, using LRFD load combinations, N

$M_r = M_u$ = required flexural strength in chord, using LRFD load combinations, N-mm.

$F_c = F_y$, MPa.

For design according to Section 10.2.3.4 (ASD):

$P_r = P_a$ = required axial strength in chord, using *ASD load combinations*, N

$M_r = M_a$ = required flexural strength in chord, using ASD load combinations, N-mm.

$F_c = 0.6F_y$, MPa.

10.11.3.2.1 Limits of applicability

The criteria herein are applicable only when the connection configuration is within the following limits of applicability:

- (a) Branch angle: $\theta \geq 30^\circ$
- (b) Chord wall slenderness: ratio of diameter to wall thickness less than or equal to 50 for T - and Y-connections; less than or equal to 40 for cross-connections
- (c) Tension branch wall slenderness: ratio of diameter to wall thickness less than or equal to 50
- (d) Compression branch wall slenderness: ratio of diameter to wall thickness less than or equal to $0.05 E/F_y$
- (e) Width ratio: $0.2 < D_b/D \leq 1.0$
- (f) Strength: $F_y \leq 360$ MPa. for chord and branches
- (g) Ductility: $F_y/F_u \leq 0.8$

10.11.3.2.2 Branches with In-plane bending moments in T-, Y- and cross-connections

The design strength, ϕM_n and the allowable strength, M_n/Ω , shall be the lowest value obtained according to the limit states of chord plastification and shear yielding (punching).

For the limit state of chord plastification,

$$M_n \sin \theta = 5.39 F_y t^2 \gamma^{0.5} \beta D_b Q_f \quad (6.10.243)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

For the limit state of shear yielding (punching),

$$M_n = 0.6 F_y t D_b^2 [(1 + 3 \sin \theta)/4 \sin^2 \theta] \quad (6.10.244)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

This limit state need not be checked when $\beta > (1 - 1/\gamma)$.

10.11.3.2.3 Branches with out-of-plane bending moments in T-, Y- and cross-connections

The design strength, ϕM_n and the allowable strength, M_n/Ω , shall be the lowest value obtained according to the limit states of chord plastification and shear yielding (punching).

For the limit state of chord plastification,

$$M_n \sin \theta = F_y t^2 D_b \left[\frac{3.0}{1 - 0.81 \beta} \right] Q_f \quad (6.10.245)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

For the limit state of shear yielding (punching),

$$M_n = 0.6 F_y t D_b^2 [(3 + \sin \theta)/4 \sin^2 \theta] Q_f \quad (6.10.246)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

This limit state need not be checked when $\beta > (1 - 1/\gamma)$.

10.11.3.2.4 Branches with combined bending moment and axial force in T-, Y- and cross-connections

Connections subject to branch axial load, branch in-plane bending moment, and branch out-of-plane bending moment, or any combination of these load effects, should satisfy the following.

For design according to Section 10.2.3.3 (LRFD):

$$\left(\frac{P_r}{\phi P_n}\right) + \left(\frac{M_{r-ip}}{\phi M_{n-ip}}\right)^2 + (M_{r-op}/\phi M_{n-op}) \leq 1.0 \quad (6.10.247)$$

Where,

$P_r = P_u$ = required axial strength in branch, using LRFD load combinations, N

ϕP_n = design strength obtained from Sec 10.11.2.2.2

M_{r-ip} = required in-plane flexural strength in branch, using LRFD load combinations, N-mm.

ϕM_{n-ip} = design strength obtained from Sec 10.11.3.2.2

M_{r-op} = required out-of-plane flexural strength in branch, using LRFD load combinations, N-mm.

ϕM_{n-op} = design strength obtained from Sec 10.11.3.2.3

For design according to Section 10.2.3.4 (ASD):

$$\left(P_r/\left(\frac{P_n}{\Omega}\right)\right) + \left(\frac{M_{r-ip}}{\frac{M_{n-ip}}{\Omega}}\right)^2 + (M_{r-op}/(M_{n-op}/\Omega)) \leq 1.0 \quad (6.10.248)$$

Where,

$P_r = P_a$ = required axial strength in branch, using ASD load combinations, N

P_n/Ω = allowable strength obtained from Sec 10.11.2.2.2

M_{r-ip} = required in-plane flexural strength in branch, using ASD load combinations, N-mm.

M_{n-ip}/Ω = allowable strength obtained from Sec 10.11.3.2.2

M_{r-op} = required out-of-plane flexural strength in branch, using ASD load combinations, N-mm.

M_{n-op}/Ω = allowable strength obtained from Sec 10.11.3.2.3

10.11.3.3 Criteria for rectangular HSS

The interaction of stress due to chord member forces and local branch connection forces shall be incorporated through the chord-stress interaction parameter Q_f .

When the chord is in tension,

$$Q_f = 1$$

When the chord is in compression,

$$Q_f = \left(1.3 - \frac{0.4U}{\beta}\right) \leq 1 \quad (6.10.249)$$

Where, U is the utilization ratio given by

$$U = \left| \frac{P_r}{A_g F_c} + \frac{M_r}{S F_c} \right| \quad (6.10.250)$$

And,

P_r = required axial strength in chord, N.

M_r = required flexural strength in chord, N-mm.

A_g = chord gross area, mm²

F_c = available stress, MPa.

S = chord elastic section modulus, mm³.

For design according to Section 10.2.3.3 (LRFD):

$P_r = P_u$ = required axial strength in chord, using LRFD load combinations, N

$M_r = M_u$ = required flexural strength in chord, using LRFD load combinations, N-mm.

$F_c = F_y$, MPa.

For design according to Section 10.2.3.4 (ASD):

$P_r = P_a$ = required axial strength in chord, using ASD load combinations, N

$M_r = M_a$ = required flexural strength in chord, using ASD load combinations, N-mm.

$F_c = 0.6F_y$, MPa.

10.11.3.3.1 Limits of applicability

The criteria herein are applicable only when the connection configuration is within the following limits:

- (a) Branch angle is approximately 90°
- (b) Chord wall slenderness: ratio of overall wall width to thickness less than or equal to 35
- (c) Tension branch wall slenderness: ratio of overall wall width to thickness less than or equal to 35
- (d) Compression branch wall slenderness: ratio of overall wall width to thickness less than or equal to $1.25(E/F_{yb})^{0.5}$ and also less than 35
- (e) Width ratio: ratio of overall wall width of branch to overall wall width of chord greater than or equal to 0.25
- (f) Aspect ratio: $0.5 \leq \text{ratio of depth to width} \leq 2.0$
- (g) Strength: $F_y \leq 360 \text{ MPa}$. for chord and branches
- (h) Ductility: $F_y/F_u \leq 0.8$
- (i) Other limits apply for specific criteria

10.11.3.3.2 Branches with In-plane bending moments in T- and cross-connections

The design strength ϕM_n and the allowable strength, M_n/Ω , shall be the lowest value obtained according to the limit states of chord wall plastification, sidewall local yielding and local yielding due to uneven load distribution.

For the limit state of chord wall plastification,

$$M_n = F_y t^2 H_b \left[\left(\frac{1}{2\eta} \right) + \frac{2}{(1-\beta)^{0.5}} + \frac{\eta}{1-\beta} \right] Q_f \quad (6.10.251)$$

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

This limit state need not be checked when $\beta > 0.85$.

For the limit state of sidewall local yielding,

$$M_n = 0.5F_y^*t(H_b + 5t)^2 \quad (6.10.252)$$

$$\phi = 1.00(\text{LRFD}) \quad \Omega = 1.50 (\text{ASD})$$

Where,

$$F_y^* = F_y \text{ for T-connections}$$

$$F_y^* = 0.8 F_y \text{ for cross-connections}$$

This limit state need not be checked when $\beta < 0.85$.

For the limit state of local yielding due to uneven load distribution,

$$M_n = F_{yb} \left[Z_b - \left(1 - \frac{b_{eoi}}{B_b} \right) B_b H_b t_b \right] \quad (6.10.253)$$

$$\phi = 0.95 (\text{LRFD}) \quad \Omega = 1.58 (\text{ASD})$$

Where,

$$b_{eoi} = \left[\frac{10}{\left(\frac{B}{t} \right)} \right] [F_y t / (F_{yb} t_b)] B_b \leq B_b \quad (6.10.254)$$

Z_b = branch plastic section modulus about the axis of bending, mm³.

This limit state need not be checked when $\beta < 0.85$.

10.11.3.3.3 Branches with out-of-plane bending moments in T- and cross-connections

The design strength, ϕM_n and the allowable strength, M_n/Ω , shall be the lowest value obtained according to the limit states of chord wall plastification, sidewall local yielding, local yielding due to uneven load distribution and chord distortional failure.

For the limit state of chord wall plastification,

$$M_n = F_y t^2 \left[\frac{0.5H_b(1+\beta)}{(1-\beta)} + \left[\frac{2BB_b(1+\beta)}{1-\beta} \right]^{0.5} \right] Q_f \quad (6.10.255)$$

$$\phi = 1.00 (\text{LRFD}) \quad \Omega = 1.50 (\text{ASD})$$

This limit state need not be checked when $\beta > 0.85$.

For the limit state of sidewall local yielding,

$$M_n = F_y^* t (B - t) (H_b + 5t) \quad (6.10.256)$$

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

Where,

$$F_y^* = F_y \text{ for T-connections}$$

$$F_y^* = 0.8 F_y \text{ for cross-connections}$$

This limit state need not be checked when $\beta < 0.85$.

For the limit state of local yielding due to uneven load distribution,

$$M_n = F_{yb} \left[Z_b - 0.5 \left(1 - \frac{b_{eoi}}{B_b} \right)^2 B_b^2 t_b \right] \quad (6.10.257)$$

$$\phi = 0.95 \text{ (LRFD)} \quad \Omega = 1.58 \text{ (ASD)}$$

Where,

$$b_{eoi} = \left[\frac{10}{\left(\frac{B}{t} \right)} \right] [F_y t / (F_{yb} t_b)] B_b \leq B_b \quad (6.10.258)$$

Z_b = branch plastic section modulus about the axis of bending, mm³.

This limit state need not be checked when $\beta < 0.85$.

For the limit state of chord distortional failure,

$$M_n = 2F_y t [H_b t + [BHt(B + H)]^{0.5}] \quad (6.10.259)$$

$$\phi = 1.00 \text{ (LRFD)} \quad \Omega = 1.50 \text{ (ASD)}$$

This limit state need not be checked for cross-connections or for T-connections if chord distortional failure is prevented by other means.

10.11.3.3.4 Branches with combined bending moment and axial force in T- and cross-connections

Connections subject to branch axial load, branch in-plane bending moment and branch out-of-plane bending moment, or any combination of these load effects, should satisfy,

For design according to Section 10.2.3.3 (LRFD):

$$\left(\frac{P_r}{\phi P_n}\right) + \left(\frac{M_{r-ip}}{\phi M_{n-ip}}\right) + \left(\frac{M_{r-op}}{\phi M_{n-op}}\right) \leq 1.0 \quad (6.10.260)$$

Where,

$P_r = P_u$ = required axial strength in branch, using LRFD load combinations, N

ϕP_n = design strength obtained from Sec 10.11.2.3.2

M_{r-ip} = required in-plane flexural strength in branch, using LRFD load combinations, N-mm.

ϕM_{n-ip} = design strength obtained from Sec 10.11.3.3.2

M_{r-op} = required out-of-plane flexural strength in branch, using LRFD load combinations, N-mm.

ϕM_{n-op} = design strength obtained from Sec 10.11.3.3.3

For design according to Section 10.2.3.4 (ASD):

$$\left(\frac{P_r}{\frac{P_n}{\Omega}}\right) + \left(\frac{M_{r-ip}}{\frac{M_{n-ip}}{\Omega}}\right) + \left(\frac{M_{r-op}}{\frac{M_{n-op}}{\Omega}}\right) \leq 1.0 \quad (6.10.261)$$

Where,

$P_r = P_a$ = required axial strength in branch, using ASD load combinations, N

P_n/Ω = allowable strength obtained from Sec 10.11.2.3.2

M_{r-ip} = required in-plane flexural strength in branch, using ASD load combinations, N-mm.

M_{n-ip}/Ω = allowable strength obtained from Sec 10.11.3.3.2

M_{r-op} = required out-of-plane flexural strength in branch, using ASD load combinations, N-mm.

M_{n-op}/Ω = allowable strength obtained from Sec 10.11.3.3.3

10.12 Design for Serviceability

This Chapter addresses serviceability performance design requirements.

10.12.1 General Provisions

Serviceability is a state in which the function of a building, its appearance, maintainability, durability, and comfort of its occupants are preserved under normal usage. Limiting values of structural behavior for serviceability (for example, maximum deflections, accelerations) shall be chosen with due regard to the intended function of the structure. Serviceability shall be evaluated using appropriate load combinations for the serviceability limit states identified.

10.12.2 Camber

Where camber is used to achieve proper position and location of the structure, the magnitude, direction and location of camber shall be specified in the structural drawings in accordance with the provisions of Chapter 1.

10.12.3 Deflections

Deflections in structural members and structural systems under appropriate service load combinations shall not impair the serviceability of the structure. Limiting values of deflections of various structural members shall be in accordance with those specified in Sec 1.4 Chapter 1.

10.12.4 Drift

Drift of a structure shall be evaluated under service loads to provide for serviceability of the structure, including the integrity of interior partitions and exterior cladding. Drift under strength load combinations shall not cause collision with adjacent structures or exceed the limiting values specified in Sec 1.5.6 Chapter 1.

10.12.5 Vibration

The effect of vibration on the comfort of the occupants and the function of the structure shall be considered. Sources of vibration to be considered include pedestrian loading, vibrating machinery and others identified for the structure. It must be shown by any rational method of analysis that the vibrations induced by any source including the above mentioned ones is within tolerable limit and shall not cause any adverse effect on the safety, stability and durability of the structure.

10.12.6 Wind-Induced Motion

The effect of wind-induced motion of buildings on the comfort of occupants shall be considered. For flexible building and structures as defined in Sec. 2.4.2 Chapter 2, it must be shown by a rational dynamic analysis that wind induced vibration does not cause any discomfort to occupants as well as the wind induced dynamic effect does not cause any adverse effect on the safety, stability and durability of the structure.

10.12.7 Expansion and Contraction

The effects of thermal expansion and contraction of a building shall be considered. Damage to building *cladding* can cause water penetration and may lead to corrosion.

10.12.8 Connection Slip

The effects of connection slip shall be included in the design where slip at bolted connections may cause deformations that impair the serviceability of the structure. Where appropriate, the connection shall be designed to preclude slip. For the design of slip-critical connections see Sections 10.10.3.8 and 10.10.3.9.

10.13 Fabrication, Erection and Quality Control

This Chapter addresses requirements for design and shop drawings, fabrication, shop painting, erection and quality control.

10.13.1 Design Drawings and Specifications

Structural Design Drawings and Specifications

Unless otherwise indicated in the contract documents, the structural design drawings shall be based upon consideration of the design loads and forces to be resisted by structural steel frame in the completed project.

The structural design drawings shall clearly show the work that is to be performed and shall give the following information with sufficient dimensions to accurately convey the quantity and nature of the structural steel to be fabricated:

- (a) The size, section, material grade and location of all members;
- (b) All geometry and working points necessary for layout;
- (c) Floor elevations;
- (d) Column centers and offsets;

- (e) The camber requirements for members;
- (f) Joining requirements between elements of built-up members; and,
- (g) The information that is required in Sections 10.13.1.1.1 to 10.13.1.1.6.

The structural steel specifications shall include any special requirements for the fabrication and erection of the structural steel.

The structural design drawings, specifications and addenda shall be numbered and dated for the purposes of identification.

10.13.1.1 Detailing of components

Permanent bracing, column stiffeners, column web doubler plates, bearing stiffeners in beams and girders, web reinforcement, openings for other trades and other special details, where required, shall be shown in sufficient detail in the structural design drawings so that the quantity, detailing and fabrication requirements for these items can be readily understood.

10.13.1.2 Designer's responsibility

The owner's designated representative for design shall indicate one of the following options for each connection:

- (1) The complete connection design shall be shown in the structural design drawings;
- (2) In the structural design drawings or specifications, the connection shall be designated to be selected or completed by an experienced steel detailer; or,
- (3) In the structural design drawings or specifications, the connection shall be designated to be designed by a licensed professional engineer working for the fabricator.

In all of the above options,

- (a) The requirements of Sec 10.13.1.1 shall apply; and,
- (b) The approvals process in Sec 10.13.2.4 shall be followed.

When option (2) above is specified:

The experienced steel detailer shall utilize tables or schematic information provided in the structural design drawings in the selection or completion of the connections. When such information is not provided, standard reference information as approved by the owner's designated representative for design, shall be used.

When option (2) or (3) above is specified:

The owner's designated representative for design shall provide the following information in the structural design drawings and specifications:

- (a) Any restrictions on the types of connections that are permitted;
- (b) Data concerning the loads, including shears, moments, axial forces and transfer forces, that are to be resisted by the individual members and their connections, sufficient to allow the selection, completion, or design of the connection details while preparing the shop and erection drawings;
- (c) Whether the data required in (b) is given at the service-load level or the factored-load level;
- (d) Whether LRFD or ASD is to be used in the selection, completion, or design of connection details; and,
- (e) What substantiating connection information, if any, is to be provided with the shop and erection drawings to the owner's designated representative for design.

When option (3) above is specified:

- (a) The fabricator shall submit in a timely manner representative samples of the required substantiating connection information to the owner's designated representatives for design and construction. The owner's designated representative for design shall confirm in writing in a timely manner that these representative samples are consistent with the requirements in the contract documents, or shall advise what modifications are required to bring the representative samples into compliance with the requirements in the contract documents. This initial submittal and review is in addition to the requirements in Sec 10.13.2.4.
- (b) The Engineer in responsible charge of the connection design shall review and confirm in writing as part of the substantiating connection information, that the shop and erection drawings properly incorporate the connection designs. However, this review by the Engineer in responsible charge of the connection design does not replace the approval process of the shop and erection drawings by the owner's designated representative for design in Sec 10.13.2.4.
- (c) The fabricator shall provide a means by which the substantiating connection information is referenced to the related connections on the shop and erection drawings for the purpose of review.

10.13.1.2.1 Levelling plates

When leveling plates are to be furnished as part of the contract requirements, their locations and required thickness and sizes shall be specified in the contract documents.

10.13.1.2.2 Non-structural elements

When the structural steel frame, in the completely erected and fully connected state, requires interaction with non-structural steel elements for strength and/or stability, those non-structural steel elements shall be identified in the contract documents as required in Sec 10.13.5.10.

10.13.1.2.3 Camber

When camber is required, the magnitude, direction and location of camber shall be specified in the structural design drawings.

10.13.1.2.4 Painting information

Specific members or portions thereof that are to be left unpainted shall be identified in the contract documents. When shop painting is required, the painting requirements shall be specified in the contract documents, including the following information:

- (a) The identification of specific members or portions thereof to be painted;
- (b) The surface preparation that is required for these members;
- (c) The paint specifications and manufacturer's product identification that are required for these members; and,
- (d) The minimum dry-film shop-coat thickness that is required for these members.

10.13.1.3 Architectural, electrical and mechanical design drawings and specifications

All requirements for the quantities, sizes and locations of structural steel shall be shown or noted in the structural design drawings. The use of architectural, electrical and/or mechanical design drawings as a supplement to the structural design drawings is permitted for the purposes of defining detail configurations and construction information.

10.13.1.4 Discrepancies

When discrepancies exist between the design drawings and specifications, the design drawings shall govern. When discrepancies exist between scale dimensions in the design drawings and the figures written in them, the figures shall govern. When discrepancies exist between the structural design drawings and the architectural, electrical or mechanical design drawings or design drawings for other trades, the structural design drawings shall govern.

When a discrepancy is discovered in the contract documents in the course of the fabricator's work, the fabricator shall promptly notify the owner's designated representative for construction so that the discrepancy can be resolved by the owner's designated representative for design. Such resolution shall be timely so as not to delay the fabricator's work. See Sections 10.13.1.5 and 10.13.7.3.

10.13.1.5 Legibility of design drawings

Design drawings shall be clearly legible and drawn to an identified scale that is appropriate to clearly convey the information.

10.13.1.6 Revisions to the design drawings and specifications

Revisions to the design drawings and specifications shall be made either by issuing new design drawings and specifications or by reissuing the existing design drawings and specifications. In either case, all revisions, including revisions that are communicated through responses to RFIs or the annotation of shop and/or erection drawings (see Sec 10.13.2.4.2), shall be clearly and individually indicated in the contract documents. The contract documents shall be dated and identified by revision number. Each design drawings shall be identified by the same drawing number throughout the duration of the project, regardless of the revision. See also Sec 10.13.7.3.

10.13.2 Shop and Erection Drawings

Shop drawings shall be prepared in advance of fabrication and give complete information necessary for the fabrication of the component parts of the structure, including the location, type and size of welds and bolts. Erection drawings shall be prepared in advance of erection and give information necessary for erection of the structure. Shop and erection drawings shall clearly distinguish between shop and field welds and bolts and shall clearly identify pretensioned and slip-critical high-strength bolted connections. Shop and erection drawings shall be made with due regard to speed and economy in fabrication and erection.

10.13.2.1 Owner responsibility

The owner shall furnish, in a timely manner and in accordance with the contract documents, complete structural design drawings and specifications that have been released for construction. Unless otherwise noted, design drawings that are provided as part of a contract bid package shall constitute authorization by the owner that the design drawings are released for construction.

Records of the meetings should be written and distributed to all parties. Subsequent meetings to discuss progress and issues that arise during construction also can be helpful, particularly when they are held on a regular schedule.

10.13.2.2 Fabricator responsibility

Except as provided in Sec 10.13.2.5, the fabricator shall produce shop and erection drawings for the fabrication and erection of the structural steel and is responsible for the following :

- (a) The transfer of information from the contract documents into accurate and complete shop and erection drawings; and,
- (b) The development of accurate, detailed dimensional information to provide for the fit-up of parts in the field.

Each shop and erection drawing shall be identified by the same drawing number throughout the duration of the project and shall be identified by revision number and date, with each specific revision clearly identified.

When the fabricator submits a request to change connection details that are described in the contract documents, the fabricator shall notify the owner's designated representatives for design and construction in writing in advance of the submission of the shop and erection drawings. The owner's designated representative for design shall review and approve or reject the request in a timely manner.

When requested to do so by the owner's designated representative for design, the fabricator shall provide to the owner's designated representatives for design and construction its schedule for the submittal of shop and erection drawings so as to facilitate the timely flow of information between all parties.

10.13.2.3 Use of CAD files and/or copies of design drawings

The fabricator shall neither use nor reproduce any part of the design drawings as part of the shop or erection drawings without the written permission of the owner's designated representative for design. When CAD files or copies of the design drawings are made available for the fabricator's use, the fabricator shall accept this information under the following conditions:

- (a) All information contained in the CAD files or copies of the design drawings shall be considered instruments of service of the owner's designated representative for design and shall not be used for other projects, additions to the project or the completion of the project by others. CAD files and copies of the design drawings shall remain the property of the owner's designated representative for design and in no case shall the transfer of these CAD files or copies of design drawings be considered a sale.

- (b) The CAD files or copies of the design drawings shall not be considered to be contract documents. In the event of a conflict between the design drawings and the CAD files or copies thereof, the design drawings shall govern;
- (c) The use of CAD files or copies of the design drawings shall not in any way obviate the fabricator's responsibility for proper checking and coordination of dimensions, details, member sizes and fit-up and quantities of materials as required to facilitate the preparation of shop and erection drawings that are complete and accurate as required in Sec 4.2; and,
- (d) The fabricator shall remove information that is not required for the fabrication or erection of the structural steel from the CAD files or copies of the design drawings.

10.13.2.4 Approval

Except as provided in Sec 10.13.2.5, the shop and erection drawings shall be submitted to the owner's designated representatives for design and construction for review and approval. The shop and erection drawings shall be returned to the fabricator within 14 calendar days.

Final substantiating connection information, if any, shall also be submitted with the shop and erection drawings. The owner's designated representative for design is the final authority in the event of a disagreement between parties regarding connection design.

Approved shop and erection drawings shall be individually annotated by the owner's designated representatives for design and construction as either approved or approved subject to corrections noted. When so required, the fabricator shall subsequently make the corrections noted and furnish corrected shop and erection drawings to the owner's designated representatives for design and construction.

10.13.2.4.1 Constituents of approval

Approval of the shop and erection drawings, approval subject to corrections noted and similar approvals shall constitute the following:

- (a) Confirmation that the fabricator has correctly interpreted the contract documents in the preparation of those submittals;
- (b) Confirmation that the owner's designated representative for design has reviewed and approved the connection details shown on the shop and erection drawings and submitted in accordance with Sections 10.13.1 and 10.13.2, if applicable; and,

- (c) Release by the owner's designated representatives for design and construction for the fabricator to begin fabrication using the approved submittals.

Such approval shall not relieve the fabricator of the responsibility for either the accuracy of the detailed dimensions in the shop and erection drawings or the general fit-up of parts that are to be assembled in the field.

The fabricator shall determine the fabrication schedule that is necessary to meet the requirements of the contract.

10.13.2.4.2 Authorization by owner

Unless otherwise noted, any additions, deletions or revisions that are indicated in responses to RFIs or on the approved shop and erection drawings shall constitute authorization by the owner that the additions, deletions or revisions are released for construction. The fabricator and the erector shall promptly notify the owner's designated representative for construction when any direction or notation in responses to RFIs or on the shop or erection drawings or other information will result in an additional cost and/or a delay. See Sections 10.13.1.5 and 10.13.7.3.

10.13.2.5 Shop and/or erection drawings not furnished by the fabricator

When the shop and erection drawings are not prepared by the fabricator, but are furnished by others, they shall be delivered to the fabricator in a timely manner. These shop and erection drawings shall be prepared, insofar as is practical, in accordance with the shop fabrication and detailing standards of the fabricator. The fabricator shall neither be responsible for the completeness or accuracy of shop and erection drawings so furnished, nor for the general fit-up of the members that are fabricated from them.

10.13.2.6 The RFI process

When requests for information (RFIs) are issued, the process shall include the maintenance of a written record of inquiries and responses related to interpretation and implementation of the contract documents, including the clarifications and/or revisions to the contract documents that result, if any. RFIs shall not be used for the incremental release for construction of design drawings. When RFIs involve discrepancies or revisions, see Sections 10.13.1.3, 10.13.1.5, and 10.13.2.4.2.

10.13.2.7 Erection drawings

Erection drawings shall be provided to the erector in a timely manner so as to allow the erector to properly plan and perform the work.

10.13.3 Materials**10.13.3.1 Mill materials**

Unless otherwise noted in the contract documents, the fabricator is permitted to order the materials that are necessary for fabrication when the fabricator receives contract documents that have been released for construction.

Unless otherwise specified by means of special testing requirements in the contract documents, mill testing shall be limited to those tests that are required for the material in the ASTM specifications indicated in the contract documents. Materials ordered to special material requirements shall be marked by the supplier as specified in ASTM A6/A6M Section 12 prior to delivery to the fabricator's shop or other point of use. Such material not so marked by the supplier, shall not be used until:

- (a) Its identification is established by means of testing in accordance with the applicable ASTM specifications; and,
- (b) A fabricator's identification mark, as described in Sec 10.13.2 has been applied. When mill material does not satisfy ASTM A6/A6M tolerances for camber, profile, flatness or sweep, the fabricator shall be permitted to perform corrective procedures, including the use of controlled heating and/or mechanical straightening, subject to the limitations in the AISC Specification.

10.13.3.2 Stock materials

If used for structural purposes, materials that are taken from stock by the fabricator shall be of a quality that is at least equal to that required in the ASTM specifications indicated in the contract documents.

Material test reports shall be accepted as sufficient record of the quality of materials taken from stock by the fabricator. The fabricator shall review and retain the material test reports that cover such stock materials. However, the fabricator need not maintain records that identify individual pieces of stock material against individual material test reports, provided the fabricator purchases stock materials that meet the requirements for material grade and quality in the applicable ASTM specifications.

Stock materials that are purchased under no particular specification, under a specification that is less rigorous than the applicable ASTM specifications or without material test reports or other recognized test reports shall not be used without the approval of the owner's designated representative for design.

10.13.4 Fabrication

10.13.4.1 Cambering, curving and straightening

Local application of heat or mechanical means is permitted to be used to introduce or correct camber, curvature and straightness. The temperature of heated areas, as measured by approved methods, shall not exceed 593⁰ C for A514/A514M and A852/A852M steel nor 1,200⁰ F (649⁰ C) for other steels.

10.13.4.2 Thermal cutting

Thermally cut edges shall meet the requirements of AWS D1.1, Sections 5.15.1.2, 5.15.4.3 and 5.15.4.4 with the exception that thermally cut free edges that will be subject to calculated static tensile stress shall be free of round-bottom gouges greater than 5 mm deep and sharp V-shaped notches. Gouges deeper than 5 mm and notches shall be removed by grinding or repaired by welding.

Reentrant corners, except reentrant corners of *beam copes* and weld access holes, shall meet the requirements of AWS D1.1, Section A5.16. If another specified contour is required it must be shown on the contract documents.

Beam copes and weld access holes shall meet the geometrical requirements of Sec 10.10.1.6. Beam copes and weld access holes in shapes that are to be galvanized shall be ground. For shapes with a flange thickness not exceeding 50 mm the roughness of thermally cut surfaces of copes shall be no greater than a surface roughness value of 50 μ m as defined in ASME B46.1 Surface Texture (Surface Roughness, Waviness, and Lay). For beam copes and weld access holes in which the curved part of the access hole is thermally cut in ASTM A6/A6M hot-rolled shapes with a flange thickness exceeding 50 mm and welded built-up shapes with material thickness greater than 50 mm, a preheat temperature of not less than 66⁰ C shall be applied prior to thermal cutting. The thermally cut surface of access holes in ASTM A6/A6M hot-rolled shapes with a flange thickness exceeding 50 mm and built-up shapes with a material thickness greater than 50 mm shall be ground and inspected for cracks using magnetic particle inspection in accordance with ASTM E709. Any crack is unacceptable regardless of size or location.

10.13.4.3 Planing of edges

Planing or finishing of sheared or thermally cut edges of plates or shapes is not required unless specifically called for in the contract documents or included in a stipulated edge preparation for welding.

10.13.4.4 Welded construction

The technique of welding, the workmanship, appearance and quality of welds, and the methods used in correcting nonconforming work shall be in accordance with AWS D1.1 except as modified in Section J2.

10.13.4.5 Bolted construction

Parts of bolted members shall be pinned or bolted and rigidly held together during assembly. Use of a drift pin in bolt holes during assembly shall not distort the metal or enlarge the holes. Poor matching of holes shall be cause for rejection.

Bolt holes shall comply with the provisions of the RCSC Specification for Structural Joints Using ASTM A325 or A490 Bolts, Sec 10.13.3.3 except that thermally cut holes shall be permitted with a surface roughness profile not exceeding 25 μm as defined in ASME B46.1. Gouges shall not exceed a depth of 2 mm.

Fully inserted finger shims, with a total thickness of not more than 6 mm within a joint are permitted in joints without changing the strength (based upon hole type) for the design of connections. The orientation of such shims is independent of the direction of application of the load.

The use of high-strength bolts shall conform to the requirements of the RCSC Specification for Structural Joints Using ASTM A325 or A490 Bolts, except as modified in Sec 10.10.3.

10.13.4.6 Compression joints

Compression joints that depend on contact bearing as part of the splice strength shall have the bearing surfaces of individual fabricated pieces prepared by milling, sawing, or other suitable means.

10.13.4.7 Dimensional tolerances

Dimensional tolerances shall be in accordance with ASTM A6/A6M.

10.13.4.8 Finish of column bases

Column bases and base plates shall be finished in accordance with the following requirements:

- (1) Steel bearing plates 50 mm or less in thickness are permitted without milling, provided a satisfactory contact bearing is obtained. Steel bearing plates over 50 mm but not over 100 mm in thickness are permitted to be straightened by pressing or, if presses are not available, by milling for bearing surfaces (except as noted in subparagraphs 2 and 3 of

this Section), to obtain a satisfactory contact bearing. Steel bearing plates over 100 mm in thickness shall be milled for bearing surfaces (except as noted in subparagraphs 2 and 3 of this Section).

- (2) Bottom surfaces of bearing plates and column bases that are grouted to ensure full bearing contact on foundations need not be milled.
- (3) Top surfaces of bearing plates need not be milled when complete-joint- penetration groove welds are provided between the column and the bearing plate.

10.13.4.9 Holes for anchor rods

Holes for anchor rods shall be permitted to be thermally *cut* in accordance with the provisions of Sec 10.13.2.2.

10.13.4.10 Drain holes

When water can collect inside HSS or box members, either during construction or during service, the member shall be sealed, provided with a drain hole at the base, or protected by other suitable means.

10.13.4.11 Requirements for galvanized members

Members and parts to be galvanized shall be designed, detailed and fabricated to provide for flow and drainage of pickling fluids and zinc and to prevent pressure build-up in enclosed parts.

10.13.5 Shop Painting

10.13.5.1 General requirements

Shop paint is not required unless specified by the contract documents.

10.13.5.2 Inaccessible surfaces

Except for contact surfaces, surfaces inaccessible after shop assembly shall be cleaned and painted prior to assembly, if required by the design documents.

10.13.5.3 Contact surfaces

Paint is permitted in bearing-type connections. For slip-critical connections, the faying surface requirements shall be in accordance with the RCSC Specification for Structural Joints Using ASTM A325 or A490 Bolts, Section 3.2.2(b).

10.13.5.4 Finished surfaces

Machine-finished surfaces shall be protected against corrosion by a rust inhibitive coating that can be removed prior to erection, or which has characteristics that make removal prior to erection unnecessary.

10.13.5.5 Surfaces adjacent to field welds

Unless otherwise specified in the design documents, surfaces within 50 mm of any field weld location shall be free of materials that would prevent proper welding or produce objectionable fumes during welding.

10.13.6 Erection

10.13.6.1 Alignment of column bases

Column bases shall be set level and to correct elevation with full bearing on concrete or masonry.

10.13.6.2 Bracing

The frame of steel skeleton buildings shall be carried up true and plumb. Temporary bracing shall be provided, wherever necessary to support the loads to which the structure may be subjected, including equipment and the operation of same. Such bracing shall be left in place as long as required for safety.

10.13.6.3 Alignment of structural elements

No permanent bolting or welding shall be performed until the adjacent affected portions of the structure have been properly aligned.

10.13.6.4 Fit of column compression joints and base plates

Lack of contact bearing not exceeding a gap of 2 mm, regardless of the type of splice used (partial-joint-penetration groove welded or bolted), is permitted. If the gap exceeds 2 mm, but is less than 6 mm, and if an engineering investigation shows that sufficient contact area does not exist, the gap shall be packed out with nontapered steel shims. Shims need not be other than mild steel, regardless of the grade of the main material.

10.13.6.5 Field welding

Shop paint on surfaces adjacent to joints to be field welded shall be wire brushed if necessary to assure weld quality.

Field welding of attachments to installed embedments in contact with concrete shall be done in such a manner as to avoid excessive thermal expansion of the embedment which could result in spalling or cracking of the concrete or excessive stress in the embedment anchors.

10.13.6.6 Field painting

Responsibility for touch-up painting, cleaning and field painting shall be allocated in accordance with accepted local practices, and this allocation shall be set forth explicitly in the design documents.

10.13.6.7 Connections

As erection progresses, the structure shall be securely bolted or welded to support dead, wind and erection loads.

10.13.7 Quality Control

The fabricator shall provide quality control procedures to the extent that the fabricator deems necessary to assure that the work is performed in accordance with this Specification. In addition to the fabricator's quality control procedures, material and workmanship at all times may be subject to inspection by qualified inspectors representing the purchaser. If such inspection by representatives of the purchaser will be required, it shall be so stated in the design documents.

10.13.7.1 Cooperation

As far as possible, the inspection by representatives of the purchaser shall be made at the fabricator's plant. The fabricator shall cooperate with the inspector, permitting access for inspection to all places where work is being done. The purchaser's inspector shall schedule this work for minimum interruption to the work of the fabricator.

10.13.7.2 Rejections

Material or workmanship not in conformance with the provisions of this Specification may be rejected at any time during the progress of the work.

The fabricator shall receive copies of all reports furnished to the purchaser by the inspection agency.

10.13.7.3 Inspection of welding

The inspection of welding shall be performed in accordance with the provisions of AWS D1.1 except as modified in Sec 10.10.2.

When visual inspection is required to be performed by AWS certified welding inspectors, it shall be so specified in the design documents.

When nondestructive testing is required, the process, extent and standards of acceptance shall be clearly defined in the design documents.

10.13.7.4 Inspection of slip-critical high-strength bolted connections

The inspection of slip-critical high-strength bolted connections shall be in accordance with the provisions of the RCSC Specification for Structural Joints Using ASTM A325 or A490 Bolts.

10.13.7.5 Identification of steel

The fabricator shall be able to demonstrate by a written procedure and by actual practice a method of material identification, visible at least to the “fit-up” operation, for the main structural elements of each shipping piece.

10.14 Direct Analysis Method

This Section addresses the direct analysis method for structural systems comprised of moment frames, braced frames, shear walls, or combinations thereof.

10.14.1 General Requirements

Members shall satisfy the provisions of Sec 10.8.1 with the nominal column strengths, P_n , determined using $K = 1.0$. The required strengths for members, connections and other structural elements shall be determined using a second-order elastic analysis with the constraints presented in Sec 10.14.3. All component and connection deformations that contribute to the lateral displacement of the structure shall be considered in the analysis.

10.14.2 Notional Loads

Notional loads shall be applied to the lateral framing system to account for the effects of geometric imperfections, inelasticity, or both. Notional loads are lateral loads that are applied at each framing level and specified in terms of the gravity loads applied at that level. The gravity load used to determine the notional load shall be equal to or greater than the gravity load associated with the load combination being evaluated. Notional loads shall be applied in the direction that adds to the destabilizing effects under the specified load combination.

10.14.3 Design-Analysis Constraints

- (1) The second-order analysis shall consider both $P-\delta$ and $P-\Delta$ effects. It is permitted to perform the analysis using any general second-order analysis method, or by the amplified first-order analysis method of Sec 10.3.2, provided that the B_1 and B_2

factors are based on the reduced stiffnesses defined in Eq. 6.10.263 and 6.10.264. Analyses shall be conducted according to the design and loading requirements specified in either Sec 10.2.3.3 (LRFD) or Sec 10.2.3.4 (ASD). For ASD, the second-order analysis shall be carried out under 1.6 times the ASD load combinations and the results shall be divided by 1.6 to obtain the required strengths.

Methods of analysis that neglect the effects of $P - \delta$ on the lateral displacement of the structure are permitted where the axial loads in all members whose flexural stiffnesses are considered to contribute to the lateral stability of the structure satisfy the following limit:

$$\alpha P_r < 0.15 P_{eL} \quad (6.10.262)$$

Where,

P_r = required axial compressive strength under LRFD or ASD load combinations, N

$P_{eL} = \pi^2 EI / L^2$ evaluated in the plane of bending

And, $\alpha = 1.0$ (LRFD) $\alpha = 1.6$ (ASD)

- (2) A notional load, $N_i = 0.002Y_i$, applied independently in two orthogonal directions, shall be applied as a lateral load in all load combinations. This load shall be in addition to other lateral loads, if any,

Where,

N_i = notional lateral load applied at level i , N

Y_i = gravity load from LRFD load combination or 1.6 times the ASD load combination applied at level i , N

The notional load coefficient of 0.002 is based on an assumed initial story out-of-plumbness ratio of 1/500. Where a smaller assumed out-of-plumbness is justified, the notional load coefficient may be adjusted proportionally.

For frames where the ratio of second-order drift to first-order drift is equal to or less than 1.5, it is permissible to apply the notional load, N_i , as a minimum lateral load for the gravity-only load combinations and not in combination with other lateral loads.

For all cases, it is permissible to use the assumed out-of-plumbness geometry in the analysis of the structure in lieu of applying a notional load or a minimum lateral load as defined above.

- (3) A reduced flexural stiffness, EI^* ,

$$EI^* = 0.8\tau_b EI \quad (6.10.263)$$

shall be used for all members whose flexural stiffness is considered to contribute to the lateral stability of the structure,

Where,

I = moment of inertia about the axis of bending, mm^4

$\tau_b = 1.0$ for $\alpha P_r/P_y \leq 0.5$

$= 4[\alpha P_r/P_y(1 - \alpha P_r/P_y)]$ for $\alpha P_r/P_y > 0.5$

P_r = required axial compressive strength under *LRFD* or *ASD load combinations*, N

$P_y = AF_y$, member yield strength, N

And $\alpha = 1.0$ (LRFD) $\alpha = 1.6$ (ASD)

In lieu of using $\tau_b < 1.0$, where, $\alpha P_r/P_y > 0.5$, $\tau_b = 1.0$ may be used for all members, provided that an additive notional load of $0.001 Y_i$ is added to the notional load required in (2).

- (4) A reduced axial stiffness, EA^* ,

$$EA^* = 0.8 EA \quad (6.10.264)$$

shall be used for members whose axial stiffness is considered to contribute to the lateral stability of the structure, where A is the cross-sectional member area.

10.15 Inelastic Analysis and Design

10.15.1 General Provisions

Inelastic analysis is permitted for design according to the provisions of Sec 10.2.3.3 (LRFD). Inelastic analysis is not permitted for design according to the provisions of Sec 10.2.3.4 (ASD) except as provided in Sec 10.15.3.

10.15.2 Materials

Members undergoing plastic hinging shall have a specified minimum yield stress not exceeding 450 MPa.

10.15.3 Moment Redistribution

Beams and girders composed of compact sections as defined in Sec 10.2.4 and satisfying the unbraced length requirements of Sec 10.15.7, including composite members, may be proportioned for nine-tenths of the negative moments at points of support, produced by the gravity loading computed by an elastic analysis, provided that the maximum positive moment is increased by one-tenth of the average negative moments. This reduction is not permitted for moments produced by loading on cantilevers and for design according to Sections 10.15.4 to 10.15.8 of this Section.

If the negative moment is resisted by a *column* rigidly framed to the beam or girder, the one-tenth reduction may be used in proportioning the column for combined axial *force* and flexure, provided that the axial force does not exceed $1.5\phi_c F_y A_g$ for LRFD or $0.15F_y A_g / \Omega_c$ for ASD,

Where,

A_g = gross area of member, mm²

F_y = specified minimum yield stress of the compression flange, MPa.

ϕ_c = resistance factor for compression = 0.90

Ω_c = safety factor for compression = 1.67

10.15.4 Local Buckling

Flanges and webs of members subject to plastic hinging in combined flexure and axial compression shall be compact with width-thickness ratios less than or equal to the limiting λ_p defined in Table 6.10.1 or as modified as follows:

- (a) For webs of doubly symmetric wide flange members and rectangular HSS in combined flexure and compression

- (i) For $P_u/(\phi_b P_y) \leq 0.125$

$$h/tw \leq 3.76 \sqrt{\frac{E}{F_y}} \left(1 - \frac{2.75 P_u}{\phi_b P_y} \right) \quad (6.10.265)$$

- (ii) For $P_u/\phi_b P_y > 0.125$

$$h/tw \leq 1.12 \sqrt{\frac{E}{F_y}} \left(2.33 - \frac{P_u}{\phi_b P_y} \right) \geq 1.49 \sqrt{\frac{E}{F_y}} \quad (6.10.266)$$

Where,

E = modulus of elasticity of steel 200 000 MPa.

F_y = specified minimum yield stress of the type of steel being used, MPa.

h = as defined in Sec 10.2.4.2, mm.

P_u = required axial strength in compression, N.

P_y = member yield strength, N.

t_w = web thickness, mm.

ϕ_b = resistance factor for flexure = 0.90

- (b) For flanges of rectangular box and hollow structural sections of uniform thickness subject to bending or compression, flange cover plates, and diaphragm plates between lines of fasteners or welds

$$b/t \leq 0.94 \sqrt{E/F_y} \quad (6.10.267)$$

Where,

b = as defined in Sec 10.2.4.2, mm.

t = as defined in Sec 10.2.4.2, mm.

- (c) For circular hollow sections in flexure

$$D/t \leq 0.045 E/F_y \quad (6.10.268)$$

Where,

D = outside diameter of round HSS member, mm.

10.15.5 Stability and Second-Order Effects

Continuous beams not subjected to axial loads and that do not contribute to lateral stability of framed structures may be designed based on a first-order inelastic analysis or a plastic mechanism analysis.

Braced frames and moment frames may be designed based on a first-order inelastic analysis or a plastic mechanism analysis provided that stability and second-order effects are taken into account.

Structures may be designed on the basis of a second-order inelastic analysis. For beam-columns, connections and connected members, the required strengths shall be determined from a second-order inelastic analysis, where equilibrium is satisfied on deformed geometry, taking into account the change in stiffness due to yielding

10.15.5.1 Braced frames

In braced frames designed on the basis of inelastic analysis, braces shall be designed to remain elastic under the design loads. The required axial strength for columns and compression braces shall not exceed $\phi_c 0.85 F_y A_g$,

Where, $\phi_c = 0.90$ (LRFD)

10.15.5.2 Moment frames

In moment frames designed on the basis of inelastic analysis, the required axial strength of columns shall not exceed $\phi_c 0.75 F_y A_g$

Where, $\phi_c = 0.90$ (LRFD)

10.15.6 Columns and Other Compression Members

In addition to the limits set in Sections 10.15.5.1 and 10.15.5.2, the required axial strength of columns designed on the basis of inelastic analysis shall not exceed the design strength, $\phi_c P_n$, determined according to the provisions of Sec 10.5.3.

Design by inelastic analysis is permitted if the column slenderness ratio, L/r , does not exceed $4.71\sqrt{E/F_y}$,

Where,

L = laterally unbraced length of a member, mm.

r = governing radius of gyration, mm.

10.15.7 Beams and Other Flexural Members

The required moment strength, M_u , of beams designed on the basis of inelastic analysis shall not exceed the design strength, ϕM_n , where

$$M_n = M_p = F_y Z < 1.6 F_y S \quad (6.10.269)$$

$$\phi_c = 0.90 \text{ (LRFD)}$$

- (a) For doubly symmetric and singly symmetric I-shaped members with the compression flange equal to or larger than the tension flange loaded in the plane of the web:

$$L_{pd} = \left[0.12 + 0.076 \left(\frac{M_1}{M_2} \right) \right] \left(\frac{E}{F_y} \right) r_y \quad (6.10.270)$$

Where,

M_1 = smaller moment at end of unbraced length of beam, N-mm

M_2 = larger moment at end of unbraced length of beam, N-mm

r_y = radius of gyration about minor axis, mm

$(M_1 \times M_2)$ is positive when moments cause reverse curvature and negative for single curvature.

- (b) For solid rectangular bars and symmetric box beams:

$$L_{pd} = \left[0.17 + 0.10 \left(\frac{M_1}{M_2} \right) \right] \left(\frac{E}{F_y} \right) r_y \geq 0.10 \left(\frac{E}{F_y} \right) r_y \quad (6.10.271)$$

There is no limit on L_b for members with circular or square cross sections or for any beam bent about its minor axis.

10.15.8 Members under Combined Forces

When inelastic analysis is used for symmetric members subject to bending and axial force, the provisions in Sec 10.8.1 apply.

Inelastic analysis is not permitted for members subject to torsion and combined torsion, flexure, shear and/or axial force.

10.15.9 Connections

Connections adjacent to plastic hinging regions of connected members shall be designed with sufficient strength and ductility to sustain the forces and deformations imposed under the required loads.

10.16 Design for Ponding

This Section provides methods for determining whether a roof system has adequate strength and stiffness to resist ponding.

10.16.1 Simplified Design for Ponding

The roof system shall be considered stable for ponding and no further investigation is needed if both of the following two conditions are met:

$$C_p + 0.9 C_s \leq 0.25 \quad (6.10.272)$$

$$I_d \geq 3940 S^4 \quad (6.10.273)$$

Where,

$$C_p = \frac{504 L_s L_p^4}{I_p}$$

$$C_s = \frac{504 S L_s^4}{I_s}$$

L_p = column spacing in direction of girder (length of primary members), m.

L_s = column spacing perpendicular to direction of girder (length of secondary members), m.

S = spacing of secondary members, m.

I_p = moment of inertia of primary members, mm⁴.

I_s = moment of inertia of secondary members, mm⁴.

I_d = moment of inertia of the steel deck supported on secondary members, mm⁴ per m.

For trusses and steel joists, the moment of inertia I_s shall be decreased 15 percent when used in the above equation. A steel deck shall be considered a secondary member when it is directly supported by the primary members.

10.16.2 Improved Design for Ponding

The provisions given below are permitted to be used when a more exact determination of framing stiffness is needed than that given in Sec 10.16.1.

For primary members, the stress index shall be

$$U_p = \left(\frac{0.8F_y - f_o}{f_o} \right)_p \quad (6.10.274)$$

For secondary members, the stress index shall be

$$U_s = \left(\frac{0.8F_y - f_o}{f_o} \right)_s \quad (6.10.275)$$

Where,

f_o = stress due to the load combination ($D + R$)

D = nominal dead load

R = nominal load due to rainwater or snow, exclusive of the ponding contribution, MPa.

For roof framing consisting of primary and secondary members, the combined stiffness shall be evaluated as follows: enter Figure 6.10.1 at the level of the computed stress index U_p determined for the primary beam; move horizontally to the computed C_s value of the secondary beams and then downward to the abscissa scale. The combined stiffness of the primary and secondary framing is sufficient to prevent ponding if the flexibility constant read from this latter scale is more than the value of C_p computed for the given primary member; if not, a stiffer primary or secondary beam, or combination of both, is required.

A similar procedure must be followed using Figure 6.10.2.

For roof framing consisting of a series of equally spaced wall-bearing beams, the stiffness shall be evaluated as follows. The beams are considered as secondary members supported on an infinitely stiff primary member. For this case, enter Figure 6.10.2 with the computed stress index U_s . The limiting value of C_s is determined by the intercept of a horizontal line representing the U_s value and the curve for $C_p = 0$.

For roof framing consisting of metal deck spanning between beams supported on columns, the stiffness shall be evaluated as follows. Employ Figure 6.10.1 or 6.10.2 using as C_s the flexibility constant for a 1 m width of the roof deck ($S = 1.0$).

10.17 Design for Fatigue

This Section applies to members and connections subject to high cyclic loading within the elastic range of stresses of frequency and magnitude sufficient to initiate cracking and progressive failure, which defines the limit state of fatigue.

10.17.1 General

The provisions of this Section apply to stresses calculated on the basis of service loads. The maximum permitted stress due to unfactored loads is $0.66F_y$.

Stress range is defined as the magnitude of the change in stress due to the application or removal of the service live load. In the case of a stress reversal, the stress range shall be computed as the numerical sum of maximum repeated tensile and compressive stresses or the numerical sum of maximum shearing stresses of opposite direction at the point of probable crack initiation.

In the case of complete-joint-penetration butt welds, the maximum design stress range calculated by Eq. 6.10.276 applies only to welds with internal soundness meeting the acceptance requirements of Sec 6.12.2 or Sec 6.13.2 of AWS D1.1.

No evaluation of fatigue resistance is required if the live load stress range is less than the threshold stress range, FTH. See Table 6.10.14.

No evaluation of fatigue resistance is required if the number of cycles of application of live load is less than 20,000.

The cyclic load resistance determined by the provisions of this Section is applicable to structures with suitable corrosion protection or subject only to mildly corrosive atmospheres, such as normal atmospheric conditions.

The cyclic load resistance determined by the provisions of this Section is applicable only to structures subject to temperatures not exceeding 150°C.

The engineer of record shall provide either complete details including weld sizes or shall specify the planned cycle life and the maximum range of moments, shears and reactions for the connections.

10.17.2 Calculation of Maximum Stresses and Stress Ranges

Calculated stresses shall be based upon elastic analysis. Stresses shall not be amplified by stress concentration factors for geometrical discontinuities.

For bolts and threaded rods subject to axial tension, the calculated stresses shall include the effects of prying action, if any. In the case of axial stress combined with bending, the maximum stresses, of each kind, shall be those determined for concurrent arrangements of the applied load.

For members having symmetric cross sections, the fasteners and welds shall be arranged symmetrically about the axis of the member, or the total stresses including those due to eccentricity shall be included in the calculation of the stress range.

For axially loaded angle members where the center of gravity of the connecting welds lies between the line of the center of gravity of the angle cross section and the center of the connected leg, the effects of eccentricity shall be ignored. If the center of gravity of the connecting welds lies outside this zone, the total stresses, including those due to joint eccentricity, shall be included in the calculation of stress range.

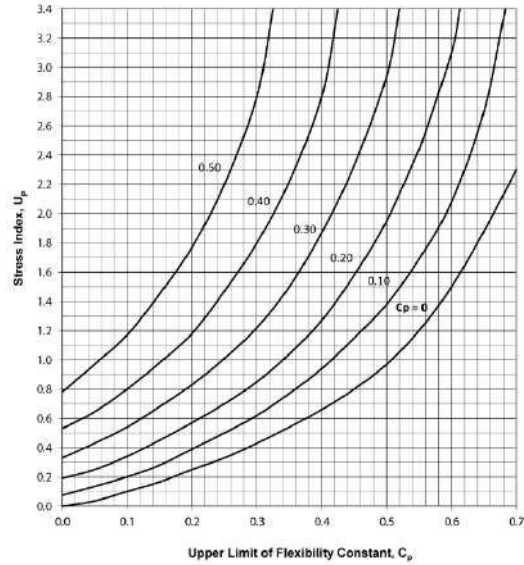


Figure 6.10.1 Limiting flexibility coefficient for the primary systems

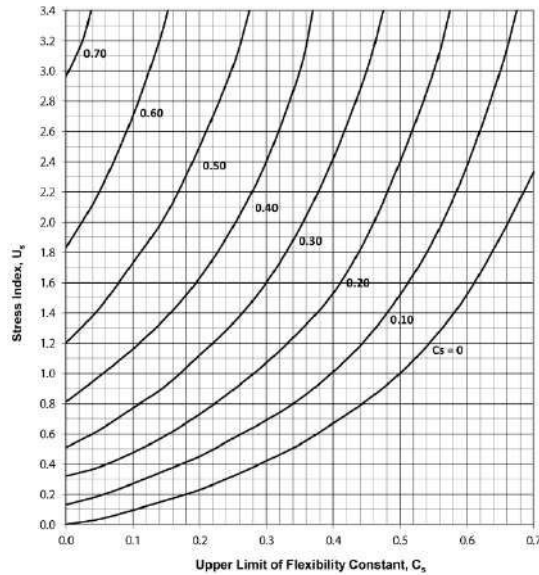


Figure 6.10.2 Limiting flexibility coefficient for the secondary systems.

10.17.3 Design Stress Range

The range of stress at service loads shall not exceed the design stress range computed as follows.

- (a) For stress categories A, B, B', C, D, E and E' (see Table 6.10.14) the design stress range, F_{SR} , shall be determined by Eq. 10.17.3.1 or 10.17.3.2.

$$F_{SR} = \left(\frac{C_f \times 329}{N} \right)^{0.333} \geq F_{TH} \quad (6.10.276)$$

Where,

F_{SR} = design stress range, MPa.

C_f = constant from Table 6.10.14 for the category

N = number of stress range fluctuations in design life

= number of stress range fluctuations per day $\times$ 365 $\times$ years of design life

F_{TH} = threshold fatigue stress (MPa) range, maximum stress range for indefinite design life from Table 6.10.14.

- (b) For stress category F, the design stress range, F_{SR} , shall be determined by Eq. 6.10.277.

$$F_{SR} = \left(\frac{C_f \times 11 \times 10^4}{N} \right)^{0.167} \geq F_{TH} \quad (6.10.277)$$

- (c) For tension-loaded plate elements connected at their end by cruciform, T, or corner details with complete-joint-penetration (CJP) groove welds or partial-joint-penetration (PJP) groove welds, fillet welds, or combinations of the preceding, transverse to the direction of stress, the design stress range on the cross section of the tension-loaded plate element at the toe of the weld shall be determined as follows:

- (i) Based upon crack initiation from the toe of the weld on the tension loaded plate element the design stress range, F_{SR} , shall be determined by Eq. 10.17.3.3 for stress category C which is equal to

$$F_{SR} = \left(\frac{14.4 \times 10^{11}}{N} \right)^{0.333} \geq 68.9 \quad (6.10.278)$$

- (ii) Based upon crack initiation from the root of the weld the design stress range, FSR, on the tension loaded plate element using transverse PJP groove welds, with or without reinforcing or contouring fillet welds, the design stress range on the cross section at the toe of the weld shall be determined by Eq. 10.17.3.4, stress category C' as follows:

$$F_{SR} = R_{PJP} \left(\frac{14.4 \times 10^{11}}{N} \right)^{0.333} \quad (6.10.279)$$

Where, R_{PJP} is the reduction factor for reinforced or nonreinforced transverse PJP groove welds determined as follows:

$$R_{PJP} = \left(\frac{1.12 - 1.01 \left(\frac{2a}{t_p} \right) + 1.24 \left(\frac{w}{t_p} \right)}{t_p^{0.167}} \right) \leq 1.0 \quad (6.10.280)$$

If $R_{PJP} = 1.0$, use stress category C.

$2a$ = the length of nonwelded root face in the direction of the thickness of the tension-loaded plate, mm.

W = the leg size of the reinforcing or contouring fillet, if any, in the direction of the thickness of the tension-loaded plate, mm.

t_p = thickness of tension loaded plate, mm.

- (iii) Based upon crack initiation from the roots of a pair of transverse fillet welds on opposite sides of the tension loaded plate element the design stress range, FSR, on the cross section at the toe of the welds shall be determined by Eq. 10.17.3.5, stress category C'' as follows:

$$F_{SR} = R_{FIL} \left(\frac{14.4 \times 10^{11}}{N} \right)^{0.333} \quad (6.10.281)$$

Where, R_{FIL} is the reduction factor for joints using a pair of transverse fillet welds only.

$$R_{FIL} = \left(\frac{0.10 + 1.24(w/t_p)}{t_p^{0.167}} \right) \leq 1.0 \quad (6.10.282)$$

If $R_{FIL} = 1.0$, use stress category C.

10.17.4 Bolts and Threaded Parts

The range of stress at service loads shall not exceed the stress range computed as follows.

- (a) For mechanically fastened connections loaded in shear, the maximum range of stress in the connected material at service loads shall not exceed the design stress range computed using Eq. 6.10.276 where C_f and FTH are taken from Section 2 of Table 6.10.14.
- (b) For high-strength bolts, common bolts, and threaded anchor rods with cut, ground or rolled threads, the maximum range of tensile stress on the net tensile area from applied axial load and moment plus load due to prying action shall not exceed the design stress range computed using Eq. 6.10.276. The factor C_f shall be taken as 3.9×10^8 (as for stress category E'). The threshold stress, FTH shall be taken as 48 MPa (as for stress category D). The net tensile area is given by Eq. 6.10.283.

$$A_t = \frac{\pi}{4} (d_b - 0.9382P)^2 \quad (6.10.283)$$

Where,

P = pitch, mm per thread

d_b = the nominal diameter (body or shank diameter), mm.

n = threads per mm.

For joints in which the material within the grip is not limited to steel or joints which are not tensioned to the requirements of Table 6.10.9, all axial load and moment applied to the joint plus effects of any prying action shall be assumed to be carried exclusively by the bolts or rods.

For joints in which the material within the grip is limited to steel and which are tensioned to the requirements of Table 6.10.9, an analysis of the relative stiffness of the connected parts and bolts shall be permitted to be used to determine the tensile stress range in the pretensioned bolts due to the total service live load and moment plus effects of any prying action. Alternatively, the stress range in the bolts shall be assumed to be equal to the stress on the net tensile area due to 20 percent of the absolute value of the service load axial load and moment from dead, live and other loads.

10.17.5 Special Fabrication and Erection Requirements

Longitudinal backing bars are permitted to remain in place, and if used, shall be continuous. If splicing is necessary for long joints, the bar shall be joined with complete penetration butt joints and the reinforcement ground prior to assembly in the joint.

Transverse joints subject to tension, backing bars, if used, shall be removed and joint back gouged and welded.

In transverse complete-joint-penetration T and corner joints, a reinforcing fillet weld, not less than 6 mm in size shall be added at re-entrant corners.

The surface roughness of flame cut edges subject to significant cyclic tensile stress ranges shall not exceed 25 μm , where ASME B46.1 is the reference standard.

Reentrant corners at cuts, *cope*s and weld access holes shall form a radius of not less than 10 mm by predrilling or subpunching and reaming a hole, or by thermal cutting to form the radius of the cut. If the radius portion is formed by thermal cutting, the cut surface shall be ground to a bright metal surface.

For transverse butt joints in regions of high tensile stress, run-off tabs shall be used to provide for cascading the weld termination outside the finished joint. End dams shall not be used. Run-off tabs shall be removed and the end of the weld finished flush with the edge of the member.

See Sec 10.10.2.2 for requirements for end returns on certain fillet welds subject to cyclic service loading.

Table 6.10.14a: Fatigue Design Parameters

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
SECTION 1 – PLAIN MATERIAL AWAY FROM ANY WELDING				
1.1 Base metal, except non-coated weathering steel, with rolled or cleaned surface. Flame-cut edges with surface roughness value of 25 μm or less, but without reentrant corners.	A	250×10^8	165	Away from all welds or structural connections
1.2 Non-coated weathering steel base metal with rolled or cleaned surface. Flame-cut edges with surface roughness value of 25 μm or less, but without reentrant corners.	B	120×10^8	110	Away from all welds or structural connections

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
1.3 Member with drilled or reamed holes. Member with reentrant corners at <i>cope</i> s, cuts, block-outs or other geometrical discontinuities made to requirements of Sec 10.17.3.5, except weld access holes.	B	120×10^8	110	At any external edge or at hole perimeter
1.4 Rolled cross sections with weld access holes made to requirements of Sec 10.10.1.6 and Sec 10.17.3.5. Members with drilled or reamed holes containing bolts for attachment of light bracing where there is a small longitudinal component of brace force.	C	44×10^8	69	At reentrant corner of weld access hole or at any small hole (may contain bolt for minor connections)
SECTION 2– CONNECTED MATERIAL IN MECHANICALLY FASTENED JOINTS				
2.1 Gross area of base metal in lap joints connected by high-strength bolts in joints satisfying all requirements for slip-critical connections.	B	120×10^8	110	Through gross section near hole
2.2 Base metal at net section of high-strength bolted joints, designed on the basis of bearing resistance, but fabricated and installed to all requirements for slip-critical connections.	B	120×10^8	110	In net section originating at side of hole
2.3 Base metal at the net section of other mechanically fastened joints except eye bars and pin plates.	D	22×10^8	48	In net section originating at side of hole
2.4 Base metal at net section of eye bar head or pin plate.	E	11×10^8	31	In net section originating at side of hole

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
SECTION 3 – WELDED JOINTS JOINING COMPONENTS OF BUILT-UP MEMBERS				
3.1 Base metal and weld metal in members without attachments built-up of plates or shapes connected by continuous longitudinal complete-joint-penetration groove welds, back gouged and welded from second side, or by continuous fillet welds.	B	120×10^8	110	From surface or internal discontinuities in weld away from end of weld
3.2 Base metal and weld metal in members without attachments built-up of plates or shapes, connected by continuous longitudinal complete-joint-penetration groove welds with backing bars not re-moved, or by continuous partial-joint-penetration groove welds.	B	61×10^8	83	From surface or internal discontinuities in weld, including weld attaching backing bars
3.3 Base metal and weld metal termination of longitudinal welds at weld access holes in connected built-up members.	D	22×10^8	48	From the weld termination into the web or flange
3.4 Base metal at ends of longitudinal intermittent fillet weld segments.	E	11×10^8	31	In connected material at start and stop locations of any weld deposit
3.5 Base metal at ends of partial length welded cover plates narrower than the flange having square or tapered ends, with or without welds across the ends of cover plates wider than the flange with welds across the ends.	E E'			In flange at toe of end weld or in flange at termination of longitudinal weld or in edge of flange with wide cover plates
Flange thickness ≤ 20 mm		11×10^8	31	
Flange thickness > 20 mm		3.9×10^8	18	

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
3.6 Base metal at ends of partial length welded cover plates wider than the flange without welds across the ends.	E'	3.9×10^8	18	In edge of flange at end of cover plate weld
SECTION 4 – LONGITUDINAL FILLET WELDED END CONNECTIONS				
4.1 Base metal at junction of axially loaded members with longitudinally welded end connections. Welds shall be on each side of the axis of the member to balance weld stresses.				Initiating from end of any weld termination extending into the base metal.
$t \leq 20$ mm	E	11×10^8	31	
$t > 20$ mm	E'	3.9×10^8	18	
SECTION 5 – WELDED JOINTS TRANSVERSE TO DIRECTION OF STRESS				
5.1 Base metal and weld metal in or adjacent to complete-joint-penetration groove welded splices in rolled or welded cross sections with welds ground essentially parallel to the direction of stress.	B	120×10^8	110	From internal discontinuities in filler metal or along the fusion boundary
5.2 Base metal and weld metal in or adjacent to complete-joint-penetration groove welded splices with welds ground essentially parallel to the direction of stress at transitions in thickness or width made on a slope no greater than 8 to 20%.				From internal discontinuities in filler metal or along fusion boundary or at start of transition when $F_y \geq 620$ MPa
$F_y < 620$ MPa	B	120×10^8	110	
$F_y \geq 620$ MPa	B'	61×10^8	83	

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
5.3 Base metal with F_y equal to or greater than 620 MPa and weld metal in or adjacent to complete-joint-penetration groove welded splices with welds ground essentially parallel to the direction of stress at transitions in width made on a radius of not less than 600 mm with the point of tangency at the end of the groove weld.	B	120×10^8	110	From internal discontinuities in filler metal or discontinuities along the fusion boundary
5.4 Base metal and weld metal in or adjacent to the toe of complete-joint-penetration T or corner joints or splices, with or without transitions in thickness having slopes no greater than 8 to 20%, when weld reinforcement is not removed.	C	44×10^8	69	From surface discontinuity at toe of weld extending into base metal or along fusion boundary.
5.5 Base metal and weld metal at transverse end connections of tension-loaded plate elements using partial-joint-penetration butt or T or corner joints, with reinforcing or contouring fillets, FSR shall be the smaller of the toe crack or root crack stress range.				Initiating from geometrical discontinuity at toe of weld extending into base metal or, initiating at weld root subject to tension extending up and then out through weld
Crack initiating from weld toe:	C	44×10^8	69	
Crack initiating from weld root:	C'	Eq. 6.10.279	None provided	

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
5.6 Base metal and filler metal at transverse end connections of tension-loaded plate elements using a pair of fillet welds on opposite sides of the plate. FSR shall be the smaller of the toe crack or root crack stress range.				Initiating from geometrical discontinuity at toe of weld extending into base metal or, initiating at <i>weld root</i> subject to tension extending up and then out through weld
Crack initiating from weld toe:	C	44×10^8	69	
Crack initiating from weld root:	C'	Eq. 6.10.280	None provided	
5.7 Base metal of tension loaded plate elements and on girders and rolled beam webs or flanges at toe of transverse fillet welds adjacent to welded transverse stiffeners.	C	44×10^8	69	From geometrical discontinuity at <i>toe of fillet</i> extending into base metal
SECTION 6 – BASE METAL AT WELDED TRANSVERSE MEMBER CONNECTIONS				
6.1 Base metal at details attached by complete joint penetration groove welds subject to longitudinal loading only when the detail embodies a transition radius R with the weld termination ground smooth.				Near point of tangency of radius at edge of member
$R \geq 600$ mm	B	120×10^8	110	
$600 \text{ mm} > R \geq 150$ mm	C	44×10^8	69	
$150 \text{ mm} > R \geq 50$ mm	D	22×10^8	48	
$50 \text{ mm} > R$	E	11×10^8	31	

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
6.2 Base metal at details of equal thickness attached by complete-joint-penetration groove welds subject to trans-verse loading with or without longitudinal loading when the detail embodies a transition radius R with the weld termination ground smooth:				
When weld reinforcement is removed:				
$R \geq 600$ mm	B	120×10^8	110	Near points of tangency of radius or in the weld or at fusion boundary or member or attachment
$600 \text{ mm} > R \geq 150$ mm	C	44×10^8	69	
$150 \text{ mm} > R \geq 50$ mm	D	22×10^8	48	
$50 \text{ mm} > R$	E	11×10^8	31	
When weld reinforcement is not removed:				
$R \geq 600$ mm	C	44×10^8	69	At toe of the weld either along edge of member or the attachment
$600 \text{ mm} > R \geq 150$ mm	C	44×10^8	69	
$150 \text{ mm} > R \geq 50$ mm	D	22×10^8	48	
$50 \text{ mm} > R$	E	11×10^8	31	
6.3 Base metal at details of unequal thickness attached by complete-joint-penetration groove welds subject to trans-verse loading with or without longitudinal loading when the detail embodies a transition radius R with the weld termination ground smooth.				

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
When weld reinforcement is removed: $R > 50$ mm	D	22×10^8	48	At toe of weld along edge of thinner material
$R \leq 50$ mm	E	11×10^8	31	In weld termination in small radius
When reinforcement is not removed: Any radius	E	11×10^8	31	At toe of weld along edge of thinner material
6.4 Base metal subject to longitudinal stress at transverse members, with or without transverse stress, attached by fillet or partial penetration groove welds parallel to direction of stress when the detail embodies a transition radius, R , with weld termination ground smooth:				In weld termination or from the toe of the weld extending into member
$R > 50$ mm	D	22×10^8	48	
$R \leq 50$ mm	E	11×10^8	31	
SECTION 7 – BASE METAL AT SHORT ATTACHMENTS¹				
7.1 Base metal subject to longitudinal loading at details attached by fillet welds parallel or transverse to the direction of stress where the detail embodies no transition radius and with detail length in direction of stress, a , and attachment height normal to the surface of the member, b :				In the member at the end of the weld

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
$a < 50$ mm	C	44×10^8	69	
$50 \text{ mm} \leq a \leq 12b$ or 100 mm	D	22×10^8	48	
$a > 12b$ or 100 mm when b is ≤ 25 mm	E	11×10^8	31	
$a > 12b$ or 100 mm when b is > 25 mm	E'	3.9×10^8	18	
7.2 Base metal subject to longitudinal stress at details attached by fillet or partial-joint-penetration groove welds, with or without transverse load on detail, when the detail embodies a transition radius, R , with weld termination ground smooth:				In weld termination ex-tending into member
$R > 50$ mm	D	22×10^8	48	
$R \leq 50$ mm	E	11×10^8	31	
¹ “Attachment” as used herein, is defined as any steel detail welded to a member which, by its mere presence and independent of its loading, causes a discontinuity in the stress flow in the member and thus reduces the fatigue resistance.				
SECTION 8 - MISCELLANEOUS				
8.1 Base metal at stud-type shear connectors attached by fillet or electric stud welding.	C	44×10^8	69	At toe of weld in base metal
8.2 Shear on throat of continuous or intermittent longitudinal or transverse fillet welds.	F	150×10^{10} (Eq. 6.10.277)	55	In throat of weld
8.3 Base metal at plug or slot welds.	E	11×10^8	31	At end of weld in base metal
8.4 Shear on plug or slot welds.	F	150×10^{10} (Eq. 6.10.277)	55	At faying surface

Description	Stress Category	Constant C_f	Threshold F_{TH} (MPa)	Potential Crack Initiation Point
8.5 Not fully tightened high-strength bolts, common bolts, threaded anchor rods and hanger rods with cut, ground or rolled threads. Stress range on tensile stress area due to live load plus prying action when applicable.	E'	3.9×10^8	48	At the root of the threads extending into the tensile stress area

Table 6.10.14b: Fatigue Design Parameters (Illustrated Typical Examples)

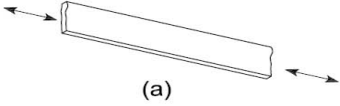
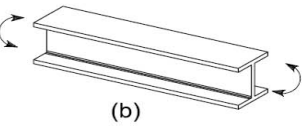
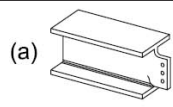
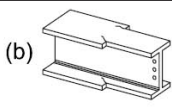
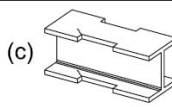
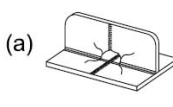
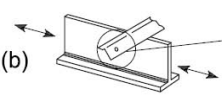
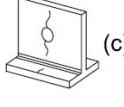
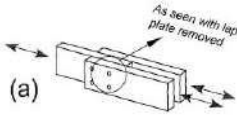
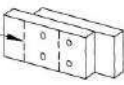
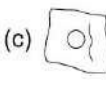
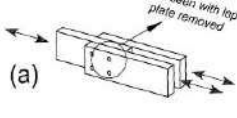
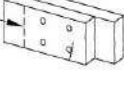
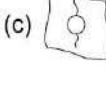
SECTION 1– PLAIN MATERIAL AWAY FROM ANY WELDING			
1.1 and 1.2			
1.3			
1.4			
SECTION 2– CONNECTED MATERIAL IN MECHANICALLY FASTENED JOINTS			
2.1			
2.2			

Table 6.10.14b: Fatigue Design Parameters (Illustrated Typical Examples)

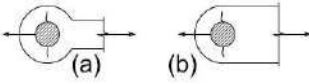
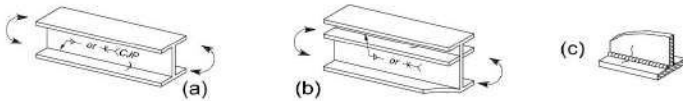
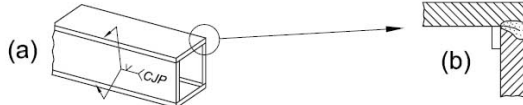
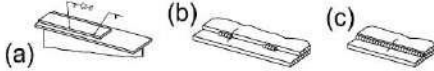
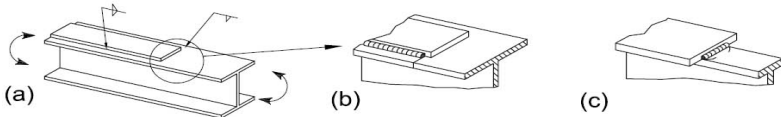
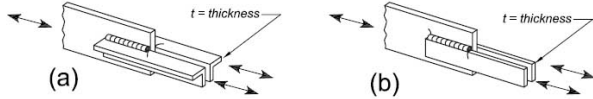
2.3	
2.4	
SECTION 3– WELDED JOINTS JOINING COMPONENTS OF BUILT-UP MEMBERS	
3.1	
3.2	
3.3	
3.4	
3.5	
3.6	
SECTION 4– LONGITUDINAL FILLET WELDED END CONNECTIONS	
4.1	

Table 6.10.14b: Fatigue Design Parameters (Illustrated Typical Examples)

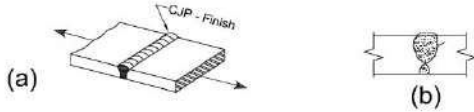
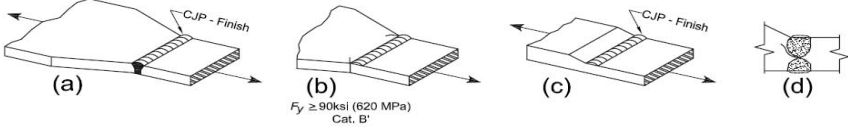
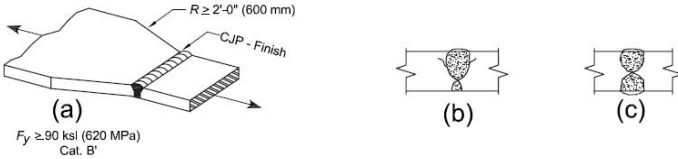
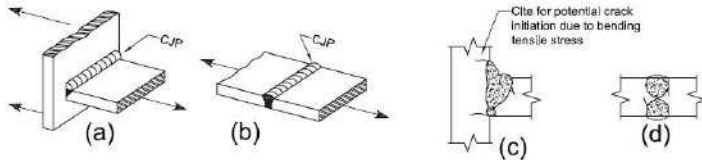
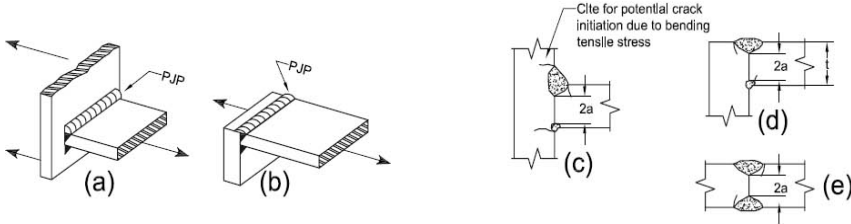
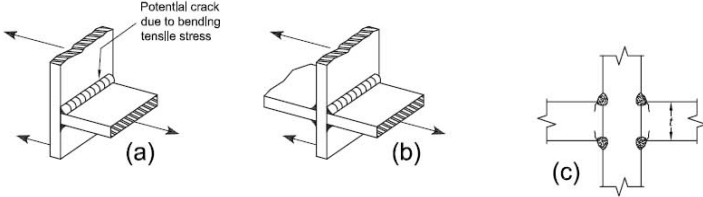
SECTION 5 – WELDED JOINTS TRANSVERSE TO DIRECTION OF STRESS	
5.1	
5.2	 $F_y \geq 90 \text{ ksi (620 MPa)}$ Cat. B'
5.3	 $F_y \geq 90 \text{ ksi (620 MPa)}$ Cat. B'
5.4	 Cite for potential crack initiation due to bending tensile stress
5.5	 Cite for potential crack initiation due to bending tensile stress 2a 2a
5.6	 Potential crack due to bending tensile stress

Table 6.10.14b: Fatigue Design Parameters (Illustrated Typical Examples)

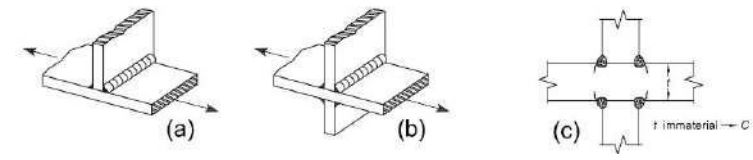
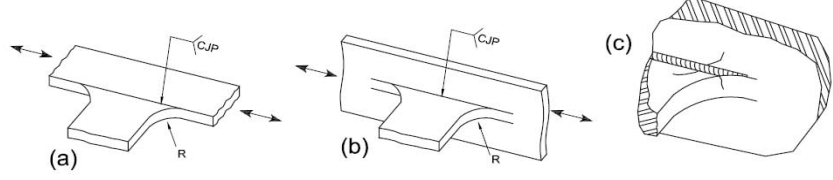
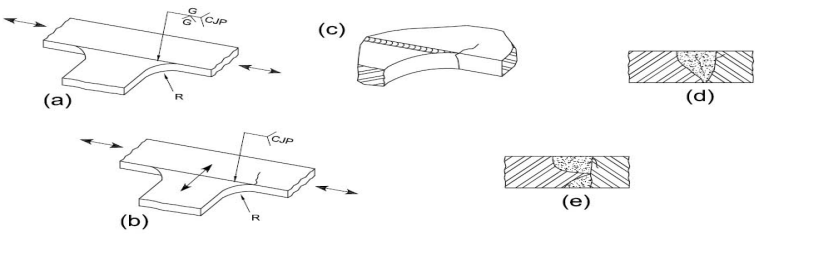
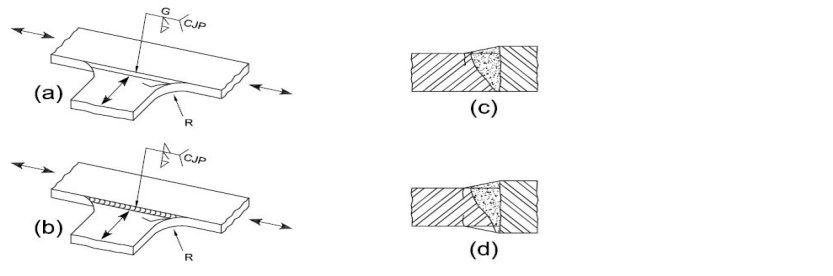
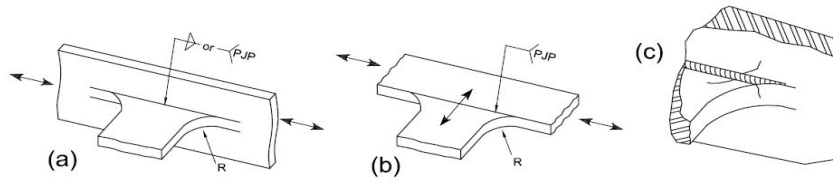
5.7	
SECTION 6- BASE METAL AT WELDED TRANSVERSE MEMBER CONNECTIONS	
6.1	
6.2	
6.3	
6.4	

Table 6.10.14b: Fatigue Design Parameters (Illustrated Typical Examples)

SECTION7- BASE METAL AT SHORT ATTACHMENTS1	
7.1	
7.2	
SECTION8- MISCELLANEOUS	
8.1	
8.2	
8.3	
8.4	
8.5	

10.18 Structural Design for Fire Conditions

This Section provides criteria for the design and evaluation of structural steel components, systems and frames for fire conditions. These criteria provide for the determination of the heat input, thermal expansion and degradation in mechanical properties of materials at elevated temperatures that cause progressive decrease in strength and stiffness of structural components and systems at elevated temperatures.

10.18.1 General Provisions

The methods contained in this Section provide regulatory evidence of compliance in accordance with the design applications outlined in this Section.

10.18.1.1 Performance objective

Structural components, members and building frame systems shall be designed so as to maintain their load-bearing function during the design-basis fire and to satisfy other performance requirements specified for the building occupancy. Deformation criteria shall be applied where the means of providing structural fire resistance, or the design criteria for fire barriers, requires consideration of the deformation of the load-carrying structure. Within the compartment of fire origin, forces and deformations from the design basis fire shall not cause a breach of horizontal or vertical compartmentation.

10.18.1.2 Design by engineering analysis

The analysis methods in Sec 10.18.2 are permitted to be used to document the anticipated performance of steel framing when subjected to design-basis fire scenarios. Methods in Sec 10.18.2 provide evidence of compliance with performance objectives established in Sec 10.18.1.1.

The analysis methods in Sec 10.18.2 are permitted to be used to demonstrate an equivalency for an alternative material or method, as permitted by the Code.

10.18.1.3 Design by qualification testing

The qualification testing methods in Sec 10.18.3 are permitted to be used to document the fire resistance of steel framing subject to the standardized fire testing protocols required by building Codes.

10.18.1.4 Load combinations and required strength

The required strength of the structure and its elements shall be determined from the following gravity load combination:

$$[0.9 \text{ or } 1.2]D + T + 0.5L \quad (6.10.284)$$

Where,

D = nominal dead load

L = nominal occupancy live load

T = nominal forces and deformations due to the design-basis fire defined in Section 4.2.1

A lateral notional load, $N_i = 0.002Y_i$, as defined in Sec 10.20, where N_i = notional lateral load applied at framing level i and Y_i = gravity load from combination 10.18.1.1 acting on framing level i , shall be applied in combination with the loads stipulated in Eq. 6.10.284. Unless otherwise stipulated by the Authority, D, L and S shall be the nominal loads specified in Chapter 2 of Part 6 of this Code.

10.18.2 Structural Design for Fire Conditions by Analysis

It is permitted to design structural members, components and building frames for elevated temperatures in accordance with the requirements of this Section.

10.18.2.1 Design-basis fire

A design-basis fire shall be identified to describe the heating conditions for the structure. These heating conditions shall relate to the fuel commodities and compartment characteristics present in the assumed fire area. The fuel load density based on the occupancy of the space shall be considered when determining the total fuel load. Heating conditions shall be specified either in terms of a heat flux or temperature of the upper gas layer created by the fire. The variation of the heating conditions with time shall be determined for the duration of the fire.

When the analysis methods in Sec 10.18.2 are used to demonstrate an equivalency as an alternative material or method as permitted by Code, the design-basis fire shall be determined in accordance with ASTM E119.

10.18.2.1.1 Localized fire

Where the heat release rate from the fire is insufficient to cause flashover, a localized fire exposure shall be assumed. In such cases, the fuel composition, arrangement of the fuel array and floor area occupied by the fuel shall be used to determine the radiant heat flux from the flame and smoke plume to the structure.

10.18.2.1.2 Post-flashover compartment fires

Where the heat release rate from the fire is sufficient to cause flashover, a post-flashover compartment fire shall be assumed. The determination of the temperature versus time profile resulting from the fire shall include fuel load, ventilation characteristics to the space (natural and mechanical), compartment dimensions and thermal characteristics of the compartment boundary.

10.18.2.1.3 Exterior fires

The exposure of exterior structure to flames projecting from windows or other wall openings as a result of a post-flashover compartment fire shall be considered along with the radiation from the interior fire through the opening. The shape and length of the flame projection shall be used along with the distance between the flame and the exterior steelwork to determine the heat flux to the steel. The method identified in Sec 10.18.2.1.2 shall be used for describing the characteristics of the interior compartment fire.

10.18.2.1.4 Fire duration

The fire duration in a particular area shall be determined by considering the total combustible mass, in other words, fuel load available in the space. In the case of either a localized fire or a post-flashover compartment fire, the time duration shall be determined as the total combustible mass divided by the mass loss rate, except where determined from Sec 10.18.2.1.2.

10.18.2.1.5 Active fire protection systems

The effects of active fire protection systems shall be considered when describing the design-basis fire.

Where automatic smoke and heat vents are installed in nonsprinklered spaces, the resulting smoke temperature shall be determined from calculation.

10.18.2.2 Temperatures in structural systems under fire conditions

Temperatures within structural members, components and frames due to the heating conditions posed by the design-basis fire shall be determined by a heat transfer analysis.

10.18.2.3 Material strengths at elevated temperatures

Material properties at elevated temperatures shall be determined from test data. In the absence of such data, it is permitted to use the material properties stipulated in this Section. These relationships do not apply for steels with a yield strength in excess of 448 MPa or concretes with specified compression strength in excess of 55 MPa.

Table 6.10.15: Properties of Steel at Elevated Temperatures

Steel Temperature °C	$k_E = E_m/E$	$k_y = F_{ym}/F_y$	$k_u = F_{um}/F_y$
20	*	*	*
93	1.00	*	*
204	0.90	*	*

Steel Temperature °C	$k_E = E_m/E$	$k_y = F_{ym}/F_y$	$k_u = F_{um}/F_y$
316	0.78	*	*
399	0.70	1.00	1.00
427	0.67	0.94	0.94
538	0.49	0.66	0.66
649	0.22	0.35	0.35
760	0.11	0.16	0.16
871	0.07	0.07	0.07
982	0.05	0.04	0.04
1093	0.02	0.02	0.02
1204	0.00	0.00	0.00
*Use Ambient Properties			

10.18.2.3.1 Thermal elongation

Thermal expansion of structural and reinforcing steels: For calculations at temperatures above 65°C, the coefficient of thermal expansion shall be 1.4×10^{-5} per °C.

Thermal expansion of normal weight concrete: For calculations at temperatures above 65°C, the coefficient of thermal expansion shall be 1.8×10^{-5} per °C.

Thermal expansion of lightweight concrete: For calculations at temperatures above 65°C, the coefficient of thermal expansion shall be 7.9×10^{-6} per °C.

10.18.2.3.2 Mechanical properties at elevated temperatures

The deterioration in strength and stiffness of structural members, components, and systems shall be taken into account in the structural analysis of the frame.

The values F_{ym} , F_{um} , E_m , f'_{cm} , E_{cm} and ϵ_{cu} at elevated temperature to be used in structural analysis, expressed as the ratio with respect to the property at ambient, assumed to be 20°C, shall be defined as in Tables 6.10.15 and 6.10.16. It is permitted to interpolate between these values.

10.18.2.4 Structural design requirements

10.18.2.4.1 General structural integrity

The structural frame shall be capable of providing adequate strength and deformation capacity to withstand, as a system, the structural actions developed during the fire within the prescribed limits of deformation. The structural system shall be designed to sustain local damage with structural system as a whole remaining stable.

Continuous load paths shall be provided to transfer all forces from the exposed region to the final point of resistance. The foundation shall be designed to resist the forces and to accommodate the deformations developed during the design-basis fire.

Table 6.10.16: Properties of Concrete at Elevated Temperatures

Concrete Temperature °C	$k_c = f'_{cm}/f'_c$		E_{am}/E_c	ϵ_{cu} (%)
	NWC	LWC		LWC
20	1.00	1.00	1.00	0.25
93	0.95	1.00	0.93	0.34
204	0.90	1.00	0.75	0.46
288	0.86	1.00	0.61	0.58
316	0.83	0.98	0.57	0.62
427	0.71	0.85	0.38	0.80
538	0.54	0.71	0.20	1.06
649	0.38	0.58	0.092	1.32
760	0.21	0.45	0.073	1.43
871	0.10	0.31	0.055	1.49
982	0.05	0.18	0.036	1.50
1093	0.01	0.05	0.018	1.50
1204	0.00	0.00	0.00	-

10.18.2.4.2 Strength requirements and deformation limits

Conformance of the structural system to these requirements shall be demonstrated by constructing a mathematical model of the structure based on principles of structural mechanics and evaluating this model for the internal forces and deformations in the members of the structure developed by the temperatures from the design-basis fire.

Individual members shall be provided with adequate strength to resist the shears, axial forces and moments determined in accordance with these provisions.

Connections shall develop the strength of the connected members or the forces indicated above. Where the means of providing fire resistance requires the consideration of deformation criteria, the deformation of the structural system, or members thereof, under the design-basis fire shall not exceed the prescribed limits.

10.18.2.4.3 Methods of analysis

(a) Advanced methods of analysis

The methods of analysis in this Section are permitted for the design of all steel building structures for fire conditions. The design-basis fire exposure shall be that determined in Sec 10.18.2.1. The analysis shall include both a thermal response and the mechanical response to the design-basis fire.

The thermal response shall produce a temperature field in each structural element as a result of the design-basis fire and shall incorporate temperature-dependent thermal properties of the structural elements and fire-resistive materials as per Sec 10.18.2.2.

The mechanical response results in forces and deflections in the structural system subjected to the thermal response calculated from the design-basis fire. The mechanical response shall take into account explicitly the deterioration in strength and stiffness with increasing temperature, the effects of thermal expansions and large deformations. Boundary conditions and connection fixity must represent the proposed structural design. Material properties shall be defined as per Sec 10.18.2.3.

The resulting analysis shall consider all relevant limit states, such as excessive deflections, connection fractures, and overall or local buckling.

(b) Simple Methods of Analysis

The methods of analysis in this Section are applicable for the evaluation of the performance of individual members at elevated temperatures during exposure to fire. The support and restraint conditions (forces, moments and boundary conditions) applicable at normal temperatures may be assumed to remain unchanged throughout the fire exposure.

(1) Tension members

It is permitted to model the thermal response of a tension element using a one-dimensional heat transfer equation with heat input as directed by the design-basis fire defined in Sec 10.18.2.1.

The design strength of a tension member shall be determined using the provisions of Sec 10.4, with steel properties as stipulated in Sec 10.18.2.3 and assuming a uniform temperature over the cross section using the temperature equal to the maximum steel temperature.

(2) Compression members

It is permitted to model the thermal response of a compression element using a one-dimensional heat transfer equation with heat input as directed by the design-basis fire defined in Sec 10.18.2.1.

The design strength of a compression member shall be determined using the provisions of Sec 10.5 with steel properties as stipulated in Sec 10.18.2.3.

(3) Flexural members

It is permitted to model the thermal response of flexural elements using a one-dimensional heat transfer equation to calculate bottom flange temperature and to assume that this bottom flange temperature is constant over the depth of the member. The design strength of a flexural member shall be determined using the provisions of Sec 10.6 with steel properties as stipulated in Sec 10.18.2.3.

(4) Composite floor members

It is permitted to model the thermal response of flexural elements supporting a concrete slab using a one-dimensional heat transfer equation to calculate bottom flange temperature. That temperature shall be taken as constant between the bottom flange and mid-depth of the web and shall decrease linearly by no more than 25 percent from the mid-depth of the web to the top flange of the beam.

The design strength of a composite flexural member shall be determined using the provisions of Chapter 13 Part 6 of this Code, with reduced yield stresses in the steel consistent with the temperature variation described under thermal response.

10.18.2.4.4 Design strength

The design strength shall be determined as in Sec 10.2.3.3. The nominal strength, R_n shall be calculated using material properties, as stipulated in Sec 10.18.2.3, at the temperature developed by the design-basis fire.

10.18.3 Design by Qualification Testing**10.18.3.1 Design strength**

Structural members and components in steel buildings shall be qualified for the rating period in conformance with ASTM E119. It shall be permitted to demonstrate compliance with these requirements using the procedures specified for steel construction in Section 5 of ASCE/SFPE 29.

10.18.3.2 Restrained construction

For floor and roof assemblies and individual beams in buildings, a restrained condition exists when the surrounding or supporting structure is capable of resisting actions caused by thermal expansion throughout the range of anticipated elevated temperatures.

Steel beams, girders and frames supporting concrete slabs that are welded or bolted to integral framing members (in other words, columns, girders) shall be considered restrained construction.

10.18.3.3 Unrestrained construction

Steel beams, girders and frames that do not support a concrete slab shall be considered unrestrained unless the members are bolted or welded to surrounding construction that has been specifically designed and detailed to resist actions caused by thermal expansion.

A steel member bearing on a wall in a single span or at the end span of multiple spans shall be considered unrestrained unless the wall has been designed and detailed to resist effects of thermal expansion.

10.19 Stability Bracing for Columns and Beams

This Section addresses the minimum brace strength and stiffness necessary to provide member strengths based on the unbraced length between braces with an effective length factor, K , equal to 1.0.

10.19.1 General Provisions

Bracing is assumed to be perpendicular to the members to be braced; for inclined or diagonal bracing, the brace strength (force or moment) and stiffness (force per unit displacement or moment per unit rotation) shall be adjusted for the angle of inclination. The evaluation of the stiffness furnished by a brace shall include its member and geometric properties, as well as the effects of connections and anchoring details.

Two general types of bracing systems are considered, relative and nodal. A relative brace controls the movement of the brace point with respect to adjacent braced points. A nodal brace controls the movement at the braced point without direct interaction with adjacent braced points. The available strength and stiffness of the bracing shall equal or exceed the required limits unless analysis indicates that smaller values are justified by analysis.

A second-order analysis that includes an initial out-of-straightness of the member to obtain brace strength and stiffness is permitted in lieu of the requirements of this Section.

10.19.2 Columns

It is permitted to brace an individual column at end and intermediate points along its length by either relative or nodal bracing systems. It is assumed that nodal braces are equally spaced along the column.

10.19.2.1 Relative bracing

The required brace strength is

$$P_{br} = 0.004 P_r \quad (6.10.285)$$

The required brace stiffness is

$$\beta_{br} = \frac{1}{\phi} \left(\frac{2P_r}{L_b} \right) \text{ (LRFD)} \quad \beta_{br} = \Omega \left(\frac{2P_r}{L_b} \right) \text{ (ASD)} \quad (6.10.286)$$

Where,

$$\phi = 0.75 \text{ (LRFD)} \quad \Omega = 2.00 \text{ (ASD)}$$

For design according to Sec 10.2.3.3 (LRFD)

P_r = required axial compressive strength using LRFD load combinations, N

For design according to Sec 10.2.3.4 (ASD)

P_r = required axial compressive strength using ASD load combinations, N

10.19.2.2 Nodal bracing

The required brace strength is

$$P_{br} = 0.01 P_r \quad (6.10.287)$$

The required brace stiffness is

$$\beta_{br} = \frac{1}{\phi} \left(\frac{8P_r}{L_b} \right) \text{ (LRFD)} \quad \beta_{br} = \Omega \left(\frac{8P_r}{L_b} \right) \text{ (ASD)} \quad (6.10.288)$$

Where, $\phi = 0.75 \text{ (LRFD)}$ $\Omega = 2.00 \text{ (ASD)}$

For design according to Sec 10.2.3.3 (LRFD)

P_r = required axial compressive strength using LRFD load combinations, N

For design according to Sec 10.2.3.4 (ASD)

P_r = required axial compressive strength using ASD load combinations, N

When, L_b is less than L_q , where L_q is the maximum unbraced length for the required column force with K equal to 1.0, then L_b in Eq. 6.10.288 is permitted to be taken equal to L_q .

10.19.3 Beams

At points of support for beams, girders and trusses, restraint against rotation about their longitudinal axis shall be provided. Beam bracing shall prevent the relative displacement of the top and bottom flanges, in other words, twist of the section. Lateral stability of beams shall be provided by lateral bracing, torsional bracing or a combination of the two. In members subjected to double curvature bending, the inflection point shall not be considered a brace point.

10.19.3.1 Lateral bracing

Bracing shall be attached near the compression flange, except for a cantilevered member, where an end brace shall be attached near the top (tension) flange. Lateral bracing shall be attached to both flanges at the brace point nearest the inflection point for beams subjected to double curvature bending along the length to be braced.

(a) Relative bracing

The required brace strength is

$$P_{br} = 0.008 M_r C_d / h_o \quad (6.10.289)$$

The required brace stiffness is

$$\beta_{br} = \frac{1}{\phi} \left(\frac{4M_r C_d}{L_b h_o} \right) \text{ (LRFD)} \quad \beta_{br} = \Omega \left(\frac{4M_r C_d}{L_b h_o} \right) \text{ (ASD)} \quad (6.10.290)$$

Where, $\phi = 0.75$ (LRFD) $\Omega = 2.00$ (ASD)

h_o = distance between flange centroids, mm.

$C_d = 1.0$ for bending in single curvature; 2.0 for double curvature; $C_d = 2.0$ only applies to the brace closest to the inflection point.

L_b = laterally unbraced length, mm.

For design according to Section 10.2.3.3 (LRFD):

M_r = required flexural strength using LRFD load combinations, N-mm

For design according to Section 10.2.3.4 (ASD):

M_r = required flexural strength using ASD load combinations, N-mm

(b) Nodal bracing

The required brace strength is

$$P_{br} = 0.02 M_r C_d / h_o \quad (6.10.291)$$

The required brace stiffness is

$$\beta_{br} = \frac{1}{\phi} \left(\frac{10 M_r C_d}{L_b h_o} \right) \text{ (LRFD)} \quad \beta_{br} = \Omega \left(\frac{10 M_r C_d}{L_b h_o} \right) \text{ (ASD)} \quad (6.10.292)$$

Where, $\phi = 0.75$ (LRFD) $\Omega = 2.00$ (ASD)

For design according to Section 10.2.3.3 (LRFD):

M_r = required flexural strength using LRFD load combinations, N-mm

For design according to Section 10.2.3.4 (ASD):

M_r = required flexural strength using ASD load combinations, N-mm

When L_b is less than L_q , the maximum unbraced length for M_r , then L_b in Eq. 6.10.292 shall be permitted to be taken equal to L_q .

10.19.3.2 Torsional bracing

It is permitted to provide either nodal or continuous torsional bracing along the beam length. It is permitted to attach the bracing at any cross-sectional location and it need not be attached near the compression flange. The connection between a torsional brace and the beam shall be able to support the required moment given below.

(a) Nodal bracing

The required bracing moment is

$$M_{br} = \frac{0.024 M_r L}{n C_b L_b} \quad (6.10.293)$$

The required cross-frame or diaphragm bracing stiffness is

$$\beta_{Tb} = \frac{\beta_T}{\left(1 - \frac{\beta_T}{\beta_{sec}}\right)} \quad (6.10.294)$$

Where,

$$\beta_T = \frac{1}{\phi} \left(\frac{2.4LM_r^2}{nEI_y C_b^2} \right) \text{ (LRFD)}$$

$$\beta_T = \Omega \left(\frac{2.4LM_r^2}{nEI_y C_b^2} \right) \text{ (ASD)} \quad (6.10.295)$$

$$\beta_{sec} = \frac{3.3 E}{h_o} \left(\frac{1.5h_o t_w^3}{12} + \frac{t_s b_s^3}{12} \right) \quad (6.10.296)$$

Where, $\phi = 0.75$ (LRFD) $\Omega = 3.00$ (ASD)

L = span length, mm

n = number of nodal braced points within the span

E = modulus of elasticity of steel 200000 MPa

I_y = out-of-plane moment of inertia, mm⁴

C_b = modification factor defined in Sec 10.6

t_w = beam web thickness, mm

t_s = web stiffener thickness, mm

b_s = stiffener width for one-sided stiffeners (use twice the individual stiffener width for pairs of stiffeners), mm.

β_T = brace stiffness excluding web distortion, N-mm/radian

β_{sec} = web distortional stiffness, including the effect of web transverse stiffeners, if any, N-mm/radian

For design according to Sec 10.2.3.3 (LRFD)

M_r = required flexural strength using LRFD load combinations, N-mm

For design according to Sec 10.2.3.4 (ASD)

M_r = required flexural strength using ASD load combinations, N-mm

If $\beta_{sec} < \beta_T$, Eq. 6.10.294 is negative, which indicates that torsional beam bracing will not be effective due to inadequate web distortional stiffness.

When required, the web stiffener shall extend the full depth of the braced member and shall be attached to the flange if the torsional brace is also attached to the flange. Alternatively, it shall be permissible to stop the stiffener short by a distance equal to $4t_w$ from any beam flange that is not directly attached to the torsional brace. When L_b is less than L_q then L_b in Eq. 6.10.293 shall be permitted to be taken equal to L_q .

(b) Continuous torsional bracing

For continuous bracing, use Equations 6.10.293, 6.10.294 and 6.10.296 with L/n taken as 1.0 and L_b taken as L_q ; bracing moment and stiffness are given per unit span length. The distortional stiffness for an unstiffened web is

$$\beta_{sec} = \frac{3.3 Et_w^3}{12h_o} \quad (6.10.297)$$

10.19.4 Slenderness Limitations

The slenderness ratio, L/r , of any stability bracing shall not exceed 180 unless a comprehensive analysis including second order effects justifies a higher value.

10.20 Seismic Provisions for Structural Steel Buildings

10.20.1 Scope

The Seismic Provisions for Structural Steel Buildings, hereinafter referred to as these Provisions as outline in this Sec 10.20, shall govern the design, fabrication and erection of structural steel members and connections in the seismic load resisting systems (SLRS) and splices in columns that are not part of the SLRS, in buildings and other structures, where other structures are defined as those structures designed, fabricated and erected in a manner similar to buildings, with building-like vertical and lateral load-resisting-elements.

These Provisions shall apply when the seismic response modification coefficient, R , (as specified in Chapter 2 of Part 6) is taken greater than 3, regardless of the seismic design category. When the seismic response modification coefficient, R , is taken as 3 or less, the structure is not required to satisfy the Provisions this Sec 10.20, unless specifically required by the applicable authority.

These Provisions shall be applied in conjunction with the specification set forth in Sections 10.1 to 10.19 whichever is applicable.

Loads, load combinations, system limitations and general design requirements shall be those in Chapter 2 Part 6 of this Code as well as those mentioned in Sec 10.2.

10.20.2 Referenced Specifications, Codes and Standards

The documents referenced in these Provisions shall include those listed in Sec 10.1.2 with the following additions and modifications:

American Institute of Steel Construction (AISC):

- Specification for Structural Steel Buildings, ANSI/AISC 360-05
- Prequalified Connections for Special and Intermediate Steel Moment Frames for Seismic Applications, ANSI/AISC 358-05

American Society for Nondestructive Testing (ASNT):

- Recommended Practice for the Training and Testing of Nondestructive Testing Personnel, ASNT SNT TC-1a-2001
- Standard for the Qualification and Certification of Nondestructive Testing Personnel, ANSI/ASNT CP-189-2001

American Welding Society (AWS):

- Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding, AWS A4.3-93R
- Standard Methods for Mechanical Testing of Welds-U.S. Customary, ANSI/AWS B4.0-98
- Standard Methods for Mechanical Testing of Welds-Metric Only, ANSI/AWS B4.0M:2000
- Standard for the Qualification of Welding Inspectors, AWS B5.1:2003
- Oxygen Cutting Surface Roughness Gauge and Wall Chart for Criteria Describing Oxygen-Cut Surfaces, AWS C4.1

Federal Emergency Management Agency (FEMA)

- Recommended Seismic Design Criteria for New Steel Moment-Frame Buildings, FEMA 350, July 2000

10.20.3 General Seismic Design Requirements

The required strength and other seismic provisions and the limitations on height and irregularity are specified in Chapter 2 Part 6 of this Code.

The design story drift shall be in accordance with the requirements set forth in Chapter 2 Part 6 of this Code.

10.20.4 Loads, Load Combinations, and Nominal Strengths**10.20.4.1 Loads and load combinations**

The loads and load combinations shall be as stipulated in Chapter 2 Part 6 of this Code. Where amplified seismic loads are required by these Provisions, the horizontal portion of the earthquake load E (as defined in Chapter 2 Part 6) shall be multiplied by the system over strength factor, Ω_o . The magnitude of over strength factor shall generally be obtained from Table 6.2.19 of Chapter 2 Part 6 of this Code. Reference may be made to Table 12.2-1 of ASCE 7-05 if data for a particular structure type is not found in Table 6.2.19. The value of Ω_o for a structural system shall be taken as 2.0 if it is not specified elsewhere.

10.20.4.2 Nominal strength

The nominal strength of systems, members and connections shall comply with the Specification, except as modified throughout these Provisions.

10.20.5 Structural Design Drawings and Specifications, Shop Drawings, and Erection Drawings**10.20.5.1 Structural design drawings and specifications**

Structural design drawings and specifications shall show the work to be performed, and include items required by the Specification and the following, as applicable:

- (1) Designation of the seismic load resisting system (SLRS)
- (2) Designation of the members and connections that are part of the SLRS
- (3) Configuration of the connections
- (4) Connection material specifications and sizes
- (5) Locations of demand critical welds
- (6) Lowest Anticipated Service Temperature (LAST) of the steel structure, if the structure is not enclosed and maintained at a temperature of 10° C or higher.
- (7) Locations and dimensions of protected zones
- (8) Locations where gusset plates are to be detailed to accommodate inelastic rotation
- (9) Welding requirements as specified in Appendix S, Sec S.2.1.

10.20.5.2 Shop drawings

Shop drawings shall include items required by the Specification and the following, as applicable:

- (1) Designation of the members and connections that are part of the SLRS
- (2) Connection material specifications
- (3) Locations of demand critical shop welds
- (4) Locations and dimensions of protected zones
- (5) Gusset plates drawn to scale when they are detailed to accommodate inelastic rotation
- (6) Welding requirements as specified in Appendix S, Sec S.2.2.

10.20.5.3 Erection drawings

Erection drawings shall include items required by the Specification and the following, as applicable:

- (1) Designation of the members and connections that are part of the SLRS
- (2) Field connection material specifications and sizes
- (3) Locations of demand critical field welds
- (4) Locations and dimensions of protected zones
- (5) Locations of pretensioned bolts
- (6) Field welding requirements as specified in Appendix S, Sec S.2.3

10.20.6 Materials

10.20.6.1 Material specifications

Structural steel used in the seismic load resisting system (SLRS) shall meet the requirements of Sec 10.1.3.1a, except as modified in present Sec 10.20. The specified minimum yield stress of steel to be used for members in which inelastic behavior is expected shall not exceed 345 MPa for systems defined in Sections 10.20.9, 10.20.10, 10.20.12, 10.20.13, 10.20.15, 10.20.16, and 10.20.17 nor 380 MPa for systems defined in Sections 10.20.11 and 10.20.14, unless the suitability of the material is determined by testing or other rational criteria. This limitation does not apply to columns for which only expected inelastic behavior is yielding at column base.

The structural steel used in the SLRS described in Sections 10.20.9 to 10.20.17 shall meet one of the following ASTM Specifications: A36/ A36M, A53/A53M, A500 (Grade B or C), A501, A529/A529M, A572/A572M [Grade 290, 345 or 380], A588/A588M, A913/A913M [Grade 345, 415 or 450], A992/A992M, or A1011 HSLAS Grade 380. The structural steel used for column base plates shall meet one of the preceding ASTM specifications or ASTM A283/A283M Grade D.

Other steels and non-steel materials in buckling-restrained braced frames are permitted to be used subject to the requirements of Sec 10.20.16 and Appendix R.

10.20.6.2 Material properties for determination of required strength of members and connections

The required strength of an element (a member or a connection) shall be determined from the expected yield stress, $R_y F_y$, of an adjoining member, where F_y is the specified minimum yield stress of the grade of steel to be used in the adjoining members and R_y is the ratio of the expected yield stress to the specified minimum yield stress, F_y , of that material.

The available strength of the element, ϕR_n for LRFD and R_n/Ω for ASD, shall be equal to or greater than the required strength, where R_n is the nominal strength of the connection. The expected tensile strength, $R_t F_u$, and the expected yield stress, $R_y F_y$, are permitted to be used in lieu of F_u and F_y , respectively, in determining the nominal strength, R_n , of rupture and yielding limit states within the same member for which the required strength is determined.

The values of R_y and R_t for various steels are given in Table 6.10.17. Other values of R_y and R_t shall be permitted if the values are determined by testing of specimens similar in size and source conducted in accordance with the requirements for the specified grade of steel.

10.20.6.3 Heavy section CVN requirements

For structural steel in the SLRS, in addition to the requirements of Sec 10.1.3.1c, hot rolled shapes with flange thickness 38 mm and thicker shall have a minimum Charpy V-Notch toughness of 27 J at 21°C, tested in the alternate core location as described in ASTM A6 Supplementary Requirement S30. Plates 50 mm thick and thicker shall have a minimum Charpy V-Notch toughness of 27 J at 21°C, measured at any location permitted by ASTM A673, where the plate is used in the following:

- (1) Members built-up from plate
- (2) Connection plates where inelastic strain under seismic loading is expected
- (3) As the steel core of buckling-restrained braces.

10.20.7 Connections, Joints and Fasteners**10.20.7.1 Scope**

Connections, joints and fasteners that are part of the Seismic Load Resisting System (SLRS) shall comply with Sec 10.10, and with the additional requirements of this Section.

The design of connections for a member that is a part of the SLRS shall be configured such that a ductile limit state in either the connection or the member controls the design.

10.20.7.2 Bolted joints

All bolts shall be pre-tensioned high strength bolts and shall meet the requirements for slip-critical faying surfaces in accordance with Sec 10.10.3.8 with a Class A surface. Bolts shall be installed in standard holes or in short-slotted holes perpendicular to the applied load. For brace diagonals, oversized holes shall be permitted when the connection is designed as a slip-critical joint, and the oversized hole is in one ply only. Alternative hole types are permitted if determined in a connection prequalification in accordance with Appendix N, or if determined in a program of qualification testing in accordance with Appendix Q or R. The available shear strength of bolted joints using standard holes shall be calculated as that for bearing-type joints in accordance with Sections 10.10.3.7 and 10.10.3.10, except that the nominal bearing strength at bolt holes shall not be taken greater than $2.4dtF_u$.

Exception: The faying surfaces for end plate moment connections are permitted to be coated with coatings not tested for slip resistance, or with coatings with a slip coefficient less than that of a Class A faying surface.

Bolts and welds shall not be designed to share force in a joint or the same force component in a connection.

Table 6.10.17: R_y and R_t Values for Different Member Types

Application	R_y	R_t
Hot-rolled structural shapes and bars:		
• ASTM A36/A36M	1.5	1.2
• ASTM A572/572M Grade 42 (290)	1.3	1.1
• ASTM A572/572M Grade 50 (345) or 55 (380), ASTM A913/A913M Grade 50 (345), 60 (415), or 65 (450), ASTM A588/A588M, ASTM A992/A992M, A1011 HSLAS Grade 55 (380)	1.1	1.1

Application	R_y	R_t
• ASTM A529 Grade 50 (345)	1.2	1.2
• ASTM A529 Grade 55 (380)	1.1	1.2
Hollow structural sections (HSS):		
• ASTM A500 (Grade B or C), ASTM A501	1.4	1.3
Pipe:		
• ASTM A53/A53M	1.6	1.2
Plates:		
• ASTM A36/A36M	1.3	1.2
• ASTM A572/A572M Grade 50 (345), ASTM A588/A588M	1.1	1.2

10.20.7.3 Welded Joints

Welding shall be performed in accordance with Appendix S. Welding shall be performed in accordance with a welding procedure specification (WPS) as required in AWS D1.1. The WPS variables shall be within the parameters established by the filler metal manufacturer.

10.20.7.3.1 General requirements

All welds used in members and connections in the SLRS shall be made with a filler metal that can produce welds that have a minimum Charpy V-Notch toughness of 27 J at minus 18°C, as determined by the appropriate AWS A5 classification test method or manufacturer certification. This requirement for notch toughness shall also apply in other cases as required in these Provisions.

10.20.7.3.2 Demand critical welds

Where welds are designated as demand critical, they shall be made with a filler metal capable of providing a minimum Charpy V-Notch (CVN) toughness of 27 J at 29°C as determined by the appropriate AWS classification test method or manufacturer certification, and 54 J at 21°C as determined by Appendix T or other approved method, when the steel frame is normally enclosed and maintained at a temperature of 10°C or higher. For structures with service temperatures lower than 10°C, the qualification temperature for Appendix T shall be 11°C above the lowest anticipated service temperature, or at a lower temperature.

SMAW electrodes classified in AWS A5.1 as E7018 or E7018-X, SMAW electrodes classified in AWS A5.5 as E7018-C3L or E8018-C3, and GMAW solid electrodes are exempted from production lot testing when the CVN toughness of the electrode equals or exceeds 27 J at a temperature not exceeding 29°C as determined by AWS classification test methods. The manufacturer's certificate of compliance shall be considered sufficient evidence of meeting this requirement.

10.20.7.3.3 Protected zone

Where a protected zone is designated by these Provisions, it shall comply with the following:

- (1) Within the protected zone, discontinuities created by fabrication or erection operations, such as tack welds, erection aids, air-arc gouging and thermal cutting shall be repaired as required by the engineer of record.
- (2) Welded shear studs and decking attachments that penetrate the beam flange shall not be placed on beam flanges within the protected zone. Decking arc spot welds as required to secure decking shall be permitted.
- (3) Welded, bolted, screwed or shot-in attachments for perimeter edge angles, exterior facades, partitions, duct work, piping or other construction shall not be placed within the protected zone.

Exception:

Welded shear studs and other connections shall be permitted when determined in accordance with a connection prequalification in accordance with Appendix N, or as determined in a program of qualification testing in accordance with Appendix Q.

Outside the protected zone, calculations based upon the expected moment shall be made to demonstrate the adequacy of the member net section when connectors that penetrate the member are used.

10.20.7.3.4 Continuity plates and stiffeners

Corners of continuity plates and stiffeners placed in the webs of rolled shapes shall be clipped as described below. Along the web, the clip shall be detailed so that the clip extends a distance of at least 38 mm beyond the published k detail dimension for the rolled shape. Along the flange, the clip shall be detailed so that the clip does not exceed a distance of 12 mm beyond the published k_1 detail dimension. The clip shall be detailed to facilitate suitable weld terminations for both the flange weld and the web weld. If a curved clip is used, it shall have a minimum radius of 12 mm.

At the end of the weld adjacent to the column web/flange juncture, weld tabs for continuity plates shall not be used, except when permitted by the engineer of record. Unless specified by the engineer of record that they be removed, weld tabs shall not be removed when used in this location.

10.20.8 Members**10.20.8.1 Scope**

Members in the seismic load resisting system (SLRS) shall comply with the specifications of Sections 10.1 to 10.11 and Sec 10.20.8. For columns that are not part of the SLRS, see Sec 10.20.8.4.2.

10.20.8.2 Classification of sections for local buckling**10.20.8.2.1 Compact**

When required by these Provisions, members of the SLRS shall have flanges continuously connected to the web or webs and the width-thickness ratios of its compression elements shall not exceed the limiting width-thickness ratios, λ_p , from Specification Table 6.10.1.

10.20.8.2.2 Seismically compact

When required by these Provisions, members of the SLRS must have flanges continuously connected to the web or webs and the width-thickness ratios of its compression elements shall not exceed the limiting width-thickness ratios, λ_{ps} , from Provisions Table 6.10.1.

10.20.8.3 Column strength

When $\frac{P_u}{\phi_c P_n}$ (LRFD) > 0.4 or $\frac{\Omega_c P_a}{P_n}$ (ASD) > 0.4 , as appropriate, without consideration of amplified seismic load,

Where,

$$\phi_c = 0.90 \text{ (LRFD)} \quad \Omega_c = 1.67 \text{ (ASD)}$$

P_a = required axial strength of a column using ASD load combinations, N

P_n = nominal axial strength of a column, N

P_u = required axial strength of a column using LRFD load combinations, N

The following requirements shall be met:

- (1) The required axial compressive and tensile strength, considered in the absence of any applied moment, shall be determined using load combinations stipulated by the Code including amplified seismic load.
- (2) The required axial compressive and tensile strength shall not exceed either of the following:
 - (a) The maximum load transferred to the column considering $1.1R_y$ (LRFD) or $\left(\frac{1.1}{1.5}\right)R_y$ (ASD), as appropriate, times the nominal strengths of the connecting beam or brace elements of the building.
 - (b) The limit as determined from the resistance of the foundation to over-turning uplift.

10.20.8.4 Column splices

10.20.8.4.1 General

The required strength of column splices in the seismic load resisting system (SLRS) shall equal the required strength of the columns, including that determined from Sections 10.20.8.3, 10.20.9.9, 10.20.10.9, 10.20.11.9, 10.20.13.5 and 10.20.16.5.2.

In addition, welded column splices that are subject to a calculated net tensile load effect determined using the load combinations stipulated by the Code including the amplified seismic load, shall satisfy both of the following requirements:

- (1) The available strength of partial-joint-penetration (PJP) groove welded joints, if used, shall be at least equal to 200 percent of the required strength.
- (2) The available strength for each flange splice shall be at least equal to $0.5R_yF_yA_f$ (LRFD) or $(0.5/1.5)R_yF_yA_f$ (ASD), as appropriate, where R_yF_y is the expected yield stress of the column material and A_f is the flange area of the smaller column connected.

Beveled transitions are not required when changes in thickness and width of flanges and webs occur in column splices where PJP groove welded joints are used.

Column web splices shall be either bolted or welded, or welded to one column and bolted to the other. In moment frames using bolted splices, plates or channels shall be used on both sides of the column web.

The centerline of column splices made with fillet welds or partial-joint-penetration groove welds shall be located 1.2 m or more away from the beam-to-column connections. When the column clear height between beam-to-column connections is less than 2.4 m, splices shall be at half the clear height.

10.20.8.4.2 Columns not part of the seismic load resisting system

Splices of columns that are not a part of the SLRS shall satisfy the following:

- (1) The splices shall be located 1.2 m or more away from the beam-to-column connections. When the column clear height between beam-to-column connections is less than 2.4 m, splices shall be at half the clear height.
- (2) The required shear strength of column splices with respect to both orthogonal axes of the column shall be M_{pc}/H (LRFD) or $M_{pc}/(1.5H)$ (ASD), as appropriate, where M_{pc} is the lesser nominal plastic flexural strength of the column sections for the direction in question, and H is the story height.

10.20.8.5 Column bases

The required strength of column bases shall be calculated in accordance with Sections 10.20.8.5.1, 10.20.8.5.2, and 10.20.8.5.3. The available strength of anchor rods shall be determined in accordance with Specification Sec 10.10.3.

The available strength of concrete elements at the column base, including anchor rod embedment and reinforcing steel, shall be in accordance with Appendix D Chapter 6 Part 6 of this Code.

Exception:

The special requirements in Appendix D Chapter 6 Part 6, for “regions of moderate or high seismic risk, or for structures assigned to intermediate or high seismic performance or design categories” need not be applied.

Table 6.10.18: Limiting Width-Thickness Ratios for Compression Elements

	Description of Element	Width-Thickness Ratio	Limiting Width- Thickness Ratios
			λ_{ps} (Seismically Compact)
Unstiffened Elements	Flexure in flanges of rolled or built-up I-shaped sections ^{[a], [c], [e], [g], [h]}	b/t	$0.30\sqrt{(E/F_y)}$
	Uniform compression in flanges of rolled or built-up I-shaped sections ^{[b], [h]}	b/t	$0.30\sqrt{(E/F_y)}$
	Uniform compression in flanges of rolled or built-up I-shaped sections ^[d]	b/t	$0.38\sqrt{(E/F_y)}$

	Description of Element	Width- Thickn ess Ratio	Limiting Width- Thickness Ratios
			λ_{ps} (Seismically Compact)
Unstiffened Elements	Uniform compression in flanges of channels, outstanding legs of pairs of angles in continuous contact, and braces ^[c] , ^[g]	b/t	$0.30\sqrt{(E/F_y)}$
	Uniform compression in flanges of H-pile sections	b/t	$0.45\sqrt{(E/F_y)}$
	Flat bars ^[f]	b/t	2.5
	Uniform compression in legs of single angles, legs of double angle members with separators, or flanges of tees ^[g]	b/t	$0.30\sqrt{(E/F_y)}$
	Uniform compression in stems of tees ^[g]	b/t	$0.30\sqrt{(E/F_y)}$
Stiffened Elements	Webs in flexural compression in beams in SMF, Sec 10.20.9, unless noted otherwise	h/t_w	$2.45\sqrt{(E/F_y)}$
	Webs in flexural compression or combined flexure and axial compression ^[a] , ^[c] , ^[g] , ^[h] , ^[i] , ^[j]	h/t_w	For $C_a > 0.125$ [k] $3.14\sqrt{(E/F_y)} (1 - 1.54C_a)$ For $C_a \leq 0.125$ [k] $1.12\sqrt{(E/F_y)} (2.33 - C_a)$ $\geq 1.49\sqrt{E/F_y}$
	Round HSS in axial and/or flexural compression ^[c] , ^[g]	D/t	$0.044\sqrt{(E/F_y)}$
	Rectangular HSS in axial and/or flexural compression ^[c] , ^[g]	b/t or h/t_w	$0.64\sqrt{(E/F_y)}$
	Webs of H-Pile sections	h/t_w	$0.94\sqrt{(E/F_y)}$

Notes:

- [a] Required of beams in SMF, Sec 10.20.9 and SPSW, Sec 10.20.17
- [b] Required of columns in SMF, Sec 10.20.9, unless the ratios from Eq. 6.10.300 are greater than 2.0 where it is permitted to use λ_p in Specification Table 6.10.1
- [c] Required for braces and columns in SCBF, Sec 10.20.13 and braces in OCBF, Sec 10.20.14
- [d] it is permitted to use λ_p in Specification Table 6.10.1 for columns in STMF, Sec 10.20.12 and columns in EBF, Sec 10.20.15
- [e] Required for link in EBF, Sec 10.20.15, except it is permitted to use λ_p in Table 6.10.1 of the Specification for flanges of links of length $1.6 M_p/V_p$ or less, where M_p and V_p are defined in Sec 10.20.15
- [f] Diagonal web members within the special segment of STMF, Sec 10.20.12
- [g] Chord members of STMF, Sec 10.20.12
- [h] Required for beams and columns in BRBF, Sec 10.20.16
- [i] Required for columns in SPSW, Sec 10.20.17
- [j] For columns in STMF, Sec 10.20.15; or EBF webs of links of length $1.6 M_p/V_p$ or less, it is permitted to use following for λ_p
 For $C_a \leq 0.125$, $\lambda_p = 3.76 \sqrt{E/F_y} (1 - 275C_a)$
 For $C_a > 0.125$, $\lambda_p = 1.12 \sqrt{E/F_y} (2.33 - C_a) \geq 1.49 \sqrt{E/F_y}$
- [k] For LFRD, $C_a = P_u/\phi_b P_y$; For ASD, $C_a = \Omega_b P_a/P_y$

Where,

P_a = required compressive strength (ASD), N

P_u = required compressive strength (LRFD), N

P_y = axial yield strength, N

$\phi_b = 0.90$

$\Omega_b = 1.67$

10.20.8.5.1 Required axial strength

The required axial strength of column bases, including their attachment to the foundation, shall be the summation of the vertical components of the required strengths of the steel elements that are connected to the column base.

10.20.8.5.2 Required shear strength

The required shear strength of column bases, including their attachments to the foundations, shall be the summation of the horizontal component of the required strengths of the steel elements that are connected to the column base as follows:

- (1) For diagonal bracing, the horizontal component shall be determined from the required strength of bracing connections for the seismic load resisting system (SLRS).
- (2) For columns, the horizontal component shall be at least equal to the lesser of the following:
 - (a) $2R_y F_y Z_x / H$ (LRFD) or $(2/1.5)R_y F_y Z_x / H$ (ASD), as appropriate, of the column

Where, H = height of story, which may be taken as the distance between the centerline of floor framing at each of the levels above and below, or the distance between the top of floor slabs at each of the levels above and below, mm.

- (b) The shear calculated using the load combinations of the applicable building Code, including the amplified seismic load.

10.20.8.5.3 Required flexural strength

The required flexural strength of column bases, including their attachment to the foundation, shall be the summation of the required strengths of the steel elements that are connected to the column base as follows:

- (1) For diagonal bracing, the required flexural strength shall be at least equal to the required strength of bracing connections for the SLRS.
- (2) For columns, the required flexural strength shall be at least equal to the lesser of the following:
 - (a) $1.1R_y F_y Z$ (LRFD) or $(1.1/1.5)R_y F_y Z$ (ASD), as appropriate, of the column
or
 - (b) moment calculated using the load combinations of the Code, including the amplified seismic load.

10.20.8.6 H-piles

10.20.8.6.1 Design of H-piles

Design of H-piles shall comply with the provisions of the Specification regarding design of members subjected to combined loads. H-piles shall meet the requirements of Sec 10.20.8.2.2.

10.20.8.6.2 Battered H-piles

If battered (sloped) and vertical piles are used in a pile group, the vertical piles shall be designed to support the combined effects of the dead and live loads without the participation of the battered piles.

10.20.8.6.3 Tension in H-piles

Tension in each pile shall be transferred to the pile cap by mechanical means such as shear keys, reinforcing bars or studs welded to the embedded portion of the pile. Directly below the bottom of the pile cap, each pile shall be free of attachments and welds for a length at least equal to the depth of the pile cross-section.

10.20.9 Special Moment Frames (SMF)

10.20.9.1 Scope

Special moment frames (SMF) are expected to withstand significant inelastic deformations when subjected to the forces resulting from the motions of the design earthquake. SMF shall satisfy the requirements in this Section.

10.20.9.2 Beam-to-column connections

10.20.9.2.1 Requirements

Beam-to-column connections used in the seismic load resisting system (SLRS) shall satisfy the following three requirements:

- (1) The connection shall be capable of sustaining an interstory drift angle of at least 0.04 radians.
- (2) The measured flexural resistance of the connection, determined at the column face, shall equal at least $0.80M_p$ of the connected beam at an interstory drift angle of 0.04 radians.

- (3) The required shear strength of the connection shall be determined using the following quantity for the earthquake load effect E :

$$E = 2[1.1R_yM_p]/L_h \quad (6.10.298)$$

Where,

R_y = ratio of the expected yield stress to the specified minimum yield stress, F_y

M_p = nominal plastic flexural strength, (N-mm)

L_h = distance between plastic hinge locations, (mm)

When E as in Eq. 6.10.298 is used in ASD load combinations that are additive with other transient loads and that are based on Chapter 2 Part 6, the 0.75 combination factor for transient loads shall not be applied to E .

Connections that accommodate the required interstory drift angle within the connection elements and provide the measured flexural resistance and shear strengths specified above are permitted. In addition to satisfying the requirements noted above, the design shall demonstrate that any additional drift due to connection deformation can be accommodated by the structure. The design shall include analysis for stability effects of the overall frame, including second-order effects.

10.20.9.2.2 Conformance demonstration

Beam-to-column connections used in the SLRS shall satisfy the requirements of Sec 10.20.9.2.1 by one of the following:

- (a) Use of SMF connections designed in accordance with ANSI/AISC 358.
- (b) Use of a connection prequalified for SMF in accordance with Appendix N.
- (c) Provision of qualifying cyclic test results in accordance with Appendix Q.

Results of at least two cyclic connection tests shall be provided and are permitted to be based on one of the following:

- (i) Tests reported in the research literature or documented tests performed for other projects that represent the project conditions, within the limits specified in Appendix Q.
- (ii) Tests that are conducted specifically for the project and are representative of project member sizes, material strengths, connection configurations, and matching connection processes, within the limits specified in Appendix Q.

10.20.9.2.3 Welds

Unless otherwise designated by ANSI/AISC 358, or otherwise determined in a connection prequalification in accordance with Appendix N, or as determined in a program of qualification testing in accordance with Appendix Q, complete-joint-penetration groove welds of beam flanges, shear plates, and beam webs to columns shall be demand critical welds as described in Sec 10.20.7.3.2.

10.20.9.2.4 Protected zones

The region at each end of the beam subject to inelastic straining shall be designated as a protected zone, and shall meet the requirements of Sec 10.20.7.4. The extent of the protected zone shall be as designated in ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or as determined in a program of qualification testing in accordance with Appendix Q.

10.20.9.3 Panel zone of beam-to-column connections (beam web parallel to column web)

10.20.9.3.1 Shear strength

The required thickness of the panel zone shall be determined in accordance with the method used in proportioning the panel zone of the tested or prequalified connection. As a minimum, the required shear strength of the panel zone shall be determined from the summation of the moments at the column faces as determined by projecting the expected moments at the plastic hinge points to the column faces. The design shear strength shall be $\phi_v R_v$ and the allowable shear strength shall be R_v/Ω_v .

Where, $\phi_v = 1.0$ (LRFD) $\Omega_v = 1.50$ (ASD)

And, the nominal shear strength, R_v , according to the limit state of shear yielding, is determined as specified in Specification Sec 10.10.10.6.

10.20.9.3.2 Panel zone thickness

The individual thicknesses, t , of column webs and doubler plates, if used, shall conform to the following requirement:

$$t \geq (d_z + w_z)/90 \quad (6.10.299)$$

Where,

t = thickness of column web or doubler plate, mm

d_z = panel zone depth between continuity plates, mm

w_z = panel zone width between column flanges, mm

Alternatively, when local buckling of the column web and doubler plate is prevented by using plug welds joining them, the total panel zone thickness shall satisfy Eq. 6.10.299.

10.20.9.3.3 Panel zone doubler plates

Doubler plates shall be welded to the column flanges using either a complete-joint-penetration groove-welded or fillet-welded joint that develops the available shear strength of the full doubler plate thickness. When doubler plates are placed against the column web, they shall be welded across the top and bottom edges to develop the proportion of the total force that is transmitted to the doubler plate. When doubler plates are placed away from the column web, they shall be placed symmetrically in pairs and welded to continuity plates to develop the proportion of the total force that is transmitted to the doubler plate.

10.20.9.4 Beam and column limitations

The requirements of Sec 10.20.8.1 shall be satisfied, in addition to the following.

10.20.9.4.1 Width-thickness limitations

Beam and column members shall meet the requirements of Sec 10.20.8.2.2, unless otherwise qualified by tests.

10.20.9.4.2 Beam flanges

Abrupt changes in beam flange area are not permitted in plastic hinge regions. The drilling of flange holes or trimming of beam flange width is permitted if testing or qualification demonstrates that the resulting configuration can develop stable plastic hinges. The configuration shall be consistent with a prequalified connection designated in ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or in a program of qualification testing in accordance with Appendix Q.

10.20.9.5 Continuity plates

Continuity plates shall be consistent with the prequalified connection designated in ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or as determined in a program of qualification testing in accordance with Appendix Q.

10.20.9.6 Column-beam moment ratio

The following relationship shall be satisfied at beam-to-column connections:

$$\frac{\sum M_{pc}^*}{\sum M_{pb}^*} > 1.0 \quad (6.10.300)$$

$\sum M_{pc}^*$ = the sum of the moments in the column above and below the joint at the intersection of the beam and column centerlines. $\sum M_{pc}^*$ is determined by summing the projections of the nominal flexural strengths of the columns (including haunches where used) above and below the joint to the beam centerline with a reduction for the axial force in the column. It is permitted to take $\sum M_{pc}^* = \sum Z_c (F_{yc} - P_{uc}/A_g)$ (LRFD) or $\sum Z_c (F_{yc}/1.5 - P_{ac}/A_g)$ (ASD), as appropriate. When the centerlines of opposing beams in the same joint do not coincide, the mid-line between centerlines shall be used.

$\sum M_{pb}^*$ = the sum of the moments in the beams at the intersection of the beam and column centerlines. $\sum M_{pb}^*$ is determined by summing the projections of the expected flexural strengths of the beams at the plastic hinge locations to the column centerline. It is permitted to take $\sum M_{pb}^* = \sum (1.1R_y F_{yb} Z_b + M_{uv})$ (LRFD) or $\sum M_{pb}^* = \sum [(1.1/1.5)R_y F_{yb} Z_b + M_{av}]$ (ASD), as appropriate. Alternatively, it is permitted to determine $\sum M_{pb}^*$ consistent with a prequalified connection design as designated in ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or in a program of qualification testing in accordance with Appendix Q. When connections with reduced beam sections are used, it is permitted to take $\sum M_{pb}^* = \sum (1.1R_y F_{yb} Z_{RBS} + M_{uv})$ (LRFD) or $\sum M_{pb}^* = \sum [(1.1/1.5)R_y F_{yb} Z_{RBS} + M_{av}]$ (ASD), as appropriate.

Where,

A_g = gross area of column, mm

F_{yc} = specified minimum yield stress of column, MPa

M_{av} = the additional moment due to shear amplification from the location of the plastic hinge to the column centerline, based on ASD load combinations, N-mm.

M_{uv} = the additional moment due to shear amplification from the location of the plastic hinge to the column centerline, based on LRFD load combinations, N-mm

P_{ac} = required compressive strength using ASD load combinations, (positive number) N.

P_{uc} = required compressive strength using LRFD load combinations, (positive number) N

Z_b = plastic section modulus of the beam, mm³

Z_c = plastic section modulus of the column, mm³

Z_{RBS} = minimum plastic section modulus at the reduced beam section, mm³

Exception:

This requirement does not apply if either of the following two conditions is satisfied:

- (a) Columns with $P_{rc} < 0.3P_c$ for all load combinations other than those determined using the amplified seismic load that satisfy either of the following:
 - (i) Columns used in a one-story building or the top story of a multistory building.
 - (ii) Columns where: (1) the sum of the available shear strengths of all exempted columns in the story is less than 20 percent of the sum of the available shear strengths of all moment frame columns in the story acting in the same direction; and (2) the sum of the available shear strengths of all exempted columns on each moment frame column line within that story is less than 33 percent of the available shear strength of all moment frame columns on that column line. For the purpose of this exception, a column line is defined as a single line of columns or parallel lines of columns located within 10 percent of the plan dimension perpendicular to the line of columns.

Where,

For design according to Specification Sec 10.2.3.3 (LRFD),

$$P_c = F_{yc}A_g, \text{ N}$$

$$P_{rc} = P_{uc}, \text{ required compressive strength, using LRFD load combinations, N}$$

For design according to Specification Sec 10.2.3.4 (ASD),

$$P_c = F_{yc}A_g/1.5, \text{ N}$$

$$P_{rc} = P_{ac}, \text{ required compressive strength, using ASD load combinations, N}$$

- (b) Columns in any story that has a ratio of available shear strength to required shear strength that is 50 percent greater than the story above.

10.20.9.7 Lateral bracing at beam-to-column connections

10.20.9.7.1 Braced connections

Column flanges at beam-to-column connections require lateral bracing only at the level of the top flanges of the beams, when the webs of the beams and column are co-planar, and a column is shown to remain elastic outside of the panel zone. It shall be permitted to assume that the column remains elastic when the ratio calculated using Eq. 6.10.300 is greater than 2.0.

When a column cannot be shown to remain elastic outside of the panel zone, following requirements shall apply:

The column flanges shall be laterally braced at the levels of both the top and bottom beam flanges. Lateral bracing shall be either direct or indirect.

Each column-flange lateral brace shall be designed for a required strength that is equal to 2 percent of the available beam flange strength $F_y b_f t_{bf}$ (LRFD) or $F_y b_f t_{bf}/1.5$ (ASD), as appropriate.

10.20.9.7.2 Unbraced connections

A column containing a beam-to-column connection with no lateral bracing transverse to the seismic frame at the connection shall be designed using the distance between adjacent lateral braces as the column height for buckling transverse to the seismic frame and shall conform to Specification Sec 10.8, except that:

- (1) The required column strength shall be determined from the appropriate load combinations, except that E shall be taken as the lesser of:
 - (a) The amplified seismic load.
 - (b) 125 percent of the frame available strength based upon either the beam available flexural strength or panel zone available shear strength.

- (2) The slenderness L/r for the column shall not exceed 60.
- (3) The column required flexural strength transverse to the seismic frame shall include that moment caused by the application of the beam flange force specified in Sec 10.20.9.7.1.(2) in addition to the second-order moment due to the resulting column flange displacement.

10.20.9.8 Lateral bracing of beams

Both flanges of beams shall be laterally braced, with a maximum spacing of $L_b = 0.086r_y E/F_y$. Braces shall meet the provisions of Eq. 6.10.291 and 6.10.292 of Sec 10.19, where $M_r = M_u = R_y Z F_y$ (LRFD) or $M_r = M_a = R_y Z F_y / 1.5$ (ASD), as appropriate, of the beam and $C_d = 1.0$.

In addition, lateral braces shall be placed near concentrated forces, changes in cross-section, and other locations where analysis indicates that a plastic hinge will form during inelastic deformations of the SMF. The placement of lateral bracing shall be consistent with that documented for a prequalified connection designated in ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or in a program of qualification testing in accordance with Appendix Q.

The required strength of lateral bracing provided adjacent to plastic hinges shall be $P_u = 0.06M_u/h_o$ (LRFD) or $P_a = 0.06M_a/h_o$ (ASD), as appropriate, where h_o is the distance between flange centroids; and the required stiffness shall meet the provisions of Eq. 6.10.292 of Sec 10.19.

10.20.9.9 Column splices

Column splices shall comply with the requirements of Sec 10.20.8.4.1. Where groove welds are used to make the splice, they shall be complete-joint-penetration groove welds that meet the requirements of Sec 10.20.7.3.2. Weld tabs shall be removed. When column splices are not made with groove welds, they shall have a required flexural strength that is at least equal to $R_y F_y Z_x$ (LRFD) or $R_y F_y Z_x / 1.5$ (ASD), as appropriate, of the smaller column. The required shear strength of column web splices shall be at least equal to $\sum M_{pc} / H$ (LRFD) or $\sum M_{pc} / (1.5H)$ (ASD), as appropriate, where $\sum M_{pc}$ is the sum of the nominal plastic flexural strengths of the columns above and below the splice.

Exception:

The required strength of the column splice considering appropriate stress concentration factors or fracture mechanics stress intensity factors need not exceed that determined by inelastic analyses.

10.20.10 Intermediate Moment Frames (IMF)

10.20.10.1 Scope

Intermediate moment frames (IMF) are expected to withstand limited inelastic deformations in their members and connections when subjected to the forces resulting from the motions of the design earthquake. IMF shall meet the requirements in this Section.

10.20.10.2 Beam-to-column connections

10.20.10.2.1 Requirements

Beam-to-column connections used in the seismic load resisting system (SLRS) shall satisfy the requirements of Sec 10.20.9.2.1, with the following exceptions:

- (1) The required interstory drift angle shall be a minimum of 0.02 radian.
- (2) The required strength in shear shall be determined as specified in Sec 10.20.9.2.1, except that a lesser value of V_u or V_a , as appropriate, is permitted if justified by analysis. The required shear strength need not exceed the shear resulting from the application of appropriate load combinations using the amplified seismic load.

10.20.10.2.2 Conformance demonstration

Conformance demonstration shall be as described in Sec 10.20.9.2.2 to satisfy the requirements of Sec 10.20.10.2.1 for IMF, except that a connection prequalified for IMF in accordance with ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or as determined in a program of qualification testing in accordance with Appendix Q.

10.20.10.2.3 Welds

Unless otherwise designated by ANSI/AISC 358, or otherwise determined in a connection prequalification in accordance with Appendix N, or as determined in a program of qualification testing in accordance with Appendix Q, complete joint penetration groove welds of beam flanges, shear plates, and beam webs to columns shall be demand critical welds as described in Sec 10.20.7.3.2.

10.20.10.2.4 Protected zone

The region at each end of the beam subject to inelastic straining shall be treated as a protected zone, and shall meet the requirements of Sec 10.20.7.4. The extent of the protected zone shall be as designated in ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or as determined in a program of qualification testing in accordance with Appendix Q.

10.20.10.3 Panel zone of beam-to-column connections (beam web parallel to column web)

No additional requirements beyond the specification.

10.20.10.4 Beam and column limitations

The requirements of Sec 10.20.8.1 shall be satisfied, in addition to the following.

10.20.10.4.1 Width-thickness limitations

Beam and column members shall meet the requirements of Sec 10.20.8.2.1, unless otherwise qualified by tests.

10.20.10.4.2 Beam flanges

Abrupt changes in beam flange area are not permitted in plastic hinge regions. Drilling of flange holes or trimming of beam flange width is permitted if testing or qualification demonstrates that the resulting configuration can develop stable plastic hinges. The configuration shall be consistent with a prequalified connection designated in ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or in a program of qualification testing in accordance with Appendix Q.

10.20.10.5 Continuity plates

Continuity plates shall be provided to be consistent with the prequalified connections designated in ANSI/AISC 358, or as otherwise determined in a connection prequalification in accordance with Appendix N, or as determined in a program of qualification testing in accordance with Appendix Q.

10.20.10.6 Column-beam moment ratio

No additional requirements beyond the Specification.

10.20.10.7 Lateral bracing at beam-to-column connections

No additional requirements beyond the Specification.

10.20.10.8 Lateral bracing of beams

Both flanges shall be laterally braced directly or indirectly. The unbraced length between lateral braces shall not exceed $0.17r_y E / F_y$. Braces shall meet the provisions of Equations 6.10.291 and 6.10.292 of Sec 10.19, where $M_r = M_u = R_y Z F_y$ (LRFD) or $M_r = M_a = R_y Z F_y / 1.5$ (ASD), as appropriate, of the beam, and $C_d = 1.0$.

In addition, lateral braces shall be placed near concentrated loads, changes in cross-section and other locations where analysis indicates that a plastic hinge will form during inelastic deformations of the IMF. Where the design is based upon assemblies tested in accordance with Appendix Q, the placement of lateral bracing for the beams shall be consistent with that used in the tests or as required for prequalification in Appendix N. The required strength of lateral bracing provided adjacent to plastic hinges shall be $P_u = 0.06M_u / h_o$ (LRFD) or $P_a = 0.06M_a / h_o$ (ASD), as appropriate, where h_o = distance between flange centroids; and the required stiffness shall meet the provisions of Eq. 6.10.292 of Sec 10.19.

10.20.10.9 Column splices

Column splices shall comply with the requirements of Sec 10.20.8.4.1. Where groove welds are used to make the splice, they shall be complete-joint-penetration groove welds that meet the requirements of Sec 10.20.7.3.2.

10.20.11 Ordinary Moment Frames (OMF)

10.20.11.1 Scope

Ordinary moment frames (OMF) are expected to withstand minimal inelastic deformations in their members and connections when subjected to the forces resulting from the motions of the design earthquake. OMF shall meet the requirements of this Section. Connections in conformance with Sections 10.20.9.2.1 and 10.20.9.5 or Sections 10.20.10.2.1 and 10.20.10.5 shall be permitted for use in OMF without meeting the requirements of Sections 10.20.11.2.1, 10.20.11.2.3, and 10.20.11.5

10.20.11.2 Beam-to-column connections

Beam-to-column connections shall be made with welds and/or high-strength bolts. Connections are permitted to be fully restrained (FR) or partially restrained (PR) moment connections as follows.

10.20.11.2.1 Requirements for FR moment connections

FR moment connections that are part of the seismic load resisting system (SLRS) shall be designed for a required flexural strength that is equal to $1.1R_yM_p$ (LRFD) or $(1.1/1.5)R_yM_p$ (ASD), as appropriate, of the beam or girder, or the maximum moment that can be developed by the system, whichever is less.

FR connections shall meet the following requirements.

- (1) Where steel backing is used in connections with complete-joint-penetration (CJP) beam flange groove welds, steel backing and tabs shall be removed, except that top-flange backing attached to the column by a continuous fillet weld on the edge below the CJP groove weld need not be removed. Removal of steel backing and tabs shall be as follows:
 - (i) Following the removal of backing, the root pass shall be back gouged to sound weld metal and back welded with a reinforcing fillet. The reinforcing fillet shall have a minimum leg size of 8 mm.
 - (ii) Weld tab removal shall extend to within 3 mm of the base metal surface, except at continuity plates where removal to within 6 mm of the plate edge is acceptable. Edges of the weld tab shall be finished to a surface roughness value of $13\ \mu\text{m}$ or better. Grinding to a flush condition is not required. Gouges and notches are not permitted. The transitional slope of any area where gouges and notches have been removed shall not exceed 1:5. Material removed by grinding that extends more than 2 mm below the surface of the base metal shall be filled with weld metal. The contour of the weld at the ends shall provide a smooth transition, free of notches and sharp corners.
- (2) Where weld access holes are provided, they shall be as shown in Figure 6.10.3. The weld access hole shall have a surface roughness value not to exceed $13\ \mu\text{m}$, and shall be free of notches and gouges. Notches and gouges shall be repaired as required by the engineer of record. Weld access holes are prohibited in the beam web adjacent to the end-plate in bolted moment end-plate connections.
- (3) The required strength of double-sided partial-joint-penetration groove welds and double-sided fillet welds that resist tensile forces in connections shall be $1.1R_yF_yA_g$ (LRFD) or $(1.1/1.5)R_yF_yA_g$ (ASD), as appropriate, of the connected element or part. Single-sided partial-joint-penetration groove welds and single-sided fillet welds shall not be used to resist tensile forces in the connections.

- (4) For FR moment connections, the required shear strength, V_u or V_a , as appropriate, of the connection shall be determined using the following quantity for the earthquake load effect E :

$$E = 2(1.1R_y M_p)/L_b \quad (6.10.301)$$

Where this E is used in ASD load combinations that are additive with other transient loads and that are based on Chapter 2 Part 6, the 0.75 combination factor for transient loads shall not be applied to E .

Alternatively, a lesser value of V_u or V_a is permitted if justified by analysis. The required shear strength need not exceed the shear resulting from the application of appropriate load combinations in the Code using the amplified seismic load.

10.20.11.2.2 Requirements for PR moment connections

PR moment connections are permitted when the following requirements are met:

- (1) Such connections shall be designed for the required strength as specified in Sec 10.20.11.2.1 above.
- (2) The nominal flexural strength of the connection, M_n , shall be no less than 50 percent of M_p of the connected beam or column, whichever is less.
- (3) The stiffness and strength of the PR moment connections shall be considered in the design, including the effect on overall frame stability.
- (4) For PR moment connections, V_u or V_a , as appropriate, shall be determined from the load combination above plus the shear resulting from the maximum end moment that the connection is capable of resisting.

10.20.11.2.3 Welds

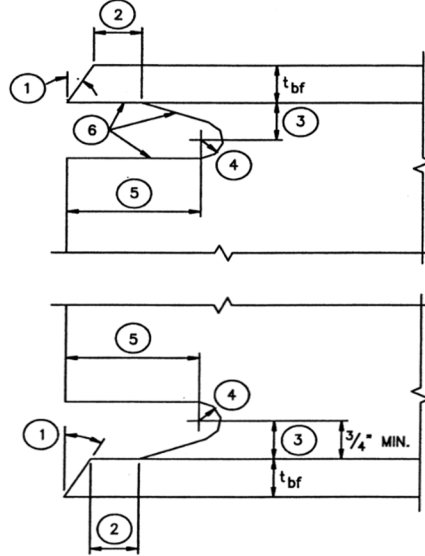
Complete-joint-penetration groove welds of beam flanges, shear plates, and beam webs to columns shall be demand critical welds as described in Sec 10.20.7.3.2.

10.20.11.3 Panel zone of beam-to-column connections (beam web parallel to column web)

No additional requirements beyond the Specification.

10.20.11.4 Beam and column limitations

No requirements beyond Sec 10.20.8.1.



Notes:

1. Bevel as required for selected groove weld.
2. Larger of t_{bf} or 13 mm (plus $\frac{1}{2} t_{bf}$, or minus $\frac{1}{4} t_{bf}$)
3. $\frac{3}{4} t_{bf}$ to t_{bf} , 19 mm minimum (± 6 mm)
4. 10 mm minimum radius (plus not limited, minus 0)
5. $3 t_{bf}$ (± 13 mm)

Tolerances shall not accumulate to the extent that the angle of the access hole cut to the flange surface exceeds 25° .

Figure 6.10.3. Weld access hole detail (FEMA 350)

10.20.11.5 Continuity plates

When FR moment connections are made by means of welds of beam flanges or beam-flange connection plates directly to column flanges, continuity plates shall be provided in accordance with Sec J10 of the Specification. Continuity plates shall also be required when:

$$t_{cf} < 0.54(b_f t_{bf} F_{yb} / F_{yc})^{1/2}$$

Or, when,

$$t_{cf} < b_f / 6$$

Where continuity plates are required, the thickness of the plates shall be determined as follows:

- (a) For one-sided connections, continuity plate thickness shall be at least one half of the thickness of the beam flange.
- (b) For two-sided connections the continuity plates shall be at least equal in thickness to the thicker of the beam flanges.

The welded joints of the continuity plates to the column flanges shall be made with either complete-joint-penetration groove welds, two-sided partial-joint-penetration groove welds combined with reinforcing fillet welds, or two-sided fillet welds. The required strength of these joints shall not be less than the available strength of the contact area of the plate with the column flange. The required strength of the welded joints of the continuity plates to the column web shall be the least of the following:

- (a) The sum of the available strengths at the connections of the continuity plate to the column flanges.
- (b) The available shear strength of the contact area of the plate with the column web.
- (c) The weld available strength that develops the available shear strength of the column panel zone.
- (d) The actual force transmitted by the stiffener.

10.20.11.6 Column-beam moment ratio

No requirements.

10.20.11.7 Lateral bracing at beam-to-column connections

No additional requirements beyond the Specification.

10.20.11.8 Lateral bracing of beams

No additional requirements beyond the Specification.

10.20.11.9 Column splices

Column splices shall comply with the requirements of Sec 10.20.8.4.1.

10.20.12 Special Truss Moment Frames (STMF)

10.20.12.1 Scope

Special truss moment frames (STMF) are expected to withstand significant inelastic deformation within a specially designed segment of the truss when subjected to the forces from the motions of the design earthquake. STMF shall be limited to span lengths between

columns not to exceed 20 m and overall depth not to exceed 1.8 m. The columns and truss segments outside of the special segments shall be designed to remain elastic under the forces that can be generated by the fully yielded and strain-hardened special segment. STMF shall meet the requirements in this Section.

10.20.12.2 Special segment

Each horizontal truss that is part of the seismic load resisting system (SLRS) shall have a special segment that is located between the quarter points of the span of the truss. The length of the special segment shall be between 0.1 and 0.5 times the truss span length. The length-to-depth ratio of any panel in the special segment shall neither exceed 1.5 nor be less than 0.67.

Panels within a special segment shall either be all Vierendeel panels or all X-braced panels; neither a combination thereof nor the use of other truss diagonal configurations is permitted. Where diagonal members are used in the special segment, they shall be arranged in an X pattern separated by vertical members. Such diagonal members shall be interconnected at points where they cross. The interconnection shall have a required strength equal to 0.25 times the nominal tensile strength of the diagonal member. Bolted connections shall not be used for web members within the special segment. Diagonal web members within the special segment shall be made of flat bars of identical sections.

Splicing of chord members is not permitted within the special segment, nor within one-half the panel length from the ends of the special segment. The required axial strength of the diagonal web members in the special segment due to dead and live loads within the special segment shall not exceed $0.03F_yA_g$ (LRFD) or $(0.03/1.5)F_yA_g$ (ASD), as appropriate. The special segment shall be a protected zone meeting the requirements of Sec 10.20.7.4.

10.20.12.3 Strength of special segment members

The available shear strength of the special segment shall be calculated as the sum of the available shear strength of the chord members through flexure, and the shear strength corresponding to the available tensile strength and 0.3 times the available compressive strength of the diagonal members, when they are used. The top and bottom chord members in the special segment shall be made of identical sections and shall provide at least 25 percent of the required vertical shear strength. The required axial strength in the chord members, determined according to the limit state of tensile yielding, shall not exceed 0.45 times ϕP_n (LRFD) or P_n/Ω (ASD), as appropriate,

$$\phi = 0.90 \text{ (LRFD)}$$

$$\Omega = 1.67 \text{ (ASD)}$$

Where,

$$P_n = F_y A_g$$

The end connection of diagonal web members in the special segment shall have a required strength that is at least equal to the expected yield strength, in tension, of the web member, $R_y F_y A_g$ (LRFD) or $R_y F_y A_g / 1.5$ (ASD), as appropriate.

10.20.12.4 Strength of non-special segment members

Members and connections of STMF, except those in the special segment specified in Sec 10.20.12.2, shall have a required strength based on the appropriate load combinations in the Code, replacing the earthquake load term E with the lateral loads necessary to develop the expected vertical shear strength of the special segment V_{ne} (LRFD) or $V_{ne} / 1.5$ (ASD), as appropriate, at mid-length, given as:

$$V_{ne} = \frac{3.75 R_y M_{nc}}{L_s} + 0.075 EI \frac{L - L_s}{L_s^3} + R_y (P_{nt} + 0.3 P_{nc}) \sin \alpha \quad (6.10.302)$$

Where,

M_{nc} = nominal flexural strength of a chord member of the special segment,
N-mm

EI = flexural elastic stiffness of a chord member of the special segment, N-mm²

L = span length of the truss, mm

L_s = length of the special segment, mm

P_{nt} = nominal tensile strength of a diagonal member of the special segment, N

P_{nc} = nominal compressive strength of a diagonal member of the special segment,
N

α = angle of diagonal members with the horizontal

10.20.12.5 Width-thickness limitations

Chord members and diagonal web members within the special segment shall meet the requirements of Sec 10.20.8.2.2.

10.20.12.6 Lateral bracing

The top and bottom chords of the trusses shall be laterally braced at the ends of the special segment, and at intervals not to exceed L_p according to Specification Sec 10.6 along the entire length of the truss. The required strength of each lateral brace at the ends of and within the special segment shall be

$$P_u = 0.06R_yP_{nc} \text{ (LRFD) or } P_a = (0.06/1.5)R_yP_{nc} \text{ (ASD), as appropriate.}$$

Where, P_{nc} is the nominal compressive strength of the special segment chord member. Lateral braces outside of the special segment shall have a required strength of

$$P_u = 0.02R_yP_{nc} \text{ (LRFD) or } P_a = (0.02/1.5)R_yP_{nc} \text{ (ASD), as appropriate.}$$

The required brace stiffness shall meet the provisions of Eq. 6.10.288 of Sec 10.19.

Where, $P_r = P_u = R_yP_{nc}$ (LRFD) or $P_r = P_a = R_yP_{nc}/1.5$ (ASD), as appropriate.

10.20.13 Special Concentrically Braced Frames (SCBF)

10.20.13.1 Scope

Special concentrically braced frames (SCBF) are expected to withstand significant inelastic deformations when subjected to the forces resulting from the motions of the design earthquake. SCBF shall meet the requirements in this Section.

10.20.13.2 Members

10.20.13.2.1 Slenderness

Bracing members shall have $KI/r \leq 4\sqrt{(E/F_y)}$.

Exception :

Braces with $4\sqrt{(E/F_y)} < KI/r \leq 200$ are permitted in frames in which the available strength of the column is at least equal to the maximum load transferred to the column considering R_y (LRFD) or $(1/1.5)R_y$ (ASD), as appropriate, times the nominal strengths of the connecting brace elements of the building. Column forces need not exceed those determined by inelastic analysis, nor the maximum load effects that can be developed by the system.

10.20.13.2.2 Required strength

Where the effective net area of bracing members is less than the gross area, the required tensile strength of the brace based upon the limit state of fracture in the net section shall be greater than the lesser of the following:

- (a) The expected yield strength, in tension, of the bracing member, determined as $R_y F_y A_g$ (LRFD) or $R_y F_y A_g / 1.5$ (ASD), as appropriate.
- (b) The maximum load effect, indicated by analysis that can be transferred to the brace by the system.

10.20.13.2.3 Lateral force distribution

Along any line of bracing, braces shall be deployed in alternate directions such that, for either direction of force parallel to the bracing, at least 30 percent but no more than 70 percent of the total horizontal force along that line is resisted by braces in tension, unless the available strength of each brace in compression is larger than the required strength resulting from the application of the appropriate load combinations stipulated by the Code including the amplified seismic load. For the purposes of this provision, a line of bracing is defined as a single line or parallel lines with a plan offset of 10 percent or less of building dimension perpendicular to line of bracing.

10.20.13.2.4 Width-thickness limitations

Column and brace members shall meet the requirements of Sec 10.20.8.2.2.

10.20.13.2.5 Built-up members

The spacing of stitches shall be such that the slenderness ratio l/r of individual elements between the stitches does not exceed 0.4 times the governing slenderness ratio of the built-up member.

The sum of the available shear strengths of the stitches shall equal or exceed the available tensile strength of each element. The spacing of stitches shall be uniform. Not less than two stitches shall be used in a built-up member. Bolted stitches shall not be located within the middle one-fourth of the clear brace length.

Exception:

Where the buckling of braces about their critical buckling axis does not cause shear in the stitches, the spacing of the stitches shall be such that the slenderness ratio l/r of the individual elements between the stitches does not exceed 0.75 times the governing slenderness ratio of the built-up member.

10.20.13.3 Required strength of bracing connections

10.20.13.3.1 Required tensile strength

The required tensile strength of bracing connections (including beam-to-column connections if part of the bracing system) shall be the lesser of the following:

- (a) The expected yield strength, in tension, of the bracing member, determined as $R_y F_y A_g$ (LRFD) or $R_y F_y A_g / 1.5$ (ASD), as appropriate.
- (b) The maximum load effect, indicated by analysis that can be transferred to the brace by the system.

10.20.13.3.2 Required flexural strength

The required flexural strength of bracing connections shall be equal to $1.1 R_y M_p$ (LRFD) or $(1.1/1.5) R_y M_p$ (ASD), as appropriate, of the brace about the critical buckling axis.

Exception:

Brace connections that meet the requirements of Sec 10.20.13.3.1 and can accommodate the inelastic rotations associated with brace post-buckling deformations need not meet this requirement.

10.20.13.3.3 Required compressive strength

Bracing connections shall be designed for a required compressive strength based on buckling limit states that is at least equal to $1.1 R_y P_n$ (LRFD) or $(1.1/1.5) R_y P_n$ (ASD), as appropriate, where P_n is the nominal compressive strength of the brace.

10.20.13.4 Special bracing configuration requirements

10.20.13.4.1 V-type and inverted-V-type bracing

V-type and inverted V-type SCBF shall meet the following requirements:

- (1) The required strength of beams intersected by braces, their connections, and supporting members shall be determined based on the load combinations of the Code assuming that the braces provide no support for dead and live loads. For load combinations that include earthquake effects, the earthquake effect, E , on the beam shall be determined as follows:
 - (a) The forces in all braces in tension shall be assumed to be equal to $R_y F_y A_g$.
 - (b) The forces in all adjoining braces in compression shall be assumed to be equal to $0.3P_n$.
- (2) Beams shall be continuous between columns. Both flanges of beams shall be laterally braced, with a maximum spacing of $L_b = L_{pd}$, as specified by Equations 6.10.270 and 6.10.271 of Sec 10.15. Lateral braces shall meet the provisions of Equations 6.10.291 and 6.10.292 of Sec 10.19, where $M_r = M_u = R_y Z F_y$ (LRFD) or $M_r = M_a = R_y Z F_y / 1.5$ (ASD), as appropriate, of the beam and $C_d = 1.0$.

As a minimum, one set of lateral braces is required at the point of intersection of the V-type (or inverted V-type) bracing, unless the beam has sufficient out-of-plane strength and stiffness to ensure stability between adjacent brace points.

10.20.13.4.2 K-type bracing

K-type braced frames are not permitted for SCBF.

10.20.13.5 Column splices

In addition to meeting the requirements in Sec 10.20.8.4, column splices in SCBF shall be designed to develop 50 percent of the lesser available flexural strength of the connected members. The required shear strength shall be $\sum M_{pc} / H$ (LRFD) or $\sum M_{pc} / 1.5H$ (ASD), as appropriate, where $\sum M_{pc}$ is the sum of the nominal plastic flexural strengths of the columns above and below the splice.

10.20.13.6 Protected zone

The protected zone of bracing members in SCBF shall include the center one-quarter of the brace length, and a zone adjacent to each connection equal to the brace depth in the plane of buckling. The protected zone of SCBF shall include elements that connect braces to beams and columns and shall satisfy the requirements of Sec 10.20.7.4.

10.20.14 Ordinary Concentrically Braced Frames (OCBF)**10.20.14.1 Scope**

Ordinary concentrically braced frames (OCBF) are expected to withstand limited inelastic deformations in their members and connections when subjected to the forces resulting from the motions of the design earthquake. OCBF shall meet the requirements in this Section. OCBF above the isolation system in seismically isolated structures shall meet the requirements of Sections 10.20.14.4 and 10.20.14.5 and need not meet the requirements of Sections 10.20.14.2 and 10.20.14.3.

10.20.14.2 Bracing members

Bracing members shall meet the requirements of Sec 10.20.8.2.2.

Exception:

HSS braces that are filled with concrete need not comply with this provision.

Bracing members in K, V, or inverted-V configurations shall have $KL/r \leq 4 \sqrt{(E/F_y)}$.

10.20.14.3 Special bracing configuration requirements

Beams in V-type and inverted V-type OCBF and columns in K-type OCBF shall be continuous at bracing connections away from the beam-column connection and shall meet the following requirements:

- (1) The required strength shall be determined based on the load combinations of the Code assuming that the braces provide no support of dead and live loads. For load combinations that include earthquake effects, the earthquake effect, E , on the member shall be determined as follows:
 - (a) The forces in braces in tension shall be assumed to be equal to $R_y F_y A_g$. For V-type and inverted V-type OCBF, the forces in braces in tension need not exceed the maximum force that can be developed by the system.
 - (b) The forces in braces in compression shall be assumed to be equal to $0.3P_n$.
- (2) Both flanges shall be laterally braced, with a maximum spacing of $L_b = L_{pd}$, as specified by Equations 6.10.270 and 6.10.271 of Sec 10.15.. Lateral braces shall meet the provisions of Equations 6.10.291 and 6.10.292 of Sec 10.19, where $M_r = M_u = R_y Z F_y$ (LRFD) or $M_r = M_a = R_y Z F_y / 1.5$ (ASD), as appropriate, of the beam

and $C_d = 1.0$. As a minimum, one set of lateral braces is required at the point of intersection of the bracing, unless the member has sufficient out-of-plane strength and stiffness to ensure stability between adjacent brace points.

10.20.14.4 Bracing connections

The required strength of bracing connections shall be determined as follows.

- (1) For the limit state of bolt slip, the required strength of bracing connections shall be that determined using the load combinations stipulated by the Code, not including the amplified *seismic load*.
- (2) For other limit states, the required strength of bracing connections is the *expected yield strength*, in tension, of the brace, determined as $R_y F_y A_g$ (LRFD) or $R_y F_y A_g / 1.5$ (ASD), as appropriate.

Exception:

The required strength of the brace connection need not exceed either of the following:

- (a) The maximum force that can be developed by the system
- (b) A load effect based upon using the amplified seismic load

10.20.14.5 OCBF above seismic isolation systems

10.20.14.5.1 Bracing members

Bracing members shall meet the requirements of Sec 10.20.8.2.2 and shall have

$$KL/r \leq 4\sqrt{(E/F_y)}.$$

10.20.14.5.2 K-type bracing

K-type braced frames are not permitted.

10.20.14.5.3 V-type and inverted-V-type bracing

Beams in V-type and inverted V-type bracing shall be continuous between columns.

10.20.15 Eccentrically Braced Frames (EBF)**10.20.15.1 Scope**

Eccentrically braced frames (EBFs) are expected to withstand significant inelastic deformations in the links when subjected to the forces resulting from the motions of the design earthquake. The diagonal braces, columns, and beam segments outside of the links shall be designed to remain essentially elastic under the maximum forces that can be generated by the fully yielded and strain-hardened links, except where permitted in this Section. In buildings exceeding five stories in height, the upper story of an EBF system is permitted to be designed as an OCBF or a SCBF and still be considered to be part of an EBF system for the purposes of determining system factors in the Code. EBF shall meet the requirements in this Section.

10.20.15.2 Links**10.20.15.2.1 Limitations**

Links shall meet the requirements of Sec 10.20.8.2.2.

The web of a link shall be single thickness. Doubler-plate reinforcement and web penetrations are not permitted.

10.20.15.2.2 Shear strength

Except as limited below, the link design shear strength, $\phi_v V_n$ and the allowable shear strength, V_n/Ω_v , according to the limit state of shear yielding shall be determined as follows:

V_n = nominal shear strength of the link, equal to the lesser of V_p or $2M_p/e$, N

$$\phi_v = 0.90 \text{ (LRFD)} \quad \Omega_v = 1.67 \text{ (ASD)}$$

Where,

$$M_p = F_y Z, \text{ N-mm}$$

$$V_p = 0.6 F_y A_w, \text{ N}$$

$$e = \text{link length, mm}$$

$$A_w = (d - 2t_f)t_w$$

The effect of axial force on the link available shear strength need not be considered if

$$P_u \leq 0.15P_y \quad (\text{LRFD}) \quad \text{or} \quad P_a \leq (0.15/1.5)P_y \quad (\text{ASD}),$$

as appropriate.

Where,

P_u = required axial strength using LRFD load combinations, N

P_a = required axial strength using ASD load combinations, N

P_y = nominal axial yield strength = $F_y A_g$, N

If $P_u > 0.15P_y$ (LRFD) or $P_a > (0.15/1.5)P_y$ (ASD), as appropriate, the following additional requirements shall be met:

- (1) The available shear strength of the link shall be the lesser of

$$\phi_v V_{pa} \text{ and } 2\phi_v M_{pa}/e \quad (\text{LRFD})$$

Or,

$$V_{pa}/\Omega_v \text{ and } 2(M_{pa}/e)/\Omega_v \quad (\text{ASD}), \text{ as appropriate,}$$

Where,

$$\phi_v = 0.90 \quad (\text{LRFD}), \quad \Omega_v = 1.67 \quad (\text{ASD})$$

$$V_{pa} = V_p \sqrt{(1 - (P_r/P_c)^2)} \quad (6.10.303)$$

$$M_{pa} = 1.18M_p[1 - (P_r/P_c)] \quad (6.10.304)$$

$$P_r = P_u \quad (\text{LRFD}) \text{ or } P_a \quad (\text{ASD}), \text{ as appropriate}$$

$$P_c = P_y \quad (\text{LRFD}) \text{ or } P_y/1.5 \quad (\text{ASD}), \text{ as appropriate}$$

(2) The length of the link shall not exceed:

$$(a) \quad [1.15 - 0.5\rho'(A_w/A_g)]1.6M_p/V_p \text{ when } \rho'(A_w/A_g) \geq 0.3$$

(6.10.305)

Nor,

$$(b) \quad 1.6M_p/V_p \text{ when } \rho'(A_w/A_g) < 0.3$$

(6.10.306)

Where,

$$A_w = (d - 2t_f)t_w$$

$$\rho' = P_r/V_r$$

And where,

$$V_r = V_u \text{ (LRFD) or } V_r = V_a \text{ (ASD), as appropriate}$$

$$V_u = \text{required shear strength based on LRFD load combinations.}$$

$$V_a = \text{required shear strength based on ASD load combinations.}$$

10.20.15.2.3 Link rotation angle

The link rotation angle is the inelastic angle between the link and the beam outside of the link when the total story drift is equal to the design story drift, Δ . The link rotation angle shall not exceed the following values:

- (a) 0.08 radians for links of length $1.6M_p/V_p$ or less.
- (b) 0.02 radians for links of length $2.6M_p/V_p$ or greater.
- (c) The value determined by linear interpolation between the above values for links of length between $1.6M_p/V_p$ and $2.6M_p/V_p$.

10.20.15.3 Link stiffeners

Full-depth web stiffeners shall be provided on both sides of the link web at the diagonal brace ends of the link. These stiffeners shall have a combined width not less than $(b_f - 2t_w)$ and a thickness not less than $0.75t_w$ or 10 mm, whichever is larger, where b_f and t_w are the link flange width and link web thickness, respectively.

Links shall be provided with intermediate web stiffeners as follows:

- (a) Links of lengths $1.6M_p/V_p$ or less shall be provided with intermediate web stiffeners spaced at intervals not exceeding $(30t_w - d/5)$ for a link rotation angle of 0.08 radian or $(52t_w - d/5)$ for link rotation angles of 0.02 radian or less. Linear interpolation shall be used for values between 0.08 and 0.02 radian.
- (b) Links of length greater than $2.6M_p/V_p$ and less than $5M_p/V_p$ shall be provided with intermediate web stiffeners placed at a distance of 1.5 times b_f from each end of the link.
- (c) Links of length between $1.6M_p/V_p$ and $2.6M_p/V_p$ shall be provided with intermediate web stiffeners meeting the requirements of (a) and (b) above.
- (d) Intermediate web stiffeners are not required in links of lengths greater than $5M_p/V_p$.
- (e) Intermediate web stiffeners shall be full depth. For links that are less than 635 mm in depth, stiffeners are required on only one side of the link web. The thickness of one-sided stiffeners shall not be less than t_w or 10 mm, whichever is larger, and the width shall be not less than $(b_f/2)t_w$. For links that are 635 mm in depth or greater, similar intermediate stiffeners are required on both sides of the web.

The required strength of fillet welds connecting a link stiffener to the link web is $A_{st}F_y$ (LRFD) or $A_{st}F_y/1.5$ (ASD), as appropriate, where A_{st} is the area of the stiffener. The required strength of fillet welds connecting the stiffener to the link flanges is $A_{st}F_y/4$ (LRFD) or $A_{st}F_y/4(1.5)$ (ASD).

10.20.15.4 Link-to-column connections

Link-to-column connections must be capable of sustaining the maximum link rotation angle based on the length of the link, as specified in Sec 10.20.15.2.3. The strength of the connection measured at the column face shall equal at least the nominal shear strength of

the link, V_n , as specified in Sec 10.20.15.2.2 at the maximum link rotation angle. Link-to-column connections shall satisfy the above requirements by one of the following:

- (a) Use a connection prequalified for EBF in accordance with Appendix N.
- (b) Provide qualifying cyclic test results in accordance with Appendix Q. Results of at least two cyclic connection tests shall be provided and are permitted to be based on one of the following:
 - (i) Tests reported in research literature or documented tests performed for other projects that are representative of project conditions, within the limits specified in Appendix Q.
 - (ii) Tests that are conducted specifically for the project and are representative of project member sizes, material strengths, connection configurations, and matching connection processes, within the limits specified in Appendix Q.

Exception:

Where reinforcement at the beam-to-column connection at the link end precludes yielding of the beam over the reinforced length, the link is permitted to be the beam segment from the end of the reinforcement to the brace connection. Where such links are used and the link length does not exceed $1.6M_p/V_p$, cyclic testing of the reinforced connection is not required if the available strength of the reinforced section and the connection equals or exceeds the required strength calculated based upon the strain-hardened link as described in Sec 10.20.15.6. Full depth stiffeners as required in Sec 10.20.15.3 shall be placed at the link-to-reinforcement interface.

10.20.15.5 Lateral bracing of link

Lateral bracing shall be provided at both the top and bottom link flanges at the ends of the link. The required strength of each lateral brace at the ends of the link shall be $P_b = 0.06M_r/h_o$, where, h_o is the distance between flange centroids, in mm.

For design according to Specification Sec 10.2.3.3 (LRFD)

$$M_r = M_{u,exp} = R_y Z F_y$$

For design according to Specification Sec 10.2.3.4 (ASD)

$$M_r = M_{u,exp}/1.5$$

The required brace stiffness shall meet the provisions of Eq. 6.10.292 of Sec 10.19, where M_r is defined above, $C_d = 1$, and L_b is the link length.

10.20.15.6 Diagonal brace and beam outside of link

10.20.15.6.1 Diagonal brace

The required combined axial and flexural strength of the diagonal brace shall be determined based on load combinations stipulated by the Code. For load combinations including seismic effects, a load Q_1 shall be substituted for the term E , where Q_1 is defined as the axial forces and moments generated by at least 1.25 times the expected nominal shear strength of the link $R_y V_n$, where V_n is as defined in Sec 10.20.15.2.2. The available strength of the diagonal brace shall comply with Specification Sec 10.10. Brace members shall meet the requirements of Sec 10.20.8.2.1.

10.20.15.6.2 Beam outside link

The required combined axial and flexural strength of the beam outside of the link shall be determined based on load combinations stipulated by the Code. For load combinations including seismic effects, a load Q_1 shall be substituted for the term E where Q_1 is defined as the forces generated by at least 1.1 times the expected nominal shear strength of the link, $R_y V_n$, where V_n is as defined in Sec 10.20.15.2.2. The available strength of the beam outside of the link shall be determined by the Specification, multiplied by R_y .

At the connection between the diagonal brace and the beam at the link end of the brace, the intersection of the brace and beam centerlines shall be at the end of the link or in the link.

10.20.15.6.3 Bracing connections

The required strength of the diagonal brace connections, at both ends of the brace, shall be at least equal to the required strength of the diagonal brace, as defined in Sec 10.20.15.6.1. The diagonal brace connections shall also satisfy the requirements of Sec 10.20.13.3.3.

No part of the diagonal brace connection at the link end of the brace shall extend over the link length. If the brace is designed to resist a portion of the link end moment, then the diagonal brace connection at the link end of the brace shall be designed as a fully-restrained moment connection.

10.20.15.7 Beam-to-column connections

If the EBF system factors in the Code require moment resisting connections away from the link, then the beam-to-column connections away from the link shall meet the requirements

for beam-to-column connections for OMF specified in Sections 10.20.11.2 and 10.20.11.5.

If EBF system factors in the Code do not require moment resisting connections away from the link, then the beam-to-column connections away from the link are permitted to be designed as pinned in the plane of the web.

10.20.15.8 Required strength of columns

In addition to the requirements in Sec 10.20.8.3, the required strength of columns shall be determined from load combinations as stipulated by the Code, except that the seismic load E shall be the forces generated by 1.1 times the expected nominal shear strength of all links above the level under consideration. The expected nominal shear strength of a link is $R_y V_n$, where V_n is as defined in Sec 10.20.15.2.2. Column members shall meet the requirements of Sec 10.20.8.2.2.

10.20.15.9 Protected zone

Links in EBFs are a protected zone, and shall satisfy the requirements of Sec 10.20.7.4. Welding on links is permitted for attachment of link stiffeners, as required in Sec 10.20.15.3.

10.20.15.10 Demand critical welds

Complete-joint-penetration groove welds attaching the link flanges and the link web to the column are demand critical welds, and shall satisfy the requirements of Sec 10.20.7.3.2.

10.20.16 Buckling-Restrained Braced Frames (BRBF)

10.20.16.1 Scope

Buckling-restrained braced frames (BRBF) are expected to withstand significant inelastic deformations when subjected to the forces resulting from the motions of the design earthquake. BRBF shall meet the requirements in this Section. Where the Code does not contain design coefficients for BRBF, provisions of Appendix P shall apply.

10.20.16.2 Bracing members

Bracing members shall be composed of a structural steel core and a system that restrains the steel core from buckling.

10.20.16.2.1 Steel core

The steel core shall be designed to resist the entire axial force in the brace.

The brace design axial strength, $\phi P_{y_{sc}}$ (LRFD), and the brace allowable axial strength, $P_{y_{sc}}/\Omega$ (ASD), in tension and compression, according to the limit state of yielding, shall be determined as follows:

$$P_{y_{sc}} = F_{y_{sc}} = A_{sc} \quad (\text{Sec 10.20.16.1})$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

Where,

$F_{y_{sc}}$ = specified minimum yield stress of the steel core, or actual yield stress of the steel core as determined from a coupon test, MPa

A = net area of steel core, mm²

Plates used in the steel core that are 50 mm thick or greater shall satisfy the minimum notch toughness requirements of Sec 10.20.6.3.

Splices in the steel core are not permitted.

10.20.16.2.2 Buckling-restraining system

The buckling-restraining system shall consist of the casing for the steel core. In stability calculations, beams, columns, and gussets connecting the core shall be considered parts of this system.

The buckling-restraining system shall limit local and overall buckling of the steel core for deformations corresponding to 2.0 times the design story drift. The buckling-restraining system shall not be permitted to buckle within deformations corresponding to 2.0 times the design story drift.

10.20.16.2.3 Testing

The design of braces shall be based upon results from qualifying cyclic tests in accordance with the procedures and acceptance criteria of Appendix R. Qualifying test results shall consist of at least two successful cyclic tests: one is required to be a test of a brace sub-assembly that includes brace connection rotational demands complying with Appendix R, Sec R.4 and the other shall be either a uniaxial or a sub-assembly test complying with

Appendix R, Sec R.5. Both test types are permitted to be based upon one of the following:

- (a) Tests reported in research or documented tests performed for other projects.
- (b) Tests that are conducted specifically for the project.

Interpolation or extrapolation of test results for different member sizes shall be justified by rational analysis that demonstrates stress distributions and magnitudes of internal strains consistent with or less severe than the tested assemblies and that considers the adverse effects of variations in material properties. Extrapolation of test results shall be based upon similar combinations of steel core and buckling-restraining system sizes. Tests shall be permitted to qualify a design when the provisions of Appendix R are met.

10.20.16.2.4 Adjusted brace strength

Where required by these Provisions, bracing connections and adjoining members shall be designed to resist forces calculated based on the adjusted brace strength.

The adjusted brace strength in compression shall be $\beta\omega R_y P_{y_{sc}}$. The adjusted brace strength in tension shall be $\omega R_y P_{y_{sc}}$.

Exception:

The factor R_y need not be applied if $P_{y_{sc}}$ is established using yield stress determined from a coupon test. The compression strength adjustment factor, β , shall be calculated as the ratio of the maximum compression force to the maximum tension force of the test specimen measured from the qualification tests specified in Appendix R, Sec R.6.3 for the range of deformations corresponding to 2.0 times the design story drift. The larger value of β from the two required brace qualification tests shall be used. In no case shall β be taken as less than 1.0. The strain hardening adjustment factor, ω , shall be calculated as the ratio of the maximum tension force measured from the qualification tests specified in Appendix R, Sec R.6.3 (for the range of deformations corresponding to 2.0 times the design story drift) to $F_{y_{sc}}$ of the test specimen. The larger value of ω from the two required qualification tests shall be used. Where the tested steel core material does not match that of the prototype, ω shall be based on coupon testing of the prototype material.

10.20.16.3 Bracing connections

10.20.16.3.1 Required strength

The required strength of bracing connections in tension and compression (including beam-to-column connections if part of the bracing system) shall be 1.1 times the adjusted brace strength in compression (LRFD) or $(1.1/1.5)$ times the adjusted brace strength in compression (ASD).

10.20.16.3.2 Gusset plates

The design of connections shall include considerations of local and overall buckling. Bracing consistent with that used in the tests upon which the design is based is required.

10.20.16.4 Special requirements related to bracing configuration

V-type and inverted-V-type braced frames shall meet the following requirements:

- (1) The required strength of beams intersected by braces, their connections, and supporting members shall be determined based on the load combinations of the Code assuming that the braces provide no support for dead and live loads. For load combinations that include earthquake effects, the vertical and horizontal earthquake effect, E , on the beam shall be determined from the adjusted brace strengths in tension and compression.
- (2) Beams shall be continuous between columns. Both flanges of beams shall be laterally braced. Lateral braces shall meet the provisions of Equations 6.10.291 and 6.10.292 of Sec 10.19, where, $M_r = M_u = R_y Z F_y$ (LRFD) or $M_r = M_a = R_y Z F_y / 1.5$ (ASD), as appropriate, of the beam and $C_d = 1.0$. As a minimum, one set of lateral braces is required at the point of intersection of the V-type (or inverted V-type) bracing, unless the beam has sufficient out-of-plane strength and stiffness to ensure stability between adjacent brace points.

For purposes of brace design and testing, the calculated maximum deformation of braces shall be increased by including the effect of the vertical deflection of the beam under the loading defined in Sec 10.20.16.4(1).

K-type braced frames are not permitted for BRBF.

10.20.16.5 Beams and columns

Beams and columns in BRBF shall meet the following requirements.

10.20.16.5.1 Width-thickness limitations

Beam and column members shall meet the requirements of Sec 10.20.8.2.2.

10.20.16.5.2 Required strength

The required strength of beams and columns in BRBF shall be determined from load combinations as stipulated in the Code. For load combinations that include earthquake effects, the earthquake effect, E, shall be determined from the adjusted brace strengths in tension and compression.

The required strength of beams and columns need not exceed the maximum force that can be developed by the system.

10.20.16.5.3 Splices

In addition to meeting the requirements in Sec 10.20.8.4, column splices in BRBF shall be designed to develop 50 percent of the lesser available flexural strength of the connected members, determined based on the limit state of yielding. The required shear strength shall be $\sum M_{pc}/H$ (LRFD) or $\sum M_{pc}/1.5H$ (ASD), as appropriate, where $\sum M_{pc}$ is the sum of the nominal plastic flexural strengths of the columns above and below the splice.

10.20.16.6 Protected zone

The protected zone shall include the steel core of bracing members and elements that connect the steel core to beams and columns, and shall satisfy the requirements of Sec 10.20.7.4.

10.20.17 Special Plate Shear Walls (SPSW)

10.20.17.1 Scope

Special plate shear walls (SPSW) are expected to withstand significant inelastic deformations in the webs when subjected to the forces resulting from the motions of the design earthquake. The horizontal boundary elements (HBEs) and vertical boundary elements (VBEs) adjacent to the webs shall be designed to remain essentially elastic under the maximum forces that can be generated by the fully yielded webs, except that plastic hinging at the ends of HBEs is permitted. SPSW shall meet the requirements of this Section. Where the Code does not contain design coefficients for SPSW, the provisions of Appendix P shall apply.

10.20.17.2 Webs

10.20.17.2.1 Shear strength

The panel design shear strength, ϕV_n (LRFD), and the allowable shear strength, V_n/Ω (ASD), according to the limit state of shear yielding, shall be determined as follows:

$$V_n = 0.42F_y t_w L_{cf} \sin 2\alpha \quad (6.10.307)$$

$$\phi = 0.90 \text{ (LRFD)} \quad \Omega = 1.67 \text{ (ASD)}$$

Where,

t_w = thickness of the web, mm.

L_{cf} = clear distance between VBE flanges, mm

α = angle of web yielding in radians, as measured relative to the vertical, and it is given by:

$$\tan^4 \alpha = \frac{1 + \frac{t_w L}{2A_c}}{1 + t_w h \left(\frac{1}{A_b} + \frac{h^3}{360I_c L} \right)} \quad (6.10.308)$$

h = distance between HBE centerlines, mm

A_b = cross-sectional area of a HBE, mm²

A_c = cross-sectional area of a VBE, mm²

I_c = moment of inertia of a VBE taken perpendicular to the direction of the web plate line, mm⁴

L = distance between VBE centerlines, mm

10.20.17.2.2 Panel aspect ratio

The ratio of panel length to height, L/h , shall be limited to $0.8 < L/h \leq 2.5$.

10.20.17.2.3 Openings in webs

Openings in webs shall be bounded on all sides by HBE and VBE extending the full width and height of the panel, respectively, unless otherwise justified by testing and analysis.

10.20.17.3 Connections of webs to boundary elements

The required strength of web connections to the surrounding HBE and VBE shall equal the expected yield strength, in tension, of the web calculated at an angle α , defined by Eq. 6.10.308.

10.20.17.4 Horizontal and vertical boundary elements

10.20.17.4.1 Required strength

In addition to the requirements of Sec 10.20.8.3, the required strength of VBE shall be based upon the forces corresponding to the expected yield strength, in tension, of the web calculated at an angle α .

The required strength of HBE shall be the greater of the forces corresponding to the expected yield strength, in tension, of the web calculated at an angle α or that determined from the load combinations in the Code assuming the web provides no support for gravity loads.

The beam-column moment ratio provisions in Sec 10.20.9.6 shall be met for all HBE/VBE intersections without consideration of the effects of the webs.

10.20.17.4.2 HBE-to-VBE connections

HBE-to-VBE connections shall satisfy the requirements of Sec 10.20.11.2. The required shear strength, V_u , of a HBE-to-VBE connection shall be determined in accordance with the provisions of Sec 10.20.11.2, except that the required shear strength shall not be less than the shear corresponding to moments at each end equal to $1.1R_yM_p$ (LRFD) or $(1.1/1.5)R_yM_p$ (ASD), as appropriate, together with the shear resulting from the expected yield strength in tension of the webs yielding at an angle α .

10.20.17.4.3 Width-thickness limitations

HBE and VBE members shall meet the requirements of Sec 10.20.8.2.2.

10.20.17.4.4 Lateral bracing

HBE shall be laterally braced at all intersections with VBE and at a spacing not to exceed $0.086r_y E/F_y$. Both flanges of HBE shall be braced either directly or indirectly. The required strength of lateral bracing shall be at least 2 percent of the HBE flange nominal strength, $F_y b_f t_f$. The required stiffness of all lateral bracing shall be determined in accordance with Eq. 6.10.292 of Sec 10.19. In these Equations, M_r shall be computed as $R_y Z F_y$ (LRFD) or M_r shall be computed as $R_y Z F_y / 1.5$ (ASD), as appropriate, and $C_d = 1.0$.

10.20.17.4.5 VBE splices

VBE splices shall comply with the requirements of Sec 10.20.8.4.

10.20.17.4.6 Panel zones

The VBE panel zone next to the top and base HBE of the SPSW shall comply with the requirements in Sec 10.20.9.3.

10.20.17.4.7 Stiffness of vertical boundary elements

The VBE shall have moments of inertia about an axis taken perpendicular to the plane of the web, I_c not less than $0.00307 t_w h^4 / L$.

10.20.18 Quality Assurance Plan**10.20.18.1 Scope**

When required by the Code or the Engineer, a quality assurance plan shall be provided. The quality assurance plan shall include the requirements of Appendix O.

10.21 List of Related Appendices

Appendix N	Prequalification of Beam-Column and Link-to-Column Connections
Appendix O	Quality Assurance Plan
Appendix P	Seismic Design Coefficients and Approximate Period Parameters
Appendix Q	Qualifying Cyclic Tests of Beam-to-Column and Link-to-Column Connections
Appendix R	Qualifying Cyclic Tests of Buckling-restrained Braces
Appendix S	Welding Provisions
Appendix T	Weld Metal/Welding Procedure Specification Notch Toughness Verification Test

PART VI**Chapter 11****TIMBERS STRUCTURES****11.1 scope**

11.1.1 This Section relates to the use of structural timber in structures or elements of structures connected together by fasteners/fastening techniques.

11.1.2 This shall not be interpreted to prevent the use of material or methods of design or construction not specifically mentioned herein; and the methods of design may be based on analytical and engineering principles, or reliable test data, or both, that demonstrate the safety and serviceability of the resulting structure. Nor is the classification of timber into strength groups to be interpreted as preventing the use of design data desired for a particular timber or grade of timber on the basis of reliable tests.

11.2 Terminology

11.2.1 This Section provides an alphabetical list of the terms used in this Chapter of the Code. In case of any conflict or contradiction between a definition given in this Section and that in Part 1, the meaning provided in this Section shall govern for interpretation of the provisions of this Chapter.

11.2.2 Structural Purpose Definitions

BUILT-UP-LAMINATED BEAM	A beam made by joining layers of timber together with mechanical fastenings, so that the grain of all layers is essentially parallel.
DURATION OF LOAD	Period during which a member or a complete structure is stressed as a consequence of the loads applied.
EDGE DISTANCE	The distance measured perpendicular to grain from the centre of the connector to the edge of the member.
END DISTANCE	The distance measured parallel to grain of the member from the centre of the connector to the closest end of timber.
FINGER JOINT	Joint produced by connecting timber members end-to-end by cutting profiles (tapered projections) in the form of V-shaped grooves to the ends of timber planks or scantlings to be joined, gluing the interfaces and then mating the two ends together under pressure.

GLUED-LAMINATED BEAM	A beam made by bonding layers of veneers or timber with an adhesive, so that grain of all laminations is essentially parallel.
INSIDE LOCATION	Position in buildings in which timber remains continuously dry or protected from weather.
LAMINATED VENEER LUMBER	A structural composite made by laminating veneers, 1.5 mm to 4.2 mm thick, with suitable adhesive and with the grain of veneers in successive layers aligned along the longitudinal (length) dimension of the composite.
LOADED EDGE DISTANCE	The distance measured from the centre to the edge towards which the load induced by the connector acts, and the unloaded edge distance is the one opposite to the loaded edge.
LOCATION	A term generally referred to as exact place where a timber is used in building.
OUTSIDE LOCATION	Position in buildings in which timbers are occasionally subjected to wetting and drying as in the case of open sheds and outdoor exposed structures.
PERMANENT STRUCTURE	Structural units in timber which are constructed for a long duration and wherein adequate protection and design measures have initially been incorporated to render the structure serviceable for the required life.
PERMISSIBLE STRESS	Stress obtained by applying factor of safety to the ultimate stress.
SPACED COLUMN	Two column sections adequately connected together by glue, bolts, screws or otherwise.
STRUCTURAL DIAPHRAGM	A structural element of large extent placed in a building as a wall, or roof, and made use of to resist horizontal forces such as wind or earthquakes-acting parallel to its own plane.
STRUCTURAL SANDWICH	A layered construction comprising a combination or relatively high-strength facing material intimately bonded to and acting integrally with a low density core material.
STRUCTURAL ELEMENT	The component timber members and joints which make up a resulting structural assembly.
STRUCTURAL GRADES	Grades defining the maximum size of strength reducing natural characteristics (knots, sloping grain, etc.) deemed permissible in any piece of structural timber within designated structural grade classification.
STRUCTURAL TIMBER	Timber in which strength is related to the anticipated in-service use as a controlling factor in grading and selection and/or stiffness.

TEMPORARY STRUCTURE	Structures which are erected for a short period, such as hutments at project sites, for rehabilitation, temporary defence constructions, exhibition structures, etc.
TERMITE	An insect of the order Isopteran which may burrow in the wood or wood products of a building for food or shelter.
ULTIMATE STRESS	The stress which is determined on small clear specimen of timber, in accordance with good practice; and does not take into account the effect of naturally occurring characteristics and other factors. Also known as Fundamental Stress
WET LOCATION	Position in buildings in which timbers are almost continuously damp or wet in contact with the earth or water, such as piles and timber foundations.

11.2.3 Definitions Related to Defects in Timber

CHECK	A separation of fibres extending along the grain which is confined to one face of a piece of wood.
COMPRESSION WOOD	Abnormal wood which is formed on the lower sides of branches and inclined stems of coniferous trees. It is darker and harder than normal wood but relatively low in strength for its weight. It can be usually identified by wide eccentric growth rings with abnormally high proportion of growth latewood.
DEAD KNOT	A knot in which the layers of annual growth are not completely inter-grown with those of the adjacent wood. It is surrounded by pitch or bark. The encasement may be partial or complete.
DECAYED KNOT	A knot softer than the surrounding wood and containing decay.
DIAMETER OF KNOT	The maximum distance between the two points farthest apart on the periphery of a round knot, on the face on which it becomes visible. In the case of a spike or a splay knot, the maximum width of the knot visible on the face on which it appears shall be taken as its diameter.
DISCOLORATION	A change from the normal colour of the wood which does not impair the strength of the wood.
KNOT	A branch base or limb embedded in the tree or timber by natural growth.
KNOT HOLE	A hole left as a result of the removal of a knot.
LIVE KNOT	A knot free from decay and other defects, in which the fibres are firmly intergrown with those of the surrounding wood. Syn. 'Intergrown knot'; <i>cf.</i> 'Dead Knot'.

LOOSE GRAIN	A defect on a 6 flat sawn surface caused by the separation or raising of wood fibres along the growth rings; also known as Loosened Grain. <i>cf</i> 'Raised Grain'.
LOOSE KNOT	A knot that is not held firmly in place by growth or position, and that cannot be relied upon to remain in place; <i>cf</i> 'Tight Knot'.
MOULD	A soft vegetative growth that forms on wood in damp, stagnant atmosphere. It is the least harmful type of fungus, usually confined to the surface of the wood.
PITCH POCKET	Accumulation of resin between growth rings of coniferous wood as seen on the cross section
ROT	Disintegration of wood tissue caused by fungi (wood destroying) or other microorganisms. Also known as Decay.
SAP STAIN	Discoloration of the sapwood mainly due to fungi.
SAPWOOD	The outer layer of log, which in the growing tree contain living cells and food material. The sapwood is usually lighter in colour and is readily attacked by insects and fungi.
SHAKE	A partial or complete separation between adjoining layers of tissues as seen in end surfaces.
SLOPE OF GRAIN	The inclination of the fibres to the longitudinal axis of the member.
SOUND KNOT	A tight knot free from decay, which is solid across its face, and at least as hard as the surrounding wood.
SPLIT	A crack extending from one face of a piece of wood to another and running along the grain of the piece.
TIGHT KNOT	A knot so held by growth or position as to remain firm in position in the piece of wood; <i>cf</i> 'Loose Knot'.
WANE	The original rounded surface of a tree remaining on a piece of converted timber.
WARP	A deviation in sawn timber from a true plane surface or distortion due to stresses causing departure from a true plane.
WORM HOLES	Cavities caused by worms.

11.3 Symbols

11.3.1 For the purpose of this Section, the following symbols shall have the meaning indicated against each:

B = Width of the beam, mm

C = Concentrated load, N

D = Depth of beam, mm

D_1 = Depth of beam at the notch, mm

D_2 = Depth of notch, mm

E = Modulus of elasticity in bending, N/mm²

F = Load acting on a bolt at an angle to grain, N

H = Horizontal shear stress, N/mm²

I = Moment of inertia of a section, mm⁴

K = Coefficient in deflection depending upon type and criticality of loading on beam

K_1 = Modification factor for change in slope of grain

K_2 = Modification factor for change in duration of loadings

$\left. \begin{array}{l} K_3, \\ K_4, \\ K_5, \\ \text{and} \\ K_6 \end{array} \right\}$ = Form factors

K_7 = Modification factor for bearing stress

K_8 = Constant equal to $0.584 \sqrt{\frac{E}{f_{cp}}}$

K_9	=	Constant equal to $\frac{\pi}{2} \sqrt{\frac{UE}{5qf_{cp}}}$
K_{10}	=	Constant equal to $0.584 \sqrt{\frac{2.5E}{f_{cp}}}$
L	=	Span of a beam or truss, mm
M	=	Maximum bending moment in beam N/mm ²
N	=	Total number of bolts in the joint
P	=	Load on bolt parallel to grain, N
P_1	=	Ratio of the thickness of the compression flange to the depth of the beam
Q	=	Moment of area about neutral axis, mm ³
R	=	Load on bolt perpendicular (normal) to grain, N
S	=	Unsupported overall length of column, mm
U	=	Constant for a particular thickness of the plank
V	=	Vertical end reaction or shear at a section, N
W	=	Total uniform load, N
Z	=	Section modulus of beam, mm ³
A	=	Projected area of bolt in main member ($t' \times d_3$), mm ²
D	=	Dimension of least side of column, mm
d_1	=	Least overall width of box column, mm
d_2	=	Least overall dimension of core in box column, mm
d_3	=	Diameter of bolt, mm
d_f	=	Bolt-diameter factor

e	=	Length of the notch measured along the beam span from the inner edge of the support to the farthest edge of the notch, mm
f_{ab}	=	Calculated bending stress in extreme fibre, N/mm ²
f_{ac}	=	Calculated average axial compressive stress, N/mm ²
f_{at}	=	Calculated axial tensile stress, N/mm ²
f_b	=	Permissible bending stress on the extreme fibre, N/mm ²
f_c	=	Permissible stress in axial compression, N/mm ²
f_{cn}	=	Permissible stress in compression normal (perpendicular) to grain, N/mm ²
f_{cp}	=	Permissible stress in compression parallel to grain, N/mm ²
f_c^θ	=	Permissible compressive stress in the direction of the line of action of the load, N/mm ²
f_t	=	Permissible stress in tension parallel to grain, N/mm ²
n	=	Shank diameter of the nail, mm
q	=	Constant for particular thickness of plank
q_1	=	Ratio of the total thickness of web or webs to the overall width of the beam
t	=	Nominal thickness of planks used in forming box type column, mm
t'	=	Thickness of main member, mm
x	=	Distance from reaction to load, mm
γ	=	A factor determining the value of form factor K_4
δ	=	Deflection at middle of beam, mm
θ	=	Angle of load to grain direction
λ_1	=	Percentage factor for t'/d_3 ratio, parallel to grain
λ_2	=	Percentage factor for t'/d_3 ratio, perpendicular to grain

11.4 Materials

11.4.1 Species of Timber

For construction purposes, species of timber are classified in three groups on the basis of their strength properties, namely, modulus of elasticity (E) and extreme fibre stress in bending and tension (f_b). The species of timber for structural purposes and their properties are given in Table 6.11.1.

Table 6.11.1: Safe Permissible Stresses for the Species of Timber

Species		Average Density at 12% Moisture Content, Kg/m ³	Modulus of Elasticity × 10 ³ N/mm ²	Permissible Stress in N/mm ² for Grade I											Preservative Characters		
Botanical Name	Trade Name			Bending and Tension Along Grains, Extreme Fibre Stress			Shear all Location		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class	Treatability Grade	Refracteriness to All
				Inside Location	outside Location	wet Location	Horizontal	Along Grain	Inside Location	outside Location	wet Location	Inside Location	outside Location	wet Location			
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
<i>Acacia nilotica</i>	Babla	797	-	-	12.9	10.3	1.4	2.1	8.9	7.9	6.4	5.2	4.0	3.3	I	b	B
<i>Aglaia odulis</i>	Aglaia	815	12.56	18.2	15.2	12.1	1.4	2.0	10.1	8.9	7.3	4.4	3.4	2.8	-	-	A
<i>Ailantahus grandis</i>	Gokul	404	7.94	8.3	6.9	5.5	0.6	0.8	5.3	4.7	3.9	1.1	0.9	0.7	III	-	C
<i>Altingia excelsa</i>	Jutili	795	11.37	17.1	14.3	11.4	1.2	1.8	11.0	9.8	8.0	6.8	5.3	4.4	II	e	A
<i>Amoora rehituka</i>	Pitraj	668	8.98	12.3	10.2	8.2	1.1	1.5	8.0	7.1	5.8	4.0	3.1	2.6	I	-	B
<i>Amoora wallichii</i>	Lali	583	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
<i>Amoora spp.</i>	Arnari	625	1.05	13.4	1.1	9.2	0.9	1.3	8.4	7.4	6.0	3.7	2.9	2.4	II	d	B
<i>Anisoplera glabra</i>	Boilam	573	-	-	-	-	-	-	-	-	-	-	-	-	III	b	-
<i>Aphenamixis polystachya</i>	Pitraj	583	-	-	-	-	-	-	-	-	-	-	-	-	III	e	B
<i>Artocarpus chaplasha</i>	Chapalish	515	9.11	13.2	11.0	8.8	0.9	1.2	8.5	7.5	6.2	3.6	2.8	2.3	III	d	B
<i>Artocarpus integrifolia</i>	Kanthali	537	-	-	-	-	-	-	-	-	-	-	-	-	III	c	B
<i>Azadirachta indica</i>	Neem	836	8.52	14.6	12.1	9.7	1.3	1.8	10.0	8.9	7.3	5.0	3.9	3.2	-	-	-
<i>Betula Inoides</i>	Birch	625	9.23	9.6	8.0	6.4	0.8	1.1	5.7	5.0	4.1	2.2	1.7	1.4	-	-	B
<i>Bischofia javanica</i>	Bhadi	769	8.84	9.6	8.2	6.5	0.8	1.1	5.9	5.3	4.3	3.6	2.8	2.3	III	-	A
<i>Bruguiera conjugata</i>	Kankra	879	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A
<i>Bucklandia populnea</i>	Plpli	672	9.89	12.8	10.7	8.6	1.1	1.5	7.9	7.0	5.7	3.5	2.7	2.2	III	e	C
<i>Canarium strictum</i>	White dhup	569	10.54	10.1	8.4	6.7	0.7	1.1	6.2	5.5	4.5	2.1	1.6	1.3	III	-	C
<i>Cassia fistula</i>	Sonalu	865	11.80	19.2	16.0	12.8	1.4	2.0	12.3	10.9	8.9	7.2	5.6	4.6	I	-	A
<i>Castanopsis hystrix</i>	Chestanut	624	9.85	10.6	8.8	7.0	0.8	1.2	6.4	5.7	4.6	2.7	2.1	1.7	II	b	B
<i>Carallia lucida</i>	Maniawaga	748	12.60	18.4	15.3	12.3	1.2	1.7	11.4	10.1	8.3	5.9	4.6	3.8	-	-	-

Species		Average Density at 12% Moisture Content, Kg/m ³	Modulus of Elasticity × 10 ³ N/mm ²	Permissible Stress in N/mm ² for Grade I											Preservative Characters		
Botanical Name	Trade Name			Bending and Tension Along Grains, Extreme Fibre Stress			Shear all Location		Compression Parallel to Grain			Compression Perpendicular to Grain			Durability Class	Treatability Grade	Refracteriness to All
				Inside Location	outside Location	wet Location	Horizontal	Along Grain	Inside Location	outside Location	wet Location	Inside Location	outside Location	wet Location			
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
<i>Cassia siamea</i>	Minjiri	695	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
<i>Chukrasia tabularis</i>	Chickrassy	666	8.35	11.8	9.8	7.9	1.1	1.5	7.1	6.3	5.2	3.9	3.1	2.5	II	c	B
<i>Dalbergia sissoo</i>	Sissoo	808	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B
<i>Dillemia indica</i>	Dillenia	617	8.61	12.1	10.0	8.0	0.8	1.2	7.3	6.5	5.3	2.7	2.1	1.7	III	a	B
<i>Dillenia pentagyne</i>	Dillenia	622	7.56	11.8	9.9	7.9	0.9	1.3	7.1	6.3	5.2	3.5	2.7	2.2	III	d	B
<i>Dipterocarpus alatus</i>	Garjan	721	-	-	-	-	-	-	-	-	-	-	-	-	III	a	B
<i>Dipterocarpus macrocarpus</i>	Hollong	726	13.34	14.5	12.0	9.6	0.8	1.1	8.8	7.9	6.4	3.5	2.7	2.2	III	a	B
<i>Duabanga sonneratioides</i>	Banderhol	485	8.38	9.8	8.2	6.5	0.6	0.9	6.4	5.7	4.7	1.8	1.4	1.1	III	c	C
<i>Garuga pinnata</i>	Garuga	571	7.58	11.7	9.7	7.8	1.0	1.5	7.2	6.4	5.3	3.4	2.6	2.1	I	e	B
<i>Geriops roxbarghiana</i>	Goran	869	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
<i>gGmeline arborea</i>	Garnar	501	7.02	9.8	8.2	6.6	0.8	1.4	5.7	5.0	4.1	4.2	3.2	2.7	I	e	B
<i>Grewia veslita</i>	Dhaman	758	12.00	15.4	12.6	10.3	1.4	2.0	9.1	8.1	6.6	4.1	3.2	2.6	III	d	B
<i>Heritiera spp.</i>	Sundri	872	13.37	17.9	14.9	11.9	1.3	1.8	11.0	9.8	8.0	6.5	5.0	4.1	I	-	A
<i>Hopea odorata</i>	Telsur	711	-	-	-	-	-	-	-	-	-	-	-	-	III	a	B
<i>Kayea floribund</i>	Karal	813	10.88	16.8	14.0	1.1	1.1	1.6	10.1	9.0	7.3	4.4	3.4	2.8	III	-	-
<i>Lagerstrocmia spp.</i>	Jarul	654	-	-	-	-	-	-	-	-	-	-	-	-	III	e	B
<i>Machilus macrantha</i>	Machilus	692	10.00	12.4	10.3	8.3	1.0	1.5	8.2	7.3	6.0	3.5	2.7	2.2	III	e	B/C
<i>Manglietia insignia</i>		449	10.37	10.9	9.1	7.3	0.7	1.4	8.0	7.1	5.8	3.4	2.6	2.1	-	-	-
<i>Manilota polyandra</i>	Ping	903	13.20	19.1	15.9	12.7	1.3	1.8	1.2	10.0	8.5	5.7	4.4	3.6	III	b	A
<i>Mesua assamica</i>	Keyea	842	12.83	17.4	14.5	11.6	1.0	1.4	11.7	10.4	8.5	5.3	4.1	3.3	II	e	-
<i>Mesua ferrea</i>	Mesua	965	16.30	23.3	19.4	15.5	1.2	1.8	15.5	13.8	11.3	5.9	4.6	3.7	I	-	A
<i>Michelia champaca</i>	Champa	644	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B
<i>Michelia montana</i>	Champ	512	8.25	10.9	9.1	7.3	0.7	1.0	6.6	5.9	4.8	2.8	2.2	1.8	I	-	B
<i>Michelia excelsa</i>	Champ	513	10.12	9.8	8.2	6.5	0.7	1.0	6.1	5.5	4.5	1.6	1.3	1.0	II	e	B
<i>Mitragyna pervifolia</i>	Dakroom	651	7.82	12.6	10.5	8.4	1.0	1.5	7.9	7.0	5.7	3.7	2.9	2.4	III	b	B
<i>Palaquium polyanthum</i>	Tali	734	11.24	14.9	12.4	10.0	1.1	1.6	9.9	8.8	7.2	4.7	3.7	3.0	-	-	B
<i>Phoebe hainesiana</i>	Bonsum	566	9.5	13.2	11.0	8.8	0.8	1.2	8.8	7.8	6.4	2.8	2.1	1.8	II	c	B

Species		Average Density at 12% Moisture Content, Kg/m ³	Modulus of Elasticity × 10 ³ N/mm ²	Permissible Stress in N/mm ² for Grade I											Preservative Characters		
Botanical Name	Trade Name			Bending and Tension Along Grains, Extreme Fibre Stress			Shear all Location		Compression Parallel to Grain		Compression Perpendicular to Grain			Durability Class	Treatability Grade	Refracteriness to All	
				Inside Location	outside Location	wet Location	Horizontal	Along Grain	Inside Location	outside Location	wet Location	Inside Location	outside Location				wet Location
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
<i>Phoebe goalperansis</i>	Bonsum	511	7.65	9.7	8.1	6.5	0.7	1.0	6.6	5.9	4.8	2.2	1.7	1.4	II	c	B
<i>Plerygota alata</i>	Narikel	593	10.95	13.4	11.8	8.9	0.8	1.2	8.2	7.3	6.0	2.7	2.1	1.7	III	-	C
<i>Prunus napeulensis</i>	Arupati	548	9.41	4.4	8.7	69.6	0.9	1.2	6.7	6.0	4.9	2.4	1.9	1.6	-	-	-
<i>Pterespermum acerifolium</i>	Hattipaila	607	9.55	13.5	11.3	9.0	0.9	1.2	8.7	7.7	6.3	3.2	2.5	2.0	III	C	B
<i>Quercus lineate</i>	Oak	874	12.63	15.2	12.7	10.1	1.2	1.7	9.6	8.6	7.0	5.3	4.1	3.4	II	c	A
<i>Quercus lamellosa</i>	Oak	87	12.44	14.5	12.1	9.7	1.2	1.7	8.7	7.8	6.4	3.8	2.9	2.4	II	c	A
<i>Schima wallichii</i>	Chilauni	693	9.57	11.1	9.3	7.4	0.9	1.3	6.6	5.9	4.8	2.3	1.8	1.4	III	d	B
<i>Seritiera fomes</i>	Sundri	1073	-	-	-	-	-	-	-	-	-	-	-	-	III	b	B
<i>Shotea assamica</i>	Makai	548	9.27	11.1	9.2	7.4	0.9	1.3	7.1	6.3	5.2	2.9	2.2	1.8	III	c	B
<i>Shorea robusta</i>	Sal	889	-	-	-	-	-	-	-	-	-	-	-	-	III	e	B
<i>Sonneralia apetale</i>	Keora	617	8.63	12.8	10.7	8.5	0.9	1.3	7.4	6.6	5.4	4.8	3.7	3.0	II	-	B
<i>Swintonia floribunda</i>	Civit	665	-	-	-	-	-	-	-	-	-	-	-	-	III	a	C
<i>Syzygium cumini</i>	Jamun	841	10.55	14.8	12.4	9.9	1.1	1.6	9.0	8.0	6.5	6.9	5.4	4.4	II	e	A
<i>Syzygium spp.</i>	Jam	823	-	-	-	-	-	-	-	-	-	-	-	-	III	e	A
<i>Taxus buccata</i>	Yew	705	7.79	14.3	11.9	9.5	1.2	1.7	8.7	7.8	6.4	4.7	3.7	3.0	-	-	-
<i>Tectona grandis</i>	Teak	660	9.97	15.5	12.9	10.3	1.2	1.6	9.4	8.3	6.8	4.5	3.5	2.8	I	e	B
<i>Toena ciliata</i>	Toon	487	6.40	8.7	7.3	5.8	0.7	1.0	5.4	4.8	3.9	2.4	1.8	1.5	II	c	B
<i>Terminalia citrma</i>		755	11.89	17.1	14.3	11.4	1.1	1.6	10.8	9.6	7.9	5.0	3.9	3.2	-	-	-
<i>Terminalia myriocarpa</i>	Hollock	615	9.62	11.9	9.9	8.0	0.9	1.2	7.6	6.7	5.5	2.9	2.2	1.8	III	a	B
<i>Xylia dolabriformis</i>	Lohakat	1007	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
<i>Xylocarpus rolloensis</i>	Passur	757	-	-	-	-	-	-	-	-	-	-	-	-	-	-	B
<i>Zanthoxylum budranga</i>	Mulliam	587	10.65	14.7	12.2	9.8	0.9	1.2	9.5	8.4	6.9	3.4	2.6	2.1	I	e	B

† Classification for preservation based on durability tests, etc.

Class I – Average life more than 120 months;

Class II – Average life 60 months or above but less than 120 months; and

Class III – Average life less than 60 months.

‡ Treatability Grades

a – Heartwood easily treatable;

b – Heartwood treatable, but complete penetration not always obtained; in case where least dimension is more than 60 mm;

c – Heartwood only partially treatable;

d – Heartwood refractory to treatment; and

e – Heartwood very refractory to treatment, penetration of preservative being practically nil even from the ends.

Data based on strength properties at three years of age of tree.

§ Classifications based on seasoning behavior of timber and refractoriness w.r.t. cracking, splitting and drying rate.

A – Highly refractory (slow and difficulty to season free from surface and end cracking);

B – Moderately refractory (may be seasoned free from surface and end cracking within reasonably short periods, given a little protection against rapid drying conditions); and

C – Non-refractory (may be rapidly seasoned free from surface and end-cracking even in the open air and sun. If not rapidly dried, they develop blue stain and mould on the surface.

Grouping of Timber

Characteristics of the three groups of species of timber are given below:

Group A: E above $12.6 \times 10^3 \text{ N/mm}^2$ and f_b above 18.0 N/mm^2 .

Group B: E above $9.8 \times 10^3 \text{ N/mm}^2$ and upto $12.6 \times 10^3 \text{ N/mm}^2$ and f_b above 12.0 N/mm^2 and upto 18.0 N/mm^2 .

Group C: E above $5.6 \times 10^3 \text{ N/mm}^2$ and upto $9.8 \times 10^3 \text{ N/mm}^2$ and f_b above 8.5 N/mm^2 and upto 12.0 N/mm^2 .

Modulus of elasticity given above is applicable for all locations and extreme fibre stress in bending is for inside location.

11.4.2 The general characteristics like durability and treat ability of the species are also given in Table 6.11.1. Species of timber other than those recommended in Table 6.11.1 may be used, provided the basic strength properties are determined and found in accordance with Sec 11.5.1.

11.4.3 The permissible lateral strength (in double shear) of mild steel wire shall be as given in Table 6.11.2 and Table 6.11.3 for different species of timber.

11.4.4 Moisture Content in Timber The permissible moisture content of timber for various positions in buildings shall be as given in Table 6.11.4.

Tolerances

Permissible tolerances in measurements of cut sizes of structural timber shall be as follows:

- (a) For width and thickness:
 - (i) Upto and including 100 mm: +3 to 0 mm
 - (ii) Above 100 mm: +6 to -3 mm
- (b) For length: +10 to 0 mm

11.4.5 Grading of Structural Timber

Cut sizes of structural timber shall be graded, after seasoning, into three grades namely (a) Select Grade, (b) Grade I and (c) Grade II, based on permissible defects given in Table 6.11.8.

11.4.6 Sawn Timber

11.4.6.1 Sizes: Preferred cut sizes of timber for use in structural components shall be as given in Tables 6.11.5 to 6.11.7.

11.4.6.2 The prohibited defects given in Sec 11.4.6.2.1 and permissible defects given in Sec 11.4.6.2.2 shall apply to structural timber.

11.4.6.2.1 Prohibited defects

Loose grains, splits, compression wood in coniferous species, heartwood rot, sap rot, crookedness, worm holes made by powder post beetles and pitch pockets shall not be permitted in all the three grades.

11.4.6.2.2 Defects to the extent specified in Table 6.11.8 shall be permissible.

Wanes are permitted provided they are not combined with knots and the reduction in strength on account of the wanes is not more than the reduction with maximum allowable knots.

11.4.6.3 Location of defects

The influence of defects in timber is different for different locations in the structural element. Therefore, these should be placed during construction in such a way so that they do not have any adverse effect on the members.

Table 6.11.2: Permissible Lateral Strengths (in Double Shear) of Nails 3.55 mm Diameter, 80 mm Long

Species of Wood		For Permanent Construction Strength per Nail		For Temporary Structures Strength per Nail (for Both Lengthening Joints and Node Joints) Nx10 ²
Botanical Name	Trade name	Lengthening Joints, Nx10 ²	Node Joints Nx10 ²	
<i>Acacia nilotica</i>	Babla	15	11	34
<i>Aphenamixis polystachya</i>	Pitraj	19	9	19
<i>Canarium strictum</i>	White dhup	9	8	10.5
<i>Castanopsis hystrix</i>	Chestanut	18	10.5	23.5
<i>Chukrasia tabularis</i>	Chickrassy	24	8	27
<i>Dillenia pentagyne</i>	Dillenia	16.5	12	16
<i>Dipterocarpus rnacropus</i>	Hollong	17	7	20
<i>Grewia veslita</i>	Dhaman	13	5	24
<i>Hopea odorata</i>	Telsur	31.5	13	28.5
<i>Lagerstrocmia spp.</i>	Jarul	24.5	21.5	22.5
<i>Maniltoa polyandra</i>	Ping	26	23.5	32
<i>Mesua ferrea</i>	Mesua	26	8	41
<i>Michelia excelsa</i>	Champ	13	9	20
<i>Phoebe hainesiana</i>	Bonsum	12	6	13
<i>Shorea robusta</i>	Sal	23	15.5	19.5
<i>Syzygium spp.</i>	Jam	15	12	25
<i>Tectona grandis</i>	Teak	14	8	13
<i>Terminalia myriocarpa</i>	Hollock	13	10	19
<i>Toona ciliata</i>	Toon	16	9	21

Note: Nails of 3.55 mm diameter are most commonly used. The above values can also be used for 4 mm diameter 100 mm long nails.

Table 6.11.3: Permissible Lateral Strengths (in Double Shear) of Nails 5.00 mm Diameter, 125 mm and 150 mm Long

Species of Wood		For Permanent Construction Strength per Nail		For Temporary Structures Strength per Nail (for Both Lengthening Joints and Node Joints)
Botanical Name	Trade name	Lengthening Joints, Nx10 ²	Node Joints Nx10 ²	Nx10 ²
<i>Acacia nilotica</i>	Babla	27	13.5	53
<i>Dalbergia sissoo</i>	Sissoo	17	15	43
<i>Mesua ferrea</i>	Mesua	24	15.5	57.5
<i>Michelia excelsa</i>	Champ	26	12.5	39
<i>Phoebe hainesiana</i>	Bonsum	20	7.5	30
<i>Shorea robusta</i>	Sal	19.5	17	37
<i>Syzygium spp.</i>	Jam	18	14.5	38.5
<i>Tectona grandis</i>	Teak	28	13	30
<i>Terminalia myriocarpa</i>	Hollock	27.5	9	41

Table 6.11.4: Permissible Percentage Moisture Content Values

Usage	Permissible Moisture Content
Structural elements	17%
Doors and windows:	
50 mm and above in thickness	14%
Thinner than 50 mm	12%
Flooring strips for general purposes	10%

Table 6.11.5: Preferred Cut Sizes of Structural Timbers for Roof Trusses (Span: 3 m to 20 m)

Thickness (mm)			Width (mm)					
20	40	50	60	80	100	-	-	-
25	40	50	60	80	100	120	160	180
30	40	50	60	80	100	120	160	180
35	-	-	60	80	100	120	160	180
40	-	-	60	80	100	120	160	180

Thickness (mm)			Width (mm)					
50	-	-	60	80	100	120	160	180
60	-	-	-	80	100	120	160	180
80	-	-	-	-	100	120	160	180

Notes:

1. For truss spans marginally above 20 m, preferred cut sizes of structural timber may be allowed.

2 Preferred lengths of timber 1, 1.5, 2, 2.5 and 3 m.

Table 6.11.6: Preferred Cut Sizes of Structural Timber for Roof Purlins, Rafters, Floor Beams and other Elements

Thickness (mm)			Width (mm)					
50	80	100	120	140	-	-	-	-
60	80	100	120	140	160	-	-	-
80	-	100	120	140	160	-	-	-
100	-	-	-	140	160	180	200	-

Note: Preferred lengths of timber: 1.5, 2, 2.5 and 3 m.

Table 6.11.7: Preferred Cut Sizes of Structural Timbers for Partition Framing and Covering, and for Centering

Thickness (mm)			Width (mm)						
10	40	50	60	80	-	-	-	-	-
15	40	50	60	80	100	-	-	-	-
20	40	50	60	80	100	120	160	200	-
25	40	50	60	80	100	120	160	200	240
30	40	50	60	80	100	120	160	200	240
40	40	-	60	80	100	120	160	200	240
50	-	50	-	80	100	120	160	200	240
60	-	-	60	80	100	120	160	200	240
80	-	-	-	80	100	120	160	200	240

11.4.7 Suitability

11.4.7.1 Suitability in respect of durability and treatability for permanent structures

There are two choices as given in Sections 11.4.7.1.1 and 11.4.7.1.2.

11.4.7.1.1 First choice

The species shall be any one of the following:

- (a) Untreated heartwood of high durability. Heartwood if containing more than 15 percent sap wood, may need chemical treatment for protection;
- (b) Treated heartwood of moderate and low durability and class 'a' and class 'b' treatability;
- (c) Heartwood of moderate durability and class 'c' treatability after pressure impregnation, and
- (d) Sapwood of all classes of durability after thorough treatment with preservative.

11.4.7.1.2 Second choice

The species of timber shall be heartwood of moderate durability and class 'd' treatability.

11.4.7.2 Choice of load bearing temporary structures or semi-structural components at construction site

- (a) Heartwood of low durability and class 'e' treatability; or
- (b) The species whose durability and/or treatability are yet to be established, as listed in Table 6.11.1.

Table 6.11.8: Permissible Defects for Cut Sizes of Timber for Structural Use

Sl. No.	Defects	Select Grade	Grade I	Grade II
1	2	3	4	5
(i)	WANE	Shall be permissible at its deepest portion up to a limit of 1/8 of the width of the surface on which it occurs	Shall be permissible at its deepest portion up to a limit of 1/6 of the width of surface on which it occurs	Shall be permissible at its deepest portion up to a limit of 1/4 of the width of the surface on which it occurs
(ii)	WORM HOLES	Other than those due to powder post beetles are permissible	Other than those due to powder post beetles are permissible	Other than those due to powder post beetles are permissible
(iii)	SLOPE OF GRAIN	Shall not be more than 1 in 20	Shall not be more than 1 in 15	Shall not be more than 1 in 12
(iv)	LIVE KNOTS			

	Max. width of Wide Faces of Cut Sizes of Timber	Permissible Maximum Size of Live Knot on		Permissible Maximum Size of Live Knot on		Permissible Maximum Size of Live Knot on	
		Narrow faces and 1/4 of the width face close to edges of cut size of timber	Remaining central half of the width of the wide faces	Narrow faces and 1/4 of the width face close to edges of cut size of timber	Remaining central half of the width of the wide faces	Narrow faces and 1/4 of the width face close to edges of cut size of timber	Remaining central half of the width of the wide faces
	1	2	3	4	5	6	7
	75	10	10	19	19	29	30
	100	13	13	25	25	38	39
	150	19	19	38	38	57	57
	200	22	25	44	50	66	75
	250	25	29	50	57	66	87
	300	27	38	54	75	81	114
	350	29	41	57	81	87	123
	400	32	44	63	87	96	132
	450	33	47	66	93	99	141
	500	35	50	69	100	105	150
	550	36	52	72	103	108	156
	600	38	53	75	106	114	159
(v)	CHECKS AND SHAKES						
	Max. width of Face of Timber	Max. Permissible Depth		Max. Permissible Depth		Max. Permissible Depth	
	1	2		3		4	
	75	12		25		36	
	100	18		35		54	
	150	25		50		75	
	200	33		65		99	
	250	40		81		120	
	300	50		100		150	
	350	57		115		171	
	400	66		131		198	
	450	76		150		225	
	500	83		165		249	
	550	90		181		270	
	600	100		200		300	

11.4.8 Fastenings

All structural members shall be framed, anchored, tied and braced to develop the strength and rigidity necessary for the purposes for which they are used.

Allowable stresses or loads on joints and fasteners shall be determined in accordance with recognized principles. Common mechanical fastenings are of bar type such as nails and spikes, wood screws and bolts, and timber connectors including metallic rings or wooden disc-dowels. Chemical fastenings include synthetic adhesives for structural applications.

11.5 Permissible Stresses

11.5.1 The permissible stresses for Groups A, B and C for different locations applicable to Grade I structural timber shall be as given in Table 6.11.9 provided that the following conditions are satisfied:

- (a) The timbers should be of high or moderate durability and be given the suitable treatment where necessary.
- (b) Timber of low durability shall be used after proper preservative treatment and
- (c) The loads should be continuous and permanent and not of impact type.

11.5.2 The permissible stresses (excepting E) given in Table 6.11.9 shall be multiplied by the following factors to obtain the permissible stresses for other grades provided that the conditions laid down in Sec 11.5.1 are satisfied:

- | | |
|-----------------------------|------|
| (a) For Select Grade Timber | 1.16 |
| (b) For Grade II Timber | 0.84 |

When low durability timbers are to be used [see Sec 11.5.1(b)] on outside locations, the permissible stresses for all grades of timber, arrived at by Sections 11.5.1 and 11.5.2 shall be multiplied by 0.80.

11.5.3 Modification Factors for Permissible Stresses

11.5.3.1 Change in slope of grain

When the timber has not been graded and has major defects like slope of grain, knots and checks or shakes but not beyond permissible value, the permissible stress given in Table 6.11.1 shall be multiplied by modification factor K_1 for different slopes of grain as given in Table 6.11.10.

11.5.3.2 Duration of load

For different durations of design load, the permissible stresses given in Table 6.11.1 shall be multiplied by the modification factor K_2 given in Table 6.11.11.

11.5.3.2.1 The factor K_2 is applicable to modulus of elasticity when used to design timber columns, otherwise they do not apply thereto.

Table 6.11.9: Minimum Permissible Stress Limits (N/mm²) in Three Groups of Structural Timbers (for Grade I Material)

Sl No.	Strength Character	Location of Use	Group A	Group B	Group C
(i)	Bending and tension along grain	Inside ⁽¹⁾	18.0	12.0	8.5
(ii)	Shear ⁽²⁾	All locations	1.05	0.64	0.49
	Horizontal				
	Along grain	All locations	15	0.91	0.70
(iii)	Compression perpendicular to grain	Inside ⁽¹⁾	11.7	7.8	4.9
(iv)	Compression perpendicular to grain	Inside ⁽¹⁾	4.0	2.5	1.1
(v)	Modulus of elasticity ($\times 10^3$ N/mm ²)	All locations and grade	12.6	9.8	5.6

⁽¹⁾ For working stresses for other locations of use, that is, outside and wet, generally factors of 5/6 and 2/3 are applied.

⁽²⁾ The values of horizontal shear to be used only for beams. In all other cases shear along grain to be used.

Table 6.11.10: Modifications Factor K_I to Allow for Change in Slope of Grain

Slope	Modification Factor K_I	Strength of Posts or Columns
	Strength of Beams, Joists and Ties	
1 in 10	0.80	0.74
1 in 12	0.90	0.82
1 in 14	0.98	0.8
1 in 15 and flatter	1.00	1.00

Note: For intermediary slopes of grains, values of modification factor may be obtained by interpolation.

Table 6.11.11: Modifications Factor K_2 , for Change in Duration of Loading

Duration of Loading	Modification Factor K_2
Continuous (Normal)	1.0
Two months	1.15
Seven days	1.25
Wind and earthquake	1.33
Instantaneous or impact	2.00

Note: The strength properties of timber under load are time dependent.

11.5.3.2.2 If there are several duration of loads (in addition to the continuous) to be considered, the modification factor shall be based on the shortest duration load in the combination, that is, the one yielding the largest increase in the permissible stresses, provided the designed section is found adequate for a combination of other larger duration loads.

Explanation: In any structural timber design for dead loads, snow loads and wind or earthquake forces, members may be designed on the basis of total of stresses due to dead, snow and wind loads using $K_2 = 1.33$, factor for the permissible stress (of Table 6.11.1) to accommodate the wind load, that is, the shortest of duration and giving the largest increase in the permissible stresses. The section thus found is checked to meet the requirements based on dead loads alone with modification $K_2 = 1.00$.

11.5.3.2.3 Modification factor K_2 shall also be applied to allowable loads for mechanical fasteners in design of joints, when the wood and not the strength of metal determine the load capacity.

11.6 Design Considerations

11.6.1 All structural members, assemblies or framework in a building, in combination with the floors, walls and other structural parts of the building shall be capable of sustaining, with due stability and stiffness the whole dead and imposed loadings as per Chapters 1 and 2 of Part 6, without exceeding the limits of relevant stresses specified in this Section.

11.6.2 Buildings shall be designed for all dead and imposed loads or forces assumed to come upon them during construction or use, including uplifts or horizontal forces from wind and forces from earthquakes or other loadings. Structural members and their

11.6.3 connections shall be proportioned to provide a sound and stable structure with adequate strength and stiffness. Wooden components in construction generally include panels for sheathing and diaphragms, siding, beams, girder, columns, light framings, masonry wall and joist construction, heavy-frames, glued laminated structural members, structural sandwiches, prefabricated panels, lamella arches, portal frames and other auxiliary constructions.

11.6.4 Net Section

11.6.4.1 The net section is obtained by deducting from the gross sectional area of timber the projected area of all material removed by boring, grooving or other means at critical plane. In case of nailing, the area of the prebored hole shall not be taken into account for this purpose.

11.6.4.2 The net section used in calculating load carrying capacity of a member shall be at least net section determined as above by passing a plane or a series of connected planes transversely through the members.

11.6.4.3 Notches shall be in no case remove more than one quarter of the section.

11.6.4.4 In the design of an intermediate or a long column, gross section shall be used in calculating load carrying capacity of the column.

11.6.5 Loads

11.6.5.1 The loads shall conform to those given in Chapter 2 Part 6 of this Code.

11.6.5.2 The worst combination and location of loads shall be considered for design. Wind and seismic forces shall not be considered to act simultaneously.

11.6.6 Flexural Members

11.6.6.1 Such structural members shall be investigated for the following:

- (a) Bending strength,
- (b) Maximum horizontal shear,
- (c) Stress at the bearings, and
- (d) Deflection.

11.6.6.2 Effective span

The effective span of beams and other flexural members shall be taken as the distance from face of supports plus one-half of the required length of bearing at each end except that for continuous beams and joists the span may be measured from centre of bearing at those supports over which the beam is continuous.

11.6.6.3 Usual formula for flexural strength shall apply in design:

$$f_{ab} \frac{M}{z} \leq f_b \quad (6.11.1)$$

11.6.6.4 Form factors for flexural members

The following form factors shall be applied to the bending stress:

- (a) Rectangular Section - For rectangular sections, for different depths of beams, the form factor K_3 shall be taken as:

$$K_3 = 0.81 \left(\frac{D^2 + 89400}{D^2 + 55000} \right) \quad (6.11.2)$$

Form factor (K_3) shall not be applied for beams having depth less than or equal to 300 mm.

- (b) Box Beams and I-Beams - For box beams and I-beams, the form factor K_4 obtained by using the formula:

$$K_4 = 0.8 + 0.8y \left(\frac{D^2 + 89400 - 1}{D^2 + 55000} \right) \quad (6.11.3)$$

Where,

$$y = p_1^2 + (6 - 8p_1 + 3p_1^2)(1 - q_1) + q_1 \quad (6.11.4)$$

- (c) Solid Circular Cross-Sections - For solid circular cross sections the form factor K_5 shall be taken as 1.18.
- (d) Square Cross-Sections - For square cross-sections where the load is in the direction of diagonal, the form factor K_6 shall be taken as 1.414.

11.6.6.5 Width

The minimum width of the beam or any flexural member shall not be less than 50 mm or 1/50 of the span, whichever is greater.

11.6.6.6 Depth

The depth of beam or any flexural member shall not be taken more than three times of its width without lateral stiffening.

11.6.6.6.1 Stiffening

All flexural members having a depth exceeding three times its width or a span exceeding 50 times its width or both shall be laterally restrained from twisting or buckling and the distance between such restraints shall not exceed 50 times its width.

11.6.6.7 Shear

11.6.6.7.1 The following formulae shall apply:

- (a) The maximum horizontal shear, when the load on a beam moves from the support towards the centre of the span, and the load is at a distance of three to four times the depth of the beam from the support, shall be calculated from the following general formula:

$$H = \frac{VQ}{Ib} \quad (6.11.5a)$$

- (b) For rectangular beams:

$$H = \frac{3V}{2bD} \quad (6.11.5b)$$

- (c) For notched beams, with tension notch at supports (Figure 6.11.1a):

$$H = \frac{3VD}{2bD_1^2} \quad (6.11.5c)$$

- (d) For notched at upper (compression) face, where $e > D$ (Figure 6.11.1b):

$$H = \frac{3V}{2bD_1} \quad (6.11.5d)$$

- (e) For notched at upper (compression) face, where $e < D$ (Figure 6.11.1b)

$$H = \frac{3V}{2b\left[D - \left(\frac{D_2}{D}\right)e\right]} \quad (6.11.5e)$$

11.6.6.7.2 For concentrated loads:

$$V = \frac{10C(l-x)(x/D)^2}{9I[2+(x/D)^2]} \quad (6.11.6a)$$

and, for uniformly distributed loads,

$$V = \frac{w}{2} \left(1 - \frac{2D}{l} \right) \quad (6.11.6b)$$

After arriving at the value of V , its value will be substituted in the formula:

$$H = \frac{VQ}{Ib} \quad (6.11.5a)$$

11.6.6.7.3 In determining the vertical reaction following deductions in loads may be made:

- Consideration shall be given to the possible distribution of load to adjacent parallel beams, if any;
- All uniformly distributed loads within a distance equal to the depth of the beam from the edge of the nearest support may be neglected except in case of beam hanging downwards from a particular support, and
- All concentrated loads in the vicinity of the supports may be reduced by the reduction factor applicable according to Table 6.11.12.

Table 6.11.12: Reduction Factor for Concentrated Loads in the Vicinity of Supports

Distance of Load from the Nearest Support	LSD or Less	2D	2.5D	3D or More
Reduction factor	0.6	0.4	0.2	No Reduction

Note : For intermediate distances, factor may be obtained by linear interpolation.

11.6.6.7.4 Unless the local stress is calculated and found to be within the permissible stress, flexural member shall not be cut, notched or bored except as follows:

- Notches may be cut in the top or bottom neither deeper than one-fifth of the depth of the beam nor farther from the edge of the support than one-sixth of the span;
- Holes not larger in diameter than one quarter of the depth may be bored in the middle third of the depth and length; and

- (c) If holes or notches occur at a distance greater than three times the depth of the member from the edge of the nearest support, the net remaining depth (Figure 6.11.1c) shall be used in determining the bending strength.

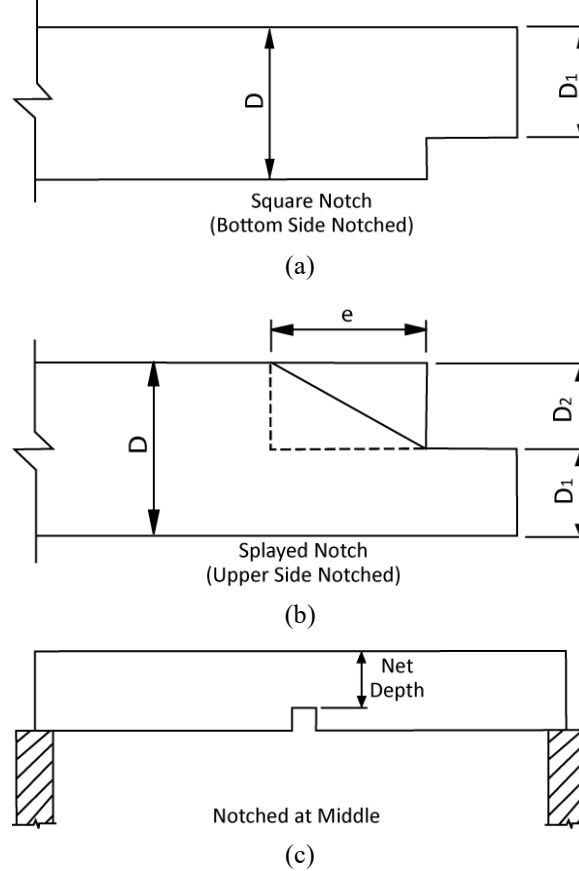


Figure 6.11.1 Notched beams

11.6.6.8 Bearing

11.6.6.8.1 The ends of flexural members shall be supported in recesses which provide adequate ventilation to prevent dry rot and shall not be enclosed. Flexural members except roof timbers that are supported directly on masonry or concrete shall have a length of bearing not less than 75 mm. Members supported on corbels, offsets and roof timbers on a wall shall bear immediately on and be fixed to wall plate not less than 75 mm x 40 mm.

11.6.6.8.2 Timber joists or floor planks shall not be supported on the top flange of steel beams unless the bearing stress, calculated on the net bearing as shaped to fit the beam, is less than the permissible compressive stress perpendicular to the grain.

11.6.6.8.3 Bearing stress

Length and position of bearing

- At any bearing on the side grain of timber, the permissible stress in compression perpendicular to the grain, f_{cn} , is dependent on the length and position of the bearing.
- The permissible stresses given in Table 6.11.1 for compression perpendicular to the grain are also the permissible stresses for any length at the ends of a member and for bearings 150 mm or more in length at any other position.
- For bearings less than 150 mm in length located 75 mm or more from the end of a member as shown in Figure 6.11.2, the permissible stress may be multiplied by the modification factor K_7 given in Table 6.11.13.
- No allowance need be made for the difference in intensity of the bearing stress due to bending of a beam.
- The bearing area should be calculated as the net area after allowance for the amount of wane.
- For bearings stress under a washer or a small plate, the same coefficient specified in Table 6.11.13 may be taken for a bearing with a length equal to the diameter of the washer or the width of the small plate.
- When the direction of stress is at angle to the direction of the grain in any structural member, then the permissible bearing stress in that member shall be calculated by the following formula:

$$f_c^\theta = \frac{f_{cp} \times f_{cn}}{f_{cp} \sin^2 \theta + f_{cn} \cos^2 \theta} \quad (6.11.7)$$

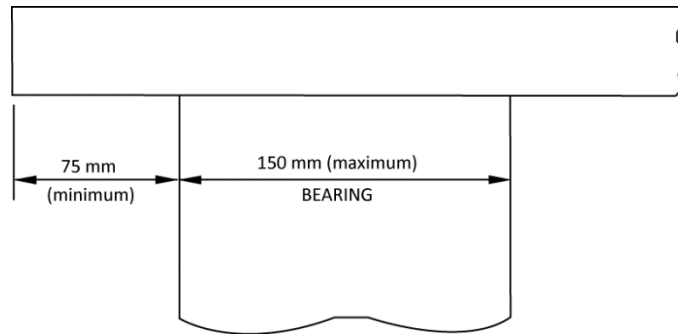


Figure 6.11.2 Position of end bearings

Table 6.11.13: Modification Factor K_7 for Bearing Stresses

Length of bearing in mm	15	25	40	50	75	100	150 or more
Modification factor K_7	1.67	1.40	1.25	1.20	1.13	1.10	1.00

11.6.6.9 Deflection

The deflection in the case of all flexural members supporting brittle materials like gypsum ceilings, slates, tiles and asbestos sheets shall not exceed 1/360 of the span. The deflection in the case of other flexural members shall not exceed 1/240 of the span and 1/150 of the freely hanging length in the case of cantilevers.

11.6.6.9.1 Usual formula for deflection shall apply (ignoring deflection due to shear strain):

$$\delta = \frac{KWL^3}{EI} \quad (6.11.8)$$

K-values = 1/3 for cantilevers with load at free end,

= 1/8 for cantilevers with uniformly distributed load,

= 1/48 for beams supported at both ends with point load at centre, and

= 5/384 for beams supported at both ends with uniformly distributed load.

11.6.6.9.2 In order to allow the effect of long duration loading on E, for checking deflection in case of beams and joists the effective loads shall be twice the dead load if timber is initially dry.

11.6.6.9.3 Self-weight of beam shall be considered in design.

11.6.7 Columns

The formulae given are for columns with pin end conditions and the length shall be modified suitably with other end conditions.

11.6.7.1 Solid columns

Solid columns shall be classified into short, intermediate and long columns depending upon their slenderness ratio S/d as follows:

- (a) Short columns - where S/d does not exceed 11.
- (b) Intermediate columns - where S/d is between 11 and K_g , and
- (c) Long columns - where S/d is greater than K_g .

11.6.7.1.1 For short columns, the permissible compressive stress shall be calculated as follows:

$$f_c = f_{cp} \quad (6.11.9)$$

11.6.7.1.2 For intermediate columns, the permissible compressive stress is calculated by using the following formula:

$$f_c = f_{cp} \left[1 - \frac{1}{3} \left(\frac{S}{K_g d} \right)^4 \right] \quad (6.11.10)$$

11.6.7.1.3 For long columns, the permissible compressive stress shall be calculated by using the following formula:

$$f_c = \frac{0.329E}{(S/d)^2} \quad (6.11.11)$$

11.6.7.1.4 In case of solid columns of timber, S/d ratio shall not exceed 50.

11.6.7.1.5 The permissible load on a column of circular cross-section shall not exceed that permitted for a square column of an equivalent cross-sectional area.

11.6.7.1.6 For determining S/d ratio of a tapered column, its least dimension shall be taken as the sum of the corresponding least dimensions at the small end of the column and one-third of the difference between this least dimension at the small end and the corresponding least dimension at the large end, but in no case shall the least dimension for the column be taken as more than one and a half times the least dimension at the small end. The induced stress at the small end of the tapered column shall not exceed the permissible compressive stress in the direction of grain.

11.6.7.2 Built-up columns

11.6.7.2.1 Box column

Box columns shall be classified into short, intermediate and long columns as follows:

- (a) Short columns - where $\frac{s}{\sqrt{d_1^2 + d_2^2}}$ is less than 8;
- (b) Intermediate columns - where $\frac{s}{\sqrt{d_1^2 + d_2^2}}$ is between 8 and K_9 ; and
- (c) Long columns - where $\frac{s}{\sqrt{d_1^2 + d_2^2}}$ is greater than K_9

11.6.7.2.2 For short columns, the permissible compressive stress shall be calculated as follows:

$$f_c = qf_{cp} \quad (6.11.12)$$

11.6.7.2.3 For intermediate columns, the permissible compressive stress shall be obtained using the following formula

$$f_c = qf_{cp} \left[1 - \frac{1}{3} \left(\frac{s}{\sqrt{d_1^2 + d_2^2}} \right)^4 \right] \quad (6.11.13)$$

11.6.7.2.4 For long columns, the permissible compressive stress shall be calculated by using the following formula:

$$f_c = \frac{0.329UE}{\left(\frac{s}{\sqrt{d_1^2 + d_2^2}} \right)^2} \quad (6.11.14)$$

11.6.7.2.5 The following values of U and q , depending upon plank thickness (t) in Sections 11.6.6.2.3 and 11.6.6.2.4, shall be used:

<u>t (mm)</u>	<u>U</u>	<u>q</u>
25	0.80	1.00
30	0.60	1.00

11.6.7.3 Spaced columns

11.6.7.3.1 The formulae for solid columns as specified in Sec 11.6.6.1 are applicable to spaced columns with a restraint factor of 2.5 or 3, depending upon distances of end connectors in the column.

A restrained factor of 2.5 for location of centroid group of fasteners at $S/20$ from end and 3 for location at $S/10$ to $S/20$ from end shall be taken.

11.6.7.3.2 For intermediate spaced column, the permissible compressive stress shall be:

$$f_c = f_{cp} \left[1 - \frac{1}{3} \left(\frac{S}{k_{10}d} \right)^4 \right] \quad (6.11.15)$$

11.6.7.3.3 For long spaced columns, the formula shall be:

$$f_c = \frac{0.329E \times 2.5}{(S/d)^2} \quad (6.11.16)$$

11.6.7.3.4 For individual members of spaced columns, S/d ratio shall not exceed 80.

11.6.7.4 Compression members shall not be notched. When it is necessary to pass services through such a member, this shall be effected by means of a bored hole provided that the local stress is calculated and found to be within the permissible stress specified. The distance from the edge of the hole to the edge of the member shall not be less than one quarter of width of the face.

11.6.8 Structural Members Subject to Bending and Axial Stresses

11.6.8.1 Structural members subjected both to bending and axial compression shall be designed to comply with the following formula:

$$\frac{f_{ac}}{f_c} + \frac{f_{ab}}{f_b} \quad (6.11.17)$$

11.6.8.2 Structural members subjected both to bending and axial tension shall be designed to comply with the following formula:

$$\frac{f_{at}}{f_t} + \frac{f_{ab}}{f_b} \quad (6.11.18)$$

11.7 Design of Common Steel Wire Nail Joints

11.7.1 General

Nail jointed timber construction is suitable for light and medium timber framings (trusses, etc) up to 15 m spans. With the facilities of readily available materials and simpler workmanship in mono-chord and split chord constructions, this type of fabrication has a large scope.

11.7.2 Dimensions of Members

11.7.2.1 The dimension of art individual piece of timber (that is, any single member) shall be within the range given below:

- (a) The minimum thickness of the main members in mono-chord construction shall be 30 mm.
- (b) The minimum thickness of an individual piece of members in split-chord construction shall
- (c) The space between two adjacent pieces of timber shall be restricted to a maximum of 3 times the thickness of the individual piece of timber of the chord member. In case of web members, it may be greater for joining facilities.

11.7.3 No lengthening joint shall preferably be located at a panel point. Generally not more than two, but preferably one, lengthening joint shall be permitted between the two panel points of the members.

11.7.4 Specification and Diameter of Nails

11.7.4.1 The nails used for timber joints shall conform to Part 5 'Building Materials'. The nails shall be diamond pointed.

11.7.4.2 The diameter of nail shall be within the limits of one-eleventh to one-sixth of the least thickness of members being connected.

11.7.4.3 Where the nails are exposed to be saline conditions, common wire nails shall be galvanized.

11.7.5 Arrangement of Nails in the Joints

The end distances, edge distances and spacing of nails in a nailed joint should be such as to avoid undue splitting of the wood and shall not be less than those given in Sections 11.7.5.1 and 11.7.5.2.

11.7.5.1 Lengthening joints

The requirement of spacing of nails in a lengthening joint shall be as follows, Table 6.11.14 (see also Figure 6.11.3):

Table 6.11.14: Requirement for Spacing of Nail

Sl. No.	Spacing of Nails	Type of Stress in the Joint	Minimum Requirement
(i)	End distance	Tension	12n
		Compression	10n
(ii)	In direction of grain	Tension	10n
		Compression	5n
(iii)	Edge distance	-	5n
(iv)	Between row of nails perpendicular to the grain	-	5n

Notes:

1. n is shank diameter of nails

2. The 5n distance between the rows of nails perpendicular to the grain may be increased subject to the availability of width of the member keeping edge distance constant.

11.7.5.2 Node joints

The requirement for spacing of nails in node joints shall be as specified in Figure 6.11.4 where the members are at right angle and as in Figure 6.11.5 where the members are inclined to one another at angles other than 90° and subjected to either pure compression or pure tension.

11.7.6 Penetration of Nails

11.7.6.1 For a lap joint when the nails are driven from the side of the thinner member, the length of penetration of nails in the thicker member shall be one and a half times the thickness of the thinner member subject to maximum of the thickness of the thicker member.

11.7.6.2 For butt joints the nails shall be driven through the entire thickness of the joint.

11.7.7 Design Considerations

11.7.7.1 Where a number of nails are used in a joint, the allowable load in lateral resistance shall be the sum of the allowable loads for the individual nails, provided that the centroid of the group of these nails lies on the axis of the member and the spacing conform to Sec 11.7.5. Where a large number of nails are to be provided at a joint, they should be so arranged that there are more of rows rather than more number of nails in a row.

11.7.7.2 Nails shall, as far as practicable, be arranged so that the line of force in a member passes through the centroid of the group of nails. Where this is not practicable, allowance shall be made for any eccentricity in computing the maximum load on the fixing nails as well as the loads and bending moment in the member.

11.7.7.3 Adjacent nails shall preferably be driven from opposite faces, that is, the nails are driven alternatively from either face of joint.

11.7.7.4 For a rigid joint, a minimum of 2 nails for nodal joints and 4 nails for lengthening joint shall be driven.

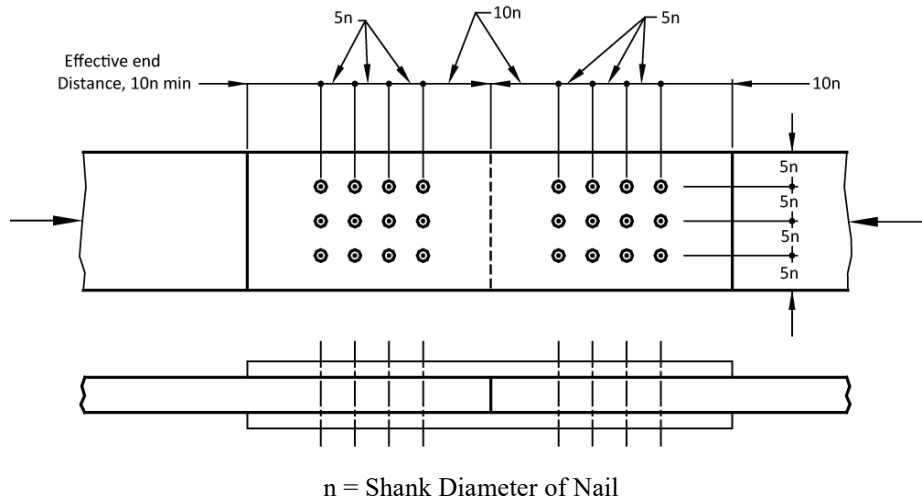
11.7.7.5 Two nails in a horizontal row are better than using the same number of nails in a vertical row.

11.7.8 Special Consideration in Nail-Jointed Truss Construction

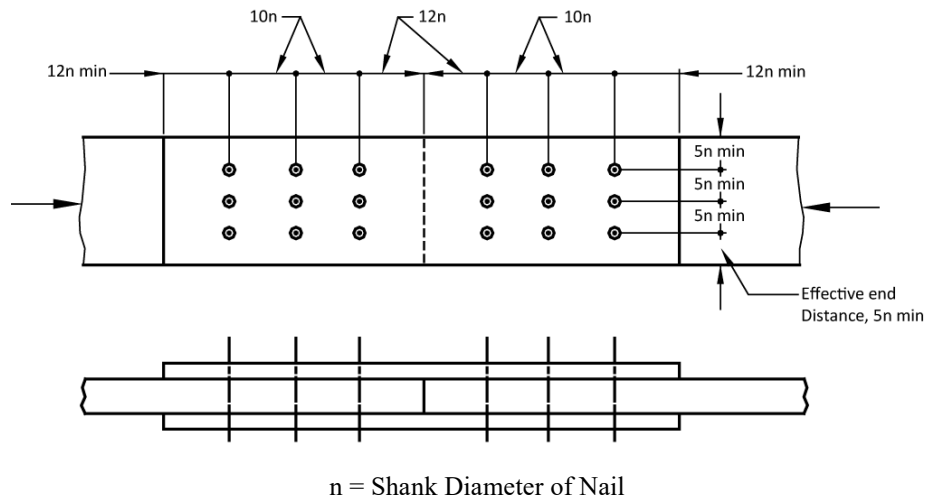
11.7.8.1 The initial upward camber provided at the centre of the lower chord of nail-jointed timber trusses shall be not less than $1/200$ of the effective span for timber structures using seasoned wood and $1/100$ for unseasoned or partially seasoned wood.

11.7.8.2 The total combined thickness of the gusset or splice plates on either side of the joint in a mono-chord type construction shall not be less than one and a half times the thickness of the main members subject to a minimum thickness of 25 mm of individual gusset plate.

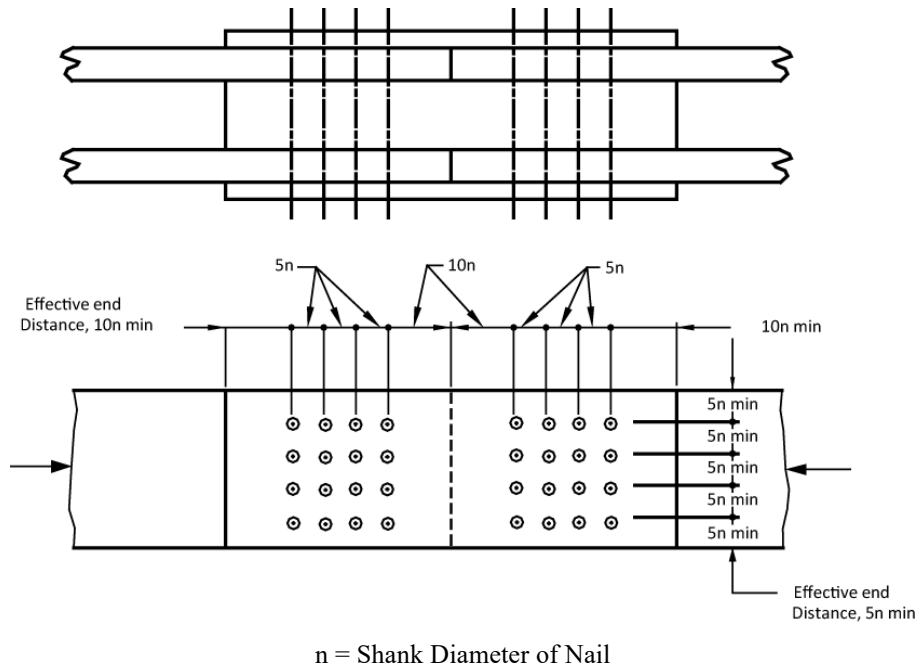
11.7.8.3 The total combined thickness of all spacer blocks or plates or both including outer splice plates, at any joint in a split-chord type construction shall not be less than one and a half times the total thickness of all the main members at that joint.



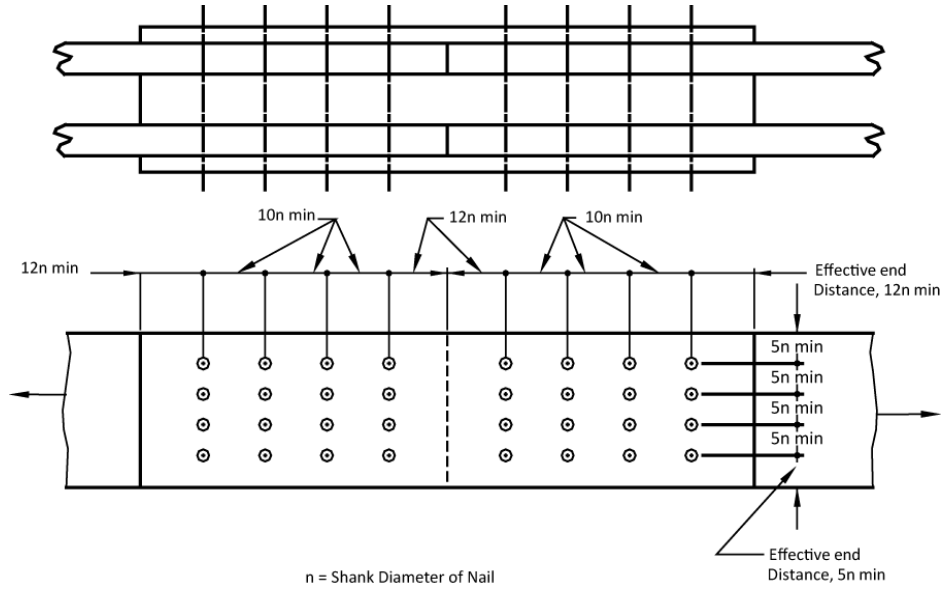
(a) Monochord type butt joint subject to compression



(b) Monochord type butt joint subject to tension

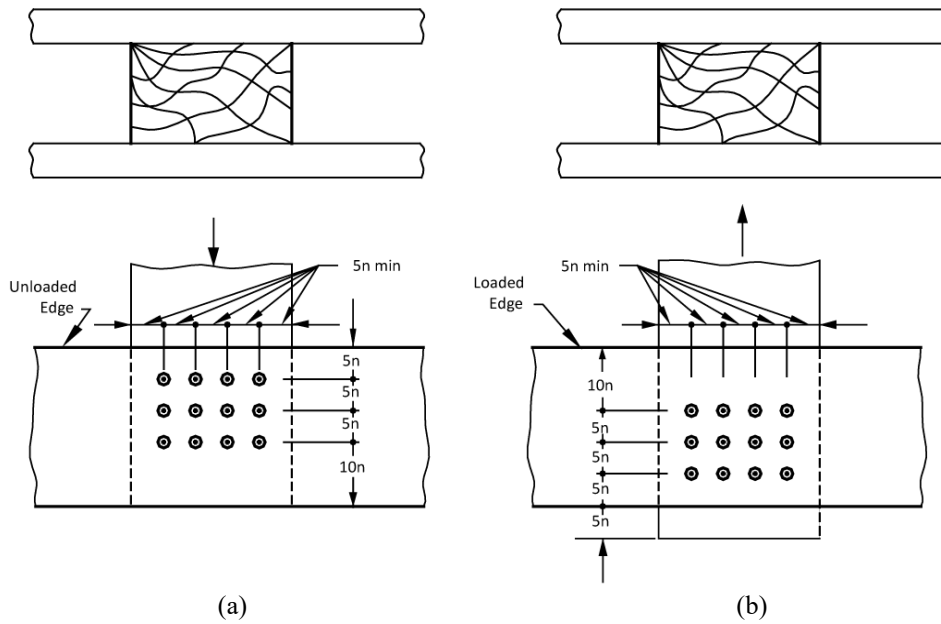
Figure 6.11.3 Spacing of nails In a lengthening joint

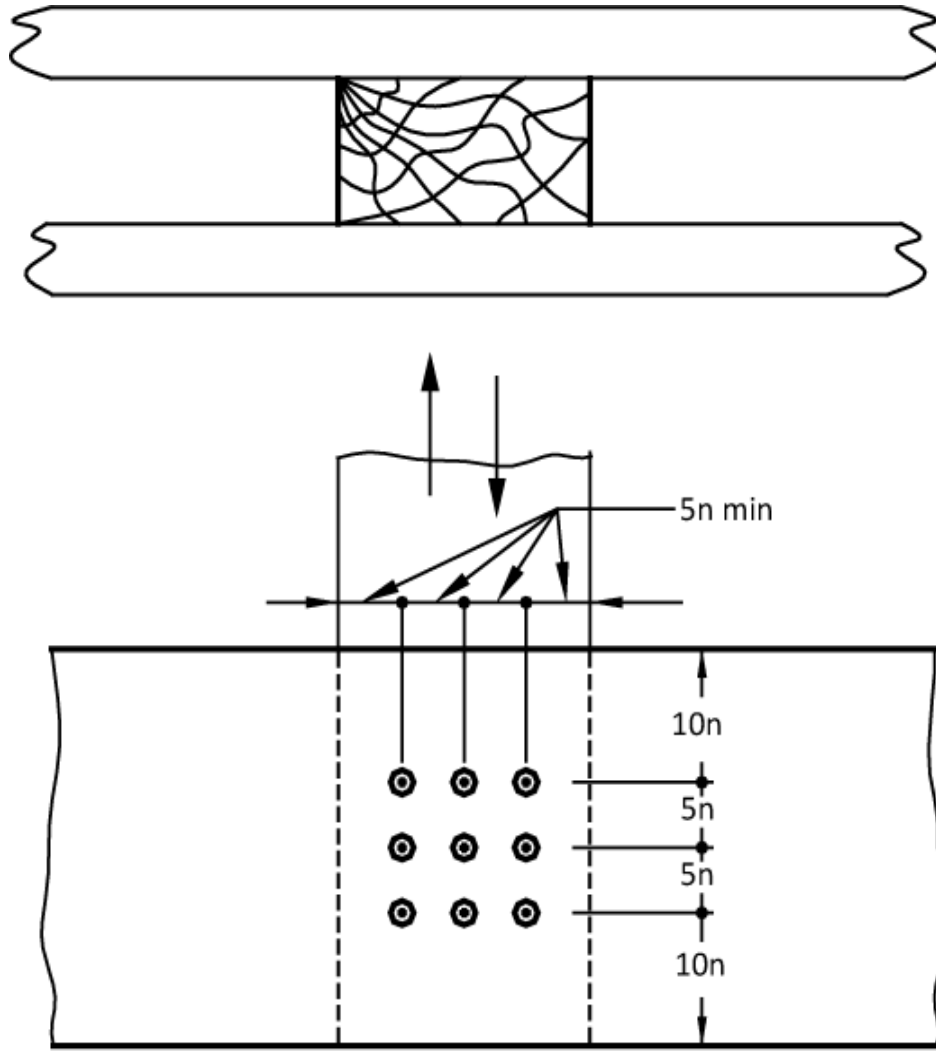
(c) Split – Chord type butt joint subject to compression



(d) Split – Chord type butt joint subject to tension

Figure 6.11.3 Spacing of nails In a lengthening joint (contd.)





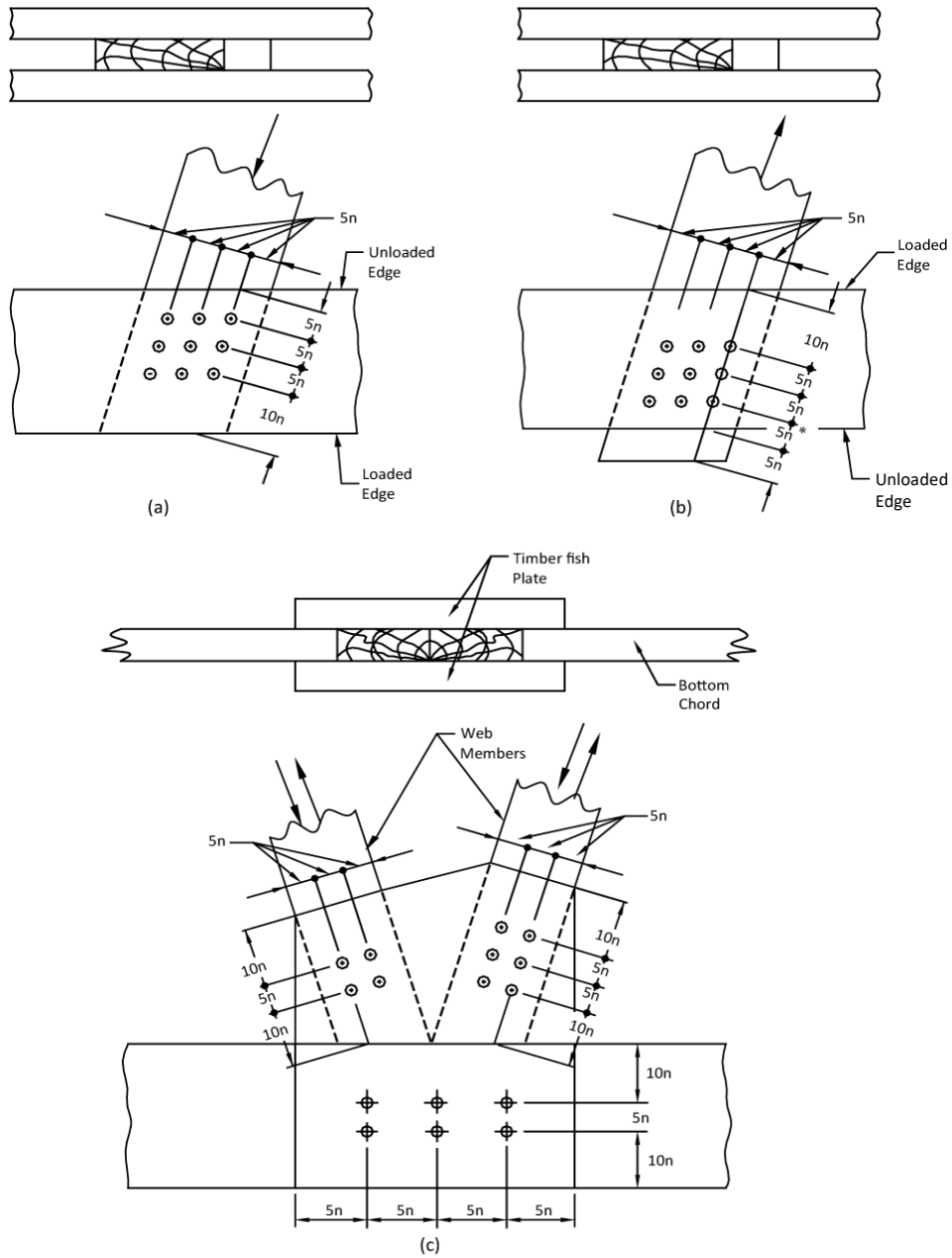
(c)

*5n may be increased to 10n, if the designed width of cord member permits.

Otherwise, the end of the loaded web member may be extended by 5n min

n = Shank diameter of nail

Figure 6.11.4 Spacing of nails where members are at right angles to one another



*5n may be increased to 10n, if the designed width of cord member permits.
 Otherwise, the end of the loaded web member may be extended by 5n min
 n = Shank diameter of nail

Figure 6.11.5 Spacing of nails at node where members are inclined to one another

11.8 DESIGN OF NAIL LAMINATED TIMBER BEAMS

11.8.1 Method of Arrangement

11.8.1.1 The beam is made up of 20 mm to 30 mm thick planks placed vertically with joints staggered in the adjoining planks with a minimum distance of 300 mm. The planks are laminated with the help of wire nails at regular intervals to take up horizontal shear developed in the beam besides keeping the planks in position (see Figure 6.11.6).

11.8.1.2 The advantage in laminations lies in dimensional stability, dispersal of defects and better structural performance.

11.8.2 Sizes of Planks and Beams

11.8.2.1 The plank thickness for fabrication of nailed laminated beams recommended are 20, 25 and 30 mm.

11.8.2.2 In case of nailed laminated timber beam the maximum depth and length of planks shall be limited to 250 mm and 2000 mm, respectively.

11.8.2.3 In order to obtain the overall width of the beam, the number and thickness of planks to form vertical nailed laminated beams, and also type and size of wire nail shall be as mentioned in Table 6.11.15. The protruding portion of the nail shall be cut off or clenched across the grains.

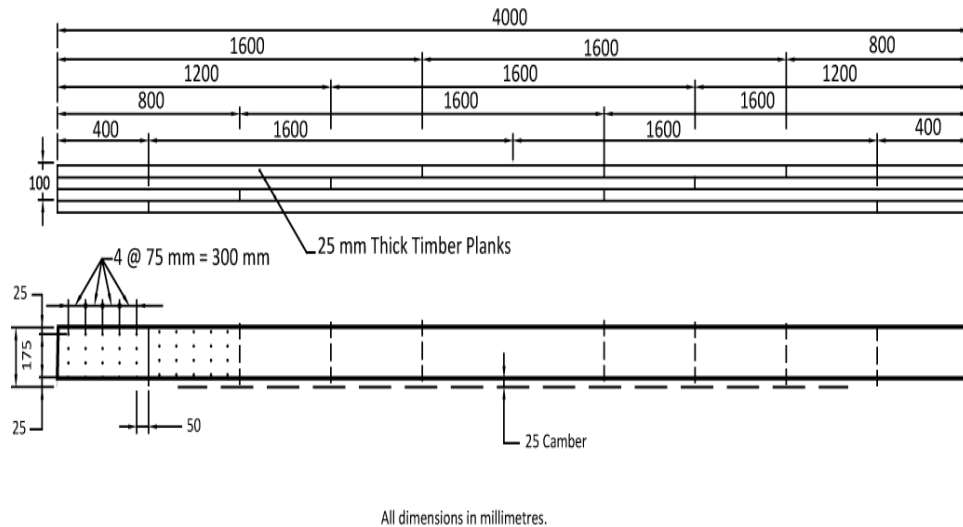


Figure 6.11.6 Plan and elevation of a typical nailed laminated timber beam (all dimensions in mm)

Table 6.11.15: Number and Size of Planks and Nails for Nailed Laminated Beams

Sl. No.	Overall Width of Beam (mm)	No. of Planks	Thickness of each Plank (mm)	Size of Nail to be used	
				Length (mm)	Diameter (mm)
(i)	50	2	25	80	3.55
(ii)	60	3	20	80	3.55
(iii)	70	3	(2x25) + (1x20)	80	3.55
(iv)	80	4	20	100	4.0
(v)	90	3	30	100	4.0
(vi)	100	4	25	125	5.0
(vii)	110	4	(3x30) + (1x20)	125	5.0
(viii)	120	4	30	125	5.0
(ix)	150	5	30	150	5.0

Notes: A number of combinations of different thickness of planks may be adopted as long as the minimum and maximum thickness of the planks are adhered to.

11.8.3 Design Considerations

11.8.3.1 Nail laminated beams shall be designed in accordance with Sec 11.6.

11.8.3.1.1 The deflection in the case of nailed laminated timber beams, joists, purlins, battens and other flexural members supporting brittle materials like gypsum, ceiling slates, tiles and asbestos sheets shall not exceed $1/480$ of the span. The deflection in case of other flexural members shall not exceed $1/360$ of the span in the case of beams and joists, and $1/225$ of the freely hanging length in case of cantilevers.

11.8.3.2 Permissible lateral strength of mild steel wire nails shall be as given in Tables 6.11.2 and 6.11.3 for Indian Species of timber, which shall apply to nails that have their points cut flush with the faces. For nails clenched across the grains strength may be increased by 20% over the values for nails with points cut flush.

11.8.3.3 Arrangement of nails

11.8.3.3.1 A minimum number of four nails in a vertical row at regular interval not exceeding 75 mm to take up horizontal shear as well as to keep the planks in position shall be used. Near the joints of the planks this distance may, however, be limited to 5 cm instead of 75 mm.

11.8.3.3.2 Shear shall be calculated at various points of the beam and [the number of nails required shall be accommodated within the distance equal to the depth of the beam, with a minimum of 4 nails in a row at a standard spacing as shown in Figure 6.11.7.

11.8.3.3.3 If the depth of the beam is more, then the vertical intermediate spacing of nails may be increased proportionately.

11.8.3.3.4 If the nails required at a point are more than that can be accommodated in a row, then these shall be provided lengthwise of the beam within the distance equal to depth of beam at standard lengthwise spacing.

11.8.3.3.5 For nailed laminated beam minimum depth of 100 mm for 3.55 mm and 4 mm diameter nails, and 125 mm for 5 mm diameter nails shall be provided.

11.9 Design of Bolted Construction Joints

11.9.1 General

Bolted joints suit the requirements of prefabrication in small and medium span timber structures for speed and economy in construction. Bolt jointed construction units offer better facilities as regards to workshop ease, mass production of components, transport convenience and re-assembly at site of work particularly in defence sector for high altitudes and far off situations.

Designing is mainly influenced by the species, size of bolts, moisture conditions and the inclination of loadings to the grains. In principle bolted joints follow the pattern of rivetted joints in steel structures.

11.9.2 Design Considerations

11.9.2.1 Bolted timber construction shall be designed in accordance with 6. The concept of critical section, that is, the net section obtained by deducting the projected area of bolt-holes from the cross-sectional area of member is very important for the successful design and economy in timber.

11.9.2.2 Bolt bearing strength of wood

The allowable load for a bolt in a joint consisting of two members (single shear) shall be taken as one half the allowable loads calculated for a three member joint (double shear) for the same t'/d_3 ratio. The percentage of safe working compressive stress of timber on bolted joints for different t'/d_3 ratios shall be as in Table 6.11.16.

11.9.2.2.1 Where a number of bolts are used in a joint, the allowable loads shall be the sum of the allowable loads for the individual bolts.

11.9.2.2.2 The factors for different bolt diameter used in calculating safe bearing stress perpendicular to grain in the joint shall be as given in Table 6.11.17.

11.9.2.2.3 Dimensions of members

- The minimum thickness of the main member in mono-chord construction shall be 40 mm.
- The minimum thickness of side members shall be 20 mm and shall be half the thickness of main members.
- The minimum individual thickness of spaced member in split-chord construction shall be 20 mm and 25 mm for webs and chord members respectively.

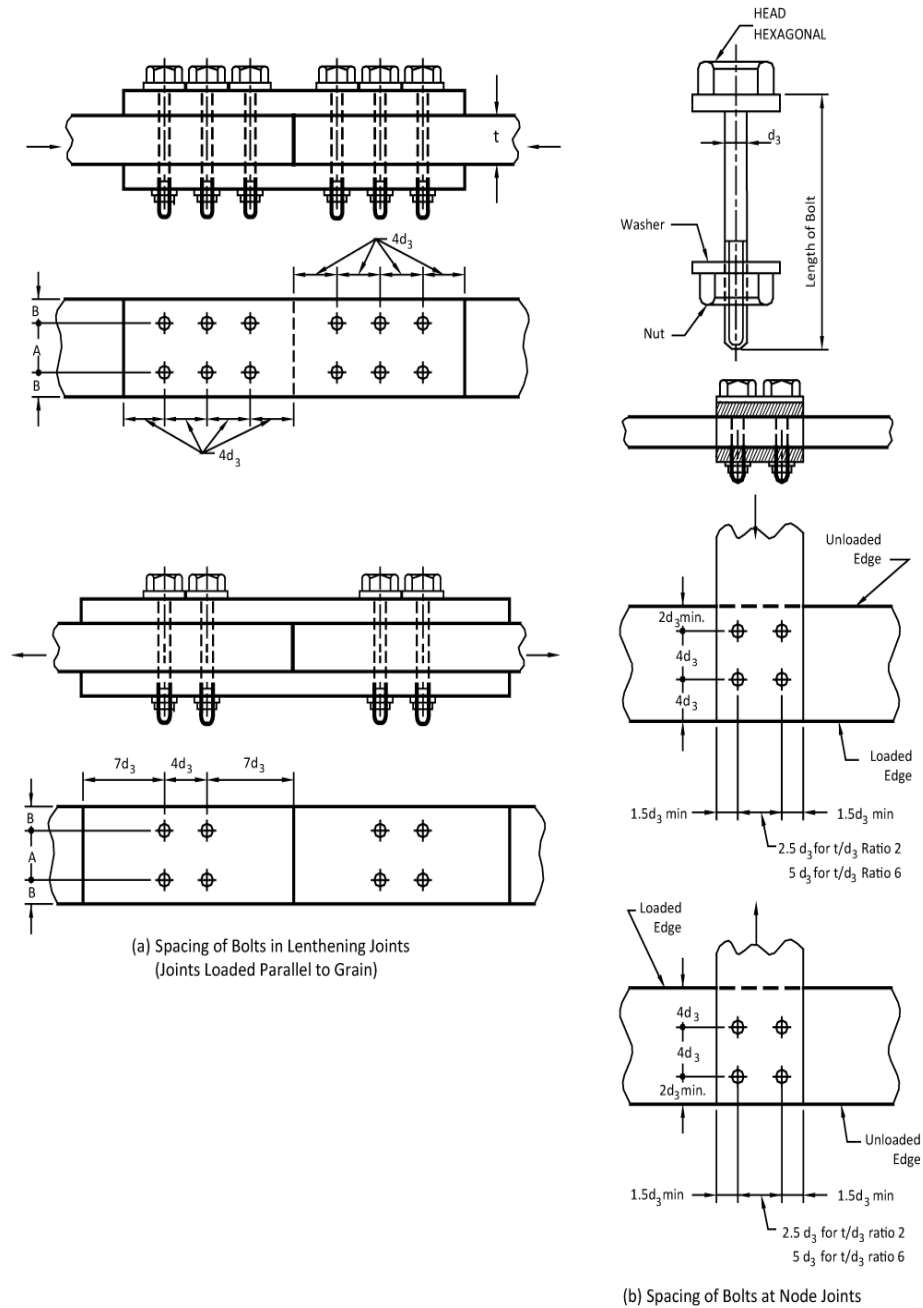


Figure 6.11.7 Typical spacing of bolts in structural joints

Table 6.11.16: Percentage of Safe Working Compressive Stress of Timber for Bolted Joints in Double Shear

t'/d_3 ratio	Stress Percentage		t'/d_3 ratio	Stress Percentage	
	Parallel to Grain	Perpendicular to Grain		Parallel to Grain	Perpendicular to Grain
	λ_1	λ_2		λ_1	λ_2
1.0	100	100	7.0	52	40
1.5	100	96	7.5	46	39
2.0	100	88	8.0	40	38
2.5	100	80	8.5	36	36
3.0	100	72	9.0	34	34
3.5	100	66	9.5	32	33
4.0	96	60	10.0	30	31
4.5	90	56	10.5	-	31
5.0	80	52	11.0	-	30
5.5	72	49	11.5	-	30
6.0	65	46	12.0	-	28
6.5	58	43			

Table 6.11.17: Bolt Diameter Factor

Diameter of Bolt (mm)	Diameter Factor (d_r)	Diameter of Bolt (mm)	Diameter Factor (d_r)
6	5.70	20	3.05
10	3.60	22	3.00
12	3.35	25	2.90
16	3.15		

11.9.2.3 Bolts and bolting

- (a) The diameter of bolt in the main member shall be so chosen to give larger slenderness t'/d_3 ratio of bolt.
- (b) There shall be more number of small diameter bolts rather than small number of large diameter in a joint.
- (c) A minimum of two bolts for nodal joints and four bolts for lengthening joints shall be provided.
- (d) There shall be more number of rows rather than more bolts in a row.
- (e) The bolt holes shall be of such diameter that the bolt can be driven easily.
- (f) Washers shall be used between the head of bolt and wood surface as also between the nut and wood.

11.9.3 Arrangement of Bolts

11.9.3.1 The following spacing in bolted joints shall be followed (see Figure 6.11.8):

- (a) Spacing of Bolts in a Row - For parallel and perpendicular to grain loading $= 4d_3$
- (b) Spacing Between Rows of Bolts
 - (i) For perpendicular to grain loading - $2.5d_3$, to $5d_3$ ($2.5d_3$, for t'/d_3 ratio of 2 and $5d_3$ for t'/d_3 ratio of 6 or more. For ratios between 2 to 6 the spacing shall be obtained by interpolation.
 - (ii) For parallel to grain loading - At least $(N-4)d_3$, with a minimum of $2.5d_3$. Also governed by net area at critical section which should be 80 percent of the total area in bearing under all bolts.
- (c) End Distance - $7d_3$ for soft woods in tension, $5d_3$ for hardwoods in tension and $4d_3$ for all species in compression.
- (d) Edge Distance
 - (i) For parallel to grain loading $1.5d_3$ or half the distance between rows of bolts, whichever is greater.

- (ii) For perpendicular to grain loading, (loaded edge distance) shall be at least $4d_3$.

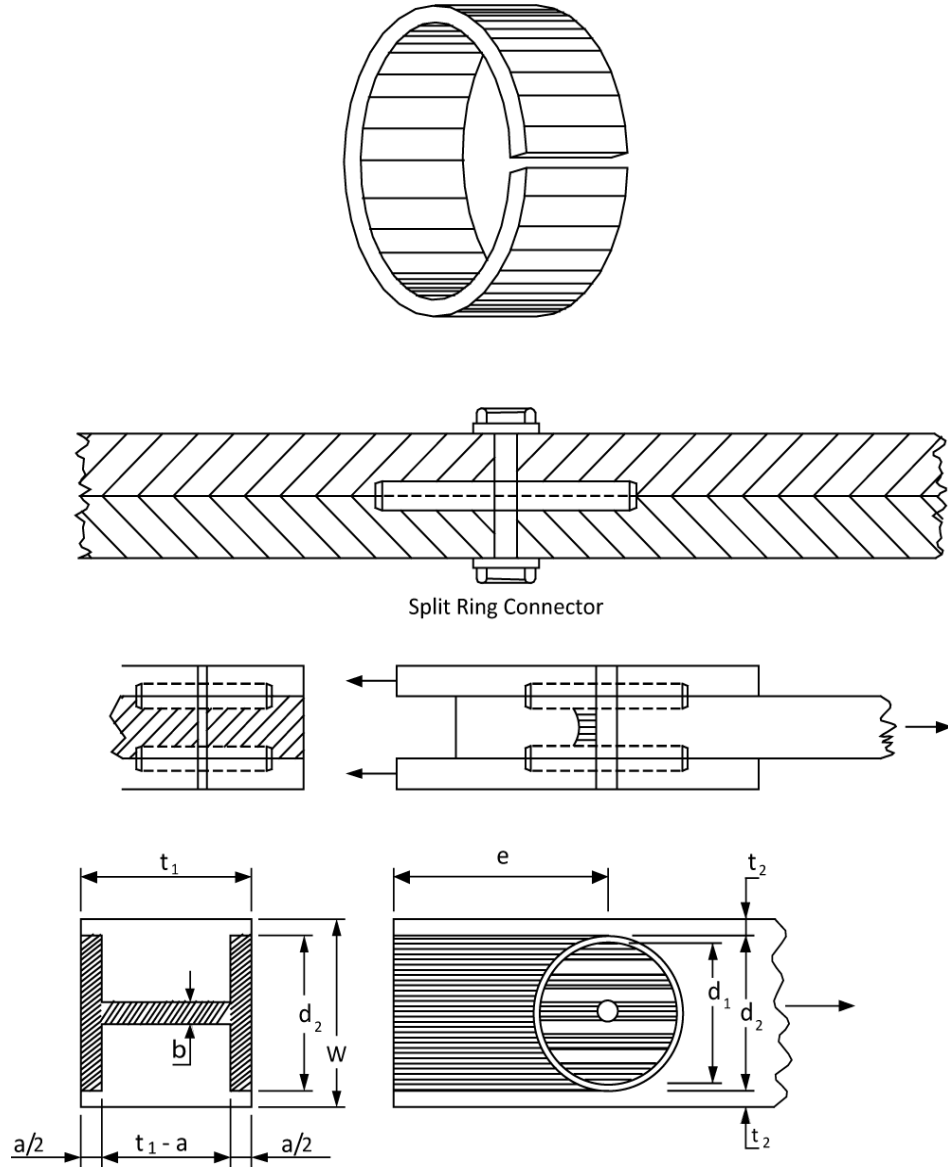


Figure 6.11.8 Stress Distribution In A Split Ring Connector

11.9.3.2 For inclined members, the spacing given above for perpendicular and parallel to grain of wood may be used as a guide and bolts arranged at the joint with respect to loading direction.

11.9.3.3 The bolts shall be arranged in such a manner so as to pass the centre of resistance of bolts through the inter-section of the gravity axis of the members.

11.9.3.4 Staggering of bolts shall be avoided as far as possible in case of members loaded parallel to grain of wood. For loads acting perpendicular to grain of wood, staggering is preferable to avoid splitting due to weather effects.

11.9.3.5 Bolting

The bolt holes shall be bored or drilled perpendicular to the surface involved. Forcible driving of the bolts shall be avoided which may cause cracking or splitting of members. A bolt hole of 1.0 mm oversize may be used as a guide for preboring.

11.9.3.5.1 Bolts shall be tightened after one year of completion of structure and subsequently at an interval of two to three years.

11.9.4 Outline for Design of Bolted Joints

Allowable load on one bolt (unit bearing stress) in a joint with wooden splice plates shall not be greater than value of P, R, F as determined by one of the following equations:

(a) For Loads Parallel to Grain

$$P = f_{cp} a \lambda_1 \quad (6.11.19)$$

(b) For Loads Perpendicular to Grain

$$R = f_{cp} a \lambda_2 d_f \quad (6.11.20)$$

(c) For Loads at an Angle to Grain

$$F = \frac{PR}{P \sin^2 \theta + R \cos^2 \theta} \quad (6.11.21)$$

11.10 Design of Timber Connector Joints

11.10.1 In large span structures, the members have to transmit very heavy stresses requiring stronger jointing techniques with metallic rings or wooden disc-dowels. Improvised metallic ring connector is a split circular band of steel made from mild steel pipes. This is placed in the grooves cut into the contact faces of the timber members to be joined, the assembly being held together by means of a connecting bolt.

11.10.1.1 Dimensions of Members Variation of thickness of central (main) and side members affect the load carrying capacity of the joint.

11.10.1.2 The thickness of main member shall be at least 57 mm and that of side member 38 mm with length and width of members governed by placement of connector at joint.

11.10.1.3 The metallic connector shall be so placed that the loaded edge distance is not less than the diameter of the connector and the end distance not less than 1.75 times the diameter on the loaded side.

11.10.1.4 Design Considerations

Figure 11.10.1 illustrates the primary stresses in a split ring connector joint under tension. The shaded areas represent the part of wood in shear, compression and tension. Related formulae for the same are indicated in Figure 6.11.9.

For fabrication of structural members, a hole of the required size of the bolt is drilled into the member and a groove is made on the contact faces of the joint.

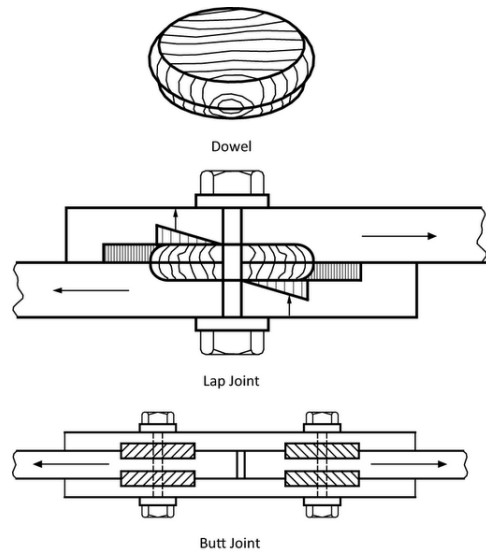
11.10.2 Wooden Disc-Dowel

11.10.2.1 It is a circular hardwood disc generally tapered each way from the middle so as to form a double conical frustum. Such a disc is made to fit into preformed holes (recesses), half in one member and the other half in another, the assembly being held by one mild steel bolt through the centre of the disc to act as a coupling for keeping the jointed wooden members from spreading apart.

11.10.2.2 Dimensions of members

The thickness of dowel may vary from 25 mm to 35 mm and diameter from 50 mm to 150 mm. The diameter of dowel shall be 3.25 to 3.50 times the thickness.

The edge clearance shall range from 12 mm to 20 mm as per the size of the dowel. The end clearance shall be at least equal to the diameter of dowel for joints subjected to tension and three-fourth the diameter for compression joints. Disc-dowel shall be turned from quarter sawn planks of seasoned material.



Lap Joint

But in simple tension due to clockwise turning moment on dowel.

Butt Joint

No tilting moment in dowel due to balancing effect [dowels are in shear (no bending, shearing and tensile stress on bolts)]

Size of dowel for equal strength in both shearing and bearing.

$$\frac{\pi d^2}{4} \times S = d \times \frac{t}{2} \times c$$

Where

d = Mid diameter of the dowel

t = Thickness of dowel

S = Safe working stress in shear along grain, and

c = Safe compressive stress along grain.

Figure 6.11.9 Distribution of forces in dowel joint

11.10.2.3 Choice of species

Wood used for making dowels shall be fairly straight grained, free from excessive liability to shrink and warp, and retain shape well after seasoning species recommended include: (a) Babul, (b) Sissoo, (c) Pyinkado and (d) Yon. Data on the above species as per Table 6.11.1 except for the species Pyinkado, which is not an indigenous species.

11.10.2.4 Design considerations

Figure 11.10.2 illustrates the forces on dowel in a lap joint and butt joint. Dowel is subjected to shearing at the mid-section, and compression along the grain at the bearing surfaces. For equal strength in both the forces, formula equations are given in Figure 6.11.10 to determine the size of dowel.

The making of wooden discs may present some problems in the field, but they may be made in small workshop to the specifications of the designer. This is also economically important. Once the wood fittings are shop tailored and made, the construction process in the field is greatly simplified. Theoretical safe loads in design shall be confirmed through sample tests.

11.11 Glued Laminated Construction and Finger Joints

11.11.1 Developments in the field of synthetic adhesive have brought gluing techniques within the range of engineering practice. Timber members of larger cross-sections and long lengths can be fabricated from small sized planks by the process of gluelam. The term glued laminated timber construction as applied to structural members refers to various laminations glued together, either in straight or curved form, having grain of all laminations essentially parallel to the lengths of the member.

11.11.1.1 Choice of glue

The adhesive used for glued laminated assembly are 'gap filling' type. A 'filler' in powder form is introduced in the adhesive. Structural adhesives are supplied either in powder form to which water is added or in resin form to which a hardener or catalyst is added. However, it is important that only boiling water proof (BWP) grade adhesives shall be used for fabrication of gluelam in tropical, high humid climates.

11.11.1.2 Manufacturing schedule

In absence of a systematic flow-line in a factory, provisions of intermediate technology shall be created for manufacturing structural elements. The schedule involves steps:

- (a) Drying of planks;
- (b) Planning;
- (c) End-jointing by scarfs or fingers;
- (d) Machining of laminations;
- (e) Setting up dry assembly of structural unit;
- (f) Application of glue;
- (g) Assembly and pressing the laminations;
- (h) Curing the glue lines, as specified; and
- (i) Finishing, protection and storage.

11.11.2 Finger joints are glued joints connecting timber members end-to-end (Figure 6.11.11). Such joints shall be produced by cutting profiles (tapered projections) in the form of V-shaped grooves to the ends of timber planks or scantling to be joined, gluing the interfaces and then meeting the two ends together under pressure. Finger joints provide long lengths of timber, ideal for upgrading timber by permitting removal of defects, minimizing warping and reducing wastage by avoiding short off-cuts.

11.11.2.1 In finger joints the glued surfaces are on the side grain rather than on the end grain and the glue line is stressed in shear rather in tension.

11.11.2.2 The fingers can be cut from edge-to-edge or from face-to-face. The difference is mainly in appearance, although bending strength increases if several fingers share the load. Thus a joist is slightly stronger with edge-to-edge finger joints and a plank is stronger with face-to-face finger joint.

11.11.2.3 For structural finger jointed members for interior dry locations, adhesives based on melamine formaldehyde cross linked polyvinyl acetate (PVA) are suited. For high humid and exterior conditions, phenol formaldehyde and resorcinol formaldehyde type adhesives are recommended. Proper adhesives should be selected in consultation with the designer and adhesive manufacturers.

11.11.2.4 Manufacturing process

In the absence of sophisticated machinery, the finger joints shall be manufactured through intermediate technology with the following steps:

- (a) Drying of wood,
- (b) Removal of knots and other defects,
- (c) Squaring the ends of the laminating planks,
- (d) Cutting the profile of finger joint in the end grain,

- (e) Applying adhesives on the finger interfaces,
- (f) Pressing the joint together at specified pressure,
- (g) Curing of adhesive line at specified temperature, and
- (h) Planning of finger-jointed planks for smooth surface.

11.11.2.5 Strength

Strength of finger joints depends upon the geometry of the profile for structural purpose; this is generally 50 mm long, 12 mm pitch.

11.11.2.5.1 End joints shall be scattered in adjacent laminations, which shall not be located in very highly stressed outer laminations.

11.11.2.6 Tip thickness will be as small as practically possible.

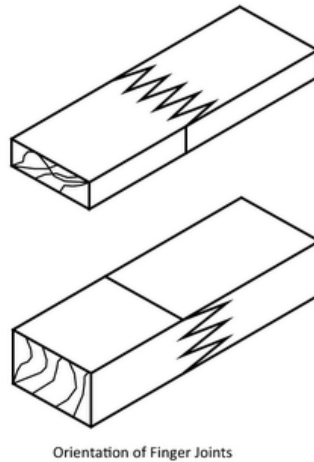
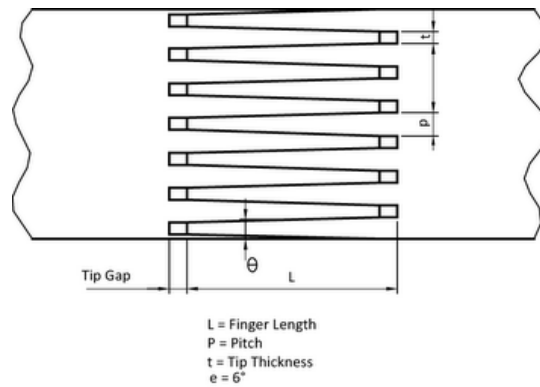


Figure 6.11.10 Typical finger joint geometry

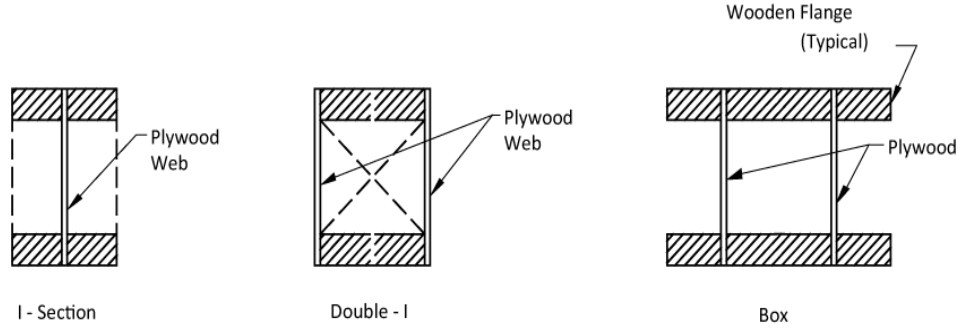


Figure 6.11.11 Typical cross - section of web beams

11.12 Laminated Veneer Lumber

11.12.1 Certain reconstituted lignocellulosic products with fibre oriented along a specific direction have been developed and are being adopted for load bearing applications. Laminated veneer lumber is one such product developed as a result of researches in plantation grown species of wood. Density of laminated veneer lumber ranges from 0.6 to 0.75.

11.12.1.1 Dimensions

Sizes of laminated veneer lumber composite shall be inclusive of margin for dressing and finishing unless manufactured to order. The margin for dressing and finishing shall not exceed 3mm in the width and thickness and 12mm in the length.

11.12.1.2 Permissible defects

- (a) Jointing gaps: Not more than 3 mm wide, provided they are well staggered in their spacing and position between the successive plies.
- (b) Slope of grain: Not exceeding 1 in 10 in the face layers.
- (c) Tight knot: Three numbers up to 25 mm diameter in one square metre provided they are spaced 300 mm or more apart.
- (d) Warp: Not exceeding 1.5 mm per metre length.

11.12.1.3 Strength requirements

The strength requirements for laminated veneer lumber shall be as per Table 6.11.18.

Table 6.11.18: Requirements of Laminated Veneer Lumber

Sl. No.	Properties	Requirement
(i)	Modulus of rupture (N/mm ²), Min	50
(ii)	Modulus of elasticity (N/mm ²), Min	7500
(iii)	Compressive strength:	
	a) Parallel to grain (N/mm ²), Min	35
	b) Perpendicular to grain (N/mm ²), Min	50
(iv)	Horizontal shear:	
	a) Parallel to laminac (N/mm ²), Min	6
	b) Perpendicular to laminac (N/mm ²), Min	8
(v)	Tensile strength parallel to grain (N/mm ²), Min	55
(vi)	Screw holding power:	
	a) Edge (N), Min	2300
	b) Face (N), Min	2700
(vii)	Thickness swelling in 2 h water soaking (percent), Max	3

11.13 Design of Glued Laminated Beams

11.13.1 General

Glued laminated structural members shall be fabricated only where there are adequate facilities for accurate sizing and surfacing of planks, uniform application of glue, prompt assembly, and application of adequate pressure and prescribed temperature for setting and curing of the glue. Design and fabrication shall be in accordance with established engineering principles and good practice. A glued laminated beam is a straight member made from a number of laminations assembled both ways either horizontally or vertically. While vertical laminations have limitations in restricting the cross-section of a beam by width of the plank, horizontally laminated section offers wider scope to the designer in creating even the curved members.

Simple straight beams and joists are used for many structures from small domestic rafters or ridges to the light industrial structures.

11.13.2 Design

The design of glue laminated wood elements shall be in accordance with good engineering practice and shall take into consideration the species and grade of timber used, presence of defects, location of end joints in laminations, depth of beams and moisture contents

expected while in service. Beams of large spans shall be designed with a suitable camber to assist in achieving the most cost effective section where deflection governs the design. The strength and stiffness of laminated beams is often governed by the quality of outer laminations. Glued laminated beams can be tapered to follow specific roof slopes across a building and/or to commensurate with the varying bending moments.

11.13.3 Material

Laminating boards shall not contain decay, knots or other strength reducing characteristics in excess of those sizes or amounts permitted by specifications. The moisture content shall approach that expected in service and shall in no case exceed 15 percent at the time of gluing. The moisture content of individual laminations in a structural member shall not differ by more than 3 percent at the time of gluing. Glue shall be of type suitable for the intended service of a structural member.

11.13.4 Fabrication/Manufacture

In order to assure a well-bonded and well-finished member of true shape and size, all equipment, end-Jointing, glue spread, assembly, pressing, curing or any other operation in connection with the manufacture of glued structural members shall be in accordance with the available good practices and as per glue manufacturers' instructions as applicable.

11.14 Structural Use of Plywood

Unlike sawn timber, plywood is a layered panel product comprising veneers of wood bonded together with adjacent layers usually at right angles. As wood is strongest when stressed parallel to grain, and weak perpendicular to grain, the lay up or arrangement of veneers in the panel determines its properties. When the face grain of the plywood is parallel to the direction of stress, veneers parallel to the face grain carry almost all the load. Some information/guidelines for structural use of plywood are given in Sections 11.14.1 to 11.14.3.

11.14.1 The plywood has a high strength to weight ratio, and is dimensionally stable material available in sheets of a number of thicknesses and construction. Plywood can be sawn, drilled and nailed with ordinary wood working tools. The glues used to bond these veneers together are derived from synthetic resins which are set and cured by heating. The properties of adhesives can determine the durability of plywood.

11.14.2 In glued plywood construction, structural plywood is glued to timber resulting in highly efficient and light structural components like web beams (I and box sections), (Figures 6.11.12 and 6.11.13) stressed skin panels (Figure 6.11.14) used for flooring and walling and pre-fabricated houses, cabins, etc.

Gluing can be carried out by nail gluing techniques with special clamps. High shear strength of plywood in combination with high flexural strength and stiffness of wood result in structures characterized by high stiffness for even medium spans. Plywood can act as web transmitting shear stress in web bearing or stressed skin or sandwich construction. The effective moment of inertia of web beam and stressed skin construction depends on modular ratio that is, E of wood to E of plywood.

11.14.3 Structural plywood is also very efficient as cladding material in wood frame construction, such as houses. This type of sheathing is capable of resisting racking due to wind and quack forces. Structural plywood has been widely used as diaphragm (horizontal) as in roofing and flooring in timber frame construction. It has been established that 6 mm thick plywood can be used for sheathing and even for web and stressed skin construction, 9-12 mm thick plywood is suitable for beams, flooring diaphragms, etc. Phenol formaldehyde (PF) and PRF adhesive are suitable for fabrication of glued plywood components. 6 mm-12 mm thick structural plywood can be very well used as nailed or bolted gussets in fixing members of trusses or lattice girders or trussed rafters.

Normally, scarf joints are used for fixing plywood to required length and timber can be joined by using either finger or scarf joints. Arch panels, folded plates, shelves are other possibilities with this technique.

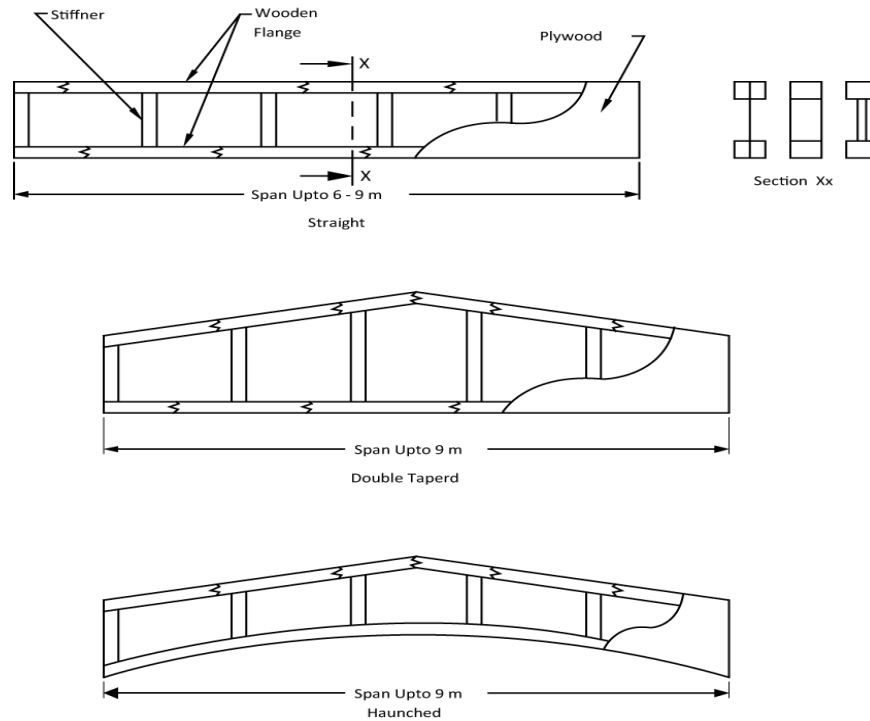


Figure 6.11.12 Web beam configurations

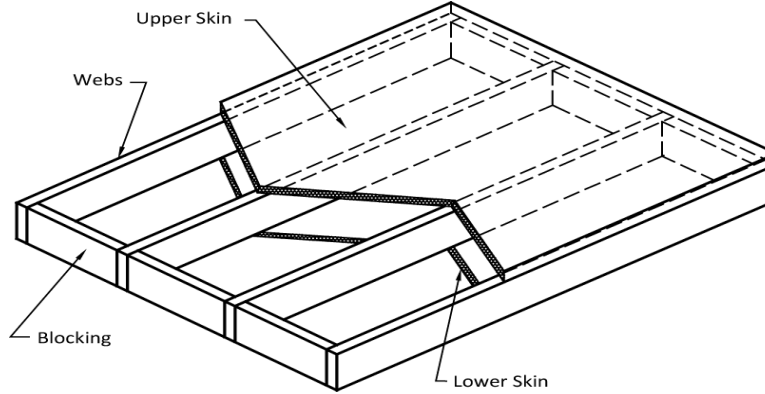


Figure 6.11.13 Stressed skin panel construction (Single Skin Or Double Skin)

11.15 Trussed Rafter

11.15.1 General

A roof truss is essentially a plane structure which is very stiff in the plane of the members, that is, the plane in which it is expected to carry loads, but very flexible in every other direction. Thus it can virtually be seen as a deep, narrow girder liable to buckling and twisting under loads. In order, therefore, to reduce this effect, eccentricity of loading and promote prefabrication for economy, low-pitched trussed rafters are designed with bolt ply/nail ply joints. Plywood as gussets, besides being simple have inherent constructional advantage of grain over solid wood for joints, and a better balance is achievable between the joint strength and the member strength.

Trussed rafters are light weight truss units spaced at close centres for limited spans to carry different types of roof loads. They are made from timber members of uniform thickness fastened together in one plane. The plywood gussets may be nailed or glued to the timber to form the joints. Conceptually a trussed rafter is a triangular pin jointed system, traditionally meant to carry the combined roof weight, cladding services and wind loads. There is considerable scope for saving timber by minimizing the sections through proper design without affecting structural and functional requirements.

Trussed rafters require to be supported only at their ends so that there is no need to provide load bearing internal walls, purlins, etc. and in comparison with traditional methods of construction they use less timber and considerably reduces site labour. Mass production of reliable units can be carried out under workshop controls.

11.15.2 Design

Trussed rafter shall be designed to sustain the dead and imposed loads specified in Chapter 2 Part 6 and the combinations expected to occur. Extra stresses/deflections during handling, transportation and erection shall be taken care of. Structural analysis, use of load-slip and moment, rotation characteristics of the individual joints may be used if feasible. Alternatively the maximum direct force in a member maybe assessed to be given by an idealized pin-jointed framework, fully loaded with maximum dead and imposed load in the combination in which they may reasonably be expected to occur.

11.15.3 Timber

The species of timber including plantation grown species which can be used for trussed rafter construction and permissible stresses thereof shall be in accordance with Table 6.11.1. Moisture contents to be as per zonal requirements in accordance with Sec 11.4.4.

11.15.4 Plywood

Boiling water resistant (BWR) grade preservative treated plywood shall be used. Introduction of a plywood gusset simplifies the jointing and in addition provides rigidity to the joint. Preservation of plywood and other panel products shall be done in accordance with good practice prescribed by Bangladesh Forest Research Institute, Chittagong.

11.16 Structural Sandwiches

11.16.1 General

Sandwich constructions are composites of different materials including wood based materials formed by bonding two thin facings of high strength material to a light weight core which provides a combination of desirable properties that are not attainable with the individual constituent materials (Figure 6.11.14). The thin facings are usually of strong dense material since that are the principal load carrying members of the construction. The core must be stiff enough to ensure the faces remain at the correct distance apart. The sandwiches used as structural elements in building construction shall be adequately designed for their intended services and shall be fabricated only where there are adequate facilities for gluing or otherwise bonding cores to facings to ensure a strong and durable product. The entire assembly provides a structural element of high strength and stiffness in proportion to its mass.

Non-structural advantages can also be derived by proper selection of facing and core material for example, an impermeable facings can be used to serve as a moisture barrier for walls and roof panels and core may also be selected to provide thermal and/or acoustic insulation, fire resistance, etc., besides the dimensional stability.

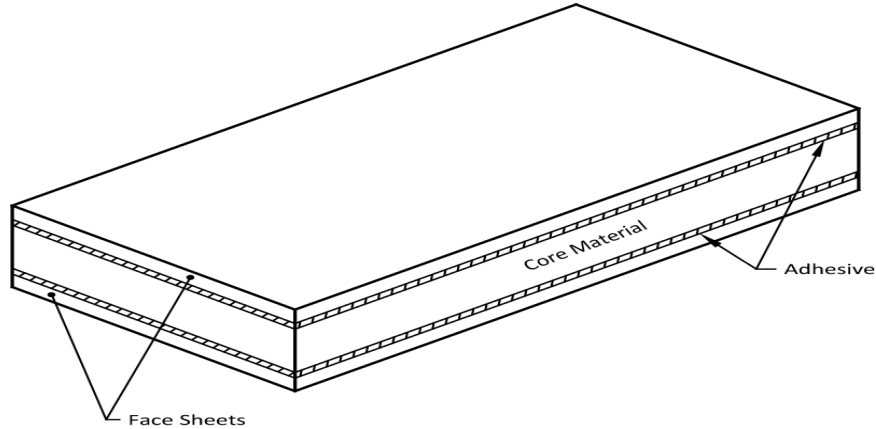


Figure 6.11.14 Sandwich construction in structural applications

11.16.2 Cores

Sandwich cores shall be of such characteristics as to give to the required lateral support to the stressed facings to sustain or transmit the assumed loads or stresses. Core generally carries shearing loads and to support the thin facings due to compressive loads. Core shall maintain the strength and durability under the conditions of service for which their use is recommended. A material with low E and small shear modulus may be suitable.

11.16.3 Facings

Facings shall have sufficient strength and rigidity to resist stresses that may come upon them when fabricated into a sandwich construction. They shall be thick enough to carry compressive and tensile stresses and to resist puncture or denting that maybe expected in normal usages.

11.16.4 Designing

Structural designing may be comparable to the design of I-beams, the facings of the sandwich represent the flanges of the I-beam and the sandwich core I-beam web.

11.16.5 Tests

Tests shall include, as applicable, one or more of the following:

- (a) Flexural strength and stiffness,
- (b) Edge-wise compressions,
- (c) Flat-wise compression,
- (d) Shear in flat-wise plane,
- (e) Flat-wise tensions,
- (f) Flexural creep (creep behaviour of adhesive),
- (g) Cantilever vibrations (dynamic property), and
- (h) Weathering for dimensional stability.

11.17 Lamella Roofing**11.17.1 General**

The Lamella roofing offers an excellent architectural edifice in timber, amenable to prefabrication, light weight structure with high central clearance. It is essentially an arched structure formed by a system of intersecting skewed arches built-up of relatively short timber planks of uniform length and cross-section. Roof is designed as a two hinged arch with a depth equal to the depth of an individual lamella and width equal to the span of the building. The curved lamellas (planks) are bevelled and bored at the ends and bolted together at an angle, forming a network (grid) pattern of mutually braced and stiffened members (Figure 6.11.15). The design shall be based on the balanced or unbalanced assumed load distribution used for roof arches. Effect of deformation or slip of joints under load on the induced stresses shall be considered in design. Thrust components in both transverse and longitudinal directions of the building due to skewness of the lamella arch shall be adequately resisted. Thrust at lamella joints shall be resisted by the moment of inertia in the continuous lamella and roof sheathing (decking) of lamella roofing. The interaction of arches in two directions adds to the strength and stability against horizontal forces. For design calculations several assumption tested and observed derivations, long-duration loading factors, seasoning advantages and effects of defects are taken into account.

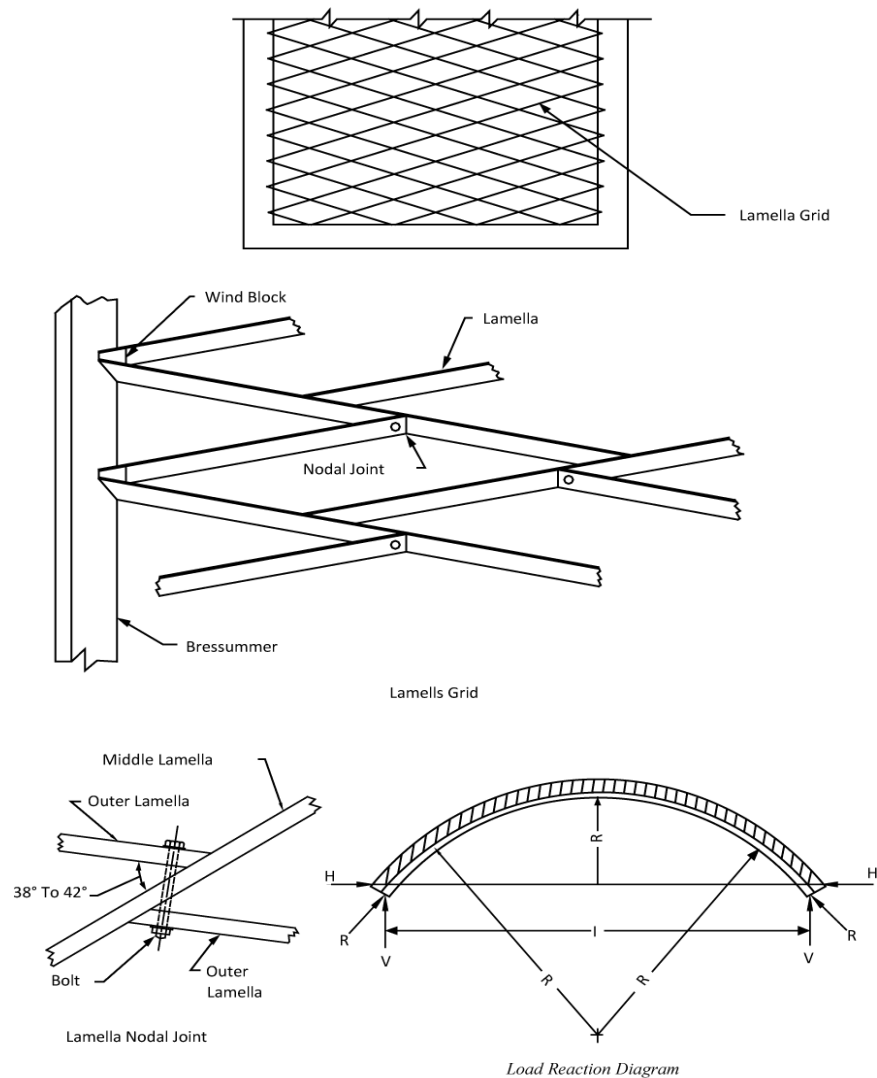


Figure 6.11.15: Typical arrangement of lamella roofing

11.17.2 Lamellas

Planking shall be of a grade of timber that is adequate in strength and stiffness to sustain the assumed loads, forces, thrust and bending moments generated in Lamella roofing. Lamella planks shall be seasoned to a moisture content approximating that they will attain in service. Lamella joints shall be proportioned so that allowable stresses at bearings of the non-continuous lamellas on the continuous lamellas or bearings under the head or washer of bolts are not exceeded.

11.17.3 Construction

Design and construction of lamella roofs in India assumes the roof surfaces to be cylindrical with every individual lamella an elliptic segment of an elliptical arch of constant curved length but of different curvature. Lamella construction is thus more of an art than science as there is no analytical method available for true generation of schedule of cutting lengths and curvature of curved members forming the lamella grid. Dependence of an engineer on the practical ingenuity of master carpenter is almost final. All the lamella joints shall be accurately cut and fitted to give full bearing without excessive deformation or slip. Bolts at lamella splices shall be adequate to hold the members in their proper position and shall not be over tightened to cause bending of the lamellas or mashing of wood under the bolt heads. Connection of lamellas to the end arches shall be adequate to transmit the thrust or any other force. Sufficient false work or sliding jig shall be provided for the support of lamella roof during actual construction/erection.

11.18 Nail and Screw Holding Power of Timber

11.18.1 General

One of the most common ways of joining timber pieces to one another is by means of common wire nails and wood screws. Timber is used for structural and nonstructural purposes in form of scantlings, rafters, joists, boarding, crating and packing cases, etc. needing suitable methods of joining them. Nevertheless it is the timber which holds the nails or screws and as such pulling of the nails/screws is the chief factor which come into play predominantly. In structural nailed joints, nails are essentially loaded laterally, the design data for which is already available as standard code of practice. Data on holding power of nails/screws in different species is also useful for common commercial purposes. The resistance of mechanical fastenings is a function of the specific gravity of wood, direction of penetration with respect to the grain direction, depth of penetration and the diameter of fastener assuming that the spacing of fasteners should be adequate to preclude splitting of wood.

11.18.2 Nails

Nails are probably the most common and familiar fastener. They are of many types and sizes in accordance with the accepted standards.

In general nails give stronger joints when driven into the side grain of wood than into the end grain. Nails perform best when loaded laterally as compared to axial withdrawal so the nailed joints should be designed for lateral nail bearing in structural design. Information on withdrawal resistance of nails is available and joints may be designed for that kind of loading as and when necessary.

11.18.3 Screw

Next to the hammer driven nails, the wood screw may be the most commonly used fastener. Wood screws are seldom used in structural work because of their primary advantage is in withdrawal resistance, for example, for fixing of ceiling “boards to joists, purlin cleats, besides the door hinges etc. They are of considerable structural importance in fixture design and manufacture. Wood screws are generally finished in a variety of head shapes and manufactured in various lengths for different screw diameters or gauges.

The withdrawal resistance of wood screws is a function of screw diameter, length of engagement of the threaded portion into the member, and the specific gravity of the species of wood. Withdrawal load capacity of wood screws are available for some species and joints may be designed accordingly. End grain load on wood screws are unreliable and wood screws shall not be used for that purpose.

11.19 Protection Against Termite Attack in Buildings

11.19.1 Two groups of organisms which affect the mechanical and aesthetic properties of wood in houses are fungi and insects. The most important wood destroying insects belong to termites and beetles. Of about 250 species of wood destroying termites recorded in India, not more than a dozen species attack building causing about 90 percent of the damage to timber and other cellulosic materials. Subterranean termites are the most destructive of the insects that infest wood in houses justifying prevention measures to be incorporated in the design and construction of buildings.

11.19.1.1 Control measures consist in isolating or sealing off the building from termites by chemical and non-chemical construction techniques. It is recognized that 95 percent damage is due to internal travel of the termites from ground upwards rather than external entry through entrance thus calling upon for appropriate control measures in accordance with good practices.

11.19.2 Chemical Methods

Termites live in soil in large colonies and damage the wooden structure in the buildings by eating up the wood or building nests in the wood. Poisoning the soil under and around the building is a normal recommended practice. Spraying of chemical solution in the trenches of foundations in and around walls, areas under floors before and after filling of earth, etc. In already constructed building the treatment can be given by digging trenches all around the building and then giving a liberal dose of chemicals and then closing the trenches.

11.19.3 Wood Preservatives

Natural resistance against organisms of quite a few wood species provides durability of timber without special protection measure. It is a property of heartwood while sapwood is normally always susceptible to attack by organisms. Preservatives should be well applied with sufficient penetration into timber. For engineers, architects and builders, the following are prime considerations for choice of preservatives:

- (a) Inflammability of treated timber is not increased and mechanical properties are not decreased;
- (b) Compatibility with the glue in laminated wood, plywood and board material;
- (c) Water repellent effect is preferred;
- (d) Possible suitability for priming coat;
- (e) Possibility of painting and other finishes;
- (f) Non-corrosive nature fasteners; and
- (g) Influence on plastics, in case of metal rubber, tiles and concrete.

11.19.4 Constructional Method

Protection against potential problem of termite attack can simply be carried out by ordinary good construction which prevents a colony from gaining access by:

- (a) periodic visual observations on termite galleries to be broken off;
- (b) specially formed and properly installed metal shield at plinth level; and
- (c) continuous floor slabs, apron floors and termite grooves on periphery of buildings.

PART VI
Chapter 12
Ferrocement Structures

12.1 Scope

This Chapter covers selection, standards and testing of ferrocement materials, design criteria and approaches, construction methods, and maintenance and repair procedures of ferrocement structures. The provisions of this Chapter are consistent with those of Chapter 6, except for the special requirements of ferrocement, such as reinforcement cover and limits on deflection.

12.2 Terminology

12.2.1 Reinforcement Parameters

For the purpose of this Chapter, the following parameters characterizing the reinforcement in ferrocement shall have the definitions given:

ARMATURE	The total reinforcement system or skeletal reinforcement and mesh for a ferrocement element.
EFFECTIVE MODULUS OF THE REINFORCEMENT	For welded steel meshes, effective modulus of the reinforcing system, E_r shall be taken equal to the elastic modulus of the steel wires. For other meshes, E_r shall be determined from tensile tests on the ferrocement composite as specified in Sec 12.8.
LONGITUDINAL DIRECTION	The roll direction (longer direction) of the mesh as produced in plant (see Figure 6.12.1).
SKELETAL REINFORCEMENT	A planar framework or widely spaced tied steel bars that provides shape and support for layers of mesh or fabric attached to either side.
SPECIFIC SURFACE OF REINFORCEMENT	Specific Surface of Reinforcement S_r is the total bonded area of reinforcement (interface area or area of the steel that comes in contact with the mortar) divided by the volume of the composite. For a ferrocement plate of width b and depth h , the specific surface of reinforcement can be computed from $S_r = \frac{\sum o}{bh}$, in which $\sum o$ is the total surface area of bonded reinforcement per unit length.

The relation between S_r and V_f when square grid wire meshes are used is $S_r = \frac{4V_f}{d_b}$, Where, d_b is the diameter of the wire.

For other types of reinforcement, such as expanded metal, S_{rl} and S_{rt} may be unequal.

SPRITZING	Spraying or squirting a mortar onto a surface.
TRANSVERSE DIRECTION	Direction of mesh normal to its longitudinal direction; also width direction of mesh as produced in plant (see Figure 6.12.1)
VOLUME FRACTION OF REINFORCEMENT (V_f)	Volume fraction of reinforcement is the total volume of reinforcement divided by the volume of composite (reinforcement and matrix). For a composite reinforced with meshes with square openings, V_f shall be equally divided into V_{fl} and V_{ft} for the longitudinal and transverse directions, respectively. For other types of reinforcement, such as expanded metal, V_{fl} and V_{ft} may be unequal. Procedures for computation of V_f are shown in Appendix U.

12.2.2 Illustration of Terminologies

Figure 6.12.1 illustrates the various terminologies used in Sec 12.2.1.

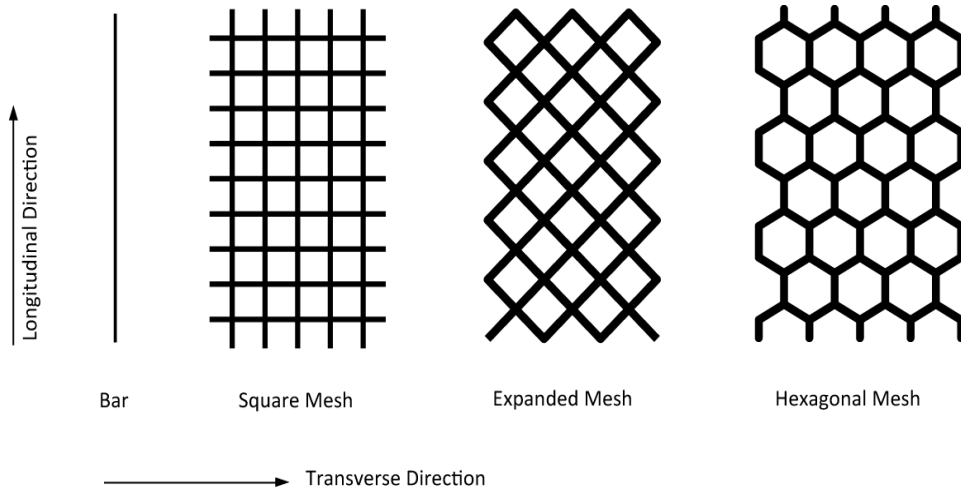


Figure 6.12.1 Assumed Longitudinal and Transverse Directions of Reinforcement

12.2.3 Notation and Symbols

A_c = Cross-sectional area of ferrocement composite

A_s	=	Total effective cross-sectional area of reinforcement in the direction considered, $A_s = \sum_{i=1}^N A_{si}$
A_{si}	=	Effective cross-sectional area of reinforcement of mesh layer i in the direction considered
C_c	=	Resultant of the compressive stress block in ferrocement
C_{si}	=	Compressive force in ferrocement layer i
E_c	=	Elastic modulus of mortar matrix
E_r	=	Effective modulus of the reinforcing system
M_n	=	Nominal moment strength
N_n	=	Nominal tensile strength
N	=	Number of layers of mesh; nominal resistance
S_r	=	Specific surface of reinforcement
S_{rl}	=	Specific surface of reinforcement in the longitudinal direction
S_{rt}	=	Specific surface of reinforcement in the transverse direction
T_{si}	=	Tensile force in the ferrocement layer i
V_f	=	Volume fraction of reinforcement
U	=	Minimum required design strength
V_{fi}	=	Volume fraction of reinforcement for mesh layer i
V_{fl}	=	Volume fraction of reinforcement in the longitudinal direction
V_{ft}	=	Volume fraction of reinforcement in the transverse direction
b	=	Width of ferrocement section
c	=	Distance from extreme compression fibre to neutral axis
d_c	=	Clear cover of mortar over first layer of mesh
d_b	=	Diameter or equivalent diameter of reinforcement used

d_i	=	Distance from extreme compression fibre to centroid of reinforcing layer i
f'_c	=	Specified compressive strength of ferrocement mortar
f_y	=	Yield strength of mesh reinforcement or reinforcing bars
h	=	Thickness of ferrocement section
n_r	=	Modular ratio of reinforcement
s	=	Mesh opening or size
β_1	=	Factor defining depth of rectangular stress block
η	=	Global efficiency factor of embedded reinforcement in resisting tension or tensile bending loads
η_l	=	Value of η when the load or stress is applied along the longitudinal direction of the mesh system or rod reinforcement
η_t	=	Value of η when the load or stress is applied along the transverse direction of the mesh reinforcement system or rod reinforcement
η_θ	=	Value of η when the load or stress is applied along a direction forming an angle with the longitudinal direction
ϵ_{si}	=	Strain of mesh reinforcement at layer i
ϵ_y	=	Nominal yield strain of mesh reinforcement = $\frac{f_y}{E_r}$
Σ_o	=	Total surface area of bonded reinforcement per unit length
ϕ	=	Strength reduction factor.

12.3 Materials

The material used in ferrocement consists primarily of mortar made of Portland cement, water and aggregate and the reinforcing mesh.

12.3.1 Cement

The cement shall comply with BDS EN 197-1 or an equivalent standard. The cement shall be fresh, of uniform consistency, and free of lumps and foreign matter. It shall be stored under dry conditions for as short a duration as possible.

The choice of a particular cement shall depend on the service conditions. Service conditions can be classified as electrochemically passive or active. Land based structures such as ferrocement silos, bins, and water tanks can be considered as passive structures, except when in contact with sulphate bearing soils, in which case the use of sulphate resistant cement, such as ASTM Type II or Type V, may be necessary.

Blended hydraulic cement conforming to ASTM C595 Type 1 (PM), IS, 1 (SM), IS-A, IP, or IP-A can also be used.

Mineral admixtures, such as fly ash, silica fumes, or blast furnace slag, may be used to maintain a high volume fraction of fine filler material. When used, mineral admixtures shall comply with ASTM C618 and C989. In addition to the possible improvement of flow ability, these materials also benefit long term strength gain, lower mortar permeability, and in some cases improved resistance to sulphates and chlorides.

12.3.2 Aggregates

Aggregate used in ferrocement shall be normal weight fine aggregate (sand). It shall comply with ASTM C33 requirements (for fine aggregate) or an equivalent standard. It shall be clean, inert, free of organic matter and deleterious substances, and relatively free of silt and clay.

The grading of fine aggregate shall be in accordance with the guidelines of Table 6.12.1. However, the maximum particle size shall be controlled by construction constraints such as mesh size and distance between layers. A maximum particle size passing sieve No. 16 (1.18 mm) may be considered appropriate in most applications. The sand shall be uniformly graded unless trial testing of mortar workability permits the use of a gap graded sand.

Aggregates that react with the alkalis in cement shall be avoided. When aggregates may be reactive, they shall be tested in accordance with ASTM C227. If proven reactive, the use of a pozzolan to suppress the reactivity shall be considered and evaluated in accordance with ASTM C441.

Table 6.12.1: Guidelines for Grading of Sand

Sieve Size U.S. Standard Square Mesh	Percent Passing by Weight
No. 8 (2.36 mm)	80 - 100
No. 16 (1.18 mm)	50 - 85
No. 30 (0.60 mm)	25 - 60
No. 50 (0.30 mm)	10 - 30
No. 100 (0.15 mm)	2 - 10

12.3.3 Water

The mixing water shall be fresh, clean, and potable. The water shall be relatively free from organic matter, silt, oil, sugar, chloride, and acidic material. It shall have a $\text{pH} \geq 7$ to minimize the reduction in pH of the mortar slurry. Salt water is not acceptable, but chlorinated drinking water can be used.

12.3.4 Admixtures

Conventional and high range water reducing admixtures (super plasticizers) shall conform to ASTM C494. Water reducing admixtures may be used to achieve an increase in sand content for the same design strength or a decrease in water content for the same workability. Decreases in water content result in lower shrinkage and less surface crazing. Retarders may be used in large time consuming plastering projects, especially in hot weather conditions.

If water tightness is important, such as in water or liquid retaining structures, special precautions shall be taken. To achieve water tightness, the water cement ratio shall preferably be kept below 0.4, crack widths limited (see Sec 12.4) and, if necessary, waterproofing coatings applied (see Sec 12.6.3).

Mineral admixtures such as fly ash (ASTM C618) can be added to the cement to increase workability and durability. Normally, 15 percent of the cement can be replaced with mineral admixtures without appreciably reducing the strength. Pozzolanic admixtures may be added to replace part of the fine aggregates to improve plasticity. The tendency for some natural pozzolans to absorb water and thus adversely affect hydration of the cement phase shall be checked by measuring the water of absorption.

A quality matrix can be obtained without using any admixtures if experience has shown its applicability. Admixtures not covered in ASTM standards shall not be used.

12.3.5 Mix Proportioning

The ranges of mix proportions for common ferrocement applications shall be sand cement ratio by weight, 1.5 to 2.5, and water cement ratio by weight, 0.35 to 0.5. The higher the sand content, the higher the required water content to maintain the same workability. Fineness modulus of the sand, water cement ratio, and sand cement ratio shall be determined from trial batches to ensure a mix that can infiltrate (encapsulate) the mesh and develop a strong and dense matrix.

The moisture content of the aggregate shall be considered in the calculation of required water. Quantities of materials shall preferably be determined by weight. The mix shall be as stiff as possible, provided it does not prevent full penetration of the mesh. Normally the slump of fresh mortar shall not exceed 50 mm. For most applications, 28 day compressive strength of 75×150 mm moist cured cylinders shall not be less than 35 N/mm^2 .

12.3.6 Reinforcement

The reinforcement shall be clean and free from deleterious materials such as dust, loose rust, coating of paint, oil, or similar substances.

Wire mesh with closely spaced wires is the most commonly used reinforcement in ferrocement. Expanded metal, welded wire fabric, wires or rods, prestressing tendons, and discontinuous fibers may also be used in special applications or for reasons of performance or economy.

12.3.6.1 Wire mesh

Reinforcing meshes for use in ferrocement shall be evaluated for their susceptibility to take and hold shape as well as for their strength performance in the composite system. Common types and sizes of steel meshes that may be used in ferrocement are provided in Appendix U.

12.3.6.2 Welded wire fabric

Welded wire fabric may be used in combination with wire mesh to minimize the cost of reinforcement. The fabric shall conform to ASTM A496 and A497. The minimum yield strength of the wire measured at a strain of 0.035 shall be 410 N/mm^2 . Welded wire fabric normally contains larger diameter wires (2 mm or more) spaced at 25 mm or more.

12.3.6.3 Expanded metal mesh reinforcement

Expanded mesh reinforcement (metal lath), formed by slitting thin gauge steel sheets and expanding them in a direction perpendicular to the slits may be used in ferrocement. Punched or otherwise perforated sheet products may also be used. Expanded mesh is suitable for tanks if proper construction procedures are adopted.

12.3.6.4 Bars, wires and prestressing strands

Reinforcing bars and prestressing wires or strands may be used in combination with wire meshes in relatively thick ferrocement elements or in the ribs of ribbed or T-shaped elements.

Reinforcing bars shall conform to ASTM A615, A616 or A617. Reinforcing bars shall be steel with a minimum yield strength of 410 N/mm² and a tensile strength of about 615 N/mm². Prestressing wires and strands, whether prestressed or not shall conform to ASTM A421 and A416, respectively.

12.3.6.5 Discontinuous fibres and nonmetallic reinforcement

Fibre reinforcement consisting of irregularly arranged continuous filaments of synthetic or natural organic fibres such as jute and bamboo may be used in ferrocement. If organic materials are used, care shall be taken to conduct appropriate investigations to ensure the strength and durability of the finished ferrocement product.

12.4 Design

12.4.1 General Principles and Requirements

12.4.1.1 The analysis of a ferrocement cross-section subject to either bending, or to bending and axial load, whether based on strength or working stresses, is similar to the analysis of a reinforced concrete beam or column having several layers of steel (Figure 6.12.2).

12.4.1.2 In the design of ferrocement structures, members shall be proportioned for adequate strength as per the provisions of this Chapter using load factors and strength reduction factors specified in Chapter 6.

12.4.1.3 Ferrocement members may alternatively be designed using service loads and permissible service load stresses in accordance with the provisions of Sec 12.4.3.

12.4.1.4 All members shall also be designed to satisfy serviceability criteria in accordance with the provisions of Sec 12.4.4.

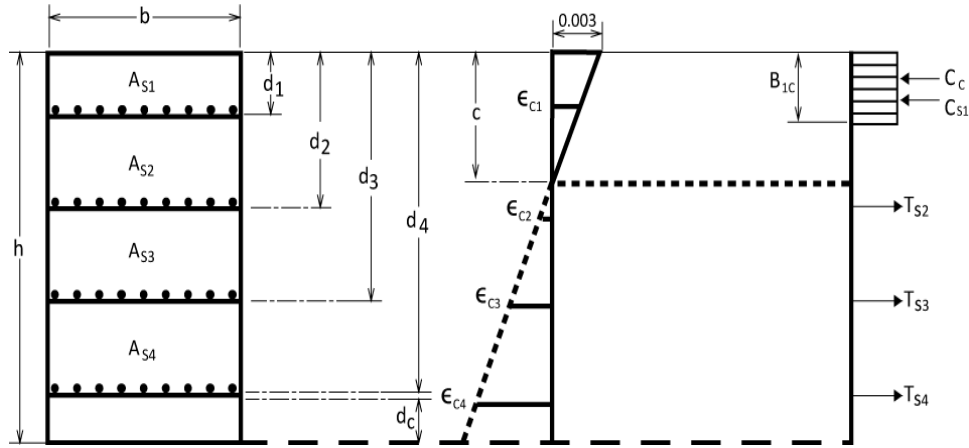


Figure 6.12.2 Strain and force distribution at ultimate in a ferrocement section under bending

12.4.2 Strength Requirements

Ferrocement structures and structural members shall have a design strength at all sections at least equal to the required strengths for the factored load and load combinations stipulated in Chapter 1, General Design Requirements. Required strength U to resist dead load D and live load L shall be determined in accordance with the provisions of Sec 2.7.3.1 Chapter 2 of this Part.

Design strength provided by a member or cross-section in terms of axial load, bending moment, shear force, or stress shall be taken as the nominal strength calculated in accordance with requirements and assumptions of Sec 6.1.4 Chapter 6 multiplied by the strength reduction factor ϕ to satisfy the general relationship.

$$U \leq \phi N \quad (6.12.1)$$

Where, U is the factored load (equal to the minimum required design strength), N is the nominal resistance, and ϕ is a strength reduction factor defined in Sec 6.2.3.1.

Design strength for the mesh reinforcement shall be based on the yield strength f_y of the reinforcement but shall not exceed 690 N/mm^2 . Design yield strengths of various mesh reinforcement shall be in accordance with Table 6.12.2. These shall be used for design only when test data are not available. When tests for determination of yield strength are needed, they shall be conducted in accordance with Sections 12.8.2.3 and 12.8.2.4.

Table 6.12.2: Minimum Values of Yield Strength and Effective Modulus for Steel Meshes and Bars Recommended for Design

Strength and Effective Modulus		Woven Square Mesh	Welded Square Mesh	Hexagonal Mesh	Expanded Metal Mesh	Longitudinal Bars
Yield Strength	f_y N/mm ²	450	450	310	310	410
Effective Modulus	$(E_r)_{long}$ (N/mm ²)	138000	200000	104000	138000	200000
	$(E_r)_{tran}$ (N/mm ²)	165000	200000	69000	69000	-

12.4.2.1 Flexure

The strain distribution at nominal moment resistance shall be assumed to be linear, and a rectangular stress block shall be used in computing the resultant compressive force acting on the concrete.

- (a) Assumptions - Strength design of ferrocement members for flexure and axial loads shall be based on the following assumptions and on satisfaction of equilibrium and compatibility of strains.
- (i) Strain in reinforcement and mortar (concrete) shall be assumed directly proportional to the distance from the neutral axis.
 - (ii) Maximum strain at extreme mortar (concrete) compression fibre shall be assumed equal to 0.003.
 - (iii) Stress in reinforcement below specified yield strength f_y shall be taken as E_r times steel strain. For strains greater than that corresponding to f_y stress in reinforcement shall be considered independent of strain and equal to f_y .
 - (iv) Tensile strength of mortar (concrete) shall be neglected in flexural strength calculations.
 - (v) Relationship between mortar (concrete) compressive stress distribution and mortar (concrete) strain may be considered satisfied by the use of the equivalent rectangular concrete stress distribution.

- (b) Effective area of reinforcement - The area of reinforcement per layer of mesh considered effective to resist tensile stresses in a cracked ferrocement section shall be determined as follows:

$$A_{si} = \eta V_{fi} A_c \quad (6.12.2)$$

Where,

- A_{si} = effective area of reinforcement for mesh layer i
- η = global efficiency factor of mesh reinforcement in the loading direction considered
- V_{fi} = volume fraction of reinforcement for mesh layer i
- A_c = gross cross-sectional area of mortar (concrete) section.

The global efficiency factor η when multiplied by the volume fraction of reinforcement, gives the equivalent volume fraction (or equivalent reinforcement ratio) in the loading direction considered. In effect, it leads to an equivalent (effective) area of reinforcement per layer of mesh in that loading direction.

For square meshes, $\eta = 0.5$ when loading is applied in one of the principal directions. For a reinforcing bar loaded along its axis, $\eta = 1.0$.

In the absence of values derived from tests for a particular mesh system, the values of η given in Table 6.12.3 for common types of mesh and loading direction may be used. The global efficiency factor shall apply whether the reinforcement is in the tension zone or in the compression zone.

The value of $\eta = 0.2$ for expanded metal mesh (Table 6.12.3) may not always be conservative, particularly in thicker sections in flexure with the mesh oriented in the short way diamond. The values in Table 6.12.3 shall be used for sections 50 mm or less in thickness, and tests conducted for global efficiency values for sections more than 50 mm in thickness.

Table 6.12.3: Recommended Design Values of the Global Efficiency Factor of Reinforcement for a Member in Uniaxial Tension or Bending

Global Efficiency Factor	Woven Square Mesh	Welded Square Mesh	Hexagonal Mesh	Expanded Metal Mesh	Longitudinal Bars
Longitudinal, η_1	0.50	0.50	0.45	0.65	1
Transverse η_t	0.50	0.50	0.30	0.20	0
At 45°, $\eta_\theta = 45$	0.35	0.35	0.30	0.30	0.70

12.4.2.2 Tension

The nominal resistance of cracked ferrocement elements subject to pure tensile loading shall be approximated by the load carrying capacity of the mesh reinforcement alone in the direction of loading by the following equation:

$$N_n = A_s f_y \quad (6.12.3)$$

Where,

N_n = nominal tensile load resistance in direction considered

A_s = effective cross-sectional area of reinforcement in direction considered

f_y = yield stress of mesh reinforcement.

The value of A_s is given by

$$A_s = \sum_{i=1}^N A_{si} \quad (6.12.4)$$

Where,

N = number of mesh layers

A_{si} = effective area of reinforcement for mesh layer i .

12.4.2.3 Compression

As a first approximation, the nominal resistance of ferrocement sections subject to uniaxial compression shall be derived from the load carrying capacity of the unreinforced mortar (concrete) matrix assuming a uniform stress distribution of $0.85f'_c$ where f'_c is the design compressive strength of the mortar matrix. However, the transverse component of the

reinforcement can contribute additional strength when square or rectangular wire meshes are used. Expanded mesh contributes virtually no strengthening beyond that achieved by the mortar alone. Slenderness effects of thin sections, which can reduce the load carrying capacity below that based on the design compressive strength shall be considered.

12.4.3 Service Load Design

12.4.3.1 Flexure

For investigation of stresses at service loads, straight line theory (for flexure) shall be used with the following assumptions.

- (a) Strains vary linearly with distance from the neutral axis.
- (b) Stress strain relationships of mortar (concrete) and reinforcement are linear for stresses less than or equal to permissible service load stresses.
- (c) Mortar (concrete) resists no tension.
- (d) Perfect bond exists between steel and mortar (concrete).

To compute stresses and strains for a given loading, the cracked transformed section shall be used. The effective area of each layer of mesh reinforcement shall be determined from Eq. 6.12.2. The same value of modular ratio $n_r = E_r/E_c$, may be used for both tensile and compressive reinforcement. Recommended design values of E_r are given in Table 6.12.2. Once the neutral axis is determined, the analysis shall proceed as for reinforced concrete beams or columns having several layers of steel and subject to pure bending.

12.4.3.2 Allowable tensile stress

The allowable tensile stress in the mesh reinforcement under service conditions shall be taken as $0.60f_y$ where f_y is the yield strength. For liquid retaining and sanitary structures, the allowable tensile stress shall be limited to 200 N/mm^2 . Consideration shall be given to increase the allowable tensile stresses if crack width measurements on a model test indicate that a higher stress will not impair performance.

12.4.3.3 Allowable compressive stress

The allowable compressive stress in either the mortar (concrete) or the ferrocement composite shall be taken as f_c where f_c is the specified compressive strength of the mortar. Measurements of the mortar compressive strength shall be obtained from tests on $75 \text{ mm} \times 150 \text{ mm}$ cylinders.

12.4.4 Serviceability Requirements

Ferrocement structures shall generally satisfy the intent of the serviceability requirements of Chapter 6 except for the concrete cover.

12.4.4.1 Crack width limitations

The maximum value of crack width under service load conditions shall be less than 0.10 mm for noncorrosive environments and 0.05 mm for corrosive environments and/or water retaining structures.

12.4.4.2 Fatigue stress range

For ferrocement structures to sustain a minimum fatigue life of two million cycles, the stress range in the reinforcement shall be limited to 200 N/mm². A stress range of 350 N/mm² shall be used for one million cycles. Higher values may be considered if justified by tests.

12.4.4.3 Corrosion durability

Particular care shall be taken to ensure a durable mortar matrix and optimize the parameters that reduce the risk of corrosion.

12.4.4.4 Deflection limitation

Since ferrocement in thin sections is very flexible and its design is very likely to be controlled by criteria other than deflection, no particular deflection limitation is recommended.

12.4.5 Particular Design Parameters

12.4.5.1 The cover of the reinforcement shall be about twice the diameter of the mesh wire or thickness of other reinforcement used. A smaller cover is acceptable provided the reinforcement is not susceptible to rapid corrosion, the surface is protected by an appropriate coating, and the crack width is limited to 0.05 mm. For ferrocement elements of thickness less than 25 mm, a cover of the order of 2 mm shall be provided.

12.4.5.2 For a given ferrocement cross-section of total thickness h , the mesh opening shall not be larger than h .

12.4.5.3 For nonprestressed water retaining structures the total volume fraction of reinforcement shall not be less than 3.5 percent and the total specific surface of reinforcement shall not be less than 0.16 mm²/mm³.

12.4.5.4 In computing the specific surface of the reinforcement, the contribution of fibres added to the matrix shall be considered, while the fibre contribution may be ignored in computing the volume fraction of reinforcement.

12.4.5.5 If skeletal reinforcement is used, the skeletal reinforcement shall not occupy more than 50 percent of the thickness of the ferrocement composite.

12.4.5.6 For a given volume fraction of reinforcement, better performance in terms of crack widths, water tightness, and ductility may be achieved by uniform distribution of the reinforcement throughout the thickness and by increasing its specific surface. A minimum of two layers of mesh shall be acceptable, but more than two layers of reinforcement are preferable.

12.4.6 Design Aids

The following nondimensional equation can be used to predict the nominal moment strength of ferrocement beams subjected to pure bending:

$$\frac{M_n}{f'_c b h^2 \eta} = 0.005 + 0.422(V_f f_y / f'_c) - 0.0772(V_f f_y / f'_c) \quad (6.12.5)$$

A design graph representing Eq. 6.12.5 is given in Figure 6.12.3.

12.5 Fabrication

12.5.1 General Requirements

This Section specifies the requirements for the mixing, placing, and handling of materials used in ferrocement construction.

12.5.1.1 Planning

Plastering for fabrication with ferrocement shall be continuous through the completion of the job. This requires a large number of workers involved in plastering and in maintaining a constant supply of materials during work, most often in confined work spaces. Adequate bond at cold joints may be achieved through surface roughness or treatment with bonding agents. Retarders may be useful in large time consuming plastering projects, especially in hot weather conditions. Planning for the job shall take into account all these requirements.

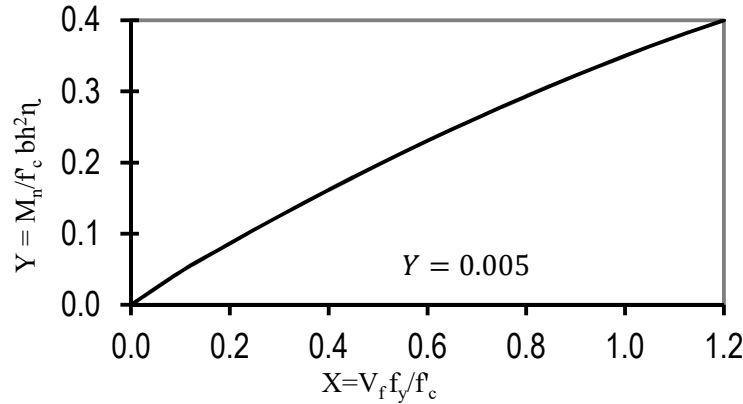


Figure 6.12.3 Chart for strength design of ferrocement in bending

12.5.1.2 Mixing

Any method, including hand mixing, which assures a homogeneous mixture of ingredients shall be satisfactory. Mixing may be accomplished in a mortar mixer with a spiral blade or paddles inside a stationary drum or in a pan type mixer. The use of rotating drum mixers with fins affixed to the sides shall not be permitted. Mix ingredients shall be carefully batched by weight, including the water, and added or charged in the mixer so that there is no caking. Mix water shall be accurately weighed so that the water cement ratio is controlled. The water cement ratio shall be as low as possible but the sand cement ratio shall be adjusted to provide a fluid mix for initial penetration of the armature followed by a stiffer more heavily sanded mix at the finish. Mortar shall be mixed in batches so that mortar is plastered within an hour after mixing. Retempering of the mortar shall be prohibited.

12.5.1.3 Mortar placement

Mortar shall generally be placed by hand plastering. In this process, the mortar is forced through the mesh. Alternatively, the mortar may be shot through a spray gun device.

12.5.1.4 Finishing

Surfaces shall be finished to assure proper cover to the last mesh layer. The surface finish shall be slightly roughened if a surface coating is to be bonded later.

Surfaces that are too smooth shall be mechanically abraded by sandblasting or other means of mechanical abrasion. Alternatively, such surfaces may be etched with phosphoric acid, provided the residue left by it will not interfere with specified finishes. Mild solutions of muriatic acid may be applied with proper attention to corrosion potential. Additional care shall be taken when plastering around openings.

12.5.1.5 Curing

Moist or wet curing is essential for ferrocement concrete construction. The low water cement ratio and high cement factors create a demand for large quantities of free water in the hydration process, and the amount permitted to evaporate into the air shall be kept to an absolute minimum. The use of fogging devices under a moisture retaining enclosure is desirable. A double layer of soaked burlap covered with polyethylene or a soaker hose may also be used. Continuous wetting of the surface or of wet burlap or the like shall be maintained to avoid dry spots. Curing shall start within a reasonable time after application of the finishing layer.

12.5.2 Construction Methods

All methods shall have high level quality control criteria to achieve the complete encapsulation of several layers of reinforcing mesh by a well compacted mortar or concrete matrix with a minimum of entrapped air. The most appropriate fabrication technique shall be decided on the basis of the nature of the particular ferrocement application, the availability of mixing, handling, and placing machinery, and the skill and cost of available labour. Several recommended construction methods are outlined in the following subsections.

12.5.2.1 Armature system

The armature system is a framework of tied reinforcing bars (skeletal steel) to which layers of reinforcing mesh are attached on each side. Mortar is then applied from one side and forced through the mesh layers towards the other side, as shown in Figure 6.12.4.

The skeletal steel can assume any shape. Diameter of the steel bars depends on the size of the structure. Skeletal steel shall be cut to specified lengths, bent to the proper profile, and tied in proper sequence. Sufficient embedment lengths shall be provided to ensure continuity. For bar sizes 6 mm or less, lap lengths from 230 to 300 mm may be sufficient. The required number of layers of mesh shall be tied to each side of the skeletal steel frame.

12.5.2.2 Closed-mould system

The mortar is applied from one side through several layers of mesh or mesh and rod combinations that have been stapled or otherwise held in position against the surface of a closed mould, i.e. a male mould or a female mould. The mould may remain as a permanent part of the finished ferrocement structure. If removed, treatment with release agents may be needed. The use of the closed mould system represented in Figure 6.12.5 tends to eliminate the use of rods or bars, thus permitting an essentially all mesh reinforcement. It requires that plastering be done from one side only.

12.5.2.3 Integral-mould system

An integral mould is first constructed by application of mortar from one or two sides onto a semi-rigid framework made with a minimum number of mesh layers. This forms, after mortar setting, a rigid but low quality ferrocement mould onto which further layer of reinforcing mesh and mortar shall be applied on both sides. Alternatively, the integral mould may be formed using rigid insulation materials, such as polystyrene or polyurethane, as the core. A schematic description of this system is given in Figure 6.12.6.

12.5.2.4 Open-mould system

In the open-mould system, mortar is applied from one side through layers of mesh or mesh and rods attached to an open mould made of a lattice of wood strips. The form, Figure 6.12.7, is coated with a release agent or entirely covered with polyethylene sheeting (thereby forming a closed but nonrigid and transparent mould) to facilitate mould removal and to permit observation and/or repair during the mortar application process.

This system is similar to the closed-mould system in which the mortar is applied from one side, at least until the mould can be removed. It enables at least part of the underside of the mould to be viewed and repaired, where necessary, to ensure complete and thorough impregnation of the mesh.

12.6 Maintenance

12.6.1 General

Terrestrial structures are susceptible to deterioration from pollutants in ground water and those that precipitate from the air (acid rain). Environmental temperature and humidity variations also affect ferrocement durability and maintenance procedures.

Maintenance shall involve detecting and filling voids, replacing spalled cover, providing protective coatings, and cosmetic treatment of surface blemishes. Due to the thin cover in ferrocement, muriatic acid (hydrochloric acid) shall be used with extreme caution. Phosphoric acid and other nonchloride cleaners shall be the specified alternative (see Sec 12.5.1.4).

Repairs not involving large quantities of materials shall be accomplished by hand. Emphasis shall be placed on the ability of the repair material to penetrate the mesh cage, to fully coat the reinforcing to inhibit corrosion, and to bond to the substrate. Rapid set and strength gain shall be the overriding considerations for emergency repairs. Protective coatings shall bond well and be alkali tolerant, thermally compatible, and resistant to environmental pollutants and ultraviolet radiation, if exposed.

12.6.2 Blemish and Stain Removal

12.6.2.1 General

Since ferrocement is usually less porous than conventional concrete, stains do not penetrate very deep in the mortar matrix. Care shall be taken when preparing the surface not to diminish the thin cover of mortar over ferrocement reinforcement.

12.6.2.2 Construction blemishes

Construction blemishes are often caused by improper selection or use of materials, faulty workmanship, uneven evaporation, and uneven curing. Care shall be exercised to minimize these and the following causes of blemishes in ferrocement.

- (a) Cement from different mills will cause colour variation, although most of the colour in mortar is due to the sand component. Where appearance is critical, care shall be taken to obtain sand from a single source and have it thoroughly washed.
- (b) Mottling results from the use of calcium chloride or high alkali cement combined with uneven curing.
- (c) The use of polyethylene sheet material to cover surfaces promotes uneven curing.
- (d) The water cement ratio affects tone and surface appearance. Low water cement ratio will result in a darker appearance.
- (e) Hard steel toweling densifies the surface, causing more rapid drying and also leaving a darkened surface.

12.6.2.3 Stain removal

Treatment of stains shall be done promptly after discoloration appears. Thorough flushing and brushing with a stiff bristle brush and detergent is the first approach. If this is ineffective, a dilute (about three percent) solution of phosphoric or acetic acid shall be applied. Another chemical treatment that may be considered safe and effective is a 20 to 30 percent solution of di-ammonium citrate, a mild acid that attacks calcium carbonates and calcium hydroxides. This treatment makes the surface more porous and promotes hydration.

When a stain has penetrated too deeply to be removed by surface chemical application and scrubbing, a poultice or a bandage may be needed. A poultice is intended to dissolve the stain and absorb it into the poultice. The poultice is made by mixing one or more chemicals such as a solution of phosphoric acid with a fine inert power such as talc, whiting, hydrated

lime, or diatomaceous earth to form a paste. The paste is spread in a thick layer over the stain and allowed to dry. A bandage may consist of a few layers of cloth or paper toweling soaked in a chemical solution. More than one application of a poultice or bandage may be needed for stubborn stains.

Caution:

Most of the chemicals used to remove stains are toxic and require safeguards against skin contact and inhalation. Whenever acids are used, surfaces shall first be saturated with water or the dissolved stain material may migrate deeper into the concrete and reappear at a later date as efflorescence.

12.6.2.4 Efflorescence:

Efflorescence is caused by deposition of salts on the surface due to the evaporation of migrating water bearing salts from within ferrocement; it is typically associated with a porous ferrocement. Water cement ratio shall be limited to within 0.4 and the mortar well compacted to minimize efflorescence. Voids, if present, may be treated by breaking into with a hammer and replastering. Alternatively, voids may be drilled into with a masonry bit and repaired by injecting a non-shrinking cement grout.

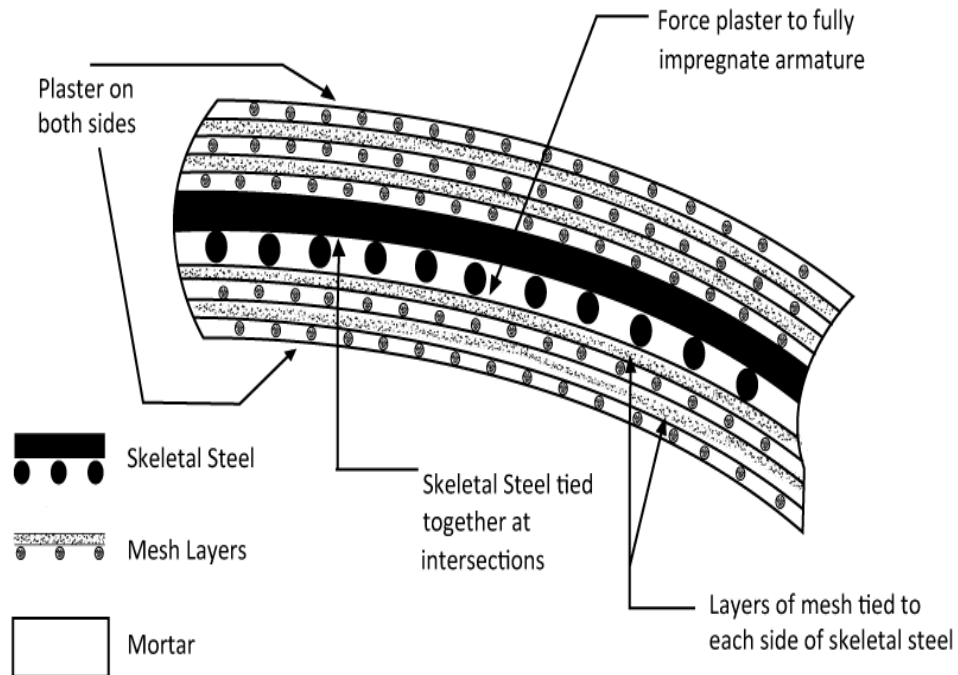


Figure 6.12.4 Armature system

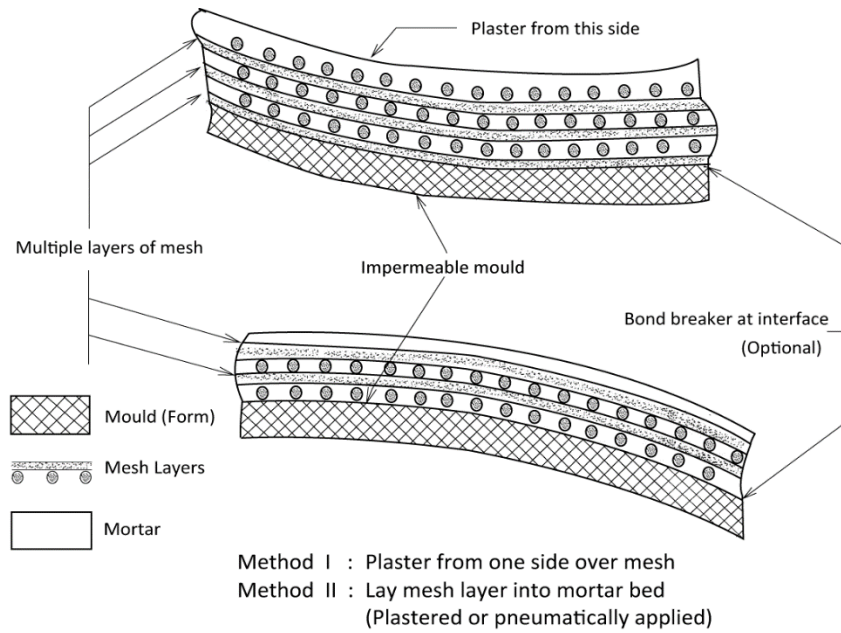


Figure 6.12.5 Closed-mould System

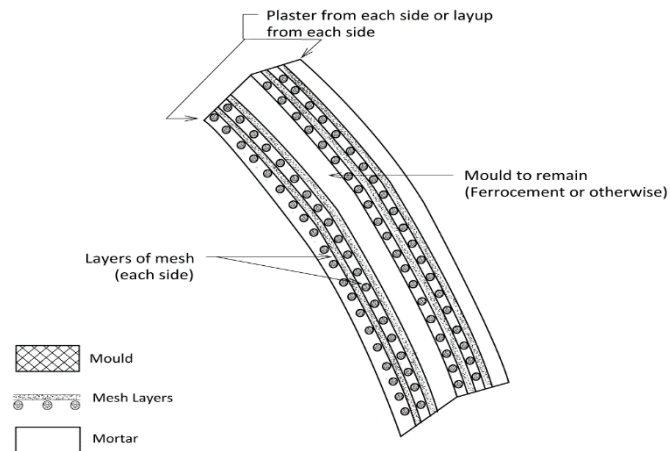


Figure 6.12.6 Integral-mould System

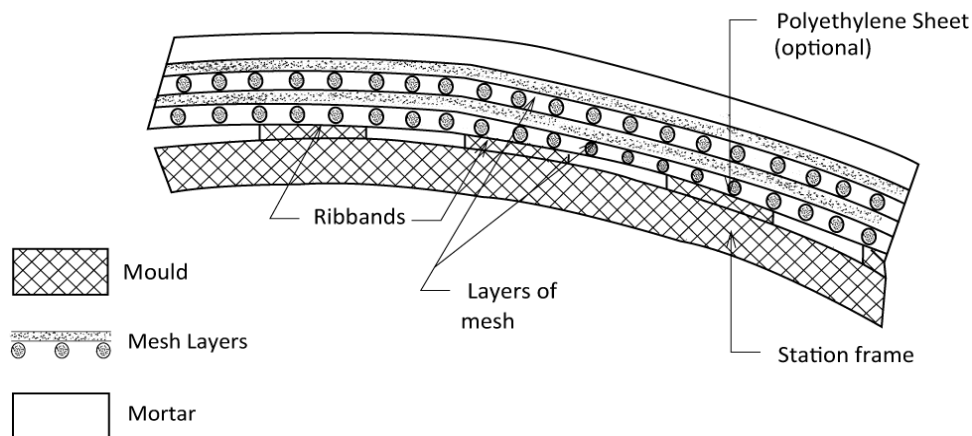


Figure 6.12.7 Open-mould System

12.6.3 Protective Surface Treatments

12.6.3.1 General

Good quality mortar has excellent resistance to weathering. The application of protective surface treatments can improve the performance of ferrocement and extend its useful service life. Surface treatments shall be used to improve appearance, harden the surface, and reduce permeability, thus guarding against the corrosive action of acids, alkaline salts, and organic substances.

12.6.3.2 Hardeners

Hardeners may be used to protect the ferrocement surface or to seal and prepare it for application of paints. When a sodium silicate hardener is used, it shall be diluted with water. The actual proportion of water to be used shall depend on the manufacturer's recommendation. The hardener shall be applied in multiple coats with the first coat being more dilute than the subsequent ones. Each coat must be completely dry before the next coat is applied.

Other hardeners that seal and prepare the surface for application of oil base paints are magnesium fluorosilicate and zinc fluorosilicate. The treatment shall consist of two or more applications. A solution containing about 1 kg of fluorosilicate crystals per 10 litres of water shall be used for the first application; and a solution containing 2.4 kg per 10 litres of water shall be used for subsequent applications. After the last application has dried, the surface shall be brushed and washed with water to remove any crystals that may have formed.

12.6.3.3 Coatings

When resistance to abrasion is desired, ferrocement surfaces may be coated with polyurethanes, especially those furnished in two part mixtures. Coatings formulated from acrylics may be used to provide resistance to sunlight and weathering. Water based acrylic

latex house paints may be used for application to damp surfaces. For any surface opposite a surface sealed with an impermeable coating, an acrylic coating formulated to allow the escape of water vapour shall be specified.

12.7 Damage Repair

12.7.1 Common Types of Damage

Ferrocement structures shall be inspected, as part of a regular maintenance programme, to detect any of the following types of damage. Appropriate repair measures shall then be taken.

12.7.1.1 Delaminations

Delaminations occur when ferrocement splits between layers in laminated constructions due to springing back or bridging of the mesh during construction. Delamination sometimes occurs at or near the neutral axis under impact or flexure when there are many voids in the interior layers. Such areas give off a hollow sound when tapped with a hammer or stroked with a steel bar.

12.7.1.2 Spalls

A spall is defined as a depression resulting when a fragment is detached from a larger mass by a blow, by the action of weather, by pressure, or by expansion within the mass. Spalls shall be considered large when their size exceeds approximately 20 mm in depth or 150 mm in any dimension, and shall be repaired by replastering.

Spalls are usually caused by corrosion of steel, which causes an expansive pressure within the ferrocement. Chlorides in the concrete greatly increase the potential for corrosion of the steel. Under such conditions, continued spalling is likely and the repair of local spall areas may even promote the deterioration of the concrete because of the presence of dissimilar materials.

An area of steel corrosion and chloride contaminated concrete may be considerably larger than the area of spalled concrete, and the full area of contamination rather than the spall itself shall be broken and replastered.

12.7.1.3 Fire damage

Ferrocement may be more susceptible to fire damage than conventional concrete because of the thin cover. If the fire were intense enough to release the amount of chemically bound water in the cement, destroy the bond between the cement and the aggregate, or oxidize the reinforcement, the surface would be charred and spalled so that the damage could be easily identified. Full scale removal and repair shall then be required.

12.7.1.4 Cracks and local fractures

Hairline cracks and crazing due to temperature changes or drying shrinkage in the cover coat do not require repair. Continuous wet curing will cause autogenous healing, and a flexible coating will conceal the crack from view. If cracks are caused by continuing overloads or are due to structural settlement and the cause cannot be removed, replacement or a structural overlay shall be required. Cracks due to occasional impact or overload may be repaired. Local fractures are cracks in which displacement of section has occurred as a result of impact.

12.7.2 Evaluation of Damage

12.7.2.1 Evaluation of damage shall take into consideration its extent, cause, and likelihood of the cause still being active. The method of repair shall be dictated by the type of damage, the availability of special equipment and repair materials, and the level of skill of the workers employed. Economic factors may influence the decision as to whether the repair shall be extensive and permanent, or limited in scope in response to an immediate problem.

12.7.2.2 Repair materials shall bond to the original structure, resist pollutants in the surrounding soil, water or air, and respond the same way to changes in temperature, moisture, and loads. Removal of deteriorated or chloride contaminated mortar trapped within the reinforcing mesh requires a large amount of hand labour, so it may be economical (and better for long term durability) to reconstruct or replace an entire area using the original structure as a form that can be left in place or removed after the overlaid structure has cured. Complete reconstruction shall be undertaken when chloride contamination, mesh corrosion, and deterioration of the mortar are extensive.

12.7.2.3 Testing for damage in ferrocement may be done by tapping with a hammer to break into any voids under the surface, or by drawing a metal bar over the surface and listening for sounds indicating voids or the presence of deteriorated concrete. A high quality ferrocement should produce a bell like sound and resist moderately severe hammer blows without damage.

12.7.3 Surface Preparation for Repair of Damage

12.7.3.1 General

The primary objective is to remove any deteriorated mortar or mortar contaminated with corrosive agents and to provide a surface to which the repair materials can be bonded properly. The rougher the surface, the greater is the area available for bonding.

12.7.3.2 Removal of deteriorated concrete

As a first step in any repair all disintegrated, unsound, and contaminated mortar shall be removed. Saws and chipping hammers used for conventional concrete shall not be used for ferrocement unless large sections are to be completely removed.

Small areas shall be prepared by hand hammering just hard enough to pulverize deteriorated or cracked mortar, but not to the point of damaging the reinforcing mesh.

A pneumatic needle gun may be used for cleaning out broken ferrocement, opening out cracks, and roughening the surface.

Particles of sound mortar embedded in the mesh need not be removed provided they are small enough not to interfere with the penetration of new mortar and they will not project from the finished surface.

12.7.3.3 Reinforcement

Any loose, scaly corrosion revealed on cleaning out the mortar shall be removed by sandblasting, water jet, air blasting, or vacuum methods.

An alternative method for removing rust is to brush naval jelly or spray dilute phosphoric acid over the repair area and flush thoroughly.

Where the mesh cage has been displaced but is still intact, it may be pushed or jacked back in place and supported securely to withstand the pressure of applying the repair material. Where the reinforcement has been torn, the old mesh shall be laced back to close the opening.

When rods supporting the mesh cage are torn they shall be spliced by a 15 diameter overlap of the partner rod or anchored by hooks.

12.7.3.4 Cleaning

Loose particles and dust residue from hammering or sandblasting shall be air jetted or vacuum cleaned if epoxy or methymethacrylate (MMA) is the repair material. Water jetting may be used if the repair is to be made with hydraulic cement or latex modified mortar.

If an air jet is used, the compressor shall be equipped with an oil trap to prevent contamination of the surface. Surface oil or dirt shall be removed by trisodium phosphate or other strong detergents.

12.7.3.5 Cracks

Cracks may be cleaned by hammering out the mortar on each side of the crack and replastered with latex mortar.

If opening the crack is not feasible, epoxy or MMA injection systems shall be attempted in accordance with the product directions. The crack shall be cleaned first with oil free compressed air, and small (about 2 to 3 mm) drill holes shall be made at the highest and lowest points in the crack. The surface between the holes shall be sealed with strong coatings or a pressure pad. Catalyzed epoxy or MMA shall be injected at the lower hole until it comes out at the upper hole. Where latex cement grout is to be used, the interior of the crack shall be thoroughly saturated with water and allowed to drain.

12.7.4 Repair Materials

12.7.4.1 Portland cement and sand

Portland cement used for repair shall conform to the requirements of Sec 12.3.1.

Sand which matches that used in the original construction may be used unless the need for the repair arose because of reactive or contaminated sand. Neat Portland or blended cement paste shall be used to fill small cracks, and a mortar with fine sand shall be used to fill larger cracks or voids. Both shall be used in combination with latex for thin patches and overlays. Larger cracks shall be coated with neat cement slurry, and then dry packed with a very low water cement ratio mortar.

The addition of latex to Portland cement mortar markedly improves bond to the substrate and the tensile strength of the patch. Of the synthetic latexes, polyvinyl acetate and polyvinylidene are unsuitable for wet environments. Acrylics may be used as admixtures to improve bonding and as curing compounds. Acrylic latex in concentrated form shall be diluted to 10 to 20 percent solids and then used as the mixing water for the mortar. Latex mortars may be applied to a damp surface, but the patch shall be allowed to dry thoroughly before being immersed in water.

12.7.4.2 Polymer mortars

Nonlatex polymer mortars shall require the use of surface dried and, preferably, oven dried sand. The monomers have very low viscosity and so shall be mixed with thickening agents to be placed in any area that cannot be sealed tightly. Epoxy resins that are moisture tolerant may be used on damp surfaces.

Care shall be exercised in applying polymers or the promoters and hardeners used with them which are toxic.

12.7.4.3 Admixture

Accelerators may be employed where cement alone is the repair material. Since chloride compounds may promote corrosion, nonchloride accelerators shall be preferred for all ferrocement. Emergency repairs of small areas below the waterline with hot plug, which is neat cement moistened to a putty consistency with a concentrated solution of calcium chloride may be permitted. The hot plug may be carried in the hand or in a plastic bag to the site of the leak, pressed into the hole, and held a few minutes until set. Permanent repair shall be accomplished as soon as possible using materials without chlorides.

12.7.5 Repair Procedure

12.7.5.1 Mixing

Small quantities of materials required for ferrocement repairs may be hand mixed on flat surface or in a tray using premixed dry ingredients. For large quantities, a plaster or pan mixer rather than a rotating drum type mixer shall be used. For machine mixing water shall be put in first; then the cement, to form slurry; then the pozzolan, if used; and finally, enough sand to bring the mortar to the desired degree of workability.

The consistency of the mortar shall be selected according to the nature of the repair. A slurry of cream consistency shall be used first to paint the moistened edges of the repair area, fill cracks or small voids, and thoroughly coat all the interior mesh and rods. After this, more sand shall be added until the mortar is stiff enough to hold its shape when brought out flush with the finished surface.

To avoid excessive amounts of entrained air, mortars containing acrylics or epoxies shall not be mixed longer than two minutes. They shall be applied within thirty minutes of mixing.

12.7.5.2 Full depth repair

When both faces are accessible, a fluid mortar shall be pushed through the mesh cage from one side until an excess appears on the opposite face. This excess shall then be pushed back and finished flush. A vibrating float or trowel may be used to place and finish a very stiff mortar. Pencil type vibrators shall not be used.

12.7.5.3 Partial depth patches

The area to be patched shall first be saturated with water, then air blown or blotted free of standing water until only surface-moist. Cement slurry of not more than 0.4 water cement ratio and of paint like consistency shall be brushed over the whole area and into any

openings in the mesh. This shall be immediately followed by a heavily sanded mortar of the same water cement ratio, which shall be vibrated or tamped into the patch and finished flush.

12.7.5.4 Overlays

The substrate shall be prepared in the manner prescribed in Sec 12.7.5.3 for patches. The old surface shall be thoroughly cleaned or scarified by mechanical means and the repair materials shall match the thermal characteristics of the substrate.

Chemical etching shall be followed by mechanical abrasion, unless the surface is flushed with high pressure water jet equipment.

For thin overlays, velocity placement such as spritzing or casting by hand, and shotcreting, shall be used.

12.7.5.5 Shotcrete

Shotcrete may be used in ferrocement repair when a large area is involved. Small, low cost portable plaster pumps operating on the Moyno progressive cavity principle with a rotor inside a stator tube shall be adequate for both original ferrocement construction and repair.

Shotcrete or plastering equipment may be used for large overlays incorporating additional layers of reinforcing mesh by laminating techniques. Existing surfaces shall be scarified or sandblasted, then saturated with water and allowed to damp dry just before the shotcrete or mortar spray is applied. An initial application of cement slurry is not needed with shotcrete but a latex or wet to dry epoxy bonding compound may be used to advantage with repairs made with plastering equipment.

12.7.5.6 Curing

All Portland cement patches and overlays shall be thoroughly cured unless latex compounds are used to seal the surface and furnish water for hydration. Curing shall be instituted immediately for thin patches and overlays. Several layers of paper or cloth soaked in water and covered with a plastic sheet that is well secured at the edges may be used on patches. A full plastic film covering overlays may be used but it may produce discoloration where it touches the surface.

12.8 Testing

12.8.1 Test Requirement

Tests and observations that are commonly made during the design, construction, and subsequent service life of concrete structures shall also be applicable to ferrocement structures. The test programme shall include

- (a) tests on physical, chemical and mechanical properties of the ferrocement ingredients, such as water purity, sieve analysis, mesh strength etc.,
- (b) control tests for fresh mortar mix, such as slump, air content etc.,
- (c) tests on the mechanical properties of the hardened ferrocement, such as bending, cracking and fatigue strengths, permeability etc., and
- (d) in-service condition tests, such as potential for corrosion, cracking, durability etc.

For predicting the mechanical properties of ferrocement, the tests specified in Sections 12.8.2.1 to 12.8.2.4 shall be conducted.

12.8.2 Test Methods**12.8.2.1 Compressive strength and static modulus of elasticity of mortar**

The compressive strength and static modulus of elasticity of the mortar used for the fabrication of ferrocement shall be determined from 75 mm x 150 mm cylinders tested in accordance with ASTM C39 and C469, respectively.

12.8.2.2 Flexural strength of ferrocement

Ferrocement specimens shall be tested as a simply supported beam with third point loading. The span to depth ratio of the beam specimen shall not be less than 20 and its width shall not be less than six times the mesh opening or wire spacing measured at right angles to the span direction.

12.8.2.3 Tensile properties of the mesh reinforcement

Square or rectangular meshes may be tested directly in tension; hexagonal meshes and expanded metal meshes shall be tested only while encapsulated in mortar. In the latter case

the tensile test shall be performed on the ferrocement material as described in Sec 12.8.2.4 below.

For square and rectangular meshes, the yield strength, elastic modulus, and ultimate tensile strength shall be obtained from direct tensile tests on samples of wires or flat coupons cut from the mesh. The test shall be in accordance with the following guidelines (see also Figure 6.12.8).

- (a) The test specimen shall be prepared by embedding both ends of a rectangular coupon of mesh in mortar over a length at least equal to the width of the sample. The mortar embedded ends shall serve as pads for gripping. The free (not embedded) portion of the mesh shall represent the test sample.
- (b) The width of the test sample shall be not less than six times the mesh opening or wire spacing measured at right angles to the loading direction.
- (c) The length of the test sample shall be not less than three times its width or 150 mm, whichever is larger.
- (d) Measurements of elongations (from which strains are to be computed) shall be recorded over half the length of the mesh sample.
- (e) Yield strain of mesh reinforcement shall be taken as the strain at the intersection of the best straight line fit of the initial portion of the stress strain curve and the best straight line fit of the yielded portion of the stress strain curve, as shown in Figure 6.12.8. The yield stress shall be taken as the stress point on the original stress strain curve at the yield strain found above. The procedure is demonstrated in Figure 6.12.8.

12.8.2.4 Tensile Test of Ferrocement:

Direct tensile tests of ferrocement elements shall be made using rectangular specimens satisfying the same minimum size requirement as those set in Sec 12.8.2.3 for the mesh reinforcement. The test specimens shall be additionally reinforced at their ends for gripping. The middle half of the nongripped (free) portion of the test specimen shall be instrumented to record elongations. A plot of the load elongation curve up to failure shall be used to estimate the effective modulus of the mesh system as well as its yield strength,

ultimate strength, and efficiency factor. The yield strain and corresponding stress shall be determined in accordance with the procedure described in Sec 12.8.2.3.

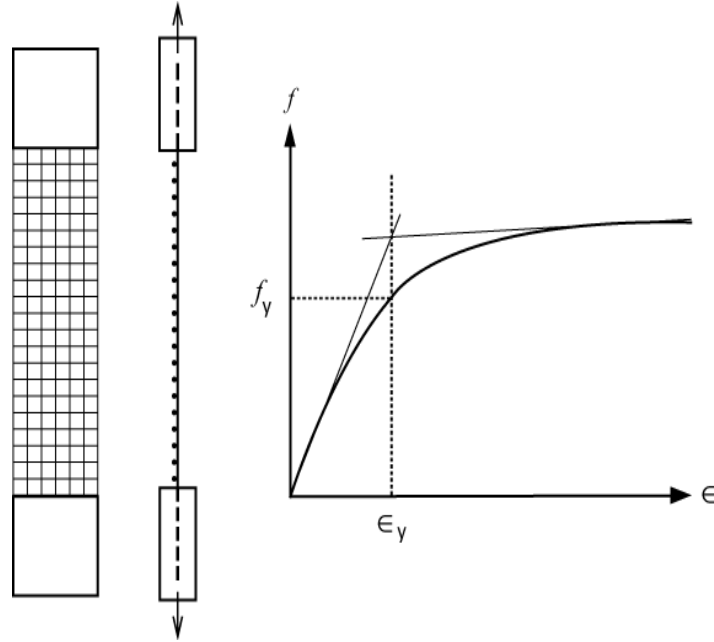


Figure 6.12.8 Schematic description of mesh tensile test sample and corresponding stress-strain curve

12.9 Related Appendix

Appendix U Volume Fraction of Reinforcement and Types of Steel Meshes Used in Ferrocement

PART VI
Chapter 13
Steel-Concrete Composite Structural Members

13.1 General

This Section states the scope of the specification, summarizes referenced specifications, codes and standard documents and provide requirements for materials for steel-concrete composite members. General provisions for composite sections and shear connectors are also included.

13.1.1 Scope

The guidelines included in Chapter 13 of part 6 of this Code presents the design guidelines for steel concrete composite members frequently used in medium to high rise buildings. This Chapter mainly addresses composite columns composed of rolled or built-up structural steel shapes or HSS, and structural concrete acting together, and steel beams supporting a reinforced concrete slab so interconnected that the beams and the slab act together to resist bending. Simple and continuous composite beams with shear connectors and concrete-encased beams, constructed with or without temporary shores, are included. Seismic provisions for steel-concrete composite members are also provided.

13.1.2 Material Limitations

Concrete and steel reinforcing bars in composite systems shall be subject to the following limitations.

- (a) For the determination of the available strength, concrete shall have a compressive strength f'_c of not less than 21 MPa nor more than 70 MPa for normal weight concrete and not less than 21 MPa nor more than 42 MPa for lightweight concrete.
- (b) The specified minimum yield stress of structural steel and reinforcing bars used in calculating the strength of a composite column shall not exceed 525 MPa.

Higher material strengths are permitted when their use is justified by testing or analysis.

13.1.3 General Provisions

In determining load effects in members and connections of a structure that includes composite members, consideration shall be given to the effective sections at the time each increment of load is applied. The design, detailing and material properties related to the concrete and reinforcing steel portions of composite construction shall comply with the reinforced concrete and reinforcing bar design specifications stipulated by the provisions in Part 6 Chapter 6.

13.1.3.1 Resistance prior to composite action

The factored resistance of the steel member prior to the attainment of composite action shall be determined in accordance with Chapter 10 Part 6.

13.1.3.2 Nominal strength of composite sections

Two methods are provided for determining the nominal strength of composite sections: the plastic stress distribution method and the strain-compatibility method. The tensile strength of the concrete shall be neglected in the determination of the nominal strength of composite members.

13.1.3.2.1 Plastic stress distribution method

For the plastic stress distribution method, the nominal strength shall be computed assuming that steel components have reached a *stress* of F_y in either tension or compression and concrete components in compression have reached a *stress* of $0.85f'_c$. For round *HSS* filled with concrete, a *stress* of $0.95f'_c$ is permitted to be used for concrete components in uniform compression to account for the effects of concrete confinement.

13.1.3.2.2 Strain-compatibility method

For the strain compatibility method, a linear distribution of strains across the section shall be assumed, with the maximum concrete compressive strain equal to 0.003 mm/mm (in/in). The stress-strain relationships for steel and concrete shall be obtained from tests or from published results for similar materials.

13.1.3.2.3 Shear connectors

Shear connectors shall be headed steel studs not less than four stud diameters in length after installation, or hot-rolled steel channels. Shear stud design values shall be taken as per Sections 13.2.1.7 and 13.3.2.4. Stud connectors shall conform to the requirements of Sec 13.3.2.4(3) Channel connectors shall conform to the requirements of Sec 13.3.2.4(4).

13.2 Design of Composite Axial Members

This section states the design guidelines for two types of composite axial members. These include—encased composite columns and concrete filled hollow structural sections.

13.2.1 Encased Composite Columns

13.2.1.1 Scope

This section applies to doubly symmetric steel columns encased in concrete, provided that

- (a) the steel shape is a compact or non-compact section
- (b) the cross-sectional area of the steel core comprises at least 1 percent of the total composite cross section
- (c) concrete encasement of the steel core is reinforced with continuous longitudinal bars and lateral ties or spirals. The minimum transverse reinforcement shall be at least 6 mm² per mm of tie spacing
- (d) The minimum reinforcement ratio for continuous longitudinal reinforcing, ρ_{sr} , shall be 0.004, where ρ_{sr} is given by:

$$\rho_{sr} = \frac{A_{sr}}{A_g} \quad (6.13.1)$$

Where,

A_{sr} = area of continuous reinforcing bars, mm²

A_g = gross area of composite member, mm²

13.2.1.2 Compressive strength

The design compressive strength, $\phi_c P_n$, and allowable compressive strength, P_n/Ω_c , for axially loaded encased composite columns shall be determined for the limit state of flexural buckling based on column slenderness as follows:

$$\phi_c = 0.75 \text{ (LRFD)} \quad \Omega_c = 2.00 \text{ (ASD)}$$

- (a) When, $P_e \geq 0.44P_o$

$$P_n = P_o \left[0.658^{\left(\frac{P_o}{P_e}\right)} \right] \quad (6.13.2)$$

- (b) When, $P_e < 0.44P_o$

$$P_n = 0.877P_e \quad (6.13.3)$$

Where,

$$P_o = A_s F_y + A_{sr} F_{yr} + 0.85 A_c f'_c \quad (6.13.4)$$

$$P_e = \frac{\pi^2 (EI_{eff})}{(KL)^2} \quad (6.13.5)$$

And where,

A_s = area of the steel section, mm²

A_c = area of concrete, mm²

A_{sr} = area of continuous reinforcing bars, mm²

E_c = modulus of elasticity of concrete = $0.043 w_c^{1.5} \sqrt{f'_c}$ MPa

E_s = modulus of elasticity of steel = 210 GPa

f'_c = specified compressive strength of concrete, MPa

F_y = specified minimum yield stress of steel section, MPa

F_{yr} = specified minimum yield stress of reinforcing bars, MPa

I_c = moment of inertia of the concrete section, mm⁴

I_s = moment of inertia of steel shape, mm⁴

I_{sr} = moment of inertia of reinforcing bars, mm⁴

K = the effective length factor determined in accordance with Chapter 10
Part 6

L = laterally unbraced length of the member, mm

w_c = weight of concrete per unit volume $1500 \leq w_c \leq 2500$ kg/m³

Where,

EI_{eff} = effective stiffness of composite section, N-mm²

$$EI_{eff} = E_s I_s + 0.5 E_s I_{sr} + C_1 E_c I_c \quad (6.13.6)$$

Where,

$$C_1 = 0.1 + 2 \left(\frac{A_s}{A_c + A_s} \right) \leq 0.3 \quad (6.13.7)$$

13.2.1.3 Tensile strength

The design tensile strength, $\phi_t P_n$, and allowable tensile strength, P_n/Ω_t , for encased composite columns shall be determined for the limit state of yielding as

$$P_n = A_s F_y + A_{sr} F_{yr} \quad (6.13.8)$$

$$\phi_t = 0.90 \text{ (LRFD)} \quad \Omega_t = 1.67 \text{ (ASD)}$$

13.2.1.4 Shear strength

The available shear strength shall be calculated based on either the shear strength of the steel section alone as specified in Sec 10.7, plus the shear strength provided by tie reinforcement, if present, or the shear strength of the reinforced concrete portion alone.

13.2.1.5 Load transfer

Loads applied to axially loaded encased composite columns shall be transferred between the steel and concrete in accordance with the following requirements:

- (a) When the external force is applied directly to the steel section, shear connectors shall be provided to transfer the required shear force, V' , as follows:

$$V' = V \left(1 - \frac{A_s F_y}{P_o} \right) \quad (6.13.9)$$

Where

V = required shear force introduced to column, N

A_s = area of steel cross section, mm²

P_o = nominal axial compressive strength without consideration of length effects, N

- (b) When the external force is applied directly to the concrete encasement, shear connectors shall be provided to transfer the required shear force, V' , as follows:

$$V' = V \left(\frac{A_s F_y}{P_o} \right) \quad (6.13.10)$$

- (c) When load is applied to the concrete of an encased composite column by direct bearing the design bearing strength, $\phi_B P_p$, and the allowable bearing strength, P_p/Ω_B , of the concrete shall be:

$$P_p = 1.7f'A_B \quad (6.13.11)$$

Where,

$$\phi_B = 0.65 \text{ (LRFD)} \quad \Omega_B = 2.31 \text{ (ASD)}$$

Where,

A_B = loaded area of concrete, mm²

13.2.1.6 Detailing requirements

13.2.1.6.1 Longitudinal bars

The concrete encasement shall be reinforced with longitudinal bars and lateral ties extending completely around the structural steel core. The clear cover shall not be less than 40 mm.

The longitudinal bars shall

- (a) Be continuous at framed levels when considered to carry load;
- (b) Have an area not less than 0.01 times the total gross cross-sectional area;
- (c) Be located at each corner; and
- (d) Spaced on all sides not further apart than $525t/f_y$ times one-half the least dimension of the composite section.

13.2.1.6.2 Lateral ties

The lateral ties shall

- (a) Be 15M bars, except that 10M bars may be used when no side dimension of the composite section exceeds 500 mm; and
- (b) Have a vertical spacing not exceeding the least of the following:
 - (i) Two-thirds of the least side dimension of the cross-section;
 - (ii) 16 longitudinal bar diameters; or
 - (iii) 500 mm.

Where required, shear connectors transferring the required shear force shall be distributed along the length of the member at least a distance of 2.5 times the width of a rectangular HSS or 2.5 times the diameter of a round HSS both above and below the load transfer region. The maximum connector spacing shall be 405 mm.

13.2.1.6.3 Shear connectors

Shear connectors shall be provided to transfer the required shear force Section specified in Sec 13.2.2.5. The shear connectors shall be distributed along the length of the member at least a distance of 2.5 times the depth of the encased composite column above and below the load transfer region. The maximum connector spacing shall be 405 mm. Connectors to transfer axial load shall be placed on at least two faces of the steel shape in a configuration symmetrical about the steel shape axes.

13.2.1.6.4 Columns with multiple built-up shapes

If the composite cross section is built up from two or more encased steel shapes, the shapes shall be interconnected with lacing, tie plates, batten plates or similar components to prevent buckling of individual shapes due to loads applied prior to hardening of the concrete.

13.2.1.7 Strength of stud shear connectors

The nominal strength of one stud shear connector embedded in solid concrete is:

$$Q_n = 0.5A_{sc}\sqrt{f'_c E_c} \leq A_{sc}F_u \quad (6.13.12)$$

Where,

A_{sc} = cross-sectional area of stud shear connector, mm²

F_u = specified minimum tensile strength of a stud shear connector, MPa

13.2.2 Concrete Filled Hollow Structural Section

13.2.2.1 Scope

Section 13.2.2 applies to composite members consisting of steel hollow structural sections (HSS) completely filled with concrete, provided that

- The cross-sectional area of the steel HSS shall comprise at least 1 percent of the total composite cross-section.
- The width-to thickness ratio of the walls of rectangular hollow structural sections does not exceed $\frac{1350}{\sqrt{F_y}}$.
- The outside diameter-to-thickness ratio of circular hollow structural sections does not exceed $\frac{28000}{F_y}$.
- The concrete strength is between 20 and 80 MPa for axially loaded columns and between 20 and 40 MPa for columns subjected to axial compression and bending.

13.2.2.2 Compressive strength

The design compressive strength, $\phi_c P_n$ and allowable compressive strength, P_n/Ω_c , for axially loaded filled composite columns shall be determined for the limit state of flexural buckling based on Section 13.2.1.2 with the following modifications:

$$P_o = A_s F_y + A_{sr} F_{yr} + C_2 A_c f'_c \quad (6.13.13)$$

$C_2 = 0.85$ for rectangular sections and 0.95 for circular sections

$$EI_{eff} = E_s I_s + E_s I_{sr} + C_3 E_c I_c \quad (6.13.14)$$

$$C_3 = 0.6 + 2 \left(\frac{A_s}{A_c + A_s} \right) \leq 0.9 \quad (6.13.15)$$

13.2.2.3 Tensile strength

The design tensile strength, $\phi_t P_n$, and allowable tensile strength, P_n/Ω_t for filled composite columns shall be determined for the limit state of yielding as:

$$P_n = A_s F_y + A_{sr} F_{yr} \quad (6.13.16)$$

$$\phi_t = 0.90 \text{ (LRFD)} \quad \Omega_t = 1.67 \text{ (ASD)}$$

13.2.2.4 Shear strength

The available shear strength shall be calculated based on either the shear strength of the steel section alone as specified in Chapter 10 or the shear strength of the reinforced concrete portion alone. The shear strength of reinforced concrete portion may be determined according to Chapter 6 of Part 6.

13.2.2.5 Load transfer

Loads applied to filled composite columns shall be transferred between the steel and concrete. When the external force is applied either to the steel section or to the concrete infill, transfer of force from the steel section to the concrete core is required from direct bond interaction, shear connection or direct bearing. The force transfer mechanism providing the largest nominal strength may be used. These force transfer mechanisms shall not be superimposed.

When load is applied to the concrete of an encased or filled composite column by direct bearing the design bearing strength, $\phi_B P_p$, and the allowable bearing strength, P_p/Ω_B , of the concrete shall be:

$$P_p = 1.7f'_c A_E \quad (6.13.17)$$

$$\phi_B = 0.65 \text{ (LRFD)} \quad \Omega_B = 2.31 \text{ (ASD)}$$

Where, A_E is the loaded area, mm².

13.2.2.6 Detailing requirements

Where required, shear connectors transferring the required shear force shall be distributed along the length of the member at least a distance of 2.5 times the width of a rectangular HSS or 2.5 times the diameter of a round HSS both above and below the load transfer region. The maximum connector spacing shall be 405 mm.

13.3 Design of Composite Flexural Members

This section applies to composite beams consisting of steel sections interconnected with either a reinforced concrete slab or a steel deck with a concrete cover slab. The steel beams and the reinforced concrete slab are so interconnected that the beams and the slab act together to resist bending. Simple and continuous composite beams with shear connectors and concrete-encased beams, constructed with or without temporary shores, are included. Design philosophy for composite columns subjected to bending moments is also stated.

13.3.1 General

13.3.1.1 Deflections

Calculation of deflections shall take into account the effects of creep of concrete, shrinkage of concrete, and increased flexibility resulting from partial shear connection and from interfacial slip. These effects shall be established by test or analysis, where practicable. Consideration shall also be given to the effects of full or partial continuity in the steel beams and concrete slabs in reducing calculated deflections.

In lieu of tests or analysis, the effects of partial shear connection and interfacial slip, creep, and shrinkage may be assessed as follows:

- (a) For increased flexibility resulting from partial shear connection and interfacial slip, the deflections shall be calculated using an effective moment of inertia given by

$$I_e = I_s + 0.85p^{0.25}(I_t - I_s) \quad (6.13.18)$$

Where,

I_s = moment of inertia of a steel beam, or of a steel joist or truss adjusted to include the effect of shear deformations, which may be taken into account by decreasing the moment of inertia based on the cross-sectional areas of the top and bottom chords by 15% or by a more detailed analysis

p = fraction of full shear connection

= 1.00 for full shear connection

I_t = transformed moment of inertia of composite beam based on the modular ratio $n = E/E_c$.

- (b) For creep, elastic deflections caused by dead loads and long-term live loads, as calculated in Item (a), need to be increased by 15% and
- (c) For shrinkage of concrete, using a selected free shrinkage strain, strain compatibility between the steel and concrete, and an age-adjusted effective modulus of elasticity of concrete as it shrinks and creeps, the deflection of a simply supported composite beam, joist, or truss shall be calculated as follows:

$$\Delta_s = \frac{L^2}{8} \psi = \frac{L^2}{8} c \frac{\epsilon_f A_c \gamma}{n_s I_{es}} \quad (6.13.19)$$

Where,

L = span of the beam, joist, or truss

ψ = curvature along length of the beam, joist, or truss due to shrinkage of concrete

c = empirical coefficient used to match theory with test results (accounting for cracking of concrete in tension, the non-linear stress-strain relationship of concrete, and other factors) free shrinkage strain of concrete

A_c = effective area of concrete slab

y = distance from centroid of effective area of concrete slab to centroidal axis of the composite beam, joist, or truss

n_s = modular ratio, $\frac{E}{E'_c}$

Age-adjusted effective modulus of elasticity of concrete E'_c is given by

$$E'_c = \frac{E_c}{1 + \chi\phi} \quad (6.13.20)$$

Where,

χ = aging coefficient of concrete

ϕ = creep coefficient of concrete

Effective moment of inertia of composite beam, truss, or joist based on the modular ratio n_s is given by

$$I_{es} = I_s + 0.85p^{0.25}(I_{ts} - I_s) \quad (6.13.21)$$

Where, I_{ts} = transformed moment of inertia based on the modular ratio n_s

13.3.1.2 Design effective width of concrete

The effective width of the concrete slab is the sum of the effective widths for each side of the beam centerline, each of which shall not exceed:

- (a) one-eighth of the beam span, center-to-center of supports;
- (b) one-half the distance to the centerline of the adjacent beam; or
- (c) the distance to the edge of the slab.

13.3.1.3 Shear strength

The available shear strength of composite beams with shear connectors shall be determined based upon the properties of the steel section alone in accordance with Sec 10.7 Chapter 10 Part 6. The available shear strength of concrete-encased and filled composite members shall be determined based upon the properties of the steel section alone in accordance with Sec 10.7 Chapter 10 Part 6 or based upon the properties of the concrete and longitudinal steel reinforcement.

13.3.1.4 Strength during construction

When temporary shores are not used during construction, the steel section alone shall have adequate strength to support all loads applied prior to the concrete attaining 75 percent of its specified strength f'_c . The available flexural strength of the steel section shall be determined according to Sec 10.6 Chapter 10 Part 6.

13.3.2 Strength of Composite Beams with Shear Connectors

13.3.2.1 Positive flexural strength

The design positive flexural strength, $\phi_b M_n$ and the allowable positive flexural strength, M_n/Ω_b , shall be determined for the limit state of yielding as follows:

$$\phi_b = 0.90 \text{ (LRFD)} \quad \Omega_b = 1.67 \text{ (ASD)}$$

$$(a) \text{ For } \frac{h}{t_w} \leq 3.76 \sqrt{\frac{E}{F_y}}$$

M_n shall be determined from the plastic stress distribution on the composite section for the limit state of yielding (plastic moment).

$$(b) \text{ For } \frac{h}{t_w} > 3.76 \sqrt{\frac{E}{F_y}}$$

M_n shall be determined from the superposition of elastic stresses, considering the effects of shoring, for the limit state of yielding (yield moment).

13.3.2.2 Negative flexural strength

The design negative flexural strength, $\phi_b M_n$, and the allowable negative flexural strength, M_n/Ω_b , shall be determined for the steel section alone, in accordance with the requirements of Sec 10.6 Chapter 10 Part 6.

Alternatively, the available negative flexural strength shall be determined from the plastic stress distribution on the composite section, for the limit state of yielding (plastic moment), with

$$\phi_b = 0.90 \text{ (LRFD)} \quad \Omega_b = 1.67 \text{ (ASD)}$$

Provided that:

- (a) The steel beam is compact and is adequately braced according to Sec 10.6.
- (b) Shear connectors connect the slab to the steel beam in the negative moment region.
- (c) The slab reinforcement parallel to the steel beam, within the effective width of the slab, is properly developed.

13.3.2.3 Strength of composite beams with formed steel deck

13.3.2.3.1 General

The available flexural strength of composite construction consisting of concrete slabs on formed steel deck connected to steel beams shall be determined by the applicable portions of Sections 13.3.2.1 and 13.3.2.2, with the following requirements:

- (a) This section is applicable to decks with *nominal rib height* not greater than 75 mm. The average width of concrete rib or haunch, w_r , shall be not less than 50 mm, but shall not be taken in calculations as more than the minimum clear width near the top of the steel deck.
- (b) The concrete slab shall be connected to the steel beam with welded stud shear connectors 19 mm or less in diameter (AWS D1.1). Studs shall be welded either through the deck or directly to the steel cross section. Stud shear connectors, after installation, shall extend not less than 38 mm above the top of the steel deck and there shall be at least 13 mm of concrete cover above the top of the installed studs.
- (c) The slab thickness above the steel deck shall be not less than 50 mm.
- (d) Steel deck shall be anchored to all supporting members at a spacing not to exceed 460 mm. Such anchorage shall be provided by stud connectors, a combination of stud connectors and arc spot (puddle) welds, or other devices specified by the designer.

13.3.2.3.2 Deck ribs oriented perpendicular to steel beam

Concrete below the top of the steel deck shall be neglected in determining composite section properties and in calculating A_c for deck ribs oriented perpendicular to the steel beams.

13.3.2.3.3 Deck ribs oriented parallel to steel beam

Concrete below the top of the steel deck may be included in determining composite section properties and shall be included in calculating A_c .

Formed steel deck ribs over supporting beams may be split longitudinally and separated to form a concrete haunch.

When the nominal depth of steel deck is 38 mm or greater, the average width, w_r , of the supported haunch or rib shall be not less than 50 mm for the first stud in the transverse row plus four stud diameters for each additional stud.

13.3.2.4 Shear connectors

13.3.2.4.1 Load transfer for positive moment

The entire horizontal shear at the interface between the steel beam and the concrete slab shall be assumed to be transferred by shear connectors, except for concrete-encased beams as defined in Section 10.9.3.3. For composite action with concrete subject to flexural compression, the total horizontal shear force, V' , between the point of maximum positive moment and the point of zero moment shall be taken as the lowest value according to the limit states of concrete crushing, tensile yielding of the steel section, or strength of the shear connectors.

Concrete crushing:

$$V' = 0.85f'_c A_c \quad (6.13.22a)$$

Tensile yielding of the steel section:

$$V' = F_y A_s \quad (6.13.22b)$$

Strength of shear connectors:

$$V' = \sum Q_n \quad (6.13.22c)$$

Where,

A_c = area of concrete slab within effective width, mm²

A_s = area of steel cross section, mm²

$\sum Q_n$ = sum of nominal strengths of shear connectors between the point of maximum positive moment and the point of zero moment, N

13.3.2.4.2 Load transfer for negative moment

In continuous composite beams where longitudinal reinforcing steel in the negative moment regions is considered to act compositely with the steel beam, the total horizontal shear force between the point of maximum negative moment and the point of zero moment shall be taken as the lower value according to the limit states of yielding of the steel reinforcement in the slab, or strength of the shear connectors:

- (a) Tensile yielding of the slab reinforcement

$$V' = A_r F_{yr} \quad (6.13.23a)$$

Where,

A_r = area of adequately developed longitudinal reinforcing steel within the effective width of the concrete slab, mm²

F_{yr} = specified minimum yield stress of the reinforcing steel, MPa

- (b) Strength of shear connectors

$$V' = \sum Q_n \quad (6.13.23b)$$

13.3.2.4.3 Strength of stud shear connectors

The nominal strength of one stud shear connector embedded in solid concrete or in a composite slab is

$$Q_n = 0.5 A_{sc} \sqrt{f'_c E_c} \leq R_g R_p A_{sc} F_u \quad (6.13.24)$$

Where,

A_{sc} = cross-sectional area of stud *shear connector*, mm²

E_c = modulus of elasticity of concrete = $0.043 w_c^{1.5} \sqrt{f'_c}$, MPa

F_u = specified minimum tensile strength of a stud shear connector

$R_g = 1.0$

- (a) for one stud welded in a steel deck rib with the deck oriented perpendicular to the steel shape;
- (b) for any number of studs welded in a row directly to the steel shape;
- (c) for any number of studs welded in a row through steel deck with the deck oriented parallel to the steel shape and the ratio of the *average rib* width to rib depth ≥ 1.5

- $R_g = 0.85$ (a) for two studs welded in a steel deck rib with the deck oriented perpendicular to the steel shape;
- (b) for one stud welded through steel deck with the deck oriented parallel to the steel shape and the ratio of the average rib width to rib depth < 1.5
- $R_g = 0.7$ for three or more studs welded in a steel deck rib with the deck oriented perpendicular to the steel shape
- $R_p = 1.0$ for studs welded directly to the steel shape (in other words, not through steel deck or sheet) and having a haunch detail with not more than 50 percent of the top flange covered by deck or sheet steel closures
- $R_p = 0.75$ (a) for studs welded in a composite slab with the deck oriented perpendicular to the beam and $e_{mid-ht} \geq 50$ mm;
- (b) for studs welded through steel deck, or steel sheet used as girder filler material and embedded in a composite slab with the deck oriented parallel to the beam
- $R_p = 0.60$ for studs welded in a composite slab with deck oriented perpendicular to the beam and $e_{mid-ht} < 50$ mm

e_{mid-ht} = distance from the edge of stud shank to the steel deck web, measured at mid-height of the deck rib, and in the *load* bearing direction of the stud (in other words, in the direction of maximum moment for a simply supported beam), mm

w_c = weight of concrete per unit volume ($1500 \leq w_c \leq 2500$ kg/m³)

13.3.2.4.4 Strength of channel shear connectors

The nominal strength of one channel shear connector embedded in a solid concrete slab is

$$Q_n = 0.3(t_f + 0.5t_w)L_c\sqrt{f'_c E_c} \quad (6.13.25)$$

Where,

t_f = flange thickness of channel shear connector, mm

t_w = web thickness of channel shear connector, mm

L_c = length of channel shear connector, mm

The strength of the channel shear connector shall be developed by welding the channel to the beam flange for a force equal to Q_n , considering eccentricity on the connector.

13.3.2.4.5 Required number of shear connectors

The number of shear connectors required between the section of maximum bending moment, positive or negative, and the adjacent section of zero moment shall be equal to the horizontal shear force as determined in Sections 10.9.3.2d(1) and 10.9.3.2d(2) divided by the nominal strength of one shear connector as determined from Sec 10.9.3.2d(3) or Sec 10.9.3.2d(4).

13.3.2.4.6 Shear connector placement and spacing

Shear connectors required on each side of the point of maximum bending moment, positive or negative, shall be distributed uniformly between that point and the adjacent points of zero moment, unless otherwise specified. However, number of shear connectors placed between any concentrated load and the nearest point of zero moment shall be sufficient to develop the maximum moment required at the concentrated load point.

Shear connectors shall have at least 25 mm of lateral concrete cover, except for connectors installed in the ribs of formed steel decks. The diameter of studs shall not be greater than 2.5 times the thickness of the flange to which they are welded, unless located over the web. The minimum center-to-center spacing of stud connectors shall be six diameters along the longitudinal axis of the supporting composite beam and four diameters transverse to the longitudinal axis of the supporting composite beam, except that within the ribs of formed steel decks oriented perpendicular to the steel beam the minimum center-to-center spacing shall be four diameters in any direction. The maximum center-to-center spacing of shear connectors shall not exceed eight times the total slab thickness nor 900 mm.

13.3.3 Slab Reinforcement

13.3.3.1 General

Slabs shall be adequately reinforced to support all loads and to control both cracking transverse to the composite beam span and longitudinal cracking over the steel section. Reinforcement shall not be less than that required by the specified fire-resistance design of the assembly.

13.3.3.2 Parallel reinforcement

Reinforcement parallel to the span of the beam in regions of negative bending moment of the composite beam shall be anchored by embedment in concrete that is in compression. The reinforcement of slabs that are to be continuous over the end support of steel sections or joists fitted with flexible end connections shall be given special attention. Reinforcement at the ends of beams supporting ribbed slabs perpendicular to the beam shall be not less than two 16 mm bars or equivalent.

13.3.3.3 Transverse reinforcement-concrete slab on metal deck

Unless it is known from experience that longitudinal cracking caused by composite action directly over the steel section is unlikely, additional transverse reinforcement or other effective means shall be provided. Such additional reinforcement shall be placed in the lower part of the slab and anchored so as to develop the yield strength of the reinforcement. The area of such reinforcement shall be not less than 0.002 times the concrete area being reinforced and shall be uniformly distributed.

13.3.3.4 Transverse reinforcement- ribbed slabs

- (a) Where the ribs are parallel to the beam span, the area of transverse reinforcement shall be not less than 0.002 times the concrete cover slab area being reinforced and shall be uniformly distributed.
- (b) Where the ribs are perpendicular to the beam span, the area of transverse reinforcement shall be not less than 0.001 times the concrete cover slab area being reinforced and shall be uniformly distributed.

13.3.4 Flexural Strength of Concrete-Encased and Filled Members

The nominal flexural strength of concrete-encased and filled members shall be determined using one of the following methods:

- (a) The superposition of elastic stresses on the composite section, considering the effects of shoring, for the limit state of yielding (yield moment), where:

$$\phi_b = 0.90 \text{ (LRFD)} \quad \Omega_b = 1.67 \text{ (ASD)}$$

- (b) The plastic stress distribution on the steel section alone, for the limit state of yielding (plastic moment), where:

$$\phi_b = 0.90 \text{ (LRFD)} \quad \Omega_b = 1.67 \text{ (ASD)}$$

- (c) If shear connectors are provided and the concrete meets the requirements of Sec 10.9.1.2, the nominal flexural strength shall be computed based upon the plastic stress distribution on the composite section or from the strain compatibility method, where:

$$\phi_b = 0.85 \text{ (LRFD)} \quad \Omega_b = 1.76 \text{ (ASD)}$$

13.3.5 Combined Axial Force and Flexure

The interaction between axial forces and flexure in composite members shall account for stability as required by Chapter 10. The design compressive strength, $\phi_c P_n$, and allowable compressive strength, P_n/Ω_c and the design flexural strength, $\phi_b M_n$ and allowable flexural strength, M_n/Ω_b , are determined as follows:

$$\phi_c = 0.75 \text{ (LRFD)} \quad \Omega_c = 2.00 \text{ (ASD)}$$

$$\phi_b = 0.90 \text{ (LRFD)} \quad \Omega_b = 1.67 \text{ (ASD)}$$

13.3.5.1 The nominal strength of the cross section of a composite member subjected to combined axial compression and flexure shall be determined using either the plastic stress distribution method or the strain-compatibility method.

13.3.5.2 To account for the influence of length effects on the axial strength of the member, the nominal axial strength of the member shall be determined by Section 10.9 with P_o taken as the nominal axial strength of the cross section determined in Section 10.9.4(13) above.

13.3.6 Special Cases

When composite construction does not conform to the requirements of Sections 13.2 and 13.3, the strength of shear connectors and details of construction shall be established by testing.

13.4 Composite Connections

This Section is applicable to connections in buildings that utilize composite or dual steel and concrete systems. Composite connections shall be demonstrated to have Design Strength, ductility and toughness that is comparable to that exhibited by similar structural steel or reinforced concrete connections that meet the requirements in Chapters 5 and 10, Part 6 of this Code. Methods for calculating the connection strength shall meet the requirements in this Section.

13.4.1 General

Connections shall have adequate deformation capacity to resist the critical Required Strengths at the Design Story Drift. Additionally, connections that are required for the lateral stability of the building under seismic forces shall meet the requirements in Sec

13.5 based upon the specific system in which the connection is used. When the Required Strength is based upon nominal material strengths and nominal member dimensions, the determination of the required connection strength shall account for any effects that result from the increase in the actual Nominal Strength of the connected member.

13.4.2 Nominal Strength of Connections

The Nominal Strength of connections in composite Structural Systems shall be determined on the basis of rational models that satisfy both equilibrium of internal forces and the strength limitation of component materials and elements based upon potential limit states. Unless the connection strength is determined by analysis and testing, the models used for analysis of connections shall meet the following requirements:

13.4.2.1 When required, force shall be transferred between structural steel and reinforced concrete through direct bearing of headed shear studs or suitable alternative devices, by other mechanical means, by shear friction with the necessary clamping force provided by reinforcement normal to the plane of shear transfer, or by a combination of these means. Any potential bond strength between structural steel and reinforced concrete shall be ignored for the purpose of the connection force transfer mechanism.

13.4.2.2 The nominal bearing and shear-friction strengths shall meet the requirements in Chapters 6 and 10, Part 6 except that the strength reduction (resistance) factors shall be as given in Chapter 6 Part 6. Unless a higher strength is substantiated by cyclic testing, the nominal bearing and shear-friction strengths shall be reduced by 25 percent for the composite seismic systems.

13.4.2.3 The Design Strengths of structural steel components in composite connections, as determined in Sections 13.2 and 13.3 and the LRFD Specification, shall equal or exceed the Required Strengths. Structural steel elements that are encased in confined reinforced concrete are permitted to be considered to be braced against out of plane buckling. Face Bearing Plates consisting of stiffeners between the flanges of steel beams are required when beams are embedded in reinforced concrete columns or walls.

13.4.2.4 The nominal shear strength of reinforced-concrete-encased steel Panel Zones in beam-to-column connections shall be calculated as the sum of the Nominal Strengths of the structural steel and confined reinforced concrete shear elements as determined in Chapters 6 and 10, Part 6 of this Code. The strength reduction (resistance) factors for reinforced concrete shall be as given in Chapter 6 Part 6.

13.4.2.5 Reinforcement shall be provided to resist all tensile forces in reinforced concrete components of the connections. Additionally, the concrete shall be confined with transverse reinforcement. All reinforcement shall be fully developed in tension or compression, as appropriate, beyond the point at which it is no longer required to resist the forces. Development lengths shall be determined in accordance with Chapter 6 Part 6. Connections shall meet the following additional requirements:

- (a) When the slab transfers horizontal diaphragm forces, the slab reinforcement shall be designed and anchored to carry the in-plane tensile forces at all critical sections in the slab, including connections to collector beams, columns, braces and walls.
- (b) For connections between structural steel or Composite Beams and reinforced concrete or Reinforced-Concrete-Encased Composite Columns, transverse hoop reinforcement shall be provided in the connection region to meet the requirements in Chapter 6 Part 6 except for the following modifications:
 - (i) Structural steel sections framing into the connections are considered to provide confinement over a width equal to that of face bearing stiffener plates welded to the beams between the flanges.
 - (ii) Lap splices are permitted for perimeter ties when confinement of the splice is provided by Face Bearing Plates or other means that prevents spalling of the concrete cover.
- (c) The longitudinal bar sizes and layout in reinforced concrete and Composite Columns shall be detailed to minimize slippage of the bars through the beam-to-column connection due to high force transfer associated with the change in column moments over the height of the connection.

13.5 Seismic Provisions for Composite Structural Systems

These Provisions are intended for the design and construction of composite structural steel and reinforced concrete members and connections in the Seismic Load Resisting Systems in buildings for which the design forces resulting from earthquake motions have been determined on the basis of various levels of energy dissipation in the inelastic range of response.

13.5.1 Scope

Provisions shall be applied in conjunction with the AISC Load and Resistance Factor Design (LRFD) Specification for Structural Steel Buildings, hereinafter referred to as the LRFD Specification. All members and connections in the Seismic Load Resisting System shall have a Design Strength as required in the LRFD Specification and shall meet the requirements in these Provisions. The applicable requirements in Chapter 10 Part 6 shall be used for the design of structural steel components in composite systems. Reinforced-concrete members subjected to seismic forces shall meet the requirements in Chapters 6 and 10, Part 6 except as modified in these provisions. When the design is based upon elastic analysis, the stiffness properties of the component members of composite systems shall reflect their condition at the onset of significant yielding of the building.

13.5.2 Seismic Design Categories

The Required Strength and other seismic provisions for Seismic Design Categories, Seismic Use Groups or Seismic Zones and the limitations on height and irregularity shall be as stipulated in Chapter 2 Part 6.

13.5.3 Loads, Load Combinations, and Nominal Strengths

The loads and load combinations shall be as stipulated by the Applicable Building Code. Where Amplified Seismic Loads are required by these provisions, the horizontal earthquake load E (as defined in Chapter 2 Part 6) shall be multiplied by the over strength factor Ω_o prescribed by Chapter 2 Part 6.

13.5.4 Materials**13.5.4.1 Structural Steel**

Structural steel used in composite Seismic Load Resisting Systems shall meet the requirements in Sec 10.20 Chapter 10 Part 6 in addition Sec 13.1 of this Chapter. The structural steels that are explicitly permitted for use in seismic design have been selected based upon their inelastic properties and weld ability. In general, they meet the following characteristics: (1) a ratio of yield stress to tensile stress not greater than 0.85; (2) a pronounced stress-strain plateau at the yield stress; (3) a large inelastic strain capability (for example, tensile elongation of 20 percent or greater in a 2-in. (50 mm) gage length); and (4) good weldability. Other steels should not be used without evidence that the above criteria are met.

13.5.4.2 Concrete and steel reinforcement

Concrete and steel reinforcement used in composite Seismic Load Resisting Systems shall meet the requirements in Chapter 5 Part 6, and the following requirements:

- (a) The specified minimum compressive strength of concrete in composite members shall equal or exceed 2.5 ksi (17 MPa).
- (b) For the purposes of determining the Nominal Strength of composite members, f'_c shall not be taken as greater than 10 ksi (69 MPa) for normal-weight concrete nor 4 ksi (28 MPa) for lightweight concrete.

Concrete and steel reinforcement used in the composite Seismic Load Resisting Systems described shall also meet the requirements in Chapter 8 Part 6.

13.5.5 Composite Members

13.5.5.1 Composite floor and roof slabs

The design of composite floor and roof slabs shall meet the requirements of ASCE 3-91. Composite slab diaphragms shall meet the requirements in this Section.

Details shall be designed to transfer forces between the diaphragm and Boundary Members, Collector Elements, and elements of the horizontal framing system.

The nominal shear strength of composite diaphragms and concrete-filled steel deck diaphragms shall be taken as the nominal shear strength of the reinforced concrete above the top of the steel deck ribs in accordance with Chapter 6 Part 6. Alternatively, the composite diaphragm design shear strength shall be determined by in-plane shear tests of concrete-filled diaphragms.

13.5.5.2 Composite beams

Composite Beams shall meet the requirements in Sec 13.3. Composite Beams that are part of C-SMF shall also meet the following requirements:

- (a) The distance from the maximum concrete compression fiber to the plastic neutral axis shall not exceed:

$$\frac{Y_{con} + d_b}{1 + \left(\frac{1700 F_y}{E_s} \right)} \quad (6.13.26)$$

Where,

Y_{con} = distance from the top of the steel beam to the top of concrete, mm

d_b = depth of the steel beam, mm

F_y = specified minimum yield strength of the steel beam, MPa

E_s = modulus of elasticity of the steel beam, MPa

- (b) Beam flanges shall meet the requirements in Part 6 Sec 10.20.9.4.2, except when fully reinforced-concrete-encased compression elements have a reinforced concrete cover of at least 2 in. (50 mm) and confinement is provided by hoop reinforcement in regions where plastic hinges are expected to occur under seismic deformations. Hoop reinforcement shall meet the requirements in Chapter 8 Part 6.

13.5.5.3 Reinforced concrete encased composite columns

This Section is applicable to columns that: (i) consist of reinforced-concrete encased structural steel sections with a structural steel area that comprises at least 4 percent of the total composite-column cross-section; and (ii) meet the additional limitations in Sec 13.2.2.1. Such columns shall meet the requirements in Sec 13.2.2, except as modified in this Section. Additional requirements, as specified for intermediate and special seismic systems in Sections 13.5.5.3.2 and 13.5.5.3.3, shall apply as required.

Columns that consist of reinforced-concrete-encased structural steel sections with a structural steel area that comprises less than 4 percent of the total composite column cross-section shall meet the requirements for reinforced concrete columns in Chapter 6 Part 6 except as modified for:

- (a) The steel shape shear connectors in Sec 13.5.4.3.1 (2).
- (b) The contribution of the reinforced-concrete-encased structural steel section to the strength of the column as provided in Chapter 6 Part 6.
- (c) The seismic requirements for reinforced concrete columns as specified in the description of the composite seismic systems in Sections 13.5.5.3.1 to 13.5.5.3.3.

13.5.5.3.1 Ordinary seismic system requirements

The following requirements for Reinforced-Concrete-Encased Composite Columns are applicable to all composite systems:

- (a) The nominal shear strength of the column shall be determined as the nominal shear strength of the structural shape plus the nominal shear strength that is provided by the tie reinforcement in the reinforced-concrete encasement. The nominal shear strength of the structural steel section shall be determined in accordance with Sec 10.20 Chapter 10 Part 6. The nominal shear strength of the tie reinforcement shall be determined in accordance with Chapter 6 Part 6. In Chapter 6 Part 6, the dimension b_w shall equal the width of the concrete cross-section minus the width of the structural shape measured perpendicular to the direction of shear. The nominal shear strength shall be multiplied by ϕ_v equal to 0.75 to determine the design shear strength.
- (b) Composite Columns that are designed to share the applied loads between the structural steel section and reinforced concrete shall have shear connectors that meet the following requirements:
 - (i) If an external member is framed directly to the structural steel section to transfer a vertical reaction V_u , shear connectors shall be provided to transfer the force $V_u(1 - A_s F_y / P_n)$ between the structural steel section and the reinforced concrete, where A_s is the area of the structural steel section, F_y is the specified minimum yield strength of the structural steel section, and P_n is the nominal compressive strength of the Composite Column.
 - (ii) If an external member is framed directly to the reinforced concrete to transfer a vertical reaction V_u , shear connectors shall be provided to transfer the force $V_u A_s F_y / P_n$ between the structural steel section and the reinforced concrete, where A_s , F_y and P_n are as defined above.
 - (iii) The maximum spacing of shear connectors shall be 16 in. (406 mm) with attachment along the outside flange faces of the embedded shape.
- (c) The maximum spacing of transverse ties shall be the least of the following:
 - (i) one-half the least dimension of the section
 - (ii) 16 longitudinal bar diameters
 - (iii) 48 tie diameters

Transverse ties shall be located vertically within one-half the tie spacing above the top of the footing or lowest beam or slab in any story and shall be spaced as provided herein within one-half the tie spacing below the lowest beam or slab framing into the column.

Transverse bars shall have a diameter that is not less than one-fiftieth of greatest side dimension of the composite member, except that ties shall not be smaller than No. 3 bars and need not be larger than No. 5 bars. Alternatively, welded wire fabric of equivalent area is permitted as transverse reinforcement except when prohibited for intermediate and special systems.

- (d) All Load-Carrying Reinforcement shall meet the detailing and splice requirements in Chapter 8 Part 6. Load-Carrying Reinforcement shall be provided at every corner of a rectangular cross-section. The maximum spacing of other load carrying or restraining longitudinal reinforcement shall be one-half of the least side dimension of the composite member.
- (e) Splices and end bearing details for reinforced-concrete-encased structural steel sections shall meet the requirements in Chapter 8 Part 6. If adverse behavioral effects due to the abrupt change in member stiffness and nominal tensile strength occur when reinforced-concrete encasement of a structural steel section is terminated, either at a transition to a pure reinforced concrete column or at the Column Base, they shall be considered in the design.

13.5.5.3.2 Intermediate seismic system requirements

Reinforced-Concrete-Encased Composite Columns in intermediate seismic systems shall meet the following requirements in addition to those in Sec 13.5.5.3.1:

- (a) The maximum spacing of transverse bars at the top and bottom shall be the least of the following:
 - (i) one-half the least dimension of the section
 - (ii) 8 longitudinal bar diameters
 - (iii) 24 tie bar diameters
 - (iv) 12 in. (305 mm)

These spacing shall be maintained over a vertical distance equal to the greatest of the following lengths, measured from each joint face and on both sides of any section where flexural yielding is expected to occur:

- (i) one-sixth the vertical clear height of the column
- (ii) the maximum cross-sectional dimension
- (iii) 18 in. (457 mm)

- (b) Tie spacing over the remaining column length shall not exceed twice the spacing defined above.
- (c) Welded wire fabric is not permitted as transverse reinforcement in intermediate seismic systems.

13.5.5.3.3 Special seismic system requirements

Reinforced-concrete-encased columns for special seismic systems shall meet the following requirements in addition to those in Sections 13.5.4.3.2 and 13.5.4.4.3:

- (a) The required axial strength for Reinforced-Concrete-Encased Composite Columns and splice details shall meet the requirements in Sec 13.2.
- (b) Longitudinal Load-Carrying Reinforcement shall meet the requirements in Chapter 6 Part 6.
- (c) Transverse reinforcement shall be hoop reinforcement as defined in Chapter 6 Part 6 and shall meet the following requirements:

The minimum area of tie reinforcement A_{sh} shall meet the following requirement:

$$A_{sh} = 0.09h_{cc}s \left(1 - \frac{F_y A_s}{P_n} \right) \left(\frac{f'_c}{F_{yh}} \right) \quad (6.13.27)$$

Where,

h_{cc} = cross-sectional dimension of the confined core measured center-to-center of the tie reinforcement, mm

s = spacing of transverse reinforcement measured along the longitudinal axis of the structural member, mm

F_y = specified minimum yield strength of the structural steel core, MPa

A_s = cross-sectional area of the structural core, mm²

P_n = nominal axial compressive strength of the Composite Column calculated in accordance with the LRFD Specification, N

f'_c = specified compressive strength of concrete, MPa

F_{yh} = specified minimum yield strength of the ties, MPa

Equation 6.13.27 need not be satisfied if the Nominal Strength of the reinforced concrete encased structural steel section alone is greater than $1.0D + 0.5L$.

The maximum spacing of transverse reinforcement along the length of the column shall be the lesser of 6 longitudinal load-carrying bar diameters and 152 mm (6 in.).

When specified in Sec 13.5.5.3.3, the maximum spacing of transverse reinforcement shall be the lesser of one-fourth the least member dimension and 102 mm (4 in.). For this reinforcement, cross ties, legs of overlapping hoops, and other confining reinforcement shall be spaced not more than 355 mm (14 in.) on center in the transverse direction.

- (d) Reinforced-Concrete-Encased Composite Columns in Braced Frames with axial compression forces that are larger than 0.2 times P_0 shall have transverse reinforcement as specified in Sec 13.5.5.3.3, over the total element length. This requirement need not be satisfied if the Nominal Strength of the reinforced-concrete-encased steel section alone is greater than $1.0D + 0.5L$.
- (e) Composite Columns supporting reactions from discontinued stiff members, such as walls or Braced Frames, shall have transverse reinforcement as specified in Sec 13.5.5.3.3 over the full length beneath the level at which the discontinuity occurs if the axial compression force exceeds 0.1 times P_0 . Transverse reinforcement shall extend into the discontinued member for at least the length required to develop full yielding in the reinforced-concrete-encased structural steel section and longitudinal reinforcement. This requirement need not be satisfied if the Nominal Strength of the reinforced-concrete-encased structural steel section alone is greater than $1.0D + 0.5L$.
- (f) Reinforced-Concrete-Encased Composite Columns that are used in C-SMF shall meet the following requirements:
 - (i) Transverse reinforcement shall meet the requirements in 13.5.5.3.3 at the top and bottom of the column over the region specified in Sec 6.4b.
 - (ii) The strong-column/weak-beam design requirements shall be satisfied. Column Bases shall be detailed to sustain inelastic flexural hinging.
 - (iii) The minimum required shear strength of column shall meet the requirements in Chapter 6 Part 6.

- (g) When the column terminates on a footing or mat foundation, the transverse reinforcement as specified in this section shall extend into the footing or mat at least 305 mm (12 in.). When the column terminates on a wall, the transverse reinforcement shall extend into the wall for at least the length required to develop full yielding in the reinforced-concrete-encased structural steel section and longitudinal reinforcement.
- (h) Welded wire fabric is not permitted as transverse reinforcement for special seismic systems.

13.5.5.4 Concrete filled composite columns

This Section is applicable to columns that: (i) consist of concrete-filled steel rectangular or circular hollow structural sections (HSS) with a structural steel area that comprises at least 4 percent of the total composite-column cross-section; and (ii) meet the additional limitations in Sec 13.2. Such columns shall be designed to meet the requirements in Sec 13.2, except as modified in this Section.

The design shear strength of the Composite Column shall be the design shear strength of the structural steel section alone.

In the special seismic systems described in, members and column splices for Concrete-Filled Composite Columns shall also meet the requirements in Sec 10.20 Chapter 10 Part 6.

Concrete-Filled Composite Columns used in C-SMF shall meet the following additional requirements:

- (a) The minimum required shear strength of the column shall meet the requirements in Chapter 5 Part 6.
- (b) The strong-column/weak-beam design requirements shall be met. Column Bases shall be designed to sustain inelastic flexural hinging.
- (c) The minimum wall thickness of concrete-filled rectangular HSS shall equal $b\sqrt{F_y/(2E_s)}$ for the flat width b of each face, where b is as defined in Table 6.10.1 Chapter 10 Part 6.

13.5.6 Composite Steel Plate Shear Walls (C-SPW)

13.5.6.1 Scope

This Section is applicable to structural walls consisting of steel plates with reinforced concrete encasement on one or both sides of the plate and structural steel or composite Boundary Members. C-SPW shall meet the requirements of this section.

13.5.6.2 Wall elements

13.5.6.2.1 Nominal shear strength

The nominal shear strength of C-SPW with a stiffened plate conforming to Section 13.5.4.2.2 shall be determined as:

$$V_{ns} = 0.6A_{sp}F_y \quad (6.13.28)$$

Where,

V_{ns} = nominal shear strength of the steel plate, N

A_{sp} = horizontal area of stiffened steel plate, mm²

F_y = specified minimum yield strength of the plate, MPa

The nominal shear strength of C-SPW with a plate that does not meet the stiffening requirements in Sec 13.5.4.2.2 shall be based upon the strength of the plate, excluding the strength of the reinforced concrete, and meet the requirements in the Chapter 10 Part 6, including the effects of buckling of the plate.

13.5.6.2.2 Detailing requirements

The steel plate shall be adequately stiffened by encasement or attachment to the reinforced concrete if it can be demonstrated with an elastic plate buckling analysis that the composite wall can resist a nominal shear force equal to V_{ns} . The concrete thickness shall be a minimum of 100 mm (4 in.) on each side when concrete is provided on both sides of the steel plate and 200 mm (8 in.) when concrete is provided on one side of the steel plate. Headed shear stud connectors or other mechanical connectors shall be provided to prevent local buckling and separation of the plate and reinforced concrete. Horizontal and vertical reinforcement shall be provided in the concrete encasement to meet the detailing requirements in Chapter 8 Part 6. The reinforcement ratio in both directions shall not be less than 0.0025; maximum spacing between bars shall not exceed 450 mm (18 in.).

The steel plate shall be continuously connected on all edges to structural steel framing and Boundary Members with welds and/or slip-critical high-strength bolts to develop the nominal shear strength of the plate. The Design Strength of welded and bolted connectors shall meet the additional requirements in Chapter 10 Part 6.

13.6 Referenced Specifications, Codes and Standards

The documents referenced in these provisions shall include those listed in Sec 10.1.2.3 Chapter 10 Part 6 with the following additions and modifications:

American Society of Civil Engineers

ASCE 3-91 Standard for the Structural Design of Composite Slabs

American Welding Society

AWS D1.1-04 Structural Welding Code- Steel

AWS D1.4-98 Structural Welding Code-Reinforcing Steel

Canadian Standards Association

CSA S16-01 Design of Steel Structures

PART VI

Appendix A

Equivalence of Nonhomogenous Equations in SI-Metric,
MKS-Metric, and U.S. Customary Units

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
General	1 MPa	10 kgf/cm ²	142.2 psi
	$f'_c = 21 \text{ MPa}$	$f'_c = 210 \text{ kgf/cm}^2$	$f'_c = 3000 \text{ psi}$
	$f'_c = 28 \text{ MPa}$	$f'_c = 280 \text{ kgf/cm}^2$	$f'_c = 4000 \text{ psi}$
	$f'_c = 35 \text{ MPa}$	$f'_c = 350 \text{ kgf/cm}^2$	$f'_c = 5000 \text{ psi}$
	$f'_c = 40 \text{ MPa}$	$f'_c = 420 \text{ kgf/cm}^2$	$f'_c = 6000 \text{ psi}$
	$f_y = 280 \text{ MPa}$	$f_y = 2800 \text{ kgf/cm}^2$	$f_y = 40,000 \text{ psi}$
	$f_y = 420 \text{ MPa}$	$f_y = 4200 \text{ kgf/cm}^2$	$f_y = 60,000 \text{ psi}$
	$f_{pu} = 1725 \text{ MPa}$	$f_{pu} = 17,600 \text{ kgf/cm}^2$	$f_{pu} = 250,000 \text{ psi}$
	$f_{pu} = 1860 \text{ MP}$	$f_{pu} = 19,000 \text{ kgf/cm}^2$	$f_{pu} = 270,000 \text{ psi}$
	$\sqrt{f'_c} \text{ in MPa}$	$3.18 \sqrt{f'_c} \text{ in kgf/cm}^2$	$12 \sqrt{f'_c} \text{ in psi}$
	$0.313 \sqrt{f'_c} \text{ in MPa}$	$\sqrt{f'_c} \text{ in kgf/cm}^2$	$3.77 \sqrt{f'_c} \text{ in psi}$
	$0.083 \sqrt{f'_c} \text{ in MPa}$	$0.27 \sqrt{f'_c} \text{ in kgf/cm}^2$	$\sqrt{f'_c} \text{ in psi}$
	$0.17 \sqrt{f'_c} \text{ in MPa}$	$0.53 \sqrt{f'_c} \text{ in kgf/cm}^2$	$2 \sqrt{f'_c} \text{ in psi}$
	$f'_{cr} = f'_c + 2.33S_s - 35$	$f'_{cr} = f'_c + 2.33S_s - 35$	$f'_{cr} = f'_c + 2.33S_s - 35$
	$f'_{cr} = f'_c + 7.0$	$f'_{cr} = f'_c + 70$	$f'_{cr} = f'_c + 1000$
	$f'_{cr} = f'_c + 8.3$	$f'_{cr} = f'_c + 84$	$f'_{cr} = f'_c + 1200$
	$f'_{cr} = 1.10f'_c + 5.0$	$f'_{cr} = 1.10f'_c + 50$	$f'_{cr} = 1.10f'_c + 700$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Eq. 6.6.3	$f_r = 0.62\lambda\sqrt{f'_c}$	$f_r = 2\lambda\sqrt{f'_c}$	$f_r = 7.5\lambda\sqrt{f'_c}$
Eq. 6.6.5	$h = \frac{\ell_n \left(0.8 + \frac{f_y}{1400}\right)}{36 + 5\beta(\alpha_{fm} - 0.2)} \geq 125mm$	$h = \frac{\ell_n \left(0.8 + \frac{f_y}{14,000}\right)}{36 + 5\beta(\alpha_{fm} - 0.2)} \geq 125mm$	$h = \frac{\ell_n \left(0.8 + \frac{f_y}{200,000}\right)}{36 + 5\beta(\alpha_{fm} - 0.2)} \geq 5in$
Eq. 6.6.6	$h = \frac{\ell_n \left(0.8 + \frac{f_y}{1400}\right)}{36 + 9\beta} \geq 90mm$	$h = \frac{\ell_n \left(0.8 + \frac{f_y}{14,000}\right)}{36 + 9\beta} \geq 9cm$	$h = \frac{\ell_n \left(0.8 + \frac{f_y}{200,000}\right)}{36 + 9\beta} \geq 3.5in$
Eq. 6.6.10	$A \frac{0.25\sqrt{f'_c}}{f_y} \frac{1.4}{w} \frac{1}{f_y w_{s,min}}$	$A \frac{0.8\sqrt{f'_c}}{f_y} \frac{14}{w} \frac{1}{f_y w_{s,min}}$	$A \frac{3\sqrt{f'_c}}{f_y} \frac{200}{w} \frac{1}{f_y w_{s,min}}$
Eq. 6.6.11	$s = 380 \left(\frac{280}{f_s}\right) - 2.5c_c \leq 300 \left(\frac{280}{f_s}\right)$	$s = 38 \left(\frac{2800}{f_s}\right) - 2.5c_c \leq 30 \left(\frac{2800}{f_s}\right)$	$s = 15 \left(\frac{40,000}{f_s}\right) - 2.5c_c \leq 12 \left(\frac{40,000}{f_s}\right)$
Eq. 6.6.24	$M_{2,min} = P_u (15 + 0.03h)$	$M_{2,min} = P_u (1.5 + 0.03h)$	$M_{2,min} = P_u (0.6 + 0.03h)$
Eq. 6.6.49	$V_c = 0.17\lambda\sqrt{f'_c}b_wd$	$V_c = 0.53\lambda\sqrt{f'_c}b_wd$	$V_c = 2\lambda\sqrt{f'_c}b_wd$
Eq. 6.6.50	$V_c = 0.17 \left(1 + \frac{N_u}{14A_g}\right) \lambda\sqrt{f'_c}b_wd$	$V_c = 0.53 \left(1 + \frac{N_u}{140A_g}\right) \lambda\sqrt{f'_c}b_wd$	$V_c = 2 \left(1 + \frac{N_u}{2000A_g}\right) \lambda\sqrt{f'_c}b_wd$
Eq. 6.6.51	$V_c = \left(0.16\lambda\sqrt{f'_c} + 17\rho_w \frac{V_u d}{M_u}\right) b_wd \leq 0.29\lambda\sqrt{f'_c}b_wd$	$V_c = \left(0.5\lambda\sqrt{f'_c} + 176\rho_w \frac{V_u d}{M_u}\right) b_wd \leq 0.93\lambda\sqrt{f'_c}b_wd$	$V_c = \left(1.9\lambda\sqrt{f'_c} + 25006\rho_w \frac{V_u d}{M_u}\right) b_wd \leq 3.5\lambda\sqrt{f'_c}b_wd$
Eq. 6.6.53	$V_c = 0.29\lambda\sqrt{f'_c}b_wd \sqrt{1 + \frac{0.29N_u}{A_g}}$	$V_c = 0.93\lambda\sqrt{f'_c}b_wd \sqrt{1 + \frac{N_u}{35A_g}}$	$V_c = 3.5\lambda\sqrt{f'_c}b_wd \sqrt{1 + \frac{N_u}{500A_g}}$
Eq. 6.6.54	$V_c = 0.17 \left(1 + \frac{0.29N_u}{A_g}\right) \lambda\sqrt{f'_c}b_wd \geq 0$	$V_c = 0.53 \left(1 + \frac{N_u}{35A_g}\right) \lambda\sqrt{f'_c}b_wd \geq 0$	$V_c = 2 \left(1 + \frac{N_u}{500A_g}\right) \lambda\sqrt{f'_c}b_wd \geq 0$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Eq. 6.6.55	$A\sqrt{f'_c}\frac{b_ws}{f_{yt}} \geq 0.35\frac{b_ws}{f_{yt}}$	$A\sqrt{f'_c}\frac{b_ws}{f_{yt}} \geq 3.5\frac{b_ws}{f_{yt}}$	$A\sqrt{f'_c}\frac{b_ws}{f_{yt}} \geq 50\frac{b_ws}{f_{yt}}$
Eq. 6.6.58	$V_s = A_v f_y \sin \alpha \leq 0.25\sqrt{f'_c}b_w d$	$V_s = A_v f_y \sin \alpha \leq 0.8\sqrt{f'_c}b_w d$	$V_s = A_v f_y \sin \alpha \leq 3\sqrt{f'_c}b_w d$
Eq. 6.6.59	$\sqrt{\left(\frac{V_u}{b_w d}\right)^2 + \left(\frac{T_u p_h}{1.7A_{oh}^2}\right)^2} \leq \phi \left(\frac{V_c}{b_w d} + 0.66\sqrt{f'_c}\right)$	$\sqrt{\left(\frac{V_u}{b_w d}\right)^2 + \left(\frac{T_u p_h}{1.7A_{oh}^2}\right)^2} \leq \phi \left(\frac{V_c}{b_w d} + 2\sqrt{f'_c}\right)$	$\sqrt{\left(\frac{V_u}{b_w d}\right)^2 + \left(\frac{T_u p_h}{1.7A_{oh}^2}\right)^2} \leq \phi \left(\frac{V_c}{b_w d} + 8\sqrt{f'_c}\right)$
Eq. 6.6.60	$\left(\frac{V_u}{b_w d}\right) + \left(\frac{T_u p_h}{1.7A_{oh}^2}\right) \leq \phi \left(\frac{V_c}{b_w d} + 0.66\sqrt{f'_c}\right)$	$\left(\frac{V_u}{b_w d}\right) + \left(\frac{T_u p_h}{1.7A_{oh}^2}\right) \leq \phi \left(\frac{V_c}{b_w d} + 2\sqrt{f'_c}\right)$	$\left(\frac{V_u}{b_w d}\right) + \left(\frac{T_u p_h}{1.7A_{oh}^2}\right) \leq \phi \left(\frac{V_c}{b_w d} + 8\sqrt{f'_c}\right)$
Eq. 6.6.64	$(A_v + 2A_t) = 0.062\sqrt{f'_c}\frac{b_ws}{f_{yt}} \geq \frac{0.35b_ws}{f_{yt}}$	$(A_v + 2A_t) = 0.2\sqrt{f'_c}\frac{b_ws}{f_{yt}} \geq \frac{3.5b_ws}{f_{yt}}$	$(A_v + 2A_t) = 0.75\sqrt{f'_c}\frac{b_ws}{f_{yt}} \geq \frac{50b_ws}{f_{yt}}$
Eq. 6.6.65	$A\frac{0.42\sqrt{f'_c}A_{cp}\left(\frac{A_t}{s}\right)_h}{f_y} \frac{f_{yt}}{f_{y\min}} \geq \frac{0.175b_w}{f_{yt}}$	$A\frac{1.33\sqrt{f'_c}A_{cp}\left(\frac{A_t}{s}\right)_h}{f_y} \frac{f_{yt}}{f_{y\min}} \geq \frac{1.75b_w}{f_{yt}}$	$A\frac{5\sqrt{f'_c}A_{cp}\left(\frac{A_t}{s}\right)_h}{f_y} \frac{f_{yt}}{f_{y\min}} \geq \frac{25b_w}{f_{yt}}$
Eq. 6.6.68	$V_c = 0.27\lambda\sqrt{f'_c}h + \frac{N_u d}{4\ell_w}$	$V_c = 0.88\lambda\sqrt{f'_c}hd + \frac{N_u d}{4\ell_w}$	$V_c = 3.3\lambda\sqrt{f'_c}hd + \frac{N_u d}{4\ell_w}$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Eq. 6.6.69	$V_c = \left[0.05\lambda\sqrt{f'_c} + \frac{l_w \left(0.1\lambda\sqrt{f'_c} + \frac{0.2N_u}{l_w h} \right)}{\frac{M_u}{V_u} - \frac{l_w}{2}} \right] h d$	$V_c = \left[0.16\lambda\sqrt{f'_c} + \frac{l_w \left(0.33\lambda\sqrt{f'_c} + \frac{0.2N_u}{l_w h} \right)}{\frac{M_u}{V_u} - \frac{l_w}{2}} \right] h d$	$V_c = \left[0.6\lambda\sqrt{f'_c} + \frac{l_w \left(1.25\lambda\sqrt{f'_c} + \frac{0.2N_u}{l_w h} \right)}{\frac{M_u}{V_u} - \frac{l_w}{2}} \right] h d$
Eq. 6.6.72	$V_c = 0.17 \left(1 + \frac{2}{\beta} \right) \lambda \sqrt{f'_c} b_o d$	$V_c = 0.53 \left(1 + \frac{2}{\beta} \right) \lambda \sqrt{f'_c} b_o d$	$V_c = \left(2 + \frac{4}{\beta} \right) \lambda \sqrt{f'_c} b_o d$
Eq. 6.6.73	$V_c = 0.083 \left(\frac{\alpha_s d}{b_o} + 2 \right) \lambda \sqrt{f'_c} b_o d$	$V_c = 0.27 \left(\frac{\alpha_s d}{b_o} + 2 \right) \lambda \sqrt{f'_c} b_o d$	$V_c = \left(\frac{\alpha_s d}{b_o} + 2 \right) \lambda \sqrt{f'_c} b_o d$
Eq. 6.6.74	$V_c = 0.33\lambda\sqrt{f'_c} b_o d$	$V_c = \lambda\sqrt{f'_c} b_o d$	$V_c = 4\lambda\sqrt{f'_c} b_o d$
Eq. 6.8.1	$e_d = \frac{f_y}{1.1\lambda\sqrt{f'_c}} \frac{\psi_t \psi_e \psi_s}{\left(\frac{c_b + K_{tr}}{d_b} \right)} d_b$	$e_d = \frac{f_y}{3.5\lambda\sqrt{f'_c}} \frac{\psi_t \psi_e \psi_s}{\left(\frac{c_b + K_{tr}}{d_b} \right)} d_b$	$e_d = \frac{3f_y}{40\lambda\sqrt{f'_c}} \frac{\psi_t \psi_e \psi_s}{\left(\frac{c_b + K_{tr}}{d_b} \right)} d_b$
Eq. 6.8.4	$e_d = 3.3 \left(\frac{A_b}{s} \right) \left(\frac{f_y}{\lambda\sqrt{f'_c}} \right)$	$e_d = \left(\frac{A_b}{s} \right) \left(\frac{f_y}{\lambda\sqrt{f'_c}} \right)$	$e_d = 0.27 \left(\frac{A_b}{s} \right) \left(\frac{f_y}{\lambda\sqrt{f'_c}} \right)$
Eq. 6.8.9	$e_{dh} = \frac{f_y d_b}{5.4\sqrt{f'_c}}$	$e_{dh} = \frac{f_y d_b}{17.2\sqrt{f'_c}}$	$e_{dh} = \frac{f_y d_b}{65\sqrt{f'_c}}$
Eq. 6.8.10	$V_n = A_{cv} (0.17\lambda\sqrt{f'_c} + \rho_n f_y)$	$V_n = A_{cv} (0.53\lambda\sqrt{f'_c} + \rho_n f_y)$	$V_n = A_{cv} (2\lambda\sqrt{f'_c} + \rho_n f_y)$
Eq. 6.8.11	$V_n = 2A_{vd} f_y \sin \alpha$ $\leq 0.83\sqrt{f'_c} A_{cw}$	$V_n = 2A_{vd} f_y \sin \alpha$ $\leq 2.65\sqrt{f'_c} A_{cw}$	$V_n = 2A_{vd} f_y \sin \alpha$ $\leq 10\sqrt{f'_c} A_{cw}$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Eq. 6.8.14	$V_n = A_{cv}(\alpha_c \lambda \sqrt{f'_c} + \rho_n f_y)$ $\alpha_c = 0.25 \text{ for } \frac{h_w}{\ell_w} \leq 1.5$ $\alpha_c = 0.17 \text{ for } \frac{h_w}{\ell_w} \geq 2.0$	$V_n = A_{cv}(\alpha_c \lambda \sqrt{f'_c} + \rho_n f_y)$ $\alpha_c = 0.80 \text{ for } \frac{h_w}{\ell_w} \leq 1.5$ $\alpha_c = 0.53 \text{ for } \frac{h_w}{\ell_w} \geq 2.0$	$V_n = A_{cv}(\alpha_c \lambda \sqrt{f'_c} + \rho_n f_y)$ $\alpha_c = 3.0 \text{ for } \frac{h_w}{\ell_w} \leq 1.5$ $\alpha_c = 2.0 \text{ for } \frac{h_w}{\ell_w} \geq 2.0$
Eq. D.7	$N_b = k_c \lambda \sqrt{f'_c} h_{ef}^{1.5}$ $k_c = 10 \text{ or } 7$	$N_b = k_c \lambda \sqrt{f'_c} h_{ef}^{1.5}$ $k_c = 10 \text{ or } 7$	$N_b = k_c \lambda \sqrt{f'_c} h_{ef}^{1.5}$ $k_c = 24 \text{ or } 17$
Eq. D.8	$N_b = 3.9 \lambda \sqrt{f'_c} h_{ef}^{5/3}$	$N_b = 5.8 \lambda \sqrt{f'_c} h_{ef}^{5/3}$	$N_b = 16 \lambda \sqrt{f'_c} h_{ef}^{5/3}$
Eq. D.17	$N_{sb} = 13 c_{a1} \sqrt{A_{brg} \lambda \sqrt{f'_c}}$	$N_{sb} = 42.5 c_{a1} \sqrt{A_{brg} \lambda \sqrt{f'_c}}$	$N_{sb} = 160 c_{a1} \sqrt{A_{brg} \lambda \sqrt{f'_c}}$
Eq. D.24	V_b $= 0.6 \left(\frac{\lambda_e}{d_a} \right)^{0.2} \sqrt{d_a} \lambda \sqrt{f'_c} (c_{a1})^{1.5}$	$V_b = 1.9 \left(\frac{\lambda_e}{d_a} \right)^{0.2} \sqrt{d_a} \lambda \sqrt{f'_c} (c_{a1})^{1.5}$	$V_b = 7 \left(\frac{\lambda_e}{d_a} \right)^{0.2} \sqrt{d_a} \lambda \sqrt{f'_c} (c_{a1})^{1.5}$
Eq. D.25	V_b $= 0.66 \left(\frac{\lambda_e}{d_a} \right)^{0.2} \sqrt{d_a} \lambda \sqrt{f'_c} (c_{a1})^{1.5}$	$V_b = 2.1 \left(\frac{\lambda_e}{d_a} \right)^{0.2} \sqrt{d_a} \lambda \sqrt{f'_c} (c_{a1})^{1.5}$	$V_b = 8 \left(\frac{\lambda_e}{d_a} \right)^{0.2} \sqrt{d_a} \lambda \sqrt{f'_c} (c_{a1})^{1.5}$
Sec 6.1.8.1	$E_c = w_c^{1.5} 0.043 \sqrt{f'_c}$ $E_c = 4700 \sqrt{f'_c}$	$E_c = w_c^{1.5} 0.14 \sqrt{f'_c}$ $E_c = 15,100 \sqrt{f'_c}$	$E_c = v_c^{1.5} 33 \sqrt{f'_c}$ $E_c = 57,000 \sqrt{f'_c}$
Sec 6.1.9.1	$\lambda = \frac{f_{ct}}{(0.56 \sqrt{f'_c})} \leq 1.0$	$\lambda = \frac{f_{ct}}{(1.78 \sqrt{f'_c})} \leq 1.0$	$\lambda = \frac{f_{ct}}{(6.7 \sqrt{f'_c})} \leq 1.0$
Sec 6.4.1.2	$\sqrt{f'_c} \leq 8.3 \text{ MPa}$	$\sqrt{f'_c} \leq 27 \text{ kgf/cm}^2$	$\sqrt{f'_c} \leq 100 \text{ psi}$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Sec 6.4.3.4.3	$0.33\sqrt{f'_c}b_wd$	$1.1\sqrt{f'_c}b_wd$	$4\sqrt{f'_c}b_wd$
Sec 6.4.3.5.1(f)	$0.17\phi\sqrt{f'_c}b_wd$	$0.53\phi\sqrt{f'_c}b_wd$	$2\phi\sqrt{f'_c}b_wd$
Sec 6.4.3.6.9	$0.66\sqrt{f'_c}b_wd$	$2.2\sqrt{f'_c}b_wd$	$8\sqrt{f'_c}b_wd$
Sec 6.4.4.1(a)	$T_u < 0.083\phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)$	$T_u < 0.27\phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)$	$T_u < \phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)$
Sec 6.4.4.1(b)	$T_u < 0.083\phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{N_u}{0.33A_g\lambda\sqrt{f'_c}}}$	$T_u < 0.27\phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{N_u}{A_g\lambda\sqrt{f'_c}}}$	$T_u < \phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{N_u}{4A_g\lambda\sqrt{f'_c}}}$
Sec 6.4.4.2.2(a)	$T_u = \phi 0.33\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)$	$T_u = \phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)$	$T_u = \phi 4\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)$
Sec 6.4.4.2.2(b)	$T_u = 0.33\phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{N_u}{0.33A_g\lambda\sqrt{f'_c}}}$	$T_u = \phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{N_u}{A_g\lambda\sqrt{f'_c}}}$	$T_u = 4\phi\lambda\sqrt{f'_c}\left(\frac{A_{cp}^2}{p_{cp}}\right)\sqrt{1 + \frac{N_u}{4A_g\lambda\sqrt{f'_c}}}$
Sec 6.4.5.5	$(3.3 + 0.08f'_c)A_c$ $11A_c$ $5.5A_c$	$(34 + 0.08f'_c)A_c$ $110A_c$ $55A_c$	$(480 + 0.08f'_c)A_c$ $1600A_c$ $800A_c$
Sec 6.4.6.3	$0.83\sqrt{f'_c}b_wd$	$2.65\sqrt{f'_c}b_wd$	$10\sqrt{f'_c}b_wd$
Sec 6.4.7.3.2.1	$(3.3 + 0.08f'_c)b_wd$ $11b_wd$	$(34 + 0.08f'_c)b_wd$ $110b_wd$	$(480 + 0.08f'_c)b_wd$ $1600b_wd$
Sec 6.4.7.3.2.2	$(5.5 - 1.9\frac{a_v}{d})b_wd$	$(55 - 20\frac{a_v}{d})b_wd$	$(800 - 280\frac{a_v}{d})b_wd$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Sec 6.4.8.3	$0.83\sqrt{f'_c}hd$	$2.65\sqrt{f'_c}hd$	$10\sqrt{f'_c}hd$
Sec 6.4.8.5	$0.17\lambda\sqrt{f'_c}hd$	$0.53\lambda\sqrt{f'_c}hd$	$2\lambda\sqrt{f'_c}hd$
Sec 6.4.10.3.1	$0.17\lambda\sqrt{f'_c}b_o d$	$0.53\lambda\sqrt{f'_c}b_o d$	$2\lambda\sqrt{f'_c}b_o d$
Sec 6.4.10.3.2	$0.5\sqrt{f'_c}b_o d$	$1.6\sqrt{f'_c}b_o d$	$6\sqrt{f'_c}b_o d$
Sec 6.4.10.4.8	$0.33\sqrt{f'_c}b_o d$ $0.58\sqrt{f'_c}b_o d$	$1.1\sqrt{f'_c}b_o d$ $1.9\sqrt{f'_c}b_o d$	$4\sqrt{f'_c}b_o d$ $7\sqrt{f'_c}b_o d$
Sec 6.4.10.5.1	$0.25\lambda\sqrt{f'_c}b_o d$ $0.66\sqrt{f'_c}b_o d$ $0.17\sqrt{f'_c}$	$0.8\lambda\sqrt{f'_c}b_o d$ $2.1\sqrt{f'_c}b_o d$ $0.53\sqrt{f'_c}$	$3\lambda\sqrt{f'_c}b_o d$ $8\sqrt{f'_c}b_o d$ $2\sqrt{f'_c}$
Sec 6.4.10.5.2	$0.5\phi\sqrt{f'_c}$	$1.6\phi\sqrt{f'_c}$	$6\phi\sqrt{f'_c}$
Sec 6.4.10.5.4 & Sec 6.4.10.7.2	$0.17\phi\lambda\sqrt{f'_c}$	$0.53\phi\lambda\sqrt{f'_c}$	$2\phi\lambda\sqrt{f'_c}$
Sec 6.4.10.7.3 & Sec 6.9.4.10	$0.33\phi\lambda\sqrt{f'_c}$	$1.1\phi\lambda\sqrt{f'_c}$	$4\phi\lambda\sqrt{f'_c}$
Sec 6.12.5.3.1 & Sec 6.12.5.3.2	$0.55b_v d$	$5.6b_v d$	$80b_v d$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Sec 6.12.5.3.3	$(1.8 + 0.6\rho_v f_y) \lambda b_v d \leq 3.5 b_v d$	$(18 + 0.6\rho_v f_y) \lambda b_v d \leq 35 b_v d$	$(260 + 0.6\rho_v f_y) \lambda b_v d \leq 500 b_v d$
Sec 8.1.11.2(a)	$\frac{0.0018 \times 420}{f_y}$	$\frac{0.0018 \times 4200}{f_y}$	$\frac{0.0018 \times 60000}{f_y}$
Sec 8.2.2	$\sqrt{f'_c} \leq 8.3 \text{ MPa}$	$\sqrt{f'_c} \leq 26.5 \text{ kgf/cm}^2$	$\sqrt{f'_c} \leq 100 \text{ psi}$
Sec 8.2.3.2	$e_d = \left(\frac{f_y \psi_t \psi_e}{2.1 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{f_y \psi_t \psi_e}{1.7 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{f_y \psi_t \psi_e}{1.4 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{f_y \psi_t \psi_e}{1.1 \lambda \sqrt{f'_c}} \right) d_b$	$e_d = \left(\frac{f_y \psi_t \psi_e}{6.6 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{f_y \psi_t \psi_e}{5.3 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{f_y \psi_t \psi_e}{4.4 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{f_y \psi_t \psi_e}{3.5 \lambda \sqrt{f'_c}} \right) d_b$	$e_d = \left(\frac{f_y \psi_t \psi_e}{25 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{f_y \psi_t \psi_e}{20 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{3 f_y \psi_t \psi_e}{50 \lambda \sqrt{f'_c}} \right) d_b$ $e_d = \left(\frac{3 f_y \psi_t \psi_e}{40 \lambda \sqrt{f'_c}} \right) d_b$
Sec 8.2.4.2	$\left(\frac{0.24 f_y}{\lambda \sqrt{f'_c}} \right) d_b \geq (0.043 f_y) d_b$	$\left(\frac{0.075 f_y}{\lambda \sqrt{f'_c}} \right) d_b \geq (0.0044 f_y) d_b$	$\left(\frac{0.02 f_y}{\lambda \sqrt{f'_c}} \right) d_b \geq (0.0003 f_y) d_b$
Sec 8.2.6.2	$e_{dh} = \left(\frac{0.24 \psi_e f_y}{\lambda \sqrt{f'_c}} \right) d_b$	$e_{dh} = \left(\frac{0.075 \psi_e f_y}{\lambda \sqrt{f'_c}} \right) d_b$	$e_{dh} = \left(\frac{0.02 \psi_e f_y}{\lambda \sqrt{f'_c}} \right) d_b$
Sec 8.2.7.5(b)	$0.41 \frac{b_w s}{f_{yt}}$	$4.2 \frac{b_w s}{f_{yt}}$	$60 \frac{b_w s}{f_{yt}}$
Sec 8.2.10.2(b)	$0.17 \frac{d_b f_{yt}}{\lambda \sqrt{f'_c}}$	$0.053 \frac{d_b f_{yt}}{\lambda \sqrt{f'_c}}$	$0.014 \frac{d_b f_{yt}}{\lambda \sqrt{f'_c}}$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Sec 8.2.14.1	$0.071f_y d_b$ $(0.13f_y - 24)d_b$	$0.0073f_y d_b$ $(0.013f_y - 24)d_b$	$0.0005f_y d_b$ $(0.0009f_y - 24)d_b$
Sec 8.2.17.2	$\ell_{dt} = \left(\frac{0.19\psi_e f_y}{\sqrt{f'_c}} \right) d_b$	$\ell_{dt} = \left(\frac{0.06\psi_e f_y}{\sqrt{f'_c}} \right) d_b$	$\ell_{dt} = \left(\frac{0.016\psi_e f_y}{\sqrt{f'_c}} \right) d_b$
Sec 8.2.18.2	$\left(\frac{f_y - 240}{f_y} \right)$	$\left(\frac{f_y - 2460}{f_y} \right)$	$\left(\frac{f_y - 35,000}{f_y} \right)$
Sec 8.3.4.2(a)	$\frac{1.4b_w d}{f_y}$	$\frac{14b_w d}{f_y}$	$\frac{200b_w d}{f_y}$
Sec 8.3.5.4 & Figure 6.8.8	$s_o = 100 + \left(\frac{350 - h_x}{3} \right)$ $100 \text{ mm} \leq s_o \leq 150 \text{ mm}$	$s_o = 10 + \left(\frac{35 - h_x}{3} \right)$ $10 \text{ cm} \leq s_o \leq 15 \text{ cm}$	$s_o = 4 + \left(\frac{14 - h_x}{3} \right)$ $4 \text{ in.} \leq s_o \leq 6 \text{ in.}$
Sec 8.3.6.2(a)	$0.083A_{cv}\lambda\sqrt{f'_c}$	$0.27A_{cv}\lambda\sqrt{f'_c}$	$A_{cv}\lambda\sqrt{f'_c}$
Sec 8.3.6.2(b)	$0.17A_{cv}\lambda\sqrt{f'_c}$	$0.53A_{cv}\lambda\sqrt{f'_c}$	$2A_{cv}\lambda\sqrt{f'_c}$
Sec 8.3.6.6	$2.8/f_y$	$28/f_y$	$400/f_y$
Sec 8.3.6.7 (b)	$0.33\lambda\sqrt{f'_c}A_{cw}$	$\lambda\sqrt{f'_c}A_{cw}$	$4\lambda\sqrt{f'_c}A_{cw}$
Sec 8.3.7.3	$1.7\sqrt{f'_c}A_j$ $1.2\sqrt{f'_c}A_j$ $1.0\sqrt{f'_c}A_j$	$5.3\sqrt{f'_c}A_j$ $4\sqrt{f'_c}A_j$ $3.2\sqrt{f'_c}A_j$	$20\sqrt{f'_c}A_j$ $15\sqrt{f'_c}A_j$ $12\sqrt{f'_c}A_j$

Reference	SI-metric stress in MPa	MKS-metric stress in kgf/cm ²	U.S. Customary unit stress in pounds per square inch (psi)
Sec 8.3.8.3(f)	$0.67A_{cv}\sqrt{f'_c}$ $0.83A_{cp}\sqrt{f'_c}$	$2.12A_{cv}\sqrt{f'_c}$ $2.65A_{cp}\sqrt{f'_c}$	$8A_{cv}\sqrt{f'_c}$ $10A_{cp}\sqrt{f'_c}$
Sec 8.3.8.3(g)	$0.83A_{cp}\sqrt{f'_c}$	$2.65A_{cp}\sqrt{f'_c}$	$10A_{cp}\sqrt{f'_c}$
Sec 8.3.12.4	$0.29\sqrt{f'_c}b_o d$	$0.93\sqrt{f'_c}b_o d$	$3.5\sqrt{f'_c}b_o d$
Table 6.6.1	$(1.65 - 0.0003w_c) \geq 1.09$ $\left(0.4 + \frac{f_y}{700}\right)$	$(1.65 - 0.0003w_c) \geq 1.09$ $\left(0.4 + \frac{f_y}{7000}\right)$	$(1.65 - 0.005w_c) \geq 1.09$ $\left(0.4 + \frac{f_y}{100,000}\right)$

PART VI
Appendix B

Local Geology, Tectonic Features and Earthquake Occurrence in the Region

B.1 Seismotectonics and Earthquake Occurrence in and Around Bangladesh

B.1.1 General

Earthquakes are closely related to plate tectonics. Bangladesh is located in a tectonically active region close to the plate boundaries of the Indian plate and the Eurasian plate. The plate boundaries lie to the north and east of Bangladesh. The collision of the north-east moving (around 4 cm or more annually) Indian Plate with the Eurasian plate (Figure 6.B.1) is the cause of frequent earthquakes in the region comprising North East India, Nepal, Bhutan, Bangladesh and Myanmar.

Major thrust faults exist to the north (at the foot of the Himalayas) and to the east (in the Indo-Burma mountain ranges) of Bangladesh. Geologic evolution of the Bengal Basin is related to this collision. Bangladesh constitutes the major portion of the Bengal Basin. The collision perhaps started in the late Cretaceous and the Bay of Bengal and the Bengal Basin attained its present configuration at the end of middle Miocene orogeny.

Tectonically Bangladesh is divided broadly into three divisions: (i) Stable Shelf (in the northwest) (ii) Bengal Foredeep (in the Central) and (iii) Chittagong-Tripura Folded Belt (in the east). In addition there is a SW-NE trending 25 km wide hinge zone separating the Bengal Foredeep from the Stable Shelf.

The Stable Pre-Cambrian Shelf in the northwest consists of relatively thin sedimentary strata over bedrock. In Madhyapara area of Dinajpur the basement is only 130 m deep from the ground surface and is overlain by Dupi Tila Sandstone and Madhupur Clay of Plio-Pliocene age. The basement plunges gently from Madhyapara towards the southeast up to the Hinge Zone. Seismic contours on top of limestone in Bogra show regional dip of 2-3° besides revealing a number of NE-SW trending faults. The hinge zone is characterized by the sharp change in the dip of the basement rocks associated with deep-seated displacements in faults and is reflected on the gravity and magnetic anomalies. In the hinge zone, the depth of the limestone increases from 4000m to 9000m within a narrow zone of 25-km. Hinge Zone is connected with Bengal Foredeep by deep basement faults that probably started with the breakup of Gondwanaland. The SW-NE trending Hinge Zone turns to the east near Indian border in Jamalpur and seems to be connected with the Dauki Fault, probably by a series of east-west trending faults. Bengal Foredeep occupies the vast area between Hinge Line and Arakan Yoma Folded System in the east. The Bengal

Foredeep consists of very thick basin-fill that overlies deeply subsided basement of undetermined origin in the south and east. Thickness of sediments increase toward the south and east to more than 16 km. The huge thickness of sediments in the basin is a result of tectonic mobility or instability of the areas causing rapid subsidence and sedimentation in a relatively short span of geologic time. The Bengal Foredeep consists of some Troughs and some relatively high lands. Faridpur Trough situated adjacent to Hinge Zone is separated from the Sylhet Trough (in the northeast) by the Madhupur High of Pleistocene origin where the basement is relatively uplifted. East-west trending Dauki Fault with 5 km wide fault zone near Bangladesh-India border forms the contact between Shillong Massif (India) and Sylhet Trough. Eastern part of the country is represented by the Chittagong-Tripura Folded Belt. The folded belt in the east consists of narrow, elongated N-S trending folds in Sylhet and Chittagong Divisions of Bangladesh, Tripura, southern Assam and Mizoram states in India and also Myanmar territory. The elevation of these elongated anticlinal folds in Bangladesh ranges from 100 -1,000m. Some of the structures are faulted and thrust and the intensity of folding increases gradually from west to east.

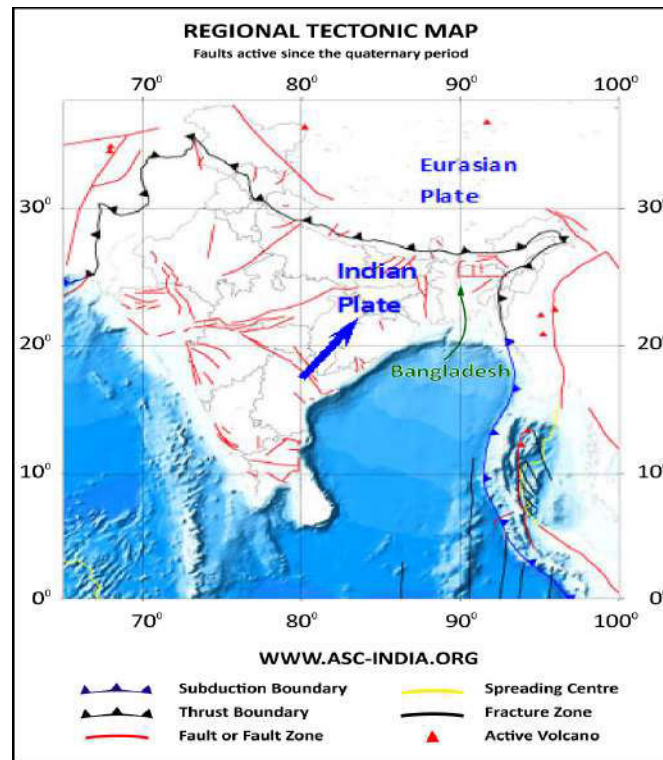


Figure 6.B.1 Movement of Indian plate relative to Eurasian plate

B.1.2 Earthquake Occurrence

Historically Bangladesh has been affected by five earthquakes of large magnitude ($M \geq 7.0$) during 1869 to 1930. One of the strongest earthquakes of the world, the 12 June 1897 Great Indian earthquake in Shillong, Assam had an epicentral distance of about 230 km from Dhaka, its magnitude has recently been reestimated to be 8.1. Two of these earthquakes had their epicenters within the country. The 14 July 1885 Bengal earthquake, measured as $M = 7.0$, originated near Bogra in Bangladesh. The 8 July 1918 Srimongal earthquake occurred in Sylhet region of Bangladesh and its magnitude was determined as $M_s = 7.6$. In addition, some historical reporting points out to the occurrence of a major earthquake ($M > 7$) near Chittagong in 1762. Table 6.B.1 gives brief information about these major earthquakes.

In recent years, the occurrence and damage caused by a number of earthquakes (magnitude between 4 and 6) inside the country or near the country's border, has raised an alarm. The Nov. 21, 1997 magnitude 6.0 earthquake at the Bangladesh-Myanmar border triggered collapse of an under-construction reinforced concrete frame building that killed several people in the port city of Chittagong. The July 22, 1999 magnitude 5.1 earthquake with its epicenter very near the island of Moheshkhali, near Cox's Bazar, caused extensive damage and collapse of rural mud-walled houses, as well as damaging column of cyclone shelter. The Dec. 2001 magnitude 4.0+ earthquake with its epicenter very near Dhaka city caused panic and injuries to prison inmates at the Dhaka central Jail. In the July 27, 2003 magnitude 5.6 Rangamati earthquake, in the village of Kolabunia, brick masonry buildings as well as mud-walled houses were severely damaged.

Table 6.B.1: List of Major Regional Earthquakes

Date	Earthquake name	Epicentre	Magnitude, M	Effects
2 April, 1762	---	Near Chittagong	>7.0	Some changes in landforms in the coastal area and liquefaction
10 Jan. 1869	Cachar	Cachar, Assam	7.5	Some damage occurred in Sylhet
14 Jul. 1885	Bengal	Bogra	7.0	Severe damage occurred to houses in Sirajganj and Sherpur (Bogra)
12 Jun. 1897	Great Indian	Shillong, Assam	8.1	Greatest damage in Rangpur including railway line and buildings; intense ground fissures and vents in Mymensingh, Jamalpur, Sylhet; damages to masonry buildings covering a major portion of Bangladesh including Dhaka

Date	Earthquake name	Epicentre	Magnitude, M	Effects
8 Jul. 1918	Srimongal	Srimongal, Sylhet	7.6	Collapse/ severe damage of buildings in Srimongal, damage to buildings in Habiganj, Moulvibazar
2 Jul. 1930	Dhubri	Garo hills	7.1	Damage to railway track in Lalmonirhat, damage to buildings in Lalmonirhat and Rangpur

Figure 6.B.2 shows distribution of earthquake ($M \geq 3.5$) epicenters for the period 1845-Feb.2010 in and around Bangladesh. This is indicative of the significant seismic activity in the region. There are earthquakes distributed all over Bangladesh. However the cluster of earthquakes appears to be quite dense in Chittagong. Although few in number, there are earthquakes occurring in south western Bangladesh including the sea. The large magnitude ($M \geq 7.0$) earthquakes have taken place within Bangladesh in Sylhet, Bogra, and Chittagong. Outside Bangladesh but close enough to cause damage in Bangladesh, major earthquakes ($M \geq 7.0$) have occurred in India to the north, northeast and northwest of Bangladesh particularly affecting Sylhet, Mymensingh and Rangpur region.

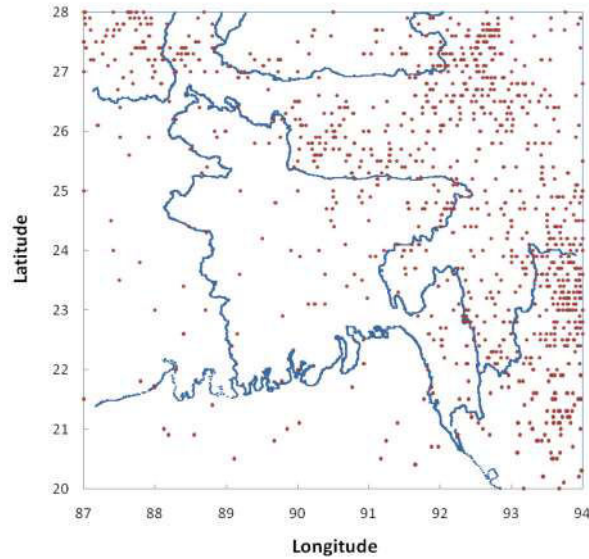


Figure 6.B.2 Earthquake ($M \geq 3.5$) occurrence in and around Bangladesh (1845-Feb.2010).

PART VI
Appendix C

Seismic Design Parameters for Alternative Method of Base Shear Calculation

Table 6.C.1: Spectral Response Acceleration Parameter S_s and S_1 for Different Seismic Zone

Parameters	Zone-1	Zone-2	Zone-3	Zone-4
S_s	0.3	0.5	0.7	0.9
S_1	0.12	0.2	0.28	0.36

Table 6.C.2: Site Coefficient F_a for Different Seismic Zone and Soil Type

Soil Type	Zone-1	Zone-2	Zone-3	Zone-4
SA	1.0	1.0	1.0	1.0
SB	1.2	1.2	1.2	1.2
SC	1.15	1.15	1.15	1.15
SD	1.35	1.35	1.35	1.35
SE	1.4	1.4	1.4	1.4

Table 6.C.3: Site Coefficient F_v for Different Seismic Zone and Soil Type

Soil Type	Zone-1	Zone-2	Zone-3	Zone-4
SA	1.0	1.0	1.0	1.0
SB	1.5	1.5	1.5	1.5
SC	1.725	1.725	1.725	1.725
SD	2.7	2.7	2.7	2.7
SE	1.75	1.75	1.75	1.75

Table 6.C.4: Spectral Response Acceleration Parameter S_{DS} for Different Seismic Zone and Soil Type

Soil Type	Zone-1	Zone-2	Zone-3	Zone-4
SA	0.2	0.333	0.466	0.6
SB	0.24	0.4	0.56	0.72
SC	0.23	0.383	0.536	0.69
SD	0.27	0.45	0.63	0.81
SE	0.28	0.466	0.653	0.84

Table 6.C.5 Spectral Response Acceleration Parameter S_{D1} for Different Seismic Zone and Soil Type

Soil Type	Zone-1	Zone-2	Zone-3	Zone-4
SA	0.08	0.133	0.186	0.24
SB	0.12	0.2	0.28	0.36
SC	0.138	0.23	0.322	0.414
SD	0.216	0.36	0.504	0.648
SE	0.14	0.233	0.326	0.42

PART VI
Appendix D**Methods of Soil Exploration, Sampling and Groundwater Measurements****D.1 Methods of Soil Exploration**

The detailed methods of soil investigation usually include collecting undisturbed samples and or performing field tests. Listed below are some of the common methods of subsoil exploration.

D.1.1 Open Trial Pits

In this method trial pits are excavated exposing the subsoil thoroughly. Undisturbed samples are taken from intact sides and bottom of the trial pits. This is suitable for all types of formation but for cuts which cannot stand below water table, proper bracing shall be provided. This method is normally used for shallow depths (up to 3 m). Test pits are usually prepared by hand excavation which allows access for a full observation and description of the soil profile. Hand-cut samples known as block or chunk samples can be obtained from the test pit. In stiff clays it provides fairly accurate idea of the depth of open excavations or vertical cuts. It also provide better picture of the patchy ground where the soil lies in pockets

D.1.2 Auger Boring

Augers, hand or power operated, are rotated and forced into soil. Augers are withdrawn and emptied when full. Soil cuttings obtained are used to interpret stratification and soil type. The method is unsatisfactory for cohesionless soils above or below ground water.

D.1.3 Shell and Auger Boring

Manual or mechanized rigs are used for vertical boring. The tools consist of auger for soft to stiff clays, shells for very stiff to hard clays, shells or sand pumps attached to sectional boring rods for sandy strata. Hand operated auger can be used up to depths of 18 m to 21 m in stiff clay. Mechanically operated tools are required for higher depths. An auger is used for boring holes to a depth of about 6 m in soft soil. Soft to stiff clays are removed by a cylindrical auger. Hard clays and cohesionless deposits are removed by a shell. The soil recovered contains all constituent. Bailers are be used to remove soil cuttings. Care is taken while withdrawing the shell to avoid sand boiling. Special care is taken to minimize soil disturbance below borehole. The advantages of shell and auger boring are as follows:

- (i) Easy to identify the soil
- (ii) Easy to note changes in strata
- (iii) Soil profile and depths undisturbed sampling can be determined with greater accuracy
- (iv) Boring in partially saturated materials above the ground water level
- (v) Determination of ground water level is relatively easy

D.1.4 Wash Boring

In this method, soil is loosened by chopping and cutting by impact and twisting action of a lightweight bit. Soil is removed from the borehole by a stream of water or drilling mud from lower end of the wash pipe which is worked up and down or rotated into the borehole. The water or mud flow carries the soil through the annular space between the wash pipe and casing and is overflowed at ground level. The soil in suspension is allowed to settle in a pond or tank and the fluid is recirculated as required. The soil brought to surface by the wash water can be used for identification purposes but is not representative of the character and consistency of the material penetrated and the flushing water may disturb the surrounding ground. Subsoil can be identified thoroughly if field tests (viz. Standard Penetration Test) are performed and or undisturbed samples are collected frequently.

D.1.5 Sounding/Probing

A number of sounding methods are available. The most common is the Standard Penetration Test (SPT). The SPT test is specified in ASTM D1586 and ASTM D6066. Other methods include procedures like Cone Penetration Test (CPT) and Dynamic Probing (DP). Sounding/probing may be done in conjunction with inhole tests such as "Field Vane Shear Test in Cohesive Soil", (ASTM D2573), bore-hole shear (Iowa Bore-hole Shear) Test, Flat Dilatometer Test (DMT) or "Prebored Pressuremeter Testing in Soils", (ASTM D4719).

D.1.6 Geophysical Methods

Geophysical survey techniques are based on determining variations in physical properties, such as electrical conductivity (resistivity), variation in density (gravimetric), magnetic susceptibility (magnetic) or velocity of sonic waves (seismic). Anomalies such as near surface disturbance (often known as noise) are common in urban environment and may limit the usefulness of geophysics in these areas. Moreover, a geophysical anomaly does not always match an engineering or geological boundary, and often there is a transition zone at a boundary. These may lead to a margin of uncertainty.

D.1.7 Percussion Boring

In percussion drilling method borehole is advanced by chopping action of a heavy bit driven by power. Drilling is performed raising and dropping a heavy drilling bit. The borehole is generally kept dry. Water is added at the bottom of the borehole during chopping action, if ground water is not already struck. Slurry formed at the bottom of hole is removed by bailer or sand pump. Casing may be needed. This method is Simple to operate and suitable for drilling bore holes in deposits of gravels and boulders. However, using this method, determination of the changes in the soil strata is very difficult. The method is relatively slow and not economical for boring diameter less than 100 mm

D.1.8 Rotary Drilling

In rotary drilling, borehole is advanced by power rotation of drilling bit. The particles from inside is removed by circulating fluids which may be water, bentonite slurry or mud slurry in a manner similar to that in wash boring. Casing may or may not be needed during drilling. This method is particularly suitable for very hard formation and rock. Rate of progress is fast. Difficult to detect changes in strata

D.2 Choice of Method

The choice of a method of soil exploration depends upon:

- (i) The topography, type of ground to be investigated and ground water conditions;
- (ii) The type of building envisaged and technical requirements;
- (iii) Amount of existing information;
- (iv) Expected variability of soil;
- (v) External constraints such as availability of plant, access, cost and time.

The technical requirements of the investigation rather than cost should be the overriding factor in the selection of exploration method. In clayey soils, borings are suitable for deep exploration and pit for shallow exploration. In sandy soils special equipment are needed for taking representative samples below the water table. Ground investigation is normally done by boreholes, but where only shallow depths are to be investigated, and where ground water problems are not envisaged, trial pits may prove more versatile and economical. Boreholes may be necessary on waterlogged sites where it is impracticable to excavate trial pits without dewatering.

Safety aspects must be considered when selecting and carrying out exploration. Precautions relating to safety, health and welfare of workmen, hazards from underground services, contaminated ground and inspection pits or shafts shall be undertaken. Overhead power lines are a hazard if ground investigation rigs are to operate in the vicinity.

D.3 Sample Disturbance and Sampling Methods

The major purposes of retrieving soil sample from ground are as follows:

- (i) Inspection of the material and to describe its fabric
- (ii) To classify the material by index tests
- (iii) Determination of mechanical properties, such as, stress-strain-strength-stiffness, compressibility and expansibility, and permeability

Samples shall contain all the mineral constituents of the strata from which they have been taken. They shall not be contaminated by any material from other strata or from additives used during the sampling procedure. For index tests, the sample should be intact in terms of its constituents, and changes in fabric and stress state are acceptable. To measure most mechanical properties the sample needs to be undisturbed and representative. Material submitted for testing should be representative of the mass which will be affected by construction and the level of sample disturbance should be acceptable.

The physical process of obtaining samples has been recognized as a prime cause of sample disturbance. The main causes of sampling disturbance are as follows:

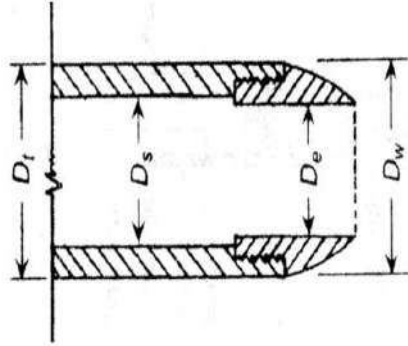
- (i) Disturbance of the soil to be sampled before the beginning of sampling as a result of poor drilling operation leading to swelling, compaction, base heave, piping and caving
- (ii) Mechanical distortion during the penetration of the sampling tube into the soil. This is referred as tube penetration disturbance
- (iii) Mechanical distortion and suction effects during the retrieval of the sampling tube.
- (iv) Release of total in situ stresses.
- (v) Disturbance of the soil during transportation, storage and sample preparation.

The first cause can be reduced by sampling with properly cleaned boreholes advanced by using bentonite slurry. The second and third causes are directly associated with sampler design and can be controlled to certain extent. The fourth cause is unavoidable even though its effects may be different depending on the depth of sampling and soil properties. The fifth cause can be reduced by storing samples for minimum time in controlled atmosphere and careful handling of samples during transportation and preparation.

The design of a sampler is one of the most important factors that should be considered for quality sampling. The amount of disturbance varies considerably depending upon the dimensions of the sampler and the precise geometry of the cutting shoe of the sampler. Hvorslev (1949) discussed at length the importance of the design of a sampler and introduced the concepts of area ratio, inside and outside clearance ratio and cutting edge taper angle in controlling sampling disturbance. Terms used to define geometry of cutting shoe of a tube sampler is shown in Figure 6.D.1.

Increasing area ratio gives increased soil disturbance and remoulding. The penetration resistance of the sampler and the possibility of the entrance of excess soil also increase with increasing area ratio. For soft clays, area ratio is kept to a minimum by employing thin-walled tubes. For composite samplers, the area ratio, however⁶, is considerably higher. In these cases, sample disturbance is reduced by tapering the outside of the sampler tube very gradually from a sharp cutting edge (Hvorslev, 1949, recommended a

maximum 10°), so that the full wall thickness is far removed from the point where the sample enters the tube. The combined requirements for area ratio and cutting edge taper angle to cause low degree of disturbance and the optimum length to diameter ratios for clays of different sensitivities were proposed by ISSMFE's sub-committee on Problems and Practices of Soil Sampling.



$$\text{Area ratio, } Ar = \frac{D_w^2 - D_e^2}{D_e^2}$$

$$\text{Inside Clearance ratio, } ICR = \frac{D_s - D_e}{D_e}$$

$$\text{Outside Clearance Ratio, } OCR = \frac{D_w - D_t}{D_t}$$

Figure 6.D.1 Dimensions of a tube sampler and terms used to define geometry of cutting

Tube sampler characteristics suggested by ISSMFE (1965) is presented in Table 6.D.1. Inside wall friction is one of the principal causes of disturbance of the sample (Hvorslev, 1949). One of the methods of reducing or eliminating wall friction between the soil and sampler is to provide inside clearance by making the diameter of the cutting edge, D_e , slightly smaller than the inside diameter of the sampler tube, D_s . Inside clearance gives the soil sample room for some swelling and lateral strain due to horizontal stress reduction. Although neither of these types of behaviour is desirable, they are less undesirable than the consequences of adhesion between the soil and the inside of the sampler tube (Clayton et al, 1982).

Table 6.D.1: Tube Sampler Characteristics (after ISSMFE, 1965)

Area ratio/cutting-edge taper*	
Leading edge	
60° taper angle:	
• up to a thickness of 0.3 mm for cohesive soils • up to a thickness of D_{10} in granular soils	
Cutting shoe	
Area ratio (%)	Taper angle (degrees)
5	15
10	12
20	9
40	5
80	4
Inside clearance ratio/length-to-diameter (L/D) ratio ⁺	

Soil type	Greatest permissible L/D ratio
Clay (sensitivity > 30)	20
Clay (sensitivity 5-30)	12
Clay (sensitivity <5)	10
Loose frictional soil	12
Medium loose frictional soil	6

* Suggested for samplers of about 75 mm diameter.

+ For samplers with smooth and clean inside surfaces, and with an inside clearance ratio of 0.5-1.0%.

Inside clearance should be large enough to allow partial swelling and lateral stress reduction but it should not allow excessive soil swelling or loss of the sample when withdrawing from the sampling tube. Hvorslev (1949) suggests an inside clearance ratio of 0.75 to 1.5% for long samplers and 0 to 0.5% for very short samplers. Kallstenius (1958) on the basis of Swedish clays sampled by six different piston samplers and also recommends that a sampler have a moderate inside clearance. The clearance reduces the wall friction and probably counteracts to a certain extent the disturbance from displacement of soil caused by the edge and sampler wall during the driving operation. If the inside clearance and the edge angle are moderate, the above positive effects outweigh the disturbance caused by deformation when the sample tends to fill the clearance. In order to reduce outside wall friction, samplers are often provided with outside clearance. An outside clearance ratio of a few per cent may decrease the penetration resistance of samplers in cohesive soils. Although outside clearance increases the area ratio, a clearance of 2 to 3% can be advantageous in clay (Hvorslev, 1949).

The recommended criteria for high quality sampling areas follow:

- (i) Area ratio : less than 10%
- (ii) ICR = 0 to 0.5 % for very short samplers; ICR = 0.75 to 1.5 % for long samplers
- (iii) OCR : not more than 2 to 3%

For 50 to 75 mm diameter samplers:

- (i) L/D_i ratio: not more than 5 to 10 for loose to dense cohesionless soils (D_i = Internal diameter of sampling tube)
- (ii) L/D_i ratio: not more than 10 to 20 for very soft to stiff cohesive soils

Samples retrieved from can be classified into the following types:

- (i) Representative
- (ii) Non-representative
- (iii) Undisturbed
- (iv) Disturbed

A representative sample is one in which the fabric and structure of the soil in situ is fully represented. A non-representative sample is which does not the grading, fabric and structure of the soil in situ.

An undisturbed sample is one, which represents as practicable the true in situ structure and moisture content of the soil. Truly, undisturbed samples cannot be taken from boreholes, and in practice there are only differing levels of disturbed samples. Material may be secured in open tube samplers for clays except of firm or of stiff consistency. For softer clays stationary piston samplers of low area ratio, shall be used and careful boring and sample preservation technique shall be employed. The minimum diameter of undisturbed sample shall be 40 mm with minimum length/diameter ratio of 3.

A disturbed sample is one which preserves the grading of the in situ soil but in which the soil structure is significantly damaged or completely destroyed and the moisture content also changed considerably from the in situ value

Sample quality is dependent on type of soil being sampled, type and condition of equipment and the skill with which it is used. The weaker material is the most significant in an investigation, and is usually difficult to secure in an undisturbed condition. It is rarely possible to sample granular (non-cohesive) materials in undisturbed condition, unless special techniques are used. Granular soil conditions are usually assessed by in-situ tests and confirmed by disturbed samples which permit classification and grading analysis and visual inspection. Cohesive soils may be tested both in-situ and in laboratory on undisturbed or relatively undisturbed samples.

The sample and/or test locations must be such that all changes of stratum are recorded. A number of extra samples and test results are usually required to assess variation of the properties of a stratum with depth. The record of all borings shall include the following information:

- (a) Size of casing (if used),
- (b) Number of blows per 300 mm required to drive the sampling spoon,
- (c) The elevation of the ground surface referred to an established datum,
- (d) Location and depth of boring and its relation to the proposed construction,
- (e) Elevation at which samples were taken,

- (f) Elevation of the boundaries of soil strata,
- (g) Description of the soil strata encountered and any particular unusual or special condition such as loss of water in the earth and rock strata, presence of boulders, cavities and obstructions, use of special type of samplers, traps, etc., and,
- (h) The level of ground water together with a description of how and when it was observed.

All abandoned and unsuccessful attempts of borings or drillings shall also be reported. In complex formations, details of sampling are necessary and, therefore, separate holes may be employed purely for sampling or testing, termed as “double hole sampling”.

Care shall be taken in protecting, handling, labelling and subsequently transporting the samples, so that samples can be received in a fit state for examination and testing, and can be correctly recognized as coming from a specific trial pit or boring.

Soil samples for laboratory tests can be divided in five quality classes with respect to the soil properties that are assumed to remain unchanged during sampling and handling, transportation and storage. The classes are described in Table 6.D.2. Class 1 and Class 2 samples listed in Table G-2 are generally referred to as 'undisturbed' while Classes 3, 4, and 5 as 'disturbed' samples. The amount of sample generally required for testing purposes is given in Table 6.D.3.

Table 6.D.2: Categories of Soil Samples Based on Quality

Quality	Recommended use of Sample
Class 1	Stratigraphy, Stratification, Atterberg limits, particle size, moisture content, organic content, unit weight, relative density/density index, porosity, permeability, shear strength and compressibility.
Class 2	Stratigraphy, Stratification, Atterberg limits, particle size, moisture content, organic content, unit weight, relative density/density index, porosity and permeability.
Class 3	Stratigraphy, Stratification, Atterberg limits, particle size, moisture content and organic content.
Class 4	Stratigraphy, Stratification, Atterberg limits, particle size and organic content.
Class 5	Stratigraphy and Stratification.

Table 6.D.3: Weight of Soil Sample Required for Laboratory Tests

Purpose of Sample	Type of Soil	Weight of Sample Required, kg.
Soil identification, natural moisture content test, mechanical analysis and index properties	Cohesive soil	1
	Sand and gravel	3
Chemical test	Cohesive soil	2
	Sand and gravel	3
Compaction test	Cohesive soil and sand	12.5
	Gravelly soil	25
Comprehensive examination of construction materials including stabilization	Cohesive soil and sand	25 to 50
	Gravelly soil	50 to 100

D.4 Ground Water Measurements in Soil

Groundwater investigations shall provide all relevant information on groundwater needed for geotechnical design and construction. Groundwater investigations should provide, when appropriate, information on:

- (i) the depth, thickness, extent and permeability of water-bearing strata in the ground
- (ii) the elevation of the groundwater surface or piezometric surface of aquifers and their variation over time and actual groundwater levels including possible extreme levels and their periods of recurrence
- (iii) the pore water pressure distribution
- (iv) the chemical composition and temperature of groundwater.

The information obtained should be sufficient to assess the following aspects, where relevant:

- (i) the scope for and nature of groundwater-lowering work
- (ii) possible harmful effects of the groundwater on excavations or on slopes (e.g. risk of hydraulic failure, excessive seepage pressure or erosion)
- (iii) any measures necessary to protect the structure (e.g. waterproofing, drainage and measures against aggressive water)
- (iv) the effects of groundwater lowering, desiccation, impounding etc. on the surroundings
- (v) the capacity of the ground to absorb water injected during construction work

The determination of the groundwater table or pore water pressures in soils and rocks shall be made by installing open or closed groundwater measuring systems into the ground. The type of equipment to be used for groundwater measurements shall be selected according to the type and permeability of ground, the purpose of the measurements, the required observation time, the expected groundwater fluctuations and the response time of the equipment and ground.

There are two main methods for measuring the groundwater pressure; open systems and closed systems. In open systems the piezometric groundwater head is measured by an observation well, usually provided with an open pipe. In closed systems the groundwater pressure at the selected point is directly measured by a pressure transducer. Open systems are best suited for soils and rock with a relatively high permeability (aquifers and aquitards), e.g. sand and gravel or highly fissured rock. With soils of low permeability they may lead to erroneous interpretations, due to the time lag for filling and emptying the pressure pipe. The use of filter tips connected to a small diameter hose in open systems, decreases the time lag. Closed systems can be used in all types of soil. They should be used in very low permeability soils. Closed systems are also recommended when dealing with high artesian water pressure. When short term variations or fast pore water fluctuations are to be monitored, continuous recording shall be used by means of transducers and data loggers.

In cases where open water is situated within or close to the investigation area, the water level shall be considered in the interpretation of the groundwater measurements. The water level in wells, the occurrence of springs and artesian water shall also be noted.

The number, location and depth of the measuring stations shall be chosen considering the purpose of the measurements, the topography, the stratigraphy and the soil conditions, especially the permeability of the ground or identified aquifers.

For monitoring projects e.g. groundwater lowering, excavations, fillings and tunnels, the location shall be chosen with respect to the expected changes to be monitored. For reference purposes, measurement of the natural fluctuations in ground water should be made, if possible, outside the area affected by the actual project. The number and frequency of readings and the length of the measuring period for a given project shall be planned considering the purpose of the measurements and the stabilization period.

If it is intended to assess groundwater fluctuations, measurements shall be taken at intervals smaller than the natural fluctuations to be characterized and over a long period of time.

During the drilling process, the observation of the water level at the end of the day and the start of the following day (before the drilling is resumed) is a good indication of the groundwater conditions and should be recorded. Any sudden inflow or loss of water during drilling should also be recorded, since it can provide additional useful information during the first phases of site investigations, some of the boreholes may be equipped with open perforated pipes protected with filters. The water level readings obtained during the following days yield a preliminary indication of groundwater conditions.

The evaluation of groundwater measurements shall take into account the geological and geotechnical conditions of the site, the accuracy of individual measurements, the fluctuations of pore water pressures with time, the duration of the observation period, the season of measurements and the climatic conditions during and prior to that period. The evaluated results of groundwater measurements shall comprise the observed maximum and minimum elevations of the water table, or pore pressures and the corresponding measuring period. If applicable, upper and lower bounds for both extreme and normal circumstances shall be derived from the measured values, by adding or subtracting the expected fluctuations or a reduced part of them, to the respective extreme or normal circumstances. The frequent lack of reliable data for extended periods of time of this type of measurements will necessitate the derived values being a cautious estimate based on the limited available information. The need for making further measurements or installing additional measuring stations should be assessed during the field investigations and in the geotechnical investigation report.

D.5 Standard Penetration Test

The Standard Penetration Test (SPT) is widely used to determine the in-situ properties of soil. The test is especially suited for cohesionless soils as the correlation between the SPT value and ϕ is now well established. The standard test methods are provided in ASTM D156 and ASTM D6066.

The test consists of driving a split spoon sampler, Figure 6.D.2, into soil through a borehole of 55 to 100 mm in diameter at the desired depth. It is done by a hammer weighing 65 kg (140 lb) dropping onto a drill rod from a height of 750 mm (30 inch). The number of blows N required to produce a penetration of 300 mm (12 inches) is regarded as the penetration resistance. To avoid seating errors, the blows for the first 150 mm (6 inches) of penetration are not taken into account; those required to increase the penetration from 150 mm to 450 mm constitute the N -value. A demonstration of Standard Penetration Test is shown in Figure 6.D.3.

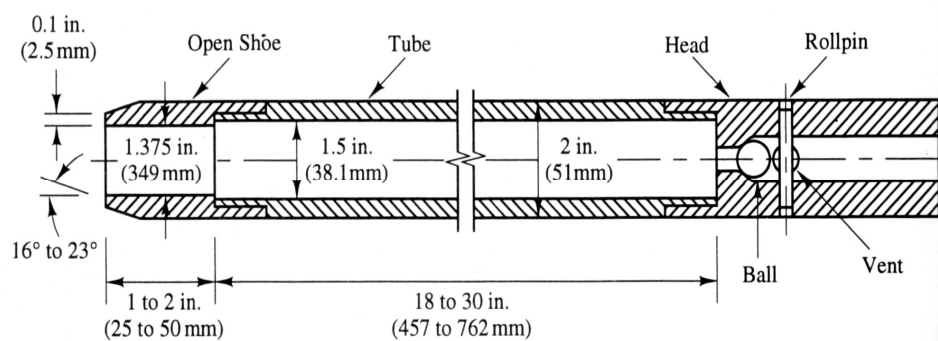


Figure 6.D.2 Schematic diagram of split spoon sampler

It is important to point out that several factors contribute to the variation of the standard penetration number N at a given depth for similar soil profiles. Among these factors are the SPT hammer efficiency, borehole diameter, sampling method, rod length, water table and overburden pressure important. The most two common types of SPT hammers used in the field are the safety hammer and donut hammer. They are usually dropped using a rope with two wraps around a pulley. The hammers are shown in Figure 6.D.4.

Usually SPT is conducted at every 1.5 m or 2 m depth or at the change of stratum. In hard formations, the testing is discontinued if N value is found to be over 100 and it is termed refusal.

D.5.1 Corrections/Standardization of SPT Value for Field Procedures for all Soil Types

On the basis of field observations, it appears reasonable to standardize the field SPT number as a function of the input driving energy and its dissipation around the sampler around the surrounding soil. The variations in testing procedures may be at least partially compensated by converting the measured N to N_{60} as follows.

$$N_{60} = \frac{E_H C_B C_S C_R N}{0.60} \quad (6.D.1)$$

Where,

N_{60} = Corrected SPT N-value for field procedures

E_H = Hammer efficiency (Table 6.D.4)

C_B = Borehole diameter correction (Table 6.D.5)

C_S = Sampler correction (Table 6.D.5)

C_R = Rod length correction (Table 6.D.5)

N = Measured SPT N-value in field

This correction is to be done irrespective of the type of soil encountered.

Table 6.D.4: SPT Hammer Efficiencies

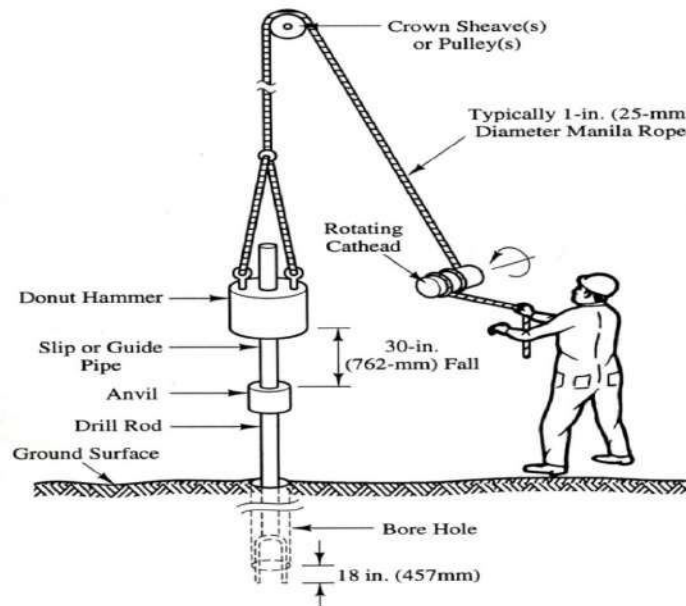
Hammer Type	Hammer Mechanism	Release	Efficiency, E_H
Automatic	Trip		0.70
Donut	Hand dropped		0.60
Donut	Cathead + 2 turns		0.50
Safety	Cathead + 2turns		0.55-0.60
Drop/ Pin	Hand dropped		0.45

Table 6.D.5: Borehole, Sampler and Rod Correction Factors

Factor	Equipment Variables	Correction Factor
Borehole Diameter Factor, C_B	65 mm – 115 mm	1.00
	150 mm	1.05
	200 mm	1.15
Sampling Method Factor, C_S	Standard sampler	1.00
	Sampler without liner	1.20
Rod Length Factor, C_R	3 m – 4 m	0.75
	4 m – 6 m	0.85
	6 m – 10 m	0.95
	> 10 m	1.00

D.5.2 Corrections of SPT Value for Overburden Pressure for all Types of Cohesionless Soils

In cohesionless soils, the overburden pressure affects the penetration resistance. For SPT made at shallow levels, the values are usually too low. At a greater depth, the same soil at the same density index would give higher penetration resistance. It was only as late as in 1957 that Gibbs & Holtz (1957) suggested that corrections should be made for field SPT values for depth.

**Figure 6.D.3 Demonstrations of standard penetration test**

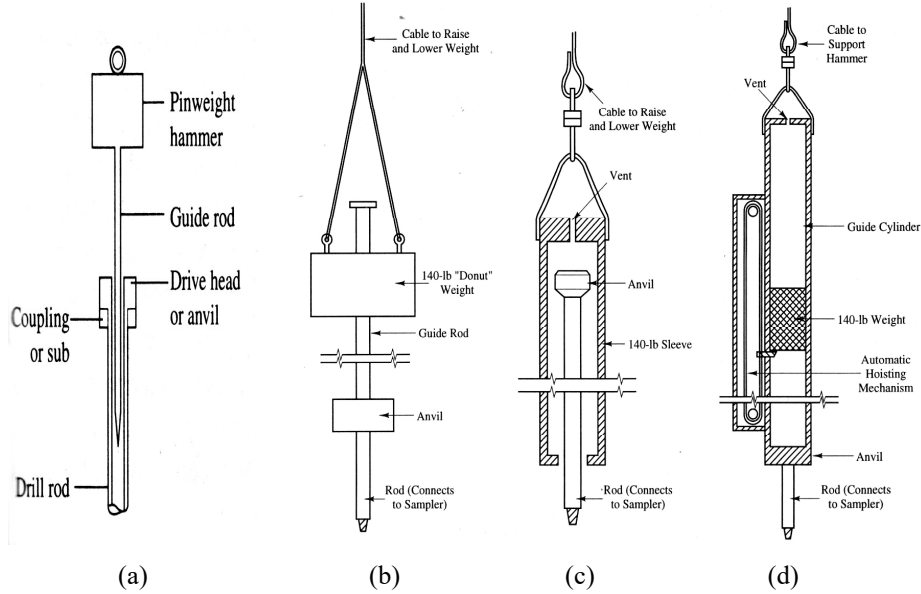


Figure 6.D.4 Configurations of various SPT hammers; (a) Pin (Drop) hammer; (b) Donut hammer; (c) Safety hammer; (d) Automatic hammer

As the correction factor came to be considered only after 1957, all empirical data published before 1957 like those by Terzaghi is for uncorrected values of SPT. Since then a number of investigators have suggested overburden correction. Gibbs & Holtz took standard pressure of 280 kN/m² (corresponding to a depth of 14 m) and duly made overburden correction for other overburdens. Thornburn suggested a standard pressure of 138 kN/m² (corresponding to a depth of 7 m). Finally, Peck et. al. (1974) suggested a standard pressure of 100 kN/m² (Equivalent to 1 tsf or 1 kg/cm² overburden corresponding to a depth of 5 m). As such, all field SPT values are to be corrected by the correction factor given by them as:

$$C_N = 0.77 \log \frac{(2000)}{\sigma'_0} \quad (6.D.2)$$

Where, σ'_0 is the effective overburden pressure. Thus,

$$(N_1)_{60} = C_N \times N_{60} \quad (6.D.3)$$

While using the correction factor given by Equations 6.D.2 and 6.D.3, the corrected N_{60} value is termed as $(N_1)_{60}$ as this correction corresponds to an overburden pressure of 1 ton/ft² or 1 kg/cm². The maximum value of correction factor C_N is 2.

D.5.3 Corrections of SPT Value for Water Table (Dilatancy) in case of Fine Sand and Silty Sand

In addition to corrections of overburden, investigators suggested corrections of SPT-value for water table in the case of fine sand or silt below water table. Apparently, high N-values may be observed especially when observed value is higher than 15 due to dilatancy effect. In such cases, following correction is recommended (Terzaghi and Peck, 1948).

$$(N_1)_{60(CORR)} = 15 + \frac{1}{2}[(N_1)_{60} - 15] \quad (6.D.4)$$

Where, $(N_1)_{60(CORR)}$ is the corrected $(N_1)_{60}$ for water table. For coarse sand this correction is not required. In applying this correction, overburden correction is applied first and then this dilatancy correction is used.

D.5.4 SPT Value and Density Index Relations

Although, the SPT is not considered as a refined and a completely reliable method of investigation, it gives useful information with regard to relative density of cohesionless soil and consistency of cohesive soils. Terzaghi and Peck give the following correlation (Tables 6.D.6 and 6.D.7) between SPT value and other soil parameters.

Several investigators presented average relations between SPT value and peak angle of internal friction, ϕ . The following relations may be noted.

$$\phi^\circ = 27 + 0.3(N_1)_{60} \quad (6.D.5)$$

$$\phi^\circ = \sqrt{20(N_1)_{60}} + 15 \quad (6.D.6)$$

The relations are also presented in Figure 6.D.5. The $(N_1)_{60}$ values used in Equations 6.D.5 and 6.D.6 are corrected values of $(N_1)_{60}$

Table 6.D.6: Penetration Resistance and Soil Properties on the Basis of SPT (Cohesionless Soil: Fairly reliable)

N-value	Soil Condition	Relative Density, D_r	Angle of internal friction, ϕ
0 - 4	Very loose	0 – 15%	< 28°
4 - 10	Loose	15 – 35%	28° – 30°
10 - 30	Medium	35 – 65%	30° – 36°
30 - 50	Dense	65 – 85%	36° – 42°
> 50	Very Dense	> 85%	> 42°

Table 6.D.7: Penetration Resistance and Soil Properties on the Basis of SPT (Cohesive Soil)

N- value	Consistency	UC Strength (q_{uv}), kN/m ²
0 - 2	Very soft	< 25
2 - 4	Soft	25 - 50
4 - 8	Medium	50 - 100
8 - 15	Stiff	100 - 200
15 - 30	Very stiff	200 - 400
> 30	Hard	> 400

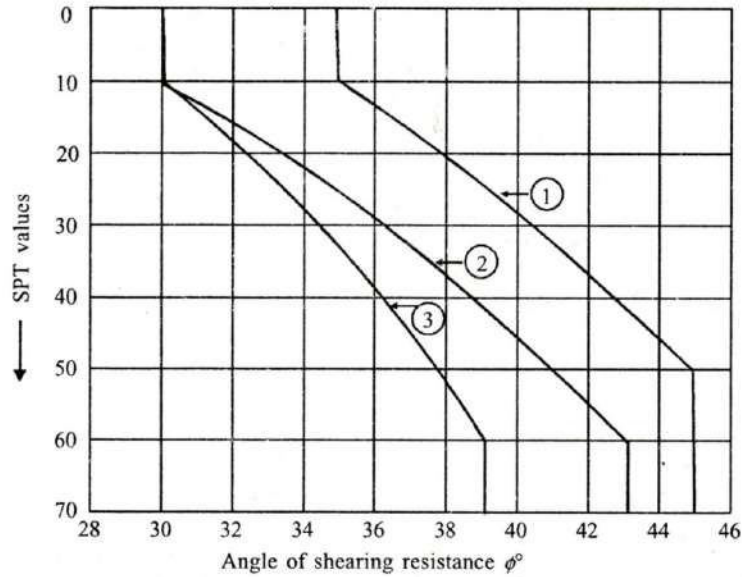


Figure 6.D.5 SPT- ϕ relations for granular soils; (1) Well graded sand and gravel; (2) Uniform fine sand (Average value); and (3) Silty Sand

The correlation for clays with SPT value is not fully established. Hence, vane shear test is recommended for more reliable information. Even though, SPT values are not considered as a good measure of the strength of clays, it is used extensively as a measure of the consistency of clays. The consistency is then related to its approximate strengths. They may be expressed as:

$$q_u \text{ (in kN per m}^2\text{)} = 12.5 N_{60} \quad (6.D.7)$$

Where, q_u is unconfined compressive strength. For highly plastic clays ($PI > 30$) the relation may be:

$$q_u \text{ (in kN per m}^2\text{)} = 10 N_{60} \quad (6.D.8)$$

The relations (q_u in kg/cm²) are also expressed in Figure 6.D.6.

D.6 Dynamic Cone Penetration Test

The dynamic cone penetration test can be considered only as a variation of SPT. Instead of the spoon sampler used for SPT, a special solid cone of 60° and 50 mm diameter is used as a penetrometer. This probe can be used either in the borehole as an SPT test or without a borehole as a continuous penetration test. The latter may be of the recoverable type or the expendable type. Experience shows that in most soils, the solid cone penetrometer test tends to give a slightly higher value than SPT. These tests are to be used along with SPT tests so that a correlation between the two can be worked out for each site under investigation. It is then used to determine the nature of deposits in other locations at the same site without putting an expensive borehole. As this test is very much cheaper than SPT tests in boreholes, a large number of dynamic cone penetration tests can be made at various locations at nominal cost along with SPT tests.

The blow count for every 100 mm penetration due to a 65 kg weight falling through 750 mm is taken. The total blows for one foot (300 mm) penetration is the dynamic cone value N_{dc} . To save the equipment from damage, driving may be stopped when the number of blows exceeds 35 for 100 mm penetration. Typical cone used dynamic cone penetration test is shown in Figure 6.D.7.

Dynamic cone test can be used with or without bentonite (mud) slurry. But when depth of investigation is more than 6 m, use of bentonite or mud slurry is recommended as otherwise friction on the rods would be tremendous. Data from dynamic cone test is plotted as a curve of penetration resistance, N_{dc} number of blows per 30 cm of penetration, versus depth. The N_c values from dynamic cone penetration tests needed to be corrected for overburden pressure in cohesionless soil like N-values of SPT.

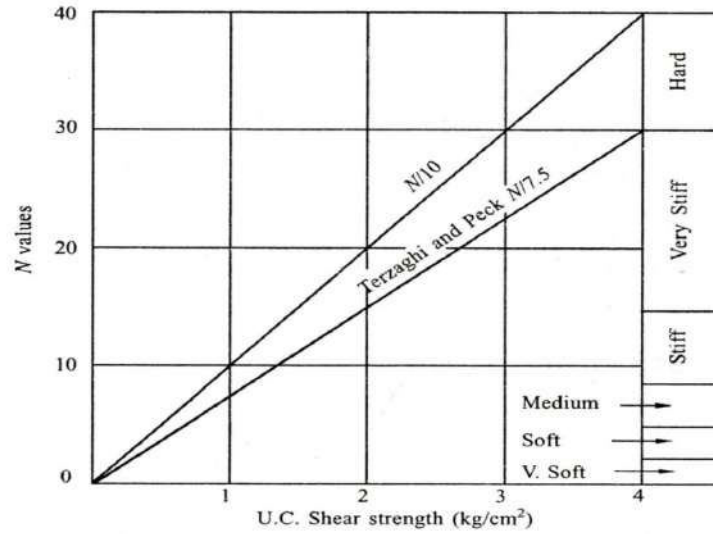


Figure 6.D.6 Relation between SPT and unconfined compressive strength (q_u) of clays

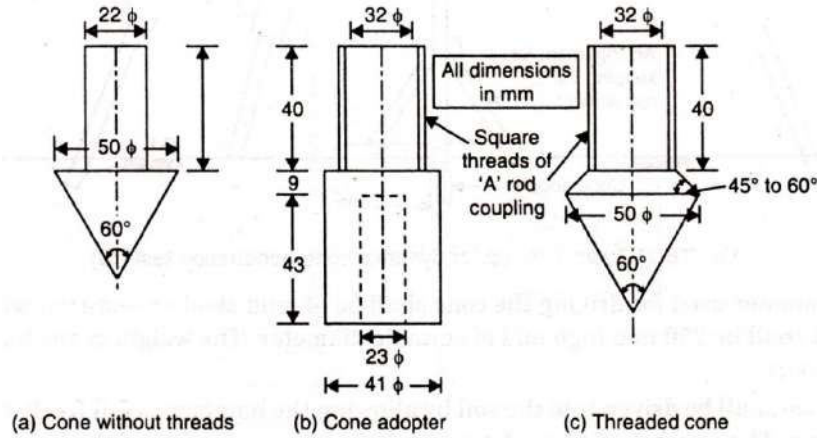


Figure 6.D.7 Typical cone details for dynamic cone penetration test

D.7 Static Cone Penetration Test (Cpt)

The static cone penetration test normally called the Dutch cone penetration test (CPT). It has gained acceptance rapidly in many countries. The method was introduced nearly 50 years ago. The test can be now performed using both mechanical and electrical cones. These test procedures can be found in ASTM D3441 and ASTM D5778 respectively. This test is widely used in Holland, Belgium, Britain, Indonesia, Malaysia, Singapore, West Indies etc. It is finding increasing use in India.

The equipment consists essentially of a steel cone with an apex angle of 60° and overall base diameter of 35.7 mm giving a cross sectional area of 10 cm^2 . The cone is attached to rod which is in turn connected to other rods as necessary. These rods are protected by sleeves known as mantle tubes.

Immediately above the cone a friction jacket, of outside diameter greater than mantle tube, is fitted. The cone and the friction jacket in combination or separately are pushed into the ground by hydraulic cylinder of a machine of capacities presently varying from 20 kN to 100 kN. The necessary reaction is obtained by anchors and sometimes by surcharge loading.

One of the greatest values of the CPT consists of its function as a scale model pile test. Empirical correlations established over many years permit the calculation of pile bearing capacity directly from the CPT results without the use of conventional soil parameters.

The CPT has proved valuable for soil profiling as the soil type can be identified from the combined measurement of end resistance of cone and side friction on a jacket. The test lends itself to the derivation of normal soil properties such as density, friction angle and cohesion. Various theories have been developed for foundation design. The popularity of the CPT can be attributed to the following three important factors:

- (i) General introduction of the electric penetrometer providing more precise measurements, and improvements in the equipment allowing deeper penetration.
- (ii) The need for the penetrometer testing in-situ technique in offshore foundation investigations in view of the difficulties in achieving adequate sample quality in marine environment.
- (iii) The addition of other simultaneous measurements to the standard friction penetrometer such as pore pressure and soil temperature.

D.7.1 The Penetrometer

There are a variety of shapes and sizes of penetrometers being used. The one that is standard in most countries is the cone with an apex angle of 60° and a base area of 10 cm^2 . The sleeve (jacket) has become a standard item on the penetrometer for most applications. On the 10 cm^2 cone penetrometer, the friction sleeve should have an area of 150 cm^2 as per standard practice. The ratio of side friction and bearing resistance, the friction ratio, enables identification of the soil type and provides useful information in particular when no borehole data are available. Even when borings are made, the friction ratio supplies a check on the accuracy of the boring logs.

Two types of penetrometers are used which are based on the method used for measuring cone resistance and friction. They are: (i) The Mechanical Type; (ii) The Electrical Type.

D.7.2 Mechanical Penetrometer

The Begemann Friction Cone Mechanical type penetrometer is shown in Figure 6.D.8. It consists of a 60° cone with a base diameter of 35.6 mm (sectional area 10 cm²). A sounding rod is screwed to the base. Additional rods of one metre length each are used. These rods are screwed or attached together to bear against each other. The sounding rods move inside mantle tubes. The inside diameter of the mantle tube is just sufficient for the sounding rods to move freely whereas the outside diameter is equal to or less than the base diameter of the cone. All dimensions in Figure 6.D.8 is in mm.

The rigs used for pushing down the penetrometer consist basically of a hydraulic system. The thrust capacity for cone testing on land varies from 20 to 30 kN for hand operated rigs and 100 to 200 kN for mechanically operated rigs as shown in Figure 6.D.9.

Bourden gauges are provided in the driving mechanism for measuring the pressures exerted by the cone and friction jacket either individually or collectively during the operation. The rigs may be operated either on the ground or mounted on heavy duty trucks. In either case, the rig should take the necessary up thrust. For ground based rigs screw anchors are provided to take up the reaction thrust.

D.7.3 Operation of Mechanical Penetrometer

The sequence of operation of the penetrometer shown in Figure 6.D.10 is explained as follows:

Position 1: The cone and friction jacket assembly in a collapsed position.

Position 2: The cone is pushed down by the inner sounding rods to a depth until a collar engages the cone. The pressure gauge records the total force Q_c to the cone. Normally, $a = 40$ mm.

Position 3: The sounding rod is pushed further to a depth b . This pushes the friction jacket and the cone assembly together; the force is Q_f . Normally, $b=40$ mm.

Position 4: The outside mantle tube is pushed down a distance $(a + b)$ which brings the cone assembly and the friction jacket to position 1. The total movement $= a + b = 80$ mm.

The process of operation illustrated is continued until the proposed depth is reached. The cone is pushed at a standard rate of 20 mm per second. The mechanical penetrometer has its advantage as it is simple to operate and the cost of maintenance is low. The quality of the work depends on the skill of the operator. The depth of CPT is measured by recording the length of the sounding rods that have been pushed into the ground.

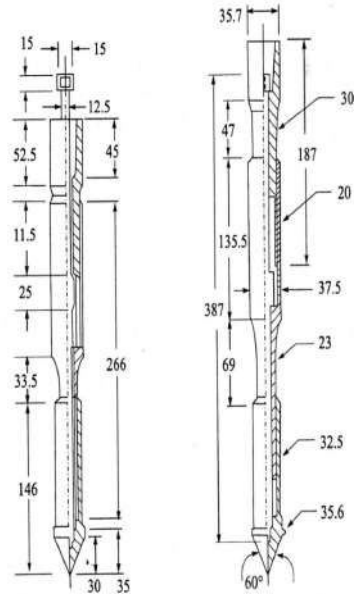


Figure 6.D.8 Friction cone mechanical type penetrometer jacking system (all dimensions are in mm)

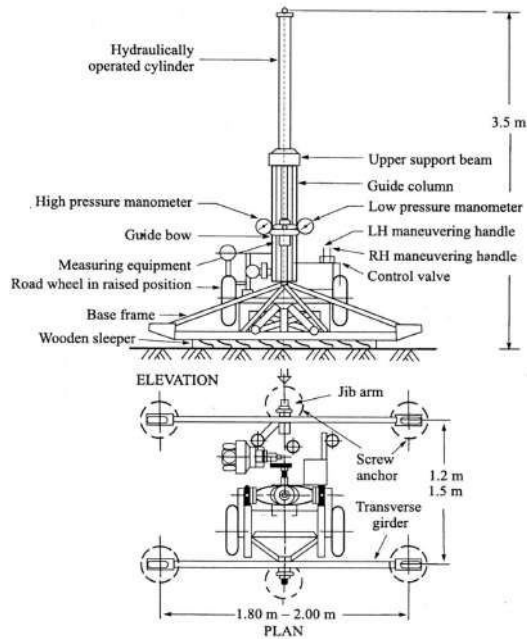


Figure 6.D.9 Assembly of cone penetration rig

D.7.4 The Electric Penetrometer

The electric penetrometer is an improvement over the mechanical one. Mechanical penetrometers operate incrementally whereas the electric penetrometer is advanced continuously.

Figure 6.D.11 shows an electric-static penetrometer with the friction sleeve just above the base of the cone. The sectional area of the cone and the surface area of the friction jacket remain the same as those of a mechanical type. The penetrometer has built in load cells that record separately the cone bearing and side friction. Strain gauges are mostly used for the load cells. The load cells have a normal capacity of 50 to 100 kN for end bearing and 7.5 to 15 kN for side friction, depending on the soils to be penetrated. An electric cable inserted through the push rods (mantle tube) connects the penetrometer with the recording equipment at the surface which produces graphs of resistance versus depth.

The electric penetrometer has many advantages. The repeatability of the cone test is very good. A continuous record of the penetration results reflects better the nature of the soil layers penetrated. However, electronic cone testing requires skilled operators and better maintenance. The electric penetrometer is indispensable for offshore soil investigation.

D.7.5 Operation of Electric Penetrometer

The electric penetrometer is pushed continuously at a standard rate of 20 mm per second. A continuous record of the bearing resistance q_c and frictional resistance f_s against depth is produced in the form of a graph at the surface in the recording unit.

D.7.6 Piezocone

A piezometer element included in the cone penetrometer is called a piezocone, Figure 6.D.12. There is now a growing use of the piezocone for measuring pore pressures at the tips of the cone. The porous element is mounted normally midway along the cone tip allowing pore water to enter the tip. An electric pressure transducer measures the pore pressure during the operation of the CPT. The pore pressure record provides a much more sensitive means to detect thin soil layers. This could be very important in determining consolidation rates in a clay soil within the sand seams.

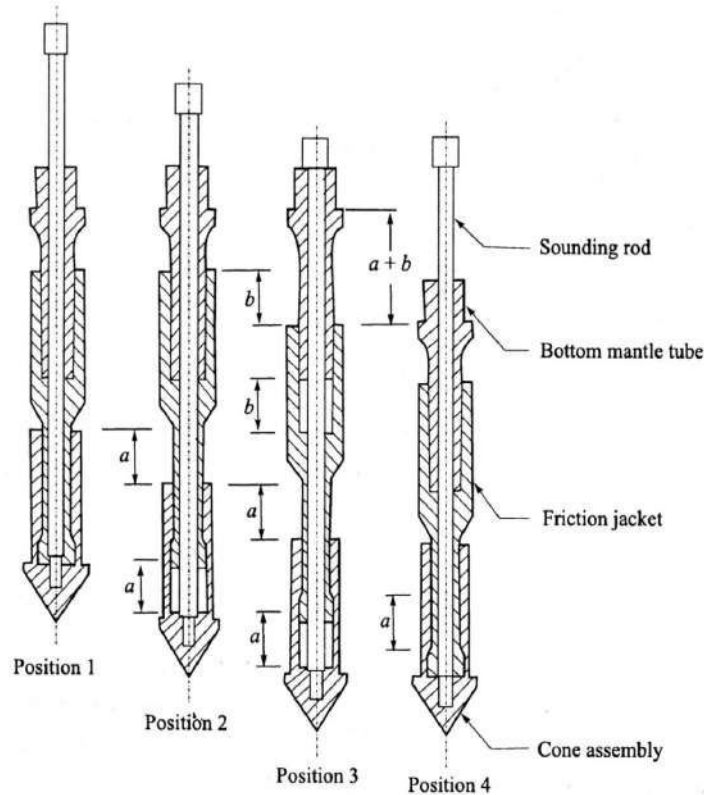


Figure 6.D.10 Steps in cone penetration testing

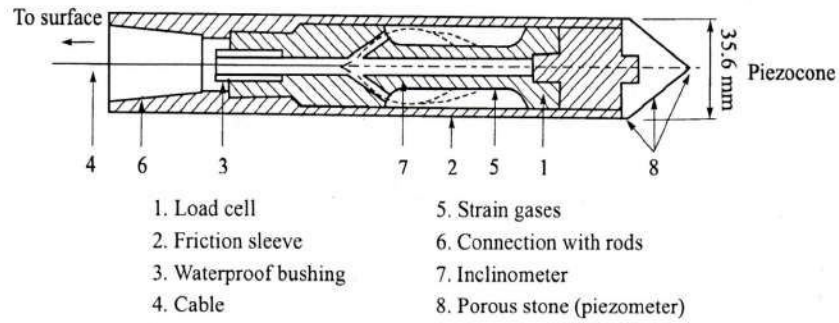


Figure 6.D.11 Electrical static cone penetrometer

D.7.7 Temperature Cone

The temperature of a soil is required at certain localities to provide information about environmental changes. The temperature gradient with depth may offer possibilities to calculate the heat conductivity of the soil. Measurement of the temperature during CPT is possible by incorporating a temperature sensor in the electric penetrometer. Temperature measurements have been made in permafrost, under blast furnaces, beneath undercooled tanks, along marine pipe lines, etc.

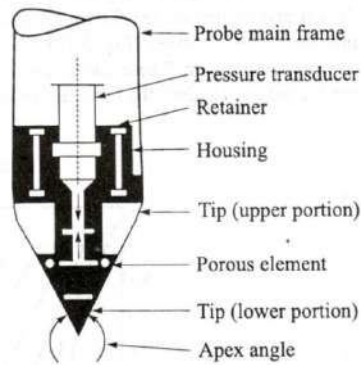


Figure 6.D.12 Details of piezocone

D.7.8 Effect of Rate of Penetration

Several studies have been made to determine the effect of the rate of penetration on cone bearing and side friction. Although the values tend to decrease for slower rates, the general conclusion is that the influence is insignificant for speeds between 10 and 30 mm per second. The standard rate of penetration has been generally accepted as 20 mm per second.

D.7.9 Cone Resistance q_c and Local Side Friction f_c

Cone penetration resistance q_c is obtained by dividing the total force Q_c acting on the cone by the base area A_c .

$$q_c = \frac{Q_c}{A_c} \quad (6.D.9)$$

In the same way, the local side friction f_c is:

$$f_c = \frac{Q_f}{A_f} \quad (6.D.10)$$

Where, $Q_f = Q_r - Q_c$ = force required for pushing the friction jacket

Q_t = total force required to push the cone and friction jacket together in the case of a mechanical penetrometer,

A_f = surface area of the friction jacket.

D.7.10 Friction Ratio, R_f

Friction ratio, R_f is expressed as:

$$R_f = \frac{f_c}{q_c} \quad (6.D.11)$$

Where, f_c and q_c are measured at the same depth. R_f is expressed as a percentage.

Friction ratio is an important parameter for classifying soil.

D.7.11 Relationship between q_c , Relative Density D_r and Friction Angle ϕ for Sand

Research carried out by many indicates that a unique relationship between cone resistance, relative density and friction angle valid for all sands does not exist. Robertson and Campanella (1983a) have provided a set of curves (Figure 6.D.13) which may be used to estimate D_r based on q_c and effective overburden pressure. These curves are supposed to be applicable for normally consolidated clean sand. Figure 6.D.14 gives the relationship between q_c and ϕ (Robertson and Campanella, 1983b).

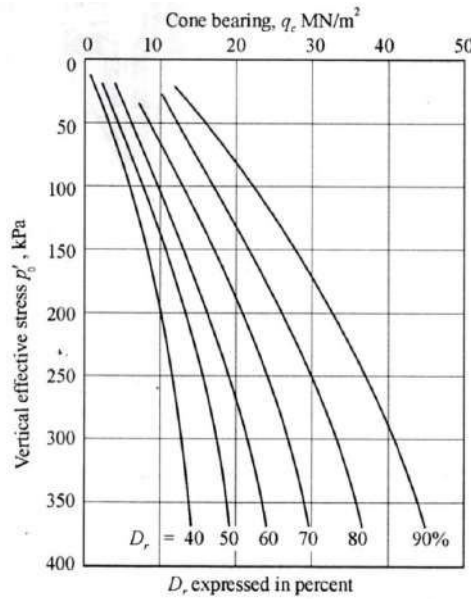


Figure 6.D.13 Relationship between relative density, D_r and cone resistance, q_c for quartz sand (Robertson and Campanella, 1983a)

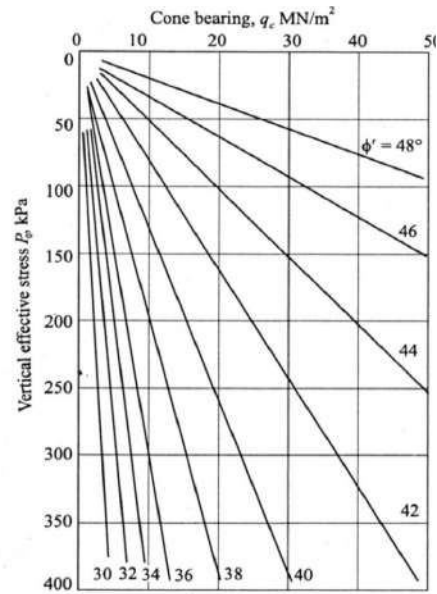


Figure 6.D.14 Relationship between cone resistance, q_c and relative density, D_r for quartz sand (Robertson and Campanella, 1983b)

D.7.12 Relationship between q_c and Undrained Shear Strength, c_u of Clay Soil

The cone penetration resistance q_c and c_u may be related as:

$$q_c = N_k c_u + p_o \text{ or, } c_u = \frac{q_c - p_o}{N_k} \quad (6.D.12)$$

Where, N_k = cone factor,

$p_o = \gamma Z$ = overburden pressure.

Lune and Kelven (1981), investigated the value of the cone factor N_k for both normally consolidated and overconsolidated clays. The values of N_k as obtained are given below in Table 6.D.7.

Table 6.D.7: Soil Type and q_c Value

Type of Clay	Cone Factor, N_k
Normally Consolidated	10 to 19
Over Consolidated:	
At shallow depth	15 to 20
At deep depth	12 to 18

Possibly a value of 20 for N_k for both types of clay may be satisfactory. Sanglerat (1972) recommends the same value for all cases where an overburden correction is of negligible value.

D.7.13 Soil Classification based on CPT Results

One of the basic uses of CPT is to identify and classify soils. A CPT-Soil Behaviour Type Prediction System has been developed by Douglas and Olsen (1981) using an electric and friction cone penetrometer. The classification is based on the friction ratio f_c/q_c . The ratio f_c/q_c varies greatly depending on whether it applies to clays or sands. Their findings have been confirmed by hundreds of tests.

For clay soils, it has been found that the friction ratio decreases with increasing liquidity index, I_L . Therefore, the friction ratio is an indicator of the soil type penetrated. It permits approximate identification of soil type though no samples are recovered.

Douglas (1984), presented a simplified classification chart shown in Figure 6.D.15. His chart uses cone resistance normalized (q_{cn}) for overburden pressure using the equation:

$$q_{cn} = q_c (1 - 1.25 p'_o) \quad 6.D.13$$

Where, p'_o = effective overburden pressure in tsf, and q_c = cone resistance in tsf.

The CPT data provides a repeatable index of the aggregate behavior of in-situ soil. The CPT classification method provides a better picture of overall subsurface conditions than is available with most other methods of exploration. A typical sounding log is given in Figure 6.D.16.

The friction ratio R_f varies greatly with the type of soil. The variation of R_f for the various types of soils is generally of the order given in Table 6.D.8.

Table 6.D.8: Soil Classification Based on Friction Ratio

Friction Ratio, R_f (%)	Type of Soil
0.0 - 0.5	Loose gravel fill
0.5 - 2.0	Sands and gravels
2.0 – 5.0	Clay sand mixtures and silts
> 5.0	Clays, peats etc.

Meyerhof (1965), presented comparative data between SPT and CPT. For fine or silty medium loose to medium dense sands, he presents the correlation as:

$$q_c = 0.4 N \quad (\text{in MN/m}^2) \quad (6.D.14)$$

Meyerhof's findings are as given in Table 6.D.9. The lowest values of the angle of ϕ given in Table 6.D.9 are conservative estimates for uniform, clean sand and they should be reduced by at least 5° for clayey sand. These values, as well as the upper values of ϕ that apply to well graded sand, may be increased by 5° for gravelly sand.

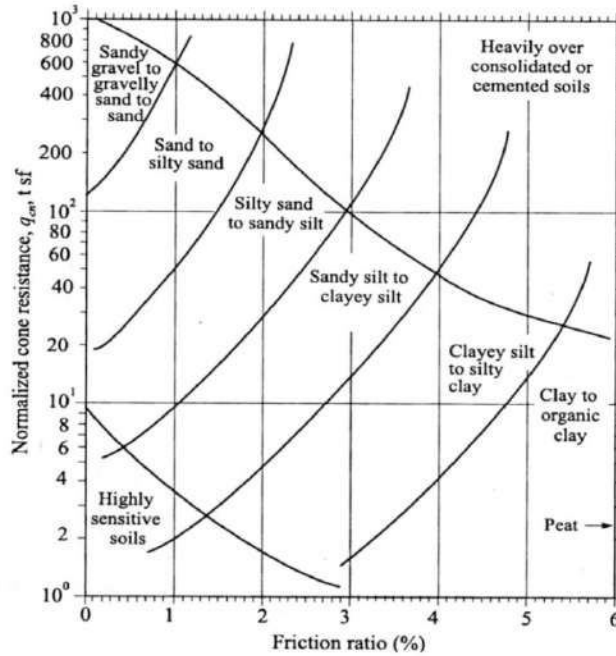


Figure 6.D.15 A simplified classification chart (after Douglas, 1984)

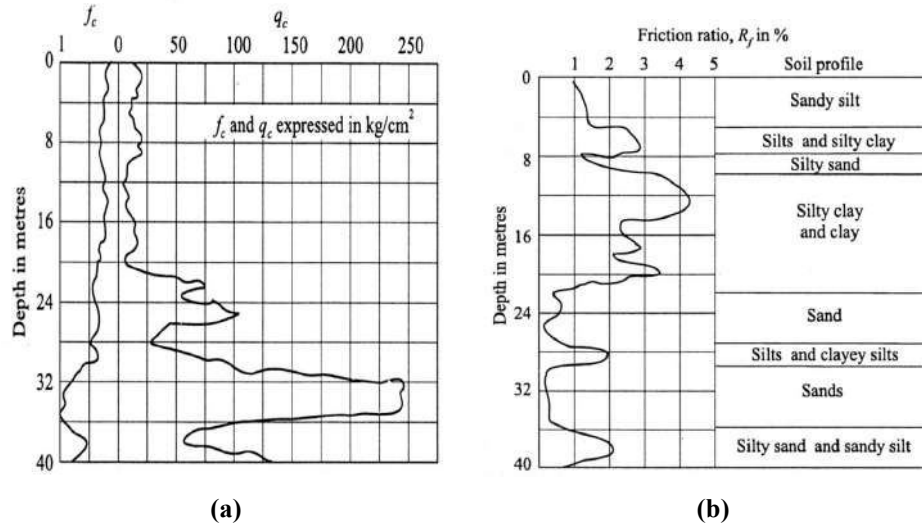


Figure 6.D.16 Typical Sounding Log; (a) Cone resistance with depth; (b) Friction ratio with depth

Table 6.D.9: Soil Classification Based on Friction Ratio

State of Sand	D_r	$(N_1)_{60}$	q_c (MPa)	ϕ^a
Very loose	< 0.2	< 4	< 2.0	< 30
Loose	0.2-0.4	4-10	2-4	30-35
Medium dense	0.4-0.6	10-30	4-12	35-40
Dense	0.6-0.8	30-50	12-20	40-45
Very dense	0.8-1.0	> 50	> 20	> 45

D.7.14 Correlation between SPT and CPT

Figure D.17 shows correlations presented by Robertson and Campanella (1983a), and Kuhawy and Mayne (1990) between the ratio of q_c/N and mean grain size, D_{50} . It can be seen from the Figure 6.D.17 that the ratio varies from 1 at $D_{50}=0.001$ mm to a maximum value of 8 at $D_{50}=1.0$ mm. The soil type also varies from clay to sand.

It is clear from the above discussions that the value of $n = q_c/N$ is not a constant for any particular soil. Designers must use their own judgement while selecting a value for n for a particular type of soil.

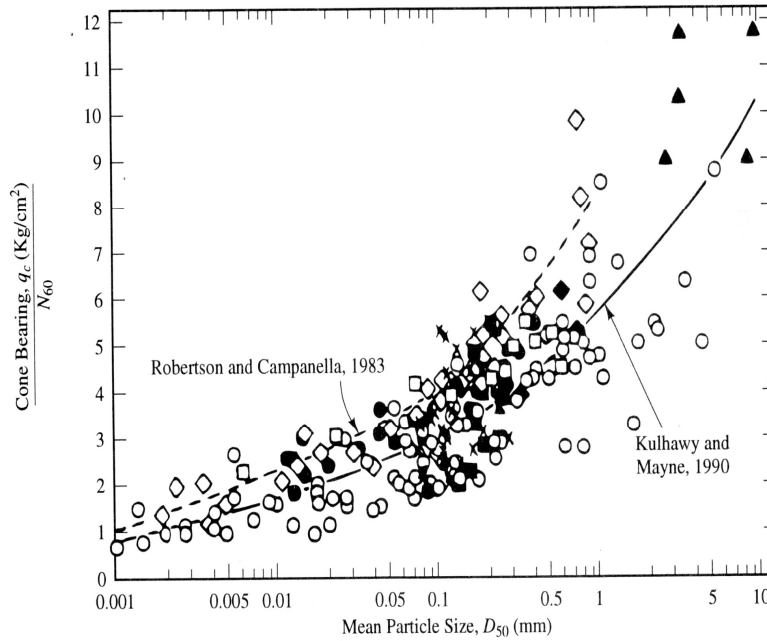


Figure 6.D.17 Relation between q_c/N and mean grain size D_{50}

D.8 Geophysical Methods of Exploration

Several types of geophysical exploration techniques permit a rapid evaluation of subsoil characteristics. These methods also allow rapid coverage of large areas and are less expensive than conventional exploration by drilling. However, in many cases, definitive interpretation of the results is difficult. For that reason, such techniques should be used for preliminary work only. Three types of geophysical exploration are common: the seismic refraction survey, cross-hole seismic survey and electrical resistivity survey.

D.8.1 Seismic Refraction Survey

This method is useful in obtaining preliminary information about the thickness of the layering of various soils and the depth to rock or hard soil at a site. The test method is described in ASTM D5777. Refraction surveys are conducted by impacting the surface, such as point A in Figure 6.D.18a, and observing the first arrival of the disturbance (stress waves) at several other points (e.g. B, C, D, ...). The impact can be created by a hammer blow or by a small explosive charge. The first arrival of disturbance waves at various points can be recorded by geophones.

The impact on the ground surface creates two types of stress wave: 'P' waves (or plane waves) and S waves (or shear waves). P waves travel faster than S waves; hence the first arrival of disturbance waves will be related to the velocities of the 'P' waves in various layers. The velocity of P waves in a medium is:

$$v = \sqrt{\frac{E_s (1 - \mu)}{\left(\frac{\gamma}{g}\right) (1 - \mu_s) (1 + \mu_s)}} \quad (6.D.15)$$

Where, E_s = modulus of elasticity of the medium

γ = unit weight of the medium

g = acceleration due to gravity

μ_s = Poisson's ratio of soil

To determine the velocity v of P waves in various layers and thicknesses of those layers, the following procedure is used.

The first arrival times $t_1, t_2, t_3 \dots$ at various distances $x_1, x_2, x_3 \dots$ from the point of impact is obtained. A graph of time t against distance x is plotted, as shown in Figure 6.D.18b. The slopes of the line ab, bc, cd , are determined.

$$\text{Slope of } ab = \frac{1}{v_1}$$

$$\text{Slope of } bc = \frac{1}{v_2}$$

$$\text{Slope of } cd = \frac{1}{v_3}$$

Here $v_1, v_2, v_3, \dots$ are the P-wave velocities in layers I, II, III, ... respectively (Figure 6.D.18a). The thickness of the top layer is determined as:

$$Z_I = \frac{x_c}{2} \sqrt{\frac{v_2 - v_1}{v_2 + v_1}} \quad (6.D.16)$$

The value of x_c can be obtained from Figure 6.D.17b. The thickness of the second layer can be determined from

$$z_2 = \frac{I}{2} \left[T_{i2} - 2Z_I \frac{\sqrt{v_3^2 - v_1^2}}{v_3 v_1} \right] \frac{v_3 v_2}{\sqrt{v_3^2 - v_2^2}} \quad (6.D.17)$$

Where, T_{i2} is the time intercept of the line cd (Figure 6.D.18a). Velocities of P-waves in various layers indicate the types of soil or rock that are present below the ground surface. They are listed in Table 6.D.10.

In analyzing the results of a refraction survey, two limitations need to be kept in mind.

- (i) The basic equations for the survey that is, Equations 6.D.16 and 6.D.17 are based on the assumption that the P-wave velocity $v_1 < v_2 < v_3 \dots \dots$.
- (ii) When a soil is saturated below the water table, the P-wave velocity may be deceptive. P-wave can travel with a velocity of about 1500 m/sec through water. For dry, loose soils the velocity may be well below. If the presence of groundwater has not been detected, the P-wave velocity may be erroneously interpreted to indicate a stronger material (sandstone). In general, geophysical interpretations should always be verified by the results obtained from borings.

D.8.2 Cross Hole Survey

The test method can be found in ASTM D4428-07. The velocity of shear waves created as a result of an impact to a given layer of soil can be effectively determined by the cross-hole seismic survey. The principle of this technique is illustrated in Figure 6.D.19. Two holes are drilled into the ground at a distance L apart. A vertical impulse is created at the bottom of one borehole by means of an impulse rod. The shear waves thus generated are recorded by means of a vertically sensitive transducer. The velocity of shear waves can be calculated as:

$$v_s = \frac{L}{t} \quad (6.D.18a)$$

Where, t is the travel time of the wave. The shear modulus G_s of the soil at the depth at which the test is done can be determined from the relation:

$$v_s = \sqrt{\frac{G_s}{(\gamma/g)}} \text{ or } G_s = \frac{v_s^2 \gamma}{g} \quad (6.D.18b)$$

Where, v_s = velocity of shear waves

γ = unit weight of soil

g = acceleration due to gravity

Shear modulus is useful in the design of foundations to support vibrating machinery and the like.

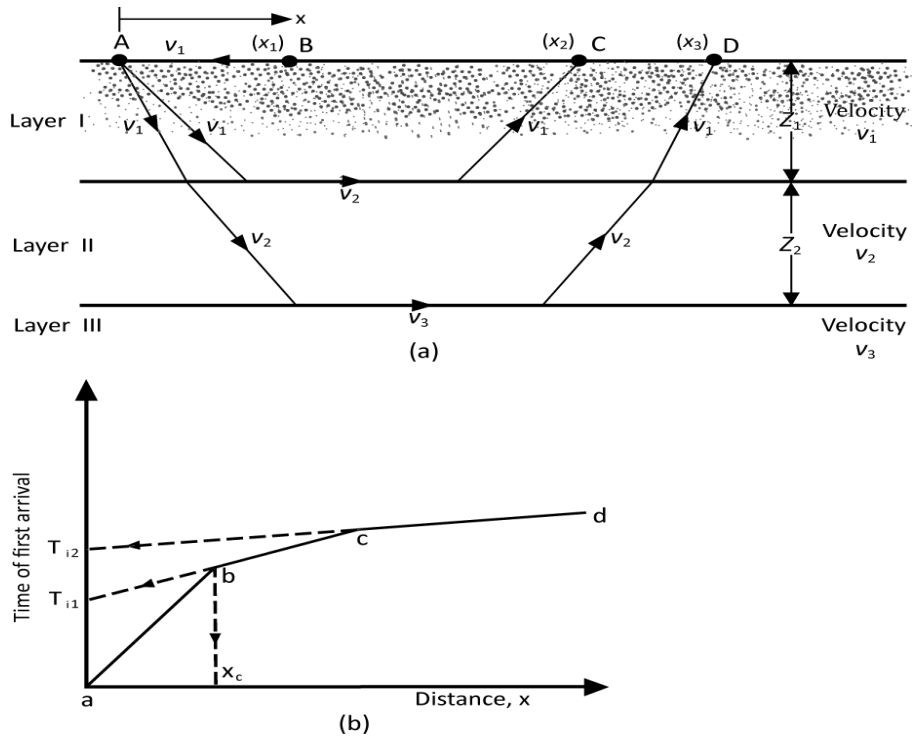


Figure 6.D.18 Seismic refraction survey

Table 6.D.10 : Range of P-Wave Velocity in Various Soils and Rocks

Type of soil or rock	P-wave velocity	
	M/sec	Ft/sec
Soil :		
Sand, dry silt, and fine grained top soil	200—1000	650—3300
Alluvium	500—2000	1650—6600
Compacted clays, clayey gravel and dense clayey sand	1000—2500	3300—8200
Loess	250—750	800—2450
Rock :		
Slate and shale	2500—5000	8200—16400
Sandstone	1500—5000	4900—16400
Granite	4000—6000	13100—19700
Sound limestone	5000—10000	16400—32800

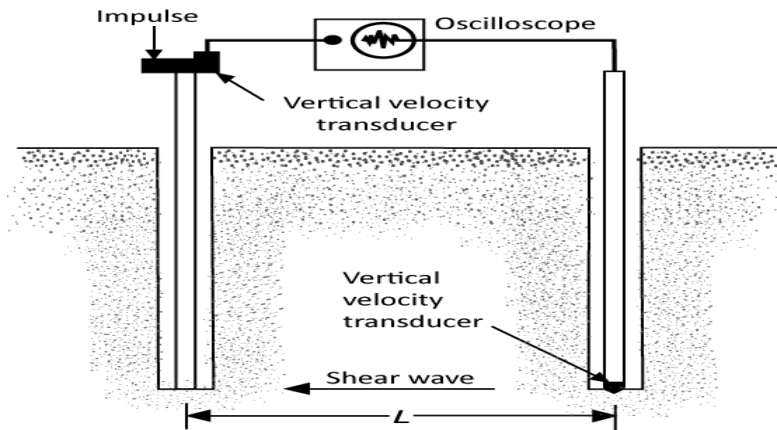


Figure 6.D.19 Cross-hole method of seismic survey

D.8.3 Resistivity Survey

The electrical resistivity of any conducting material having length L , cross sectional area A and electrical resistance R can be defined as :

$$\rho = \frac{RA}{L} \quad (6.D.19)$$

The unit of resistivity is ohm-metre. The resistivity of various soils depends primarily on their moisture content and also on the concentration of dissolve ions in them. The range of resistivity of various soils and rocks are given in Table 6.D.11.

The most common procedure of measuring the electrical resistivity of a soil profile makes use of four electrodes driven into the ground spaced equally along a straight line. The procedure is generally known as Wenner method. The two outside electrodes are used to send an electrical current I (using dc current) into the ground. The current is typically in the range of 50-100 milliamperes. The voltage drop V is measured between the two inside electrodes. The test arrangements are shown in Figure 6.D.20a. If the soil profile is homogeneous, its electrical resistivity is given by :

$$\rho = \frac{2\pi d v}{I} \quad (6.D.20)$$

In most cases, the soil profile may consists of various layers with different resistivities and Eq. 6.D.20 will yield the apparent resistivity. To obtain actual resistivity of various layers and their thicknesses, an empirical method may be used. It involves conducting test at various electrode spacing. Thus, the sum of the apparent resistivities $\Sigma\rho$ obtained is plotted against the spacing d as shown in Figure 6.D.20b. The plot thus obtained has relatively straight segments, the slopes of which give the resistivity of individual layers. The determination of thickness of layers is illustrated in Figure 6.D.20b. The resistivity survey is particularly useful in locationg gravel deposits within a fine grained soil.

Table 6.D.11 Electrical Resistivity of Various Types of Soils

Soil type	Resistivity (ohm-m)
Sand	500-1500
Clays, saturated silts	0-100
Clayey sand	200-500
Gravel	1500-4000
Weathered rock	1500-2500
Sound rock	> 5000

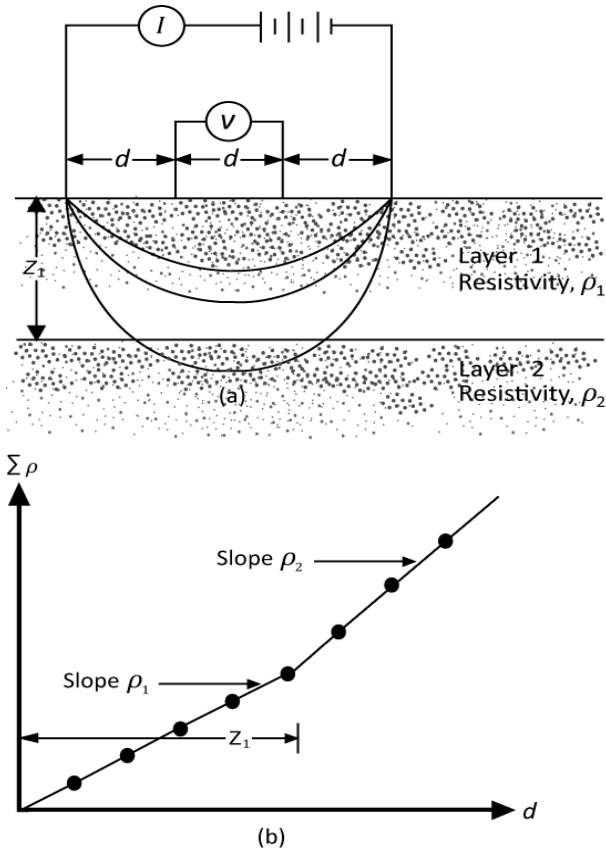


Figure 6.D.20 Electrical resistivity survey; (a) Wenner method of placing electrode; (b) Empirical method of determining resistivity and thickness of soil layer.

PART VI
Appendix E

Recommended Criteria for Identification and Classification of Expansive Soil

The following criteria may be adopted to identify and classify expansive soils:

- (1) Based on the values of plasticity index and shrinkage limit, United States Bureau of Reclamation (USBR) suggests the following classification criteria for expansive soil:

Plasticity Index	Shrinkage Limit	Degree of Expansion
>35	<10	Very High
25-41	6-12	High
15-28	8-18	Medium
<18	>13	Low

- (2) On the basis of previous data for linear shrinkage of Bangladesh soils, criteria for the degree of expansion proposed by Hossain (1983) is as follows:

Linear Shrinkage (%)	Degree of Expansion
> 14	High
10-14	Medium
0-10	Low

- (3) On the basis of the values of free swell, Indian standard (IS: 1948, 1970) recommends criteria of expansion is as follows:

Free Swell (%)	Degree of Expansion	Danger of Severity
<50	Low	Non-critical
50-100	Medium	Marginal
100-200	High	Critical
>200	Very High	Severe

- (4) Based on the value of free swell index, Indian Standard (IS: 2911, Part III, 1980) suggests the following criteria for the degree of expansion of soils:

Free Swell Index (%)	Degree of Expansion	Danger of Severity
<20	Low	Non-critical
20 to 35	Medium	Marginal
35 to 50	High	Critical
>50	Very high	Severe

- (5) Based on the values of liquid limit, plasticity index and shrinkage limit, Indian Standard (IS: 2911, Part 3, 1980) suggests the following criteria for the degree of expansiveness of soils:

Liquid Limit (%)	Plasticity Index	Degree of Expansion	Danger of Severity
20-35	<12	Low	Non-critical
35-50	12-23	Medium	Marginal
50-70	23-32	High	Critical
>70	>32	Very high	Severe

- (6) Based on the values of swelling potential, Seed et al. (1962) proposed the following four categories of expansion characteristics:

Swelling Potential (%)	Degree of Expansion
0-1.5	Low
1.5-5	Medium
5-25	High
> 25	Very high

- (7) Based on the values of swelling pressure, Chen (1965) proposed the following criteria for degree of expansion:

Swelling Pressure kg/cm ²	Degree of Expansion
0-1.5	Low
1.5-2.5	Medium
2.5-9.8	High
>9.8	Very high

- (8) Based on the values of volume change from air dry to saturated condition, Seed et al. (1962) proposed the following four categories of expansion characteristics:

Volume Change from Air Dry to Saturated Condition (%)	Degree of Expansion
0 – 10	Low
10 – 20	Medium
20 – 30	High
> 30	Very high

- (9) Look (2007) reports that the plasticity index by itself can be misleading, as the test is carried out on the percent passing the 425 micron sieve, i.e. any sizes greater than 425 μm is discarded. There have been cases when a predominantly “rocky/granular” site has a high PI test results with over 75 percent of the material discarded. The weighted plasticity index (WPI) considers the percent of material used in the test, where $WPI = PI \times \% \text{ passing the } 425 \text{ micron sieve}$. Degree of expansion with weighted plasticity index is presented as under.

Weighted Plasticity Index (%)	Degree of Expansion
< 1200	Very Low
1200 – 2200	Low
2200 – 3200	Moderate
3200 – 5000	High
> 5000	Very high

PART VI
Appendix F
Construction of Pile Foundation

F.1 Introduction

The pile driving process needs to fulfill assumptions and goals of the design engineer just as much as the design process has to foresee the conception and installation of the pile at the site. This is only possible through the selection of the right driving equipment especially hammer with proper assembly mounted on the most suitable leader, operated according to the specified practices of installation that consists of a series of principle and subsidiary procedures.

There are three methods of driving piles: jacking, vibratory driving, and driving. The first two, jacking and vibratory driving are comparatively rare. The reaction needed to push a pile into the ground is equal to the limit pile capacity, which can be a very large load. Until recently, this made jacking suitable only for small piles; large, heavy rigs are now available that can jack normal size piles for onshore applications.

Vibratory driving is only suitable for loose sands, particularly if saturated, because liquefaction of the sand results from the vibration, making it easy to drive the pile into the ground. Vibratory driving is routinely used to drive sheet piles and less frequently used to install relatively small steel H-piles.

The most common method of installing displacement piles is by driving the piles into the ground by blows of an impact hammer. Piles installed in this manner are referred to as driven piles. In order to understand this method of installation, we need to examine first the equipment that is required. A brief description of the driving equipment and procedures as given by Salgado (2011) are presented as under.

F.2 Piling Driving Equipment

Pile driving equipment are broadly classified into three groups; the leader for positioning the pile for driving, the hammer for delivering energy for driving and the driving system components for better and safe distribution of energy on the top of the piles. Hammers are used to install the driven piles. The leader, the pile and the hammer are often carried by a special crawler rig or crane.

F.3 Installation of Driven Piles

Driven piles are installed by the kinetic energy developed through the

- (a) Ramming action of piling hammer which can be
 - Drop Hammers
 - Single-Acting Air/Steam Hammers
 - Double-Acting Air/Steam Hammers
 - Differential-Acting Air/Steam Hammers
 - Hydraulic Hammers
 - Diesel Hammers (Single acting & Double acting)
- (b) Vibration and weight of vibro-hammers which can be
 - Electric Motor Vibrohammers
 - Hydraulic Vibrohammers
 - Sonic Resonance Vibrohammers

As such, there are three methods of installing displacement piles: jacking, vibratory driving, and driving. The first two, jacking and vibratory driving are comparatively rare. In jacking, the reaction needed to push a pile into the ground is equal to the limit pile capacity, which can be a very large load. Until recently, this made jacking suitable only for small piles. However, large, heavy rigs are now available that can jack normal size piles for onshore applications. Vibratory driving is only suitable for loose sands, particularly if saturated, because liquefaction of the sand results from the vibration, making it easy to drive the pile into the ground. Vibratory driving is routinely used to drive sheet piles and less frequently used to install relatively small steel H-piles.

The most common method of installing displacement piles is by driving the piles into the ground by blows of an impact hammer. Piles installed in this manner are referred to as driven piles. In order to understand this method of installation, we need to examine first the equipment that is required and described as under.

Sometimes, to penetrate the pile through a compact ground layer or a rock layer, predrilling is used. By predrilling, it is ensured that driven piles reach their designed minimum tip depths, and also, their risk of being tip or head damaged due to increased hammering impact loads are avoided. Another benefit of predrilling is increased driving speeds and much lower ratios of early refusal.

In addition, water jetting is used to penetrate piles through dense granular layers and to guarantee their penetration to minimum penetration depths without getting damaged. Both predrilling and water jetting need to be stopped several meters (about 1.5 – 3 m) before the final penetration point and driving refusal must be obtained in the undisturbed soil.

F.4 Pile Driving Leads

Pile driving leads or leaders are steel frames used to correctly position the pile for driving and to keep the pile head and hammer aligned concentrically during driving. Leads, with length exceeding that of the pile to be driven by 5-7 m, are attached to a crane in one of the two ways shown in Figure 6.F.1. Fixed leads are connected near the top with a horizontal hinge at the tip of the boom. A hydraulically operated horizontal brace allows the operator to adjust the inclination of the lead to install battered piles and to adjust verticality. Hanging leads are suspended from the crane boom by a cable. Stabbing points at the base of the lead allow the operator to adjust position and inclination, but it is more difficult to position the pile with hanging leads than with fixed leads. If hanging leads are to be used to drive piles that require a high degree of positioning accuracy, a suitable template should be provided to maintain the leads in a steady or fixed position. Construction tolerances on positioning depend to some extent on the diameter of the piles and whether they are isolated piles or are part of a group. For group piles, pile location may be off by as much as 75 mm and deviation from vertical as large as 1 in 25 may be acceptable.

Leads that are not properly restrained may cause pile damage, particularly to concrete piles. When driving long slender piles, the use of intermediate pile supports in the leads may be necessary to prevent pile damage that may be caused by long unbraced pile lengths. Leads are not absolutely necessary for every pile driving operation, but they are normally used to maintain concentric alignment of the pile and hammer and to obtain the required accuracy of pile position and alignment while driving the pile, especially for battered piles. Even if leads are not used, it is highly advisable to use a template to maintain the pile at the right location throughout driving.

F.5 Piling Hammers

Pile driving hammers are devices used to deliver blows to the head of a pile in order to drive it into the ground. The type of hammers can be highly varied. Figure 6.F.2 taken from Hannigan et al (1998) gives a classification of Piling Hammer Types. Simple drop and steam hammers are still being used but single, differential or double acting diesel, and hydraulic hammers are more common. Also, vibrating and sonic hammers are being used more and more often due to environmental or urban impact factors.

Pile hammers vary in the rate at which blows are delivered, the maximum amount of energy delivered in any one single blow, and the duration of the blow. To a large extent, these performance parameters reflect the mechanism of operation of the hammer. There are basically four types of hammers: gravity or drop hammers, single-acting hammers, double-acting hammers, and differential hammers.

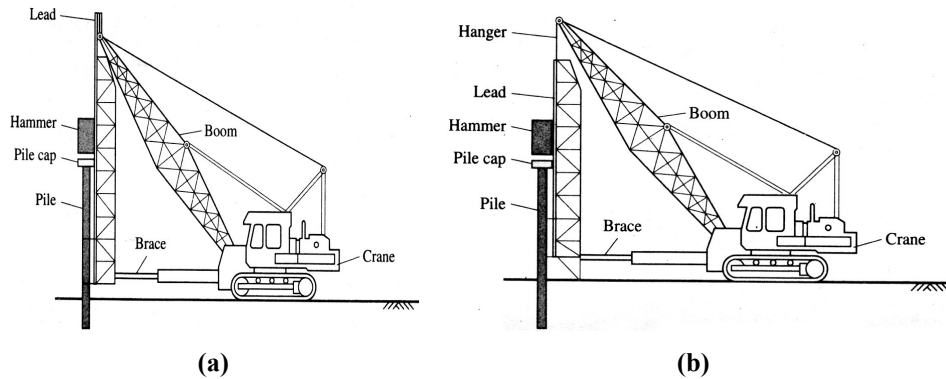


Figure 6.F.1 Crane mounted leads; (a) Fixed and (b) Hanging

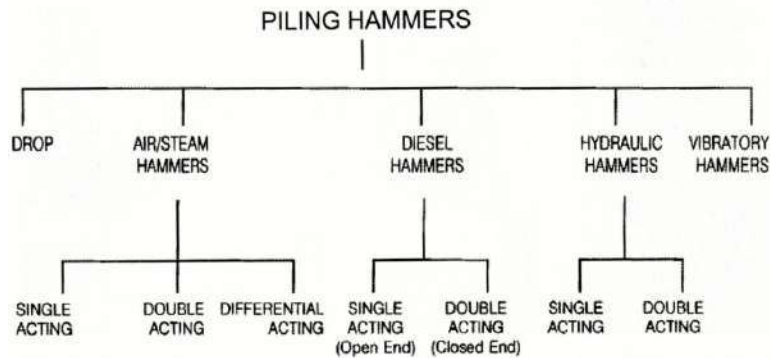


Figure 6.F.2 Pile driving hammer classification (Hannigan et al, 1998)

Table 6.F.1 summarizes the main features of each type of pile driving hammer. Drop hammers are the simplest, relying solely on gravity for delivering the blows to the pile head. A drop hammer is usually made of a single block or a system of steel blocks, which may be removed or added as needed. Drop hammer weights are typically in the range of 10-50 kN (1-5 tons). Because the weight of the hammer is usually fixed during the driving of any given pile, the only variable available to the operator for adjusting the energy delivered by hammer blows is the drop height. There is an implied danger when driving concrete piles through hard, strong soil (under so called hard driving conditions). The operator may drop the hammer from too large a height, generate excessive large stresses in the pile, and damage it.

In single acting hammers, the ram is connected to a piston located within a cylinder [Figure 6.F.3(a)]. The piston is lifted by either steam or compressed air (in what is called the upstroke) and then allowed to fall by the action of gravity (the downstroke). When it does, the ram impacts the head of the pile, driving it some distance into the ground. The weight of a single acting hammer (in the 20 to 150 kN range) is much larger than that of a drop hammer, but the fall height (stroke) is much smaller (up to 1.5 m, typically).

In a double-acting hammer [Figure 6.F.3(b)], either steam or compressed air is used to increase the pressure on the piston, both on its way up and on its way down, so that the ram impacts the pile with greater force and higher velocity and does so more times per minute. This is possible because these hammers are closed at the top. The stroke is typically less than that for the single acting hammer, resulting in higher production rate. The stroke cannot be controlled visually because the hammer is closed. The differential hammer is much like the double-acting hammer, but it relies on the different areas of the upper and lower parts of the piston to generate the repeating up and down strokes.

Table 6.F.1: General Characteristics of Different Types of Piling Hammers

Hammer	Ram weight (kN)	Stroke (m)	Max. strike rate (bpm)	Conditions under which use indicated	Caution
Drop	10-50 ($\frac{1}{2}$ to 2 times pile weight)	Wide range	5-10	Noise restrictions	Possible damage during hard driving of concrete piles
Single acting	20-150	< 1.5	40-60		
Double acting / Differential	0.5-180		90-300	Underwater operations; Sheet pile driving	
Diesel	10-150		40-100	All types of piles (with diameter up to 2.2m) in most soil conditions	Soft clays (where combustion may not occur)

Hydraulic hammers are moved by oil pressure and can be of single or double acting varieties. Their principle of operation Intake is essentially the same as that of other single- and double-acting hammers. Diesel hammers, such as the one shown in Figure

6.F.3(d), also come in both the single and double acting varieties, but they differ from other hammers in one important aspect. In single acting hammers, extra "zip" is added to the blow by combustion of fuel injected before the down stroke is completed. In a double acting hammer, a bounce chamber is present in the upper part of the hammer, providing quicker and stronger rebound from the upstroke. These hammers tend to be smaller and lighter than double acting hammer. Other hammers, as the extra energy and blow duration obtained from the fuel combustion makes them very efficient.

The amount of fuel injection into the chamber of diesel hammers can be controlled, allowing adjustment for lighter or harder driving conditions. However, in soils alternating loose/soft layers with extremely hard layers, the bounce of the ram will vary from low to high, which may be damaging to concrete piles. Diesel hammers may be attached directly to the pile head, not strictly requiring the use of leads for their operation.

F.6 Driving System Components

The components of a driving system are the hammer (Impact block) itself and a number of additional components that may or may not be present, as shown in Figure 6.F.4. Each of these components is referred to by various names, the most common being:

- (i) Anvil (striker plate)
- (ii) Cap block or hammer cushion
- (iii) Driving head (helmet, cap, anvil block)
- (iv) Follower
- (v) Pile cushion (used for driving precast concrete piles)

All of these elements, except the follower, aim to diffuse some of the energy from the hammer blow in order to avoid damage to the pile or any of the driving system components. The driving head goes on top of the pile; it is shaped in a way that allows it to slide along the leads, forcing the alignment of the pile and the hammer. The follower is an extension used when a pile needs to be driven to a level below the level of operation of the rig, such as when the heads of the piles for a bridge, for example, will be located under water. The pile cushion is used to further diffuse and better distribute the energy on top of concrete piles, which are more susceptible to damage during driving.

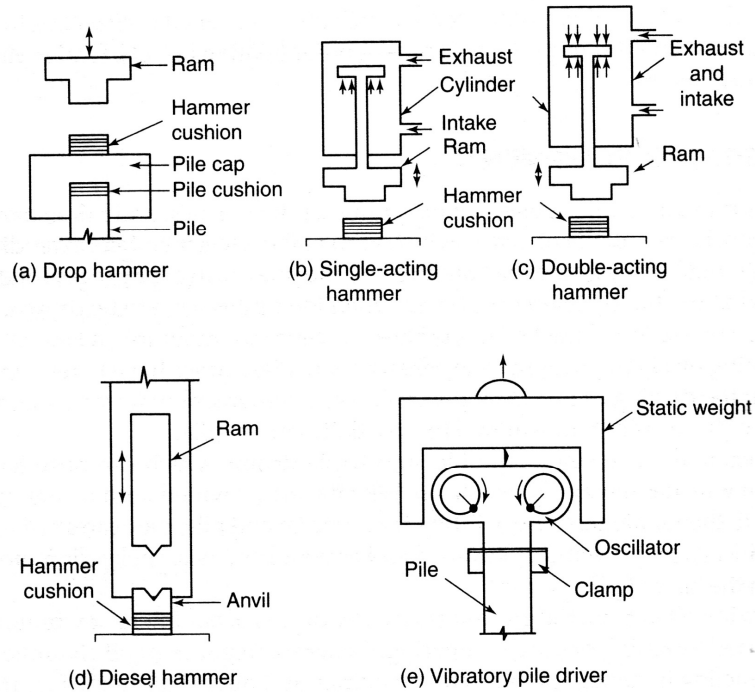


Figure 6.F.3 Principles of pile driving hammers

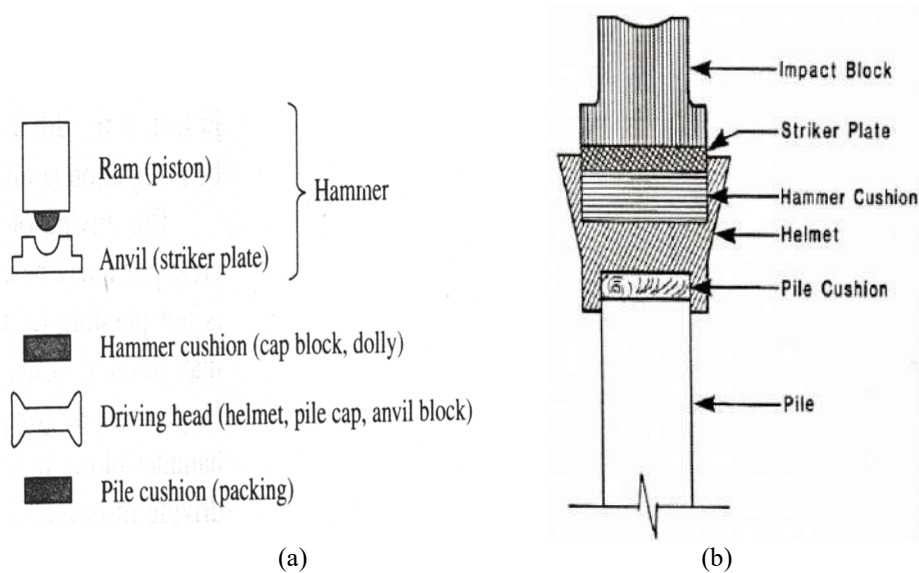


Figure 6.F.4 (a) Driving system components; (b) Positioning of the components

F.7 Driving Procedures

As mentioned earlier, the key to efficient pile driving is a good match of the pile with the hammer and other driving system components. When this is done, the operator will not be forced to try anything out of the ordinary to drive the piles to the required depth. For example, if an excessively light drop hammer is used, the operator may feel compelled to raise the hammer to excessive heights, which may in turn damage concrete piles. Mismatches of this sort quite often result either in inability to drive the pile as specified or in pile damage.

When using drop hammers, the key decision is the weight of the hammer. A drop hammer is typically a cylindrical weight that is raised to a certain height using a winch and dropped on top of the pile. The weight of drop hammer ranges from one half to twice the pile weight. The corresponding drop heights are in the 0.2 to 2m range. It is usually preferable to select heavier rather than lighter hammers, as the drop heights are then smaller and the likelihood of damage to concrete piles, in particular, is much lower. The ratio of hammer weight to pile weight for other types of hammer lies in the range of 0.25 to 1.0, but the selection of a suitable hammer also depends on other factors and is best done with the aid of computer-based drivability analysis. For example, the energy delivered by diesel hammers to piles increases with the driving resistance; in fact, if the driving is too easy, as in the first few meters in soft clays, there may be no ignition at all in the hammer, which would make it very inefficient.

The driving of precast concrete piles is probably the most challenging. Concrete piles may be damaged when driven through soft or loose soil, something that is not possible for either timber or steel piles. This is so because tensile stresses may develop in the pile under the conditions stated and, concrete is very weak in tension. In general, in going through soft/loose soil layers, the operator should use light hammer blows to avoid this. As a general rule, light blows are always used when driving resistance is small.

Immediately after driving, the pile resistance may be either higher or lower than the resistance it will ultimately have. The process by which pile resistance increases with time after driving is referred to as setup or freeze. When pile resistance decreases with time after driving, the process is referred to as relaxation. At least approximate estimation of the rate at which these processes take place is important to plan continuing pile driving around previously driven piles, to plan and perform load tests, and to take account in design of the real, long term resistance of the pile.

F.8 Pile Hammer Selection Guidelines

Selection of pile and pile hammer is usually done using wave equation program. If wave analysis is not done following Tables 6.F.2(a) and 6.F.2(b) may be used as an approximate guide. (Tables 6.F.2(a) and 6.F.2(b) were prepared by adapting the Table presented in “Pile Driving Equipment”, US Army Corps of Engineers, July 1997.)

Table 6.F.2(a): Guidelines for Selection of Pile Hammers: Sandy Soils

SPT (N) Value	Soil Density	Open End Pipe Piles	Closed End Pipe Piles	H-Piles	Sheet Piles	Concrete Piles
0 - 3	Very Loose	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA (A, S, H)
4 - 10	Loose	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA (A, S, H)
10 - 30	Medium	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA (A, S, H)
30 - 50	Dense	DA, SA, V (A, S, H)	SA, V (A, S, H)	SA, V (A, S, H)	DA, SA, V (A, S, H)	SA (A, S, H)
Over 50	Very Dense	SA (A, S, H)	SA (A, S, H)	SA (A, S, H)	DA, SA, V (A, S, H)	SA (A, S, H)

LEGEND: DA = Double Acting; SA = Single Acting; A = Air/Diesel; S = Steam; H = Hydraulic; V = Vibratory

Table 6.F.2(b): Guide lines for Selection of Pile Hammers: Clayey Soil

SPT (N) Value	Soil Density	Open End Pipe Piles	Closed End Pipe Piles	H-Piles	Sheet Piles	Concrete Piles
0 - 4	Soft	DA, SA, V (A, S, H)	DA, SA (A, S, H)	DA, SA, V (A, S, H)	DA, SA, V (A, S, H)	DA, SA (A, S, H)
4 - 8	Medium	DA, SA, V (A, S, H)	SA (A, S, H)	DA, V (A, S, H)	DA, SA, V (A, S, H)	SA (A, S, H)
8 - 15	Stiff	DA, SA (A, S, H)	SA (A, S, H)	DA, SA (A, S, H)	DA, SA (A, S, H)	SA (A, S, H)
15 - 30	Very Stiff	SA (A, S, H)	SA (A, S, H)	SA (A, S, H)	SA (A, S, H)	SA (A, S, H)
Over 30	Hard	SA (A, S, H)	SA (A, S, H)	SA (A, S, H)	SA (A, S, H)	SA (A, S, H)

General Guidelines for Selecting a Pile Hammer

Single Acting Steam and Air/Diesel Hammers

- * Dense sands and stiff clays need heavy hammers with low blow counts. This makes single acting hammers ideal for such situations.

Double Acting Steam and Air Hammers

- * Double acting hammers have light hammers compared to single acting hammers with same energy level. Light hammers with high velocity blows are ideal for medium dense sands and soft clays.

Vibratory Hammers

- * Avoid vibratory hammers for concrete and timber piles. Vibratory hammers could create cracks in concrete.
- * Avoid vibratory hammers for clayey soils. Vibratory hammers are best suited for loose to medium sands.
- * Vibratory hammers are widely used for sheet piles since it may be necessary to extract and reinstall piles. Extraction of piles can be readily done with vibratory hammers.
- * In loose to medium soil conditions, sheet piles can be installed at a much faster rate by vibratory hammers.

Hydraulic Hammers:

- * Hydraulic hammers provide an environmental friendly operation. Unfortunately rental cost is high for these hammers.

F.9 Noise Level In Pile Driving

Another important aspect of selection of pile driving method is the noise level. Following Table 6.F.3 is provided by White et. al. (2000) as a guide for noise produced during pile driving by selected methods.

Table 6.F.3 Guidelines for Noise Level during Pile Driving

Installation Method	Observed Noise Level (dB)	Distance of Observation (m)
Pressing (Jacking)	61	7
Vibratory (Med. Fre.)	90	1
Drop Hammer	98-107	7
Light Diesel Hammer	97	18

F.10 Construction of Bored Cast-In-Situ Pile/ Drilled Shaft

Bored Cast-in-situ piles/shafts are very much preferred in cities as they do not cause any disturbance to surroundings. In bored cast in-place piles, the holes are first bored with a temporary or permanent steel casing or by using bentonite slurry to stabilize the sides of the bore. Holes can also be formed by augers. A prefabricated steel cage is then lowered into the hole and concreting is carried by the tremie method. Boring holes by using bentonite mud is much more popular than using casing pipes. One of the great advantages of this method is that large diameter piles (up to 5 m in diameter) can be installed by this method. Hence, these piles are very much used for bridges and other heavy structures. However, it should be clearly remembered that bored piles smaller than 400 mm in diameter are not normally recommended for use in practice. The sides of cast in-situ small diameter piles are liable to cave in. In such cases, there will be no continuity in the length of piles. In the following Sections methods of advancing the holes, choice of tools to be used and other related topics described by Varghese (2005) are reproduced.

F.10.1 Method of Advancing the Hole for Bored Pile/Drilled Shaft

There are various methods of advancing bore holes with the circulation of bentonite. Some of these methods are now discussed.

Method 1: Piles installed by bailer and cutting tools: This is crude but the simplest method of advancing the hole when using the chisel and bailer bucket to advance the hole. The slurry is formed by simply adding the bentonite into the hole and mixing it in the hole, with the level of the suspension inside the bore hole always kept about 1 m above the ground water level, or if necessary to the top of the level of the casing. When meeting cohesionless materials, the slurry may be thickened. The aim is to help the stabilization of hole by forming an impermeable thin film around the bore hole. The bentonite suspension is assumed to penetrate into the sides under positive pressure. After a while it forms a jelly, thus making the sides impervious by producing a plastering effect. This is described as a crude method of installing bored piles because adding bentonite in the hole as in this method does not give us the full benefit of piling. The up and down movements of the bailer cause the soil from the sides and bottom to flow in. It is also very difficult to remove all the loose materials that collect at the bottom in the end. Hence as far as possible, the bailer method should not be used on important works.

Method 2: Continuous mud circulation (CMC) method (Figure 6.F.5). This method is a more refined one than the above method. In this case, bentonite of sufficient viscosity and velocity (as delivered by a mud pump) is maintained in continuous circulation so that particles are suspended in the mud and brought to the surface by the flow of

bentonite. The level of the bentonite suspension is kept constant. For this purpose, a mud pump of sufficient capacity (depending on diameter and depth of hole) is employed for continuous circulation. Material in the bore hole is loosened (spoil formed) by means of a suitably designed chopper or reamer or drilling bit. The bentonite solution is circulated by pumping. It serves the two purposes of (a) stabilizing the bore hole and (b) conveying the spoil from bottom of the hole to the top. The mud pump capacity should be able to maintain the volume and velocity to lift up the spoil from the bore hole. It will depend on the diameter and depth of the hole to be bored. The mud pump may be used in the following four different ways:

- (i) Direct mud circulation (DMC): In this method, the bentonite suspension is pumped into the bottom of the hole through the drill rods and it overflows at the top of the casing. The mud pump should have the capacity to maintain a velocity of 0.41 to 0.76 metres per second to float the cuttings.
- (ii) Reverse mud circulation (RMC): For large diameter holes, the pump is more efficient if the bentonite suspension is fed directly at the top of the hole and it is pumped out from bottom of the hole with suitable rotary pump fitted at the bottom of the drill rods. This method is called the reverse mud circulation method. Whereas borehole sizes in direct circulation are limited by the mud pump capacity, in reverse circulation method even a medium sized pump can create enough bailing velocity to bring cuttings up and the inner diameter of the drill pipe need not be large.
- (iii) Rapid direct mud circulation (RDMC): This is an improved version of the DMC where a tube carrying compressed air is also sent to the bottom of the bore. The air helps in mixing up the loosened soil with the bentonite slurry more effectively so that even heavy particles are forced out of the bore suspended in the bentonite. However, in all cases where rapid excavation of the bore is planned, the tendency of sides to cave in should be carefully examined.
- (iv) Air lift reverse mud circulation drilling (ARMC): This method of drilling is used for large diameter holes. Compressed air is used in this method to circulate the drilling fluid and cuttings to the surface.

It has also been observed in the field that with bentonite clay there is more caving in during the time there is no work than during the working period. This may perhaps be due to thickening of the bentonite into a gel when not in agitation. This gel may exert less lateral pressure than bentonite in liquid mud form. Hence concreting of holes should be planned immediately after circulation of bentonite and never in a hole, in which work was suspended overnight.

F.10.2 Limitations of Bentonite Method

The bentonite method has some limitations. A brief list of these is as follows:

- (i) Pile diameters should not be small. Normally, they should be 400 mm to 5 in in diameter.
- (ii) It will have potential danger if used in artesian conditions.
- (iii) It is difficult to use this method in soils with permeability greater than 1 in per second or in soft clays with shear strength less than 20 kN/m².
- (iv) It is difficult to clean the bottom of the hole when boring ends in coarse materials, disintegrated rocks, etc. which do not come up easily along with the suspension.
- (v) It is difficult to install raker piles by this method.
- (vi) In non-cohesive soils or fine sands, the rate of progress of work should be slow enough for the bentonite to penetrate into the soil and produce the plastering effect. The rate of progress should be suitably adjusted. Otherwise side collapse may occur.
- (vii) If subsoil or ground water contains salts, it will adversely affect the action of bentonite. Protection of sides from caving in may be found to be difficult.

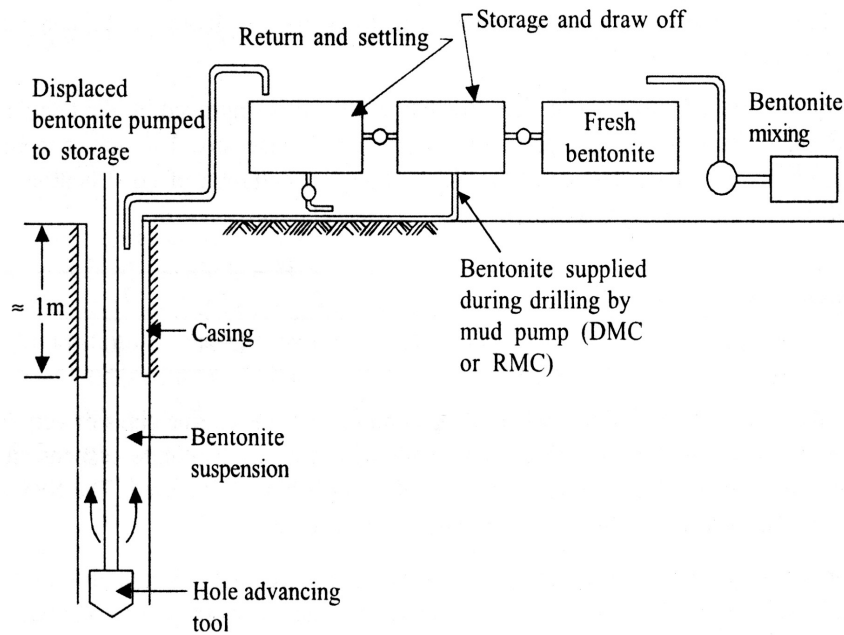


Figure 6.F.5 Schematic diagram and layout of equipment for boring using bentonite suspension (after Varghese, 2005).

F.10.3 Actions to be Taken Prior to Concreting

The following precautions are very important for success in the construction by bentonite stabilization:

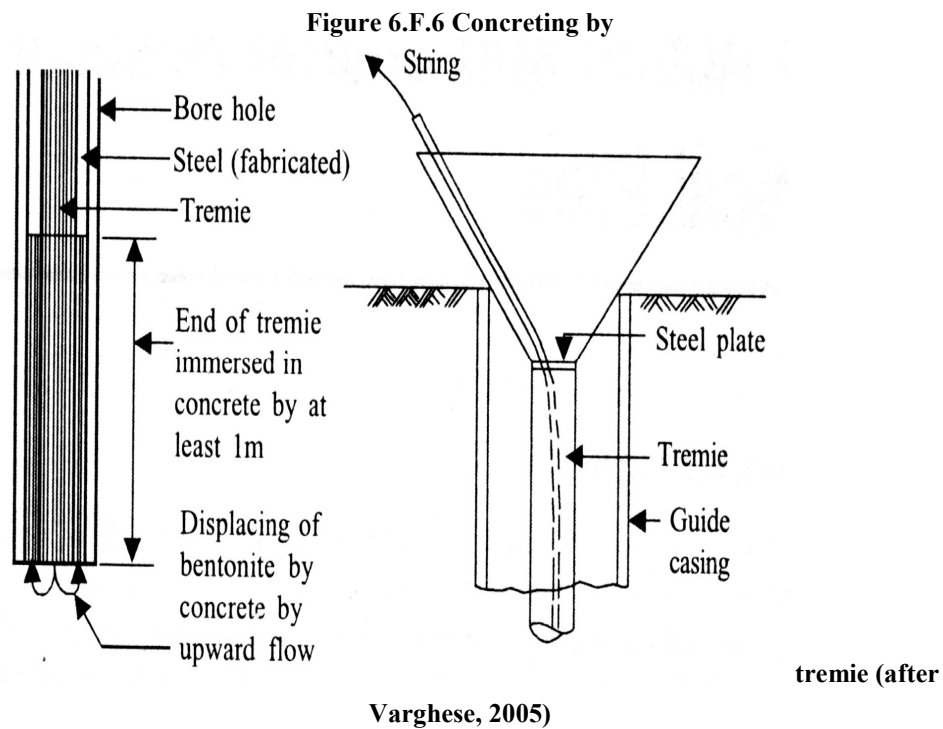
- (i) The specific gravity of bentonite should be checked at intervals by taking samples from the bottom. If it exceeds 1.25, replacement of bentonite at the bottom may be necessary without decreasing the level of bentonite in the hole. The density should be brought down to about 1.12 by flushing before concreting.
- (ii) As the tendency of caving in is more, when the bentonite is not in circulation, operations for final concreting should always start as soon as the hole is completed and cleaned.
- (iii) Before the steel cage is lowered, the hole should be flushed with fresh bentonite slurry for at least 15 minutes (in direct circulation by the mud circulation chisel resting at the bottom) so that it is completely cleared. Accumulated debris at the bottom can considerably increase the settlement when the piles are loaded. This aspect is very important in construction of bored cast in-situ piles. Many load tests (especially in bearing piles) have produced inconsistent results due to carelessness in cleaning the bottom of the hole before concreting.

F.10.4 Concreting of Piles

The precautions to be taken in the use of tremie concrete in piles. After cleaning the holes, the reinforcement cage is lowered into the hole. The bore is once again flushed and concrete poured through a tremie pipe of 200 mm in diameter. Concrete of slump 150 mm, cement content not less than 400 kg/m³, water cement ratio > 0.5, maximum size of aggregate 20 to 25 mm with suitable plasticizer is recommended for use. The procedure is as follows:

First, a guide casing, if not already provided, is placed over the hole for proper seating of the tremie funnel. The tremie is lowered to the bottom of the hole. To start with, the bottom of concreting funnel is closed with a steel plate. After filling the funnel to its full capacity the steel plate is removed and concrete discharged. The bottom of the tremie should always be at least 2 m within the concrete so that the bentonite is replaced from bottom upwards. Only the initially poured concrete is in contact with the bentonite as shown in Figure 6.F.6. Concreting is carried out to at least 60-90 cm above the cut off level. If the cut off level is at the ground level the top concrete is allowed to spill over till good concrete is visible.

When bentonite piling was introduced before its final adoption, much doubt was raised about the strength of concrete placed in bentonite (which is a suspension of clay) as well as about the bond characteristics of steel that have been coated with bentonite. However, tests have shown that placing concrete by displacing bentonite suspension from bottom (in contrast to pouring concrete into bentonite suspension) does not affect concrete strength. Similarly the bond between steel and concrete is also not very much reduced in this process. Hence the importance of properly placing concrete by tremie by displacement of bentonite from bottom up should be strictly followed in the field.



PART VI
Appendix G

Other Methods of Estimating Ultimate Axial Capacity of Piles and Drilled Shafts, and Design Charts for Settlement

G.1 Method Based on the Standard Penetration Test (Canadian Foundation Engineering Manual):

This method is based on N-values obtained from Standard Penetration Test (SPT). This method has been described in Canadian Foundation Engineering Manual published by Canadian Geotechnical Society (CGS, 1985).

The capacity of a single pile in granular soils can be estimated from the results of SPT using the following expression as suggested by Meyerhof (1976).

$$R = mNA_t + nN'DA_s \quad (6.G.1)$$

Where,

- R = pile capacity (N)
- m, n = constants depending on type of pile (driven or cast-in-situ bored piles)
- N = SPT index at the pile toe
- A_t = pile toe area
- N' = average SPT index along the pile
- D = pile embedment length
- A_s = pile unit shaft area

The Standard Penetration Test is subject to a multitude of errors, and a lot of care must be exercised when using test results. For this reason, in this method a minimum factor of safety of 4 should be applied to calculate allowable capacity of a drilled shaft or bored pile.

G.2 Method Based on the Theory of Plasticity (Canadian Foundation Engineering Manual)

In this method the capacity of a single pile may be determined from the friction angle of the soil by use of the theory of plasticity (or bearing-capacity theory).

The capacity of a pile in a soil of uniform density increases in a linear manner with increase in effective overburden pressure at least to a certain depth called the critical depth. Investigations of single piles indicate that there is very little increase in toe resistance or unit shaft resistance below the critical depth. The ratio of the critical depth to the pile diameter increases with increase in the angle of shearing resistance. For most applications, the ratio ranges between a value of 7 at $\phi' = 30^\circ$ to a value of 22 at $\phi' = 45^\circ$.

The ultimate static resistance, R of a single pile is a function of the sum of the toe and shaft resistance, R_t and R_s , as follows:

$$R = R_t + R_s \quad (6.G.2)$$

Where for toe resistance:

$$R_t = A_t r_t = A_t \sigma'_D N_t \quad (6.G.3)$$

Where,

A_t = cross-sectional area of pile at toe

r_t = unit toe resistance = $\sigma'_D N_t$

σ'_D = unit effective vertical stress at the pile toe = $\gamma'D$ (below the critical depth, $D = D_c$)

γ' = submerged unit weight of soil

D = embedment length of the pile in soil

N_t = bearing-capacity coefficient as recommended by Canadian Foundation Engineering Manual

The expression for shaft resistance is as follows:

$$R_s = \sum_{Z=0}^D A_s r_s = \sum_{Z=0}^D A_s \beta \sigma'_z = \sum_{Z=0}^D A_s M K_s \tan \phi' \sigma'_z \quad (6.G.4)$$

Where,

A_s = shaft area per unit length of pile

r_s = unit shaft resistance along the pile

σ'_z = effective vertical stress at depth z (below the critical depth, $z = D_c$, use σ'_z)

β = shaft resistance coefficient = $K_s M \tan \phi'$

K_s = ratio between the horizontal effective soil stress to the vertical effective soil stress at the pile shaft

$M = \tan \delta' / \tan \phi'$

$\tan \phi'$ = soil friction

$\tan \delta'$ = soil-pile friction

The value of K_s is influenced by the angle of shearing resistance, the method of installation, the compressibility and original state of stress in the ground, and the size and shape of the pile. It increases with the in-situ density and angle of shearing resistance of the soil and with the amount of displacement. It is higher for displacement-type piles than for low-displacement-type piles such as H-piles. For bored piles, the value of K_s is usually assumed equal to the coefficient of earth pressure at rest, K_0 . For driven displacement-type piles, the value of K_s is normally assumed to be twice the value of K_0 .

The value of M ranges from 0.7 to 1.0, depending on the pile material (steel, concrete, wood) and method of installation (Bozozuk et al., 1978). The combined shaft resistance coefficient, β , is generally assumed to range from 0.3 to 0.8, where the lower value is used in clay and silt, and the higher value in coarse and dense soils (Burland, 1973).

Terzaghi and Peck (1967) reported typical values of angle of internal friction for different types of sands which are shown in Table 6.G.1. In the absence of test loading, this method recommends a factor of safety of at least 3 in order to calculate the allowable capacity.

Table 6.G.1: Typical Values of Angle of Internal Friction for Different Types of Sand (after Terzaghi and Peck, 1967)

Type of Sand	Angle of Internal Friction, ϕ'	
	Loose	Dense
Uniform sand, rounded particles	27°	35°
Well graded sand, angular particles	33°	45°
Sandy gravels	35°	50°
Silty sands	27° to 30°	30° to 34°
Inorganic silts	27° to 30°	30° to 35°

G.3 Tomlinson Method

This method of estimating ultimate axial load capacity of a single pile has been described by Tomlinson (1995). In this method, the design ultimate unit skin friction of an individual pile is given by the following expression:

$$q_z = K_s \sigma'_{vo} \tan \delta \quad (6.G.5)$$

Where, K_s = coefficient of horizontal soil stress,

σ'_{vo} = average effective overburden pressure over the length of the soil layer

δ = angle of wall friction

The value of coefficient K_s is related to the coefficient of earth pressure at rest (K_o) and also to the method of installation of the piles. Values of coefficient of horizontal soil stress (K_s) are shown in Table 6.G.2 while values of the angle of pile to soil friction (δ) for various interface conditions are shown in Table 6.G.3.

Table 6.G.2: Values of Coefficient of Horizontal Soil Stress, K_s (after Kulhawy, 1984)

Installation method	K_s/K_o
Driven piles, large displacement	1-2
Driven piles, small displacement	0.75-1.75
Bored and cast-in-place piles	0.71-1.0
Jetted piles	0.5-0.7

Table 6.G.3: Values of Soil Pile Friction Angle (δ) for Various Interface Conditions (after Kulhawy, 1984)

Pile/soil interface condition	Angle of pile to soil friction (δ)
Smooth (coated) steel/sand	$0.5\phi'$ to $0.7\phi'$
Rough (corrugated) steel/sand	$0.7\phi'$ to $0.9\phi'$
Precast concrete/sand	$0.8\phi'$ to $1.0\phi'$
Cast-in-place concrete/sand	$1.0\phi'$
Timber/sand	$0.8\phi'$ to $0.9\phi'$

The equation for estimating ultimate skin friction implies that in a uniform cohesionless soil the unit skin friction continues to increase linearly with increasing depth. This is not the case. Vesic (1970) showed that at some penetration depth between 10 and 20 pile diameters, a peak value of unit skin friction is reached which is not exceeded at greater penetration depths. Research has not yet established whether the peak value is a constant in all conditions, or is related to factors such as soil grain size or angularity. A peak value of 110 kN/m² has been recommended by Tomlinson (1995) for straight-sided piles.

The base resistance is obtained from the following equation:

$$Q_b = q_b A_b = N_q \sigma'_{vo} A_b \quad (6.G.6)$$

Where, N_q = bearing capacity coefficient

σ'_{vo} = average effective overburden pressure over the length of the soil layer

A_b = base area of pile

Comparisons of observed base resistances of piles by Nordlund (1963) and Vesic (1964) have shown that N_q values established by Berezantsev (1961) which take into account the depth to width ratio of the pile most nearly conform to practical criteria of pile failure.

G.4 Design Charts for Estimation of Settlement for Drilled Shafts

Reese and O'Neill (1988) developed the following charts, (Figures 6.G.1 to 6.G.4), to estimate the settlement of drilled shaft under service loads. These charts express the settlement in terms of the ratio of the mobilized resistance to the actual resistance. If the computed settlement is too large, these charts may be used to modify the design accordingly. It is important to mention that the notation used in these charts, in several instances, differ from that mentioned in the Code. They are indicated as under and should be carefully considered.

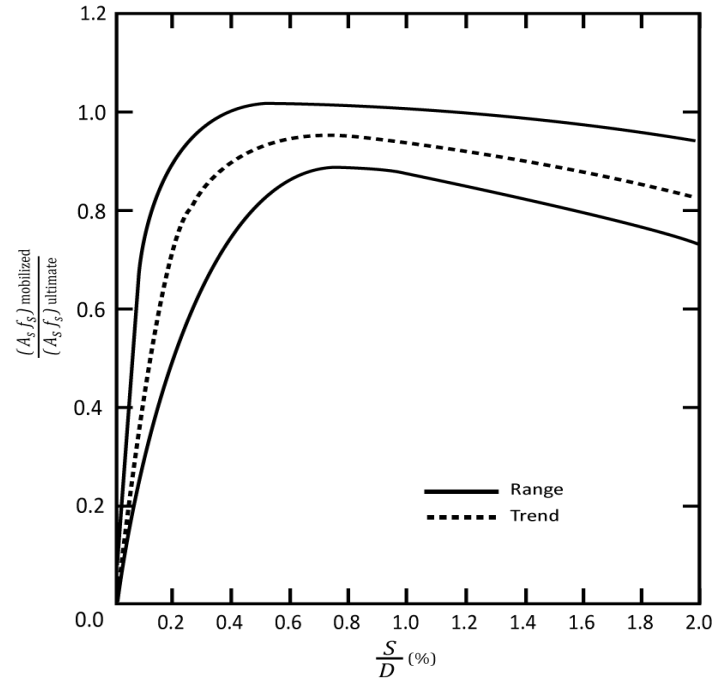


Figure 6.G.1 Normalised Curves for Load Transfer in Skin Friction vs Settlement for Drilled Shafts in Cohesive Soils (after Reese and O'Neill, 1988)

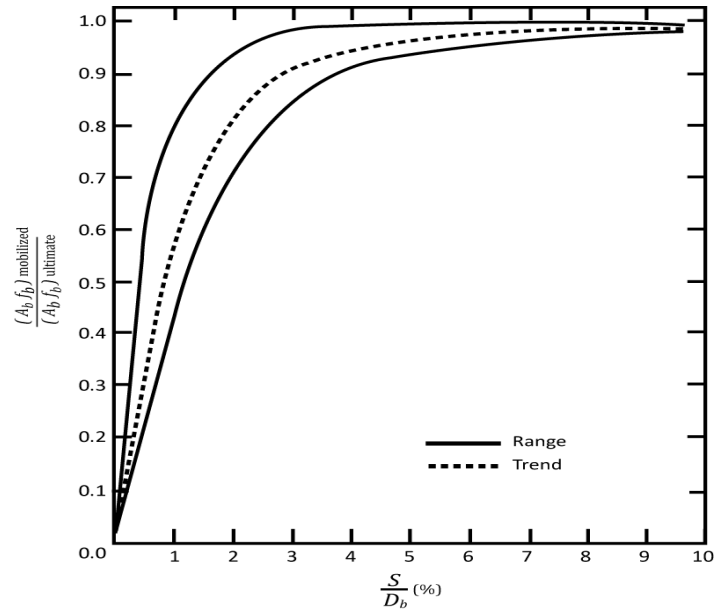


Figure 6.G.2 Normalised Curves for Load Transfer in End Bearing vs Settlement for Drilled Shafts in Cohesive Soils (after Reese and O'Neill, 1988)

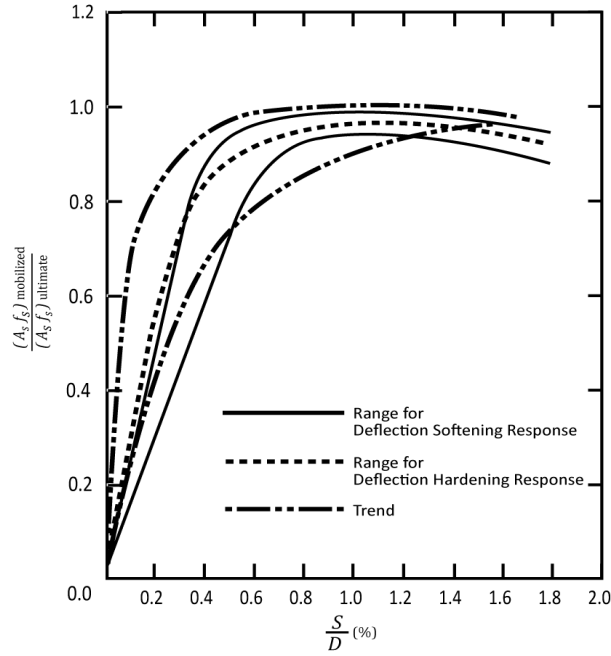


Figure 6.G.3 Normalised Curves for Load Transfer in Skin Friction vs Settlement for Drilled Shafts in Cohesionless Soils (after Reese and O'Neill, 1988)

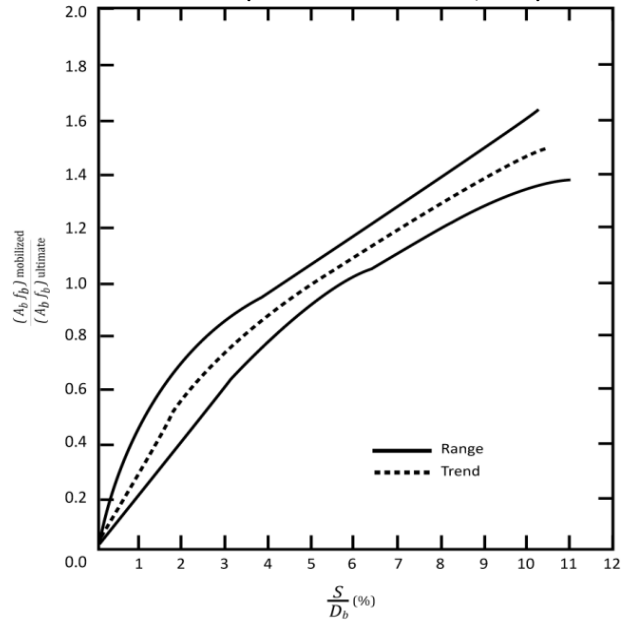


Figure 6.G.4 Normalised Curves for Load Transfer in End Bearing vs Settlement for Drilled Shafts in Cohesionless Soils (after Reese and O'Neill, 1988)

**PART VI
Appendix H****References of Chapter 3 Part 6 (Soils and Foundations)**

ASTM D1143-07 (2012), Standard Test Method for Deep Foundations Under Static Axial Compressive Load, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 108-122.

ASTM D1557-09 (2012), Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 143-173.

ASTM D1586-11 (2012), Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 161-167.

ASTM D2487-11 (2012), Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System), Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 268-279.

ASTM D2573-08 (2012), Standard Test Method for Field Vane Shear Test in Cohesive Soil, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 291-298.

ASTM D2974-07a (2012), Standard Test Methods for Moisture, Ash, and Organic Matter of Peat and Other Organic Soils, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 339-342.

ASTM D3441-05 (2012), Standard Test Method for Mechanical Cone Penetration Tests of Soil, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 395-400.

ASTM D3689-07 (2012), Standard Test Method for Deep Foundations Under Static Axial Tensile Load, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 408-420.

ASTM D3966-07 (2012), Standard Test Method for Deep Foundations Lateral Load, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 434-451.

ASTM D4318-10 (2012), Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 592-607.

ASTM D4380-12 (2012), Standard Test Method for Density of Bentonitic Slurries, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 617-618.

ASTM D4428-07 (2012), Standard Test Methods for Crosshole Seismic Testing, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 660-670.

ASTM D4719-07 (2012), Standard Test Methods for Prebored Pressuremeter Testing in Soils, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 916-925.

ASTM D4972-07 (2012), Standard Test Method for pH of Soils, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 1024-2026.

ASTM D5777-00 (2012), Standard Guide for Using the Seismic Refraction Method for Subsurface Investigation, Annual Book of ASTM Standards, Vol. 04.08, ASTM International, pp. 1572-1585.

ASTM D5778-12 (2012), Standard Test Method for Electronic Friction Cone and Piezocone Penetration Testing of Soils, Annual Book of ASTM Standards, Vol. 04.09, ASTM International, pp. 1586-1604.

ASTM D5882-07 (2012), Standard Test Method for Low Strain Impact Integrity Testing of Deep Foundations, Annual Book of ASTM Standards, Vol. 04.09, ASTM International, pp. 69-74.

ASTM D6066-11 (2012), Standard Practice for Determining the Normalized Penetration Resistance of Sands for Evaluation of Liquefaction Potential, Annual Book of ASTM Standards, Vol. 04.09, ASTM International, pp. 328-343.

ASTM D6910-09 (2012), Standard Test Method for Marsh Funnel Viscosity of Clay Construction Slurries, Annual Book of ASTM Standards, Vol. 04.09, ASTM International, pp. 1049-1051.

ASTM G57-06 (2012), Standard Test Method for Field Measurement of Soil Resistivity Using the Wenner Four-Electrode Method, Annual Book of ASTM Standards, Vol. 03.02, ASTM International, pp. 223-227.

BDS 819:1975 (1975), Code of Practice for Preservation of Timber, Bangladesh Standards and Testing Institution.

Berezantzev, V.G., Kristofornov, V. and Golubkov, V. (1961), Load Bearing Capacity and Deformation of Pile Foundation, Proceedings of the 5th International Conference on Soil Mechanics, Paris, Vol. 2, pp.11-12.

Borden, R. H., and Gabr, M.A. (1987), Analysis of Compact Pole-Type Footing-LT Base: Computer Program for Laterally Loaded Pier Analysis Including Base and Slope Effect, Raleigh, N.C., North Carolina Department of Transportation, USA.

Bowles, J.E. (1988), Foundation Analysis and Design, 4th Edition, McGraw Hill Book Company, Singapore.

Bozozuk, M., Fellenius, B.H. and Samson, L., (1978), Soil disturbance from Pile Driving in Sensitive Clay, Canadian Geotechnical Journal, Vol. 15, No. 3, pp. 346-361.

Brinch-Hansen, J. (1963), Hyperbolic Stress Strain Response: Cohesive Soils, Discussion, JSMFED, ASCE, Vol. 89, No. SM4, pp. 241-242.

BS 8004: 1986 (1986), Code of Practice for Foundations, British Standards Institution (current version: BS EN 1997-1:2004, Euro Code 7 Geotechnical Design General Rules).

Burland, J.B. (1973), "Shaft Friction of Piles in Clay : A simple Fundamental Approach", Ground Engineering, Foundation Publications Ltd., London, Vol. 6, No. 3, pp. 30-42.

Butler, H.D. and Hoy, H.E. (1977), User's Manual for the Texas Quick Load Method for Foundation Load Testing, FHWA-IP-77-8, Federal Highway Administration, Office of Development, Washington, pp.59 .

CGS (1985), Canadian Foundation Engineering Manual, 2nd Edition, Canadian Geotechnical Society, Toronto

Chellis, R.R. (1961), Pile Foundation, 2nd Edition, McGraw Hill, New York.

Chen, F.H. (1975), Foundation on Expansive Soils, Developments in Geotechnical Engineering Vol. 12, 1st Edition, Elsevier Science Publishers B.V., Netherlands.

Clayton, C.R.I., Simons, N.E. and Matthews, M.C. (1982), Site Investigation, Granada Publishing Limited, London.

Coduto, D.P. (1994), Foundation Design Principles and Practice, Prentice Hall, Englewood Cliff, New Jersey, USA, pp. 322.

Davisson, M.T. (1973), High Capacity Piles: Proc. Of Lecture Series on Innovation in Foundation Construction, Soil Mechanics Division, Illinois Section, ASCE, Department of Civil Engineering, Illinois Institute of Technology, Chicago, IL, 1973.

Douglas, B.J. and Olsen, R.S. (1981), Soil Classification using the Electric Cone Penetrometer Test, Cone Penetration Testing and Experience, ASCE Fall Convention, 1981.

Douglas, B.J. (1984), The Electric Cone Penetrometer Test: A User's Guide to Contracting for Services, Quality Assurance, Data Analysis, The Earth Technology Corporation, Long Beach, CA, USA.

- Edil, T.B. (1997), Construction over Peats and Organic Soils, Proc. Conf. On Recent Advances in Soft Soil Engineering, Kuching, Sarawak, Malaysia, March, 1997, pp. 85-108.
- Finn, W.D.L., Ledbetter, R.H., and Wu, G. (1994), Liquefaction in Silty Soils: Design and Analysis, Ground Failures under Seismic Conditions, Geotechnical Special Publication, ASCE, 1994, pp. 51-76.
- Fuller, F.M. (1983), Engineering of Pile Installation, McGraw Hill Book Co., New York, USA, pp. 286.
- Gibbs, H.J. and Holtz, W.G. (1957), Research on Determining the Density of Sands by Spoon Penetration Testing, Proc. 4th ICSMFE, Vol. 1, London, UK, pp. 35-39.
- Hannigan, P. J., Goble, G.G., Thendean, G., Likens, G.E. and Rauche, F. (1998), Design and Construction of Driven Pile Foundations, Vols. I & II, FHWA H 97-013 & FHWA H 97-014, DOT, USA.
- Hossain, M. M. (1983), Swelling Properties of Selected Local Soils, M. Sc. Engineering thesis, Department of Civil Engineering, Bangladesh University of Engineering and Technology, Dhaka, Bangladesh.
- Hvorslev, M.J. (1949), Subsurface Exploration and Sampling of Soils for Civil Engineering Purposes, Waterways Experimental Station, Vicksburg, Mississippi.
- ISSMFE (1965), Report of the Subcommittee on Problems and Practices of Soil Sampling, Proc., 6th ICSMFE, Montreal, Vol. 3, Appendix II, pp. 64-71.
- IS: 1948 (1972), Classification and Identification of Soils for General Engineering Purposes, Bureau of Indian Standards, New Delhi.
- IS: 2911, Part 3 (1980), Code of Practice for Design and Construction of Pile Foundations: Under reamed Piles (First Revision), Bureau of Indian Standards, New Delhi.
- IS: 2911 – Part 1 (1979), Design and Construction of Pile Foundations- Driven Cast In Situ Concrete Piles, Bureau of Indian Standards, 1979.
- IS: 2911 – Part 4 (1979), Load Test on Piles, Bureau of Indian Standards, 1979.
- IS: 2974-Part 1 (1982), Foundations for Reciprocating Type Machines, Bureau of Indian Standards, 1982.
- Kallstenius, T. (1958), Mechanical Disturbances in Clay Samples Taken With Piston Samplers, Proc., Royal Swedish Geotechnical Institute, No. 16, pp. 1-75.

Klingmuller, O. (1993), *Dynamische Integritätsprüfung und Qualitätssicherung bei Bohrpfehlen*, Geotechnik 16, Verlag Gluckauf (in German).

Kulhawy, F.H. (1984), Limited tip and side Resistance : Fact or Fallacy, *Proceedings of the Symposium on Analysis and Design of Pile Foundations*, ASCE, San Francisco, pp. 80-98.

Kulhawy, F.H. and Jackson, C.S. (1989), Some Observations on Undrained Side Resistance of Drilled Shafts, *Foundation Engineering: Current Principles and Practices*, pp. 1011-1014, ASCE.

Kulhawy, F.H. and Mayne, P.W.. (1990), *Manual on Estimating Soil Properties for Foundation Design*, Report No. EL-6800, Electric Power Research Institute, Palo Alto, CA, USA.

Look, B.G. (2007), *Handbook of Geotechnical Investigation and Design Tables*, First Edition, Taylor & Francis, London, pp. 81.

Meyerhof, G.G. (1976), Bearing capacity and Settlement of Pile Foundations, The Eleventh Terzaghi Lecture, *Journal of Geotechnical Engineering Division*, ASCE, Vol. 102, GT3, pp. 195-228.

NBCI (2005), *National Building Code of India*, Bureau of Indian Standards, New Delhi.

Nordlund, R.L. (1963), Bearing Capacity of Piles in Cohesionless Soils, *Journal of Soil Mechanics and Foundation Division*, ASCE, Vol. 89, No. SM3, pp. 1-35.

Peck, R.B., Hanson, W.E. and Thornburn, T.H. (1974), *Foundation Engineering*, 2nd Edition, John Wiley & Sons, Inc., New York, USA.

Poulos, H. G. and Davis, E.H. (1980), *Pile Foundation Analysis and Design*, John Wiley and Sons, NY, USA.

Reese, L. C., Cooley, L. A., and Radhakrishnan, N. (1984), *Laterally Loaded Piles and Computer Program COM624G*, Technical Report K-84-2, U.S. Army Engineer Division, Lower Mississippi Valley, Vicksburg, MS, 1984.

Reese, L.C. and O'Neill, M.W. (1988), *Drilled Shafts: Construction Procedures and Design Methods*, Report No. FHWA-HI-88-042, Federal Highway Administration, USA.

Richart, F.E. Jr. (1962), *Foundation Vibrations*, Trans ASCE, Vol. 127, pp. 863-898, 1962.

Robertson, P.K. and Campanell, R.G. (1983a), Interpretation of Cone Penetration Tests, Part- Sand, CBJ, Ottawa, Vol. 20, No. 4.

- Robertson, P.K. and Campanell, R.G. (1983b), SPC-CPT Correlations, JSMFED, ASCE, Vol. 109.
- Salgado, R. (2011), The Engineering of Foundations, Tata McGraw Hill Edition, 2011, Tata McGraw Hill Education Private Limited, New Delhi.
- Sanglerat, G. (1972), The Penetrometer and Soil Exploration, , Elsevier Publishing Co., Amsterdam, Netherlands.
- Seed, H.B., Woodward, R.J. and Lundgren, R. (1962), Prediction of Swelling Pressure for Compacted Clays, JSMFED, ASCE, Vol. 88.
- Sherard, J.L, Dunnigan, L.P., Decker, R.S. and Steele, E.F.(1976), Pinehole Test for Identifying Dispersive Soils, JGED, ASCE, Vol. 102 (GT 1), pp. 69-85.
- Terzaghi, K. (1942), Discussion of the Progress Report of the Committee on the Bearing Value of Pile Foundations, Proc. ASCE, Vol. 68, pp. 311-323.
- Terzaghi, K. and Peck, R.B. (1967), Soil Mechanics in Engineering Practice, 2nd Edition, John Wiley, New York, USA.
- Tomlinson, M.J. (1995), Pile Design and Construction Practice, E & FN Spon, London.
- Tomlinson, M.J. (1995), "Foundation Design and Construction", Sixth Edition, Wesley Longman Singapore Publishers (Private) Limited, Singapore.
- Varghese, P.C. (2005), Foundation Engineering, Prentice – Hall of India, New Delhi.
- Vesic, A.S. (1964), Investigations on Bearing Capacity of Piles in Sand, Duke University Soil Mechanics Laboratory Publication No. 3.
- Vesic, A.S. (1970), Tests on Instrumented Piles, Ogeechee River Site, JSMFED, ASCE, Vol. 96, No. SM2, pp. 561-584.
- Whitaker, T. (1963), The Constant Rate of Penetration Test for the Determination of the Ultimate Bearing Capacity of a Pile, Proceedings, Institution of Civil Engineers, Vol. 26, London, UK, 1963, pp. 119-123
- Whitaker, T. (1976), Design of Piled Foundation, 2nd Edition, Pergamon Press, Oxford, UK.
- White, D.J., Sidhu, H.K., Finlay, T.C.R., Bolton, M. D. and Nagayama, T. (2000), Press in Piling: The Influence of Plugging on Driveability, 8th International Conference of the Deep Foundations Institute, New York. pp 299-310.

PART VI
Appendix I
Strut-and-Tie Models

I.1 Scope and Definitions

Scope: Strut-and-Tie model is elaborated in this Appendix. Sec. I.1 introduces the basic definitions. The design procedure by Strut-and-Tie model is described in Sec. I.2. Strength of struts, ties and nodal zones are given in Sections I.3, I.4 and I.5 respectively. Definitions and clauses in the above sections are followed by clarifications as necessary.

B-region: A region of a member where the plane sections assumption of flexure theory from Sec 6.3.2.2 can be applied. In general, B-region is any portion of a member outside of a D-region.

Discontinuity: An abrupt change in loading or geometry.

Clarification for Discontinuity: It is assumed that discontinuities extend a distance h from the section where the load or change in geometry occurs. Figure 6.I.1(a) shows typical geometric discontinuities, and Figure 6.I.1(b) shows combined geometrical and loading discontinuities.

D-region: The region of a member within a distance, h , from a force discontinuity or a geometric discontinuity.

Clarification for D-region: Typical D-regions are the shaded regions in Figure 6.I.1(a) and Figure 6.I.1(b). The plane sections assumption of Sec 6.3.2.2 is not applicable in such regions. Each shear span of the beam in Figure 6.I.2(a) is a D-region.

When two D-regions overlap or meet as shown in Figure 6.I.2(b), they can be considered as a single D-region for design purposes. The maximum length-to-depth ratio of such a D-region would be approximately 2. Thus, the smallest angle between the strut and the tie in a D-region is $\arctan \frac{1}{2} = 26.5^\circ$, rounded to 25° .

When there is a B-region between the D-regions in a shear span, as shown in Figure 6.I.2(c), the strength of the shear span is governed by the strength of the B-region if the B- and D-regions have similar geometry and reinforcement, as because the shear strength of a B-region is less than the shear strength of a comparable D-region. Shear spans containing B-regions-the usual case in beam design-are designed for shear using the traditional shear design procedures from Sections 6.4.1 to 6.4.4 ignoring D-regions.

Deep Beam: See Sections 6.3.7.1 and 6.4.6.1.

Clarification for Deep Beam: See Figures 6.I.2(a), 6.I.2(b), and 6.I.3, and Sections 6.3.7 and 6.4.6.

Nodal Zone: The volume of concrete around a node assumed to transfer strut-and-tie forces through the node.

Clarification for Nodal Zone: Hydrostatic nodal zones as shown in Figure 6.I.4 were used traditionally. These were largely superseded by what are called extended nodal zones, shown in Figure 6.I.5.

A hydrostatic nodal zone has equal stresses on the loaded faces which are perpendicular to the axes of the struts and ties acting on the node. A C-C-C nodal zone is shown in Figure 6.I.4(a). If the stresses on the face of the nodal zone are the same in all three struts, the ratios of the lengths of the sides of the nodal zone, w_{n1} : w_{n2} : w_{n3} are in the same proportions as the three forces C_1 : C_2 : C_3 . The faces of a hydrostatic nodal zone are perpendicular to the axes of the struts and ties acting on the nodal zone.

As the in-plane stresses are the same in all directions, these nodal zones are called hydrostatic nodal zones. This terminology, strictly speaking, is incorrect because the in-plane stresses are not equal to the out-of-plane stresses.

A C-C-T nodal zone can be represented as a hydrostatic nodal zone if the tie is assumed to extend through the node to be anchored by a plate on the far side of the node, as shown in Figure 6.I.4(b), provided that the size of the plate results in bearing stresses that are equal to the stresses in the struts. The bearing plate on the left side of Figure 6.I.4(b) is used to represent an actual tie anchorage. The tie force can be anchored by a plate, or through development of straight or hooked bars, as shown in Figure 6.I.4(c).

Portion of a member bounded by the intersection of the effective strut width, w_s , and the effective tie width, w_t (see Sec I.4.2) is an extended nodal zone. The shaded areas in Figures 6.I.5(a) and (b) are extended nodal zones.

The reaction R equilibrates the vertical components of the forces C_1 and C_2 in the nodal zone shown in Figure 6.I.6(a). Frequently, calculations are easier if the reaction R is divided into R_1 , which equilibrates the vertical component of C_1 and R_2 , which equilibrates the vertical component of the force C_2 , as shown in Figure 6.I.6(b).

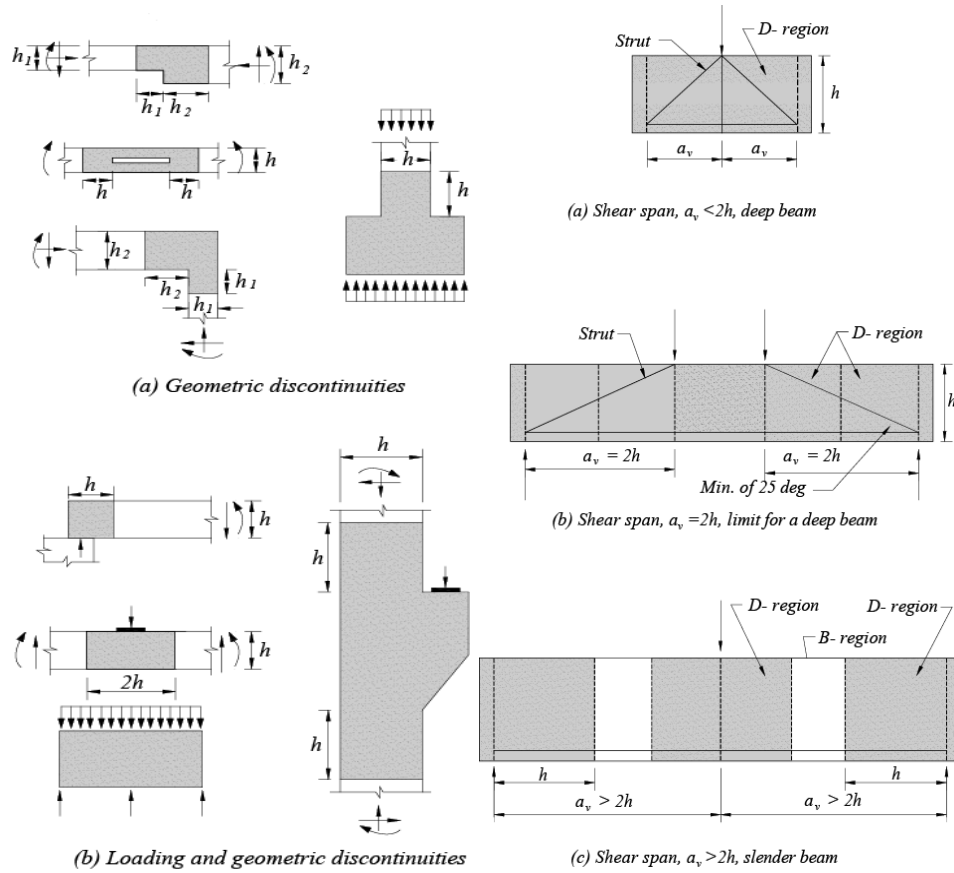


Figure 6.1.1 D-Regions and discontinuities

Figure 6.1.2 Description of deep and slender beams

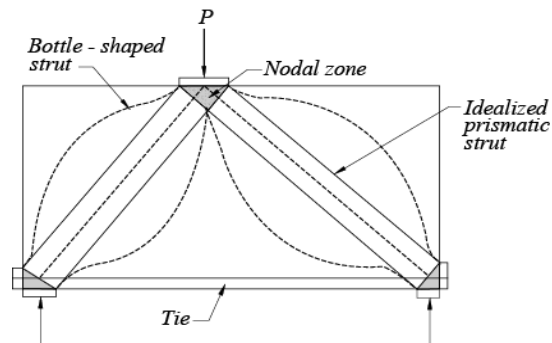


Figure 6.1.3 Description of strut-and-tie model

Note: The point in a joint in a strut-and-tie model where the axes of the struts, ties, and concentrated forces acting on the joint intersect.

Clarification for Node: To be in equilibrium, at least three forces should act on a node in a strut-and-tie model, as shown in Figure 6.I.7. Classification of nodes is based on signs of these forces. A C-C-C node resists three compressive forces, a C-C-T node resists two compressive forces and one tensile force, and so on.

Strut: A member that is in compression in a strut-and-tie model. A strut represents the resultant of a parallel or a fan-shaped compression field.

Clarification for Strut: Struts are usually idealized in design as prismatic compression members, as shown by the straight line outlines of the struts in Figures 6.I.2 and 6.I.3. The strut is idealized as a uniformly tapered compression member, if the effective compression strength f_{ce} differs at the two ends of a strut, due either to different nodal zone strengths at the two ends, or to different bearing lengths.

Bottle-shaped Strut: A strut that is wider at mid-length than at its ends.

Clarification for Bottle-shaped Struts: A strut located in a part of a member where the width of the compressed concrete at mid-length of the strut can spread laterally is a bottle-shaped strut. The curved dashed outlines of the struts in Figure 6.I.3 and the curved solid outlines in Figure 6.I.8 approximate the boundaries of bottle-shaped struts. A split cylinder test is an example of a bottle-shaped strut. The internal lateral spread of the applied compression force in such a test leads to a transverse tension that splits the specimen.

Bottle-shaped struts are idealized either as prismatic or as uniformly tapered in design, and crack-control reinforcement from Sec I.3.3 is provided to resist the transverse tension. The amount of confining transverse reinforcement can be computed using the strut-and-tie model shown in Figure 6.I.8(b) with the struts that represent the spread of the compression force acting at a slope of 1:2 to the axis of the applied compressive force. Alternatively for f_c' not exceeding 40 MPa, Eq. (I.4) can be used. The cross-sectional area A_c of a bottle-shaped strut is taken as the smaller of the cross-sectional areas at the two ends of the strut. See Figure 6.I.8(a).

Strut-and-tie Model: A truss model of a structural member, or of a D-region in such a member, made up of struts and ties connected at nodes, capable of transferring the factored loads to the supports or to adjacent B-regions.

Clarification for Strut-and-tie Model: In Figure 6.I.3, the components of a strut-and-tie model of a single-span deep beam loaded with a concentrated load are identified. The thickness and width, both perpendicular to the axis of the strut or tie are designated as the cross-sectional dimensions of a strut or tie. Thickness is perpendicular to the plane of the truss model, and width is in the plane of the truss model.

Tie: A member that is in tension in a strut-and-tie model.

Clarification for Tie: A tie is a member consisting of reinforcement or pre-stressing steel plus a portion of the surrounding concrete that is concentric with the axis of the tie. The surrounding concrete is included to define the zone in which the forces in the struts and ties are to be anchored. The concrete in a tie is not used to resist the axial force in the tie. Although not considered in design, the surrounding concrete will reduce the elongations of the tie, especially at service loads.

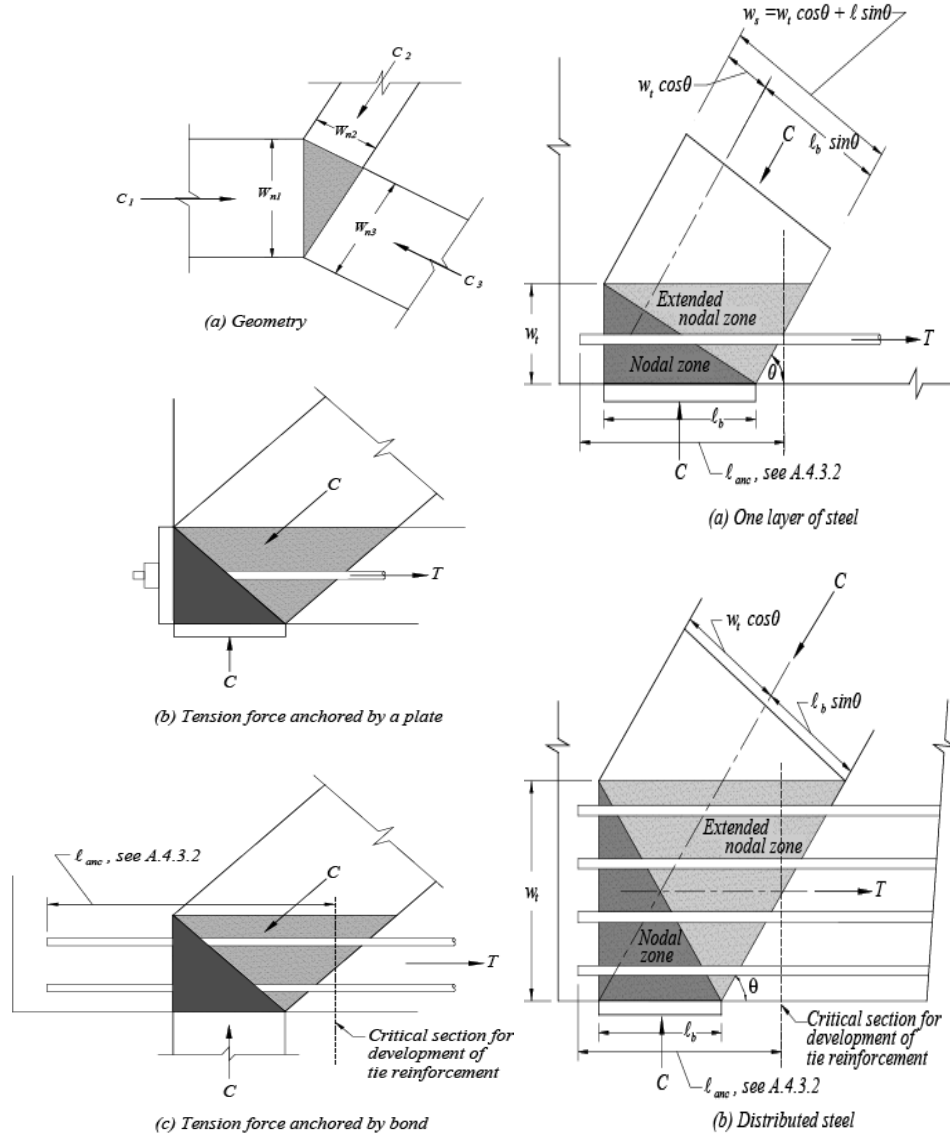


Figure 6.I.4 Hydrostatic nodes

Figure 6.I.5 Extended nodal zone showing the effect of the distribution of the force

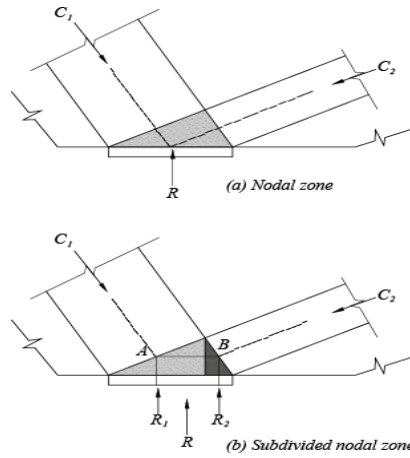


Figure 6.I.6 Subdivision of nodal zone

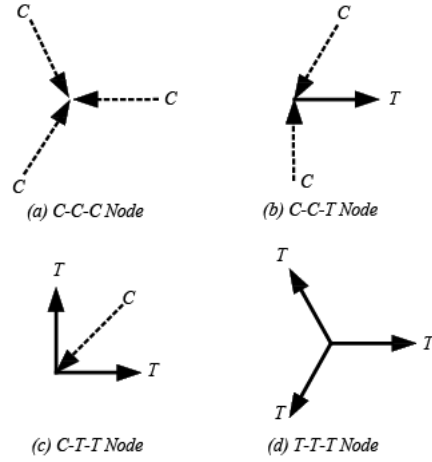


Figure 6.I.7 Classification of nodes

I.2 Design Procedure for Strut-and-Tie Model

I.2.1 It shall be allowed to design structural concrete members, or D-regions in such members, by modeling the member or region as an idealized truss. The truss model shall contain struts, ties, and nodes as defined in Sec I.1. The truss model shall be capable of transferring all factored loads to the supports or adjacent B-regions.

Clarification for Section I.2.1: The truss model described in Sec I.2.1 is what is referred to as a strut-and-tie model. Details of the use of strut-and-tie models are available in References I.1 to I.7 of Sec I.6. The design of a D-region includes the following four steps:

- First, define and isolate each D-region;
- Then, compute resultant forces on each D-region boundary;
- Then, a truss model is to be selected to transfer the resultant forces across the D-region. The axes of the struts and ties, respectively, are chosen to approximately coincide with the axes of the compression and tension fields. The forces in the struts and ties are computed.
- Finally, the effective widths of the struts and nodal zones are determined considering the forces from Step 3 and the effective concrete strengths defined in Sections I.3.2 and I.5.2, and reinforcement is provided for the ties considering the steel strengths defined in Sec I.4.1. The reinforcement should be anchored in the nodal zones.

Code requirements for serviceability should be satisfied for strut-and-tie models representing strength limit states. Deflections of deep beams or similar members can be estimated using an elastic analysis to analyze the strut-and-tie model. In addition, the crack widths in a tie can be controlled using Sec 6.3.6.4, assuming the tie is encased in a prism of concrete corresponding to the area of tie from Sec I.4.2.

I.2.2 Under the applied loads and the reactions, the strut-and-tie model shall be in equilibrium.

I.2.3 The dimensions of the struts, ties, and nodal zones shall be taken into account in determining the geometry of the truss,

Clarification for Section I.2.3: The components of the strut-and-tie model, i.e. the struts, ties, and nodal zones, all have finite widths that should be taken into account in selecting the dimensions of the truss. Figure 6.I.9(a) shows a node and the corresponding nodal zone. The vertical and horizontal forces equilibrate the force in the inclined strut. If the stresses are equal in all three struts, a hydrostatic nodal zone can be used and the widths of the struts will be in proportion to the forces in the struts.

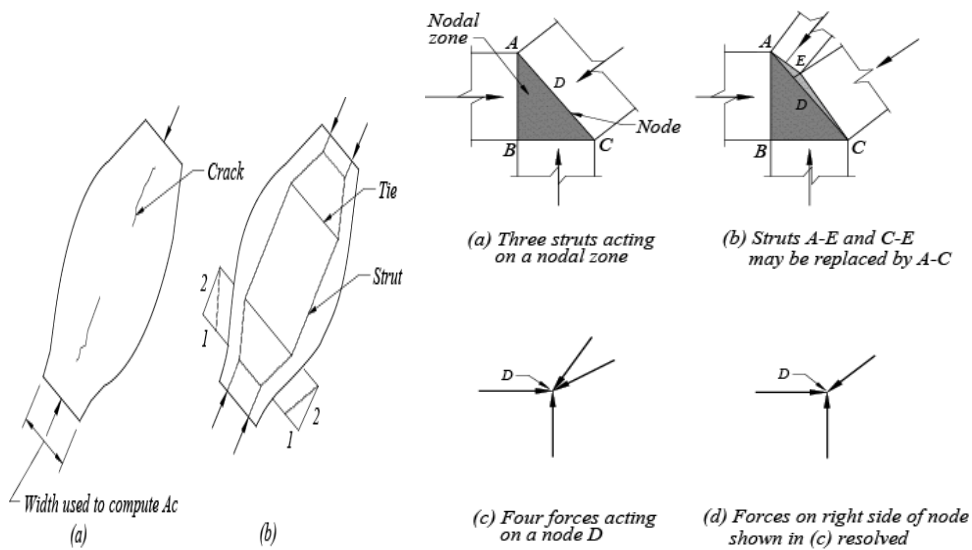


Figure 6.I.8 Bottle-shaped strut:
(a) cracking of a bottle-shaped strut; and (b) strut-and-tie model of a bottle-shaped strut

Figure 6.I.9 Resolution of forces on a nodal zone

When more than three forces act on a nodal zone in a two-dimensional structure, as shown in Figure 6.I.9(b), it is generally necessary to resolve some of the forces to end up with three intersecting forces. The strut forces acting on Faces A-E and C-E in Figure 6.I.9(b) can be replaced with one force acting on Face A-C. This force passes through the node at D.

Alternatively, the strut-and-tie model could be analyzed assuming all the strut forces acted through the node at D, as shown in Figure 6.I.9(c). In this case, the forces in the two struts on the right side of Node D can be resolved into a single force acting through Point D, as shown in Figure 6.I.9(d).

Transverse reinforcement may be required to restrain vertical splitting in the plane of the node when the width of the support in the direction perpendicular to the member is less than the width of the member. This can be modeled using a transverse strut-and-tie model.

I.2.4 Ties shall be permitted to cross struts. Struts shall cross or overlap only at nodes.

I.2.5 The angle, θ , between the axes of any strut and any tie entering a single node shall not be taken as less than 25° .

Clarification for Section I.2.5: In order to mitigate cracking and to avoid incompatibilities due to shortening of the struts and lengthening of the ties occurring in almost the same directions, the angle between the axes of struts and ties acting on a node should be large enough. This limitation on the angle prevents modeling the shear spans in slender beams using struts inclined at less than 25° from the longitudinal steel (Reference I.6 of Sec I.6).

I.2.6 Struts, ties, and nodal zones shall be designed based on

$$\phi F_n \geq F_u \quad (6.I.1)$$

Where, F_u is the factored force acting in a strut, in a tie, or on one face of a nodal zone; F_n is the nominal strength of the strut, tie, or nodal zone; and ϕ is specified in Sec 6.2.3.2.6.

Clarification for Section I.2.6: The forces in all the struts, ties, and nodal zones are computed after applying factored loads to the strut-and-tie model. If several loading cases exist, each should be investigated. The strut-and-tie model, or models, are analyzed for the loading cases and, for a given strut, tie, or nodal zone, F_u is the largest force in that element for all loading cases.

I.3 Strength of Struts

I.3.1 For a strut without longitudinal reinforcement, the nominal compressive strength, F_{ns} , shall be taken as the smaller value of

$$F_{ns} = f_{ce} A_{cs} \quad (6.I.2)$$

at the two ends of the strut, where A_{cs} is the cross-sectional area at one end of the strut, and f_{ce} is the smaller of (a) and (b):

- (a) the effective compressive strength of the concrete in the strut given in Sec I.3.2;
- (b) the effective compressive strength of the concrete in the nodal zone given in Sec I.5.2.

Clarification for Section I.3.1: The smaller dimension perpendicular to the axis of the strut at the ends of the strut is taken as the width of strut w_s used to compute A_{cs} . This strut width is illustrated in Figures 6.I.4(a), 6.I.5(a) and 6.I.5(b). The thickness of the struts may be taken as the width of the member in two-dimensional structures, such as deep beams.

I.3.2 The effective compressive strength of the concrete, f_{ce} , in a strut shall be taken as

$$f_{ce} = 0.85\beta_s f'_c \quad (6.I.3)$$

Clarification for Section I.3.2: The strength coefficient, $0.85f'_c$, in Eq. 6.I.3 represents the effective concrete strength under sustained compression, similar to that used in Equations 6.6.8 and 6.6.9.

I.3.2.1 When a strut has uniform cross-sectional area over its length, $\beta_s = 1.0$

Clarification for Section I.3.2.1: The value of β_s in Sec I.3.2.1 applies to a strut equivalent to the rectangular stress block in a compression zone in a beam or column.

I.3.2.2 For struts located such that the width of the midsection of the strut is larger than the width at the nodes (bottle-shaped struts):

- (a) With reinforcement satisfying Sec I.3.3, $\beta_s = 0.75$
- (b) Without reinforcement satisfying Sec I.3.3, $\beta_s = 0.60\lambda$

Where the value of λ is defined in Sec 6.1.8.1.

Clarification for Section I.3.2.2: The value of β_s given in Sec I.3.2.2 applies to bottle-shaped struts as shown in Figure 6.I.3. The internal lateral spread of the compression forces can lead to splitting parallel to the axis of the strut near the ends of the strut, as shown in Figure 6.I.8. Reinforcement placed to resist the splitting force restrains crack width, allows the strut to resist more axial load, and permits some redistribution of force.

The value given for β_s in Sec I.3.2.2(b) includes the correction factor, λ , for lightweight concrete because the strength of a strut without transverse reinforcement is assumed to be limited to less than the load at which longitudinal cracking develops.

I.3.2.3 For struts in tension members, or the tension flanges of members, $\beta_s = 0.40$

Clarification for Section I.3.2.3: The value given for β_s in Sec I.3.2.3 applies, for example, to compression struts in a strut-and-tie model used to design the longitudinal and transverse reinforcement of the tension flanges of beams, box girders, and walls. The low value of β_s reflects that these struts need to transfer compression across cracks in a tension zone.

I.3.2.4 For all other cases, $\beta_s = 0.60\lambda$

Clarification for Section I.3.2.4: The value given for β_s in Sec I.3.2.4 applies to strut applications not included in Sections I.3.2.1, I.3.2.2, and I.3.2.3. Examples are struts in a beam web compression field in the web of a beam where parallel diagonal cracks are likely to divide the web into inclined struts, and struts are likely to be crossed by cracks at an angle to the struts [see Figures 6.I.10(a) and (b)]. Sec I.3.2.4 gives a reasonable lower limit on β_s except for struts described in Sections I.3.2.2(b) and I.3.2.3.

I.3.3 If the specified value of β_s in Sec I.3.2.2(a) is used, the axis of the strut shall be crossed by reinforcement proportioned to resist the transverse tensile force resulting from the compression force spreading in the strut. It shall be permitted to assume the compressive force in the strut spreads at a slope of 2 longitudinal to 1 transverse to the axis of the strut.

Clarification for Section I.3.3: The reinforcement necessary from Sec I.3.3 is related to the tension force in the concrete due to the spreading of the strut, as shown in the strut-and-tie model in Figure 6.I.8(b). Sec I.3.3 allows the use of local strut-and-tie models to compute the amount of transverse reinforcement needed in a given strut. The compressive forces in the strut may be assumed to spread at a 2:1 slope, as shown in Figure 6.I.8(b). For specified concrete compressive strengths not exceeding 40 MPa, the amount of reinforcement required by Eq. 6.I.4 is deemed to satisfy Sec I.3.3.

Figure 6.I.11 shows two layers of reinforcement crossing a cracked strut. If the crack opens without shear slip along the crack, bars in layer i in the figure will cause a stress perpendicular to the strut of $\frac{A_{si}f_{si}}{b_s s_i} \sin \alpha_i$.

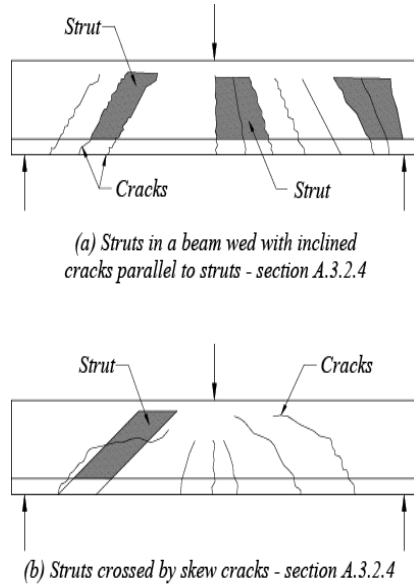


Figure 6.I.10 Types of struts

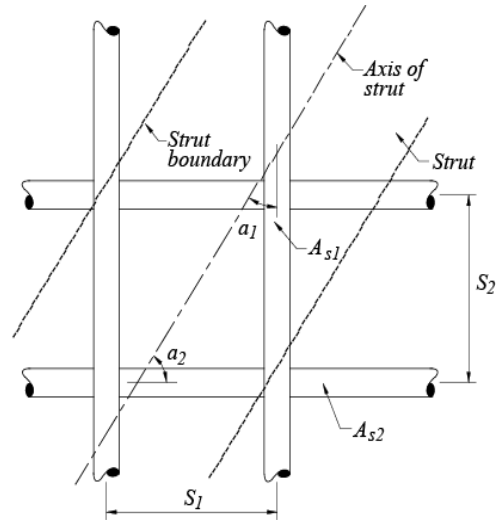


Figure 6.I.11 Reinforcement crossing a strut

Where the subscript i shall have the values of 1 and 2 for the vertical and horizontal bars, respectively, as shown in Figure 6.I.11. Eq. 6.I.4 is written in terms of a reinforcement ratio rather than a stress to simplify the calculation.

The confinement reinforcement given in Sec I.3.3 is difficult to place in three-dimensional structures (e.g. pile caps) most of the time. If this reinforcement is not provided, the value of f_{ce} given in Sec I.3.2.2(b) is used.

I.3.3.1 For f_c' not exceeding 40 MPa, the requirement of Sec I.3.3 shall be permitted to be satisfied by the axis of the strut being crossed by layers of reinforcement that satisfy Eq. 6.I.4.

$$\sum \frac{A_{si}}{b_s s_i} \sin \alpha_i \geq 0.003 \quad (6.I.4)$$

Where A_{si} is the total area of surface reinforcement at spacing s_i in the i -th layer of reinforcement crossing a strut at an angle α_i to the axis of the strut.

I.3.3.2 The reinforcement necessary in Sec I.3.3 shall be placed in either two orthogonal directions at angles α_1 and α_2 to the axis of the strut, or in one direction at an angle α to the axis of the strut. If the reinforcement is in one direction only, α shall not be less than 40° .

Clarification for Section I.3.3.2: The confinement reinforcement required to satisfy Sec I.3.3 is usually provided in the form of horizontal stirrups crossing the inclined compression strut in a corbel with a shear span-to-depth ratio less than 1.0, as shown in Figure 6.6.13 Chapter 6.

I.3.4 It shall be permitted to use an increased effective compressive strength of a strut due to confining reinforcement, if supported by tests and analyses.

Clarification for Section I.3.4: The design of tendon anchorage zones for pre-stressed concrete sometimes uses confinement to enhance the compressive strength of the struts in the local zone. Confinement of struts is discussed in References I.4 and I.8 of Sec I.6.

I.3.5 It shall be permitted to use compression reinforcement to increase the strength of a strut. Compression reinforcement shall be properly anchored, parallel to the axis of the strut, located within the strut, and enclosed in ties or spirals satisfying Sec 8.1.10 Chapter 8. In such cases, the nominal strength of a longitudinally reinforced strut is

$$F_{ns} = f_{ce} A_{cs} + A_s' f_s' \quad (6.I.5)$$

Clarification for I.3.5 - The last term in Eq. 6.I.5 gives the strength added by the reinforcement. The stress f_s' in the reinforcement in a strut at nominal strength can be obtained from the strains in the strut when the strut crushes. For Grade 40 or 60 reinforcement, f_s' can be taken as f_y .

I.4 Strength of Ties

I.4.1 The nominal strength of a tie, F_{nt} , shall be taken as

$$F_{nt} = A_{ts}f_y + A_{tp}(f_{se} + \Delta f_p) \quad (6.I.6)$$

Where $(f_{se} + \Delta f_p)$ shall not exceed f_{py} , and A_{tp} is zero for nonprestressed members.

In Eq. (I.6), it shall be permitted to take Δf_p equal to 420 MPa for bonded prestressed reinforcement, or 70 MPa for unbonded prestressed reinforcement. Other values of Δf_p shall be permitted when justified by analysis.

I.4.2 In the strut-and-tie model, the axis of the reinforcement in a tie shall coincide with the axis of the tie.

Clarification for Section I.4.2: The effective tie width w_t assumed in design can vary between the following limits, depending on the distribution of the tie reinforcement:

- (a) If the bars in the tie are in one layer, the effective tie width can be taken as the diameter of the bars in the tie plus twice the cover to the surface of the bars, as shown in Figure 6.I.5(a); and
- (b) A practical upper limit of the tie width can be taken as the width corresponding to the width in a hydrostatic nodal zone, calculated as

$$w_{t,max} = F_{nt}/(f_{ce}b_s)$$

Where f_{ce} is computed for the nodal zone in accordance with Sec I.5.2. If the tie width exceeds the value from (a), the tie reinforcement should be distributed approximately uniformly over the width and thickness of the tie, as shown in Figure 6.I.5(b).

I.4.3 Anchorage of tie reinforcement by mechanical devices, post-tensioning anchorage devices, standard hooks, or straight bar development as required by Sections I.4.3.1 to I.4.3.4 shall be ensured.

Clarification for Section I.4.3: Special attention is often required for anchorage of ties in nodal zones of corbels or in nodal zones adjacent to exterior supports of deep beams. The reinforcement in a tie should be anchored before it leaves the extended nodal zone at the point defined by the intersection of the centroid of the bars in the tie and the extensions of the outlines of either the strut or the bearing area. This length is l_{anc} . In Figure 6.I.5(a) and (b), this occurs where the outline of the extended nodal zone is crossed by the centroid of the reinforcement in the tie. Some of the anchorage may be achieved by extending the reinforcement through the nodal zone, as shown in Figure 6.I.4(c), and developing it beyond the nodal zone. If the tie is anchored using 90-degree

hooks, the hooks should be confined within the reinforcement extending into the beam from the supporting member to avoid cracking along the outside of the hooks in the support region.

In deep beams, hairpin bars spliced with the tie reinforcement can be used to anchor the tension tie forces at exterior supports, provided the beam width is large enough to accommodate such bars.

Figure 6.I.12 shows two ties anchored at a nodal zone. Development is required where the centroid of the tie crosses the outline of the extended nodal zone.

The development length of the tie reinforcement can be reduced through hooks, mechanical devices, additional confinement, or by splicing it with several layers of smaller bars.

I.4.3.1 Nodal zones shall develop the difference between the tie force on one side of the node and the tie force on the other side.

I.4.3.2 At nodal zones anchoring one tie, the tie force shall be developed at the point where the centroid of the reinforcement in a tie leaves the extended nodal zone and enters the span.

I.4.3.3 At nodal zones anchoring two or more ties, the tie force in each direction shall be developed at the point where the centroid of the reinforcement in the tie leaves the extended nodal zone.

I.4.3.4 The transverse reinforcement required by Sec I.3.3 shall be anchored in accordance with Sec 8.2.10 Chapter 8.

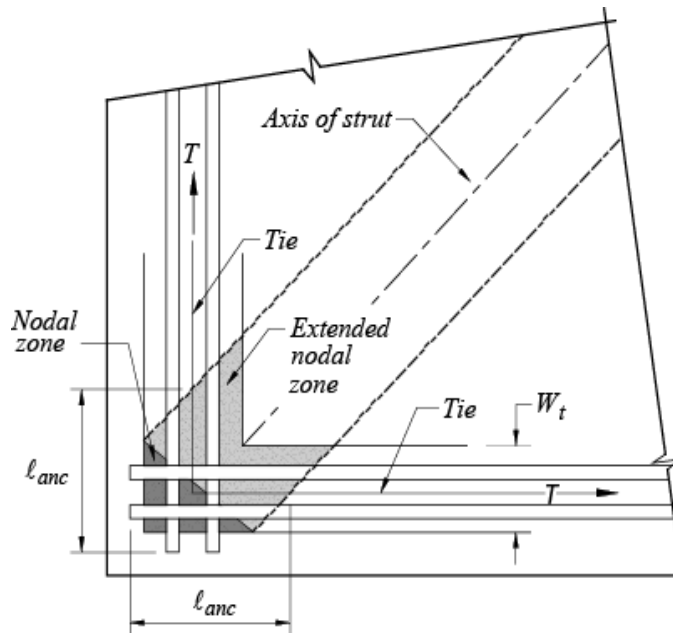


Figure 6.I.12 Extended nodal zone anchoring two ties

I.5 Strength of Nodal Zones

I.5.1 The Nominal compression strength of a nodal zone, F_{nn} , shall be

$$F_{nn} = f_{ce} A_{nz} \quad (6.1.7)$$

Where f_{ce} is the effective compressive strength of the concrete in the nodal zone as given in Sec I.5.2, and A_{nz} is the smaller of (a) and (b):

- (a) The area of the face of the nodal zone on which F_u acts, taken perpendicular to the line of action of F_u ;
- (b) The area of a section through the nodal zone, taken perpendicular to the line of action of the resultant force on the section.

Clarification for Section I.5.1: A hydrostatic nodal zone can be used, if the stresses in all the struts meeting at a node are equal. The faces of such a nodal zone are perpendicular to the axes of the struts, and the widths of the faces of the nodal zone are proportional to the forces in the struts.

Assuming the principal stresses in the struts and ties act parallel to the axes of the struts and ties, the stresses on faces perpendicular to these axes are principal stresses, and Sec I.5.1(a) is used. If, as shown in Figure 6.1.5(b), the face of a nodal zone is not perpendicular to the axis of the strut, there will be both shear stresses and normal stresses on the face of the nodal zone. Typically, these stresses are replaced by the normal (principal compression) stress acting on the cross-sectional area A_c of the strut, taken perpendicular to the axis of the strut as given in Sec I.5.1(a).

Sec I.5.1(b) requires in some cases that the stresses be checked on a section through a subdivided nodal zone. The stresses are checked on the least area section which is perpendicular to a resultant force in the nodal zone. In Figure 6.1.6(b), the vertical face which divide the nodal zone into two parts is stressed by the resultant force acting along A-B. The design of the nodal zone is governed by the critical section from Sec I.5.1(a) or Sec I.5.1(b), whichever gives the highest stress.

I.5.2 The calculated effective compressive stress, f_{ce} , on a face of a nodal zone due to the strut-and-tie forces shall not exceed the value given by

$$f_{ce} = 0.85 \beta_n f'_c \quad (6.1.8)$$

Unless confining reinforcement is provided within the nodal zone and its effect is supported by tests and analysis. The value of β_n for Eq. 6.1.8 is given in Sections I.5.2.1 to I.5.2.3.

I.5.2.1 In nodal zones bounded by struts or bearing areas, or both, $\beta_n = 1.0$

I.5.2.2 In nodal zones anchoring one tie, $\beta_n = 0.80$ or

I.5.2.3 $\beta_n = 0.80, \beta_n = 0.60$

Clarification for I.5.2 - In two-dimensional members, such as deep beams, the nodes can be classified as C-C-C if all the members intersecting at the node are in compression; as C-C-T nodes if one of the members acting on the node is in tension; and so on, as shown in Figure 6.I.7. The effective compressive strength of the nodal zone is given by Eq. 6.I.8, as modified by Section I.5.2.1 to I.5.2.3 apply to C-C-C nodes, C-C-T nodes, and C-T-T or T-T-T nodes, respectively.

The increasing degree of disruption of the nodal zones due to the incompatibility of tension strains in the ties and compression strains in the struts is reflected by the β_n values. The stress on any face of the nodal zone or on any section through the nodal zone should not exceed the value given by Eq. 6.I.8, as modified by Sections I.5.2.1 to I.5.2.3.

I.5.3 In a three-dimensional strut-and-tie model, the area of each face of a nodal zone shall not be less than that given in I.5.1, and the shape of each face of the nodal zones shall be similar to the shape of the projection of the end of the struts onto the corresponding faces of the nodal zones.

Clarification for Sections I.5.3: In order to simplify the calculations of the geometry of a three-dimensional strut-and-tie model, this description of the shape and orientation of the faces of the nodal zones is introduced.

I.6 Related References to Appendix I

I.1. Schlaich, J.; Schäfer, K.; and Jennewein, M., "Toward a Consistent Design of Structural Concrete," PCI Journal, V. 32, No. 3, May-June 1987, pp. 74-150.

I.2. Collins, M. P., and Mitchell, D., Prestressed Concrete Structures, Prentice Hall Inc., Englewood Cliffs, NJ, 1991, 766 pp.

I.3. MacGregor, J. G., Reinforced Concrete: Mechanics and Design, 3rd Edition., Prentice Hall, Englewood Cliffs, NJ, 1997, 939 pp.

I.4. FIP Recommendations, Practical Design of Structural Concrete, FIP-Commission 3, "Practical Design," Pub.: SETO, London, Sept. 1999.

I.5. Menn, C., Prestressed Concrete Bridges, Birkhäuser, Basle, 535 pp.

I.6. Muttoni, I.; Schwartz, J.; and Thürlimann, B., Design of Concrete Structures with Stress Fields, Birkhauser, Boston, MA, 1997, 143 pp.

I.7. Joint ACI-ASCE Committee 445, "Recent Approaches to Shear Design of Structural Concrete (ACI 445R-99)," American Concrete Institute, Farmington Hills, MI, 1999, 55 pp.

I.8. Bergmeister, K.; Breen, J. E.; and Jirsa, J. O., "Dimensioning of the Nodes and Development of Reinforcement," IABSE Colloquium Stuttgart 1991, International Association for Bridge and Structural Engineering, Zurich, 1991, pp. 551-556.

PART VI
Appendix J

Working Stress Design Method for Reinforced Concrete Structures

J.1 Analysis and Design - General Consideration

J.1.1 Notation

A_s	=	Area of tension reinforcement
A'_s	=	Area of compression reinforcement
A_v	=	Area of shear reinforcement perpendicular to flexural tension reinforcement within a distance s , mm ²
A_{vh}	=	Area of shear reinforcement parallel to flexural tension reinforcement within a distance s_l , mm ²
E_c	=	Modulus of elasticity of concrete, N/mm ²
E_s	=	Modulus of elasticity of reinforcement, N/mm ²
H	=	Total lateral force acting in any storey
M	=	Moment at the section acting simultaneously with P
M_n	=	Flexural moment capacity
M_r	=	Resisting moment capacity based on f'_c
N	=	Axial load normal to cross section occurring simultaneously with V , to be taken as positive for compression, negative for tension and to include effects of tension due to creep and shrinkage
P	=	Working axial load at the section
R	=	Constant, $\frac{1}{2} f_c k j$
T	=	Torsional moment at section
T_c	=	Torsional moment strength provided by concrete
T_s	=	Torsional moment strength provided by torsion reinforcement
V	=	Shear at section
V_c	=	Shear strength provided by concrete
V_s	=	Shear strength provided by shear reinforcement
V_n	=	Shear strength
a	=	Shear span, distance between concentrated load and face of support, mm

b	=	Width of rectangular beam, or effective width of compression flange for T-beam
b_w	=	Web width, or diameter of circular section
d, d'	=	Distance of extreme compression fibre to centroid of compression reinforcement
d_c	=	Thickness of concrete cover measured from the extreme tension fibre to centre of bar or wire located closest thereto
f_c	=	Allowable stress in concrete
f'_c	=	Specified compressive strength of concrete, N/mm ²
f_s	=	Allowable/Permissible tensile stress in reinforcement, N/mm ²
f_y	=	Specified yield strength of reinforcement, N/mm ²
h	=	Overall thickness of members, mm.
j, k	=	Beam constants defined in Sec J.2.6.1
l	=	Effective span, mm
l_n	=	Clear span measured face-to-face of supports, mm
n	=	Modular ratio, E_s/E_c
r	=	Stress ratio, f_s/f_c
s	=	Spacing of shear or torsion reinforcement in direction parallel to longitudinal reinforcement, mm
s_1	=	Spacing of shear or torsion reinforcement in direction perpendicular to longitudinal reinforcement, mm
t	=	Thickness of compression flange of T-beams
v	=	Design shear stress, N/mm ²
v_c	=	Permissible shear stress carried by concrete, N/mm ²
z	=	lever arm used in Sec J.8.3
β_c	=	Ratio of long side to short side of concentrated load or reaction area.
Δ	=	Elastically computed first order lateral deflection due to H at the top of the storey relative to the bottom of the storey.
ρ	=	Ratio of tension reinforcement, $= A_s/bd$
ρ_w	=	$A_s/b_w d$

For all other symbols reference shall be made to Sec 6.1.1.

J.1.2 Design Methods

In the design of reinforced concrete structures using working stress design method, members shall be proportioned for adequate capacity in accordance with the provisions of this Chapter using working loads and permissible stresses. The working stress design method may be used as an alternative method with the requirement that provisions of Chapter 6, except Sec 6.1.5, shall apply to members designed by this method.

J.1.3 Design Assumptions

The design of reinforced concrete structures by the working stress design method is based on the following assumptions.

- J.1.3.1 At any cross section, plane sections before bending remain plane after bending; strains vary with the distance from the neutral axis.
- J.1.3.2 All tensile stresses are taken up by reinforcement and none by concrete, except otherwise specifically permitted.
- J.1.3.3 The stress-strain relation for concrete is a straight line under working loads within the allowable working stresses. Stresses vary linearly with the distance from the neutral axis except for deep beams.
- J.1.3.4 The tension reinforcement area is replaced in design computations with a concrete tension area equal to n times that of the reinforcement steel, where n is the modular ratio E_s/E_c .
- J.1.3.5 In doubly reinforced beams the compression reinforcement shall be transformed to an equivalent concrete area which is $2n$ times that of the reinforcement steel.
- J.1.3.6 The modular ratio $n = \frac{E_s}{E_c}$ may be taken as the nearest whole number, but not less than 6.
- J.1.3.7 The compressive stress developed in compression reinforcement of doubly reinforced beams shall not exceed the permissible tensile stress for such steel.

J.1.4 Loading

- J.1.4.1 Design provisions of this Chapter based on the assumption that structures shall be designed to resist all applicable loads.
- J.1.4.2 Service loads shall be in accordance with Chapter 2, Loads, with such live load reductions as are permitted therein.

- J.1.4.3 In the design for wind and earthquake loads, integral structural parts shall be designed to resist the total lateral loads.
- J.1.4.4 Consideration shall be given to effects of forces due to crane loads, vibration, impact, shrinkage, temperature changes, creep and unequal settlement of supports.
- J.1.4.5 When dead load reduces effects of other loads, members shall be designed for 85 percent of the dead load in combination with the other loads.

J.1.5 Stiffness

- J.1.5.1 Use of any consistent set of assumptions is permitted for computing relative flexural and torsional stiffness of columns, walls, floors, and roof systems.
- J.1.5.2 In computing the value of I for relative flexural stiffness of slabs, beams, girders, and columns, contribution of the reinforcement may be neglected. In T-shaped sections allowance shall be made for the effect of flange.
- J.1.5.3 If the total torsional stiffness in the plane of a continuous system at a joint does not exceed 20 percent of the flexural stiffness at the joint, the torsional stiffness need not be taken into consideration in the analysis.
- J.1.5.4 Effect of haunches shall be considered both in determining the moments and in the design of members.

J.1.6 Span Length

- J.1.6.1 Span length of members not built integrally with supports shall be considered as the clear span plus depth of member but need not to exceed distance between centres of supports.
- J.1.6.2 In determining moments in frames or continuous construction, span lengths shall be taken as the centre-to-centre distance of supports.
- J.1.6.3 For design of beams built integrally with supports, the use of moments at faces of support is permitted.
- J.1.6.4 Solid or ribbed slabs built integrally with supports, with clear span not more than 3.0 m, are permitted to be analysed as continuous slabs on knife edge supports, with spans equal to the clear spans of the slab, the width of beams being otherwise neglected.
- J.1.6.5 Effective span of cantilevered beams or slabs shall be taken as its span to the face of support plus half its effective depth, except where it is an overhang of a continuous beam, the length to the centre of the support shall be used.

J.1.7 Arrangement of Live Loads

For continuous beams and frames the arrangement of live load may be limited to the combination of:

- (a) Service dead load on all spans with full service live load on two adjacent spans, and
- (b) Service dead load on all spans with full service live load on alternate spans.

J.1.8 Floor Finish

J.1.8.1 A floor finish shall not be included as part of a structural member unless placed monolithically with the floor slab or designed in accordance with requirements of composite concrete flexural members.

J.1.8.2 It is allowed to consider all concrete floor finishes as part of required cover or total thickness for non-structural considerations.

J.1.9 Allowable Stresses in Concrete

Allowable stresses in concrete shall not exceed the following:

- (a) Flexure:

Extreme fibre stress in compression $0.45 f'_c$

- (b) Shear:

Beams, one-way slabs and footings :

Shear stress carried by concrete, v_c $0.091\sqrt{f'_c}$

Maximum shear stress carried by concrete plus shear reinforcement $0.457\sqrt{f'_c}$

Ribs: $0.10\sqrt{f'_c}$

Shear stress carried by concrete, v_c

Two-way slabs and footings : $(0.083 + 0.17/\beta_c)\sqrt{f'_c} \leq 0.17\sqrt{f'_c}$

Shear stress carried by concrete, v_c

- (c) Bearing stress on loaded area :

When the loaded area (area of column, pier or base plate) and the supporting area (area of the top of footing) are equal $0.3\sqrt{f'_c}$

When the supporting area is larger than the loaded area on all sides $0.3\sqrt{\left(\frac{A_1}{A_2}\right)}f'_c \leq 0.6f'_c$

Where,

A_1 = Area of the lower base of the largest frustum of a pyramid, cone, or tapered wedge contained wholly within the footing and having for its upper base, the area actually loaded, and having side slopes of 1 vertical to 2 horizontal, and

A_2 = Loaded area of the column base.

J.1.10 Allowable Stresses in Reinforcement

Allowable tensile stresses in reinforcement f_s , shall be those as specified below:

(a) Except as specified in (b) below, f_s shall be determined as follows :

(i) For $250 \text{ N/mm}^2 \leq f_y < 275 \text{ N/mm}^2$: $f_s = 125 \text{ N/mm}^2$

(ii) For $275 \text{ N/mm}^2 \leq f_y < 420 \text{ N/mm}^2$: $f_s = 138 \text{ N/mm}^2$

(iii) For $f_y \geq 420 \text{ N/mm}^2$: $f_s = 165 \text{ N/mm}^2$

(b) For flexural reinforcement, 100 mm or less in diameter in one-way slabs of not more than 3.5 m span: $f_s = 0.5f_y$ but not greater than 200 N/mm^2

J.1.11 Allowable Stresses for Wind and Earthquake Forces

Members subject to stresses produced by wind or earthquake forces combined with other loads may be proportioned for stresses 33 percent greater than those specified in Sections J.1.9 and J.1.10, provided that the section thus required is not less than that required for the combination of dead and live load.

J.1.12 Development and Splices of Reinforcement

J.1.12.1 Development and splices of reinforcement shall be in accordance with Chapter 8, Detailing of Reinforcement in Concrete Structures.

J.1.12.2 In satisfying requirements of Sec 8.2.8.3, M_n shall be taken as computed moment capacity assuming all positive moment tension reinforcement at the section to be stressed to the permissible tensile stress f_s and V_u shall be taken as unfactored shear force at the section.

J.2 Beams and One-Way Slabs

J.2.1 Notation

All the notation used this Section are provided in Sec. J.1.1

J.2.2 Span Length

Determination of span length shall be in accordance with Sec J.1.6.

J.2.3 Design Assumptions

Design assumptions shall be in accordance with Sec J.1.3.

J.2.4 General Principles and Requirements

J.2.4.1 Design of cross section subject to flexural or combined flexure and axial loads shall be based on design assumptions of Sec J.1.3.

J.2.4.2 Compression reinforcement in conjunction with additional tension reinforcement may be used to increase flexural strength of the members.

J.2.4.3 The effective depth, d , of a beam or slab shall be taken as the distance from the centroid of its tensile reinforcement to its compression face.

J.2.4.4 The effects of lateral eccentricity of load shall be taken into account in determining the spacing of lateral supports for a beam. The spacing shall never exceed 50 times the least width b of compression flange or face.

J.2.4.5 Requirements of T-beams

- (a) In T-beam construction the slab and beam shall be built integrally or otherwise effectively bonded together.
- (b) The effective flange width to be used in the design of symmetrical T-beams shall not exceed one-fourth of the span length of the beam, and its overhanging width on either side of the web shall not exceed eight times the thickness of the slab nor one-half the clear distance to the next beam.
- (c) Isolated beams in which the T-form is used only for the purpose of providing additional compression area, shall have a flange thickness not less than one-half the width of the web and a total flange width not more than four times the width of the web.
- (d) For beams having a flange on one side only, the effective overhanging flange width shall not exceed $\frac{1}{12}$ th of the span length of the beam, nor six times the thickness of the slab, nor one-half the clear distance to the next web.

- (e) The overhanging portion of the flange of the beam shall not be considered effective in computing the shear and diagonal tension resistance of T-beams.
- (f) Provision shall be made for the compressive stress at the support in continuous T-beam construction.

J.2.5 Continuous Beams

Continuous beams shall be analysed in accordance with Sec J.2.5.2 and designed and detailed according to Sec J.2.6 and J.2.7 to resist moments and shear forces.

J.2.5.1 Arrangement of Live Loads: Arrangement of live loads shall be in accordance with Sec J.1.7.

J.2.5.2 Methods of analysis

- (a) All members of frames or continuous construction shall be designed for the maximum effects of working loads as determined by the theory of elastic analysis.
- (b) In lieu of exact analysis, the approximate moments and shears given in Sec 6.1.4.3 may be used for design of continuous beams and one way slabs (slab reinforced to resist flexural stresses in only one direction), provided that the quantity w_u in the expressions in Sec. 6.1.4.3 is replaced by the working load w .
- (c) No redistribution of negative moment shall be permitted for working stress design.

J.2.6 Design for Flexure

J.2.6.1 The following equations are applicable to singly and doubly reinforced rectangular beams :

When, the stress ratio, r is known

$$k = \frac{n}{n + r} \quad (6.J.1)$$

When, steel ratio, ρ is known

$$k = \sqrt{2n\rho + (n\rho)^2} - n\rho \quad (6.J.2)$$

$$j = 1 - k / 3 \quad (6.J.3)$$

$$R = \frac{1}{2} f_c k j \quad (6.J.4)$$

$$M_r = R b d^2 \quad (6.J.5)$$

J.2.6.2 Formulae for Singly Reinforced Rectangular Beams: If external bending moment M is less than resisting moment M_r , the area of tensile reinforcement shall be calculated using the following formula :

$$A_s = \frac{M}{f_s j d} \quad (6.J.6)$$

J.2.6.3 Formulae for Doubly Reinforced Beams : If $M > M_r$ the beam shall be designed for tensile and compressive reinforcements using the following formulae :

$$A'_s = \frac{M - M_r}{f'_s (d - d')} \quad (6.J.7)$$

Where,

$$f'_s = \frac{2n-1}{n} \left(\frac{k - d'/d}{1-k} \right) f_s \leq f_s \quad (6.J.8)$$

$$A_s = \frac{M_r}{f_s j d} + \frac{(M - M_r)}{f_s (d - d')} \quad (6.J.9)$$

J.2.6.4 Design of T-beams: A T-beam, where the flange is on the compression side, shall be treated as a rectangular beam if $M \leq \frac{1}{2} f_c b t (d - t/3)$. Otherwise, the beam shall be considered as a T-beam, in which case the following formulae shall be applicable :

$$k = \frac{n\rho + \frac{1}{2}(t/d)^2}{n\rho + (t/d)} \quad (6.J.10)$$

Where,

$$\rho = A_s / b d$$

$$j = 1 - \left[\frac{3k - 2(t/d)}{2k - (t/d)} \right] (t/d) \quad (6.J.11)$$

$$A_s = \frac{M}{f_s j d} \quad (6.J.6)$$

Actual stress in concrete, f_{ca} can be obtained from the relation:

$$f_{ca} = \frac{M}{(1 - t/2 k d) b t j d} \quad (6.J.12)$$

While using Eq. 6.J.10, if ρ is not known, it may be initially estimated as $\rho = M / [(d - t/2) b d f_s]$

J.2.7 Shear and Torsion

J.2.7.1 The design shear force V shall not exceed the sum of the shear strength provided by concrete, V_c and that provided by shear reinforcement, V_s .

$$V \leq V_c + V_s \quad (6.J.13)$$

J.2.7.2 When the reaction, in the direction of applied shear, introduces compression into the end regions of a member, sections located less than a distance d from face of support may be designed for the same shear force V as that computed at a distance d .

J.2.7.3 Shear strength provided by concrete

- (a) For members subject to shear and flexure, shear strength provided by concrete, V_c shall not exceed $0.091\sqrt{f'_c}b_wd$ unless a more detailed calculation is made in accordance with (d) below.
- (b) For members subject to shear and axial compression, shear strength provided by concrete V_c shall not exceed $0.091\sqrt{f'_c}b_wd$ unless a more detailed calculation is made in accordance with (e) below.
- (c) For members subject to significant axial tension, shear reinforcement shall be designed to carry total shear, unless a more detailed calculation is made using

$$V_c = 0.091 \left(1 + 0.58 \frac{N}{A_g} \right) \sqrt{f'_c} b_w d \quad (6.J.14)$$

Where, N is design axial load normal to cross-section occurring simultaneously with V and is negative for tension.

(d) For members subject to shear and flexure only, V_c may be computed by :

$$V_c = \left(0.083\sqrt{f'_c} + 9\rho_w \frac{Vd}{M} \right) b_w d \leq 0.16\sqrt{f'_c}b_wd \quad (6.J.15)$$

Quantity Vd/M shall not be taken greater than 1.0, where M is design moment occurring simultaneously with V at section considered, and $\rho_w = A_s/b_wd$.

(e) For members subject to axial compression, V_c may be computed by :

$$V_c = 0.091 \left(1 + 0.09 \frac{N}{A_g} \right) \sqrt{f'_c} b_w d \quad (6.J.16)$$

- (f) For members subjected to torsional moment T exceeding $(0.023\sqrt{f'_c})\sum x^2y$, V_c may be computed by

$$V_c = \frac{0.091\sqrt{f'_c}b_wd}{\sqrt{1+(2.5C_t T/V)^2}} \quad (6.J.17)$$

For calculation of $\sum x^2y$, the following conditions shall apply:

- (i) For members with rectangular or flanged sections, the sum $\sum x^2y$ shall be taken for the component rectangles of the section, but the overhanging flange-width used in design shall not exceed three times the flange thickness.
- (ii) A rectangular box section shall be taken as solid section provided the wall thickness h is at least $x/4$. A box section with wall thickness less than $x/4$ but greater than $x/10$ shall be taken as solid section except that $\sum x^2y$ shall be multiplied by $4h/x$. When h is less than $x/10$, the stiffness of the wall shall be considered. Fillets shall be provided at interior corners of box sections.
- (g) In determining shear strength provided by concrete V_c , whenever applicable, effects of axial tension due to creep and shrinkage in restrained members shall be considered and effects of inclined flexural compression in variable-depth members may be included.

J.2.7.4 Shear strength provided by shear reinforcement

- (a) Types of shear reinforcement

Shear reinforcement may consist of:

- (i) stirrups perpendicular to axis of member,
- (ii) bent up longitudinal reinforcement with bent portion making an angle of 30° or more with longitudinal tension reinforcement,
- (iii) combination of stirrups and bent longitudinal reinforcement,
- (iv) spirals.

- (b) Design yield strength of shear reinforcement shall not exceed 420 N/mm^2 .

- (c) Stirrups shall extend to a distance d from extreme compression fibre and shall be anchored at both ends in accordance with Sec 8.2.

(d) Spacing limits for shear reinforcement

- (i) Spacing of shear reinforcement perpendicular to member axis shall not exceed $d/2$, nor 600 mm.
- (ii) Bent longitudinal bars shall have a maximum spacing of $0.375d (1 + \cot \alpha)$, but not greater than 600 mm, where, α is the acute angle between the bent bar and the horizontal.
- (iii) When $(V - V_c)$ exceeds $0.17\sqrt{f'_c}b_wd$ maximum spacing given in (i) and (ii) above shall be reduced by one-half.

(e) Minimum shear reinforcement

- (i) A minimum area of shear reinforcement shall be provided in all reinforced concrete flexural members where design shear force V is greater than one-half the permissible shear strength V_c provided by concrete, except slabs, footings, ribbed construction and beams with total depth not exceeding the largest of 2.5 times thickness of flange, one-half the width of web, and 250 mm.
- (ii) Where shear reinforcement is required by (i) above or by analysis, minimum area of shear reinforcement shall be computed by

$$A_v = 0.35 \frac{b_w s}{f_y} \quad (6.J.18)$$

- (iii) Where torsional moment T exceeds $(0.023\sqrt{f'_c})\sum x^2y$ and where web reinforcement is required by (i) above or by analysis, the minimum area of closed stirrups shall be computed by

$$A_v + 2A_t = 0.35 \frac{b_w s}{f_y} \quad (6.J.19)$$

Where, A_t is the area of one leg of closed stirrup.

(f) Design of Shear Reinforcement

- (i) Where design shear force V exceeds shear strength provided by concrete V_c , shear reinforcement shall be provided in accordance with (ii) to (viii) below.
- (ii) When shear reinforcement perpendicular to axis of member is used,

$$A_v = \frac{(V - V_c) s}{f_s d} \quad (6.J.20)$$

- (iii) When inclined stirrups are used as shear reinforcement,

$$A_v = \frac{(V - V_c) s}{f_s d (\sin \alpha + \cos \alpha)} \quad (6.J.21)$$

- (iv) When shear reinforcement consists of a single bar or a single group of parallel bars, all bent up at the same distance from the support,

$$A_v = \frac{(V - V_c) s}{f_s d \sin \alpha} \quad (6.J.22)$$

Where $(V - V_c)$ shall not exceed $0.133\sqrt{f'_c}b_w d$

- (v) When shear reinforcement consists of a series of parallel bent-up bars or groups of parallel bent-up bars at different distances from the support, required area shall be computed by Eq. 6.J.21.
- (vi) Only the centre three-quarters of the inclined portion of any longitudinal bent bar shall be considered effective for shear reinforcement.
- (vii) When more than one type of shear reinforcement is used to reinforce the same portion of member, required area shall be computed as the sum of the various types separately. In such computations, V_c shall be included only once.
- (viii) Value of $(V - V_c)$ shall not exceed $0.365\sqrt{f'_c}b_w d$

J.2.7.5 Combined shear and torsion

- (a) Torsion effects shall be included with shear and flexure where torsional moment T exceeds $(0.023\sqrt{f'_c})\sum x^2 y$. Otherwise, torsion may be neglected. For calculation of $\sum x^2 y$, see Sec J.2.7.3(f).
- (b) If torsional moment T in a member is required to maintain equilibrium, the member shall be designed to carry that torsional moment in accordance with (c) to (j) below.
- (c) In a statically indeterminate structure where reduction of torsional moment in a member can occur due to redistribution of internal forces, maximum torsional moment may be reduced to $(0.06\sqrt{f'_c})\sum x^2 y$.
- (i) In such case the corresponding adjusted moments and shears in adjoining members shall be used in design.
- (ii) In lieu of exact analysis, torsional loading from a slab shall be taken uniformly distributed along the member.
- (d) Sections located less than a distance d from face of support may be designed for the same torsional moment T as that computed at a distance d .
- (e) Torsional Moment Strength

Design of cross-section subject to torsion shall be based on

$$T = T_c + T_s \quad (6.J.23)$$

Where,

T = is torsional moment at section,

T_c = is torsional moment strength provided by concrete in accordance with (f) below,

T_s = is torsional moment strength provided by torsion reinforcement in accordance with (j) below.

(f) Torsional moment strength provided by concrete

(i) Torsional moment strength T_c shall be computed by

$$T_c = \frac{(0.036\sqrt{f'_c})\sum x^2y}{\sqrt{1 + \left(\frac{0.4V}{C_tT}\right)^2}} \quad (6.J.24)$$

(ii) For members subject to significant axial tension, torsion reinforcement shall be designed to carry the total torsional moment, unless a more detailed calculation is made, in which T_c given by Eq. 6.J.24 and V_c given by Eq. 6.J.17 shall be multiplied by $\left(1 + \frac{0.3N}{A_g}\right)$, where N is negative for tension.

(g) Torsion Reinforcement Requirements

- (i) Torsion reinforcement, where required, shall be provided in addition to reinforcement required to resist shear, flexure and axial forces.
- (ii) Reinforcement required for torsion shall be combined with that required for other forces, provided the area furnished is the sum of individually required areas and the most restrictive requirements for spacing and placement are met.
- (iii) Torsion reinforcement shall consist of closed stirrups, closed ties or spirals, combined with longitudinal bars.
- (iv) Design yield strength for torsion reinforcement shall not exceed 420 N/mm².
- (v) Stirrups used as torsion reinforcement shall extend to a distance d from extreme compression fibre and shall be anchored in accordance with Sec 8.2.
- (vi) Torsion reinforcement shall be provided at least a distance $(bt + d)$ beyond the point theoretically required.

(h) Design of Torsion Reinforcement

- (i) Where torsional moment T exceeds torsional moment strength T_c , torsion reinforcement shall be provided to satisfy Eq. 6.J.23, where torsional moment strength T_s shall be computed by

$$T_s = 0.55 \frac{A_t x_1 y_1 f_y}{s} \quad (6.J.25)$$

Where A_t is the area of one leg of closed stirrup resisting torsion within a distance s and $\alpha_t = (2 + y_1/x_1)/3$, but not more than 1.5. Longitudinal bars distributed around the perimeter of the closed stirrup A_t shall be provided in accordance with (iii) below.

- (ii) A minimum area of closed stirrup shall be provided in accordance with Sec J.2.7.4(e).
- (iii) Required area of longitudinal bar A_l distributed around the perimeter of the closed stirrup A_t shall be computed by :

$$A_l = 2A_t \left(\frac{x_1 + y_1}{s} \right) \quad (6.J.26)$$

$$\text{or, } A_l = \left[\frac{2.8xs}{f_y} \left(\frac{T}{T + \frac{V}{3C_t}} \right) - 2A_t \right] \left(\frac{x_1 + y_1}{s} \right) \quad (6.J.27)$$

$$\text{or, } A_l = \left[\frac{2.8xs}{f_y} \left(\frac{T}{T + \frac{V}{3C_t}} \right) - \frac{b_w s}{3f_y} \right] \left(\frac{x_1 + y_1}{s} \right) \quad (6.J.28)$$

Whichever is the greatest

- (iv) Torsional moment strength T_s shall not exceed $4T_c$

(i) Spacing Limits for Torsion Reinforcement

- (i) Spacing of closed stirrups shall not exceed the smaller of $\left(\frac{x_1 + y_1}{4} \right)$, or 300 mm.
- (ii) Spacing of longitudinal bars, not less than 10 mm diameter, distributed around the perimeter of the closed stirrup shall not exceed 300 mm. At least one longitudinal bar shall be placed in each corner of the closed stirrups.

J.2.8 Reinforcement

J.2.8.1 At any section of a beam or one-way slab, except as provided in Sec J.2.8.2 and J.2.8.3 below, where positive reinforcement is required by analysis, the ratio ρ provided shall not be less than that given by

$$\rho_{\min} = \frac{1.38}{f_y} \quad (6.J.29)$$

In flanged beams where the web is in tension, the ratio ρ shall be computed for this purpose using the width of web.

J.2.8.2 Alternatively, area of reinforcement provided at every section, positive or negative, shall be at least one-third greater than that required by analysis.

J.2.8.3 For structural slabs of uniform thickness, minimum area and maximum spacing of reinforcement in the direction of the span shall be as required for shrinkage and temperature according to Sec 8.1.11.

J.2.8.4 Where the principal reinforcement in a slab which is considered as the flange of a T-beam (not ribbed floor) is parallel to the beam, transverse reinforcement shall be provided in the top of the slab. This reinforcement shall be designed to carry the load on the portion of the slab assumed to act as the flange of the T-beam. For isolated beams, the full width of overhanging flange shall be considered. The flange shall be assumed to act as a cantilever. The spacing of the bars shall not exceed five times the thickness of the flange, nor 450 mm. This reinforcement need not be additive to any other reinforcements required.

J.2.9 Crack Control

J.2.9.1 This section prescribes rules for distribution of flexural reinforcement to control flexural cracking in beams and in one-way slabs (slabs reinforced to resist flexural stresses in only one direction).

J.2.9.2 Flexural tension reinforcement shall be well distributed within the maximum flexural tension zone of a member cross-section as required by Sec J.2.9.3 below.

J.2.9.3 When design yield strength f_y for tension reinforcement exceeds 275 N/mm², cross-section of maximum positive and negative moment shall be so proportioned that the quantity z given by

$$z = f_s (d_c A)^{1/3} \quad (6.J.30)$$

does not exceed 30 kN/mm for interior exposure and 25 kN/mm for exterior exposure. Calculated stress in reinforcement at working load, f_s , shall be computed as the moment divided by the product of steel area and internal moment arm. In lieu of such computations, it is permitted to take f_s as 60 percent of specified yield strength of f_y .

J.2.9.4 Provisions of Sec J.2.8.3 are not sufficient for structures subject to very aggressive exposure or designed to be watertight. For such structures, special investigation and precautions are required.

J.2.9.5 When flanges of T-beam construction are in tension, part of the flexural tension reinforcement shall be distributed over an effective flange width as defined in Sec J.2.4.5 or a width equal to $\frac{1}{10}$ the span, whichever is smaller. If the effective flange width exceeds $\frac{1}{10}$ the span, some longitudinal reinforcement shall be provided in the outer portion of the flange.

J.2.9.6 If the depth of the web exceeds 900 mm, longitudinal skin reinforcement shall be uniformly distributed along both side faces of the member for a distance $d/2$ from the nearest flexural tension reinforcement. The area of skin reinforcement A_{sk} on each side face shall be at least $(d - 750) \text{ mm}^2$ per metre height. The maximum spacing of the skin reinforcement shall not exceed the lesser of $d/6$ and 300 mm. Such reinforcement may be included in strength computation if a strain compatibility analysis is made to determine stresses in the individual bars. The total area of longitudinal skin reinforcement in both faces need not exceed one-half of the required flexural tensile reinforcement.

J.2.10 Deflection

J.2.10.1 Beams and one-way slabs shall be designed to have adequate stiffness to limit deflections or any deformations that affect strength or serviceability of a structure adversely at working load.

J.2.10.2 Minimum thickness stipulated in Table 6.2.5.1 of Chapter 6 shall apply for beams and one-way slabs not supporting or attached to partitions or other construction likely to be damaged by large deflections, unless computation of deflection indicates a lesser thickness can be used without adverse effects.

J.2.10.3 Deflections, when computed, shall be those which occur immediately on application of the load evaluated by the usual methods or formulae for elastic deflections, considering the effects of cracking and reinforcement on member stiffness.

J.2.10.4 Unless stiffness values are obtained by a more comprehensive analysis, immediate deflection shall be computed with the modulus of elasticity E_c for concrete as specified in Sec 6.1.7, and with the effective moment of inertia I_e computed by Eq (6.2.1) of Chapter 6, but not greater than I_g .

J.2.10.5 For continuous members, effective moment of inertia may be taken as the average of values obtained from Eq (6.2.1) for the critical positive and negative moment sections. For prismatic members, effective moment of inertia may be taken as the value obtained from Eq (6.2.1) at mid-span for simple and continuous spans, and at support for cantilevers.

J.2.10.6 Unless values are obtained by a more comprehensive analysis, additional long-term deflection resulting from creep and shrinkage of flexural members shall be determined by multiplying the immediate deflection caused by the sustained load considered, by the factor λ_{Δ} as determined from Eq (6.2.4) of Chapter 6.

J.2.10.7 Deflections computed in accordance with Sec J.2.10.3 through J.2.10.6 shall not exceed the limits stipulated in Table 6.2.5.2 of Chapter 6.

J.3 Columns

Sections J.3.1 to J.3.5 as detailed hereunder along with the Sec 6.3, except Sections 6.3.2.1 to 6.3.2.7, and 6.3.3, shall form part of this section. In case of any conflict, the provisions of this Appendix B shall prevail.

In using the provisions of Sec 6.3, the word factored shall be read as working or working load whichever is applicable.

J.3.1 Definitions and Notation

J.3.1.1 Notation

All the notation used this Section are provided in Sec. J.1.1. For other symbols used in this section but not provided in Sec J.1.1, the notation given in Sec 6.1.1 shall be applicable.

J.3.1.2 Definitions: The definitions given in Chapter 6 shall apply to this section. In applying the provision of Chapter 6, the terms P_u and Δ_u shall be replaced by their working load counterparts P and Δ respectively.

J.3.2 Design Assumptions

J.3.2.1 The design assumptions specified in Sec J.1.3 are valid for this section.

J.3.2.2 The provisions of Sec 6.3.8.2 and 6.3.8.3 shall apply to this section.

J.3.3 General Principles and Requirements

J.3.3.1 Design of cross-section subject to flexure, or to axial loads, or to combined flexure and axial loads shall be based on design assumptions of Sec J.1.3.

J.3.3.2 All compression members, with or without flexure, shall be proportioned using the ultimate strength design method.

J.3.3.3 Combined flexure and axial load capacity of compression members shall be taken as 40 percent of that computed in accordance with the provisions of Chapter 6 of this part.

J.3.3.4 Design axial load P of compression members shall not be taken greater than the following :

- (a) For members with spiral reinforcement conforming to Sec 8.1.9.3 or composite compression member conforming to Sec 6.3.13 :

$$P_{\max} = 0.289f'_c A_g + (0.34f_y - 0.289f'_c) A_{st} \quad (6.J.31)$$

- (b) For members with tie reinforcement conforming to Sec 8.1.9.4

$$P_{\max} = 0.272f'_c A_g + (0.32f_y - 0.272f'_c) A_{st} \quad (6.J.32)$$

J.3.3.5 Members subject to compressive axial load shall be designed for maximum moment that can accompany the axial load. The axial load P at given eccentricity shall not exceed that given in Sec J.3.3.4 above. The maximum moment M shall be magnified for slenderness effects in accordance with Sec J.3.4.

J.3.4 Slenderness Effects

J.3.4.1 Slenderness effects shall be included in accordance with the requirements of Sec. 6.3.10.

J.3.4.2 In applying the provisions of Sec. 6.3.10, the following convention and modification shall be used:

- (a) the term factored shall be replaced by working or working load as the context implies,
- (b) the value of strength reduction factor ϕ shall be taken as unity, and
- (c) the term P_u shall be replaced by 2.5 times the design axial working load P when gravity loads govern the design, and by 1.875 times P when gravity loads combined with wind or earthquake forces govern the design.

J.3.5 Reinforcement

Column reinforcements shall comply with the requirements of Sec 6.3.9.

J.4 Flat Plates, Flat Slabs and Edge-Supported Slabs

J.4.1 General

General requirements for the design of slabs by working stress design method shall be the same as those specified in Sec 6.5 of Chapter 6.

The provisions of Sec 6.5 except those for nominal strength evaluation shall also be applicable along with the provisions of this section.

In using Sec 6.5, the word factored shall be read as working or working load whichever is applicable and the factor ϕ shall be taken as unity.

J.4.2 The shear strength of slabs in the vicinity of columns, concentrated loads or reactions is governed by the more severe of the following two conditions:

- (a) Beam action for slab, with critical section extending in a plane across the entire width and located at a distance d from the face of columns, concentrated loads or reaction. For this condition, the slab shall be designed in accordance with Sec J.2.7.1 through J.2.7.4.
- (b) Two way action for slab, with a critical section perpendicular to plane of slab and located so that its perimeter is a minimum, but need not approach closer than $d/2$ to:
 - (i) edges or corners of columns, concentrated loads or reaction areas or
 - (ii) change in slab thickness such as edges of capitals or drop panels.

For two way action, the slab shall be designed in accordance with Sec J.4.3 and J.4.4.

J.4.3 Design shear stress shall be computed by

$$v = \frac{V}{b_o d} \quad (6.J.33)$$

Where V and b_o shall be taken at the critical section defined in Sec J.4.2(b) above.

J.4.4 Design shear stress v shall not exceed v_c given by Eq. 6.J.34 unless shear reinforcement is provided.

$$v_c = 0.083 \left(1 + \frac{2}{\beta_c} \right) \sqrt{f'_c} \leq 0.17 \sqrt{f'_c} \quad (6.J.34)$$

Where β_c is the ratio of long side to short side of concentrated load or reaction area.

- J.4.5** If shear reinforcement consisting of bars or wires is used in accordance with Sec 6.4.3, v_c shall not exceed $0.083\sqrt{f'_c}$, and v shall not exceed $0.25\sqrt{f'_c}$.
- J.4.6** If shear reinforcement in the form of shear heads is used in accordance with Sec 6.4.10.4, v on the critical section, as defined in Sec J.4.2.(b) above, shall not exceed $0.29\sqrt{f'_c}$ and v on the critical section, as defined in Sec 6.4.10.4.7, shall not exceed $0.17\sqrt{f'_c}$. In using Equations 6.6.75 and 6.6.76, the quantity v_u shall be replaced by 2 times the design working shear force V .

J.5 Alternative Design of Two-Way Edge-Supported Slabs

- J.5.1** The provisions of this section may be used as alternative to those of Sec J.4 for two-way slabs supported on all four edges by walls, steel beams or monolithic concrete beams having a total depth not less than 3 times the slab thickness.
- J.5.2** The provisions of Sec 6.5.8 (except as may be superseded by the provisions of Appendix B), shall also form a part of this section. In using the provisions of Sec 6.5.8, the word factored shall be read as working or working load as the context implies, and the factor ϕ shall be taken as unity.

J.5.3 Analysis by the Coefficient Method

The slab may be analysed for the determination of negative moments and dead and live load positive moments in accordance with the provisions of Sec. 6.5.8.3.

J.5.4 Flexural Design of Slabs

The flexural design of slabs shall be performed in accordance with the provisions of Sec J.2.6.1.

J.5.5 Shear Strength of Slabs

The shear strength of slabs shall be provided in accordance with the requirements of Sec J.4.2 through J.4.6.

J.6 Ribbed and Hollow Slabs

General requirements for the design of ribbed and hollow slabs by the working stress design method shall be in accordance with Sec 6.5.9 Chapter 6. The provisions of Sec 6.5.9 except Sec 6.5.9.3 shall also form a part of this section.

J.6.1 In applying the provisions of Sec 6.5.9, the word factored shall be read as working or working load as the context implies, and the factor ϕ shall be taken as unity.

J.6.2 Ribbed and hollow slabs shall be designed for flexure in accordance with Sec J.2.6.

J.6.3 The shear strength of ribbed and hollow slabs shall be provided to satisfy the requirements of Sec J.4.2 through J.4.6, except as specified in Sec J.6.4 below.

J.6.4 For one-way ribbed and hollow slab construction, contribution of concrete to shear strength V_c is permitted to be 10 percent more than that specified in Sec J.2.7. It is allowed to increase shear strength using shear reinforcement or by widening the ends of ribs.

J.7 Framed Structures

J.7.1 Scope

The provisions of this section shall apply to rigidly jointed RC framed structures subject to lateral loads in addition to gravity loads.

J.7.2 Continuity

All intersections of members in a framed structure shall be continuous, with the steel reinforcements continued through the joints into the adjacent members to provide adequate development length. At construction joints, special care shall be taken to bond the new concrete to the old by carefully cleaning the latter, by extending the reinforcement through the joint and by other means.

J.7.3 Placement of Loads

All individual members and joints of the framed structure shall be designed for the worst combination of loads as provided in Sec 2. Gravity live loads in different bays and in different storeys of a framed structure shall be so arranged as to produce the maximum moment and shear at all critical sections.

J.7.4 Idealization

J.7.4.1 For the purpose of analysis, the members of the frame shall be represented by straight lines coincident with their centroidal axes. When the centroidal axes of the members meeting at a joint do not coincide at a single point, the effect of offset from the point representing the joint shall be taken into consideration.

J.7.4.2 Use of any set of reasonable assumptions is permitted for computing relative flexural and torsional stiffness of columns, walls, floors, and roof systems. The assumptions adopted shall be consistent throughout the analysis.

J.7.4.3 The moment of inertia of the frame members shall be based on the gross concrete cross-section.

J.7.4.4 Effect of haunches shall be considered both in determining moments and in the design of members.

J.7.4.5 Columns having their bases monolithically cast in a substantial foundation, which may be anchored to a solid rock mass or supported on piles with their tops encased in pile cap, or which is a continuous raft or mat, may be assumed to be fixed at their bases. Otherwise, the column bases shall be assumed to permit rotation. In either case, the foundation shall be designed to resist any moment that may be transferred to it from the structure in view of the assumptions made and the detailing used at the base.

J.7.5 Method of Analysis

J.7.5.1 Gravity Loads: For building frames with reasonably regular outline, not involving unusual asymmetry of loading or shape, moments due to gravity loads may be determined by dividing the entire frame into simpler sub-frames. Each sub-frame shall consist of one continuous beam, plus the top and bottom columns framing into that particular beam. The far ends of the columns, built integrally with the structure, shall be considered fixed. For the sub-frame at the bottom of the structure, the column end conditions at the base shall be dictated by the soil and foundation considerations in accordance with Sec J.7.4.5 above.

The arrangement of live load on the sub-frame may be limited to the combinations, (a) dead load on all spans with full live load on two adjacent spans, and (b) dead load on all spans with full live load on alternate spans.

For building frames not satisfying the requirements above, a full frame analysis using elastic method shall be carried out for gravity loads.

J.7.5.2 Lateral Loads: Any method of elastic analysis that satisfies equilibrium and compatibility requirements may be used for framed structures. Approximate methods that reduce the frame to a statically determinate structure by making simplifying assumptions shall not be used except for preliminary proportioning of sections for subsequent more accurate analysis.

J.7.6 Design

The frame members shall be designed for the shear, moment, torsion and axial force obtained from the elastic analysis. All members of frames shall be designed for the maximum effects of working loads using allowable working load stresses. The critical section for design for negative moment in beams may be assumed to be at the face of the support.

J.8 Deep Beams

J.8.1 Notation

All the notation used this Section are provided in Sec. J.1.1.

J.8.2 General

J.8.2.1 Flexural members with overall depth to clear span ratio greater than 0.4 for continuous spans, or 0.5 for simple spans, shall be designed as deep beams taking into account nonlinear distribution of strain and lateral buckling (See also Sec 8.2.7.6).

J.8.2.2 Shear strength of deep beams shall be provided in accordance with Sec J.8.4 below.

J.8.2.3 Minimum flexural tension reinforcement shall conform to Sec J.2.8.

J.8.2.4 Minimum horizontal and vertical reinforcement in the side faces of deep beams shall satisfy the requirements of Sec J.8.4.8, J.8.4.9 and J.8.4.10 below, but the reinforcement shall not be less than that required for walls in Sec 6.6.3.2 and 6.6.3.3.

J.8.3 Flexure

Deep flexural members shall be designed as beams. The lever arm, z , shall be computed in compatibility with Sec. 6.3.7, 6.4.6, and Appendix I.

J.8.4 Shear

J.8.4.1 The provisions of this section shall apply to members with $\frac{l_n}{d}$ less than 5 that are loaded on one face and supported on the opposite face so that compression stress can develop between the loads and the supports.

J.8.4.2 The design of simply supported deep beams for shear shall be based on Sec J.2.7.1. The shear strength provided by concrete, V_c , shall be computed in accordance with Sec J.8.4.6 or J.8.4.7 and that provided by steel, V_s , in accordance with Sec J.8.4.8.

J.8.4.3 The design of continuous deep beams for shear shall be based on Sections J.2.7.1 to J.2.7.5 or on any method satisfying equilibrium, compatibility and strength requirements. In either case the design shall also satisfy Sec J.8.4.4, J.8.4.9 and J.8.4.10 below.

J.8.4.4 Shear strength V_n for deep beams shall not be taken greater than $0.37\sqrt{f'_c}b_wd$ when $\frac{l_n}{d}$ is less than 2. When $\frac{l_n}{d}$ lies between 2 and 5,

$$V_n = 0.31 \left(10 + \frac{l_n}{d} \right) \sqrt{f'_c} b_w d \quad (6.J.35)$$

J.8.4.5 Critical section for shear shall be taken at a distance of $0.15 \ell_n$ for uniformly loaded beams and $0.50a$ for beams with concentrated loads, measured from the face of support, but in either case not greater than d .

J.8.4.6 Unless a more detailed calculation is made in accordance with Sec J.8.4.7, V_c shall be taken as

$$V_c = 0.091\sqrt{f'_c}b_wd \quad (6.J.36)$$

J.8.4.7 Shear strength V_c may be computed more accurately by

$$V_c = \left(1.93 - 1.38 \frac{M}{Vd} \right) \left(0.16\sqrt{f'_c} + 17.2\rho_w \frac{Vd}{M} \right) b_w d \quad (6.J.37)$$

Except that the term $\left(1.93 - 1.38 \frac{M}{Vd} \right)$ shall not exceed 1.38 and V_c shall not to be taken greater than $0.275\sqrt{f'_c}b_wd$.

J.8.4.8 Where shear force V exceeds shear strength V_c , shear reinforcement shall be provided to satisfy the requirement of Sec J.2.7.1. The shear strength, V_s , contributed by shear reinforcement shall be computed by

$$V_s = \left[\frac{A_v}{s} \left(\frac{1 + \frac{l_n}{d}}{12} \right) + \frac{A_{vh}}{s_1} \left(\frac{11 - \frac{l_n}{d}}{12} \right) \right] f_s d \quad (6.J.38)$$

Where A_v is the area of shear reinforcement perpendicular to flexural tension reinforcement within a distance s , and A_{vh} is the area of shear reinforcement parallel to flexural reinforcement within a distance s_1 .

J.8.4.9 Area of shear reinforcement A_v shall not be less than $0.0015 b_w s$, and s shall not exceed $d/5$, nor 450 mm.

J.8.4.10 The area of horizontal shear reinforcement A_{vh} shall not be less than $0.0025 b_w s_1$ and s_1 shall not exceed $d/3$, nor 450 mm.

J.8.4.11 Shear reinforcement required at the critical section defined in Sec J.8.4.5 shall be used throughout the span.

J.9 Reinforced Concrete Walls

J.9.1 General requirements for and analysis of reinforced concrete walls for design by the working stress design method shall be the same as those specified in Sec. 6.4.8 and Sec 6.6.

In applying the provision of Sec. 6.4.8 and Sec 6.6, the word factored shall be read as working or working load as the context implies.

J.9.2 Walls shall be designed in accordance with Sec 6.6 with flexural and axial load capacities taken as 40 percent of that computed using Sec 6.6. Strength reduction factor ϕ shall be taken equal to 1.0.

J.9.3 In computing the effect of slenderness, the quantity P_u shall be taken as $2.5P$ when gravity loads govern the design and as $1.875P$ when lateral loads combined with gravity loads govern the design, where P is the design working axial load in the wall.

J.9.4 Design of walls for shear shall be in accordance with the provisions of Sec 6.4.8 except the following :

J.9.4.1 Shear strengths provided by concrete and the limiting maximum strengths for shear shall be taken as 55 percent of the values given in Sec 6.4.8.

J.9.4.2 In Sec 6.4.8.6, N_u shall be replaced by 2 times the design axial load for tension and 1.2 times the design axial load for compression.

J.9.4.3 The terms V_u and M_u shall be replaced by their working load values V and M respectively.

J.10 Footings

J.10.1 General requirements for the design of footings by the working stress design method shall be the same as those specified in Sec 6.4.10 and 6.8.

J.10.2 In using the provisions of Sec 6.4.10 and 6.8, the word factored shall be read as working or working load as the context implies, and the value of strength reduction factor ϕ shall be taken as 1.0.

J.10.3 Footings (combined or isolated), mats or pile caps shall be designed to resist the service loads and induced reactions in accordance with the appropriate design requirements of this chapter.

J.10.4 For flexural design of footings, the provisions of Sec 6.8.4 shall be applicable.

J.10.5 Development of reinforcement shall be provided in accordance with Sec 6.8.6.

J.10.6 The requirements of Sec 6.8.8 for transfer of force at base shall be applicable except the following:

J.10.6.1 The limiting bearing stress in Sec 6.8.8.1.1 and 6.3.14.1 shall be $0.3f'_c$ instead of $0.85\phi f'_c$.

J.10.6.2 When supporting surface is wider on all sides than the loaded area, the limiting bearing stress in Sec 6.8.8.1.1 and 6.3.14.1 shall be $0.3f'_c\sqrt{A_2/A_1}$ instead of $0.85\phi f'_c\sqrt{A_2/A_1}$. Where, the root of the area ratio is not to be taken larger than 2.

J.10.7 The provisions of Sec 6.8.9 for sloped or stepped footings and Sec 6.8.10 for combined footings and mats shall be applicable.

J.10.8 Shear in Footings

J.10.8.1 Shear capacity of footings in the vicinity of concentrated loads or reactions is governed by the more severe of the following two conditions :

- (a) Beam action for footing, with a critical section extending in a plane across the entire width and located at a distance d from face of concentrated load or reaction area. For this condition, the footing shall be designed in accordance with Sec J.2.7.1 through J.2.7.4.
- (b) Two-way action for footing, with a critical section perpendicular to plane of footing and located so that its perimeter is a minimum, but the critical section need not approach closer than $d/2$ to perimeter of concentrated load or reaction area. For this condition, the footing shall be designed in accordance with Sec J.10.2.2 and J.10.2.3.

J.10.8.2 Design shear stress v shall be computed by

$$v = \frac{V}{b_o d} \quad (6.J.39)$$

Where V and b_o shall be taken at the critical section defined in J.10.8.1(b) above.

J.10.8.3 Design shear stress v shall not exceed v_c given by Eq. 6.J.40 unless shear reinforcement is provided

$$v_c = \left(0.083 + \frac{0.17}{\beta_c} \right) \sqrt{f'_c} \leq 0.17 \sqrt{f'_c} \quad (6.J.40)$$

Where β_c is the ratio of long side to short side for concentrated load or reaction area.

J.10.8.4 If shear reinforcement consisting of bars or wires is provided in the footings, v_c shall not exceed $0.083\sqrt{f'_c}$, and v shall not exceed $0.25\sqrt{f'_c}$. The required area of shear reinforcement A_v shall be calculated in accordance with Sec J.2.7.4 and anchored in accordance with Sec 8.2.

J.10.9 Pile Caps

J.10.9.1 Pile caps shall be designed either by bending theory or by truss analogy.

J.10.9.2 Truss analogy method

- (a) When truss method is used, the truss shall be of triangulated form, with a node at the centre of loaded area. The lower nodes of the truss shall lie at the intersections of the centre lines of the piles with the tensile reinforcement.
- (b) Where the truss method is used with widely spaced piles (spacing exceeding three times the pile diameter), only the reinforcement within a band width of 1.5 times the pile diameter from the centre of a pile shall be considered to constitute a tension member of the truss.

J.10.9.3 Beam shear in pile cap shall be checked at critical sections extending across the full width of the cap. Critical sections shall be assumed to be located at 20% of the diameter of the pile inside the face of the pile. The total force from all the piles with centres lying outside this line shall be considered to constitute the shear force on this section.

The shear force V on the critical section shall not exceed V_c , where

$$V_c = 0.4\sqrt{f'_c}bd(2d/a_v) \quad (6.J.41)$$

in which $2d/a_v$ shall be greater than or equal to 1.0, a_v is the distance from the face of the column to the critical section as defined above, and b shall be taken as the full width of the critical section if the spacing of the piles is less than or equal to 3 times the pile diameter d_p , otherwise b shall be equal to 3 times the pile diameter.

J.10.9.4 Punching Shear: A check shall be made to ensure that the shear stress calculated at the perimeter of the column for the working loads does not exceed $0.4\sqrt{f'_c}$ or 2.5 N/mm^2 , whichever is the smaller. In addition, if the spacing of the piles is greater than 3 times the pile diameter, punching shear shall be checked on the perimeter defined in Sec. J.10.9.3, in accordance with Sec 6.4.10.

J.10.9.5 Anchorage: The tension reinforcement shall be provided with full anchorage in accordance with Sec 8.2.

J.11 Stairs

Requirements for the design of stairs by the working stress design method shall be in accordance with Sec 6.7 except the following:

- (a) Staircases shall be designed to support design working loads in accordance with the provisions of Sec J.1.4.
- (b) The provisions for beams and one-way slabs given in Sec J.2 shall apply for the design of stairs.

J.12 Shells and Folded Plates

Requirements for the design of shells and folded plates by the working stress design method shall be in accordance with Sec 6.9 Chapter 6 except the following:

- (a) All provisions of Sections J.1 and J.2 shall apply to thin-shell structures.
- (b) A portion of the shell equal to the flange width specified in Sec J.2.4.5 may be assumed to act with the auxiliary member. In such portions of the shell, the reinforcement perpendicular to the auxiliary member shall be at least equal to that required for the flange of a T-beam by Sec J.2.8.4.
- (c) Reinforcement required to resist shell membrane forces shall be provided so that the design strength in every direction shall be at least equal to the component of the principal membrane forces in the shell in the same direction due to working loads.
- (d) Where the principal membrane tensile stress on the gross concrete area due to working loads exceeds $0.17\sqrt{f'_c}$ reinforcement shall not be spaced farther apart than three times the shell thickness.
- (e) Design for flexure shall be in accordance with Sec J.2.6.

J.13 Precast and Composite Construction

Requirements for the design of precast and composite construction by the working stress design method shall be in accordance with Sections 6.10, 6.2.5.4, 6.3.13, and 6.12 except the following:

J.13.1 For design of composite concrete flexural members, allowable horizontal shear strength V_h shall not exceed 55 percent of the horizontal shear strengths V_{nh} given in Sec 6.12.5.3.

J.13.2 When an entire composite member is assumed to resist vertical shear, design shall be in accordance with requirements of Sec J.2.7 as for a monolithically cast member of the same cross-sectional shape.

J.13.3 Design for flexure shall be in accordance with Sec J.2.6.

J.13.4 Shear-friction provision of Sec 6.4.5 shall be applied with limiting maximum stress for shear taken as 55 percent of that given. Allowable stress in shear friction reinforcement shall be that given in Sec J.1.10.

PART VI
Appendix K
Anchoring to Concrete

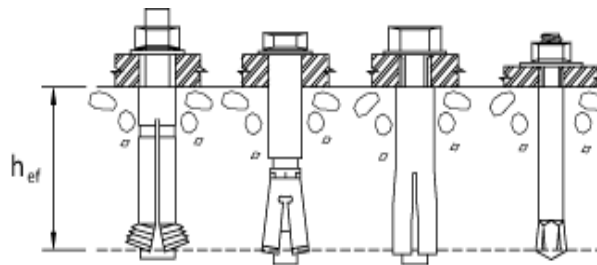
K.1 Definitions

ANCHOR	A steel element used to transmit applied loads either by casting into concrete or post-installed into a hardened concrete member. An anchor includes headed bolts, hooked bolts (J- or L-bolt), headed studs, expansion anchors, or undercut anchors.
ANCHOR GROUP	Anchor group is formed by a number of anchors of approximately equal effective embedment depth with each anchor spaced at less than $3h_{ef}$ from one or more adjacent anchors when subjected to tension, or $3c_{a1}$ from one or more adjacent anchors when subjected to shear. Only those anchors susceptible to the particular failure mode under investigation shall be included in the group.
CLARIFICATION FOR ANCHOR GROUP	Only those anchors susceptible to a particular failure mode out of all potential modes (steel, concrete breakout, pullout, side-face blowout, and pryout) should be considered when evaluating the strength associated with that failure mode.
ANCHOR PULLOUT STRENGTH	The strength corresponding to the anchoring device or a major component of the device sliding out from the concrete without breaking out a substantial portion of the surrounding concrete.
ANCHOR REINFORCEMENT	It is the reinforcement used to transfer the full design load from the anchors into the structural member. See Sec K.5.2.9 or Sec K.6.2.9.
CLARIFICATION FOR ANCHOR REINFORCEMENT	The design and detailing of anchor reinforcement is done specifically for the purpose of transferring anchor loads from the anchors into the structural member. Hairpins are generally used for this purpose (see Clarification for Sections K.5.2.9 and K.6.2.9); however, other configurations that can be shown to effectively transfer the anchor load are acceptable.
ATTACHMENT	The structural assembly, external to the surface of the concrete, that transmits loads to or receives loads from the anchor.

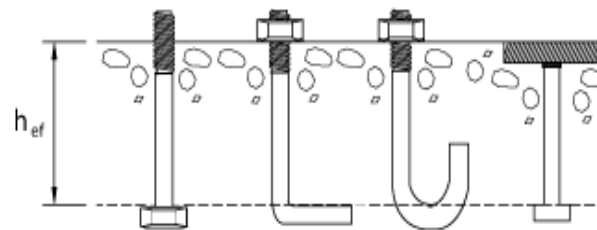
BRITTLE STEEL ELEMENT	An element having a tensile test elongation of less than 14 percent, or reduction in area of less than 30 percent, or both.
CLARIFICATION FOR BRITTLE STEEL ELEMENT AND DUCTILE STEEL ELEMENT	The gauge length specified in the appropriate ASTM standard for the steel shall be used for measuring the 14 percent elongation.
CAST-IN ANCHOR	An anchor either a headed bolt, headed stud, or hooked bolt installed before placing concrete.
CONCRETE BREAKOUT STRENGTH	The strength corresponding to which a volume of concrete surrounding the anchor or group of anchors separates from the member.
CONCRETE PRYOUT STRENGTH	The strength corresponding to formation of a concrete spall behind short, stiff anchors displaced in the direction opposite to the applied shear force.
DISTANCE SLEEVE	A sleeve that encases the center part of an undercut anchor, a torque-controlled expansion anchor, or a displacement-controlled expansion anchor without expanding.
DUCTILE STEEL ELEMENT	An element with a tensile test elongation of not less than 14 percent and reduction in area of at least 30 percent. A steel element meeting the requirements of ASTM A307 shall be considered ductile.
EDGE DISTANCE	The distance to the edge of the concrete surface from center of the anchor closest to edge.
EFFECTIVE EMBEDMENT DEPTH	It is the overall depth through which the anchor transfers force to or from the surrounding concrete. The effective embedment depth will normally be the depth of the concrete failure surface in tension applications. For cast-in headed anchor bolts and headed studs, the effective embedment depth is measured from the bearing contact surface of the head.
CLARIFICATION FOR EFFECTIVE EMBEDMENT DEPTH	Figure 6.K.1 illustrates the effective embedment depths for a variety of anchor types.

EXPANSION ANCHOR	It is a post-installed anchor, inserted into hardened concrete and it transfers loads to or from the concrete by direct bearing or friction or both. Expansion anchors may be torque-controlled, where the expansion is achieved by a torque acting on the screw or bolt; or displacement-controlled, where the expansion is achieved by impact forces acting on a sleeve or plug and the expansion is controlled by the length of travel of the sleeve or plug.
EXPANSION SLEEVE	The outer part of an expansion anchor that is forced outward by the center part, either by applied torque or impact, to bear against the sides of the predrilled hole.
FIVE PERCENT FRACTILE	Statistically it means 90 percent confidence that there is 95 percent probability of the actual strength exceeding the nominal strength.
CLARIFICATION FOR FIVE PERCENT FRACTILE	The coefficient K_{05} associated with the 5 percent fractile, $X_m - K_{05}S_s$ is determined depending on the number of tests, n , used to compute the sample mean, X_m , and sample standard deviation, S_s . Values of K_{05} range, for example, from 1.645 for $n = \infty$, to 2.010 for $n = 40$, and 2.568 for $n = 10$. With this definition of the 5 percent fractile, the nominal strength in K.4.2 is the same as the characteristic strength in ACI 355.2.
HEADED STUD	A steel anchor conforming to the requirements of AWS D1.1 and affixed to a plate or similar steel attachment by the stud arc welding process before casting.
HOKED BOLT	A cast-in anchor, which is anchored mainly by bearing of the 90° bend (L-bolt) or 180° bend (J-bolt) against the concrete, at its embedded end, and having a minimum e_h of $3d_a$.
POST-INSTALLED ANCHOR	An anchor installed in concrete, which is hardened. Expansion anchors and undercut anchors are examples of post-installed anchors.
PROJECTED AREA	The area on the free surface of the concrete member that is used to represent the larger base of the assumed rectilinear failure surface.
SIDE-FACE BLOWOUT STRENGTH	The strength of anchors with deeper embedment but thinner side cover corresponding to concrete spalling on the side face around the embedded head while no major breakout occurs at the top concrete surface.

SPECIALTY INSERT	Predesigned and prefabricated cast-in anchors specifically designed for attachment of bolted or slotted connections. Specialty inserts are often used for handling, transportation, and erection, but are also used for anchoring structural elements. Specialty inserts are not within the scope of this Appendix.
SUPPLEMENTARY REINFORCEMENT	Reinforcement which acts to restrain potential concrete breakout which is not designed to transfer full design load from the anchors into the structural member.
CLARIFICATION FOR SUPPLEMENTARY REINFORCEMENT	Supplementary reinforcement has a configuration and placement similar to anchor reinforcement but is not specifically designed to transfer loads from the anchors into the structural member. Stirrups for shear reinforcement may fall into this category.
UNDERCUT ANCHOR	It is a post-installed anchor that develops its tensile strength from the mechanical interlock provided by undercutting of the concrete at the embedded end of the anchor. The undercutting is achieved with a special drill before installing the anchor or alternatively by the anchor itself during its installation.



(a) Post-installed anchors



(b) Cast-in-place anchors

Figure 6.K.1 Types of anchors.

K.2 Scope

K.2.1 In this Appendix design requirements are provided for anchors in concrete used to transmit structural loads by means of tension, shear, or a combination of tension and shear between: (a) connected structural elements; or (b) safety-related attachments and structural elements. Safety levels specified are intended for in-service conditions, rather than for short-term handling and construction conditions.

Clarification for Section K.2.1: The scope of Appendix K is restricted to structural anchors that transmit structural loads related to strength, stability, or life safety. Two types of applications are envisioned. The first one is connections between structural elements where the failure of an anchor or an anchor group could result in loss of equilibrium or stability of any portion of the structure. The second one is where safety-related attachments that are not part of the structure (such as sprinkler systems, heavy suspended pipes, or barrier rails) are attached to structural elements. The levels of safety defined by the combinations of load factors and ϕ factors are appropriate for structural applications. Other standards may require more stringent safety levels during temporary handling.

K.2.2 This Appendix is applicable to both cast-in anchors and post-installed anchors. Specialty inserts, through bolts, multiple anchors connected to a single steel plate at the embedded end of the anchors, adhesive or grouted anchors, and direct anchors such as powder or pneumatic actuated nails or bolts, are not included. Reinforcement used as part of the embedment shall be designed in accordance with other parts of this Code.

Clarification for Section K.2.2: It is difficult to prescribe generalized tests and design equations for many insert types because of the wide variety of shapes and configurations of specialty inserts. Hence, they have been excluded from the scope of Appendix K. Adhesive anchors are widely used and can perform adequately. However, such anchors are outside the scope of this Appendix at this time.

K.2.3 Headed bolts and headed studs having a geometry that has been demonstrated to result in a pullout strength in uncracked concrete equal or exceeding $1.4N_p$ (where N_p is given by Eq. 6.K.15) are included. Hooked bolts that have a geometry that has been demonstrated to result in a pullout strength without the benefit of friction in uncracked concrete equal or exceeding $1.4N_p$ (where N_p is given by Eq. 6.K.16) are included. Post-installed anchors that meet the assessment requirements of ACI 355.2 are included. The suitability of the post-installed anchor for use in concrete shall have been demonstrated by the ACI 355.2 prequalification tests.

Clarification for Section K.2.3: Typical cast-in headed bolts and headed studs with geometries consistent with ANSI/ASME B1.1,^{K.1} B18.2.1,^{K.2} and B18.2.6^{K.3} have been tested and proven to behave predictably, so calculated pullout values are acceptable. Post-installed anchors do not have predictable pullout capacities, and therefore are required to be tested. For a post-installed anchor to be used in conjunction with the requirements of this Appendix, the results of the ACI 355.2 tests have to indicate that pullout failures exhibit an acceptable load-displacement characteristic or that pullout failures are precluded by another failure mode.

K.2.4 This Appendix does not cover the load applications that are predominantly high cycle fatigue or impact loads.

Clarification for Section K.2.4: The exclusion of load applications producing high cycle fatigue or extremely short duration impact (such as blast or shock wave), however, does not mean that seismic load effects are excluded from the scope. K.3.3 presents additional requirements for design when seismic loads are included.

K.3 General Requirements

K.3.1 Anchors and anchor groups shall be designed for critical effects of factored loads determined through elastic analysis. Plastic analysis approaches are permitted where nominal strength is controlled by ductile steel elements, provided that deformational compatibility is taken into account.

Clarification for Section K.3.1: If the strength of an anchor group is governed by breakage of the concrete, the behavior is brittle and there is limited redistribution of the forces between the highly stressed and less stressed anchors. In this case, the theory of elasticity is required to be used assuming the attachment that distributes loads to the anchors is sufficiently stiff. The forces in the anchors are considered to be proportional to the external load and its distance from the neutral axis of the anchor group.

If anchor strength is governed by ductile yielding of the anchor steel, significant redistribution of anchor forces can occur. In this case, an analysis based on the theory of elasticity will be conservative. References K.4 to K.6 discuss nonlinear analysis, using theory of plasticity, for the determination of the capacities of ductile anchor groups.

K.3.2 The design strength of anchors shall equal or exceed the largest required strength calculated from the applicable load combinations in Sec 6.2.2.

K.3.3 When anchor design includes earthquake forces for structures assigned to Seismic Design Category C, or D, the additional requirements of Sections K.3.3.1 to K.3.3.6 shall apply.

Clarification for Section K.3.3: Post-installed structural anchors are required to be qualified for Seismic Design Categories C, or D, by demonstrating the ability to undergo large displacements through several cycles as specified in the seismic simulation tests of ACI 355.2. Because ACI 355.2 excludes plastic hinge zones, Appendix K is not applicable to the design of anchors in plastic hinge zones under seismic forces. In addition, the design of anchors for earthquake forces is based on a more conservative approach by the introduction of 0.75 factor on the design strength ϕN_n and ϕV_n for the concrete failure modes, and by requiring the system to have adequate ductility. Anchor strength should be governed by ductile yielding of a steel element. If the anchor cannot meet these ductility requirements, then either the attachment is designed to yield or the calculated anchor strength is substantially reduced to minimize the possibility of a brittle failure. In designing attachments for adequate ductility, the ratio of yield to design strength should be considered. A connection element could yield only to result in a secondary failure as one or more elements strain harden and fail if the design strength is excessive when compared to the yield strength.

The full shear force should be assumed in any direction for a safe design as the direction of shear may not be predictable under seismic conditions.

K.3.3.1 The provisions of Appendix K are not applicable to the design of anchors in plastic hinge zones of concrete structures under earthquake forces.

Clarification for Section K.3.3.1: Section 3.1 of ACI 355.2 specifically states that the seismic test procedures do not simulate the behavior of anchors in plastic hinge zones. The possible higher level of cracking and spalling in plastic hinge zones are beyond the damage states for which Appendix K is applicable.

K.3.3.2 Post-installed structural anchors shall be qualified for use in cracked concrete and shall have passed the Simulated Seismic Tests in accordance with ACI 355.2. Pullout strength N_p and steel strength of the anchor in shear V_{sa} shall be based on the results of the ACI 355.2 Simulated Seismic Tests.

Clarification for Section K.3.3.2: Anchors that are not suitable for use in cracked concrete should not be used to resist seismic loads.

K.3.3.3 The design strength of anchors associated with concrete failure modes shall be taken as $0.75\phi N_n$ and $0.75\phi V_n$, where ϕ is given in K.4.4, and N_n and V_n are determined in accordance with Sections K.5.2, K.5.3, K.5.4, K.6.2, and K.6.3, assuming the concrete is cracked unless it can be demonstrated that the concrete remains uncracked.

Clarification for Section K.3.3.3: The anchor strength associated with concrete failure modes is to account for increased damage states in the concrete resulting from seismic actions. Because seismic design generally assumes that all or portions of the structure are loaded beyond yield, it is likely that the concrete is cracked throughout for the purpose of determining the anchor strength unless it can be demonstrated that the concrete remains uncracked.

K.3.3.4 Anchors shall be designed to be governed by the steel strength of a ductile steel element as determined in accordance with Sections K.5.1 and K.6.1, unless either Sec K.3.3.5 or Sec K.3.3.6 is satisfied.

Clarification for Section K.3.3.4: Ductile steel anchor elements are required to satisfy the requirements of K.1, Ductile Steel Element. For anchors loaded with a combination of tension and shear, the strength in all loading directions must be controlled by the steel strength of the ductile steel anchor element.

K.3.3.5 Instead of K.3.3.4, the attachment that the anchor is connecting to the structure shall be designed so that the attachment will undergo ductile yielding at a force level corresponding to anchor forces no greater than the design strength of anchors specified in K.3.3.3.

K.3.3.6 Alternative to K.3.3.4 and K.3.3.5, it shall be allowed to take the design strength of the anchors as 0.4 times the design strength determined in accordance with K.3.3.3. For the anchors of stud bearing walls, it shall be allowed to take the design strength of the anchors as 0.5 times the design strength determined in accordance with K.3.3.3.

Clarification for Section K.3.3.6: As a matter of desirable practice, a ductile failure mode in accordance with K.3.3.4 or K.3.3.5 should be provided for in the design of the anchor or the load should be transferred to anchor reinforcement in the concrete. Where geometric or material constraints do not permit, K.3.3.6 allows the design of anchors for nonductile failure modes at a reduced permissible strength to minimize the possibility of a brittle failure. The attachment of light frame stud walls typically involves multiple anchors that allow for load redistribution. This justifies the use of a less conservative factor for this case.

K.3.4 In this Appendix, modification factor λ for lightweight concrete shall be in accordance with Sec 6.1.8.1 unless specifically noted otherwise.

K.3.5 The values of f'_c used for calculation purposes in this Appendix shall not be greater than 70 MPa for cast-in anchors, and 55 MPa for post-installed anchors. Testing is required for post-installed anchors when used in concrete with f'_c exceeding 55 MPa.

Clarification for Section K.3.5: Limited tests of cast-in-place and post-installed anchors in high-strength concrete^{K.7} indicate that the design procedures contained in this Appendix become unconservative, particularly for cast-in anchors in concrete with compressive strengths in the range of 75 to 85 MPa. Until adequate test results are available, an upper limit on f'_c of 70 MPa has been imposed in the design of cast-in-place anchors. This is consistent with Sec. 6.4 and 8.2. The companion ACI 355.2 does not require testing of post-installed anchors in concrete with f'_c greater than 55 MPa because some post-installed anchors may have difficulty expanding in very high-strength concretes. Because of this, f'_c is limited to 55 MPa in the design of post-installed anchors unless testing is performed.

K.4 General Requirements for Strength of Anchors

K.4.1 Strength design of anchors shall be based either on computation using design models that satisfy the requirements of K.4.2, or on test evaluation using the 5 percent fractile of test results for the following:

- (a) Steel strength of anchor in tension (K.5.1);
- (b) Steel strength of anchor in shear (K.6.1);
- (c) Concrete breakout strength of anchor in tension (K.5.2);
- (d) Concrete breakout strength of anchor in shear (K.6.2);
- (e) Pullout strength of anchor in tension (K.5.3);
- (f) Concrete side-face blowout strength of anchor in tension (K.5.4); and
- (g) Concrete pryout strength of anchor in shear (K.6.3).

Anchors shall also have to satisfy the required edge distances, spacings, and thicknesses to preclude splitting failure, as required in K.8.

Clarification for Section K.4.1: This Section gives the requirements for establishing the strength of anchors to concrete. The various types of steel and concrete failure modes for anchors are shown in Figures K.4.1(a) and K.4.1(b). Comprehensive discussions of anchor failure modes are included in References K.8 to K.10. Any model that complies with the requirements of Sections K.4.2 and K.4.3 can be used to establish the concrete related strengths. For anchors such as headed bolts, headed studs, and post-installed anchors, the concrete breakout design methods of Sections K.5.2 and K.6.2 are acceptable. The anchor strength is also dependent on the pullout strength of Sec K.5.3, the side-face blowout strength of Sec K.5.4, and the minimum spacings and edge distances of Sec K.8. The design of anchors for tension recognizes that the strength of anchors is sensitive to appropriate installation; installation requirements are included in Sec K.9. Some post-installed anchors are less sensitive to installation errors and tolerances. This is reflected in varied ϕ factors based on the assessment criteria of ACI 355.2.

Test procedures can also be used to determine the single anchor breakout strength in tension and in shear. The test results, however, are required to be evaluated on a basis statistically equivalent to that used to select the values for the concrete breakout method “considered to satisfy” provisions of Sec K.4.2. The basic strength cannot be taken greater than the 5 percent fractile. The number of tests has to be sufficient for statistical validity and should be considered in the determination of the 5 percent fractile.

K.4.1.1 Except as required in Sec K.3.3, the design of anchors shall satisfy,

$$\phi N_n \geq N_{ua} \quad (6.K.1)$$

$$\phi V_n \geq V_{ua} \quad (6.K.2)$$

K.4.1.2 In Equations 6.K.1 and 6.K.2, ϕN_n and ϕV_n are the lowest design strengths determined from all appropriate failure modes. ϕN_n is the lowest design strength in tension of an anchor or group of anchors as determined from consideration of ϕN_{sa} , $\phi n N_{pn}$, either ϕN_{sb} or ϕN_{sbg} , and either ϕN_{cb} or ϕN_{cbg} . ϕV_n is the lowest design strength in shear of an anchor or a group of anchors as determined from consideration of: ϕV_{sa} , either ϕV_{sb} or ϕV_{sbg} , and either ϕV_{cb} or ϕV_{cbg} .

K.4.1.3 Interaction effects shall be considered in accordance with Sec K.4.3, when both N_{ua} and V_{ua} are present.

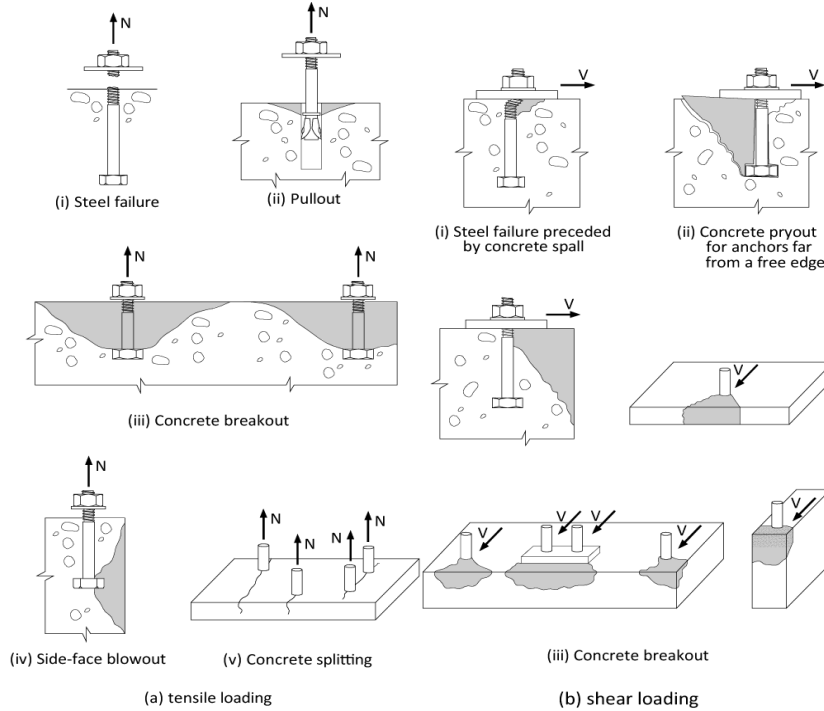


Figure 6.K.2 Failure modes for anchors.

K.4.2 For any anchor or group of anchors, the nominal strength shall be based on design models that result in predictions of strength in substantial agreement with results of comprehensive tests. The materials used in the tests shall be compatible with the materials used in the structure. The nominal strength shall be based on the 5 percent fractile of the basic individual anchor strength. For nominal strengths related to concrete strength, modifications for size effects, the number of anchors, the effects of close spacing of anchors, proximity to edges, depth of the concrete member, eccentric loadings of anchor groups, and presence or absence of cracking shall be taken into account. Limits on edge distances and anchor spacing in the design models shall be consistent with the tests that verified the model.

K.4.2.1 The effect of reinforcement provided to restrain the concrete breakout shall be permitted to be included in the design models used to satisfy Sec K.4.2. Where anchor reinforcement is provided in accordance with Sections K.5.2.9 and K.6.2.9, calculation of the concrete breakout strength in accordance with Sections K.5.2 and K.6.2 is not required.

Clarification for Section K.4.2.1: The addition of reinforcement in the direction of the load to restrain concrete breakout can greatly enhance the strength and deformation capacity of the anchor connection. Such enhancement is practical with cast-in anchors such as those used in precast sections.

References K.8, K.11, K.12, K.13, and K.14 provide information regarding the effect of reinforcement on the behavior of anchors. The effect of reinforcement is not included in the ACI 355.2 anchor acceptance tests or in the concrete breakout calculation method of Sections K.5.2 and K.6.2. The beneficial effect of supplementary reinforcement is recognized by the Condition A ϕ -factors in Sec K.4.4. Anchor reinforcement may be provided instead of calculating breakout strength using the provisions of Sec. 8.2 in conjunction with Sections K.5.2.9 and K.6.2.9.

The breakout strength of an unreinforced connection can be taken as an indication of the load at which significant cracking will occur. Such cracking can represent a serviceability problem if not controlled. (See Clarification for Sec K.6.2.1.)

K.4.2.2 When anchor diameters are not greater than 50 mm, and tensile embedments are not greater than 635 mm in depth, the concrete breakout strength requirements shall be considered satisfied by the design procedure of Sections K.5.2 and K.6.2.

Clarification for Section K.4.2.2: The method for concrete breakout design included as “considered to satisfy” K.4.2 was developed from the Concrete Capacity Design (CCD) Method,^{K.9,K.10} which was an adaptation of the κ Method^{K.15,K.16} and is considered to be accurate, relatively easy to apply, and capable of extension to irregular layouts. The CCD Method predicts the strength of an anchor or group of anchors by using a basic equation for tension, or for shear for a single anchor in cracked concrete, and multiplied by factors that account for the number of anchors, edge distance, spacing, eccentricity, and absence of cracking. The limitations on anchor size and embedment length are based on the current range of test data.

The breakout strength calculations are based on a model suggested in the κ Method. It is consistent with a breakout prism angle of approximately 35° [Figure 6.K.3(a) and (b)].

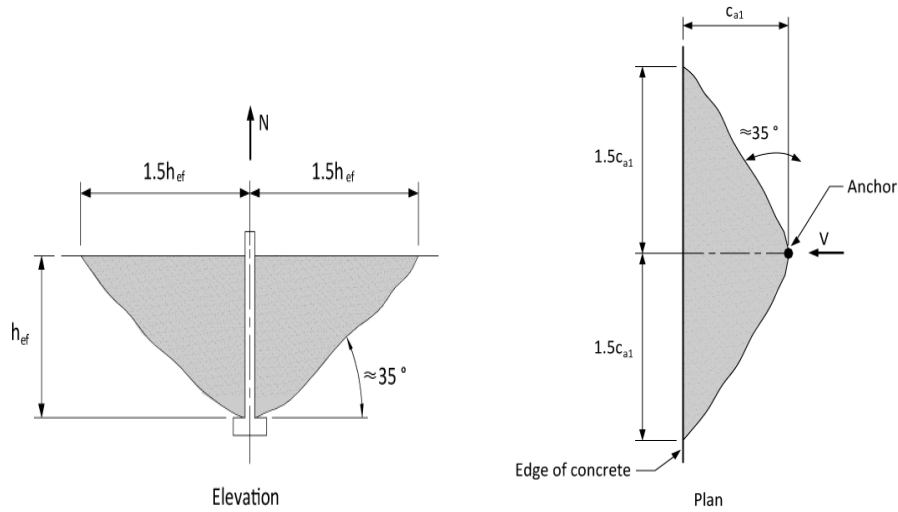


Figure 6.K.3(a) Breakout cone for tension **Figure 6.K.3(b) Breakout cone for shear**

K.4.3 Resistance to combined tensile and shear loads shall be considered in design using an interaction expression that results in computation of strength in substantial agreement with results of comprehensive tests. This requirement shall be considered satisfied by Sec K.7.

Clarification for Sections K.4.2 and K.4.3: Sections K.4.2 and K.4.3 establish the performance factors for which anchor design models are required to be verified. Many possible design approaches exist and the user is always permitted to “design by test” using Sec K.4.2 as long as sufficient data are available to verify the model.

K.4.4 For anchors in concrete, strength reduction factor ϕ shall be as follows when the load combinations of Sec 6.2.2 are used:

- (a) When strength of a ductile steel element governs anchor
 - (i) Tension loads 0.75
 - (ii) Shear loads 0.65
- (b) When strength of a brittle steel element governs anchor
 - (i) Tension loads 0.65
 - (ii) Shear loads 0.60

- (c) When concrete breakout, side-face blowout, pullout, or pryout strength governs anchor

		Condition A	Condition B
(i)	Shear Loads	0.75	0.70
(ii)	Tension loads		
	Cast-in headed studs, headed bolts, or hooked bolts	0.75	0.70
	Post-installed anchors with category as determined from ACI 355.2		
	Category 1	0.75	0.65
	(Low sensitivity to installation and high reliability)		
	Category 2	0.65	0.55
	(Medium sensitivity to installation and medium reliability)		
	Category 3	0.55	0.45
	(High sensitivity to installation and lower reliability)		

Condition A applies where supplementary reinforcement is present except for pullout and pryout strengths. Condition B applies where supplementary reinforcement is not present, and for pullout or pryout strength.

Clarification for Section K.4.4: The ϕ factors for steel strength are based on using f_{uta} to determine the nominal strength of the anchor (see Sections K.5.1 and K.6.1) rather than f_{ya} as used in the design of reinforced concrete members. Although the ϕ factors for use with f_{uta} appear low, they result in a level of safety consistent with the use of higher ϕ factors applied to f_{ya} . The smaller ϕ factors for shear than for tension do not reflect basic material differences but rather account for the possibility of a non-uniform distribution of shear in connections with multiple anchors. It is acceptable to have a ductile failure of a steel element in the attachment if the attachment is designed so that it will undergo ductile yielding at a load level corresponding to anchor forces no greater than the minimum design strength of the anchors specified in Sec K.3.3. (See Sec K.3.3.5.)

Two conditions are recognized for anchors governed by the more brittle concrete breakout or blowout failure. If supplementary reinforcement is present (Condition A), greater deformation capacity is provided than in the case where such supplementary reinforcement is not present (Condition B). An explicit design of supplementary

reinforcement is not required. However, the arrangement of supplementary reinforcement should generally conform to that of the anchor reinforcement shown in Figures 6.K.7 and 6.K.11(b). Full development is not required. The strength reduction factors for anchor reinforcement are given in Sections 6.K.7 and K.6.2.9.

The ACI 355.2 tests for sensitivity to installation procedures determine the category appropriate for a particular anchoring device. In the ACI 355.2 tests, the effects of variability in anchor torque during installation, tolerance on drilled hole size, energy level used in setting anchors, and for anchors approved for use in cracked concrete, increased crack widths are considered. The three categories of acceptable post-installed anchors are:

Category 1: low sensitivity to installation and high reliability;

Category 2: medium sensitivity to installation and medium reliability; and

Category 3: high sensitivity to installation and lower reliability.

The capacities of anchors under shear loads are not as sensitive to installation errors and tolerances. Therefore, for shear calculations of all anchors, $\phi = 0.75$ for Condition A and $\phi = 0.70$ for Condition B.

K.5 Design Requirements for Tensile Loading

K.5.1 Steel strength of anchor in tension

K.5.1.1 The nominal strength of an anchor in tension as governed by the steel, N_{sa} , shall be evaluated by calculations based on the properties of the anchor material and the physical dimensions of the anchor.

K.5.1.2 The nominal strength of a single anchor or group of anchors in tension, N_{sa} , shall not be greater than

$$N_{sa} = nA_{se,N}f_{uta} \quad (6.K.3)$$

Where, n is the number of anchors in the group, $A_{se,N}$ is the effective cross-sectional area of a single anchor in tension, mm², and f_{uta} shall not be taken greater than the smaller of $1.9f_{ya}$ and 860 MPa.

Clarification for Section K.5.1.2: The nominal strength of anchors in tension is best represented as a function of f_{uta} rather than f_{ya} because the large majority of anchor materials do not exhibit a well-defined yield point. The American Institute of Steel Construction (AISC) has based tension strength of anchors on $A_{se,N}f_{uta}$ since the 1986 edition of their specifications. The use of Eq. 6.K.3 with Sec 6.2.2 load factors and the ϕ -factors of K.4.4 give design strengths consistent with the AISC Load and Resistance Factor Design Specifications.^{K.19}

The limitation of $1.9f_{ya}$ on f_{uta} is to ensure that, under service load conditions, the anchor does not exceed f_{ya} . The limit on f_{uta} of $1.9f_{ya}$ was determined by converting the LRFD provisions to corresponding service level conditions. For Section 6.2.2, the average load factor of 1.4 (from $1.2D + 1.7L$) divided by the highest ϕ -factor (0.75 for tension) results in a limit of f_{uta}/f_{ya} of $1.4/0.75 = 1.87$. The serviceability limitation of f_{uta} was taken as $1.9f_{ya}$. If the ratio of f_{uta} to f_{ya} exceeds this value, the anchoring may be subjected to service loads above f_{ya} under service loads. Although not a concern for standard structural steel anchors (maximum value of f_{uta}/f_{ya} is 1.6 for ASTM A307), the limitation is applicable to some stainless steels. The effective cross-sectional area of an anchor should be provided by the manufacturer of expansion anchors with reduced cross-sectional area for the expansion mechanism. For threaded bolts, ANSI/ASME B1.1^{K.1} defines $A_{se,N}$ as

$$A_{se,N} = \frac{\pi}{4} \left(d_a - \frac{0.9743}{n_t} \right)^2, \text{ Where } n_t \text{ is the number of threads per mm.}$$

K.5.2 Concrete Breakout Strength of Anchor in Tension

K.5.2.1 The nominal concrete breakout strength, N_{cb} or N_{cbg} , of a single anchor or group of anchors in tension shall not exceed

(a) For a single anchor

$$N_{cb} = \frac{A_{NC}}{A_{NCO}} \psi_{ed,N} \psi_{c,N} \psi_{cp,N} N_b \quad (6.K.4)$$

(b) For a group of anchors

$$N_{cbg} = \frac{A_{NC}}{A_{NCO}} \psi_{ec,N} \psi_{ed,N} \psi_{c,N} \psi_{cp,N} N_b \quad (6.K.5)$$

Factors $\psi_{ec,N}$, $\psi_{ed,N}$, $\psi_{c,N}$, and $\psi_{cp,N}$ are defined in Sections K.5.2.4, K.5.2.5, K.5.2.6, and K.5.2.7, respectively. A_{NC} is the projected concrete failure area of a single anchor or group of anchors that shall be approximated as the base of the rectilinear geometrical figure that results from projecting the failure surface outward $1.5h_{ef}$ from the centerlines of the anchor, or in the case of a group of anchors, from a line through a row of adjacent anchors. A_{NC} shall not exceed nA_{NCO} , where, n is the number of tensioned anchors in the group. A_{NCO} is the projected concrete failure area of a single anchor with an edge distance equal to or greater than $1.5h_{ef}$.

$$A_{NCO} = 9h_{ef}^2 \quad (6.K.6)$$

Clarification for Section K.5.2.1: The effects of multiple anchors, spacing of anchors, and edge distance on the nominal concrete breakout strength in tension are included by applying the modification factors A_{NC}/A_{NCO} and $\psi_{ed,N}$ in Eq. 6.K.4 and 6.K.5. Figure 6.K.4(a) shows A_{NCO} and the development of Eq. 6.K.6. A_{NCO} is the maximum projected area for a single anchor. Figure 6.K.4(b) shows examples of the projected areas for various single-anchor and multiple-anchor arrangements. Because A_{NC} is the total

projected area for a group of anchors, and A_{Nco} is the area for a single anchor, there is no need to include n , the number of anchors, in Eq. 6.K.4 or 6.K.5. If anchor groups are positioned in such a way that their projected areas overlap, the value of A_{Nc} is required to be reduced accordingly.

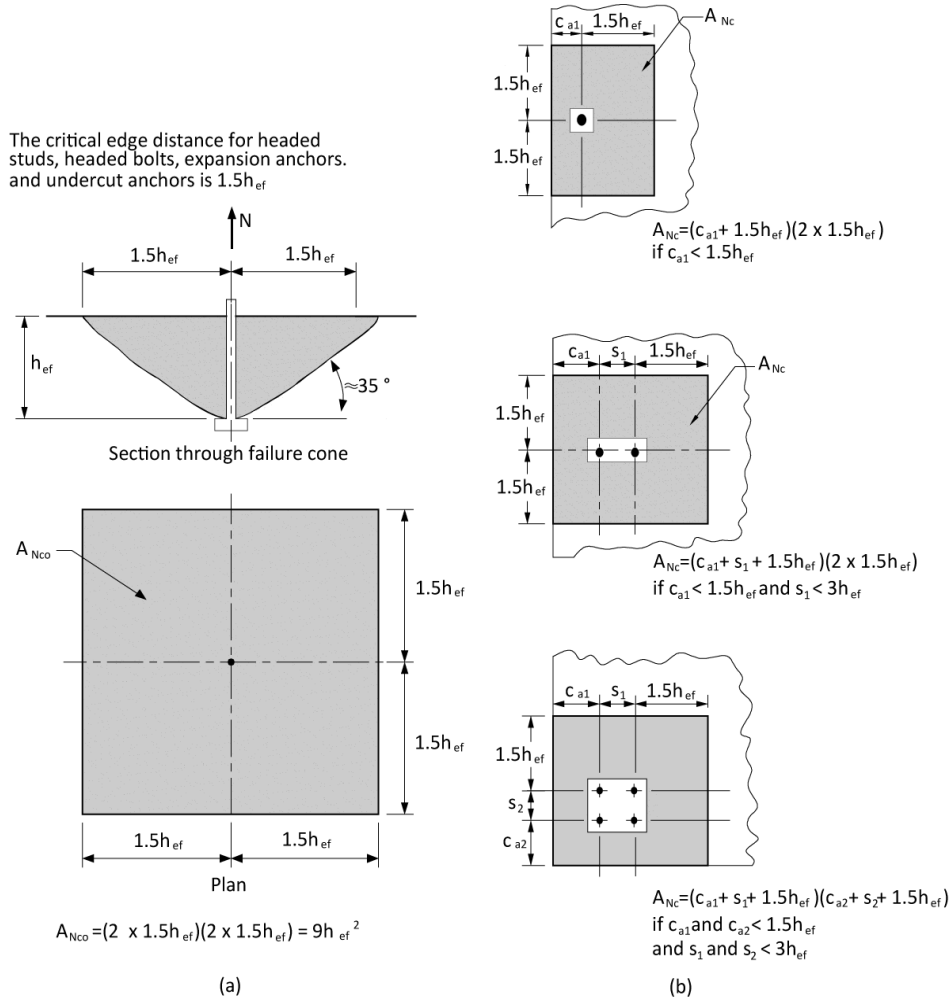


Figure 6.K.4(a) Calculation of A_{Nco} ; and (b) calculation of A_{Nc} for single anchors and groups of anchors.

K.5.2.2 The basic concrete breakout strength of a single anchor in tension in cracked concrete, N_b , shall not exceed

$$N_b = k_c \lambda \sqrt{f'_c} (h_{ef})^{1.5} \quad (6.K.7)$$

Where,

$k_c = 10$ for cast-in anchors; and

$k_c = 7$ for post-installed anchors.

The value of k_c for post-installed anchors shall be permitted to be increased above 7 based on ACI 355.2 product-specific tests, but shall in no case exceed 10. Alternatively, for cast-in headed studs and headed bolts with $280 \text{ mm} \leq h_{ef} \leq 635 \text{ mm}$, N_b shall not exceed

$$N_b = 3.9\lambda\sqrt{f'_c}(h_{ef})^{5/3} \quad (6.K.8)$$

Clarification for Section K.5.2.2: The basic equation for anchor strength was derived^{K.9-K.11, K.16} assuming a concrete failure prism with an angle of about 35° , considering fracture mechanics concepts.

The values of k_c in Eq. 6.K.7 were determined from a large database of test results in uncracked concrete^{K.9} at the 5 percent fractile. The values were adjusted to corresponding k_c values for cracked concrete.^{K.10, K.20} Higher k_c values for post-installed anchors may be permitted, provided they have been determined from product approval testing in accordance with ACI 355.2. For anchors with a deep embedment ($h_{ef} > 280 \text{ mm}$) test evidence indicates the use of $(h_{ef})^{1.5}$ can be overly conservative for some cases. Often, such tests have been with selected aggregates for special applications. An alternative expression (Eq. 6.K.8) is provided using $(h_{ef})^{5/3}$ for evaluation of cast-in anchors with $280 \text{ mm} \leq h_{ef} \leq 635 \text{ mm}$. The limit of 635 mm corresponds to the upper range of test data. This expression can also be appropriate for some undercut post-installed anchors. However, for such anchors, the use of Eq. 6.K.8 should be justified by test results in accordance with Sec K.4.2.

K.5.2.3 If anchors are located less than $1.5h_{ef}$ from three or more edges, the value of h_{ef} used in Eq. 6.K.4 to 6.K.11 shall be the greater of $C_{a, \max}/1.5$ and one-third of the maximum spacing between anchors within the group.

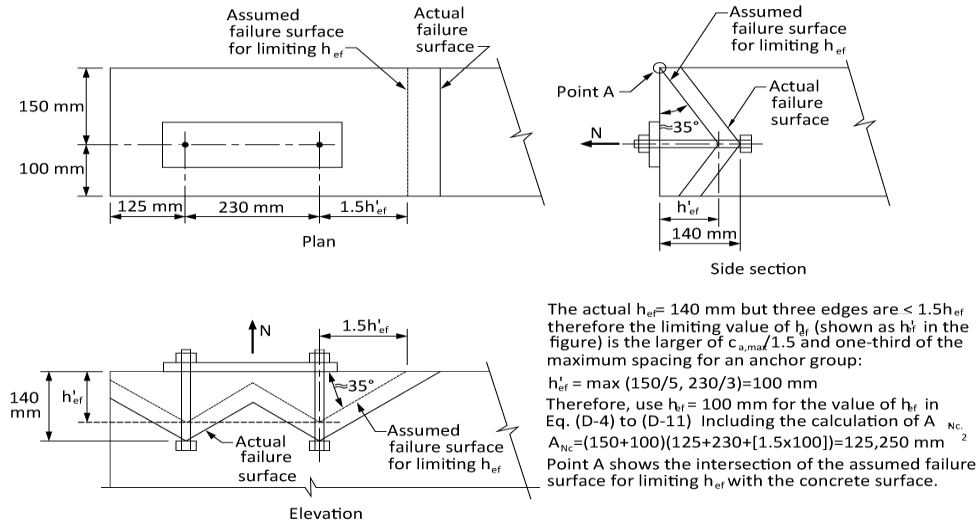


Figure 6.K.5 Tension in narrow members.

Clarification for Section K.5.2.3: For anchors located less than $1.5h_{ef}$ from three or more edges, the tensile breakout strength computed by the CCD Method, which is the basis for Eq. 6.K.4 to 6.K.11, gives overly conservative results.^{K.21} This occurs because the ordinary definitions of A_{Nc}/A_{Nco} do not correctly reflect the edge effects. This problem is corrected by limiting the value of h_{ef} used in Eq. 6.K.4 through 6.K.11 to $C_{a,max}/1.5$, where $C_{a,max}$ is the largest of the influencing edge distances that are less than or equal to the actual $1.5h_{ef}$. In no case should $C_{a,max}/1.5$ be taken less than one-third of the maximum spacing between anchors within the group. The limit on h_{ef} of at least one-third of the maximum spacing between anchors within the group prevents the use of a calculated strength based on individual breakout prisms for a group anchor configuration.

This approach is illustrated in Figure 6.K.5. In this example, the proposed limit on the value of h_{ef} to be used in the computations where, $h_{ef} = C_{a,max}/1.5$, results in $h_{ef} = 100$ mm. For this example, this would be the proper value to be used for h_{ef} in computing the resistance even if the actual embedment depth is larger. The requirement of K.5.2.3 may be visualized by moving the actual concrete breakout surface, which originates at the actual h_{ef} , toward the surface of the concrete parallel to the applied tension load. The value of h_{ef} used in Eq. 6.K.4 to 6.K.11 is determined when either: (a) the outer boundaries of the failure surface first intersect a free edge; or (b) the intersection of the breakout surface between anchors within the group first intersects the surface of the concrete. For the example shown in Figure 6.K.5, Point “A” defines the intersection of the assumed failure surface for limiting h_{ef} with the concrete surface.

K.5.2.4 The modification factor for anchor groups loaded eccentrically in tension, $\psi_{ec,N}$, shall be computed as

$$\psi_{ec,N} = \frac{1}{\left(1 + \frac{2e'_N}{3h_{ef}}\right)} \quad (6.K.9)$$

But, $\psi_{ec,N}$ shall not be taken greater than 1.0. If the loading on an anchor group is such that only some anchors are in tension, only those anchors that are in tension shall be considered when determining the eccentricity e'_N for use in Eq. 6.K.9 and for the calculation of N_{cbg} in Eq. 6.K.5.

In the case where eccentric loading exists about two axes, the modification factor, $\psi_{ec,N}$, shall be computed for each axis individually and the product of these factors used as $\psi_{ec,N}$ in Eq. 6.K.5.

Clarification for Section K.5.2.4: Figure 6.K.6(a) shows a group of anchors that are all in tension but the resultant force is eccentric with respect to the centroid of the anchor group. Groups of anchors can also be loaded in such a way that only some of the anchors are in tension (Figure 6.K.6(b)). In this case, only the anchors in tension are to be considered in the determination of e'_N . The anchor loading has to be determined as the resultant anchor tension at an eccentricity with respect to the center of gravity of the anchors in tension.

K.5.2.5 For single anchors or anchor groups loaded in tension, the modification factor for edge effects, $\psi_{ed,N}$, shall be computed as

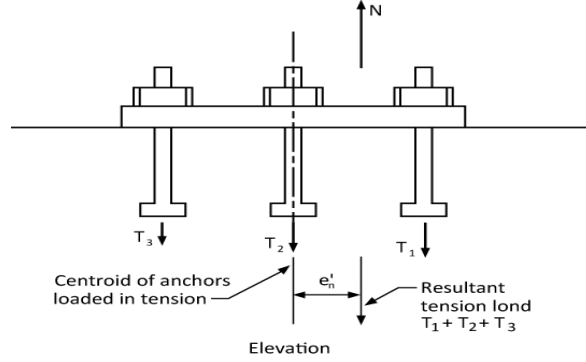
For, $c_{a,min} \geq 1.5h_{ef}$

$$\psi_{ed,N} = 1.0 \quad (6.K.10)$$

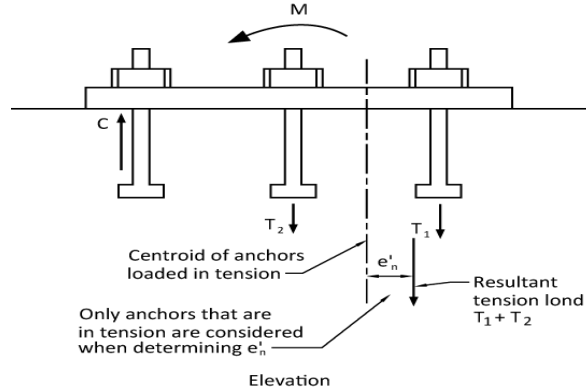
For, $c_{a,min} < 1.5h_{ef}$

$$\psi_{ed,N} = 0.7 + 0.3 \frac{c_{a,min}}{1.5h_{ef}} \quad (6.K.11)$$

Clarification for Section K.5.2.5: When anchors are located close to an edge so that there is not enough space for a complete breakout prism to develop, the strength of the anchor is further reduced beyond that reflected in A_{Nc}/A_{Nco} . If the smallest side cover distance is greater than or equal to $1.5h_{ef}$, a complete prism can form and there is no reduction ($\psi_{ec,N} = 1$). If the side cover is less than $1.5h_{ef}$, the factor $\psi_{ed,N}$ is required to adjust for the edge effect.^{K.9}



(a) Where all anchors in a group are in tension



(b) Where only some anchors in a group are in tension

Figure 6.K.6 Definition of e'_N for a group of anchors.

K.5.2.6 When anchors are located in a region of a concrete member where analysis indicates no cracking at service load levels, the following modification factor shall be permitted:

$\psi_{c,N} = 1.25$ for cast-in anchors; and

$\psi_{c,N} = 1.4$ for post-installed anchors, where the value of k_c used in Eq. 6.K.7 is 7.

Where the value of k_c used in Eq. 6.K.7 is taken from the ACI 355.2 product evaluation report for post-installed anchors qualified for use in both cracked and uncracked concrete, the values of k_c and $\psi_{c,N}$ shall be based on the ACI 355.2 product evaluation report.

Where the value of k_c used in Eq. 6.K.7 is taken from the ACI 355.2 product evaluation report for post-installed anchors qualified for use in uncracked concrete, $\psi_{c,N}$ shall be taken as 1.0.

When analysis indicates cracking at service load levels, $\psi_{c,N}$ shall be taken as 1.0 for both cast-in anchors and post-installed anchors. Post-installed anchors shall be qualified for use in cracked concrete in accordance with ACI 355.2. The cracking in the concrete shall be controlled by flexural reinforcement distributed in accordance with 10.6.4, or equivalent crack control shall be provided by confining reinforcement.

Clarification for Section K.5.2.6: Post-installed and cast-in anchors that have not met the requirements for use in cracked concrete according to ACI 355.2 should be used in uncracked regions only. The analysis for the determination of crack formation should include the effects of restrained shrinkage (see Sec. 8.1). The anchor qualification tests of ACI 355.2 require that anchors in cracked concrete zones perform well in a crack that is 0.3 mm wide. If wider cracks are expected, confining reinforcement to control the crack width to about 0.3 mm should be provided.

The concrete breakout strengths given by Eq. 6.K.7 and 6.K.8 assume cracked concrete (that is, $\psi_{c,N} = 1.0$) with $\psi_{c,N}k_c = 10 = 10$ for cast-in-place, and 7 for post-installed (cast-in 40 percent higher). When the uncracked concrete $\psi_{c,N}$ factors are applied (1.25 for cast-in, and 1.4 for post-installed), the results are $\psi_{c,N}k_c$ factors of 13 for cast-in and 10 for post-installed (25 percent higher for cast-in). This agrees with field observations and tests that show cast-in anchor strength exceeds that of post-installed for both cracked and uncracked concrete.

K.5.2.7 The modification factor for post-installed anchors designed for uncracked concrete in accordance with Sec K.5.2.6 without supplementary reinforcement to control splitting, $\psi_{cp,N}$, shall be computed as follows using the critical distance c_{ac} as defined in Sec K.8.6.

For, $c_{a,min} \geq c_{ac}$

$$\psi_{cp,N} = 1.0 \quad (6.K.12)$$

For, $c_{a,min} < c_{ac}$

$$\psi_{cp,N} = \frac{c_{a,min}}{c_{ac}} \quad (6.K.13)$$

But $\psi_{cp,N}$ determined from Eq. 6.K.13 shall not be taken less than $1.5h_{ef}/C_{ac}$, where the critical distance C_{ac} is defined in Sec K.8.6. For all other cases, including cast-in anchors, $\psi_{cp,N}$ shall be taken as 1.0.

Clarification for Section K.5.2.7: The design provisions in K.5 are based on the assumption that the basic concrete breakout strength can be achieved if the minimum edge distance, $C_{a,min}$, equals $1.5h_{ef}$. However, test results^{K.22} indicate that many torque-controlled and displacement-controlled expansion anchors and some undercut anchors require minimum edge distances exceeding $1.5h_{ef}$ to achieve the basic concrete breakout strength when tested in uncracked concrete without supplementary reinforcement to control splitting. When a tension load is applied, the resulting tensile stresses at the embedded end of the anchor are added to the tensile stresses induced due to anchor installation, and splitting failure may occur before reaching the concrete breakout strength defined in Sec K.5.2.1. To account for this potential splitting mode of failure, the basic concrete breakout strength is reduced by a factor $\psi_{cp,N}$ if $C_{a,min}$ is less than the critical edge distance C_{ac} . If supplementary reinforcement to control splitting is present or if the anchors are located in a region where analysis indicates cracking of the concrete at service loads, then the reduction factor $\psi_{cp,N}$ is taken as 1.0. The presence of supplementary reinforcement to control splitting does not affect the selection of Condition A or B in Sec K.4.4.

K.5.2.8 Where an additional plate or washer is added at the head of the anchor, it shall be permitted to calculate the projected area of the failure surface by projecting the failure surface outward $1.5h_{ef}$ from the effective perimeter of the plate or washer. The effective perimeter shall not exceed the value at a section projected outward more than the thickness of the washer or plate from the outer edge of the head of the anchor.

K.5.2.9 Where anchor reinforcement is developed in accordance with Sec. 8.3 on both sides of the breakout surface, the design strength of the anchor reinforcement shall be permitted to be used instead of the concrete breakout strength in determining ϕN_n . A strength reduction factor of 0.75 shall be used in the design of the anchor reinforcement.

Clarification for Section K.5.2.9: For conditions where the factored tensile force exceeds the concrete breakout strength of the anchor(s) or where the breakout strength is not evaluated, the nominal strength can be that of anchor reinforcement properly anchored as illustrated in Figure 6.K.7. Care needs to be taken in the selection and positioning of the anchor reinforcement. The anchor reinforcement should consist of stirrups, ties, or hairpins placed as close as practicable to the anchor. Only reinforcement spaced less than $0.5h_{ef}$ from the anchor centerline should be included as anchor reinforcement. The research^{K.14} on which these provisions is based was limited to anchor reinforcement with maximum diameter similar to a No. 16 bar. It is beneficial for the anchor reinforcement to enclose the surface reinforcement. In sizing the anchor reinforcement, use of a 0.75 strength reduction factor ϕ is recommended as is used for strut-and-tie models. As a practical matter, use of anchor reinforcement is generally limited to cast-in-place anchors.

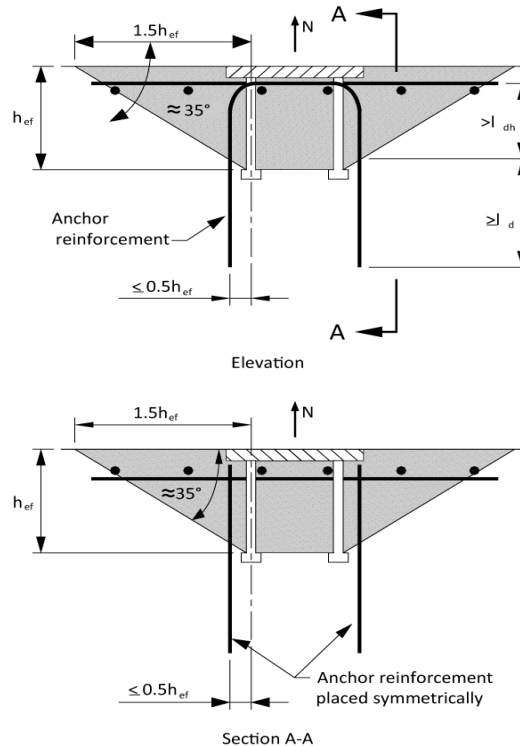


Figure 6.K.7 Anchor reinforcement for tension.

K.5.3 Pullout Strength of Anchor in Tension

K.5.3.1 The nominal pullout strength of a single anchor in tension, N_{pn} , shall not exceed

$$N_{pn} = \psi_{c,p} N_p \quad (6.K.14)$$

Where, $\psi_{c,p}$ is defined in K.5.3.6.

K.5.3.2 The values of N_p for post-installed expansion and undercut anchors shall be based on the 5 percent fractile of results of tests performed and evaluated according to ACI 355.2. It is not permissible to calculate the pullout strength in tension for such anchors.

Clarification for Section K.5.3.2: The pullout strength equations given in Sections K.5.3.4 and K.5.3.5 are only applicable to cast-in headed and hooked anchors;^{K.8.K.23} they are not applicable to expansion and undercut anchors that use various mechanisms for end anchorage unless the validity of the pullout strength equations are verified by tests.

K.5.3.3 It shall be permitted to evaluate the pullout strength in tension for single cast-in headed studs and headed bolts using Sec K.5.3.4. For single J- or L-bolts, it shall be permitted to evaluate the pullout strength in tension using K.5.3.5. Alternatively, it shall be permitted to use values of N_p based on the 5 percent fractile of tests performed and evaluated in the same manner as the ACI 355.2 procedures but without the benefit of friction.

Clarification for Section K.5.3.3: The pullout strength in tension of headed studs or headed bolts can be increased by providing confining reinforcement, such as closely spaced spirals, throughout the head region. This increase can be demonstrated by tests.

K.5.3.4 For a single headed stud or headed bolt, the pullout strength in tension, N_p , for use in Eq. 6.K.14, shall not exceed

$$N_p = 8A_{brg}f'_c \quad (6.K.15)$$

Clarification for K.5.3.4: The value computed from Eq. 6.K.15 corresponds to the load at which crushing of the concrete occurs due to bearing of the anchor head.^{K.8.K.13} It is not the load required to pull the anchor completely out of the concrete, so the equation contains no term relating to embedment depth. Local crushing of the concrete greatly reduces the stiffness of the connection, and generally will be the beginning of a pullout failure.

K.5.3.5 For a single hooked bolt, the pullout strength in tension, N_p , for use in Eq. 6.K.14 shall not exceed

$$N_p = 0.9f'_c e_h d_a \quad (6.K.16)$$

Where, $3d_a \leq e_h \leq 4.5d_a$

Clarification for Section K.5.3.5: Eq. 6.K.16 for hooked bolts was developed by Lutz based on the results of Reference K.23. Reliance is placed on the bearing component only, neglecting any frictional component because crushing inside the hook will greatly reduce the stiffness of the connection, and generally will be the beginning of pullout failure. The limits on e_h are based on the range of variables used in the three tests programs reported in Reference K.23.

K.5.3.6 When an anchor is located in a region of a concrete member where analysis indicates no cracking at service load levels, the following modification factor shall be permitted $\psi_{c,p} = 1.4$ Where analysis indicates cracking at service load levels, $\psi_{c,p}$ shall be taken as 1.0.

K.5.4 Concrete Side-Face Blowout Strength of a Headed Anchor in Tension

Clarification for Section K.5.4: Concrete side-face blowout strength of a headed anchor in tension. For side-face blowout, the design requirements are based on the recommendations of Reference K.24. These requirements are applicable to headed anchors that usually are cast-in anchors. Splitting during installation rather than side-face blowout generally governs post-installed anchors, and is evaluated by the ACI 355.2 requirements.

K.5.4.1 With deep embedment close to an edge ($h_{ef} > 2.5c_{a1}$), the nominal side-face blowout strength, $h_{ef} > N_{sb}$, of a single headed anchor shall not exceed

$$N_{sb} = (13c_{a1}\sqrt{A_{brg}})\lambda\sqrt{f'_c} \quad (6.K.17)$$

If c_{a2} for the single headed anchor is less than $3c_{a1}$, the value of N_{sb} shall be multiplied by the factor $\frac{1+\frac{c_{a2}}{c_{a1}}}{4}$, where, $1.0 \leq c_{a2}/c_{a1} \leq 3.0$.

K.5.4.2 For multiple headed anchors with deep embedment close to an edge ($h_{ef} > 2.5hc_{a1}$) and anchor spacing less than $6c_{a1}$, the nominal strength of those anchors susceptible to a side-face blowout failure N_{sbg} shall not exceed

$$N_{sbg} = \left(1 + \frac{s}{6c_{a1}}\right)N_{sb} \quad (6.K.18)$$

Where, s is the distance between the outer anchors along the edge, and N_{sb} is obtained from Eq. 6.K.17 without modification for a perpendicular edge distance.

Clarification for Section K.5.4.2: Only those anchors close to an edge ($h_{ef} > 2.5c_{a1}$), that are loaded in tension should be considered when determining nominal side-face blowout strength for multiple headed anchors. Their strength should be compared to the proportion of the tensile load applied to those anchors.

K.6 Design Requirements for Shear Loading

K.6.1 Steel Strength of Anchor in Shear

K.6.1.1 For an anchor in shear, the nominal strength governed by steel, V_{sa} , shall be evaluated by calculations based on the properties of the anchor material and the physical dimensions of the anchor.

K.6.1.2 For a single anchor or group of anchors in shear, the nominal strength, V_{sa} , shall not exceed (a) to (c):

(a) For cast-in headed stud anchor

$$V_{sa} = nA_{se,V}f_{uta} \quad (6.K.19)$$

Where, n is the number of anchors in the group, $A_{se,V}$ is the effective cross-sectional area of a single anchor in shear, mm^2 , and f_{uta} shall not be taken greater than the smaller of $1.9f_{ya}$ and 860 MPa.

(b) For cast-in headed bolt and hooked bolt anchors and for post-installed anchors where sleeves do not extend through the shear plane

$$V_{sa} = 0.6nA_{se,V}f_{uta} \quad (6.K.20)$$

Where, n is the number of anchors in the group, $A_{se,V}$ is the effective cross-sectional area of a single anchor in shear, mm^2 , and f_{uta} shall not be taken greater than the smaller of $1.9f_{ya}$ and 860 MPa.

(c) For post-installed anchors where sleeves extend through the shear plane, V_{sa} shall be based on the results of tests performed and evaluated according to ACI 355.2. Alternatively, Eq. 6.K.20 shall be permitted to be used.

Clarification for Section K.6.1.2: The nominal shear strength of anchors is best represented as a function of f_{uta} rather than f_{ya} because the large majority of anchor materials do not exhibit a well-defined yield point. Welded studs develop a higher steel shear strength than headed anchors due to the fixity provided by the weld between the studs and the base plate. The use of Eq. 6.K.19 and 6.K.20 with 6.2.2 load factors and the ϕ -factors of K.4.4 give design strengths consistent with the AISC Load and Resistance Factor Design Specifications.^{K.19}

The limitation of $1.9f_{ya}$ on f_{uta} is to ensure that, under service load conditions, the anchor stress does not exceed f_{ya} . The limit on f_{uta} of $1.9f_{ya}$ was determined by converting the LRFD provisions to corresponding service level conditions as discussed in Clarification for K.5.1.2.

The effective cross-sectional area of an anchor should be provided by the manufacturer of expansion anchors with reduced cross-sectional area for the expansion mechanism. For threaded bolts, ANSI/ASME B1.1^{K.1} defines $A_{se,V}$ as

$$A_{se,V} = \frac{\pi}{4} \left(d_a - \frac{0.9743}{n_t} \right)^2$$

Where, n_t is the number of threads per mm.

K.6.1.3 Where anchors are used with built-up grout pads, the nominal strengths of Sec K.6.1.2 shall be multiplied by a 0.80 factor.

K.6.2 Concrete Breakout Strength of Anchor in Shear

K.6.2.1 The nominal concrete breakout strength, V_{cb} or V_{cbg} , in shear of a single anchor or group of anchors shall not exceed:

- (a) For shear force perpendicular to the edge on a single anchor

$$V_{cb} = \frac{A_{vc}}{A_{vco}} \psi_{ed,V} \psi_{ec,V} \psi_{ch,V} V_b \quad (6.K.21)$$

- (b) For shear force perpendicular to the edge on a group of anchors

$$V_{cbg} = \frac{A_{vc}}{A_{vco}} \psi_{ec,V} \psi_{ed,V} \psi_{ch,V} V_b \quad (6.K.22)$$

- (c) For shear force parallel to an edge, V_{cb} or V_{cbg} shall be permitted to be twice the value of the shear force determined from Eq. 6.K.21 or 6.K.22, respectively, with the shear force assumed to act perpendicular to the edge and with $\psi_{ed,V}$ taken equal to 1.0.
- (d) For anchors located at a corner, the limiting nominal concrete breakout strength shall be determined for each edge, and the minimum value shall be used.

Factors $\psi_{ec,V}$, $\psi_{ed,V}$, $\psi_{ch,V}$, and $\psi_{cb,V}$ are defined in Sections K.6.2.5, K.6.2.6, K.6.2.7, and K.6.2.8, respectively. V_b is the basic concrete breakout strength value for a single anchor. A_{vc} is the projected area of the failure surface on the side of the concrete member at its edge for a single anchor or a group of anchors. It shall be permitted to evaluate A_{vc} as the base of a truncated half pyramid projected on the side face of the member where the top of the half pyramid is given by the axis of the anchor row selected as critical. The value of c_{a1} shall be taken as the distance from the edge to this axis. A_{vc} shall not exceed nA_{vco} , where n is the number of anchors in the group.

A_{vco} is the projected area for a single anchor in a deep member with a distance from edges equal or greater than $1.5c_{a1}$ in the direction perpendicular to the shear force. It shall be permitted to evaluate A_{vco} as the base of a half pyramid with a side length parallel to the edge of $3c_{a1}$ and a depth of $1.5c_{a1}$

$$A_{vco} = 4.5(c_{a1})^2 \quad (6.K.23)$$

Where anchors are located at varying distances from the edge and the anchors are welded to the attachment so as to distribute the force to all anchors, it shall be permitted to evaluate the strength based on the distance to the farthest row of anchors from the edge. In this case, it shall be permitted to base the value of c_{a1} on the distance from the edge to the axis of the farthest anchor row that is selected as critical, and all of the shear shall be assumed to be carried by this critical anchor row alone.

Clarification for Section K.6.2.1: The shear strength equations were developed from the CCD Method. They assume a breakout cone angle of approximately 35° (see Figure 6.K.3(b)), and consider fracture mechanics theory. The effects of multiple anchors, spacing of anchors, edge distance, and thickness of the concrete member on nominal concrete breakout strength in shear are included by applying the reduction factor of A_{vc}/A_{vco} in Eq. 6.K.21 and 6.K.22, and $\psi_{ec,V}$ in Eq. 6.K.22. For anchors far from the edge, Sec K.6.2 usually will not govern. For these cases, Sections K.6.1 and K.6.3 often govern.

Figure 6.K.8(a) shows A_{vco} and the development of Eq. 6.K.23. A_{vco} is the maximum projected area for a single anchor that approximates the surface area of the full breakout prism or cone for an anchor unaffected by edge distance, spacing, or depth of member. Figure 6.K.8(b) shows examples of the projected areas for various single-anchor and multiple-anchor arrangements. A_{vc} approximates the full surface area of the breakout cone for the particular arrangement of anchors. Because A_{vc} is the total projected area for a group of anchors, and A_{vco} is the area for a single anchor, there is no need to include the number of anchors in the equation.

When using Eq. 6.K.22 for anchor groups loaded in shear, both assumptions for load distribution illustrated in examples on the right side of Figure 6.K.8(b) should be considered because the anchors nearest the edge could fail first or the whole group could fail as a unit with the failure surface originating from the anchors farthest from the edge. If the anchors are welded to a common plate, when the anchor nearest the front edge begins to form a failure cone, shear load would be transferred to the stiffer and stronger rear anchor. For this reason, anchors welded to a common plate do not need to consider the failure mode shown in the upper right figure of Figure 6.K.8(b). The PCI Design Handbook approach^{K.18} suggests in Sec 6.5.2.2 that the strength of the anchors away from the edge be considered. Because this is a reasonable approach, assuming that the anchors are spaced far enough apart so that the shear failure surfaces do not intersect,^{K.11} Sec K.6.2 allows such a procedure. If the failure surfaces do not intersect, as would generally occur if the anchor spacing s is equal to or greater than $1.5c_{a1}$, then after

formation of the near-edge failure surface, the higher strength of the farther anchor would resist most of the load. As shown in the bottom right example in Figure 6.K.8(b), it would be appropriate to consider the shear strength to be provided entirely by this anchor with its much larger resisting failure surface. No contribution of the anchor near the edge is then considered. Checking the near-edge anchor condition is advisable to preclude undesirable cracking at service load conditions. Further discussion of design for multiple anchors is given in Reference K.8.

For the case of anchors near a corner subjected to a shear force with components normal to each edge, a satisfactory solution is to check independently the connection for each component of the shear force. Other specialized cases, such as the shear resistance of anchor groups where all anchors do not have the same edge distance, are treated in Reference K.11.

The detailed provisions of 6.K.8(a) apply to the case of shear force directed toward an edge. When the shear force is directed away from the edge, the strength will usually be governed by Sec K.6.1 or Sec K.6.3.

The case of shear force parallel to an edge is shown in Figure 6.K.8(c). A special case can arise with shear force parallel to the edge near a corner. In the example of a single anchor near a corner (see Figure 6.K.8(d)), the provisions for shear force applied perpendicular to the edge should be checked in addition to the provisions for shear force applied parallel to the edge.

K.6.2.2 For a single anchor in cracked concrete, the basic concrete breakout strength in shear, V_b shall not exceed

$$V_b = 0.6 \left(\frac{l_e}{d_a} \right)^{0.2} \sqrt{d_a} \lambda \sqrt{f'_c} (c_{a1})^{1.5} \quad (6.K.24)$$

Where, l_e is the load-bearing length of the anchor for shear ($l_e = h_{ef}$) for anchors with a constant stiffness over the full length of embedded section, such as headed studs and post-installed anchors with one tubular shell over full length of the embedment depth, $l_e = 2d_a$ for torque-controlled expansion anchors with a distance sleeve separated from expansion sleeve, and $l_e \leq 8d_a$ in all cases.

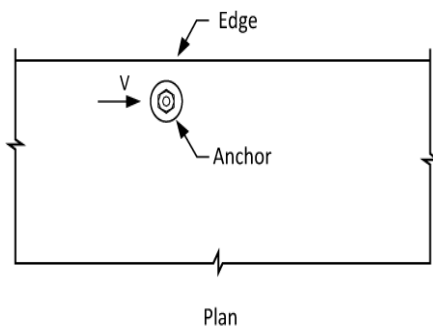
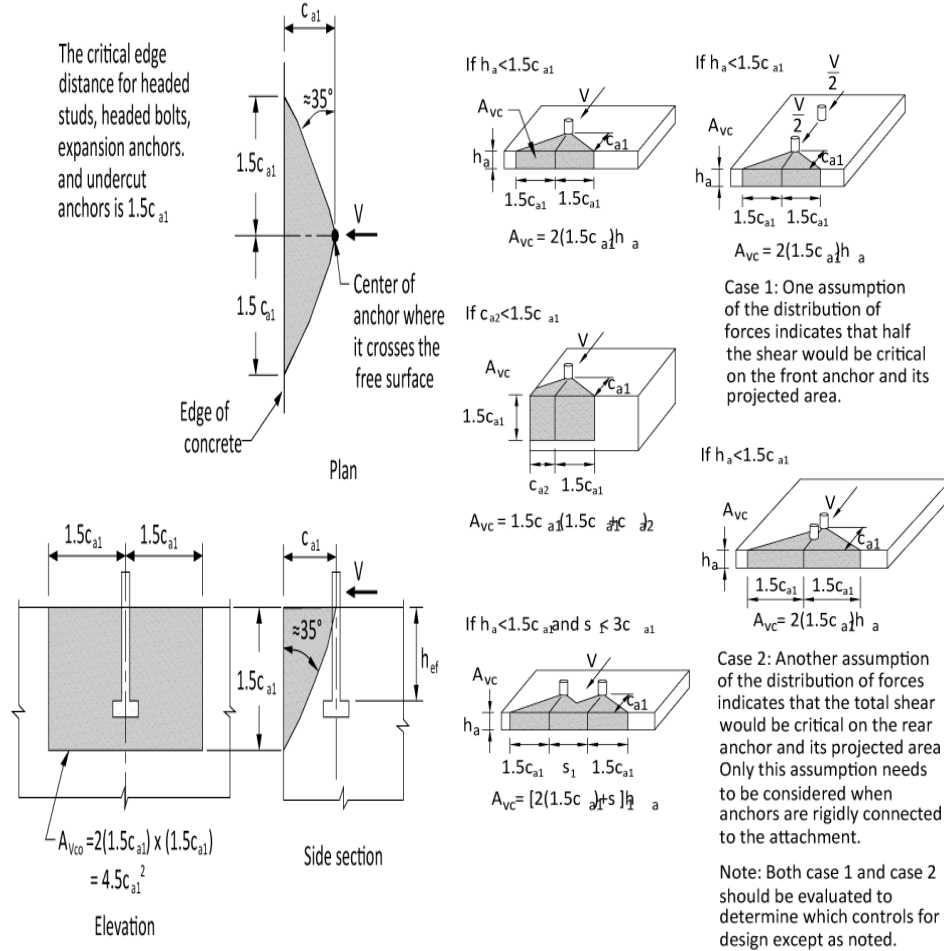


Figure 6.K.8(c) Shear force parallel to an edge.

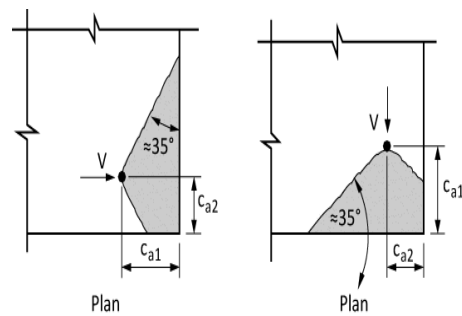
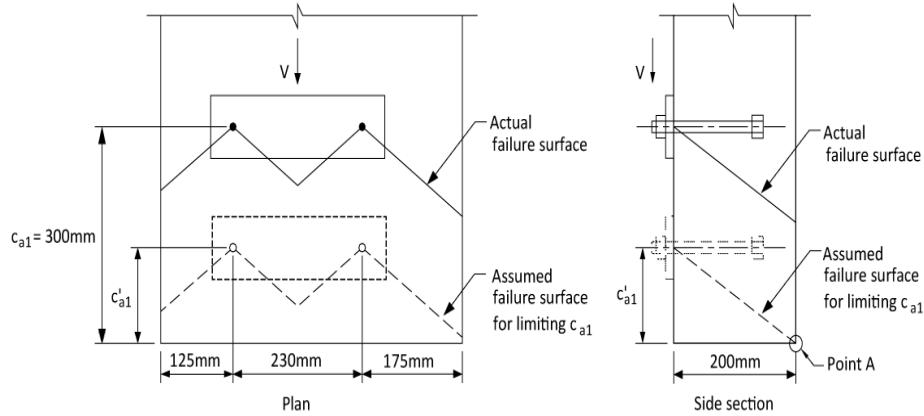


Figure 6.K.8(d) Shear force near a corner.



The actual $c_{a1} = 300\text{mm}$ but two orthogonal edges c_{a2} and h_a are $< 1.5c$ therefore the limiting value of c_{a1} (shown as c'_{a1} in the figure) is the larger of $c_{a2}/1.5$, $h_a/1.5$ and one-third of the maximum spacing for an anchor group:

$$c'_{a1} = \max(175/1.5, 200/1.5, 230/3) = 133\text{mm}$$

Therefore, use $c'_{a1} = 133\text{mm}$ in Eq. (D-21) to (D-28) including the calculation of A_{vc} :

$$A_{vc} = (125+230+175)(1.5 \times 133) = 105,735\text{mm}^2$$

Point a shows the intersection of the assumed failure surface for limiting c_{a1} with the concrete surface.

Figure 6.K.9 Shear when anchors are influenced by three or more edges.

Clarification for Section K.6.2.2: Like the concrete breakout tensile strength, the concrete breakout shear strength does not increase with the failure surface, which is proportional to $(C_{a1})^2$. Instead, the strength increases proportionally to $(C_{a1})^{1.5}$ due to size effect. The strength is also influenced by the anchor stiffness and the anchor diameter.^{K.9-K.11,K.16}

The constant, 0.6, in the shear strength equation was determined from test data reported in Reference K.9 at the 5 percent fractile adjusted for cracking.

K.6.2.3 For cast-in headed studs, headed bolts, or hooked bolts that are continuously welded to steel attachments having a minimum thickness equal to the greater of 10 mm and half of the anchor diameter, the basic concrete breakout strength in shear of a single anchor in cracked concrete, V_b , shall not exceed

$$V_b = 0.66 \left(\frac{I_e}{d_a} \right)^{0.2} \sqrt{d_a} \lambda \sqrt{f'_c} (c_{a1})^{1.5} \quad (6.K.25)$$

Where, I_e is defined in Sec K.6.2.2, provided that:

- for groups of anchors, the strength is determined based on the strength of the row of anchors farthest from the edge;
- anchor spacing, s , is not less than 65 mm; and
- reinforcement is provided at the corners if $C_{a2} \leq 1.5h_{ef}$.

Clarification for Section K.6.2.3: For the case of cast-in headed bolts continuously welded to an attachment, test data^{K.25} show that somewhat higher shear strength exists, possibly due to the stiff welding connection clamping the bolt more effectively than an attachment with an anchor gap. Because of this, the basic shear value for such anchors is increased. Limits are imposed to ensure sufficient rigidity. The design of supplementary reinforcement is discussed in References K.8, K.11, and K.12.

K.6.2.4 If anchors are influenced by three or more edges, the value of C_{a1} used in Eq. 6.K.23 to 6.K.29 shall not exceed the greatest of: $C_{a2}/1.5$ in either direction, $h_a/1.5$; and one-third of the maximum spacing between anchors within the group.

Clarification for Section K.6.2.4: If anchors are influenced by three or more edges where any edge distance is less than $1.5C_{a1}$, the shear breakout strength computed by the basic CCD Method, which is the basis for Equations 6.K.21 to 6.K.29, gives safe but overly conservative results. These cases were studied for the κ Method^{K.16} and the problem was pointed out by Lutz.^{K.21} Similarly, the approach used for tensile breakouts in Sec K.5.2.3, strength is correctly evaluated if the value of c_{a1} used in Equations 6.K.21 to 6.K.29 is limited to the maximum of $C_{a2}/1.5$ in each direction, $h_a/1.5$, and one-third of the maximum spacing between anchors within the group. The limit on c_{a1} of at least one-third of the maximum spacing between anchors within the group prevents the use of a calculated strength based on individual breakout prisms for a group anchor configuration.

This approach is illustrated in Figure 6.K.9. In this example, the limit on the value of c_{a1} is the largest of $C_{a2}/1.5$ in either direction, $h_a/1.5$, and one-third the maximum spacing between anchors for anchor groups results in $C_{a1} = 133$ mm. For this example, this would be the proper value to be used for C_{a1} in computing V_{cb} or V_{cbg} , even if the actual edge distance that the shear is directed toward is larger. The requirement of Sec K.6.2.4 may be visualized by moving the actual concrete breakout surface originating at the actual c_{a1} toward the surface of the concrete in the direction of the applied shear load. The value of C_{a1} used in Equations 6.K.21 to 6.K.29 is determined when either: (a) the outer boundaries of the failure surface first intersect a free edge; or (b) the intersection of the breakout surface between anchors within the group first intersects the surface of the concrete. For the example shown in Figure 6.K.9, Point “A” shows the intersection of the assumed failure surface for limiting C_{a1} with the concrete surface.

K.6.2.5 The modification factor for anchor groups loaded eccentrically in shear, $\psi_{ec,V}$, shall be computed as

$$\psi_{ec,V} = \frac{1}{\left(1 + \frac{2e'_V}{3c_{a1}}\right)} \quad (6.K.26)$$

But $\psi_{ec,V}$ shall not be taken greater than 1.0.

If the loading on an anchor group is such that only some anchors are loaded in shear in the same direction, only those anchors that are loaded in shear in the same direction shall be considered when determining the eccentricity of e_v' for use in Eq. 6.K.26 and for the calculation of V_{cbg} in Eq. 6.K.22.

Clarification for Section K.6.2.5: This section provides a modification factor for an eccentric shear force toward an edge on a group of anchors. If the shear force originates above the plane of the concrete surface, the shear should first be resolved as a shear in the plane of the concrete surface, with a moment that may or may not also cause tension in the anchors, depending on the normal force. Figure 6.K.10 defines the term e_v' for calculating the $\psi_{ec,V}$ modification factor that accounts for the fact that more shear is applied to one anchor than others, tending to split the concrete near an edge.

K.6.2.6 The modification factor for edge effect for a single anchor or group of anchors loaded in shear, $\psi_{ed,V}$, shall be computed as

For, $c_{a2} \geq 1.5c_{a1}$

$$\psi_{ed,V} = 1.0 \quad (6.K.27)$$

For, $c_{a2} < 1.5c_{a1}$

$$\psi_{ed,V} = 0.7 + 0.3 \frac{c_{a2}}{1.5c_{a1}} \quad (6.K.28)$$

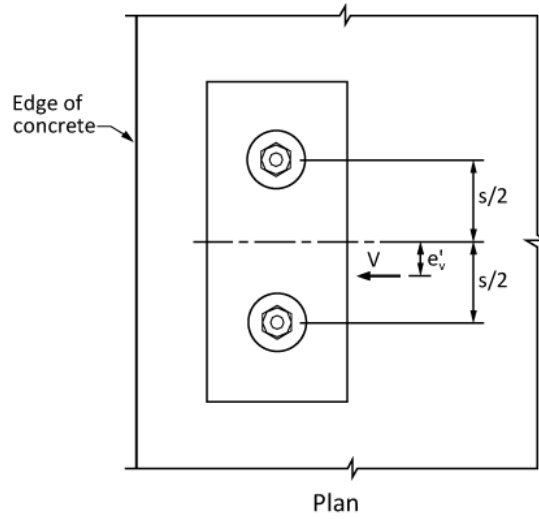


Figure 6.K.10 Definition of e_v' for a group of anchors.

K.6.2.7 For anchors located in a region of a concrete member where analysis indicates no cracking at service loads, the modification factor shall be permitted as $\psi_{c,V} = 1.4$.

For anchors located in a region of a concrete member where analysis indicates cracking at service load levels, the following modification factors shall be permitted:

$\psi_{c,V} = 1.0$ for anchors in cracked concrete with no supplementary reinforcement or edge reinforcement smaller than a No. 13 bar;

$\psi_{c,V} = 1.2$ for anchors in cracked concrete with reinforcement of a No. 13 bar or greater between the anchor and the edge; and

$\psi_{c,V} = 1.4$ for anchors in cracked concrete with reinforcement of a No. 13 bar or greater between the anchor and the edge, and with the reinforcement enclosed within stirrups spaced at not more than 100 mm.

Clarification for Section K.6.2.7: Torque-controlled and displacement-controlled expansion anchors are permitted in cracked concrete under pure shear loadings.

K.6.2.8 The modification factor for anchors located in a concrete member where $h_a < 1.5c_{a1}$, $\psi_{h,V}$ shall be computed as

$$\psi_{h,V} = \sqrt{\frac{1.5c_{a1}}{h_a}} \quad (6.K.29)$$

But, $\psi_{h,V}$ shall not be taken less than 1.0.

Clarification for Section K.6.2.8: For anchors located in a concrete member where, $h_a < 1.5c_{a1}$ tests^{K.8.K.14} have shown that the concrete breakout strength in shear is not directly proportional to the member thickness h_a . The factor $\psi_{h,V}$ accounts for this effect.

K.6.2.9 Where anchor reinforcement is either developed in accordance with Sec 8.2 on both sides of the breakout surface, or encloses the anchor and is developed beyond the breakout surface, the design strength of the anchor reinforcement shall be permitted to be used instead of the concrete breakout strength in determining ϕV_n . A strength reduction factor of 0.75 shall be used in the design of the anchor reinforcement.

Clarification for Section K.6.2.9: For conditions where the factored shear force exceeds the concrete breakout strength of the anchor(s) in shear, or where the breakout strength is not evaluated, the nominal strength can be that of anchor reinforcement properly anchored as shown in Figure 6.K.11(a) and (b). To ensure yielding of the anchor reinforcement, the enclosing anchor reinforcement in Figure 6.K.11(a) should be in contact with the anchor and placed as close as practicable to the concrete surface. The research^{K.14} on which the provisions for enclosing reinforcement (see Figure 6.K.11(a)) are based was limited to anchor reinforcement with maximum diameter similar to a No. 16 bar. The larger bend radii associated with larger bar diameters may significantly reduce the effectiveness of the anchor reinforcement, and therefore anchor reinforcement with a diameter larger than No. 19 is not recommended.

The reinforcement could also consist of stirrups and ties (as well as hairpins) enclosing the edge reinforcement embedded in the breakout cone and placed as close to the anchors as practicable (see Figure 6.K.11(b)). Only reinforcement spaced less than the lesser of $0.5c_{a1}$ and $0.3c_{a2}$ from the anchor centerline should be included as anchor reinforcement. In this case, the anchor reinforcement must be developed on both sides of the breakout surface. For equilibrium reasons, an edge reinforcement must be present. The research on which these provisions are based was limited to anchor reinforcement with maximum diameter similar to a No. 19 bar.

Because the anchor reinforcement is placed below where the shear is applied (see Figure 6.K.11(b)), the force in the anchor reinforcement will be larger than the shear force. In sizing the anchor reinforcement, use of a 0.75 strength reduction factor ϕ is recommended as used for shear and for strut-and-tie models. As a practical matter, the use of anchor reinforcement is generally limited to cast-in-place anchors.

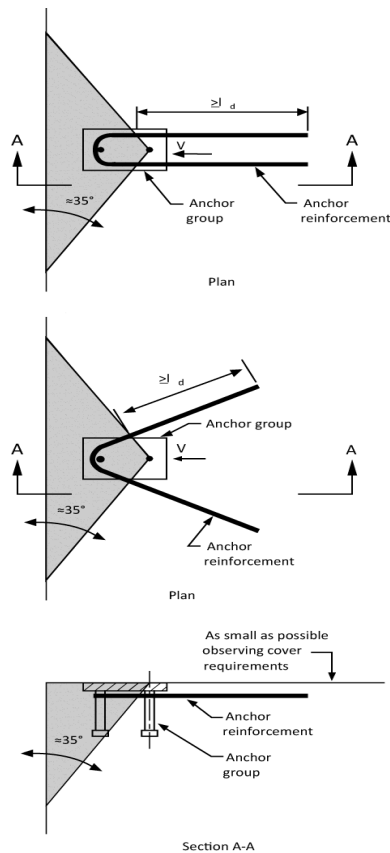


Figure 6.K.11(a) Hairpin anchor reinforcement for shear.

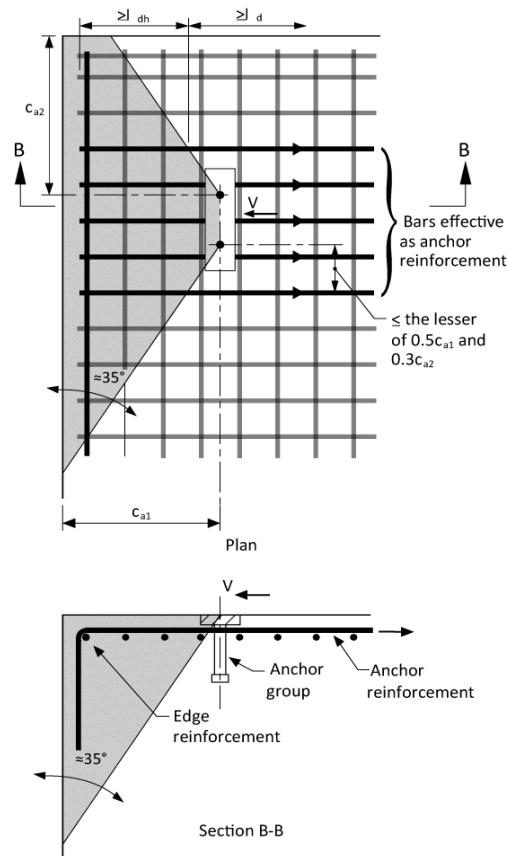


Figure 6.K.11(b) Edge reinforcement and anchor reinforcement for shear.

K.6.3 Concrete Pryout Strength of Anchor in Shear

Clarification for Section K.6.3: Concrete pryout strength of anchor in shear

Reference K.9 indicates that the pryout shear resistance can be approximated as one to two times the anchor tensile resistance with the lower value appropriate for h_{ef} less than 65 mm.

K.6.3.1 The nominal pryout strength, V_{cp} or V_{cpg} shall not exceed:

(a) For a single anchor

$$V_{cp} = k_{cp} N_{cb} \quad (6.K.30)$$

(b) For a group of anchors

$$V_{cpg} = k_{cp} N_{cbg} \quad (6.K.31)$$

Where,

$k_{cp} = 1.0$ for $h_{ef} < 65$ mm < 65 mm; and $k_{cp} = 2.0$ for $h_{ef} \geq 65$ mm.

N_{cb} and N_{cbg} shall be determined from Eq. 6.K.4 and 6.K.5, respectively.

K.7 Interaction of Tensile and Shear Forces

Unless determined in accordance with Sec K.4.3, anchors or groups of anchors that are subjected to both shear and axial loads shall be designed to satisfy the requirements of Sections K.7.1 to K.7.3. The value of ϕN_n shall be as required in Sec K.4.1.2. The value of ϕV_n shall be as defined in Sec K.4.1.2.

Clarification for Section K.7: Interaction of tensile and shear forces

The shear-tension interaction expression has traditionally been expressed as

$$\left(\frac{N_{ua}}{N_n} \right)^\zeta + \left(\frac{V_{ua}}{V_n} \right)^\zeta \leq 1.0$$

Where, ζ varies from 1 to 2. The current trilinear recommendation is a simplification of the expression where, $\zeta = 5/3$ (Figure 6.K.12). The limits were chosen to eliminate the requirement for computation of interaction effects where very small values of the

second force are present. Any other interaction expression that is verified by test data, however, can be used to satisfy Sec K.4.3.

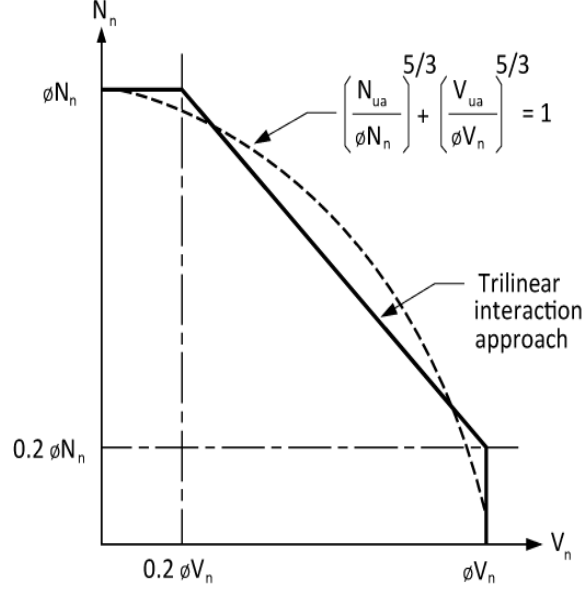


Figure 6.K.12 Shear and tensile load interaction equation.

K.7.1 If $V_{ua} \leq 0.2\phi V_n$, then full strength in tension shall be permitted: $\phi N_n \geq N_{ua}$.

K.7.2 If $N_{ua} \leq 0.2\phi N_n$, then full strength in tension shall be permitted: $\phi V_n \geq V_{ua}$.

K.7.3 If $V_{ua} > 0.2\phi V_n$ and $N_{ua} > 0.2\phi N_n$, then

$$\frac{N_{ua}}{\phi N_n} + \frac{V_{ua}}{\phi V_n} \leq 1.2 \quad (6.K.32)$$

K.8 Edge Distance, Spacing & Thickness Required Against Splitting Failure

Minimum spacing and edge distances for anchors and minimum thicknesses of members shall conform to Sections K.8.1 to K.8.6, unless supplementary reinforcement is provided to control splitting. Lesser values from product-specific tests performed in accordance with ACI 355.2 shall be permitted.

Clarification for Section K.8: Required edge distances, spacing, and thicknesses to preclude splitting failure

The minimum spacing, edge distances, and thicknesses are very dependent on the anchor characteristics. Installation forces and torques in post-installed anchors can cause splitting of the surrounding concrete. Such splitting also can be produced in subsequent torquing during connection of attachments to anchors including cast-in anchors. The primary source of values for minimum spacing, edge distances, and thicknesses of post-installed anchors should be the product-specific tests of ACI 355.2. In some cases, however, specific products are not known in the design stage. Approximate values are provided for use in design.

K.8.1 Unless determined in accordance with Sec K.8.4, minimum center-to-center spacing of anchors shall be $4d_a$ for untorqued cast-in anchors, and $6d_a$ for torque cast-in anchors and post-installed anchors.

K.8.2 Unless determined in accordance with Sec K.8.4, minimum edge distances for cast-in headed anchors that will not be torqued shall be based on specified cover requirements for reinforcement in Sec 8.1.7. For cast-in headed anchors that will be torqued, the minimum edge distances shall be $6d_a$.

Clarification for Section K.8.2: Because the edge cover over a deep embedment close to the edge can have a significant effect on the side face blowout strength of Sec K.5.4, in addition to the normal concrete cover requirements, it may be advantageous to use larger cover to increase the side-face blowout strength.

K.8.3 Unless determined in accordance with Sec K.8.4, minimum edge distances for post-installed anchors shall be based on the greater of specified cover requirements for reinforcement in Sec 8.1, or minimum edge distance requirements for the products as determined by tests in accordance with ACI 355.2, and shall not be less than 2.0 times the maximum aggregate size. In the absence of product-specific ACI 355.2 test information, the minimum edge distance shall be taken as not less than:

Undercut anchors	$6d_a$
Torque-controlled anchors	$8d_a$
Displacement-controlled anchors	$10d_a$

Clarification for Section K.8.3: Drilling holes for post-installed anchors can cause microcracking. The requirement for a minimum edge distance twice the maximum aggregate size is to minimize the effects of such microcracking.

K.8.4 For anchors where installation does not produce a splitting force and that will remain untorqued, if the edge distance or spacing is less than those specified in Sections K.8.1 to K.8.3, calculations shall be performed by substituting for d_a a smaller value d'_a that meets the requirements of Sections K.8.1 to K.8.3. Calculated forces applied to the anchor shall be limited to the values corresponding to an anchor having a diameter of d'_a .

Clarification for Section K.8.4: In some cases, it may be desirable to use a larger diameter anchor than the requirements on K.8.1 to K.8.3 permit. In these cases, it is permissible to use a larger-diameter anchor provided the design strength of the anchor is based on a smaller assumed anchor diameter, d'_a .

K.8.5 The value of h_{ef} for an expansion or undercut post-installed anchor shall not exceed the greater of 2/3 of the member thickness and the member thickness minus 100 mm.

Clarification for Section K.8.5: This minimum thickness requirement is not applicable to through-bolts because they are outside the scope of Appendix K. In addition, splitting failures are caused by the load transfer between the bolt and the concrete. Because through-bolts transfer their load differently than cast-in or expansion and undercut anchors, they would not be subject to the same member thickness requirements. Post-installed anchors should not be embedded deeper than 2/3 of the member thickness.

K.8.6 Unless determined from tension tests in accordance with ACI 355.2, the critical edge distance, C_{ac} , shall not be taken less than:

Undercut anchors	$2.5h_{ef}$
Torque-controlled anchors	$4h_{ef}$
Displacement-controlled anchors	$4h_{ef}$

Clarification for Section K.8.6: The critical edge distance C_{ac} is determined by the corner test in ACI 355.2. Research has indicated that the corner-test requirements are not met with $C_{a,min} = 1.5h_{ef}$ for many expansion anchors and some undercut anchors because installation of these types of anchors introduces splitting tensile stresses in the concrete that are increased during load application, potentially resulting in a premature splitting failure. To permit the design of these types of anchors when product-specific information is not available, conservative default values for C_{ac} are provided.

K.8.7 Project drawings and project specifications shall specify use of anchors with a minimum edge distance as assumed in design.

K.9 Installation of Anchors

Clarification for Section K.9: Installation of anchors

Many anchor performance characteristics depend on proper installation of the anchor. Anchor strength and deformation capacity can be assessed by acceptance testing under ACI 355.2. These tests are carried out assuming that the manufacturer's installation directions will be followed. Certain types of anchors can be sensitive to variations in hole diameter, cleaning conditions, orientation of the axis, magnitude of the installation torque, crack width, and other variables. Some of this sensitivity is indirectly reflected in the assigned ϕ values for the different anchor categories, which depend in part on the results of the installation safety tests. Gross deviations from the ACI 355.2 acceptance testing results could occur if anchor components are incorrectly exchanged, or if anchor installation criteria and procedures vary from those recommended. Project specifications should require that anchors be installed according to the manufacturer's recommendations.

K.9.1 Anchors shall be installed in accordance with the project drawings and project specifications.

K. 10 References of Appendix K

- K.1. ANSI/ASME B1.1, "Unified Inch Screw Threads (UN and UNR Thread Form)," ASME, Fairfield, NJ, 1989.
- K.2. ANSI/ASME B18.2.1, "Square and Hex Bolts and Screws, Inch Series," ASME, Fairfield, NJ, 1996.
- K.3. ANSI/ASME B18.2.6, "Fasteners for Use in Structural Applications," ASME, Fairfield, NJ, 1996.
- K.4. Cook, R. A., and Klingner, R. E., "Behavior of Ductile Multiple-Anchor Steel-to-Concrete Connections with Surface-Mounted Baseplates," *Anchors in Concrete: Design and Behavior*, SP-130, American Concrete Institute, Farmington Hills, MI, 1992, pp. 61-122.
- K.5. Cook, R. A., and Klingner, R. E., "Ductile Multiple-Anchor Steel-to-Concrete Connections," *Journal of Structural Engineering*, ASCE, V. 118, No. 6, June 1992, pp. 1645-1665.
- K.6. Lotze, D.; Klingner, R. E.; and Graves III, H. L., "Static Behavior of Anchors under Combinations of Tension and Shear Loading," *ACI Structural Journal*, V. 98, No. 4, July-Aug. 2001, pp. 525-536.

-
- K.7. Primavera, E. J.; Pinelli, J.-P.; and Kalajian, E. H., "Tensile Behavior of Cast-in-Place and Undercut Anchors in High-Strength Concrete," *ACI Structural Journal*, V. 94, No. 5, Sept.-Oct. 1997, pp. 583-594.
 - K.8. Design of Fastenings in Concrete, Comité Euro-International du Béton (CEB), Thomas Telford Services Ltd., London, Jan. 1997.
 - K.9. Fuchs, W.; Eligehausen, R.; and Breen, J., "Concrete Capacity Design (CCD) Approach for Fastening to Concrete," *ACI Structural Journal*, V. 92, No. 1, Jan.-Feb. 1995, pp. 73-93. Also discussion, *ACI Structural Journal*, V. 92, No. 6, Nov.-Dec. 1995, pp. 787-802.
 - K.10. Eligehausen, R., and Balogh, T., "Behavior of Fasteners Loaded in Tension in Cracked Reinforced Concrete," *ACI Structural Journal*, V. 92, No. 3, May-June 1995, pp. 365-379.
 - K.11. "Fastenings to Concrete and Masonry Structures, State of the Art Report," Comité Euro-International du Béton (CEB), Bulletin No. 216, Thomas Telford Services Ltd., London, 1994.
 - K.12. Klingner, R.; Mendonca, J.; and Malik, J., "Effect of Reinforcing Details on the Shear Resistance of Anchor Bolts under Reversed Cyclic Loading," *ACI JOURNAL*, Proceedings V. 79, No. 1, Jan.-Feb. 1982, pp. 3-12.
 - K.13. ACI Committee 349, "Code Requirements for Nuclear Safety Related Concrete Structures (ACI 349-01)," American Concrete Institute, Farmington Hills, MI, 2001, 134 pp.
 - K.14. Eligehausen, R.; Mallee, R.; and Silva, J., *Anchorage in Concrete Construction*, Ernst & Sohn (J. T. Wiley), Berlin, May 2006, 380 pp.
 - K.15. Eligehausen, R.; Fuchs, W.; and Mayer, B., "Load Bearing Behavior of Anchor Fastenings in Tension," *Betonwerk + Fertigteiltechnik*, 12/1987, pp. 826-832, and 1/1988, pp. 29-35.
 - K.16. Eligehausen, R., and Fuchs, W., "Load Bearing Behavior of Anchor Fastenings under Shear, Combined Tension and Shear or Flexural Loadings," *Betonwerk + Fertigteiltechnik*, 2/1988, pp. 48-56.
 - K.17. Farrow, C. B., and Klingner, R. E., "Tensile Capacity of Anchors with Partial or Overlapping Failure Surfaces: Evaluation of Existing Formulas on an LRFD Basis," *ACI Structural Journal*, V. 92, No. 6, Nov.-Dec. 1995, pp. 698-710.

-
- K.18. PCI Design Handbook, 5th Edition, Precast/Prestressed Concrete Institute, Chicago, IL, 1999.
- K.19. "AISC Load and Resistance Factor Design Specifications for Structural Steel Buildings," Dec. 1999, 327 pp.
- K.20. Zhang, Y.; Klingner, R. E.; and Graves III, H. L., "Seismic Response of Multiple-Anchor Connections to Concrete," ACI Structural Journal, V. 98, No. 6, Nov.-Dec. 2001, pp. 811-822.
- K.21. Lutz, L., "Discussion to Concrete Capacity Design (CCD) Approach for Fastening to Concrete," ACI Structural Journal, Nov.-Dec. 1995, pp. 791-792. Also authors' closure, pp. 798-799.
- K.22. Asmus, J., "Verhalten von Befestigungen bei der Versagensart Spalten des Betons (Behavior of Fastenings with the Failure Mode Splitting of Concrete)," dissertation, Universität Stuttgart, Germany, 1999.
- K.23. Kuhn, D., and Shaikh, F., "Slip-Pullout Strength of Hooked Anchors," Research Report, University of Wisconsin-Milwaukee, submitted to the National Codes and Standards Council, 1996.
- K.24. Furche, J., and Eligehausen, R., "Lateral Blow-out Failure of Headed Studs Near a Free Edge," Anchors in Concrete—Design and Behavior, SP-130, American Concrete Institute, Farmington Hills, MI, 1991, pp. 235-252.
- K.25. Shaikh, A. F., and Yi, W., "In-Place Strength of Welded Studs," PCI Journal, V. 30, No. 2, Mar.-Apr. 1985.

PART VI
Appendix L
Information on Steel Reinforcement

Information on sizes, areas, and weights of various steel reinforcement is presented below to facilitate the use of Bangladesh National Building Code.

Table 6.L.1: Sizes, Areas, and Weights of Reinforcing Bars (ASTM Standard)

Bar Size, No.*	Nominal Diameter, mm	Nominal Area, mm ²	Nominal Mass, kg/m
10	9.5	71	0.560
13	12.7	129	0.994
16	15.9	199	1.552
19	19.1	284	2.235
22	22.2	387	3.042
25	25.4	510	3.973
29	28.7	645	5.060
32	32.3	819	6.404
36	35.8	1006	7.907
43	43.0	1452	11.38
57	57.3	2581	20.24

**Bar numbers approximate the number of millimeters of the nominal diameter of the bar. Within the text in Chapters, the referred ϕ (diameter in mm) of bars correspond to the figures in the first column (Bar size, no.)*

Table 6.L.2: Sizes, Areas, and Weights of Prestressing Tendons (ASTM Standard)

Type*	Nominal Diameter, mm	Nominal Area, mm ²	Nominal Mass, kg/m
Seven-wire strand (Grade 1725)	6.4	23.2	0.182
	7.9	37.4	0.294
	9.5	51.6	0.405
	11.1	69.7	0.548
	12.7	92.9	0.730
	15.2	139.4	1.094
Seven-wire strand (Grade 1860)	9.53	54.8	0.432
	11.1	74.2	0.582
	12.70	98.7	0.775
	15.24	140.0	1.102
Prestressing wire	4.88	18.7	0.146
	4.98	19.5	0.149
	6.35	31.7	0.253
	7.01	38.6	0.298
Prestressing bars (plain)	19	284	2.23
	22	387	3.04
	25	503	3.97
	29	639	5.03
	32	794	6.21
	35	955	7.52
Prestressing bars (deformed)	15	181	1.46
	20	271	2.22
	26	548	4.48
	32	806	6.54
	36	1019	8.28

*Depends on availability of some tendon sizes.

Table 6.L.3: Sizes, Areas, and Weights of Wire Reinforcement* (WRI Standard)

MW & MD size		Nominal Diameter, mm	Nominal Mass, kg/m	Area, mm ² /m of Width for Various Spacings							
				Center-to-center spacing, mm							
Plain	Deformed			50	75	100	150	200	250	300	
MW290	MD290	19.22	2.27	5800	3900	2900	1900	1450	1160	970	
MW200	MD200	15.95	1.5700	4000	2700	2000	1300	1000	800	670	
MW130	MD130	12.90	1.0204	2600	1700	1300	870	650	520	430	
MW120	MD120	12.40	0.9419	2400	1600	1200	800	600	480	400	
MW100	MD100	11.30	0.7849	2000	1300	1000	670	500	400	330	
MW90	MD90	10.70	0.7064	1800	1200	900	600	450	360	300	
MW80	MD80	10.10	0.6279	1600	1100	800	530	400	320	270	
MW70	MD70	9.40	0.5494	1400	930	700	470	350	280	230	
MW65	MD65	9.10	0.5102	1300	870	650	430	325	260	220	
MW60	MD60	8.70	0.4709	1200	800	600	400	300	240	200	
MW55	MD55	8.40	0.4317	1100	730	550	370	275	220	180	
MW50	MD50	8.00	0.3925	1000	670	500	330	250	200	170	
MW45	MD45	7.60	0.3532	900	600	450	300	225	180	150	
MW40	MD40	7.10	0.3140	800	530	400	270	200	160	130	
MW35	MD35	6.70	0.2747	700	470	350	230	175	140	120	
MW30	MD30	6.20	0.2355	600	400	300	200	150	120	100	
MW25	MD25	5.60	0.1962	500	330	250	170	125	100	83	
MW20		5.00	0.1570	400	270	200	130	100	80	67	
MW15		4.40	0.1177	300	200	150	100	75	60	50	
MW10		3.60	0.0785	200	130	100	70	50	40	33	
MW5		2.50	0.0392	100	67	50	33	25	20	17	

Table 6.L.4: Dimensions, Mass per Unit Length and Permissible Deviations (BDS ISO_6935-1)

Nominal Bar Diameter, d mm	Nominal Cross-Sectional Area ^a A_n mm ²	Mass per Unit Length	
		Requirement ^b kg/m	Permissible deviation ^c %
6	28,3	0,222	± 8
8	50,3	0,395	± 8
10	78,5	0,617	± 5
12	113	0,888	± 5
14	154	1,21	± 5
16	201	1,58	± 5
20	314	2,47	± 5
22	380	2,98	± 5

^a $A_n=0,785\ 4\times d^2$

^b Mass per unit length $=7,85\times 10^{-3}\ A_n$

^c Permissible deviation refers to a single bar.

Table 6.L.5: Dimensions, Mass per Unit Length and Permissible Deviations (BDS ISO_6935-2)

Nominal Bar Diameter ^a , d mm	Nominal Cross-Sectional Area ^b A_n mm ²	Mass per unit length	
		Requirement ^c Kg/m	Permissible deviation ^d %
6	28,3	0,222	± 8
8	50,3	0,395	± 8
10	78,5	0,617	± 6
12	113	0,888	± 6
14	154	1,21	± 5
16	201	1,58	± 5
20	314	2,47	± 5
25	491	3,85	± 4
28	616	4,84	± 4
32	804	6,31	± 4
40	1257	9,86	± 4
50	1964	15,42	± 4

^a Diameters larger than 50mm should be agreed between the manufacturer and purchaser. The permissible deviation on such bars shall be ± 4 %

^b $A_n=0,785\ 4\times d^2$

^c Mass per unit length $=7,85\times 10^{-3}\ A_n$

^d Permissible deviation refers to a single bar.

Staircases shall, generally, be designed to support the design uniformly distributed load according to the load combinations specified in Chapter 2, loads. For common free standing stairs, it generally sufficient to design for gravity loading only, using the relevant load combinations.

M.1.3 Design

M.1.3.1 Empirical expressions for forces and moments (Fig.6.M.1) at critical locations of stair in terms of stair geometric and material parameters under service load are given in Table 6.M.1.

Table 6.M.1: Expressions for Free Standing Stair Forces and Moments under Service Load

Sl.	Description	Expression	Unit
1.	Deflection at landing corner, δ	$0.002 \times [1 + 3.6(A - 0.125)^{0.94}] \times [1 + 2.275(B - 0.915)^{1.1}] \times [1 + 1.017(C - 0.914)^{0.93}] \times [1 - 7.87 \times 10^{-3}(L - 2.03)] \times [1 - 0.2(H - 2.44)] \times [1 - 1.617(T - 0.1)^{0.334}] \times [1 - 1.074 \times 10^{-6}(f'_c - 14)^{0.93}]$	m
2.	Support moment, M_S	$4.712 \times [1.555 + 0.787(A - 0.05)] \times [1.06 - 0.22(B - 0.86)] \times [1.2 + 2.76(C - 0.864)] \times [1 + 0.748(L - 2.03)] \times [1 + 5.9 \times 10^{-3}(H - 2.44)] \times [0.39 + 1.73(T - 0.09)]$	kN-m
3.	Flight mid-span moment, M_F	$1.526 \times [1.1 - 1.143(A - 0.15)^{1.52}] \times [1 - 0.872(B - 0.915)^{1.365}] \times [1 + 12.22(L - 2.03)^{2.66}] \times [1 + 0.184(H - 2.44)^{2.77}] \times [1 - 5.34(T - 0.1)^{1.17}]$	kN-m
4.	Moment at flight-landing junction, M_K	$3.447 \times [1.23 + 0.512(A - 0.125)] \times [1.01 + 3.23(B - 0.915)] \times [0.85 + 0.709(C - 0.915)] \times [0.95 + 5.5(T - 0.1)^{1.03}]$	kN-m
5.	Moment at mid-landing section, M_o	$6.14 \times [1 + 0.303(A - 0.15)] \times [1 + 1.18(B - 0.915)] \times [1 + 1.06(C - 0.915)] \times [1 + 0.409(L - 2.03)] \times [1 + 0.02637(H - 2.04)] \times [1 + 1.85(T - 0.1)]$	kN-m
6.	Axial force in flights, A_F	$34.69 \times [1 + 0.236(A - 0.125)] \times [1 + 0.787(B - 0.915)] \times [1 + 0.827(C - 0.915)] \times [1 + 0.354(L - 2.03)] \times [1 - 0.157(H - 2.44)] \times [1 + 2.76(T - 0.1)]$	kN
7.	Torsion in flights, T_F	$2.312 \times [1 + 1.77(A - 0.125)] \times [1 + 0.63(B - 0.915)] \times [1 + 2.68(C - 0.915)] \times [1 - 1.423 \times 10^{-3}(L - 2.03)^{0.75}] \times [1 + 3.58(T - 0.1)]$	kN-m
8.	In-plane moment in flights, M_I	$14.35 \times [1.1 + 0.866(A - 0.15)] \times [1 + 0.984(B - 0.915)] \times [1 + 1.57(C - 0.915)] \times [1 + 0.59(L - 2.03)] \times [1 - 0.197(H - 2.44)] \times [1 + 2.6(T - 0.1)]$	kN-m
9.	Lateral shear at mid-landing section, V_o	$30.17 \times [1 - 0.276(A - 0.15)] \times [1 + 1.38(B - 0.915)] \times [1 + 0.709(C - 0.915)] \times [1 + 0.669(L - 2.03)] \times [1 - 0.24(H - 2.44)] \times [1 + 6.092(T - 0.1)^{1.3}]$	kN

In the above expressions, the unit of length is meter and the unit of force is kilo-Newton. The unit for f'_c is in MPa. The ranges of parameters for which the above expressions are applicable are as follows:

$0.15 \leq A \leq 1.0$, $0.92 \leq B \leq 1.9$, $0.92 \leq C \leq 1.9$, $2.0 \leq L \leq 3.6$, $2.44 \leq H \leq 4.32$, $0.1 \leq T \leq 0.28$, $14 \leq f'_c \leq 40$.

M.1.3.2 The empirical expressions of forces and moments given above provide working/service values corresponding to 5.0 kN/m^2 service live load and appropriate service dead load of slab and steps based on unit weight of 23.56 kN/m^3 . Forces and moments for other values of live load, L_L , may be calculated by simple proportioning. This can be reasonably done by multiplying the quantities of Table 6.M.1 by a factor $(23.56 T + L_L + 2.36)/(23.56 T + 7.36)$ where L_L is in kN/m^2 and T is in meter. Once the working/service values are obtained for desired load as described, the ultimate design values (factored) for USD method of reinforcement design may be obtained by multiplying the working values by a conversion factor equal to $(1.2D_L + 1.6L_L)/(D_L + L_L)$ where D_L is the dead load in kN/m^2 . For 5.0 kN/m^2 live load and concrete unit weight of 23.56 kN/m^3 , this conversion factor can safely be approximated as $(42.41 T + 8.0)/(35.34 T + 5.0)$. Once the values of design moments and forces are obtained, design procedure of Chapter 6 (Part 6) may be employed for the reinforcement design of flights and landing.

M.1.3.3 Moments M_S , M_K and M_o are such that they produce tension in the top fiber requiring longitudinal slab reinforcement to be placed near the top face of slab. Moment M_F produce tension in the bottom fiber of flight slab requiring longitudinal reinforcement to be placed near the bottom face of slab. Axial force A_F produce tension in upper flight and compression in lower flight. This requires tension reinforcement to be provided only in the upper flight. In-plane moment M_I acts in such a manner that it produces tension along the inner edge of upper flight and outer edge of lower flight.

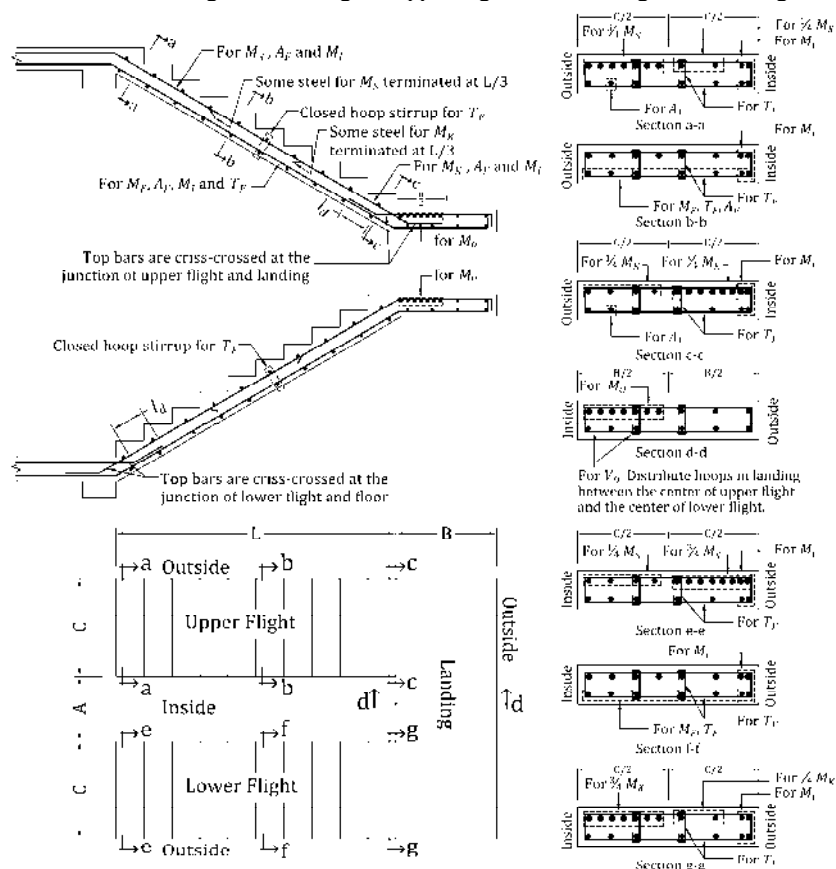


Figure 6.M.2 Recommended reinforcement layout details for free standing stair

M.1.3.4 Apart from maintaining the standard code provisions in detailing the reinforcement as stipulated elsewhere in this Code, additional detailing shall be done to take care of the features which are special to the free standing stairway. To account for the non-uniform distribution of the bending moment M_S at support across the width of the slab, 75 percent of the total negative steel shall be distributed across the outer half of the width (Zone – O in Fig. 6.M.1) of support section for both the flights and the rest of the negative steel shall be distributed within the inner half of the width of support section (Zone – I). For moment M_K , 75 percent of the total negative steel shall be distributed across the inner half of the width (Zone – I in Fig. 6.M.1) of flight-landing junction for both the flights and the rest of the negative steel shall be distributed within the outer half of the width of section (Zone – O at flight-landing junction). For moment M_o , the total negative steel shall be distributed across the inner half of the width (Landing Zone – I in Fig. 6.M.1) and the rest of the section shall be provided with nominal reinforcement as per provisions of Chapter 6. At mid-span of flights, the positive steel required for M_F shall be distributed uniformly across the section. Longitudinal steel required to resist flight in-plane moment M_l shall be placed near the inner edge in Zone-I of upper flight and near the outer edge in Zone-O of lower flight. Longitudinal steel required to resist tensile axial force A_F in upper flight may be distributed across the section of upper flight. Closed rectangular hoop reinforcements accompanied by cross ties shall be designed to resist the action of torsion in flights and lateral shear at mid-landing section. The suggested bar detailing for the free standing stairway is shown in Figure 6.M.2.

M.2 Sawtooth Stair

M.2.1 Loading

Sawtooth (slab less) stair shall be designed to support the design ultimate load according to the load combinations specified in Chapter 2, Loads.

M.2.2 Distribution of Loading

Where flights or landing are embedded at least 110 mm into the walls and are designed to span in the direction of the flight, a 150 mm strip may be deducted from the loaded area and the effective breadth of the section may be increased by 75 mm for the purpose of design (Figure 6.M.3).

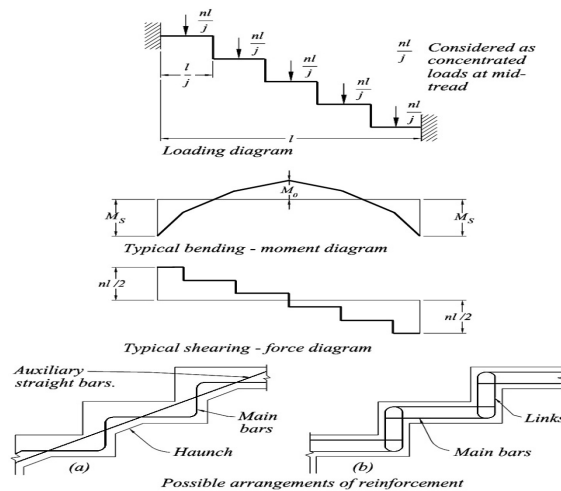


Figure 6.M.3 Elements of saw-tooth stair and typical reinforcement arrangements

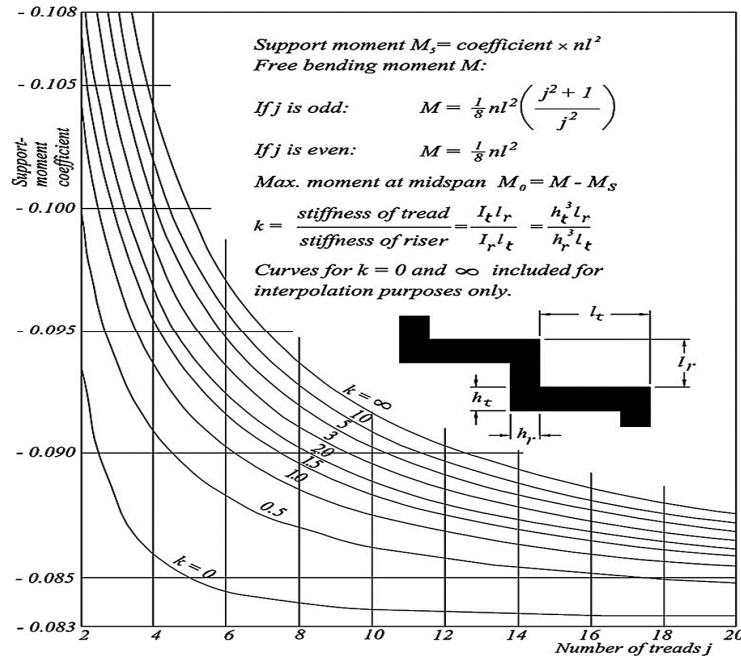


Figure 6.M.4 Support moment coefficients for saw-tooth stair

M.2.3 Effective Span

Sawtooth stairs shall be supported with stringer beams or walls at landing levels (Figure 6.M.3). The effective span for the stair shall be the going of the stair measured horizontally (Figure 6.M.3) from the face of the stringer beam or wall.

M.2.3.1 Design

The support moments for sawtooth stairs are given by:

$$M_s = \frac{nl^3 (k_{11} + k_0 k_{13})}{j^3 (k_{13} + k_0 k_{14})} \quad (6.M.1)$$

Where, k_o = stiffness of tread/stiffness of riser and j is the number of treads and n is the width of flight (Figure 6.M.3).

If j is odd: $k_{11} = \frac{1}{16} j^2 + \frac{1}{48} j(j-1)(j-2),$

$$k_{12} = \frac{1}{16} (j-1)^2 + \frac{1}{48} j(j-1)(j-2)(j-3),$$

$$k_{13} = \frac{1}{2} j, \quad k_{14} = \frac{1}{2} (j-1).$$

If j is even: $k_{11} = \frac{1}{48} j(j-1)(j-2), \quad k_{12} = \frac{1}{48} (j-1)(j-2)(j-3),$

$$k_{13} = \frac{1}{2} (j-1),$$

$$k_{14} = \frac{1}{2} (j-2)$$

The chart on Figure 6.M.4 gives the support-moment coefficients for various ratios of stiffness of tread/stiffness of riser and numbers of treads. Having found the support moment, the maximum mid-span bending moment can be alternatively determined by using the appropriate expression on the Figure 6.M.4 and subtracting the support moment.

M.2.4 Detailing

Typical bending-moment and shearing-force diagrams for a stair are shown on Figure 6.M.3 together with suggested arrangements of reinforcement. The re-entrant corners of the stair-profile shall be designed for stress concentrations. This has to be facilitated by providing twice of the reinforcements calculated from Eq. 6.M.1 and Figure 6.M.4. Fillets or haunches can also be incorporated in lieu at these junctions. The method of reinforcing the stair shown in diagram (a) of Figure 6.M.3 is very suitable but is generally only practicable if haunches are provided. Otherwise the arrangement shown in diagram (b) should be adopted.

M.3 Helicoidal Stair

M.3.1 Loading

Helicoidal stair shall be designed to support the design ultimate load according to the load combinations specified in Chapter 2, Loads.

M.3.2 Geometry

The pertinent geometry of the Helicoidal stair is given at Figure 6.M.5 where:

- b : Width of stair slab
- h : Thickness of the stair slab
- n : Total load per unit length projected along centre-line of load
- R_l : Radius of centre-line of loading = $(2/3)(R_0^3 - R_i^3)/(R_0^2 - R_i^2)$
- R_2 : Radius of centre-line of steps = $(1/2)(R_i + R_o)$, where R_i and R_o are the internal and external radii of the stair, respectively
- θ : Angle subtended in plan between point considered and midpoint of stair
- β : Total angle subtended by helix in plan
- ϕ : Slope of tangent to helix center-line measured from horizontal
- M_o : Bending moment at midpoint of stair.
- H : Lateral shear at the midpoint of stair.
- I_1, I_2 : Second moment of area of stair section about horizontal axis and axis normal to slope, respectively

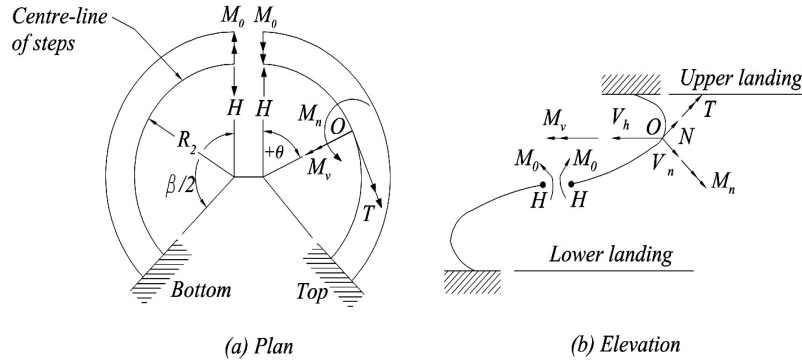


Figure 6.M.5 Elements of helicoidal stair

M.3.3 Effective Span

Helicoidal stairs shall be supported with stringer beams at landing levels (Figure 6.M.5). The effective span for the stair shall be βR , where β is the total angle subtended by helix in plan measured horizontally (Figure 6.M.5) from the face of the stringer beams.

M.3.4 Depth of Section

The depth of the section shall be taken as the minimum thickness perpendicular to the soffit of the stair unless otherwise the large geometric dimensions warrant calculating the deflections through a suitable numerical analysis.

M.3.5 Design

The moments, thrust, torsion and shear forces shall be obtained from the following equations:

Lateral moment (in-plane moment):

$$M_n = M_o \sin \theta \sin \phi - HR_2 \theta \tan \phi \cos \theta \sin \phi - HR_2 \sin \theta \cos \phi + nR_1 \sin \phi (R_1 \sin \theta - R_2 \theta) \quad (6.M.2)$$

Torsional moment:

$$T = (M_o \sin \theta - HR_2 \theta \cos \theta \tan \phi + nR_1^2 \sin \theta - nR_1 R_2 \theta) \cos \phi + HR_2 \sin \theta \sin \phi \quad (6.M.3)$$

Vertical moment (bending moment):

$$M_y = M_o \cos \theta + HR_2 \theta \tan \phi \sin \theta - nR_1^2 (1 - \cos \theta) \quad (6.M.4)$$

Thrust (axial force):

$$N = -H \sin \theta \cos \phi - nR_1 \theta \sin \phi \quad (6.M.5)$$

Lateral shearing force across stair:

$$V_n = nR_1 \theta \cos \phi - H \sin \theta \sin \phi \quad (6.M.6)$$

Radial horizontal shearing force:

$$V_h = H \cos \theta \quad (6.M.7)$$

Where,

$$M_o = \text{redundant moment acting tangentially at mid-span} = k_1 n R_2^2 \quad (6.M.8)$$

$$H = \text{horizontal redundant force at mid-span} = k_2 n R_2 \quad (6.M.9)$$

$$M_{vs} = \text{vertical moment at supports} = k_3 n R_2^2 \quad (6.M.10)$$

The values of coefficients k_1 , k_2 and k_3 can be obtained from the charts provided in Figures 6.M.6 to 6.M.9 for different combinations of R_1/R_2 and b/h ratios. To determine values of coefficients k_1 , k_2 and k_3 for other intermediate values of R_1/R_2 and b/h ratios, interpolations may be performed.

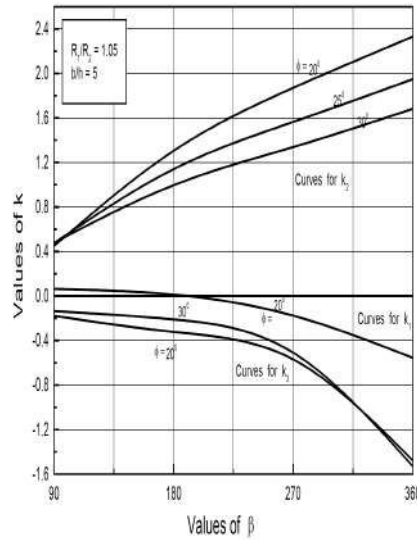


Figure 6.M.6 Design charts for helicoidal stair slabs for $R_1/R_2 = 1.05$ and $b/h = 5$.

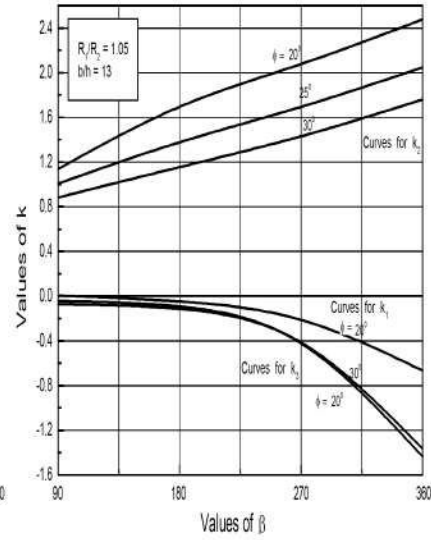


Figure 6.M.7 Design charts for helicoidal stair slabs for $R_1/R_2 = 1.05$ and $b/h = 13$.

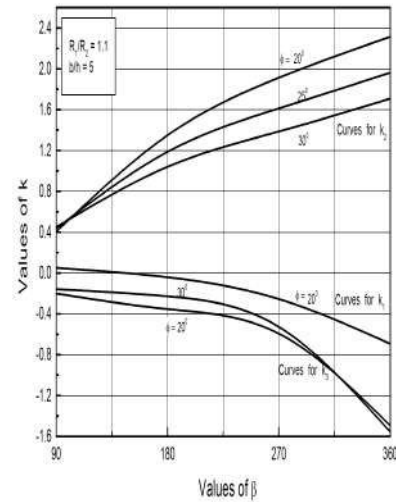


Figure 6.M.8 Design charts for helicoidal stair slabs for $R_1/R_2 = 1.1$; $b/h = 5$.

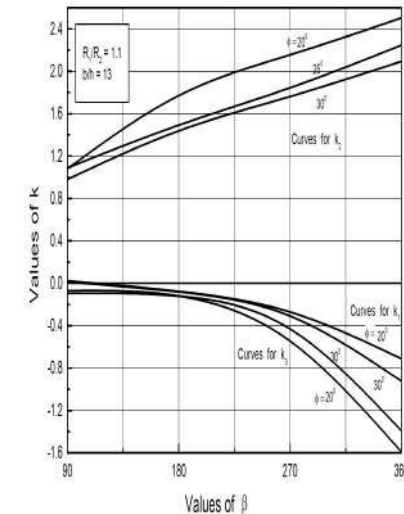


Figure 6.M.9 Design charts for helicoidal stair slabs for $R_1/R_2 = 1.1$; $b/h = 13$.

PART VI
Appendix N
Prequalification of Beam-Column and
Link-to-Column Connections

N.1 Scope

This Appendix contains minimum requirements for prequalification of beam-to-column moment connections in special moment frames (SMF), intermediate moment frames (IMF), and link-to-column connections in eccentrically braced frames (EBF). Prequalified connections are permitted to be used, within the applicable limits of prequalification, without the need for further qualifying cyclic tests. When the limits of prequalification or design requirements for prequalified connections conflict with the requirements of these Provisions, the limits of prequalification and design requirements for prequalified connections shall govern.

N.2 General Requirements

N.2.1 Basis for Prequalification

Connections shall be prequalified based on test data satisfying Section N.3, supported by analytical studies and design models. The combined body of evidence for prequalification must be sufficient to assure that the connection can supply the required interstory drift angle for SMF and IMF systems, or the required link rotation angle for EBF, on a consistent and reliable basis within the specified limits of prequalification. All applicable limit states for the connection that affect the stiffness, strength and deformation capacity of the connection and the seismic load resisting system (SLRS) must be identified. These include fracture related limit states, stability related limit states, and all other limit states pertinent for the connection under consideration. The effect of design variables listed in Section N.4 shall be addressed for connection prequalification.

N.2.2 Authority for Prequalification

Prequalification of a connection and the associated limits of prequalification shall be established by a connection prequalification review panel (CPRP) approved by the authority having jurisdiction.

N.3 Testing Requirements

Data used to support connection prequalification shall be based on tests conducted in accordance with Appendix Q. The CPRP shall determine the number of tests and the variables considered by the tests for connection prequalification. The CPRP shall also provide the same information when limits are to be changed for a previously prequalified connection. A sufficient number of tests shall be performed on a sufficient number of nonidentical specimens to demonstrate that the connection has the ability and reliability to undergo the required interstory drift angle for SMF and IMF and the required link rotation angle for EBF, where the link is adjacent to columns. The limits on member sizes for prequalification shall not exceed the limits specified in Appendix Q, Section Q.5.2.

N.4 Prequalification Variables

In order to be prequalified, the effect of the following variables on connection performance shall be considered. Limits on the permissible values for each variable shall be established by the CPRP for the prequalified connection.

- (1) Beam or link parameters:
 - (a) Cross-section shape: wide flange, box, or other.
 - (b) Cross-section fabrication method: rolled shape, welded shape, or other.
 - (c) Depth.
 - (d) Weight per foot.
 - (e) Flange thickness.
 - (f) Material specification.
 - (g) Span-to-depth ratio (for SMF or IMF), or link length (for EBF)
 - (h) Width thickness ratio of cross-section elements.
 - (i) Lateral bracing.
 - (j) Other parameters pertinent to the specific connection under consideration.
- (2) Column parameters:
 - (a) Cross-section shape: wide flange, box, or other.
 - (b) Cross-section fabrication method: rolled shape, welded shape, or other.
 - (c) Column orientation with respect to beam or link: beam or link is connected to column flange, beam or link is connected to column web, beams or links are connected to both the column flange and web, or other.
 - (d) Depth.
 - (e) Weight per foot.
 - (f) Flange thickness.
 - (g) Material specification.
 - (h) Width-thickness ratio of cross-section elements.
 - (i) Lateral bracing.
 - (j) Other parameters pertinent to the specific connection under consideration.
- (3) Beam (or link)—column relations:
 - (a) Panel zone strength
 - (b) Doubler plate attachment details
 - (c) Column-beam (or link) moment ratio

(4) Continuity plates:

- (a) Identification of conditions under which continuity plates are required
- (b) Thickness, width and depth
- (c) Attachment details

(5) Welds:

- (a) Location, extent (including returns), type (CJP, PJP, fillet, etc.) and any reinforcement or contouring required
- (b) Filler metal classification strength and notch toughness
- (c) Details and treatment of weld backing and weld tabs
- (d) Weld access holes: size, geometry and finish
- (e) Welding quality control and quality assurance beyond that described in Appendix O, including the nondestructive testing (NDT) method, inspection frequency, acceptance criteria and documentation requirements

(6) Bolts:

- (a) Bolt diameter
- (b) Bolt grade: ASTM A325, A490, or other
- (c) Installation requirements: pretensioned, snug-tight, or other
- (d) Hole type: standard, oversize, short-slot, long-slot, or other
- (e) Hole fabrication method: drilling, punching, sub-punching and reaming, or other
- (f) Other parameters pertinent to the specific connection under consideration

(7) Workmanship: All workmanship parameters that exceed AISC, RCSC and AWS requirements, pertinent to the specific connection under consideration, such as:

- (a) Surface roughness of thermal cut or ground edges.
- (b) Cutting tolerances.
- (c) Weld reinforcement or contouring.
- (d) Presence of holes, fasteners or welds for attachments.

(8) Additional connection details:

All variables pertinent to the specific connection under consideration, as established by the CPRP.

N.5 Design Procedure

A comprehensive design procedure must be available for a prequalified connection. The design procedure must address all applicable limit states within the limits of prequalification.

N.6 Prequalification Record

A prequalified connection shall be provided with a written prequalification record with the following information:

- (a) General description of the prequalified connection and drawings that clearly identify key features and components of the connection.
- (b) Description of the expected behavior of the connection in the elastic and inelastic ranges of behavior, intended location(s) of inelastic action, and a description of limit states controlling the strength and deformation capacity of the connection.
- (c) Listing of systems for which connection is prequalified: SMF, IMF, or EBF
- (d) Listing of limits for all prequalification variables listed in Section N.4.
- (e) Listing of demand critical welds.
- (f) Definition of the region of the connection that comprises of the protected zone.
- (g) Detailed description of the design procedure for the connection, as required in Section N.5.
- (h) List of references of test reports, research reports and other publications that provided the basis for prequalification.
- (i) Summary of quality control and quality assurance procedures.

PART VI
Appendix O
Quality Assurance Plan

O.1 Scope

Quality control (QC) and quality assurance (QA) shall be provided as specified in this Section.

O.2 Inspection and Nondestructive Testing Personnel

Visual welding inspection and nondestructive testing (NDT) shall be conducted in accordance with a written practice by personnel qualified in accordance with Appendix S.

Bolting inspection shall be conducted in accordance with a written practice by qualified personnel.

O.3 Contractor Documents

The following documents shall be submitted for review by the engineer of record or designee, prior to fabrication or erection, as applicable:

- (1) Shop drawings.
- (2) Erection drawings.
- (3) Welding Procedure Specifications (WPS), which shall specify all applicable essential variables of AWS D1.1 and the following, as applicable.
 - (a) power source (constant current or constant voltage).
 - (b) for demand critical welds, electrode manufacturer and trade name.
- (4) Copies of the manufacturer's typical certificate of conformance for all electrodes, fluxes and shielding gasses to be used. Certificates of conformance shall satisfy the applicable AWS A5 requirements.
- (5) For demand critical welds, applicable manufacturer's certifications that the filler metal meets the supplemental notch toughness requirements, as applicable. Should the filler metal manufacturer not supply such supplemental certifications, the contractor shall have the necessary testing performed and provide the applicable test reports.
- (6) Manufacturer's product data sheets or catalog data for SMAW, FCAW and GMAW composite (cored) filler metals to be used. The data sheets shall describe the product, limitations of use, recommended or typical welding parameters, and storage and exposure requirements, including baking, if applicable.

The following documents shall be available for review by the engineer of record or designee prior to fabrication or erection, as applicable, unless specified to be submitted:

- (1) Material test reports for structural steel, bolts, shear connectors, and welding materials.
- (2) Inspection procedures.
- (3) Nonconformance procedure.
- (4) Material control procedure.
- (5) Bolt installation procedure.
- (6) Welder performance qualification records (WPQR), including any supplemental testing requirements.
- (7) QC Inspector qualifications.

O.4 Quality Assurance Agency Documents

The agency responsible for quality assurance shall submit the following documents to the authority having jurisdiction, the engineer of record, and the owner or owner's designee:

- (1) QA agency's written practices for the monitoring and control of the agency's operations. The written practice shall include:
 - (a) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualification and certification of inspection personnel, and
 - (b) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.
- (2) Qualifications of management and QA personnel designated for the project.
- (3) Qualification records for Inspectors and NDT technicians designated for the project.
- (4) NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.
- (5) Daily or weekly inspection reports.
- (6) Nonconformance reports.

O.5 Inspection Points and Frequencies

Inspection points and frequencies of quality control (QC) and quality assurance (QA) tasks and documentation for the seismic load resisting system (SLRS) shall be as provided in the following Tables.

The following entries are used in the Tables:

Observe (O) - The inspector shall observe these functions on a random, daily basis. Welding operations need not be delayed pending observations.

Perform (P) - These inspections shall be performed prior to the final acceptance of the item. Where a task is noted to be performed by both QC and QA, it shall be permitted to coordinate the inspection function between QC and QA so that the inspection functions need be performed by only one party. Where QA is to rely upon inspection functions performed by QC, the approval of the engineer of record and the authority having jurisdiction is required.

Document (D) - The inspector shall prepare reports indicating that the work has been performed in accordance with the contract documents. The report need not provide detailed measurements for joint fit-up, WPS settings, completed welds, or other individual items listed in the Tables in Sections O.5.1, O.5.3, or O.5.4. For shop fabrication, the report shall indicate the piece mark of the piece inspected. For field work, the report shall indicate the reference grid lines and floor or elevation inspected. Work not in compliance with the contract documents and whether the noncompliance has been satisfactorily repaired shall be noted in the inspection report.

O.5.1 Visual Welding Inspection

Visual inspection of welding shall be the primary method used to confirm that the procedures, materials, and workmanship incorporated in construction are those that have been specified and approved for the project. As a minimum, welding inspection tasks shall be as follows:

Table 6.O.1: Visual Welding Inspection Tasks

Visual Inspection Tasks Before Welding	QC		QA	
	Task	Doc.	Task	Doc.
Material identification (Type/Grade)	O	–	O	–
Fit-up of Groove Welds (including joint geometry)	P/O**	–	O	–
– Joint preparation				
– Dimensions (alignment, root opening, root face, bevel)				
– Cleanliness (condition of steel surfaces)				
– Tacking (tack weld quality and location)				
– Backing type and fit (if applicable)				
Configuration and finish of access holes	O	–	O	–
Fit-up of Fillet Welds	P/O**	–	O	–
– Dimensions (alignment, gaps at root)				
– Cleanliness (condition of steel surfaces)				
– Tacking (tack weld quality and location)				
** Following performance of this inspection task for ten welds to be made by a given welder, with the welder demonstrating adequate understanding of requirements and possession of skills and tools to verify these items, the Perform designation of this task shall be reduced to Observe, and the welder shall perform this task. Should the inspector determine that the welder has discontinued adequate performance of this task, the task shall be returned to Perform until such time as the Inspector has reestablished adequate assurance that the welder will perform the inspection tasks listed.				

Visual Inspection Tasks During Welding	QC		QA	
	Task	Doc.	Task	Doc.
WPS followed	O	–	O	–
– Settings on welding equipment				
– Travel speed				
– Selected welding materials				
– Shielding gas type/flow rate				
– Preheat applied				
– Interpass temperature maintained (min./max.)				
– Proper position (F, V, H, OH)				
– Intermix of filler metals avoided unless approved				
Use of qualified welders	O	–	O	–
Control and handling of welding consumables	O	–	O	–
– Packaging				
– Exposure control	O	–	O	–
Environmental conditions				
– Wind speed within limits				
– Precipitation and temperature	O	–	O	–
Welding techniques				
– Interpass and final cleaning				
– Each pass within profile limitations				
– Each pass meets quality requirements				
No welding over cracked tacks	O	–	O	–

Visual Inspection Tasks After Welding	QC		QA	
	Task	Doc.	Task	Doc.
Welds cleaned	O	–	O	–
Welder identification legible	O	–	O	–
Verify size, length, and location of welds	O	–	O	–
Visually inspect welds to acceptance criteria	P	D	P	D
– Crack prohibition				
– Weld/base-metal fusion				
– Crater cross-section				
– Weld profiles				
– Weld size				
– Undercut				
– Porosity				
Placement of reinforcement fillets	P	D	P	D
Backing bars removed and weld tabs removed and finished (if required)	P	D	P	D
Repair activities	P	–	P	D

O.5.2 Nondestructive Testing (NDT) of Welds

Nondestructive testing of welds shall be performed by quality assurance personnel.

(1) Procedures

Ultrasonic testing shall be performed by QA according to the procedures prescribed in Appendix S, Sec S.4.1.

Magnetic particle testing shall be performed by QA according to procedures prescribed in Appendix S, Sec S.4.2.

(2) Required NDT**(a) k-Area NDT**

When welding of doubler plates, continuity plates, or stiffeners has been performed in the k-area, the web shall be tested for cracks using magnetic particle testing (MT). The MT inspection area shall include the k-area base metal within 75 mm of the weld.

(b) CJP Groove Weld NDT

Ultrasonic testing shall be performed on 100 percent of CJP groove welds in materials 8 mm thick or greater. Ultrasonic testing in materials less than 8 mm thick is not required. Magnetic particle testing shall be performed on 25 percent of all beam-to-column CJP groove welds.

(c) Base Metal NDT for Lamellar Tearing and Laminations

After joint completion, base metal thicker than 38 mm loaded in tension in the through thickness direction in tee and corner joints, where the connected material is greater than 19 mm and contains CJP groove welds, shall be ultrasonically tested for discontinuities behind and adjacent to the fusion line of such welds. Any base metal discontinuities found within $t/4$ of the steel surface shall be accepted or rejected on the basis of criteria of AWS D1.1 Table 6.2, where t is the thickness of the part subjected to the through-thickness strain.

(d) Beam Cope and Access Hole NDT

At welded splices and connections, thermally cut surfaces of beam copes and access holes shall be tested using magnetic particle testing or penetrant testing, when the flange thickness exceeds 38 mm for rolled shapes, or when the web thickness exceeds 38 mm for built-up shapes.

(e) Reduced Beam Section Repair NDT

Magnetic particle testing shall be performed on any weld and adjacent area of the reduced beam section (RBS) plastic hinge region that has been repaired by welding, or on the base metal of the RBS plastic hinge region if a sharp notch has been removed by grinding.

(f) Weld Tab Removal Sites

Magnetic particle testing shall be performed on the end of welds from which the weld tabs have been removed, except for continuity plate weld tabs.

(g) Reduction of Percentage of Ultrasonic Testing

The amount of ultrasonic testing is permitted to be reduced if approved by the engineer of record and the authority having jurisdiction. The nondestructive testing rate for an individual welder or welding operator may be reduced to 25 percent, provided the reject rate is demonstrated to be 5 percent or less of the welds tested for the welder or welding operator. A sampling of at least 40 completed welds for a job shall be made for such reduction evaluation. Reject rate is the number of welds containing rejectable defects divided by the number of welds completed. For evaluating the reject rate of continuous welds over 1 m in length where the effective throat thickness is 25 mm or less, each 300 mm increment or fraction thereof shall be considered as one weld. For evaluating the reject rate on continuous welds over 1 m in length where the effective throat thickness is greater than 25 mm, each 150 mm of length or fraction thereof shall be considered one weld.

(h) Reduction of Percentage of Magnetic Particle Testing

The amount of MT on CJP groove welds is permitted to be reduced if approved by the engineer of record and the authority having jurisdiction. The MT rate for an individual welder or welding operator may be reduced to 10 percent, provided the reject rate is demonstrated to be 5 percent or less of the welds tested for the welder or welding operator. A sampling of at least 20 completed welds for a job shall be made for such reduction evaluation. Reject rate is the number of welds containing rejectable defects divided by the number of welds completed. This reduction is not permitted on welds in the k-area, at repair sites, weld tab and backing removal sites and access holes.

(3) Documentation

All NDT performed shall be documented. For shop fabrication, the NDT report shall identify the tested weld by piece mark and location in the piece. For field work, the NDT report shall identify the tested weld by location in the structure, piece mark, and location in the piece.

O.5.3 Inspection of Bolting

Observation of bolting operations shall be the primary method used to confirm that the procedures, materials, and workmanship incorporated in construction are those that have been specified and approved for the project. As a minimum, the tasks shall be as follows:

Inspection Tasks Prior to Bolting	QC		QA	
	Task	Doc.	Task	Doc.
Proper bolts selected for the joint detail	O	–	O	–
Proper bolting procedure selected for joint detail	O	–	O	–
Connecting elements are fabricated properly, including the appropriate faying surface condition and hole preparation, if specified, meets applicable requirements	O	–	O	–
Pre-installation verification testing conducted for fastener assemblies and methods used	P	D	O	D
Proper storage provided for bolts, nuts, washers, and other fastener components	O	–	O	–

Inspection Tasks During Bolting	QC		QA	
	Task	Doc.	Task	Doc.
Fastener assemblies placed in all holes and washers (if required) are properly positioned	O	–	O	–
Joint brought to the snug tight condition prior to the pretensioning operation	O	–	O	–
Fastener component not turned by the wrench prevented from rotating	O	–	O	–
Bolts are pretensioned progressing systematically from most rigid point toward free edges	O	–	O	–

Inspection Tasks After Bolting	QC		QA	
	Task	Doc.	Task	Doc.
Document accepted and rejected connections	P	D	P	D

O.5.4 Other Inspections

Where applicable, the following inspection tasks shall be performed:

Other Inspection Task	QC		QA	
	Task	Doc.	Task	Doc.
Reduced beam section (RBS) requirements, if applicable	P	D	P	D
– contour and finish				
– dimensional tolerances				
Protected zone – no holes and unapproved attachments made by contractor	P	D	P	D

PART VI
Appendix P

Seismic Design Coefficients and Approximate Period Parameters

P.1 Scope

This appendix contains design coefficients, system limitations and design parameters for seismic load resisting systems (SLRS) that are included in these provisions but not yet defined in the applicable building code for buckling-restrained braced frames (BRBF) and special plate shear walls (SPSW). The values presented in Tables 6.P.1 and 6.P.2 in this Appendix shall only be used where neither the applicable building code nor SEI/ASCE 7 contain such values.

P.2 Symbols

The following symbols are used in this appendix.

C_d	Deflection amplification factor
C_r, x	Parameters used for determining the approximate fundamental period
Ω_o	System overstrength factor
R	Response modification coefficient

P.3 Coefficients and Factors for Basic Seismic Load Resisting Systems

TABLE 6.P.1 Design Coefficients and Factors for Basic Seismic Load Resisting Systems

Basic Seismic Load Resisting System	Response Modification/ Reduction Coefficient R	System Overstrength Factor Ω_o	Deflection Amplification Factor C_d	Height Limit (m)	
				Seismic Design Category	
				B & C	D
Building Frame Systems					
Buckling-Restrained Braced Frames, non-moment-resisting beam-column connections	7	2	5½	NL	48
Special Plate Shear Walls	7	2	6	NL	48
Buckling-Restrained Braced Frames, moment-resisting beam-column connections	8	2½	5	NL	48
Dual Systems with Special Moment Frames Capable of Resisting at Least 25% of the Prescribed Seismic Forces					
Buckling-Restrained Braced Frame	8	2½	5	NL	NL
Special Plate Shear Walls	8	2½	6½	NL	NL

(NL = Not Limited)

P.4 VALUES OF APPROXIMATE PERIOD PARAMETERS

Table 6.P.2 Values of Approximate Period Parameters C_r and x

Structure Type	C_r	x
Buckling-Restrained Braced Frames	0.03	0.75
Special Plate Shear Walls	0.02	0.75

PART VI
Appendix Q
Qualifying Cyclic Tests of Beam-to-Column and
Link-to-Column Connections

Q.1 Scope

This Appendix includes requirements for qualifying cyclic tests of beam-to-column moment connections in special and intermediate moment frames and link-to-column connections in eccentrically braced frames, when required in these Provisions. The purpose of the testing described in this Appendix is to provide evidence that a beam-to-column connection or a link-to-column connection satisfies the requirements for strength and interstory drift angle or link rotation angle in these Provisions. Alternative testing requirements are permitted when approved by the engineer of record and the authority having jurisdiction.

This Appendix provides minimum recommendations for simplified test conditions.

Q.2 Symbols

The numbers in parentheses after the definition of a symbol refers to the Section number in which the symbol is first used.

θ Interstory drift angle (Q.6)

γ_{total} Total link rotation angle (Q.6)

Q.3 Definitions

Complete loading cycle: A cycle of rotation taken from zero force to zero force, including one positive and one negative peak.

Interstory drift angle: Interstory displacement divided by story height, radians.

Inelastic rotation: The permanent or plastic portion of the rotation angle between a beam and the column or between a link and the column of the test specimen, measured in radians. The inelastic rotation shall be computed based on an analysis of test specimen deformations. Sources of inelastic rotation include yielding of members, yielding of connection elements and connectors, and slip between members and connection elements. For beam-to-column moment connections in special and intermediate moment frames, inelastic rotation is computed based upon the assumption that inelastic action is concentrated at a single point located at the intersection of the centerline of the beam with the centerline of the column. For link-to-column connections in eccentrically braced frames, inelastic rotation shall be computed based upon the assumption that inelastic action is concentrated at a single point located at the intersection of the centerline of the link with the face of the column.

Prototype : The connections, member sizes, steel properties, and other design, detailing, and construction features to be used in the actual building frame.

Test specimen : A portion of a frame used for laboratory testing, intended to model the prototype.

Test setup : The supporting fixtures, loading equipment, and lateral bracing used to support and load the test specimen.

Test subassemblage : The combination of the test specimen and pertinent portions of the test setup.

Total link rotation angle : The relative displacement of one end of the link with respect to the other end (measured transverse to the longitudinal axis of the undeformed link), divided by the link length. The total link rotation angle shall include both elastic and inelastic components of deformation of the link and the members attached to the link ends.

Q.4 Test Subassemblage Requirements

The test subassemblage shall replicate as closely as is practical the conditions that will occur in the prototype during earthquake loading. The test subassemblage shall include the following features :

- (1) The test specimen shall consist of at least a single column with beams or links attached to one or both sides of the column.
- (2) Points of inflection in the test assemblage shall coincide approximately with the anticipated points of inflection in the Prototype under earthquake loading.
- (3) Lateral bracing of the test subassemblage is permitted near load application or reaction points as needed to provide lateral stability of the test subassemblage. Additional lateral bracing of the test subassemblage is not permitted, unless it replicates lateral bracing to be used in the prototype.

Q.5 Essential Test Variables

The test specimen shall replicate as closely as is practical the pertinent design, detailing, construction features, and material properties of the prototype. The following variables shall be replicated in the test specimen.

Q.5.1 Sources of Inelastic Rotation

Inelastic rotation shall be developed in the test specimen by inelastic action in the same members and connection elements as anticipated in the prototype (in other words, in the beam or link, in the column panel zone, in the column outside of the panel zone, or in connection elements) within the limits described below. The percentage of the total inelastic rotation in the test specimen that is developed in each member or connection element shall be within 25 percent of the anticipated percentage of the total inelastic rotation in the prototype that is developed in the corresponding member or connection element.

Q.5.2 Size of Members

The size of the beam or link used in the test specimen shall be within the following limits:

- (1) The depth of the test beam or link shall be no less than 90 percent of the depth of the prototype beam or link.
- (2) The weight per foot of the test beam or link shall be no less than 75 percent of the weight per foot of the prototype beam or link.

The size of the column used in the test specimen shall properly represent the inelastic action in the column, as per the requirements in Section Q.5.1. In addition, the depth of the test column shall be no less than 90 percent of the depth of the prototype column.

Extrapolation beyond the limitations stated in this Section shall be permitted subject to qualified peer review and approval by the authority having jurisdiction.

Q.5.3 Connection Details

The connection details used in the test specimen shall represent the prototype connection details as closely as possible. The connection elements used in the test specimen shall be a full-scale representation of the connection elements used in the prototype, for the member sizes being tested.

Q.5.4 Continuity Plates

The size and connection details of continuity plates used in the test specimen shall be proportioned to match the size and connection details of continuity plates used in the prototype connection as closely as possible.

Q.5.5 Material Strength

The following additional requirements shall be satisfied for each member or connection element of the test specimen that supplies inelastic rotation by yielding:

- (1) The yield stress shall be determined by material tests on the actual materials used for the test specimen, as specified in Section Q.8. The use of yield stress values that are reported on certified mill test reports are not permitted to be used for purposes of this Section.

The yield stress of the beam shall not be more than 15 percent below R_y F_y for the grade of steel to be used for the corresponding elements of the prototype. Columns and connection elements with a tested yield stress shall not be more than 15 percent above or below R_y F_y for the grade of steel to be used for the corresponding elements of the prototype. R_y F_y shall be determined in accordance with Section Q.6.2.

Q.5.6 Welds

Welds on the test specimen shall satisfy the following requirements :

- (1) Welding shall be performed in strict conformance with Welding Procedure Specifications (WPS) as required in AWS D1.1. The WPS essential variables shall meet the requirements in AWS D1.1 and shall be within the parameters established by the filler-metal manufacturer. The tensile strength of the welds used in the tested assembly and the Charpy V-Notch (CVN) toughness used in the tested assembly shall be determined by material tests as specified in Section Q.8.3. The use of tensile strength and CVN toughness values that are reported on the manufacturer's typical certificate of conformance is not permitted to be used for purposes of this Section, unless the report includes results specific to Appendix T requirements.
- (2) The specified minimum tensile strength of the filler metal used for the test specimen shall be the same as that to be used for the corresponding prototype welds. The tested tensile strength of the test specimen weld shall not be more than 125 MPa above the tensile strength classification of the filler metal specification specified for the prototype.
- (3) The specified minimum CVN toughness of the filler metal used for the test specimen shall not exceed the specified minimum CVN toughness of the filler metal to be used for the corresponding prototype welds. The tested CVN toughness of the test specimen weld shall not be more than 50 percent, nor 34 kJ, whichever is greater, above the minimum CVN toughness that will be specified for the prototype.
- (4) The welding positions used to make the welds on the test specimen shall be the same as those to be used for the prototype welds.
- (5) Details of weld backing, weld tabs, access holes, and similar items used for the test specimen welds shall be the same as those to be used for the corresponding prototype welds. Weld backing and weld tabs shall not be removed from the test specimen welds unless the corresponding weld backing and weld tabs are removed from the prototype welds.
- (6) Methods of inspection and nondestructive testing and standards of acceptance used for test specimen welds shall be the same as those to be used for the prototype welds.

Q.5.7 Bolts

The bolted portions of the test specimen shall replicate the bolted portions of the prototype connection as closely as possible. Additionally, bolted portions of the test specimen shall satisfy the following requirements :

- (1) The bolt grade (for example, ASTM A325, A325M, ASTM A490, A490M, ASTM F1852) used in the test specimen shall be the same as that to be used for the prototype, except that ASTM A325 bolts may be substituted for ASTM F1852 bolts, and vice versa.

- (2) The type and orientation of bolt holes (standard, oversize, short slot, long slot, or other) used in the test specimen shall be the same as those to be used for the corresponding bolt holes in the prototype.
- (3) When inelastic rotation is to be developed either by yielding or by slip within a bolted portion of the connection, the method used to make the bolt holes (drilling, sub-punching and reaming, or other) in the test specimen shall be the same as that to be used in the corresponding bolt holes in the prototype.
- (4) Bolts in the test specimen shall have the same installation (pretensioned or other) and faying surface preparation (no specified slip resistance, Class A or B slip resistance, or other) as that to be used for the corresponding bolts in the prototype.

Q.6 LOADING HISTORY

Q.6.1 General Requirements

The test specimen shall be subjected to cyclic loads according to the requirements prescribed in Section Q.6.2 for beam-to-column moment connections in special and intermediate moment frames, and according to the requirements prescribed in Section Q.6.3 for link-to-column connections in eccentrically braced frames.

Loading sequences other than those specified in Sections Q.6.2 and Q.6.3 may be used when they are demonstrated to be of equivalent or greater severity.

Q.6.2 Loading Sequence for Beam-to-Column Moment Connections

Qualifying cyclic tests of beam-to-column moment connections in special and intermediate moment frames shall be conducted by controlling the interstory drift angle, θ , imposed on the test specimen, as specified below:

- (1) 6 cycles at $\theta = 0.00375$ rad
- (2) 6 cycles at $\theta = 0.005$ rad
- (3) 6 cycles at $\theta = 0.0075$ rad
- (4) 4 cycles at $\theta = 0.01$ rad
- (5) 2 cycles at $\theta = 0.015$ rad
- (6) 2 cycles at $\theta = 0.02$ rad
- (7) 2 cycles at $\theta = 0.03$ rad
- (8) 2 cycles at $\theta = 0.04$ rad

Continue loading at increments of $\theta = 0.01$ radian, with two cycles of loading at each step.

Q.6.3 Loading Sequence for Link-to-Column Connections

Qualifying cyclic tests of link-to-column moment connections in eccentrically braced frames shall be conducted by controlling the total link rotation angle, γ_{total} , imposed on the test specimen, as follows :

- (1) 6 cycles at $\gamma_{total} = 0.00375$ rad
- (2) 6 cycles at $\gamma_{total} = 0.005$ rad
- (3) 6 cycles at $\gamma_{total} = 0.0075$ rad
- (4) 6 cycles at $\gamma_{total} = 0.01$ rad
- (5) 4 cycles at $\gamma_{total} = 0.015$ rad
- (6) 4 cycles at $\gamma_{total} = 0.02$ rad
- (7) 2 cycles at $\gamma_{total} = 0.03$ rad
- (8) 1 cycle at $\gamma_{total} = 0.04$ rad
- (9) 1 cycle at $\gamma_{total} = 0.05$ rad
- (10) 1 cycle at $\gamma_{total} = 0.07$ rad
- (11) 1 cycle at $\gamma_{total} = 0.09$ rad

Continue loading at increments of $\gamma_{total} = 0.02$ radian, with one cycle of loading at each step.

Q.7 Instrumentation

Sufficient instrumentation shall be provided on the test specimen to permit measurement or calculation of the quantities listed in Section Q.9.

Q.8 Materials Testing Requirements**Q.8.1 Tension Testing Requirements for Structural Steel**

Tension testing shall be conducted on samples of steel taken from the material adjacent to each test specimen. Tension-test results from certified mill test reports shall be reported but are not permitted to be used in place of specimen testing for the purposes of this Section. Tension-test results shall be based upon testing that is conducted in accordance with Section Q.8.2. Tension testing shall be conducted and reported for the following portions of the test specimen:

- (1) Flange(s) and web(s) of beams and columns at standard locations
- (2) Any element of the connection that supplies inelastic rotation by yielding

Q.8.2 Methods of Tension Testing for Structural Steel

Tension testing shall be conducted in accordance with ASTM A6/A6M, ASTM A370, and ASTM E8, with the following exceptions:

- (1) The yield stress, F_y , that is reported from the test shall be based upon the yield strength definition in ASTM A370, using the offset method at 0.002 strain.
- (2) The loading rate for the tension test shall replicate, as closely as practical, the loading rate to be used for the test specimen.

Q.8.3 Weld Metal Testing Requirements

The tensile strength of the welds used in the tested assembly and the CVN toughness used in the tested assembly shall be determined by material tests as specified in Appendix T. The use of tensile strength and CVN toughness values that are reported on the manufacturer's typical certificate of conformance is not permitted to be used for purposes of this section, unless that report includes results specific to Appendix T requirements.

A single test plate may be used if the WPS for the test specimen welds is within plus/minus 0.8 kJ/mm of the WPS for the test plate.

Tensile specimens and CVN specimens shall be prepared in accordance with ANSI/AWS B4.0 Standard Methods for Mechanical Testing of Welds.

Q.9 Test Reporting Requirements

For each test specimen, a written test report meeting the requirements of the *authority having jurisdiction* and the requirements of this Section shall be prepared. The report shall thoroughly document all key features and results of the test. The report shall include the following information :

- (1) A drawing or clear description of the test subassembly, including key dimensions, boundary conditions at loading and reaction points, and location of lateral braces.
- (2) A drawing of the connection detail showing member sizes, grades of steel, the sizes of all connection elements, welding details including filler metal, the size and location of bolt holes, the size and grade of bolts, and all other pertinent details of the connection.
- (3) A listing of all other essential variables for the test specimen, as listed in Section Q.5.
- (4) A listing or plot showing the applied load or displacement history of the test specimen.
- (5) A listing of all demand critical welds.

- (6) Definition of the region of the connection that comprises the protected zones.
- (7) A plot of the applied load versus the displacement of the test specimen. The displacement reported in this plot shall be measured at or near the point of load application. The locations on the test specimen where the loads and displacements were measured shall be clearly indicated.
- (8) A plot of beam moment versus interstory drift angle for beam-to-column moment connections; or a plot of link shear force versus link rotation angle for link-to-column connections. For beam-to-column connections, the beam moment and the interstory drift angle shall be computed with respect to the centerline of the column.
- (9) The interstory drift angle and the total inelastic rotation developed by the test specimen. The components of the test specimen contributing to the total inelastic rotation due to yielding or slip shall be identified. The portion of the total inelastic rotation contributed by each component of the test specimen shall be reported. The method used to compute inelastic rotations shall be clearly shown.
- (10) A chronological listing of significant test observations, including observations of yielding, slip, instability, and fracture of any portion of the test specimen as applicable.
- (11) The controlling failure mode for the test specimen. If the test is terminated prior to failure, the reason for terminating the test shall be clearly indicated.
- (12) The results of the material tests specified in Section Q.8.
- (13) The Welding Procedure Specifications (WPS) and welding inspection reports.

Additional drawings, data, and discussion of the test specimen or test results are permitted to be included in the report.

Q.10 Acceptance Criteria

The test specimen must satisfy the strength and interstory drift angle or link rotation angle requirements of these Provisions for the special moment frame, intermediate moment frame, or eccentrically braced frame connection, as applicable. The test specimen must sustain the required interstory drift angle or link rotation angle for at least one complete loading cycle.

PART VI Appendix R

Qualifying Cyclic Tests of Buckling-restrained Braces

R.1 Scope

This Appendix includes requirements for qualifying cyclic tests of individual buckling-restrained braces and buckling-restrained brace subassemblages, when required in these provisions. The purpose of the testing of individual braces is to provide evidence that a buckling-restrained brace satisfies the requirements for strength and inelastic deformation by these provisions; it also permits the determination of maximum brace forces for design of adjoining elements. The purpose of testing of the brace subassemblage is to provide evidence that the brace-design can satisfactorily accommodate the deformation and rotational demands associated with the design. Further, the subassemblage test is intended to demonstrate that the hysteretic behavior of the brace in the subassemblage is consistent with that of the individual brace elements tested uniaxially.

Alternative testing requirements are permitted when approved by the engineer of record and the authority having jurisdiction. This Appendix provides only minimum recommendations for simplified test conditions.

R.2 Symbols

The numbers in parentheses after the definition of a symbol refers to the Section number in which the symbol is first used.

- Δ_b Deformation quantity used to control loading of the test specimen (total brace end rotation for the subassemblage test specimen; total brace axial deformation for the brace test specimen) (Section R.6).
- $\Delta_{b,m}$ Value of deformation quantity, Δ_b , corresponding to the design story drift (Section R.6).
- $\Delta_{b,y}$ Value of deformation quantity, Δ_b , at first significant yield of test specimen (Section R.6).

R.3 Definitions

BRACE TEST SPECIMEN	A single buckling-restrained brace element used for laboratory testing intended to model the brace in the Prototype.
DESIGN METHODOLOGY	A set of step-by-step procedures, based on calculation or experiment, used to determine sizes, lengths, and details in the design of buckling-restrained braces and their connections.

INELASTIC DEFORMATION	The permanent or plastic portion of the axial displacement in a buckling-restrained brace.
PROTOTYPE	The brace, connections, members, steel properties, and other design, detailing, and construction features to be used in the actual building frame.
SUBASSEMBLAGE TEST SPECIMEN	The combination of the brace, the connections and testing apparatus that replicate as closely as practical the axial and flexural deformations of the brace in the prototype.
TEST SPECIMEN	Brace test specimen or subassembly test specimen.

R.4 Subassembly Test Specimen

The subassembly test specimen shall satisfy the following requirements :

- (1) The mechanism for accommodating inelastic rotation in the subassembly test specimen brace shall be the same as that of the prototype. The rotational deformation demands on the subassembly test specimen brace shall be equal to or greater than those of the prototype.
- (2) The axial yield strength of the steel core, F_{yc} , of the brace in the subassembly test specimen shall not be less than that of the prototype where both strengths are based on the core area, A_{gc} , multiplied by the yield strength as determined from a coupon test.
- (3) The cross-sectional shape and orientation of the steel core projection of the subassembly test specimen brace shall be the same as that of the brace in the prototype.
- (4) The same documented design methodology shall be used for design of the subassembly as used for the prototype, to allow comparison of the rotational deformation demands on the subassembly brace to the prototype. In stability calculations, beams, columns, and gussets connecting the core shall be considered parts of this system.
- (5) The calculated margins of safety for the prototype connection design, steel core projection stability, overall buckling and other relevant subassembly test specimen brace construction details, excluding the gusset plate, for the prototype, shall equal or exceed those of the subassembly test specimen construction.
- (6) Lateral bracing of the subassembly test specimen shall replicate the lateral bracing in the prototype.
- (7) The brace test specimen and the prototype shall be manufactured in accordance with the same quality control and assurance processes and procedures.

Extrapolation beyond the limitations stated in this section shall be permitted subject to qualified peer review and approval by the authority having jurisdiction.

R.5 Brace Test Specimen

The brace test specimen shall replicate as closely as is practical the pertinent design, detailing, construction features, and material properties of the prototype.

R.5.1 Design of Brace Test Specimen

The same documented design methodology shall be used for the brace test specimen and the prototype. The design calculations shall demonstrate, at a minimum, the following requirements :

- (1) The calculated margin of safety for stability against overall buckling for the prototype shall equal or exceed that of the brace test specimen.
- (2) The calculated margins of safety for the brace test specimen and the prototype shall account for differences in material properties, including yield and ultimate stress, ultimate elongation, and toughness.

R.5.2 Manufacture of Brace Test Specimen

The brace test specimen and the prototype shall be manufactured in accordance with the same quality control and assurance processes and procedures.

R.5.3 Similarity of Brace Test Specimen and Prototype

The brace test specimen shall meet the following requirements:

- (1) The cross-sectional shape and orientation of the steel core shall be the same as that of the prototype.
- (2) The axial yield strength of the steel core, P_{JSC} , of the brace test specimen shall not vary by more than 50 percent from that of the prototype where both strengths are based on the core area, A_{SC} , multiplied by the yield strength as determined from a coupon test.
- (3) The material for, and method of, separation between the steel core and the buckling restraining mechanism in the brace test specimen shall be the same as that in the prototype.

Extrapolation beyond the limitations stated in this section shall be permitted subject to qualified peer review and approval by the authority having jurisdiction.

R.5.4 Connection Details

The connection details used in the brace test specimen shall represent the prototype connection details as closely as practical.

R.5.5 Materials

- (1) Steel core: The following requirements shall be satisfied for the steel core of the brace test specimen:
 - (a) The specified minimum yield stress of the brace test specimen steel core shall be the same as that of the prototype.

- (b) The measured yield stress of the material of the steel core in the brace test specimen shall be at least 90 percent of that of the prototype as determined from coupon tests.
- (c) The specified minimum ultimate stress and strain of the brace test specimen steel core shall not exceed those of the prototype.

(2) Buckling-restraining mechanism

Materials used in the buckling-restraining mechanism of the brace test specimen shall be the same as those used in the prototype.

R.5.6 Connections

The welded, bolted, and pinned joints on the test specimen shall replicate those on the prototype as close as practical.

R.6 Loading History

R.6.1 General Requirements

The test specimen shall be subjected to cyclic loads according to the requirements prescribed in Sections R.6.2 and R.6.3. Additional increments of loading beyond those described in Section R.6.3 are permitted. Each cycle shall include a full tension and full compression excursion to the prescribed deformation.

R.6.2 Test Control

The test shall be conducted by controlling the level of axial or rotational deformation, Δ_b , imposed on the test specimen. As an alternate, the maximum rotational deformation may be applied and maintained as the protocol is followed for axial deformation.

R.6.3 Loading Sequence

Loads shall be applied to the test specimen to produce the following deformations, where the deformation is the steel core axial deformation for the test specimen and the rotational deformation demand for the subassembly test specimen brace:

- (1) 2 cycles of loading at the deformation corresponding to $\Delta_b = \Delta_{by}$
- (2) 2 cycles of loading at the deformation corresponding to $\Delta_b = 0.5\Delta_{bm}$
- (3) 2 cycles of loading at the deformation corresponding to $\Delta_b = 1.0\Delta_{bm}$
- (4) 2 cycles of loading at the deformation corresponding to $\Delta_b = 1.5\Delta_{bm}$
- (5) 2 cycles of loading at the deformation corresponding to $\Delta_b = 2.0\Delta_{bm}$
- (6) Additional complete cycles of loading at the deformation corresponding to $\Delta_b = 1.5\Delta_{bm}$ as required for the brace test specimen to achieve a cumulative inelastic axial deformation of at least 200 times the yield deformation (not required for the subassembly test specimen).

The design story drift shall not be taken as less than 0.01 times the story height for the purposes of calculating Δ_{bm} . Other loading sequences are permitted to be used to qualify the test specimen when they are demonstrated to be of equal or greater severity in terms of maximum and cumulative inelastic deformation.

R.7 Instrumentation

Sufficient instrumentation shall be provided on the *test specimen* to permit measurement or calculation of the quantities listed in Section R.9.

R.8 Materials Testing Requirements

R.8.1 Tension Testing Requirements

Tension testing shall be conducted on samples of steel taken from the same material as that used to manufacture the steel core. Tension test results from certified mill test reports shall be reported but are not permitted to be used in place of specimen testing for the purposes of this Section. Tension-test results shall be based upon testing that is conducted in accordance with Section R.8.2.

R.8.2 Methods of Tension Testing

Tension testing shall be conducted in accordance with ASTM A6, ASTM A370, and ASTM E8, with the following exceptions:

- (1) The yield stress that is reported from the test shall be based upon the yield strength definition in ASTM A370, using the offset method of 0.002 strain.
- (2) The loading rate for the tension test shall replicate, as closely as is practical, the loading rate used for the test specimen.
- (3) The coupon shall be machined so that its longitudinal axis is parallel to the longitudinal axis of the steel core.

R.9 Test Reporting Requirements

For each test specimen, a written test report meeting the requirements of this Section shall be prepared. The report shall thoroughly document all key features and results of the test. The report shall include the following information:

- (1) A drawing or clear description of the test specimen, including key dimensions, boundary conditions at loading and reaction points, and location of lateral bracing, if any.
- (2) A drawing of the connection details showing member sizes, grades of steel, the sizes of all connection elements, welding details including filler metal, the size and location of bolt or pin holes, the size and grade of connectors, and all other pertinent details of the connections.

- (3) A listing of all other essential variables as listed in Section R.4 or R.5, as appropriate.
- (4) A listing or plot showing the applied load or displacement history.
- (5) A plot of the applied load versus the deformation, Δ_b . The method used to determine the deformations shall be clearly shown. The locations on the test specimen where the loads and deformations were measured shall be clearly identified.
- (6) A chronological listing of significant test observations, including observations of yielding, slip, instability, transverse displacement along the test specimen and fracture of any portion of the test specimen and connections, as applicable.
- (7) The results of the material tests specified in Section R.8.
- (8) The manufacturing quality control and quality assurance plans used for the fabrication of the test specimen. These shall be included with the welding procedure specifications and welding inspection reports.

Additional drawings, data, and discussion of the test specimen or test results are permitted to be included in the report.

R.10 Acceptance Criteria

At least one subassembly test that satisfies the requirements of Section R.4 shall be performed. At least one brace test that satisfies the requirements of Section R.5, shall be performed. Within the required protocol range all tests shall satisfy the following requirements:

- (1) The plot showing the applied load vs. displacement history shall exhibit stable, repeatable behavior with positive incremental stiffness.
- (2) There shall be no fracture, brace instability or brace end connection failure.
- (3) For brace tests, each cycle to a deformation greater than Δ_{by} the maximum.
- (4) tension and compression forces shall not be less than the nominal strength of the core.
- (5) For brace tests, each cycle to a deformation greater than Δ_{by} the ratio of the maximum compression force to the maximum tension force shall not exceed 1.3.

Other acceptance criteria may be adopted for the brace test specimen or subassembly test specimen subject to qualified peer review and approval by the authority having jurisdiction.

PART VI
Appendix S
Welding Provisions

S.1 Scope

This Appendix provides additional details regarding welding and welding inspection, and is included on an interim basis pending adoption of such criteria by AWS or other accredited organization.

**S.2 STRUCTURAL DESIGN DRAWINGS AND SPECIFICATIONS,
SHOP DRAWINGS, AND ERECTION DRAWINGS**

S.2.1 Structural Design Drawings and Specifications

Structural design drawings and specifications shall include, as a minimum, the following information:

- (1) Locations where backup bars are required to be removed.
- (2) Locations where supplemental fillet welds are required when backing is permitted to remain.
- (3) Locations where fillet welds are used to reinforce groove welds or to improve connection geometry.
- (4) Locations where weld tabs are required to be removed.
- (5) Splice locations where tapered transitions are required.
- (6) The shape of weld access holes, if a special shape is required.
- (7) Joints or groups of joints in which a specific assembly order, welding sequence, welding technique or other special precautions are required.

S.2.2 Shop Drawings

Shop drawings shall include, as a minimum, the following information:

- (1) Access hole dimensions, surface profile and finish requirements.
- (2) Locations where backing bars are to be removed.
- (3) Locations where weld tabs are to be removed.
- (4) NDT to be performed by the fabricator, if any.

S.2.3 Erection Drawings

Erection drawings shall include, as a minimum, the following information:

- (1) Locations where backing bars to be removed.
- (2) Locations where supplemental fillets are required when backing is permitted to remain.
- (3) Locations where weld tabs are to be removed.
- (4) Those joints or groups of joints in which a specific assembly order, welding sequence, welding technique or other special precautions are required.

S.3 Personnel

S.3.1 QC Welding Inspectors

QC welding inspection personnel shall be associate welding inspectors (AWI) or higher, as defined in AWS B5.1 Standard for the Qualification of Welding Inspectors, or otherwise qualified under the provisions of AWS D1.1 Section 6.1.4 and to the satisfaction of the contractor's QC plan by the fabricator/erector.

S.3.2 QA Welding Inspectors

QA welding inspectors shall be welding inspectors (WI), or senior welding inspectors (SWI), as defined in AWS B5.1, except AWIs may be used under the direct supervision of WIs, on site and available when weld inspection is being conducted.

S.4 Nondestructive Testing Technicians

NDT technicians shall be qualified as follows:

- (1) In accordance with their employer's written practice which shall meet or exceed the criteria of the American Society for Nondestructive Testing, Inc. SNT TC-1A Recommended Practice for the Training and Testing of Nondestructive Personnel, or of ANSI/ASNT CP-189, Standard for the Qualification and Certification of Nondestructive Testing Personnel.
- (2) Ultrasonic testing for QA may be performed only by UT technicians certified as ASNT Level III through examination by the ASNT, or certified as Level II by their employer for flaw detection. If the engineer of record approves the use of flaw sizing techniques, UT technicians shall also be qualified and certified by their employer for flaw sizing.
- (3) Magnetic particle testing (MT) and dye penetrant testing (PT) for QA may be performed only by technicians certified as Level II by their employer, or certified as ASNT Level III through examination by the ASNT and certified by their employer.

S.5 Nondestructive Testing Procedures

S.5.1 Ultrasonic Testing

Ultrasonic testing shall be performed according to the procedures prescribed in AWS D1.1 (Section 6, Part F) following a written procedure containing the elements prescribed in paragraph K3 of Annex K, Section 6, Part F (AWS) procedures shall be qualified using weld mock-ups having 1.5 mm diameter side drilled holes similar to Annex K, Figure K-3 (AWS).

S.5.2 Magnetic Particle Testing

Magnetic particle testing shall be performed according to procedures prescribed in AWS D1.1, following a written procedure utilizing the Yoke Method that conforms to ASTM E709.

S.6 Additional Welding Provisions

S.6.1 Intermixed Filler Metals

When FCAW-S filler metals are used in combination with filler metals of other processes, including FCAW-G, a test specimen shall be prepared and mechanical testing shall be conducted to verify that the notch toughness of the combined materials in the intermixed region of the weld meets the notch toughness requirements of Section 10.20.7.3.1 and, if required, the notch toughness requirements for demand critical welds of Section 10.20.7.3.2.

S.6.2 Filler Metal Diffusible Hydrogen

Welding electrodes and electrode-flux combinations shall meet the requirements for H16 (16 mL maximum diffusible hydrogen per 100 grams deposited weld metal) as tested in accordance with AWS A4.3 Standard Methods for Determination of the Diffusible Hydrogen Content of Martensitic, Bainitic, and Ferritic Steel Weld Metal Produced by Arc Welding. (Exception: GMAW solid electrodes.) The manufacturer's typical certificate of conformance shall be considered adequate proof that the supplied electrode or electrode-flux combination meets this requirement. No testing of filler metal samples or of production welds shall be required.

S.6.3 Gas-Shielded Welding Processes

GMAW and FCAW-G shall not be performed in winds exceeding 5 km/hr. Windscreens or other shelters may be used to shield the welding operation from excessive wind.

S.6.4 Maximum Interpass Temperatures

Maximum interpass temperatures shall not exceed 290 °C, measured at a distance not exceeding 75 mm from the start of the weld pass. The maximum interpass temperature may be increased by qualification testing that includes weld metal and base metal CVN testing using AWS D1.1 Annex III. The steel used for the qualification testing shall be of the same type and grade as will be used in production.

The maximum heat input to be used in production shall be used in the qualification testing. The qualified maximum interpass temperature shall be the lowest interpass temperature used for any pass during qualification testing. Both weld metal and HAZ shall be tested. The weld metal shall meet all the mechanical properties required by Sec 10.20.7.3.1 or those for demand critical welds of Sec 10.20.7.3.2, as applicable. The heat affected zone CVN toughness shall meet a minimum requirement of 27 J at 21 °C with specimens taken at both 1 and 5 mm from the fusion line.

S.6.5 Weld Tabs

Where practicable, weld tabs shall extend beyond the edge of the joint a minimum of one inch or the thickness of the part, whichever is greater. Extensions need not exceed 50 mm.

Where used, weld tabs shall be removed to within 3 mm of the base metal surface, except at *continuity plates* where removal to within 6 mm of the plate edge is acceptable, and the end of the weld finished. Removal shall be by air carbon arc cutting (CAC-A), grinding, chipping, or thermal cutting. The process shall be controlled to minimize errant gouging. The edges where weld tabs have been removed shall be finished to a surface roughness of 13 μm or better. Grinding to a flush condition is not required. The contour of the weld end shall provide a smooth transition, free of notches and sharp corners. At T-joints, a minimum radius in the corner need not be provided. The weld end shall be free of gouges and notches. Weld defects not greater than 2 mm deep shall be faired to a slope not greater than 1:5. Other weld defects shall be excavated and repaired by welding in accordance with an applicable WPS.

S.6.6 Bottom Flange Welding Sequence

When using weld access holes to facilitate CJP groove welds of beam bottom flanges to column flanges or continuity plates, the groove weld shall be sequenced as follows:

- (1) As far as is practicable, starts and stops shall not be placed directly under the beam web.
- (2) Each layer shall be completed across the full width of the flange before beginning the next layer.
- (3) For each layer, the weld starts and stops shall be on the opposite side of the beam web, as compared to the previous layer.

S.7 Additional Welding Provisions for Demand Critical Welds Only

S.7.1 Welding Processes

SMAW, GMAW (except short circuit transfer), FCAW and SAW may be used to fabricate and erect members governed by this specification. Other processes may be used, provided that one or more of the following criteria is met:

- (a) The process is part of the prequalified connection details, as listed in Appendix N,
- (b) The process was used to perform a connection qualification test in accordance with Appendix Q, or
- (c) The process is approved by the engineer of record.

S.7.2 Filler Metal Packaging

Electrodes shall be provided in packaging that limits the ability of the electrode to absorb moisture. Electrode from packaging that has been punctured or torn shall be dried in accordance with the manufacturer's recommendations, or shall not be used for *demand critical welds*. Modification or lubrication of the electrode after manufacture is prohibited, except that drying is permitted as recommended by the manufacturer.

S.7.3 Exposure Limitations on FCAW Electrodes

After removal from protective packaging, the permissible atmospheric exposure time of FCAW electrodes shall be limited as follows:

- (1) Exposure shall not exceed the electrode manufacturer's guidelines.
- (2) In the absence of manufacturer's recommendations, the total accumulated exposure time for FCAW electrodes shall not exceed 72 hours. When the electrodes are not in use, they may be stored in protective packaging or a cabinet. Storage time shall not be included in the accumulated exposure time. Electrodes that have been exposed to the atmosphere for periods exceeding the above time limits shall be dried in accordance with the electrode manufacturer's recommendations, or shall not be used for demand critical welds. The electrode manufacturer's recommendations shall include time, temperature, and number of drying cycles permitted.

S.7.4 Tack Welds

Tack welds attaching backing bars and weld tabs shall be placed where they will be incorporated into a final weld.

PART VI
Appendix T
Weld Metal/Welding Procedure Specification Notch Toughness
Verification Test

T.1 Scope

This Appendix provides a standard method for qualification testing of weld filler metals required to have specified notch toughness for service in joints designated as demand critical.

Testing of weld metal to be used in production shall be performed by filler metal manufacturer's production lot, as defined in AWS A5.01, Filler Metal Procurement Guidelines, as follows:

- (1) Class C3 for SMAW electrodes,
- (2) Class S2 for GMAW-S and SAW electrodes,
- (3) Class T4 for FCAW and GMAW-C, or
- (4) Class F2 for SAW fluxes.

Filler metals produced by manufacturers audited and approved by one or more of the following agencies shall be exempt from these production lot testing requirements, provided a minimum of 3 production lots of material, as defined above, are tested in accordance with the provisions of this appendix:

- (1) American Bureau of Shipping (ABS),
- (2) Lloyds Register of Shipping,
- (3) American Society of Mechanical Engineers (ASME),
- (4) ISO 9000,
- (5) US Department of Defense, or
- (6) A quality assurance program acceptable to the engineer of record.

Under this exemption from production lot testing, the filler metal manufacturer shall repeat the testing prescribed in this appendix at least every three years on a random production lot.

T.2 Test Conditions

Tests shall be conducted at the range of heat inputs for which the weld filler metal will be qualified under the welding procedure specification (WPS). It is recommended that tests be conducted at the low heat input level and high heat input level indicated in Table 6.T.1.

Table 6.T.1 WPS Toughness Verification Test Welding and Preheat Conditions

Cooling Rate	Heat Input	Preheat (°C)	Interpass (°C)
Low heat input test	31.2 kJ/mm	21 ± 14	93 ± 28
High heat input test	3.1 kJ/mm	149 ± 14	260 ± 28

Alternatively, the filler metal manufacturer or contractor may elect to test a wider or narrower range of heat inputs and interpass temperatures. The range of heat inputs and interpass temperatures tested shall be clearly stated on the test reports and user data sheets. Regardless of the method of selecting test heat input, the WPS, as used by the contractor, shall fall within the range of heat inputs and interpass temperatures tested.

T.3 Test Specimens

Two test plates, one for each heat input, shall be welded following Table 6.T.1. Five CVN specimens and one tensile specimen shall be prepared per plate. Each plate shall be steel, of any AISC-listed structural grade. The test plate shall be 19 mm thick with a 13 mm root opening and 45° included groove angle. The test plate and specimens shall be as shown in Figure 2A in AWS A5.20, or as in Figure 5 in AWS A5.29. Except for the root pass, a minimum of two passes per layer shall be used to fill the width.

All test specimens shall be taken from near the centerline of the weld at the mid-thickness location, in order to minimize dilution effects. CVN and tensile specimens shall be prepared in accordance with AWS B4.0, Standard Methods for Mechanical Testing of Welds. The test assembly shall be restrained during welding, or preset at approximately 5° to prevent warpage in excess of 5°. A welded test assembly that has warped more than 5° shall be discarded. Welded test assemblies shall not be straightened.

The test assembly shall be tack welded and heated to the specified preheat temperature, measured by temperature indicating crayons or surface temperature thermometers one inch from the center of the groove at the location shown in the figures cited above. Welding shall continue until the assembly has reached the interpass temperature prescribed in Table 6.T.1. The interpass temperature shall be maintained for the remainder of the weld. Should it be necessary to interrupt welding, the assembly shall be allowed to cool in air. The assembly shall then be heated to the prescribed interpass temperature before welding is resumed.

No thermal treatment of weldment or test specimens is permitted, except that machined tensile test specimens may be aged at 93 °C to 104 °C for up to 48 hours, then cooled to room temperature before testing.

T.4 Acceptance Criteria

The lowest and highest Charpy V-Notch (CVN) toughness values obtained from the five specimens from a single test plate shall be disregarded. Two of the remaining three values shall equal, or exceed, the specified toughness of 54 J energy level at the testing temperature. One of the three may be lower, but not lower than 41 J, and the average of the three shall not be less than the required 54 J energy level. All test samples shall meet the notch toughness requirements for the electrodes as provided in Section 10.20.7.3.2.

For filler metals classified as E70, materials shall provide a minimum yield stress of 400 MPa, a minimum tensile strength of 480 MPa, and a minimum elongation of 22 percent. For filler metals classified as E80, materials shall provide a minimum yield stress of 470 MPa, a minimum tensile strength of 550 MPa, and a minimum elongation of 19 percent.

PART VI**Appendix U****Volume Fraction of Reinforcement and Types of Steel Wire Meshes
Used in Ferrocement****U.1 Calculation of Volume Fraction of Reinforcement**

The volume fraction of reinforcement in a ferrocement section can be readily calculated if the density of the mesh material and the weight of mesh per unit area are known.

For ferrocement section reinforced with expanded metal mesh, the volume fraction of mesh reinforcement may be calculated from the following relationship.

$$V_f = \frac{\text{Volume of mesh}}{\text{Volume of ferrocement section}} = \frac{\omega_m N}{\gamma_m h} \times 100 \text{ per cent}$$

For ferrocement reinforced with square or rectangular mesh, the volume fraction of mesh reinforcement may be calculated from the following relationship:

$$V_f = \frac{N \pi d_b^2}{4 h} \left(\frac{1}{D_t} + \frac{1}{D_l} \right) \times 100 \text{ per cent}$$

Where,

N	=	number of layers of mesh reinforcement
d_b	=	diameter of mesh wire
h	=	thickness of ferrocement
D_t	=	centre to centre spacing of wires aligned transversely in reinforcing mesh, mm
D_l	=	centre to centre spacing of wires aligned longitudinally in reinforcing mesh, mm
ω_m	=	weight of mesh per unit area, N/mm ²
γ_m	=	unit weight of steel, N/mm ²

U.2 COMMON TYPES AND SIZES OF STEEL MESHES USED IN FERROCEMENT

Type	Shape	Fabrication	Mesh Size*	Wire Gauge*	Wire Spacing (mm)	Wire Diameter or Sheet Thickness (mm)
Wire Mesh	Square	Woven or Welded	$\frac{3}{4} \times \frac{3}{4}$	No. 16	19.0	1.60
			2 x 2	No. 19	13.0	1.00
		Welded	3 x 3	No. 22	8.5	0.72
			4 x 4	No. 23	6.4	0.64
	Rectangular	Welded	1 x 1	No. 14	25.0	2.00
		Welded	2 x 1	No. 14	50 x 25	2.00
		Twisted	1	No. 18	25.0	1.20
			1	No. 20	25.0	0.88
Expanded Metal Mesh	Diamond	Slit and Drawn	$\frac{1}{2}$	No. 22	13.0	0.72
				Gauge No. 18		1.00
				Gauge No. 20		0.76
				18 N/m ²		0.58

*American wire gauge

PART VII**Chapter 1****Constructional Responsibilities and Practices****1.1 Introduction**

This part of the Code provides the minimum requirements for safe constructional operations, constructional planning, management and practices in buildings; as well as for storage, stacking and handling of materials and resources used in buildings. It describes precautionary measures to be adopted to ensure the safety of public, environment and infrastructure, property, workmen, materials, services, plant and equipment. It also covers guidelines relating to maintenance management, repairs, retrofitting and strengthening of buildings.

1.2 Scope

The regulations stated in this part cover the constructional responsibilities and practices in building sites; safe storing, stacking and handling of materials, equipment and other resources; and safety of personnel during construction operations. The provisions of this part shall apply to all construction operations viz. erection, alteration, repair, removal or demolition of buildings and structures.

Nothing herein contained shall be construed to nullify any rules, regulations, safety standards or those contained in the various act of the Government of Bangladesh, statutes governing the protection of the public or workers from any hazard involved in manufacturing, mining and other processes and operations which generate toxic gases, dust or other elements dangerous to the respiratory system, eye sight or health.

1.3 Terminology

This section provides an alphabetical list of the terms used in and applicable to this part of the Code. In case of any conflict or contradiction between a definition given in this section and that in any other part, without prejudice to provisions arising from laws, statutes and recourses provided under such laws, statutes and covenants of GOB and trade bodies. The meaning provided in this part shall govern for interpretation of the provisions of this part. References shall be made to other part of this Code for terms not defined in this section.

AUTHORITY Same as defined in Chapter 2 Part 1.

AUTHORIZED OFFICER Same as defined in Chapter 2 Part 1.

BLAST AREA The area in which danger may arise during or prior to demolition including the potential area affected by preparation, handling and use of explosives.

BLASTING	The operation of disintegrating rock, structure etc. by firing an explosive charge.
CARTRIDGE	A wrapped or otherwise protected cylinder of defined size of a homogeneous explosive material.
CONSULTANT	The “Consultant” is the organisation/person whose proposal to perform the Services like design, supervision or other technical and/or management services has been accepted by the Owner/ Client and has a Contract Agreement to execute the service.
CONSTRUCTION EQUIPMENT	All equipment, machineries, tools and temporary retaining structures and working platforms, such as derricks, concrete pump, staging, scaffolds, runways, ladders and all material handling equipment including safety devices.
CONTRACTORS	Contractor means the natural person, private or government enterprise, or a combination of the above, whose Tender to carry out the Works has been accepted by the Employer and is named as such in the Contract Agreement, and includes the legal successors or permitted assigns of the Contractor.
DETONATOR	An instantaneous or delay initiator for explosive materials and containing a charge of high explosive fired by means of a flame, spark or electric current.
EMPLOYER	The Employer is the party named who employs the Consultant and/or Contractor to carry out the Works.
EXPLOSIVE	Any substance, whether or not contained in a device, used or manufactured with a view to producing an effect by explosion.
FLOOR HOLE	An opening in any floor, platform, pavement, or yard, measuring less than 300 mm but more than 25 mm in its least dimension, through which materials but not persons may fall; e.g. a belt hole, pipe opening or slot opening.
FLOOR OPENING	An opening in any floor, platform, pavement or yard bigger than a floor hole measuring 300 mm or more in its least dimension, through which a person may fall; e.g. hatchway, stair or ladder opening, hopper mouth pit or large manhole.
GUARD RAILING	A barrier erected along exposed edges of an open side, floor opening, wall opening, ramp, platform or catwalk or balcony, etc. to prevent the fall of persons.

HOISTS	A platform, bucket or similar enclosure made of steel frames, struts and timber planks used for the lifting or lowering of construction material and workmen, the hoists being operated from a point outside the conveyance.
MAGAZINE	Any building or structure used for the storage of explosives with approval of the Authority.
PILE RIG	The complete pile driving equipment comprising piling frame, leader, hammer, extractor, winch and power unit. Complete pile driving rig may be mounted on rafts or pontoon or rails. Pile rig may also be a mobile unit mounted on trailers or trucks, or a special full revolving rig for raking piles.
PLATFORM	A working space for persons, elevated above the surrounding floor or ground, e.g. balcony or platform for the operation of machinery and equipment.
PRIMER	A cartridge cord or container of explosive into which a detonator or detonating cord is inserted or attached and is designed to initiate a larger charge.
PROFESSIONALS	“Professionals” means technical personnel and support staff provided by the Consultant or by any Sub-Consultant and assigned for supervising the execution and completion of the Works and administering the Contract.
SALVAGE	An act of saving and utilization of reusable scrap materials conforming to the requirements of this Code.
SCAFFOLD	A temporary erection of timber or metal work used to support or to allow the hoisting and lowering of workmen, tools and materials during construction, alteration or demolition of a building.
SHOTFIRER	The person in immediate control of the use of explosives.
TOE BOARD	A vertical barrier erected along exposed edge of a floor opening, wall opening, platform, catwalk or ramp at floor level to prevent fall of materials or persons.
WALL HOLE	An opening in any wall or partition having a height of more than 25 mm to less than 750 mm and having no restriction in width.
WALL OPENING	An opening in any wall or partition having a height of at least 750 mm and a width of at least 450 mm.
WORKMEN/ LABOURERS	“Workmen” means any natural person who has a Contract with the Contractor to carry out a part of the work in the Contract, which includes work on the site.

1.4 Planning

Construction planning aspects aim to identify and develop various stages of project execution on site which shall be consistent with the management considerations. Planning aspects evolve out of the objectives of project and requirements of the final completed constructed facility. These objectives can relate to the final constraints, cost considerations, quality standards, safety standards as well as both environmental and health considerations. Construction practices shall, then have to satisfy these objectives during construction phase of the project. Having established objectives of the construction phase, planning determines processes, resources (including materials, equipment, human and environmental) and monitoring system to ensure that the practices are appropriately aligned. Adequate knowledge about pre-construction phase evolution of project, especially related to customer's requirements, is an essential prerequisite for construction planning.

1.4.1 Responsibilities

In a construction or demolition work, the terms of contract between the owner and the contractor, and between a consultant and the owner, shall be clearly defined and put in writing. These, however, will not absolve the owner from any of his responsibilities under the various provisions of this Code, and other applicable regulations and bye-laws.

The terms of contract between the owner and the contractor will determine the responsibilities and liabilities of either party in the concerned matters, within the provisions of the relevant acts and codes (e.g. the Employer's Liability Act 1938, the Factories Act 1965, the Fatal Accident Act 1955 and Workmen's Compensation Act 1923).

The owner, or the professional appointed by him to supervise the work, shall ensure the quality of materials used, soundness of the work and observance of all precautionary measures.

1.4.2 First Aid Attendant

Depending on the scope and nature of the work, at least one person trained in first aid for every 100 workers shall be available at work site to render and direct first aid to casualties. The first aid attendant shall have a refresher course every five years and certificates renewed.

A telephone shall be made available to first aid assistant with emergency telephone numbers prominently displayed. Record/reports of all accidents and actions taken thereon shall be kept by the first aid attendant and forwarded to appropriate authorities when asked.

1.4.3 Temporary Construction

Plan, layout, design and specification of all temporary constructions, e.g. workers' shed, toilet, site store, site office, runway, trestle, foot bridge, guard shed etc., which are likely to interfere with right-of-way or utility services provided by various agencies, shall be submitted to the respective authorities for approval before commencement of any construction operation.

Temporary structures shall be constructed from inflammable materials, but they shall be so located as not to cause any fire hazard to adjoining structures or works and neighboring properties.

1.4.4 Preconstruction Phase

Besides the design aspects, preconstruction phase should also address all the issues related to the implementation of the design at the site through suitable construction strategy. During the design stage, the site conditions should be fully understood with anticipated difficulties and avoid the risk of subsequent delays and changes after the construction has started.

The selection of construction methods, building systems and materials, components, manpower and equipment and techniques are best done in the preconstruction phase. Such selection is influenced by the local conditions like terrain, climate, vulnerability for disasters, etc.

Construction in busy localities of cities needs special considerations and meticulous planning due to restricted space, adjoining structures, underground utilities, traffic restrictions, noise and other environmental pollution and other specific site constraints,

The constructability aspects of the proposed construction methods needs to be carefully evaluated at the planning stage to ensure ease of construction besides optimizing the construction schedule and achieving quality, reliability and maintainability of the constructed facilities.

Constructional practices in hilly regions needs to take into considerations the problem of landslides, slope stability, drainage, etc, besides ensuring no adverse impact on the fragile environmental conditions.

Durability of constructions in corrosive atmospheric conditions like coastal regions and aggressive ground situations with high chlorides and sulphates should also be taken care of with appropriate constructional practices.

Constructional practices in disaster prone areas need specific planning. The type of construction, use of materials, construction techniques require special considerations in such areas.

Adverse weather conditions have strong bearing on construction phase. Situations wherein constructions are to be carried out in adverse weather conditions, such as heavy and continuous rain fall, extreme hot or cold weather, dust storms, etc, the practices have to address the relevant aspects. Accordingly, suitable design and field operations should be adapted or redefined in anticipation of these aspects. Some of these aspects are.

- (a) Site layout which enables accessibility in adverse weather.
- (b) Adequate protected storage for weather sensitive materials/equipment.
- (c) Protections to personnel from extreme hot/cold conditions.
- (d) Scheduling to allow maximization of outdoor activities during fair weather conditions.
- (e) Special design and construction provisions for activities in extreme temperature conditions like hot or cold weather concreting, staple of false work in extreme wind conditions (gusts).
- (f) Adequate lighting for shorter days in winter/night work.
- (g) Design for early enclosure; and
- (h) Adjacent historically important structure shall be given highest care against any damage during construction process.

All sanitary facilities shall be kept in a hygienic condition. Temporary toilets shall be enclosed, screened and weather proofed and shall be installed and maintained in accordance with the relevant part of the Code.

1.4.5 Construction Phase

1.4.5.1 Organizational structure

The site management should be carried out through suitable site organization structure with roles and responsibilities assigned to the construction personnel for various construction related functions. Safety management is one of the important components of site management.

1.4.5.2 Site layout

The layout of the construction site shall be carefully planned keeping in view the various requirements to construction activities and the specific constraints in terms of its size, shape, topography, traffic and other restrictions, in public interest. The site layout shall take into considerations the following factors.

- (a) Easy access and exit, with proper parking of vehicle and equipment during construction.
- (b) Properly located material stores for easy handling and storage.

- (c) Adequate stack areas for bulk construction materials.
- (d) Optimum location of plants and equipment (batching plants etc.).
- (e) Layout of temporary services (water, power, power suppression unit, hoists, cranes, elevators etc.).
- (f) Adequate yard lighting and lighting for night shifts.
- (g) Temporary buildings; site office and shelter for workforce with use of non-combustible materials as far as possible including emergency medical aids.
- (h) Roads for vehicular movement with effective drainage plan.
- (i) Construction safety with emergency access and evacuations and security measures.
- (j) Fabrication yards for reinforcement assembly, concrete casting and shattering materials; and
- (k) Fencing, barricades and signage.

1.4.5.3 Access for firefighting equipment vehicles

Access for firefighting equipment shall be provided to the construction site at the start of construction and maintained until all construction work is completed.

Free access from the street to fire hydrants/static water tanks, where available, shall be provided and maintained at all times.

No materials for construction shall be placed within 3 m of hydrants/static water tanks.

During building operations, free access to permanent, temporary or portable first-aid firefighting equipment shall be maintained at all times.

In all buildings over two stories high, at least one stairway shall be provided in usable condition at all times. This stairway shall be extended upward as each floor is completed. There shall be a handrail on the staircase.

1.4.5.4 Construction strategy and construction sequence

Construction strategy and construction methods are to be evolved at the planning and design stage specific to the conditions and constraints of the project site and implemented by the site management personnel to ensure ease of construction and smooth flow of construction activities.

Sites of high water table conditions with aggressive chemical contents of subsoil needs special design considerations. Buildings with basement in sites of high water table should be planned with dewatering scheme with appropriate construction sequence, Duration of

dewatering shall continue till sufficient dead loads are achieved to stabilize the buoyancy loads with adequate factor of safety. The construction sequence should be planned taking into consideration the following aspects.

- (a) Availability of resources (men, material and equipment).
- (b) Construction methods employed including prefabrication.
- (c) Planned construction time.
- (d) Design requirements and load transfer mechanism.
- (e) Stability of ground like in hilly terrain.
- (f) Ensuring slope stability with retaining structure before the main construction.
- (g) Installation and movement of heavy equipment like cranes and piling equipment.
- (h) Effect of weather.
- (i) Minimum time to be spent below ground level working; and
- (j) Protection against ground water seepage.

1.5 Construction Control

1.5.1 General

All construction including extension, alteration and demolition shall require a permit from the Authority. Permits shall also be obtained from relevant organizations for service connections and other facilities. The construction work shall conform to the plan approved by the Authority.

The owner shall make arrangements for obtaining the required approvals.

All new work or alteration shall be planned, designed, supervised and executed by competent professionals of relevant discipline.

1.5.2 Professional Services and Responsibilities

The responsibility of professionals with regard to planning designing and supervision of building construction work, etc. and that of the owner shall be in accordance with the relevant part of the Code and professional practice. Employment of trained workers shall be encouraged for building construction activity.

1.5.3 Construction of all Elements

Construction of all elements of a building shall be in accordance with good practice. It shall also be ensured that the elements of structure satisfy the appropriate fire resistance requirements as specified in Part 4 'Fire Protection', and quality of building materials/components used shall be in accordance with Part 5 'Building Materials'.

1.5.4 Construction Using Bamboo

Bamboo being a versatile resource characterized by high strength, low mass and ease of working with simple tools, it is desirable to increasingly make appropriate use of this material. Design of structures using bamboo shall be done in accordance with Chapter 4, Part 6.

For construction using bamboo, some of the important constructional provisions given below shall be followed.

Bamboo can be cut and split easily with very simple hand tools. Immature bamboos are soft, pliable and can be molded to desired shape. It takes polish and paint well.

While it is possible to work with bamboo simply using a machete, a few basic tools, such as, machete, hack saw, axe, hatchet, sharpening tools, adze, chisel (20 mm), chill, wood rasps, steel rod, and pliers, will greatly increase the effectiveness of the construction process.

For providing safety to the structure against fire, bamboo may be given fire retardant treatment using following chemicals; a few drops of concentrated HCL shall be added to the solution to dissolve the precipitated salts:

Ammonium phosphate	3 parts	Zinc chloride	5 parts
Boric acid	3 parts	Sodium dichromate	3 parts
Copper sulphate	1 part	Water	100 parts

Bamboo indirect contact with ground, bamboo on rock or preformed concrete footing, bamboo incorporated into concrete or bamboo piles may form the foundation structure.

The floor of bamboo may be at ground level with covering of bamboo matting, etc. In elevated floors, bamboo members become an integral part of structural framework of building. The floor will comprise structural bamboo elements and bamboo decking.

The jointing techniques in construction using bamboo shall be in accordance with Chapter 4, Part 6.

1.5.5 Low Income Housing

For low income housing, appropriate planning and selection of building materials and techniques of construction have to be judiciously done and applied in practice. Requirements of low income housing specified in Part 3 shall be followed. However, all requirements regarding structural safety, health safety and fire safety shall be in accordance with this Code.

1.5.6 Site Preparation

While preparing the site for construction, bush and other wood, debris, etc. shall be removed and promptly disposed of so as to minimize the attendant hazards. Temporary buildings for construction offices and storage shall be so located as to cause the minimum fire hazards and shall be constructed from noncombustible materials as far as possible.

1.5.7 Use of New /Alternative Construction Techniques

The provisions of this part are not intended to prevent use of any construction techniques including any alternative materials, nonspecifically prescribed by the Code, provided any such alternative has been approved. The Authority may approve any such alternative such as ferrocement construction, row-lock (rat trap) bond in masonry, stretcher bond in filler slab and filler slab provided; that the proposed alternative is satisfactory and conforms to the provisions of relevant parts regarding material, design and construction of this Code. The material or method or work offered as alternative is, for the purpose intended, at least equivalent to that prescribed in the Code in quality, strength, compatibility, effectiveness, fire and water resistance, durability and safety.

1.5.8 Permits

The owner of a building shall obtain permission from the Authority for the work to be undertaken in accordance with the provisions of the relevant part of this Code.

Special permits shall be obtained from relevant authorities before commencement of a particular construction work for the following items and for any other item as decided by the Building Official.

- (a) Storing materials on roads and sidewalks.
- (b) Using water, electricity, gas, sewerage or other public utilities.
- (c) Digging roads or interfering with the drainage system.
- (d) Storing and handling of explosives; and
- (e) Affecting any structure having historical association and antiquity.

1.5.9 Tests and Inspections

The Authority shall notify both the owner and the contractor of any unsafe, unlawful or unethical situation discovered during inspection and direct them to take necessary remedial measures to remove the hazard or rectify the violation.

Where the strength or adequacy of any scaffold or other device or construction equipment is in doubt, or where any complaint is lodged, the Authority shall inspect such equipment and shall prohibit its use until tested safe or until all danger is removed.

1.6 Construction Management

Employer and Management shall be responsible for optimizing Construction Planning, resource utilization, and scope, time, quality, health, safety and environment and cost for implementation, monitoring and control for their effectiveness. This may be preferably in line with proven National/International documentation system covering all aspects of monitoring and controls. Various parameters to be managed during construction are as below.

1.6.1 Time Management

The project shall be completed in the defined time schedule to get its fruitful benefits. The system planned shall cover total schedule of completion with one or more construction agencies, number of vendors, identification of total resources, timely availability of all inputs, including critical ones, its processing during construction of a project. The system shall include a periodic review of a project with all parameters as well as catch up plans in case of delay identified for controls and reporting from time to time. The system planned shall preferably be computer friendly and simple to follow for implementation, monitoring and controls and for reporting from time-to-time.

1.6.2 Quality Management

Quality of a project shall be planned for all activities from inception to completion. It is desirable that the system planned gives adequate assurance and controls that it shall meet project quality objectives. The system shall cover review of existing requirements, subcontracting, materials, processes and controls during process, auditing, training of personnel, final inspection and acceptance. All activities shall be planned and controlled. Quality systems approach may be referred for planning, suitable to a particular project for implementation.

1.6.3 Health, Safety and Environment

Each project affects the safety and health of the workmen and surroundings during construction. Various activities having impact on health, safety and environment need to be identified with their likely effect and proposed preventive corrective actions, together with the concerned statutory obligations. The system planned for health, safety and environment shall address and cover the above including use of personnel protective equipment by all concerned and reporting on their monitoring and controls during project implementation.

1.6.4 Cost Management

To keep the project under viable proposition, it is desired that cost of the project during construction are monitored and controlled through a documentation system. The various parameters which may affect the basic cost, escalations, cost due to variation in scope and quantities, etc. need to be monitored at a defined frequency. The system planned shall be in line with a proven cost control method or similar in nature and cost incurred vis-a-vis cost sanctioned and cost anticipated to be reported and controlled from time to time.

1.7 Protection of Public and Workers

1.7.1 General

Erection, alteration, renovation, remodeling, repairing, removal or demolition of a building or structure shall be conducted in a safe manner. Suitable protection for the general public and workers employed thereon shall be provided according to the various provisions of this Code.

All existing and adjoining public and private property shall be protected from any damage due to construction operations. Whenever requested, site plans, construction details, and specification shall be submitted for review by the concerned agency.

All equipment and safeguard required for the construction work such as temporary stair, ladder, ramp, scaffold, hoist, runway, barricade, chute, lift etc. shall be substantially constructed and erected so as not to create any unsafe situation for the workmen using them or the workmen and general public passing under, on or near them.

Public walkway shall not be occupied to carry out work under a building permit unless the pedestrians are protected as specified in this section. Any material or structure temporarily occupying public property, including fences and walkways, shall be adequately lighted at night.

1.7.2 Adjoining Property

The owner of the building shall preserve all adjoining structures and walls from damage. He shall support the adjoining building or structure by proper foundations to comply with the Code.

Necessary permissions to preserve and protect the adjoining plot, building or structure shall be obtained by the owner of the building to be constructed. Adjoining property shall be completely protected from any damage due to the building operation when the owner of the adjoining property permits free access to the adjoining site and building.

If required, the owner of the adjoining plot, building or structure shall be granted necessary permission to enter the construction site to make his own property safe.

No part of any structure, except signs, shall project beyond the property line of the site. Sidewalk sheds, underpinning and other temporary protective guards and devices may project beyond the property lines if approved by the Authority. Where necessary, the permission of the adjoining property owner shall also be obtained.

Where a construction or demolition is undertaken at a level higher than the adjacent structure, the roof, roof outlets, skylights and other roof structures of adjoining buildings shall be protected against damage. This shall be ensured by the owner of the construction site at his own expense.

Where the grade of the adjoining plot is lower than the site level, a retaining wall shall be erected, if necessary, at the owner's expense and on his site. Design and construction of retaining wall shall conform to the structural requirements for such walls, and may have a railing or fence at the top to provide a total height of not less than 1 m above the finished grade of the higher plot.

If the owner, lessee or tenant of the adjoining building refuses permission to have the roofs and skylights of the adjoining building protected, the responsibility and expense for the said protection shall transfer to the person refusing such permission.

During any demolition or excavation work, the structure or the wall shall be maintained structurally safe by adequate temporary props and lateral supports.

1.7.3 Protective Fences and Railings

Pedestrian traffic on the adjacent road or footpath, or the walkway constructed shall be protected by a railing or fence. Protective railing or fence shall also be placed adjacent to excavations. Railings shall be at least 1m in height and when adjacent to excavations, shall be provided with a mid-rail.

All construction work within 1.5 m from the road shall be enclosed with a fence not less than 2.4 m high from the grade. If the work is more than 1.5 m away from the road, a fence or other barriers shall be erected at least on the side of the site nearest to the footpath/road. The fence shall extend over the entire length of the side. Openings in fences may have doors which normally shall be kept closed.

All fences shall be of adequate strength to resist wind pressure and other load as specified in relevant part of the Code. All fences shall be well braced. The side of any fence/handrail adjacent to a road or sidewalk shall be kept smooth. Fences, barriers, or temporary structures of any kind located on public roads shall not obstruct vision at the intersection of streets.

1.7.4 Canopies, Overhangs and Platforms

Protective canopy shall have a clear height of 2.4 m over the walkway. Walkways under the canopy shall be not less than 1.2 m wide in the clear. However, the Building Regulatory Authority may instruct differently regarding the clear width in congested areas.

Every canopy shall have a fence built along its entire length on the construction side. If materials are stored or work is done on the roof of the canopy, edges of the canopy roof shall have a tight curb board not less than 200 mm high and a railing not less than 1 m high. The entire structure shall be designed to carry the loads to be imposed.

The posts or other supporting members of any temporary structure on the road side shall be designed for the load due to vibration generated by the street traffic. The framework supporting the covering shall be well braced and designed to support at least 7 kPa. However the top deck shall be designed to carry load not less than 10 kPa.

The roof covering shall be of a width sufficient to cover the entire walkway or side walk and shall be made watertight. Covered walkways shall be provided with adequate lights at all times.

Cantilevered platforms or other substitute protection in lieu of sidewalk sheds shall not be used unless approved by the authority and deemed adequate to insure public safety.

Materials shall not be stored on overhangs unless these are designed for the load. Such storage shall in no case exceed a day's supply. All materials shall be piled in an orderly manner and height to permit removal without endangering the stability of the pile and canopy.

1.7.5 Protection of Utilities

Protective frame and boarding shall be built around and over every street lamp, utility box, fire and police alarm box, fire hydrant, catch basin and manhole that may be damaged by any construction work. The protection shall be maintained while such work is being done; and shall not obstruct the normal functioning of the device.

Building material, fence, shed etc. shall not obstruct free access to any fire hydrant, lamppost, manhole, fire alarm box, or catch basin, or interfere with the drainage of the site. Protective covers shall be provided to such utility fixtures during the progress of the work without obscuring their identity.

Precaution shall be taken during construction to prevent concrete, mortar washing or any other material from entering and blocking a sewer.

1.7.6 Use of Road and Footpath

Road and footpath spaces may be used only temporarily during construction subject to the following conditions.

- (a) Permissions shall be obtained from relevant authorities for all such uses.
- (b) The allocated space or any portion thereof shall be more than 1.5 m away from a railway track.
- (c) A walkway shall be constructed in the outer portion of the road space permitted to be occupied in conformity with Sec 1.4.2 and 1.4.3.
- (d) One (1) metre clear passage shall be maintained along the building site.
- (e) Person(s) who has been issued a permit to use road and footpath spaces shall furnish a bond with the relevant authority of such type and amount as may be deemed advisable by the authority as protection from all liabilities.

- (f) The permittee shall repair any damages done to the adjacent road due to its use for construction work at his own expense; the bond money shall stand forfeited if the permittee fails to comply with this requirement; and
- (g) It shall be used in a manner that will not deface it or create a nuisance. The owner, upon the completion of the building, shall immediately remove all temporary walkways, debris and all other obstruction and leave such public property in as good a condition as it was before such work commenced.

1.7.7 Protective Devices

No structure, fire protection or sanitary safeguard or device shall be removed or made inoperative unless instructed by the Authority. Pedestrian protection required by all relevant regulations shall be maintained in place and kept in good order as long as pedestrians may be endangered. Every protection, fence, canopy and other protective devices shall be removed within 7 days after such protection is no longer required.

1.7.8 Notices and Signs

Every walkway adjacent to a construction, demolition or excavation site shall be kept well-lighted at night. The outer edge of the occupied space of the street or footpath shall have red lights placed thereon which shall flash continuously day and night.

Boards with caution signs, along with safety regulations and emergency instructions painted in bright colour, preferably red, shall be erected near the entry and in prominent places of the site. It shall describe appropriate measures for the elimination or control of the danger and the conduct and course of action to be taken. Red caution marks shall also be placed on the building, equipment and utility connections.

1.7.9 Watchman and Auditory Signal

A watchman shall be employed to warn the general public when intermittent hazardous operations are conducted. Audible signal shall be used in case of extreme danger. It shall be such that any person in the reception area can recognize and react to the signal as intended. An auditory emergency evacuation signal shall take precedence concerning recognition over all auditory signals.

1.7.10 Safe Load

No structure, temporary support, scaffolding, sidewalk, footpath and drain covers, shed, other devices and construction equipment shall be loaded in excess of its safe working capacity.

Whenever the structural quality or strength of scaffolding plank or other construction equipment is in doubt, these shall be replaced or be subject to a strength test to two and half times the superimposed live load; the member may be used if it sustains the test load without failure. Requirements of Sec 3.8 shall be observed regarding design loads in scaffolds.

1.8 Environmental Protection

The following provisions shall be met during construction for environmental protection. The construction and operation of the work/project shall comply with relevant national environmental legislation including environmental quality standards. The basic responsibility of the contractor/owner towards the environment shall be:

- (a) To take all reasonable steps to protect the environment and avoid damage and nuisance arising because of his/her operations.
- (b) To comply with all status and regulations concerning the execution of works.
- (c) To familiarize himself with all legislation relating to environmental protection that is relevant to his activities. Reference to national environmental quality guidelines shall be made.
- (d) Responsible for the costs of cleaning up any environmental pollution resulting from his/her activities during construction.

1.8.1 Protection of Existing Drainage Systems and Utilities

During construction work all excavation and/or filling work shall be taken as to ensure safety to the existing underground utility lines and drainage system. The Contractor/Owner shall obtain written permission from the respective authorities before excavation or filling in such areas.

1.8.2 Protection of Soil, Aquifers, and Water Channels against Pollution

Construction activities are likely to generate waste in various forms. This shall be dealt with adequately to avoid pollution. The following measures shall be taken during construction of work.

- (a) The Contractor/Owner shall, all times, maintain all sites under his control in a clean and tidy condition and shall provide appropriate and adequate facilities for the temporary storage of all wastes before disposal.
- (b) The Contractor/Owner shall be responsible for the safe transportation and disposal of all wastes generated because of his/her activities in such a manner as to not cause environmental pollution or hazards to health in any form. In the event of any third party being employed to dispose of wastes, the Contractor/Owner shall be considered to have discharged his/her responsibilities from the time the wastes leave sites under his/her control, providing that he/she has exercised due diligence in ascertaining that the proposed transport and disposal arrangements are such as to not cause pollution or health hazards.

- (c) The Contractor/Owner shall not allow waste oils or other petroleum derived wastes to be used as dust suppressants and that all reasonable precautions shall be taken to prevent accidental spillage of petroleum products, their contact with soil or discharge into water courses.
- (d) The Contractor/Owner shall be responsible for the provision of adequate sanitary facilities for the construction workforce (including those employed under subcontracts) at all construction and camp sites. The Contractor/Owner shall not knowingly allow the discharge of any untreated sanitary wastes to ground water or surface water. Before mobilization of the construction workforce, the Contractor/Owner shall provide details of sanitary arrangements. The detail shall include maintenance and operation plans and generally be sufficient to assess whether the proposed facilities are adequate.
- (e) Where abstraction from a borehole by the Contractor/Owner results in adverse effects on groundwater, which at the time of commencement of the contract was being used by local people, the Contractor/Owner shall arrange supplies of equivalent quality and quantity of water to that previously available.

1.8.3 Protection of Air Quality from Obnoxious Emissions

To cover the unlikely event that dust blow becomes a nuisance, to the following effect shall be taken.

The Contractor/Owner shall take all reasonable measures to minimize dust-blow arising from any sites under his/her control by regular watering of any stockpiles, bare soil, and haul roads. Unsurpassed traffic areas and any sources of fugitive dust, when conditions require dust suppression.

1.8.4 Protection from Sound Pollution

The management shall be responsible for confining all construction and transportation activities in Residential and mixed Residential Areas strictly to normal business hours, so as not to cause emission of such sound and noise which is considered detrimental to human health. Such noisy activities shall not be carried out from 1800 hours in the evening to 0600 hours in the morning and on non-working days and holidays.

Noise nuisance shall be minimized through adequate machine maintenance and good site practices. However, a degree of unavoidable noise nuisance from construction is expected. Control vibration from piling operations is not possible without incurring an unreasonable financial cost.

The following noise control measures shall be taken during construction work.

- (a) All vehicles and plant operated by the contractor or (including subcontractors) shall be maintained according to the original manufacturer's specifications and manuals, with particular regard to the control of noise emissions. The Consultant/Authority shall have the right to require the Contractor to replace or rectify any vehicle or plant that he thinks emits excessive noise, within 48 hours of notice in writing.
- (b) The contractor shall make every reasonable effort to reduce noise nuisance caused by construction activities, including using of crusher and ancillary plant in locations where the distance between them and residential areas is such that it results in attenuation of noise at existing residential areas.

1.8.5 Site Reinstatement

The construction sites shall be reinstated to an acceptable level to the following effect shall be included.

Upon completion of construction the Contractor/Owner shall remove equipment, surplus material, rubbish and temporary works of every kind, and the site in clean condition to the satisfaction of the appropriate Authority.

PART VII**Chapter 2****Storage, Stacking and Handling Practices****2.1 General Practices****2.1.1 General Requirements and Restrictions on Storage and Handling**

Materials required in construction operations shall be stored, and handled in a manner to prevent deterioration and damage to the materials, ensure safety of workmen in handling operations and non-interference with public life including safety of public, prevention of damage to public property and natural environment.

Materials shall be stored and placed so as not to endanger the public, the workers or the adjoining property. Materials shall be stacked on well-drained, flat and unyielding surface. Material stacks shall not impose any undue stresses on walls or other structures.

Materials shall be separated according to kind, size and length and placed in neat, orderly piles. High piles shall be staggered back at suitable intervals in height. Piles of materials shall be arranged so as to allow a minimum 800 mm wide passageway in between for inspection and removal. All passageways shall be kept clear of dry vegetation, greasy substance and debris.

For any site, there should be proper planning of the layout for stacking and storage of different materials, components and equipment with proper access and proper maneuverability of the vehicles carrying the material. While planning the layout, the requirements of various materials, components and equipment at different stages of construction shall be considered.

Stairways, passageways and gangways shall not become obstructed by storage of building materials, tools or accumulated rubbish.

Materials stored at site, depending upon the individual characteristics, shall be protected from atmospheric actions, such as rain, sun, winds and moisture, to avoid deterioration.

Special and specified care should be taken for inflammable and destructive chemicals and explosive during storage.

2.1.2 Manual Handling

When heavy materials have to be handled manually each workman shall be instructed by his foreman or supervisor for the proper method of handling such materials. Each workman shall be provided with suitable equipment for his personal safety as necessary. Supervisors shall also take care to assign enough men to each such job depending on the weight and the distance involved.

2.1.3 Protection against Fire

Timber, Bamboo, coal, paints and similar combustible materials shall be kept separated from each other. A minimum of two dry chemical powder (DCP) type fire extinguishers shall be provided at both open and covered locations where combustible and flammable materials are stored.

Flammable liquids like petrol, thinner etc., shall be stored in conformity with relevant regulations.

Explosives like detonators, gun powder etc. shall be stored in conformity with the fire protection provisions set forth in this Code so as to ensure desire safety during storage. Stacks shall not be piled so high as to make them unstable under fire fighting conditions and in general they shall not be more than 4.5 m in height.

Materials which are likely to be affected by subsidence of soil like precast beams, slabs and timber of sizes shall be stored by adopting suitable measures to ensure unyielding supports.

Materials liable to be affected by floods, tides, etc. shall be suitably stored to prevent their being washed away or damaged due to floods, tides, etc.

2.1.4 Housekeeping

Stairways, walkways, scaffolds, gangways and access ways shall be kept free of building material, tools, accumulated rubbish and obstructions.

Materials or equipment stored on the street, footpath and other public places with permission from the proper Authority, and conforming to Sec 1.5.3, shall not interfere with vehicular traffic or pedestrians on the highway or street. The piles shall be arranged to leave a safe walkway unobstructed for its full length, and adequately lighted at night and at all other necessary times.

Material and equipment shall not be located within 7.5 m of a street intersection. These shall neither be so placed as to obstruct normal observation of traffic signals nor to hinder the use of public transit loading platforms.

2.2 Storage Requirement by Classification of Materials

Stored materials shall be separately stored under following classifications, with appropriate care necessary precautions to each classification.

- a) Climatically Sensitive Materials.
- b) Durable Materials.
- c) Materials Vulnerable to Rough Handling.
- d) Inflammable and/or Fire Sensitive Materials.
- e) Hazardous Materials.

Under each classification a list of commonly used materials are listed below. Other materials used but not mentioned here shall be treated under one or more of the above listed classifications which most closely match the unlisted material.

2.2.1 Climatically Sensitive Materials

Such material shall be stored in properly constructed sheds which must be stored in cool dry and well ventilated and confines, ensuring its storage without deterioration and without contact to ground and structural members, without exposure to moisture and heat, and away from direct sun.

Materials requiring breathing, such as timber and other natural products, shall be allowed ample air flow between successive layers of stacking.

Materials subject to deformation under stress shall be supported uniformly so as not to subject it to bending load or excessive vertical load.

Materials subject to loss of quality through moisture shall be kept within impermeable wrapping, if not used within a reasonable period.

2.2.1.1 Cement

Cement shall be stored at the work site in a building or a shed which is dry, leak proof and moisture proof. The building or shed shall have minimum number of windows and close fitting doors which shall be kept closed at all times except during loading and unloading.

Cement received in bags shall be prevented from coming into contact with any dampness or moisture. Cement bags shall be stacked on wooden planks maintaining a minimum clearance of 200 mm from the floor. A minimum clear space of 450 mm shall be provided between the stacks and any exterior wall.

Maximum height of the stack shall be 15 bags and the width not more than four bags or 3m. In stacks more than 8 bags high, the bags shall be arranged alternate length and crosswise. The bags shall be stacked closely as to minimize the surface area exposed to air.

During monsoon and for storage for more than 2 months, the stack shall be kept completely enclosed by a waterproofing membrane such as polyethylene sheet which shall close on top of the sack. Care should be taken to see that waterproofing membrane is not damaged any time during the use.

Heavy containers of cement shall not be stacked more than two tiers high. Cement shall be used in the order they are received; storage shall facilitate this requirement.

Hooks shall not be used for handling cement bags unless permitted by the supervisor. Workers handling cement shall put on protective hand and face coverings and use skin protective. They shall be instructed to the need of cleanliness from time to time.

When entering a silo or bin for any purpose, the workman shall wear a lifeline attended by another workman outside. The ejection system shall be shut down and locked out during such operation.

In case cement is received in silos, the silos shall be placed near the concrete batching plan. Proper access shall be provided for the replacement of silos.

Different types of cements shall be stacked and stored separately. In similar manner cements in gunny bags, paper bags and polythene sheets shall be stored separately.

2.2.1.2 Lime

Quicklime shall be slaked as soon as possible. If unavoidable, it may be stored in compact heaps having only the minimum of exposed area. The heaps shall be stored on a suitable platform under a roof protected from rain and wind. A minimum space of 300 mm shall be provided all-round the heaps to avoid bulging of walls.

Unslaked lime shall be stored in a watertight place and shall be separated from combustible materials.

Hydrated lime shall be supplied either in containers or sacks, such as jute bags lined with polyethylene or high density polyethylene woven bags lined with polyethylene or craft paper bags.

It shall be stored in a dry room to protect the lime from dampness and to minimize warehouse deterioration. The building should be with a concrete floor and having least ventilation to eliminate draughts through the walls and roof. In general, the recommendations given in storing of cement shall be applicable for hydrated lime. When air movement is reduced to a practical minimum, hydrated lime can be stored for up to three months without appreciable change.

When dry slaked lime is to be used within a few days, it shall be stored on a covered platform and protected from rain and wind. It shall be kept in a dry and air-tight go down when immediate use is not required. However, it shall never be stored for more than two months.

Handling of Cement and Lime

Bulk cement stored in silos or bins may fail to feed to the ejection system. When necessary to enter a silo or bin for any purpose, the ejection system employed shall be shut down and locked out electrically as well as mechanically, when necessary for a workman to enter such storage area, he shall wear a life-line, with another workman outside the silo or hopper attending the rope.

Workmen, handling bulk cement or lime shall wear protective clothing, respirators, and goggles; shall be instructed in the need of cleanliness to prevent dermatitis, and shall be provided with hand cream, petroleum jelly, or similar preparation for protection of exposed skin.

2.2.1.3 Timber

Timber shall be stored in stacks on well treated and even surfaced beams, sleepers or brick pillars so as to be at least 200 mm above the ground level. Contact with water shall be avoided under all circumstances. Members shall be stored separately in layers according to lengths and materials of equal lengths shall be piled together in layers with crossers or wooden battens of sound wood, straight and uniform thickness.

In any layer a 25 mm air space shall be kept between adjacent members. The longer pieces shall be placed in the bottom layers and shorter pieces in the top layers. At least one end of the stack shall be in true vertical alignment. The crossers themselves in different layer shall be in vertical alignment.

The recommended width and height of a stack are 1.5 m and 2.0 m respectively. Minimum distance between two stacks shall be 800 mm. In case stacking with battens is not possible, the timber may be close piled in heaps, and the precautions specified above observed.

All timbers to be stored for a year or more, the ends of members shall be coated with coal tar, aluminum leaf paints (hardened gloss oil), microcrystalline wax or other suitable material.

The stacks of timbers shall be protected from hot dry wind, direct sun and rain. Heavy weights may be placed on top of the stacks to prevent warping of timber. Nails, metal straps, etc. attached to used timber, particularly planks and formwork for shuttering shall be removed before stacking.

Care must be taken that handler or workmen are not injured by rails, straps, etc, attached to the used timber. This applies particularly to planks and formwork for shuttering.

2.2.1.4 Bamboo

The site shall be properly inspected and termite colonies or mounds if detected shall be destroyed. All refuse and useless cellulosic materials shall be removed from the site. The ground may then be disinfected by suitable insecticides. The area should have good drainage.

Bamboo may preferably be stacked on high skids or raised platform at least 300 mm above ground, Storage under cover reduces the liability to fungal attack. Good ventilation and frequent inspection are important.

Bamboo dries by air-seasoning under cover in the storage yards within 6 to 12 weeks time.

Prophylactic treatment of bamboo during storage prevents losses due to fungi and insects even under open storage. Following chemicals are found suitable at the coverage rate of 24 liters per ton.

Sodium Pentachlorophenate [1% solution]

Boric acid + Borax (1:1) [2% solution]

Sodium Pentachlorophenate + Boric acid + Borax (5:1:1) [2.5% solution]

Note: A mixture of these compounds yields the best results. For better protection of structural bamboo, (if stored outside) repetition of the treatment after four to six months is desirable.

2.2.1.5 Particle Board

See Sec 2.2.3.9.

2.2.2 Durable Materials

2.2.2.1 Steel Bars and Sections

Steel reinforcement bars and structural steel shall be stored in a way to prevent distortion, corrosion, scaling and rusting. Reinforcement bars and structural steel sections shall be coated with cement wash before stacking, especially in humid areas. In case of long time storage or storage in coastal areas, reinforcement bars and steel sections shall be stacked at least 200 mm above ground level.

Steel sections shall be stacked upon platforms, skids or any other suitable supports. Bars of different types, sizes and lengths and structural steel sections shall be stored separately to facilitate issues in required sizes and lengths without cutting from standard lengths. Ends of bars and sections of each type shall be painted with separate designated colors.

Tag lines shall be used to control the load in handling reinforcing bars or structural steel when a crane is used. Heavy steel sections and bundles of reinforcing bars shall be lifted and carried with the help of slings and tackles.

2.2.2.2 Bricks and Masonry Blocks

Bricks shall be stacked on dry firm ground in regular tiers. For proper inspection of quality and ease in counting, the stacks shall be 50 bricks long and 10 bricks high and not more than 4 bricks in width, being placed on edge two at a time along the width of the stack. Clear distance between adjacent stacks shall be not less than 800 mm.

Bricks of each truckload shall be put in one stack. Bricks of different types, such as, clay bricks, clay fly ash bricks, fly ash lime bricks, sand lime (calcium silicate) bricks shall be stacked separately.

Bricks of different classifications from strength consideration and size consideration (such as, conventional and modular) shall be stacked separately. Also bricks of different types, such as, solid, hollow and perforated shall be stacked separately.

Bricks made of clay containing lime shall be thoroughly soaked in water (docked) while in stack.

Bricks of different types shall be stacked separately. Concrete blocks, stone blocks and other masonry blocks shall be stored in stacks of such height as not to damage the blocks in the lower layers or topple.

Bricks shall be loaded or unloaded with care, and shall not be thrown or dumped. They shall be carried from the stack to the site of placement in small batches as and when necessary.

Brick stacks shall be placed close to the site of work so that least effort is required to unload and transport the bricks again by loading on pallets or in barrows. Unloading of building bricks or handling in any other way likely to damage the corners or edges or other parts of bricks shall not be permitted.

Blocks are available as hollow and solid concrete blocks, hollow and solid light weight concrete blocks, autoclave aerated concrete blocks, concrete stone masonry blocks and soil based blocks. Blocks shall be unloaded one at a time and stacked in regular tiers to minimize breakage and defacement. These shall not be dumped at site. The height of the stack shall not be more than 1.2 m, the length of the stack shall not be more than 3.0 m, as far as possible and the width shall be of two or three blocks. Normally blocks cured for 28 days only should be received at site. In case blocks cured for less than 28 days are received, these shall be stacked separately. All blocks should be water cured for 10 to 14 days and air cured for another 15 days; thus no blocks with less than 28 days curing shall be used in building construction. Blocks shall be placed close to the site of work so that least effort is required for their transportation. The date of manufacture of the blocks shall be suitably marked on the stacks of blocks manufactured at factory or site.

2.2.2.3 Stones

Stones of different sizes, types and classification shall be stored separately. Stones shall be stacked on dry firm ground in a regular heap not more than 1 m in height.

Veneering stones shall be stacked against vertical support on a firm dry ground in tiers up to a height of 1.2 m. A distance of about 0.8 m shall be kept between two adjacent stacks.

2.2.2.4 Aggregates

Aggregates shall be stored at site on a hard, dry and level ground. If such a surface is not available, a platform of planks or old corrugated iron sheets, or a floor of bricks, or a thin layer of lean concrete shall be used. Contact with clay, dust, vegetable and other foreign matters shall be avoided.

Fine and coarse aggregates shall either be stored separately or heaps be separated by dividing walls. Fine aggregate shall be stored in a place and manner where loss due to the effect of wind is minimum, viz. in the leeward side behind a wall, or by covering with a polyethylene sheet.

On a large job it is desirable to construct dividing walls to give each type of aggregates its own compartment. Fine aggregates shall be stacked in a place where loss due to the effect of wind is found minimum.

When withdrawals are made from heaps, no overhang in the original heap shall be permitted. Employees required to enter hoppers shall be equipped with safety belts and life-lines, attended by another person. Machine driven hoppers, feeders, and loaders shall be locked in the off position prior to entry electrically as well as mechanically.

2.2.2.5 Water

Water to be used in construction shall be stored in tanks, bottom and the sides of which shall be constructed with brick or concrete. Contact with any organic impurities shall be prevented. The total capacity of the storage tank shall be determined after taking into account the water required for firefighting. Also see Sec 4.2 of Chapter 4 Part 4.

The tank shall be so located as to facilitate easy storage and filling in, and supply both for construction work and for firefighting. Passage of water to the water tank shall not be blocked at any time.

2.2.3 Materials Vulnerable to Rough Handling

2.2.3.1 Aluminum Sections

Aluminum sections of different classification, sizes and lengths shall be stored separately, on a level platform under cover.

The aluminum sections shall not be pulled or pushed from the stack nor shall be slid over each other, to protect the anodizing layer.

2.2.3.2 Pulverized Fuel Ash/Fly Ash

Fly ash shall be stored in such a manner as to permit easy access for proper inspection and identification of each consignment. Fly ash in bulk quantities shall be stored in stack similar to fine aggregates, avoiding any intrusion of foreign matter. Fly ash in bags shall be stored in stacks not more than 10 bags high. For handling see Sec 2.2.1.2.

2.2.3.3 Cinder

Cinder shall be stored in bulk quantities in stacks similar to coarse aggregates avoiding any extrusion of foreign matter.

2.2.3.4 Pipes and Tubing

Pipes shall be stored in stacks with stoppers provided at the bottom layer to keep the pipe stack stable. The stack, particularly of smaller diameter pipes, shall be in pyramid shape. Pipes shall not be stacked more than 1.5 m high.

Each stack shall have pipes of the same type and size only. Removal of pipes shall start from the top layer and by pulling from one end. A pipe shall not be stored inside another pipe. The pipes may also be placed alternately length and crosswise.

Pipe shall be carried one at a time on shoulders by at least two workmen. Pipe fittings and joints shall be handled individually.

Black polyethylene pipes may be stored either under cover or in the open. However, natural coloured polyethylene pipes shall be stored under cover only and protected from direct sunlight.

Coils of tubing shall be stored either on edge or stacked flat one on top of the other; in either case they shall not be allowed to come into contact with hot water or steam pipes and should be kept away from hot surface.

Straight lengths of unplasticized PVC pipes shall be stored on horizontal racks supported throughout their lengths on a reasonably flat surface free from stones and sharp projections. Pipes shall not be stacked in large piles, especially under warm conditions. Socket and spigot pipes shall be stacked in layers with sockets placed at alternate ends of the stack to avoid top sided stack.

PVC pipes shall be stored in a shaded area. The ends of pipe, particularly those specially prepared for jointing, shall be protected from abrasion. Damaged portion of a pipe shall be cut out completely.

Pipes of conducting materials shall be stacked on solid level sills and contained in a manner to prevent spreading or rolling of the pipe. For storage in large quantity, suitable packing shall be placed between the layers. During transportation, the pipes shall be so secured as to prevent displacement/rolling.

In stacking and handling of pipes and other conducting materials, the following minimum vertical safety distances from overhead power lines shall be provided

11 KV and below	1.40 m
Above 11 KV and below 33 KV	3.60 m
Above 33 KV and below 132 KV	4.70 m
Above 132 KV and below 230 KV	5.70 m
Above 275 and below 400 KV	6.50 m

Handling: Removal of pipes from a pile shall be accomplished by working from the ends of the pipe. During transportation, the pipes shall be so secured as to ensure against displacement.

2.2.3.5 Timber Piles and Poles

Piles and poles shall be stacked on solid and level sills so as to prevent rolling or spreading of the stack. The storage area shall be maintained free of vegetation and flammable materials.

Removal of piles and poles shall start from the top layer and by pulling from one end. Tag lines shall be used to control movement of piles and poles. In stacking and handling of piles and poles, precautions as laid down in Sec 2.2.3.4 shall be followed.

2.2.3.6 Sanitary Appliances

All sanitary appliances shall be stored under cover to prevent damage. In receiving and storing appliances consideration shall be given to the sequence of removal from the store to the assembly positions. Vitreous fittings shall be stacked separately from the metal ones.

Bigger sanitary appliances shall be handled one at a time. Traps, water seals and gullies shall be handled separately. Sanitary fittings shall be protected from any oil spillages; hands of the workers shall be free of any oily substance. The supporting brackets, pedestals etc. shall be checked before lowering the appliances in their position.

2.2.3.7 Doors, Windows, Ventilators and Grilles

Metal and plastic doors, windows and ventilators shall be stacked upright (on their sills) on level ground preferably on wooden battens and shall not come in contact with dirt or ashes. If received in crates they shall be stacked according to manufacturer's instructions and removed from the crates as and when required for the work. Metal and plastic frames of doors, windows and ventilators shall be stacked upside down with the kick plates at the top. These shall not be allowed to stand for long in this manner before being fixed so as to avoid the door frames getting out of shape and hinges being strained and shutters drooping. During the period of storage of aluminum doors, windows and ventilators, these shall be protected from loose cement and mortar by suitable covering, such as tarpaulin. The tarpaulin shall be hung loosely on temporary framing to permit circulation of air to prevent moisture condensation. All timber and other lignocellulosic material based frames and shutters shall be stored in a dry and clean covered space away from any infestation and dampness. The storage shall preferably be in well-ventilated dry rooms. The frames shall be stacked one over the other distances to keep the stack vertical and straight. These cross battens should be of uniform thickness and placed vertically one above the other. The door shutters shall be stacked in the form of clean vertical stacks

one over the other and at least 80 mm above ground on pallets or suitable beams or rafters. The top of the stack shall be covered by a protecting cover and weighted down by means of scantlings or other suitable weights. The shutter stack shall rest on hard and level surface. If any timber or other lignocellulosic material based frame or shutter becomes wet during transit, it shall be kept separate from the undamaged material. The wet material may be dried by stacking in shade with battens in between adjacent boards with free access of dry air. Separate stacks shall be built up for each size, each grade and each type of material. When materials of different sizes, grades and types are to be stacked in one stack due to shortage of space, the bigger size shall be stacked in the lower portion of the stacks. Suitable pallets or separating battens shall be kept in between the two types of material. Precast concrete door and window frames shall be stored in upright position adopting suitable measures against risk of subsidence of soil support.

While unloading, shifting, handling and stacking timber or other lignocellulosic material based, metal and plastic door and window frames and shutters, care shall be taken that the pieces are not dragged one over the other as it may cause damage to their surface particularly in case of the decorative shutters. The pieces should be lifted and carried preferably flat avoiding damage to corners or sides.

Metal frames of doors, windows and ventilators shall be stacked with the kick plates at the top. They shall not be kept in this manner for long, and should be taken to the fixing position as soon as possible.

2.2.3.8 Floors, Wall and Roof Tiles

Floor, wall and clay roof tiles of different types, such as, cement concrete tiles (plain, colored and terrazzo) and ceramic tiles (glazed and unglazed) shall be stacked on regular platform as far as possible under cover in proper layers and in tiers and they shall not be dumped in heaps. In the stack, the tiles shall be so placed that the mould surface of one faces that of another. Height of the stack shall not more than 1000 mm. Tiles of different quality, size and thickness shall be stacked separately to facilitate easy removal for use in work. Tiles when supplied by manufacturers packed in wooden crates shall be stored in crates. The crates shall be opened one at a time as and when required for use.

Ceramic tiles and roof tiles are generally supplied in cartons which shall be handled with care to avoid breakage. It is preferable to transport these at the site on platform trolleys.

2.2.3.9 Sheets and Boards

For storing and handling of sheets and boards, such as CGI sheets, particle boards, gypsum boards etc., the following requirements shall be fulfilled:

- (a) sheets and boards shall be stacked to a height of not more than 1 m on dry, clean, firm and level ground with timber or other packing beneath them;

- (b) bottom of the stack shall be raised adequately from the ground level where there is a risk of water coming on the floor;
- (c) sheets and boards shall be stacked under cover and protected from damage due to wind, rain and sun;
- (d) at least one edge of the stack shall be in true vertical alignment; the top sheet in each stack shall be suitably weighed down;
- (e) damage to the corners and surface of sheets and boards shall be prevented and damaged sheets shall not be stacked with sound materials;
- (f) sheets shall not be pushed forward against the lower sheet for more than one-fourth of the sheet length;
- (g) they shall be lifted into position by two workmen, if necessary;
- (h) sheets and boards shall be lowered or raised gently and not thrown; and suitable hand protection like gloves, jelly etc. shall be provided to the workmen wherever necessary.

CGI sheets shall be stacked in not more than 100 bundles per stack built solidly. Corrugations of sheets in one stack shall run in the same direction. One end of the stack shall be raised by at least 100 mm to drain accumulated water, if any. Sheets not for immediate use shall be stacked under roof.

Plywood, fiber board, particle board, block board etc. shall be stacked on a flat dunnage on top of which a wooden frame shall be constructed with battens of suitable size in such a way that it supports all four corners and edges of the boards. For boards up to a length of 2 m, minimum of one intermediate batten and for boards longer than 2 m, at least two intermediate battens shall be provided to avoid warping.

Decorative plywood and laminated and decorative boards shall be stacked in pairs facing each other. Sheets shall not be dragged one over another.

Specification laid out in BDS 1159 shall be followed for packaging of plywood, particle board, hard board and flush doors.

2.2.3.10 Cast Iron, Galvanized Iron and Asbestos Cement Pipes and Fittings

The pipes shall be unloaded where they are required, when the trenches are ready to receive them. Storage shall be provided at the bottom layer to keep the stack stable. The stack shall be in pyramid shape or the pipes placed length-wise and cross-wise in alternate layers. The pyramid stack is advisable in smaller diameter pipes for conserving space in storing them. The height of the stack shall not exceed 1.5 m. Each stack shall contain only pipes of the same class and size. Each stack shall contain only pipes of same

class and size, with consignment or batch number marked on it with particulars or suppliers wherever possible. Cast iron detachable joints and fittings shall be stacked under cover. Rubber rings shall be kept clean, away from grease, oil, heat and light.

Pipes in the top layer shall be handled first. At a time only one pipe shall be handled by two laborers while conveying to the actual site and shall be carried on shoulders. Fittings shall be handled individually.

2.2.3.11 Glass Sheets

All glass sheets shall be kept dry and stored in a covered space. Glass sheets shall be lifted and stored upright on their long edges and put into stacks of not more than 25 sheets. They shall be supported at two points at about 300 mm from each end by fillets of wood.

The bottom of each stack shall be about 25 mm clear from the base of the wall and other support against which the stack rests. The whole stack shall be as close to upright as possible. Smooth floors shall be covered with gunny bags.

Workmen handling glass sheets, remnants and waste glass pieces, and fibre glass shall be provided with gloves, jelly and other suitable hand protections. In removing glass sheets from crates, great care shall be taken to avoid damages from breakage. Glass edges shall be covered or protected to prevent injuries to workmen.

2.2.4 Inflammable and/or Fire-Sensitive Materials

Materials under this classification shall be stored within fire-preventive confines, furnished with firefighting provisions. Buckets containing sand shall be kept ready for use. A 5 kg dry powder fire extinguisher conforming to accepted standards shall be kept at an easily accessible position. Besides the areas shall be close to fire hydrants.

2.2.4.1 Plastic and Rubber sheets

Plastic and rubber sheets shall be stored within fire proof confines according to manufacturer's instructions. Sheets shall be stored in the coolest of the store rooms available. The room shall be well ventilated and kept dark; direct sun light shall not be allowed to fall on the stored sheets.

The sheets shall be stored away from electric generators, electric motors, switchgears and other such electrical equipment.

Contamination of the sheets with vegetable and mineral oil, grease, organic solvents, acid and their fumes, alkalis, dust and grit shall be prevented. All greasy contamination shall be removed immediately with kerosene or similar liquid, and the sheets thoroughly wiped dry and dusted with French chalk.

Undue stretch and strain, kinks, sharp bends or folds of the sheets shall be avoided in case of long time storage. The sheets shall be turned over periodically and treated with fresh chalk.

In addition, safety precautions common for all types of sheets, as laid down in Sec 2.2.3.9, shall be followed.

2.2.4.2 Paints, Varnishes, Thinners, Bitumen and Road Tar

Paints, varnishes, lacquers, thinners and other inflammable materials shall be kept in properly sealed or closed containers. The containers shall be kept in a well-ventilated location, free from excessive heat, smoke, sparks or flame. The floor of the paint store shall have 100 mm thick loose sand on it.

Paint materials in quantities other than required for daily use shall be kept stocked in the regular storage place. The manner of storage shall facilitate removal and use of lots in the same order in which they are received.

Temporary electrical wiring and fittings shall not be installed in the paint store. When electric lights, switches or electrical equipment are necessary to be stored or used in the same room, the room shall be designed in a way to reduce explosion risk.

Sources of ignition, such as open flame and exposed heating elements, shall not be permitted in paint store, nor shall smoking be allowed there.

Drums or containers containing bitumen, road tar, asphalt, etc. shall be stacked vertically on their bottoms in up to 3 tiers. Leaky drums shall be either totally removed or separated. Empty drums shall be stored in pyramidal stacks neatly in rows.

Bituminous roofing felts shall be stored away from other combustible or flammable materials. They shall be handled gently to prevent cracking and damages.

Workers engaged on jobs involving handling of hot bitumen, tar, and bituminous mixtures shall use protective wears, such as boots and gloves of rubber, goggles and helmet. No workers shall be permitted to handle such materials without wearing the needed protective covering.

Bitumen/tar shall not be heated beyond the temperature recommended by the manufacturer of the product. While discharging heated binder from the boiler, workers shall not stand opposite to the jet so as to avoid the possibility of hot binder falling on them. The container shall be handled only after closing the control valve. While handling hot bitumen/tar, workers shall exercise scrupulous care to prevent accidental spillage thereof. The buckets and cans in which the hot material is carried from boiler shall be checked before use to ensure that they are intact and safe. Mops and other applicators contaminated with bituminous materials shall not be stored inside buildings.

Outdoor storage of drums containing flammable materials like hydraulic brake and transmission fluid, gasoline and lubricants shall be such that contamination from moisture and dirt is avoided.

The storage shall be free of spilled products, debris and other hazardous material.

Compressed gases and petroleum products shall not be stored in the same building or close to each other. Proper identification by markings, tags etc. shall be used for petroleum products delivered to the job site and stored there in drums.

Highly flammable liquids shall be stored in fire resisting containers in a special store room secluded from the main working site. For uses of up to 50 litres, liquids can be stored in the workroom in fire resistant cupboards or bins. Stores of liquids shall be clearly marked highly flammable. All empty containers shall be returned to the store.

The workmen shall dispose of any clothing or apparel spilled over by or soaked in flammable materials immediately. They shall not be allowed to continue work unless affected clothing and apparels are changed.

2.2.5 Hazardous Materials

Materials under this category are (a) those posing health hazard through breathing, such as asbestos, glass fibre, etc. or injurious and/or intoxicating fluids of various kinds, (b) materials corrosive to living bodies and (c) materials likely to explode under heat or pressure. These should be stored in a manner specific to its properties, so as to prevent hazards of all kinds.

2.2.5.1 Asbestos-based Materials

Whenever possible, materials which do not contain asbestos shall be used. Special precautions as specified by the following sub-sections shall be taken while handling asbestos containing materials to minimize the risk of inhaling asbestos. Handling shall be limited to as few workers as possible.

(a) Handling of Asbestos-based Materials

When cutting, sawing or machining takes place in confined place efficient local dust extraction equipment shall be installed. Alternatively, a wet method of machining by water type dust suppressed powered tools shall be used.

The best standards of good housekeeping and hygiene shall apply to cutting areas which shall be segregated and used for no other purpose. Waste materials and dust shall not be allowed to accumulate in working area or store.

A vacuum cleaning device with a high efficiency filter shall be used to keep floors, walls and fixtures free from dust accumulation. Alternatively all surfaces shall be cleaned with a wet rag and floors washed by gently spraying water. Dry sweeping or compressed air blowing shall never be used.

Asbestos insulation boards shall preferably be supplied precut and drilled from the workshop using a suitable dust control equipment. On-site preparations shall be performed in the open.

Polyethylene sheet shall be used to screen a work area in an enclosed space. Only authorized workers shall be allowed access to such areas. Appropriate signs shall mark an asbestos working area and warn against inhaling asbestos dust.

A guillotine or knife die cutter shall be used to cut sheets. The use of hammer and chisel shall be avoided.

At the end of each work shift, dust shall be either collected by a vacuum cleaner or swept up after being wetted. The dust shall then be put into a sealable container. Any rejected material shall also be placed in an impermeable bag.

(b) Removal and disposal Asbestos-based Materials

Spray method shall be used for removal of asbestos-based materials which is not covered or coated by other materials. For removing thick asbestos-based materials, soaking method with total saturation shall be used. Dry method shall only be used where the spray or soaking method cannot be used.

All moveable furniture and fittings shall be removed from the work area and other non-removable items covered with plastic sheets. Air conditioning systems shall either be isolated from the asbestos removal area or closed down.

Before removal or stripping the asbestos, insulation coatings shall be thoroughly soaked with water or steam. In case of dry demolition of asbestos, a portable exhaust extraction plant shall be used.

Transport and storage containers shall be labeled of the contents. Waste shall be kept in strong enclosed containers or in strong sealed impervious bags. These shall not be overfilled; care shall be taken to avoid damage or spillage before disposal.

The filter bags used in a dust extracting system shall be impermeable and capable of being readily sealed and disposed of without further treatment.

(c) Protective Clothing and Equipment

Workmen engaged in works using asbestos-based material, shall wear a full body coveralls with pockets, and close fitted cuffs and necks together with a head cover. Protective clothing shall also be worn by all persons in an area into which asbestos dust is liable to escape.

The clothing shall be made of synthetic fiber. Wet weather overalls which can be hosed down may be used.

The use of suitable working clothing shall not be necessary when minor handling of asbestos containing insulation is carried out provided adequate dust control techniques are employed.

Whenever, work methods create asbestos dust, suitable protective respirator shall be used.

Respiratory protective equipment shall be properly maintained and regularly cleaned and serviced.

Every person required to use protective equipment shall be fully instructed and trained in its use.

Protective clothing and equipment shall be regarded as the means of last resort and used as a back-up of other techniques, or where effective asbestos dust control cannot be achieved by other means.

(d) Personal Hygiene

Changing room and shower facilities shall be provided for the exclusive use of persons working in an asbestos working area. Locker accommodation shall be provided for every person required to wear respirators and coveralls.

Lockers for work clothes shall be separated from others. Contaminated clothing shall be placed in a dustproof container immediately on removal. Contaminated clothing or belongings shall not be shaken or brushed. These shall be superficially cleaned by vacuum cleaning or hosing down with water.

Food and drinks shall not be handled, stored or consumed in the asbestos work area. Smoking shall be prohibited. Workmen shall take shower before changing back into their own clothing; work clothing shall not be taken home. Parts of the body exposed to asbestos dust shall be thoroughly washed after completion of the job or before taking any meal.

Asbestos workers shall have a full size chest X-ray before commencement of work and also yearly. The reports shall be kept properly by the contractor for ready reference.

2.2.5.2 Acids and Other Corrosive Materials Working with Acid/Chemicals

When working with acids, bases, or other chemicals, one shall wear the proper clothing. The following are the five clothing items that shall be used while working with chemicals.

- (a) Safety glasses/goggles: Should completely cover your eye at all times.
- (b) Safety face shield: Wear over the top of any safety glasses or goggles.

- (c) Full-length acid smock: Wear over the clean-room clothing.
- (d) Rubber gloves: Wear with a two-inch cuff. This prevents acid from running down your arm. Also, inflate with nitrogen and submerge in water to check for pinhole leaks before using.
- (e) Hard leather or other non-porous shoes.

2.2.5.2.1 Transporting Acid/Chemical

The acid/chemicals used in work shall be stored in glass or plastic bottles. Transport of these chemicals shall be made by hand in a rubber or plastic bucket. If the bottle breaks or the lid leaks, the chemical will be contained in the bucket.

While transporting Acid/Chemical following rules shall be followed:

Actions to be done	Prohibitions
✓ Use the appropriate size of container for the job.	☒ Do not reuse containers (adverse chemical reaction may occur).
✓ Get help when needed.	☒ Do not eat, drink, smoke, or touch any part of the body before washing your hands when working with chemicals.
✓ Clean containers after use with deionized water.	☒ Do not be afraid to ask questions.
✓ Work under a fume hood.	☒ Do not pour leftover chemicals back in its source container; that may result in contamination.
✓ Use a funnel when pouring chemicals into a small container.	☒ Do not put your face close to the bottle when pouring.
✓ Open bottles slowly to avoid spilling and allow vapors to escape.	☒ Do not puncture cap or lid of any bottle.
✓ Know what type of reactions to expect.	
✓ Remember to triple-A (AAA): Always Add Acid to water.	

2.2.5.2.2 Storage of Acid/Chemical

Proper storage of the acid/chemicals will ensure everyone's safety. Therefore when storing acid/chemical the following care shall be taken:

- (a) Store acids and bases in separate cabinets.
- (b) Keep acids and solvents in different cabinets.
- (c) Label shelves for quick chemical identification.

- (d) Make sure that incompatibles are not stored on the same shelf.
- (e) Keep same shaped bottles on the same shelf to conserve shelf space.
- (f) Never store chemical containers anywhere except in designated cabinets.

When need to use the acid/chemicals:

- (a) Take the oldest container whose shelf life has not expired.
- (b) Make sure the container is sealed when you return it.
- (c) Always return the container to its labeled shelf.

2.2.5.3 Explosives

(a) Transportation of Explosive

Loading, unloading and handling of explosives will be supervised by competent personnel. The safety provisions of Sections 4.1 and 4.3, Chapter 4 of this Part shall also be applicable.

Where the magazine is located near the construction site and blasting operations continue daily, actual requirements of explosives shall be issued from the magazine and transported to the site. Any leftovers shall be returned to the magazine after every use.

For carrying up to 5 kg of explosives, insulated containers constructed of minimum 50 mm thick finished wood or 6 mm thick plastic or 10 mm thick pressed fiber shall be used. The containers shall have no metal parts, be waterproof and provided with a lid and nonconductive carrying device.

Vehicles transporting explosives shall have a wooden or non-sparking metal floor with high sides and ends. In open-bodied vehicles, the explosives shall be covered with a waterproof and fire-resistant tarpaulin. Electric wiring in vehicle shall be fully insulated. The nature of cargo in the vehicle shall be properly indicated on its body.

Metal, flammable, or corrosive substances shall not be transported with explosives. Explosive and detonators or blasting caps shall not be transported in the same vehicle; they shall be transported in original containers or in securely locked separate nonmetallic containers.

Smoking shall be prohibited in the vehicle carrying explosives.

(b) Storage of Explosives

Explosives shall only be stored in remote and isolated structures of substantial construction and blast-release isolated yards. The storage area shall be clean, dry, well ventilated, and cool. The material shall not be stored near oil, gasoline, cleaning solutions, radiators, steam pipes, or other sources of heat.

Storage shall require bullet and fire-resistant magazine. Blasting caps or primers shall not be stored with explosives.

Smoking, matches, fire or flame shall not be allowed near a magazine. No leaves, grass, bush or debris shall be allowed to accumulate within 8 m of an explosive magazine. No sparking metal or tools shall be stored in a magazine. Persons shall put off shoes with metal nails before entering a magazine.

If nitroglycerine leaks down on the floor, the floor shall be immediately desensitized by washing thoroughly with an agent obtained beforehand from the supplier of the explosives.

(c) Handling of Explosives

No package containing explosives shall be dragged, dropped or handled roughly. These shall be opened only at a safe distance and properly shielded from the packages of explosives in bulk storage. The covers of the explosive cases or packages shall be replaced every time after taking out part of the contents.

Sparking metal tools shall not be used to open kegs or cases of explosives. Smoking or carrying matches, fire, flame or devices capable of producing fire or flame, shall not be permitted while handling or using explosives. Explosives shall not be carried in the pockets of any clothing or on any person.

(d) Disposal of Explosives

No explosives shall be abandoned. They shall be disposed of in accordance with the approved methods; manufacturers or the appropriate authority shall be consulted in this matter.

Explosives caps or packing shall not be left lying around. Paper or fiber materials used in packing explosives shall not be put in any subsequent use. Such materials shall be destroyed by burning.

2.3 Miscellaneous

Small articles like screws, bolts, nuts, door and window fittings, polishing stones, protective clothing, spare parts of machinery, linings, packing, water supply and sanitary fittings, and electrical fittings, insulation board etc. shall be kept in suitable and properly protected containers, boxes or store rooms. Valuable small materials shall be kept under lock and key.

Polymeric materials such as coating, sheeting, reflective surfacing/sheeting, etc. shall be stored as per the manufacturers' instructions. Special precautions shall be taken in case of storage, handling and usage of toxic materials.

2.4 Special Considerations

Materials constantly in use shall be relatively nearer the place of use.

Heavy units like precast concrete members shall be stacked near the hoist or the ramp.

Materials which normally deteriorate during storage shall be kept constantly moving, by replacing old materials with fresh stocks. Freshly arrived materials shall never be placed over materials which had arrived earlier.

Appropriate types of fire extinguishers shall be provided at open sites where combustible materials are stored and for each storage shed room where flammable/combustible materials are stored. For guidance regarding selection of the appropriate types of fire extinguishers reference may be made to good practice. It is desirable that a minimum of two extinguishers are provided at each such location.

Workers handling excavated earth from foundation, particularly if the site happens to be reclaimed area or marshy area or any other infected area, shall be protected against infection affecting their exposed body portions.

2.5 Loading and Unloading of Materials

(a) Loading and Unloading Railway Wagons and Motor Vehicles

Each workman shall be instructed for the proper method of loading and unloading from rail wagons and motor vehicles, and provided with necessary equipment for safety. Supervisors shall ensure that the required number of workmen based on the weight and the distance involved in each job is available and engaged for the particular job.

Warning signals shall be displayed to indicate that the rail-wagons must not be coupled or moved while loading and unloading are carried out. The wheels of wagons and vehicles shall always be sprigged or chained while these are being unloaded; brakes alone shall not be relied upon.

Special lever bars, rather than ordinary crowbars, shall be used for moving rail wagons. Where gangplanks are used, either cleats at lower end of gangplank or pin through end of gangplanks shall be used to prevent sliding and slipping. If the gangplank is on a slope, cleats or abrasive surface shall be provided for the entire length.

When rail road wagons and motor vehicles are being loaded or unloaded near passageways or walkways, adequate warning signals shall be placed on each end of the way.

(b) Manual Handling

Loading and unloading of heavy items shall be done with cranes or gantries, if available. The workmen shall stand clear of the path of the material being moved by mechanical equipment. The slings and the ropes used shall be of adequate load carrying capacity.

For loading heavy and long components manually into motor vehicles, rail wagons, trailer etc., either wooden sleepers or steel rails of sufficient length and properly secured in position shall be put against the body of the wagon/vehicle at three or four places. The slope of such makeshift ramp shall be less than 30° with horizontal.

Long items shall be dragged, one by one, gently and uniformly along the ramps by means of ropes (tag). Workmen pulling long items shall anchor their feet against a firm surface.

Loaded items may be shifted by crowbars and other suitable leverage mechanism in their right position. These shall not be pushed or moved by hand. Similar procedures as outlined above shall be followed for manual unloading of long or heavy items.

For regular and frequent handling, the maximum load a single workman is subject to carry shall be limited to 20 kg. Workmen to carry heavier loads shall be specially selected, and if necessary, trained.

While lifting a load, the body shall be kept upright; weight shall be distributed evenly and supported on the bone structure, and held close to the body. Advantage shall be taken of any device provided for assistance.

PART VII
Chapter 3
Safety During Construction

3.1 General**3.1.1 Scope**

The provisions of this Chapter shall apply to the safety of life and property during construction, erection and alteration of various parts of a building or any other structures. Nothing stated herein shall be construed to nullify any rules, regulations, safety standards or statutes of the local authority, Corporations, or those contained in the various Acts of the Government of Bangladesh. The specific rules, regulations and acts pertaining to the protection of the public or workmen from health and other hazards wherever specified by the local Authority/Corporation etc. or by the Act/Ordinance of the Government shall take precedence over whatever is herein specified.

3.1.2 Safety Management

The safety of personnel engaged in building construction shall be ensured through a well-planned and well organized mechanism. For this, depending on the size and complexity of building construction project, safety committee shall be constituted to efficiently manage all safety related affairs. The site in-charge or his nominee of a senior rank shall head the committee and a safety officer shall act as Member Secretary. The safety committee shall be organized a training program for the personals and workers to train up them about safety issues involved in the construction process and also organize meeting of the committee regularly say fortnightly or monthly depending on the nature of the project, however, emergency meetings shall also be called as and when required. The safety committees shall deal with all the safety related issues through well-structured agenda, in the meetings and all safety related measures installed at the site and implementation thereof shall be periodically reviewed.

3.2 Terminology

For the purpose of this Part the following definitions shall apply.

AUTHORITY HAVING JURISDICTION	The Authority which has been created by a statute and which for the purpose of administering the Code/Part, shall authorize a committee or an official to act on its behalf; hereinafter called the 'Authority'.
CONSTRUCTION EQUIPMENT	All equipment, machinery, tools and temporary retaining structures and working platforms, that is, tools, derricks, staging, scaffolds, runways, ladders and all material, handling equipment including safety devices.

FLOOR HOLE	An opening measuring less than 300 mm but more than 25 mm in its least dimension, in any floor, platform, pavement, or yard, through which materials but not persons may fall; such as, a belt hole, pipe opening or slot opening.
FLOOR OPENING	An opening measuring 300 mm or more in its least dimension, in any floor, platform, pavement or yard through which person may fall; such as hatch way, stair or ladder opening, pit or large manhole.
GUARD RAILING	A barrier erected along exposed edges of an open side floor opening, wall opening, ramp, platform, or catwalk or balcony, etc, to prevent fall of persons.
MATERIALS HANDLING HOISTS	A platform, bucket or similar enclosure exclusively meant for the lifting or lowering of construction material the hoists being operated from a point outside the conveyance.
PILE RIG	The complete pile driving equipment comprising piling frame, leader, hammer, extractor winch and power unit. Complete pile driving rig shall be mounted on rafts or pontoon or rails. Pile rig shall also be a mobile unit mounted on trailers or trucks, or a special full revolving rig for raking piles.
PLATFORM	A working space for persons, elevated above the surrounding floor or ground, such as balcony or platform for the operation of machinery and equipment.
SCAFFOLD	A temporary erection of timber, bamboo or metal framework used in the construction, alteration or demolition of a building, to support or to allow the hoisting and lowering of workmen, their tools and materials.
TOE BOARD	A vertical barrier erected along exposed edge of a floor opening, wall opening, platform, catwalk or ramp to prevent fall of materials or persons.
WALL HOLE	An opening in any wall or partition having height of less than 750 mm but more than 25 mm and width unrestricted.
WALL OPENING	An opening in any wall or partition having both height of at least 750 mm and width of at least 450 mm.

3.2.1 Safety of Workmen

Helmets conforming to BDS 1265 and BDS 1266 shall be worn by the workmen and other personnel at all times during the work. Safety goggles of accepted standard (BDS 1360) shall be used by individuals engaged in drilling, cutting, welding and all such works which cause hazard to the eye. The welders and gas cutters shall be equipped with proper protective equipment like gloves, safety boots, aprons and hand shields having filter glass of accepted standard and suitable to the eyes of the particular worker.

3.2.2 Site Precautions

Construction site shall be delineated, in absence of boundary walls, by fences. During the erection of tall buildings, nylon net shall be put around the building periphery 3 m to 4 m below the working level.

Warning signs shall be displayed, where necessary, to indicate hazardous areas like high voltage zone, area of no smoking etc. Hand lamps shall be of low voltage, preferably 24V. All electrically operated hand tools shall be provided with double earthing.

3.2.3 Site Amenities

Toilet facilities shall be provided at all construction sites. If sewer connection is not available, temporary wells shall be used. The wells shall be provided with proper covers, bad smell protector and have to clean regularly.

Men and women workers shall be provided with separate sanitary and washing facilities.

The toilet facilities shall be located at a corner of the site so as to avoid any obstruction. Protection from bad weather and falling object, and proper privacy shall be provided to the toilet users.

Temporary toilets shall be dismantled, all wells filled up, and the whole area made level, dressed and restored back to proper grade at the end of the project. All temporary sewer connections shall be removed and the sewer capped.

Washing facilities provided at the site shall be connected to the available running water supply.

Drinking water shall be supplied to the site. In absence of any water supply facility at the site, hand tube wells shall be sunk to meet the requirements of drinking and washing.

Numbers of the sanitary and plumbing facilities required in a construction site shall be regulated by the 1965 Factories Act, and Part 8 Chapters 5 and 6.

Proper accommodation for taking meals and for taking shelter during interruption of work in night time and due to adverse weather condition with amenities of sleeping bed including provision for lights and fans have to provide.

3.3 Excavation and Foundation Work

3.3.1 General

The requirements of this Section shall be satisfied in addition to those of Sec 3.12 Part 6 for all excavation and foundation works.

The distribution of the supporting foundation shall be such as to avoid any harmful differential settlement of the structure. The type and design of the foundation adopted shall ensure safety to workmen during construction and residents of the neighboring property. Sufficient care shall be taken in areas, where withdrawal of ground water from surrounding areas could result in damages to such foundations. During the construction of the foundation, it shall be ensured that the adjoining properties are not affected by any harmful effects.

The process of excavation, filling in, pumping etc. shall avoid endangering the strength or stability of the partially completed structure. The partially completed structure shall be capable of carrying loads previously taken by temporary works which, as part of the construction procedure, have to be transferred before the completion of the work.

Excavation with intervals on any site shall be avoided. If such excavation is unavoidable, the excavated site shall be properly fenced and warning signals.

Excavation of interrupted or temporarily suspended construction shall be either backfilled or barricaded.

During construction, inspection shall be made by the engineer-in-charge to ensure that all protective works carried out to safe-guard the adjoining property are sufficient and in good order to ensure safety.

Arrangements for safe movement of workers and inspectors in the trench have to be planned and provided.

Before carrying out any excavation work/pile driving, the position, depth and size of underground structures, such as water pipes, mains, cables or other services in the vicinity to the proposed work, shall be obtained from the appropriate Authority to prevent accidents to workmen engaged in excavation work and calamities for the general public. Prior to commencement of excavation detailed data of the type of soils that are likely to be met with during excavation shall be obtained and the type of protective works by way of shoring timbering, etc, shall be decided upon for the various strata that are likely to be encountered during excavation. For detailed information regarding safety requirements during excavation reference shall be made to good practice.

3.3.2 Excavating Machinery and Tools

Heavy equipment, such as excavating machinery, shall be kept away from the trenches by a distance at least equal to the depth of trench to a maximum of 6 meters. All excavating tools shall be kept far away from the edge of trench.

3.3.3 Excavated Materials and Surcharges

Excavated materials shall be kept away from the edges of the trench to provide a clear berm of safe width. Where this is not feasible, the design of protection for the trenches shall include the additional load due to the materials.

Proximity of buildings, piles of lumber, crushed rocks, sand and other construction materials, large trees, etc. may impose surcharges on the side of the trench to cause bulging, sliding, etc.

Additional protective measures shall be taken to support the sides of the trenches under these conditions. The objects creating such threat shall be removed if possible before excavation starts.

3.3.4 Ground Water

Where deep excavation is required, the location of water-bearing strata shall be determined and the water pressure observed to take necessary precautions. Direction of natural drainage shall be determined to facilitate the design of intercepting drains to prevent the influx of ground water.

In areas where the ground water or soil contains constituents in amounts sufficient to cause damage to cement or buried metals, a chemical analysis of samples of ground water and soil shall be obtained and necessary precautions taken.

Basements or pits below ground water level, which rely on the weight the superstructure for their stability against floatation, shall be pumped day and night. Protective filters shall be used during heavy pumping in excavations. The water shall be drawn away from the excavation rather than through the ground towards the excavation.

3.3.5 Ground Condition

Adequate precautions, depending upon the type of strata met with during excavation (like quick sand, loose fills and loose boulder) shall be taken to protect the workmen during excavation. Effect of climatic variations and moisture content variations on the materials under excavation shall be constantly watched and precautions taken, where necessary, immediately to prevent accidents at work site.

Where portions of the foundation are underlain by soft materials or where the layers of such materials vary in thickness, the assessment of allowable bearing pressure shall require a settlement analysis.

Site investigations shall be sufficiently extensive to ensure that significant variations in strata thickness are detected. If required, either the resistance of the inclined or jointed strata shall be increased or the foundations shall be carried deep enough to prevent sliding.

Precautions, against pockets of poisonous/dangerous gases including protection to the workmen, shall be taken during deep excavation. Effect of climatic variations and variation in moisture content of the soil shall be constantly monitored and precautions taken immediately, when necessary.

3.3.6 Overhang, Slopes and Cavities

Overhangs in the trenches shall be supported by props. Use of heavy machinery shall be avoided under or over this area.

Where climatic or other conditions may result in deterioration of the sides of excavation, consideration shall be given to their support and protection. During excavation, adequate protections justified by established method of analysis shall be taken to prevent slope instability.

3.3.7 Blasting and Vibration

Blasting for foundation of buildings is prohibited unless special permission is obtained from the Authority. Where blasting technique is to be used, an analysis for the stability of slopes shall be carried out and steps be taken accordingly.

Attention shall be given to the geological strata of the site to ensure that it is not liable to transmission of ground vibration to areas where it may cause damage to property or the ground.

After blasting, overhangs or loose boulders shall be cleared off the site. In all excavation works, precautions shall be taken to eliminate/reduce vibration generated by adjacent machinery, vehicles, railroads, blasting, piling and other sources.

Appropriate authorities shall be notified in advance of any blasting operations when these are to take place close to public roads and railways. Also see Sec 4.3.

3.3.8 Health Hazards during Excavation

Mechanical ventilation shall be provided where gases or fumes are likely to be present in trenches.

All personnel working there shall be provided with protective respiratory equipment. All trenches/tunnel shall be provided with emergency exits (Sections 3.11.2 and 3.11.3).

The precautionary measures provided shall meet the requirements of the local health authority.

The owner shall ensure that all precautionary measures have been taken and been inspected by the appropriate Authority prior to commencement of such work.

3.3.9 Safety of Materials

Materials required for excavation, like ropes, planks for gangways and walkways, ladders, etc. shall be inspected by the Engineer-in-charge who shall ensure that no accident shall occur due to the failure of such materials (see Part 5 of this Code).

(a) Fencing, Warning Signs and Watchman

Where excavation is going on, for the safety of public and the workmen, fencing shall be erected.

Sufficient number of notice boards and danger sign lights shall be provided in the area to avoid any member of public from inadvertently falling into the excavation. When excavations are being done on roads, diversion of the roads shall be provided with adequate notice board and lights indicating the diversion well ahead. Where necessary, recourse shall be had for additional precautionary measures by way of watchmen to prevent accident to the general public, especially during hours of darkness. If necessary, watchmen shall be employed as an additional precautionary measure to prevent any accident, especially during the night.

(b) Vibrations from Nearby Sources

Vibration due to adjacent machinery, vehicles, railroads, blasting, piling and other sources require additional precautions to be taken.

(c) Precautions While Using Petroleum Powered Equipment

At the site of excavation, where petroleum powered equipment is used, petroleum vapors are likely to accumulate at lower levels and may cause fire explosion under favorable circumstances. Care shall, therefore, be taken to avoid all sources of ignition in such places.

3.3.10 Piling and Deep Foundation

All piling and deep foundation operations shall be supervised by a competent Geotechnical Engineer. He shall also be responsible for the precautionary measures to be taken.

For work during night, lighting of at least 100 lux intensity shall be provided at the work site. In excavations deeper than 1.5 m, ladders, ramps or other means of escape, and staging shall be provided.

Every crane driver or hoisting appliance operator shall be competent to the satisfaction of the engineer-in-charge and no person under the age of 21 years shall be in-charge of any hoisting machine including any scaffolding winch, or giving signals to operators.

3.3.11 Working in Compressed Air

Working in compressed air, in case of deep foundations, requires several precautions to be observed to safeguard the workmen against severe hazards to life, compressed air disease and related ailments.

Filtered compressed air shall be supplied to a working chamber sufficient to provide 0.3 m³ of fresh air per minute per person at the pressure in the chamber. Means for the escape of foul air, as well as circulation of fresh air in the chamber, shall be ensured.

Hot drinks shall be supplied to workmen employed in compressed air after leaving the chamber. No person shall carry any flammable materials inside the air-lock and nobody shall be allowed to smoke inside. Only approved type of lamps and torches shall be used. Lighting of at least 4.5 lux intensity shall be provided.

Methanometer shall be used to detect hazardous gases. Samples of air inside the well shall be taken every eight hours and tested for the presence of hazardous gases and for deficiency of oxygen. In case any hazardous gas is detected, it shall be immediately reported to the engineer and the work in the compressed air stopped.

The pressure in the chamber, in the first minute, after starting compression shall be increased to 35 kPa. It shall not be further increased until the lock attendant has checked whether or not there are complaints of discomfort. The pressure shall then be increased at a rate of 65 kPa/min. If any person complains of discomfort, the proceeding compression shall be immediately stopped and the person evacuated unless he feels comfortable again in a reduced pressure.

In case of airlocks where blasting is done, the workmen shall be permitted to start work only after an inspection by a competent professional found it to be safe. Air required for pneumatic tools shall be cooled and purified in the same way as air for working chamber.

Every man lock shall have a minimum head room of 1.8 m and at least 0.85 m³ of space per person. It shall be suitably equipped with an accurate pressure gauge, clocks, and efficient means to convey visible or nonverbal signals to the lock attendant outside. All electrical installations inside the airlock shall be of flame proof type.

All equipment shall be thoroughly inspected after every 45 days of working and every time it is shifted and reinstalled, and certified to be in a safe working condition by a competent person. A record of all such inspections shall be kept in a register.

The receiver shall be capable of maintaining the working pressure for at least four hours. Adequate access through the bulk heads and sufficient ladders shall be provided. Escape routes in tunnels shall be in the corner. Whilst any person is in a working chamber, the door between such chamber and any man-lock providing egress towards a lower pressure shall be kept open.

No person shall be in a working chamber under pressure where the wet bulb temperature exceeds 29°C measured by a thermometer using nontoxic materials.

No person shall be employed on work in compressed air unless under the supervision of a person experienced in such work. No person shall be employed where the pressure exceeds 120 kPa unless he has, within the previous four weeks been examined and certified to be fit for employment in compressed air. If a person is suffering from cold in head, sore throat, earache etc., he/she shall not be employed in compressed air. Finally work in compressed air shall be carried out only by workers whose physical aptitude for such work has been established by a medical examination and when competent person is present to supervise the conduct of the operations.

Where the pressure exceeds, a suitably constructed medical-lock shall be provided. It shall have two chambers, and doors fitted with bulls' eyes and air valve. The lock shall have couch, blanket, dry woolen garments, food etc. The medical lock shall be supplied with air, free of oil and carbon monoxide, and capable of raising the pressure from 0 to 520 kPa in 5 minutes.

3.3.12 Adjoining Properties and Service Lines

Where bored or driven piling works are to be carried out in the vicinity of old structures which are likely to be damaged, tell-tales shall be fixed on such structures to monitor their behavior while piling is in progress; timely precautions shall be taken against any adverse effect.

Steps shall be taken, if necessary, to increase the general stability of the construction site or the adjoining site(s), before new structures are erected. In all cases, the possible effect of slopes and excavation of foundation stability shall be carefully investigated.

Before excavation or pile driving, information on the location of underground utility connections shall be obtained from the relevant authorities. Probable extent of all damages due to pile driving to adjoining structures or service lines shall be ascertained in advance of operation; pile driving shall be planned accordingly, especially in the case of pre-cast pile driving.

If excavation involves cutting through existing land drains, they shall be carefully diverted into the ground drainage system. All other precautionary measures required by Sec 1.5 of this Chapter shall also be taken.

3.4 Pile Rig

3.4.1 Erection of Pile Rig

The frame of the rigs shall be structurally safe for all anticipated dead, live and wind loads.

Whenever the structural strength is in doubt, suitable test shall be carried out by the engineer and the results recorded. No pile driving equipment shall be used until it has been inspected and found safe.

When two or more pile drivers are used at the same location, they shall be separated by a distance at least equal to the longest leg of either rig.

Pile drivers shall be firmly supported on heavy timber sills, concrete beds or other secure foundations. If necessary, pile drivers shall be adequately guyed. Rigs not in use shall be supported by at least three guys to withstand wind, storm, gales and earthquake.

3.4.2 Operation of Pile Rig

Access to working platforms and top of pulley shall be provided by ladders. Working platforms shall be protected from wind and rain. Ladder in regular use in tall driven piling rigs, or rigs of similar nature, shall be securely fastened and extended for the full height of the rig.

Exposed gears, flywheels, etc. shall be fully enclosed. Motor gearing, transmission, electrical wiring and other parts of a hoisting machine which are sources of hazard shall have proper safeguards.

To operate energized electrical installations, insulating mats and wearing apparel, such as gloves, etc. shall be used. Sheaves on pile drivers shall be guarded against workers drawn into them accidentally.

No steam or air driven equipment shall be repaired while it is in operation or under pressure.

Steam and air lines shall be controlled by easily accessible shut-off valves. These lines shall consist of armoured hose or its equivalent.

The hose of steam and air hammers shall be securely lashed to the hammer so as to prevent it from whipping if a connection breaks. Couplings of sections of hose shall be additionally secured by ropes or chains. When not in use, the hammer shall remain in dropped position held in place by a cleat, timber or other suitable means.

Hoisting appliances shall be provided with means to reduce the risk of accidental descent of the load. Adequate precautions shall also be taken to reduce the risk of any part of suspended load becoming accidentally displaced. Care shall be taken to prevent the hammer from missing the pile.

Loads shall be adequately counter-balanced, and the tilting device secured against slipping.

Precautions in the form of securing the legs shall be taken to prevent a pile driver from overturning if a wheel breaks. Stirrups or other means shall be provided to prevent the rope from coming out of the top pulley or wheel. Hoisting ropes on pile drivers shall be made of galvanized steel.

Pile drivers shall not be erected in proximity to electric conductors. When electricity is used as power for piling rig, only armored cable conforming to BDS 901 and other relevant standards shall be used. The cable shall be thoroughly waterproofed.

3.4.3 Piles

Piles shall be prepared at a distance at least equal to twice the length of the longest pile, from the pile driver. Workers employed in the vicinity of pile drivers shall wear helmets conforming to BDS 1265. No steam or air shall be released until all workers are at a safe distance.

Piles shall be so slung that they do not swing or whip round. A hand rope shall be fastened to a pile hoisted to control its movement. Long piles and heavy sheet piling shall be secured against falling. While a pile is being guided into position in the leads, workers shall not put their hands or arms between the pile and the inside guide or on top of the pile. Inclined piles shall rest in a guide while driven.

The maximum length of wooden piles (ballies) shall be limited to 9 m. Ballies shall not be less than 50 mm in diameter at any place and shall spread to 75-200 mm in diameter at the top depending on the class of ballies. Each ballie shall be legibly and indelibly marked with information on the species of timber, suppliers name, class of ballie etc. Whenever required, butt ends of ballies shall be preserved with creosote-fuel oil mixture 50:50.

The driving end of a ballie post shall be provided with an iron ring or cap. When creosoted ballies are driven, adequate precautions, such as the provision of personal protective equipment and barrier creams, shall be taken to prevent injury from splashes of creosote.

3.4.4 Inspection and Tests

Pile driving equipment shall be inspected by an engineer at regular intervals not exceeding three months. A register shall be maintained at the site for recording the results of such inspection. Pile lines and pulley blocks shall be inspected by the foreman before the beginning of each shift for any excess wear or other defects.

Defective parts of pile drivers, such as sheaves, mechanism slings and hose shall be repaired by only competent technicians and duly inspected by foreman in-charge of the rig. The findings of such inspection shall be recorded in the register.

For every hoisting machine, chain, rig, hook, shackle, swivel and pulley block used in hoisting or suspending, the safe working loads shall be ascertained. Every hoisting machine and all gears shall be marked with the safe working loads and the conditions under which it is applicable.

Tests shall be performed in case of doubt and half of the tested load shall be taken as the safe working load. No part of any machine or any gear shall be loaded beyond the safe working load.

3.5 Construction of Walls

3.5.1 General

The height of wall constructed per day shall be restricted to ensure that the newly constructed wall does not collapse due to the lack of strength in the lower layers. Adequate number of expansion joints shall be provided in long walls to prevent crumpling.

3.5.2 Scaffold

Properly designed and constructed scaffolding built by competent workmen shall be provided during the construction of the walls to ensure the safety of workers. The scaffolding shall be of timber, metal or bamboo sections and the materials in scaffolding shall be inspected for soundness, strength, etc. at site by the Engineer-in-charge prior to erection of scaffolds. Steel scaffolds intended for use in normal building construction work shall conform to accepted BDS standards. Bamboo and timber scaffolds shall be properly tied to the junctions with coir ropes of sufficient strength or mechanical joints to ensure that joints do not give way due to the load of workmen and material. Joining the members of scaffolds only with nails shall be prohibited as they are likely to get loose under normal weathering conditions. The scaffold has to be checked after every 15 days in rainy season and 30 days in dry season. In the erection or maintenance of tall buildings, scaffoldings shall be of noncombustible material especially when the work is being done on any building in occupation. After initial construction of the scaffolding, frequent inspections of scaffolding shall be made by the Engineer-in-charge. The platforms, gangways and runways provided on the scaffoldings shall be of sufficient strength and width to ensure safe passage for the workmen working on the scaffolding. The joints provided in these gangways, platforms, etc. shall be such as to ensure a firm foot-hold to the workmen. Where necessary cross bars shall be provided to the full width of gangway or runway to facilitate safe walking.

The Engineer-in-charge shall ensure by frequent inspections that gangways of scaffolding have not become slippery due to spillage of material. Loose materials shall not be allowed to remain on the gangways. Where necessary, because of height or restricted width, hand-rails shall be provided on both sides. Workers shall not be allowed to work on the scaffolding during bad weather and high winds.

In the operations involved in the erection or maintenance of outside walls, fittings, etc. of tall buildings, it is desirable to use one or more net(s) for the safety of the workmen when the workmen are required to work on scaffoldings.

3.5.3 Ladders

Setting of Ladders: Rails of ladders shall extend at least 1m above the landing and shall be secured at the upper end. As an alternative, there shall be adequate handhold at landing or side guys with anchorage at the bottom.

To prevent slipping, a ladder shall be secured at the bottom end or held by a person at the time of use. A lean-to-ladder shall have a maximum angle of 75° with the horizontal. Ladders shall be provided with nonslip bases on slippery or sloping floors. Ladders used in strong wind shall be securely lashed in position.

A ladder shall neither be placed against window pane, sashes or such other fragile or easy yielding objects, nor in front of doors opening towards it. If set up in driveways, passageways or public walkways, it shall be protected by barricades. Ladders shall not be supported on any insecure base, e.g. scaffold, planking over trenches etc.

Use of Ladders: All ladders shall be constructed of sound material, and shall be capable of carrying the design loads. No ladder with a missing or defective rung, or supported on nails only, shall be used. A dropped ladder shall be inspected prior to reuse.

Ladders shall not be used as guys, braces or skids or in horizontal position as runways and catwalk. They shall not be generally overcrowded. Ladders shall not be spliced; when unavoidable, splicing shall be done only under the supervision of a foreman.

A user shall place his feet near the ends of the rungs rather than near the middle, and face the ladder when using it. Both the hands shall be used in climbing a ladder.

Leaning more than 300 mm from the side in order to reach another area from a single setting of the ladder shall not be allowed; the ladder shall be shifted to the required position.

All joints in the ladder shall be properly constructed. Where necessary, handrails shall be provided to the ladders. A brace shall be attached at the middle and supported from a non-yielding fixed object if a ladder shows tendency to spring. Excessive deflection of ladders shall be prevented by stiffeners.

Metal ladder shall not be used close to electrical equipment or circuits. They shall be marked with “CAUTION: DO NOT USE NEAR ELECTRICAL EQUIPMENT” signs. Overhead protection shall be provided for workers working under a ladder.

Wooden ladders shall be inspected at least once in 6 weeks for damage and deterioration. Close visual inspection is recommended in preference to load testing. This condition is particularly applicable to rope and bamboo ladders where fraying of ropes and damage to bamboo is likely to occur.

3.5.4 Opening in Walls

Before making an opening in an existing wall, adequate supports against the collapse or cracking of the wall portion above the opening or roof or adjoining walls shall be provided. Staging shall be of full length of the wall opening.

Wall opening barriers and screens shall be capable of withstanding the intended load. Every chute, wall opening or any other wall opening from which there is a vertical drop of more than 1200 mm shall be guarded by barriers.

The guard shall be removable, hinged or otherwise mounted. The guards shall be kept in position regardless of the use of the opening. In addition, a grab handle shall be provided on each side of the opening. The opening shall have a minimum 25 mm high sill.

3.5.5 Projection from Walls

Form work provided for horizontal projections out of the wall shall not be removed till walls, or other stabilizing construction, over the supporting edge of the projecting slabs providing protection against overturning are constructed.

3.5.6 Common Hazards During Walling

3.5.6.1 Lifting of Materials for Construction

Implements used for carrying materials to the top of scaffoldings shall be of adequate strength and shall not be overloaded during the work. Where workmen have to work below scaffoldings or ladder, overhead protection against the falling materials shall be provided. Care shall be taken in carrying large bars, rods, etc. during construction of the walls to prevent any damage to property or injury to workmen.

3.5.6.2 Haulage of Materials

In case of precast columns, steel beams, etc. proper precautions shall be taken to correctly handle, use and position them with temporary arrangement of guys till grouting of the base.

Manila or sisal rope shall not be used in rainy season for hoisting of heavy materials as they lose their strength with alternate wetting and drying.

3.5.6.3 Electrical Hazards

No scaffolding, ladder, working platform, gangway runs, etc. shall exist within 3 m from any uninsulated electric wire. The distance from high tension line for those features would be as per specifications of BPDB.

3.5.6.4 Fire Hazards

Gangways and the ground below the scaffolding shall be kept free from readily combustible materials including waste and dry vegetation at all times.

Where extensive use of blow torch or other flame is anticipated scaffoldings, gangways, etc. shall be constructed with fire resistant materials. A portable dry powder extinguisher of 3 kg capacity shall be kept handy.

3.5.6.5 Mechanical Hazards

Care shall be taken to see that no part of scaffolding or walls is struck by truck or heavy moving equipment and no material shall be dumped against them to prevent any damage. When such scaffoldings are in or near a public thoroughfare, sufficient warning lights and boards shall be provided on the scaffoldings to make them clearly visible to the public.

3.5.6.6 Fragile Materials

During glazing operations, adequate precautions shall be taken to ensure that the fragments of fragile materials do not cause any injury to workmen or general public in that area by way of providing covering to such material, side protection at work site, etc.

3.6 Construction of Floors

3.6.1 General

Platforms, catch ropes, nets etc. shall be provided during the construction of roofs. Precautions shall be taken to employ the correct technique of hoisting materials, to use hoists of sufficient strength for the quantity of stores to be hoisted, and to prevent overloading and overturning of hoists or buckets, etc.

Where, the floor of one storey is to be used for storage of materials for the construction of roof, it shall be ensured that the total load does not exceed the capacity of the floor.

3.6.2 Use of Sheets

It shall be ensured that joints in corrugated galvanized iron or asbestos cement sheets are kept secured in position and sheets do not slip. Walking on asbestos cement sheets shall not be allowed.

Tiles shall not be left loose on the roof.

Injury to passers-by due to breakage of glass or plastic sheets shall be prevented. During wet conditions, work on sloped roof shall not be allowed unless the foreman decides that the roof is not as slippery as to pose any risk. In slopes of more than 30° to the horizontal, ladders, waist-tie etc. shall be used.

3.6.3 Platforms

Working platform required according to the type of roof shall be provided. Additional precaution shall be taken to construct the platform with sound material secured and fixed, and checked from time to time throughout the period of construction.

3.6.4 Flat Roof

Formwork provided for flat concrete roof shall be designed and constructed for the anticipated loads.

During the construction of the roof, the formwork shall be frequently inspected for defects. Enough walking platforms shall be provided in the reinforcement area to facilitate safe walking to the concreting area. Loose wires and unprotected rod ends shall be avoided.

Formwork supporting cast-in-place reinforced and pre stressed concrete floors and roofs shall be adequately tied or braced together to withstand all loads until the new construction has attained the required strengths.

3.6.5 Openings and Holes

Every temporary floor opening shall either have railing of at least 900 mm height, or shall be constantly attended. Every floor hole shall be guarded by either a railing with toe board, or a hinged cover. Alternatively, the hole shall be constantly attended or protected by a removable railing.

Every stairway floor opening shall be guarded by a railing at least 900 mm high on the exposed sides except at entrance to stairway. Every ladder way floor opening or platform shall be guarded by a guard railing with toe board except at entrance to opening.

Every open sided floor or platform 1.2 m or more above adjacent floor or ground level shall be guarded by a railing on all open sides, except where there is entrance to ramp, stairway or fixed ladder. Such entrances shall be either guarded with a swinging gate, or so offset that a person is prevented from walking directly into the opening. The railing shall be accompanied by a toe board at least 200 mm high.

The above precautions shall also be taken near the open edges of floors and roofs. Requirements of Sections 1.7.3 and 1.7.4 shall also be met.

3.6.6 Skeleton Construction

Temporary flooring of skeleton construction shall be provided with tightly planked timber over timber supports to withstand all loads. The temporary flooring can also be made of metal sheet supported on timber or tubular steel frame. No end of the timber plank or metal sheet shall remain unsupported.

A temporary safety platform or tier shall be maintained within two stories or 6 m, whichever is less, below and directly under the portion where erection of steel or precast concrete member is required. Tiers shall extend 2.5 m beyond the edge of the work area.

3.7 Concrete Work**3.7.1 General**

All workmen involved in concrete work shall be provided with helmet and hand gloves, especially when concrete pumps, concrete trucks or concrete precast elements are used. Precast piles shall be lifted and driven by skilled workmen under the supervision of a foreman.

Temporary fencing, either with bamboo or C.I. sheet, shall be erected around heavy equipment delineating the danger zone. All centering and shuttering materials shall be kept stacked at site before and after use.

3.7.2 Prestressed Concrete

Operating, maintenance and replacement instructions of the supplier of the prestressing equipment shall be strictly adhered to in all relevant operations. During the jacking of any tension element, the anchor shall be kept turned up close to anchor plate.

Thread on bolts and nuts shall be frequently checked for deterioration; choked units shall be cleaned. Hydraulic jacks/rams, pulling-headers and other temporary anchoring devices shall be inspected before use. The prestressing jacks shall be periodically examined for wear and tear.

No person shall stand in line with the tensioning elements and jacking equipment during the tensioning operation. Also no one shall be directly over the jacking equipment when deflection is done. Workmen shall be prevented from working behind the jacks when the tensioning operation is in progress by putting signs, barriers, or protective shields.

3.7.3 Concrete Mixers

All gears, chains and rollers of mixer plants shall be guarded. If the mixer has a charging skip, the operator shall ensure that the workmen are at safe distance before the skip is lowered. Barriers shall be provided to prevent walking under the skip while it is being lowered.

All cables, clamps, hooks, wire ropes, gears, clutches, etc. of the mixer shall be checked and serviced once a week. A trial run of the mixer shall be made and defects rectified before using a mixer.

While cleaning inside of the mixing drums, the power shall be shut and fuses removed.

3.7.4 Concrete Truck and Buckets

A reasonably smooth traffic surface shall be provided for concrete trucks. If possible, a loop road shall be provided to allow continuous operation. An easy turnout shall be provided if a loop is not possible to provide. Workmen and moving plants shall not cross the truck lines as far as practicable.

Concrete buckets conveyed by crane or overhead cableway shall be suspended from deep throated hooks, preferably equipped with swivel and safety latch. Closing and locking of the exit door of the concrete bucket shall always be checked.

3.8 Formwork and Scaffold

3.8.1 Scaffold and Centering Materials

Scaffolds shall be made from strong bamboo poles, wooden posts, steel pipes or any other suitable materials. They shall be adequately tied to vertical members resting on firm floor. Strong ropes shall be used to tie up bamboo poles. In addition, cross-bracing with bamboo or wooden posts shall be provided along with ties or guys of steel wire or rod not less than 6 mm in diameter.

Wooden planks or steel sheets shall be placed across horizontal poles to provide suitable footrest and carry construction materials. The whole assembly shall be securely lashed together.

Deterioration of tying ropes and rotting planks shall be checked from time to time during the construction period and changed if required.

Scaffold shall be dismantled after use piece by piece. Holes in the wall shall be filled up with the same materials as that of the wall. Filled up holes shall have uniformity in texture and color with the surrounding surface. Crash striking shall not be allowed.

Triangular wooden wedges shall be put under the posts for easy dismantling of the members.

Timber planks or steel sheets covering several posts at a time shall be placed below the vertical or inclined posts.

Horizontal and inclined bracings shall be provided for posts higher than 3 m. Spans of beam bottoms shall be supported by posts at most 1 m apart if steel is used; instructions from the manufacturer/supplier shall be strictly followed. Spacing of props under beams shall consider the increased load, and shall be posted closer than those under the floor slab.

All scaffolding exceeding 20 m or six stories in height shall be constructed of noncombustible or fire-retardant materials. Centering layout shall be planned by the Engineer, bearing capacity of the soil and the effect of weather shall be considered in the planning.

All nails and similar projecting objects shall be removed or hammered down into the timber component of the centering and shuttering materials immediately after stripping off.

3.8.2 Formwork for Concrete

The formwork shall be strong and rigidly braced so as not to bulge or sag when concrete is placed.

It shall be constructed in such a way that it can be dismantled without causing damage to the concrete or disturbing the centering and shuttering of other elements.

Forms shall not be removed until the concrete has developed sufficient strength to support all predicted loads. Workers removing formwork shall wear helmets, gloves, heavy soled safety shoes and belts if adequate footing is not available above 2 m. In case of removal of roof shuttering, staging has to provide below the roof. While cutting any tying wires in tension, care shall be taken against backlash.

Bolts and nuts in vertical concrete walls shall be loosened and withdrawn before initial setting of concrete. The resulting hole shall be filled with rich mortar. The supports shall be dismantled in the order instructed by the Engineer.

All walls, columns, slabs etc. shall have plastic or mortar spacers (round for vertical structures and flat for slab) to be placed with the reinforcement to provide clear cover as per design. Top layers of slab reinforcement shall be held in position by steel chairs.

The formwork shall be water-tight especially for the roof slab. Bamboo matting shall be placed on planks or steel sheets to provide a rough surface after stripping of the formwork. Alternatively, ceilings shall be roughened up by chiseling immediately after stripping off the formwork.

Suitable camber shall be provided in the formwork for horizontal members. The camber for beams and slabs shall be 1 in 250, and for cantilevers, 1 in 50 of the projected length.

Half-seasoned soft-wood, laminated board or other smooth sheet shall be used for formwork for a Fair-faced finish. The upper surface of the formwork shall be covered with oiled soft building board or veneered particle board. Oiled paper or polythene sheet shall never be used.

The formwork made of materials liable to absorb water shall always be sprinkled with water before laying concrete. Water shall not be profusely used; the formwork shall be in a saturated surface dry condition.

All the forms shall be tested both individually and in combination before final use to detect any flaw or defect. Measures shall be taken immediately to remedy any faults, if detected, before the formwork is ready for use.

The frame and its joints shall be checked from time to time for the decay in ropes, bamboos, planks etc. The defective parts shall be replaced before the formwork is used.

3.8.3 Load Capacity

Scaffolds, formwork and components thereof shall be capable of supporting without failure, at least two times the maximum intended load. The following information shall be considered in designing the formwork:

- (a) Weight of wet concrete: 20 kN/m³.
- (b) Live load due to workmen and impact of ramming or vibrating: 1.5-4.0 kPa (light duty for carpenter and stone setters, medium duty for bricklayers and plasterers, heavy duty for stone masons).
- (c) Allowable bending stress (flexural tensile stress) in soft timbers: 8,000 kPa.

The sizes for formwork elements specified in Table 7.3.1 are applicable for spans of up to 5 m and height of up to 4 m. In case of longer span and height, formwork and support sizes shall be determined by calculating the load and approved by the engineer before use.

All formworks and scaffolds shall be strong, substantial and stable. All centering and props shall be adequately braced to ensure lateral stability against all construction and incidental loads, especially in the case of floor height more than 3.3 m.

The space under the scaffold or formwork shall not be used as a working or living space. The space shall not be used as a shelter or refuge during inclement weather or at any other time.

3.8.4 Bamboos

Good, sound and uniform bamboo shall be collected in sufficient quantities for providing scaffolding, propping, temporary staging, ramp etc. The bamboos shall be free from any defects, firmly tied to each other and joints made smooth. Joining members only with nails shall be prohibited.

Bamboos for vertical support shall not be less than 75 mm in diameter, and shall be straight as far as possible. Bamboos shall be used as vertical support for up to a height of 4 m, if horizontal bracings are provided at the centre. Splicing shall be avoided.

After stripping the formwork, the bamboo posts shall be cleaned and stacked vertically in shade protected from rain and sun. Defective or damaged bamboo posts shall be removed from the site.

3.8.5 Timber Posts

Timber posts shall be used in supporting formwork up to a height of 6 m. The posts shall not be less than 80 mm in diameter at any place and shall spread to at least 150 mm in diameter at the top.

The timber posts shall be supported on timber planks at the bottom. Either the bottom or the top of the posts shall be wedged with a piece of triangular wood peg for easy removal. Adequate horizontal and inclined braces shall be used for all timber centering.

All timber posts shall be carefully inspected before use and members with cracks and excessive knots and crookedness shall be discarded. The joints shall normally be made with bolts and nuts. No rusted or spoilt threaded bolts and nuts shall be used.

Table 7.3.1 Sizes of Timber and other Sections for Formwork

Types of Formwork	Members Size in mm
Flat sheetings for slab bottoms, columns and beam side	25 - 50
Beam bottoms	75 × 100 - 150 × 150
Vertical posts	75 × 100 - 150 × 150
Bamboo posts	Minimum 75 dia
Ballies	Diameter not less than 100 at mid-length and 80 at thin end
Joist and ledgers supporting sheetings of slab	50 × 100 - 75 × 200
Studs for supporting vertical wall sheetings	50 × 100 - 150 × 150
Columns yokes-horizontal cross pieces supporting vertical sheetings	50 × 100 - 100 × 100

3.8.6 Steel Centering

Steel centering shall be used for any height. In case of patented material, the instructions of the manufacturer regarding the load carrying capacities shall be followed.

Post to post supports shall be provided with wooden planks. When tubular steel and timber centering is to be used in combination, necessary precautions shall be taken to avoid any unequal settlement.

Tubular steel centering shall be thoroughly inspected before erection. Defective members shall be discarded and coupling pins aligned to frames. Adjustment screws shall be set to their approximate final adjustment after assembling the basic unit, and the unit shall be level and plumb.

The centering frames shall be braced to make a rigid and solid unit. Struts and diagonal braces shall be in proper position and secured. As erection progresses, all connecting devices shall be in place, and fastened for full stability of joints and units.

3.9 Erection Operations

3.9.1 Erection and Hoisting

The erection and striking off, especially of steel structural frame, shall be done by skilled workers. Built-up, swinging and suspended scaffolds shall also be erected by competent workers.

Care shall be taken to keep fire alarms, hydrants, cable tunnels etc. unobstructed during the construction of scaffolding and placement of ladders etc.

Anchors for guys or ties shall be checked for proper placement. The weight of concrete in which the anchors are embedded shall be checked for uplift and sliding. In a tall and heavy guy derrick, tension in guys shall be controlled by hand winches.

Enough number of bolts shall be used in connecting each piece using a minimum of two bolts in a pattern to ensure that the joint will not fail. All splice connections in columns, crane girders etc. shall be completely bolted or riveted or welded before erection as specified in the drawings.

The top flange of a truss, girder or long beam shall be temporarily reinforced with a flat bar on top of the member. On deep girders and large trusses, a safety bar running their full length shall be provided. The bar can be a single 16 mm diameter wire rope through vertical stiffeners of each member about one meter above the bottom flange and clamped at the ends with wire rope clamps. If holes cannot be provided, short eye bolts can be welded to the webs of the girder at intervals. The bolts shall be removed, and the surface chipped to leave it smooth after the erection is completed.

The first load lifted by a guy derrick shall be hanged at a low height for 10 minutes and the anchor inspected for any signs or indications of failure. No load shall be allowed to rest on wire ropes.

Ropes in operation shall not be touched. Each truss or deep girder loaded in a vehicle shall be tied back or braced together with other trusses or girders already loaded.

The ropes shall be chemically treated to resist dew and rotting. They shall not be tied on sharp edges of steel structures. They shall not be tied beyond the reach of safety belts complying to BDS 1359.

The proper size, number and spacing of wire rope clamps, depending on the diameter of the wire rope, shall be used. They shall be properly fixed and checked as soon as the rope has been stretched, particularly if new. The clamps shall be promptly tightened when expansion in rope is detected. Clamps and ropes shall be inspected frequently to be sure that they are secured at place.

3.9.2 Small Articles

Adequate supply of bolts, washers, rivets, pins etc. of required sizes shall be maintained at all times. Foot boxes on a guy derrick or climbing crane, shall be moved to the new working floor each time the rig is changed. On a mobile crane, the boxes shall be moved as soon as the crane is moved.

Bolt baskets or similar containers with handles shall be provided on floats or scaffolds where small material, such as bolts and drift pins are used. Small tools shall be gathered up and put away in tool boxes when not in use. Rivet heaters shall have safe containers or buckets for unused hot rivets.

Materials shall not be dumped overboard when a scaffold is to be moved.

3.9.3 Hoist Protection

A material hoist shall not be used to transport workers; temporary elevators shall be installed, if necessary. Proper protection by way of railing, footboard etc. shall be provided to the hoists.

Railing shall have a minimum height of 1 m while the toe board shall be at least 200 mm high.

Where erected on the outside of a building over 20 m or six stories in height, the hoist structure shall be built of noncombustible or fire retardant materials. Interlocking or any other safety device shall be installed at all stopping points of the hoists. The hoists shaft way shall be fenced in accordance with Sec 3.6.5.

No part of scaffolding or walls and openings shall be hit by crane, truck or heavy moving equipment.

3.9.4 Lifting Gear

Lifting gears shall be of good construction, sound material and adequate strength. Lifting gears must be tested and examined by a competent person. Chains, ropes and lifting tackle shall be thoroughly examined by a competent person every 6 months.

Special devices like cleats and hooks shall be used in erecting girders and other heavy structural members. These shall be shop-assembled, bolted, riveted or welded to the piece and left permanently in place after the work. A balance beam shall be used to lift laterally imbalanced pieces. Alternatively, a pair of bridle slings shall be used at safe lifting points.

Table of safe working loads shall be posted in the tackle store and in prominent positions. No chain, rope or lifting tackle shall be used for loads exceeding the safe working load. Wrought iron gear shall be effectively heat treated.

All lifting gear shall be obtained from reliable manufacturers. No home-made or improvised gear shall be used.

3.9.5 Cranes

All parts of a crane must be of good construction, free from defects, and properly maintained. Before the crane is used for the first time, it must be thoroughly examined and tested by a competent person.

Crane rails shall be installed and secured on firm ground. In tower cranes, the level difference between the two rails shall remain within the limits prescribed by the manufacturer.

The safe working load shall be clearly shown on the crane; no crane shall be loaded beyond this limit. Nobody shall be allowed to work on the wheel tracks within 6 m of a crane, or under crane where he might be struck, unless effective steps are taken to warn him.

Electrical wires within the site which can possibly touch the crane or any member being lifted shall be removed or made dead. Cranes shall not be operated in proximity to a live overhead power line.

If it becomes necessary to operate the crane crossing the safe clearance from power line, the overhead power lines shall be shut off.

Cranes shall be thoroughly examined, at least once in 9 months and the results entered in a register. The crane operator shall not violate the safe reach limit of the crane as specified by the manufacturer. Cranes shall not be operated at a speed which causes the boom to swing.

No person shall be lifted or transported by the crane on its hook or boom. Toe boards and limit stops shall be provided for wheel barrows on the loading and unloading platforms. Material shall be loaded securely on the platform with no projection.

Every crane driver or hoisting machine operator shall be competent to the satisfaction of the engineer and no person under the age of 21 years shall be allowed to operate any hoisting machine and scaffolding winch, or give signals to the operator. The crane driver shall have the full knowledge of controls, signals, loading, misuse, ground and emergency regulations.

When the bucket or other members being lifted are out of sight of the crane operator, a signalman shall be posted in clear view of the loading and unloading areas, and the crane operator. Standard hand signals shall be used in controlling the movements of the crane; both the operator and the signalman shall be familiar with the signals.

The crane operator shall respond to signals only from the assigned signalman but shall obey stop signal at any time from anybody both inside and outside the site.

If a gantry crane is used, a warning bell which sounds automatically during the movement of the crane shall be given to avoid accidents to workmen crossing or standing in the path of the moving loads.

3.9.6 Slings

Idle and loaded slings shall not be carried together on the crane hook. In multi-legged slings, each leg shall be evenly loaded. The slings shall be of sufficient length to avoid wide angle between the legs.

Chains shall not be joined by bolting or wiring links together. Shortening the chains by tying knots shall be prohibited. The chain shall be made free of twists and kinks. Proper eye splices shall be used to attach the chain hooks.

Chains with locked or stretched links and which do not move freely shall not be used. Ropes shall move freely in the sheave grooves. Sharp bends in wire ropes shall be avoided; pulley shall be used for these.

3.9.7 Inspection

Materials and joints in scaffolding shall be inspected from time to time both before and after erection for the soundness, strength, damage due to weathering etc. Inspections shall be made for spillage of material or liquids, loose material lying on the gangways, and proper access to the platform.

The scaffold shall be secured to the building at enough places; no ties shall be removed. Warning sign prohibiting the use of any defective or incomplete scaffold and working in bad weather and high wind shall be posted in a prominent place. Inspections shall be made for the observance of these requirements.

3.10 Electrification, Equipment and Operations

3.10.1 Wiring System

All temporary and permanent wiring systems shall be designed by an engineer. All temporary wiring shall be done by an electrician holding relevant license.

No scaffolding, ladder, working platform, gangway, runway, etc. shall be placed within 3 m of an un-insulated live electric wire. Overhead wires/cables shall be so laid that clearances as required by Sec 2.2.3.4 are maintained.

Protection shall be provided for all electrical wiring laid on floor which shall have to be crossed over. All flexible wiring connecting the electrical appliances shall preferably be enclosed in a flexible metal sheath. Frayed and bare wires shall not be used for any temporary or permanent electrical connection.

All electrical circuits, other than those required for illuminating the site at night, shall be switched off daily at the end of the work. The main switch board shall be located in an easily accessible and prominent place. No clothing or stores shall be kept near it. One CO₂ extinguisher (3 to 4.5 kg), or one 5-kg dry powder extinguisher, shall be provided near the switch board.

3.10.2 Guarding of Cables

All cables and signal cords shall be guarded wherever such cables and cords pass through or cross working spaces. Location of underground cables, if any, as well as overhead cables, shall be identified and the scaffolds, hoists etc. shall be installed after providing proper guards to such cables.

Respective agencies shall be consulted for the proper method of providing protection to such cables, distance to be maintained to avoid all hazards etc. Cables, especially underground, and their routes shall be marked for future reference and use.

3.10.3 Lifts

Lifts shall be installed as per instruction of the manufacturer and under proper guidance. If necessary, guards shall be stationed at the installation site. Building materials shall preferably not be carried in a lift.

Entry to the empty lift well shall be blocked; the blockade shall be capable of withstanding bumping of an individual against it. Notices/signs shall be displayed in the lift lobby when the lift is not in operation.

3.10.4 Construction Machinery

Construction machinery shall conform to standards specified in the specification of works, or determined as required on site and approved by the engineer. They shall be in running condition without any defect.

The machinery shall be operated by competent operators only. The machinery will be checked thoroughly for any defect periodically, as well as each day before use.

Every moving part of or prime mover, and every part of electric generators, motors and rotary converters shall be securely fenced. Fencing shall be of substantial construction, maintained in efficient working order, and kept in position when the machine is in motion.

If machines need to be examined, oiled or adjusted while in motion, it shall be approached by certified mechanics only. Approach to unfenced machinery is allowed only when examination, lubrication etc. cannot be done with machinery at rest, or when machinery cannot be stopped without serious interference with the ongoing process.

Exhaust of petrol or diesel powered air compressors, hoists, derricks, pumps and all such machinery shall be well away from combustible materials. Exhausts opening outside the building shall have a minimum clearance of 200 mm from combustible materials. All sources of ignition like naked flame shall be banned near petroleum- fired equipment.

3.10.5 Heating of Bitumen and Tar

3.10.5.1 Bitumen and Tar Vessels

Tanks, vats, kettles, pots, drums and other vessels for heating tar, bitumen and other bituminous materials shall be made resistant to damage due to transportation, excessive heating etc. All such vessels shall be capable of holding a full load without danger of collapse, bursting or distortion. They shall be provided with a close-fitting cover suitable for smothering a fire in the vessel preventing spillage or protecting the bituminous material from rain.

Buckets for hot bitumen, bituminous material or tar shall have the bail or handle firmly secured, and a second handle near the bottom for tipping. Bitumen or tar boilers shall be mounted on wheels for easy transportation or towing, and provided with hand pumps for spraying purposes.

Heated vessels shall not be left unattended. Only vessels using electricity for heating shall be used inside buildings. Tar boilers shall never be used on a roof constructed of combustible materials.

Bituminous material shall not be thrown into the hot vessels. Vessels shall be kept closed when not in use. Containers shall not be filled to the brim with hot bitumen or tar. Enough space shall be left in vessels for expansion of heated binder.

The vessel shall be leak-proof, and provided with controllable outlets. The buckets and cans in which the hot material is carried shall be checked for any defect before use.

3.10.5.2 Heating of Bitumen and Tar

Gas and oil-fired bitumen and tar kettles or pots shall be equipped with burners, regulators, and safety devices. Heating appliances for vessels shall distribute the heat uniformly over the heating surface. If bituminous mixtures have mineral aggregate filler, some means for stirring shall be provided.

Vessels filled with bituminous materials shall be kept at a distance from combustible materials. When vessels are used in confined spaces, the gases, fumes and smoke generated shall be removed by exhaust or forced ventilation.

No naked light shall be used near heated boilers. If a burner stops burning, the fuel supply shall be cut-off immediately and the heating tube shall be thoroughly blown out by the fan.

Cutbacks shall not be heated over an open flame unless a water jacket is used. While they are being heated, the vessel shall be kept open. Blow-lamps or similar devices shall be used for warming pipes instead of burning rags. Bitumen and tar shall not be heated beyond the temperature recommended by the manufacturer of the product.

3.10.5.3 Other Precautions

Indicator gauges shall be used to ascertain level and temperature of the material in the boiler; nobody shall be allowed to peep into the boiler to ascertain the level. In small plants, dipstick shall be used to gauge the levels in the boiling pot.

Bitumen and tar shall be kept dry. Boiler shall either have a device that prevents foam from reaching the burners, or anti-foaming agents shall be used to control foaming. The heating shall be at low temperature till the water entrapped, if any, is completely evaporated. Any water present in the boiler shall also be drained out before using it.

Bitumen or tar spilled around boilers shall be promptly cleaned up. When tanks are cleaned by steam, building-up of pressure shall be prevented. No inspection shall be made while the boiler is under use, or is pressurized.

While discharging heated binder from the boiler, workers shall not stand opposite to the jet. The container shall be handled only after closing the valve. Bitumen and tar shall be handled in a way as not to spill.

Mops and other applicators covered with bituminous materials shall not be stored inside buildings.

3.10.6 Flame Cutting and Welding

For all arc welding work, either a helmet or a hand-held face shield conforming to BDS 1360 shall be used. See also Sec 3.2.1.

All welding and flame-cutting operations shall be performed in protected areas; closed spaces shall be properly ventilated. Suitable protection against the rays of the electric arc shall be provided where arc welding operations might be viewed within normal range by persons other than the welding operators and inspectors.

When working on aluminum structures, or close to other welders, protection for the back of the head shall be arranged. When slag is being removed from weld by clipping, the eyes shall be protected by goggles conforming to BDS 1360.

Leather gauntlet gloves with canvas or leather cuffs, shall be worn by welders. Any visible foam near the arc shall be rapidly dispersed. Where argon or carbon dioxide is being used as the shielding gas, particularly in confined spaces, breathing apparatus of the airline type shall be worn.

Gas cylinders shall be kept in the upright position, and conveyed in trolleys. While being carried by cranes, the gas cylinders shall be put in cages. The cylinder shall be marked 'full' or 'empty' as the case may be.

Gas cylinders shall be stored away from open flames and other sources of fire. Oxygen cylinders shall not be stored near oil, grease, sources of gas and similar combustible materials.

When the cylinders are in use, cylinder valve key or wrench shall be placed in position. Cylinder valve shall be closed before a cylinder is moved, when the torches are being replaced or welding is stopped for some reason. The cylinder valve and connection shall not be lubricated.

A 5 Kg CO₂ or Dry Chemical Powder (DCP) type fire extinguisher must be kept where gas cutting and welding works are done. Acetylene cylinder which has been subject to heat must be kept completely submerged in water at least for 12 hours before further use.

Gas cutting and welding torches shall be lighted by special lighters, not with matches. The cables from welding equipment shall not be run over by traffic. Double earthing shall be provided to the welding machines.

If welding is to be done near combustible materials, suitable blanket shall be provided and fire extinguishers kept nearby. Welding shall not be done in areas where flammable liquids and gases are stored.

Gas lines and compressed air lines shall be marked differently by suitable color codes. Facilities shall be provided in approved closed containers for housing the necessary vision, respiratory and protective equipment required in welding operations.

3.10.7 Riveting Operation

Rivets shall be carefully handled to prevent accidental fall; wooden bottom shall be provided in rivet catchers. Chains shall not be used in riveting dollies; leather, canvas or rope sling shall be used.

Snap and plunger shall be prevented from dropping out of place by securing the pneumatic riveting hammer. Nozzle of the hammer shall be inspected from time to time. Torn or worn wire attachment shall be renewed. Water shall be kept ready for putting out fire during riveting operations.

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3.11 Construction Hazards

3.11.1 General

Implements used for carrying materials to the top of scaffolding shall be of adequate strength and shall not be overloaded during the work. Overhead protection against falling materials shall be provided under scaffoldings and ladders. Care shall be taken in carrying long and heavy bars, rods, angles and other such materials.

Precautions shall be taken to correctly handle, use and position precast RC columns, piles, steel beams, joists, angles and other heavy elements. Temporary supports with guys and props shall be provided in handling heavy elements till the member is properly and permanently secured in position. Manila or Sisal rope shall not be used in rainy season for hoisting heavy materials.

People suffering from asthma, chronic bronchitis, pulmonary fibrosis, or pneumoconiosis shall be screened out from being employed in works involving the use of paints, varnishes, plastic foam, rubber, adhesives, etc. Those having impaired lung function, hay fever, eczema, dermatitis etc. shall also be advised to avoid such work.

All construction sites shall have sufficient general and local ventilation unless otherwise required.

Adequate number of Absorptive respirators shall be provided to sites with inhalation hazard. Full breath apparatus shall be used for works of limited period in dangerous situations.

The workers shall be made aware of personal hygiene. Regular health checkup shall be arranged for works requiring high physical fitness for prolonged period.

3.11.2 Fire Hazards

Gangways and the ground below the scaffolding shall be kept free from readily combustible materials including waste, debris and any vegetation at all times.

Scaffoldings, gangways, etc. shall be constructed with fire resistant materials when blow torch or other equipment producing flame is extensively used near it. A portable dry powder extinguisher of 3 kg capacity shall be kept near all flame producing equipment. Sections 2.1.3, 2.2.4 and 2.2.5.3 of Chapter 2 of this Part shall also be followed in addition to the following requirements.

3.11.2.1 Fire Protection

Fire extinguishers, preferably of water type, shall be placed at strategic points.

Extinguishers shall always be placed in cranes, hoists, compressors and similar places. Where electrical equipment is used, CO₂ or dry powder extinguishers shall be provided.

In addition to fire extinguishers, other fire extinguishing equipment, e.g. sprinklers and hydrants shall also be provided and conveniently located both within the building under construction and at the building site. All extinguishers shall be maintained in a usable condition at all times in accordance to the instructions of the manufacturer.

All workmen and supervisory staff shall be clearly briefed on the use of fire extinguishers provided at the construction site. Free access shall be provided and maintained at all times to all firefighting equipment including fire hose, extinguishers, sprinkler valves and hydrants.

Where the project itself requires the installation of fixed firefighting equipment, such as hydrants, stand pipes, sprinklers and underground water mains or other suitable arrangements for the provision of water, it shall be installed and made available for permanent use as soon as possible, in no case later than the scheduled time.

A permanent hydrant system shall be made available before the building has reached the height of 20 m. This shall be extended with every increase in the number of floors, and securely capped at the top. Top hose outlets shall be at all times not more than one floor below the floor under construction. All construction sites with a fire risk shall have at least two exits.

Temporary stand pipes with required pumps shall be provided in place of permanent systems if they are designed to furnish 400 liters of water per minute at 450 kPa pressure with a standpipe size of not less than 100 mm.

A metal box of substantial size preferably to be kept open, shall be provided and maintained near each hose outlet. It shall contain adequate length of hose fitted with 12 or 20 mm nozzle to reach all parts of the floor.

Free access from the street to such stand pipe shall be maintained at all times. Materials shall not be stored within 1.5 m of any fire hydrant or in the roadway between such hydrant and the centre line of the street.

Contact shall be established and maintained with the local fire authority during construction of all buildings above 20 m in height and buildings of special occupancies like educational, assembly, institutional, industrial, storage, hazardous and mixed occupancies having areas in excess of 500 m² on each floor.

Telephone or other means of inter-communication system within the site shall be provided during the construction of all buildings over 20 m in height or buildings with a plinth area in excess of 1000 m².

All waste, such as scrap timber, wood shavings, sawdust, paper, packing materials and oily substance, particularly in or near vertical shaft openings like stairways, lift shaft etc. shall be collected and disposed of safely at the end of each day's work.

An independent water storage facility shall be provided before the commencement of construction operations for fire-fighting purposes. The tank shall be kept filled up at all times. Sec 2.2.5 shall also be followed.

3.11.2.2 Flammable Materials and Explosives

Highly flammable materials, such as gasoline, oil, paints etc. shall be stored in approved containers. Storage of large quantities shall not be allowed unless stored in separate compartments or enclosures of noncombustible construction.

Where cellulose or other highly flammable paint is sprayed, flame-proof exhaust ventilation equipment shall be provided. Smoking shall be strictly controlled where highly flammable liquids are used.

Explosives like detonators, gunpowder etc. shall be stored in conformity with relevant regulations for storage and handling of explosives. Combustible materials shall not be stored on any floor under construction until all combustible form works are removed from the tier immediately above.

3.11.2.3 Temporary Heating

When temporary heating is used, all regulations as to the maximum temperature, distance from combustible materials, spark arrestors, removal of noxious gases and other similar requirements shall be fully observed. Temporary enclosure shall be provided where the source of temporary heat includes open-flame devices.

3.11.2.4 Steam Boiler

All temporary or permanent high pressure steam boilers shall be operated only by licensed operators. Where located within a building or within 3 meters of combustible materials or electric power lines, all such boilers shall be enclosed with approved noncombustible covers. Safety valves shall be adjusted to exactly 70 kPa in excess of working pressure. Two Dry Chemical Powder (DCP) type fire extinguishers of 5 kg capacity each shall be kept at easily accessible locations.

3.11.2.5 House Keeping

Rubbish, trash, nuts, bolts and small tools shall not be allowed to accumulate on the site and shall be removed as soon as conditions warrant. Combustible rubbish shall be removed daily. Rubbish shall not be burnt on the premises or in the immediate vicinity. The entire premises and area adjoining and around the construction site shall be kept in a safe and sanitary condition.

3.11.2.6 Fire Exits

All construction sites with a fire-risk shall have at least two clearly marked fire exits.

Other means of escape as required by various sections of this Code shall be provided in a construction site. Fire exits shall be easily operable; stores, packing materials or rubbish shall not obstruct the exit.

Fire walls and exit stairways required for a building shall be given priorities in construction schedule. Where fire doors, with or without automatic closing devices, are to be set in the building, they shall be hung as soon as practicable, and before fire risk is increased by way of greater use of combustible material.

3.11.3 Health Hazards

3.11.3.1 Emission

Precautionary measures shall be taken against the emission of dust, small particles, toxic gases and other harmful substances in quantities hazardous to health. Such measures shall include local ventilation, use of protective devices, medical check-up etc. Exhaust ventilation shall be employed in enclosed spaces.

3.11.3.2 Clothing

Clothes worn by the workmen shall not be of such nature and materials as to increase the chances of inflicting injuries to themselves or others. Wearing of loose garments shall be strictly avoided.

Workmen using naked flames (such as in welding) shall not wear clothing of synthetic fibre or similar materials which increases the risk of fire hazards.

3.11.3.3 Removal of Dust

Spread of dust, sand blasts and other harmful materials and chemical agents shall be controlled at or near the source to prevent overspill to adjoining premises or streets.

Proper gear and protection as required by regulations shall be provided to the workmen.

Proper methods of handling and transportation shall be followed. Places prone to generate dust shall be frequently cleaned. Machinery and plants shall be designed for easy cleaning.

3.11.3.4 First Aid and Ambulance

A copy of all pertinent regulations and notices concerning accidents, injury and first aid shall be prominently displayed at the work site.

A first aid box or cupboard shall be provided for every 150 workmen and be accessible. The provision shall also include a stretcher and cot with accessories for every 300 workmen.

In case of a site where more than 600 workmen are employed at any one time, or in which more than 300 workmen are employed at any one time and is 15 km from the nearest health service facility, provision of an ambulance shall be made.

3.11.4 Skin Hazard

Workmen engaged in works which may splash liquid or other materials liable to injure the skin shall have enough protective clothing to cover the body and limbs.

Whenever epoxy resins are mixed indoor, the place shall be adequately ventilated. Damaged protective gears shall not be used, and shall be replaced. Containers of hazardous chemicals shall be kept in a clearly marked-off area of the work space.

Spillage on and contamination of tools, equipment, or the outside of the containers shall be avoided. If spillage or contamination occurs, the affected area shall be cleaned up immediately.

Contaminated skin/part of the body shall be washed immediately with warm soapy water. Proper barrier creams shall be used. All contamination on part of the body shall be regularly and efficiently removed during breaks and after finishing time.

3.11.5 Noise Hazard

Noise shall be controlled, if possible, by soundproof shields, baffles or absorbent lined booths being fitted near or around the source. Other general methods of control shall include silencing of machine exhaust, choice of quiet machines etc.

Protective measures shall be taken if the continuous noise level at the construction site exceeds 90 dB. For levels up to 110 dB, properly fitted ear plugs of plastic, rubber or glass wool shall be provided. For levels up to 120 dB, ear muffs shall be used; for levels exceeding 120 dB, noise protection helmets shall be provided.

All noise control equipment shall be regularly inspected and maintained by adequately trained personnel. Care shall be taken to prevent noise becoming a nuisance to neighbouring property.

Other precautions as specified in Chapter 3 of Part 8 shall also apply.

3.12 Additional Safety Requirements for Erection of Concrete Framed Structures (High Rise Buildings)

Workmen working in any position where there is a falling hazard shall wear safety belts or other adequate protection shall be provided.

3.12.1 Handling of Plant

3.12.1.1 Mixers

All gears, chains and rollers of mixers shall be properly guarded. If the mixer has a charging skip the operator shall ensure that the workmen are out of danger before the skip is lowered. Railings shall be provided on the ground to prevent anyone walking under the skip while it is being lowered.

All cables, clamps, hooks, wire ropes, gears and clutches, etc. of the mixer, shall be checked and cleaned, oiled and greased, and serviced once a week. A trial run of the mixer shall be made and defects shall be removed before operating a mixer.

When workmen are cleaning the inside of the drums, operating power of the mixer shall be locked in the off position and all fuses shall be removed and a suitable notice hung at the place.

3.12.1.2 Cranes

See Sec 3.9.5 of this Chapter.

3.12.1.3 Trucks

When trucks are being used on the site, traffic problems shall be taken care of. A reasonably smooth traffic surface shall be provided. If practicable, a loop road shall be provided to permit continuous operation of vehicles and to eliminate their backing. If a continuous loop is not possible, a turnout shall be provided. Backing operations shall be controlled by a signalman positioned so as to have a clear view of the area behind the truck and to be clearly visible to the truck driver. Movement of workmen and plant shall be routed to avoid crossing, as much as possible, the truck lanes.

3.12.2 Formwork

The Formwork shall conform to the shape, lines and dimensions as shown on the plans, and be so constructed as to remain sufficiently rigid during the placing and compacting of the concrete, and shall be sufficiently tight to prevent loss of liquid from the concrete.

Formwork shall be designed after taking into consideration spans, setting temperature of concrete, dead load and working load to be supported and safety factor for the materials used for formwork.

All timber formwork shall be carefully inspected before use and members having cracks and excessive knots shall be discarded.

As timber centering usually takes an initial set when vertical load is applied, the design of this centering shall make allowance for this factor.

The vertical supports shall be adequately braced or otherwise secured in position that these do not fall when the load gets released or the supports are accidentally hit.

Tubular steel centering shall be used in accordance with the manufacturer's instructions. When tubular steel and timber centering is to be used in combination necessary precautions shall be taken to avoid any unequal settlement under load.

A thorough inspection of tubular steel centering is necessary before its erection and members showing evidence of excessive resting, kinks, dents or damaged welds shall be discarded. Buckled or broken members shall be replaced. Care shall also be taken that locking devices are in good working order and that coupling pins are effectively aligned to frames.

After assembling the basic unit, adjustment screws shall be set to their approximate final adjustment and the unit shall be level and plumb so that when additional frames are installed the tower shall be in level and plumb. The centering frames shall be tied together with sufficient braces to make a rigid and solid unit. It shall be ensured that struts and diagonals braces are in proper position and are secured so that frames develop full load carrying capacity. As erection progresses, all connecting devices shall be in place and shall be fastened for full stability of joints and units.

In case of timber posts, vertical joints shall be properly designed. The connections shall normally be with bolts and nuts. Use of rusted or spoiled threaded bolts and nuts shall be avoided.

Unless the timber centering is supported by a manufacturer's certificate about the loads it can stand, centering shall be designed by a competent engineer.

Centering layout shall be made by a qualified engineer and shall be strictly followed. The bearing capacity of the soil shall be kept in view for every centering job. The effect of weather conditions shall be considered as dry clay may become very plastic after a rainfall and show marked decrease in its bearing capacity.

Sills under the supports shall be set on firm soil or other suitable material in a pattern which assures adequate stability for all props. Care shall be taken not to disturb the soil under the supports.

Adequate drainage shall be provided to drain away water coming due to rains, washing of forms or during the curing of the concrete to avoid softening of the supporting soil strata.

All centering shall be finally, inspected to ensure that :

- (a) Footings or sills under every post of the centering are sound.
- (b) All lower adjustment screws or wedges are snug against the legs of the panels.
- (c) All upper adjustment screws or heads of jacks are in full contact with the formwork.
- (d) Panels are plumb in both directions.
- (e) All cross braces are in place and locking devices are in closed and secure position.
- (f) In case of balconies, the props shall be adequate to transfer the load to the supporting point.

During pouring of the concrete, the centering shall be constantly inspected and strengthened, if required, wedges below the vertical supports tightened and adjustment screws properly adjusted as necessary. Adequate protection of centering shall be secured from moving vehicles or swinging loads.

Forms shall not be removed earlier than as laid down in the specifications and until it is certain that the concrete has developed sufficient strength to support itself and all loads that will be imposed on it. Only workmen actually engaged in removing the formwork shall be allowed in the area during these operations. Those engaged in removing the formwork shall wear helmets, gloves and heavy soled shoes and approved safety belts if adequate footing is not provided above 2 m level. While cutting any tying wires in tension, care shall be taken to prevent backlash which might hit a workman.

The particular order in which the supports are to be dismantled shall be followed according to the instructions of the site engineer.

3.12.3 Ramps and Gangways

Ramps and gangways shall be of adequate strength and evenly supported. They shall either have a sufficiently flat slope or shall have cleats fixed to the surface to prevent slipping of workmen.

Ramps and gangways shall be kept free from grease, mud, snow or other slipping hazards or, other obstructions leading to tripping and accidental fall of a workman.

Ramps and gangways meant for transporting materials shall have even surface and be of sufficient width and provided with skirt boards on open sides.

3.12.4 Materials Hoists

The hoist shall be erected on a firm base, adequately supported and secured. All materials supporting the hoist shall be appropriately designed and strong enough for the work intended and free from defects.

The size of the drum shall match the size of the rope. Not less than two full turns of rope shall remain on the drum at all times. Ropes shall be securely attached to the drum.

All ropes, chains and other lifting gear shall be properly made of sound materials, free from defects and strong enough for the work intended. They shall be examined by a competent person who shall clearly certify the safe working load on each item and the system.

Hoist ways shall be protected by a substantial enclosure at ground level, at all access points and wherever persons may be struck by any moving part.

Gates at access points shall be at least 2 m high wherever possible. Gates shall be kept closed at all times except when required open for immediate movement of materials at that landing place.

All gates shall be fitted with electronic or mechanical interlocks to prevent movement of the hoist in the event of a gate being opened.

Winches used for hoists shall be so constructed that a brake is applied when the control lever or switch is not held in the operating position (dead-man's handle).

The hoist tower shall be tied to a building or structure at every floor level or at least every 3 m. The height of the tower shall not exceed 6 m after the last tie or a lesser height as recommended by the manufacturer. All ties on a hoist tower shall be secured using right angled couples.

The hoist shall be capable of being operated only from one position at a time. It shall not be operated from the cage. The operator shall have a clear view of all levels or, if he has not, a clear and distinct system of signaling shall be employed.

All hoist platforms shall be fitted with guards and gates to a height of at least 1 m, to prevent materials rolling/falling from the platform.

Where materials extend over the height of the platform guards, a frame shall be fitted and the materials secured to it during hoisting/lowering. (Care shall be taken to ensure that neither the frame nor materials interfere or touch any part of the hoisting mechanism.)

The platform of a goods hoist shall carry a notice stating :

- (a) The safe working load; and
- (b) That passengers shall not ride on the hoist.

All hoist operators shall be adequately trained and competent, and shall be responsible for ensuring that the hoist is not overloaded or otherwise misused.

All hoists shall be tested and thoroughly examined by a competent person before use on a site, after substantial alteration, modification or repair of hoists, and at least every 6 months.

Every hoist shall be inspected at least once each week by a competent person and a record of these inspections kept.

3.12.5 Prestressed Concrete

In pre-stressing operations, operating, maintenance and replacement instructions of the supplier of the equipment shall be strictly adhered to.

Extreme caution shall be exercised in all operations involving the use of stressing equipment as wires/strands under high tensile stresses become a lethal weapon.

During the jacking operation of any tensioning element(s) the anchor shall be kept turned up close to anchor plate, wherever possible, to avoid serious damage if a hydraulic line fails.

Pulling-headers, bolts and hydraulic jacks/rams shall be inspected for signs of deformation and failure. Threads on bolts and nuts shall be frequently inspected for diminishing cross section.

Choked units shall be carefully cleaned.

Care shall be taken that no one stands in line with the tensioning elements and jacking equipment during the tensioning operations and that no one is directly over the jacking equipment when deflection is being done. Signs and barriers shall be provided to prevent workmen from working behind the jacks when the stressing operation is in progress.

Necessary shields shall be put up immediately behind the prestressing jacks during stressing operations.

Wedges and other temporary anchoring devices shall be inspected before use.

The pre-stressing jacks shall be periodically examined for wear and tear.

3.12.6 Erection of Prefabricated Members

A spreader beam shall be used wherever possible so that the cable can be as perpendicular to the members being lifted as practical. The angle between the cable and the members to be lifted shall not be less than 60°.

The lifting wires shall be tested for double the load to be handled at least once in six months. The guy line shall be of adequate strength to perform its function of controlling the movement of members being lifted,

Temporary scaffolding of adequate strength shall be used to support precast members at predetermined supporting points while lifting and placing them in position and connecting them to other members.

After erection of the member, it shall be guyed and braced to prevent it from being tipped or dislodged by accidental impact when setting the next member.

Precast concrete units shall be handled at specific picking points and with specific devices. Girders and beams shall be braced during transportation and handled in such a way as to keep the members upright.

Methods of assembly and erection specified by the designer shall be strictly adhered to at site.

Immediately on erecting any unit in position, temporary connections or supports as specified shall be provided before releasing the lifting equipment. The permanent structural connections shall be established at the earliest opportunity.

3.12.7 Heated Concrete

When heaters are being used to heat aggregates and other materials and to maintain proper curing temperatures, the heaters shall be frequently checked for functioning and precautions shall be taken to avoid hazards in using coal, liquid, gas or any other fuel.

3.12.8 Structural Connections

When reliance is placed on bond between precast and in-situ concrete the contact surface of the precast units shall be suitably prepared in accordance with the specifications.

The packing of joints shall be carried out in accordance with the assembly instructions.

Leveling devices, such as wedges and nuts which have no load bearing function in the completed structure shall be released or removed as necessary prior to integrating the joints.

If it becomes necessary to use electric power for in-situ work, the same shall be stepped down to a safe level as far as possible.

3.13 Miscellaneous

3.13.1 Stair, Ramp and Gangway

Buildings higher than two stories shall have at least one stair in usable condition at all times. This shall be extended upward with each completed floor. Till the permanent handrails are provided, temporary provisions like ropes, bamboo poles etc. shall be provided on stair.

Suitable precautions by way of support, formworks, etc. shall be taken to prevent any collapse of the stair during its construction. No person shall be allowed to use such stair until they are tested by the engineer and found fit for usage.

Where a building has been constructed to a height greater than 14 m or four stories, or where an existing building higher than 14 m is altered, at least one temporary lighted stairway shall be provided unless one or more of the permanent stairways are erected as the construction progresses.

Ramps and gangways shall be of adequate strength and evenly supported. They shall either have a sufficiently flat slope (maximum 15° to horizontal), or shall have cleats fixed to the surface. They shall be kept free from slipping hazards and obstructions.

Ramps for transporting materials shall have even surfaces, be of sufficient width and provided with 200 mm high toe boards on open sides.

Requirements as set in Sec 3.11.2 of this Chapter shall also be observed.

3.13.2 Fragile Fixture

It shall be ensured that sufficient number of workmen and equipment are provided to carry the fragile fixtures in the site like sanitary fittings, glass sheets, etc. Fragile fixtures shall be stored in a safe place away from the normal circulation path of people, equipment and vehicle (see Sec 2.2 Chapter 2 of this Part for additional requirements of safe handling of fragile fixtures and materials).

3.13.3 Hand Tools

Correct tools in good condition shall be used for each type of job. All tools, particularly at heights, shall be stowed. Wooden handles shall be made of good quality straight-grained materials. Hand tools shall be issued through a tool room where they are stored safely and inspected periodically by competent people.

Hammer head shall be securely attached to the shaft. The head shall be in good condition and the face free from chipped edges and not rounded from wear. The hammer shall not be used if the shaft is split, broken or loose.

Set spanners with splayed jaws, or box spanners showing signs of splitting shall not be used. A fixed spanner of correct size shall be preferred over an adjustable spanner. A tube shall not be used to obtain extra leverage; end of a spanner shall never be hammered. A spanner shall not be used as a hammer, nor as a wedge.

A chisel with a mushroom head shall never be used. A chisel shall be used to cut in a direction away from the body. Screwdriver handle shall be properly secured. A screwdriver shall never be used as a chisel.

Use of files with an exposed tang shall be avoided. Files shall not be used as levers or toggle-bar.

When a knife is used to cut greasy materials, the handle shall be such that it offers a firm grip and a shield shall be fitted between the handle and the blade. The cut shall always be made away from the body.

3.13.4 Steel Structure

Riding on trusses while hauling them to their final position, shall not be allowed. The hauling ropes shall be load tested before use.

Once in position, the trusses shall be kept secured with adequate temporary measures till the final fixing is carried out. Standard safety belts conforming to BDS 1359 shall be used while fixing purlins on the trusses.

In steel construction, the entire tier of iron or steel beams shall be planked over, with the exception of necessary hoist ways and permanent openings. Steelwork shall not advance more than six floors ahead of the permanent floor construction.

The proposed erection scheme of a steel work shall be analyzed and checked for safety measures undertaken; the scheme shall cover safety aspects at all stages.

3.13.5 Finish Works

3.13.5.1 Painting

The quantity of paint and thinner required only for the day's work shall be issued from the store. All unused containers of paint and thinner shall be closed with tight-fitting lids, and kept at a safe place away from the work site.

Metal receptacles with pedal operated metal lids shall be kept at the work site for depositing used cotton rags and waste. The contents of such receptacles shall be disposed of daily at a safe place, preferably by burning under proper supervision.

All containers of paint shall be deposited in the paint store after use. Used paint brushes shall be cleaned and deposited in the store. A 5 kg dry powder fire extinguisher shall be kept near the paint store (see Sec 2.2.16 Chapter 2 of this Part).

Adequate ventilation to prevent the accumulation of flammable vapour to hazardous level of concentration shall be provided in all areas where painting is done. When painting is done in confined spaces where flammable or explosive vapour may develop, required heat and power shall only be provided through covered ducts remote from the likely source of flame.

Sources of ignition, such as open flame and exposed heating elements, shall not be permitted in areas or rooms where spray painting is done, nor shall smoking be allowed there.

3.13.5.2 Polishing

Extra care shall be taken while handling polish consisting of acid and other chemical ingredients. Only the quantity of polish required for the day's work shall be kept at the work spot.

All containers of polish shall be kept closed with tight fitting lids in a safe place.

Protective clothing, gloves, respiratory equipment, etc. shall be provided to the workmen applying polishes. See also Sec 2.2.4 Chapter 2 of this Part.

3.13.5.3 Pavements

Pavement risers shall not be higher than 225 mm. All undulating surfaces shall be smoothed. At least a 1 m x 0.5 m area of the pavement adjacent to a vehicular road crossing shall have a checkered surface preferably of a texture and colour different from those of the surrounding surface.

3.13.5.4 Terracing

Protective clothing, gloves and shoes shall be used in terracing work, especially while handling lime and other ingredients. Lime and mortar stuck on the body shall be thoroughly cleaned. Other requirements for handling lime are specified in Sec 2.2.1.2 Chapter 2 of this Part.

PART VII
Chapter 4
Demolition Work

4.1 Preliminary Procedure

4.1.1 General

The safety provisions specified in this chapter shall apply to demolition and dismantling of all types of buildings and structures in addition to the safety requirements mentioned in Chapter 3.

4.1.2 Planning

Before commencing the demolition work, a detailed survey and study shall be made of the structure to be demolished and the structures in its surroundings. This shall include the manner in which the various parts of the building to be demolished are supported and how far the demolition will affect the safety of the surrounding structures. Planning for demolition and safety of adjoining structures shall be made accordingly.

The sequence of operations shall be planned by an Engineer-in-charge recognized by the Authority as having experience in demolition work of similar magnitude. No deviation from the approved plan shall be permitted without the approval of the Engineer-in-charge. Before the commencement of each stage of demolition, the foreman shall brief the workmen in detail regarding the safety aspects to be kept in view.

Demolition of buildings and structures shall be carried out under supervision of qualified Engineer and with prior notification to the Authority as prescribed by the latter.

The Authority may require the permittee to submit the plans and a schedule of demolition. Neighbors and public shall be notified of the intended demolition through newspaper or other media. The extent, duration and time of the demolition shall be clearly specified in the notice.

4.1.3 Protection of Adjoining Property

A written notice shall be delivered to the owner of each potentially affected plot, building or structure at least a week in advance of the commencement of work. The notice shall request written permission to enter the plot, building or structure prior to the commencement of work and as and when required during the work to inspect and preserve them from damage.

Owner of the structure to be demolished or dismantled shall under all circumstances preserve and protect the adjoining lot, building or structure from damage or injury. This shall be done at his own expense.

In case damage to the adjoining property is imminent, the demolition operation shall be stopped forthwith and shall not be restarted until the necessary measures to prevent such damage have been taken. All waste materials and debris from the demolition shall be removed immediately.

If the owner of the property to be demolished is denied entry to an adjoining structure, he shall immediately notify the Authority in writing of such denial. In this situation, the Authority may hold the adjoining property owner fully responsible for any damage to his property.

4.1.4 Precautions prior to Demolition

Demolition of any building shall not commence until the required pedestrian protection structures in accordance with Sec 4.1.5 of this Chapter have been built. Building or structure damaged by fire, flood, explosion or earthquake, shall be protected from collapse by way of bracing, shoring etc. before demolition is commenced.

Permission shall be secured from the Authority for using explosives. General public and owners of the adjoining properties shall be notified beforehand of such use. All precautions as required by Sec 2.2.5.3 Chapter 2 and Sec 4.3 Chapter 4 of this Part shall have to be ensured before, during and after the use of the explosives.

Danger signs shall be posted round the property; this shall conform to the relevant sections of Part 10. All entrances shall be barricaded or manned. At least two independent exits shall be provided at night; warning lights shall be placed above all barricades during the night and dark hours. Even when work is not in progress, watchmen shall be provided to prevent unauthorized entry of the public in the danger zone.

4.1.5 Protection of Public

Safe distances shall be clearly marked and prominent signs posted. Every sidewalk and road adjacent to the site shall be either closed or protected. All public roads shall be kept open and unobstructed at all times unless unavoidable circumstances arise.

If a covered walk is not necessary in the opinion of the Engineer-in-charge he shall issue a permit to block off part of the sidewalk and have a temporary walk provided. Pedestrians shall be provided with diversion roads or alternate protection as specified in Sec 1.7 Chapter 1 and Table 7.4.1.

All utility lines shall be disconnected upon the approval of the concerned Authorities. Temporary service connection for the demolition work shall be taken separately. See Sections 3.3.9 and 3.3.12 of Chapter 3 for other requirements.

Workmen shall be provided with all necessary safety appliances as specified in the following sections and in Chapter 3 prior to the start of work. Safety precautions for fire shall be provided.

The site shall be thoroughly cleaned of combustible materials and debris before commencement of demolition.

4.1.6 Sidewalk Shed and Canopies

A toe board at least 1 m high above the roof of the shed shall be provided on the outside edge and ends of the sidewalk shed. Such boards may be vertical or inclined outward at no more than 45° angle with the vertical. The side of the shed adjacent to the building shall be completely blocked by planking/sheeting.

The roof of sidewalk sheds shall be capable of sustaining a load of 7 kPa. Impact of falling debris shall be considered in designing and constructing the shed. Maximum load on the roof of the shed shall be maintained below 12 kPa.

The flooring of the sidewalk shed shall consist of closely laid planks with a minimum thickness of 50 mm made watertight. Only in exceptional cases, temporary storage on the sidewalk shed may be permitted; in such situation, the roof of the shed shall be designed for sustaining 14 kPa.

Entrances to the building shall be protected by canopies extending at least 2.5 m from the building facade. Such overhead protection shall be at least 600 mm wider than the entrance, and 2.5 m in height.

4.2 Precautions During Demolition

4.2.1 General

The owner shall provide protection against all damages or loss of life and property during demolition. Constant supervision shall be provided during a demolition work by a competent and experienced engineer.

The demolition site shall be provided with sufficient natural and artificial lighting and ventilation.

All existing features required during demolition operations shall be well protected with substantial covering to the entire satisfaction of the rules and regulations of the undertakings or they shall be temporarily relocated.

For a building or structure more than 8 m or two stories high, all windows and exterior wall openings that are within 6 m of floor opening used for the passage of debris from floors above, shall be solidly boarded. Openings in floors below the level of demolition, not used for removal of materials or debris, shall be barricaded or covered by planks.

4.2.2 Sequence of Demolition Operation

The demolition shall proceed in descending order and storey by storey. All work in the upper floor shall be completed and approved by the engineer prior to disturbing any supporting member on the lower floor. Demolition of the structure in sections may be permitted in exceptional cases only if necessary precautions are ensured. The demolition work shall proceed within such a way that:

it causes the least damage and nuisance to the adjoining building and the members of the public, and it satisfies all safety requirements to avoid any accidents.

Table 7.4.1: Type of Protection Required for Pedestrians near a Demolition Site

Horizontal Distance from inside of the Sidewalk to the Structure	Height* to Horizontal Distance Ratio	Type of Minimum Protection Required
Less than 3 m	6:1 or more	Type A
	4:1 - 6:1	Type B
	3:1 - 4:1	Type C
	2:1 - 3:1	Type D
	up to 2:1	Type E
3 m - 4.5 m	10:1 or more	Type A
	6:1 - 10:1	Type B
	4:1 - 6:1	Type C
	3:1 - 4:1	Type D
	up to 3:1	Type E
4.5 m - 7.5 m	15:1 or more	Type A
	10:1 - 15:1	Type B
	6:1 - 10:1	Type C
	4:1 - 6:1	Type D
	up to 4:1	Type E
7.5 m - 12 m	15:1 or more	Type B
	10:1 - 15:1	Type C
	6:1 - 10:1	Type D
	up to 6:1	Type E
12 m and more	10:1 or more	Type D
	up to 10:1	Type E

* Height of the building or portion thereof to be demolished

Type A: Total blockade of the road.

Type B: Temporary diversion over the entire length of the footpath adjacent to the structure.

Type C: A sidewalk shed for the entire length, in accordance with Sec 4.1.6.

Type D: A fence of tightly seated 25 mm planks, minimum height 2.5 m.

Type E: A railing at least 1.5 m high with mid rail and cross bracing.

4.2.3 Wall

Walls shall be removed part by part in reasonably level courses. No wall or any part of the structure shall be left in a condition that may collapse or be toppled by wind, vibration etc.

Fall of the demolished wall in large chunks, which endangers the adjoining property or exceeds the safe load capacity of the floor below, shall be avoided. Debris shall be removed at frequent intervals to avoid piling up and overloading of any structural member.

Platforms shall be provided for demolition of walls less than one and half brick thick. Lateral bracing shall be provided for sections of walls having a height more than 22 times its thickness, or otherwise considered unsound. No workman shall stand on any wall to remove materials; staging or scaffold shall be provided at a maximum of 3.5 m below the top of the wall.

At the end of each day's work, all walls shall be left stable to avoid any danger of getting overturned. Foundation walls which serve as retaining walls shall not be demolished until the adjoining structure has been underpinned or braced and the earth removed.

4.2.4 Floor

Support/centering shall be provided prior to removal of masonry or concrete floor. Planks of sufficient strength shall be used in shuttering. No person shall be allowed to work in an area underneath a floor being removed; such areas shall be barricaded.

The total area of a hole cut in any intermediate floor for dropping debris shall not exceed 25% of that floors' area. No barricades or rails for guarding the floor hole shall be removed until the storey immediately above has been demolished down to the floor line and all debris cleared from the floor.

In cutting holes in a floor which spans in one direction, at first, a maximum 300 mm wide slit shall be cut along the entire length of the slab; the slit shall be increased gradually thereafter.

Planks of sufficient width, not less than 50 mm thick, 250 mm wide and 2 m long shall be provided at spacing not greater than 400 mm for the workmen to work. These shall be so spaced as to firmly support the workmen against any floor collapse.

4.2.5 Special Elements**4.2.5.1 Catch Platform**

Catch platform shall be provided during demolition of exterior walls of structures more than 20 m in height. These shall be constructed and maintained not more than three storeys below the storey from which exterior wall is being demolished.

Catch platform shall not be used for storage or dumping of materials. These shall be capable of sustaining a minimum live load of 7 kPa. The out-riggers shall not be placed more than 3 m apart.

Additional requirements of Sec 1.4.3 Chapter 1 and Sec 4.1.6 Chapter 4 of this Part shall also be followed.

4.2.5.2 Stairs, Passageways and Ladders

Make-shift stairs with railings, passageways, and ladders shall be left in place as long as possible, and maintained in a safe condition. They shall not be removed from their position unless instructed by the foreman. See also Sections 3.5.3 and 3.13.1 of Chapter 3 of this Part for additional requirements.

4.2.5.3 Roof Trusses and Steel Structures

Structural frame of a pitched roof shall be removed to wall plate level by hand methods. Sufficient purlins and bracing shall be retained to ensure stability of the remaining roof truss while each individual truss is removed progressively. The bottom tie of roof trusses shall not be cut until the principal rafters are secured against making outward movement.

Temporary bracing shall be provided, where necessary, to maintain stability. All trusses except the one being dismantled shall be independently and securely guyed in both directions before work starts.

Hoisting gear suitable for the loads to be lifted shall be provided. A truss or a part thereof shall not be put on a floor; it may be allowed to rest only temporarily on the floor below if it can be ensured that the floor is capable of taking the load.

The steel frame may be left in place during demolition of masonry work. All steel beams/girders shall be cleared of all loose materials as the demolition of masonry work progresses downward provided it is still strong enough to stand as an independent structure.

4.2.5.4 Heavy Floor Beam

Heavy timber and steel beams shall be supported before cutting at the extremities. Beams shall be lowered gently and kept in a distant place without obstructing any passageway.

4.2.5.5 Jack Arch

Arches shall be demolished by standing on scaffolding clear of the arch. Tie rods between main supporting beams shall not be cut until the arch or series of arches have been removed. The floor shall be demolished in strips parallel to the span of the arch rings at right angles to the main floor beam.

4.2.5.6 Brick Arch

Abutments shall not be removed before the dead load of the spandrel wall and the arch rings are removed. A single span arch can be demolished by hand cutting narrow segments progressively from each springing parallel to the span of the arch until its width has been reduced to a minimum. The remainder of the arch can then be collapsed.

The crown may be demolished by the demolition ball method progressively from edges to the centre. Explosives may be used for a complete collapse of the structure by inserting charges into bore holes drilled in both the arch and the abutments.

In multi-span arches, lateral restraint shall be provided at the springing level before individual arches are removed. Demolition procedures as for single span may then be applied. Special temporary support shall be provided in the case of skew bridges.

No partial demolition leaving unstable portion standing shall be allowed. Where debris cannot be allowed to fall to the ground, centering capable of carrying load of the debris shall be designed and provided accordingly.

4.2.5.7 Cast-in-Situ RC

Before commencing demolition, the nature and condition of concrete and position of reinforcement and the possibility of lack of continuity of reinforcement shall be ascertained.

Demolition of cast-in-situ RC members shall start by removing partitions and external non-load bearing cladding and other decorative features.

Reinforced concrete beams shall be demolished one at a time after the slabs have been removed.

Ties shall be attached to the beam to support the beam when suspended.

The reinforcement near the supports shall first be exposed by drilling with pneumatic drill and removing the concrete. The reinforcement shall then be cut at both supports in such a way as to allow the beam to be lowered to the floor or the ground under control.

RC columns and any other supporting columns of one level shall only be demolished after all other building elements of that level have been completely removed.

The reinforcement in columns shall be exposed at the base after restraining wire guy ropes have been placed round the member at the top. The reinforcement shall then be cut in a way to allow it to be pulled down to the floor or the ground under control.

Reinforced concrete walls shall be cut into strips and demolished in the same way as concrete columns.

4.2.5.8 Precast Reinforced Concrete

Precautions in the form of providing temporary supports or balancing weights shall be taken to avoid toppling over of prefabricated units or any other part of the structure.

4.2.5.9 Suspended Floor, Roof and Cantilevered Structure

Suspended floor and roof slabs shall be cut into strips parallel to the main reinforcement and demolished strip by strip. For ribbed floors, the principle of design and method of construction shall be considered and procedures determined accordingly.

Ribs and beams shall never be cut at their mid-span and without securing by ties. Cantilevered portions, canopies, cornices, staircases and balconies shall be demolished after providing support to the portion before demolition of the main structure.

4.2.6 Mechanical Demolition

Mechanical demolition shall be restricted to a height of 25 m. When mechanical devices, such as weight ball and power shovels are used in demolition work, the area shall be barricaded up to a minimum distance of one and a half times the height of the wall in addition to the requirements laid out in Table 7.4.1.

While the mechanical device is in operation, no person shall be allowed to enter the building.

Location of the devices shall be such that it is neither hit by falling debris nor it causes any damage to adjacent structure, power line, etc.

4.2.7 Miscellaneous

No demolition work shall be carried out at night, or during storm or heavy rain. If demolition has to be done at night, precautions in the form of red warning signals, sirens, working lights and watchmen shall be provided. Auditory warning devices shall be installed at the demolition site.

Safety devices like industrial safety helmets (BDS 1265, BDS 1266), boots, gloves, goggles made of celluloid lens (BDS 1360), safety belts (BDS 1359) etc. shall be used by the workmen.

First aid box shall be made available at all demolition sites. In fire-risk area, appropriate portable firefighting appliances shall be kept at hand. See also Sections 3.11.2 and 3.11.3 Chapter 3 of this Part.

4.3 Blasting Operation and Use of Explosives

4.3.1 General

Before any work involving the use of explosives is started, a detailed survey and examination of the site, buildings or structures and adjoining areas and property shall be made. Due care shall be taken to avoid disruptions or damage to underground wells, tunnels, storage tanks etc.

Proximity of underground and overground services shall be carefully considered before blasting operations are carried out. Relevant authorities responsible for concealed underground works shall be duly consulted. Special attention shall be paid to the presence of power cables, radio and television transmitting stations sited within 3 km of the site.

Experts shall be consulted before proceeding with any work where sources of danger like flammable gases or liquids, sewage and drainage, unexploded missiles or mines, waste, explosive etc. are likely to be found. Also see Sec 4.1.

4.3.2 Code of Signal

Before any blasting commences on the construction or demolition site, both audible and visual signaling systems giving warning of blasting operations shall be established. These shall be such that they can be clearly heard and seen by site personnel working within the site areas, and also by the general public who may be affected.

Audible warnings shall comprise a series of readily recognizable signals with a distinctive tone. The Code of signals, once established for a particular site, shall not be altered without good reason and adequate warning to personnel.

Visual signs shall comprise clearly painted notices posted on all access roads to the site. Sentries shall be posted at the entries at blasting times with clear instructions; if necessary, they shall warn personnel who failed to hear warning signals or see signs.

4.3.3 Supervision and Responsibility

Only competent persons shall be employed as shotfirers. When subcontractors are taking part in the work on same site, the main contractor shall ensure a close liaison and collaboration with other contractors.

All site personnel present during blasting operations shall come under the control of the shotfirer.

All explosives shall be under the control of the shotfirer.

The handling of explosives on the site shall be restricted to personnel who are required to do so in the discharge of their duties and who are authorized in writing by the engineer. All site personnel shall be warned against maltreatment of explosives and blasting accessories.

4.3.4 Protection of site Personnel and Installations

The contractor shall provide all tools and equipment used in charging and firing blasts. The shotfirer shall inform the engineer the necessity of replacing any item. Shotfiring cables shall be examined before use for cuts or abraded insulation.

Circuit testers and exploders shall be handled with care and used and maintained according to the manufacturer's instructions; any malfunction shall immediately be reported and repair shall be carried out only by a competent person.

The area where explosives are to be used shall be defined before the charging of blasts. Vehicles and other mobile equipment shall be prohibited from entering the defined blast area, except as required to deliver or remove explosives.

All personnel shall be instructed as to what places of shelter they are to take up during blasting operations. Mobile plant and equipment shall be moved to a place of shelter and switched off when a blast is to be fired.

After a blast, no personnel shall be allowed to return to the danger area until the shotfirer has conducted a general examination and declared the site safe. The shotfirer shall not return to the blasting site until at least 5 min has elapsed after firing.

Electric detonators shall only be carried in boxes made of non-conducting materials, with a lid and catch. The shotfirer shall maintain a check on the number of detonators used against number issued. The boxes shall be kept locked until detonators are needed.

Blasting shall not be carried out in confined spaces without adequate ventilation; positive ventilation at the working face shall be maintained at all times.

No members shall be cut until precautions have been taken to prevent it from swinging freely. All structural steel members shall be lowered from the building and shall not be allowed to drop.

4.3.5 Safety of Third Parties

The safety of persons who reside or work in the vicinity of the site shall be considered. Where necessary, they shall be advised to vacate their homes or offices during blasting operations. In addition to notices giving warning of blasting on all roads and paths approaching the site, sentries shall be posted to maintain surveillance around the site when blasting is in progress.

Blasts shall normally be fired during the hours of daylight. The blasting technique and period shall be chosen so that any annoyance to the general public from noise, ground vibration, dust etc. is reduced to a minimum.

In heavily built-up areas, small-scale short delay blasting techniques employing light charges in small diameter holes shall be adopted. In such situations, short holes shall be carefully placed and charge weights correctly chosen. Sand bags, blasting mats or other screening material of suitable construction shall be placed over the top of each hole.

4.3.6 Use of Explosives

A sketch plan with sufficient duplicate copies shall be prepared for each blast. Before the explosive is deposited at the point of use, a check shall first be made of the depth of each shot hole. The engineer shall be informed of any departure from the planned arrangement.

Exposure to any compressive action or severe effect of a similar kind shall be avoided and grinding, scouring or rubbing actions eliminated. The vigorous use of stemming rods to force explosives into a hole shall be avoided. There shall be adequate clearance to allow easy insertion of the cartridges into the shot holes. The wrapping of the explosive cartridge shall not be removed, nor the cartridge be cut.

Primers shall not be made up in a magazine, or near excessive quantity of explosives, or in excess of immediate use. No attempt shall be made to use fuses, blasting caps, or explosives which have been water soaked. No attempt shall be made to soften hard set explosives by heating or rolling.

A bore hole shall not be loaded with explosives after springing unless it is cool and does not contain any hot metal. Temperature in excess of 65° C is dangerous. A bore hole near another hole loaded with explosives shall not be sprung.

No attempt shall be made to slit, drop, deform or abuse the primer. No metallic device shall be used in tamping. Wooden tamping tools with no exposed metal parts except non-sparking metal connectors for jointed poles shall be used.

4.3.7 Blasting Accessories

No person shall attempt to uncoil the wires and open out the short circuited bare leading wires of the electric blasting cap during approach of dust storm, or near sources of large charge of static electricity or near a radio transmitter. Firing circuit shall be kept completely insulated from the ground, other conductors, paths or stray current.

Except at the time and for the purpose of firing the blast, there shall be no electric live wires or cables near electric blasting caps or other explosives. All wire ends to be connected shall be bright and clean. The electric cap wires or leading wires shall be kept short-circuited until ready to fire.

All electric blasting caps shall be tested both singly and when connected to a circuit. Electrical blasting caps made by more than one manufacturer or electric blasting caps of different design or function, even if made by the same manufacturer, shall not be used in the same circuit. These shall not be fired by less than the minimum current specified by the manufacturer.

Where energy for blasting is taken from power circuits, the voltage shall not exceed 220 V. A safety switch, the same type as the blasting switch, shall be installed between the blasting switch and the firing circuit and lead lines at a distance not exceeding 1800 mm from the blasting switch.

Both safety switch and blasting switch shall be locked in the open position immediately after firing the shot. Key to the switches shall remain with the shotfirer at all times. Blasting shall be carried out using suitable exploder with 25% excess capacity.

Rubber covered or other adequately insulated copper wires shall be used for firing lines; sufficient firing line shall be provided. Single conductor lead lines shall be used. All holes loaded on a shift shall be fired on the same shift.

In very cold weather, the safety fuse shall be slightly warmed before using. Short fuse shall not be used. The length of a fuse shall be at least 1200 mm and the maximum burning rate 600 mm/min.

A fuse shall not be cut until the blasting cap is ready. The fuse shall be cut squarely across about 50 mm with a clean and sharp blade to ensure a dry end.

The fuse shall not be twisted after it has been seated lightly against the cap charge. Blasting caps shall not be crimped except by a cap crimper designed for the purpose. The cap shall be squarely crimped to the face.

The fuse shall be lighted with a fuse lighter designed for the purpose. It shall not be lighted until sufficient stemming has been placed over the explosives. The explosives shall not be held in hands when lighting the fuse.

In case of firing with safety fuse, the number of loud reports shall be counted; in the event of misfire, no person shall be allowed to the blasting site for at least 30 minutes. An inspection for remaining of un-detonated explosives shall be made; all misfired shot holes shall be marked.

If the misfire is due to faulty wiring or connection, the defect shall be remedied and the shotfired.

The stemming shall be floated out by using hose water until the hole has been opened to within 600 mm of the charge; the water shall be siphoned out thereafter and a new charge placed or, a new hole drilled 600 mm away from the old bore and parallel to it and about 300 mm less in depth and the new hole charged and duly fired.

4.4 Lowering, Removal and Disposal of Materials

4.4.1 General

No material shall be dropped or thrown on the ground or outside the exterior walls. They shall be lowered either in containers or by ropes, tackles, properly designed wood or metal chutes etc.

When the removal of any material causes an excessive amount of dust, it shall be wet before lowering or dropping, if feasible. Tag lines shall be used on all materials being lowered or hoisted up and a standard signal system shall be used and the workmen instructed on the signals. No person shall be permitted to ride the load line.

4.4.2 Use of Chutes

Chutes, if provided, shall be at the centre of the building. It shall have an angle of more than 45° with the horizontal, and shall be entirely closed on all sides except at the opening for receiving the material. The chute opening shall be kept locked. The top opening of chute shall be protected with guard rails.

Debris may be dropped through holes in the floor, if absolutely necessary. Precautions shall be taken to avoid overloading of the floor with debris. The debris dropping area shall be protected by rails.

4.4.3 Removal of Debris

Temporary stacking of demolished materials at the site shall be done in a manner ensuring fire prevention and orderly removal. Debris shall be removed from the site as soon as possible. Materials like glass, nails, etc. shall not be strewn about. Standard precautions to prevent fire from debris shall be taken.

4.4.4 Disposal of Materials

Demolished materials shall be disposed of according to their salvage value. Materials, which can be re-used, shall be salvaged and re-used with the approval of the owner.

Rubbish having no salvage value shall be removed from the site and disposed of according to the local statutory rules and regulations. Rubbish of combustible materials shall be disposed of immediately. All such operations shall have the approval of the owner.

4.4.5 Regularization of Plots

If there is no immediate construction planned on the plot vacant after demolition, it shall be filled, graded and maintained in conformity to the established street grades at curb level. The plot shall be maintained free from the accumulation of rubbish and water, and all other unsafe and hazardous conditions.

Provisions shall be made to prevent damage to any foundation on the premises or on the adjoining property. All previous service connections shall be capped.

PART VII
Chapter 5
Maintenance Management, Repairs, Retrofitting
and Strengthening of Buildings

5.1 Maintenance Management

Maintenance management of building is the art of preserving over a long period what has been constructed. Whereas construction stage lasts for a short period, maintenance continues for comparatively very large period during the useful life of building. Inadequate or improper maintenance adversely affects the environment in which people work, thus affecting the overall output and also the overall service life of the building. In the post construction stage the day to day maintenance or upkeep of the building shall certainly delay the decay of the building structure. Though the building shall be designed to be very durable it needs maintenance to keep it in good condition.

5.2 Terminology

For the purpose of this Section, the following definitions shall apply.

BUILDING FABRIC	Elements and components of a building other than furniture and services.
BUILDING MAINTENANCE	Work undertaken to maintain or restore the performance of the building fabric and its services to provide an efficient and acceptable operating environment to its users.
CONFINED SPACE	Space which is inadequately ventilated for any reason and may result in a deficiency of oxygen, or a build-up of toxic gases, e.g. closed tanks, sewers, ducts, closed and unventilated rooms, and open topped tanks particularly where heavier than air gases or vapors may be present.
HOUSEKEEPING	The routine recurring work which is required to keep a structure in good condition so that it can be utilized at its original capacity and efficiency along with proper protection of capital investment, throughout its economic life.
MAINTENANCE	The combination of all technical and associated administrative actions intended to retain an item in or restore it to a state in which it can perform its required function.
MAINTENANCE MANAGEMENT	The organization of maintenance within an agreed policy. Maintenance can be seen as a form of 'steady state' activity.

OWNER

Person or body having a legal interest in a building. This includes freeholders, leaseholders or those holding a sub-lease which both bestows a legal right to occupation and gives rise to liabilities in respect of safety or building condition.

In case of lease or sub-leaseholders, as far as ownership with respect to the structure is concerned, the responsibility of structure of a flat or structure on a plot belongs to the allottee/lessee during the leasehold.

Provisions of Sections 8.1 and 8.2 of Chapter 8 Part 6 shall apply for detailing of reinforced concrete members, in general. For reinforced concrete structures, subject to earthquake loadings in zone 2 and zone 3, special provisions contained in Sec 8.3 of this chapter shall apply.

5.3 Building Maintenance

5.3.1 General

Any building (including its services) when built has certain objectives and during its total economic life, it has to be maintained. Maintenance is a continuous process requiring a close watch and taking immediate remedial action. It is interwoven with good quality of housekeeping. It is largely governed by the quality of original construction. The owners, engineers, constructors, occupants and the maintenance agency are all deeply involved in this process and share a responsibility. Situation in which all these agencies merge into one is ideal and most satisfactory.

There are two processes envisaged, that is, the work carried out in anticipation of failure and the work carried out after failure. The former is usually referred to as preventive maintenance and the latter as corrective maintenance. The prime objective of maintenance is to maintain the performance of the building fabric and its services to provide an efficient and acceptable operating environment to its users.

Maintenance in general term can be identified in the following broad categories.

- (a) Cleaning and servicing - This is largely of preventive type, such as checking the efficacy of rain water gutters and servicing the mechanical and electrical installations. This covers the house keeping also.
- (b) Rectification and repairs - This is also called periodical maintenance work undertaken by, say, annual contracts and including external re-plastering, internal finishing etc.
- (c) Replacements - This covers major repair or restoration such as reproofing or re-building defective building parts.

5.3.2 Factors Affecting Maintenance

Maintenance of the buildings is influenced by the following factors:

- (a) Technical factors - These include age of building, nature of design, material specifications, past standard of maintenance and cost of postponing maintenance.
- (b) Policy - A maintenance policy ensures that value for money expended is obtained in addition to protecting both the asset value and the resource value of the buildings concerned and owners.
- (c) Environmental - All buildings are subject to the effects of a variety of external factors such as air, wind precipitation, temperature etc. which influence the frequency and scope of maintenance.

The fabric of building can be adversely affected as much by the internal environment as by the elements externally. Similar factors of humidity, temperature and pollution shall be considered. Industrial buildings can be subject to many different factors subject to processes carried out within. Swimming pool structures are vulnerable to the effects of chlorine used in water.

- (d) User - The maintenance requirements of buildings and their various parts are directly related to the type and intensity of use they receive.

5.3.2.1 Influence of design

The physical characteristics, the life span and the aesthetic qualities of any building depend on the considerations given at the design stage. All buildings, however well designed and conscientiously built, will require repair and renewal as they get older. However, for better performance of the building envelop, the following are the ways to minimize troubles at the later stage:

- (a) Minimize defects during construction and design.
- (b) Detail and choose materials during construction so that the job of maintenance is lessonerous.

In addition to designing a building for structural adequacy, consideration shall also be given to environmental factors such as moisture, natural weathering, corrosion and chemical action, user wear and tear, pollution, flooding, subsidence, earthquake, cyclones etc.

5.3.3 Maintenance Policy

The policy shall cover such items as the owner's anticipated future requirement for the building taking account of the building's physical performance and its functional suitability. This shall lead to decisions regarding:

- (a) the present use of the building anticipating any likely upgrading and their effect on the life cycles of existing components or engineering services; and
- (b) A change of use for the building and the effect of any conversion work on the life cycles of existing components or engineering services.

5.3.4 Maintenance Work Programmes

The programming of maintenance work can affect an owner or his activities in the following ways:

- (a) Maintenance work shall be carried out at such times as are likely to minimize any adverse effect on output or function and with due consideration to the comforts of the occupants and public and Third Party stakeholders.
- (b) Programme shall be planned to obviate as far as possible any abortive work. This may arise if upgrading or conversion work is carried out after maintenance work has been completed or if work such as rewiring is carried out after redecoration.
- (c) Any delay in rectifying a defect shall be kept to a minimum only if such delay is likely to affect output or function. The cost of maintenance increases with shortening response times.
- (d) Maintenance work, completed or being carried out shall comply with all statutory and other legal requirements.

5.3.5 Maintenance Guides

An owner responsible for a large number of buildings has to established procedures for maintenance. When an owner is responsible for the maintenance of only one building or a small number of buildings, the preparation of a guideline manual tailored to suit each particular building, can offer significant advantages. Such a manual shall take into account the following.

- (a) Type of construction and residual life of the building, and
- (b) Environment and intensity of use (see Sec 5.3.2).

The guide shall form part of a wider manual covering operational matters.

5.3.6 Planning of Maintenance Work

Work shall take account of the likely maintenance cycle of each building element and be planned logically, with inspections being made at regular intervals. Annual plans shall take into account subsequent years' programmed to incorporate items and to prevent additional costs. It shall be stressed that the design of some buildings can lead to high indirect costs in maintenance contracts and therefore, careful planning can bring financial benefits. Decisions to repair or replace shall be taken after due consideration.

5.3.7 Feed Back

Feed back is normally regarded as an important procedure of providing information about the behaviour of materials and detailing for the benefit of the architect and engineer designing new buildings, which will result in lessening maintenance costs. It is an equally valuable source of information for the persons responsible for maintenance. Every maintenance organization shall develop a sample way of communicating its know-how, firstly for benefit of others in the organization and secondly for the benefit of the building industry as a whole. There shall be frank and recorded dialogue on an on-going basis between those who occupy and care for buildings and those who design and construct them.

Feedback shall aim at the following:

- (a) User satisfaction.
- (b) Continuous improvement, and
- (c) Participation by all.

The information on feed back can be obtained from the following:

- (a) Occupants,
- (b) Inspections,
- (c) Records, and
- (d) Discussions.

5.3.8 Means of Effecting Maintenance

Some maintenance work will be carried out by the occupier of a building or by the occupier's representative. In the case of leasehold or similar occupation not all maintenance shall be the responsibility of occupier. The regular maintenance shall be clearly mentioned in the tenancy agreement. Responsibility of common areas shall be clearly defined.

Maintenance work sub-divided into major repair, restoration, periodical and routine or day-to-day operations will be undertaken by one of the following.

- (a) Directly employed labour.
- (b) Contractors, and
- (c) Specialist contractors under service agreement or otherwise.

The merits of each category for typical maintenance work must be considered because optimum use of resources appropriate to tasks in a given situation is an important element of policy.

The success of contracting out depends on the nature of the services, conditions in which contracting is undertaken (the tendering process), how the contract is formulated and subsequent monitoring of service quality. The important consideration in the decision to contract out is whether a contractor can ensure a socially desirable quantity and quality of service provision at, a reasonable cost to the consumers.

5.4 Access

5.4.1 General

All maintenance activities including any preliminary survey and inspection work require safe access and in some situations this will have to be specially designed. Maintenance policy, and maintenance costs, will be much influenced by ready or difficult access to the fabric and to building services. Special precautions and access provisions shall also need to be taken for roof work or for entry into confined spaces such as ducts or voids.

5.4.2 Access Facilities

Permanent accessibility measures shall be provided at the design stage only for all the areas for safe and proper maintenance. It is a matter on which those experienced in the case of the building can make an important contribution at design stage in the interest of acceptable maintenance costs.

A wide variety of temporary access equipment shall appropriately be provided for maintenance work, ranging from ladders to scaffoldings or powered lift platforms.

Wherever possible it is better to provide permanent access facilities such as fixed barriers, ladders, and stairways. When such permanent access facilities are provided necessary arrangement shall be included in maintenance plans for their regular inspection, maintenance and testing.

All personnel employed for carrying out maintenance shall be provided with the necessary protective clothing and equipment and instructed in its use.

When physical access is not possible in situations such as wall cavities, drains etc. inspections shall be made with the aid of closed circuit television or optical devices such as endoscopes.

5.4.3 Access to Confined Spaces**5.4.3.1 Ventilation**

Good ventilation shall be necessary in order that maintenance work can be carried out safely. This is especially important in confined spaces. When the normal ventilation is inadequate it shall be supplemented by temporary and forced ventilation installations. These shall provide general and spot ventilation as appropriate.

Special precautions need to be taken when entering a confined space. Such confined spaces shall be adequately ventilated and trapped gas removed, particularly before being entered, to ensure that they are free from harmful concentrations of gases, vapors other airborne substances and that the air is not deficient in oxygen.

5.4.3.2 Lighting

Good lighting is necessary in order that maintenance work can be carried out satisfactorily. This is particularly important in confined spaces. When the normal lighting is inadequate it shall be supplemented by temporary installations. These shall provide general and spot illumination as appropriate.

5.5 Records**5.5.1 General**

Good records can save owners and users/occupiers much unnecessary expense and reduce potential hazards in exploration work when faults arise.

5.5.2 Use of Building Records

All personnel involved in the maintenance of the building shall be made aware of the existence of the building records.

Known hazardous areas shall be explicitly marked on the records as well as being marked on site and shall be pointed out to such personnel together with any system of work adopted for use in such areas.

Records are of value only if they are kept up to date and arrangements for this shall be included in any provision that may be made for records.

Records shall be readily accessible for use and the place of storage shall take into account the form of the records and the conditions needed to keep them from damage of any kind. It is recommended that a duplicate set of records is kept in a secure place other than building itself and is kept up to date.

Following shall be typical contents of the maintenance records:

- (a) A brief history of property, names and addresses of consultants and contractors.
- (b) Short specifications, constructional processes, components, material finishes, hidden features, special features etc.
- (c) “As built” plans and as subsequently altered with sections, elevations and other detailed drawings.
- (d) Foundation and structural plans/sections such as concrete reinforcement drawings.
- (e) Detail specification of all materials incorporated, for example, concrete mix, species and grades of timber etc. Potentially hazardous materials and types or methods of construction that under some circumstances may become hazardous shall be identified.
- (f) Information on housekeeping and routine maintenance with details of internal and external surfaces and decorations, schedule of cleaning, inspection and maintenance.
- (g) Means of operating mechanical, electrical and plumbing installations.
- (h) Description of renovations, extensions, adaptations and repair to each element.
- (i) All plant, machinery and propriety articles including manufacturers trade literature and instructions for installation, use and maintenance.
- (j) Methods of work used in construction such as assembly of prefabricated units.
- (k) All information related to fire such as:
 - Location and service arrangements of all fire alarm and call points;
 - Location and service arrangements of all extinguishers, hose reels and other firefighting installations;
 - Location of all fire compartment walls, doors, floors and screens;
 - Location of all areas of exceptional fire hazard;
 - Fire escape routes;
 - Details of application of any fire protection treatment; and
 - Location details and description of any installation for smoke control or protection of escape routes.
- (l) There shall be a wall chart showing at a glance the various operations which have to be undertaken. Line drawings of buildings are always to be there.
- (m) Records of security measures shall be known to Authorized personnel only.
- (n) Where no records exist, information shall be slowly built up as it becomes available during the course of maintenance work.
- (o) Use of computers for storing information shall be compulsory.

5.5.3 Mechanical Records

5.5.3.1 Documentation

Documentation shall record the following as installed.

- (a) The location, including level if buried, of all public service connections (for example, fuel gas and cold water supplies) together with the points of origin and termination, size and materials of pipes, line pressure and other relevant information.
- (b) The layout, location and extent of all piped services showing pipe sizes, together with all valves for regulation, isolation and other purposes as well as the results of all balancing, testing and commissioning data.
- (c) The location, identity, size and details of all apparatus and all control equipment served by, or associated with, each of the various services together with copies of any test certificates for such apparatus where appropriate. The information with respect to size and details shall be presented in schedule form.
- (d) The layout, location and extent of all air ducts showing dampers and other equipment, acoustic silencers, grilles, diffusers or other terminal components. Each duct and each terminal component shall be marked with its size, the air quantity flowing and other relevant balancing data, and
- (e) The location and identity of each room or space housing plant, machinery or apparatus.

5.5.3.2 Drawings

Drawings shall record the following as installed.

- (a) Detailed general arrangements of boiler houses, machinery spaces, air handling plants, tank rooms and other plant or apparatus, including the location, identity, size and rating of each apparatus, The information with respect to the size and rating can be presented in schedule form;
- (b) Isometric or diagrammatic views of boiler houses, plant rooms, tank rooms and similar machinery, including valve identification charts. It is useful to frame and mount a copy of such drawings on the wall of the appropriate room, and
- (c) Comprehensive diagrams that show power wiring and control wiring and /or pneumatic or other control piping including size, type or conductor or piping used and identifying the terminal points of each.

5.5.4 Electrical Records

Documentation shall record the following including locations, as installed.

- (a) Main and sub main cables, showing origin, route, termination, size and type of each cable; cables providing supplies to specialist equipment, for example, computers, shall be identified separately; and
- (b) Lighting conduits and final sub circuit cables, showing origin, route, termination and size of each, together with the number and size of cables within each conduit. The drawings shall indicate for each conduit or cable, whether it is run on the surface or concealed, for example, in a wall chase, in a floor screed, cast in-situ, above a false ceiling etc.

These drawings shall also indicate the locations of lighting fittings, distribution boards, switches, draw-in-boxes and point boxes, and shall indicate circuitry.

- (a) Location and purpose of each emergency lighting fitting including an indication of the circuit to which it is connected.
- (b) Single and three phase power conduits and final sub circuit cables showing locations of power distribution boards, motors, isolators, starters, remote control units, socket outlets and other associated equipment.
- (c) Other miscellaneous equipment, conduits and cables.
- (d) Lightning conductor, air terminals, conductors, earth electrodes and test clamps.
- (e) Location of earth tapes, earth electrodes and test points other than those in(f); and
- (f) Cables providing earth circuits for specialist equipment, for example computers, shall be identified separately.

Documentation shall also include, when applicable.

- (a) Distribution diagrams or schedules to show size, type and length (to within 1 m) of each main and sub main cable, together with the measured earth continuity resistance of each.
- (b) Schedule of lighting fittings installed stating location, manufacturer and type or catalogue number together with the type or manufacturer's reference, voltage and wattage of the lamp installed.

- (c) Schedule of escape and emergency lighting fittings installed stating location, manufacturer, type or catalogue number together with the type or manufacturer's reference, voltage and wattage of the lamp installed. For battery systems the position of the battery, its ampere hour rating and battery system rated endurance in hours shall be stated.
- (d) Records of smoke detectors, sprinklers, fire precautions.
- (e) Incoming supply details; the type of system, voltage, phases, frequency, rated current and short circuit level, with the details of the supply protection and time of operation as appropriate.
- (f) Main switchgear details; for purpose made equipment this shall include a set of manufacturers' drawings and the site layout.
- (g) Transformer, capacitor and power plant details; the leading details shall be given, for example, for transformers the V.A rating, voltages and type of cooling; and
- (h) Completion certificate, according to the Bangladesh Electricity Act.

5.6 Inspections

5.6.1 General

Regular inspections are actual part of the procedures for the maintenance of buildings. They are needed for a variety of purposes and each purpose requires a different approach if it is to be handled with maximum economy and efficiency. A more detailed inspection covering all parts of a building is needed to determine what work shall be included in cyclic and planned maintenance programme.

5.6.2 Frequency of Inspection

Inspection shall be carried out at the following frequencies:

- (a) Routine - Continuous regular observations shall be undertaken by the building user as part of the occupancy of building. Feedback resulting from this type of observation shall be recorded in record book.
- (b) General - Visual inspections of main elements shall be made annually under the supervision of suitably qualified personnel at appropriate times.
- (c) Detailed - The frequency of full inspection of the building fabric by suitably qualified personnel shall not normally exceed a 5 year period.

5.6.2.1 Inspection schedule

The preparation of a specific inspection schedule shall be formulated at the beginning. Once prepared, it shall be used for subsequent inspections.

5.6.3 Inspection of Engineering Services

Engineering services generally have a shorter life expectancy than building fabric and because of their dynamic function shall be subjected to more frequent inspections and maintenance.

Inspection of services shall be carried out for three purposes as follows.

- (a) To check if maintenance work is required,
- (b) To check if maintenance work is being adequately carried out, and
- (c) For safety reasons to comply with statutory requirements and if required, with recommendations of other relevant organizations.

The frequency of inspections for the purpose:

- (a) will depend upon types of plant and system manufacturer's recommendations and subjective judgment.
- (b) shall be carried out on an annual basis.

5.6.3.1 Method of inspection

The limited life of building services means it is important to record their residual life so that their replacement can be budgeted for, and inspection methods shall be arranged accordingly.

A checklist of items of plant to be inspected shall be considered. Detailed specifications of how inspections shall be carried out are necessary because a simple visual inspection is unlikely to show whether plant is operating correctly and efficiently.

Inspections frequently necessitate the use of appropriate instruments by competent persons. An example of this is the inspections carried out to check compliance with statutory requirements.

When instruments are used it is important that adequate training is provided in the use of the instruments and the interpretation of the results.

Records of all inspections shall be kept in suitable locker.

5.7 Maintenance of Electrical Appliances

5.7.1 Planning of Maintenance Work

If the authorized person has complete knowledge of the electrical appliances to be worked upon, then safety will be more assured. If the person attending to the job is not technically competent to handle the job then more careful planning is required before hand.

Repetitive nature of jobs involves little or no pre-planning whereas infrequent nature of jobs shall need careful planning even if the person attending the job is technically competent.

Planned routine maintenance will facilitate continued safe and acceptable operation of an electrical system with a minimum risk of breakdown and consequent interruption of supply.

As far as the electrical equipment/installations are concerned, it is not possible to lay down precise recommendations for the interval between the maintenance required. The recommendation for frequency of maintenance in this regard from the manufacturer is more relevant. The manufacturer shall be requested to specify minimum maintenance frequency under specified conditions. These intervals depend greatly upon the design of the equipment, the duty that it is called on to perform and the environment in which it is situated.

Following two types of maintenance are envisaged.

5.7.1.1 Routine maintenance

Routine maintenance of the electrical equipment goes along with the regular inspections of the equipment. Inspections shall reveal the undue damage and excessive wear to the various components. Examination of the equipment shall reveal any need for conditioning of the contact system, lubrication and adjustment of the mechanisms.

5.7.1.2 Post fault maintenance

When there is a breakdown in the system and certain parts are identified for the replacement and then the maintenance/repair of the defective part away from the operating environment is covered under post fault maintenance.

5.7.2 Guidelines for the Maintenance of Electrical Appliances

Uninterrupted and hazard free functioning of the electrical installations are the basic parameters of maintenance. The equipment shall be restored to correct working conditions. Special attention shall be paid to the items and settings that might have been disturbed during the operational phase. Loose and extraneous equipment or wiring gives rise to potential safety hazards. All covers and locking arrangements shall be properly checked and secured to achieve original degree of protection.

Guidelines to be followed for the maintenance of electrical equipment to ensure their smooth functioning are given in Appendix A of this Part.

5.8 Operating and Maintenance Manuals

The engineering services within buildings frequently are dynamic, involving complex systems of integrated plant items. Operation of such plant can require detailed knowledge and direction. Maintenance can also require extensive information to be available. It is, therefore, important to have suitable operating and maintenance manuals to provide the necessary guidance. These shall be included as part of the contractual requirements for new installations and shall ideally be prepared as reference documents for existing installations where no such information exists.

5.9 Prevention of Cracks

Cracks in buildings are of common occurrence. A building component develops cracks whenever stress in the component exceeds its strength. Stress in a building component could be caused by externally applied forces, such as dead, imposed, wind or seismic loads, or foundation settlement or it could be induced internally due to thermal movements, moisture changes, chemical action, etc.

Cracks could be broadly classified as structural or non-structural. Structural cracks are those which are due to incorrect design, faulty construction or overloading and these may endanger the safety of a building. Extensive cracking of an RCC beam is an instance of structural cracking. Non-structural cracks are mostly due to internally induced stresses in building materials and these generally do not directly result in structural weakening. In course of time, however, sometime non-structural cracks may, because of penetration of moisture through cracks or weathering action, result in corrosion of reinforcement and thus may render the structure unsafe. Vertical cracks in a long compound wall due to shrinkage or thermal movement is an instance of non-structural cracking. Non-structural cracks, normally do not endanger the safety of a building, but may look unsightly, or may create an impression of faulty work or may give a feeling of instability. In some situations, cracks may, because of penetration of moisture through them, spoil the internal finish, thus adding to cost of maintenance. It is, therefore, necessary to adopt measures of prevention or minimization of these cracks.

5.10 Repairs and Seismic Strengthening of Buildings

5.10.1 Non-structural/Architectural Repairs

The buildings affected by earthquake may suffer both non-structural and structural damages. Nonstructural repairs may cover the damages to civil and electrical items including the services in the building. Repairs to non-structural components need to be taken up after the structural repairs are carried out. Care shall be taken about the

connection details of architectural components to the main structural components to ensure their stability,

Non-structural and architectural components get easily affected/dislocated during the earthquake. These repairs involve one or more of the following.

- (a) Patching up of defects such as cracks and fall of plaster.
- (b) Repairing doors, windows, replacement of glass panes.
- (c) Checking and repairing electric conduits/ wiring.
- (d) Checking and repairing gas pipes, water pipes and plumbing services.
- (e) Re-building non-structural walls, smoke chimneys, parapet walls, etc.
- (f) Re-plastering of walls as required.
- (g) Rearranging disturbed roofing tiles.
- (h) Relaying cracked flooring at ground level; and
- (i) Redecoration - whitewashing, painting, etc.

The architectural repairs as stated above do not restore the original structural strength of structural components in the building and any attempt to carry out only repairs to architectural/non-structural elements neglecting the required structural repairs may have serious implications on the safety of the building. The damage would be more severe in the event of the building being shaken by the similar shock because original energy absorption capacity of the building would have been reduced.

5.10.2 Structural Repairs

Prior to taking up of the structural repairs and strengthening measures, it is necessary to conduct detailed damage assessment to determine:

- (a) the structural condition of the building to decide whether a structure is amendable for repair; whether continued occupation is permitted; to decide the structure as a whole or a part require demolition, if considered dangerous;
- (b) if the structure is considered amendable for repair then detailed damage assessment of the individual structural components (mapping of the crack pattern, distress location; crushed concrete, reinforcement bending/yielding, etc.). Non-destructive testing techniques could be employed to determine the residual strength of the members; and
- (c) to work out the details of temporary supporting arrangement of the distressed member so that they do not undergo further distress due to gravity loads.

After the assessment of the damage of individual structural elements, appropriate repair methods are to be carried out component wise depending upon the extent of damage. The repair shall consist of the following:

- (a) Removal of portions of cracked masonry walls and piers and rebuilding them in richer mortar. Use of non-shrinking mortar will be preferable.
- (b) Addition of reinforcing mesh on both faces of the cracked wall, holding it to the wall through spikes or bolts and then covering it, suitably, with cement mortar or micro-concrete.
- (c) Injecting cement or epoxy like material which is strong in tension, into the cracks in walls.
- (d) The cracked reinforced cement elements shall be repaired by epoxy grouting and could be strengthened by epoxy or polymer mortar application like shotcreting, jacketing, etc.

5.10.3 Seismic Strengthening

The main purpose of the seismic strengthening is to upgrade the seismic resistance of a damaged building while repairing so that it becomes safer under future earthquake occurrences. This work shall involve some of the following actions:

- (a) Increasing the lateral strength in one or both directions by increasing column and wall areas or the number of walls and columns.
- (b) Giving unity to the structure, by providing a proper connection between its resisting elements, in such a way that inertia forces generated by the vibration of the building can be transmitted to the members that have the ability to resist them. Typical important aspects are the connections between roofs or floors and walls, between intersecting walls and between walls and foundations.
- (c) Eliminating features that are sources of weakness or that produce concentration of stresses in some members. Asymmetrical plan distribution of resisting members, abrupt changes of stiffness from one floor to the other, concentration of large masses and large openings in walls without a proper peripheral reinforcement are examples of defects of this kind.
- (d) Avoiding the possibility of brittle modes of failure by proper reinforcement and connection of resisting members.

5.10.4 Seismic Retrofitting

Many existing buildings do not meet the seismic strength requirements of present earthquake codes due to original structural inadequacies and material degradation due to time or alterations carried out during use over the years. Their earthquake resistance can be upgraded to the level of the present day codes by appropriate seismic retrofitting techniques, such as mentioned in Sec 5.10.3 of this Chapter.

5.10.5 Strengthening or Retrofitting Versus Reconstruction

Replacement of damaged buildings or existing unsafe buildings by reconstruction is, generally, avoided due to a number of reasons, the main ones among them being:

- (a) higher cost than that of strengthening or retrofitting.
- (b) preservation of historical architecture, and
- (c) maintaining functional social and cultural environment.

In most instances, however, the relative cost of retrofitting to reconstruction cost determines the decision. As a thumb rule, if the cost of repair and seismic strengthening is less than about 50 percent of the reconstruction cost, the retrofitting is adopted. This shall also require less working time and much less dislocation in the living style of the population. On the other hand reconstruction may offer the possibility of modernization of the habitat and may be preferred by well-to-do communities.

Cost-wise the building construction including the seismic code provisions in the first instance, works out the cheaper in terms of its own safety and that of the occupants. Retrofitting an existing inadequate building may involve as much as 4 to 5 times the initial extra expenditure required on seismic resisting features. Repair and seismic strengthening of a damaged building may even be 5 to 10 times as expensive. It is, therefore, very much safe as well as cost-effective to construct earthquake resistant buildings at the initial stage itself according to the relevant seismic codes.

5.11 Maintenance Manual

The Consultant/Engineer shall prepare a “Maintenance Manual” prior to handing over of the completed project and furnish this to the Employer, which must contain following items.

- (a) As-built drawings for the completed project.
- (b) Sources of all items of work including materials, furnishes equipment and fixed furniture, containing names and addresses of suppliers, catalogue numbers, technical information & specifications and warranty documents for respective items.
- (c) Frequency of routine preventive maintenance and the procedure thereof, including information on available local and foreign after-sales service sources.
- (d) Corrective maintenance procedure and sources of available.

5.12 Related Appendix

Appendix A Guidelines for Maintenance of Electrical Equipment

PART VII
Appendix A**Guidelines for Maintenance of Electrical Equipment**

- A.1** In case of electrical appliances, manufacturer's instructions for the usage and maintenance of the equipment should be strictly followed.
- A.2** The detailed/working drawings of all the components of electrical installations should always be available with the maintenance unit. Following records should be available.
- (a) Manufacturer's name.
 - (b) Nameplate of the equipment and its salient features such as capacity, rating etc.
 - (c) Manufacturer's recommendations regarding availability/usage of spare parts.
 - (d) Manufacturer's recommendations for periodical maintenance and post fault maintenance.
 - (e) Details of the maintenance operations performed in the past.
- A.3** Care should be taken while selecting replacement parts. The spare parts should be correct and suitable, preferably as recommended by the manufacturer of the installation. During the placement of order for the supply of spare parts, nameplate particulars and serial number should be quoted.
- A.4** The space where the equipment is kept should be clean and properly ventilated. Equipment should not be disturbed needlessly. Before cleaning, the equipment should be made dead. For internal cleaning a section cleaner should be used.
- A.5** Covers and doors should not be left open unnecessarily during maintenance. Afterwards they should be promptly and correctly closed and locked.
- A.6** Before removing the covers and connections, all covers and cable terminations should be marked to ensure correct replacements. Disturbed connections and temporary connections should be marked to facilitate re-connection. Temporary connections and markings should be removed before the installation is put to use.
- A.7** Those connections which have not been disturbed should also be checked for soundness and overheating.
- A.8** All insulations should be regularly checked. Solid insulations should be checked for cracks and other defects. Fibrous and organic insulations should be checked for sign of blistering, delamination and mechanical damage. For insulating oils the interval between tests should be carried out as per the recommendations of the manufacturer and keeping the adverse environmental conditions in mind.

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- A.9** It should be ensured that the earthing connections are sound and all contact screws are tight.
- A.10** During the examination of interlocks it is necessary to take precautions to prevent danger to plant or persons in the event of malfunction or inadvertent operation. A person responsible for checking and maintaining any interlock system should have thorough knowledge of the extent, nature and function of the interlock.
- A.11** If the equipment is ventilated then it should be ensured that the airflow is smooth and not restricted. If filters are provided, they should be cleaned or replaced as necessary.
- A.12** The standby system for tripping and closing supplies should always be kept in good order. Indicators and alarms should be maintained in time with the manufacturer's instructions.
- A.13** Tools, spares and instruments should be stored near to the installation. These should be regularly checked against an inventory.
- A.14** Before the start of maintenance of the circuit switches it should be ensured that all incoming and outgoing main auxiliary circuits are dead and remain so during the maintenance. Overheating of the circuit switches is the root cause for faults. Overheating may be caused by inadequate ventilation, overloading, loose connection, insufficient contact force and malalignment.
- A.15** Some circuit breakers are not intended to be maintained, such as miniature circuit breakers (MCBs). Such items should not be dismantled for maintenance. These should be renewed periodically.
- A.16** For the maintenance of fuses periodical inspection should be done for correct rating, security, overheating and correct location/orientation. Element of renewable fuses should be renewed when the deterioration is apparent. The availability and correct replacement of fuse links should be ensured.
- A.17** If a fuse link of certain rating has failed and is replaced, then all fuse-links of same rating apparently subjected to the fault should be destroyed and replaced by new fuse links.
- A.18** In order to be reasonably sure that circuit breaker is capable of operation when required, these should be tripped and reclosed at regular intervals. Tripping should be proved manually and where possible electrically via the protective relay contacts. The leakage of oil, sign of corrosion, and any unusual smell which may indicate over-heating should be detected through inspections.

- A.19** Timing devices are mostly designed for specialist maintenance. These should not be dismantled for maintenance or overhaul purposes unless specifically recommended by the manufacturers'. Actual timing periods should be verified with set values and application requirements.
- A.20** In case of cable boxes and terminations, security of mounting and earthing should be examined. Exposed tails should be inspected for good conditions of insulation and freedom from moisture.
- A.21** Battery cells should be inspected for shedding of active material, sedimentation and buckling of plates. Level of electrolyte should be regularly checked and the level should be corrected with distilled water.

PART VIII**Chapter 1****Electrical and Electronic Engineering Services for Buildings****1.1 Introduction****1.1.1 Scope**

The provisions of the Code presented in this Chapter, cover the Electrical and Electronic Engineering Services for Buildings to ensure that the related installation work becomes perfect and safe for the persons residing in and around the building. The term safe means safe for the persons and safe for the properties.

Provisions of the Specifications are to set minimum standards for Electrical and Electronic Engineering Installations in various Occupancy categories of buildings, as described in Part 3 of this Code, including annexes and premises. All the systems and equipment intended for the supply of normal power and standby power to all these places are covered by the provisions of this Code.

The provisions of the Code for various Electrical and Electronic Engineering systems and/or installations for the buildings include, but not limited to:

- (a) Lighting and illumination.
- (b) Fans, cooling and heating.
- (c) Normal and standby power supply.
- (d) Supply system and feeder for lifts/escalator/moving walk, including protection.
- (e) Cable television distribution.
- (f) Electronic access control.
- (g) Burglar alarm/CCTV monitoring/security.
- (h) Electrical cables/conductors and equipment.
- (i) Switches, sockets, other accessories.
- (j) Cables and conductors in a building that connect to the supply of electricity.
- (k) Electrical protection system.
- (l) Earthing system of an electrical installation.
- (m) Lightning protection of a building and its electrical installation.
- (n) Fire alarm.
- (o) Multi-media communications, data communications and telecommunications.

Electrical wiring/cabling form a major part in the above mentioned installation works. Electrical wiring/cabling must be reasonably safe to persons and property. Installations, alteration, or extension of Electrical wiring/cabling systems conforming to the provisions of this Code shall be deemed to be reasonably safe to persons and property.

The provisions of the Code in this Chapter do not cover Installations in ship, water craft, railway rolling stock, aircraft, or automotive vehicles and recreational vehicles,

1.1.2 Designing an Electrical and Electronic Engineering Installations in Buildings and Related Structures

The provisions of the Code presented in this Section are not meant to provide adequate information to design Electrical and Electronic Engineering Installations and Systems in Buildings and related structures. These should not be taken to be adequate or complete for the efficient design work of installations.

Such design work, the required features, detailed technical specifications, schedule of items etc., should be obtained through the services of an engineer adequately qualified in this area. Applications of energy efficient appliances should be kept in mind while preparing electrical design of a building or related installations.

1.1.3 Terminology and Definitions

This Section provides an alphabetical list of the terms used in and applicable to this Chapter of the Code. In case of any conflict or contradiction between a definition given in this Section and that in Part 1, the meaning provided in this Section shall govern for interpretation of the provisions of this Chapter.

ACCESSORY	A device associated with current using equipment or with the wiring of an installation; for example, a switch, a plug, a socket outlet, a lamp holder, or a ceiling rose.
ALIVE	See LIVE.
APPARATUS	Apparatus means Energy Efficient Apparatus. Electrical apparatus including all machines, appliances and fittings in which conductors are used or of which they form a part.
APPLIANCE	Appliance means Energy Efficient Appliance. An item of electric current using equipment other than a luminaries or an independent motor.
BDB	Branch-Distribution Board located in the same floor of a building and connected to one of the SDBs in the same floor.

BRANCH CIRCUIT, APPLIANCE	A branch circuit supplying energy to one or more outlets to which appliances are to be connected; such branch circuits do not have any permanently connected lighting fixtures except those that are integral parts of the appliances themselves.
BRANCH CIRCUIT, GENERAL PURPOSE	A branch circuit that supplies a number of outlets for lighting and/or appliance.
BRANCH CIRCUIT, INDIVIDUAL	A branch circuit that supplies only one utilization equipment.
BUNCHED	Cables are said to be bunched when two or more are either contained within a single conduit, duct, ducting, or trunking or, if not enclosed, are not separated from each other.
CABLE	PVC insulated copper cables having copper cross section of 1 mm ² and above. A length of single insulated conductor (solid or stranded), or two or more such conductors, each provided with its own insulation. The insulated conductor or conductors may or may not be provided with an overall mechanical protective covering.
CEILING ROSE	A ceiling rose is used for terminating the point wiring for a Light or a Fan in the ceiling. It has brass terminals in which incoming cables are terminated using brass screws on the terminals and the outgoing flexible cables get connection through the screw connections.
CIRCUIT	An assembly of electrical equipment supplied from the same origin and protected against overcurrent by the same protective device.
SUB CIRCUIT, FINAL CIRCUIT	An outgoing circuit connected to one way of a distribution board or a fuse board and intended to supply electrical energy, to one or more points, to current using appliances without the intervention of a further distribution fuse board other than a one-way board. It includes all branches and extensions derived from that particular way in the distribution board or fuse board.

CIRCUIT BREAKER	A device designed to open and close a circuit by non-automatic means and to open the circuit automatically on a predetermined overcurrent, without injury to itself when properly applied within its rating.
CIRCUIT BREAKER	A device used to break a circuit during over current or short circuit condition. An LV Circuit Breaker is used in a low voltage distribution system and an HV Circuit Breaker is used in a high voltage distribution system.
CORD, FLEXIBLE CABLE	A flexible cable having large number of strands of conductors of small cross-sectional area with a soft PVC insulation. Two flexible cords twisted together may be termed as twin flexible cord. However, some flexible cords are made following the style of a twin core PVC insulated copper cables but much soft and flexible.
CUTOUT	Any appliance for automatically interrupting the transmission of energy through a conductor when the current rises above some predetermined value. A cutout contains a part for holding either fuse wire (rectangular cross section type) or a part for holding tubular fuse (cylindrical body rectangular cross section type). (see FUSE)
DB	Distribution Board. This may be the box where the main incoming cable enters and terminates from the main service feed connection. The SDBs get feed from a DB.
DEMAND FACTOR	The ratio of the maximum demand of a system, or part of a system, to the total connected load of the system or the part of the system under consideration.
DUCT	A closed passageway formed underground or in a structure and intended to receive one or more cables which may be drawn in.
EARTH	The conductive mass of the earth, whose electric potential at any point is conventionally taken as zero.
EARTH ELECTRODE	A metal plate, pipe or other conductor electrically connected to the general mass of the earth.
EARTH LEAD WIRE	The final conductor by which the connection to the earth electrode is made.

EARTH CONTINUITY CONDUCTOR (ECC)	The conductor, including any clamp, connecting to the earthing lead or to each other, those parts of an installation which are required to be earthed. It may be in whole or in part the metal conduit or the metal sheath or armour of the cables, or the special continuity conductor of a cable or flexible cord incorporating such a conductor. ECCs of appropriate size must run from an MDB to its DBs, from a DB to its corresponding SDBs, from an SDB to the Switch Boards under this SDB, from an SDB to the BDBs if there are any, from a BDB to the Switch Boards under this BDB, from an SDB or a BDB to the Sockets under this SDB or BDB.
EDB	Emergency Distribution Board. This may be the box where the main incoming cable from the Emergency or Standby Generator Panel enters and. The ESDBs get feed from a EDB.
EFDB	Emergency Floor Distribution Board located in each of the floors of a multistoried building. The EDBs get feed from EFDB.
ENGINEER-IN- CHARGE	An engineer responsible for implementation/execution of the work of a building or a project. Such an engineer is expected to have significant knowledge in Electrical Engineering, Electrical Construction, Measurement, Codes and Practices of such work and availability of different materials needed for the construction.
FDB	Floor Distribution Board located in each of the floors of a multistoried building. The DBs get feed from FDB.
FUSE	A device that, by the fusion of one or more of its specially designed and proportioned components, opens the circuit in which it is inserted when the current through it exceeds a given value for a sufficient time. Fuse is generally made of fusible wires of appropriate ratings which is either mounted inside glass tubes or porcelain tubes or on a two terminal cutout.
FUSE SWITCH	A composite unit, comprising a switch with the fuse contained in, or mounted on, the moving member of the switch.
LIGHTING FITTING	A device for supporting or containing a lamp or lamps (for example, fluorescent or incandescent) together with any holder, shade, or reflector; for example, a bracket, a pendant with ceiling rose, or a portable unit.

INSULATION	Suitable non-conducting material, enclosing, surrounding or supporting a conductor. Usually PVC, polymer, specially treated rubber.
LIVE	Electrically charged so as to have a potential different from that of earth. Also known as ALIVE.
LUMINAIRE	A complete light fitting consisting of lamp, holder, starting gears, reflectors, housing and mounting accessories.
LT / LV and HT/ HV	LT or LV in this document indicates 230 Volt single phase and 400 volt 3 phase. HT or HV in this document indicates 11 kV Line to line 3 phase system.
MDB	Main Distribution Board. This is the distribution box where the main incoming cable enters and terminates from the main service feed connection of a large building. The FDBs get feed from MDB.
OVER-CURRENT	A current exceeding the rated current. For conductors, the rated value is the nominal current carrying capacity.
PANEL BOARD	A single panel or a group of panel units designed for assembly in the form of a single panel including buses, automatic overcurrent devices, and with or without switches for the control of light, heat, or power circuits, designed to be placed in a cabinet or cutout box placed in or against a wall or partition and accessible only from the front.
PLUG	A device carrying metallic contacts in the form of pins intended for engagement with corresponding socket contacts and arranged for attachment to a flexible cord or cable. A plug may contain tubular fuse inside it although some plugs do not contain fuse.
POINT (in wiring)	A termination of the fixed wiring intended for the connection of current using equipment e.g., a Light, a fan, an exhaust fan.
SDB	Sub- Distribution Board located in the same floor of a building and connected to the DB. The BDBs get feed from SDB.
SERVICE	The conductors and equipment required for delivering energy from the electric supply system to the wiring system of the premises served.

SWITCH	A manually operated device for closing and opening or for changing the connection of a circuit. A 5A SPST switch is used for the control of a Light or Fan point. A 5A SPDT switch is also used for the control of a Light or Fan point.
SWITCHBOARD	An assemblage of switchgear with or without instruments; the term, however, does not apply to a group of local switches on a final sub-circuit where each switch has its own insulating base.
SWITCHGEAR	Main switches cutouts or fuses, conductors and other apparatus in connection therewith, used for the purpose of controlling or protecting electrical circuits or machines or other current using appliances.

1.1.4 Voltage Ratings

The provisions of the Code specified in this Chapter covers installations utilizing nominal voltage not exceeding 415 V AC between conductors or 240 V AC to earth. The nominal voltage in Bangladesh is 230 volts AC single phase and 400 volts AC 3 phase.

1.2 Lighting and Illumination

1.2.1 Determination of Illumination Levels for Different Application (Principle of Lighting)

The essential features of an efficient lighting system are:

- (a) Visual comfort through adequate illumination of the working surface,
- (b) Prevention of glare,
- (c) Avoidance of shadows, and
- (d) Ease of maintenance.

The design of a lighting system shall involve:

- (a) careful planning of the brightness and colour pattern within both the working areas and the surroundings so that attention is drawn naturally to the important areas, so that details can be seen quickly and accurately, and so that the appearance inside the room is free from any sense monotony,
- (b) use of directional lighting to assist perception of task detail,
- (c) controlling direct and reflected glare from light sources to eliminate visual discomfort,

- (d) minimizing flicker from certain types of lamps and paying attention to the colour rendering properties of the light,
- (e) the correlation of lighting throughout the building to prevent excessive differences between adjacent areas, so as to reduce the risk of accidents, and
- (f) the installation of emergency lighting systems, wherever necessary.

The general impressions associated with different illuminance and colour appearances of light are shown in Table 8.1.1. The various colour rendering groups with examples of use are presented in Table 8.1.2.

Table 8.1.1: General Impressions Associated with Different Illuminance and Colour Appearances

Illuminance (lux)	Associated Impression (Colour Appearance)		
	Warm	Intermediate	Cool
≤ 500	Pleasant	Neutral	Cool
500 – 1000	Pleasant to Stimulating	Neutral to Pleasant	Cool to Neutral
1000 – 2000	Stimulating	Pleasant	Neutral
2000 – 3000	Stimulating to Unnatural	Pleasant to Stimulating	Neutral to Pleasant
≥ 3000	Unnatural	Stimulating	Pleasant

Table 8.1.2: Lamp Colour Rendering Groups

Colour rendering Group	Range of Index R_a	Colour Appearance	Examples of Use
1	$R_a \geq 85$	Cool Intermediate Warm	Textile industries, paint and printing industries Shops, hospitals Homes, hotels, restaurants
2	$70 \leq R_a < 85$	Intermediate	Offices, schools, department store, fine industrial work
3	$40 \leq R_a < 70$		Interiors where colour rendering is of comparatively minor importance

Note: Certain applications, e.g. colour matching, may be extremely critical with regard to the colour rendering properties of the lamps used. Here, the minimum colour rendering index used shall be 90.

1.2.2 Planning the Brightness Pattern

The brightness pattern seen within an interior is composed of three parts.

- (a) Brightness of the task itself.
- (b) Brightness of the immediate background of the task and
- (c) Brightness of the general surroundings of walls, ceiling, floor, equipment, furnishing etc.

1.2.2.1 The illumination of all work areas within a building shall be a minimum of 150 lux.

1.2.2.2 Where work takes place over the whole utilizable area of a room, the general illumination over that area shall be reasonably uniform and the diversity ratio of minimum to maximum illumination shall not be less than 0.7. This diversity ratio does not however take into account of the effects of any local lighting provided for specific tasks.

1.2.2.3 When the brightness appropriate to an occupation has been determined, the brightness of the other parts of the room shall be planned to give proper emphasis to visual comfort and interest. The recommended brightness ratios are shown in Table 8.1.3.

Table 8.1.3: Brightness Ratios between Task, Adjacent Sources and Surroundings

For high task brightness (above 100 cd/m²) Maximum ratio between task brightness and the adjacent sources like table tops Maximum ratio between task brightness and illumination of the remote areas of the room not being used as work areas	 3 to 1 10 to 1
For low and medium task brightness (below 100 cd/m²)	The task must be brighter than both the background and the surroundings; the lower the task brightness, the less critical is the relationship.

1.2.3 Lighting Calculations

1.2.3.1 In order to determine the necessary number of lamps and luminaires for a specified illumination level or the average illuminance obtained from a particular lighting design, the Lumen Method of calculation shall be employed.

1.2.3.2 Unless the reflection factors are known to the lighting designer, the triplet 0.7/0.5/0.3 for the reflectance of ceiling, walls and working plane respectively shall be used for offices and the triplet 0.7/0.5/0.1 for other premises. Typical reflection factors of smooth coloured surfaces are given in Table 8.1.4.

Table 8.1.4: Reflection Factors of Smooth Coloured Surfaces

Colour	Reflection Factor	Colour	Reflection Factor
Flat white	0.75 – 0.85	Light green	0.40 – 0.50
Ivory	0.70 – 0.75	Grey	0.30 – 0.50
Buff	0.60 – 0.70	Blue	0.25 – 0.35
Yellow	0.55 – 0.65	Red	0.15 – 0.20
Light tan	0.45 – 0.55	Dark brown	0.10 – 0.15

1.2.4 Recommended Illumination Values

The recommended values of illumination required for buildings of different occupancies, based on activity, are given in Tables 8.1.5 to 8.1.14. The initial illuminance should be higher than the recommended value as the illuminance drops below this value by the end of the cleaning and replacing period. A gradual transition (rather than a sudden change) of brightness from one portion to another within the field of vision is recommended to avoid or minimize glare discomfort.

1.2.5 Artificial Lighting to Supplement Daylight

Supplementary lighting shall be used when illumination from daylight falls below 150 lux on the working plane. For supplementary artificial lighting when daylight availability becomes insufficient, cool daylight fluorescent tubes with semi-direct luminaires are recommended. To ensure a good distribution of illumination, the mounting height should be between 1.5 and 2.0 m above the work plane with a separation of 2.0 to 3.0 m between the luminaires.

1.2.6 Selection of Appropriate Light Fittings

1.2.6.1 Light fitting

An electric lamp and its fitting accessories, reflector, diffuser, mounting brackets, suspenders etc., shall be regarded as one unit. During design, an appropriate type of light fitting shall be selected to match the requirement of desired distribution of light. While selecting light fittings having focus or aiming arrangements which enable the light distribution to be varied by adjustment of the lamp position, care should be taken to select the appropriate type of fitting with appropriate beam to serve the aimed lighting applications.

1.2.6.2 Classification of light fittings

Light fittings may be classified into five categories according to the proportion of the total light output in the lower hemisphere. These are:

- Direct fittings, giving 90-100 percent light downwards
- Semi-direct fittings, giving 60-90 percent downwards
- General diffusing fittings, giving 40-60 percent light downwards
- Semi-indirect fittings, giving 10-40 percent light downwards
- Indirect fittings, giving 0-10 percent light downwards

(a) Direct fittings: Direct fittings shall be used in situations where efficiency of illumination is the major criterion, while contrast of the light source with the surroundings, shadows, and direct/reflected glare may be considered to be of relatively minor importance.

(b) Semi-direct fittings: Semi-direct fittings shall be used in areas where it felt that the reduction of contrast resulting from the small indirect component of light directed towards the ceiling shall be sufficient for the purpose.

(c) General diffusing fittings: General diffusing fittings shall be used where, in addition to a substantial indirect component of light aiding materially to the diffused character of the general illumination, an upward component providing a brighter background against which to view the luminance, especially for interiors with light-colored ceiling and walls, is desirable.

Table 8.1.5: Recommended Values of Illumination for Residential Buildings

Area or Activity	Illuminance (lux)	Area or Activity	Illuminance (lux)
Dwelling Houses		Hotels	
Bedrooms		Entrance halls	150
General	70	Reception and accounts	300
Bed-head, Dressing table	250	Dining rooms (tables)	150
Kitchens	200	Lounges	150
Dining rooms (tables)	150	Bedrooms	
Bathrooms		General	100
General	100	Dressing tables, bed heads, etc.	250
Shaving, make-up	300	Writing rooms (tables)	300
Stairs	100	Corridors	70
Lounges	100	Stairs	100
Garages & Porches	100	Laundries	200

Area or Activity	Illuminance (lux)	Area or Activity	Illuminance (lux)
Basement Car Park	100	Kitchens	
Porches, Entrances	70	Food stores	100
Sewing and darning	600	Working areas	250
Reading (casual)	150	Goods and passenger lifts	70
Home work and sustained reading	300	Cloak-rooms and toilets	100
		Bathrooms	100
		Above mirror in bathrooms	300

Table 8.1.6: Recommended Values of Illumination for Educational Buildings

Area or Activity	Illuminance (lux)	Area or Activity	Illuminance (lux)
School and College Assembly halls		Offices	300
General	150	Staff rooms and common rooms	150
When used for examinations	300	Corridors	100
Platforms	300	Stairs	100
Class and Lecture Rooms		Gymnasium	100
Desks	300	General	150
Black boards	300	Matches	300
Embroidery and sewing rooms	500	Library	see Table 8.1.8
Laboratories	350	Living quarters	see Table 8.1.5
Art rooms	400		

Table 8.1.7: Recommended Values of Illumination for Health Care Buildings

Area or Activity	Illuminance (lux)	Area or Activity	Illuminance (lux)
Hospitals and Clinics		Hospitals and Clinics (contd.)	
Reception and waiting rooms	150	Doctor's examination rooms	150
Outpatient department	150	Radiology departments	100
Wards		Casualty	150
General	150	Stairs and corridors	100
Beds	150	Dispensaries	250
Operating theatres			
General	300		
Tables (with adjustable operation lamp lighting)			
Minor	2000		
Major	5000		

Table 8.1.8: Recommended Values of Illumination for Assembly Buildings

Area or Activity	Illuminance (lux)	Area or Activity	Illuminance (lux)
Cinemas		Theatres	
Foyers	150	Foyers	150
Auditorium	100	Auditorium	70
Corridors	100	Corridors	90
Stairs	150	Stairs	150
Libraries		Indoor Sports Centre	
Shelves (stacks)	150	Halls	200
Reading rooms (newspapers and magazines)	200	Swimming pools	250
Reading tables	300	Lawn or table tennis, badminton, volley ball	
Book repair and binding	300	Tournament	300
Cataloguing, sorting and stock rooms	150	Club	200
Museums and Art Galleries		Recreational	150
Museums		Shooting ranges	
General	200	On target	300
Displays	special lighting	Firing point	200
Art galleries		Range	100
General	250	Football	500
Paintings	250		
Restaurant			
Dining rooms	150		
Cash desks	300		
Self-carrying counters	300		
Kitchens	200		
Cloak-rooms and toilets	100		

Table 8.1.9: Recommended Values of Illumination for Business and Commercial Buildings

Area or Activity	Illuminance (lux)	Area or Activity	Illuminance (lux)
Airport Building		Doctor's Surgeries	
Reception areas (desks)	300	Waiting rooms and consulting rooms	150
Baggage, customs and immigration halls	300	Corridors	70
Circulation areas, lounges	200	Stairs	100
Banks		Eyesight testing (acuity) wall charts and near vision types	450
Counter, typing and accounting book areas	300	Jewellery and Watch-Making	
Public areas, lobby	150	Fine processes	700
Offices	200	Minute processes	3000
Book Binding		Gem cutting, polishing and setting	1500
Pasting, punching and stitching	200	Laundries and Dry-Cleaning Works	
Binding and folding and miscellaneous machines	300	Receiving, sorting, washing, drying, ironing (calendaring) and dispatch	200
Finishing, blocking and inlaying		Dry-cleaning and bulk machine work	200
Dental Surgeries		Fine hand ironing, pressing, inspection, mending and spotting	300
Waiting rooms	300		
Surgeries	150		
General	300		
Chairs	special lighting		
Laboratories	300		
Offices		Offices (contd.)	
Entrance lobby and reception areas	150	Stairs	100
Conference rooms and executive offices	300	Lift landings	150
General offices	300	Telephone exchanges	
Business machine operation	450	Manual exchange rooms (on desk)	200
Drawing office		Main distribution frame room	150
General	300	Shops and Stores	
Boards and tracing	450	General areas	150 to 300
Corridors and lift cars	70	Stock rooms	200
		Display windows	500

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes

Area or Activity	Illuminance (lux)
Aircraft Factories and Maintenance Hangars	
Stock parts productions	450
Drilling, riveting, screw fastening, sheet aluminium layout and template work, wing sections, coving, welding, sub-assembly, final assembly and inspection	300
Maintenance and repair (hangars)	300
Assembly Shops	
Rough work, for example frame assembly and assembly of heavy machinery	150
Medium work, for example machined parts, engine assembly	300
Fine work, for example radio and telephone equipment, typewriter and office machinery assembly	700
Very fine work, for example assembly of very small precision mechanisms and instruments	1500
Automobile Manufacturing	
Frame assembly	200
Chassis assembly line	300
Final assembly and inspection line	600

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Body Manufacturing	
Parts	200
Assembly	300
Finishing and inspection	700

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Automobile Service Garages	
Repairs	250
Active traffic areas	100
Storage	25
Bakeries	
General working area	150
Decorating and icing	250
Breweries and Distilleries	
General working areas	150
Brew house, bottling and canning plants	200
Bottle inspection	special lighting
Carpet Factories	
Winding and beaming	200
Designing, Jacquard card cutting, setting pattern, tufting, topping, cutting, hemming and fringing	300
Weaving, mending and inspection	450
Chemical Works	
Hand furnaces, boiling tanks, stationary driers, stationary and gravity crystallizers	150
Mechanical furnaces, evaporators, filtration, mechanical crystallizers, bleaching	200
Tanks for cooking, extractors, percolators	200

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Chocolate and Confectionery Factories	
Mixing, blending and boiling	150
Chocolate husking, winnowing, fat extraction, crushing and refining, feeding, bean cleaning, sorting, milling and cream making	200
Hand decorating, inspection, wrapping and packing	300
Clay Products and Cements	
Grinding, filter presses, kiln rooms moulding, pressing, cleaning and trimming	150
Enameling	150
Colour and glazing - rough work	400
Colour and glazing - fine work	750
Clothing Factories	
Matching-up	450
Cutting, sewing	
Light	300
Medium	450
Dark	700
Inspection	
Light	450
Medium	1000
Dark	1500
Hand Tailoring	
Light	450
Medium	1000
Dark	1500
Pressing	300

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Dairies	
General working areas	200
Filling and bottle inspection	450
Cooling equipment	150
Laboratories	450
Pasteurizers	150
Separators	150
Electrical Industries	
Impregnating	250
Winding and insulating	500
Assembly works	
Fine	500
Very fine	750
Testing	500
Electricity Generating Stations (Indoor Locations)	
Turbine halls	150
Auxiliary equipment, battery rooms, blowers, auxiliary generators, switchgear and transformer chambers	150
Boiler house (including operating floors) platforms, coal conveyors, pulverizers, feeders, precipitators, soot and slag	100 to 150
Boiler house and turbine house	150
Basements	100
Conveyor house, conveyor gantries and junction towers	80 to 100
Emergency lighting - all areas	30
Control rooms	
Vertical control panels	200 to 300
Control desks	300
Rear of control panels	150
Switch houses	150

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Electricity Generating Stations (Outdoor Locations)	
Switchyard	70
Conveyors	70
Fuel oil delivery headers	70
Oil storage tanks	70
Cat-walks	70
Platforms, boiler and turbine decks	70
Transformer and outdoor switchgear	100
Emergency lighting - all areas	50
Flour Mills	
Rolling	150
Sifting	150
Packing	150
Purifying	150
Product control	300
Cleaning screens, man lifts, aisleways and walkways, bin checking	100
Forge Shops and Foundries	
Forge shop	150
Annealing (furnaces)	150
Cleaning	100
Core making (fine)	300
Core making (medium)	150
Grinding and chipping	200
Inspection (fine)	1000
Inspection (medium)	300
Moulding (medium)	300
Moulding (large)	150
Pouring	150
Sorting	200
Cupola	100
Shake out	150

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Garages	
Parking areas (interior)	70
Washing and polishing, greasing, general servicing and pits	200
Gas Works	
Retort houses, oil gas plants, purifiers, coke screening and coke handling plants (indoor)	70
Governor, meter, compressor, booster and exhaustor houses	100
Open type plants	
Cat-walks	20
Platforms	50
Glass Works	
Furnace rooms, bending, annealing lehrs	100
Mixing rooms, forming (blowing, drawing, pressing and rolling)	150
Cutting to size, grinding, polishing and toughening	200
Finishing (bevelling, decorating, etching and silvering)	300
Brilliant cutting	
General	200
Fine	500
Inspection, etching and decorating	500
Glove Making	
Pressing, knitting, sorting and cutting	300
Sewing	
Light	300
Medium	450
Dark	700
Inspection	
Light	450
Medium	1000
Dark	1500

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Hosiery and Knitwear	
Circular and flat knitting machines, universal winders, cutting out, folding and pressing	300
Lock-stitch and overlocking machines	
Light	300
Medium	450
Dark	700
Mending	1500
Examining and hand finishing, light, medium and dark	700
Linking or running on	450
Iron and Steel Works	
Manufacturing by open hearth	
Stock yard	20
Charging floor	100
Slag pits	100
Control platforms	100
Mould yard	25
Hot top	100
Hot top storage	100
Stripping yard	100
Scrap stockyard	20
Mixer building	100
Calcining building	50
Rolling mills	
Blooming, slabbing, hot strip, hot sheet	100
Cold strip, plate	150
Pipe, rod, tube, wire drawing	200
Merchant and sheared plate	100

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Tin plate mills	
Tinning and galvanizing	200
Cold strip rolling	200
Motor room, machine room	150
Sheet metal works	
Miscellaneous machines, ordinary bench work	200
Pressing, folding, stamping, shearing, punching and medium bench work	200
Tin plate and galvanized sheet inspection	500
Structural Steel Fabrication	
Fabrication and general work	150
Marking and cutting	300
Plating shops	
Vat, baths, buffing and polishing	200
Final buffing and polishing	500
Leather Manufacturing	
Cleaning, tanning and stretching, vats	150
Cutting, fleshing and stuffing	200
Finishing and scarfing	200
Machine Shops	
Rough bench and machine work	150
Medium bench and machine work, ordinary automatic machines, rough grinding medium buffing and polishing	300
Fine bench and machine work, fine automatic machines, medium grinding, fine buffing and polishing	700
Extra fine bench and machine work, grinding fine work	1000

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Paint Works	
General, automatic processes	200
Special batch mixing	450
Colour matching	700
Paper Manufacturing	
Beaters, grinding, calendaring	150
Finishing, cutting, trimming, paper making machines	200
Hand counting, wet end of paper machine	350
Paper machine reel, paper inspection and laboratories	500
Rewinder	500
Paper box manufacturing	200
Pharmaceuticals and Fine Chemical Works	
Raw material storage	200
Grinding, granulating, mixing and drying, tableting, sterilizing, preparation of solutions, filling, labelling, capping, wrapping and cartoning	300
Control laboratories and testing	300
Fine chemical processing	200
Fine chemical finishing	300
Printing Industries	
Photo-engraving	
Block-making, etching and staging	200
Finishing, routing and proofing	300
Masking and tint laying	300
Colour Printing	
Inspection area	700

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Type foundries	
Matrix making, dressing type	250
Front assembly and sorting	200
Hand casting	300
Machine casting	200
Printing plants	
Machine composition and imposing stones	200
Presses	300
Composition room	450
Proof reading	300
Colour inspection and appraisal	1000
Electrotyping	
Block-making, electroplating, washing and baking	200
Moulding, finishing and routing	300
Rubber Tyre and Tube Manufacturing	
Stock preparation	
Plasticating, milling	100
Calendering	150
Fabric preparation	
Stock cutting, bead building	250
Tube tubing machines	250
Tread tubing machines	250
Tyre building	
Solid tyre	150
Pneumatic tyre	250
Curing department	

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Tubing curing, casing curing	350
Final Inspection	
Tube, casing	1000
Wrapping	200
Shoe Manufacturing (Leather)	
Cutting and stitching	
Cutting tables	450
Marking, buttonholing skiving, sorting and counting	450
Stitching	
Light materials	300
Dark materials	1000
Making and finishing	
Nailers, sole layers, welt beaters and scarfers, trimmers, welters, lasters, edge setters, slugers, randers, wheelers, treers, cleaning, spraying, buffing, polishing, embossing	600
Shoe Manufacturing (Rubber)	
Washing, coating, mill run compounding	100
Varnishing, vulcanizing, calendering, upper and sole cutting	300
Sole rolling, lining, making and finishing process	500
Soap Factories	
Kettle houses and ancillaries, glycerine evaporation and distillation and continuous indoor soap making	
General areas	150
Control panels	200 to 300
Batch or continuous soap cooling, cutting and drying, soap milling and plodding	

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
General areas	150
Control panels and key equipment	200 to 300
Soap stamping, wrapping and packing, granules making, granules storage and handling, filling and packing granules	
General areas	150
Control panels and machines	200 to 300
Edible products processing and packing	200
Textile Mills (Cotton)	
Bale breaking and picking	150
Carding and drawing	200
Slubbing, roving, spinning, spooling	200
Beaming and slashing on comb	
Grey goods	200
Denims	300
Weaving	
Patterned cloth and fine counts, light	300
Patterned cloth and fine counts, dark	500
Plain grey cloth	200
Cloth inspection	700
Textile Mills (Silk and Synthetics)	
Manufacturing	
Soaking, fugitive tinting, conditioning, setting or twist	200

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Textile Mills (Silk and Synthetics) (contd.)	
Winding, twisting, rewinding and coining, quilting and slashing	
Light thread	200
Dark thread	300
Warping (silk or cotton system) on creel, on running ends, on reel, on beam, on warp at beaming	300
Healding (drawing-in)	700
Weaving	300 - 500
Inspection	1000
Textile Mills (Woollen and Worsted)	
Scouring, carbonizing, testing, preparing, raising, brushing, pressing, back-washing, gilling, crabbing and blowing	150
Blending, carding, combing(white), tentering, drying and cropping	200
Spinning, roving, winding, warping, combing (coloured) and twisting	450
Healding (drawing-in)	700
Weaving	
Fine worsteds	700
Medium worsteds and fine woollens	450
Heavy woollens	300
Burling and mending	700
Perching	
Grey	700
Final	2000

Table 8.1.10: Recommended Values of Illumination for Industrial Buildings and Processes (contd.)

Area or Activity	Illuminance (lux)
Wood Working	
Rough sawing and bench work	150
Sizing, planing, rough sanding, medium machine and bench work glueing, veneering	200
Fine bench and machine work, fine sanding and finishing	300

Table 8.1.11: Recommended Values of Illumination for Storage Buildings

Area or Activity	Illuminance (lux)
Storage Rooms of Ware House	
Inactive	50
Rough bulky	50
Medium	100
Fine	250

Table 8.1.12: Recommended Values of Illumination for Outdoor Stadiums Colour TV broadcasting

Area or Activity	Illuminance (lux)
Football Stadium	1700
Cricket Stadium	2200

Table 8.1.13: Recommended Values of Illumination for Outdoor open yards

Area or Activity	Illuminance (lux)
Outdoor Car Parking Lot	100
Airport Apron	200
Container Yard	200
Jetty	250

Table 8.1.14: Recommended Values of Illumination for Roads

Area or Activity	Illuminance (lux)
Roads inside a Housing Area	50- 100
Roads in a Congested Town / City Area	50- 100
Wide Roads with dividers	100 - 120
Avenues	100 - 120

- (d) Semi-indirect fittings: Semi-indirect fittings shall be used when a comfortable brightness ratio between the ceiling and the luminaire is desirable but an efficiency of illumination, higher than that obtainable from indirect fittings is required.
- (e) Indirect fittings: Indirect fittings shall be used in situations where an environment of evenly distributed illumination is to be achieved.
- (f) Angle lighting: Lighting on vertical surfaces shall be done avoiding shadows using interior or exterior light fitting of appropriate type concentrated source light fitting depending upon the place and the color tone required. However, if creating shadows is necessary then appropriate type concentrated source light fitting should be chosen depending upon the place.

1.2.7 Illumination of Exit Signs and Means of Escape

1.2.7.1 Exit signs

- (a) All required exit signs shall be illuminated at night, or during dark periods within the area served.
- (b) Exit signs may be illuminated either by lamps external to the sign or by lamps contained within the sign. The source of illumination shall provide not less than 50 lux at the illuminated surface with a contrast of not less than 0.5. Approved self-luminous signs which provide evenly illuminated letters having a minimum luminance of 0.2 cd/m^2 may also be used.
- (c) Exit signs within an area where the normal lighting may be deliberately dimmed or extinguished, such as places of entertainment, shall be illuminated either by lamps contained within the sign or by approved self-luminous signs.

1.2.7.2 Means of Escape Lighting

- (a) The means of escape and exit access in buildings requiring more than one exit shall be equipped with artificial lighting. The lighting facilities so installed shall provide the required level of illumination continuously during the period when the use of the building requires the exits to be available.
- (b) The intensity of illumination at floor level by means of escape lighting shall not be less than 10 lux, except that the minimum required floor level illumination of aisles in assembly halls, theatres and cinema during projection of motion or still pictures by directed light shall not be less than 2 lux.
- (c) The illumination of exit signs and the lighting of the means of escape and exit access shall be powered by an alternate or emergency electrical system to ensure continued illumination for a duration of not less than 30 minutes after the failure of primary power supply.

1.2.8 Selection of Appropriate Type of Lamp

It is important to select appropriate types of lamps for each purpose. The lamps which are used for various purposes are:

(i) General Service Lamps (GLS)/Incandescent Lamps

General Service Lamps (GLS) are well known Incandescent Lamps. These are available in a number of watt ratings. However, most commonly used ratings are 40 W, 60 W, 100 W, 150 W and 200 W rated lamps are also used for special applications. These types of lamps produce heat and should be minimized while selecting for a particular design because other better choices are available nowadays. For kitchen, cooking areas of a hotel, serving counters of a food shop or hotel, porch these are often essential because of the color temperature. For living room, toilet, corridor, veranda, bed room these have been used for long but from energy saving point of view other lamps which perform better in terms of light output to watts ratio may be used. In general, the GLS type of lamp may be used for almost all interior and exterior applications but from energy saving point of view other lamps which perform better in terms of light output to watts ratio should be used as much as possible.

(ii) Fluorescent Lamps (FL):

These are available in 20 W and 40 W ratings. These lamps are strongly recommended for reading room, educational buildings, laboratories, office room, commercial space applications, factory illumination, illumination of areas around industrial plant and machineries, exterior lighting applications.

40 W FL should be used wherever possible because a 40 W FL is more energy efficient compared to a 20 W FL. These are long life lamps, have wide applications and are advantageous in many respects.

(iii) Compact Fluorescent Lamp(CFL) Energy Saving Lamps:

CFL Lamps are available in a number of watts ratings e.g., 4 W, 7 W, 11 W, 14 W and 24 W. CFLs have been finding wide application for almost all applications because of their high Light output to watts ratio and also because of the attractive light color. CFL lamps, therefore, should be widely used for energy saving purpose.

However, for reading areas, library areas, educational buildings, laboratories fluorescent lights give better service and thus should be selected for these purposes. It is worthwhile mentioning that Fluorescent lamps with high quality ballasts closely meet the energy saving purpose.

(iv) LED Lights:

Compact light fitting formed using a cluster of white LED is currently used to replace a conventional lamp. An LED operates at very small amount of voltage. These are good for lighting, energy efficient, have almost negligible heat dissipation. These are good for relaxed environment interior lighting. LED lights are becoming more and more popular because of much lower power consumption compared to other lamps.

(v) Halogen lamp:

Halogen lamps are used for spot lights, decorative lights in shops and commercial spaces, inside show cases, stage lighting, and projection lights. Due to high temperature rise and UV light output these should be avoided for interior lighting unless needed.

(vi) Mercury Vapour Lamp

These have been widely used for shops, streets, for high bay lighting, warehouse lighting and similar special lighting. Most likely, this type of lamp will be discontinued within next five to six years due some of its ill effects. Metal halide lamp is coming up as a better alternative to mercury vapour lamp.

(vii) Metal Halide Lamp:

These are available in a number of watts ratings e.g., 150 W, 200 W, 250 W, 500 W, 1000 W, 2000 W. Good for exterior lighting, indoor and outdoor athletic facilities, for high bay lighting, warehouse lighting. These are required where massive flood lighting is required from high altitudes for coverage of large areas.

(viii) HP Sodium Lamp:

These are available in a number of watts ratings e.g., 40 W, 50 W, 70 W, 100 W, 150 W, 250 W, 400 W, 1000 W. Good for exterior lighting, Lighting for areas where higher concentration of vehicles and people exist e.g., Street lighting, building exterior lighting, security lighting.

(ix) Low Pressure Sodium Lamp:

For outdoor lighting such as street lights and security lighting where faithful color rendition is considered unimportant. This type of lamps may be used for street lights, observatory, parking lot and similar types of areas.

(x) Solar PV Cell Powered LED Lights:

These fittings require a solar panel, a storage battery system apart from the cluster of LEDs. For outdoor lighting such as street lights, security lighting, outdoor parking area, this type of light fitting may be used.

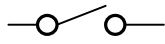
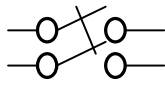
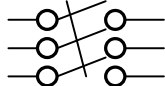
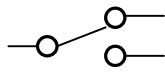
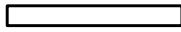
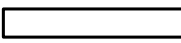
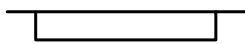
1.3 ELECTRICAL AND ELECTRONIC INSTALLATIONS IN BUILDINGS

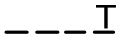
1.3.1 List of Symbols used for Electrical Drawings

A list of general graphical symbols used for electrical drawings is given in Table 8.1.15. These are given as guideline. In case of justified reasons a designer may modify certain symbol.

Table 8.1.15: Symbols used for Electrical Drawings

Serial No.	Description	Symbol
1	Main Distribution Board (MDB)	
2	Floor Distribution Board (FDB)	
3	Distribution Board (DB)	
4	Sub-distribution Board (SDB)	
5	Branch Distribution Board (BDB)	
6	Switch Board (SB)	
7	Telephone Outlet (PSTN)	
8	Telephone Outlet (PABX)	
9	Change over switch	
10	Energy meter	
11	Ammeter	
12	Voltmeter	
13	Power factor meter	
14	Circuit breaker	
15	Fuse	
16	Ceiling mounted Incandescent light fitting	
17	Wall mounted bracket light fitting	

Serial No.	Description	Symbol
18	Ceiling fan	
19	Exit light pendant	
20	Exit light-wall mounted	
21	2 pin socket Outlet (single phase)	
22	3 pin 13A switched socket Outlet (single phase)	
23	Weatherproof and waterproof socket outlet	
24	SPST Single – pole, one-way switch	
25	DPST Two - pole, one-way switch	
26	TPST Three - pole, one-way switch	
27	SPDT Two – way switch	
28	Push button switch	
29	Buzzer	
30	Single fluorescent lamp on ceiling	
31	Double fluorescent lamp on ceiling	
32	Double fluorescent lamp on wall	
33	Spot light	

Serial No.	Description	Symbol
34	Wall mounted bracket fan	
35	Exhaust fan	
36	Pull box	
37	TV socket outlet	
38	Fire Alarm bell	
39	Fire detector	
40	Smoke detector	
41	Speaker	
42	Microphone	
43	Conduit, concealed in ceiling or in wall	
44	Conduit, concealed in floor or through under ground	
45	Telephone conduit	
46	Television antenna conduit	
47	Earth Electrode	

1.3.2 Estimating the Load of a Building/Complex

Estimating the total load of a building has to be started with the listing of the connected loads in a building. The steps are to list the loads in each of the rooms, in each of the flats/offices of a floor, in each of the floors and the load of the total building. In this way an account of the total building area/the total complex has to be prepared. Loads of the Lift(s), water pump(s), bulk ventilating system in the basement and any other equipment installed in the building must also be added. For completing the load calculation, practical value of appropriate diversity factors will have to be applied at each stage.

Estimating the total load of a complex consisting of a number of buildings has to be started with the listing of the connected load of each of the buildings, they are lighting load, water pump and any other equipment installed in the complex. For completing the load calculation, practical value of appropriate diversity factors among the buildings will have to be applied.

1.3.2.1 Maximum demand and diversity

Two items need to be determined, which are: (i) Maximum demand and (ii) Diversity factor. These are needed in completing the load calculation and in the computation of current.

In determining the maximum demand of an installation or parts thereof, diversity shall be taken into account. Appendix A gives some information on the determination of the maximum demand of an installation and includes the current demand to be assumed for commonly used equipment together with guidance on the application of allowances for diversity.

1.3.2.2 Estimation of load in kW, in kVA and in Amperes

An estimation of loads is necessary initially for design purposes and later for keeping a track of the growth of load. Estimation of loads means estimation of watts or kilowatts in small scale. In bigger scale the kVA is assessed together with the power factor. A calculation of current is then to be performed for the selection of breakers/fuses and the current carrying cables.

1.3.2.3 Estimation of electrical load in Watts

Energy efficient and energy saving should be considered in estimating the electrical load, the watts rating of individual equipment/fittings connected to the system need to be listed and added. Typical watt ratings of some of the equipment/fittings are shown in Table 8.1.16 which may be used for estimation if the actual values are not known or specified.

Table 8.1.16: Estimated Load for Different Fittings/Fixtures

Type of Fitting/Fixture	Ratings in Watts	Type of Fitting/Fixture	Ratings in Watts
CFL	5-65	15 A Socket outlets	1500
LED and Solar Panel Powered LED Security /Street Lights	10-60	Microwave Oven (domestic)	1200- 1500
Fluorescent lamp with accessories:		Washing machine (domestic)	350-500
Nominal length 600 mm	20	Television (medium size)	120-200

Type of Fitting/Fixture	Ratings in Watts	Type of Fitting/Fixture	Ratings in Watts
Nominal length 1200 mm	40	Computer (without printer)	200
Photo copiers	1200- 1500	Computer with printer	700-800
Ceiling fans	100 (Max)	Window type A.C. machine (12000 BTU/hr)	1500
Electric	1500	Split type A.C. machine (12000 BTU/hr)	1300
Table fans	85 (Max)	Geyser (water heater, domestic)	1000- 1200
Pedestal fans	120 (Max)	Toaster (domestic)	800-1000
Exhaust fans	100 (Max)	Electric calendar	700-1000
5A socket outlets	300		

1.3.2.4 Calculation of current

For the calculation of current (for the selection of cables and breakers) of the fluorescent lamps the ratings are to be multiplied by a factor of 1.65 to take care of the power factor and the starting current situation.

For the calculation of current (for the selection of cables and breakers) of the ceiling fans, table fans, pedestal fans, exhaust fans the ratings are to be multiplied by a factor of 1.65 to take care of the power factor and the starting current situation.

For the calculation of current (for the selection of cables and breakers) of the small inductive loads (up to 1.0 kW) the ratings are to be multiplied by a factor of 1.65 to take care of the power factor and the starting current situation. The factor shall be higher for higher rated motors.

1.3.2.5 Minimum load densities

While estimating the electrical load, the minimum load densities to be considered are those shown in Table 8.1.17.

1.3.3 Fittings, Fixtures and Accessories

Switch boards with back boxes and cover plates, ceiling roses, socket outlets with back boxes, plugs, light fittings, fans, pull boxes with cover plates have been put in this category, although there may be other items which may be included under electrical accessories related to electrical and electronic installations in buildings.

1.3.3.1 Switch boards

Tumbler switches have been used for surface wiring and piano switches have been used for concealed wiring. Now a day piano switches are also used with surface wiring. Piano switches are mounted on either a plastic back box or a metal back box. These piano switches are available in gangs. The other alternative is to have piano switches mounted on a Perspex or Ebonite sheet which is then mounted on a metal back box.

The Switches must conform to the relevant BS standard. The minimum ampere rating of switch shall not be below 5 A.

Switches may be Single Pole Single Throw (SPST) or Single Pole Double Throw (SPDT) depending on the operation. For some application Double Pole Single Throw (DPST) and Double Pole Double Throw (DPDT) are also available. Usually the DPST switches are made for 10 A, 15 A and 20 A rating.

The phase (Live) wire (Brown PVC insulated cable) connection to the point must go through the switch.

The metal / sheet steel back boxes of a switch board must have an earthing terminal to terminate the Earth Continuity Conductor (ECC) coming from a BDB or an SDB.

Table 8.1.17: Minimum Load Densities

Type of Occupancy	Unit Load (Watts/m ²)	
	Non A/C	A/C
Residence/ Dwelling : Single family	20	75
Residence/ Dwelling : Multi-family (other than hotels)	20	75
Hospitals	32	80
Hotels, including apartment house (excluding any provisions for electric cooking)	24	75
Office and commercial multi-storeyed buildings	28	75
Industrial building (excluding the loads for machines)	16	-

Type of Occupancy	Unit Load (Watts/m ²)	
	Non A/C	A/C
Departmental stores	28	75
Banks	20	75
Restaurants (excluding any provisions for electric cooking)	16	75
Barber shops and beauty parlours	32	75
Schools and colleges	12	70
Parking area in commercial buildings	4	-
Warehouses, large storage areas	2	-

1.3.3.2 Socket outlets and plugs

In general, all socket outlets must be switched (combined) and shuttered.

(a) General requirements of socket outlets

Socket outlets shall be 13 A switched shuttered 3 pin flat pin type. All socket outlets must be switched (combined) and shuttered and shall be for 3 pin Flat pin type (rectangular cross section) 13 A plugs fitted with tubular fuse.

The corresponding plugs must be fitted with fuse. The maximum fuse rating shall be 13A for 13A Sockets. The fuse rating may be smaller depending upon the current rating of the appliances used.

The phase wire (Brown cable) shall be connected to the L terminal of the socket outlet through the combined switch and the neutral wire (Blue cable) shall be directly connected to the N terminal of the socket. Earth Continuity Conductor (ECC) (Yellow + Green bi-colour cable) for such a socket outlet shall be connected to the Earth terminal of the socket.

The plug for each 13A socket outlet provided in a building for the use of domestic appliances shall be provided with its own individual fuse. The feed cables for such a circuit must have fuse or miniature circuit breaker (MCB) at the originating point in the Distribution Board or Sub-Distribution Board or Branch Distribution Board. For some high current applications, additional fuses/ circuit breakers adjacent to the sockets are recommended.

Each socket outlet shall also be controlled by a switch which shall normally be located immediately adjacent thereto or combined therewith.

The phase (Live) wire (Brown PVC insulated cable) connection to the socket outlet must be through the switch.

Copper size of the Earth Continuity Conductor (ECC) for such a socket outlet shall not be smaller in size than 1.5 mm² PVC insulated cable.

(b) 15 A/20 A rated socket outlets

(c) Round pin socket outlets of 15 A/20 A rating may be used for air conditioner outlets and water heater outlets under special circumstances, for air conditioner outlets (requiring 15 A or 20 A), 15 A/20 A rated socket outlets for round pin plugs may be used along with a circuit breaker or fuse protection in a box adjacent to the sockets..

Each 15 A/20 A socket outlet provided in a building for the use of domestic appliances such as air-conditioner, water cooler, etc. shall be provided with its own individual fuse. The feed cables for such a circuit must have fuse or miniature circuit breaker (MCB) at the originating point in the Distribution Board or Sub-Distribution Board or Branch Distribution Board. For some high current applications, additional fuses/circuit breakers adjacent to the sockets are recommended.

Each socket outlet shall also be controlled by a switch which shall normally be located immediately adjacent to the Socket or shall be combined with the Socket.

The corresponding plugs for 15 A should be fitted with fuse. The maximum fuse rating shall be 15 A for 15 A Sockets. For a 15 A rated socket outlet a 15 A rated fuse or a 15 A circuit breaker must be placed adjacent to the socket.

For a 20 A rated socket outlet a 20 A rated fuse or a 20 A circuit breaker must be placed adjacent to the socket.

Wiring for sockets shall be radial type of wiring. However, ring type wiring may be used by strictly following the rules given in IEE Wiring regulations BS 7671 and by using appropriate size of cable.

(d) Earth Continuity Conductor (ECC) for a socket

The ECC for a socket outlet shall not be smaller in size than 1.5 mm² PVC insulated annealed copper cable. The colour of the ECC cable insulation shall be Yellow + Green bi-colour.

(e) Mounting height of a three pin switched socket outlet

Three pin switched shuttered socket outlets shall be mounted on a wall at a height 250 mm above floor level. Switched shuttered socket outlets are essential for safety in particular for the safety of infants.

For certain applications like computers, printers, UPS, IPS such sockets may be mounted at a higher level for the ease of operation.

(f) Restriction on mounting socket outlets in wet places

No socket outlets shall be provided inside bath rooms/toilets or any other place where floor may remain wet.

(g) 5 A rated 2 pin socket outlets

5 A rated 2 pin socket outlets may be used along with the light and fan switch boards only. Such sockets shall not be used as socket outlets at the skirt level.

(h) Number of socket outlets in a room/in a building

The number of socket outlets in a building depends upon the specific requirements of occupants and the type of building. Adequate number of 13 A switched flat pin (rectangular cross section pin) shuttered socket outlets shall be provided and arranged around the building to cater to the actual requirements of the occupancy.

15 A round pin (rectangular cross section pin) socket outlets shall be provided for specially Air-conditioners and water heaters of such ratings only.

For residential buildings, the minimal guidelines given in Table 8.1.18 shall be used to determine the required number of 13 A switched flat pin (rectangular cross section pin) shuttered socket outlets, when actual requirements cannot be ascertained. All socket outlets shall conform to BDS 115.

Table 8.1.18: Minimum Number of 13 A flat pin Socket Outlets

Location	No. of Switch Socket Outlets
Bed room	2
Living room	3
Drawing room	3
Dining room	1
Toaster/Snack toaster	1
Kitchen	1
Bathroom	0
Verandah	1
Refrigerator	1
Air-conditioner	one for each room

(i) Restriction on installation of two socket outlets in room fed from two phases

Installation of two socket outlets in a room fed from two different phases should be avoided as far as possible. However, in unavoidable cases, the minimum distance between two such socket outlets in a room fed from two different phases must not be less than 2 m under any circumstances.

(j) Exterior/outdoor sockets

Socket outlets in exposed places where chances of dripping/falling rain water exist should not be placed. In case of necessity, weather proof/waterproof covered socket outlets may be mounted with appropriate precautions. In such a case the back box should preferably be of bakelite or Acrylic or plastic material.

(k) Exterior/outdoor switches

Switches in exposed places where chances of dripping/falling rain water exist should not be placed. In case of necessity, weather proof/waterproof covered switches may be mounted with appropriate precautions. In such a case the back box should preferably be of bakelite or Acrylic or plastic material.

1.3.3.3 Ceiling rose

A ceiling rose is needed for terminating the point wiring for a Light or a Fan in the ceiling.

- (a) A ceiling rose shall not be installed in any circuit operating at a voltage normally exceeding 250 volts.
- (b) Normally, a single pendant be suspended from only one ceiling rose using a flexible cord. A ceiling rose shall not be used for the attachment of more than one outgoing flexible cord unless it is specially designed for multiple pendants.
- (c) A ceiling rose shall not contain a fuse terminal as an integral part of it.
- (d) The ceiling rose shall conform to BS 67.
- (e) Luminaire supporting couplers are designed specifically for the mechanical support as well as for the electrical connection of luminaires and shall not be used for the connection of any other equipment.

1.3.3.4 Light fitting

Switches shall be provided for the control of every light fitting. A switch may control an individual light point or a group of light points.

Where control at more than one position is necessary for a lighting fitting or a group of lighting fittings, as many two-way or intermediate switches may be provided as the required number of control positions.

In industrial premises light fittings shall be supported by suitable pipe/conduits, brackets fabricated from structural steel, steel chains or similar materials depending upon the type and weight of the fittings. Where a lighting fitting is to be supported by one or more flexible cords, the maximum weight to which the twin flexible cords may be subject are shown in Table 8.1.19.

Table 8.1.19: Maximum Permissible Weight to which Twin Flexible Cords may be Subject

Nominal Cross-sectional Area of Twin Flexible Cord (mm ²)	Number and Diameter (mm) of Wires	Maximum Permissible Weight (kg)
0.5	16/0.2	2
0.75	24/0.2	3
1.0	32/0.2	5
1.5	48/0.2	5.3
2.5	80/0.2	8.8
4	128/0.2	14

For a Light fitting with shade, no flammable shade shall form part of the light fitting and the shade shall be well protected against all risks of fire. Celluloid shade or lighting fitting shall not be used under any circumstances.

(a) Lighting point

At each fixed lighting point one of the following accessories shall be used

- (i) one ceiling rose conforming BS 67
- (ii) one luminaire supporting coupler conforming BS 6972 or BS 7001
- (iii) one batten lamp holder conforming BS 7895, BS EN 60238 or BS EN 61184
- (iv) one luminaire designed to be connected directly to the circuit wiring
- (v) one suitable socket-outlet
- (vi) one connection unit conforming BS 5733 or BS 1363-4.

A lighting installation shall be appropriately controlled e.g., by a switch or combination of switches to BS 3676 and/or BS 5518, or by a suitable automatic control system, which where necessary shall be suitable for discharge lighting circuits.

(b) Wires/cables used inside light fittings and any other fitting

Wires/cables used inside a light fitting or any other fittings are mostly flexible types. In some cases single core PVC insulated wiring cables mostly 1.5 mm² are used. In such cases the cables must be of high quality in terms of insulation and must have appropriate copper cross section. Such cables are usually terminated in a ceiling rose.

1.3.3.5 Fans

(a) Ceiling fan

Ceiling fans including their suspension shall conform to BDS 818.

With respect to the position of a lighting fitting, the positioning of a fan shall be such so that it does not throw any shadow on the working plane is not acceptable. The unit module area shall be so chosen that the required number of fans could be suitably located, to avoid creation of pockets receiving little or no air circulation.

In general, fans in large halls may be spaced at 3 to 3.5 m in both the directions in the horizontal plane. If building modules do not lend themselves to proper positioning of the required number of ceiling fans, other types of fans, such as air circulators or wall mounted bracket fans shall have to be installed for the areas uncovered by the ceiling fans. In such cases, necessary electrical outlets shall have to be provided for the purpose. Table 8.1.20 gives the recommended areas to be served by different sizes of ceiling fans where the height of fan blades is at 2.5 m above the finished floor level.

Table 8.1.20: Recommended Fan Sizes in Rooms

Room Area (m ²)	Fan Sweep
Up to 6	915 mm
Over 6 to 9	1220 mm
Over 9 to 12	1442 mm

Wiring for a ceiling fan outlet from the switch board up to the ceiling fan outlet shall be done through pre-laid 18 mm dia PVC conduits using 1.5 mm² PVC insulated 2 cables of Brown and Blue insulation. A high quality ceiling rose is to be installed at the ceiling fan point for the termination of the wiring and the connection of the two wires of the Fan.

A fan hook is required to be placed during casting of the roof. The fan hook is to be made using a 12 mm dia MS rod having at least 600 mm on both sides and shall be placed above the MS rod mesh of the roof slab.

(b) Wall mounted bracket fan

For Wall mounted bracket fans shall be mounted on the wall using appropriate rowel bolts. Wiring for a Wall mounted bracket fan outlet from the switch board up to the Wall mounted bracket fan outlet shall be done through pre-laid 18 mm dia PVC conduits using 1.5 mm² PVC insulated 2 cables of Brown and Blue insulation. A high quality ceiling rose is to be installed at the ceiling fan point for the termination of the wiring and the connection of the two wires of the Fan.

(c) Pedestal fans and table fans

These items are movable and no fixed connections are necessary. Sockets will be used to energize these fans.

(d) Installation/mounting of ventilating fans or exhaust fans

Exhaust fans are necessary for spaces, such as toilets, kitchens, canteens and godowns to provide the required air changes. Since the exhaust fans are located generally on the outer walls of a room, appropriate openings in such walls shall be provided right from the planning stage. The sizes and the rpm of the exhaust fans will vary according to the application and the volume for which a fan used. In some applications (such as some industries, big size gas generator room etc.) high rpm fans are essential. In all cases appropriate types of fan need to be chosen and appropriate arrangement need to be made so that rain water cannot get inside the rooms.

(e) Installation/mounting of ceiling fans

Ceiling Fans shall be suspended from Fan hooks that are to be placed in position during casting of the Roof.

(f) Fan hooks

Fan hooks may be concealed (hidden) or may be exposed type. Fan hooks shall be made using MS rods of 12 mm diameter. The diameter of this rod shall not be below 10 mm under any circumstances.

(g) Ceiling roses for fan points

Appropriate type of ceiling roses must be provided at the fan points for the termination of the Fan point wiring cables. Connection to the Ceiling Fans will go from the ceiling roses.

(h) Cutout box/circuit breaker box

If the BDB or the SDB from which a 3-pin switched shuttered socket receives power is at a significant distance away and the load connected to the socket needs special care an additional cutout box or a circuit breaker box may be placed adjacent to the socket. Such a cutout Box or a Circuit Breaker box shall be placed inside a 18 SWG Sheet Steel (coated with two coats of synthetic enamel paint) of appropriate size with appropriate Perspex cover plate. Such a box may be surface fitted or may be concealed fitted. The box shall have a brass terminal for the termination of the ECC.

1.3.4 Distribution Wiring in a Building

1.3.4.1 General

Loads are separated into known and unknown loads.

General illumination is a known load, whether derived from detailed lighting layout, or developed from watts per square meter calculation. Similarly fans are also known loads. Besides these two types, there may be some other known loads.

Number, rating and layout of outlets for general illumination, fans and other known loads should accurately be distributed among a number of branch circuits. These branch circuits should then be carefully loaded with due regard to voltage drop, operating voltage and possible increase in lighting levels in future. On the other hand the sockets are unknown loads. Socket loads will be determined from projections based on the utility of the building and type of applications.

Every installation shall be divided into small circuits (following the rules given in this document) to avoid danger in case of a fault, and to facilitate safe operation, inspection, maintenance and testing. For the establishment of the circuits appropriate type of wiring is needed and appropriate terminations/connections/junctions of these circuits are needed. At the same time appropriate types of protection against faults must be given at different levels. These are to be achieved through installation of appropriate distribution wiring in the building.

1.3.4.2 Distribution board

A Distribution Board is the junction point of the incoming line and the outgoing lines for the distribution of Electricity throughout the building. The incoming as well as the outgoing lines must have Circuit Breaker protection or Fuse protection. The junctions and terminations of the incoming and outgoing cables are made through copper bars containing bolts and nuts for cable lugs known as bus-bars. A Distribution board may be named as MDB or FDB or DB or SDB or BDB.

- (a) MDB stands for Main Distribution Board. This is the distribution box where the main incoming cable enters and terminates from the main service feed connection of a large building. The FDBs get feed from MDB.
- (b) FDB stands for Floor Distribution Board located in each of the floors of a multistoried building. The DBs get feed from FDB. Usually, more than one FDB are needed.
- (c) DB is the abbreviation for Distribution Board. This may be the box where the main incoming cable enters and terminates from the main service feed connection. The SDBs get feed from a DB.

- (d) SDB is used to represent Sub- Distribution Board. This board is located in the same floor of a building and connected to the DB. Usually more than one SDB are needed. The BDBs get feed from SDB.
- (e) BDB stands for Branch-Distribution Board located in the same floor of a building and connected to the SDB. Usually more than one BDB are needed.
- (f) EDB, EFDF, ESDB, EBDB Sections of DB, FDB, SDB, BDB receiving feed from the Emergency Bus-bar which in turn is getting feed from standby generator through changeover switch. These may be separate DBs placed by the corresponding normal supply DBs.

Each of these distribution boards must have busbars for Line, neutral and earthing for a single phase box. A 3-phase distribution board must have busbars for Line 1, Line 2, and Line 3, neutral and earthing.

These boxes shall be made with sheet steel of not less than 18 SWG thicknesses and must be appropriately paint finished to match the wall paint.

1.3.4.3 Circuit wiring

(a) Separate branch circuits for separate control

Separate branch circuits shall be provided for different parts of a building area which need to be separately controlled. A branch circuit should be independently working and should not be affected due to the failure of another branch circuit.

The number of final circuits (also termed as sub-circuits or circuits) required and the points supplied by any final circuits shall comply with

- (i) the requirement of over-current protection,
- (ii) the requirement for isolation and switching, and
- (iii) the selection of cables and conductors.

All final circuits shall be wired using loop wiring system; no joint box shall be used.

Sufficient number of 18 SWG sheet steel made (painted with two coats of grey synthetic enamel paint) pull boxes, with ebonite/perspex sheet cover plate, must be given on the walls near the ceiling. If brick walls are not available, pull boxes must be given in the ceilings.

(b) For domestic and office buildings

5A Light/Fan Circuits must be used for all Domestic and Residential buildings. 5A Light/Fan Circuits are also to be used for Office and commercial Buildings. The corresponding circuit wire in the BDB/ SDB/ DB then shall be not less than 1.5 mm².

(c) For office and commercial buildings having large open floor areas

Under unavoidable circumstances, in case of difficulties in forming 5 A light/fan circuits for office and commercial buildings having large open floor areas, 10 A light/fan circuits may be used. The corresponding circuit wire in the BDB/SDB/DB then shall be not less than 2.5 mm². However, use of 5 A light/fan circuits is still emphasized.

(d) For industrial/factory buildings having large open floor areas

For industrial/factory buildings having large open floor areas, 10 A light/fan circuits may be used.

(e) For industrial/factory buildings/warehouses having too large open floor areas

For industrial/factory buildings/warehouses having large open floor areas, efforts should be given to use circuits not exceeding 10 A. The corresponding circuit wire in the BDB/SDB/DB then shall be not less than 2.5 mm².

For Industrial/Factory Buildings having very large open floor areas, 15 A light/fan circuits may be used as exceptional cases only. The corresponding circuit breaker in the BDB/SDB/DB then shall be not less than 4 mm².

Increase in the sizes of the above mentioned cables may be required if the distance is too long. Voltage drop calculation will give the guidance in that case.

(f) Separate branch circuits from Miniature Circuit Breaker (MCB)

Separate branch circuits shall be provided from miniature circuit breaker (MCB) of a BDBD/SDB or fuse of the fuse distribution boards (FDB) for light/fan.

Separate branch circuits shall be provided from miniature circuit breaker (MCB) of a BDBD/SDB or fuse of the Fuse distribution boards (FDB) for automatic and fixed appliances with a load of 500 watt or more and socket outlets. Each automatic or fixed appliance shall be served by a protected socket circuit.

(g) Less than 50% loading of circuits with more than one outlet

Circuits with more than one outlet shall not be loaded in excess of 50% of their current carrying capacity.

(h) Branch circuits must have spare capacity to permit at least 20% increase in load

Each branch circuit running between a DB and a SDB, between a SDB and a BDB must have spare capacity to permit at least 20% increase in load before reaching the level of maximum continuous load current permitted for that circuit

- (i) One spare circuit must be allowed in the distribution board for each five circuits in use.

At least one spare circuit must be allowed in the distribution board for each five circuits in use. Additional space for a circuit breaker along with the provision for connecting a pair of outgoing cables shall be kept.

- (j) Each final circuit shall be connected to a separate way in a distribution board

Where an installation comprises more than one final circuit, each final circuit shall be connected to a separate way in a distribution board. The wiring of each final circuit shall be electrically separate from that of every other final circuit, so as to prevent unwanted energization of a final circuit.

- (k) Size of cables in a branch circuit shall be at least one size larger than that needed for the computed load current

Size of cables to be used in a branch circuit shall be at least one size larger than that computed from the loading if the distance from the over-current protective device to the first outlet is over 15 m.

- (l) 4 mm² (7/0.036) and 6 mm² (7/0.044) wiring cable for a 15 A socket outlet branch circuit

The minimum size of wiring cable used for a 15 A socket outlet branch circuit shall be 4 mm² (7/0.036). When the distance from the over-current protective device to the first socket outlet on a receptacle circuit is over 30 m the minimum size of wire used for a 15 A branch circuit shall be 6 mm² (7/0.044).

- (m) Length of a lighting circuit

The length of a lighting circuit shall be limited to a maximum of 30 m, unless the load on the circuit is so small that voltage drop between the over-current protective device and any outlet is below 1 percent.

- (n) Use of common neutral for more than one circuit is prohibited

Each circuit must have its own neutral cable. Use of common neutral cable for more than one circuit is not permitted.

- (o) Following the appropriate new colour codes of cables

During wiring, correct colour codes of the insulation of the cables must be used.

Previously, for a single phase circuit red colour insulation was used for the live wire and the black colour insulation for the neutral and green + yellow bi-colour insulation was used for the ECC. Previously, for a three phase circuit red colour was used for the live (L1), Yellow colour for the live (L2), Blue colour for the live (L3) cable and the black

colour for the neutral and green + yellow bi-colour for the ECC. This colour code of cables shall now be replaced by the current IEC cable colour code standards, Table 8.1.21 and Figure 8.1.1. The current IEC colour code is recommended to be followed in Bangladesh.

Table 8.1.21: New introduced Colour Codes of Cables Following IEC Standards

Item	Pre-1977 IEE	Pre-2004 IEE	Current IEC
Protective earth (PE)	Green	Green/yellow bi-colour	Green/yellow bi-colour
Neutral (N)	Black	Black	Blue
Single phase: Line (L) Three-phase: L1	Red	Red	Brown
Three-phase: L2	Yellow	Yellow	Black
Three-phase: L3	Blue	Blue	Grey

Existing

Harmonised

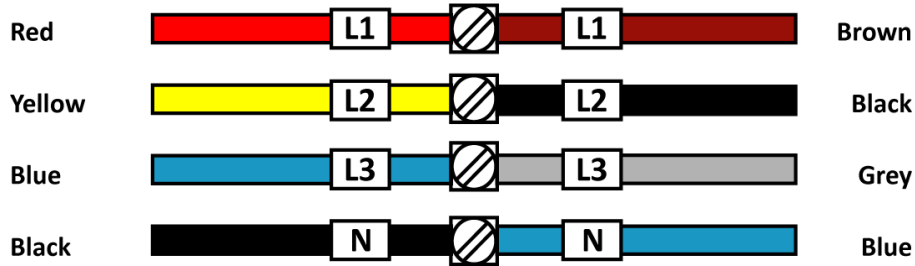


Figure 8.1.1 Existing and harmonised colour code by IEC recommended for use in Bangladesh.

The above mentioned colour coding must be indicated in the design drawing. This should also be mentioned in the specification.

(p) Balancing of circuits in three phase SDBs, DBs, FDBs, and MDBs.

In a 3 phase distribution system special care must be taken during wiring to obtain balancing of loads among the three phases.

In a 3 phase SDB, DB, FDB, MDB connections of the circuits to the bus-bars must be made in such a way so that the load current remains balanced among the three lines during low load as well as full load. After completing the installation balancing should be checked by clamp meter current measurement of each phase.

The above mentioned current balancing must be indicated in the SDB (if 3 phase), DB, FDB, and MDB circuit diagram of design drawing. This should also be mentioned in the specification.

1.3.5 Electrical Layout and Installation Drawings

An electrical layout drawing shall be prepared after proper locations of all outlets for lamps, fans, fixed and transportable appliances, motors etc. have been selected. This is the beginning of the electrical distribution design work. This job must be done with due importance prior to starting the construction and installation work. Strong emphasis is given on this work in this document.

1.3.5.1 Locating positions of the points on the plan of the building

At the beginning, the Light points, Fan points, Socket points, Switch Boards, BDBs, SDBs, FDBs. DBs and MDBs shall be located on each plan based on convention, suitability, application and safety view point.

Conduit layout and cable layout shall then be shown on the drawing.

1.3.5.2 Light and fan circuits must not be mixed with the socket circuits

In designing the wiring layout, power (socket) and heating (socket) sub-circuits shall be kept separate and distinct from light and fan sub-circuits.

All wiring shall be done on the distribution system with main and branch distribution boards placed at convenient positions considering both physical aspects and electrical load centres. All types of wiring whether concealed or surface, shall be as near the ceiling as possible. In all types of wiring due consideration shall be given to neatness and good appearance.

1.3.5.3 Balancing of circuits in three phase distribution boxes is a must

Balancing of circuits in three phase installations shall be arranged in the drawing and also must be done during physical connection.

1.3.5.4 Single phase socket outlets receiving connection from two different phases

Single phase socket outlets receiving connection from two different phases located in the same room is to be avoided. However, if it is essential to have such socket connection these must be located 2 m or more apart.

1.3.5.5 Electrical Layout drawings for industrial premises

Electrical layout drawings for industrial premises shall indicate the relevant civil structure/barrier/duct and mechanical equipment/duct.

1.3.5.6 Preparation of detailed circuit diagram

Circuit diagrams of each of the Light and Fan circuits must first be prepared based on the selection whether it is 5A or 10A circuit. The cable size of each of the circuit's size of the ECC must be shown in the drawing. The circuit diagrams of the BDBs, SDBs, DBs, FDBs, and MDBs etc. are then to be prepared and presented in the form of single line drawings indicating the cable sizes of each interconnection and the sizes of the ECCs. The distribution of BDBs, SDBs, DBs, FDBs, MDBs etc. are to be shown in a distribution drawing indicating the cable sizes of each interconnection and the sizes of the ECCs.

1.3.5.7 Preparation of electrical distribution and wiring design drawing by an experienced Engineer

Electrical Distribution and Wiring Design drawing of building must be prepared by an eligible Engineer as mentioned in Table 2.3.4 Chapter 3 Part 2.

1.3.6 Electrical Wiring in the Interior of Buildings

1.3.6.1 Surface wiring or exposed wiring

Wiring run over the surface of walls and ceilings, whether contained in conduits or not, is termed as surface wiring or exposed wiring.

Single core PVC insulated copper through PVC channels or through PVC conduits or through GI pipes of approved quality may be used for surface wiring.

Surface wiring using twin core flat PVC insulated copper on wooden battens used to be used long back. This is almost discontinued and discouraged now a day.

PVC conduits or GI pipes, when used for surface wiring, shall be clamped with saddles at a spacing not exceeding 600 mm, to the wall or ceiling using plastic rowel plugs with countersunk galvanized screws.

(a) Surface wiring using wood battens

The wood batten used in this method shall be of good quality wood with a minimum thickness of 12 mm. They shall be installed exposed and run straight on the ceiling or wall surfaces. Battens on walls shall be run either horizontally or vertically, and never at an angle. Battens on ceilings shall run parallel to the edges in either orthogonal direction, and not at an angle, they shall be fixed to the wall or ceiling by rowel plugs and countersunk galvanized screws. Cables shall be fixed to the battens by using galvanized steel clips or brass link clips or PVC clips of required size at a spacing not exceeding 100 mm.

(b) Surface wiring using PVC conduits

PVC conduits or GI pipes, when used for surface wiring, shall be clamped with saddles at a spacing not exceeding 600 mm, to the wall or ceiling using plastic rowel plugs with countersunk galvanized screws.

The conduits placed concealed inside roof or in wall must have 20 SWG GI pull wires placed during laying of the pipes for pulling the cables later.

(c) Surface wiring using PVC channels

Surface wiring may be done using single core PVC insulated cables placed inside surface fixed PVC channels of appropriate size. Fixing of channels must be done using screws in rowel plugs inserted into drilled holes on the walls/ceilings. The channels must be placed in a straight line with adequate number of screws so that no sag is observed. Cables must not be stressed in the bends. Adequate space must exist inside the channel to put the cables in position without difficulty.

Surface wiring using flexible chords, clips and nails shall not be used in general.

(d) Surface wiring using Round core flexible cable with plastic clips and nails

Surface wiring using exposed Round core flexible cable with plastic clips and long nails have been used for extending a point wiring, for extending a socket wiring due to shift, for add a circuit wiring.

This is not recommended for regular wiring. Instead of using this method, one should go for the recommended surface wiring using single core PVC cables with PVC channels or single core PVC cables with PVC conduits as mentioned above in this document.

For a length of not exceeding 1 m this may be used only for shifting an existing Light/Fan point or for shifting an existing socket point only under unavoidable circumstances.

1.3.6.2 Concealed wiring

The wires in this type of wiring shall be placed inside GI conduits or PVC conduits that are buried in roofs and in brick/concrete walls. The conduits in the walls shall be run horizontally or vertically, and not at an angle.

Conduits in concrete slabs shall be placed at the centre of thickness and supported during casting by mortar blocks or 'chairs' made of steel bars or any other approved means. All conduits shall be continuous throughout their lengths.

Appropriate planning should be made in which there shall be adequate spare capacity in the conduits placed in roof slabs so that unforeseen situation during execution of the installation can be taken care of. Conduits will run through the roof and then bend downward for going up to the outlets, DBs, switch boards, sockets.

In a column structure building having no permanent walls, switch boards and socket boards, pull boxes shall be placed in columns and must be done during the casting of columns.

Concealed wiring through floors and upward mounting of PVC/GI pipes from the floor is strongly discouraged because of the occurrence of condensation and accumulation of water from condensation eventually leading to damaging of the simple PVC insulated cable insulation. This method should not be followed as a general practice.

Underground cables for electrical distribution in the premises/garden/compound of the building shall be encased in GI or PVC pipes and laid in earth trenches of sufficient depth. Armoured cables need not be encased in conduits except for crossings under road, footpath, walkway or floors.

The conduits placed concealed inside roof or in wall must have 20 SWG GI pull wires placed during laying of the pipes for pulling the cables later.

1.3.6.3 Wiring inside suspended ceilings (false ceilings)

Wiring inside suspended ceilings (false ceilings) shall be surface wiring through conduits or through PVC channels mentioned under the heading of surface wiring methods.

Cables shall not be placed loosely and haphazardly on the suspended ceilings. Placing naked cables inside the suspended ceiling is not permitted.

Cable joints with PVC tape wrapping is not allowed for connection of a fitting from the ceiling rose or from a junction box inside the gap space.

1.3.6.4 Wiring through cable tray

Wiring for connections to some machines may be carried through a cable tray suspended from the ceiling. This is very rare for a domestic building. However in a commercial / office or industrial building this technique may be needed. In special circumstances Cables may be pulled through pre laid GI/ PVC pipes under the floor where there will be no chances of water accumulation in the floor or condensation.

1.3.6.5 Mounting height of light and fan switch boards

Light and fan switch boards shall be placed 1220 mm above floor level in the residential buildings (i.e, the clearance between the floor and the bottom of the switch board shall be 1220 mm).

This above mentioned height shall be 1300 mm above floor level in the office buildings, commercial buildings and industrial buildings. However, the minimum height shall not be below 1220 mm.

1.3.6.6 Restriction on the use of plastic/PVC insulated flexible chords/cables

Plastic/PVC insulated flexible chords/cables shall not be used for wiring of light/fan points or for wiring of sockets, or for wiring of any sub circuits.

1.3.6.7 Cable joints and cable joint boxes in concealed and surface wiring

Both the Brown (L) and Blue (N) cables of a final circuit shall run from a BDB/SDB up to the switch board without a joint. Similarly, both the Brown (L) and Blue (N) cables of a point shall run from the point up to the switch board. Cable joints are to be made in the switch board back box. Where the above methods are not implementable, joints shall be made using approved cable joint methods.

1.3.7 Methods of Point Wiring and Circuit Wiring

1.3.7.1 Methods of Point Wiring

Wiring between a light/fan point and its corresponding switch board is termed as Point Wiring. The load of such a point is not in excess of 100 watts in general, and in special cases this may be up to 200 watts. Wiring for a light/fan point shall be made using one of the following two methods: (i) Surface wiring or (ii) Concealed wiring. For wiring of a point one brown and one blue PVC insulated copper cable shall run between a point and its switch board. Cable joints inside conduits or within channels are forbidden. The current carrying capacity for such a circuit shall not be more than 5 A for a residential or a commercial (business/mercantile) building. The minimum size of a cable for such wiring shall be 1.5 mm².

Common neutral shall not be used under any circumstances.

1.3.7.2 Methods of Circuit Wiring

Wiring between a switch board and a BDB/SDB/DB will be called Circuit Wiring. Circuit wiring shall be done with a live cable a neutral cable and an ECC cable for a single phase circuit. Sometimes this circuit is also referred to as sub-circuit.

An ECC must be provided with each circuit. The ECC at the switch board end shall be terminated in the earth terminal of the metal part of the switch board using a brass screw/bolt and a nut. The BDB/SDB/DB end of the ECC shall be terminated in the earthing busbar of the BDB/SDB/DB.

The ECC in this case shall be PVC insulated copper cable of appropriate size but with yellow + green bi-colour insulation.

For each circuit, the live cable must be drawn using brown colour insulated PVC cable and the neutral cable shall be drawn using blue colour insulated PVC cable.

Common neutral shall not be used under any circumstances.

The minimum sizes of cable for various uses shall be as follows:

- (a) For a 5 A circuit protected by a 5 A circuit breaker or fuse shall not be below 1.5 mm²
- (b) For a 10 A circuit protected by a 10 A circuit breaker or fuse shall not be below 2.5 mm².
- (c) For a 15 A circuit protected by a 15 A circuit breaker or fuse shall not be below 4 mm².
- (d) For a 20 A circuit protected by a 20 A circuit breaker or fuse shall not be below 6 mm².

The above mentioned sizes must be increased for long cables as mentioned elsewhere in this document.

In general, the minimum size of cable for a particular circuit shall depend on the rating of the fuse or circuit breaker used for the protection of that circuit. A voltage drop check is to be made for each length of the circuit to ensure that the voltage drop at the farthest end of the load from the main distribution point does not exceed 2.5 percent.

Sockets shall get direct connection from the BDB/SDB through breaker/fuse protection. Depending on the assessed requirements sockets may be grouped/looped at the socket end. Such grouping shall not exceed 3 numbers of sockets in one circuit.

1.3.8 Feeder Wiring between SDB and BDB, DB and SDB, FDB to DB, MDB to FDB etc.

Wiring between a BDB and an SDB, an SDB and a DB, a DB and an FDB, an FDB and an MDB needs special attention and the rules are similar to Circuit Wiring. ECC must be present for each of the feed connections. The ECC in this case also shall be PVC insulated copper cable of appropriate size but with Green + Yellow bi-colour insulation.

At both ends the ECC must be terminated at the earthing busbar.

Appropriate cable lugs/cable sockets must be used for terminating the L1, L2, L3, N and E connections on the busbars of both the boards. The sizes of the cables must be chosen to match with the rating of the circuit breaker/fuse ratings as mentioned above.

Circuit breakers/fuses must be provided at the outgoing and incoming sides of each of the busbars of each BDB/SDB/DB/FDB boxes.

1.3.9 Conduits, Channels, Cables, Conductors and related Accessories

Conduits, Cables, Conductors and Accessories are important parts of an electrical distribution installation.

1.3.9.1 Conduits and conduit fittings

Cables of an electrical distribution installation are drawn through electrical conduits. For the installation of conduits various types of fittings are needed. For the two types of commonly used conduits, PVC and Metal, fittings should be as under.

(a) PVC conduits

- (i) PVC conduits and conduit fittings shall be of heavy wall water grade type. All bends shall be large radius bends formed by heat or by mechanical bending machine. The cross-section of the conduit shall remain circular at the bend and the internal diameter shall not be reduced due to bending. PVC pipe fittings shall be sealed with PVC solvent cement or adhesive for PVC of approved quality.
- (ii) Conduits installed in floors, if installed, shall have a slope of at least 1:1000 towards floor mounted pull box or cable duct.
- (iii) Conduits placed concealed inside roof or in wall must have 20 SWG GI pull wires placed during laying of the pipes for pulling the cables later.
- (iv) Water grade PVC conduits must be used for both concealed and surface wiring. Water grade PVC conduits of different diameters shall be used as per necessity.
- (v) Appropriate high grade bends and circular boxes must be used with the PVC pipes.
- (vi) 18SWG metal sheet made and synthetic enamel paint coated quality boxes of matching sizes shall be used as pull boxes and junction boxes. Appropriate pull-box covers of ebonite or perspex sheet shall be fitted with GI machine screw and washer.
- (vii) The PVC conduits placed concealed inside roof or in wall must have 20 SWG GI pull wires placed during laying of the pipes for pulling the cables later.

(b) PVC channels

PVC channels should be used only for extension work in an already installed building. A design drawing should not show use of such wiring except inside a false ceiling. Even inside the false ceiling this item should be used for lengths. For long distances PVC conduits should be used. High quality PVC channels of sufficient thickness should be used and fixed in a neat manner. For large number of cables and for thick cables PVC channels should not be used.

(c) PVC flexible pipes/conduits

PVC flexible conduits shall be used with surface wiring only and only in places where PVC bends cannot be used. Except special circumstances flexible PVC conduits shall not be used.

(d) Metal/steel conduits

Galvanized Iron (GI) conduits shall be made using at least 16 SWG sheet. The conduits shall have seamless joint along the length and must be suitable for making bends. No projections are allowed inside the conduits. Metal conduits must be threaded for end to end joints using sockets. In case of necessity, threads will be cut at the end of short pieces. Sharp edges at the ends must be properly treated so that cable injury does not take place during cable pulling.

(e) Pull boxes

- (i) Pull boxes/Joint boxes must be placed closed to the ceiling where conduits from the ceiling are going downward toward a switch box or are going toward a socket box or are going toward a BDB/ SDB/ DB / FDB.
- (ii) Pull boxes are extremely essential for pulling the cables without injuring the cables and thus should not be avoided under any circumstances. These are also essential for future maintenance and extension work.
- (iii) Pull boxes/Joint boxes must be placed in the ceiling of office/factory building where conduits are running over a long distance between two walls (terminal points) and where fixed walls are not available and also where heavy beams are used. In case of big cross section beams pull boxes/joint boxes shall be placed closed to the beams.
- (iv) Pull boxes/Joint boxes must be made with 18 SWG GI sheet or with 18 SWG MS sheet but coated with two coats of Grey Synthetic Enamel paint.
- (v) Covers of pull boxes should be ebonite or perspex sheet of not less than 1/8 inch thickness.

(f) Metal Boxes for Switch Boards

Metal Boxes for Switch Boards must be made with 18 SWG GI sheet or with 18 SWG MS sheet but coated with two coats of Grey Synthetic Enamel paint. A Switch Board Metal Box must have a small Copper / Brass earthing busbar for terminating the ECCs.

(g) Switches for operating light and fan points

Switches for operating Light and Fan points must be of 5A rating. These switches are usually SPST type. However, for special applications like stairs and some other places these may be SPDT type.

Switches for operating Light and Fan points may be of Gang type or may be isolated type. The isolated types are to be mounted on an ebonite top plate which is again fitted on the above mentioned Metal boxes for Switch Boards.

(h) Mounting regulators of ceiling fans

Metal Boxes for Mounting Inductor Regulators of Ceiling Fans must be made with 18 SWG GI sheet or with 18 SWG MS sheet but coated with two coats of Grey Synthetic Enamel paint. Metal Boxes for mounting regulators of ceiling fans must have a small copper/brass earthing busbar for terminating the ECCs.

However, such regulators may be placed inside the 18 SWG GI sheet or MS sheet made Metal Boxes for Switch Boards. In such a case arrangements must be made so that the PVC insulated point and circuit wiring cables and their joints inside the switch board do not touch a regulator. This may be done by appropriately dressing the cables and fastening the cables by using polymer cable fasteners.

1.3.9.2 Cables and conductors

For application in building wiring, PVC insulated stranded cables shall be used for Live and Neutral Wires for single phase and 3-lines (L1, L2, L3) and one neutral for 3-phase. For ECC also PVC insulated stranded cables shall be used. As a result, use of bare conductors is non-existent.

(a) Cables

Conductors of a PVC insulated cable, thin or thick, shall be copper. Cable containing Aluminum conductors may be used for thick cable of size more than 35 mm² but copper is always preferred.

Cables for power and lighting circuits shall be of adequate size to carry the designed circuit load without exceeding the permissible thermal limits for the insulation. The voltage drop shall also be within the specified limit of 2.5 percent from a distribution point up to their farthest end of the load point. Recommended sizes (in mm²) of copper conductors are as follows:

1, 1.5, 2.5, 4, 6, 10, 16, 25, 35, 50, 70, 95, 120, 150, 185, 240, 300, 400, 500, 630, 800, 1000

For final circuit/sub-circuit and for Light/fan point wiring the cable nominal cross-section of the cable shall not be less than 1.5 mm² for copper conductors.

Standard copper conductor sizes of cables which should be used for electrical installations/distribution in buildings are given below. Conductors of sizes other than the sizes listed below are not recommended.

(b) Phase and neutral cables shall be of the same size

In the wiring of the sub-circuit/circuit and all other circuits inside a building the Phase cable and the neutral cable shall be of the same size.

(c) Flexible cables/flexible cords

The minimum cross-sectional area of conductors of flexible cables/flexible cords shall be 0.5 mm² for copper conductors. Flexible cable or cords shall not be used as fixed wiring unless contained in an enclosure affording mechanical protection.

Flexible cables/flexible cords may be used for connections to portable equipment. For the purpose of this regulation an electric cooker of rated input exceeding 3 kW is not considered to be portable. The flexible cord shall be of sufficient length so as to avoid undue risk of damage to the outlet, cord or equipment and of being a hazard to personnel.

(d) Treatment of cable ends/cable terminations

All stranded conductors must be provided with cable sockets/cable lugs of appropriate size fitted using appropriate hand press tool or hand crimp tool or hydraulic press tool depending on the size of the cable. This is necessary for termination of the cable ends on bus-bars.

(e) Jointing of cables in wiring

Cable joints for the PVC insulated cables used in circuit wiring (thin cables) are to be made through porcelain/PVC connectors with PIB tape wound around the connector before placing the cable inside the box joint/pull box.

1.3.10 Conduits through the Building Expansion Joints

Conduits shall not normally be allowed to cross expansion joints in a building. Where such crossing is found to be unavoidable, special care must be taken to ensure that the conduit runs and wiring are not in any way put to strain or are not damaged due to expansion/contraction of the building structure. In unavoidable situations, PVC conduit through an oversize flexible PVC conduit may be used with pull boxes on both sides of expansion joints.

1.3.11 Types of Electrical Wiring for Exterior Lighting and other exterior purposes

1.3.11.1 Electrical wiring for garden lighting

For garden lighting PVC insulated PVC sheathed underground cables shall be used. For protection purpose these may be drawn through PVC pipe of appropriate dimension so that adequate clearance remains for the ease of pulling. In general, no junction of cables shall be provided in underground level. However, in case of necessity, metal sleeve cable ferrule joints using Crimp Tool or hydraulic press and heat shrink insulated sleeve shall be used on top.

1.3.11.2 Electrical wiring for street lighting

For street lighting PVC insulated PVC sheathed underground cables shall be used. For protection purpose these may be drawn through PVC pipe of appropriate dimension so that adequate clearance remains for the ease of pulling. In general, no junction of cables shall be provided in underground level. However, in case of necessity, metal sleeve cable ferrule joints using Crimp Tool or hydraulic press and heat shrink insulated sleeve shall be used on top. Joining the cables at the bottom of a street pole must be done inside a metal joint box located sufficiently above the street level so that water cannot reach the box even during the worst rain/flood situation.

1.3.11.3 Electrical wiring for boundary light

For boundary lighting PVC insulated PVC sheathed underground cables shall be used. For protection purpose these may be drawn through PVC pipe of appropriate dimension so that adequate clearance remains for the ease of pulling. In general, no junction of cables shall be provided in underground level. However, in case of necessity, metal sleeve or cable ferrule joints using Crimp Tool or hydraulic press and heat shrink insulated sleeve shall be used on top. However, for the portion of the cable running concealed through a wall, PVC insulated cables through PVC conduits may be used.

1.3.12 Branch Distribution Boards, Sub-distribution Boards, Distribution Boards, FDBs and MDBs

1.3.12.1 Enclosure/box

Enclosures for sub-distribution boards located inside the building shall be dust-proof and vermin-proof using sheet steel fabrication of a minimum thickness of 20 SWG. The boards shall be safe in operation and safe against spread of fire due to short circuit.

1.3.12.2 Size of the enclosure of a BDB/SDB/DB/FDB/MDB

Table 8.1.22 provides a guidance of sizes of enclosures for SDB containing miniature circuit breakers or fuses. However, the size will depend on the number and size of the circuit breakers or the fuses the number of outgoing cables and their sizes, the size of the busbars and the type of insulators used for the busbars.

Table 8.1.22: Recommended Enclosure Sizes for MCB's and Fuses

Dimensions (mm)			No. of MCB's or Fuses
Height	Width	Depth	
350	390	120	up to 12
480	390	120	up to 24
610	390	120	up to 36
740	390	120	up to 48

1.3.12.3 Location

A Sub-distribution board (SDB) shall be located as close as possible to the electrical load centre for that SDB. This is also applicable for determining the locations of FDBs, DB and BDBs. These boards shall never be located on water soaked or damp walls.

1.3.12.4 Wiring of sub-distribution boards

- (a) In wiring a sub-distribution board, total load of the consuming devices shall be distributed, as far as possible, evenly between the numbers of ways of the board, leaving the spare way(s) for future extension.
- (b) All connections between pieces of apparatus or between apparatus and terminals on a board shall be neatly arranged in a definite sequence, following the arrangements of the apparatus mounted thereon, avoiding unnecessary crossings.
- (c) Cables shall be connected to terminals only by soldered or welded lugs, unless the terminals are of such form that it is possible to securely clamp them without cutting away the cable strands.

1.3.13 Electrical Services Shafts, Bus Ducts, L.T. Riser Cables and L.T. Busbar Trunking

1.3.13.1 Vertical service shaft for electrical risers

For buildings over six-storey or 20 m high there shall, in general, be a minimum of one vertical electrical service shaft of (200 mm x 400 mm) size for every 1500 m² floor area. The electrical service shaft shall exclusively be used for the following purposes:

- (a) Electric supply feeder cables or riser mains
- (b) Busbar Trunking
- (c) telephone cables
- (d) Data Cables
- (e) fire alarm cables
- (f) CCTV cables
- (g) Other signal cables
- (h) Area fuse/circuit breakers
- (i) Floor Distribution board/sub-distribution boards for individual floors.

The construction of the floors of the duct area shall be constructed in such a way so that the remaining empty open space after putting the cables/busbar trunking/ pipes/ conduits in position is filled up with RCC slab(s) or any other non-inflammable material so that fire or molten PVC cannot fall from one floor to the next lower floor(s). For this purpose arrangements need to be made during floor casting.

Free and easy access to the electrical shaft room in each floor must be available for operation, maintenance and emergency shut downs.

Vertical cables other than electrical cables shall be placed at a sufficient distance from the nearest electrical cable. A vertical separating brick wall between electrical and non-electrical wall is preferable.

Vertical Service Shaft for Electrical Risers as mentioned above must not be placed adjacent to the Sanitary Shafts. They should be placed at significant separation in order to ensure that the Vertical Service Shaft for Electrical Risers remains absolutely dry.

1.3.13.2 LT Riser main cables

- (a) For low rise building riser main cables will serve to bring L.T. connection to the floor distribution boards (FDBs) of each floor from the main distribution board. For a 5 storied building or lesser having a floor space of less than 600 m² in each floor the riser cables may be PVC insulated cables through PVC or GI pipes.
- (b) For bringing the riser main cables a common vertical wall and holes or slots in the floors must be given by the building construction people.
- (c) However, for larger floor area or for higher buildings PVC insulated PVC sheathed underground cables must be used with protection and spacing.
- (d) For more than 9 storied building Busbar preferably sandwiched copper busbar trunking should be used for safety reasons.
- (e) PVC insulated PVC Sheathed underground cables must be used as Riser Main Cables. These cables shall be placed in or pulled through a PVC pipe of higher diameter so that the cable can be easily pulled through it. The PVC pipes must be fixed vertically in a straight line on the wall of the shaft using appropriate saddles. However, in some cases PVC insulated PVC Sheathed underground cables may be directly fixed on the wall using appropriate saddles with 37mm spacing between two adjacent cables. Sheet metal made Joint Boxes (with ebonite cover plates) must be placed at each floor tapping point.
- (f) The cable work shall be done neatly so that no suspended cables are seen around the place and no suspended flexible pipes are seen.
- (g) Each riser cable must have appropriate fuse or circuit breaker protection at the source busbar junction and also at the tap off point.

1.3.13.3 LT Busbar Trunking

For high rise buildings, LT (0.4KV TP&N) busbar trunking system is used instead of riser main cables to minimize space in the vertical electrical shaft, to minimize the risk of spreading of fire from one floor to another due to electrical short circuit in one of the cables or sparks, to have a neat distribution system. Most part of the busbar trunking shall be installed vertically. The horizontal portion of the busbar trunking shall usually connect the vertical portion with the Substation LT panel.

- (a) Busbar trunking are specially useful to minimize space and to minimize risks of spreading fire (during accidents) which may happen with bundles of insulated cables. The conductors supported by insulators inside the busbar trunking shall be copper of solid rectangular cross-section. The copper bars are insulated. A busbar trunking system shall be laid with minimum number of bends for distribution system. Typical rating of feeder busbar trunking for 3-phase- 3-wire or 3-phase- 4-wire system shall range from 200 amperes to 3000 amperes although lower amperes are not impossible.
- (b) Horizontal busbar trunking of suitable size may be provided along the roads for a group of buildings to be fed by a single substation but with heavy weather (moisture and water) protection and covered with appropriate weather resistant water proof material. Extreme care need to be taken in these cases for protection against moisture, water and outside weather.
- (c) Busbar trunking shall be placed in a dry place and must not be installed in a place which is even slightly exposed to weather/moisture/ spray or sprinkle of water.

1.3.13.4 LT Busducts

In certain applications, especially in factory lighting and factory power distribution of large area factories Busducts are used. In most cases, these Busducts are suspended from ceiling. Busducts offer safe, reliable, neat distribution system in these cases. The choice will depend on the floor area, type of machineries, type of jobs and other factors.

Appropriate circuit protection using adequate number of circuit breakers of appropriate rating is needed. In most cases these busducts are horizontally mounted/suspended. The busbars shall be copper. The rating shall depend on the current on each segment and the current carried by each segment.

1.3.14 L T Main Incoming Cable and Service Connection

- (a) Overhead service connection to a building shall be achieved with PVC insulated Cables with GI support wire (similar to catenary) or catenary wire (mainly for single phase consumers). The overhead service connection shall be led into buildings via roof poles or service masts made of GI pipe at least 38 mm in diameter having a goose neck bend at the top and installed on the outer wall. The alternative is to have underground cable connection.

- (b) Underground PVC insulated PVC sheathed water proof cables shall be placed in underground cable trench or pulled through a PVC pipe of higher diameter placed in a cable trench so that the cable can be easily pulled through it. PVC insulated stranded annealed copper ECC cables matching with the main cable size shall run along the Main incoming cable with termination at the earthing busbar at both end.

Each of the PVC pipes must have 18 SWG GI pull wires placed during laying of the pipes for pulling the cables later.

- (c) For main incoming thick underground cables joints are strongly discouraged and should be avoided as far as possible. However, for unavoidable cases joints must be made through sleeve or ferrule of appropriately matched size fitted with hydraulic press following neat processing of the cable ends. Appropriate fusible heat shrink cover must be used over such junction.

For thick cables running through conduits as vertical risers, these joints must be put inside metal joint/pull boxes with covers.

- (d) Special forms of construction, such as flame proof enclosures, shall be adopted where risk of fire or explosion exists near a place where thick incoming cable or riser cables are placed.
- (e) The Underground service cable shall be laid in conformity with the requirements of Sec 1.3.24 titled “Laying of LT underground Cables”, of this Chapter.
- (f) The power and telecommunication or antenna cables must be laid separately maintaining sufficient distance.
- (g) The fire alarm and emergency lighting circuits shall be segregated from all other cables and from each other in accordance with BS 5839 and BS 5266. Telecommunication circuits shall be segregated in accordance with BS 6701 as appropriate.
- (h) Where a wiring system is located in close proximity to a non-electrical service both the following conditions shall be met:
- The wiring system shall be suitably protected against the hazards likely to arise from the presence of the other service in normal use, and
 - Appropriate protection against indirect contact shall be taken.

- (i) A wiring system shall not be installed in the vicinity of a service which produces heat, smoke or fume likely to be detrimental to the wiring, unless protected from harmful effects by shielding arranged so as not to affect the dissipation of heat from the wiring.
- (j) Where a wiring system is routed near a service liable to cause condensation (such as water, steam or gas services) precautions shall be taken to protect the wiring system from deleterious effects.
- (k) No cable shall be run in a lift (or hoist) shaft unless it forms part of the lift installation as defined in BS 5655.

1.3.15 Design for Electrical Wiring

Design of Electrical wiring must be done following the provisions provided in this Chapter. Detailed design drawings must be prepared by eligible Engineer for complete execution of the electrical works mentioned in this document and any other new items arising because of the evolution of new technologies in the near future.

Typically, there must be conduit layout drawing(s) indicating the conduit layouts, the locations of the switch boards, locations of the sockets, locations of the BDBs, locations of the SDBs, locations of the DBs, locations of the FDBs, location of the MDB, location of the Main incoming cable.

A distribution diagram of the BDBs, SDBs up to MDBs as applicable indicating the ampere rating of the incoming MCB/MCCB, interlinking cable sizes and the ECCs must be presented.

Detailed circuit diagrams of the circuits and the BDBs, SDBs, MDBs as applicable must be presented.

Detailed drawings of earthing and earth inspection pits and any other complicated parts must be presented. The contractor shall prepare as built drawings after completing a project.

1.3.15.1 Design for electrical wiring in bedrooms and drawing rooms

The location of a switch board must be near the entrance door of a bedroom like any other room. The location of the wall mounted light fittings must be chosen based on the possible locations of furniture which is also needed in other rooms. Sufficient number of 3-pin 13 A switched shuttered flat pin sockets must be provided in a bed room. The same principles are applicable for a Living room.

Design must be made in such a way that sufficient clearance (space) is left inside the concealed conduits (i) for the ease of pulling the cables and also for adding few more cables in case of necessity during future modification.

For bedrooms and drawing rooms the light plus fan sub circuits for shall not be of more than 5A rating.

Generally, single core PVC insulated stranded electrolytic annealed copper cables shall be used in concealed wiring technique or in the other methods.

1.3.15.2 Design for electrical wiring in a kitchen.

The sensitive item in a kitchen is placing 3-pin 13 A switched shuttered flat pin sockets on wall of the kitchen side table near the wall. Good distance must be maintained between the kitchen water tap and the socket. The socket for the refrigerator (if any) shall also be a 3-pin 13 A switched shuttered flat pin socket, and may be placed at the same level as the other socket. For the ease of operation a 3-pin 13 A switched shuttered flat pin socket for this purpose may be placed at the bottom level height of a switch board provided this is acceptable in terms of aesthetics.

For kitchens, the light plus fan sub circuits for shall not be of more than 5A rating.

1.3.15.3 Switches for toilets and bath rooms

Switches for toilet lights and toilet ventilating fans must be placed outside the toilets adjacent to the entrance door but must not be placed inside the toilet. The same rule should be followed for bath rooms. Using ceiling mounted chord switch at the entrance path of the door of a toilet is a good idea for small toilets attached to bed rooms. Ceiling mounted chord switches may be used with a chord suspended from the ceiling near the opening of the door.

1.3.15.4 Design for electrical wiring in office rooms

The location of a switch board must be near the entrance door of an office room. The location of the light fittings must be chosen based on the possible locations of work table, furniture. Sufficient number of 3-pin 13 A switched shuttered flat pin sockets must be provided in each office room. In this regard special consideration needs to be given on the possible location of computers and other office equipment.

Sufficient conduits and cables must be left for future modification as often rearrangement of cables needs to be made.

Generally, single core PVC insulated stranded electrolytic annealed copper cables shall be used for wiring by using the concealed wiring technique or the other two methods mentioned in the wiring section.

In case of special requirements, PVC insulated PVC sheathed Stranded Electrolytic Annealed Copper Cables may be used for wiring through conduits or other methods.

For Offices the sub circuits for shall not be of more than 5 A rating.

1.3.16 Temporary Electrical Connection for a Building Construction Site

Temporary connections are needed for a building construction site. A fuse distribution board containing incoming cut out fuse, outgoing cutout fuses plus busbars or distribution boards containing incoming circuit breakers, outgoing circuit breakers plus busbars of appropriate rating must be installed for such connections. Such boards shall be installed in a dry place so that rain water or waters coming from a construction zone cannot reach such boards.

1.3.17 Temporary Electrical Connection for an Outdoor Concert

Temporary connections are needed for an outdoor concert stage for special lighting, for various display systems, for high power audio systems. A fuse distribution board containing incoming cut out fuse, outgoing cutout fuses plus busbars or a distribution board containing incoming circuit breakers, outgoing circuit breakers plus busbars of appropriate rating must be installed for such connections. Such boards shall be installed in a dry place and shall be mounted at a safe height above ground so that rain water or waters coming from anywhere cannot reach such boards. Such boards shall not be installed near flammable materials.

Cables of appropriate types and appropriate ratings must be used for such applications.

Appropriate type of sockets, preferably flat 3-pin switched shuttered 13 A sockets should be used for distribution.

1.3.18 11 kV/ 0.4 kV Electrical Substation in a Building**1.3.18.1 General**

According to the rule of the distribution companies of Bangladesh, 11 kV/ 0.4 kV Electrical substations shall be required for a building if the load requirement of the building exceeds 50 kW. In most cases, substations are required for Multi-storied residential, Multi-storied Commercial buildings, Multi-storied Office building and Industries.

To determine the rating of the substation required, a load factor of at least 80% shall be applied to the estimated load of the building. The future expansion requirements should definitely be taken into consideration.

1.3.18.2 Location of an electrical substation

In a multi-storied building, the substation shall preferably be installed on the lowest floor level, but direct access from the street for installation or removal of the equipment shall be provided. The floor level of the substation or switch room shall be above the highest flood level of the locality. Suitable arrangements should exist to prevent the entrance of storm or flood water into the substation area.

The location of a substation will depend on (i) the feed point of the 11 kV Supply Authority line and (ii) the location of the LT vertical riser cables.

It is preferable to locate the air-conditioning plant room (if any) adjacent to the electrical substation in such a way that the distance from the controlling switchboard of the air-conditioning plant rooms and corresponding switches in the electrical substation are kept minimum.

In case of a building complex, or a group of buildings belonging to the same organization, the substation should preferably be located in a separate building and should be adjacent to the generator room, if any. Location of substation in the basement floor and on the floors above ground floor level (GFL) preferably be avoided. If Sub-Station it to be installed on the basement floor or the floors above ground floor level (GFL) special safety measures is to be taken by the user or owner. Measures are as follows:

- (i) No objection certificate stating the Sub-Station safe by the Fire Service and Civil Defense Department.
- (ii) Certification of the building consultant stating safe, proper ventilation, easy entrance and exit and safe load bearing capacity of the floors above the ground floor level (GFL).
- (iii) Proper undertaking of the Sub-Station user or owner as the case may be, Stating safety and liability will be ensured by them.

In case the electric substation has to be located within the main building itself for unavoidable reasons, it should be located on ground floor or assessment floor or the floors above the ground floor (GFL) with easy access from outside.

1.3.18.3 Height, area, floor level and other requirements of a substation room

- (a) The minimum height of a substation room should be 3.0 m to 3.6 m depending upon the size of the transformer.
- (b) The recommended area required for substation and transformer rooms for different capacities are given in Table 8.1.23 for general guidance. Minimum recommended spacing between the transformer periphery and walls should be :
 - (i) 0.75 m for Transformer installed in a room with wall on two sides.
 - (ii) 1.0 m for Transformer installed in a room with wall on three sides.
 - (iii) 1.25 m for Transformer installed in an enclosed room.
 - (iv) 1.5 m distance from one to another transformer for multiple transformers in room for 11 kV voltage level and 2.5 m distance for higher level of voltage.

- (c) For transformers having large oil content (more than 2000 litres), soak pits are to be provided.

The areas given in Table 8.1.23 hold good if they are provided with windows and independent access doors in accordance with local regulations.

All the rooms shall have significant ventilation. Special care should be taken to ventilate the transformer rooms and where necessary louvers at lower level and exhaust fans at higher level shall be provided at suitable locations in such a way that cross ventilation is maintained. Fans should be provided so that the transformer gets air supply from the fans.

The floor level of the substation should be high. Arrangement shall be made to prevent storm water entering the transformer and switch rooms through the soak pits, if floor level of the substation is low.

Substation of higher voltage may also be considered to the basement floor having proper and safe building design.

Table 8.1.23: Recommended Area for Transformer and Substation of Different Capacities

Capacity of Transformer (kVA)	Transformer Area (m ²)	Total Substation Area (with HT, LT Panels & Transformer Room but without Generators), (m ²)
1 × 150	12	45
1 × 250	13	48
2 × 250	26	100
1 × 400	13	48
2 × 400	30	100
3 × 400	40	135
2 × 630	26	100
3 × 630	40	190
2 × 1000	40	180
3 × 1000	45	220

1.3.18.4 11kV/0.4kV Distribution transformer for the substation of a building

An 11 kV/0.4kV indoor distribution Transformer is a major part of an indoor substation. These Substations may be installed inside the building itself or may be housed in a separate building adjacent to the building.

For small to moderate power rating up to 2 MW, two types of indoor transformers have been widely used in recent years. These are (i) Oil Type Natural Cooled transformer and (ii) Cast Resin Dry Type Natural Cooled transformers.

In most cases Oil Type Natural Cooled transformer may be used for substations if adequate space is available to accommodate the transformer.

Cast Resin Dry Type Natural Cooled transformers should be used (i) in places where stringent protection against spread of fire is needed and (ii) in places where space saving is of utmost importance.

Choice of oil type or dry type transformers

Dry type transformer should be installed where risk of spreading of fire is high and where flammable materials are to be kept around the substation.

For Hospital buildings, Multistoried Shopping Centers Dry type transformers should be used to for minimizing fire risks.

An industrial buildings containing inflammable materials, chemical and having the substation in the same building dry type transformers should be used for minimizing fire risks.

1.3.18.5 Type of connection between a substation transformer and its LT panel

Connection between a substation transformer and its LT panel can be established a) by using NYY underground LT Cables or b) by using ceiling suspended busbar trunking. For small size transformers the first method should be used although there is no restriction in using the second method. However, for big substations the second method is safer and at the same time gives a neat solution.

1.3.18.6 Ventilation of a substation

In an electrical substation significant amount of forced ventilation is very much needed apart from natural ventilation. Exhaust fans (minimum 450 mm dia) must be provided in sufficient numbers on all sides of the substation above the lintel level. Grill fitted windows having window panes must be provided on all sides for natural ventilation. The windows must have sun sheds so that no rain water can enter inside the substation.

If due to space constraint or due to any other difficulties, sufficient number of windows and ventilating fans cannot be installed, high velocity forced ventilation using ducts must be provided.

1.3.18.7 Layout of a substation

- (a) In general, substation HT to LT transformer shall be placed in one corner of the room so that the HT side remains away from the passage of the persons.
- (b) The HT metering panel shall be located near the exterior of the substation room near the exit gate and also shall be convenient for the HT cable entry.
- (c) The HT Panel shall be located near the exterior, just after or adjacent to the HT panel.
- (d) LT panel shall remain at a sufficient distance from the transformer but not too far away from the transformer. On the other hand, the location of the LT panel should be such that the riser main cable can have their way upward or outward within very short distance.
- (e) In allocating the areas within a substation, it is to be noted that the flow of electric power is from supply company network to HT room, then to transformer and finally to the low voltage switchgear room. The layout of the rooms shall be in accordance with this flow.
- (f) All the rooms shall have significant ventilation. Special care should be taken to ventilate the transformer rooms and where necessary louvers at lower level and sufficient number of high speed exhaust fans at higher level shall be provided at suitable locations in such a way that cross ventilation is maintained. Sufficient numbers of ceiling fans must be provided so that the transformer gets air supply from ceiling fans.
- (g) The 11 kV/0.4 kV substation shall not be placed in a basement.
- (h) The substation shall preferably be placed in ground floor. Placing a substation on any other floor other than ground floor shall be avoided.
- (i) The substation room and the areas adjacent to cable routes must have adequate fire alarm and fire extinguishing/fighting system appropriate for extinguishing fire due to electrical system, cable burning and oil burning.

1.3.19 Standby Power Supply

1.3.19.1 Provision for standby power supply

Provision should be made for standby power supply, in buildings, where interruption of electrical power supply would cause significant discomfort, result in interruption of activities, major production loss, cause hazard to life and property and cause panic. The standby power supply may be a petrol engine or diesel engine or gas engine generator or an IPS or a UPS.

1.3.19.2 Capacity of a standby generating set

The capacity of standby generating set shall be chosen on the basis of essential light load, essential air-conditioning load, essential equipment load and essential services load, essential lift (s), one or all water pumps and other loads required as essential load. Table 8.1.24 shows recommended room area for different sizes of generators as a general guidance. Minimum recommended spacing between the generator periphery and walls need to be included:

- (i) 1.0 m for generator installed in an enclosed room.
- (ii) 1.25 m distance from one to another generator for multiple generators in the room.

1.3.19.3 Generator room

The generating set should preferably be housed in the substation building or should be placed adjacent to the substation room to enable transfer of electrical load (Change over) with negligible voltage drop as well as to avoid transfer of vibration and noise to the main building. The generator room should have significant amount of ventilation and fitted with a number of ceiling fans. Appropriate type and number of firefighting equipment must be installed inside the generator room. The generator engine exhaust should be appropriately taken out of the building and should preferably be taken out through any other side except South. The generator oil tank should be placed away from the control panel side. In case of gas engine generator extra precaution must be taken regarding ventilation, leakage to prevent explosion.

The standby generator room should preferably be located outside the building. In the case of a gas engine driven generator, the generator must be located outside the building with adequate ventilation and windows. In general the generator room must have adequate ventilation and fans for continuous cooling.

The generator shall be placed either on the ground floor or in the first basement. Location of generator in the basement floor preferably be avoided. If generator is to be installed on the basement floor special safety measures is to be taken by the user or owner as per manufacturer's safety specification.

A continuous running generator must be located outside the building. Other rules mentioned above for standby generator are strictly applicable for this case.

For both the standby and continuous running generators the generator room and the areas adjacent to cable routes must have adequate fire alarm and fire extinguishing/fighting system appropriate for extinguishing fire due to cable burning and fuel burning. Generators must be installed on shock absorbing mounting bases.

1.3.19.4 Changeover switch of a generator

A standby generator, if needed, is to be connected at the supply input point after the energy meter and after the main incoming switch or the main incoming circuit breaker, but through a changeover switch of appropriate rating. The rating of such a switch shall be at least 1.25 times the rating of the main incoming circuit breaker. The changeover switch shall be of such a type so that when moved to the mains position, there is no chance that the generator will be connected and vice versa.

The Changeover Switch may be manual type with knife switch type switching or may be automatic type with magnetic contactors. In both the cases the Changeover Switch shall be properly made so that there is no chance of loose connection or spark.

The wiring for this purpose shall be made following the standard practices mentioned under the heading of wiring using cables of appropriate size.

Table 8.1.24: Recommended Area for Standby Generator Room

Capacity (kW)	Area (m ²)
1 × 25	20
1 × 48	24
1 × 100	30
1 × 150	36
1 × 300	48
1 × 500	56

1.3.19.5 Installation of an IPS or a UPS

- For safety purpose size of a UPS should be kept as small as possible.
- For the installation of a 200 - 600 VA IPS a 5A circuit must be made with the light points and fan points of different rooms to be brought under the control of the IPS. This circuit must have 3A Fuse protection using fuse cutout box. Wiring and connection has to be made following the wiring rules given in the wiring sections of this document. Cables of appropriate size must be used for wiring.
- For the installation of a 600 - 700 VA IPS a 5A circuit must be made with the light points and fan points of different rooms to be brought under the control of the IPS. This circuit must have 5A fuse protections or 5A circuit breaker protection. Wiring and connection has to be made following the wiring rules given in the wiring sections of this document. Cables of appropriate size must be used for wiring.

- (d) For the installation of an IPS of higher capacity, a BDB with multiple outgoing circuits each not exceeding 5 A shall be used with cutout - fuse protection at both incoming and outgoing sides. Cables of appropriate size must be used for wiring of each circuit.
- (e) Battery maintenance (checking water level, temperature rise and the condition of the terminals) should be done at least every 15 days. Connection of the Battery terminals should be made properly and checked periodically for loose connection and deposition of sulphate. Battery of an IPS must be kept in a safe place so that short circuit between the battery terminals does not occur. Inflammable materials must not be kept in the vicinity of the IPS or battery.
- (f) Safety issues must be taken into consideration in placing an IPS in a room. Same points shall apply for the installation of an UPS.

1.3.19.6 Installation of a solar photovoltaic system on top of a building

Building should be provided with solar photovoltaic system. For installation of a solar photovoltaic system, necessary precaution needs to be taken. Separate wiring and protection system must also be ensured.

Installation of solar water heaters on the roof tops of the residential and commercial buildings:

Buildings or apartments where hot water will be required, use of solar water heaters instead of electric and gas water heaters should be made mandatory. Flat plate heat collectors or vacuum tube solar water heaters of various capacities are available in the market.

The integral parts of a conventional solar photovoltaic system are:

- (a) Solar photovoltaic panel(s)
- (b) Battery charge controller
- (c) Inverter
- (d) Cables between the solar photovoltaic panel(s) and the battery charge controller
- (e) Cables between the battery and the battery charge controller
- (f) Cables between the inverter and the distribution board (DB/SDB/BDB)
- (g) Other cables and accessories.

For the installation of a solar photovoltaic system of higher capacity, a DB with multiple outgoing circuits each not exceeding 5 A shall be used with cutout - fuse protection at both incoming and outgoing side. Copper cables of appropriate size must be used for wiring of each circuit.

Battery maintenance (checking water level, temperature rise and the condition of the terminals) should be done at least every 15 days. Connection of the battery terminals should be made properly and checked periodically for loose connection and deposition of sulphate.

Batteries of a solar photovoltaic system must be kept in a safe place so that short circuit between the battery terminals does not occur. Inflammable materials must not be kept in the vicinity of the IPS or battery. In most cases for roof top solar panels, the battery room shall be placed inside a roof top room with adequate natural ventilation and forced cooling using ceiling fans. Because of the roof top location of the Solar panels, the room temperature is expected to be higher.

Safety issue must be taken into consideration in placing the batteries of a solar photovoltaic system.

For a residential flat system building, one or two circuits for each flat shall come from the DB of the photo-voltaic source at roof top to each flat depending on the requirement. Connection to load in each flat will be done through a changeover switch for each circuit.

For a commercial/office building, one or two circuits for each office/office area shall come from the DB of the photo-voltaic source at roof top to each flat depending on the requirement. Connection to load in each flat will be done through a changeover switch for each circuit.

Conduit based riser system must carefully be installed, separately for this system only, during the construction of the building to bring down the cables from the roof top DB room up to each flat/office/office area. Special care must be taken during installation so that rain water can under no circumstances get into the conduit and cable system.

1.3.19.7 Installation of a Solar Photovoltaic System on the exterior Glass of a Building having Large Glass area Facade

For semitransparent solar panels mounted on exterior glass of multistoried building similar process and precautions mentioned above must be followed.

1.3.20 Electrical Distribution System

1.3.20.1 Design, selection and choice of the type of connection

- (a) In the planning and design of an electrical wiring installation, due consideration shall be given to prevailing conditions. Advice of a knowledgeable and experienced electrical design engineer must be sought from the initial stage up to the completion of the installation with a view to have an installation that will prove adequate for its intended purpose, and which will be safe and will be efficient.
- (b) All electrical apparatus shall be suitable for the voltage and frequency of supply of this country mentioned earlier.

- (c) The number and types of connection required e.g., single-phase two-wire AC or three-phase four-wire AC shall be assessed, both for the supply source and for the internal circuits needed within the installation.
- (d) The following characteristics of the supply shall be ascertained :
 - (i) nominal voltage(s)
 - (ii) current and frequency
 - (iii) prospective short circuit current at the origin of the installation
 - (iv) type and rating of the over-current protective device acting at the origin of the installation
 - (v) suitability for the requirements of the installation, including the maximum demand
 - (vi) expected maximum value of the earth loop impedance of that part of the system external to the installation.
- (e) In case of connected loads of 50 kW and above, HT 11 kV three-phase supply line with substation must be installed because of the requirement of the distribution companies although the use of HT supply will involve higher expenses due to installation of a distribution transformer, HT metering Panel, HT panel and LT Panel at the consumer's premises.

In this respect, the rules of the electrical distribution authorities will be the ultimate deciding factor.

1.3.20.2 Equipment and accessories

(a) High Voltage Switchgear

The selection of the type of high voltage switchgear for any installation should consider the following:

- (i) voltage of the supply system,
 - (ii) the prospective short circuit current at the point of supply,
 - (iii) the size and layout of electrical installation,
 - (iv) the substation room available, and
 - (v) the types machineries of the industry (if applicable).
- (b) Guidelines on Various Types of Switchgear Installation
- (i) Banks of switchgears shall be segregated from each other by means of fire resistant barriers in order to prevent the risk of damage by fire or explosion arising from switch failure. Where a bus-section switch is installed, it shall also be segregated from adjoining banks in the same way.

- (ii) In the case of duplicate or ring main supply, switches with interlocking arrangement shall be provided to prevent simultaneous switching of two different supply sources.

(c) Low Voltage Switchgear

- (i) Switchgear and fusegear must have adequate breaking capacity in relation to the capacity of the transformers.
- (ii) Isolation and protection of outgoing circuits forming the main distribution system may be effected by means of circuit breakers, or fuses or switch fuse units mounted on the main switchboard. The choice between alternative types of equipment will take the following points into consideration:
- (iii) In certain installations supplied with electric power from remote transformer substations, it may be necessary to protect main circuits with circuit breakers operated by earth leakage trips, in order to ensure effective earth fault protection.
- (iv) Where large electric motors, furnaces or other heavy electrical equipment are installed, the main circuits shall be protected by metal clad circuit breakers or conductors fitted with suitable instantaneous and time delay overcurrent devices together with earth leakage and backup protection where necessary.
- (v) In installations other than those mentioned above or where overloading of circuits may be considered unlikely, HRC type fuses will normally afford adequate protection for main circuits separately as required; the fuses shall be mounted in switch fuse units or with switches forming part of the main switch boards.
- (vi) Where it is necessary to provide suitable connection for power factor improvement capacitors at the substation bus, suitable capacitors shall be selected in consultation with the capacitor and switchgear manufacturer and necessary switchgear/feeder circuit breaker shall be provided for controlling the capacitor bank(s).

1.3.21 Transformers

- (a) Where two or more transformers are to be installed in a substation to supply an LT distribution system, the distribution system shall be divided into separate sections each of which shall normally be fed from one transformer only unless the LT switchgear has the requisite short circuit capacity.
- (b) Provision may, however, be made to interconnect busbar sections through bus couplers to cater for the failure or disconnection of one transformer which need to be executed with much care using locking system.

- (c) The transformers, which at any time operate in parallel, shall be so selected as to share the load in proportion to their respective ratings. Appropriate protection must be provided and appropriate arrangements need to be made.
- (d) When a step-up transformer is used, a linked switch shall be provided for disconnecting the transformer from all poles of the supply, including the neutral conductor.

1.3.22 Precautions regarding Rotating Machines

- (a) All equipment including cables, of every circuit carrying the starting, accelerating and load currents of motors, shall be suitable for a current at least equal to the full load current rating of the motor. When the motor is intended for intermittent duty and frequent stopping and starting, account shall be taken of any cumulative effects of the starting periods upon the temperature rise of the equipment of the circuit.
- (b) The rating of circuits supplying the rotors of slip ring or commutator of a motor or an induction motor shall be suitable for both the starting and loaded conditions.
- (c) Every electric motor having a rating exceeding 0.376 kW shall be provided with control equipment incorporating means of protection against overcurrent.
- (d) Every motor shall be provided with means to prevent automatic restarting after a stoppage due to drop in voltage or failure. This requirement does not apply to any special cases where the failure of the motor to start after a brief interruption of the supply would be likely to cause greater danger. It also does not preclude arrangements for starting a motor at intervals by an automatic control device, where other adequate precautions are taken against danger from unexpected restarting.

1.3.23 LT Energy Meters

LT energy meters shall be installed in residential buildings at such a place which is readily accessible to the owner of the building and the Authority. Installation of digital energy meters at the users' premises is a requirement of the distribution Companies.

LT energy meters should be installed at a height where it is convenient to note the meter reading but should not be installed at a level less than 1.5 meter above the ground.

The energy meters should either be provided with a protective covering, enclosing it completely except the glass window through which the readings are noted, or shall be mounted inside a completely enclosed panel provided with hinged or sliding doors with arrangement for locking. Earthing terminal must be provided if a metal box is used. Such an earthing terminal must be connected to the ECC.

1.3.24 Laying of LT underground Cables

PVC-PVC NYY underground LT cables shall be laid using one of the three methods.

- (a) In the first method, brick wall prepared 900 mm deep trenches with cover plates shall be used for placing the cables at the bottom of the trench.
- (b) In the second method, 900 mm deep trenches prepared by ground excavation (underground direct burial method) shall be used for placing the cables on top of a 75 mm sand layer. In this second method (underground direct burial method), two layers of brick on top, marking tape and then back filling the trench will have to be done. The depth of the trench in general shall be 900 mm.
- (c) In the third method, pre-laid PVC pipes having sufficient clearance compared to the cable size (s) may be required at places. The PVC pipes must be laid in trenches of the 900 mm depth. For pre-laid PVC pipe ducts, brick wall made underground inspection pits will be required at an interval of at least 10 m for cable pulling and future extensions or alterations.

1.3.25 Laying of HT Underground Cables

The HT underground armoured cables shall be laid using one of the three methods.

In the first method (i) brick wall prepared 900 mm deep trenches with cover plates shall be used for placing the cables at the bottom of the trench.

In the second method, 900 mm deep trenches prepared by ground excavation (underground direct burial method) shall be used for placing the cables on top of a 75 mm sand layer. In this second method (underground direct burial method), two layers of brick on top, marking tape and then back filling the trench will have to be done. The depth of the trench in general shall be 900 mm.

In the third method, pre-laid PVC pipes having sufficient clearance compared to the cable size(s) may be required at places. The PVC pipes must be laid in trenches of the 900 mm depth.

For pre-laid PVC pipe ducts, brick wall made underground inspection pits will be required at an interval of at least 10 m for cable pulling and future extensions or alterations.

PVC pipe having sufficient clearance may be used for bringing the cable up to the trench of the metering panel or HT panel.

The PVC pipes must have 18 SWG GI pull wires placed during laying of the pipes for pulling the cables later.

Methods of installation of cables and conductors in common use are specified in Table 8.1.25.

1.3.26 Main Switch and Switchboards**1.3.26.1 Metal clad enclosed type**

All main switches shall be either metal clad enclosed type or of any other insulated enclosed type and the circuit breakers shall be fixed at close proximity.

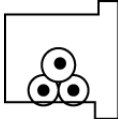
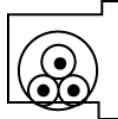
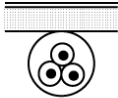
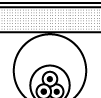
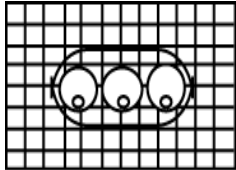
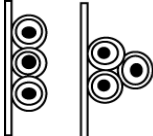
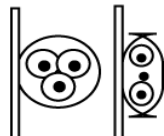
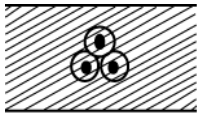
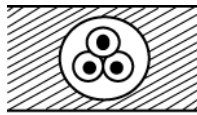
1.3.26.2 Circuit breakers on each live conductor

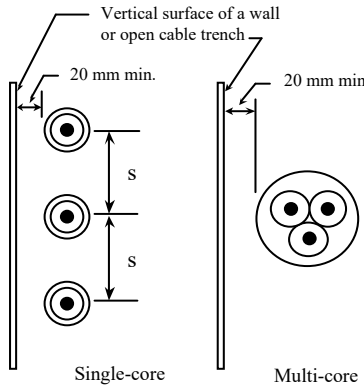
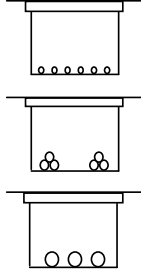
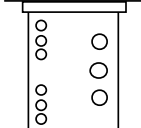
There shall be circuit breakers or miniature circuit breakers or load break switch fuses on each live conductor of the supply mains at the point of entry. The wiring throughout the installation shall be such that there is no break in the neutral wire in the form of a switch or fuse unit or otherwise.

1.3.26.3 Location

- (a) The location of the main board shall be such that it is easily accessible for firemen and other personnel to quickly disconnect the supply in case of emergencies.
- (b) Main switchboards shall be installed in boxes or cupboards so as to safeguard against operation by unauthorized personnel.
- (c) Open type switchboards shall be placed only in dry locations and in ventilated rooms and they shall not be placed in the vicinity of storage batteries or exposed to chemical fumes.
- (d) In damp situation or where inflammable or explosive dust, vapour or gas is likely to be present, the switchboard shall be totally enclosed or made flame proof as may be necessitated by the particular circumstances.
- (e) Switchboards shall not be erected above gas stoves or sinks, or within 2.5 m of any washing unit in the washing rooms or laundries.
- (f) In case of switchboards being unavoidable in places likely to be exposed to weather, to drip, or in abnormally moist atmosphere, the outer casing shall be weather proof and shall be provided with glands or bushings or adapted to receive screwed conduit.
- (g) Adequate illumination shall be provided for all working spaces about the switchboards, when installed indoors.

Table 8.1.25: Different ways of Installation of Cables and Conductors in Common Use

Type	Description	Example	
A	Cables enclosed in conduit	 Single-core	 Multi-core
B	Cables enclosed in trunking	 Single Core	 Multi-Core
C	Cables enclosed in underground conduit, ducts, and cable ducting.	 Single-core	 Multi-core
D	Two or more single-core cables contained in separate bores of a multi-core conduit and intended to be solidly embedded in concrete or plaster or generally incorporated in the building structure.		
E	Sheathed cables clipped direct to a nonmetallic surface.	 Single-core	 Multi-core
F	Sheathed cables on a cable tray.	 Single-core	 Multi-core
G	Sheathed cables embedded direct in plaster.	 Single-core	 Mutli-core

Type	Description	Example
H	Sheathed cables suspended from or incorporating a catenary wire.	 Single-core Multi-core
J	Sheathed cables in free air. For cables in which the conductor cross-sectional area does not exceed 185 mm^2 , S is equal to twice the overall diameter of the cable. For cables in which the conductor cross-sectional area exceeds 185 mm^2 , S is about 90 mm. For two cables in horizontal formation on brackets fixed to a wall, S may have any lesser value.	 Single-core Multi-core
K	Single and multi-core cables in enclosed trench 450 mm wide by 600 mm deep (minimum dimensions) including 100 mm cover.	Two single-core cables with surfaces separated by a distance equal to one diameter; three single-core cables in trefoil and touching throughout. Multi-core cables or groups of single-core cables separated by a minimum distance of 50 mm. 
L	Single and multi-core cables in enclosed trench 450 mm wide by 600 mm deep (minimum dimensions) including 100 mm cover.	Single-core cables arranged in flat groups of two or three on the vertical trench wall with surfaces separated by a distance equal to one diameter with a minimum separation of 50 mm between groups. Multi-core cables installed singly separated by a minimum* distance of 75 mm. All cables spaced at least 25 mm from the trench wall. 

Type	Description	Example
M	Single and multi-core cables in enclosed trench 600 mm wide by 760 mm deep (minimum dimensions) including 100 mm cover.	Single-core cables arranged in groups of two or three in flat formation with the surfaces separated by a distance equal to one diameter or in trefoil formation with cables touching. Groups separated by a minimum* distance of 50 mm either horizontally or vertically. Multi-core cables installed singly separated by a minimum* distance of 75 mm either horizontally or vertically. All cables spaced at least 25 mm from the trench wall.

* Larger spacing to be used where practicable.

1.3.27 Mounting of Metal clad switchgear

A metal clad switchgear shall be mounted on hinged type metal boards or fixed type metal boards.

- Hinged type metal boards shall consist of a box made of sheet metal not less than 2 mm thick and shall be provided with a hinged cover to enable the board to swing open for examination of the wiring at the back. The joints shall be welded. The board shall be securely fixed to the wall by means of rag bolt plugs, or wooden plugs and shall be provided with locking arrangement and an earthing stud. All wires passing through the metal board shall be protected by a rubber or wooden bush at the entry hole. The earth stud should be commensurate with the size of the earth lead(s).
- Fixed type metal boards shall consist of an angle or channel steel frame fixed on the wall at the top, if necessary.
- There shall be a clearance of one meter at the front of the switchboards.

1.3.28 Wooden Boards as Main Boards or Sub-Boards Containing Fused Cutouts and Main Switches

Use of Wooden Board is discouraged because of the fear of break out of fire from a spark or from an overheated cable. However, for small installations, not exceeding 15A SP, connected to a single-phase 230 V supply, wooden boards may be used as main boards or sub-boards containing fused cutouts and main switches of appropriate ratings may be used. Such a board shall be made using seasoned teak or other approved quality timber.

1.3.29 Location of Distribution Boards

The distribution boards shall be located as near as possible to the centre of the load they are intended to control.

- (a) They shall be fixed on suitable stanchion or wall and shall be accessible for replacement of fuses. All switches and circuit breakers used as switches shall be located so that they may be operated from a readily accessible place. They shall be installed such that the center of the grip of the operating handle of the switch or circuit breaker, when in its highest position, is not more than 2.0 m and the bottom of the panel shall be more than 0.45m above the floor or working platform.
- (b) They shall be either metal clad type, or all insulated type. But if exposed to weather or damp situations, they shall be of the weather proof type and if installed where exposed to explosive dust, vapour or gas, they shall be of flame proof type. In corrosive atmospheres, they shall be treated with anticorrosive preservative or covered with suitable plastic compounds.
- (c) Where two or more distribution fuse boards feeding low voltage circuits are fed from a supply of medium voltage, these distribution boards shall be:
 - (i) fixed not less than 2 m apart, or
 - (ii) arranged so that it is not possible to open two at a time, namely, they are interlocked, and the metal case is marked "Danger 415 Volts" and identified with proper phase marking and danger marks, or
 - (iii) installed in rooms or enclosures accessible to authorized persons only.
- (d) All distribution boards shall be marked "Lighting" or "Power", as the case may be, and also be marked with the voltage and number of phases of the supply. Each shall be provided with a circuit list giving diagram of each circuit which it controls and the current rating for the circuit and size of fuse element.
- (e) Distribution boards must be easily accessible for the ease of maintenance and switching off during accidents.

1.3.30 Over-current and Short Circuit Protection of Circuits

- (a) Appropriate protection shall be provided at the distribution boards for all circuits and sub-circuits against short circuit and over-current. The installed protective devices shall be capable of interrupting any short circuit current that may occur, without causing any danger. The ratings and settings of fuses and the protective devices shall be coordinated so as to obtain absolute certain discrimination of the faulty area only during a fault.
- (b) Where circuit breakers are used for protection of main circuit and the sub-circuits, discrimination in operation shall be achieved by adjusting the protective devices of the sub-main circuit breakers to operate at lower current settings and shorter time-lag than the main circuit breaker.
- (c) A fuse carrier shall not be fitted with a fuse element larger than that for which the carrier is designed.
- (d) The current rating of fuses shall not exceed the current rating of the smallest cable in the circuit protected by the fuse.

1.3.31 Fire Alarm and Emergency Lighting Circuits

Fire alarm and emergency lighting circuits shall be segregated from all other cables and from each other in accordance with BS 5839 and BS 5266. Telecommunication circuits shall be segregated in accordance with BS 6701 as appropriate.

1.3.32 Earthing**1.3.32.1 General**

Earthing refers to connecting the exposed conductive part of electrical equipment and also the extraneous conductive parts of earthed bodies like water pipe to the general mass of the earth to carry away safely any fault current that may arise due to ground faults. The object of an earthing system is to provide a system of conductors, as nearly as possible at a uniform and zero, or earth, potential. The purpose of this is to ensure that, in general, all parts of equipment and installation other than live parts shall be at earth potential, thus ensuring that persons coming in contact with these parts shall also be at earth potential at all times.

1.3.32.2 Earthing used in electrical installation for buildings

The usual method of earthing is to join the exposed metal work to earth via a system of earth continuity conductors (ECC) connected to an earth electrode buried in the ground through a system of earth lead wires. In conjunction with a fuse, or other similar device, this then forms a protective system.

Thus, if a live conductor accidentally comes into contact with an exposed metal, the fuse or protective device operates. As long as the overall resistance of the protective system is low, a large fault current flows which blows the fuse. This cuts off the supply and isolates the faulty circuit, preventing risk of shock, fire, or damage to equipment/ installation.

In Electrical installation for buildings, following types of earthing systems are required to be installed:

- (i) L.T. circuit/system earthing,
- (ii) Equipment earthing (LT side),
- (iii) Substation neutral earthing,
- (iv) Substation LT system earthing, and
- (v) H.T. circuit earthing for a substation.

The purpose of L.T. circuit/system earthing is to limit excessive voltage from line surges, from cross-overs with higher voltage lines, or from lightning, and to keep noncurrent carrying enclosures and equipment at zero potential with respect to earth.

Earthing the system helps facilitate the opening of overcurrent protection devices in case of earth faults. Earthing associated with current carrying conductors is normally essential for the protection and safety of the system and is generally known as circuit/ system earthing, while earthing of non-current carrying metal work and conductor is essential for the safety of human life, animals, and property and it is generally known as equipment earthing.

1.3.32.3 Arrangements of earthing systems:

- (a) The value of resistance from the consumer's main earthing terminal to the earthed point of the supply, or to earth, is in accordance with the protective and functional requirements of the installation, and expected to be continuously effective.
- (b) Earth fault currents and earth leakage currents likely to occur are carried without danger, particularly from the point of view of thermal, thermo-mechanical and electromechanical stresses.
- (c) Where a number of installations have separate earthing arrangements, protective conductors running between any two of the separate installations shall either be capable of carrying the maximum fault current likely to flow through them, or be earthed within one installation only and insulated from the earthing arrangements of any other installation.

1.3.32.4 Integral parts of an earthing system

The integral parts of an Earthing System are:

- (a) Earth electrode(s) buried under the ground
- (b) Earth lead cables/wires connecting the earth electrode(s) with the earthing busbar system. Earth lead cables/wires are also need to interconnect the earth electrodes when there are more than one earth electrode.
- (c) Earth continuity conductors (ECC) for linking earthing busbar at the substation LT panel or main distribution DB of a building.
- (d) Earth electrode clamp.

Connections of (i) Earth continuity conductors (ECC), (ii) Earth lead cables/wires and (iii) Earth electrode(s) must be made in appropriate and long lasting manner because poor connection or loss of connection will render the earthing system ineffective.

1.3.32.5 Earth continuity conductors (ECC)

ECC runs along the circuits/sub-circuits, socket circuits, interlinking circuits between a BDB and a SDB, between a SDB and a DB, between a DB and a FDB, between a FDB and a MDB, between a MDB and the LT panel earthing busbar of the substation. At each point an ECC shall be terminated in a copper earthing busbar. In metal switch boards back boxes and in metal socket back boxes appropriate copper or brass bolt nut termination shall be provided.

ECC of an earthing system joins or bonds together all the metal parts of an installation. PVC insulated wiring copper cables of appropriate size having Green + Yellow bi-colour insulation shall be used as ECC.

The minimum size of the ECC shall be same as cross-sectional area of the phase conductor for the PVC insulated wiring copper cables of appropriate size having Green + Yellow bi-colour insulation.

1.3.32.6 Earth lead cable/wire

Earth Lead cable/wire runs between an earth electrode and the earthing busbar of the MDB/DB or between an earth electrode and the LT panel earthing busbar of the substation.

Often more than one earth electrodes are needed. In such a case duplicate earth lead cables/wires from each earth electrode must be brought to the MDB/DB or to the LT panel earthing busbar of the substation and properly terminated. In addition, in the case of multiple earth electrodes, they must be interlinked by additional earth lead cables/wires.

PVC insulated wiring copper cables of appropriate size having Green + Yellow bi-colour insulation shall be used as earth lead wire. At both ends of the earth lead cable/wire, copper cable lugs must be fitted using crimp tools or hydraulic press.

The minimum size of the earth lead wire shall be 2 numbers of 1.5 mm² PVC insulated wiring copper cables of appropriate size having Green + Yellow bi-colour insulation.

The ends of the earth lead wires shall be terminated using crimp tool fitted cable lugs for fitting on the busbar or with the Earth Electrode Clamp.

- (a) An earth lead cable/wire establishes connection between the main earthing busbar and the earth electrode(s). The earth lead wire shall be brought to one or more connecting points, according to size of installation; the copper wire earthing leads shall run from there to the electrodes. Usually more than one earth lead wires are needed for one earth electrode to make sure that this link never fails.
- (b) Earth lead cable/ wires shall one of the following types:
 - (i) PVC insulated cable
 - (ii) stranded copper cables without insulation
 - (iii) copper strips (copper bars)
 - (iv) PVC insulated cable is preferable in most cases.
 - (v) Earth lead wires shall run through PVC pipe from the earth electrode up to the earthing busbar of the MDB/DB or LT Panel.
- (c) Earth lead cables/ wires shall run, at least, 2 in parallel (at least) down to the earth electrode so as to increase the safety factor of the installation. The two cables shall be terminated in two separate cable lugs and bolts at both ends. Copper wire used as earthing lead must not be smaller than single core stranded 2 × 4 mm² PVC insulated cables (i.e. 2 nos. of single core 4 mm² PVC insulated cables in parallel). Depending on the current capacity of the Main incoming line the size will have to be raised.

Earth lead cables/wires shall be pulled from the earth electrode up to the terminating earthing busbar through PVC conduits or GI pipes of appropriate dimension.

Table 8.1.26: Minimum Cross-sectional Area of Copper ECCs in Relation to the Area of Associated Phase Conductors

Cross-sectional Area of Phase Conductor(s) (mm ²)	Minimum Cross-sectional Area of the Corresponding Earth Conductor (mm ²)
Less than 16	Same as cross-sectional area of phase conductor
16 or greater but less than 35	16 mm ²
35 or greater	Half the cross-sectional area of phase conductor

1.3.32.7 Earth electrodes and their installation

The earth electrode shall, as far as practicable, penetrate into moist soil (which will remain moist even during the dry season) preferably below ground water table. The resistance of an earthing system after measured after the installation of earth electrodes (individually or combined as a single group) shall be around one ohm.

The types of earth electrodes are to be used for earthing of electrical installations of a building and their sizes shall be as under:

- (a) Copper rod earth electrode: shall have a minimum diameter of 12.5 mm of minimum length of 3.33 m. Multiple copper rod earth electrodes may have to be installed to achieve an acceptable value of earthing resistance of around 1 ohm.
- (b) Copper plate earth electrodes: shall be 600 mm x 600 mm x 6 mm minimum in size. The copper plate shall be buried at least 2 m below the ground level. Multiple Copper plate earth electrodes may have to be installed to achieve an acceptable value of earthing resistance of around 1 ohm.
- (c) Galvanized Iron (GI) pipes: GI pipe earthing shall have a minimum diameter of 38 mm and of minimum length of 6.5m. Multiple GI pipes Earth Electrode may have to be installed to achieve an acceptable value of earthing resistance of around 1 ohm.

Schematic drawings of typical earthing systems are shown in Figures 8.1.2 to 8.1.4. For the installation of the earthing system the following points shall be considered.

- (a) For installing a copper rod earth electrode, a 38 mm GI pipe shall be driven below ground up to a depth of 5 m and shall be withdrawn. The 12 mm dia copper rod earth electrode of 4 m length shall then be easily driven into that hole up to a depth of 3.6 m and 0.33 m shall be left for placing inside the earthing pit described below.
- (b) For installing a 600 mm x 600 mm x 6 mm Copper plate 2 m below the ground level earth excavation will have to be done. The earth lead wire shall come via an earthing pit.
- (c) GI pipe earth electrodes driven by tube well sinking method are suggested. For this purpose 38 mm dia GI pipes are recommended for domestic buildings. For large plinth area buildings and multi-storied buildings 50 mm dia GI pipes are recommended. The length of GI pipe to be driven below the ground level depends on the earthing resistance which in turn depends on the availability of water table during the dry season in this country. However, except the high land and mountains, this depth varies between 12 m to 25 m.

- (d) Multiple numbers of GI pipe earth electrodes need to be used and connected in parallel in order to lower the earthing resistance measured with an earth resistance measuring meter. This is applicable for copper rod earthing and plate earthing also.
- (e) If multiple rod, pipe, or plate electrodes of one grounding system are installed to meet the earth resistance requirements, they shall not be less than 1.8 m apart. Two or more grounding electrodes that are bonded together shall be considered a single grounding electrode system.
- (f) If multiple rod, pipe, or plate electrodes of one grounding system are installed to meet the earth resistance requirements, they shall not be less than 1.8 m apart. Two or more grounding electrodes that are bonded together shall be considered a single grounding electrode system.

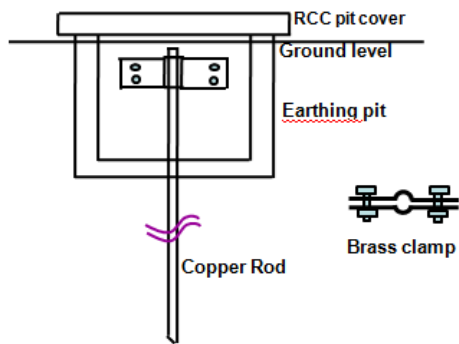


Figure 8.1.2 Copper Rod Earthing

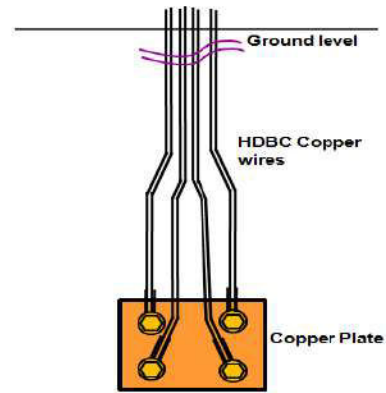
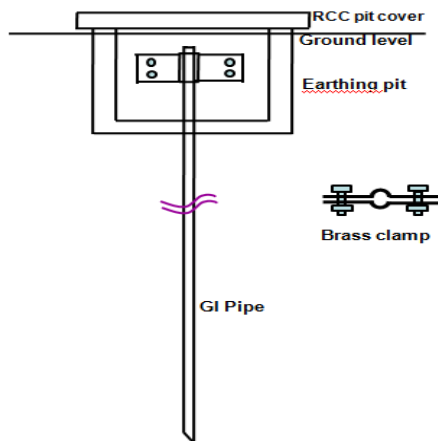
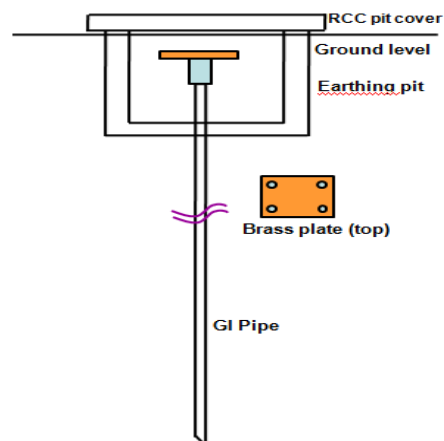


Figure 8.1.3 Copper Plate Earthing



(a)



(b)

Figure 8.1.4 Pipe Earthing; (a) Type 1; (b) Type 2

1.3.32.8 Brass clamps/terminals on earth electrodes (earth electrode clamp)

- (a) A brass clamp must be fitted on top of a GI pipe earth electrode to terminate the earth lead wire and to maintain electrical contact with the earth electrode and also to terminate the earth lead wire coming from the earthing busbar of the LT panel/ MDB/DB. This is needed to establish long lasting and firm connection between the earth electrode and earth lead wire, which in turn means connection between the earth electrode and earthing busbar of an LT panel or MDB/DB.
- (b) The Brass clamp shall be made using at least 9.5 mm thick and at least 50 mm wide Brass plate bent and shaped properly to fit tightly around the GI pipe earth electrode and shall have sufficient length (at least 35 mm) on both sides for fixing bolts and cable lugs. This clamp shall have two hexagonal head 9.5 mm bolts on one side and two hexagonal head 9.5 mm bolts on the other side, Figure 8.1.4(a). Sufficient space should be available for fixing the cable lugs of the earth lead wires. After fitting the lugs of the earth lead cables the brass clamp and the GI pipe head should be coated with two coats of synthetic enamel paint on top of one undercoat paint layer.
- (c) An alternative to this clamp is to use a 9.5 mm (at least) thick brass plate having 4 holes for fitting four hexagonal brass bolts on the four corners for fitting the cable lugs of the earth lead wires as shown in Figure 8.1.4(b). The brass plate is welded to a GI pipe socket and threaded on top of the earth electrode (pipe).

1.3.32.9 Earthing busbars

A copper earthing busbar shall be provided inside the LT Panel or MDB/DB of a building. The earth lead wire coming from the Earth Electrode (s) shall be terminated on this busbar using cable lugs (cable lugs must be fitted using crimp tools or hydraulic press) and brass bolts and nuts.

Copper earthing busbar shall also be provided inside the DBs, FDBs, SDBs and BDBs. Hexagonal head brass screw, nuts and washers are needed for fixing the ECC and earth lead cables with this busbar

1.3.32.10 Earthing pit

An earthing pit shall be constructed around the top of the Earth Electrode, below the ground level using 250 mm brick walls on a CC floor with a 150 mm thick RCC slab cover on top having lifting hooks. The top of the earth electrode (in case of pipe earthing) shall remain 375 mm above the top of the bottom CC floor of the pit. The minimum inside dimension of the earthing pit shall be 600 mm × 600 mm × 600 mm. The outside as well as the inside of the walls of the pit and the floor of the pit shall be cement mortar plastered. The inside shall be net cement finished. The top of the RCC slab pit cover shall remain 38 mm above the ground level. The pit shall be made in such a way that water cannot get in to the pit. One earthing pit is needed for one earth electrode.

1.3.33 Lightning Protection of Buildings

Whether a building needs protection against lightning depends on the probability of a stroke and acceptable risk levels. Assessment of the risk and of the magnitude of the consequences needs to be made. As an aid to making a judgment, a set of indices is given in Table 8.1.27 below for the various factors involved.

Table 8.1.27 (a): Index Figures Associated with Lightning Protection Design

Index A: Use of Structure	Index	Index B: Type of Construction	Index
Houses and similar buildings	2	Steel framed encased with nonmetal roof ^a	1
Houses and similar buildings with outside aerial	4	Reinforced concrete with nonmetal roof	2
Small and medium size factories, workshops and laboratories	6	Brick, plain concrete, or masonry with nonmetal roof	4
Big industrial plants, telephone exchanges, office blocks, hotels, blocks of flats	7	Steel framed encased or reinforced concrete with metal roof	5
Places of assembly, for example, places of workshop, halls, theatres, museums, exhibitions, department stores, post offices, stations, airports, stadiums	8	Timber formed or clad with any roof other than metal or thatch	7
Schools, hospitals, children's homes and other such structures	10	Any building with a thatched roof	10

^a A structure of exposed metal which is continuous down to ground level is excluded from the table as it requires no lightning protection beyond adequate earthing arrangements.

Table 8.1.27 (b): Index Figures Associated with Lightning Protection Design

Index C : Contents or Consequential Effects	Index	Index D : Degree of Isolation	Index
Ordinary domestic or office building, factories and workshops not containing valuable materials	2	Structure located in a large area having structures or trees of similar or greater height, e.g. a large town or forest	2
Industrial and agricultural buildings with specially susceptible ^b contents	5	Structure located in an area with a few other structures or trees of similar height	5
Power stations, gas works, telephone exchanges, radio stations	6	Structure completely isolated or exceeding at least twice the height of surrounding structures or trees	10
Industrial key plants, ancient monuments, historic buildings, museums, art galleries	8	Index E : Type of Terrain	Index
Schools, hospitals, children's and other homes, places of assembly	10	Flat terrain at any level	2
^b This means specially valuable plant or materials vulnerable to fire or the results of fire.		Hilly terrain	6
		Mountainous terrain 300 m and above	8

Table 8.1.27 (c) : Index Figures Associated with Lightning Protection Design

Index F : Height of Structure	Index	Index G : Lightning Prevalence	Index
Up to 9 m	2	Number of thunderstorm days per year:	
9-15 m	4	Up to 3	2
15-18 m	5	4-6	5
18-24 m	8	7-9	8
24-30 m	11	10-12	11
30-38 m	16	13-15	14
38-46 m	22	16-18	17
46-53 m ^c	30	19-21	20
^c Structures higher than 53 m require protection in all cases		Over 21	21

1.3.33.1 Degree of Isolation

The relative exposure of a particular building will be an element in determining whether the expense of lightning protection is warranted. In closely built-up towns and cities, the hazard is not as great as in the open country.

1.3.33.2 Type of terrain

In hilly or mountainous areas, buildings are more susceptible to damage due to lightning than buildings in the plains or flat terrain. In hilly areas, a building upon high ground is usually subject to greater hazard than one in a valley or otherwise sheltered area.

1.3.33.3 Height of structure

Height of the structure is an important factor for the purpose of lightning protection. Taller structures are subject to greater hazards than smaller structures and, therefore, lightning protection is more desirable for tall structures.

1.3.33.4 Lightning prevalence

The number of thunderstorm days in a year varies in different parts of a country. However, the severity of lightning storms, as distinguished from their frequency of occurrence, is usually much greater in some locations than others. Hence, the need for protection varies from place to place, although not necessarily in direct proportion to the thunderstorm frequency.

1.3.33.5 Risk assessment

"Risk Index" is the sum of the indices for all the factors, as given in Table 8.1.27. A few examples of calculation of Risk Index are given in Table 8.1.28, based on a marginal Risk Index of 40.

1.3.33.6 Integral parts of a lightning protection system

A smallest complete lightning protection system shall consist of (i) An air spike or air terminal, (ii) A down conductor, (iii) A roof conductor and (iv) An earth electrode.

An air spike or air terminal is that part which is intended to intercept lightning discharges. It consists of a vertical thick conductor of round cross section mounted on the highest part of the building to protect the required area. However, in general there may be more than one air spike or air terminal. In such a case roof conductors (made with copper strips or PVC insulated Annealed Stranded copper cables) need to be used to interconnect the Air Spikes or Air Terminals. Usually, for each Air Spike or Air Terminal there shall be one down conductor (made with copper strips or PVC insulated Annealed Stranded copper cables) going down up to the Earth Electrode pit and connected to the Earth Electrode. In all junctions, appropriate type of copper or brass junction plates or brass clamps must be used to ensure low resistance, firm and long lasting connection.

Table 8.1.28: Example of Calculation of Risk Index

Example	A	B	C	D	E	F	G	Total Index Figure	Recommendations
Small residential building in a thickly populated locality (height less than 10 m)	2	4	2	2	2	2	21	35	No protection needed, in general
Office building in a locality (height 20 m)	7	2	2	2	2	5	21	41	As the figure is around 40, need of protection will depend upon the importance of the building
Hotel building (height 31m) exceeding twice the height of surrounding structures	7	2	2	10	2	16	21	60	Protection essential
Building of historical importance completely isolated (height > 55 m)	8	4	8	10	2	30	21	83	Protection essential
Structure of high historical importance (height > 55 m)	-	-	-	-	-	-	-	-	Protection essential as the height exceeds 53 m
Structure, such as hydro-electric power stations, sufficiently protected by means of surrounding structures, for example, high vertical cliffs, high metallic structures or earth wire of transmission system (height 15 m)	7	2	6	2	6	4	21	48	Protected by surroundings

(a) Air spike/air terminal

An air spike or air terminal shall be made with copper rod of minimum 12 mm diameter with tin coating on top. The terminal shall have a copper/brass base plate for mounting on top of roof, column, parapet wall using rowel bolts. The minimum dimension of such a base plate shall be 152 mm x 152 mm x 13 mm. The length and width may need to be increased depending on the number of connection of the down conductors and the roof conductors. Such connections are to be made using hexagonal head brass bolts and nuts of 10mm diameter with brass washers. Minimum height of the air terminal shall be 300 mm above the highest point of the building fulfilling all the conditions of NFPA 780.

(b) Down conductor

A Down Conductor shall be made with copper strip or Stranded PVC insulated annealed copper cable.

(c) Roof conductor

A Roof Conductor shall be made with copper strip or Stranded PVC insulated annealed copper cable. This shall run along the periphery of the roof to link all air spikes and all down conductors installed on top of a building. The joints shall be made using clamps made of copper strips (of 1/8 inch minimum thickness) and appropriate brass bolts and washers of 3/8 inch minimum diameter.

(d) Earth electrode

The Earth Electrode is exactly of the same type as the Earth Electrode of the Electrical Distribution (Electrical Installation for Buildings) system described earlier in this document. Considering the practical situation in this country and Pipe Earth Electrodes are suggested. For each Air spike one Earth Electrode is an ideal solution.

(e) Earth inspection boxes

A 18 SWG GI sheet made Earth Inspection Box must be provided for each down conductor 1000 mm above the plinth level of the building (concealed inside the wall) which will contain a copper strip made clamp on the insulation peeled down conductor to check the continuity of the Earth Lead Down Conductor and the Earth Electrode and also to measure the Earth Resistance of the system. The box shall have a GI sheet made cover plate.

(f) Earthing pit

Earthing pits shall be provided as described in the earthing topic above.

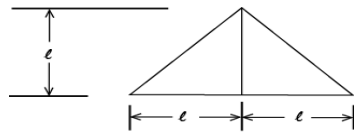
1.3.33.7 Number of lightning arrestors required and their installation

Number of Lightning Protection Air Spikes in a building will depend on the nature of the roof top, on the total area of the roof top, on the height of the building, height of the adjacent buildings, height of the nearby towers or other similar structures. However, as a thumb rule, for every 80 m² area at least one air spike should be chosen at the beginning. During placement of the air spikes the total number may have to be increased or adjusted.

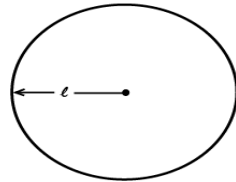
1.3.33.8 Protection zone

The zone of protection is the space within which an air spike provides protection by attracting the stroke to itself. It has been found that a single vertical conductor attracts to itself strokes of average or above average intensity which in the absence of the conductor would have struck the ground within a circle having its centre at the conductor and a radius equal to twice the height of the conductor. For weaker than average discharges the protected area becomes smaller. For practical design it is therefore assumed that statistically satisfactory protection can be given to a zone consisting of a cone with its apex at the top of the vertical conductor and a base radius equal to the height of the conductor. This is illustrated in Figure 8.1.5. A horizontal conductor can be regarded as a series of apexes coalesced into a line, and the zone of protection thus becomes a tent-like space (Figure 8.1.6).

- (a) When there are several parallel horizontal conductors the area between them has been found by experience to be better protected than one would expect from the above considerations only. The recommended design criterion is that no part of the roof should be more than 9 m from the nearest horizontal conductor except that an additional 0.3 m may be added for each 0.3 m or part thereof by which the part to be protected is below the nearest conductor.
- (b) The earth termination is that part which discharges the current into the general mass of the earth. In other words, it is one or more earth electrodes. Earth electrodes for lightning protection are no different from earth electrodes for short circuit protection systems. The total resistance of an electrode for a lightning protection system must not exceed 10 ohms for buildings up to 10 storied and 2 ohms for high rise buildings.



Elevation

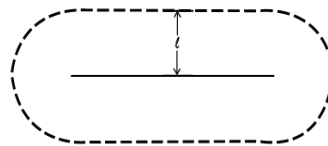


Plan

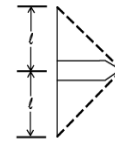
Figure 8.1.5 Protected zone for vertical conductors



Elevation



Plan



Slide View

Figure 8.1.6 Protected zone for horizontal conductors

- (c) The down conductor is the conductor which runs from the air termination to the earth termination. A building with a base area not exceeding 100 m² shall be provided with one down conductor. For a larger building, there shall be one down conductor for the first 80m² plus a further one for every 100 m² or part thereof in excess of the first 80 m². Alternatively, for a larger building one down conductor may be provided for every 30 m of perimeter. Ideally, every air spike should have a down conductor going down up to the earth electrode.
- (d) The material used for lightning conductors must be copper. The criterion for design is to keep the resistance from air termination to earth electrode to a negligible value.
- (e) Recommended dimensions for various components of lightning arrester are given in Table 8.1.29. Larger conductors should however be used if the system is unlikely to receive regular inspection and maintenance.

Table 8.1.2 9: Sizes of the Components of Lightning Protection Systems

Components	Minimum Dimensions
Air terminals	12 mm dia
Copper strip	20 mm W x 3 mm T
Copper and phosphor bronze rods	12 mm dia
PVC insulated stranded annealed copper cable (minimum size)	19 strands of 1.8 mm dia
<u>Down Conductors</u>	
Copper strip	20 mm x 3 mm
PVC insulated stranded annealed copper cable (minimum size)	19 strands of 1.8 mm dia
<u>Earth Electrode</u>	
Hard drawn copper rods for driving into soft ground	12 mm dia
Hard drawn or annealed copper rods for indirect driving or laying in ground	12 mm dia
Phosphor bronze for hard ground	12 mm dia
Copper clad steel for hard ground	50 mm dia
GI pipe	38 mm/50 mm dia

- (f) External metal on a building should be bonded to the lightning conductor with bonds at least as large as the conductor.
- (g) When a lightning conductor carries a stroke to earth, it is temporarily raised to a potential considerably above that of earth. There is, therefore, a risk that the discharge will flash over to nearby metal and cause damage to the intervening structure. This can be prevented by either, (i) providing sufficient clearance between conductor and other metal or (ii) by bonding these together to ensure that there can be no potential difference between them. The necessary clearance is obtained from :

$$D = 0.3 R + \frac{H}{15 n} \quad (8.1.1)$$

Where,

D = Clearance in metres

R = Resistance to earth in ohms

H = Height of building in metres

n = Number of down electrodes

Since it is often impracticable to provide the necessary clearance, the alternative technique of bonding is preferred.

(h) Surge arrester selection

A surge arrester is a protective device for limiting surge voltages by discharging, or bypassing, surge current through it. It also prevents continued flow of follow-through current while remaining capable of repeating these functions. It is used to protect overhead lines, transformers and other electrical apparatus mostly in an outdoor substation from lightning voltages traveling through the overhead lines.

(i) Horn-gap lightning arresters

Horn-gap lightning arresters are commonly used for low and medium voltage overhead lines. The rating of the surge arrester shall be equal to or greater than the maximum continuous phase to ground power frequency voltage available at the point of application.

(j) Design of lightning arrester using rolling sphere method

Lightning arrester (including number and height) can also be designed and installed by using rolling sphere method following NFPA 780. Minimum height of the air terminal shall be in accordance with Sec 1.3.33.6.

1.3.34 Telecommunications in Buildings**1.3.34.1 General**

Placing concealed 2 pair indoor cables is needed to get (ii) telephone lines of the wired telephone companies inside rooms of a building and (ii) to get the PABX lines of the building /offices in the building to the respective rooms under the PABXs. In addition to this, 10/20/50 pair telephone cables are required to be brought in to the PABX room(s) of the building. Conduits are to be installed for both of these two categories. For the entry of 10/20/50 pair cables, conduits through straight and easy path (in most cases, through one side of the vertical electrical duct) need to be brought in.

1.3.34.2 Concealed telecommunication cable wiring

2 pair PVC insulated PVC sheathed annealed copper telecommunication cable shall be drawn through sufficient number of pre-laid 19/25/38 mm PVC conduits to establish telecommunication network inside a building. A clearance of at least 40 percent must be maintained inside the PVC conduits. Sufficient number of 18 SWG GI sheet made pull boxes (with Perspex sheet / ebonite sheet cover plates) at all suitable places must be placed for the ease of pulling these cables.

2 pair PVC insulated PVC sheathed annealed copper telecommunication cable shall be used for wiring between a Telephone DP/Patch panel and a telecommunication outlet. The extra pair shall remain for future maintenance. The minimum size of the copper wire of this cable shall be 0.5 mm. The copper shall be preferably tinned.

1.3.34.3 Surface telecommunication cable wiring

Surface wiring should not be a choice during designing a building wiring. However, if the building is already constructed or under compulsory conditions or for extension of an existing network one may go for surface wiring. The same 2 pair PVC insulated PVC sheathed annealed copper telecommunication cable shall be used for this purpose. Wiring shall be done either by using channels or by using PVC conduits following the power line surface wiring methods mentions earlier.

1.3.34.4 Telecommunications outlets

Wall mounted Telecommunication outlets shall contain RJ11 or RJ45 connectors/jacks (shuttered). For simple telephone connection RJ11 shuttered jacks are sufficient. The outlet box shall have a back-box which may be made of the same polymer material as the front panel or shall be made using 18 SWG GI sheet or 18 SWG MS sheet but painted with two coats of synthetic enamel paint.

1.3.34.5 Telephone DP room, patch panel room and digital PABX room

Telephone DP room, Patch Panel Room and PABX room should be located near the vertical riser duct of the building so that the incoming 50/100 pair underground telephone cable can be terminated in the DP/MDF or patch panel for distribution among the flats of a multistoried residential building or among the offices of a multistoried commercial/office building.

If a digital telephone PABX is to be installed then this can be installed in the same room. A separate earth electrode with earth lead wire will be required for the PABX.

1.3.35 Television Antennas/Cable Television system

In a multistoried residential/office building, television antennas shall be placed at one suitably sited antenna location on roof top and connect these to individual flats/residences/offices in the same building by coaxial cables through concealed conduits.

1.3.35.1 Cable work for television antennas/cable television system

Vertical duct and easy entry to each flats/ offices must be provided as sharp bending of these cables is difficult and harmful to the cables. These cables must not be placed in the same conduit with power cables. A distance of at least 350 mm must be maintained if a portion runs in parallel with the power cable conduits.

RF and Video cables shall be PVC sheathed Co-axial Cables shall be made with solid Copper centre conductor, foamed polythene insulated and further sealed Alluminium foil taped and Copper wire braided.

1.3.35.2 Television antenna outlets/cable television system outlets

Wall mounted television coaxial cable outlets shall contain high quality coaxial connectors/jacks. The outlet box shall have a back-box which may be made of the same polymer material as the front panel or 18 SWG GI sheet or 18 SWG MS sheet made but painted with two coats of synthetic enamel paint.

1.3.36 Data Communication Network for LAN and Internet Services inside a Building

Data Communication Network for LAN and Internet Services inside a Building may be installed using Cat 6 unshielded twisted pair (UTP) cables in a concealed manner following the concealed wiring power cables installation procedure mentioned in the wiring methods section of this document. Each of the cables will be terminated at one end at the 8P8C (RJ45) connector based data socket outlet board in the required rooms at the power socket level. On the other end, the cable will be terminated in the patch panel. From the patch panel up to the data socket outlet the cable shall be in one piece i.e., no

joints will be allowed. As a result the concealed conduit work needs to be done carefully to have a straight line path and without any bend in the roof slab. Sufficient pull boxes will be required in the roof slab. Pull box will also be needed close to the vertical bend near the bend and ceiling at any downward drop of the conduit. The conduits must have 20 SWG GI pull wires during laying for pulling the cables later.

Because of the nature of these cables more clearances are needed inside the PVC conduits compared to the power cables. If the conduits are running parallel to the power cables then there should be at least a distance of 410 mm between these two.

Recently Cat. 7 cables are emerging as a better choice in place of Cat. 6 cables.

1.3.37 Fire Detection and Alarm System inside a Building

The major parts of a Fire Detection and Alarm System inside a Building may be listed as

- (a) A number of different types of Fire Detectors/ detection devices wired in a number of radial circuits
- (b) Manual call points
- (c) A central control panel for fire detection
- (d) A number of alarm sounders/alarm devices wired in a number of radial circuits
- (e) Cables for wiring the fire detectors/detection devices
- (f) Cables for wiring the alarm sounders/alarm devices

Control Panel

The control panel will indicate in which detection circuit (zone) an alarm or fault condition has been generated and will operate common or zonal sounders and auxiliary commands (for example door release or fire brigade signaling).

Detectors

A number of types of detectors (smoke detectors, heat detectors, ionization smoke detectors, optical beam smoke detectors, opto-heat detectors) for the installation

Alarm Devices

Alarm devices fall into two types, audible and visual. The audible types are most common, with a variety of types being available from bells to all kinds of different electronic sounders including those containing pre-recorded spoken messages. The choice of device is dependent on local preference, legal requirement and the need to have a tone distinct from all other building audible alarms.

Speech alarms or links to PA systems overcome some of the complacent responses to warning tones and can be used to good effect when carrying out regular fire tests in buildings where there are many people unfamiliar with the regular routines - such as hotels. Finally visual alarms are to be used where the hard of hearing may be occupying a building or where the ambient noise is such (above 90 dBA) that audible warning may not be heard, where hearing protectors are in use or where the sounder levels would need to be so high that they might impair the hearing of the building occupant.

Audible and Visual Alarm Devices

The audible types are most common, with a variety of types being available from bells to all kinds of different electronic sounders including those containing pre-recorded spoken messages. The choice of device is dependent on local preference, legal requirement and the need to have a tone distinct from all other building audible alarms.

Cables for Fire Detectors

BS 5839-1 introduced more onerous requirements for the types of cables used in fire detection and alarm systems. Fireproof cables should now be used for all parts of the system and enhanced fire resistance cables should be used where there is a requirement to ensure cable integrity over a longer period of time. For example when connecting to alarm sounders or where connection between sub-panels provides any part of alarm signal path.

Fire alarm cables should be segregated from the cables of other systems; they should be clearly marked, preferably coloured red and should be routed through parts of the building that provide minimum risk. This latter point is particularly relevant where the use of the building is being changed - for example if a fuel store is being moved.

Specific Areas of Application for Fire Detection and Alarm Equipment

The BS 5839 suite of standards relate to specific areas of application for fire detection and alarm equipment. Specifically part 1 relates to public premises and part 6 relates to residential premises. BS5839-1 is a comprehensive code of practice for fire detection and alarm systems, the requirements relate to both life and property protection and the standard includes much advice and comment which is very useful in informing the building owner or system specifier of the background to the requirements.

Codes of Practice for Different Types of Fire Protection Systems

The parts of BS7273 are codes of practice for different types of fire protection systems. Generally this is considered separately to fire alarm systems but there may be occasions where a tradeoff can be made between the two systems, or where the two systems interact and must be interfaced.

Standards Related to Design and Performance of Items of Equipment that Make up a Fire Detection and Alarm System

The EN 54 suite of standards relates to the design and performance of items of equipment that make up fire detection and alarm system. Each part relates to a different piece of equipment, for example part 3 relates to alarm devices, part 11 to call points, part 4 to power supplies etc.

Fire Detection Zones

Fire detection zones are essentially a convenient way of dividing up a building to assist in quickly locating the position of a fire. BS 5839-1 has some specific recommendations with respect to detection zones.

Wiring of the fire detection and alarm system will be done using the concealed wiring and the surface wiring methods described in the power line wiring section of this document.

1.3.38 CCTV System inside a Building

Installation of cable network for CCTV System inside a Building shall be done following the guidelines given for cable work for television antennas/cable television system earlier in this document.

For wiring of the power lines required for the Installation of CCTV system inside a building will be done using the concealed wiring and the surface wiring methods described in the power line wiring section of this document.

1.3.39 Design and Installation of Access Control System

Wiring of the Installation of access control systems will be done using the concealed wiring and the surface wiring methods described in the power line wiring section of this document.

1.3.40 Installation of Electronic Security Systems

Wiring of the installation of electronic security systems will be done using the concealed wiring and the surface wiring methods described in the power line wiring section of this document.

1.3.41 Qualification of the Contractor of Electrical and Electronic Engineering Works in a Building

A Contractor who will be working with the electrical and electronic engineering works in a building must have appropriate ABC license from the electrical licensing board of government of Bangladesh.

The contractor must have sufficient number of well trained and experienced technicians to execute the job. For big volume of work, the contractor must have at least one Electrical Engineer assigned for the job.

1.3.42 Inspection and Testing

1.3.42.1 General

Every installation shall, on completion and before being energized, be inspected and tested. The methods of test shall be such that no danger to persons or property or damage to equipment occurs even if the circuit tested is defective.

1.3.42.2 Periodic inspection and testing

Periodic inspection and testing shall be carried out in order to maintain the installation in a sound condition after putting it into service. Where an addition is to be made to the fixed wiring of an existing installation, the latter shall be examined for compliance with the recommendations of the Code.

1.3.42.3 Checking the conformity with the Bangladesh Standard

The individual equipment and materials which form part of the installation shall generally conform to the relevant Bangladesh Standard (BDS) wherever applicable. If there is no relevant Bangladesh standard specification for any item, these shall be approved by the appropriate authority.

(i) Inspection of the colour identification of cables of wiring

For single phase, Brown for Live, Blue for Neutral, Green + Yellow bi-colour for ECC. For three phase, Brown for L1, Black for L2, Grey for L3, Blue for Neutral and Green + Yellow bi-colour for ECC and Earth Lead Wire.

(ii) Inspection of earthing terminal, earthing bus

Inspection should be made to check whether brass made earthing terminals have been provided inside the metal back boxes of the switchboards and socket boards (welded or screwed to the metal back box) and whether the ECCs of the sub circuit have been terminated in these terminals. Inspection should be made to check whether at least one copper earthing busbar has been provided in the BDBs, SDBs, FDBs, DBs, MDBs and the LT panel and whether ECCs have been appropriately terminated in these busbars using hexagonal head brass bolt and nuts. Also it should be checked whether the Earth Lead Wires have been properly terminated in the LT Panel / MDB / DB as appropriate.

(iii) Insulation Tests

Insulation test is one of the most important tests for Electrical Installations in a Building.

Insulation resistance test shall be made on all electrical equipment, using a self-contained instrument such as the direct indicating ohm-meter of the generator type. DC potential shall be used in these tests and shall be as follows or an appropriate Meggar:

Circuits below 230 volts 500 volts Meggar

Circuits between 230 volts to 400 volts 1000 volts Meggar

The minimum acceptable insulation resistance value is 5 mega ohms for LT lines. Before making connections at the ends of each cable run, the insulation resistance measurement test of each cable shall be made. Each conductor of a multi-core cable shall be tested individually to all other conductors of the group and also to earth. If insulation resistance test readings are found to be less than the specified minimum in any conductor, the entire cable shall be replaced.

All transformers, switchgears etc. shall be subject to an insulation resistance measurement test to ground after installation but before any wiring is connected. Insulation tests shall be made between open contacts of circuit breakers, switches etc. and between each phase and earth.

(iv) Earth Resistance Test and the Continuity Resistance Test

Earth resistance tests shall be made on the system, separating and reconnecting each earth connection using earth resistance meter.

The electrical resistance of the Earth Continuity Conductor of different segment shall be measured separately using sensitive digital Ohm meter or by means of resistance bridge instrument. The resistance of the Earth Lead Wire shall be measured from the earthing busbar of the LT Panel/MDB/DB and the earth electrode(s). The electrical resistance of any section shall not exceed 1 ohm.

Where more than one earthing sets are installed, the earth resistance between two sets shall be measured by means of sensitive digital Ohm meter or by means of resistance bridge instrument. The earth resistance between two sets shall not exceed 1 ohm.

Operation Tests

Current load measurement shall be made on equipment and on all power and lighting feeders using Clamp on Ammeters.

The current reading shall be taken in each phase wire and in each neutral wire while the circuit or equipment is operating under actual load conditions.

Clamp on Ammeters are required to take current readings without interrupting a circuit.

All light fittings shall be tested electrically and mechanically to check whether they comply with the standard specifications.

Fluorescent light fittings shall be tested so that when functioning no flickering or choke singing is felt.

(v) Inspection of the Installation

On completion of wiring a general inspection shall be carried out by competent personnel in order to verify that the provisions of this Code and that of the Electricity Act of Bangladesh have been complied with. A certificate may be issued on satisfactory completion of the work in a format as shown in Appendix C. Items to be inspected are detailed in the following sections.

Inspection of Substation Installations

In substation installations, it shall be checked whether:

- The installation has been carried out in accordance with the approved drawings;
- Phase to phase and phase to earth clearances are provided as required;
- All equipment are efficiently earthed and properly connected to the required number of earth electrodes;
- The required ground clearance to live terminals is provided;
- Suitable fencing is provided with gate with lockable arrangements;
- The required number of caution boards, firefighting equipment, operating rods, rubber mats, etc., are kept in the substation;
- In case of indoor substation sufficient ventilation and draining arrangements are made;
- All cable trenches have covers of nonflammable material;
- Free accessibility is provided for all equipment for normal operation;
- All name plates are fixed and the equipment are fully painted;
- All construction materials and temporary connections are removed;
- Oil level, busbar tightness, transformer tap position, etc. are in order;
- Earth pipe troughs and cover slabs are provided for earth electrodes/earth pits and the neutral and LA earth pits are marked for easy identification;

- Earth electrodes are of GI pipes or CI pipes or copper plates. For earth connections, brass bolts and nuts with lead washers are provided in the pipes/plates;
- Earth pipe troughs and oil sumps/pits are free from rubbish, dirt and stone jelly and the earth connections are visible and easily accessible;
- HT and LT panels and switchgears are all vermin and damp-proof and all unused openings or holes are blocked properly;
- The earth busbars have tight connections and corrosion free joint surfaces;
- Control switch fuses are provided at an accessible height from ground;
- Adequate headroom is available in the transformer room for easy topping-up of oil, maintenance, etc.;
- Safety devices, horizontal and vertical barriers, busbar covers/shrouds, automatic safety shutters/door interlock, handle interlock etc. are safe and in reliable operation in all panels and cubicles;
- Clearances in the front, rear and sides of the main HT and LT and subswitch boards are adequate;
- The switches operate freely; the 3 blades make contact at the same time, the arcing horns contact in advance; and the handles are provided with locking arrangements,
- Insulators are free from cracks, and are clean;
- In transformers, there is no oil leak;
- Connections to bushing in transformers are tight and maintain good contact;
- Bushings are free from cracks and are clean;
- Accessories of transformers like breathers, vent pipe, buchholz relay, etc. are in order;
- Connections to gas relay in transformers are in order;
- In transformers, oil and winding temperature are set for specific requirements to pump out;
- In case of cable cellars, adequate arrangements exist to pump off water that has entered due to seepage or other reasons; and
- All incoming and outgoing circuits of HT and LT panels are clearly and indelibly labeled for identifications.

Inspection of Low Tension (LT) Installation

In Low Tension (LT) or Medium Voltage (MV) Installations, it shall be checked whether:

- All blocking materials that are used for safe transportation in switchgears, contactors, relays, etc. are removed;
- All connections to the earthing system have provisions for periodical inspection;
- Sharp cable bends are avoided and cables are taken in a smooth manner in the trenches or alongside the walls and ceilings using suitable support clamps at regular intervals;
- Suitable linked switch or circuit breaker or lockable push button is provided near the motors/apparatus for controlling supply to the motor/apparatus in an easily accessible location;
- Two separate and distinct earth connections are provided for the motor apparatus;
- Control switch fuse is provided at an accessible height from ground for controlling supply to overhead travelling crane, hoists, overhead busbar trunking;
- The metal rails on which the crane travels are electrically continuous and earthed and bonding of rails and earthing at both ends are done;
- Four-core cables are used for overhead travelling crane and portable equipment, the fourth core being used for earthing, and separate supply for lighting circuit is taken;
- If flexible metallic hose is used for wiring to motors and other equipment, the wiring is enclosed to the full lengths, and the hose secured properly by approved means;
- The cables are not taken through areas where they are likely to be damaged or chemically affected;
- The screens and armours of the cables are earthed properly;
- The belts of belt driven equipment are properly guarded;
- Adequate precautions are taken to ensure that no live parts are so exposed as to cause danger;
- Installed Ammeters and voltmeters work properly and are tested; and
- The relays are inspected visually by moving covers for deposits of dusts or other foreign matter.

Inspection of Overhead Lines

For overhead lines, every care must be taken so that:

- All conductors and apparatus including live parts thereof are inaccessible;
- The types and size of supports are suitable for the overhead lines/conductors used and are in accordance with approved drawing and standards;
- Clearances from ground level to the lowest conductor of overhead lines, sag conditions, etc. are in accordance with the relevant standard;
- Where overhead lines cross the roads suitable grounded guarding shall be provided at road crossings,
- Where overhead lines cross each other or are in proximity with one another, suitable guarding shall be provided at crossings to protect against possibility of the lines coming in contact with one another;
- Every guard wire shall be properly grounded/earthed;
- The type, size and suitability of the guarding arrangement provided shall be adequate;
- Stays cables must be provided suitably with the overhead line carrying poles as required and shall be efficiently earthed at the bottom and shall be provided with suitable stay insulators of appropriate voltages;
- Anti-climbing devices and Danger Board/Caution Board Notices are provided on all HT supports;
- Clearances along the route are checked and all obstructions such as trees/branches and shrubs are cleared on the route to the required distance on either side;
- Clearance between the live conductor and the earthed metal parts are adequate; and
- For the service connections tapped off from the overhead lines, cutouts of adequate capacity are provided.

Inspection of Lighting Circuits

The lighting circuits shall be checked to see whether:

- Wooden boxes and panels are avoided in factories for mounting the lighting boards, switch controls, etc.;
- Neutral links are provided in double pole switch fuses which are used for lighting control, and no fuse is provided in the neutral;
- The plug points in the lighting circuit are all 3-pin type, the third pin being suitably earthed;

- Tamper proof interlocked switch socket and plug are used for locations easily accessible;
- Lighting wiring in factory area is enclosed in conduit and the conduit is properly earthed, or alternatively, armoured cable wiring is used;
- A separate earth wire is run in the lighting installation to provide earthing for plug points, fixtures and equipment;
- Proper connectors and junction boxes are used wherever joints are in conductors or cross over of conductors takes place;
- Cartridge fuse units are fitted with cartridge fuses only;
- Clear and permanent identification marks are painted in all distribution boards, switchboards, sub-main boards and switches as necessary;
- The polarity has been checked and all fuses and single pole switches are connected on the phase conductor only and wiring is correctly connected to socket outlets;
- Spare knockouts provided in distribution boards and switch fuses are blocked;
- The ends of conduits enclosing the wiring leads are provided with ebonite or other suitable bushes;
- The fittings and fixtures used for outdoor use are all of weatherproof construction, and similarly, fixtures, fittings and switchgears used in the hazardous area are of flameproof application;
- Proper terminal connectors are used for termination of wires (conductors and earth leads) and all strands are inserted in the terminals;
- Flat ended screws are used for fixing conductor to the accessories;
- Flat washers backed up by spring washers are used for making end connections.

Accessibility of Connections and Cable joints for Inspection

Except for the following, every connection and joint shall be accessible for inspection, testing and maintenance:

- A compound-filled or encapsulated joint
- A connection between a cold tail and a heating element (e.g. a ceiling and floor heating system, a pipe trace-heating system)
- A joint made by welding, soldering, brazing or compression tool
- A joint forming part of the equipment complying with the appropriate product standard.

1.4 Related Codes and Standards

Significant modification, upgradation and additions of the previous electrical engineering section of BNBC of 1993 have been incorporated in this updated version. While making changes and additions, the following documents/regulations/codes have been taken as reference/guiding sources:

- (a) Bangladesh Electricity Act.
- (b) IEE wiring Regulation (17th edition) BS: 7671 2008 including all parts.
- (c) British Standards (BS).

In addition to these, the following documents/regulations/codes have also been taken as references as required:

- (a) National Building Code of India - 2005.
- (b) Building Code of Pakistan - latest version.
- (c) National Electrical Code of USA.
- (d) International Electrotechnical Commission (IEC) Standards.
- (e) ISO 50001 Standard for Energy Management System.
- (f) Verband Deutscher Elektrotechniker (Association of German Electrical Engineers) (VDE).

However, efforts have been be given to accept a significant part of rules and practices mentioned in IEE wiring Regulation (17th edition) BS: 7671 2008 including all parts with necessary modifications for our system and suitable for our country.

While preparing this document the following Standards and practices are kept in mind.

- (a) For having safe domestic electrical systems, domestic electrical installations shall be designed and installed according to the "fundamental principles" given in British Standard BS 7671 Chapter 13. These are similar to the fundamental principles defined in international standard IEC 60364-1. It is necessary to apply British Standard BS 7671 (the "Wiring Regulations"), including carrying out adequate inspection and testing to this standard of the completed works.
 - To meet the above mentioned requirements the following rules and guidance shall be followed.
 - The rules of the IEE wiring regulations (BS 7671), colloquially referred to as "the regs" (BS 7671: 2008, 17th Edition).;
 - The rules of an equivalent standard approved by a member of the European Economic Area (e.g., DIN/VDE 0100);

- (b) Guidance given in installation manuals that is consistent with BS 7671, such as the IEE On-Site Guide and IEE Guidance Notes 1 to 7.
- (c) Installations in commercial and industrial premises must satisfy the requirements set in Electricity at Work Regulations 1989 (UK) and must follow recognised standards and practices, such as BS 7671 "Wiring Regulations".

Apart from these, some modifications had to be made considering the weather and other local conditions, practices and previous experiences in this country.

1.5 List Of Related Appendices

Appendix A	Maximum Demand and Diversity
Appendix B	Useful Tables Relating to Conductor Sizes
Appendix C	Completion Certificate Form

PART VIII
Chapter 2
Air-Conditioning, Heating And Ventilation

2.1 General

2.1.1 Purpose

The purpose of this Chapter is to provide minimum standards for regulating and controlling the design, construction, installation, quality of materials, location, operation, performance, maintenance and use of air conditioning, heating and ventilation systems to ensure acceptable conditions of air inside the building required for human health, safety and welfare with energy conservation.

2.2 Scope

2.2.1 The provisions of this Code shall apply to erection, installation, alteration, repair, relocation, replacement, addition to, use and maintenance of any air-conditioning, heating and ventilation systems.

2.2.2 Additions, alterations, repairs and replacement of equipment or systems shall comply with the provisions for new equipment and systems except as otherwise provided in Sec 2.2.2.1.

2.2.3 Where, in any specific case, different sections of this Code specify different materials, methods of construction or other requirements, the most restrictive one shall govern. Where there is a conflict between a general requirement and a specific requirement, the specific requirement shall be applicable.

2.2.4 The regulations of this Code are not intended, and shall not be understood to permit violation of the provisions of other ordinances, regulations or official requirements in force.

2.3 Application

It shall be unlawful to install, extend, alter, repair or maintain air-conditioning, heating and ventilation systems in or adjacent to buildings except in compliance with this Code.

2.3.1 Existing Systems

Existing Installations: Air-conditioning, heating and ventilation systems in existence at the time of adoption of this Code may have their use, maintenance or repair continued if the use, maintenance or repair is in accordance with original design and location and no hazard to life, health or property has been created by such system.

Additions, Alterations or Repairs: Additions, alterations or repairs may be made to any air-conditioning, heating or ventilation system without requiring the existing system to comply with all the requirements of this Code, provided the addition, alteration or repair conforms to the requirements of a new system. Additions, alterations or repairs shall not make an existing system unsafe, create unhealthy or overloaded conditions.

Changes in Building Occupancy: Air-conditioning, heating and ventilation systems which are a part of any building or structure undergoing a change in use or occupancy, as defined in the Building Code, shall comply with all requirements of this Code which may be applicable to the new use, or occupancy.

Maintenance: All air-conditioning, heating and ventilation systems, materials and appurtenances, both existing and new, and all parts thereof shall be maintained in proper operating condition in accordance with the original design and in a safe and hazard free condition. All devices or safeguards which are required by this Code shall be maintained in conformance with this Code. The owner or the owner's designated agent shall be responsible for maintenance of the systems and equipment.

Moved Buildings: Air-conditioning, heating and ventilation systems of a building or a structure if moved to another premises shall comply the provisions of this Code for new installations.

2.3.2 Alternative Materials and Methods of Construction

The provisions of this Code are not intended to prevent the use of any material or method of construction not specifically prescribed by this Code, provided any such alternative material and/or method of construction has been approved and the use authorized by the Authority.

The Authority shall require that sufficient evidence or proof be submitted to substantiate any claims made regarding the use of alternatives.

2.3.3 Modifications

Whenever there are practical difficulties involved in carrying out any of the provisions of this Code, the Authority, within the limitations set forth in Part 2 may allow modifications for individual cases. The modifications shall be in conformity with the intent and purpose of this Code and that such modification shall not lessen health, life and fire safety requirements.

2.4 Terminology

This Section provides an alphabetical list of the terms used in and applicable in this Chapter of the Code. In case of any conflict or contradiction between a definition given in this Section and that in Part 1, the meaning provided in this Section shall govern for interpretation of the provisions of this Chapter.

ABSORPTION	A process whereby a material extracts one or more substances present in an atmosphere or mixture of gases or liquids accompanied by the material's physical and/or chemical changes.
ABSORPTION REFRIGERATING SYSTEM	A refrigerating system in which refrigerant gas evaporated in the evaporator is absorbed in the absorber by an absorbent solution. This also includes a generator for separation of refrigerant from the absorbent solution, a condenser to liquefy the refrigerant and an expansion device.
ADSORPTION	The action, associated with the surface adherence, of a material in extracting one or more substances present in an atmosphere or mixture of gases and liquids, unaccompanied by physical or chemical change.
AIR CHANGE	Introducing new, cleansed, or recirculated air to conditioned space, measured by the number of complete changes per unit time.
AIR TERMINALS	A round, square, rectangular, or linear air outlet or inlet device used in the air distribution system.
AIR, OUTSIDE	External air; atmosphere exterior to refrigerated or conditioned space; ambient (surrounding) air.
AIR, RECIRCULATED	The part of return air passed through the air-conditioner before being resupplied to the conditioned space. Also known as AIR, RETURN.
AIR, RETURN	See AIR, RECIRCULATED.
AIR-CONDITIONING	The process of treating air so as to control simultaneously its temperature, humidity, purity, distribution, pressure and air movement to meet the thermal requirements of the space.
AIR-HANDLING UNIT	Equipment comprised of cooling and/or heating coil and a blower or fan with electric motor used for the purpose of cooling/heating and distributing supply air to a room, space or area.
BLOWER	A fan used to force air under pressure.
BOILER	A closed vessel in which a liquid is vaporized.
BRINE	Any liquid cooled by the refrigerant and used for the heat transmission without a change in its state. This also includes chilled water.

CHIMNEY	Primarily a vertical shaft enclosing at least one flue for conducting flue gases to the outdoors.
COIL	A cooling or heating element made of pipe or tubing.
CONDENSER (Refrigerant)	A heat exchanger in which the refrigerant, compressed to a suitable pressure, is condensed to liquid by rejecting heat to an appropriate external cooling medium. When the condenser rejects heat to air, the condenser is termed as air cooled condenser. When the condenser rejects heat to water, the condenser is termed as water cooled condenser. When the condenser rejects heat to glycol (brine), the condenser is termed as glycol cooled condenser.
CONDENSING UNIT	A condensing unit is a complete set consisting of compressor(s) and condenser(s) with or without receiver. It may be air cooled or water cooled.
CONTROL	Any device for regulating a system or component in normal operation, manual or automatic.
COOLING TOWER	An enclosed device for evaporatively cooling water by contact with air.
DAMPER	A device for regulating the flow of air or other fluid.
DEHUMIDIFICAT ION	Condensation of water vapour from air by cooling below the dew point.
DEW POINT TEMPERATURE	The temperature at which condensation of moisture begins when the air is cooled at same pressure.
DRY BULB TEMPERATURE	The temperature of air as registered by a thermometer, taken in such a way as to avoid errors due to radiation.
DUCT SYSTEM	A continuous passageway for the transmission of air which, in addition to the ducts, may include duct fittings, dampers, plenums, grilles and diffusers.
ENERGY EFFICIENCY RATIO	The ratio between refrigeration capacity of a complete air conditioning unit in btu/hr with the power consumption in watt.
ENTHALPY	A thermal property indicating quantity of heat in the air above an arbitrary datum, in kilo joules per kg of dry air (or btu per pound of dry air).
EVAPORATIVE AIR COOLING	The removal of sensible heat from the air by the adiabatic exchange of heat between air and a water-spray or wetted surface, wherein the evaporating water absorbs the sensible heat of air.

EVAPORATOR (refrigerant)	A heat exchanger in which liquid refrigerant, after reducing its pressure (expansion), is evaporated by absorbing heat from the medium to be cooled.
EXFILTRATION	The phenomenon of inside air leaking out of an air conditioned space.
FAN	An air moving device comprising of a wheel or blade, and housing or orifice plate.
FAN, TUBEAXIAL	A propeller or disc type wheel within a cylinder and including driving mechanism supports for either belt drive or direct connection.
FILTER	A device to remove solid particles from a fluid.
FIRE DAMPER	A closure which consists of a normally held open damper installed in an air distribution system or in a wall or floor assembly and designed to close automatically in the event of a fire in order to isolate the conditioned space from the fire zone.
FIRE SEPARATION	A construction assembly that acts as a barrier against spread of fire and may not be required to have a fire resistance rating or fire protection rating.
GLOBAL WARMING POTENTIAL (GWP)	Global warming potential of a chemical compound is its relative contribution to global warming compared to Carbon Dioxide (CO₂). Global warming can make our planet and its climate less hospitable and more hostile to human life. Thus it is necessary to reduce reduction in emission of greenhouse gases such as CO₂, SO_x, NO_x and refrigerants. Long atmospheric life time of refrigerants results in global warming unless the emissions are controlled. GWP values of some of the refrigerants are given below. The values indicated are for an integration period of 100 years.

Refrigerant	GWP values
R-11	4,000
R-12	2,400
R-22	1,700
R-123	0.02
R-134a	1,300
R-407A	2,000
R-407C	1,600
R-410A	1,890
R-744 (CO ₂)	1.00

HUMIDITY	Water vapour within a space.
HUMIDITY, RELATIVE	The ratio of partial pressure or density of water vapour in air to the saturation pressure or density, respectively, of water vapour at the same temperature.
HYDRONIC	Of, relating to, or being a system of heating or cooling that involves transfer of heat by a circulating fluid (as water or vapour) in a closed system of pipes.
INDOOR AIR QUALITY (IAQ)	Air quality that refers to the nature of conditioned air that circulates throughout the space/area where one works, lives, that is, the air one breathes when indoors. It not only refers to comfort which is affected by temperature, humidity, air movement and odors but also harmful biological contaminants and chemicals present in the conditioned space. Poor IAQ may be a cause of serious health hazard. Carbon dioxide has been recognized as the surrogate ventilation index.
INFILTRATION	The phenomenon of outside air leaking into an air conditioned space.
INSULATION, THERMAL	A material having a relatively high resistance to heat flow and used principally to retard heat flow.
INTEGRATED PART LOAD VALUE (IPLV)	It is the part-load efficiency figure of the chiller at the ARI 550/590 standard rating point, measured in kW/ton, according to the following standard formula $IPLV = \frac{1}{\frac{0.01}{A} + \frac{0.42}{B} + \frac{0.45}{C} + \frac{0.12}{D}}$ where, A = kW/ton at 100% load B = kW/ton at 75% load C = kW/ton at 50% load D = kW/ton at 25% load
MECHANICAL REFRIGERATION EQUIPMENT	A refrigerating system in which the gas evaporated in the evaporator is compressed by mechanical means usually by a compressor. This also includes condenser and expansion device.
NON-STANDARD PART LOAD VALUE (NPLV)	It is the part-load efficiency figure of the chiller at the rating conditions other than the ARI standard rating point but within prescribed limits. The rating points are actually values at which the chiller will actually be operating.
OVERALL HEAT TRANSFER COEFFICIENT (U)	The time rate of heat flow per unit area (normal to the flow) from the fluid on the warm side of a barrier to the fluid on the cold side, per unit temperature difference between the two fluids.

OZONE
DEPLETION
POTENTIAL
(ODP)

Ozone depletion potential of a chemical compound is its relative contribution to the depletion of the ozone layer compared to CFC-11. ODP values of some of the refrigerants are as follows

Refrigerant	ODP values
R-11	1.0
R-12	1.0
R-22	0.05
R-123	0.02
R-134a	0
R-407A	0
R-407C	0
R-410A	0

PACKAGED AIR
CONDITIONER

An encased assembly of equipment/machinery for thermal conditioning (cooling/heating) of air along with cleaning and circulation of air to maintain internal thermal environment of an air conditioned space. It includes a prime source of refrigeration for cooling and dehumidification with or without internal and external air distribution ducting. It may also include means for heating, humidifying and ventilating air. These units may be floor mounted, wall mounted or ceiling mounted type. They may provide free delivery or ducted delivery of conditioned air.

These machines are equipped with air cooled or water cooled condenser(s). These machines are equipped with reciprocating, rotary or scroll compressor(s).

PLENUM

An air compartment or chamber to which one or more ducts are connected and which forms part of an air distribution system.

POSITIVE
VENTILATION

The supply of outside air by means of a mechanical device, such as a fan.

PSYCHROMETRY

The science involving thermodynamic properties of moist air and the effect of atmospheric moisture on materials and human comfort. It also includes methods of controlling properties of moist air.

PSYCHROMETRIC
CHART

A chart graphically representing the thermodynamic properties of moist air.

REFRIGERANT	The fluid used for heat transfer in a refrigerating system, which absorbs heat at a low temperature and a low pressure of the fluid and rejects heat at a higher temperature and a higher pressure of the fluid, usually involving changes of phase of the fluid.
REHEATING	The process by which air, which has been cooled down in order to condense out part of the moisture it contains, is heated again in order to raise its temperature to a suitable level.
RETURN AIR GRILLE	These are fittings fixed at the openings through which air is taken out from the air-conditioned enclosure by an air-conditioning plant or unit.
ROOM AIR- CONDITIONER	A factory made, encased assembly designed as a self-contained unit primarily for mounting in a window or through the wall or as a console. It is designed to provide free delivery of conditioned air to an enclosed space, room or zone (conditioned space). It includes a prime source of refrigeration for cooling and dehumidification and means for the circulation and cleaning of air. It may also include means for heating, humidifying, ventilating or exhausting air.
SHADE FACTOR	The ration of instantaneous heat gain through fenestration with shading device to that through the fenestration without shading device.
SUPPLY AIR	The air that has been passed through the conditioning apparatus and taken through the duct system and distributed in the conditioned space.
SPLIT AIR CONDITIONER	A split package air conditioner is same as the packaged air conditioner excepting that the condenser or the condensing unit is built as a separate package for remote field installation and interconnecting refrigerant pipes between indoor unit and outdoor unit is considered as a package. The indoor units may be floor mounted, wall mounted, ceiling mounted (concealed) and may be free blow type or ducted type.
SUPPLY AIR DIFFUSERS/ GRILLES	These are fittings fixed at the openings through which air is delivered into the air-conditioned enclosure by an air-conditioning plant or unit.
TEMPERATURE, DRY BULB	The temperature of air as registered by a thermometer.
TEMPERATURE, WET BULB	The temperature at which water, by evaporating into air, may bring the air to saturation adiabatically at the same temperature. Wet-bulb temperature (without qualification) is the temperature indicated by a wet bulb psychrometer constructed and used according to specifications.

THERMAL TRANSMITTANCE	Thermal transmission per unit time through unit area of the given building unit divided by the temperature difference between the air or some other fluid on either side of the building unit in 'steady state' conditions.
THERMAL ENERGY STORAGE	Storage of thermal energy, sensible, latent or combination thereof for use in central system of air conditioning or refrigeration. It uses a primary source of refrigeration for cooling and storing thermal energy for reuse at peak demand or for backup as planned.
VARIABLE REFRIGERANT FLOW (VRF) SYSTEM	A variable refrigerant flow (VRF) air conditioning system is similar to a split air conditioning system excepting that it is of larger capacity and covers multiple zones/areas simultaneously. It is consisted of one or more outdoor condensing units, multiple indoor units, interconnected refrigerant pipes between outdoor unit(s) and indoor units, etc. all considered as a single package. The condensing units contain at least one inverter controlled rotary/scroll compressor or digital scroll compressor. The objective of using inverter controlled compressor or digital scroll compressor is to regulate capacity of air conditioning system, in response to the load requirements, by regulating refrigerant flow through the indoor units. The indoor units may be floor mounted, wall mounted, ceiling mounted (concealed) and may be free blow type or ducted type.
VENTILATION	The process of supplying and/or removing air by natural or mechanical means to or from any space. Such air may or may not have been conditioned.
WATER CONDITIONING	The treatment of water circulating in a hydronic system, to make it suitable for air conditioning system due to its effect on the economics of the air conditioning plant. Untreated water used in air conditioning system may create problems such as scale formation, corrosion and organic growth. Appraisal of the water supply source including chemical analysis and determination of dissolves solids is necessary to devise a proper water conditioning program.

2.5 General Provisions

2.5.1 Air conditioning, heating and ventilation system shall be designed, constructed, installed, operated and maintained in accordance with good engineering practice such as described in the ASHRAE (American Society of Heating, Refrigerating and Air-conditioning Engineers) Handbooks, HI (Hydraulic Institute of USA) manuals and relevant chapters of this Code.

2.5.2 All electrical work in connection with air-conditioning, heating and ventilation system shall be carried out in accordance with the provisions of latest Bangladesh Electricity Act and the provisions of any of its regulations and bye-laws, and shall also comply with the requirements of Chapter 1.

2.5.3 All plumbing work in connection with air-conditioning, heating and ventilation system shall be carried out in accordance with the provisions and guidelines of ASHRAE handbooks and HI manuals.

2.5.4 All gas and fuel piping in connection with air-conditioning, heating and ventilation system shall be carried out in accordance with the provisions of Chapter 8.

2.5.5 Fire Safety: Installations of equipment of air-conditioning, heating and ventilation system shall conform to the requirements of Part 4 of this Code.

2.6 Planning

2.6.1 General

2.6.1.1 All relevant aspects of air-conditioning, heating and ventilation system installations shall be analyzed and evaluated properly during the planning stage of the building in order to determine the necessary provisions to be kept in the building for proper and safe installation of the system machinery, equipment and other facilities.

2.6.1.2 Necessary particulars of electrical requirements of air-conditioning, heating or ventilation system shall be determined early in the planning stage to include it in the electrical provisions of the building.

2.6.1.3 Where necessary, all plans, calculations, specifications and data for air-conditioning, heating and ventilation system serving all buildings and all occupancies within the scope of the Code shall be supplied to the Authority, for review purposes.

2.6.1.4 Design air conditioning, heating and ventilation system taking consideration for energy efficiency and energy conservation. Provide data to design architect to keep provisions in the building for reduction of energy usage required for operation of air-conditioning system. Optimize the design by coordinating the design with the design architect at the early stage i.e. in the schematic design phase and continually improve design during the design development process.

2.6.2 Building Planning

2.6.2.1 Orientation of building

Effect of orientation of building and arrangement of rooms/spaces shall be analyzed in the planning stage of the building to find out the most effective plan of the building in terms of building use, application of air-conditioning, heating and ventilation system and reduction of energy consumption.

2.6.2.2 Building design and use of materials

Analysis shall be carried out in the design stage for selection of appropriate shading devices and other materials as set forth in Sec 2.7.1 so as to take advantage of reduction in energy consumption related air-conditioning, heating and ventilation system. Selection of glazing materials shall be in compliance with Sec 2.12.2.4.

2.6.2.3 Equipment space

Requirements of space for erection and installation of air-conditioning, heating and ventilation system equipment and machinery (ducting, cooling, heating and air-conditioning equipment; refrigerating machinery, boiler etc.) shall be determined during the planning stage of the building so that it can be incorporated in the building planning effectively. Requirements of equipment/machinery space shall be determined taking consideration of actual equipment and machinery space; clearance space for operation; maintenance and fire prevention requirement; access space and other requirements of this Code. Building plan shall also include adequate provisions for transportation of equipment and machinery to and from equipment/machinery room, installation of outdoor air inlets and exhaust air outlets. Planning for equipment space shall take into consideration of different parameters described in Sections 2.5 and 2.6.

2.6.2.4 Equipment space planning for central air conditioning plant

- (a) In selecting the location for plant room, the aspects of efficiency, economy and good practice should be considered and wherever possible it shall be made contiguous with the building. This room shall be located as centrally as possible with respect to the area to be air conditioned and shall be free from obstructing columns.
- (b) In case of large capacity water cooled chiller installations (500 TR and above), it is advisable to have a separate isolated equipment room where possible. The clear headroom below soffit of beam should be minimum 4.5 m for centrifugal chillers, and minimum 3.6 m for reciprocating and screw type chillers.
- (c) The floors of the equipment rooms should be light colored and finished smooth. For floor loading, the air conditioning engineer should be consulted.
- (d) Supporting of pipe within plant room spaces should be normally from the floor. However, outside plant room areas, structural provisions shall be made for supporting the water pipes from the floor/ceiling slabs. All floor and ceiling supports shall be isolated from the structure to prevent transmission of vibrations.
- (e) Equipment rooms, wherever necessary, shall have provision for mechanical ventilation. In hot climate, evaporative air-cooling may also be considered.
- (f) Plant machinery in the plant room shall be placed on plain/reinforced cement concrete foundation and provided with anti-vibration supports. All foundations should be protected from damage by providing epoxy coated angle nosing. Requirements of seismic restraint supports may also be considered.
- (g) Equipment room should preferably be located adjacent to external wall to facilitate equipment movement and ventilation.
- (h) Wherever necessary, acoustic treatment should be provided in plant room space to prevent noise transmission to adjacent occupied areas.

- (i) Air conditioning plant room should preferably be located close to main electrical panel of the building in order to avoid large cable lengths.
- (j) In case the air conditioning plant room is located in basement floor, equipment movement route shall be planned to facilitate future replacement and maintenance. Service ramps or hatch in ground floor slab should be provided in such cases.
- (k) Floor drain channels or dedicated drainpipes in slope shall be provided within plant room space for effective disposal of waste water. Fresh water connection may also be provided in the air conditioning plant room.
- (l) Thermal energy storage: In case of central plants, designed with thermal energy storage its location shall be decided in consultation with the air conditioning engineer. The system may be located in plant room, on rooftop, in open space near plant room or buried in open space near plant room. For roof top installations, structural provision shall take into account load coming due to the same. For open area surface installation horizontal or vertical system options shall be considered and approach ladders for manholes provided. Buried installation shall take into account loads due to movement above, of vehicles, etc. Provision for adequate expansion tank and its connection to thermal storage tanks shall be made.

2.6.2.5 Space planning for air cooled chillers

- (a) Air Cooled chiller shall be installed where adequate open space is available for heat transfer of air cooled condensers of the chiller.
- (b) Where such space is available on ground, it can be installed on ground provided noise and hot air from the chiller do not create any problem to the adjoining building.
- (c) Roof of the building is a suitable location for installation of air cooled chiller. When it is intended to install air cooled chiller on roof, prior planning is a must. The roof shall be structurally strong enough to withstand the dynamic load of the chiller along with chilled water pumps, pipes, valves and associated equipment required for this purpose. Advice from an air conditioning engineer shall be taken at the planning stage.
- (d) Vibration from the machine shall not transmit to the roof structure. Chiller shall be installed on seismic restraint type vibration isolators.
- (e) Noise of the air cooled chiller shall be attenuated so that it does not transmit to the occupied area. A low speed condenser fan with acoustically treated fan cylinder shall be preferable. Similarly acoustically encased compressors shall also be preferable.

2.6.2.6 Planning equipment room for air handling units and package units

- (a) This shall be located as centrally as possible to the conditioned area and contiguous to the corridors or other spaces for carrying air ducts. For floor loading, air conditioning engineer shall be consulted.

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- (b) In the case of large and multistoried buildings, independent air handling unit should be provided for each floor. The area to be served by air-handling unit should be decided depending upon the provision of fire protection measures adopted. Air handling unit rooms should preferably be located vertically one above another.
 - (c) Provision should be made for the entry of fresh air. The fresh air intake shall have louvers having rain protection profile, with volume control damper and bird screen.
 - (d) In all cases air intakes shall be so located as to avoid contamination from exhaust outlets or to the sources in concentrations greater than normal in the locality in which the building is located.
 - (e) Exterior openings for outdoor air intakes and exhaust outlets shall preferably be shielded from weather and insects.
 - (f) No air from any dwelling unit shall be circulated directly or indirectly to any other dwelling unit, public corridor or public stairway.
 - (g) All air handling unit rooms should preferably have floor drains and water supply. The trap in floor drain shall provide a water seal between the air-conditioned space and the drain line.
 - (h) Supply/return air duct shall not be taken through emergency fire staircase.
Exception: Ducts can be taken inside the fire stair provided fire isolation of ducts at wall crossings is (are) carried out.
 - (i) Waterproofing of air handling unit rooms shall be carried out to prevent damage to floor below.
 - (j) The floor should be light colored, smooth finished with terrazzo tiles or the equivalent. Suitable floor loading should also be provided after consulting with the air conditioning engineer.
 - (k) Where necessary, structural design should avoid beam obstruction to the passage of supply and return air ducts. Adequate ceiling space should be made available outside the air handling unit room to permit installation of supply and return air ducts and fire dampers at air handling unit room wall crossings.
 - (l) The air handling unit rooms may be acoustically treated, if located in close proximity to occupied areas.
 - (m) Access door to air handling unit room shall be single/double leaf type, airtight, opening outwards and should have a sill to prevent flooding of adjacent occupied areas. It is desired that access doors in air conditioned spaces should be provided with tight sealing, gaskets and self-closing devices for air conditioning to be effective.

- (n) Air handling unit rooms shall be separated from the air conditioned space by 4 hour fire rated walls and 2-hour fire rated door. Fire/smoke dampers shall be provided in supply/return air duct at air handling unit room wall crossings and the annular space between the duct and the wall should be fire-sealed using appropriate fire resistance rated material.
- (o) Fire isolation shall be provided for vertical fresh air duct, connecting several air handling units.

2.6.2.7 Planning of pipe shafts

- (a) The shafts carrying chilled water pipes should be located adjacent to air handling unit room or within the room.
- (b) Shaft carrying condensing water pipes to cooling towers located on roof/terrace should be vertically aligned.
- (c) All shafts shall be provided with fire barrier at floor crossings in accordance with the provisions of Chapter 4.
- (d) Access to shaft shall be provided at every floor.

2.6.2.8 Planning for supply air ducts and return air

- (a) Duct supports, preferably in the form of angles of mild steel supported using stud anchors shall be provided on the ceiling slab from the drilled hole. Alternately, duct supports may be fixed with internally threaded anchor fasteners and threaded rods without damaging the slabs or structural members.
- (b) If false ceiling is provided, the supports for the duct and the false ceiling shall be independent. Collars for grilles and diffusers shall be taken out only after false ceiling/boxing framework is done and frames for fixing grilles and diffusers have been installed.
- (c) Where a duct penetrates the masonry wall it shall either be suitably covered on the outside to isolate it from masonry or an air gap shall be left around it to prevent vibration transmission. Further, where a duct passes through a fire resisting compartment/barrier, the annular space shall be sealed with fire sealant to prevent smoke transmission (see also Part 4 of this Code).

2.6.2.9 Space planning for cooling tower

- (a) Cooling towers are used to dissipate heat from water cooled refrigeration, air conditioning and industrial process systems. Cooling is achieved by evaporating a small proportion of re-circulating water into outdoor air stream. Cooling towers are installed at a place where free flow of atmospheric air is available.

- (b) Cooling towers shall be installed at least 3 m above the bases of the chillers. Cooling tower shall preferably be installed on the roof of the concerned building. In special cases it may be installed on ground or on any elevated platform or on the roof of the adjacent building provided the moisture laden discharge air from the cooling towers do not pose any problem to other buildings. Cooling tower should be so located as to eliminate nuisance from drift to adjoining structures.
- (c) Any obstruction to free flow of air to the cooling tower shall be avoided.
- (d) Structural provisions for the cooling tower shall be taken into account while designing the building. Wind speed shall be taken into consideration while designing the foundations/supports for cooling towers. Vibration isolation shall be an important consideration in structural design.
- (e) Special design requirements are necessary where noise to the adjoining building is to be avoided. Special provisions shall be included in the design to reduce water droplet noise.
- (f) Provisions for make-up water tank to the cooling tower shall be made. Make-up water tank to the cooling tower shall be separate from the tank serving drinking water.
- (g) Make-up water having contaminants or hardness, which can adversely affect the refrigeration plant life, shall be treated.

2.6.2.10 Building structure

Structural design requirements viz. load on the floor or ceiling; punches in the roof, floor and walls; vertical shaft for pipe risers and duct risers; concrete ducts etc. shall be determined in the planning stage to make adequate provisions in the structural design and to keep such provisions in the building. The structural design shall consider static and dynamic loads of equipment and machinery including vibration of machinery.

2.6.2.11 Design drawings

For the purpose of effective installation of air-conditioning, heating and ventilation system, working drawings showing layout of machinery, equipment, ducts, pipes etc., details of builders' works, holes and/or punches in roof, floors, walls, supports for machinery/equipment etc. shall be prepared prior to finalization of building design drawings. Such drawings/documents shall be properly stored for future reference.

2.7 Air-Conditioning System Design

2.7.1 Building Design Requirements**2.7.1.1 Glazing**

- (a) Building design shall consider all the aspects for reduction of heat transfer through the glazing. Building orientation shall be such that, if possible, glazing in walls subject to direct and intensive sun exposure shall be avoided. In case where it is not possible to do so, necessary protective measures shall be taken to reduce heat transfer through the glazing. Such protective measures may be in the form of sun breakers, double glazing, heat resistant glass or application of other shading devices.
- (b) When sun breakers are used, it shall preferably be 1m away from the wall face, with free ventilation, particularly from bottom to top, being provided for cooling of sun breakers and window by free convection. Conduction from sun breakers to main building shall be the minimum. Sun breakers shall shade the maximum glazed area possible, especially for the altitude and azimuth angle of the sun. Sun breakers shall preferably be light and bright in colour so as to reflect back as much of the sunlight as possible.
- (c) Where the above protection is in the form of reflective surfaces, adequate care shall be taken to avoid any hazard to the traffic surrounding the building and people on the road because of the reflected light from the surfaces.
- (d) Application of any protection shall not restrict entry of light to a limit demanding artificial lights.

2.7.1.2 Roof insulation

- (a) Construction of exposed roofs shall be such that the heat transmission through the roof is not excessive. Where required the overall heat transfer coefficient (U) of the roof exposed to sun shall be reduced effectively by using appropriate construction materials and/or proper type of insulation material (s). The overall thermal transmittance from the exposed roof should be kept as minimum as possible and under normal conditions, the desirable value should not exceed $0.58 \text{ W/m}^2/^{\circ}\text{C}$.
- (b) Under-deck or over-deck insulation shall be provided for exposed roof surface using suitable Insulating materials. Over-deck insulation shall be properly waterproofed to prevent loss of insulating properties.
- (c) The ceiling surface of floors which are not to be air conditioned may be suitably insulated to give an overall thermal transmittance not exceeding $1.16 \text{ W/m}^2/^{\circ}\text{C}$.

2.7.2 Design Conditions**2.7.2.1 Inside design conditions**

- (a) For comfort air-conditioning, the inside design conditions shall be selected with an objective to reduce energy consumption in the operation of the air-conditioning system. Acceptable values of inside design conditions for summer are provided in Table 8.2.1. Unless otherwise specifically required, the design calculations shall be based on the normal practice values of Table 8.2.1.

Table 8.2.1: Inside Design Conditions of Some of Applications for Summer^a

Sl. No.	Use Category of Space	Indoor Design Conditions	
		Dry Bulb Temperature (°C)	Relative Humidity (%)
1.	Restaurants, Cafeteria and Dining Hall	23 ~ 26	55 ~ 60
2.	Kitchens	28 ~ 31	--
3.	Office buildings	23 ~ 26	50 ~ 60
4.	Bank/Insurance/Commercial building	23 ~ 26	45 ~ 55
5.	Departmental stores	23 ~ 26	50 ~ 60
6.	Hotel guest rooms	23 ~ 26	50 ~ 60
7.	Ball room/meeting room	23 ~ 26	40 ~ 60
8.	Class rooms	23 ~ 26	50 ~ 60
9.	Auditoriums	23 ~ 26	50 ~ 60
10.	Recovery rooms	24 ~ 26	45 ~ 55
11.	Patient rooms	24 ~ 26	45 ~ 55
12.	Operation theatres	17 ~ 27	45 ~ 55
13.	Delivery room	20 ~ 23	45 ~ 55
14.	ICU/CCU	20 ~ 23	30 ~ 60
15.	New born Intensive care	22.5 ~ 25.5	30 ~ 60
16.	Treatment room	23 ~ 25	30 ~ 60
17.	Trauma room	17 ~ 27	45 ~ 55
18.	Endoscopy/Bronchoscopy	20 ~ 23	30 ~ 60

Sl. No.	Use Category of Space	Indoor Design Conditions	
		Dry Bulb Temperature (°C)	Relative Humidity (%)
19.	X-ray (diagnostic & treatment)	25.5 ~ 27	40 ~ 50
20.	X-ray (surgery/critical area and catheterization)	21 ~ 24	30 ~ 60
21.	Laboratory (diagnostics)	22.5 ~ 24.5	30 ~ 60
22.	Art Galleries/Museums	17 ~ 22	40 ~ 55
23.	Libraries	20 ~ 22	45 ~ 55
24.	Radio studio/Television studio	23 ~ 26	45 ~ 55
25.	Telephone terminal rooms	22 ~ 26	40 ~ 50
26.	Airport terminal/ bus terminal	23 ~ 26	50 ~ 60

Note:

^a The room design dry bulb temperature should be reduced when hot radiant panels are adjacent to the occupant and increased when cold panels are adjacent, to compensate for the increase or decrease in radiant heat exchange from the body. A hot or cold panel may be un-shaded glass or glass block windows (hot in summer, cold in winter) and thin partitions with hot or cold spaces adjacent. Hot tanks, furnaces, or machines are hot panels.

- (b) To avoid thermal shock, the difference between the dry bulb temperatures of outdoor air and indoor air shall not exceed 11°C. If it is absolutely necessary to have a difference more than 11°C, there shall have adequate provision for ante-room to reduce the effect of thermal shock.
- (c) For air-conditioning systems other than comfort air-conditioning, design conditions required by the specific processes involved or applications may be adopted. When required, proper protective measures shall be taken for persons working therein.
- (d) Velocity of air in an air-conditioned space, in the zone between the floor level and the 1.5 m level, shall be within 0.12 m/s and 0.25 m/s for comfort applications for commercial buildings, and for other applications it shall not exceed 0.5 m/s.

2.7.2.2 Outside design conditions

- (a) The outside design conditions for summer months for different cities are provided in Table 8.2.2. Selection of outside design conditions from this table shall be based on requirements of the application and the percent of time the outside air temperature is allowed to exceed the outside design conditions.

- (b) In case of stringent design conditions a meteorologist with experience in applied climatology may be consulted to evaluate conditions such as; the formation of heat sinks in urban areas; the duration of extreme temperatures; project sites located remotely from reporting stations.

2.7.2.3 Ventilation air

- (a) Every space served by the air-conditioning system shall be provided with outside fresh air not less than the minimum amount mentioned in Table 8.2.3. If adequate temperature regulation along with efficient filtration of air and absorption of odour and gas are provided, the amount of fresh air requirement may be reduced. However, in no case the outdoor air quantity shall be lower than 2.5 l/s per person.
- (b) In hospital operation theaters, a large quantity of outdoor air supply is recommended to overcome explosion hazard of anesthetics and to maintain sterile conditions. However, if adequate filtration with efficient absorption of anesthetics and laminar flow of supply air is provided, outside air requirement may be substantially reduced. Recirculation of air shall comply with the requirements of Sec 2.11.3.6.

2.7.3 Noise and Vibration

2.7.3.1 General

Air-conditioning, heating and ventilation system design and installations shall consider all the aspects of noise and vibration control related to the system and shall conform to the requirements of Chapter 3 of this Part. Selection and installation of equipment for air-conditioning, heating and ventilation system shall be such that noise and vibration transmitted to the space served by the system shall not exceed the recommended value for the space served.

2.7.3.2 Equipment room

Equipment room for installation of air handling units, refrigeration machinery, pumps, boilers, blowers and other equipment, which produce noise and vibration, shall not preferably be located adjacent to any acoustically sensitive area. Location of the equipment room shall be such that direct transmission of noise and vibration from the equipment room to acoustically sensitive areas do not occur. Appropriately designed sound barriers shall be used to restrict transmission of noise from equipment room to any acoustically sensitive areas, wherever necessary. Similarly, adequate measures shall be taken to restrict transmission of vibration from equipment room to other rooms.

2.7.3.3 Selection of equipment

Where possible, the equipment shall be selected which produce low sound power level consistent with the required performance and ensuring operation at maximum efficiency. Noise levels shall be reduced by appropriate shrouding of the equipment, if necessary. Equipment shall be so oriented that the noise will be radiated away from the likely areas of complaint.

Table 8.2.2: Outside Design Conditions for Different Stations^a

Station	Cooling DB/MWB ^b						Evaporation, WB/MDB ^c						Range of
	0.4%		1%		2%		0.4%		1%		2%		
	DB	MWB	DB	MWB	DB	MWB	WB	MDB	WB	MDB	WB	MDB	DB ^d
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
Barisal	35.5	28.5	34.5	28.5	34.0	28.0	30.0	33.0	29.5	33.0	29.0	32.0	5.0
Bogra	36.5	28.5	35.5	28.5	34.5	28.0	31.5	33.5	30.0	33.0	29.0	33.0	5.5
Chandpur	35.5	28.5	34.5	28.5	34.0	28.0	29.5	33.0	29.5	33.0	29.0	32.5	5.0
Chittagong	34.0	28.5	33.5	28.5	33.0	28.0	29.5	32.5	29.0	32.5	28.5	32.0	4.5
Comilla	35.0	27.5	34.5	27.5	34.0	27.5	29.0	33.0	28.5	32.5	28.0	32.0	5.5
Cox's Bazar	34.0	27.5	33.5	27.5	33.0	27.5	29.0	31.0	29.0	30.0	29.0	30.0	5.0
Dhaka	35.55	27.5	35.0	27.0	34.5	27.0	29.0	33.0	28.5	29.0	28.5	29.0	5.0
Dinajpur	36.0	28.5	35.0	28.0	34.5	28.0	30.0	32.5	29.0	33.0	29.0	33.0	5.5
Faridpur	36.5	28.5	35.5	28.0	34.5	28.0	29.5	33.0	29.0	33.0	29.0	33.0	5.0
Ishurdi	37.5	27.0	36.5	27.0	35.5	28.0	30.0	34.5	29.0	33.5	29.0	33.5	6.0
Jessore	38.0	28.5	37.0	28.5	36.0	28.0	30.0	35.0	30.0	35.0	29.5	34.0	6.5
Khulna	36.5	29.0	36.0	28.0	35.5	28.0	30.0	34.0	30.0	34.0	29.5	33.5	5.0
Mongla	37.0	31.0	36.0	30.5	35.0	30.0	33.0	34.0	32.5	34.0	32.0	33.5	5.0
Mymensingh	35.0	28.0	34.0	28.0	33.5	27.5	29.5	33.0	29.0	32.5	28.5	32.0	4.5
Patuakhali	35.5	28.5	35.0	28.5	34.5	28.5	30.0	33.0	30.0	33.0	29.5	32.5	5.0
Rajshahi	35.5	29.0	34.5	28.5	34.0	29.0	31.5	33.0	31.0	32.5	30.0	31.5	5.0
Rangamati	35.5	26.0	34.5	27.0	34.0	27.5	28.5	33.5	28.0	32.5	28.0	32.5	6.0
Rangpur	35.5	28.5	34.5	28.0	34.0	28.0	29.5	30.5	29.5	30.0	29.5	30.0	5.0
Sylhet	35.5	27.0	34.5	27.0	34.0	26.5	28.5	32.5	28.0	32.5	28.0	32.5	5.5
Tangail	36.0	27.5	35.5	27.5	34.5	27.5	29.0	33.5	28.5	33.0	28.5	33.0	5.5

Notes:

a This table has been prepared by statistical analysis of weather data of ten years, recorded three hourly by trained observers of Bangladesh Meteorological Department.

b The dry bulb (DB) temperatures presented in column-2, 4 & 6 represent values which have equaled or exceeded by 0.4%, 1%, and 2% of the total hours during the summer months of May through September. The coincident mean wet bulb temperatures (MWB) listed in columns 3, 5 & 7 are the mean of all wet bulb temperatures occurring at the specific design dry bulb temperatures. These values shall be used for cooling load calculation.

c Wet bulb (WB) temperatures presented in column-8, 10 & 12 represent values which have been equaled or exceeded by 0.4%, 1% and 2% of the total hours during the summer months of May through September. The coincident mean dry bulb temperatures (MDB) listed in columns 9, 11 & 13 are the mean of all dry bulb temperatures occurring at the specific design wet bulb temperatures. These values shall be used for selection of Cooling Tower, evaporative cooling equipment, fresh air cooling and other similar equipment.

d Mean daily range temperatures presented in column-14 are the difference between the average daily maximum and average daily minimum temperatures during the warmest months at each station.

Table 8.2.3: Minimum Ventilation Rates for Air Conditioned Spaces

Occupancy Classification	Outdoor Airflow Rate in Breathing Zone <i>l/s per person</i>	Outdoor Airflow Rate in Breathing Zone ^a <i>l/s per m²</i>	Default Occupant Density ^a <i>No./100 m²</i>	Exhaust Airflow Rate ^a <i>l/s per m²</i>
Correctional facilities				
Cells without plumbing fixtures	2.5	0.6	25	-
Cells with plumbing fixtures ^g	2.5	0.6	25	5.0
Dining halls	(see food and beverage service)			
Guard stations	2.5	0.3	15	-
Day room	2.5	0.3	30	-
Booking/waiting	3.75	0.3	50	-

Occupancy Classification	Outdoor Airflow Rate in Breathing Zone l/s per person	Outdoor Airflow Rate in Breathing Zone ^a l/s per m ²	Default Occupant Density ^a No./100 m ²	Exhaust Airflow Rate ^a l/s per m ²
Dry cleaners, laundries				
Coin-operated dry cleaner	3.75	-	20	-
Coin-operated laundries	3.75	0.3	20	-
Commercial dry cleaner	15	-	30	-
Commercial laundry	12.5	-	10	-
Storage, pick up	3.75	0.6	30	-
Education				
Auditoriums	2.5	0.3	150	-
Corridors		(see public spaces)		
Media center	5	0.6	25	-
Sports locker rooms ^g	-	-	-	2.5
Music/theater/dance	5	0.3	35	-
Smoking lounges	30		70	-
Day care (through age 4)	5	0.9	25	-
Classrooms (ages 5-8)	5	0.6	25	-
Classrooms (age 9 plus)	5	0.6	35	-
Lecture classroom	3.75	0.3	65	-
Lecture hall (fixed seats)	3.75	0.3	150	-
Art classroom ^g	10	0.9	20	3.5
Science laboratories ^g	5	0.9	25	5.0
Wood/metal shops ^g	5	0.9	20	2.5
Computer lab	5	0.6	25	-
Multiuse assembly	3.75	0.3	100	-
Locker/dressing rooms ^g	-	-	-	1.25

Occupancy Classification	Outdoor Airflow Rate in Breathing Zone	Outdoor Airflow Rate in Breathing Zone^a	Default Occupant Density^a	Exhaust Airflow Rate^a
	<i>l/s per person</i>	<i>l/s per m²</i>	<i>No./100 m²</i>	<i>l/s per m²</i>
Food and beverage service				
Bars, cocktail lounges	3.75	0.9	100	-
Cafeteria, fast food	3.75	0.9	100	-
Dining rooms	3.75	0.9	70	-
Kitchens (cooking) ^b	-	-	-	3.5
Hospitals, nursing and convalescent homes				
Autopsy rooms	-	-	-	2.5
Medical procedure rooms	7.5	-	20	-
Operating rooms	15	-	20	-
Patient rooms	12.5	-	10	-
Physical therapy	7.5	-	20	-
Recovery and ICU	7.5	-	20	-
Hotels, motels, resorts and dormitories				
Multipurpose assembly	2.5	0.3	120	-
Bathrooms/toilet-private ^g	-	-	-	12.5/25 ^f
Bedroom/living room	2.5	0.3	10	-
Conference/meeting	2.5	0.3	50	-
Dormitory sleeping areas	2.5	0.3	20	-
Gambling casinos	3.75	0.9	120	-
Lobbies/pre-function	3.75	0.3	30	-

Occupancy Classification	Outdoor Airflow Rate in Breathing Zone l/s per person	Outdoor Airflow Rate in Breathing Zone ^a l/s per m ²	Default Occupant Density ^a No./100 m ²	Exhaust Airflow Rate ^a l/s per m ²
Offices				
Conference rooms	2.5	0.3	50	-
Office spaces	2.5	0.3	5	-
Reception areas	2.5	0.3	30	-
Telephone/data entry	2.5	0.3	60	-
Main entry lobbies	2.5	0.3	10	-
Private dwellings, single and multiple				
Garages, common for multiple units ^b	-	-	-	3.75
Garages, separate for each dwelling ^b	-	-	-	50 l/s per car
Kitchens ^b	-	-	-	12.5/50 ^f
Living areas ^c	0.35 Air Change per Hour (ACH) but not less than 7.5 l/s per person	-	Based upon number of bedrooms. First bedroom 2 persons; each additional bedroom 1 person	-
Toilet rooms and bath rooms ^g	-	-	-	10/25 ^f
Public spaces				
Corridors	-	0.3	-	-
Elevator car	-	-	-	5.0
Shower room (per shower head) ^g	-	-	-	25/10 ^f

Occupancy Classification	Outdoor Airflow Rate in Breathing Zone	Outdoor Airflow Rate in Breathing Zone^a	Default Occupant Density^a	Exhaust Airflow Rate^a
	<i>l/s per person</i>	<i>l/s per m²</i>	<i>No./100 m²</i>	<i>l/s per m²</i>
Smoking lounges	30	-	70	-
Toilet rooms - public ^g	-	-	-	25/30 ^e
Places of religious worship	2.5	0.3	120	-
Courtrooms	2.5	0.3	70	-
Legislative chambers	2.5	0.3	50	-
Libraries	2.5	0.6	10	-
Museums (children's)	3.75	0.6	40	-
Museums/galleries	3.75	0.3	40	-
Retail stores, sales floors and showroom floors				
Sales (except as below)	3.75	0.6	15	-
Dressing rooms	-	-	-	1.25
Mall common areas	3.75	0.3	40	-
Shipping and receiving	-	0.6	--	-
Smoking lounges ^b	30	-	70	-
Storage rooms	-	0.6	--	-
Warehouses (see storage)	-	-	-	-
Specialty shops				
Automotive motor-fuel dispensing stations ^b	-	-	-	7.5
Barber	3.75	0.3	25	2.5
Beauty and nail salons ^b	10	0.6	25	3.0
Embalming room ^b	-	-	-	10.0
Pet shops (animal areas) ^b	3.75	0.9	10	4.5
Supermarkets	3.75	0.3	8	-

Occupancy Classification	Outdoor Airflow Rate in Breathing Zone l/s per person	Outdoor Airflow Rate in Breathing Zone ^a l/s per m ²	Default Occupant Density ^a No./100 m ²	Exhaust Airflow Rate ^a l/s per m ²
Sports and amusement				
Disco/dance floors	10	0.3	100	-
Bowling alleys (seating areas)	5	0.6	40	-
Game arcades	3.75	0.9	20	-
Ice arenas without combustion engines	-	0.30	-	2.5
Gym, stadium, arena (play area)	-	0.30	-	-
Spectator areas	3.75	0.3	150	-
Swimming pools (pool and deck area)	-	2.4	--	
Health club/aerobics room	10	0.3	40	-
Health club/weight room	10	0.3	10	-
Storage				
Repair garages, enclosed parking garages ^{b,d}	-	-	-	3.75
Warehouses	-	0.3	-	-
Theaters				
Auditoriums (see education)	-	-	--	
Lobbies	2.5	0.3	150	-
Stages, studios	5	0.3	70	-
Ticket booths	2.5	0.3	60	-
Transportation				
Platforms	3.75	0.3	100	-
Transportation waiting	3.75	0.3	100	-

Occupancy Classification	Outdoor Airflow Rate in Breathing Zone l/s per person	Outdoor Airflow Rate in Breathing Zone ^a l/s per m ²	Default Occupant Density ^a No./100 m ²	Exhaust Airflow Rate ^a l/s per m ²
Workrooms				
Bank vaults/safe deposit	2.5	0.3	5	-
Darkrooms	-	-	-	5.0
Copy, printing rooms	2.5	0.3	4	2.5
Meat processing ^c	7.5	-	10	-
Pharmacy (prep. area)	2.5	0.9	10	-
Photo studios	2.5	0.6	10	-
Computer (without printing)	2.5	0.3	4	-

a Based upon net occupiable floor area.

b Mechanical exhaust required and the recirculation of air from such spaces is prohibited. All air supplied to such spaces shall be exhausted, including any air in excess of that required by this table.

c Spaces unheated or maintained below 50°F are not covered by these requirements unless the occupancy is continuous.

d Ventilation systems in enclosed parking garages shall comply with Sec 2.11.3.8.

e Rates are per water closet or urinal. The higher rate shall be provided where periods of heavy use are expected to occur, such as toilets in theaters, schools and sports facilities. The lower rate shall be permitted where periods of heavy use are not expected.

f Rates are per room unless otherwise indicated. The higher rate shall be provided where the exhaust system is designed to operate intermittently. The lower rate shall be permitted where the exhaust system is designed to operate continuously during normal hours of use.

g Mechanical exhaust is required and recirculation is prohibited except that recirculation shall be permitted where the resulting supply air-stream consists of not more than 10 percent air re-circulated from these spaces (see Sec 2.11.3.6, items a and c).

h For nail salons, the required exhaust shall include ventilation tables or other systems that capture the contaminants and odors at their source and are capable of exhausting a minimum of 25 l/s per station.

2.7.3.4 Noise control

- (a) Air Ducts: Air ducts shall be so designed and installed to avoid any transmission of noise and vibration which may be picked up by the duct system from equipment room or adjoining rooms. Duct system shall not allow cross talk or noise transfer from one occupied space to another.

Duct system shall be appropriately designed, constructed and installed to obtain adequate attenuation of noise required to maintain recommended noise level in the air-conditioned space.

Duct construction and installation shall be such that drumming effect of duct walls and noise transmission through the duct walls can be minimized to the approved level.

- (b) Plenum Chamber: If required, properly designed plenum chamber, lined with approved sound absorbed material, and/or sound attenuators shall be used for attenuation of noise.
- (c) Flow Control Devices: Air dampers and other flow control devices shall be so selected that noise generation does not exceed approved levels.
- (d) Air Terminals: Air terminals shall be selected for the approved noise generation characteristics.
- (e) Piping: Velocity of fluids in piping shall be so selected that noise generation does not exceed approved levels.
- (f) Chiller and Refrigeration Equipment: Chiller(s) and refrigeration equipment(s) shall be so selected and installed that the combined effect of noise level does not exceed 65 dBA or approved levels at the property boundary line. Where ever possible refrigerant compressors may be encased in acoustically treated enclosures to reduce noise transmission. Similarly, low speed condenser fans may be used to have reduced noise generation. Fan cylinders may be acoustically treated to reduce noise transmission.
- (g) Cooling Tower: Cooling Towers(s) shall be so selected and installed that the combined effect of noise generation does not exceed 65 dBA or approved levels at the proper boundary level. Where ever possible, fan cylinders shall be acoustically treated to reduce noise transmission. Floating mats may also be used to reduce water droplet noise.

2.7.3.5 Vibration control

- (a) Appropriately designed vibration isolators shall be installed under the machinery to restrict vibration transmission to structures. Similarly vibration isolators shall also be used between machinery and all pipe work and duct work including the supports when applicable.
- (b) Where ever necessary “Inertia Block” with spring vibration isolators shall be used to restrict vibration transmission to structures.
- (c) Spring vibration isolators shall be earthquake restraint type.

2.8 Air Distribution System ---

2.8.1 Duct Work

2.8.1.1 General

- (a) Supply air, return air and outside air for air-conditioning, heating and ventilation systems shall be conducted through duct systems. Ducts and plenums shall be of independent construction or shall be formed by parts of the building structure.
- (b) Supply and return air plenums shall be limited to uninhabited crawl spaces, areas above a ceiling or below the floor, or attic spaces. Plenums shall be limited to one fire area. Fuel-fired equipment shall not be installed within a plenum. Venting systems and exhaust ducts shall not be extended into or through ducts or plenums.
- (c) Prohibited Use: Exits and exit access corridors shall not be used as supply or return air ducts or plenums.

Exception: The restriction on the use of the space between the corridor ceiling and the floor or roof structure above as return air plenum shall not apply when the corridor is not required to be of fire resistance rated construction or is separated from the plenum by fire resistance rated construction or is located within a dwelling unit.

- (d) Flood Proofing: For building located in a flood hazard zone, plenum spaces shall be either placed above the base flood elevation or protected so as to prevent water from entering or accumulating within the plenum space during floods up to the base flood elevation.

2.8.1.2 Material

- (a) All ducts, duct connectors, associated fittings and plenums used to convey supply air, return air, and outdoor air for air-conditioning, heating and ventilation system shall be constructed of steel, aluminum alloy or some other approved metal. Ducts, plenums and fittings may be constructed of concrete, clay or ceramics when installed in the ground or in a concrete slab, provided the joints are tightly sealed.

- (b) When gypsum products are exposed in ducts or plenums, the air temperature shall neither be lower than 10°C nor be higher than 52°C and the moisture content shall be controlled so as not to adversely affect the material. Gypsum products shall not be exposed in ducts serving evaporative coolers.

2.8.1.3 Combustibles within ducts or plenums

Plenums shall be constructed with non-combustible materials. Materials exposed within ducts or plenums shall have a flame spread index of not more than 25, and smoke developed rating of not more than 50 when tested in accordance with ASTM E84.

Exceptions:

- (i) Return air and outside air ducts, plenums and concealed spaces which serve a dwelling unit may be of combustible construction.
- (ii) Air filters serving dwelling unit.
- (iii) Air filters used as water evaporation medium in an evaporative cooler.
- (iv) Charcoal filters when protected with an approved fire suppression system.
- (v) Exposed electric cables installed in concealed space used as plenums exhibit a flame propagation of not more than 1.5 m and produce smoke having a peak optical density not greater than 0.5 and average optical density not greater than 0.15 when tested in accordance with UL910.
- (vi) Nonmetallic fire sprinkler piping in the plenum exhibit a flame propagation of not more than 1.5 m and shall produce smoke having a peak optical density not greater than 0.5 and average optical density not greater than 0.15 when tested in accordance with UL1820.

2.8.1.4 Duct construction

- (a) Ducts shall be of square, rectangular, round or oval cross-section. Construction of required size of duct shall be as per good practice described in ASHRAE Handbooks and SMACNA (Sheet Metal and Air-conditioning Contractors' National Association, USA) duct construction standards.
- (b) Joints of duct systems shall be made substantially airtight by means of tapes, mastics, gasketing or other means and shall have no opening other than those required for proper operation and maintenance of the system. Access openings shall be provided in the duct system for periodic cleaning of the system. Removable grilles requiring only the loosening of catches or screws for removal may be considered as access openings. Walk in access doors shall be so constructed that the door may be readily opened from the inside without the use of keys.

- (c) Vibration isolators installed between equipment and metal ducts (or casings) or between two sections of the ducts where duct crosses building expansion joint, shall be made of an approved flame retardant fabric or shall consist of sleeve joints with packing of approved material having flame spread rating of not more than 25 and a smoke developed rating of not more than 50 when tested in accordance with ASTM E84. Vibration isolation connectors constructed of fabric shall not exceed 250 mm in length.

2.8.1.5 Duct coverings

- (a) Supply and return air ducts and plenums of a cooling or heating system shall be insulated with approved quality insulating material of adequate thickness required as per location of the duct system and temperatures of air inside and around the duct system. Insulation shall be of such quality and thickness to prevent the formation of condensation on the exterior or interior walls of any duct.
- (b) Materials used within the ducts and plenums for insulation, sound absorption or other purposes shall have a mold, humidity and erosion resistant face that meets the requirements of accepted standards. These materials when exposed to air velocities within the ducts in excess of 10 m/s shall be fastened with both adhesive and mechanical fasteners, and exposed edges shall have adequate treatment to withstand the operating velocity.
- (c) Duct coverings, duct linings, vapour barrier facings, tapes, adhesives used in duct system shall have a flame spread rating not over 25 and a smoke development rating no higher than 50 when tested as a composite installation:

Exceptions:

- (i) Duct coverings shall not be required to meet these requirements where they are located entirely outside of a building, do not penetrate a wall or roof, and do not create an exposure hazard.
- (ii) Duct covering having a flame spread index not exceeding 50 and a smoke density not greater than 100 may be used in dwelling/apartment houses where duct system serves not more than one dwelling unit.
- (d) Duct coverings, linings, including associated tapes and adhesives shall be interrupted at least 1 m from heat source in a duct system such as electric resistance heaters, fuel burning heaters or furnaces and at the area of a fire damper or fire door, where the duct penetrates a fire separation. Interior insulation and acoustical linings shall be placed so as not to interfere with positive closing of fire dampers or other closures.
- (e) Service openings shall not be concealed.

2.8.1.6 Duct installation

- (a) An air distribution system shall be designed and installed as per good practice described in ASHRAE Handbooks and SMACNA Handbook so as to meet the requirement of proper distribution of air as per provisions of this Code. The installation of an air distribution system shall not affect the fire protection requirements specified in this Code.
- (b) Ducts and all parts of the duct system shall be substantially supported and securely fastened to the structural members of the building with approved devices of noncombustible material designed to carry the required loads. Duct supports shall not lessen the fire protections of structural members. Ducts shall be braced and guyed to prevent lateral or horizontal swing.
- (c) Hangers shall have sufficient strength and durability to properly and safely support the duct work. Hangers shall have sufficient resistance to the corrosive effect of the atmosphere to which they will be exposed. Hangers shall not be used in direct contact with a dissimilar metal that would cause galvanic action in the hanger, duct, fastenings, or structure.
- (d) Ducts shall not be hung from or supported by suspended ceilings.
- (e) Metal ducts shall not usually be installed within 100 mm of the ground. Metal ducts not having an approved protective coating, when installed in or under concrete slab shall be encased in at least 50 mm of concrete. Metallic ducts having an approved protective coating and nonmetallic ducts shall be installed in accordance with the manufacturer's installation instructions.
- (f) When ducts penetrate any masonry wall, it shall either be lined with felt to isolate it from the masonry, or an air gap shall be left around it.
- (g) All underground ducts located in a flood hazard zone shall be capable of resisting hydrostatic and hydrodynamic loads and stresses, including the effects of buoyancy, during the occurrence of flooding to the base flood elevation.
- (h) Ducts installed in locations where they are subject to mechanical damage by vehicles or from other causes shall be protected by approved barriers.

2.8.1.7 Fire damper

- (a) Fire dampers shall be provided at locations where air distribution systems penetrate assemblies that are required to be fire resistance rated by this Code.

Exceptions:

- (i) Fire dampers are not required in the following cases:
 - (ii) Where an exhaust duct penetrates a fire resistance rated shaft wall and the sub-duct extends not less than 560 mm vertically upward.
 - (iii) At penetrations of tenant separation and corridor walls in buildings equipped throughout with an automatic sprinkler system installed in accordance with the Code.
 - (iv) Where the ducts are constructed of steel and are part of an engineered smoke removal system.
 - (v) At penetration of corridor walls where the ducts are constructed of steel and do not have openings which communicate the corridor with adjacent spaces or rooms.
 - (vi) At penetrations of a roof assembly where ducts are open to the atmosphere.
 - (vii) In hazardous exhaust systems.
 - (viii) Where ceiling dampers are installed in accordance with the building code.
 - (ix) In garage exhaust or supply shafts which are separated from all other building shafts by not less than 2-hour fire resistance rated fire separation assembly.
 - (x) In ducted air-conditioning, heating and ventilation systems penetrating walls with a 1 hour fire resistance rating or less. Where fire dampers will interfere with the operation of the smoke control system, approved alternative protective devices shall be utilized.
- (b) Fire dampers shall comply with UL555 and bear the label of an approved agency. Fire dampers shall be installed in accordance with the manufacturing installation instructions.
- (c) Fire dampers shall be accessible. Suitable openings with tightly fitted covers shall be provided to make fire dampers accessible for inspection and this shall be large enough to permit maintenance and resetting of the damper.
- (d) Ductwork shall be connected to fire damper sleeves or assemblies in such a way that collapse of the ductwork will not dislodge the damper.

2.8.1.8 Automatic shutoff

- (a) Each single air distribution system providing air-conditioning, heating or ventilation air in excess of 1000 l/s in various occupancies, shown below, shall be equipped with an automatic shutoff provision activated by smoke detectors. When the system serves more than one occupancy, automatic shutoff must be provided.

Type of Occupancy	Subdivision	Type of Occupancy	Subdivision
A	A5	G	G2
B	All (B1, B2 & B3)	H	H2
C	All (C1, C2, C3, C4 & C5)	I	I1, I2, & I3
D	All (D1 & D2)	J	All (J1, J2, J3 & J4)
E	E1 & E3	L	L
F	F1 & F2		

Exceptions:

- Automatic shutoff need not be installed when all rooms have direct exit to the exterior of the building.
 - Automatic shutoff need not be installed in systems specifically designed for smoke control.
- (b) Smoke Detection: Smoke detectors required by Sec 2.5.1.8 shall be installed in the main return-air duct ahead of any outside air inlet or they may be installed in each room or space served by the return air duct. Detectors shall also be installed in the supply duct, downstream of the filters. Activation of any detector shall cause the air moving equipment to automatically shut down.

2.8.2 Air Terminals**2.8.2.1 Registers, grilles and diffusers**

Supply air registers, grilles and diffusers; and return air grills shall be installed in accordance with the manufacturer's installation instructions. Selection and installation of registers, grilles and diffusers shall comply with the requirements of air distribution system.

2.8.2.2 Ventilating ceilings

Perforated ceilings may be used for air supply except in exit corridors which are required to be of fire resistive construction. Ceiling material shall be of Class-I flame spread classification on both sides in accordance with requirements of this Code. All wiring shall be in enclosures regardless of the voltage carried. Suspended ventilating ceiling supports shall be of non-combustible materials.

2.8.2.3 Visual duct openings

Duct openings in bathrooms, toilets and changing rooms shall prevent visual observation from adjoining rooms.

2.8.2.4 Capped opening

All duct openings shall be capped during construction.

2.8.2.5 Return air intake and outside air intake

Return air and outside air intake openings shall be located in accordance with the requirements of Sec 2.6.2.7.

2.8.2.6 Exhaust openings

Outside exhaust openings shall be located so as not to create nuisance. Exhaust air shall not be directed onto walkways.

2.8.2.7 Opening protection

Outside air intake and exhaust openings shall be protected with corrosion-resistant screens, louvers or grilles. Openings shall be protected against all local weather conditions. Exhaust openings shall have provision to prevent back draft under wind conditions.

2.8.3 Exhaust Air Systems

2.8.3.1 General

- (a) Exhaust air systems serving kitchens or toilets and/or bathrooms shall be independent exhaust systems and shall not be combined with exhaust air ducts serving other areas, except at immediately before the point of final delivery to the outside, such as at the base of a roof ventilator or when all interconnected systems are equipped with suitable back pressure devices to prevent passage of odours from one system to another when the fan is not in operation.
- (b) Exhaust ducts shall have provision for removal of condensates where this may be a problem, such as for swimming pools and shower exhausts and for these applications duct joints shall be water tight.
- (c) Construction and installation of exhaust air ducts for toilet, bathrooms and swimming pools shall be in accordance with the provisions of Sec 2.5.1.
- (d) Design, construction and installation of exhaust air systems for exhaust of harmful and hazardous gases and industrial/process exhaust gases shall be in accordance with the provisions of Sec 2.8.4.
- (e) Design, construction and installation of kitchen exhaust system shall be in accordance with the provisions of Sec 2.8.5.

2.9 Air-Conditioning Equipment

2.9.1 General**2.9.1.1 Scope**

Air-conditioning, heating and ventilation equipment shall conform to the requirements of this Code.

Equipment shall not be installed or altered in violation of this Code. Defective materials or parts shall be replaced in such a manner as not to invalidate any approval.

2.9.1.2 Approval

When required each appliance shall be approved by the building official for safe use or comply with applicable nationally recognized standard. For this purpose installers shall furnish satisfactory evidence that the appliance is constructed in conformity with the requirements of this Code. The permanently attached label of an approved agency may be accepted as such evidence.

2.9.1.3 Labeling

All mechanical equipment and appliances shall bear permanent and legible factory applied name plate on which shall appear construction and operation data including safety requirements.

2.9.1.4 Testing

Where required an approved agency shall test a representative sample of the mechanical equipment or appliance being labeled to the standard or standards pertinent to the equipment or appliance. The approved agency shall maintain a record of all tests performed. The records shall provide sufficient detail to verify compliance with the test standard.

2.9.1.5 Equipment installation

- (a) General: Mechanical equipment and appliances shall be installed in accordance with the manufacturer's installation instructions for the labeled equipment. Connections to mechanical equipment or appliances, such as fuel supply, electrical, hydronic piping, vent and ducts shall conform to the requirements of this Code.
- (b) Clearance: Appliances shall be installed with the minimum clearances to combustibles for which the appliance has been tested as specified by the manufacturer.
- (c) Anchorage of Appliances: Appliances designed to be fixed in position shall be securely fastened in position. Supports for appliances shall be designed and constructed to sustain vertical and horizontal loads within the stress limitations specified in the Code.

- (d) Noise and Vibration: Equipment noise and vibration transmitted to the occupied space shall not exceed the recommended value for the space. Selection and installation of equipment shall be in accordance with Sec 2.4.3.
- (e) Identification of Equipment: When more than one air-conditioning, heating, refrigerating or ventilation systems are installed on the roof of a building or within the building, each equipment shall be identified as to the area or space served by the equipment.

2.9.1.6 Access

All mechanical equipment and appliances shall be accessible for inspection, service, repair and replacement without removing permanent construction. Unless otherwise specified not less than 750 mm of working space and platform shall be provided to service the equipment or appliance.

Appliance controls, gauges, filters, blowers, motors and burners shall be accessible. The operating instructions shall be clearly displayed near the appliance where they can be read easily.

2.9.1.7 Location

- (a) Remote Location: Where an appliance is located in a remote location, a walkway having a minimum width of 600 mm shall be provided, leading from the access opening to the appliance.
- (b) Hazardous Location: Appliances installed in garages, warehouses, or other areas where they may be subject to mechanical damage shall be installed behind suitable protective barriers or at a suitable height above the floor or located out of the normal path of vehicles to guard against such damages.

Air-conditioning or heating equipment located in a garage and which generates a glow, spark or flame capable of igniting flammable vapours shall be installed in such a way that the pilots and burners or heating elements and switches are at least 450 mm above the floor level.

Where such appliances installed within a garage are enclosed in a separate approved compartment having access only from outside of the garage such appliances may be installed at floor level, provided the required combustion air is taken from and discharged to the exterior of the garage.

Heating equipment located in rooms where cellulose nitrate plastic or other explosive materials are stored or processed shall comply with the requirements of Part 4 of this Code.

- (c) Outdoor Installation: Mechanical equipment and appliance located outdoors shall be approved for outdoor installation. Mechanical equipment and appliances installed outdoors shall conform to the requirements of Sec 2.6.1.5.

Where appliances are located within 3 m of a roof edge or open side of a drop greater than 600 mm, guards shall be provided. Height of the guard shall be a minimum of 900 mm and a maximum of 1050 mm above the surface.

Equipment that are located outdoors and may be adversely affected by sun and/or water shall be adequately protected. Access shall be possible under all weather conditions. All outdoor installed equipment shall be so located that the sound level shall not be more than 65 dB when measured anywhere on the property boundary line.

2.9.1.8 Electrical installations

- (a) Equipment regulated by this Code requiring electrical connections of more than 50 volts shall have a positive means of disconnect adjacent to and in sight from the equipment served. A 230 volt AC grounding type receptacle shall be located within 8 m of the equipment for service and maintenance purposes. The receptacle need not be located on the same level as the equipment. Low voltage wiring of 50 volts or less within a structure shall be installed in a manner to prevent physical damage.
- (b) Permanent lighting shall be provided to illuminate the area in which an appliance is located. For remote locations, the light switch shall be located near the access opening leading to the appliance.

Exceptions:

Lighting fixtures need not be installed when the fixed lighting for the building will provide sufficient light for safe servicing of the equipment.

2.9.1.9 Condensate wastes

Condensates from air cooling coils, fuel burning condensing appliances and the overflow from evaporative coolers and similar water supplied equipment shall be collected and discharged to an approved plumbing fixture and disposal area. The waste pipe shall have a slope of not less than 1 in 100 and shall be of approved corrosion resistant material and approved size. Condensate or waste water shall not drain over a public way.

2.9.1.10 Personnel Protection

A suitable and substantial metal guard shall be provided around exposed flywheels, fans, pulleys, belts and moving machinery which are portions of air-conditioning, heating and ventilation system.

2.9.2 Cooling by Refrigeration**2.9.2.1 General**

- (a) Scope: Every air cooling system and equipment using refrigerant coils, chilled water coils and brine coils shall conform to the requirements of this Section and to the applicable requirements of Sections 2.6.1 and 2.7.
- (b) Use of Group 2 Refrigerants: Direct refrigerant systems containing Group 2 refrigerants shall not serve an air-cooling or air-conditioning system used for human comfort.

2.9.2.2 Installation

- (a) Clearance from Ground: When cooling equipment other than ducts and piping is suspended from the under floor construction, a clearance of at least 150 mm shall be provided between the base of the equipment and the ground.
- (b) Exterior Wall Installation: All equipment mounted on exterior wall at a height of 6 m or more above the ground shall be provided on a platform not less than 750 mm in depth, with 1 m high handrails on operation and control side of the equipment. The platform shall be accessible through catwalk not less than 450 mm wide and handrail of 1 m high from inside the building or from roof access.

Exceptions:

Equipment located on exterior wall but removable from inside may not require platform and catwalk.

2.9.2.3 Access

- (a) Cooling Units: Except for piping, ducts and similar equipment that does not require servicing or adjusting, an unobstructed access and passageway not less than 600 mm in width and 2 m in height shall be provided to every cooling unit installed inside buildings.

Exception:

The access opening to a cooling unit located in an attic space may be reduced to 750 mm in length and width, provided the unit can be replaced from this opening or another opening into this space or area.

- (b) Attic or Furred Space Installation: Access to and working platforms for cooling units or cooling system compressors located in an attic or furred space shall be provided with a solid continuous flooring not less than 600 mm in width from the access opening to the required working space and platform in front of the equipment when access opening is located more than 1 m away from working space.

- (c) Filters, Fuel Valves and Air Handlers: An unobstructed access space not less than 600 mm in width and 750 mm in height shall be provided to filters, fuel control valves and air handling units. Refrigerant, chilled water and brine piping control valves shall be accessible.

Exception:

An access opening from the unobstructed access space which opens directly to such equipment may be reduced to 375 mm in the least dimension if the equipment can be serviced, repaired and replaced from this opening without removing permanent construction.

- (d) Refrigeration Machinery Room Installations: Access to equipment located in a refrigeration machinery room shall comply with Sec 2.7.

- (e) Roof or Exterior Wall Installation

- (i) Equipment installed on the roof or on an exterior wall shall be accessible under all weather conditions. A portable ladder or other portable temporary means may be used for access to equipment located on the roof, or on exterior wall of a single-storey portion of the building.
- (ii) Platform: When the roof has a slope greater than 4 in 12 a level working platform at least 750 mm in depth shall be provided along the control or servicing sides of the unit. Sides of a working platform facing the roof edge below shall be protected by a substantial railing of minimum 1 m in height with vertical rails not more than 525 mm apart, except that parapets at least 600 mm in height may be utilized in lieu of rails or guards.
- (iii) Catwalk: On roofs having slopes greater than 4 in 12, a catwalk at least 400 mm in width with substantial cleats spaced not more than 400 mm apart shall be provided from the roof access to the working platform at the appliance.

2.9.2.4 Working space

Equipment requiring access thereto, as specified in Sec 2.6.2.3, shall be provided with an unobstructed space on the control or servicing side of the equipment of not less than 750 mm in depth and 2 m in height. Working space for equipment located in a machinery room shall comply with Sec 2.7.

Exception:

The height of the working space may be reduced to 750 mm for an air handling unit, air filter or refrigerant, chilled water piping and brine piping control valves.

2.9.2.5 Lighting in concealed spaces

When access is required to equipment located in an under floor space, attic or furred space, a permanent electric light outlet and lighting fixture shall be installed in accordance with Sec 2.6.1.8.

2.9.2.6 Condensate control

When a cooling coil or cooling unit is located in the attic or furred space where damage may result from condensate overflow, an additional water tight pan of corrosion resistant metal shall be installed beneath the cooling coil or unit to catch the overflow condensate due to clogged primary condensate drain, or one pan with a standing overflow and a separate secondary drain may be provided in lieu of the secondary drain pan. The additional pan or the standing overflow shall be provided with a drain pipe, minimum 19 mm nominal pipe size, discharging at a point which can be readily observed. This requirement is in addition to the requirements for condensate waste piping set forth in Sec 2.6.1.9.

2.9.2.7 Return air and outside air

- (a) Source: A cooling unit shall be provided with outside air, return air, or both. Cooling systems regulated by this Code and designed to replace required ventilation shall be arranged to deliver into the conditioned space not less than the amount of outside air specified in Building Code.
- (b) Prohibited Sources: The outside air or return air for a cooling system or cooling unit shall not be taken from the following locations:
 - (i) Closer than 3 m from an appliance vent outlet, a vent opening or a plumbing drainage system or the discharge outlet of an exhaust fan, unless the outlet is 1 m above the outside air inlet.
 - (ii) Where it will pick up objectionable odours, fumes or flammable vapours; or where it is less than 3 m above the surface of any abutting public way or driveway; or where it is in a horizontal position in a sidewalk, street, alley or driveway.
 - (iii) A hazardous or insanitary location or a refrigeration machinery room;
 - (iv) An area the volume of which is less than 25 percent of the entire volume served by such system, unless there is a permanent opening to an area the volume of which is equal to 25 percent of the entire volume served.
 - (v) A room or space having any fuel burning appliances therein, except when 75 percent of the conditioned air is discharged back into the same room or space and air inlets are not located within 3 m of firebox or draft diverter of fuel burning appliance and the room has a volume exceeding 1 m³ for each 100 watts fuel input rate of all fuel burning appliance therein.
 - (vi) A closet, bathroom, toilet or kitchen.

- (c) Return Air Limitation: Return air from one dwelling unit shall not be discharged into another dwelling unit through the cooling system.

2.9.2.8 Air velocity

Cooling systems shall be designed and constructed so that velocity through filters does not exceed the filter manufacturer's recommendation.

2.9.2.9 Screen

Required outside air inlets shall be covered with screen having 6 mm openings.

Exception:

An outside air inlet serving a nonresidential portion of a building may be covered with screen having opening larger than 6 mm but not larger than 25 mm.

2.9.2.10 Duct system

If ducts are required for circulation of air, the duct system shall be constructed and installed in accordance with Sec 2.5.1. Selection and installation of registers, diffusers and grilles shall conform to the requirements of Sec 2.5.2.

2.9.3 Evaporative Cooling

2.9.3.1 General

- (a) Scope: Where possible evaporative cooling system may be installed. Evaporative cooling systems shall comply with this Section.
- (b) Outside Air: Evaporative cooling system shall be provided with outside air as specified in Sec 2.6.2.7
- (c) Air Ducts: Air duct systems for evaporative cooling shall comply with Sec 2.5.1.

2.9.3.2 Location

Evaporative cooler shall normally be installed outdoor. It may be installed indoor if duct is provided between cooler and outside air intake.

Evaporative cooling systems shall be installed in a manner to minimize the probability of damage from an external source.

2.9.3.3 Access

Evaporative coolers shall be accessible for inspection, service and replacement without removing permanent construction.

2.9.3.4 Installation

An evaporative cooler supported by the building structure shall be installed on a substantial level base and shall be secured directly or indirectly to the building structure by suitable means to prevent displacement of the cooler.

An evaporative cooler supported directly by the ground shall rest on a level concrete slab. The upper surface of the concrete slab shall not be less than 75 mm above the adjoining ground level.

An evaporative cooler supported on an above ground platform shall be elevated at least 150 mm above the adjoining ground level.

Openings in the exterior walls shall be flushed in an approved manner in accordance with this Code.

2.9.4 Heating Equipment

2.9.4.1 General

- (a) Scope: Provisions of this Section shall apply to all electric, hot water or steam air heating systems.
- (b) Outside Air: Heating system shall be provided with outside air as specified in Sec 2.6.2.7.
- (c) Air Ducts: Air ducts for heating systems shall comply with the applicable provisions of Sec 2.5.1.

2.9.4.2 Location

- (a) Steam shall not be used in heating coil of air handling unit when it is located inside the building but not installed in a machinery room.
- (b) All fuel burning equipment such as boilers shall not be installed inside a building and shall be installed inside a machinery room.
- (c) Appliances generating a glow, spark or flame capable of igniting flammable vapours shall not be located in places where such vapours exist.

2.9.4.3 Access

All appliances shall be accessible for inspection, services, repair and replacement without removing permanent construction. On control and servicing side(s) of the appliance, an unobstructed working space of not less than 750 mm in width and 1250 mm in height shall be provided.

2.9.4.4 Installation

All heating appliances shall be installed as per applicable provisions of Sec 2.6.1.5.

2.9.4.5 Controls

- (a) In case of air-conditioning plants where heating or reheating is required, a safety device shall be incorporated in the installation to cut off automatically the source of heating, such as steam, hot water or electricity by means of a suitable thermostat or some other device, as soon as the temperature of the room reaches a predetermined high level not exceeding 44°C, unless a higher temperature is required for an industrial process carried out in the air-conditioned enclosure. In no case the outlet temperature of the heater shall exceed 90°C.
- (b) In the case of air-conditioning plants where heating or reheating by means of an electrical heater designed to operate in an air current is done, the system shall be equipped with a safety device to cut off the electricity to the heating device whenever there is failure of the air flow in which the heater is required to operate.

The surface temperature of all electrical heaters used in air-conditioning systems shall be limited preferably to 400°C, and in no case more than 538°C when measured in still air.

2.9.4.6 Boilers and Furnaces

- (a) Steam and hot water boilers and furnaces used for air-conditioning systems shall be designed, constructed and installed in conformance with the requirements of acceptable standards in this regard and the appropriate Boiler Code.
- (b) Boilers and furnaces shall be installed in a machinery room having:
 - (i) A sufficiently large floor area to permit accessibility for inspection and servicing of the appliance and to provide adequate clearance to satisfy requirements of fire safety. The volume of the room for housing central heating furnaces shall be at least 12 times the total volume of the furnace. The volume of the room for housing central heating boilers shall be at least 16 times the total volume of the boiler. If the ceiling height of the room or space is greater than 2.5 m, the volume shall be calculated on the basis of 2.5 m height.
 - (ii) A permanent opening or opening connecting with the outdoors or with some space that freely connects with outdoors,
 - (iii) A knockout panel to act as explosion relief panel to prevent damage to structure in case of any explosion in boiler rooms,
 - (iv) Boiler rooms and furnace rooms shall be protected with an automatic fire suppression system installed in accordance with the Code.
- (c) Before commissioning of the boiler a certificate of compliance from the Chief Inspector of Boiler shall be obtained.

2.9.5 Air Handling Unit**2.9.5.1 General**

Air handling units shall comply with the applicable requirements as set forth in Sections 2.6.1 and 2.6.2.

2.9.5.2 Location

Air handling unit rooms shall, as far as possible, be centrally located with the equipment room contiguous to the corridors or other spaces for running of air ducts.

Air handling unit rooms shall be located in areas where reasonable sound levels can be tolerated. Air handling unit rooms shall not preferably be located adjacent to conference rooms, sound recording studios, broadcasting studios, bed rooms and other acoustically sensitive areas. If it is absolutely necessary to locate air handling unit room adjacent to the above acoustically sensitive areas, adequate acoustic treatment in the air handling units, supply and return air ducts, air handling unit rooms shall be provided. In such case, the access door to the air handling unit room shall be of single leaf type properly acoustically treated and shall have a door sill. The door shall open outwards.

In case of multi-storied buildings and for large capacity plant, independent air handling unit room(s) shall be provided for each floor when design calls for the same. The area served by each air handling unit shall conform to the fire protection measures adopted.

2.9.5.3 Access

Floor area of the air handling unit room shall be sufficient to allow proper layout of equipment with adequate access space and working space for proper operation and maintenance.

2.9.5.4 Installation

Air handling units shall be installed on vibration isolators to restrict transmission of vibration to the building structure. The base of the air handling unit shall be minimum 75 mm above the adjoining floor level. All air handling unit rooms shall have properly installed floor drains.

2.9.6 Packaged Air-conditioners**2.9.6.1 General**

Packaged air-conditioners shall comply with the applicable requirements set forth in Sections 2.6.1 and 2.6.2.

2.9.6.2 Prohibited use

Packaged air-conditioners shall not be used for,

- (a) Operation theatres where provisions for high percentage of fresh air and high quality filtration of air are required.
- (b) Special applications like sterile rooms for hospitals and clean rooms where high efficiency filtration is required.
- (c) Sound recording studios and other areas where criteria for acoustics are stringent.

Exceptions:

Single package units when installed far away from the air-conditioned space and are provided with properly designed sound attenuators that maintains the desired sound level inside the conditioned space.

- (d) Area requiring close and independent control of temperature and relative humidity.

Exception: Computer room air-conditioning.

- (e) Internal zones where no exposed wall is available for installation of room air-conditioners or no external platform is available for installation of outdoor installed unit.
- (f) The width of the area is such that throw of air from the air-conditioner cannot cover the required area.

2.9.6.3 Installation

- (a) Wall punches for room air-conditioners shall have proper sealing and resilient pad around the body of the unit to avoid leakage of air and vibration transmission.
- (b) Outdoor units shall be installed keeping adequate space for condenser air flow. The discharge of condenser air shall not create any disturbance to the adjacent rooms or buildings.
- (c) Refrigerant pipes and condensate drain pipes shall be properly installed and shall have proper insulation to avoid condensation on pipes. Indoor installations shall comply with the requirements of Sec 2.9.2.6.

2.9.7 Accessory Equipment

2.9.7.1 Air curtain

- (a) Where Required: In super markets, departmental stores, commercial buildings and other applications where the continuous movement of people and/or equipment through the door requires that the door be remained open continuously, adequately sized air curtains may be used to restrict entry of unconditioned air to conditioned space.

- (b) Installation: Air curtains shall be installed in such a way as to cover the whole width of the door. The width and velocity of air jet shall be sufficient to restrict the entry of unconditioned air to the conditioned space. The unit shall have provisions to control the jet velocity with respect to pressure and velocity of air in the unconditioned space.

2.9.7.2 Air filters

- (a) Air supplied to any space for cooling, heating or ventilation shall be adequately filtered before its point of discharge into the space. Minimum filtration efficiency shall be in accordance with good engineering practice for the space served, as recommended in ASHRAE Handbook.
- (b) Access: Adequate access to facilitate servicing of filters shall be provided. Doors, ladders, electric lighting etc. shall be provided where necessary. A device for indicating differential pressure across the filter bank shall preferably be fitted to determine the need for filter change.
- (c) Electrostatic Filters: Electrostatic filters when used shall be electrically interlocked so that power supply is disconnected when access door is opened.

2.9.8 Piping System

2.9.8.1 Material

Piping material for air-conditioning, heating and ventilation system shall be metallic only.

Exception: Condensate drain and waste water drain piping for cooling units may be nonmetallic.

2.9.8.2 Support and anchors

Adequately designed piping supports shall be used at approved space intervals to prevent undue stress on the pipe and building structure. Piping shall also be adequately anchored. Pipes shall not be supported or hanged from another pipe.

2.9.8.3 Expansion and contraction

Piping shall be installed with provisions to take care of expansion and contraction of the piping because of temperature changes of the fluid it conveys.

2.9.8.4 Pipe covering

- (a) All pipes likely to achieve a surface temperature during normal operation exceeding 70°C and are exposed to human contact or surface temperature lower than the dew point temperature of the surrounding air, shall be insulated with approved material suitable for the operating temperature of the system. The insulating material and its thickness shall be as recommended in ASHRAE Handbook.

(b) Insulation and covering on pipes in which the temperature of the fluid exceeds 120°C:

- (i) Shall be of noncombustible material.
- (ii) Shall not produce flame and smoke, glow or smoulder when tested in accordance with the latest standard in this regard at the maximum temperature to which such insulation or covering is to be exposed in service.

Combustible insulation and covering shall have a flame spread rating throughout the material, not exceeding 25 units in buildings of noncombustible construction, when pipes run in a horizontal or vertical service space. When pipes run in a room or space other than service space, the pipe covering shall have a flame spread rating not exceeding that required for the interior finish of the ceiling of the room or space.

Exception:

Pipe coverings may have a flame spread rating more than 25 and smoke developed index more than 100 when pipes are enclosed within walls, floor slabs or non-combustible raceways or conduits.

2.9.8.5 Steam or hot water bare pipes passing through a storage space shall be protected to prevent direct contact between the surface of pipe and the material stored.

Bare pipes containing steam or fluid at temperature above 120°C and passing through a combustible floor, ceiling or wall shall have a sleeve of metal at least 50 mm larger in diameter than pipe, packed with noncombustible material.

Minimum clearance between bare pipe and combustible materials shall not be less than 15 mm when temperature of steam or water in the pipe does not exceed 120°C and shall not be less than 25 mm for temperatures exceeding 120°C.

2.9.8.6 All piping shall be marked with approved markings for type of fluid carrying with direction of flow.

2.9.9 Split Air-Conditioners

2.9.9.1 General

Split air-conditioners shall comply with the applicable requirements set forth in Sections 2.6.1 and 2.6.2.

2.9.9.2 Prohibited Use

Split air-conditioners shall not be used for,

- (a) Operation theatres where provisions for high percentage of fresh air and high quality filtration of air are required.
- (b) Special applications like sterile rooms for hospitals and clean rooms where high efficiency filtration is required.
- (c) Sound recording studios and other areas where criteria for acoustics are stringent.

Exceptions:

Split air conditioners when installed far away from the air-conditioned space and are provided with properly designed sound attenuators which can maintain the desired sound level inside the conditioned space.

- (d) Area requiring close and independent control of temperature and relative humidity.

Exception: Computer room air-conditioning.

- (e) Internal zones where no exposed wall is available for installation of room air-conditioners or no external platform is available for installation of outdoor installed unit.
- (f) The width of the area is such that throw of air from the air-conditioner cannot cover the required area.

2.9.9.3 Installation

- (a) Wall punches for room air-conditioners shall have proper sealing and resilient pad around the body of the unit to avoid leakage of air and vibration transmission.
- (b) Outdoor units shall be installed keeping adequate space for condenser air flow. The discharge of condenser air shall not create any disturbance to the adjacent rooms or buildings.
- (c) Refrigerant pipes and condensate drain pipes shall be properly installed and shall have proper insulation to avoid condensation on pipes. Indoor installations shall comply with the requirements of Sec 2.9.2.6.

2.9.10 Variable Refrigerant Flow (VRF) System

2.9.10.1 General

Variable refrigerant flow (VRF) air-conditioning system shall comply with the applicable requirements set forth in Sections 2.6.1 and 2.6.2.

2.9.10.2 Prohibited Use

VRF system shall not be used for,

- (a) Operation theatres where provisions for high percentage of fresh air and high quality filtration of air are required.
- (b) Special applications like sterile rooms for hospitals and clean rooms where high efficiency filtration is required.
- (c) Sound recording studios and other areas where criteria for acoustics are stringent.

Exceptions:

VRF system when installed far away from the air-conditioned space and are provided with properly designed sound attenuators that maintains the desired sound level inside the conditioned space.

- (d) Area requiring close and independent control of temperature and relative humidity.

2.9.10.3 Installation

- (a) Wall punches for room air-conditioners shall have proper sealing and resilient pad around the body of the unit to avoid leakage of air and vibration transmission.
- (b) Outdoor units shall be installed keeping adequate space for condenser air flow. The discharge of condenser air shall not create any disturbance to the adjacent rooms or buildings.
- (c) Refrigerant pipes and condensate drain pipes shall be properly installed and shall have proper insulation to avoid condensation on pipes. Indoor installations shall comply with the requirements of Sec 2.9.2.6.

2.10 Refrigerating Equipment

2.10.1 General

2.10.1.1 Scope

In addition to other provisions of this Code, refrigerating systems and equipment shall conform to the requirements of this Section.

2.10.1.2 Approval

All refrigerating equipment and components shall comply with relevant internationally recognized Standards. The listing and label, attached to the equipment, of an approved agency may be accepted as evidence of compliance with applicable internationally recognized Standards.

2.10.1.3 Installation

Refrigerating equipment shall be installed to conform to the provisions of Sec 2.6.1 and the manufacturer's installation instructions.

2.10.1.4 Access

Access for refrigerating units shall be provided as for cooling units and cooling systems set forth in Sections 2.6.1.6 and 2.6.2.3.

2.10.1.5 Working space and working platform

Working space and working platform shall be provided as for cooling units and cooling systems set forth in Sec 2.6.2.4.

2.10.1.6 Prohibited location

Refrigerating systems and portion thereof shall not be located in an elevator shaft, dumb waiter shaft or a shaft having moving objects therein, or in a location where it will be subject to mechanical damage.

2.10.1.7 Condensate control

Piping and fittings which convey refrigerant, brine, chilled water or coolant, which generally reach a surface temperature below the dew point of the surrounding air and which are located in spaces or areas where condensation could cause a hazard to the building occupants, structure, electrical or other equipment shall be insulated to prevent such damage.

2.10.2 Absorption Refrigerating Equipment

2.10.2.1 Location

Fuel burning absorption systems shall not be installed in the following locations:

- (a) In any room or space less than 300 mm wider than the units installed therein, with a minimum clear working space of not less than 75 mm along the sides, back and top of the unit.
- (b) In a hazardous location.
- (c) In a surgical operating room or medical treatment room.
- (d) In any occupancy group unless separated from the rest of the building by not less than a one hour fire resistive occupancy separation.

Exceptions:

A separation shall not be required for equipment serving only one dwelling unit.

- (e) In a room used or designed to be used as a bedroom, bathroom, closet or in any enclosed space with access only through such room or space.
- (f) In a room from where noise and vibration may be transmitted to acoustically sensitive areas.

Absorption systems containing Group 2 refrigerants shall not be located in any building unless installed within a refrigeration machinery room provided as per Sec 2.7.3.3.

Absorption systems containing more than 9 kg of a Group 2 refrigerant shall be located not less than 6 m from any door, window or ventilating air inlet to a building.

2.10.2.2 Installation

Fuel burning absorption systems located outside of a building shall be completely enclosed in a weather proof housing of approved materials, unless approved for outdoor installation. The housing shall not be larger than necessary to properly cover and provide a minimum 150 mm clearance around the unit or units enclosed therein, including all controls and draft diverters.

An absorption system supported from the ground shall rest on a concrete slab. The upper surface of the concrete slab shall be at least 75 mm above the adjoining ground level.

2.10.2.3 Pressure relief devices

An absorption system shall be equipped with a factory installed pressure relief device, either a fusible plug, a rupture member or a pressure relief valve.

2.10.2.4 Combustion air

A fuel burning absorption system shall be provided with adequate combustion air including venting appliances.

2.10.2.5 Steam or hot water absorption system

All absorption systems using steam or hot water as energy source shall be installed in a machinery room unless the manufacturer has certified it suitable for outdoor installation. The machinery room shall comply with the provisions of Sec 2.7.3.3.

2.10.3 Mechanical Refrigerating Equipment

2.10.3.1 General

- (a) Scope: Mechanical refrigerating equipment shall comply with the provisions of Sec 2.7.1.

Refrigerating systems and equipment, including the replacement of parts and alteration, shall comply with the provisions of this Section.

- (b) Supports: Supports for compressors, condensing units and chillers shall be designed to safely carry the equipment. Supports from buildings or parts of buildings that are of noncombustible construction shall be noncombustible.

A compressor or portion of condensing unit supported from the ground shall rest on a concrete or other approved base. The upper surface of the concrete base shall be at least 75 mm above the adjoining ground level.

- (c) Ventilation of Rooms Containing Condensing Units: Rooms or spaces other than a refrigeration machinery room complying with the requirements of this Section, in which any refrigerant containing portion of a condensing unit is located, shall be provided with one of the following means of ventilation:
- (i) Permanent gravity ventilation openings of not less than 0.2 m² net free area opening directly to the outside of the building or extending to the outside of the building by continuous ducts,
 - (ii) A mechanical exhaust system arranged to provide at least 3 complete air change per hour and to discharge to the outside of the building.

Exception:

Mechanical exhaust system shall not be required if the room or space has a volume exceeding 40 m³ per kW of the unit or where such room or space has permanent gravity ventilation openings of 0.2 m² minimum total area to the other rooms or spaces exceeding 40 m³ per kW.

- (d) Compressor Near Exits: Refrigerant compressors of more than 4 kW rating shall be located at least 3 m from an exit unless separated by a one hour fire resistive occupancy separation.

2.10.3.2 Refrigerants

- (a) Classification: Refrigerants listed in Tables 8.2.4 and 8.2.5 or other refrigerants equivalent in safety to life, limb, health or property shall only be used in refrigerating equipment.

Note: Bangladesh is a signatory to the Montreal Protocol which proclaims phasing out of the use of some refrigerants viewed as responsible for depletion of the ozone layer and/or causing global warming. If at the time of using this Code, any of the refrigerants mentioned in Tables 8.2.4 and 8.2.5 is prohibited from use by the Government, the relevant row or rows of these two tables shall be deemed to be deleted. Likewise, if any safer substitutes to these refrigerants are available and permitted by the Government, these shall be included in the list of refrigerants permitted by this Code. In general, preference shall be given to equipment using refrigerants having relatively lower Ozone Depletion Potential and Global Warming Potential.

Table 8.2.4: Group-I Refrigerant Classification, Amount and Occupational Exposure Limit (OEL)

Refrigerant Designation	Name	Refrigerant Classification ^f	Degrees of Hazard ^a	Max. Quantity in Space Intended for Human Occupancy (g/m ³)	OEL ^e
R-11 ^d	Trichlorofluoromethane	A1	2-0-0 ^b	6.2	1,000
R-12 ^d	Dichlorodifluoromethane	A1	2-0-0 ^b	90	1,000
R-13 ^d	Chlorotrifluoromethane	A1	2-0-0 ^b	-	1,000
R-13B1 ^d	Bromotrifluoromethane	A1	2-0-0 ^b	-	1,000
R-14	Tetrafluoromethane	A1	2-0-0 ^b	400	1,000
R-22	Chlorodifluoromethane	A1	2-0-0 ^b	210	1,000
R-32	Difluoromethane (Methylene chloride)	A2	-	77	1,000
R-113	Trichlorotrifluoroethane	A1	2-0-0 ^b	20	1,000
R-114	Dichlorotetrafluoroethane	A1	2-0-0 ^b	140	1,000
R-115	Chloropentafluoroethane	A1	2-0-0 ^b	760	1,000
R-123	Dichlorotrifluoroethane	B1	2-0-0 ^b	57	1,000
R-134a	Tetrafluoroethane	A1	2-0-0 ^b	210	1,000
R-407C	R-32/125/134a	A1	2-0-0 ^b	270	1,000
R-500	R-12/152a	A1	2-0-0 ^b	120	1,000
R-502	R-22/115	A1	2-0-0 ^b	330	1,000
R-717	Ammonia	B2	3-3-0 ^c	0.22	25
R-744	Carbon dioxide	A1	2-0-0 ^b	72	5,000

Notes :

- a* Degrees of hazard are for health, fire, and reactivity, in accordance with NFPA 704.
- b* Reduction to 1-0-0 is allowed if analysis satisfactory to the code official shows that the maximum concentration for a rupture or full loss of refrigerant charge would not exceed the IDLH, considering both the refrigerant quantity and room volume.
- c* For installations that are entirely outdoors, use 3-1-0.

- d* Class 1 ozone depleting substance; prohibited for new installations.
- e* Occupational Exposure Limit based on OSHA PEL.
- f* The capital letter designates the toxicity of the refrigerant at 400 ppm by volume. The number denotes the flammability of the refrigerant.
- Class A: Toxicity not identified
- Class B: Evidence of toxicity identified.
- Class 1: No flame propagation in air at 65°F and 14.7 psia.
- Class 2: Lower flammability limit (LML) greater than 0.00625 lb/ft³ at 70°F and 14.7 psia and heat of combustion less than 8174 Btu/lb.
- Class 3: Highly flammable as defined by LFL less than or equal to 0.00625 lb/ft³ at 70°F and 14.7 psia or heat of combustion greater than or equal to 8174 Btu/lb.

Table 8.2.5: Group 2 Refrigerants

Refrigerant Designation	Name
R-40	Methyl chloride
R-611	Methyl format
R-717	Ammonia
R-764	Sulphur dioxide

(b) Group 1 Refrigerants

- (i) Direct Systems: The maximum amount of Group 1 refrigerants in direct systems shall not exceed that set forth in Table 8.2.4.
- (ii) Indirect Systems: The amount of Group 1 refrigerants used in indirect systems shall be unlimited.
- (iii) General: Condensing units or combinations of refrigerant interconnected condensing units totaling 75 kW or more rating which contain a Group 1 refrigerant shall be enclosed in a refrigeration machinery room.

Exception:

The requirement shall not apply when the condensing unit is located outside of a building or on the roof of a building and not less than 6 m from a door, window or ventilating air opening in a building or when the condensing unit located in the building is exclusively used for ice making or cold storage together with the usual accessory rooms in connection therewith.

- (c) Group 2 Refrigerants: A mechanical refrigerating system or unit refrigerating system containing a Group 2 refrigerant shall not be located within a building unless all refrigerant containing portions of the system are enclosed in a refrigeration machinery room. Such system when installed outside of a building shall be located at least 6 m from an exit door, window or ventilating air inlet in a building.

Exception:

This shall not apply to a building used exclusively for ice making, cold storage or for the manufacturing or processing of food or drink, provided the occupant load does not exceed one person per 10 m² of floor area served by such system. Portions of refrigerating systems containing Group 2 refrigerants shall not be located in an exit.

Direct refrigeration systems containing Group 2 refrigerants shall not serve an air cooling or air-conditioning system used for human comfort.

2.10.3.3 Refrigeration machinery room

- (a) General: Required refrigeration machinery rooms shall be of at least one hour fire resistive construction. All doors shall be tight fitting. Every door shall be clearly labeled "Machinery Room". The room shall have no openings that will permit the passage of escaping refrigerant to the other parts of the building. There shall be no direct opening between a refrigeration machinery room containing Group 2 refrigerant and a room or space in which there is an open flame, spark producing device or heating surface hotter than 426°C. A refrigeration machinery room containing Group 2 refrigerants shall have at least two means of escape located at least one-fifth the perimeter of the room apart. Refrigeration machinery rooms containing Group 1 refrigerant shall have at least one exit door. Size of the exit door shall be at least 1 m by 2 m.

A refrigeration machinery room door shall open in the direction of escape.

An unobstructed working space of at least 750 mm in width and at least 2100 mm in height shall be provided around two adjacent sides of all moving machinery in a refrigeration machinery room.

- (b) Refrigeration Machinery Room Ventilation: Refrigeration machinery room shall be provided with either mechanical or gravity ventilation.
- (i) Mechanical exhaust system shall be a separate and individual system of ventilation serving no other area and shall exhaust air to outdoors at the rate of 12 air changes per hour. Exhaust air outlet shall not be located within 6m from any exterior door, window or ventilation air inlet in any building. Provisions shall be made for makeup air to replace that being exhausted. Control switch for exhaust system shall be located within the machine room and shall be readily accessible.

- (ii) Area of gravity ventilation openings to outside of the building shall not be less than one twentieth of the floor area of the machinery room but shall be more than 0.65 m². Approximately one-half of the openings shall be located within 300 mm of the floor and one half within 300 mm of the ceiling of the machinery room.
- (c) Equipment in a Refrigeration Machinery Room: Combustion air shall not be taken from a refrigeration machinery room. Electrical equipment, switch or control panel other than those used exclusively for air-conditioning, heating and ventilation system shall not be located in a refrigeration machinery room. This provision shall not apply to electrical lighting fixtures for machinery room and switches thereof.

A readily accessible single emergency refrigeration control switch shall be provided to shut off all electrically operated machineries in a refrigeration machinery room, except the exhaust ventilation system complying with Sec 2.7.3.3. Such switch shall be located outside the machinery room, within a distance of 3 m from the machinery room exit.
- (d) First Aid Facility: Each refrigeration machinery room shall be provided with first aid boxes. Refrigeration machinery room containing Group 2 refrigerants shall be provided with two gas masks.

2.10.3.4 Refrigerant piping and equipment

- (a) Materials: Materials used in the construction and installation of refrigerating systems shall be suitable for the refrigerant in the system, and no material or equipment shall be installed which will deteriorate due to the chemical action of the refrigerant or the compressor oil, or combination of both.
- (b) Erection of Refrigerant Piping: Refrigerant piping and tubing shall be installed in such a way so as to prevent excessive vibration and strains at joints and connections. Adequate type of supports shall be used at points as required but not exceeding 4.5 m apart.

Refrigerant piping and tubing shall be installed in such a way so that it is not subject to damage from an external source.

Copper tubing containing other than Group 1 refrigerant shall not be located in a public hallway, lobby or stairway or a building unless enclosed in iron or steel piping and fittings or in rigid metal conduit.

Iron or steel refrigerant piping placed underground shall be coated with sufficient asphalt paint or equivalent material to inhibit corrosion.

- (c) Refrigerant Containers: A refrigerant receiver or evaporator or condenser shall be constructed in accordance with approved standards.

- (d) Valves and fittings: All valves and fittings shall be of approved type rated for the maximum operating pressure of the system.
- (e) Pressure Limiting Device: A pressure limiting device shall be installed on a positive displacement refrigerant compressor which is a portion of:
- (i) A refrigerating system containing Group 2 refrigerant.
 - (ii) An air cooled refrigerating system containing Group 1 refrigerant of 7.5 kW or more rating.
 - (iii) A water cooled refrigerating system containing Group 1 refrigerant of 2.25 kW or more rating.
- A stop or shutoff valve shall not be placed between a pressure limiting device required by this Section and the compressor it serves.
- (f) Pressure Relief Valves: The following compressors of the positive displacement type shall be equipped with a pressure relief valve:
- (i) A compressor of 15 kW or more rating which is a portion of a refrigeration system containing Group 1 refrigerant and operating at a pressure exceeding 103 kPa in the high pressure side of the system.
 - (ii) A compressor which is a portion of a refrigerating system containing a Group 2 refrigerant.
- A pressure relief valve shall be connected to the refrigerant discharge side of the compressor it serves, between such compressor and a stop valve. A stop or shutoff valve shall not be located between a pressure relief valve required by this Section and the compressor it serves.
- A pressure relief valve required by this Section that terminates outside shall discharge at a location at least 4.5 m above the adjoining ground level and at least 6 m from a window, ventilating opening or exit from a building.
- (g) Pressure Relief Devices for Pressure Vessels: A pressure vessel over 150 mm diameter which may be shutoff by valves from other parts of the system shall be equipped with a pressure relief device(s) or rupture member complying with the requirements of this Code.
- (h) Manual Discharge of Group 2 Refrigerant: A refrigerating system located in a building and containing carbon dioxide or Group 2 refrigerant shall be equipped with approved means for manual discharge of the refrigerant to the atmosphere. The discharge pipe shall terminate outside of the building not less than 2 m above the highest structure on the building and at least 6 m from any window, ventilating opening or exit from a building.

2.10.3.5 Storage of refrigerants

Refrigerants not contained in refrigeration system regulated by the Code shall be stored in original containers kept in machinery room. The total amount shall not exceed 135 kg.

A portable refrigerant container shall not be connected to the refrigerating system for a period longer than is necessary to charge or discharge the refrigerating system.

2.10.4 Cooling Tower

2.10.4.1 Location

Cooling Tower shall not be located where warm and humid air discharge from cooling tower is likely to cause damage to building structure.

2.10.4.2 Installation

Cooling tower located at roof shall meet the requirements of structures as specified in this Code. Clearances for air suction and discharge shall be maintained in accordance with the recommendation of the manufacturer of the cooling tower.

Wind speed shall be taken into consideration while designing the foundation/supports for cooling tower.

Necessary vibration isolators shall be installed to restrict transmission of machine vibration to the structure.

2.10.4.3 Access

An easy access to cooling tower located at roof shall be provided.

2.10.4.4 Waste water disposal

Cooling towers or evaporative condensers which are equipped with a positive water discharge to prevent excessive build-up of alkalinity and are used for water cooled condensing units or absorption units shall discharge the water into an approved disposal system.

2.10.4.5 Piping connections

Water supply, waste water piping and other piping connections shall comply with the provisions of the Code.

2.10.4.6 Noise

Cooling tower noise shall not be more than 65 dBA or that approved by the jurisdiction at the property boundary line. If necessary, the fan cylinder may be covered with acoustic materials to attenuate noise. Similarly floating type mat may be used to reduce the water droplet noise.

2.10.4.7 Safety

Cooling tower fan shall be protected by a strong metal screen so that no external object and/or bird can come in contact with the fan blades.

An electric isolating switch shall be installed, in a locked enclosure, at a suitable location near the cooling tower to disconnect power to the cooling tower fan when maintenance works are to be carried out.

Each cooling tower shall be provided with a securely fixed ladder to facilitate maintenance works.

2.11 Ventilation Systems

2.11.1 General

2.11.1.1 Scope

The provisions of this Section shall govern the ventilation of spaces within a building intended for human occupancy.

2.11.1.2 Where required

Every space intended for human occupancy shall be provided with ventilation by natural or mechanical means during the periods when the room or space is occupied.

2.11.2 Natural Ventilation

2.11.2.1 Sources

Natural ventilation of an occupied space shall be through windows, doors, louvers, skylights or other openings to the outdoor. Such ventilating openings shall open to the sky or a public street, space, alley, park, highway, yard, court, plaza or other approved space which comply with the requirements of the building code.

2.11.2.2 Area of ventilating openings

The minimum ventilating opening to the outdoors shall be four percent of the floor area being ventilated.

- (a) **Adjoining Spaces:** Where rooms and spaces without openings to the outdoors are ventilated through an adjoining room, the unobstructed opening to the adjoining rooms shall be at least eight percent of the floor area of the interior room or space, but not less than 2.33 m². The ventilation openings to the outdoors shall be based on the total floor area being ventilated.
- (b) **Opening below Grade:** Openings below grade shall be acceptable for natural ventilation provided the outside horizontal clear space measured perpendicular to the opening is one and one-half times the depth below the average adjoining grade.

2.11.2.3 Contaminants exhausted

Naturally ventilated spaces having contaminants present shall comply with the requirements of Sec 2.8.4.

2.11.2.4 LP-gas distribution facilities

LP-gas distribution facilities shall be provided with air inlets and outlets arranged so that air movement across the floor of the facility will be uniform. The total area of both inlet and outlet openings shall be at least 0.70 percent of the floor area. The bottom of such openings shall not be more than 150 mm above the floor.

2.11.3 Mechanical Ventilation

2.11.3.1 Where required

Mechanical ventilation shall be provided in all occupiable rooms or spaces where the requirements for natural ventilation are not met; in all rooms or spaces, which because of the nature of their use or occupancy, involve the presence of dust, fumes, gases, vapours, or other noxious or injurious impurities, or substances which create a fire hazard; where space temperature is more than 40° C; where relative humidity of inside air is more than 70 percent; where job conditions require ventilation; or where required as per provisions of this Code.

2.11.3.2 Ventilation system

Mechanical ventilation shall be provided by a method of supply air and return or exhaust air. The amount of supply air shall be approximately equal to the amount of return and exhaust air; however, the system shall not be prohibited from producing a negative or positive pressure. The ventilation system ducts and equipment shall be designed and installed in accordance with Sec 2.5.

2.11.3.3 Ventilation air quantity

The minimum amount of air circulation rate for ventilation shall be determined based on the occupant load/space area and use of the building in accordance with Table 8.2.6. The air circulation rate specified in the Table 8.2.6 shall be equal to the combined total of outside air and re-circulated air. The occupant load shall be determined in accordance with the data provided in Table 8.2.3.

Table 8.2.6 Required: Minimum Air Circulation Rate for Mechanical Ventilation of Non-Air Conditioned Space

Sl. No.	Application	Air Change per Hour	Sl. No.	Application	Air Change per Hour
1	Assembly rooms	4-8	28	Hospitals-sterilizing	15-25
2	Bakeries	20-30	29	Hospital-wards	6-8
3	Banks/building societies	4-8	30	Hospital domestic	15-20
4	Bathrooms	6-10	31	Laboratories	6-15
5	Bedrooms	2-4	32	Launderettes	10-15
6	Billiard rooms	6-8	33	Laundries	10-30
7	Boiler rooms	15-30	34	Lavatories	6-15
8	Cafes and coffee bars	10-12	35	Lecture theatres	5-8
9	Canteens	8-12	36	Libraries	3-5
10	Cellars	3-10	37	Living rooms	3-6
11	Churches	1-3	38	Mushroom houses	6-10
12	Cinemas and theatres	10-15	39	Offices	6-10
13	Club rooms	12, Min	40	Paint shops (not cellulose)	10-20
14	Compressor rooms	10-12	41	Photo and X-ray darkroom	10-15
15	Conference rooms	8-12	42	Public house bars	12, Min
16	Dairies	8-12	43	Recording control rooms	15-25
17	Dance halls	12, Min	44	Recording studios	10-12
18	Dye works	20-30	45	Restaurants	8-12
19	Electroplating shops	10-12	46	School rooms	5-7
20	Engine rooms	15-30	47	Shops and Supermarkets	8-15
21	Entrance halls	3-5	48	Shower baths	15-20

SI. No.	Application	Air Change per Hour	SI. No.	Application	Air Change per Hour
22	Factories and work shops	8-10	49	Stores and Warehouses	3-6
23	Foundries	15-30	50	Squash courts	4, Min
24	Garages	6-8 ^a	51	Swimming baths	10-15
25	Glass houses	25-60	52	Toilets	6-10
26	Gymnasium	6, Min	53	Utility rooms	15-20
27	Hair dressing saloon	10-15	54	Welding shops	15-30

Note: The ventilation rates may be increased by 50 percent where heavy smoking occurs or if the room is below ground.

^a Only outdoor air and no recirculation shall be done.

2.11.3.4 Minimum outdoor air

The minimum amount of the outdoor air shall be in accordance with Table 8.2.3.

Mandatory requirement: in no case the ventilation air quantity shall be lower than 2.5 l/s per person.

2.11.3.5 Air temperature

The temperature differential between ventilation air and air in the conditioned space shall not exceed 5.5°C.

Exception:

Ventilation air that is part of the air-conditioning system.

2.11.3.6 Recirculation

- (a) Amount of Recirculation: Not more than 67 percent of the required ventilation air specified in Table 8.2.6 shall be permitted for recirculation, when the concentration of particulates is less than specified in Table 8.2.7. Air in excess of the required ventilation air shall be permitted to be completely recirculated. Air shall not be recirculated to another dwelling unit or occupancy of dissimilar use.

Not more than 85 percent of the required ventilation air shall be permitted for recirculation when the system is equipped with effective adsorption or filtering equipment so that the condition of the air supplied to the room or space is within the quality limitations of Table 8.2.7.

Table 8.2.7: Maximum Allowable Contaminant Concentrations

Contaminant	Annual Average (arithmetic mean) $\mu\text{g}/\text{m}^3$	Short Term Level (not to be exceeded more than once a year) $\mu\text{g}/\text{m}^3$	Averaging Period (hours)
Particulates	60	150	24
Sulfur oxides	80	400	24
Carbon monoxide	20,000	30,000	8
Photochemical oxidant	100	500	1
Hydrocarbons (not including methane)	1,800	4,000	3
Nitrogen oxides	200	500	24
Odour	--	Essentially unobjectionable ^a	--

Note: ^a Judged unobjectionable by 60 percent of a panel of 10 untrained subjects.

- (b) Prohibited Use of Recirculated Air: Air drawn from mortuary rooms, bathrooms or toilets or any space where an objectionable quantity of flammable vapours, dust, odours, or noxious gases is present shall not be recirculated. Air drawn from rooms that must be isolated to prevent the spread of infection shall not be recirculated.

Exception:

Air drawn from hospital operating rooms may be recirculated, if the following requirements are met:

- A minimum of twenty five total air changes per hour shall be provided, of which five air changes per hour shall be outdoor air.
- All fans serving exhaust systems are located at the discharge end of the system.
- Outdoor air intakes shall be located at least 7.5 m from exhaust outlets of ventilation systems, combustion equipment stacks, medical surgical vacuum systems, plumbing vent stacks or from areas which may collect vehicular exhaust and other noxious fumes. The bottom of outdoor air intakes serving central systems shall be located at least 2 m above ground level, or if installed above roof, at least 1 m above roof level.
- Positive air pressure shall be maintained at all times in relation to adjacent areas.

- (v) All ventilation or air-conditioning systems serving such rooms shall be equipped with a filter bed of 25 percent efficiency upstream of air-conditioning equipment and a filter bed of 99 percent efficiency downstream of the supply fan, any recirculating spray water systems and water reservoir type humidifiers. All filter efficiencies shall be average atmospheric dust spot efficiencies tested in accordance with the latest ASHRAE standard.
- (vi) Duct linings shall not be used in air-conditioning and ventilation systems serving such rooms unless terminal filters of at least ninety percent efficiency are installed downstream of linings.
- (vii) Air supplied shall be delivered at or near the ceilings and all exhaust air shall be removed near floor level, with at least two exhaust outlets not less than 75 mm above the floor.
- (c) Swimming Pool Area Recirculation: Return air from a swimming pool and deck area shall be permitted to be recirculated in accordance with Sec 2.8.3.6 when such air is dehumidified to maintain the relative humidity of the area at 60 percent or less. The return air shall only be recirculated to the area from which it was removed.

2.11.3.7 Ventilation in uninhabited spaces

Uninhabited spaces, such as crawl spaces or attics, shall be provided with natural ventilation openings as required by the Code or such spaces shall be mechanically ventilated. The mechanical ventilation system shall be a mechanical exhaust and supply air system. The exhaust rate shall be 0.1 litre per square metre of horizontal area. The ventilation system shall operate when the relative humidity exceeds 70 percent in the space.

2.11.3.8 Ventilation in enclosed parking garages

Mechanical ventilation systems for enclosed parking garages shall be permitted to operate intermittently where the system is arranged to operate automatically upon detection of vehicle operation or the presence of occupants or sensing the CO (carbon monoxide) concentration level by approved automatic detection devices.

Average concentration of CO shall not exceed 35 ppm with a maximum of 200 ppm. Automatic operation of the system shall not reduce the ventilation air flow rate below 0.25 litre per m² of the floor area and the system shall be capable of producing a ventilation air flow rate of 7.6 litre per m² of floor area. In no case the outdoor air quantity shall be lower than 5 litre per m² of floor area.

Connecting offices, waiting rooms, ticket booths and similar uses that are accessory to a public garage shall be maintained at a positive pressure.

2.11.4 Mechanical Exhaust

2.11.4.1 Where required

All rooms and areas having air with dust particles sufficiently light enough to float in the air, odours, fumes, spray, gases, vapours, smoke, or other noxious or impurities in such quantities as to be irritating or injurious to health or safety or which is harmful to building and materials or has substances which create a fire hazard, and rooms or areas as indicated in Table 8.2.6 shall have air exhausted to the outdoors in accordance with this Section.

2.11.4.2 Design of exhaust system

- (a) General: The design of the system shall be such that the emissions or contaminants are confined to the area in which they are generated by currents, hoods or enclosures and shall be exhausted by a duct system to a safe location or treated to remove contaminants. Ducts conveying explosives or flammable vapours, fumes or dusts shall extend directly to the exterior of the building without entering other spaces. Exhaust ducts shall not extend into or through ducts or plenums.

Exception:

Ducts conveying vapour or fumes having flammable constituents less than 25 percent of their lower flammability limit (LFL) may pass through other spaces.

Separate and distinct systems shall be provided for incompatible exhaust materials.

Contaminated air shall not be recirculated to occupied areas unless contaminants have been removed. Air contaminated with explosive or flammable vapours, fumes or dusts; flammable or toxic gases; or radioactive material shall not be recirculated.

- (b) Exhaust Air Inlet: The inlet to the exhaust system shall be located in the area of heaviest concentration of contaminants.
- (c) Velocity and Circulation: The velocity and circulation of air in work areas shall be such that contaminants are captured by an air stream at the area where the emissions are generated and conveyed into a product - conveying duct system. Mixtures within work areas where contaminants are generated shall be diluted below 25 percent of their lower explosive limit or lower flammability limit with air which does not contain other contaminants.

2.11.4.3 Make up air

Make up air shall be provided to replenish air exhausted by the ventilating system. Make up air intakes shall be located so as to avoid recirculation of contaminated air within enclosures.

2.11.4.4 Hoods and enclosures

Hoods and enclosures shall be used when contaminants originate in a concentrated area. The design of the hood or enclosure shall be such that air currents created by the exhaust systems will capture the contaminants and transport them directly to the exhaust duct. The volume of air shall be sufficient to dilute explosive or flammable vapours, fumes or dusts as set forth in Sec 2.11.4.2.

2.11.4.5 Exhaust outlets

The termination point for exhaust ducts discharging to the atmosphere shall not be less than the following:

- (a) Ducts conveying explosive or flammable vapours, fumes or dusts: 9 m from property line; 3 m from opening into the building; 2 m from exterior walls or roofs; 9 m from combustible walls or openings into the building which are in the direction of the exhaust discharge; 3 m above adjoining grade.
- (b) Other product conveying duct outlets: 3 m from property line; 1 m from exterior wall or roof; 3 m from openings into the buildings; 1 m above adjoining grade.
- (c) Domestic kitchen, bathroom, domestic clothes dryer exhaust duct outlets: 1 m from property line; 1 m from opening into the building.
- (d) Outlets for exhausts that exceed 80°C shall be in accordance with the relevant code.

2.11.4.6 Motors and fans

- (a) General: Motors and fans shall be sized to provide the required air movement. Motors in areas which contain flammable vapours and dusts shall be of a type approved for such environments. A manually operated remote control device shall be installed to shutoff fans or blowers in flammable vapour or dust system. Such control device shall be installed at an approved location.

Electrical equipment used in operations that generate explosive or flammable vapours, fumes or dusts shall be interlocked with the ventilation system so that the equipment cannot be operated unless the ventilation fans are in operation. Motors for fans used to convey flammable vapours or dusts shall be located outside the duct and shall be protected with approved shields and dust proofing. Motors and fans shall be accessible for servicing and maintenance.

- (b) Fans: Parts of fans in contact with explosive or flammable vapours, fumes or dusts shall be of nonferrous or non-sparking materials or their casing shall be lined or constructed of such material. When the size and hardness of materials passing through a fan could produce a spark, both the fan and the casing shall be of non-

sparking materials. When fans are required to be spark resistant, their bearings shall not be within the air stream, and all parts of the fan shall be grounded. Fans in systems handling materials that are likely to clog the blades, and fans in buffing or woodworking exhaust systems, shall be of the radial blade or tube axial type.

Equipment used to exhaust explosive or flammable vapours, fumes or dusts shall bear an identification plate stating the ventilation rate for which the system was designed.

Fans located in systems conveying corrosives shall be of materials that are resistant to the corrosion or shall be coated with corrosion resistant materials.

2.11.4.7 Exhaust systems of special areas

- (a) Motor Vehicle Operation: In areas where motor vehicles operate for a period of time exceeding 10 seconds, the ventilation return air shall be exhausted. In fuel dispensing areas, the bottom of the air inlet or exhaust opening shall be located a maximum of 450 mm above the floor.
- (b) Spray Painting and Dipping Rooms: Rooms booth for spray painting or dipping shall have mechanical exhaust systems which create a cross-sectional air velocity of 0.5 m/s. The system shall provide a uniform exhaust of air across the width and height of the room or booth. The exhaust system shall operate while spray painting or dipping is being done.
- (c) Motion Picture Projectors: Projectors equipped with an exhaust discharge shall be directly connected to a mechanical exhaust system. The exhaust system shall operate at an exhaust rate as indicated by the manufacturer's instructions.

Projectors without an exhaust shall have contaminants exhausted through a mechanical exhaust system. The exhaust rate for electric arc projectors shall be a minimum of 100 l/s per lamp. The exhaust rate for xenon projectors shall be a minimum of 150 l/s per lamp. The lamp and projection room exhaust systems, if combined or independent, shall not be interconnected with any other exhaust or return system within the building.

- (d) Dry Cleaning Equipment: Dry cleaning equipment shall be provided with an exhaust system capable of maintaining a minimum air velocity of 0.5 m/s across the face of the loading door.
- (e) LP gas Distribution Facilities: LP gas distribution facilities that are not provided with natural ventilation shall have a continuously operating exhaust system at the rate of 5 l/s per square metre of floor area. The bottom of air inlet and outlet openings shall not be more than 150 mm above the floor.

2.11.4.8 Exhaust system ducts

- (a) Construction: Ducts for exhaust systems shall be constructed of materials approved for the type of particulates conveyed and as per latest standard in this regard. Ducts shall be of substantial airtight construction and shall not have openings other than those required for operation and maintenance of the system.
- (b) Supports: Spacing of supports for ducts shall not exceed 3.7 m for 200 mm ducts and 6 m for larger ducts unless justified by the design. The design of supports shall assume that 50 percent of the duct is full of the particulate being conveyed.
- (c) Explosion Venting: Ducts conveying explosive dusts shall have explosion vents, openings protected by antflash-back swing valves or rupture diaphragms. Openings to relieve explosive forces shall be located outside the building.
- (d) Fire Protection: Fire suppression system shall be installed within ducts having a cross-sectional dimension exceeding 250 mm when the duct conveys flammable vapours or fumes.
- (e) Clearances: Ducts conveying flammable or explosive vapours, fumes or dusts shall have a clearance from combustibles of not less than 450 mm.

2.11.5 Kitchen Exhaust Equipment

2.11.5.1 Kitchen exhaust ducts

- (a) Materials: Kitchen exhaust ducts and plenums shall be constructed of at least 16 SWG steel or 18 SWG stainless steel sheet.

Joints and seams shall be made with a continuous liquid tight weld or braze made on the external surface of the duct system. A vibration isolator connector may be used, provided it consists of noncombustible packing in a metal sleeve joint of approved design. Duct bracing and supports shall be of noncombustible material securely attached to the structure and designed to carry gravity and lateral loads within the stress limitations of the Building Code. Bolts, screws, rivets and other mechanical fasteners shall not penetrate duct walls. Exhaust fan housings shall be constructed of steel.

Exception:

Kitchen exhaust ducts which are exclusively used for collecting and removing steam, vapour, heat or odour may be constructed as per provisions of Sec 2.4.1.

- (b) Corrosion Protection: Ducts exposed to the outside atmosphere or subject to a corrosive environment shall be protected against corrosion. Galvanization of metal parts, protection with noncorrosive paints and waterproof insulation are considered acceptable methods of protection.
- (c) Prevention of Grease Accumulation: Duct systems shall be so constructed and installed that grease cannot become pocketed in any portion thereof, and the system shall have a slope not less than 1 in 48 towards the hood or an approved grease reservoir. Where the horizontal ducts exceed 23 m in length the slope shall not be less than 1 in 12.
- (d) Air Velocity: The air velocity in the duct shall be a minimum of 7.62 m/s and a maximum of 12.7 m/s.
- (e) Cleanouts and Other Openings: Duct systems shall not have openings other than those required for proper operation and maintenance of the system. Any portion of such system having sections inaccessible from the duct entry or discharge shall be provided with adequate cleanout openings of approved construction spaced not more than 6 m apart. The cleanout shall be located on the side of the duct having a minimum opening dimension of 300 mm or the width of the duct when less than 300 mm.
- (f) Duct Enclosure: The duct which penetrates a ceiling, wall or floor shall be enclosed in a fire-resistant rated enclosure from the point of penetration in accordance with the Code. The duct enclosure shall be sealed around the duct at the point of penetration and vented to the exterior through weather-protected openings. The clearance between the duct enclosure and the duct shall be at least 75 mm and not more than 300 mm. Each duct enclosure shall contain only one exhaust duct. Approved fire rated access openings shall be provided at cleanout points.
- (g) Kitchen exhaust air flow rate shall be calculated based on the data provided in Table 8.2.8.

2.11.5.2 Kitchen exhaust hoods

- (a) A commercial exhaust hood shall be provided for each commercial cooking appliance.

Exceptions:

- (i) An appliance located within a dwelling unit and not used for commercial purposes.
- (ii) Completely enclosed ovens.
- (iii) Steam tables.
- (iv) Auxiliary cooking equipment that does not produce grease laden vapours, including toasters, coffee makers and egg cookers.

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- (b) Domestic cooking appliances used for commercial purposes shall be provided with a commercial exhaust hood. Domestic cooking appliances used for noncommercial purposes shall be provided with ventilation in accordance with Sec 2.8.
- (c) Hood Construction: The hood and other parts of the primary collection system shall be constructed of galvanized steel, stainless steel, copper or other material approved by the Building Official for the use intended. The minimum nominal thickness of the galvanized steel shall be 1.2 mm (No. 18 SWG). The minimum nominal thickness of stainless steel shall be 0.93 mm (No. 20 SWG). Hoods constructed of copper shall be of copper sheets weighing at least 7.33 kg/m². All external joints shall be welded liquid tight. Hoods shall be secured in place in noncombustible supports.
- (d) Interior Surface: The interior surfaces of the hood shall not have any areas that can accumulate grease.
- Exception: Grease collection systems under fitters and troughs on the perimeter of canopy hoods.
- (e) Canopy Hoods: Canopy hoods shall be designed to completely cover the cooking equipment. The edge of the hood shall extend a minimum horizontal distance of 150 mm beyond the edge of the cooking surface on all open sides and the vertical distance between the lip of the hood and the cooking surface shall not exceed 1.22 m.
- (f) Non-canopy Type Hoods: Hoods of the non-canopy type shall be located a maximum of 900 mm above the cooking surface. The edge of the hood shall be set back a maximum of 300 mm from the edge of the cooking surface.
- (g) Hood Exhaust: The hood exhaust shall create a draft from the cooking surface into the hood. Canopy hoods attached to wall shall exhaust a minimum of 500 l/s per m² of the hood area. Canopy hoods exposed on all sides shall exhaust a minimum of 750 l/s per m² of hood area. Hoods of the non-canopy type shall exhaust a minimum of 460 l/s per lineal metre of cooking surface.
- (h) Exhaust Outlet: An exhaust outlet within the hood shall be so located as to optimize the capture of particulate matter. Each outlet shall serve not more than a 3.7 m section of the hood.

Table 8.2.8: Design Exhaust Air Flow in litre per second per kW of the Kitchen Equipment

SI No.	Kitchen Equipment	Electricity based Equipment	Gas based Equipment
1	Cooking pot	8	12
2	Pressure cooker cabinet	5	-
3	Convection oven	10	-
4	Roasting oven (salamander)	33	33
5	Griddle	32	35
6	Frying pan	32	35
7	Deep fat fryer	28	-
8	Cooker/stove	32	35
9	Grill	50	61
10	Heated table/bath	30	-
11	Coffeemaker	3	-
12	Dish washer	17	-
13	Refrigeration equipment	60	-
14	Ceramic cooker/stove	25	-
15	Microwave oven	3	-
16	Pizza oven	15	-
17	Induction cooker/stove	20	-

2.11.5.3 Make up air

Make up air shall be supplied during the operation of the kitchen exhaust system. The amount of make-up air shall be approximately equal to the amount of the exhaust air. The make-up air shall be supplied in such a way as to avoid short circuiting and reducing the effectiveness of the exhaust system. Windows and doors shall not be used for the purpose of providing make up air.

2.11.5.4 Grease removal

The air exhausted in every commercial exhaust hood shall pass through approved grease filters or grease removal device designed for the specific purpose. Grease removal devices shall bear the label of an approved agency, and shall be installed in accordance with the manufacturer's instructions for the labeled equipment. All grease filters shall be accessible. Grease filters shall be installed at a minimum angle of 45° to the horizontal. The filters shall be arranged so as to capture and drain grease to a point of collection.

2.11.5.5 Motors, fans and safety devices

- (a) Motors and fans shall be of sufficient capacity to provide required air movement. Electrical equipment shall be approved for the class of use as provided in the Code. Motors and fans shall be accessible for servicing and maintenance. Motors of the exhaust fan shall not be installed within the ducts or under hoods.
- (b) Commercial exhaust system hoods and ducts shall have a minimum clearance to combustibles of 450 mm.
- (c) Fire Suppression System Required: All commercial cooking surfaces, kitchen exhaust systems, grease removal devices and hoods shall be protected with an approved automatic fire suppression system as per the Code.

2.12 Energy Conservation

2.12.1 General

Air-conditioning, heating and ventilation systems of all buildings shall be designed and installed for efficient use of energy as herein provided. Calculations of cooling and heating loads shall be based on data which lead to a system with optimum energy use.

General standards of comfort or particular environmental requirements within the building shall not be sacrificed in an endeavor to achieve low consumption of energy. For special applications, such as hospitals, laboratories, thermally sensitive equipment, computer rooms and manufacturing processes, the design concepts and parameters shall conform to the requirements of the application at minimum energy levels.

2.12.2 Design Parameters**2.12.2.1 Outdoor design conditions**

Unless specifically required, the outdoor design temperature shall be selected from columns of 2 percent value of Table 8.2.2 for cooling.

2.12.2.2 Indoor design conditions

Indoor design temperature shall not be less than 24°C for cooling unless otherwise required for specific application.

2.12.2.3 Humidity

The actual design relative humidity shall be selected from the range shown in Table 8.2.1 for the minimum total air-conditioning, heating and ventilation system energy use.

2.12.2.4 Shading co-efficient of glazing

- (a) The shading co-efficient (SC) and solar heat gain co-efficient (SHGC) shall be selected so as to reduce total heat influx through the glazing.
- (b) For any specific value of glazing to wall ratio (GWR) of any wall the shading co-efficient and solar heat gain co-efficient shall be based on the Table 8.2.9(a).

Table 8.2.9(a): SHGC and SC Values Based on GWR Value

GWR	SHGC	SC	GWR	SHGC	SC
10	0.85	0.98	60	0.33	0.38
20	0.60	0.69	70	0.31	0.36
30	0.50	0.57	80	0.30	0.34
40	0.40	0.46	90	0.27	0.31
50	0.35	0.40			

Notes:

- (i) Solar heat gain co-efficient (SHGC) = Shading Co-efficient (SC) x 0.87
- (ii) Glazing to wall ratio (GWR) = Total glazing area on any wall divided by total area of that wall including the glazing area.
- (iii) The Visible Light Transmittance (VLT) of the glazing element shall not be lower than 35 percent.

- (c) For buildings with external shading in the form of overhang and/or vertical fins a higher SHGC can be selected. The adjusted value of SHGC can be found out by using the following equation:

$$SHGC_{adj} = SHGC + A \text{ or } B \quad (8.2.1)$$

Values of A or B shall be taken from Table 8.2.9(b). Only higher of A & B shall be taken when both overhang and fins are used.

Table 8.2.9(b): Correction Factor for Overhang Shading and Vertical Fins

Value of overhang projection factor or vertical projection factor	Shading correction factor for overhang (A)	Shading correction factor for vertical fins (B)	Value of overhang projection factor or vertical projection factor	Shading correction factor for overhang (A)	Shading correction factor for vertical fins (B)
0.0	0.00	0.00	0.6	0.28	0.24
0.1	0.05	0.04	0.7	0.33	0.28
0.2	0.09	0.08	0.8	0.38	0.32
0.3	0.14	0.12	0.9	0.43	0.36
0.4	0.19	0.16	1 or higher	0.47	0.40
0.5	0.24	0.20			

Notes :

- (i) Overhead projection factor is the ratio between depth of overhang and height of window.
- (ii) Fin projection factor is the ratio between depth of fin and length of fin.
- (iii) Shading can only be counted if shade structure are placed over the window and glazing.

2.12.3 System Design

2.12.3.1 Load variation

Consideration shall be given to changes in building load and the system designed, so that maximum operational efficiency is maintained under part load conditions. The total system shall be separated into smaller zones having similar load requirements, so that each zone can be separately controlled to maintain optimum operating conditions by reducing wastage of energy.

2.12.3.2 Temperature of cooling media

The temperature of refrigerant, chilled water or brine circulated within the system shall be maintained at the level necessary to achieve the required output to match the prevailing load conditions with the minimum expenditure of energy.

2.12.3.3 Energy recovery

Energy recovery system shall be adopted, where possible.

2.12.4 Equipment and Control

2.12.4.1 General

Air-conditioning, heating and ventilation system shall be equipped with devices and controls to automatically control the capacity of the system when the building requirement reduces. The control system shall have devices to reduce energy use considering the effect of building energy storage.

2.12.4.2 Cooling with outdoor air

Each air handling system shall have facility to use up to and including 100 percent of the air handling system capacity for cooling with outdoor air automatically whenever the use of outdoor air will result in lower usage of energy than would be required under normal operation of the air handling system.

Exception:

Cooling with outdoor air is not required under any one or more of the following conditions:

- (a) Where the air handling system capacity is less than 2500 l/s or total cooling capacity is less than 40 kW.
- (b) Where the quality of outdoor air is so poor as to require extensive treatment of the air.
- (c) Where the need for humidification or dehumidification requires the use of more energy than is conserved by outdoor air cooling on an annual basis.
- (d) Where the use of outdoor air cooling would affect the operation of other systems so as to increase the overall energy consumption of the building.

2.12.4.3 Mechanical ventilation

Each mechanical ventilation system shall be equipped with a readily accessible means for either shutoff or volume reduction, and shutoff when ventilation is not required. Automatic or gravity dampers that close when the system is not operating shall be provided for outdoor air intakes and exhausts.

2.12.4.4 Maintenance

Heat exchange tubes shall be periodically cleaned to maintain its heat transfer characteristics. Maintenance of all equipment shall be periodically done to maintain its efficiency at satisfactory level.

2.12.4.5 Minimum equipment efficiencies

Cooling equipment shall meet or exceed the minimum efficiency requirements presented in Tables 8.2.10 and 8.2.11. Heating and cooling equipment not listed here shall comply with ASHRAE 90.1.

Table 8.2.10: Minimum Performance of Unitary Air Conditioning Equipment

Equipment Class and Size Category	Minimum COP	Minimum IPLV	Test Standard
Unitary air cooled air conditioner			
≥19 and <40 kW (≥5.4 and <11 tons)	3.08	--	ARI 210/240
≥40 to <70 kW (≥11 to <20 tons)	3.08	--	ARI 340/360
≥70 kW (≥20 tons)	2.93	2.99	ARI 340/360
Unitary water cooled air conditioner			
<19 kW (<5.4 tons)	4.10	--	ARI 210/240
≥19 and <40 kW (≥5.4 and <11 tons)	4.10	--	ARI 210/240
≥40 kW (≥11 tons)	3.22	3.02	ARI 210/240

Table 8.2.11: Minimum Performance of Chillers

Equipment Class and Size Category	Minimum COP	Minimum IPLV	Test Standard
Air cooled chiller, electrically operated			
<530 kW (<150 tons)	2.90	3.16	ARI 550/590
≥530 kW (≥150 tons)	3.05	3.32	ARI 550/590
Centrifugal water cooled chiller, electrically operated			
<530 kW (<150 tons)	5.80	6.09	ARI 550/590
≥530 and <1050 kW (≥150 and <300 tons)	5.80	6.17	ARI 550/590
≥1050 kW (≥300 tons)	6.30	6.61	ARI 550/590
Reciprocating compressor, water cooled chiller, electrically operated			
All Capacities	4.20	5.05	ARI 550/590
Rotary screw and scroll compressor, water cooled chiller, electrically operated			
<530 kW (<150 tons)	4.70	5.49	ARI 550/590
≥530 and <1050 kW (≥150 and <300 tons)	5.40	6.17	ARI 550/590
≥1050 kW (≥300 tons)	5.75	6.43	ARI 550/590

Equipment Class and Size Category	Minimum COP	Minimum IPLV	Test Standard
Air cooled absorption, single effect			
All Capacities	0.60	--	ARI 560
Water cooled absorption, single effect			
All Capacities	0.70	--	ARI 560
Water cooled absorption, double effect, indirect fired			
All Capacities	1.00	1.05	ARI 560
Water cooled absorption, double effect, direct fired			
All Capacities	1.00	1.00	ARI 560

2.12.4.6 Controls

- (a) All mechanical cooling and heating shall be controlled by a time clock that:
- Can start and stop the system under different schedules for three different day-types per week.
 - Is capable of retaining programming and time setting during a loss of power for a period of at least 10 hours, and
 - Includes an accessible manual override that allows temporary operation of the system for up to 2 hours.
- Exceptions: (i) Cooling systems < 28 kw (8 tons); (ii) Heating systems < 7 kw (2 tons)
- (b) All heating and cooling equipment shall be temperature controlled. Where a unit provides both heating and cooling, controls shall be capable of providing a temperature dead band of 3°C (5°F) within which the supply of heating and cooling energy to the zone is shut off or reduced to a minimum. Where separate heating and cooling equipment serve the same temperature zone, thermostats shall be interlocked to prevent simultaneous heating and cooling.
- (c) All cooling towers and closed fluid coolers shall have either two speed motors, pony motors, or variable speed drives controlling the fans.

2.12.5 System Balancing**2.12.5.1 General**

Construction documents shall require that all HVAC systems be balanced in accordance with generally accepted engineering standards.

Construction documents shall require that a written balance report be provided to the owner or the designated representative of the building owner for HVAC system servicing zones with a total conditioned area exceeding 500 m² (5000 ft²).

(a) Air System Balancing

Air systems shall be balanced in a manner to first minimize throttling losses. Then, for fans with fan system power greater than 0.75 kW (1.0 hp), fan speed shall be adjusted to meet design flow conditions.

(b) Hydronic System Balancing

Hydronic systems shall be proportionately balanced in a manner to first minimize throttling losses; then the pump impeller shall be trimmed or pump speed shall be adjusted to meet design flow conditions.

Exceptions:

- (i) Impellers need not be trimmed nor pump speed adjusted for pumps with pump motors of 7.5 kW (10 hp) or less.
- (ii) Impellers need not be trimmed when throttling results in no greater than 5% of the nameplate horse power draw, or 2.2 kW (3 hp), whichever is greater.

2.12.6 Condensers**2.12.6.1 Condenser locations**

Care shall be exercised in locating the condensers in such a manner that heat sink is free of interference from heat discharge by devices located in adjoining spaces and also does not interfere with such other systems installed nearby.

2.12.6.2 Treatment water for condensers

All high-rise buildings using centralized cooling water system shall use soft water for the condenser and chilled water-system.

2.12.7 Economizers

2.12.7.1 Air side economizer: Each individual cooling fan system that has a design supply capacity over 1200 l/s (2500 cfm) and a total mechanical cooling capacity over 22 kW (6.3 tons) shall include either:

- (a) An air economizer capable of modulating outside-air and return-air dampers to supply 100 percent of the design supply air quantity as outside-air; or

- (b) A water economizer capable of providing 100% of the expected system cooling load at outside air temperature of 10°C (50°F) dry-bulb/7.2°C (45°F) wet-bulb and below:

Exception:

- (i) Projects in the hot-dry and warm-humid climate zones are exempted.
- (ii) Individual ceiling mounted fan systems <3200 l/s (6500 cfm) are exempted.

2.12.7.2 Where required by Sec 2.12.7.1 economizers shall be capable of providing partial cooling even when additional mechanical cooling is required to meet the cooling load.

2.12.8 Variable Flow-Hydronic Systems

2.12.8.1 Chilled or hot-water systems shall be designed for variable fluid flow and shall be capable of reducing pump flow rates to no more than the larger of:

- (a) 50 percent of the design flow rate, or
- (b) The minimum flow required by the equipment manufacturer for proper operation of the chillers or boilers.

2.12.8.2 Water cooled air-conditioning or heat pump units with a circulation pump motor greater than or equal to 3.7 kW (5 hp) shall have two-way automatic isolation valves on each water cooled air-conditioning or heat pump unit that are interlocked with the compressor to shut off condenser water flow when the compressor is not operating.

2.12.8.3 Chilled water or condenser water systems that must comply with either Sec 2.12.8.1 or Sec 2.12.8.2 and that have pump motors greater than or equal to 3.7 kW (5 hp) shall be controlled by variable speed drives.

2.12.9 Variable Air Flow Systems

2.12.9.1 Air conditioning air distribution system shall be designed for variable air flow and shall be capable of reducing air flow by using any or all of the following devices:

- (a) Variable speed drives for controlling speeds of fan motors,
- (b) Variable air volume units/terminals,
- (c) Dampers for regulating air flow through ducts. Dampers may preferably be motor driven and modulating type.

2.12.9.2 Air handling units that has fan capacity greater than 7.5 kW shall be controlled by variable speed drives.

Exception: Kitchen ventilation fans.

2.13 Inspection, Testing and Commissioning

2.13.1 Inspection and Testing**2.13.1.1 General**

All air-conditioning, heating and ventilation system shall be inspected and tested by the Authority before the system is commissioned for normal operation. It should be ensured that these are carried out thoroughly and that all data and results are properly documented. It is recommended that whole inspection, testing and commissioning be done under the guidance and control of a single Authority.

2.13.1.2 Inspection

All machinery, equipment and other accessories of the air-conditioning, heating and ventilation system shall be inspected by the Authority to determine whether the system components and the system as a whole has been installed as per design and provisions of this Code; proper safety requirements have been maintained; and adequate fire protection measures have been taken.

Inspection shall also be carried out on structural supports, hangers, fastening devices, vibration isolators etc.

2.13.1.3 Testing

- (a) General: All machinery, equipment and other accessories shall be tested as per approved procedures. Tests shall be conducted to determine the strength capacity of any item and performance of any machine and equipment. All test data shall be properly documented.
- (b) Pressure Testing of Piping: All field installed refrigerant and hydronic piping system along with their valves and pipe fittings shall be tested at their approved test pressures to determine whether the piping system can withstand the test pressures.
- (c) Air Distribution System Testing: All ducting system shall be tested to determine whether the duct system has any leakage at test pressures. All air terminals and air dampers shall be tested for their flow characteristics.
- (d) Machinery and Equipment: Tests shall be conducted on machinery and equipment to determine whether these operate and function properly. All machinery and equipment shall also be tested for their electrical power consumption characteristics and overall performance. Before performance testing of the system all air distribution system and hydronic system shall be properly balanced by approved procedure.
- (e) Safety Devices and Controls: Tests shall be carried out to determine whether the safety devices and controls function properly.
- (f) All air filters shall be tested in accordance with the latest standard.

2.13.2 Commissioning

If the Authority becomes satisfied regarding satisfactory installation and performance of the air-conditioning, heating and ventilation system after testing, the system shall be commissioned following approved procedure. Before complete commissioning, all air distribution systems and hydronic systems shall be properly balanced and all the controls and their sensors shall be properly adjusted.

2.14 Operation and Maintenance

2.14.1 General

The owner of the building where the air-conditioning, heating and ventilation system is installed, shall follow a properly designed operation procedure and maintenance schedule.

2.14.2 Operation

A well sequenced operation procedure shall be followed to ensure effective operation of the air-conditioning, heating and ventilation system, safety from hazard to personnel and property. Operation procedure shall take account for saving in energy use.

All operational data of all the machinery and equipment shall be properly recorded for determination of performance of the machinery, equipment and the system. These data shall be properly preserved for future reference for maintenance purposes.

2.14.3 Maintenance

A well designed maintenance program for the air-conditioning, heating and ventilation system shall be implemented in order to achieve the following:

- (a) Optimum reliability and continuity of service.
- (b) Extended longevity and economic life.
- (c) Functional effectiveness, whereby the intended performance of mechanical equipment and system can be fully attained.
- (d) Minimum operating cost, attendant requirements, servicing and repairs.
- (e) Safety from hazard to personnel and property.

Maintenance program and procedure shall comply with the instructions of machinery/equipment manufacturers in this regard.

PART VIII
Chapter 3
Building Acoustics

3.1 Purpose

The purpose of this Chapter is to provide codes, recommendations and guidelines for fulfilling acoustical requirements in buildings.

3.2 Scope

This Chapter specifies planning, design and construction codes, recommendations and guidelines on spatial, architectural and technical aspects of acoustics within or outside buildings to ensure acoustical performance, comfort and safety. Planning and design aspects are discussed generally and also particularly for buildings with different occupancies.

3.3 Terminology

This Section provides meanings and definitions of terms used in and applicable to this Chapter of the Code. The terms are arranged in alphabetical order. In case of any contradiction between a meaning or a definition given in this Section and that in any other part of the Code, the meaning or definition specified in this Section shall govern for interpretation of the provisions of this Section.

BALANCED NOISE CRITERIA (NCB) CURVES	Balanced Noise Criteria (NCB) curves are used to specify acceptable background noise levels in occupied spaces (see Appendix F).
BEL	See sound pressure level.
CYCLE	See Frequency.
dBA	A sound pressure level measurement, when the signal has been weighted with a frequency response of the A curve. The dBA curve approximates the human ear and is therefore used most in building acoustics.
DECIBEL (dB)	See sound pressure level.
DIRECT SOUND	Sound that travels directly from a source to the listener or receiver. In a room, the sequence of arrivals is the direct sound first, followed by sound reflected from room surfaces.
ECHO	Echo is a reflection of a sound wave back to its source in sufficient strength and with a sufficient time lag to be separately distinguished. Usually, a time lag of at least 50 to 80 ms is required for hearing discrete echoes.

EFFECTIVE PERCEIVED NOISE LEVEL IN DECIBEL (EPN dB)	The number for rating the noise of an individual aircraft flying overhead is the Effective Perceived Noise Level in decibels (unit, <i>EPN dB</i>). This value takes into account the subjectively annoying effects of the noise including pure tones and duration. In principle, it is a kind of time-integrated loudness level.
FLUTTER ECHO	A quick succession of echoes; it may be present as a disturbing phenomenon in small rooms or between a pair of parallel reflectors. If the time between echoes is greater than about 30 to 50 ms, the periodicity is audible as a distinct flutter.
FREQUENCY	The frequency of sound is the number of vibrations per second of the molecules of air, generated by the vibrating body. One complete movement to and fro of the vibrating body is referred to as a 'cycle'. Frequency is expressed as the number of cycles per second (cps); it is also referred to its unit as <i>Hertz (Hz)</i> .
IMPACT ISOLATION CLASS (IIC)	Impact Isolation Class (IIC) is a single-number impact isolation rating for floor construction. Tests are made with a standard tapping machine and noise level measured in 1/3-octave bands. These are plotted and compared to a standard contour.
INTENSITY	Intensity at a point is the average rate at which sound energy is transmitted through a unit area around the point and perpendicular to the direction of propagation of sound. It is also known as sound intensity. Its units is $W m^{-2}$.
L_{10}	In environmental noise assessment, the A-weighted noise level (unit dBA), with fast (F) time weighting, that is exceeded by 10 percent of sated time period is known as L_{10} .
$L_{Aeq,T}$	Expression for Equivalent Continuous A-Weighted Sound Pressure Level for airborne sounds that are non-stationary with respect to time. It is formed by applying A-weighting to the original signal before squaring and averaging. Also known as equivalent continuous sound level.
LOUDNESS	Loudness is the sensation produced in the human ear and depends on the intensity and frequency of sound. The unit of loudness level is <i>phon</i> .

NOISE	Noise is defined as unwanted sound. Noise conditions vary from time to time and a noise which may not be objectionable during the day may be increased in annoying proportions in the silence of the night, when quiet conditions are essential.
NOISE EXPOSURE FORECAST (NEF)	Noise exposure forecast at any location is the summation of the noise levels in Effective Perceived Noise Level (unit <i>EPN dBA</i>) from all aircraft types, on all runways, suitably weighted for the number of operations during day time and night time.
NOISE MAP	A noise map is a graphic representation of the sound level distribution existing in a given region, for a defined period.
NOISE REDUCTION (NR)	Noise Reduction (<i>NR</i>) is a general term for specifying sound insulation between rooms. It is more general than Transmission Loss (<i>TL</i>). If all boundary surfaces in the receiving room are completely absorbent, the <i>NR</i> will exceed the <i>TL</i> by about 5 dB, i.e. $NR = TL + 5dB$.
PERCENTAGE SYLLABLE ARTICULATION (PSA)	The percentage of meaningless syllables correctly written by listeners is called Percentage Syllable Articulation (PSA).
REVERBERATION	The prolongation of sound, as a result of successive reflections in an enclosed space, when the source of the sound has stopped, is called reverberation.
REVERBERATION TIME (RT)	The reverberation time of a room is defined as the time required for the sound pressure level in a room to decrease by 60 dB after the sound is stopped, and is calculated by the formula $RT = \frac{0.16V}{A + xV}$ Where, RT= reverberation time, s V= room volume, m³ A= total room absorption, m² <i>sabin</i> x= air absorption coefficient
SIGNAL-TO-NOISE	Signal-to-Noise Ratio (SNR) defined as the power ratio

RATIO (SNR)

between a signal (meaningful information) and the background noise (unwanted signal), which can be expressed as

$$\text{SNR} = (P_{\text{signal}}/P_{\text{noise}})$$

where, P is average power

SNR can be obtained by calculating the square of amplitude ratio:

$$\text{SNR} = (P_{\text{signal}}/P_{\text{noise}}) = (A_{\text{signal}}/A_{\text{noise}})^2$$

where, A is root mean square (RMS) amplitude

In decibel, the SNR is defined as

$$\text{SNR}_{\text{dB}} = 10 \log_{10} (P_{\text{signal}}/P_{\text{noise}}) = P_{\text{signal,dB}} - P_{\text{noise,dB}}$$

which might be equivalently expressed in amplitude ratio as

$$\text{SNR}_{\text{dB}} = 10 \log_{10} (A_{\text{signal}}/A_{\text{noise}})^2 = 20 \log_{10} (A_{\text{signal}}/A_{\text{noise}})$$

SOUND FOCUS
AND DEAD SPOT

When a sound wave is reflected by a concave surface, large enough compared to the wavelength, it concentrates on a spot where sound pressure rises excessively. This is called a 'sound focus'. As a consequence, sounds become weak and inaudible at some other spots, called 'dead spots'.

SOUND PRESSURE
LEVEL (SPL)

Sound Pressure Level or Sound Intensity Level is measured in terms of the unit *bel* (B), which is defined as the logarithm of the ratio of the sound pressure to the minimum sound pressure audible to the average human ear. The unit *decibel* (dB) is one-tenth of a *bel* (B). Thus,

$$\text{Sound Pressure Level} = \log_{10} \frac{1}{I_0} \text{ bcls} = 10 \log_{10} \frac{1}{I_0} \text{ dccibcls}$$

where,

I = Sound Pressure in watt cm^2 , and

I_0 = Sound Pressure audible to the average human ear taken as 10^{-16} watt/ cm^2 .

SOUND TRANSMISSION CLASS (STC)	To avoid the misleading nature of an average transmission loss (TL) value and to provide a reliable single figure rating for comparing partitions, a different procedure for single figure rating, called Sound Transmission Class (STC) rating, of a partition is determined by comparing the 16 frequency TL curve with a standard reference contour, the sound transmission class contour. STC ratings of some common walls and floors are given in Appendix E.
SPEECH INTELLIGIBILITY	The percentage of correctly received phrases is called Speech Intelligibility.
TRANSMISSION LOSS	Transmission loss (TL) of a partition is a measure of its sound insulation. It is equal to the number of decibels by which sound energy is reduced in passing through the structure. Units <i>dB</i> .
WAVELENGTH	The wavelength of sound is the distance over which a complete cycle occurs. It can be found by measuring distance between the centres of compression of the sound waves. It is dependent upon the frequency of the sound.

3.4 Building Acoustics: General Considerations and Provisions

Generalised considerations and provisions for planning and design of building are furnished in this Section.

3.4.1 Classifications of Building Acoustics

3.4.1.1 Considering diversity of desired objectives and salient design features, building acoustics can be broadly classified as,

- (a) Acoustics for Speech
- (b) Acoustics for Music
- (c) Acoustics for Multipurpose

Table 8.3.1 shows classifications of acoustics with brief description and examples of spaces involved.

Table 8.3.1: Classifications of Building Acoustics, Brief Description and Examples of Spaces Involved

Classifications	Brief Description	Examples of Spaces
(a) Acoustics for Speech	Relates to speech with foremost objectives of intelligibility. A space should have relatively lower reverberation time for speech. Generally, it covers narrow range of frequency spectra in lower-mid level (about 170 to 4,000 Hz, for an average dynamic range of 42 dBA).	Classroom, lecture hall, conference hall, recital hall, assembly hall, courtroom, auditorium for speech etc.
(b) Acoustics for Music	Involves music with prime objectives of liveliness or reverberance, intimacy, fullness, clarity, warmth, brilliance, texture, blend and ensemble. Music may include instrumental and vocal melody, or either of the two. A space requires relatively higher reverberation time for music. Generally, it involves broad range of frequency spectra (about 50 to 8,500 Hz, for an average dynamic range of about 75 dBA).	Music practice booth, rehearsal room, band room, listening booth, orchestra, concert hall, symphony hall, cathedral etc.
(c) Acoustics for Multipurpose	Includes both speech and music acoustics to fulfil objectives of the both at a rationally compromised level. Acoustics design of a multipurpose space is quite challenging as the design objectives and measures vary remarkably for speech and music. For example, there is a significant variation in desired reverberation times of a space for speech and music.	Multipurpose hall, cinema, theatre, opera house, mosque (for speech and melodious recitation), church, temple etc.

3.4.1.2 A building or a building complex is usually a group of spaces or rooms intended for various functions. Those spaces may require involvement of different types of acoustics as stated in Table 8.3.1. For example, a school has spaces for speech (e.g., classroom), music (e.g., music room) and multipurpose (e.g., auditorium). Thus, a building or a building complex should not be generally classified as a whole for a particular type of acoustics, rather its spaces or rooms shall be classified individually and appropriate acoustical design shall be considered accordingly.

3.4.1.3 Spaces or rooms of a building or a building complex, if those even have different types of acoustical requirements, shall be designed in such a way, so that those can coexist and work as a whole.

3.4.2 Acoustical Planning and Design Targets

3.4.2.1 A space, involving either of the acoustical types stated in Sec 3.4.1, must achieve few design targets. Some of these important design targets are mentioned below:

- (a) Noise exceeding allowable limit should be controlled
- (b) Speech intelligibility should be satisfactory
- (c) Music should have liveliness, intimacy, fullness, clarity, warmth etc.
- (d) The desired sound level must be optimum to be heard properly
- (e) Diffusion of sound throughout the whole space
- (f) There should be no defects such as echoes, flutter echoes etc.

3.4.2.2 Necessary planning and design measures shall be taken for achieving these targets to optimum levels or standards as dictated in this Code.

3.4.3 Factors Affecting Acoustical Planning and Design

3.4.3.1 Among many, following are the most significant factors affecting acoustical planning and design; noise, reverberation time, sound level and diffusion of sound.

3.4.3.2 For various types of building acoustics, as stated in Sec 3.4.1, the effects of these factors might be different. These factors are dependent on different conditions, like noise and sound level, room volume, building materials, surface materials, sound levels, room geometry etc.

3.4.4 General Considerations and Provisions for Planning, Design, Assessment and Construction

3.4.4.1 In Appendix F, a flow diagram summarises activities required for planning, design, assessments and construction related to building acoustics.

3.4.4.2 Acoustical planning and design, including all parts and details, shall be performed during design phase of any project and must comply with standards and codes as dictated in this Code.

3.4.4.3 During planning and design phase, the expected results for acoustical performance of a space or a room or building, as dictated in different Sections of this Chapter, shall be precisely analysed and assessed through standard practice, for example, precise computational methods based on computer analysis, simulation and prediction techniques.

3.4.4.4 Acoustical planning and design targets and expected results shall be clearly specified and documented as a part of the design proposal.

3.4.4.5 Acoustical planning and design measures shall be compatible with requirements of other environmental factors including natural light, ventilation and heat for working in an overall synergy.

3.4.4.6 Acoustical planning and design measures shall be congenial to other design parameters including function, structure and aesthetics for an overall harmony in design.

3.4.4.7 The proposal for acoustical design, materials, devices, supporting structures and construction methods shall be safe for health during construction and post-construction occupancy.

3.4.4.8 Acoustical materials, devices and supporting structures shall be safe in case of disasters including earthquake and fire.

3.4.4.9 The acoustical design measures and materials shall be reasonably energy efficient and compatible with Green Building practice.

3.4.4.10 Acoustical materials shall be eco-friendly, recyclable and should require minimum maintenance. Those shall be compliant to sustainable acoustics in particular and sustainable building practice in general.

3.4.4.11 It is recommended to conduct peer supervisions and periodic assessments at different phases of construction process to rectify any drawback at its initial stage.

3.4.4.12 Post-construction and post-occupancy assessments shall be conducted and findings shall be compared with expected results. Assessment shall include instrumental measurements and opinion survey of occupants. If any discrepancy is found, the space shall be modified until it reaches reasonably close to the expected result.

3.4.4.13 Preceding provisions shall be applicable for modifications of a space to eliminate acoustical faults, retrofitting a space for acoustical performance or any other acoustical design and construction activities.

3.4.4.14 Form G, Checklist: Acoustical Planning, Design and Post-occupancy Assessments, as in Appendix G, shall be filled in and signed by the acoustical consultant for each acoustical space or room or building of any project.

3.5 Planning and Design For Noise Control

3.5.1 Types of Noise

3.5.1.1 Depending on location of source, noise might be of two types:

(a) Outdoor Noise.

Following are some common sources of outdoor noise:

- (i) Traffic noise generated from air traffic, road traffic, rail traffic etc. (see Appendix H)
- (ii) Noise from zones and buildings within built-up areas, machinery, appliances, construction activity, loudspeakers, people, animals etc.

(b) Indoor Noise.

Following are some common source of indoor noise:

- (i) Household appliances, machinery, footsteps on floor, air conditioner duct etc.
- (ii) Activities performed by occupants, like people, pets etc.

3.4.2.2 Basing on transmission path, noise can be classified as

(a) Airborne Noise

Examples noise from appliances, car horn, telephone ring etc.

(b) Structure-borne Noise

Example footsteps, slamming of door, furniture movement, vibrating mechanical equipment etc.

3.5.2 Design Sequence for Noise Control

3.5.1.1 In order to achieve noise control effectively, measures should be taken in the following order:

- (a) Suppression of noise generation at its source
- (b) Layout planning
- (c) Insulation design
- (d) Absorption design

3.5.3 Planning and Design for Outdoor Noise Control

3.5.3.1 Planning to control outdoor noise is an integral part of country and town planning ranging from regional to detailed zoning and three dimensional layouts of built form and traffic routes.

3.5.3.2 Noise causes more disturbances to people at rest than those at work. For this reason, outdoor noise levels in various zones or areas should be considered in planning and design with respect to critical hours of space occupancy (see Table 8.H.1, Appendix H).

3.5.3.3 Planning and design of buildings shall consider all sources of noise mentioned in Sec 3.5.1 and keep provisions to control those from transmitting in and around buildings. For example, the orientation of buildings might be decided in way to reduce the noise disturbance from noisy neighbourhood.

3.5.3.4 A noise survey shall be conducted and a Noise Map shall be prepared to identify source, type, intensity, frequency and other parameters of noise in and around the site of any specific project. Noise levels should be measured for pick and off pick hours of both working and holidays, and also for 'Day Time' and 'Night Time' as defined in 'Noise Pollution (Control) Rules 2006' and its subsequent amendments by the Government of the People's Republic of Bangladesh (see Table 8.H.1, Appendix H). The noise levels shall be analysed statistically for value of L_{10} , $L_{Aeq,T}$ etc.

3.5.3.5 A Noise Map shall be used to examine compliance to the permissible upper limit of noise levels set for different land use zones in the 'Noise Pollution (Control) Rules 2006' and its subsequent amendments by the Government of the People's Republic of Bangladesh (see Table 8.H.1, Appendix H). As references, intensity levels of some common noise are shown in (see Table 8.H.3, Appendix H).

3.5.3.6 The planning should be undertaken in such a manner that the noise can be kept at a distance. Quiet zones and residential zones should be placed with adequate setback from noise sources, like airports, highways, railway lines and factories. It might be useful to note that doubling the distance drops the sound pressure level by about 6 dBA.

3.5.3.7 Buildings (or parts of buildings) which are considered to be especially susceptible to noise, including hospitals, research laboratories, recording studios or the like, should not be sited near sources of noise.

3.5.3.8 It might be a preferable option to place a noise tolerant buffer zone, developing green belt, public gardens etc. between a noisy zone and a quiet zone.

3.5.3.9 Noise barriers might be provided by placing buildings and occupancies less susceptible to noise between the source and the more susceptible ones. Purpose built noise barriers made of bricks, concrete, fibreglass, fibre reinforced plastic or other materials can also be used to protect buildings from noise.

3.5.3.10 If noise barriers (as stated in Sec 3.5.3.9) is neither attainable nor adequate, a building itself should have all necessary measures to protect itself against outside noise. The following might be some options:

- (a) In zoning of spaces, noise tolerant spaces might be placed near a noise source, while placing less-tolerant spaces at a distance.
- (b) External walls or partitions should have appropriate Sound Transmission Class (STC) to reduce external noise to the acceptable indoor background noise levels (Tables 8.3.3 to 8.3.6, 8.E.1 and 8.E.2, Appendix E)
- (c) Preferably, external walls near source of noise should not have any operable window. However, to meet the demand of natural light, fixed windows allowing only light might be placed with proper noise insulation measures.
- (d) If need for operable windows allowing natural light and ventilation are inevitable in external walls near source of noise, special measures should be taken for restricting noise while allowing light and ventilation. Acoustic louvers, active noise cancellation devices etc. are examples of these types of special measures.
- (e) If natural ventilation is required but natural light is not required, ventilation ducts or chutes with lining of acoustic absorbers might be designed in a manner to absorb noise while air flows through.

3.5.3.11 Following special provisions shall be applicable for air traffic noise:

- (a) No building for human occupancy shall preferably be constructed, where NEF value due to air traffic noise exceeds 40 EPN dBA. As a reference, typical noise levels of some aircraft types are shown in Table 8.H.2, Appendix H.
- (b) Educational institutions, hospitals, auditoriums etc. shall preferably be located at places where the value of NEF is less than 25 EPN dBA.
- (c) In areas exposed to less than 90 EPN dBA, all of the windows shall be closed and properly sealed, having double glazing, in order to provide an acceptable interior noise environment.
- (d) Industrial and commercial activities generating high interior noise environments might be located in areas exposed to noise levels greater than 90 EPN dBA.
- (e) In airport areas of highest noise levels, sparsely manned installations like sewage disposal plants, utility substations and similar other facilities might be located.

3.5.3.12 Following provisions shall be applicable for road traffic noise:

- (a) For road traffic noise level, the value of L_{10} shall be limited to a maximum of 70 dBA for zoning and planning new buildings in urban areas, while dwellings are proposed to have sealed windows.
- (b) The maximum permissible upper limit of L_{10} shall be reduced to 60 dBA when the dwellings are proposed to have open windows.
- (c) Major new residential developments shall preferably be located in areas with L_{10} levels substantially lower than those specified above.
- (d) Where L_{10} is greater than 70 dBA, design solutions such as barrier blocks, noise buffers or purpose built noise barriers shall be utilized in order to reduce noise levels at least to that level.
- (e) Through traffic roads shall preferably be excluded from quiet and residential zones to avoid excessive traffic noise.
- (f) In the neighbourhood of residential, educational, institutional and health care buildings, legislative control shall be exercised for road noise particularly from vehicles as dictated in 'Noise Pollution (Control) Rules 2006' and its subsequent amendments by the Government of the People's Republic of Bangladesh.

3.5.3.13 Following special provisions shall be applicable for rail traffic noise:

- (a) No residential or public building, except for the railway station and its ancillary structures, shall preferably be connected to the railway lines.
- (b) Mercantile or commercial buildings should not abut the railway lines or the marshalling yards. Only planned industrial zones may be located beside the railway tracks.
- (c) In order to reduce the high noise levels, produced at the arrival and departure of trains, platforms in railway stations shall be treated with sound absorbing materials particularly on the ceiling.
- (d) The main platform floor shall be separated from the station building with a minimum gap of 50 mm so that the ground or structure-borne vibrations are not transmitted to the building.
- (e) Windows and other openings shall preferably be placed as less as possible in the facade along the railway tracks.
- (f) Greenbelts, landscaping or any other form of barrier might be developed along the railway lines.

3.5.3.14 Construction noise shall be controlled according to the 'Noise Pollution (Control) Rules 2006' and its subsequent amendments by the Government of the People's Republic of Bangladesh.

3.5.4 Planning and Design for Indoor Noise Control

3.5.4.1 The allowable upper limits of indoor background noise levels (in dBA) are as shown in Table 8.3.2 and Figure 8.D.1 in Appendix D. Design shall comply with recommended range of Balanced Noise Criteria (NCB) Curve for different types of activity.

Table 8.3.2: Allowable Upper Limit of Indoor Background Noise Levels and Recommended Range of NCB Curves

Type of Space	dBA	NCB Curve
Broadcast and recording studios (distant microphone used)	18	10
Concert halls, opera houses, and recital halls	18-23	10-15
Large theatres and auditoriums, mosques, temples, churches and other prayer spaces	<28	<20
Television and recording studio (close microphone used)	<33	<25
Small theatres, auditoriums, music, rehearsal rooms, large meeting and conference rooms	<38	<30
Bedrooms, hospitals, hotels, residences, apartments, etc.	33-48	25-40
Classrooms, libraries, small offices and conference rooms. Living rooms, and drawing rooms in dwellings	38-48	30-40
Large offices, receptions, retail shops and stores, cafeterias, restaurants, indoor stadiums, gymnasium, large seating-capacity spaces with speech amplification	43-53	35-45
Lobbies, laboratory, drafting rooms, and general offices	48-58	40-50
Kitchens, laundries, computer and maintenance shops	53-63	45-55
Shops, garages, etc. (for just acceptable telephone conversation)	58-68	50-60
For work spaces where speech is not required	63-78	55-70

3.5.4.2 Noise generated from within a building shall not be transmitted to neighbourhood at a noise level higher than the allowable upper limit set for that zone (see Table 8.H.1, Appendix H).

3.5.4.3 Buildings, in which there are some sources of noise, shall have buffers separating the noise producing area from the other areas. The less vulnerable areas of the building may be planned to act as noise buffers.

3.5.4.4 In the assessment of indoor noise levels, direct sound shall be separated from reverberant sound.

3.5.4.5 The reverberant sound transmitted from one room to another shall be cut down by employing suitable sound absorption materials and by structural and non-structural partitions.

3.5.5 Sound Insulation

3.5.5.1 The recommended sound insulation criteria are classified in some Grades. The STC value for airborne sound insulation is graded as stated below (see also, Figure 8.D.2, Appendix D):

- (a) Grade I STC = 55 Apply mainly to fully residential, quiet rural and suburban areas and in certain luxury apartment buildings.
- (b) Grade II STC = 52 Apply to residential spaces in relatively noisy environments, typical of urban and suburban areas.
- (c) Grade III STC = 48 Express minimal requirements applicable to very noisy locations, such as commercial or business areas, like shop houses with dwelling units on the upper floors, or downtown areas.

3.5.5.2 Transmission of sound should be controlled with appropriate material, assembly of building elements. Typical STC rating for different types of building element, like stud partitions, masonry walls, doors, windows and interior partitions are shown in Table 8.E.1, Appendix E.

3.5.5.3 Recommended STC for partitions for specific occupancies are shown in Table 8.E.2, Appendix E.

3.5.6 Control of Structure-borne Impact Noise

3.5.6.1 Impact noise problems can be controlled in following ways:

- (a) Preventing or minimising the impact by cushioning the impact with resilient materials, like floor tiles of rubber and cork, carpeting on pads with desired Impact Isolation Class (IIC). Criteria for airborne and impact sound insulation of floor-ceiling assemblies between dwelling unit, Tables 8.3.3 and 8.3.4.

- (b) Floating the floor for isolating the impacted floor from the structural floor by a resilient element is extremely effective. This element can be rubber or mineral wood pads, blankets or special spring metal sleepers.
- (c) Suspending the ceiling and using an absorber in the cavity.
- (d) Isolating all rigid structures, such as pipes, and caulking penetrations with resilient sealant.

Table 8.3.3: Airborne Sound Insulation of Partitions between Dwelling Units

Apt. A		Apt. B	Grade II STC	Apt. A		Apt. B	Grade II STC
Bedroom	to	Bedroom	52	Bathroom	to	Living room	54
Living room	to	Bedroom ^a	54	Corridor	to	Living room ^{a,c,d}	52
Kitchen ^b	to	Bedroom ^a	55	Kitchen	to	Kitchen ^e	50
Bathroom	to	Bedroom ^a	56	Bathroom	to	Kitchen	52
Corridor	to	Bedroom ^{a,c}	52	Corridor	to	Kitchen ^{a,c,d}	52
Living room	to	Living room	52	Bathroom	to	Bathroom	50
Kitchen ^b	to	Living room ^a	52	Corridor	to	Bathroom ^{a,c}	48

Notes:

For Grade I, add 3 points; for Grade III, subtract 4 points.

^a Whenever a partition wall may serve to separate several functional spaces, the highest criterion must prevail.

^b Or dining or family or recreation room.

^c It is assumed that there is no entrance door leading from the corridor to the living unit.

^d Criterion applies to the partition. Doors in corridor partition must have the rating of partition, not vice versa.

^e Double wall construction is recommended to minimise kitchen impact noise.

Table 8.3.4: Airborne and Impact Sound Insulation of Floor-Ceiling between Dwelling Units

Apt. A	Apt. B	Grade II STC	Grade II IIC	Apt. A	Apt. B	Grade II STC	Grade II IIC
Bedroom	above Bedroom	52	52	Living room	above Kitchen ^{c,e}	52	52
Living room	above Bedroom ^a	54	57	Kitchen	above Kitchen ^c	50	52
Kitchen ^b	above Bedroom ^{a,c}	55	62	Bathroom	above Kitchen ^{a,c}	52	52
Family room	above Bedroom ^{a,d}	56	62	Family room	above Kitchen ^{a,c,d}	52	58
Corridor	above Bedroom ^a	52	62	Corridor	above Kitchen ^{a,c}	48	52
Bedroom	above Living room ^e	54	52	Bedroom	above Family room ^e	56	48
Living room	above Living room	52	52	Living room	above Family room ^e	54	50
Kitchen	above Living room ^{a,c}	52	57	Kitchen	above Family room ^e	52	52
Family room	above Living room ^{a,d}	54	60	Bathroom	above Bathroom ^c	50	50
Corridor	above Living room ^a	52	57	Corridor	above Corridor	48	48
Bedroom	above Kitchen ^{c,e}	55	50				

Notes:

For Grade I, add 3 points; for Grade III, subtract 4 points.

- ^a Arrangement requires greater impact sound insulation than inverse, where a sensitive area is above less sensitive area.
- ^b Or dining or family or recreation room.
- ^c It is assumed that the plumbing fixtures, appliances and piping are insulated with proper vibration isolation.
- ^d The airborne STC criteria in this table apply as well to vertical partitions between these two spaces.
- ^e This arrangement require equivalent airborne sound insulation than the converse.

3.5.7 Control of Electro-Mechanical System Noise

3.5.7.1 Mechanical noise is generated from mechanical devices like air-conditioning and air-handling systems, lifts, escalators, pumps, electric generators etc.

3.5.7.2 Mechanical noise problems can be controlled in following ways:

- (a) Reducing the vibration of electro-mechanical equipment by damping and isolation.
- (b) Reducing the airborne noise by decoupling the vibration from efficient radiating sources.
- (c) Decoupling the vibrating source from the structure.
- (d) In air-conditioning duct system, smooth transitions at changes of duct size, large radius bends, lining with absorbing materials etc. are effective measures.
- (e) Active noise cancellation technique, by producing a synthesised signal exactly out-of-phase with the original noise signal to make the resultant signal effectively zero, might be applied in special cases.

3.5.8 Occupational Noise Exposure

3.5.8.1 Protection against the effects of noise exposure shall be provided when the sound level exceeds those shown in Table 8.3.5.

3.5.8.2 Exposure to impulsive or impact noise should not exceed 140 dBA peak sound level.

Table 8.3.5: Permissible Noise Exposure

Sound Level dBA (slow response)	Duration per Day Hour - Minute	Sound Level dBA (slow response)	Duration per Day Hour - Minute
85	16-00	98	2-50
86	13-56	99	2-15
87	12-08	100	2-00
88	10-34	101	1-44
89	9-11	102	1-31
90	8-00	103	1-19
91	6-58	104	1-09
92	6-04	105	1-00
93	5-17	106	0-52
94	4-36	107	0-46
95	4-00	108	0-40
96	3-29	109	0-34
97	3-02	110	0-30

Notes:

- (i) The sound level should be measured on *A* scale at slow response.
- (ii) When the daily noise exposure is composed of two or more periods of noise exposure of different levels, their combined effect should be considered, rather than the individual effect of each. If the sum of the following fractions: $C_1/T_1 + C_2/T_2 + \dots + C_n/T_n$ exceeds unity, then, the mixed exposure should be considered to exceed the limit value. C_n indicates the total time of exposure permitted at that time.

3.6 Reverberation Time, Sound Pressure Level and Diffusion of Sound

3.6.1 General Considerations

- (a) For an overall performing, comfortable and safe acoustical environment, along with the issues of noise, other significant aspects of acoustics should be considered. This shall include sound pressure level, reverberation time and diffusion of sound.
- (b) Speech intelligibility is a significant parameter to achieve satisfactory acoustical design. Percentage Syllable Articulation (PSA) is an index for assessing speech intelligibility. PSA can be expressed as

$$PSA = 96 k_i k_r k_n k_s (\%) \quad (\text{for English Language})$$

$$PSA = 93 k_i k_r k_n k_s (\%) \quad (\text{for Bangla Language})$$

Where,

k_i , k_r , k_n and k_s are the coefficient for average speech level, Reverberation Time, Noise level/ Speech level and room shape, respectively (see Figure 8.I.1, Appendix I).

- (c) For a PSA of 82%, almost a perfect Speech Intelligibility (nearly 100%) can be achieved. However, in reality, there are some background noise (>20 dBA) and reverberation time in different spaces, causing lower PSA. The minimum admissible PSA should be 75% for a satisfactory Speech Intelligibility.

3.6.2 Reverberation Time

Spaces for various uses should be designed for recommended optimum reverberation time to achieve a level of intelligibility and liveliness (see Figure 8.3.1).

3.6.3 Sound Pressure Level

- (a) In a space with a low background noise (<20 dBA) and a minimum Reverberation Time (close to 0.0 s), a maximum Percentage Syllable Articulation (PSA), and thus Speech Intelligibility can be achieved at sound pressure level of speech ranging from 60 dBA to 70 dBA (see Figure 8.I.1, Appendix I).
- (b) For speech halls with higher background noise (>20 dBA), the recommended Signal-to-Noise Ratio (SNR) is +15 dBA for children and at least +6 dBA for adults.

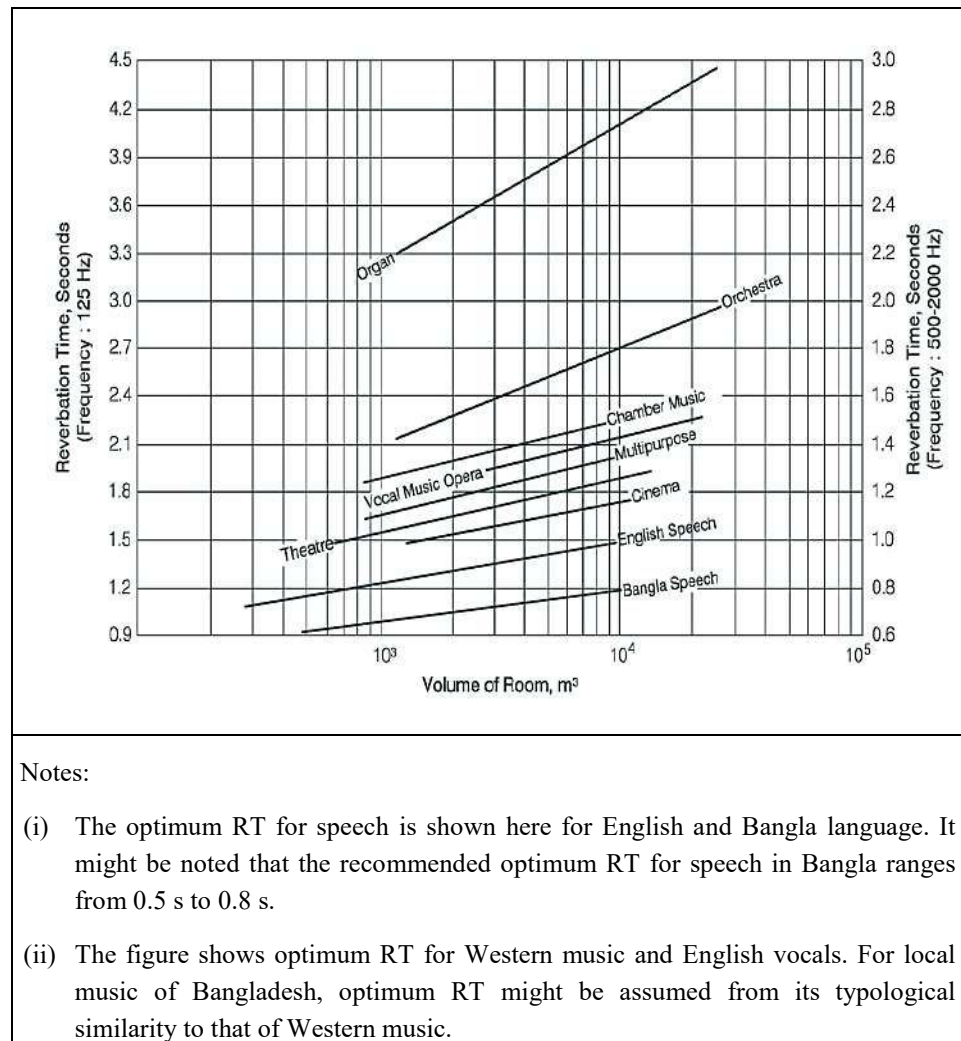


Figure 8.3.1 Recommended optimum reverberation times for spaces of various uses

3.6.4 Diffusion of Sound

- (a) Diffusion of sound should be achieved in any space, so that certain key acoustical properties, like sound pressure level, reverberation time etc. are the same anywhere in the space.
- (b) There shall not be a difference greater than 6 dBA between sound pressure levels of any two points in the audience area.
- (c) Appropriate room geometry should be chosen to achieve diffusion of sound. Figure 8.E.2, Appendix E shows recommended proportion of a space to avoid standing wave, flutter echo etc., which are obstacles to achieve diffusion of sound.

3.7 Speech Privacy

3.7.1 Principle of Speech Privacy between Enclosed Spaces

3.7.1.1 When noise carries information, productivity and noise are related inversely. When noise does not carry information, it can be annoying, counterproductive or can be useful as a masking sound, depending upon its frequency, intensity level and constancy.

3.7.1.2 The degree of speech privacy in a space is a function of following two factors:

- (a) The degree of sound isolation provided by the barriers between rooms
- (b) The ambient sound level in the receiving room

3.7.1.3 In case of an airtight barrier between two rooms, the sound intensity level of the source room (1) and the receiving room (2) are related as,

$$IL_2 = IL_1 - NR$$

where, NR is reduction, IL_2 and IL_1 are sound intensity levels in the receiving and source room respectively.

3.7.1.4 Transmitted noise level IL_2 is not annoying to a majority of adults, if a properly designed background sound is a maximum 2 dBA less than IL_2 . For example, a transmitted noise IL_2 of 40 dBA in a room with a background sound of at least 38 dBA will not cause annoyance to most people.

3.7.1.5 The upper intensity level of usable background masking sound is usually taken as about 50 dBA; any higher intensity level itself will cause annoyance.

3.7.2 Sound Isolation Descriptor

3.7.2.1 For speech sound, a descriptive scale is shown in Table 8.3.6

3.7.2.2 Relation between barrier STC and hearing condition on receiving side with background noise level at NC-25 is shown in Table 8.3.7

Table 8.3.6: Relative Quality of Sound Isolation

Ranking	Descriptor	Hearing Condition ^a
6	Total privacy	Shouting barely audible.
5	Excellent	Normal voice levels not audible. Raised voices barely audible but not intelligible.
4	Very good	Normal voice levels barely audible. Raised voices audible but largely unintelligible.
3	Good	Normal voice levels audible but generally unintelligible. Raised voices partially intelligible.
2	Fair	Normal voice levels audible and intelligible some of the time. Raised voices generally intelligible.
1	Poor	Normal voice levels audible and intelligible most of the time.
0	None	Normal voice levels always intelligible.

^a Hearing condition in the presence of ambient noise, if any.

Table 8.3.7: Barrier STC and Hearing Condition on Receiving Side with Background Noise Level at NC-25

Barrier STC	Hearing Condition	Descriptor and Ranking ^a	Application
25	Normal speech can be understood quite easily and distinctly through the wall.	Poor/1	Space divider
30	Loud speech can be understood fairly well. Normal speech can be heard but not easily understood.	Fair/2	Room divider where concentration is not essential
35	Loud speech can be heard but not easily intelligible. Normal speech can be heard only faintly, if at all.	Very Good/4	Suitable for offices next to quiet spaces

Barrier STC	Hearing Condition	Descriptor and Ranking ^a	Application
42-45	Loud speech can be faintly heard but not understood. Normal speech is inaudible.	Excellent/5	For dividing noisy and quiet areas; party wall between apartments
46-50	Very loud sounds (such as loud singing, brass musical instruments or a radio at full volume) can be heard only faintly or not at all.	Total Privacy/6	Music room, practice room, sound studio, bedrooms adjacent to noisy areas

^a See Table 8.3.6.

3.7.3 Speech Privacy Design for Enclosed Space

3.7.3.1 Figure 8.J.1, Appendix J shows a Speech Privacy Analysis Sheet, which shall be used to determine speech privacy rating number for design of enclosed space.

3.7.3.2 Following factors are involved in speech privacy rating of enclosed-space:

- (a) Space rating of source room (Room No. 1)
 - (i) Speech effort - a measure of loudness of speech
 - (ii) Source room factor - gives the approximate effect of room absorption on the speech level in the source room. The scale in Figure 8.J.1, Appendix J represents average absorption. For live rooms the factor should be raised by 2 points and for dead room the factor should be lowered by 2 points. Factors (a + b) give the approximate source-room voice level.
 - (iii) Privacy allowance-determines the measure of privacy required, such as *Normal Privacy* and *Confidential Privacy*.
- (b) Isolation rating of receiving room (Room No. 2)
 - (i) The STC rating of the barrier (see Tables 8.E.1 and 8.E.2, Appendix E)
 - (ii) Noise reduction factor A_2/S indicates receiving room absorption, that is, the difference between NR and TL , where A_2 is the area of receiving room and S is the area of the barrier between the rooms. Absorption is assumed to be average. For live rooms the factor should be lowered by 2 points and for dead room the factor should be raised by 2 points.
 - (iii) Recommended background noise level in the receiving room. As a reference, Table 8.3.2 might be used.

3.8 Sound Amplification System

3.8.1 Objectives and Design Criteria

3.8.1.1 A well designed sound amplification system should augment the natural transmission of sound from source to listener with adequate loudness and diffusion. It should never be used as a substitute for good building acoustics design, because it rarely overcomes or corrects any serious deficiency; rather, it may amplify and exaggerate the deficiency.

3.8.1.2 An ideal sound amplification system shall give the listener the desired loudness, directivity, intelligibility and other acoustical qualities.

3.8.1.3 Spaces seating less than 500 (approximately, 1400 m³ volume) should not require any sound amplification system if it is properly designed; since, a normal speaking voice can maintain speech level of 55 to 60 dBA in this volume of space.

3.8.1.4 The central type amplification system is preferred, in which a loudspeaker or a cluster of loudspeakers is placed directly above the source of sound to provide desired realism and intelligibility. In case, the ceiling height is low and sound cannot reach all listeners from a central type; a distributed system can be used with a number of loudspeaker each serving a small area with low-level amplification. A distributed system is particularly feasible in areas under the balcony.

3.8.1.5 A careful location of microphone should be chosen to avoid feedback of sound from loudspeaker to the microphone.

3.9 Occupancy A: Residential Buildings

3.9.1 Controlling Noise

Controlling measures shall have to be taken against noise coming from outdoor and indoor sources as specified in Sections 3.4 and 3.5.

3.9.2 Space Layout

- (a) Quiet and noisy quarters shall be grouped and separated horizontally and vertically from each other by rooms (or spaces) not particularly sensitive to noise such as entry, corridor, staircase, wall closets or other built-in building components.
- (b) If a living room in one apartment is located adjacent to a living room in another apartment, adequate sound insulation should be provided in separating wall.
- (c) Bedrooms shall be located in a relatively quiet part of the building.

- (d) Bathrooms must be separated acoustically from living rooms both horizontally and vertically.

If bathroom fixtures are installed along walls which separate living room and bathroom, adequate sound insulation should be provided in separating wall.

- (e) Measures should be taken to avoid transmission of footstep noise through floors.

3.9.3 Sound Insulation Factors

- (a) Separation for Sound Insulation: The sound insulation criteria in residential units are to be based on three grades:
 - (i) Grade I criteria apply mainly to fully residential, quiet rural and suburban areas and in certain cases to luxury apartment buildings or to dwelling units above the eighth floor of a high-rise building.
 - (ii) Grade II criteria apply to residential buildings built-in relatively noisy environments typical of urban or suburban areas.
 - (iii) Grade III criteria express minimal requirements applicable to very noisy locations, such as commercial or business areas (like shop houses with dwelling units on the upper floors) or downtown areas.
 - (iv) Among the above three categories, Grade II covers the majority of residential constructions and shall therefore be regarded as a basic guide.
 - (v) In all grades wall constructions and floor-ceiling assemblies between dwelling units shall have STC ratings at least equal to the values given in Tables 8.3.3 and 8.3.4.
 - (vi) An STC rating of not less than 45 dB is to be provided in walls and floors of residential buildings, between dwelling units of the same building and between a dwelling unit and any space common to two or more dwelling units.
 - (vii) Table 8.E.2, Appendix E shows STC requirements for different spaces of specific occupancies.
- (b) Reduction of Airborne Noise : In case of air borne noise (between the frequency range 100-31500 Hz), a sound insulation of 50 dB shall be provided in between the living room in one house or flat and rooms/bed rooms in another. The value shall be 35 dB in between different rooms of the same house. (See Appendix E for airborne sound insulation properties of walls, doors and windows).
- (c) Reduction of Airborne Noise Transmitted through the Structure: Exterior walls shall be rigid and massive and have good sound insulation characteristics with as few openings as possible. Windows with acoustic louvers might be used to protect noise intrusion, while allowing ventilation.

Ventilation ducts or air transfer openings (ventilators), where provided, shall be designed to minimize transmission of noise, if necessary, by installing some attenuating devices.

- (d) Construction of sound insulation doors shall be of solid core and heavy construction with all edges sealed up properly. Hollow core wooden doors and light weight construction shall be avoided because these are dimensionally unstable and can warp, destroying the seal along the perimeter of the door.
- (e) Rubber, foam rubber or foamed plastic strips, adjustable or self-aligning stops and gaskets shall be used for sealing the edges of the doors. They shall be so installed that they are slightly compressed between doors and stop when the door is in a closed position. In simple cases the bottom edges shall have a replaceable strip of felt or foam rubber attached to minimize the gap between door and floor.
- (f) Separation between the two faces of the door shall be carried through uninterruptedly from edge to edge in both directions. Damping treatments shall be inserted between individual layers of the doors. Ordinary doors with surface leather padding shall not be used.
- (g) Automatic damped door closers are to be used whenever applicable and economically feasible in order to avoid the annoying sound of doors slamming.
- (h) The difference between the TL of the wall and that of the door shall not exceed 10 dB.
- (i) The floor of a room immediately above the bedroom or a living room shall satisfy the Grade I impact sound insulation.

3.10 Occupancy B: Educational and Occupancy C: Institutional Buildings

3.10.1 Sources of Noise

3.10.1.1 Outdoor Noise: Measures shall be taken in planning and design to control noise from external sources mentioned in Sections 3.4 and 3.5.

3.10.1.2 Indoor Noise:

The following sources of indoor noise shall be taken into consideration:

- (a) Wood and metal workshops, machine shops, technical as well as engineering testing laboratories, other machine rooms, typing areas etc. which produce continuous or intermittent noises of disturbing nature,
- (b) Music rooms,

- (c) Assembly halls, particularly those which are attached to the main building,
- (d) Practical work spaces, gymnasiums and swimming pools,
- (e) School kitchen and dining spaces,
- (f) Entry lobby, foyer, lounge, corridor and other circulation spaces.

3.10.2 Planning and Design Requirements

3.10.2.1 Site Planning: The school building shall be located as far away as possible from the sources of outdoor noise such as busy roads, railways, neighbouring market places or adjacent shopping areas as well as local industrial and small scale manufacturing concerns.

Where the site permits, the building shall be placed back from the street, in order to make use of the noise reducing effect of the increased distance between street line and building line.

If adequate distance between the school/institution building and the noisy traffic route cannot be provided, rooms which do not need windows or windowless walls of classrooms shall face the noisy road.

Car parking areas shall preferably be located in remote parts of the site.

3.10.2.2 Activities and Space Layout: The minimum requirement for sound insulation in educational buildings shall be as specified in Table 8.3.5.

3.10.2.3 Halls and Circulation Areas: The lobby, lounge areas etc. or other circulation spaces and linking corridors shall be separated from teaching areas, lecture galleries or laboratories. No direct window openings shall be placed along the walls of the corridors or circulation areas.

Doors, ventilators and other necessary openings shall be designed with sufficient foam or rubber seals, so that they are noise proof when closed.

3.10.2.4 Noise Reduction within Rooms: Lecture halls of educational institutions (with a seating capacity of more than 100 persons) shall be designed in accordance with the relevant acoustical principles.

Lecture halls with volumes of up to about 550 m³ or for an audience of up to about 150 to 200, shall not require a sound amplification system, if their acoustical design is based on appropriate principles and specifications.

A diagonal seating layout shall preferably be used for rectangular lecture rooms of the capacity mentioned above as it automatically eliminates undesirable parallelism between walls at the podium and effectively utilizes the diverging front walls as sound reflectors.

3.11 Occupancy D: Health Care Buildings

3.11.1 Sources of Disturbing Noise

3.11.1.1 Outdoor Noise: Sources of outdoor noise specified in Sec 3.4 shall be taken into consideration for planning and design. Additionally, health care service facilities like ambulance, medicine and equipment vans, store deliveries, laundry and refuse collection trolleys are also frequent sources of noise. Health care buildings shall be sited away from such sources as far as practicable.

3.11.1.2 Indoor Noise: Indoor noise sources include mechanical and mobile equipment like X-ray and suction machines, drilling equipment etc. Planning and design shall take into account the following sources of noise:

- (a) The handling of sterilizing, as well as metal or glass equipment,
- (b) Wheeled trolleys used for the purpose of carrying foods and medical supplies,
- (c) Mechanical equipment like mechanical and electrical motors, machineries, boilers, pumps, fans, ventilators, transformers, elevators, air-conditioning equipment etc.
- (d) Operational facilities like refrigerators, sterilizers, autoclaves etc. ,
- (e) Patient service facilities including oxygen cylinders or tanks, saline stands, carrier carts and instrument cases, etc.
- (f) Maintenance work of engineering services like plumbing and sanitary fixtures or fittings, hot and cold water and central heating pipes, air-conditioning ducts, ventilation shafts etc., and
- (g) Audible calling systems, radio and television sets.

3.11.2 Planning and Design Requirements

3.11.2.1 Site Planning: Site shall be selected to keep adequate distance from traffic noise from highways, main roads, railroads, airports and noise originating from parking areas. In addition to the requirements of Sec 4.4.3, the following requirements shall be fulfilled:

- (a) In the selection of a site and site planning, consideration shall be given to:
 - (i) Distance from exterior noise,
 - (ii) Effect of high buildings adjacent to the site which can act as noise reflectors, and
 - (iii) Traffic conditions surrounding the site.

- (b) Parking areas might be carefully located at the farthest possible corners of the premises. If enough space is not available to provide facilities for the desired number of vehicles, parking spaces shall be provided in more than one area. Loading platforms and service entries are to be planned in such a manner as to minimize noise in areas requiring silence.
- (c) Closed courts shall preferably be avoided.

3.11.2.2 Activities and Space Layout: The following points might be given due consideration in the planning and design of health care buildings.

- (a) Rooms to be used for board meetings, conferences, counseling and instructional purposes shall be grouped near public zones of the building in such a way that spread of noise can be avoided.
- (b) Long corridors might be avoided, as it may freely spread noise.
- (c) The main kitchen might be housed in a separate building and connected to the wards only by service lifts or a service stair. If this is impracticable, it shall be planned beneath the wards, rather than above them.
- (d) Mechanical plants might preferably be placed in separate buildings.
- (e) Rooms housing equipment, operational facilities and patient service facilities shall be designed for adequate sound insulation.
- (f) Closed courts might be avoided, unless rooms facing the court are air-conditioned with completely sealed and air tight windows.
- (g) The units which are themselves potential sources of noise for example, children's wards and outpatient departments, shall be treated with special care regarding the protection against noise.

3.11.2.3 Noise Reduction in the Sensitive Area: In health care buildings, many sensitive areas such as operation theatres, doctor's consultation rooms, intensive care units and post-operative areas shall be provided with special noise control arrangements.

These rooms shall preferably be isolated in locations (or corners) surrounded by other intermediate zones which ensure protection of the core area from outdoor noise.

A sound reduction of about 45 dBA between the consulting and the waiting rooms shall be provided in order to weaken the transmission of sound.

A lobby like space in between the interconnecting and communicating doors shall be provided.

3.11.2.4 Sound Insulation Factors: The rooms and indoor spaces of a health care building shall be treated with sound absorptive materials. Different STC ratings of walls specified for separate components of buildings shall have to be considered as follows:

- (a) For airborne noise, the average STC rating of wall and floors shall be 50 dB.
- (b) An STC rating of 55 dB shall be required between rooms whose occupants are susceptible to noise.
- (c) In general an average STC of 45 dB is to be provided for corridor walls and for walls between patient rooms.
- (d) All doors shall be fitted with silent closers. Doors to opposite rooms might be positioned in a staggered manner.
- (e) For ward doors, a corresponding STC of 35 dB shall be provided.
- (f) PVC mats, rubber mats or other resilient materials and rubber shod equipment shall be used in utility rooms, ward kitchens and circulation areas as floor coverings.

Other finish materials like rubber tile, cork tile, vinyl tile or linoleum which can also help reduce the impact noise substantially shall be used alternatively.

- (g) Mobile equipment, such as trolleys and bed, oxygen cylinder carriers and stretchers shall be made relatively silent by means of non-friction wheels with rubber tyre.
- (h) Special treatments such as thin nonporous coverings or films over some soft absorbent materials shall be used for good sound absorption when a washable acoustical treatment is desired.
- (i) Door and window curtains or screens, as well as bed sheets etc. shall be used wherever the indoor openings are located to help reduce reverberation in the hard surfaced surroundings. Curtain rails, rings and runners of silent type shall be used so that they generate as little frictional noise as possible.
- (j) Ventilation ducts and conduits shall be laid out in such a way that they do not open an easy by-pass for spreading out any noise from other sources. These conduits and ducts shall be completely sealed around the pipes where they pass through walls and floors.
- (k) Special care shall be taken to reduce noise of plumbing equipment and fixtures. Specially made silencing pipes and flushing fixtures shall be used to reduce the noise of water closet and cisterns in lavatories and toilets.

Ducts carrying waste or water pipes shall be properly lined with sound insulation material to prevent noise from the pipes passing through duct walls into the patients' wards or cabins or the spaces susceptible to noise.

- (l) Wherever available, cisterns shall be used to replace the pressure operated flushing system so that the disturbance becomes less irritating.

3.12 Occupancy I: Assembly

3.12.1 General

Buildings of Occupancy I shall be designed both for transmission of noise through the walls and openings and also for internal acoustics. Public address systems installed in such buildings shall conform to the standards and specifications.

3.12.2 Sources of Noise

3.12.2.1 Outdoor Noise: The following sources of noise shall be taken into account in planning and design:

- (a) Traffic noise (air, road and rail) and noise from other outdoor sources entering through walls, roofs, doors, windows or ventilation openings,
- (b) Noise from any other gathering spaces, public meetings, outdoor activities and crowds, particularly during the time of breaking of shows and performances,
- (c) Noise produced from parking areas.

3.12.2.2 Indoor Noise: The following indoor noise sources shall be taken into account in planning and design:

- (a) Noise from other adjacent halls located within the same building used for similar performance, or for seminar, symposium or general meetings,
- (b) Noise produced from ticket counters, lobby or lounge areas, rehearsal rooms, waiting areas and corridors,
- (c) Noise generated from other ancillary services located within the building, like cafeteria or snack bar, tea shop, post office, bank or the like,
- (d) Noise generated from the mechanical or electrical equipment, air-conditioning plants, ventilation channels and ducts, plumbing and water lines etc.

3.12.3 Planning and Design Requirements

3.12.3.1 Site Planning and Acoustical Requirements: The noise control of auditoria or assembly halls shall begin with sensible site planning following the measures and precautions stated below:

- (a) The auditorium shall be effectively separated from all exterior and interior noise and vibration sources as far as practicable;
- (b) The assembly halls shall be protected from vehicular or air traffic, parking or loading areas, mechanical equipment, electrical rooms or workshops.

The following are the acoustical requirements for good hearing conditions in an auditorium which shall be ensured in planning and design:

- (a) Adequate loudness shall have to be ensured in every part of the auditorium;
- (b) The sound energy shall be uniformly distributed in the hall;
- (c) Optimum reverberation characteristics shall have to be provided;
- (d) The hall shall be free of such acoustical defects as echoes, long delayed reflections, flutter echoes, sound concentrations, distortions, sound shadow and room resonance etc.;
- (e) Noise and vibration shall be excluded or reasonably reduced in every part or the hall room.

3.12.3.2 Activities and Space Layout in Divisible and Multi-purpose Auditoria

- (a) A protective buffer zone of rooms between exterior noise source and auditorium proper shall be designed.
- (b) Rooms in the buffer zone (lobbies, vestibules, circulation areas, restaurants, ticket counters, offices etc.) shall be shut off from the auditorium proper by sound insulation doors.
- (c) The purposes of the subdivided spaces shall be clarified, in order to establish the predictable intensity of the various sound programs.

3.12.3.3 Noise Reduction within Rooms

- (a) There shall not be any use of continuous, unrecognizable and loud background noise.
- (b) The ventilating and air-conditioning system shall be so designed that the noise level created by the system is at least 10 dB below the permissible background noise level specified in noise criteria level.
- (c) In order to protect the hall from external noise the minimum sound reduction value required in an auditorium is 65 dB for a concert hall and 60 dB for a theatre. This reduction shall be provided on all sides.

3.12.3.4 Sound Insulation Factors

- (a) Rooms in the buffer zone (lobbies, vestibules, circulation areas, restaurants, counter and issue desk corners, office etc.) shall have sound absorbing ceilings and carpeted floor. If the rooms are to be used for the purposes of verbal instructions only, a moderate degree of sound insulation (STC 40 to 45 dB) shall be accomplished by the movable partitions.

- (b) If audio equipment or loudspeakers are to be used, an acoustically more effective, efficient partition system shall be used, with sound insulation of STC 45 to 50 dB.
- (c) An insulation of STC 50 to 60 dB shall be provided if any section of the space is selected for the performance of live music.
- (d) All windows shall have to be eliminated from the main auditorium walls in order to exclude excessive outdoor noises.
- (e) Suspended ceilings shall accommodate the ventilating, air-conditioning and electrical services above the room.
- (f) In order to increase the effectiveness of the suspended ceilings the following measures shall be taken :
 - (i) The ceiling membrane shall weigh not less than 25 kg/m²;
 - (ii) The ceiling membrane shall not be too rigid;
 - (iii) Noise transmission through the ceiling shall have to be avoided by the use of a solid, airtight membrane;
 - (iv) Gaps between ceiling and surrounding structure shall be sealed;
 - (v) The air space between ceiling membrane and structural floor shall be increased to a reasonable maximum;
 - (vi) An absorbent blanket is to be used in the air space above the ceiling;
 - (vii) The number of points of suspension from the structural floor above shall be reduced to a minimum;
 - (viii) Hangers made of resilient substance shall be preferable to the rigid ones.
- (g) In order to improve the airborne or impact sound insulation of a ceiling the following specifications shall be followed:
 - (i) The ceiling membrane shall have a minimum of 25 mm solid cement plaster layer with completely closed, airtight and sealed joints all around;
 - (ii) If further reduction of undesirable noise is desired within a sound insulated room, sound absorptive treatment shall be provided along the underside of the solid ceiling.

3.12.3.5 Masking Noise: The artificial noise produced by electronically created background noise for the purpose of drowning out or masking unwanted noise, shall be provided. The process shall effectively suppress minor intrusions which might interrupt the recipient's privacy.

3.12.3.6 The maximum permissible background noise levels in various occupancies are specified in terms of Balanced Noise Criteria (NCB) curves. Each of the NCB curves is expressed by the sound pressure level values in the important 1200-2400 Hz frequency band. The NCB levels shall be used to specify the desirable lowest limit under which the background noise must not fall. (See Table 8.3.1 and Figure 8.F.1, Appendix F).

Note: The general configuration of the NCB curves is quite similar to the noise rating (NR) curves established by the International Organization for Standardization, used mostly in the European practice.

3.13 Occupancy E: Business and Occupancy F: Mercantile Buildings

3.13.1 General

Buildings of Occupancy E and F shall be planned and designed to minimize noise from external and internal sources.

3.13.2 Sources of Disturbing Noise

3.13.2.1 Outdoor Noise: The following sources of outdoor noise and those specified in Sec 3.4 shall be taken into account in the planning and design of business and mercantile buildings:

- (i) Traffic,
- (ii) Playgrounds,
- (iii) Market places and shopping areas,
- (iv) Crowds grouped around the buildings for business purpose or other.

3.13.2.2 Indoor Noise: The following sources of indoor noise shall be identified for noise attenuation within buildings:

- (a) Mechanical noise, caused by heating, ventilating and air-conditioning systems, elevators, escalators and pneumatic tubes etc. ;
- (b) Noise produced by office equipment or machines such as typewriters, printers, teleprinters, reproduction, tabulating and punching machines etc.;
- (c) Noise produced by mechanical amplifiers, for example in seminar halls, conference rooms or staff training rooms or the like where public address system is used;
- (d) Machine noise generated from slide rooms, projection rooms and from electrical and mechanical machines like generators, transformers, switch rooms and electric substations etc. ;
- (e) Typical office noise created by speech, voices in circulation areas, opening and closing of doors etc. ;
- (f) Plumbing systems, ventilation plants, lift machineries, air-conditioning and cooling systems.

3.13.3 Planning and Design Requirements

3.13.3.1 Site Planning: Rooms susceptible to noise shall be located away from the sources of noise.

3.13.3.2 Activities and Space Layout: Spaces producing noise and those susceptible to noise shall be separated as far as practicable. The effective length of long corridors shall be minimized. Swing doors are to be provided at intervals.

3.13.3.3 Noise Reduction in the Sensitive Areas**(a) Open plan Offices**

- (i) The floor area may be carpeted in order to absorb airborne noise and footstep noise. The carpet shall preferably be thick and placed on top of resilient floors.
- (ii) The entire portion of the ceiling shall be treated with sound absorption materials. Such treatment shall be applied to the screens and nearby walls also.
- (iii) A highly sound absorptive ceiling with a sound absorption coefficient of 0.70 shall preferably be used to absorb 70 percent of the sound energy reflecting 30 percent of it.
- (iv) Moderately noisy office equipment (like typewriters, telephones, computers etc.) shall be distributed as uniformly as possible all over the office space.
- (v) Noisy office equipment shall be concentrated into specific areas of the office space. The space shall be treated with maximum amount of sound absorptive material and visually separated from the rest of the office.

(b) General Offices: Sound absorbent ceiling shall be provided in corridors. Hard floor finishes and batten floors in corridors shall be avoided. Floor ducts shall be planned on one side of corridors.

3.13.3.4 Reduction of Noise at Source: The following measures shall be undertaken to reduce noise at source depending on the degree of noise reduction desired.

- (a) The noise from slamming of doors shall be reduced by fitting automatic quiet action type door closers. Continuous soft, resilient strip set into the door frames as well as quiet action door latches shall be used.
- (b) Machines like typewriters, calculators, printers etc. shall be fitted or installed with resilient pads to prevent the floors or tables (on which they stand) from acting as large radiating panels.
- (c) Noises from ventilating systems, from a uniform flow of traffic or from general office activities, shall be considered to generate an artificial masking noise. In open plan offices the provision of a relatively high but acceptable degree of

background noise (from the ventilating or air-conditioning system) shall be provided, in order to mask undesirable office noises created by typewriters, telephones, office machines or loud conversation and to provide a reasonable amount of privacy.

The background noise masking system shall be introduced gradually without disturbing the feeling of the occupants.

The air-conditioning system may be used to generate background masking noise if the noise level from the ceiling fans, ducts etc. can be suitably reduced to generate the desired frequency spectrum.

3.13.3.5 Sound Insulation Factors: The acoustical performance of the partitions dividing rentable office spaces shall not exceed an STC rating of 25 to 30 dB, unless the background noise is so high that it masks the sound coming through the lightweight partition.

If lightweight partitions are employed for subdivision of large spaces into executive cabins and secretarial areas, the following measures shall be taken to increase the insulation factors:

- (a) Sound barriers shall be provided up to above the false ceiling with a noise reduction characteristic that will not be affected by ducts, conduits or other cable lines including electricity and water piping installed in the ceiling space.
- (b) Where construction of light weight partitions is considered essential, a double skin panel shall be preferred.

The panels shall be installed apart from each other either by use of separate framing or by use of elastic discontinuities in the construction. Sound absorbing materials shall be provided in the air cavity between the panels so that more insulation can be assured.

- (c) All apertures, gaps and joints at side walls, floors and ceiling junctions shall be properly sealed.
- (d) A double panel hollow floor construction shall be employed with heavy sound damping materials introduced between the panels for effective reduction of the structure-borne noise transmitted from upper floors to the floors below, particularly when lightweight floors are provided in multi-use spaces.

Lightweight materials having high natural frequencies may resonate or vibrate due to an applied vibratory force, which may be caused by mechanical equipment, road or rail traffic etc. These materials, if used for specific reasons, shall be isolated from the source of noise in order to reduce the amount of vibration transmitted to the building.

- (e) The floor surfaces surrounding the office space may be lined with a carpet of high sound absorption.

- (f) For sound adsorption with floor carpeting, the following characteristics shall be maintained:
 - (i) Fibre type carpet shall not be used, as it has practically no effect on sound absorption;
 - (ii) Hair, hair jute and foam rubber pads shall be used for higher sound absorption than the less permeable rubber coated hair jute, sponge rubber etc.;
 - (iii) To improve sound absorption the loop-pile fabrics with increased pile height (with the density held constant) shall be applied;
 - (iv) The backing shall be more permeable for higher sound absorption.

3.14 Occupancy G: Industrial Buildings

3.14.1 General Noise Levels

In the noise control of industrial buildings the following requirements are to be fulfilled:

- (a) An acceptable acoustical environment for individual workers and machine operators;
- (b) Speech communication among operators to the required degree;
- (c) Protection of other workers or office employees (either close to the noise source or at some other location within the same building);
- (d) Prevention of noise transmission into adjacent buildings or into the surrounding community.

3.14.1.1 Intermittent Noises : Intermittent noise in the form of isolated explosions, and periodic noise related to pressure relief valves, hammering, grinding and sawing operations etc. shall be identified for enforcing controlling measures.

3.14.1.2 Sources of Noise: The following sources of noise in industrial buildings and manufacturing plants shall be identified and investigated to find whether the machines are in smooth operation and producing minimal mechanical noise.

- (a) Fabrication and assembly machines;
- (b) Machines used for material transport and general plant services;
- (c) Noise caused by impact and coupled with resonant response of the structural members, connected to the impacting surfaces;
- (d) High frequency sounds generated from grinders;
- (e) Frictional noise occurring at the time of sawing, grinding or sanding, as well as during the cutting on lathe machines and in brakes or from bearings;

- (f) Noise generated from piping systems and valves;
- (g) High velocity flow of air, steam or other fluids that undergo an abrupt change in pipe diameter which give rise to turbulence and resultant noise, and noise generated by rapid variation in air pressure caused by turbulence from high velocity air, steam or gases;
- (h) Unpleasant noise identified with rotating or reciprocating machines, which is generated due to pressure fluctuation in the fluids inside the machines.

3.14.2 Hearing Damage Risk Criteria

When the sound level at a particular section in a factory or industrial building exceeds the specified level in terms of magnitude and time (as shown in Table 8.3.5), feasible engineering control shall be applied and implemented in order to reduce the sound to the limits shown. Personal hearing protection equipment shall be provided and used if such control fails to reduce sound levels.

3.14.3 Interference with Communication

In industries where the operator has to follow verbal instructions during operation of the machine the background noise shall be reduced to an acceptable level.

Precautionary measures shall be taken so that the noise generated inside may not be the cause of accidents by hindering communication or by masking warning signals.

3.14.4 Requirements for Noise Reduction

3.14.4.1 Noise Reduction by Layout and Location: Considerable noise reduction may be achieved by a sensible architectural layout in noisy industrial buildings following the steps mentioned below:

- (a) Noisy areas shall be separated from spaces requiring silence.
- (b) The office block is to be located in a separate building. If this is not possible, the office space in a factory shall be segregated from the production area as far as practicable.
- (c) The office building shall not have a common wall with the production areas. Where a common wall is unavoidable it should be of heavy construction (not less than 375 mm thick).
- (d) Electrically operated vehicles shall be used as far as practicable, since they eliminate most of the noise normally associated with combustion engines.

3.14.4.2 Noise Reduction at Source: In order to suppress the noise at the source relatively silent machines and equipment shall be installed. Additionally the following provisions shall be adhered to:

- (a) Appropriate type of manufacturing process or working method shall be selected which does not cause disturbing noise. Machine tools and equipment are to be selected carefully in order to attain lower noise levels in the machine shop.
- (b) Maintenance of vibrating and frictional machineries shall be ensured.
- (c) Impact noises in general shall be reduced; soft and resilient materials shall be applied on hard surfaces where impact noise can originate.
- (d) Rubber tyres or similar other materials shall be fixed on the areas or surfaces used for the handling and dropping of materials.
- (e) The area of the radiating surface from which a noise is radiated shall be reduced to a minimum.
- (f) Resilient flooring (carpeting, rubber tile, cork tile, etc.) shall be used adequately to reduce impact transmission onto the floor.
- (g) Flexible mountings, anti-vibration pads, floating floors etc. shall be used to prevent the transmission of vibration and shock from various machines into the building or structure.
- (h) Mechanically rigid connecting paths must be interrupted by resilient materials so that the transmission of vibration and noise is reduced.

3.14.4.3 Isolator Specifications

- (a) Isolators shall be made of resilient materials like steel (in the form of springs), soft rubber and corks.
- (b) Direct contact between the spring and the supporting structure shall be eliminated, in order to reduce transmission of high frequencies by metal springs.
- (c) Rubber or felt pads shall be inserted between the ends of the springs and the surfaces to which they are fixed.
- (d) Felt or cork shall be used under machine bases, as resilient mats or pads.
- (e) If the equipment is massive like drop hammers causing serious impact vibration (in larger manufacturing plants), it shall be mounted on massive blocks of concrete, on its own separate foundation.
- (f) The foundation shall have a weight 3 to 5 times that of the supported machines.
- (g) A sound reduction of 5 to 10 dBA shall have to be realized from the vibration isolation measures.

3.14.4.4 Noise Reduction by Enclosures and Barriers: When the plant is large in which the overall noise level results from many machines, an enclosure shall be provided.

- (a) When only one or two machines are the dominant source of disturbing noise, the noisy equipment shall be isolated in a small area of enclosure.
- (b) The enclosure shall be in the form of close fitting acoustic box around the machines. The box shall be of such character that the operator can continue with his normal work outside the box.
- (c) An enclosure around the offending unit shall be impermeable to air and lined with sound absorbing materials such that the noise generated by machines is reduced substantially.
- (d) When the industrial plant is a large one in which the resultant noise level is produced from a number of machines, enclosures shall be used either for supervisory personnel or operators who are engaged in monitoring the automatic machines. Such barriers may have inspection openings.
 - (i) Enclosures of this type shall ensure noise reduction of at least 30 dBA, and shall be made of sheet metal lined inside with an appropriate insulation material.
 - (ii) Where curtains are used to isolate the noisy equipment in a small area, they shall be of full length i.e. from ceiling to floor and shall be made of fibre glass cloth and lead or leaded vinyl.
- (e) If the size of the machine is large and asks for more working spaces, thus not permitting close fitting enclosures, the machine shall be housed in a separate room or enclosure.
The inside of the enclosure shall be lined with sound absorbing materials in order to reduce the contained noise.
- (f) If after all these measures are taken the noise level still remains above a tolerable degree, the workers shall be provided with earplugs for protection.

3.15 Acoustical Requirements of Special Occupancies

3.15.1 Susceptible Buildings

3.15.1.1 Recording and radio studios

A recording studio shall present optimum acoustical conditions. A differentiation shall be made among the numerous various purposes of studio use.

- (a) Particular attention shall be given to the following requirements:
 - (i) An optimum size and shape of the studio shall be established following the design criteria;
 - (ii) A high degree of diffusion shall be secured;
 - (iii) Ideal reverberation characteristics shall be provided;
 - (iv) Noises and vibration shall be completely eliminated and acoustical defects shall be totally prevented.

- (b) The acoustical treatments shall be uniformly and proportionately distributed over the three pairs of opposite surfaces enclosing the studio.
- (c) Portable acoustic screen and a reverberation chamber shall be provided so that the desired reverberation condition can be achieved.
- (d) Variable absorbers such as hinged or sliding panels, rotatable cylinders, adjustable drapery etc. shall be fixed on wall surfaces and ceiling areas.
- (e) All surfaces shall be carefully checked for echoes, flutter echoes etc.
- (f) Parallel surfaces shall be eliminated or treated with highly absorptive acoustical materials (throughout the frequency range between 63 and 8000 Hz).

3.15.1.2 Research laboratories

- (a) In the selection of site, care shall be taken to ensure that no noise generating installations exist in the vicinity.
- (b) Location of laboratories shall be secluded from the noisy zones within the building.
- (c) A sound insulation of at least 35 dB shall be achieved by means of acoustic partitions where offices are attached to the laboratory.
- (d) Sound absorbing screens shall be used where scientists and researchers are engaged in laboratory activities and desk work simultaneously.
- (e) Transmission of noise through service ducts, pipes, lifts and staircases shall be guarded.
- (f) Double glazed windows shall be provided in the noise sensitive areas. There shall be a minimum gap of 100 mm between the two glasses.

3.15.1.3 Music rooms

The following provisions shall apply to music rooms, including rehearsal rooms, instructional space, practice booth etc.

- (a) Acoustical conditions in practice booths and listening booths shall have a reverberation time of 0.4 to 0.5 second.
- (b) Adequate floor area, room height, room shape and volume must be established to achieve proper reverberation.
- (c) Sound absorbing materials shall be applied sufficiently so that the excessive sound generated by bands or individual instruments can be soaked up.
- (d) Parallelism between opposite surfaces shall be avoided.
- (e) Entire surfaces of at least two adjacent walls and all the ceiling area shall be treated with sound absorbing materials.

3.15.1.4 Libraries

A quiet and peaceful interior shall be maintained inside libraries. The following provisions are to be adhered to in planning and design:

- (a) Screening and sound insulation measures shall be undertaken in and around the reception/issue desk and photocopying facility areas.
- (b) Stack rooms, store rooms and administrative offices shall be planned in such a way that the audiovisual areas are properly isolated from external noises.
- (c) Walls enclosing the library shall have a sound reduction value of not less than 50 dB.
- (d) Fanlights shall be double glazed and non-openable.
- (e) Walls facing the corridors or other noisy areas shall not have fanlights or borrowed lights unless they are double glazed.

3.15.1.5 Law courts and council chambers

- (a) Entrance into court rooms and council chambers (especially from circulation areas and gathering spaces) shall be through baffle lobbies, with two sets of doors fitted with silencers.
- (b) Offices shall be planned around the court rooms or chambers for further protection against outdoor noise and the central rooms shall have a sound insulation value of not less than 50 dB (provided by 225 mm thick brick wall) to insulate against airborne noise in the corridors.
- (c) The court and chamber rooms shall have floors finished with resilient materials.
- (d) Ceiling and upper parts of the walls of lobbies and circulation areas shall have sound absorbing treatments.

3.15.2 Public Address System

3.15.2.1 Design of public address systems shall take care of equipment choice, positioning of the individual elements and other precautions to obtain optimum performance of the system.

3.15.2.2 Passenger terminals and other public places equipped with public address systems shall as far as practicable avoid the use of sound reflecting surfaces like hard walls and floors. Reverberation time shall be reduced as far as possible by using sound absorbing materials on walls and ceilings.

3.15.2.3 Reverberation built-up sound level shall not be relied upon. Direct sound shall preferably be audible in all areas to be covered by the public address system.

3.15.2.4 Sound levels of the public address system in the areas covered shall be adequately high to overcome background noise.

3.16 Related References

Maekawa Z. and Lord P. 1994. Environmental and Architectural Acoustics. E&FN SPON, UK (Table 8.3.2)

Grondzik, W. T., Kwok, A. G., Stein, B and Reynolds, J. S. 2006. Mechanical and Electrical Equipment for Buildings. John Wiley & Sons, New Jersey. (Tables 8.3.3 to 8.3.6)

3.17 List of Related Appendices

Appendix D	NC, NCB and Recommended Criteria for Sound Insulation
Appendix E	STC, Aural Field and Proportion of Space
Appendix F	Activity Flow Diagram: Planning, Design, Assessment and Construction in Building Acoustics
Appendix G	Checklist for Acoustical Planning, Design and Post-occupancy Assessments
Appendix H	Noise Levels and Subjective Evaluation
Appendix I	PSA and Liveliness
Appendix J	Speech Privacy Analysis Sheet
Appendix K	Sound Absorption Coefficients

PART VIII
Chapter 4
Lifts, Escalators and Moving Walks

4.1 General

4.1.1 Purpose

The purpose of this Chapter is to provide minimum standards for regulating and controlling the design, construction, installation, quality of materials, location, operation, maintenance and use of lifts, escalators and moving walks to ensure public safety and welfare.

4.1.2 Scope

4.1.2.1 The provisions of this Chapter shall apply to the erection, installation, alteration, repair, relocation, replacement, addition to, operation and maintenance of lifts, escalators and moving walks.

4.1.2.2 Additions, alterations, repairs and replacement of equipment or systems shall comply with the provisions for new equipment and systems.

4.1.2.3 Where, in any specific case, different sections of the Code specify different materials or other requirements, the most restrictive one shall govern. Where there is a conflict between a general requirement and a specific requirement, the specific requirement shall be applicable.

4.1.2.4 It shall be unlawful to install, extend, alter, repair or maintain lift, escalator or moving walk systems in or adjacent to buildings except in compliance with this Code.

4.1.3 Terminology

This Section provides an alphabetical list of the terms used in this Chapter of the Code. In case of any conflict or contradiction between a definition given in this Section and that in Part 1, the meaning provided in this Section shall govern for interpretation of the provisions of this Chapter.

AUTOMATIC RESCUE
DEVICE

A device meant to bring a lift stuck between floors due to loss of power, to the nearest level and open the doors in order to allow trapped passengers to be evacuated. Such a device may use some form of internal auxiliary power source for such purpose, complying with all the safety requirements of a lift during normal run. The speed of travel is usually lower than the normal speed. In the case of manual doors on reaching the level, the device shall allow the door to be opened and in case of power operated doors the device shall automatically open the door.

BALUSTER	One of the slender upright supports of a hand rail.
BALUSTRADE	A row of balusters meant for supporting moving hand rails.
BASEMENT STOREY	The lower storey of a building below or partly below the ground level.
BOTTOM CAR CLEARANCE	The clear vertical distance from the pit floor to the lowest structural or mechanical part, equipment or device installed beneath the car platform aprons or guards located within 300 mm, measured horizontally from the sides of the car platform when the car rests on its fully compressed buffers.
BOTTOM CAR RUNBY	The distance between the car buffer striker plate and the striking surface of the car buffer when the car is in level with the bottom terminal landing.
BOTTOM COUNTER WEIGHT RUNBY	The distance between the counter weight buffer striker plate and the striking surface of the counterweight buffer when the car is in level with the top terminal landing.
BUFFER	A device designed to absorb the impact of the falling car or counter weight beyond its normal limit of travel by absorbing and dissipating the kinetic energy of the car or counterweight.
BUFFER, OIL	A buffer using oil as a medium which absorbs and dissipates the kinetic energy of the descending car or counterweight. Oil buffer stroke - The oil displacing movement of the buffer plunger or piston, excluding the travel of the buffer plunger accelerating device.
BUFFER, SPRING	A buffer which stores in a spring the kinetic energy of the descending car or counterweight. • Spring buffer load rating - The load required to compress the spring by an amount equal to its stroke. • Spring buffer stroke - The distance, the contact end of the spring can move under a compressive load until the spring is compressed solid.

CALL INDICATOR	A visual and audible device in the car to indicate to the attendant the lift landings from which the calls have been made.
CAR BODY WORK	The enclosing body work of the lift car which comprises the sides and roof, and is built upon the car platform.
CAR DOOR ELECTRIC CONTACT	An electric device, the function of which is to prevent operation of the driving machine by the normal operating device unless the car door is in the closed position.
CAR FRAME	The supporting frame to which the platform of the lift car, its safety gear, guide shoes and suspension ropes are attached.
CAR PLATFORM	The part of the lift car which forms the floor and directly supports the load.
CAR SPEED	See RATED SPEED (LIFT).
COMB PLATE	A pronged plate that forms part of an escalator (or moving walk) landing and engages with the Cleats of the steps (or tread way) at the limits of travel.
CONTROL SYSTEM	The system of equipment by means of which starting, stopping, direction of motion, speed, acceleration, and retardation of the moving member are controlled.
CONTROL, SINGLE-SPEED ALTERNATING CURRENT	A control for a driving machine induction motor which is arranged to run at a single-speed.
CONTROL, TWO-SPEED ALTERNATING CURRENT	A control for a two-speed driving machine induction motor which is arranged to run at two different synchronous speeds either by pole changing of a single motor or by two different armatures.
CONTROL, RHEOSTATIC	A system of control which is accomplished by varying resistance or reactance or both in the armature or field circuit or both, of the driving machine motor.
CONTROL, VARIABLE VOLTAGE MOTOR (GENERATOR FIELD CONTROL)	A system of control which is accomplished by the use of an individual generator for each lift wherein the voltage applied to the driving machine motor is adjusted by varying the strength and direction of the generator field.

CONTROL, ELECTRONIC DEVICES	A system of control which is accomplished by the use of electronic devices for driving the lift motor at variable speed.
CONTROL, ALTERNATING CURRENT VARIABLE VOLTAGE (ACW)	A system of speed control which is accomplished by varying the driving and braking torque by way of voltage variation of the power supply to the driving machine induction motor.
CONTROL, ALTERNATING CURRENT VARIABLE VOLTAGE VARIABLE FREQUENCY (ACVVVF)	A system of speed control which is accomplished by varying the voltage and frequency of the power supply to the driving machine induction motor.
CONTROL, SOLID- STATE D.C. VARIABLE VOLTAGE	A solid-state system of speed control which is accomplished by varying the voltage and direction of the power supply to the armature of driving machine D.C. motor.
COUNTER WEIGHT	A weight or combination of weights to counterbalance the weight of the car and part of the rated load.
DETERMINING ENTRANCE LEVEL	The inside floor level at the entrance to the building.
DEFLECTOR SHEAVE	An idler pulley used to change the direction of a rope lead.
DOOR, CENTRE OPENING SLIDING	A door which slides horizontally and consists of two panels which open from the centre and are so interconnected that they move simultaneously.
DOOR, HINGED	The hinged portion of the lift well enclosure which closes the opening giving access to the landing.
DOOR, MID BAR COLLAPSIBLE	A collapsible door with vertical bars mounted between the normal vertical members.
DOOR, MULTI-PANEL	A door arrangement whereby more than one panel is used such that the panels are connected together and can slide over one another by which means the clear opening can be maximized for a given shaft width. Multi-panels are used in centre opening and two speed sliding doors.

DOOR, SINGLE SLIDE	A single panel door which slides horizontally.
DOOR, TWO SPEED	A two panel door which slides horizontally in the same direction wherein each panel has different operating speed and reaches the ends simultaneously.
DOOR, VERTICAL BI-PARTING	A door or shutter which slides vertically and consists of two panels or sets of panels that move away from each other to open and are so interconnected that they move simultaneously.
DOOR, VERTICAL LIFTING	A single panel door, which slides in the same plane vertically up to open.
DOOR, SWING	A swinging type single panel door which is opened manually and closed by means of a door closer when released.
DOOR CLOSE	A device which automatically closes a manually opened door.
DOOR OPERATOR	A power-operated device for opening and closing doors.
DRIVING MACHINERY	The motorized power unit for driving the lift, escalator or moving walks.
DUMBWAITER	A small lift with a car which moves in guides in a substantially vertical direction and has a net floor area, total inside height and capacity not exceeding 0.9 m ² , 1.25 m and 225 kg respectively, and is exclusively used for carrying materials and no person. It may or may not be provided with fixed or removable shelves.
ELECTRICAL AND MECHANICAL INTERLOCK	A device provided to prevent simultaneous operation of both up and down relays.
ELECTRO-MECHANICAL LOCK	A device which combines in one unit, electrical contact and a mechanical lock jointly used for the landing and/or car doors.
EMERGENCY STOP PUSH OR SWITCH	A push button or switch provided inside the car designed to open the control circuit to cause the lift car to stop during emergency.

ENCLOSED WELL	The lift well having enclosure walls of continuous construction without openings except for doors at landings.
ESCALATOR	A power driven, inclined, continuously moving stairway used for carrying passengers from one level to another.
ESCALATOR LANDING	The portion of the building or structure which is used to receive or discharge passengers into or from an escalator.
ESCALATOR LANDING ZONE	A space extending from a horizontal plane 40 cm below a landing to a plane 40 cm above the landing.
ESCALATOR MACHINE	The mechanism and other equipment in connection therewith used for moving the escalator
FLOOR	The lower surface in a storey on which one normally walks in a building. The general term 'floor', unless otherwise specifically mentioned shall not refer to a 'mezzanine floor'.
FLOOR LEVELING SWITCH	A switch for bringing the car to level at slow speed in case of double speed or variable speed machines.
FLOOR SELECTOR	A mechanism forming a part of the control equipment, in certain automatic lifts, designed to operate controls which cause the lift car to stop at the required landings.
FLOOR STOPPING SWITCH	A switch or combination of switches arranged to bring the car to rest automatically at or near any pre-selected landing.
GEARED MACHINE	A machine in which the power is transmitted to the sheave through a worm or, worm and spur reduction gearing.
GEARLESS MACHINE	A lift machine in which the motive power is transmitted to the driving sheave from the motor without intermediate reduction gearing and has the brake drum mounted directly on the motor shaft.
GOODS LIFT	A lift designed primarily for the transport of goods, but which may carry a lift attendant or other persons necessary for the loading or unloading of goods.

GOVERNOR	A device which automatically actuates safety devices to bring the lift car and/or counter weight to rest in the event the speed of the equipment in the descending direction exceeds a predetermined limit.
GUIDE RAILS	The members used to guide the movement of a lift car or counterweight in a vertical direction.
GUIDE RAILS FIXING	The complete assy. comprising the guide rails bracket and its fastenings.
GUIDE RAILS SHOE	An attachment to the car frame or counterweight for the purpose of guiding the lift car or counter weight frame.
HANDLING CAPACITY	The capacity of the lift system to carry passengers during a five minute peak period, expressed as the percentage of the estimated total population handled.
HOISTING BEAM	A beam, mounted immediately below the machine room ceiling, to which lifting tackle can be fixed for raising or lowering parts of the lift machine.
HOSPITAL LIFT	A lift normally installed in a hospital, nursing home or clinic and designed to accommodate one number bed/stretchers along its depth, with sufficient space all around to carry a minimum of three attendants in addition to the lift operator.
HYDRAULIC LIFT	A lift where the vertical movement of the lift car is done by hydraulic force of a hydraulic fluid. In this type of lift a plunger is attached to the bottom or top of the lift car wherein the plunger moves inside a cylinder by hydraulic force. The hydraulic lift may be direct-plunger-driven type where the cylinder extends into the ground as deep as the lift rises. Hydraulic lifts may also be roller-chain type actuated by a sheave on a vertical hydraulic cylinder, installed beside the lift shaft. Hydraulic lifts may be used for passenger and freight services. The speeds are between 0.125 mps and 1.0 mps. The load capacities are between 1,000 kg and 50,000 kg.
INTERVAL	Average time gap(s) between consecutive lifts leaving the ground floor or passing any specific floor.

LANDING	That portion of a building or structure used for the reception and discharge of passengers or goods or both into and from a lift car, escalator or moving walk.
LANDING CALL PUSH BUTTON (LIFT)	A push button fitted at a lift landing, either for calling the lift car or for actuating the call indicator.
LANDING DOOR (LIFT)	The hinged or sliding portion of a lift well enclosure, controlling access to a lift car at a lift landing.
LANDING PLATE	The portion of the landing immediately above the mechanism at either end of escalator or moving walk and constructed so as to give access to this mechanism in these areas.
LANDING ZONE	A space extending from a horizontal plane 400 mm below a landing to a plane 400 mm above the landing.
LEVELING DEVICE, LIFT CAR	Any mechanism which either automatically or under the control of the operator, moves the car within the leveling zone towards the landing only and automatically stops it at the landing.
LEVELING DEVICE, ONE WAY AUTOMATIC	A device which corrects the car level only in case of under run of the car but will not maintain the level during loading and unloading.
LEVELING DEVICE, TWO-WAY AUTOMATIC MAINTAINING	A device which corrects the car level on both under run and over-run and maintains the level during loading and unloading.
LEVELING DEVICE, TWO WAY AUTOMATIC NON-MAINTAINING	A device which corrects the car level on both under run and over run but will not maintain the level during loading and unloading.
LEVELING ZONE	The limited distance above or below a lift landing within which the leveling device may cause movement of the car towards the landing.
LIFT	A machine designed to transport persons or materials between two or more levels in a vertical or substantially vertical direction by means of a guided car or platform. The lifting force is provided by electric motor or fluid pressure. The word "elevator" is also synonymously used for "lift".

LIFT CAR	The load carrying unit with its floor or platform, car frame and enclosing bodywork.
LIFT LANDING	That portion of a building or structure used for discharge of passengers or goods or both into or from a lift car.
LIFT MACHINE	The part of the lift equipment comprising of electric motor(s) and control gear therewith, reduction gear (if any), brake(s) and winding drum or sheave, by which the lift car is raised or lowered.
LIFT PIT	The space in the lift well below the level of the lowest lift landing served.
LIFT SYSTEM	One or more lift cars serving the same building.
LIFT WELL	The unobstructed space within an enclosure provided for the vertical movement of the lift car(s) and any counter weight(s), including the lift pit and the space for top clearance.
LIFT WELL ENCLOSURE	Any structure which separates the lift well from its surroundings.
LIFTING BEAM	A beam, mounted immediately below the machine room ceiling to which lifting tackle can be fixed for raising parts of the lift machine.
MACHINE ROOM	The compartment allocated to house the lift machine and associated items.
MACHINERY SPACE	The space occupied by the driving machine and control gear of the lift, escalator or moving walk.
MEZZANINE	An intermediate floor between two floors above ground level.
MOVING WALK	A power driven, horizontal or inclined, continuously moving conveyor used for carrying passengers, horizontally or at an incline up to a maximum of 15 degree.
NEWEL	An upright support of the handrail at the landing of escalator/moving walk where the handrail reverses its direction.

OPEN TYPE WELL	A lift well having enclosure walls of wire grille or similar construction.
OPERATION	The method of actuating the control and/or functioning of any lift machine/equipment.
OPERATION, AUTOMATIC	A method of operation in which by a momentary pressure of a button the lift car is set in motion and caused to stop automatically at any required lift landing.
OPERATION, NON-SELECTIVE COLLECTIVE AUTOMATIC	Automatic operation by means of one button in the car for each landing level served and one button at each landing, wherein all stops registered by the momentary actuation of landing or car buttons are made irrespective of the number of buttons actuated or of the sequence in which the buttons are actuated. With this type of operation, the car stops at all landings for which buttons have been actuated making the stops in the order in which the landings are reached after the buttons have been actuated but irrespective of its direction of travel.
OPERATION, SELECTIVE COLLECTIVE AUTOMATIC	Automatic operation by means of one button in the car for each landing level served and by up and down buttons at the landings, wherein all stops registered by the momentary actuation of the car made as defined under non-selective collective automatic operation, but wherein the stops registered by the momentary actuation of the landing buttons are made in the order in which the landings are reached in each direction of travel after the buttons have been actuated. With this type of operation, all 'up' landing calls are answered when the car is travelling in upward direction and all 'down' landing calls are answered when the car is travelling in downward direction, except in case of the uppermost or lowermost calls which are answered as soon as they reached irrespective of the direction of travel of the car.
OPERATION, SINGLE AUTOMATIC	Automatic operation by means of one button in the car for each landing level served and one button at each landing so arranged that if any car or landing button has been actuated, the actuation of any other car or landing operation button will have no effect on the movement of the car until the response to the first button has been completed.

OPERATION, GROUP AUTOMATIC	Automatic operation of two or more non-attendant lifts equipped with power-operated car and landing doors. The operation of the cars is co-ordinated by a supervisory operation system including automatic dispatching means whereby selected cars at designated dispatching points automatically close their doors and proceed on their trips in a regulated manner. Typically, it includes one button in each car for each floor served and up and down buttons at each landing (single buttons at terminal landings). The stops set up by the momentary actuation of the car buttons are made automatically in succession as a car reaches the corresponding landings irrespective of its direction of travel or the sequence in which the buttons are actuated. The stops set up by the momentary actuation of the landing buttons may be accomplished by any lift in the group, and are made automatically by the first available car that approaches the landing in the corresponding direction.
OPERATION, CAR SWITCH	Method of operation by which the movement of lift car is directly under the operation of the attendant by means of a handle.
OPERATION, SIGNAL	Same as collective operation, except that the closing of the door is initiated by the attendant.
OPERATION, DOUBLE BUTTON (CONTINUOUS PRESSURE)	Operation by means of buttons or switches in the car and at the landings any of which may be used to control the movement of the car as long as the button or switch is manually pressed in the actuating position.
OPERATING DEVICE	A car switch, push button or other device employed to actuate the control.
OVERHEAD BEAMS (LIFT)	The members, usually of steel or reinforced concrete, which immediately support the lift equipment at the top of the lift well.
OVERHEAD PULLEY	An idler pulley used to change the direction of rope.
PASSENGER LIFT	A lift designed for the transport of passengers.

POSITION AND/OR DIRECTION INDICATOR	A device which indicates on the lift landing or in the lift car or both, the position of the car in the lift well or the direction or both in which the lift car is travelling.
POWER OPERATED DOOR	A door operated automatically by a device initiated by a momentary pressure on the push button or by operation of the control system.
RATED LOAD	The maximum load which the lift car, escalator or moving walk is designed and installed to carry safely at its rated speed.
RATED SPEED (LIFT)	The speed attained by the lift in the up direction with rated load in the lift car. Also known as CAR SPEED.
RATED SPEED (ESCALATOR)	The speed at which the escalator is designed to operate in the up direction. It is the rate of travel of the steps, measured along the angle of inclination with rated load on the steps or carriage.
RATED SPEED (MOVING WALK)	The speed at which the moving walk is designed to operate in the up direction. It is the rate of travel of the tread way, measured along the angle of inclination with rated load on the tread way.
RETIRING CAM	A device which prevents the landing doors from being unlocked by the lift car unless it stops at a landing.
ROPING MULTIPLE	A system of roping where, in order to obtain a multiplying the factor from the machine to the car, multiple falls of rope are run around sheave on the car or counterweight or both. It includes roping arrangement of 2 to 1, 3 to 1 etc.
SAFETY GEAR	A mechanical device attached to the car frame or the counter weight to stop and hold the car or counter weight to the guides in the event of a free fall. Governor operated safety gears are used to stop the car or counterweight when it travels at a speed exceeding a predetermined speed.
SERVICE LIFT	A lift designed primarily for the transport of goods, but which may carry a lift attendant or other persons necessary for the loading and unloading of goods.

SHEAVE	A rope wheel, the rim of which is grooved to receive the suspension ropes but to which the ropes are not rigidly attached and by means of which power is transmitted from the lift machine to the suspension ropes.
SLACK ROPE SWITCH	Switch provided to open the control circuit in case of slackening of rope(s)
STOREY	The space between the surface of one floor and the surface of the adjacent floor vertically above or below it. The term 'Floor' shall include 'Roof' but will exclude mezzanine floors.
STOREYS FOR SPECIFIC USE	Storeys which are named according to their functions and the specific uses they are put to. For example, a duct storey is one through which service pipes and electrical conduits may be taken.
SUBSIDIARY STOREY	A storey which occurs below the determining entrance level but above the basement storey.
SUSPENSION ROPES (LIFT)	The ropes by which the car and counter-weight are suspended.
TERMINAL SLOW DOWN SWITCH	A switch when actuated shall compulsorily cut off the high speed and switch on the circuitry to run the lift in leveling speed before reaching on terminal landings.
TERMINAL STOPPING SWITCH NORMAL	Switch for cutting all the energizing current in case of car travelling beyond the top bottom landing or a switch cuts off the energizing current so as to bring the car to stop at the top and bottom level.
TERMINAL STOPPING DEVICE FINAL	A device which automatically cause the power to be removed from an electric lift driving machine motor and brake, independent of the functioning of the normal terminal stopping device, the operating device or any emergency terminal stopping device, after the car has passed a terminal landing.
TOP CAR CLEARANCE	The shortest vertical distance between the top of the car crosshead, or between the top of the car where no crosshead is provided, and the nearest part of the overhead structure or any other obstruction when the car floor is level with the top terminal landing.

TOP COUNTERWEIGHT CLEARANCE	The shortest vertical distance between any part of the counterweight structure and the nearest part of the overhead structure or any other obstruction when the car floor is level with the bottom terminal landing.
TOTAL HEADROOM	The vertical distance from the level of the top lift landing to the floor of the machine room.
TRAVEL (LIFT)	The vertical distance between the bottom and top lift landings served by the equipment.
VENT	An opening provided in the roof or the external wall of a space for the purpose of ventilation.

4.1.4 Preliminary Design Particulars

4.1.4.1 All relevant aspects of lift, escalator or moving walk installations shall be properly evaluated during the planning stage of the building in order to design the most effective conveying system.

4.1.4.2 Appropriate steps shall be taken during the planning stage of the building to determine particulars of lift, escalator or moving walk and the necessary provisions to be kept in the building structure so as to meet the requirements of this Code. Discussion shall be carried out, during planning stage, with all concerned parties, viz. building owner, architect, consulting engineer and/or lift/escalator/moving walk manufacturer to determine the extent of necessary provisions to be kept in the building.

4.1.4.3 Minimum amount of information to be collected for lifts during such meetings shall be the following:

- (a) Number, capacity, speed and disposition of the lifts necessary to give adequate lift service in the building
- (b) Layout and sizes of lift well
- (c) Particulars of lift well enclosure, sizes of punches In the lift well enclosure
- (d) Location of lift machine room (above or below), height of lift machine room
- (e) Provision of adequate access to the lift machine room and size of machine room
- (f) Total headroom clearance
- (g) Provision of ventilation of the lift well
- (h) Depth of lift pit

- (i) Loads which the lift will impose on the building structure, and the holes to be left in the machine room floor and cut-outs for wall boxes for push buttons and signals
- (j) Necessity for and type of insulation to minimize the transmission of vibration and noise to other parts of the building
- (k) Requirements for fixing guide brackets to the building structure, hoisting beam for hoisting of lift machine
- (l) Requirements and layout of electrical power feeders for the lift.

4.1.4.4 Minimum amount of information to be collected for the escalators shall be the following :

- (a) Number and size of each escalator
- (b) Angle of escalator
- (c) Arrangement and layout of escalators with dimensions of floor punches required
- (d) Minimum floor to floor height requirement
- (e) Dimensions of top and bottom escalator landings

4.1.4.5 Minimum amount of information to be collected for the moving walks shall be the following:

- (a) Number and size of each moving walk
- (b) Angle of moving walk
- (c) Arrangement and layout of moving walks with dimensions of floor punches required
- (d) Dimensions of top and bottom moving walk landings

4.1.4.6 For the safety considerations of lift installations and effective utilization of lift installations, locations and arrangement of lifts shall be in accordance with Sec 4.3.3.

4.1.4.7 The building plan submitted with the application for seeking permission of installation of lift, escalator or moving walk from the Authority shall include layout of lift, escalator or moving walk properly identified in the drawing along with the detailed particulars as per Appendix L.

4.1.4.8 Specifications for lifts, escalators and moving walks shall include detailed particulars as per Appendix L.

4.1.4.9 For the purpose of effective installation of lifts, escalators or moving walks, working drawings showing the layout of lifts, escalators or moving walks properly identified in the drawing, details of builders works, for example, holes and/or punches in floors or, walls and supports for lifts, escalators or moving walks shall be prepared prior to the finalization of building design drawings.

4.1.4.10 Necessary particulars of electrical requirements of lifts, escalators or moving walks shall be determined early in the planning stage to include it in the electrical provisions of the building.

4.2 Essential Requirements for Lifts

4.2.1 General

4.2.1.1 Lifts shall be provided in buildings more than six storeys or 20 m in height. Installation of lifts shall be carried out in conformity with the "Lift Act" and rules there under, wherever they are in force.

4.2.1.2 Stretcher Facility in Lifts

- (a) When passenger lifts are installed in any building having more than ten storeys or a height of more than 32 m, each floor served by these lifts must have access to at least one lift with a stretcher facility in accordance with Sec 4.2.1.2(b).
- (b) A lift required to have a stretcher facility by Sec 4.2.1.2(a) shall accommodate a raised stretcher with a patient lying on it horizontally by providing a minimum inside platform area 1275 mm wide x 2000 mm long with a minimum clear opening width of 1050 mm, unless otherwise designed to provide an equivalent facility, to allow the entrance and exit of an ambulance stretcher (minimum size 600 mm wide x 2000 mm long) in its horizontal position. These lifts shall be identified by the internationally recognized symbol for emergency medical services.
- (c) In any multi-storied hospital and health care building there shall be at least one hospital lift having stretcher facility in accordance with Sec 4.2.1.2(a).

4.2.1.3 Standby power

- (a) One or more lifts shall be provided with standby power in
 - (i) A building which has more than ten storeys or a height of more than 32 m,
 - (ii) Hospital and health care buildings.

- (b) Standby power shall be provided by an approved self-contained generator set to operate automatically whenever there is a disruption of electrical power supply to the building.
- (c) The operation of the standby power system shall be as follows:
 - (i) Where only one lift is installed, the lift shall transfer to standby power within 60 seconds after failure of normal power.
 - (ii) Where two or more lifts are controlled by a common operating system, all lifts may be transferred to standby power within 60 seconds after failure of normal power, or if the standby power source is of insufficient capacity to operate all lifts at the same time, all lifts shall be transferred to standby power in sequence, shall return to the designated landing and discharge their load. After all lifts have been returned to the designated landing, at least one lift shall remain operable from the standby power.

4.2.1.4 ADA (American Disabilities Act) Approved Type Lift

- (a) At least one of the lifts of any lift bank shall have features as per requirements of ADA accessibility guidelines.
- (b) Accessible lifts shall be on an accessible route and shall comply with the ASME A17.1-1990, safety code for Elevators and Escalators.
- (c) Lift operation shall be automatic. It shall have door safeties as per clause 4.2.3.9. It shall have self-leveling feature as per clause 4.2.3.13.
- (d) Hall call buttons in the lift lobbies and halls shall be centered at 1065 mm above the floor. Such call buttons shall have visual signals to indicate when each call is registered and each call is answered. Call buttons shall be a minimum 19 mm in the smallest dimension. The button designating the up direction shall be on the top.
- (e) A visible and audible signal shall be provided at each hoist way entrance to indicate which car is answering a call. Audible signal shall sound once for the up direction and twice for the down direction or shall have verbal annunciators that say “up” or “down”. visible signals shall have the following features:
 - (i) Hall lantern fixtures shall be mounted so that their centerline is at least 1830 mm above the lobby floor.
 - (ii) Visual elements shall be at least 64 mm in the smallest dimension.
 - (iii) Signals shall be visible from the vicinity of the hall call button. In-car lanterns located in cars, visible from the vicinity of hall call buttons, and conforming to the above requirements, shall be acceptable.

- (f) All lift hoistway entrance shall have raised and Braille floor designations provided on both jambs. The centerline of the characters shall be 1525 mm above finish floor. Such characters shall be 50 mm high. Permanently applied plates are acceptable if they are permanently fixed to the jambs.
- (g) Lift doors shall open and close automatically. They shall be provided with a reopening device that will stop and reopen a car door and hoistway door automatically if the door becomes obstructed by an object or person. The device shall be capable of completing these operations without requiring contact for an obstruction passing through the opening at heights of 125 mm and 735 mm above finish floor. Door reopening devices shall remain effective for at least 20 seconds. After such an interval, doors may close in accordance with the requirements of ASME 17.1.
- (h) The minimum acceptable time from notification that a car is answering a call until the doors of that car start to close shall be calculated from the following equation:

$$T = D / (445 \text{ mm/s})$$

Where,

T = total time in seconds

D = distance in millimeters from a point in the lobby or corridor 1525 mm directly in front of the farthest call button controlling that car to the centerline of its hoistway door. For cars with in-car lanterns, T begins when the lantern is visible from the vicinity of hall call buttons and an audible signal is sounded. The minimum acceptable notification time shall be 5 seconds.

- (i) The minimum time for lift doors to remain fully open in response to a car call shall be 3 seconds.
- (j) The floor area of lift cars shall provide space for wheel-chair users to enter the car, maneuver within reach of controls, and exit from the car. The minimum width and depth of the car shall be 2000 mm and 1291 mm. The clearance between the car platform sill and the edge of any hoistway landing shall be no greater than 32 mm.
- (k) The level of illumination at the car controls, platform, and car threshold and landing sill shall be at least 53.8 lux.

(l) Lift control panels shall have the following features:

- (i) Buttons: All control buttons shall be at least 19 mm in their smallest dimensions. They shall be raised or flush.
- (ii) Tactile, Braille, and Visual Control Indicators: All control buttons shall be designated by Braille and by raised standard alphabet characters for letters, Arabic characters for numerals, or standard symbols, and as required in ASME 17.1. The call button for the main entry floor shall be designated by a raised star at left of the floor designation, and as required in ASME 17.1. All raised designations for control buttons shall be placed immediately to the left of the button to which they apply. Applied plates, permanently attached, are an acceptable means to provide raised control designations. Floor buttons shall be provided with visual indicators to show when each call is registered. The visual indicators shall be extinguished when each call is answered.

4.2.1.5 Responsibility of the owner

- (a) It is the responsibility of the owner of the premises where the lift will be installed, to obtain necessary permission from the Authority before the erection of lifts(s) and for the subsequent commissioning and operation of lift (s).
- (b) The owner shall conduct periodic inspection and maintain the installation in safe working condition at all times.
- (c) Conformity with the provisions of this Code does not relieve the owner of his responsibility to satisfy the requirements of any other Act, Regulations or Ordinances that may be in force from time to time.

4.2.1.6 Conformity with Bangladesh electricity act

All electrical work in connection with electrical lifts shall be carried out in accordance with the provisions of the latest Bangladesh Electricity Act and the provisions of any of its bye-laws and regulations, and shall also comply with the requirements of Chapter 1 of Part 8 of this Code.

4.2.1.7 For detailed specifications of lifts, escalators and moving walks reference shall be made to the latest edition of the ANSI/ASME A 17.1 code or the European EN81 code.

4.2.2 Safety Considerations

4.2.2.1 Fire protection

- (a) Necessary provisions shall be kept to prevent spread of fire through the lift well. Adequate measures shall also be taken to reduce the possibility of spread of fire from the machine room into the lift well.
- (b) Lift well enclosures and machine room shall be constructed with fire resistant materials. In case of fire, the lift well enclosure shall not give off harmful gas or fumes.
- (c) Where lift enclosures are fire rated, manually closing doors at the enclosure well shall have a fire rating equal to that of the enclosure well and automatically closing doors shall have a fire rating equal to one-half of that of the enclosure well.

4.2.2.2 Fire switch

When required fire switch shall be provided, the function of which is to enable the fire authority to take over complete control of one or more lifts in an installation by operating with a fireman's key.

4.2.2.3 Fireman's lift

For buildings having height of 15 m or more at least one lift shall meet the requirements of fireman's lift as described below:

- (a) Lift car shall have floor area of not less than 1.44 m². It shall also have a loading capacity of not less than 544 kg (8 persons).
- (b) Lift landing doors shall have minimum fire resistance of two hours.
- (c) Doors shall be of automatic operation for car and landing.
- (d) The lift speed shall be 1.0 m/s or more so as to reach the top floor from the ground (or entrance) floor within 60 seconds.

4.2.2.4 Warning signs

Warning signs against use of the lifts during a fire shall be displayed near every call button for a passenger lift in accordance with Sec 4.2.10.3.

4.2.2.5 Over speed safety

Efficient automatic devices shall be provided and maintained in each lift to stop the car by suitable braking devices and to cutoff power from the motor whenever excessive descending speed is attained.

4.2.2.6 Over travel safety

Efficient automatic devices shall be provided and maintained in each lift to cut off power from the motor if the car over travels either the top or bottom terminal landing.

4.2.2.7 Manual cranking system

There shall have standard cranking system operable from the lift machine room to move the car manually, during a power failure, to the nearest higher or lower landing for evacuation of passengers.

4.2.2.8 Emergency evacuation system

There shall have arrangement for emergency unlocking of the landing and lift door with a special key from any landing for evacuation as well as for maintenance.

4.2.2.9 Protection of rope breakage

Necessary protection shall be taken against breaking of steel rope.

4.2.2.10 Safe working environment

In order to maintain a safe work environment, and to avoid potential hazards, the following shall be provided:

- (a) caution sign shall be installed in the areas listed below where potential hazard exists:
 - (i) Trip hazard in machine room; and
 - (ii) Caution notice against unauthorized use of rescue devices (for example, brake release device).
- (b) Use of hard hats for entry in pit and car top during construction period.
- (c) Warning sign shall be provided on the controller so also to eliminate the possibility of contact with any exposed or concealed power circuit.
- (d) Car top barricade system shall be provided as primary protection against fall, on car top.
- (e) Whenever work is carried out on the lift and lift is not required to be moved on power, notice shall be put on electrical main switch indicating requirement of de-energized condition.
- (f) During lift installation/maintenance, protection against fall shall be provided with suitable barricades for all open landing entrances.

4.2.2.11 Car door safeties

Lift car doors and landing doors shall be provided with necessary safeties as per Sec 4.2.3.9.

4.2.3 Lift Cars

4.2.3.1 Lift cars shall have net maximum inside area for different loading capacities as shown in Table 8. 4.1.

4.2.3.2 Lift car frame shall be of steel construction having sufficient strength to support safely the rated load, the lift car and all requisite accessories.

4.2.3.3 There shall be provisions for elastic isolators between metal parts to ensure low vibration and low noise during car travel.

Table 8.4.1: Maximum inside Net Platform Areas for Various Rated Loads

Rated Load (mass) (kg)	Maximum Available Car Area (see note) (m ²)	Maximum Number of Passengers	Rated Load (mass) (kg)	Maximum Available Car Area (see note) (m ²)	Maximum Number of Passengers
100	0.40	1	975	2.35	14
180	0.50	2	1000	2.40	14
225	0.70	3	1050	2.50	15
300	0.90	4	1125	2.65	16
375	1.10	5	1200	2.80	17
400	1.17	5	1250	2.90	18
450	1.30	6	1275	2.95	18
525	1.45	7	1350	3.10	19
600	1.60	8	1425	3.25	20
630	1.66	9	1500	3.40	22
675	1.75	10	1600	3.56	23
750	1.90	11	1800	3.88	26
800	2.00	11	2100	4.36	30
825	2.05	12	2500	5.00	36
900	2.20	13			

Note: (i) Beyond 2500 kg, add 0.16 m² for each 100 kg extra

(ii) Maximum available car area = (W x D) + Available area near the car door(s) inside the car.

Where, W = Car inside width in metre; D = Car inside depth in metre

4.2.3.4 The car bodywork shall be of sufficient mechanical strength to resist accidental impact by users or goods. The roof, solid or perforated, shall be capable of supporting two persons or a minimum load of 150 kg. Perforations shall be sufficiently close in mesh not exceeding 40 mm to provide reasonable protection against falling articles to any person travelling in the car.

4.2.3.5 The floor shall be a smooth nonslip surface. If carpeting is used, it shall be securely attached, heavy duty, with a tight weave and low profile, installed without padding.

4.2.3.6 A handrail shall be provided on at least one wall of the car, preferably the rear. The rails shall be smooth and the inside surface at least 38 mm clear of the walls at a nominal height of 800 mm from the floor.

4.2.3.7 Height of the entrance to the lift car shall not be less than 2 m.

4.2.3.8 The lift car doors, shall be power operated horizontally sliding type (non-collapsible), opened and closed by automatic means. However, if space is limited, collapsible doors may be installed in case of buildings not exceeding 8 storeys or 26 m in height, but they shall not be power operated. Sliding doors shall be guided at top and bottom. Means shall be provided to prevent all sliding doors from jumping off the tracks and suitable stops shall be provided to prevent the hanger carriage from leaving the end of the track.

4.2.3.9 Lift door safeties

- (a) Car and landing doors shall open and close in full synchronization being mechanically connected to each other.
- (b) Doors closed by automatic means shall be provided with door reopening device(s) which will function to stop and reopen a car door and adjacent landing door in case the car door is obstructed while closing. The reopening device shall also be capable of sensing an object or person in the path of a closing door without requiring contact for activation. Door reopening devices shall remain effective for a period of not less than 20 seconds. The operating mechanism of car door shall not exert a force more than 125 N.
- (c) Car doors shall be equipped with efficient interlocking or other devices so that the door cannot be opened except when the lift car is at the landing, and that the lift car cannot be moved away from the landing until the leading edge of the single slide or double speed door is within 50 mm of the nearest face of the door jamb or the leading edges of the centre opening doors are within 50 mm of contact of each other.

4.2.3.10 Lift car doors, when closed, shall cover the opening fully except in the case of vertical biparting car doors of goods lifts.

4.2.3.11 Where the lift car has solid enclosure and doors, provision shall be made for a fan for adequate ventilation. To permit switching off the power supply to the lift without switching off the fan and light, a separate switch shall be provided for fan and light.

4.2.3.12 Any vision panel in a car door shall be fire resisting and shall be of safety wired glass or similar material. The area between division bars or other supports shall not exceed 0.1 m². The bottom rail of a framed and glazed door shall be not less than 300 mm deep. Any projections on or recesses (including vision panels) in sliding car doors shall be kept to a minimum in order to avoid finger trapping between sliding parts of the door and any fixed part of the structure.

4.2.3.13 The lift car shall be provided with a self-leveling feature that will automatically bring the car to the floor landing within a tolerance of + 13 mm under normal loading and unloading conditions. This self-leveling shall, within its zone, be entirely automatic and independent of the operating device and shall correct the over-travel or under-travel. The car shall also be maintained approximately level with the landing, irrespective of load. Where no self-leveling device is provided, the leveling difference between the car and the landing shall be within + 40 mm.

4.2.3.14 Car operating panels shall be conveniently located on the side near the door so that passengers can register calls as quickly as possible. The centre line of the alarm button and emergency stop switch shall be at a nominal height of 890 mm, and the highest floor button no higher than 1.37 m from the floor. Floor registration buttons, exclusive of border, shall be a minimum of 18 mm in size, raised, flush or recessed. Visual indication shall be provided to show each call registered and extinguished when the call is answered. Depth of flush or recessed buttons when operated shall not exceed 10 mm. Markings shall be adjacent to the controls on a contrasting colour background to the left of the controls; letters or numbers shall be a minimum of 15 mm high and raised or recessed 0.75 mm. Sign plates permanently attached shall be acceptable. Emergency controls shall be grouped together at the bottom of the panel.

4.2.3.15 A suitable battery operated alarm system shall be installed inside the lift car so as to raise an alarm at a convenient place for getting assistance for passengers trapped inside the lift car.

4.2.3.16 A car position indicator shall be provided above the car operating panel or over the opening of each car to show the position of the car in the lift well by illuminated visual indicator corresponding to the landing at which the car is stopped or through which it is passing.

4.2.3.17 In addition, an audible signal shall preferably be installed which shall sound to tell a passenger that the car is stopping at a floor served by the lift. A special button located with emergency controls may be provided, operation of which shall activate an audible signal only for the desired trip.

4.2.3.18 Each lift car shall be fitted with a light and the car shall be kept illuminated during the whole period the lift is available for use.

4.2.3.19 In installations with more than two lifts in a bank, a telephone or other device for two-way communication between each lift car and a convenient point outside the lift well shall preferably be provided. Markings or the international symbol for telephones shall be placed adjacent to the control on a contrasting colour background.

4.2.3.20 Lift well and lift well enclosures

The Lift well shall only be used for housing equipment forming part of the lift installation or for its operation and maintenance. No other equipment or services shall be accommodated therein. For this purpose, the main electric supply line for lift machine shall be deemed to be part of the lift and the electric cable, if laid along the lift well shaft, shall be properly clamped to the wall.

4.2.3.21 The lift well shall not form part of the ventilation system of the building.

4.2.3.22 In multi-story residential buildings, hotels and hospitals, lift well shall be isolated from sleeping rooms (bed rooms) by lobbies or other spaces.

4.2.3.23 Lift well shall not be located above any room, passage or thoroughfare. However, when absolutely necessary, this can only be permissible with the prior approval of the competent authority and in such case the following provisions shall be made:

- (a) The pit shall be sufficiently strong to withstand the impact of the lift car with the rated load or the impact of the counterweight when either of these is descending at the rated speed or at governor tripping speed;
- (b) Spring or oil buffers shall be provided for lift car and counterweight; and
- (c) The car and counterweight shall be provided with a governor operated safety gear.

4.2.3.24 When there are three or fewer lift cars in a building, they may be located within the same lift well enclosure. When there are four lift cars, they shall be divided in such a manner that at least two separate lift well enclosures are provided. When there are more than four lifts, not more than four lift cars may be located within a single lift well enclosure.

4.2.3.25 The lift car and its counterweight shall travel in juxtaposition to each other.

4.2.3.26 Totally enclosed wells

The enclosure of the totally enclosed wells shall be continuous and shall extend on all sides from floor to floor or stair to stair. No openings except for doors at landings and necessary access panels shall be provided. The enclosure shall be of sufficient mechanical strength to support the lift guides at appropriate intervals and to support in true alignment the landing doors with operating mechanisms and locking devices.

4.2.3.27 Open type wells

- (a) When Lift well enclosures are constructed of wire grille or similar material, the mesh opening shall not be greater than 30 mm (except for door at landings). Such enclosures shall be of sufficient strength to resist accidental impact by users of adjoining areas or by materials or vehicles being moved in the vicinity.

- (b) Where the clearance between the inside of an open type lift well enclosure and any moving or movable part of the lift equipment or apparatus is less than 50 mm, the openings in the enclosure material shall not be more than 10 mm. Larger openings up to 30 mm shall be permissible provided it is further protected by square mesh netting with aperture of not greater than 10 mm and wire not smaller than 1 mm in diameter.

4.2.3.28 There shall be no opening in the lift well enclosure for access to the lift well through the space under the counterweight.

4.2.3.29 The inside surfaces of the lift well enclosures facing any car entrance shall form a smooth continuous flush surface devoid of projections or recesses. Where projections or recesses cannot be avoided, the underside of these projections/recesses shall be beveled to an angle of 60 degrees from the horizontal by means of metal plates or other fire resistive materials as shown in Figure 8.4.1.

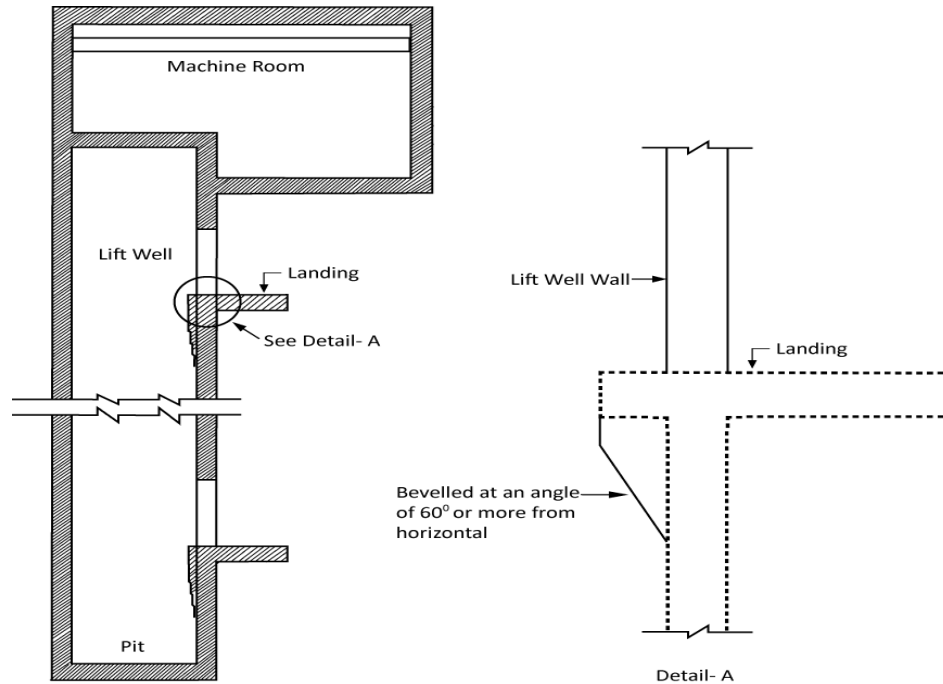


Figure 8.4.1 Beveled projections/recesses

4.2.3.30 Sufficient clearance space shall be provided between the guides for the car and the side walls of the lift well enclosure to allow safe and easy access to the parts of the safety gears for their maintenance and repairs.

4.2.3.31 Maximum clearance between the inner surface of well enclosure on the landing door side and any part of car facing the surface shall be 150 mm except that 230 mm and 200 mm clearance will be permissible when power operated vertically bi-parting landing doors or two speed horizontally sliding doors are installed respectively.

4.2.3.32 Each lift well serving more than two floors shall have vent(s) properly located at the top of the exterior wall. The vents shall be louvered with birds screens. If the well is located in such a way that no exterior wall is available for louvers, vents with connecting noncombustible ducts to an outside wall shall be provided. The area of vent shall not be less than 3.5% of the area of the lift well, provided that a minimum of 0.3 m² per lift is provided. Of the total required vent area not less than one-third shall be permanently open or automatically opened by a damper.

4.2.3.33 Bottom car clearance, passenger and service lift

When the car rests on its fully compressed buffer there shall be a vertical clearance of not less than 600 mm between the pit floor and the buffer striker plate or the **lowest** structural or mechanical part equipment or device installed. The clearance shall be available beneath the whole area of the platform except for:

- (a) Guide shoes or rollers, safety jaw blocks, platform aprons, guards of other equipment located within 300 mm measured horizontally from the sides of the car platform; and
- (b) Compensating sheaves.

Provided that:

- (i) In all the cases, including small cars, a minimum clearance of 600 mm is available over a horizontal area of 800 mm x 500 mm.
- (ii) In all the cases, when the car rests on its fully compressed buffers, there shall be a vertical clearance of not less than 50 mm between any part of the car and any obstruction of device mounted in the pit.

4.2.3.34 Top car clearance, passenger and service lift

The vertical clearance between the car cross-head and the nearest overhead obstruction within 500 mm measured horizontally to the nearest part of the crosshead when the car platform is level with the top landing, shall be not less than the sum of the following;

- (a) The bottom counterweight runby
- (b) The stroke of the counterweight buffer used.

- (c) One-half of the gravity stopping distance based on:
- 115 percent of the rated speed where oil buffers are used and no provision is made to prevent the jump of the car at counterweight buffer engagement; and
 - Governor tripping speed where spring buffers are used.

Note: The gravity stopping distance based on the gravity retardation from any initial velocity may be calculated according to the following formula

$$S = 51V^2$$

Where,

S = Free fall in mm (gravity stopping distance), and

V = Initial velocity in m/s

- (d) 600 mm.

Where, there is a projection below the ceiling of the well and the projection is more than 500 mm, measured horizontally from the centre line of the cross-head but over the roof of the car, a minimum vertical clearance not less than that calculated above shall also be available between the roof of the car and the projection.

Provided that the vertical clearance between any equipment mounted on top of the car and the nearest overhead obstruction shall be not less than the sum of the three items (a), (b) and (c) as calculated above plus 150 mm

4.2.3.35 Bottom runby for cars and counterweights, passenger and service lift

The bottom runby of cars and counterweights shall be not less than the following:

- Where oil buffers are used 150 mm
- Where spring-buffers are used:
 - 150 mm for variable voltage motor control, electronic devices, ACW control, ACVVVF control and solid state DC variable voltage control as defined in Sec 4.1.3
 - Not less than the following values for single-speed AC control, two-speed AC control and rheostatic control as defined in Sec 4.1.3.

<u>Rated speed (m/s)</u>	<u>Runby (mm)</u>
Up to 0.125	75
0.125 to 0.25	150
0.25 to 0.50	225
0.50 to 1.0	300

4.2.3.36 Maximum bottom runby, passenger and service lift

In no case the maximum bottom runby shall exceed the following:

- (i) 600 mm for cars
- (ii) 900 mm for counterweights.

4.2.3.37 Top Counterweight Clearances, passenger and service lift

The top counterweight clearance shall be not less than the sum of the following four items:

- (a) the bottom car runby;
- (b) the stroke of the car buffer used;
- (c) 150 mm; and
- (d) one-half the gravity stopping distance based on
 - (i) 115 percent of the rated speed where oil buffers are used and no provision is made to prevent jump of the counterweight at car buffer engagement; and
 - (ii) Governor tripping speed where spring buffers are used.

4.2.3.38 Top car clearance, Goods lift

The top car clearance shall be sufficient to avoid any protruding part fixed on the top of the car coming in direct contact with the ceiling or diverting sheave. The clearance shall not be less than the sum of the following four items:

- (a) The bottom counterweight runby,
- (b) The stroke of the counterweight buffer used,
- (c) The dimensions of the portion of the diverting sheave hanging underneath the ceiling in the lift well
- (d) 150 mm for compensating for gravity stopping distance and future repairs to the rope connections at counter weight and at the car or at the suspension points.

4.2.3.39 Bottom car clearance, goods lift

The bottom car clearance shall be maintained in such a way that the counterweight shall not come in contact with the ceiling or any part hanging underneath the ceiling, when the car completely rests on fully compressed buffers, provided the buffers are spring type mounted on solid concrete or steel bed.

In case of wooden buffers the bottom car clearance shall be maintained in such a way that the total downward travel of the car from the service level of the immediate floor near the pit, shall not be more than the top counterweight clearance, when the wooden buffers are completely crushed.

4.2.3.40 Top counterweight clearance, goods lift

The top clearance for the counterweight can be calculated taking into account the following and shall not be less than the sum of the following three items:

- (a) Car runby,
- (b) Compression of the buffer spring or height of the wooden block used as buffer, and
- (c) 150 mm to compensate for gravity stopping distance for counterweight and any future repairs to rope connections at the counterweight at the car ends or at the suspension points.

4.2.3.41 Runby for cars and counterweights, goods lift

The bottom runby for cars and counterweights shall not be less than 150 mm

4.2.3.42 Maximum bottom runby, goods lift

In no case the maximum bottom runby shall exceed 300 mm.

4.2.3.43 Overhead Height

The overhead height shall not be less than as shown in Table 8.4.2

4.2.3.44 Lift well dimensions

Recommended dimensions of lift well and its entrances are provided in Tables 8.4.3 to 8.4.6. These dimensions are primarily for architects and building planners for planning of lift well.

4.2.4 Landing Doors

4.2.4.1 Every landing, where there is access from the landing to the lift car, shall be fitted with a landing door. Such door shall be fitted with efficient interlocking or other devices so as to ensure that the door cannot be opened except when the lift car is at the landing and that the lift car cannot be moved away from the landing until the door is closed and locked. Where mid bar collapsible doors are used for landing entrance, they shall not be power operated.

4.2.4.2 Where landing doors are manually operated and no indicators are provided, vision panels of similar construction as in Sec 4.2.3.12 shall be provided.

4.2.4.3 No automatic fire door or shutter which operates by means of a fusible link or otherwise due to the action of heat shall be allowed in any landing opening or lift way enclosure of any lift, if such opening gives access to any exit from the building.

4.2.4.4 In case of passenger lifts, solid sliding doors shall preferably be provided for buildings above 6 storeys or 20 m in height. Solid swing doors may also be used where sliding space is not available parallel to the entrance door. Collapsible doors shall not be provided in case of buildings above 8 storeys or 26 m in height.

Table 8.4.2: Minimum Pit Depths, Overhead Heights and Machine Room sizes for Traction Lifts - Overhead Machines

Speed (m/s)	Up to 0.70	>0.70 ≤1.00	>1.00 ≤1.50	>1.50 ≤1.75	>1.75 ≤2.00	>2.00 ≤2.50	>2.50 ≤3.00	>3.00 ≤4.00
(i) Pit Depth, mm	1500	1500	1600	2150	2200	2500	3000	3200
(ii) Overhead Height, mm	4200	4250	4800	4800	5200	5400	-	-
(iii) Machine Room Depth, mm	D + 2000	D + 2000	D + 2000	D + 2500	D + 2500	D + 2500	D + 3000	D + 3000
(iv) Machine Room Width, mm	C + 1000	C + 1000	C + 1200	C + 1200	C + 1500	C + 1500	C + 1800	C + 1800

Notes:

1. C is lift well depth (mm) and D is lift well width (mm).
2. The total overhead height has been calculated on the basis of car height of 2300mm.
3. Dimensions of pit depth and overhead height may differ in practice as per individual manufacturer's design depending upon load, speed and drive. However, the pit depth and overhead height shall be such as to conform to the requirements of bottom clearance and top clearance in accordance with the accepted standard.

Table 8.4.3: Recommended Dimensions of Passenger & Service Lifts and Lift Wells

Capacity		Car Dimensions (mm)		Lift Well Dimensions (mm)		Entrance Size (mm)
Persons	Kg	Width	Depth	Width	Depth	
4	272	1100	700	1900	1300	700 (Min)
6	408	1100	1000	1900	1700	700 (Min)
8	544	1300	1100	1900	1900	800
10	680	1300	1350	1900	2100	800
13	884	2000	1100	2500	1900	900
16	1088	2000	1300	2500	2100	1000
20	1360	2000	1500	2500	2400	1000

Notes :

1. In case of manually operated doors, clear entrance will be reduced by the amount of projection of handle on the landing.
2. All dimensions given above for lifts having centre opening power operated doors with counterweight at rear, are recommended dimensions primarily for architects and building planners. Any variations, mutually agreed between the manufacturer and purchaser, are permitted. However variation in:
 - (i) Car inside dimensions shall be within the maximum area limits specified in accordance with Table 8.4.1
 - (ii) Entrance width on the higher side is permitted.
 - (iii) Entrance width may be reduced up to a maximum of 100 mm subject to a minimum of 700 mm.

Table 8.4.4: Recommended Dimensions of Goods Lifts and Lift Wells

Load (Kg)	Car Dimensions (mm)		Lift Well Dimensions (mm)		Entrance Size (mm)
	Width	Depth	Width	Depth	
500	1100	1200	1900	1500	1100
1000	1400	1800	2300	2100	1400
1500	1700	2000	2600	2300	1700
2000	1700	2500	2600	2800	1700
2500	2000	2500	2900	2800	2000
3000	2000	3000	2900	3300	2000
4000	2500	3000	3400	3300	2500
5000	2500	3600	3400	3900	2500

Notes :

- (i) The width of lift machine room shall be equal to be lift well width subject to a minimum of 2500 mm.
- (ii) Clear entrance width is based on vertical lifting car door and vertical bi-parting landing doors. For collapsible mid-bar doors the clear entrance width will be reduced by 200 mm (maximum 1800 mm).
- (iii) All dimensions given above are recommended dimensions primarily for architects and building planners. Any variations mutually agreed between the manufacturer and the purchaser are permitted. However, variation in car inside dimensions shall be within the maximum area limits in accordance with Table 8.4.1.
- (iv) For dimensions of pit depth and overhead height, consider data shown in Table 8.4.2.

Table 8.4.5: Recommended Dimensions of Hospital Lifts and Lift Wells

Capacity		Car Dimensions (mm)		Lift Well Dimensions (mm)		Entrance Size (mm)
Persons	Kg	Width	Depth	Width	Depth	
12	1020	1000	2400	1800	3000	800
20	1360	1300	2400	2200	3000	1200
26	1768	1600	2400	2400	3000	1200

Notes :

- (i) In the case of manually operated doors, clear entrance will be reduced by the amount of projection of handle on the landing door.
- (ii) Although 15 persons capacity lift is not standard one, this is included to cover lifts of smaller capacity which can be used in small hospitals.
- (iii) All dimensions given above are recommended dimensions primarily for architects and building planners. Any variations mutually agreed between the manufacturer and the purchaser are permitted. However, variation in car inside dimensions shall be within the maximum area limits in accordance with Table 8.4.1.
- (iv) For dimensions of pit depth and overhead height, consider data shown in Table 8.4.2.

Table 8.4.6: Recommended Dimensions of Dumb Waiter and Lift Wells (for speeds up to 0.5 m/s)

Load (Kg)	Car Inside Dimensions (mm)			Lift Well Dimensions (mm)		Entrance Size (mm)
	Width	Depth	Height	Width	Depth	
100	700	700	800	1200	900	700
150	800	800	900	1300	1000	800
200	900	900	1000	1400	1100	900
250	1000	1000	1200	1500	1200	1000

Notes: Entrance width is based on assumption of provision of vertical bi-parting doors (no car door is normally provided).

4.2.5 Guide Rails

4.2.5.1 Guide rails shall be made of high quality steel, straight and of proper thickness. Where the nature of processes carried on in the building gives rise to acid fumes or corrosive substances the steel rails shall be treated for corrosion resistance.

4.2.5.2 Lift car and counterweight guide rails shall be continuous throughout the entire length right from the bottom of the pit floor to the top most floor served plus additional length as may be required for operation of safety against over run. They shall be provided with adequate brackets or equivalent fixing devices of such design and spacing that the rails shall not deflect more than 4 mm under normal operations

4.2.5.3 For passenger and goods lifts having a rated speed of 0.5 m/s or more, the car guide rails shall have working surfaces machined and smooth.

4.2.6 Lift Pits

4.2.6.1 A lift pit shall be provided at the bottom of every lift well. The minimum depth of lift pit shall be as shown in Table 8.4.2.

4.2.6.2 Lift pits shall be of sound construction and shall be maintained in dry and clean condition. Where necessary, provision shall be made for permanent drainage.

4.2.6.3 Lift pits having depth more than 1.6 m shall be provided with a suitable descending arrangement to reach the lift pit.

4.2.6.4 Light points shall be provided in all lift pits for facility of repair and maintenance works.

4.2.6.5 In case of a group of two or more lift wells, arrangements shall be provided to allow inspection of a lift pit through the adjoining one.

4.2.7 Buffers

4.2.7.1 Buffers of spring or oil shall be used for safety. Buffers shall be fitted under the lift car and counterweight directly or on the pit floor with suitable concrete or steel foundation. Oil resistant rubber buffers may be used with lifts having a rated speed not exceeding 0.25 m/s. Lifts having rated speed in excess of 0.25 m/s and up to and including 1.0 m/s, spring or oil buffers shall be used. For lifts having rated speed more than 1.0 m/s, only oil buffers shall be used. Wooden blocks suitably treated may also be used for service lifts for speeds up to 0.5 m/s. Buffers shall be located symmetrically with reference to the vertical centre line of the car/counterweight with a tolerance of 50 mm.

4.2.7.2 The minimum stroke of oil buffers shall be such that the car or the counterweight on striking the buffers at 115 percent of rated speed shall be brought to rest with an average retardation of not more than 10 m/s^2 .

4.2.7.3 When buffers are struck with an initial speed of less than 115 percent of the rated speed, the peak retardation shall not exceed 25 m/s^2 for a duration of more than 0.04 second, with any load in the car ranging from 75 kg to the rated load.

4.2.8 Machine Room and Overhead Structures

4.2.8.1 The lift machine room shall only be used for housing lift machinery, controller and other associated apparatus and equipment. No other services or equipment shall be accommodated therein. If motor-generators for controlling speed of multi-voltage or variable voltage machines, secondary sheaves, pulleys, governors, floor selecting equipment and other associated equipment are installed in an adjoining room, this room shall also be reserved for exclusive use of lift equipment.

4.2.8.2 Lift machine room and other associated equipment rooms shall be fire proof, weather proof and adequately lighted. Means to prevent spread of fire or smoke from machine room into lift well shall be provided. Machine room shall have permanent ventilation opening direct to the open air having a free area not less than 0.1 m^2 per lift. Ambient temperature of machine room shall be maintained between $+5^\circ\text{C}$ and $+40^\circ\text{C}$.

4.2.8.3 The height of the machine room shall not be less than 2.30 m throughout under the lifting beam (trolley beam) to allow any portion of equipment to be accessible and removable for repair and replacement. An overhead trolley beam of steel construction of adequate strength shall be provided in the machine room, for movement of equipment during installation.

4.2.8.4 The machine room shall be adequately sized and shall have sufficient floor area required for easy access to all parts of the machines and equipment located therein for purposes of inspection, maintenance or repair. Clearance space of 1 m shall be provided on those sides of control panels where maintenance is required to be carried out while the panel is energized, otherwise 0.5 m clearance space may be provided. For planning purposes the lift machine room size can be as shown In Table 8.4.2

4.2.8.5 The room shall be kept closed, except to those who are concerned with the operation and maintenance of the equipment. When the electrical voltage exceeds 220/230 V dc, a danger notice plate shall be displayed permanently on the outside of the door and on or near the machinery.

4.2.8.6 Machine room floor shall not have holes/punches in it except for necessary small openings for passage of ropes cables etc. If any machine room floor or platform does not extend to the enclosing walls the open sides shall be provided with hand rails or otherwise suitably guarded.

4.2.8.7 All machines, pulleys, over speed governors and similar units shall be securely fixed on the machine room floor.

4.2.8.8 Adequate artificial light shall be provided in the machine room. A 15 amps 3 pin power outlet for hand operated tools and a 5 amps 2 pin electrical outlet for portable hand lamp set shall be provided in the machine room.

4.2.8.9 Access to Machine Room

- (a) The machine room shall be provided with a direct, independent and convenient access. Access to a machine room above a lift well may be either from the roof or by an internal staircase.
- (b) Machine room floor may be provided with a trap door. When access to the machine room is provided through the trap door, the size of the trap door shall not be less than 1.0 m x 1.0 m otherwise it may be 0.5 m x 0.5 m. Trap doors shall be hinged, opening into the machine room, of sound construction, balanced and tightly secured to minimize noise travel. Hand rails shall be provided around trap door opening.
- (c) Where a machine room entrance is less than 1.5 m above or below the adjacent floor or roof surface, a substantial permanently attached ladder may be used.
- (d) Where the machine room entrance is 1.5 m or more above or below the adjacent floor or roof surface, access shall be provided by means of standard stairs.
- (e) Access to a machine room in a basement may be provided from a corridor.
- (f) Access to a machine room via the lift well shall be prohibited.
- (g) Emergency exit shall be provided in case of large machine room having four or more lifts.

4.2.8.10 The space at secondary level in which the overhead pulleys, overspeed governors and similar machinery are housed shall have a clear height of at least 1.2 m. Where practicable, it shall have a substantial platform or floor and be provided with permanent and adequate artificial illumination. Safe and convenient access to secondary level shall be provided. Means of access between a secondary floor and machine room may be a ladder. Hand rails shall be provided at platform and access to floor.

4.2.9 Hall Buttons, Hall Lanterns and Special Signs

4.2.9.1 Hall buttons

- (a) Each landing shall have hall call buttons to register call for lift service for upward or downward movements. The centre line of the hall call buttons shall be at a nominal height of 1 m above the floor.

- (b) Direction buttons, exclusive of borders, shall be a minimum of 18 mm in size, raised, flush or recessed. Visual indication shall be provided to show each call registered and extinguished when the call is answered. Depth of flush or recessed button when operated shall not exceed 10 mm.

4.2.9.2 Hall lantern

- (a) Where lifts are installed in totally enclosed wells, a visual signal shall be provided at each lift well entrance indicating to the prospective passenger the car answering the call and its direction of travel. An audible signal may also be included.
- (b) The visual signal may be in the form of digital lift position indicator or directional indicator. The visual signal for each direction/lift position shall be a minimum of 62 mm in size and visible from the proximity of the hall call buttons.
- (c) The centre line of the fixture shall be located at a minimum of 1.8 m from the floor.

4.2.9.3 Special signs

- (a) Door Jamb Marking: The floor designation shall be provided at each lift well entrance on both sides of jamb visible from within the car and the lift lobby at a height of 1.5 m above the floor. Designations shall be on a contrasting background 50 mm high and raised 0.75 mm.
- (b) Applied plates permanently attached shall be acceptable. In case of a completely enclosed lift well a notice with the word 'Lift' shall be placed outside of each landing door. Electric light shall be provided such that this sign remains visible even if the surroundings are dark.
- (c) A permanent warning sign shall be installed immediately above each hall push button station on each floor reading: IN FIRE EMERGENCY, DO NOT USE LIFT. USE EXIT STAIRS. This sign shall be in letters not less than 12 mm high.

The warning sign may consist of incised, inlaid or embossed letters on a metal, wood, plastic or similar plate securely and permanently attached to the wall, or letters incised or inlaid directly into the surface of the material forming the wall.

4.2.10 Electrical Wiring and Apparatus

4.2.10.1 Construction, installation and maintenance of all electrical supply lines and apparatus in connection with lift installation shall be done with proper protection so that there may be no danger to persons there from. No bare conductor shall be used in any lift car. Installation of electrical wiring shall conform to the provisions of Chapter 1.

4.2.10.2 Electrical circuits for lights and ventilation fans, and supply to 3 pin and 2 pin socket outlets shall be controlled by a separate main switch or circuit breaker, and shall be independent of machinery power supply such that lighting circuits remain alive when power to machinery is interrupted.

4.2.10.3 Suitable cautionary notice shall be affixed near every motor or other apparatus in which energy used is at a voltage exceeding 220 volts.

4.2.10.4 Travelling cable

- (a) Circuits which supply current to the motor shall not be included in any twin or multi-core travelling cable used in connection with the control and safety devices.
- (b) For building 10 storeys (33 m) or less in height, a travelling cable which incorporates conductors for the control circuits shall be separate and distinct from that of lighting and signaling circuits. In case of buildings more than 10 storeys or 33 m in height or where high speed (1.50 m/s or more) lifts are employed, a single travelling cable for lighting and control circuits may be permitted, provided that all conductors are insulated for the maximum voltage in the cables.

4.2.10.5 Supply cables and switches

- (a) Each lift shall be provided with a main switch or circuit breaker of a capacity determined by the lift manufacturer and the incoming supply cable shall terminate in this switch. For a single lift, this switch shall be fixed adjacent to the machine room entrance. In a machine room common to more than one lift, each main switch shall be conveniently situated with respect to the lift it controls. Switches and fuses (which may form part of a distribution switch board) shall be provided for isolating the supply power to machine room.
- (b) Where a supply cable serves more than one lift, a diversity factor may be used for the determination of conductor size. The actual diversity factor to be adopted shall be decided by the lift manufacturer.

4.2.10.6 Earthing

All electrical machinery/equipment viz. electric motor, winding machine, control panel etc. which normally carry mains current shall be properly connected to the earthing system. Similarly all metallic cases, covers of door interlocks, door contacts, call and control buttons, stop buttons, car switches, limit switches, junction boxes and similar electrical fittings which normally carry only the control current shall also be properly connected to the earthing system. All earthing terminal and earthing conductors in this regard shall conform to the requirements of Chapter 1 Part 8.

4.3 Design Considerations

4.3.1 Number of Lifts and Capacity

4.3.1.1 The number of lifts, car capacity and speed of the lift shall be selected to have the most effective lift system. The lift system shall be able to handle adequate number of passengers during peak hours and at the same time Interval and Travel Time shall be within reasonable limits.

4.3.1.2 Average Interval shall not be more than shown in Table 8.4.7. Travel time shall not exceed 150 seconds.

4.3.1.3 The passenger handling capacity (H) of a lift system for different occupancies in terms of the number of passengers to be handled in the building in a five minute peak period shall not be less than that indicated in Table 8.4.7.

4.3.1.4 For the purpose of population estimation, the density of people shall be based on the actual number of occupants, but in no case less than those specified in Table 8.4.8. The occupant load of a mezzanine floor shall be taken into account for working out the population for a particular floor to which the mezzanine floor discharges its loads.

4.3.1.5 Car speed and acceleration

The car speed for the different types of lifts in different occupancies shall normally be as given in Table 8.4.9. A higher or lower speed lift may be used in special cases when conditions warrant use of such lifts. The car acceleration or deceleration shall not be more than 1.2 m/s^2 .

Table 8.4.7: Recommended Interval and 5-minute Handling Capacity for Different Occupancy

Type of Occupancy	Interval (Sec)	5-min. Handling Capacity (H) %	Type of Occupancy	Interval (Sec)	5-min. Handling Capacity (H) %
<u>Office</u>			<u>Apartments (contd.)</u>		
Diversified Tenancy			Middle Income Apartments	60-80	6-8
High Quality	15-29	12	Low Income Apartments	80-120	10-11
Standard Quality	30-39	14	Dormitories, Halls of Residence	60-80	10-11
Single Tenancy			<u>Hospitals</u>		
High Quality	15-24	14	Private Hospital	50-60	12
Standard Quality	25-35	16	General Hospital	60-70	14
<u>Hotels and Motels</u>			Long term Nursing Facilities	60-70	8
High Quality	30-50	12-15	Educational Institutions	40-50	25
Standard Quality	50-70	10-12	Buildings with Assembly Facilities	40-50	15
<u>Apartments</u>			Shops and stores	40-50	5
High Cost Apartments	50-70	5-7			

Table 8.4.8: Occupant Load for Estimation of Population

Type of Occupancy	Population Factor	Type of Occupancy	Population Factor
<u>Office</u>		<u>Apartments (contd.)</u>	
Diversified Tenancy		Middle Income Apartments	2 people per bedroom
High Quality	14-23 m ² net usable area per person ^a	Low Income Apartments	2.5-3 people per bedroom
Standard Quality	10-12 m ² net usable area per person	Dormitories, Halls of Residence	20 m ² net usable area per person
Single Tenancy		<u>Hospitals</u>	
High Quality	12-19 m ² net usable area per person	Private Hospital	3 people per bed
Standard Quality	8-10 m ² net usable area per person	General Hospital	3-4 people per bed
<u>Hotels and Motels</u>		Long term Nursing Facilities	1.75 people per bed
High Quality	1.3 people per room	Educational Institutions	4 m ² per student
Standard Quality	1.7 people per bedroom	<u>Buildings with Assembly Facilities</u>	
Conventions	1.9 people per bedroom	With fixed or movable seats and dance floor	0.60 m ² per person ^b
<u>Apartments</u>		Without seating facilities including dining rooms	1.5 m ² per person ^b
High Cost Apartments	1.5 people per bedroom	Shops and stores	2 m ² of net selling area ^c

Notes :

- (i) Net usable area = gross area less lift shaft and lobby space, mechanical space, columns, toilets, corridor around core, air-conditioning machinery space.
- (ii) Population estimation shall be based on gross area (plinth area or covered area). The gross area shall include, in addition to the main assembly room or space, any occupied connecting room or space in the same storeys or in the storey above and below, where entrance is common to such rooms and spaces and they are available for use by the occupants of the assembly place. No deductions shall be made in the gross area for corridors, closets or other subdivisions, the area shall include all space serving the particular assembly occupancy.
- (iii) Net selling area is area open to the public.

Table 8.4.9: Car Speed for Lift in Different Kinds of Usage

Building Type	Rise (m)	Minimum Car Speed (m/s)	Building Type	Rise (m)	Minimum Car Speed (m/s)
Office Building	0 to 40	2.0	Hospital (contd.)	31 to 40	1.6
	41 to 70	2.5		41 to 55	2.0
	71 to 85	3.6		56 to 75	2.5
	86 to 115	4.0	Apartments	Above 75	3.6
	Above 115	5.0		0 to 25	0.63
Hotels	0 to 40	2.0		26 to 40	1.0
	41 to 70	2.5	Stores	41 to 60	1.6
	71 to 85	3.6		Above 60	2.0
	86 to 115	4.0		0 to 30	1.0
	Above 115	5.0		31 to 45	1.6
Hospital	0 to 20	0.63		46 to 60	2.0
	21 to 30	1.0		Above 60	2.5

4.3.1.6 Handling capacity and interval

- (a) The handling capacity, for incoming/up-peak passenger, shall be calculated by the following formula:

$$H = \frac{300 \times Q \times N \times 100}{P \times T}$$

Where,

H = passenger handling capacity of the lift system during five minute peak period, expressed as the percentage of the estimated total population handled.

Q = average number of passengers carried in each car per trip,

N = number of lifts in the system,

P = total population to be handled during peak period (it is related to the area for which a particular bank of lift serves)

T = average round trip time in seconds, that is, the average time required by each lift in taking one full load of passengers from ground floor discharging them in various upper floors and coming back to ground floor for taking fresh passengers for the next trip;

The value of Q depends on the dimension of the car. It may be noted that the car is not loaded always to its maximum capacity during each trip and therefore, for calculating T and H the value of Q shall be taken as 80 percent of the maximum capacity of the car.

T is the sum of the time in seconds required in the following process:

- (i) Time for entry of passengers on the ground floor or lowest lift lobby;
- (ii) Time for exit of the passengers on each floor of discharge;
- (iii) Door operation time (opening and closing) and car start time on each floor the lift stops, including ground floor;
- (iv) Acceleration and deceleration periods;
- (v) Stopping and leveling periods;
- (vi) Periods of full rated speeds between stops going up and
- (vii) Periods of full rated speeds between stops going down.

- (a) The average Interval shall be calculated by the following formula:

$$I = \frac{T}{N}$$

Where, I = Interval of availability of lifts on the lowest lift lobby.

4.3.2 Shape and Size of Lifts

4.3.2.1 Careful analysis shall be performed during selection of shape and size of lifts so as to get full advantage of its shape for the most effective use of lifts and building space.

4.3.2.2 The dimensions of the car platform shall be such that the car will not exceed its rated load when packed full. Net inside area of the lift car shall be as per Sec 4.2.3.1.

For the same platform area, a lift having higher width to depth ratio can accommodate more passengers and takes less time for passenger transfer. The width of the car is determined by the width of the entrance and the depth of the car is regulated by the loading.

4.3.3 Location and Arrangement of Lifts

4.3.3.1 A thorough investigation shall be carried out for assessing the most suitable location for lift(s) while planning the building. It shall take into account future expansions, if any.

4.3.3.2 The lifts shall be easily accessible from all entrances to the building. For maximum efficiency, they shall be grouped near the centre of the building. Walking distance from the lift to the farthest office or suite shall not exceed 60 m.

4.3.3.3 Arrangement of lifts

- (a) When more than one lifts are installed in a group, they shall be arranged side by side or in two rows facing each other. Separation of lifts in the group shall be avoided.

- (b) The lift lobby in front of lifts shall be wide enough to allow sufficient space for waiting passengers and proper vision of hall button and hall lanterns. Figures 8.4.2 to 8.4.6 give acceptable arrangements of lifts in a group with acceptable space for waiting passengers. More space shall be allowed in front of the lifts in the main floor than in the upper floors.
- (c) It is preferable that the lift lobby is not used as a thoroughfare, but when absolutely needed the lift lobby shall be wider enough to take into account of the space for people who are moving.

4.3.4 Location of Machine Room

4.3.4.1 The machine room shall, as far as practicable, be placed immediately above the lift well.

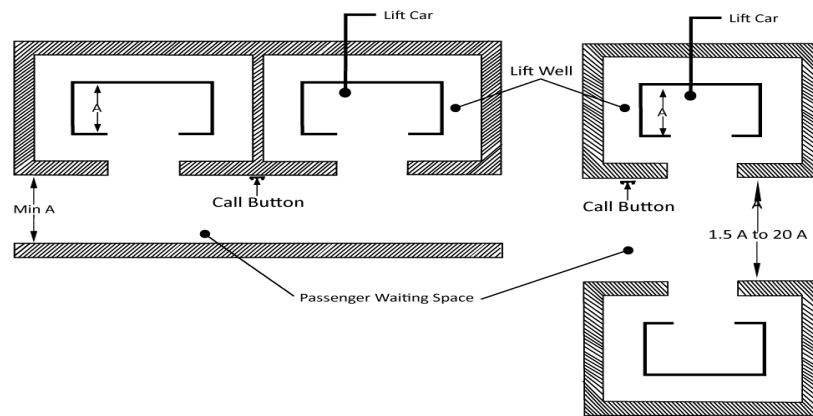


Figure 8.4.2 Two car arrangement

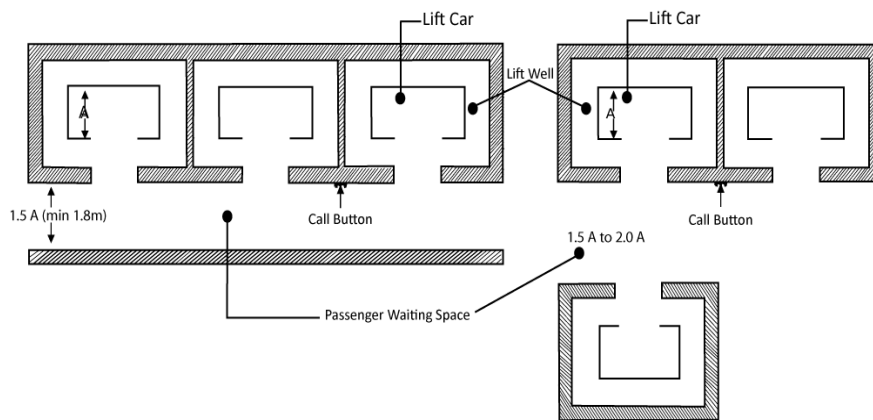


Figure 8.4.3 Three car arrangement

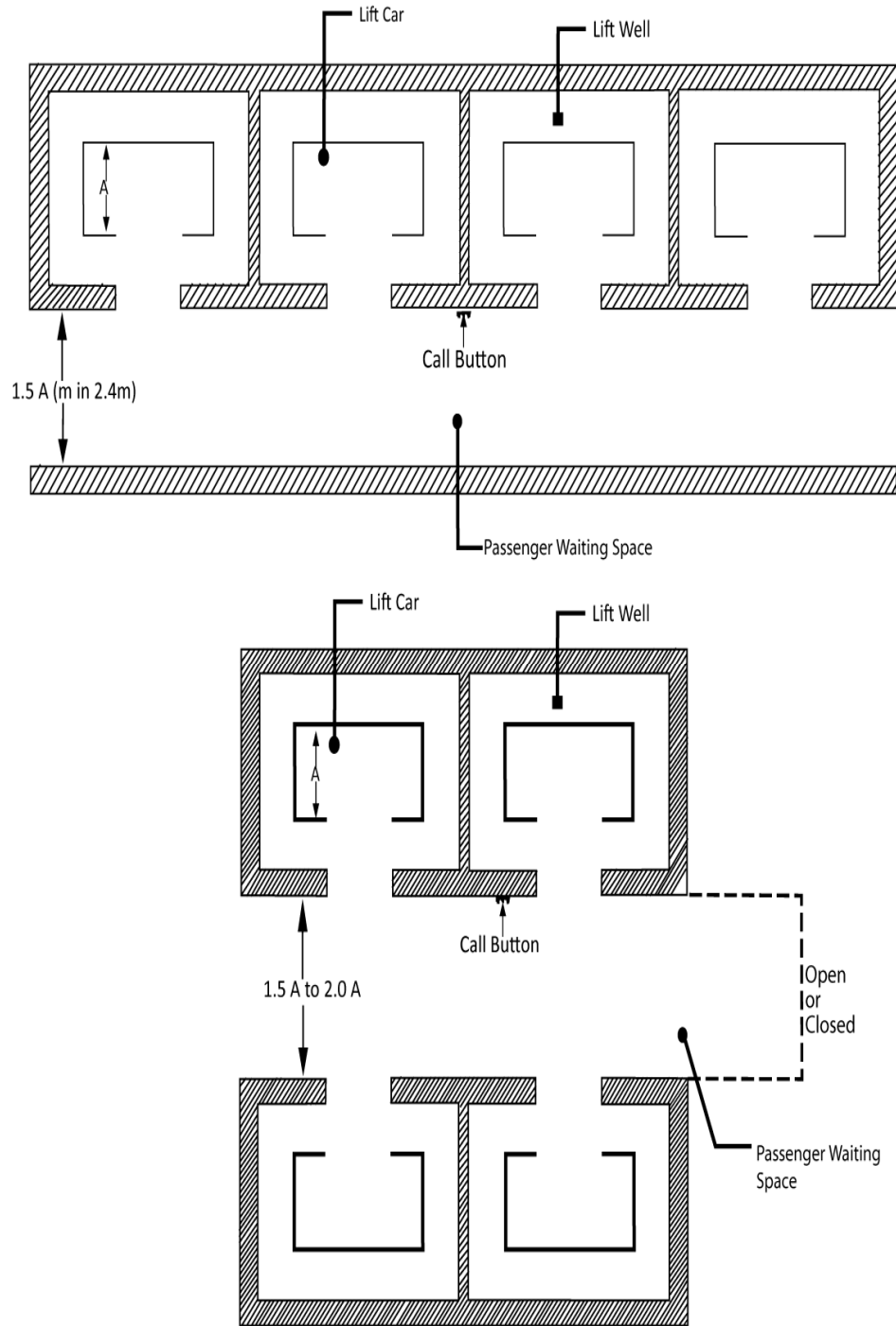


Figure 8.4.4 Four car arrangement

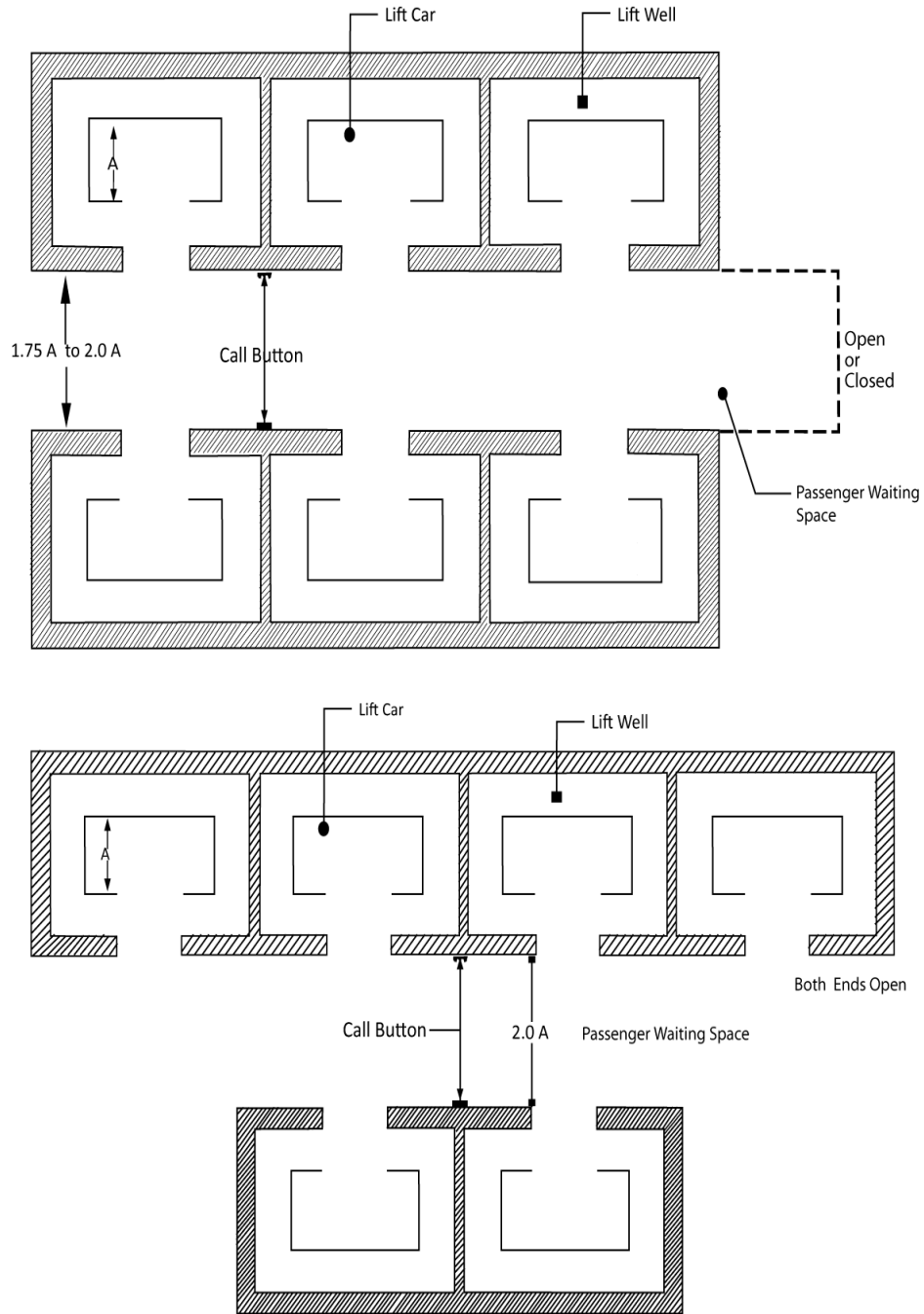


Figure 8.4.5 Six car arrangement

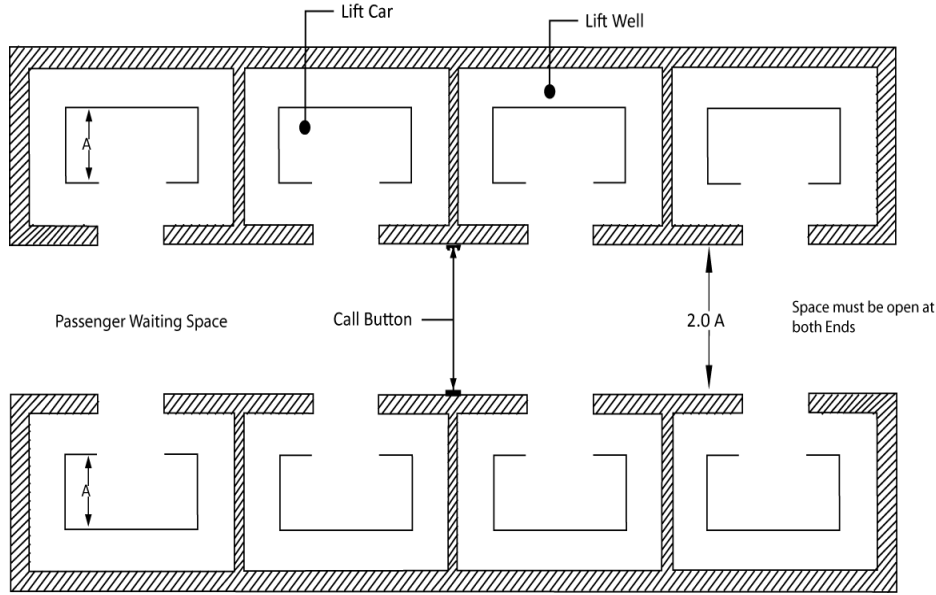


Figure 8.4.6 Eight car arrangement

4.3.4.2 If a machine room on the lift well is impracticable for architectural or other reasons, the machine room may be placed below the lift well or in the basement, keeping adequate safety provisions. If the lift machine room is located in the basement, it shall be separated from the lift well by a separation wall. Alternatively, machine room less lift can be considered for installation.

4.3.4.3 High speed lifts with gearless machine shall, in all cases, have machine room above the lift well.

4.3.4.4 Machine room shall not be located adjacent to or above sleeping rooms (bed rooms) in residential and hotel buildings; and patients' rooms, intensive care rooms and operation theatres of hospital/health care buildings.

4.3.5 Structural Considerations

4.3.5.1 Lift well enclosures, lift pits, machine rooms and machine supports, besides conforming to the essential requirements in Sec 4.2, shall form part of the building construction and comply with the lift manufacturer's drawings.

4.3.5.2 Machine room

Machine room floor shall be strong enough to support the heaviest component of lift machinery and shall be designed to carry a load of not less than 500 kg/m^2 over the whole area and also any load which may be imposed thereon by the equipment used in the machine room or by any reaction from any such equipment during periods of both normal operation and repair.

4.3.5.3 The total load on overhead beams and their supporting structural members shall be assumed to be equal to the dead load of slabs including load of all equipment resting on the beams plus twice the minimum load suspended from the beams.

4.3.5.4 The deflection of the overhead beams under the minimum static load calculated in accordance with Sec 4.3.5.3 shall not exceed $\frac{1}{1500}$ of the span.

4.3.5.5 Beams at all other floor slabs which correspond to the beam at machine room floor shall also be made stronger to take the reaction from the guides when the lift is made to stop consequent to the breaking of the wire ropes or the application of the safety device.

4.3.5.6 Suitable lifting beams may be provided immediately below the machine room ceiling for carrying the tackle to facilitate lifting of any heavy part of a heavy lift. For lower capacity lifts, suitable suspension hooks may be provided.

4.3.5.7 The roof of the machine room shall be strong enough to take up the pulley which could be used for lifting up parts of the lift machinery for inspection and repair.

4.3.5.8 The equivalent dead loads imposed upon the building by the lift installations shall be shown on the lift manufacturer's drawing so that the architect/engineer may make provisions accordingly.

4.3.6 Control System

4.3.6.1 The control of operation of the lift system, leveling, door opening and closing, response to hall calls etc. shall be fully automatic. All control equipment shall be efficient and fail-safe.

4.3.6.2 The control system shall be capable of accelerating the car smoothly to full running speed and stopping the lift with smooth retardation.

4.3.6.3 Variation in speed of the lift between no load and full load conditions shall not be more than plus or minus five percent. The control system shall be capable of correcting any tendency to over speed or under speed. The control system shall have safety device(s) to stop the lift car if its running speed exceeds its rated speed by ten percent.

4.3.6.4 It shall have facility to level or re-level the lift car within ± 13 mm. The leveling system shall be fully automatic and shall correct for over travel or under travel and rope stretch. The car stopping and leveling system shall be unaffected by external influences like variation in load, temperature, rope elongation etc.

4.3.6.5 Closing and opening of car doors and landing doors shall be fully automatic and shall operate in full synchronization with one another. Door opening and closing operations shall be so controlled as to ensure proper safety of passengers.

4.3.6.6 Door opening and closing time and door hold open time shall be automatically controlled to get minimum transfer time in any landing. For larger installations, transfer times shall be independently adjustable to suit the requirements of the building as well as the characteristics of the traffic.

4.3.6.7 Independent door closing push button shall be provided in the lift car to allow instant door closing. Similarly door opening push button shall be provided in the lift car to reverse the closing motion of the doors or hold them open.

4.3.6.8 When there are conditions that particularly affect the safety of passengers, the closing of doors shall only be effective by the continuous pressure of push buttons in the lift car or landings.

4.3.6.9 Each lift shall have key operated switch to transfer from normal passenger control to a car preference control. During car preference control the operation of the lift shall be from the car only and the doors shall remain open until a car call is registered for a floor designation. All landing calls shall be bypassed and car position indicators on the landings for this lift shall not be illuminated.

4.3.6.10 Provisions shall be made in the control system to take any car out of service still maintaining the controlled operation of the remaining cars of a group of cars required for passenger traffic. It is essential that such provision shall not stop the fireman's control from being operative in the event of the lift being designated as a fireman's lift.

4.3.6.11 When required, fire switch shall be provided in the control system as per Sec 4.2.2.2.

4.4 Escalators

4.4.1 General

4.4.1.1 Escalators shall be located in the main line of circulation and in such a way that most persons entering the building can see it. Care shall be taken to eliminate interference to the traffic movement.

4.4.1.2 Escalators shall discharge into an open area with no turns or change of direction necessary. Ample space for people must be provided at the entry and exit landings of an escalator, space between the newel and the nearest obstruction in front of the escalator shall be a minimum of 3 m.

4.4.1.3 If an unloading area is restricted, such restrictions as doors or gates shall be interlocked with the escalator to insure that the restriction is removed before the escalator can be run.

4.4.1.4 The escalator shall have provision to run in both upward and downward directions. However it shall not run in one direction for one trip and reversed for the next. Starting, stopping or reversal shall be controlled only by an attendant and with the assurance that no passenger is riding at that time.

4.4.1.5 Minimum head room above the escalator (minimum vertical clearance between the line of step nosing and lowest edge of ceiling opening) shall not be less than 2.3 m.

4.4.1.6 Near the place of escalator installation, one lift with wheel chair facility shall be installed to facilitate vertical movement of disabled persons.

4.4.2 Essential Requirements

4.4.2.1 Angle of incline of the escalator shall not be more than 30° from the horizontal. In particular cases, an angle of incline up to 35° may be permitted for escalators having a vertical rise not more than 6 m.

4.4.2.2 The speed of the escalator, measured along the incline shall not be more than 0.63 m/s for 30° angle of incline and 0.50 m/s for 35° angle of incline.

4.4.2.3 Balustrades

- (a) Escalators shall be provided on each side with solid balustrades. On the step side (interior panel), the balustrades shall be smooth and substantially flush except for protective molding parallel to the run of the steps. Vertical moldings that cover joints of panels shall be properly beveled and shall not project more than 6.5 mm. Gaps between interior panels of the balustrade shall not be wider than 4mm.
- (b) The width between balustrades, measured on the incline up to a point 680 mm vertically above the nose line of the steps, shall not be less than the width of the step. It shall not exceed the width of the step by more than 330 mm with a maximum of 165 mm on either side of the escalator.
- (c) There shall be no abrupt changes in the width between the balustrades on the two sides of the escalator. Where a change in width is unavoidable, such change shall not exceed 8 percent of the greater width. In changing the direction of the balustrades resulting from a reduction in width the maximum allowable angle of change in balustrades shall not exceed 15 degrees from the line of the escalator travel.
- (d) The balustrade interior paneling shall have adequate mechanical strength and rigidity. When a force of 500 N is applied at an angles on gap greater than 4 mm and no permanent deformation.
- (e) The use of glass for balustrade interior panelling is permitted, provided it is splinter free one layer safety (tempered) glass and has sufficient mechanical strength and rigidity. The thickness of the glass shall not be less than 6 mm.

4.4.2.4 The clearance on either side of the steps between the steps and the adjacent skirt guard shall not be more than 5 mm and the sum of the clearances on both sides shall not be more than 6 mm.

4.4.2.5 Where the intersection of the outside balustrade (deck board) and the ceiling or soffit is less than 600 mm from the centre line of the handrail, a solid guard shall be provided in the intersecting angle of the outside balustrade (deck board) and the ceiling or soffit. The vertical face of the guard shall project at least 360 mm horizontally from the apex of the angle.

4.4.2.6 Handrails

- (a) Each balustrade shall be provided with a handrail moving in the same direction and at the same speed as the steps.
- (b) Each moving handrail shall extend at normal handrail height not less than 300 mm beyond the line of points of comb plate teeth at the upper and lower landings.
- (c) Hand or finger guards shall be provided at points where the handrails enter the balustrade.
- (d) The horizontal distance between the centre lines of two handrails, measured on the incline, shall not exceed the width between the balustrades by more than 150 mm, with a maximum of 75 mm on either side of the escalator.

4.4.2.7 Step treads

- (a) The depth of any step tread in the direction of travel shall not be less than 400 mm and the rise between treads shall not be more than 220 mm.
- (b) The maximum clearance between step treads on the horizontal run shall be 4 mm.
- (c) The tread surface of each step shall be slotted in a direction parallel to the travel of the steps. Each slot shall not be more than 6.5 mm wide and not be less than 9.5 mm deep; and the distance from centre to centre of adjoining slots shall not be more than 9.5 mm.
- (d) Safety provision shall be installed in the system to stop the escalator when anything is stuck in the clearance between the step tread and the skirting.

4.4.2.8 Landings

Landings shall be made of anti-slip material.

4.4.2.9 Comb plates

There shall be comb plates at the upper and lower landings of every escalator. The comb plate teeth shall be meshed with and set into the slots of the tread surface. Comb plates shall be adjustable vertically. Safety provision shall be installed in the comb plate assembly so that the safety contact stops the escalator when anything is caught between the comb plate and the step.

4.4.2.10 Trusses

The truss shall be designed to sustain the dead and live loads of the steps and running gear in operation safely. In the event of failure of the track system it shall retain the running gear in its guides.

4.4.2.11 Step wheel tracks

These shall be designed to prevent displacement of steps and running gear if a step chain breaks.

4.4.2.12 Rated load

The escalator shall be selected in such a way that it does not exceed its rated load during operation. The rated load in kilogram on an escalator shall be computed by the following formula:

$$\text{Rated load} = 0.27WA \text{ kg}$$

Where,

W = width between the balustrades, mm ; and

A = horizontal distance between the upper and lower comb plate teeth , (m).

4.4.2.13 Design factor of safety

The factor of safety based on static load shall be at least the following:

(i) Trusses and all structural members including tracks	5
(ii) Driving machine parts:	
Made of steel or bronze	8
Made of cast iron and other materials	10
(iii) Power-transmission members	10
(iv) Step chain composed of cast-steel links thoroughly annealed	20

4.4.2.14 Driving machine, motor and brake

- (a) The driving machine shall be connected to main drive shaft by toothed gearing, a coupling, or a chain.
- (b) An electric motor shall not drive more than one escalator.
- (c) Each escalator shall be provided with an electrically released, mechanically applied brake capable of stopping the up or down travelling escalator with any load up to the rated load. The brake shall be located either on the driving machine or on the main drive shaft. Where a chain is used to connect the driving machine to the main drive shaft, a brake shall be provided on this shaft. It is not required that this brake be of the electrically released type if an electrically released brake is provided on the driving machine.

- (d) The braking system shall have provision to bring the escalator automatically to a smooth stop in the event of failure of electrical power or mechanical parts.
- (e) Speed Governor: A speed governor shall be provided, the operation of which shall automatically shut down the escalator in case of over speed or under speed, and prevent reversal of direction (up or down).
- (f) Adequate illumination shall be provided at all landings, at the comb plates and completely down all stair ways.
- (g) An emergency stop switch shall be located near the comb plate or in some obtrusive location.
- (h) All machinery spaces shall have access doors or panels for inspection and maintenance. These panels shall remain locked to prevent unauthorized access.
- (i) Reasonable ventilation shall be provided in machinery spaces.

4.4.2.15 Escalator capacity

- (a) For normal peak period, the recommended handling capacity for design purposes shall be taken as 3200 to 6400 persons per hour depending upon the width of the escalator.
- (b) The number of persons that may be theoretically carried by the escalator in 1 hour can be calculated as follows :
 - (i) For determination of theoretical capacity it is assumed that one step with an average depth of 0.4 m can carry 1 person for step width of 0.6 m, 1.5 persons for a step width of 0.8 m and two persons for step width of 1 m.
 - (ii) The theoretical capacity = $3600 \times (v \times k) / 0.4$

Where,

v = rated speed of escalator in m/s

k = 1, 1.5 or 2 for step width of 0.6 m, 0.8 m and 1 m respectively.

4.5 Moving Walks

4.5.1 Essential Requirements

4.5.1.1 Angle of incline of moving walks shall be no more than 15°. A moving walk may have sloping entrance and exit or level entrance and exit.

4.5.1.2 The operating speeds of moving walk at different inclinations and different entrance and exit conditions shall not be more than those given in Table 8.4.10

Table 8.4.10: Operating Speeds of Moving Walk (Based on 1000 mm Nominal Tread Width)*

Incline of Ramp on Slope	Maximum Speed with Level Entrance and Exit (m/s)	Maximum Speed with Sloping Entrance and Exit (m/s)
0 to 3°	0.9	0.9
Over 3 to 5°	0.9	0.8
Over 5 to 8°	0.9	0.7
Over 8 to 12°	0.7	0.65
Over 12 to 15°	0.7	0.63

* Higher tread width may be allowable on horizontal runs.

4.5.2 Balustrades

- Moving walks shall be provided on each side with solid balustrades. On the tread way side the balustrades shall be smooth and substantially flush.
- The width between balustrades, measured up to a point 680 mm vertically above the tread way, shall not be less than the width of the tread way. It shall not exceed the width of the tread way by more than 330 mm with a maximum of 165 mm on either side of the moving walk.
- There shall be no abrupt changes in width between the balustrades on the two sides of the moving walk. Where a change in width is unavoidable, such change shall not exceed 8 percent of the greater width. In changing the direction of the balustrades resulting from a reduction in width the maximum allowable angle of change in balustrades shall not exceed 15° from line of moving walk travel.

4.5.3 Handrails

4.5.3.1 Each balustrade shall be provided with a handrail moving in the same direction and at the same speed as the tread way. Only one handrail may be allowed in a moving walk when the slope of the walkway does not exceed 3°, operating speed is less than 0.35 m/s or the width is no more than 530 mm.

4.5.3.2 Each moving handrail shall extend at normal handrail height not less than 300 mm beyond the line of points of comb plate teeth at the upper and lower landings.

4.5.3.3 Hand or finger guards shall be provided at the point where the handrails enter the balustrade.

4.5.3.4 The horizontal distance between the centre lines of two handrails shall not exceed the width between the balustrades by more than 150 mm with a maximum of 75 mm on either side of the moving walk.

4.5.4 Tread Way

4.5.4.1 The tread surface of the tread way shall be slotted in a direction parallel to the direction of travel.

4.5.4.2 The clearance on either side of the tread way between the tread way and the adjacent skirt guard shall not be more than 5 mm and the sum of the clearances on both sides shall not be more than 6 mm. Safety provisions shall be kept in the system to stop the moving walk when anything is stuck in the clearance between the tread way and the adjacent skirt guard.

4.5.5 Landings

Landings shall be made of anti-slip material.

4.5.6 Comb Plates

4.5.6.1 There shall be comb plates at the entrance and exit of each moving walk. The comb plate teeth shall be meshed with and set into the slots in the tread surface. Safety provision shall be installed in the comb plate assembly so that the safety contact stops the moving walk when anything is caught between the comb plate and the tread.

4.5.6.2 An emergency stop switch shall be located near the comb plate or at some obtrusive location.

4.5.6.3 Adequate illumination shall be provided at comb plates.

4.6 Energy Conservation

4.6.1 General

Lifts, escalators and moving walks shall be designed and installed for efficient use of energy herein provided.

4.6.2 Equipment and Controls

4.6.2.1 Lift

All lifts shall be equipped with necessary sensors and controls to reduce energy usage. For this purpose following features shall be included in the lift system:

- (a) AC Variable-Voltage and Variable-Frequency (ACVVVF) drives for the car and door shall be used.
- (b) Energy efficient lamps shall be used inside the car and for displays.
- (c) The lifts shall operate in standby mode during off-peak periods.
- (d) The power side of the lift controller and other operating equipment such as car lights, display lights and ventilation fans shall be switched off when the lift has been inactive for more than five minutes.

4.6.2.2 Escalator

All Escalators shall be equipped with necessary sensors and controls to reduce energy usage. For this purpose following features shall be included in the lift system:

- (a) The escalator shall reduce speed and operate at lower speed when there is no passenger on the escalator for a period of a maximum of three (03) minutes.
- (b) The escalator shall shut down when no activity has been detected for a period of a maximum of fifteen (15) minutes.

4.7 Inspection and Certification

4.7.1 All new lifts, escalators and moving walks, after installation, shall be inspected and tested by the Authority before these are put into normal services. These shall not be brought into use unless the Authority is satisfied that the installations have been carried out as per provisions of this Code and tests indicate that all the safety devices operate satisfactorily. It shall be unlawful to operate any lift, escalator or moving walk without a current certificate of inspection issued by the Authority. Certificates shall not be issued when the conveyance is posted as unsafe pursuant to Sec 4.7.7.

4.7.2 All electrical lines, control lines and earthings of lift, escalator and moving walk systems shall be tested to determine whether these have been installed properly to meet the requirements of the machine and as per provisions of Chapter 1.

4.7.3 Testing: Tests shall be carried out to determine the operational and safety conditions of lifts, escalators and moving walks in accordance with the provisions of the sections as under:

4.7.3.1 Lift

Tests shall be conducted to ascertain that

- (a) the motor, brake control equipment and car leveling mechanism function properly,
- (b) the door operation is proper and door locking devices function properly,
- (c) the car raises and lowers rated load,
- (d) the car achieves at least the rated speed,
- (e) the lift motor can be overloaded up to a minimum of 10% above the rated capacity,
- (f) the safety gear stops the car with the rated load in case of over speed and/or over travel etc.,
- (g) the buffers function properly, and
- (h) the safety gear operate and keeps operation of the lift suspended in case of the lift car is loaded above its maximum capacity.

4.7.3.2 Escalator and moving walk

Tests on escalators and moving walks shall be conducted to ascertain that

- (a) the automatic control device functions properly and brings the escalator to a smooth stop in case of failure of any mechanical parts or electrical power,
- (b) the automatic safety protection for over speed, under speed and direction reversal functions properly
- (c) safety devices function properly and stops the escalator or moving walk when anything is caught between the comb plate and the treads or the skirting and the treads.
- (d) the handrail and steps or tread way travel at exactly the same speed.

4.7.4 A lift, escalator or moving walk, in which repair and/or maintenance work has been carried out shall also be put to the relevant tests as provided for in Sec 4.7.3.

4.7.5 After proper testing, the Authority shall issue certificate regarding suitability of the lift, escalator or moving walk for normal or regular service. A lift, escalator or moving walk shall be allowed to work only on issuance of this certificate.

4.7.6 The lift, escalator or moving walk shall be inspected periodically to ensure safety.

4.7.7 When an inspection reveals an unsafe condition and the Authority finds that the unsafe condition endangers human life, the Authority shall cause to be placed on such lift, escalator or moving walk, in a conspicuous place, a notice stating that such conveyance is unsafe. The owner shall see to it that such notice of unsafe condition is legibly maintained where placed by the Authority. The Authority shall also issue an order in writing to the owner requiring repairs or alterations to be made to such conveyance necessary to render it safe and may order the operation thereof discontinued until the repairs or alterations are made or the unsafe conditions are removed. A posted notice of unsafe conditions shall be removed only by the Authority and when satisfied that the unsafe conditions have been corrected.

4.8 Operation and Maintenance

4.8.1 The owner shall be responsible for the safe operation and maintenance of each lift, escalator or moving walk installation and shall cause periodic inspections, tests and maintenance to be made on such conveyances as required in this Section.

4.8.2 The lift, escalator or moving walk shall receive regular cleaning and lubrication of relevant parts, and adjustment and adequate servicing by authorized competent persons at such intervals as the type of equipment and frequency of service demand. In order that the lift, escalator or moving walk installation is maintained at all times in a safe condition, a proper maintenance schedule shall be drawn up in consultation with the machine manufacturer which shall be strictly followed.

4.8.3 In case of lift, periodic examination of wire ropes, components of landing and car doors, door interlocking mechanism, brakes, gears, components of safety gears, guides, rollers, channels etc. shall be carried out as recommended by the manufacturer. In no case shall the interval between such inspections exceed six months.

4.8.4 Grooves of drums, sheaves and pulleys of lifts shall also be examined when rope replacement is made. If necessary, the drums, sheaves or pulleys shall be properly re-machined.

4.8.5 In case of escalators and moving walks, periodic examination of balustrades, handrail, tread way, tread way interconnection, comb plates speed governor, drives, chains, non-reversal device, brakes, gears etc. shall be carried out as recommended by the manufacturer, but in no case the interval shall exceed six months.

4.8.6 Sundry Precautions

4.8.6.1 Adequate precaution shall be taken to guard against any possibility of a lift being operated by unauthorized persons. Precautions shall also be taken to prevent a lift from being operated by any person when it is not intended for use.

4.8.6.2 No person shall remain in the pit while the lift is working. Adequate precautions shall be taken to protect persons working in the pit from accidental contact with the counter weight.

4.8.6.3 While the lift is under examination or repairs, suitable steps shall be taken to ensure that the lift is not operated inadvertently by a person in such a manner as may endanger the safety of persons working in the lift.

4.8.6.4 No such explosive or other inflammable material shall be carried in the lift car as may endanger the safety of persons and property.

4.9 Related Appendix

Appendix L Format for Particulars of Lifts, Escalators and Moving Walks.

PART VIII
Chapter 5
Water Supply

5.1 Purpose and Scope

5.1.1 The purpose of this Chapter of the Code is to provide minimum standards for the design, installation and maintenance of water supply and distribution system within a building and its premises.

5.1.2 The regulations of this Chapter also provide guidelines for water requirements for different classes of buildings according to their occupancy classification.

5.1.3 The provisions stated herein do not cover the requirements of water supply for industrial plants and process, municipal uses, viz. street washing, street hydrant, etc.

5.2 Terminology

This Section provides an alphabetical list of the terms used in and applicable to this Chapter of the Code. In case of any conflict or contradiction between a definition given in this Section and that in Part 1 of the Code, the meaning provided in this Section shall govern the interpretation of the provisions of this Chapter.

ACCESSIBLE	When applied to a fixture, appliance or equipment shall mean having access thereto, but which may require the removal of an access panel or similar obstruction; "readily accessible" shall mean direct access without the necessity of removing any panel, door or similar obstruction.
AIRGAP	The unobstructed vertical distance through the free atmosphere between the lowest opening from any pipe or faucet supplying water to tank, plumbing fixture or other device and the flood level rim of the receptacle.
AVAILABLE HEAD	The head of water available at the point of consideration due to mains' pressure or storage tank or any other source of pressure.
BACK SIPHONAGE	The flowing back of used, contaminated, or polluted water from a plumbing fixture or vessel into a water supply pipe due to a reduced pressure in such a pipe (see BACKFLOW).
BACKFLOW	The flow of water or other liquids, mixtures or substances into the distribution pipes of a potable water supply from any source other than its intended source.

BACKFLOW CONNECTION OR CONDITION	Any arrangement whereby backflow can occur.
BACKFLOW PREVENTER	A device or means to prevent backflow.
BALL COCK	A water supply valve, opened or closed by means of a float or similar device, used to supply water to a tank forming an approved air gap or vacuum breaker and acting as an anti-siphon device. Also known as FLOAT OPERATED VALVE.
BEDPAN WASHER AND STERILIZER	A fixture designed to wash bedpans and to flush the contents into the sanitary drainage system and located adjacent to a water closet or clinical sink. Such fixtures can also be provided for disinfecting utensils by scalding with steam or hot water.
BRANCH	Any part of the piping system other than a riser or main.
BRANCH CONNECTOR	A connector between water main and branch pipes by T, Y, T-Y, double Y, and V branches according to their respective shapes.
BUILDING SUPPLY	The water supply pipe carrying potable water from the water meter or other source of water supply to a building or other point of use or distribution on the lot.
CONTAMINATION	A general term meaning the introduction into the potable water supply of chemicals, wastes or sewage which will render the water unfit for its intended purpose.
CRITICAL LEVEL	The level at which the vacuum breaker may be submerged before backflow occurs. When the critical level is not indicated on the vacuum breaker, the bottom of the device shall be considered as the critical level.
CROSS- CONNECTION	Any physical connection or arrangement between two otherwise separate piping systems, one of which contains potable water and the other either water of unknown or questionable safety or steam, gas, or chemical whereby there may be a flow from one system to the other, the direction of flow depending on the pressure differential between the two systems (See BACKFLOW).

CYLINDER	A cylindrical closed vessel capable of containing water under pressure greater than the atmospheric pressure.
DEVELOPED LENGTH	Length of a pipe along the centerline of the pipe and fittings.
DISTRIBUTION PIPE	Any pipe conveying water from a storage tank/cistern or from a hot water apparatus supplied from a feed cistern under pressure from that cistern.
EFFECTIVE OPENING	The minimum cross-sectional area at the point of water supply discharge measured or expressed in terms of; (i) diameter of a circle, (ii) if the opening is not circular, the diameter of a circle of equivalent cross-sectional area. (also applicable to AIR GAP.)
FAUCET	A valve end of a water pipe by means of which water can be drawn from or held within the pipe.
FEED CISTERN	A storage tank/cistern used for supplying cold water to a hot water apparatus.
FITTING	Anything fitted or fixed in connection with the supply, measurement, control, distribution, utilization or disposal of water. "Water fitting" includes pipes (other than mains), taps, cocks, valves, ferrules, meters, cisterns, baths, water closets, soil pans and other similar apparatus used in connection with supply and use of water.
FIXTURE	See PLUMBING FIXTURE.
FIXTURE BRANCH	Water supply pipe between the fixture supply pipe and the water distribution pipe.
FIXTURE SUPPLY	Water supply pipe connecting the fixture with the fixture branch.
FIXTURE UNIT	A quantity in terms of which the load producing effects on the plumbing system of different kinds of plumbing fixtures are expressed on some arbitrary chosen scale.
FLOAT OPERATED VALVE	See BALL COCK.
FLOOD LEVEL RIM	The top edge of a receptacle from which water overflows.

FLUSH TANK	A tank located above water closets, urinals or similar fixtures for the purpose of flushing the usable portion of the fixture. Also known as FLUSHING CISTERN and FLUSHOMETER TANK.
FLUSH VALVE	See FLUSHOMETER VALVE.
FLUSHING CISTERN	See FLUSH TANK.
FLUSHOMETER TANK	See FLUSH TANK.
FLUSHOMETER VALVE	A device located at the bottom of the tank, and which discharges a predetermined quantity of water to fixtures for flushing purposes and is closed by direct water pressure or other mechanical means. Also known as FLUSH VALVE.
FULL FACILITIES	The modern plumbing facilities allowed to the occupants of modern dwellings or, of VIP hotels and accommodations.
ULL OPEN VALVE	A shutoff valve that in the full position has a straight through flow passageway with a diameter not less than one nominal pipe size smaller than nominal pipe size of the connecting pipe.
GEYSER	An apparatus for heating water with supply control on the inlet side and delivering it from an outlet.
GRADE	The slope or fall of a line of pipe with reference to a horizontal plane.
HANGERS	See SUPPORTS.
HORIZONTAL PIPE	Any pipe or fitting which is installed in a horizontal position or which makes an angle less than 45 degrees with the horizontal.
HOT WATER TANK	A vessel for storing hot water under pressure greater than the atmospheric pressure.
INDIVIDUAL WATER SUPPLY	A supply other than an approved public water supply which serves one or more families.
LAGGING	The material used for thermal or acoustic insulation.

LIQUID WASTE	The discharge from any fixture, appliance or appurtenance in connection with a plumbing system which does not receive faecal matter.
MAIN	The principal artery of the system, to which branches may be connected, for the purpose of water supply from a supply to individual consumers. Also known as WATER MAIN.
MECHANICAL JOINT	A connection between pipes, fittings or pipes and fittings which is neither screwed, caulked, threaded, soldered, solvent cemented, brazed nor welded.
OFFSET	A combination of approved bends in a line of piping used to connect two pipes whose axes are parallel but not in line.
PLUMBING	The business, trade or work having to do with the installation, removal, alteration or repair of plumbing and drainage systems or part thereof.
PLUMBING APPLIANCES	The plumbing fixtures whose operation or control can be dependent upon one or more energized components, such as motors, controls, heating elements, or pressure or temperature sensing elements.
PLUMBING APPURTENANCE	A manufactured device or prefabricated assembly of component parts, which is an adjunct to basic piping system and plumbing fixtures, performing some useful function in the operation, maintenance, servicing, economy or safety of a plumbing system.
PLUMBING FIXTURE	A receptacle or device which is either permanently or temporarily connected to the water distribution system of the premises, and demands a supply of water there from, or discharges used water, waste materials or sewage either directly or indirectly to the drainage system of the premises, or requires both a water supply connection and a discharge to the drainage system of the premises. Also known as FIXTURE.
PLUMBING SYSTEM	A system of potable water supply and distribution pipes, plumbing fixtures and traps, soil waste and vent pipes, and sanitary and storm sewers and building drains including their respective connections, devices and a appurtenances within a building or premises.

POTABLE WATER	Water free from impurities which may cause diseases or harmful physiological effects and water which is satisfactory for drinking, culinary and domestic purposes.
PRIVATE/PRIVATE USE	Plumbing fixtures intended for the use of a family in residences, or for the restricted use of an individual in commercial establishments.
QUICK CLOSING VALVE	A valve or faucet that closes automatically when released manually or controlled by mechanical means for fast action closing.
RECEPTOR	An approved plumbing fixture or device of such material, shape and capacity as to adequately receive the discharge from indirect waste pipes, so constructed and located as to be readily cleaned.
RESIDUAL HEAD	The head available at any particular point in the distribution system.
RESTRICTED FACILITIES	The minimum plumbing facilities acceptable for the occupants of low income group.
RIM	An unobstructed open edge of a fixture.
RISER	A water supply pipe which extends vertically one full storey or more to convey water to branches or fixtures.
ROUGHING-IN	The installation of all parts of the plumbing system which can be completed prior to the installation of fixtures. This includes water supply, drainage, vent piping and necessary supports.
SERVICE PIPE	The pipe that runs between the distribution main in the street and the riser in case of a multi-storied building or the water meter in the case of an individual house and is subject to water pressure from such main.
SLIP JOINT	An adjustable tubing connection, consisting of a compression nut, a friction ring, and a compression washer, designed to fit a threaded adapter fitting, or a standard taper pipe thread.
SOLDERED JOINT	A joint obtained by the joining of metal parts with metallic mixtures of alloys which melt at a temperature below 427°C and above 149°C.

STOP VALVE	Any device (including a stopcock or stop tap) other than a draw off tap; for stopping the flow of water in a pipe at will.
STORAGE CISTERN	A container, other than a flashing cistern, having a free water surface under atmospheric pressure and used for storage of water, and is connected to the water main or tube-well by means of supply pipe. Also known as STORAGE TANK.
STORAGE TANK	See STORAGE CISTERN.
SUPPORTS	Hangers and anchors or devices for supporting and securing pipe, fixture and equipment to walls, ceilings, floors or other structural members. Also known as HANGERS.
TEMPERED WATER	The water ranging in temperature from 29°C up to 43°C.
VACUUM BREAKER	A type of backflow preventer installed on openings subject to normal atmospheric pressure.
VERTICAL PIPE	Any pipe which is installed in a vertical position or which makes an angle of not more than 45 degrees with the vertical.
WARMING PIPE	An overflow pipe so fixed that its outlet whether inside or outside a building, is in a conspicuous position where the discharge of any water there from can be readily seen.
WASHOUT VALVE	A device located at the bottom of the tank for the purpose of draining a tank for cleaning, maintenance, etc.
WATER CONDITIONING OR TREATING DEVICE	A device which conditions or treats a water supply so as to change its chemical content or remove suspended solids by filtration.
WATER HAMMER ARRESTER	A device used to absorb the pressure surge (water hammer) which occurs when water flow is suddenly stopped in a water supply system.
WATER HEATER	Any heating device that heats potable water and supplies it to the potable hot water distribution system.
WATER LINE	A line marked inside a cistern to indicate the highest water level at which the ball valve should be adjusted to shut off.
WATER MAIN	See MAIN.

WATER OUTLET	A discharge opening through which water is supplied to a fixture, into the atmosphere (except into an open tank which is part of the water supply system), to a boiler or heating system, or to any devices or equipment requiring water to operate but which are not part of the plumbing system.
WATER SUPPLY SYSTEM	A system consisting of building supply pipe, water distributing pipes, and necessary connecting pipes, fittings, control valves, and all appurtenances carrying or supplying potable water in or adjacent to the building or premises.
WELDED JOINTS OR SEAM	Any joint or seam obtained by the joining of metal parts in the plastic molten state.

5.3 Permit For Water Connection

5.3.1 Requirement of Permit

No water supply system shall be installed in a new building until a permit for such work has been issued by the Authority. The addition or alteration of the existing water supply facilities in a building shall also require a permit for their installation.

5.3.2 Application for Permit (Obtaining Public Supply Connection)

Application for a permit for water supply system shall be made in writing by the licensed plumber and the owner or his appointed person(s) or agent on a prescribed form (Appendix M).

The application shall accompany building drawings showing the water supply system with the following details:

- (a) Site plans showing the location of water main.
- (b) Typical floor plan(s) and elevations of the building with the position of different plumbing fixtures and piping.
- (c) Materials, sizes and gradients (if any) of the proposed interconnecting piping system.
- (d) Pipes (if any) conveying non-potable water (for flushing water closets and urinals) shall be marked by distinctive (durable) yellow color.
- (e) Design calculations of water requirement, indicating considerations of per capita water requirement and population.

5.3.3 Application of Permit for Bulk Water Supply

In the case of large housing colonies or where new services are so situated that it will be necessary for the Authority to lay new mains or extend an existing main, full information about the proposed housing scheme shall be furnished to the Authority; information shall also be given regarding their phased requirements of water supply with full justification. Such information shall include site plans, showing the layout of roads, footpaths, building and boundaries and indicating thereon the finished line and level of the roads or footpaths and water supply lines and appurtenances.

5.3.3.1 Application for individual (permission for DTW Installation) water supply

For private water supply facility in addition to public water main connection through installing own deep tubewell, permission must be sought submitting application to water supply Authority in a prescribed form. Necessity for such connection indicating total water requirement should be mentioned.

5.3.4 Justification of Requirement

The design calculations for water supply system of high rise and public buildings shall be submitted along with the drawings mentioned in Sec 5.3.1 above.

5.3.5 Permits and Approvals

The Building Official shall examine or cause to be examined the application for a permit and amendments thereto within 45 days from the day of receipt of such application. If the application does not conform to the provisions of this Code, it shall be rejected in writing, stating the reasons thereof. The Authority shall issue a nontransferable permit, if the proposed work satisfies the provisions of this Code (Sections 5.3.2 and 5.3.3).

5.3.6 Completion Certificate

On completion of the plumbing work for the water supply system, the licensed plumber shall give a completion certificate in the prescribed form (Appendix N) to the Authority for getting water connection from the mains.

5.4 Licensing/Registration of Plumbers

5.4.1 License Requirement

Plumbing work shall be executed only by a licensed plumber under the control of the Authority and shall be responsible to carry out all lawful directions given by the Authority. No individual, partnership, corporation or firm shall engage in the business of installation, repair or alteration of water supply system without obtaining a license from the Authority.

5.4.2 Examination and Certification of Plumber

The Authority shall establish a plumber's selection and examination board. The board will determine:

- (a) The requirements of obtaining license, i.e, (i) minimum academic qualification (ii) minimum practical vocational and other training (iii) minimum years of experience (iv) total volume of works done and
- (b) Finally, establish standards and procedures for examination of the applicants for license.

The Authority will issue license to such applicants who meet the qualifications thereof and successfully pass the examination conducted by the board.

5.4.3 Annulment of License

The license of a plumber may be annulled by the Authority, if it is proved that a plumbing work has been completed and certified by the licensed plumber violating the provisions of this Code and deliberately setting aside the approvals given in the permit or without receiving the permit from the Authority.

5.5 Water Supply Requirements

5.5.1 General

5.5.1.1 Buildings equipped with plumbing fixtures and used for human occupancy or habitation shall be provided with the supply of cold potable water in the amounts specified in Sections 5.5.2 to 5.5.4 and at the pressures specified in Sections 5.10.4.2 and 5.10.4.3. Only potable water shall be accessible to the plumbing fixtures supplying water for drinking, bathing, and culinary use and for the processing of food.

5.5.1.2 Non-potable water may be used for flushing water closets and urinals provided such water shall not be accessible for drinking or such other purposes.

5.5.2 Water Requirement for Domestic Use

According to the socio-economic status, type of habitants, population of the area and public facilities present water requirement for domestic purposes may be classified only for the purpose of this Chapter as follows:

Socio - Economic group:

- | | |
|---|---|
| x. High income group - monthly income | > 6 times monthly gross per capita income of Bangladesh |
| y. Middle income group - monthly income | < 6 times monthly gross per capita income of Bangladesh |
| z. Low income group - monthly income | < 2 times monthly gross per capita income of Bangladesh |

Type of Habitants/Population:

- (a) Metropolitan Cities/City Corporation Area/District Towns
- (b) Pourashavas/Upazilas and Urban Growth Centre
- (c) Village areas

Water requirements for daily domestic use of a building shall be assessed on the basis of the one or a combination of the following two methods:

- (a) Number of occupants according to their occupancy classification and their water requirements as specified in Tables 8.5.1(a), 8.5.1(b), 8.5.1(c) and 8.5.1(d)
- (b) Peak demand or maximum probable flow is specified in Appendix O.

Table 8.5.1(a): Water Consumption for Domestic Purposes in Residential Buildings (Cities/Big District Towns)

Category	Socio-economic group, Type of Building, Source and Other Facilities	Water Consumption	
		Full Facility (lpcd)	Restricted Facility (lpcd)
A	Metropolitan Cities/City Corporation Area/District Towns		
x	High income group:		
A1	Single family dwelling (with garden and car washing)	260	200
A3	Big multi-family apartment/flat (> 2500 sft)	200	150
y	Middle income group:		
y ₁	Officer's qtr./Colony and moderate apartment (< 2000 sft)	180	135
y ₂	Small building/staff qtr. and small apartment (< 1500 sft)	---	120
z	Low income group:		
z ₁	Junior staff qtr./flat (< 1000 sft) and temporary shade	---	80
z ₂	Stand post connection in the fringe area	---	65
z ₃	Common yard (stand post) connection in the fringe area	---	50
z ₄	Slum dwellers collection from road side public stand post	---	40

Table 8.5.1(b): Water Requirement for Domestic Purposes in Residential Buildings (Pourashavas/Upazilas/Urban growth Centers)

Category	Socio-economic group, Type of Building, Source and Other Facilities	Water Consumption	
		Full Facility (<i>lpcd</i>)	Restricted Facility (<i>lpcd</i>)
A	Pourashavas/Upazilas and Urban Growth Centre		
y	Middle income group:		
A1	Single family dwelling (with garden)	---	150
A3	Officer's qtr./colony and moderate apartment (< 2000 sft)	---	135
A3	Small building/staff qtr. and small apartment (< 1500 sft)	---	120
z	Low income group:		
z ₁	Junior staff qtr. /flat (< 1000 sft) and temporary shade	---	80
z ₂	Private stand post connection in the fringe area	---	65
z ₃	Common yard (stand post) connection in the fringe area	---	50
z ₄	Slum dwellers collection from road side public stand post	---	40

Table 8.5.1(c): Water Requirement for Domestic Purposes in Residential Buildings (Village Areas and Small Communities)

Category	Socio-economic group, Type of Building, Source and Other Facilities	Water Consumption	
		Full Facility (<i>lpcd</i>)	Restricted Facility (<i>lpcd</i>)
A	Village areas/small community from hand tubewell, dugwells, ponds and rivers (non-piped water supply system)		
z	Low income group:		
z ₁	Private source (own tubewell/dugwell and pond)	---	50 - 60
z ₂	Public sources (public tubewell/dugwell/ other sources)	---	40 - 50

Table 8.5.1(d): Domestic Water Requirements for Various other Occupancies and Facility Groups

Class of Occupancy	Occupancy Groups	For Full^a Facilities (lpcd)	For Restricted Facilities (lpcd)
A: Residential	A4: Mess, Boarding Houses, Dormitories and Hostels	135	70
	A5: Hotels and Lodging Houses (per bed)	300	135
B: Educational Facilities	B1: Educational Facilities up to Higher Secondary Levels	70	45
	B2: Facilities for Training and above Higher Secondary Education	100	70
	B3: Pre-School Facilities	50	35
C: Institutional	C1: Institution for Care of Children	180	100
	C2: Custodian Institution for Physically Capable Adults	180	100
	C3: Custodian Institution for the Incapable Adults	120	70
	C4: Penal and Mental Institutions for Children	100	60
	C5: Penal and Mental Institutions for Adults	120	70
D: Healthcare Facilities	D1: Normal Medical Facilities (Small Hospitals)	340	225
	Big Hospitals (Over 100 beds)	450	250
	D2: Emergency Medical Facilities	300	135
	Nurses & Medical Quarters	250	135
E: Business	E1: Offices	45	30
	E2: Research and Testing Laboratories	70	45
	E3: Essential Services	70	45
F: Mercantile	F1: Small Shops and Market	45	30
	F2: Large Shops and Market	45	30
	F3: Refueling Station	70	45
G: Industrial Buildings	G1: Low Hazard Industries	40	25
	G2: Moderate Hazards Industries	40	25

Class of Occupancy	Occupancy Groups	For Full ^a Facilities (lpcd)	For Restricted Facilities (lpcd)
H: Storage Buildings	H1: Low Fire Risk Storage	10	6
	H2: Moderate Fire Risk Storage	10	6
I: Assembly	I1: Large Assembly with Fixed Seats (per seat)	90	45
	I2: Small Assembly with Fixed Seats (per seat)	90	45
	I3: Large Assembly without Fixed Seats ^b	15	10
	I4: Small Assembly without Fixed Seats	15	10
	I5: Sports Facilities	15	10
J: Hazardous Building	J1: Explosion Hazard Building	15	10
	J2: Chemical Hazard Building	15	10
	J3: Biological Hazard Building	10	6
	J4: Radiation Hazard Building	10	6
K: Garage ^c	K1: Parking Garage	15	10
	K2: Private Garage	15	10
	K3: Repair Garage	15	10
L: Utility	L: Utility	10	6
M: Miscellaneous	M1: Special Structures	-- ^d	-- ^d
	M2: Fences, Tanks and Towers	---	3

Notes:

^a For full facility in occupancy classifications A, B, C and D, the water requirement value includes 25% hot water.

^b In the case of mosques, the water requirements given above shall be adequate for ablution and other uses of one devotee per prayer. The appropriate LPCD value may be calculated on this basis.

^c Water requirement for occupancy K is shown as a provision for unknown visitors only.

^d Water requirement for occupancy M1 shall be assessed considering its nature of use and the similarity in purpose with any of the occupancies mentioned above.

5.5.3 Water Requirement for Fire Fighting

5.5.3.1 The Authority shall make provision to meet the water supply requirements for firefighting in the city/area, depending on the population density and types of occupancy.

5.5.3.2 Provision shall be made by the owner of the building for water supply requirements for firefighting purposes within the building, depending upon the height and occupancy of the building, in conformity with the requirements laid down in Part 4 of this Code.

5.5.3.3 The requirements regarding water supply in storage tanks, capacity of fire pumps, arrangements of wet riser-cum-down feeder and wet riser installations for high rise buildings, depending upon the occupancy use, shall be in accordance with Sec 4.2 Part 4 of this Code.

5.5.4 Water Requirement for Special Equipment

5.5.4.1 Water supply in many buildings is also required for many other applications other than domestic use, which must be identified in the initial stages of planning so as to provide the requisite water quantity, storage capacity and pressure as required for each application.

In such instances information about the water use and the quality required may be obtained from the users. Some typical uses other than domestic use and firefighting purposes are air conditioning and air washing, swimming pools and water bodies and gardening. The water requirement for special equipment like air-conditioning or such others shall be based on the specification of the manufacturer.

5.6 Estimation of Demand Load

5.6.1 Estimates of total water supply requirements for buildings shall be based on the estimation of total present and predicted future population and per capita water requirement as mentioned in Sec 5.5.

5.6.2 In making assessment of water supply requirements of large complexes, the future occupant load shall be kept in view. Use may be made of the following methods for estimating future requirements.

- (a) Demographic method of population projection,
- (b) Arithmetic progression method,
- (c) Geometrical progression method,
- (d) Method of varying increment or incremental increase,
- (e) Logistic method,
- (f) Graphical projection method, and
- (g) Graphical comparison method.

5.6.3 For residential buildings, the requirements of water shall be based on the actual number of occupants; where this information is not available, the number of occupants for each residential unit may be based on a family size around 5 (five). For assessing the population in other occupancies, reference may be made to Part 4 of this Code.

5.6.4 To estimate office building occupancy, allow 7.4 to 9.3 m² (80 to 100 ft²) of floor space per person, depending on the type of office building, exclusive of elevator and stair space, corridors, or service areas.

5.6.5 To determine the present and future water requirement:

- (a) Classify the total population based on the basis socio-economic status (higher/middle/lower income group and slum dwellers)
- (b) Determine per capita water requirement for different categories (types) of people for different type of domestic, recreational, and commercial uses.
- (c) Multiply the population with per capita water consumption to determine the present and future water requirement.

5.7 Water Sources and Quality

5.7.1 Sources of Water

The origin of all sources of water is rainfall, Water can be collected as it falls as rain before it reaches the ground; or as surface water when it flows over the ground or is pooled in lakes or ponds; or as ground water when it percolates into the ground and flows or collects as ground water; or from the sea into which it finally flows. Surface waters are physically and microbially contaminated and cannot be used without treatment, on the other hand ground water sources are chemically contaminated and treatment may be necessary in many cases.

5.7.2 Quality of Water

The quality of water to be used for drinking shall be maintained using WHO Water Safety Plan (WSP) and shall comply with the Environment Conservation Rules (ECR, 1997) and WHO Guideline (2004) Values as presented in Appendix P. For purposes other than drinking, water if supplied separately, shall be safe from bacteriological contamination so as to ensure that there is no danger to the health of the users due to such contaminants.

5.7.3 Waste Water Reclamation

Treated sewage or other waste water of the community may be utilized for non-domestic purposes such as water for cooling, flushing, lawns, parks, firefighting and for certain industrial purposes after giving the necessary treatment to suit nature of the use. This supply system shall be allowed in residences only if proper provision is made to avoid a cross connection of treated waste water with domestic water supply system.

Whenever a building is used after long intervals, the water quality of the stored water must be checked so as to ensure that the water is safe for use as per water quality requirements specified in this Code.

5.8 Water Supply System

Each floor or unit within the water supply system shall be provided with a control valve in addition to the main control valve at the entrance of the system. One of the following public water supply systems shall be adopted for distributing water to the plumbing fixtures within the building [see Appendix O].

5.8.1 Direct Connection to Water Main

For continuous water supply system with sufficient pressure to feed all plumbing fixtures during peak demand period, the direct connection of water distribution system to the water mains may be adopted. However, direct pumping from the public water main should strictly be prohibited.

5.8.2 System Incorporating Balancing Roof Tank

For continuous water supply system with inadequate pressure only during peak demand hour or for intermittent water supply with sufficient pressure to feed balancing tank, a balancing roof tank shall be required to feed plumbing fixtures within the building. The connection to the balancing roof tank from the water main or from ground tank or from individual water sources shall be through a non-return valve.

5.8.3 System Incorporating Ground Tank

For water supply system with inadequate pressure to feed plumbing fixtures or balancing roof tank, the building premises shall have a ground (or underground) tank to store water. The water from the ground tank shall be boosted up to the roof tank to feed plumbing fixtures. The connection of water main to the ground tank shall be through a ball valve system. Installation of booster pump directly into the water main shall not be allowed. Since, this system cannot ensure protection against possible contamination (particularly during flood), disinfection system should be incorporated.

5.8.4 Individual Water Supply

In the absence of a public water supply system, or In case of need of additional supply of water, the building premises shall have individual water supply as specified in Sec 5.23.1. The water from the sources (DTW) shall be boosted up to the roof storage tank to feed plumbing fixtures. The system shall be protected as specified in Sections 5.13.3 through 5.23.7.

5.9 Storage of Water

5.9.1 Capacity of Storage Tank

5.9.1.1 The type and capacity of a storage tank shall be determined considering the following factors:

- (a) The rate and regularity of supply
- (b) The frequency of replenishment of the storage tank during 24 hours

- (c) Building occupancy classification
- (d) Hours of supply of water at sufficiently high pressure to fill up the roof storage tank in absence of a ground (or underground) storage tank
- (e) The amount of water required for firefighting and method of firefighting system (See Part 4)
- (f) The amount of water required by special equipment (Sec 5.5.4).

5.9.1.2 The size and volume of a storage tank shall be calculated considering the following factors:

- (a) The amount of storage to be provided is a function of capacity of the distribution network, the location of the service storage, and the use to which it is to be put.
- (b) To compute the required equalizing or operating storage, a mass diagram or hydrograph indicating the hourly rate of consumption is required. The procedure to be used in determining the needed storage volume follows:
 - (i) Obtain a hydrograph of hourly demands for the maximum day, through a study of available records.
 - (ii) Tabulate the hourly demand data for the maximum day and draw a cumulative demand curve,
 - (iii) The required operating storage is found by comparing (maximum deviation) the cumulative demand curve (S- Shaped Curve) with cumulative pumping curve (Straight line) plotted on it.
- (c) The required capacity of a tank varies with the capacity and running time of the house or fill pumps, however, following procedure may be followed to determine the capacity of storage tanks and pump:
 - (i) Capacity of Roof Tank = $\frac{1}{2} \times \text{Total daily demand of water (m}^3\text{)} + 1 \text{ hr. reserve (m}^3\text{)}$ for fire-fighting requirement (for tall building). Therefore, two times filling of roof tank will meet the daily requirement.
 - (ii) Capacity of Delivery Pump = Peak hourly demand of water, **lph**. Therefore, daily pumping period would be around 7 - 8 hours (2-3 hours in the morning + 3 hours in the afternoon + 2 hours in the evening).
 - (iii) Capacity of Under Ground Reservoir = $1 \times \text{Total daily demand of water (m}^3\text{)} + 1 \text{ hr. reserve (m}^3\text{)}$ for firefighting (for tall building). For emergency requirement 2-3 days daily demand of water is sometimes stored.

5.9.2 Construction of Storage Tank

5.9.2.1 General

Storage tank shall be easily accessible for inspection and cleaning. The tank shall be provided with adequate size of valved drains at its lowest point in accordance with Table 8.5.2. The water supply inlet into the storage tank shall be at an elevation that is required for an air gap in an open tank with overflow (Sec 5.18.6) or 100 mm above the overflow whichever is greater. The diameter of overflow pipe shall not be less than the size shown in Table 8.5.3 for the specific discharge into storage tank. The storage tank shall be equipped with water tight and vermin and rodent proof cover. The tank shall be provided with return bend vent pipe with an open area not less than half the area of the riser (up feed or down feed). All openings (overflow pipe and vent pipe) shall be provided with corrosion resistant screens against the entrance of insects and vermin. There must be at least two compartments/units for alternative cleaning.

Table 8.5.2: Sizes of Storage Tank Drainage Pipes

Tank Capacity (V) in Liters	Diameter of Drainage Pipe (mm)
$V \leq 2800$	25
$2800 < V \leq 5500$	38
$5500 < V \leq 11000$	50
$11000 < V \leq 19000$	63
$19000 < V \leq 28000$	75
$28000 < V$	100

Table 8.5.3: Sizes of Overflow Pipes for Storage Tank

Maximum Discharge (Q) of Water Supply Pipe into Storage Tank (l/min)	Diameter of Overflow Pipe (mm)
$Q \leq 190$	50
$190 < Q \leq 570$	63
$570 < Q \leq 760$	75
$760 < Q \leq 1500$	100
$1500 < Q \leq 2650$	125
$2650 < Q \leq 3800$	150
$3800 < Q$	200

5.9.2.2 Roof storage tank

The roof storage tank shall be constructed with pre-stressed or reinforced concrete or ferro-cement or galvanized steel or of the material that will resist any action by the plain or chlorinated water. The tank shall be made of water tight without the use of putty. Tanks made of non-galvanized metal sheets shall be coated internally with a nontoxic material which does not impart a taste or odor. The metal storage tank shall be coated externally with a good quality anticorrosive weather resistant paint. The outlet of storage tank to the distribution system shall be at least 50 mm above the tank bottom.

To provide sufficient pressure, the bottom of the tank must be elevated sufficiently above the highest floor water fixtures.

Vent pipe should be provided to avoid any air lock and should be placed where the horizontal branch pipes connect the vertical down feed pipes (not adjacent to storage tank and at interconnection place between storage tank and distribution pipes).

5.9.2.3 Ground or underground storage tank

The ground or underground storage tank shall be constructed of either pre-stressed or reinforced concrete or ferrocement. The tank shall be absolutely waterproof and have a water tight cast iron manhole cover suitable for inspection. The inside and outside of the tank may be coated with nontoxic and waterproof materials. The ground tank shall be placed at a location so as to avoid contamination by flood water or any other sources. Each compartment/units should be divided in two chambers with provision of sump for longer contact time with chlorine and easy cleaning.

5.10 Design of Distribution System

5.10.1 Rate of Flow of Water

One of the important items that need to be determined before the sizes of pipes and fittings for any part of the water piping system may be decided upon is the rate of flow in the service pipe which, in turn depends upon the number of hours for which the supply is available at sufficiently high pressure. If the number of hours for which the supply is available is less, there will be large number of fittings in use simultaneously and the rate of flow will be correspondingly large.

The data required for determining the size of the communication and service pipes are:

- (a) the maximum rate of discharge required at peak demand period (Peak hourly demand of water)

$$= \frac{\text{Total daily requirement of water}}{\text{Hour of water consumption (usually 15 to 17 hours)}} \times \text{Peak Factor (usually 2.2)} \quad (8.5.1)$$

- (b) the length of the pipe; and

- (c) the head loss by friction in pipes, fittings and meters.

For head loss calculation in piping system—

- (i) Determine the total length of pipe and calculate the Equivalent Pipe Length (Head Loss from bend, gate valves, reducer etc.) from Table O.2 of Appendix O of Part 8.

- (ii) Consider maximum permissible head loss of 1.5-1.6 m/100 m and assume the probable maximum velocity of flow (v) against tentative diameter of pipes (d) as follows;

For tentative diameter of pipe 13 mm and 50 mm: 0.30 - 0.60 m/sec

For tentative diameter of pipe 75 mm and 100 mm: 0.75 - 0.90 m/sec

For tentative diameter of pipe 150 mm and 200 mm: 1.15 - 1.30 m/sec

- (iii) Determination of total head loss (h_f) from Hazen William's Nomograph (Appendix O) or friction loss formula, $h_f = \frac{4fLV^2}{2gd}$, where, f = friction loss

factor, L = length of pipe, v = velocity of flow and d = diameter of pipe.

5.10.2 Discharge Computation

5.10.2.1 Based on fixture units

The design of the consumers' pipes or the supply pipe to the fixtures is based on:

- the number and kind of fixtures installed;
- the fixture unit flow rate; and
- the probable simultaneous use of these fixtures.

The rates at which water is desirably drawn into different types of fixtures are known. These rates become whole numbers of small size when they are expressed in fixture unit. The fixture units for different sanitary appliances or groups of appliances are given in Table 8.5.4.

5.10.2.2 Probable simultaneous demand (Hunter Curve)

The possibility of drawing water at the same time by all water supply taps in any system of domestic and commercial use is extremely remote. Designing the water mains for the gross flow will result in larger and uneconomical pipe mains and is not necessary. A probability study made by Hunter suggests the relationship as shown in Appendix O and may also be calculated from Table 8.5.5.

5.10.3 Pipe Size Computation

Commercially available standard sizes of pipes are only to be used against the sizes arrived at by actual design. Therefore, several empirical formulae are used, even though they give less accurate results. The Hazen and William's formula and the charts based on the same may be used without any risk of inaccuracy in view of the fact that the pipes normally to be used for water supply are of smaller sizes. For nomogram of Hazen and William's equation see Appendix O.

5.10.4 General Features of Distribution System Design

5.10.4.1 The water supply system shall be designed to supply minimum but requisite quantity of water to all fixtures, devices and appurtenances in every section of the building with adequate pressure. The design requirements of a water supply system are presented in Table 8.5.6.

Table 8.5.4: Fixture Unit for different Types of Fixtures with Inlet Pipe Diameter

Sl. No.	Type of Fixture	Fixture Unit (FU) as Load Factor	Minimum Size of Fixture Branch, mm
1	Ablution Tap	1	15
2	Bath tub supply with spout	3	15
3	Shower Stall Domestic	2	15
4	Shower in Group per head	3	15
5	Wash Basin (Domestic Use)	1	15
6	Wash Basin (Public Use)	2	15
7	Wash Basin (Surgical)	2	15
8	Kitchen Sink (Domestic Use)	2	15/20
9	Washing Machine	3	15/20
10	Drinking Fountain	0.5	15

Table 8.5.5: Probable Simultaneous Demand

No. of Fixture Units	System with Flush Tanks Demand (Based on Fixture Units)		System with Flush Valves Demand (After Hunter)	
	Unit Rate of Flow ¹)	Flow in Litre/Minute	Unit Rate of Flow ¹)	Flow (Litre/Minute)
20	2.0	56.6	4.7	133.1
40	3.3	93.4	6.3	178.4
60	4.3	121.8	7.4	209.5
80	5.1	144.4	8.3	235.0
100	5.7	161.4	9.1	257.7
120	6.4	181.2	9.8	277.5
140	7.1	201.0	10.4	294.5
160	7.6	215.2	11.0	311.5
180	8.2	232.2	11.6	328.5
200	8.6	243.5	12.3	348.3
220	9.2	260.5	12.7	359.6
240	9.6	271.8	13.1	370.9
300	11.4	322.8	14.7	416.2
400	14.0	396.4	17.0	481.4
500	16.7	472.9	19.0	538.0
600	19.4	549.3	21.1	597.5
700	21.4	606.0	23.0	651.3
800	24.1	682.4	24.5	693.7
900	26.1	739.0	26.1	739.0
1000	28.1	795.7	28.1	795.7

¹ Unit rate of flow= Effective fixture units.

Table 8.5.6: Water Supply System Design Requirements

Fixture	Supply Control	Minimum Size of Supply Pipe (mm)	Required Flow Pressure (kPa)	Required Flow Rate (lpm)
Bathroom group	Flush tank	-	55	--
Bathroom group	Flushometer valve	-	55	--
Bathtub	Faucet	13	55	15.1
Clothes washer		13	55	--
Combination fixture	Faucet	13	55	--
Dishwashing machine		13	55	10.4
Drinking fountain	Faucet	13	55	2.8
Kitchen sink	Faucet	13	55	9.5
Laundry tray	Faucet	13	55	15.1
Wash basin	Faucet	19	55	--
Pedestal urinal	Flush tank	13	55	56.8
Pedestal urinal	Flushometer valve	19	100	56.8
Restaurant sink	Faucet	19	55	--
Service sink	Faucet	13	55	11.4
Shower head	Mixing valve	13	55	11.4
Water closet	Flush tank	19	55	11.4
Water closet	Flushometer tank	19	55	6.1
Water closet	Flushometer valve	25	100	132
For fixture not listed here but maximum		13	55	--
supply size requirement		19	55	--
		25	100	--

5.10.4.2 For a down feed water distribution system (roof tank supply), static pressure due to gravity increases with increasing floor height (4.32 psi or 0.3 Bar per floor of 10 ft. height at non-flow condition). Therefore, water distribution pipe in a building shall be maintained at a pressure so that none of their fittings shall be subject to a water head greater than 35 m (345 kPa $\approx$ 50 psi).

5.10.4.3 The distribution system shall be maintained at a pressure not less than those specified in Table- 8.5.6 during peak demand period.

5.10.4.4 The minimum size of supply pipe for different fixtures shall be in accordance with Table 8.5.6. The fixture supply shall not terminate more than 0.75 m (2.5 ft.) from the point of connection to the fixture. A reduced size flexible water connection pipe shall be used.

5.10.4.5 The water flow velocity in the distribution system shall be controlled to minimize the possibility of water hammer.

5.10.4.6 The design of water distribution system shall conform to approve engineering practices. An alternative guide to the design of a building water distribution system is also presented in Appendix O.

Note: The sizing of water distribution piping within the building may be made either by considering the velocity of flow or by velocity of flow and pressure loss as governing parameters. The first method have limited application for one or two storey buildings provided the minimum available pressure is sufficient to operate the highest or most remote fixtures during peak demand period. The second method provides better estimate of pipe sizes for a water distribution system.

5.10.5 Design of Water Distribution Pump

The capacity of a water delivery pump can be calculated from the estimated maximum rate of flow (Q) of water in gpm, and total head ($H = h_s + h_d + h_f + h_y$) of supply of water in ft. and using the following formula:

$$\text{Break Horse Power, BHP} = \frac{H \times Q}{3960 \times E} \quad (8.5.2)$$

Where, static head (h_s) is the total suction lift of water (sum of vertical distance between the underground reservoir and pump level), delivery head (h_d) is the vertical distance between the pump level and roof tank storage point and E is the efficiency of the pump in percent.

Frictional head loss (h_f) can be determined for whole length (vertical and horizontal) of pipe flow using the procedures described in Section 5.10.1(c).

$$\text{Velocity head, } h_v = \frac{v^2}{2g} \quad (8.5.3)$$

Where, v = velocity of flow of water at discharge point in roof tank and g is the acceleration due to gravity.

5.11 Water Distribution in Tall Buildings

5.11.1 Distribution Methods

In tall buildings some of the fixtures at the lower level may be subject to excessive pressure. The sanitary appliances and fittings in tall buildings shall not be subject to a pressure of greater than 350 kPa. This shall be achieved by one or a combination of the following two methods:

- (a) Zoning Floors by Intermediate Tank: High rise buildings shall be zoned by providing intermediate tanks on different floors, each feeding a zone ranging from 5 - 6 storied so that the plumbing fixtures are not subjected to excessive pressure. Distribution in each zone shall be through independent down-take pipes from intermediate tanks as shown in Appendix O. The floors on which an intermediate tank is located shall be fed from the story above it.
 - (i) System Incorporating Intermediate Tanks Supplied by Storage Tank - Water required for the building shall be pumped from the underground tank to the storage tank. The intermediate tanks shall be fed from the storage tank through a separate down take pipe.
 - (ii) System Incorporating Intermediate Tanks Supplied by Independent Pumps - Alternatively the intermediate tanks may be supplied from the underground tank through independent pumps.
- (b) System Incorporating Pressure Reducing Valves: The excessive pressures suffered by different fixtures shall be minimized by pressure reduction valves.
- (c) Hydro-Pneumatic System: This system may be adopted where the source is independent of public water supply system. In this system the supply shall be through a pneumatic pressure vessel fitted with accessories like non-return and pressure relief valves. The pump and compressor shall be automatically controlled through an electric control panel to provide air and water as and when needed.

5.11.2 Recirculation of Waste Water

Recirculation of cooling water and/or waste water from wash basin to the cistern of water closets and urinals in the lower floor may be provisioned only through a separate tank. No connection between potable water supply line and re-circulated waste water line shall be allowed with or without any non-reflex or non-return valves.

5.12 HOT WATER SUPPLY INSTALLATION

5.12.1 Hot Water Requirements

For a residential building, hot water may be supplied to all plumbing fixtures and equipment used for bathing, washing, cleansing, laundry and culinary purposes. For a nonresidential building, hot water may be supplied for bathing and washing purposes. Water requirement for hot water supply shall be in accordance with Sec 5.5.

5.12.2 Storage Temperature

The design of hot water supply system may be based upon the following temperature requirement:

Scalding	68°C
Hot bath	41°C
Warm bath	37°C
Tepid bath	30°C
Sink	60°C

5.12.3 Storage Capacity

The capacity of hot water vessel shall be based on the maximum short time demand of the premises.

5.12.4 Hot Water Heater

The hot water heater shall conform to the following standards: BS 758, BS 779, BS 843, BS 855, BS 1250, BS 2883 and those specified in Part 5 of this Code.

5.12.5 Cold Water Supply Connection to Water Heaters

The connection of cold water supply piping into water heater shall be made near its bottom. The minimum size of cold water supply piping shall be based on the probable hot water demand of different fixtures but not less than 25 mm. The supply pipe shall deliver cold water into hot water cylinder as follows:

- The water heater (electric or gas) of less than 15 liters storage capacity may be directly connected to the water main through a non-return valve.
- The storage heater of 20 to 70 liters capacity may be connected directly to the water main through a non-return valve and with an additional device that will prevent the siphonage of hot water back to the water main.
- The enclosed water heater with storage capacity greater than 70 liters shall be fed from the storage tank. This water heater shall not be connected directly to the water main.

- (d) The water supply connection to an open vessel type water heater may be made with an air gap of at least 15 mm above the top edge of the water heater. Ball valve connection shall not be used to control the flow of water into this water heater.

5.12.6 Hot Water Distribution Piping

The connection of hot water distribution pipe to the hot water cylinder shall be at the top of the cylinder. The installation of piping shall be such as to avoid airlock. In case of hot water horizontal piping the gradient shall not be less than 1 in 250.

5.12.6.1 The procedure for sizing may be the same as that for cold water distribution system in Section- 5.10 but in no case shall the size be less than that specified in Table 8.5.7.

Table 8.5.7: Minimum Pipe Size for Hot Water Piping

Different Pipes	Minimum Diameter(mm)
The size of main distributing pipe that supplies hot water to the fixtures in the same storey of the hot water cylinder	25
The size of main distributing pipe that supplies hot water to the fixtures not in the same storey of hot water cylinder	19
Branch pipe supplying hot water to the bath tub or shower	19
Branch pipe supplying hot water to the sink	19
Branch pipe supplying hot water to the wash basin	19

5.12.6.2 The design consideration of hot water piping shall be such that hot water will appear quickly at the outlet of different fixtures. To improve the situation, a secondary circulation system with flow and return pipe from and to the hot water cylinder (Figure 8.5.1) may be adopted. The length of hot water distribution pipe measured along the pipe from the top of draw off tap to the hot water cylinder or the secondary circulation pipe shall not exceed the length prescribed in Table 8.5.8. The draw-off tap shall not be connected to the primary flow or return pipe.

Table 8.5.8: Maximum Permissible Length of Hot Water Draw Off Pipe

Largest Internal Diameter (D) of Pipe	Length of Pipe (m)
D < 19 mm	12
20 mm < D < 25 mm	7.5
25 mm < D	3

5.12.7 Vent Pipe

The pressure type hot water heater shall be provided with a vent pipe of not less than 19 mm diameter. The vent pipe shall rise vertically above the water line of cold water tank by at least 150 mm plus 1 mm for every 30 mm height of waterline above the bottom of the water heater.

The vent pipe shall be connected to the top of the hot water cylinder. The vent pipe may be used to supply hot water to the point in between the cold water tank and the hot water cylinder. The vent pipe shall not be provided with any valve or check valves.

The termination of vent pipe shall be such as not to cause any accidental discharge to hurt or scald any passerby or person in the vicinity.

5.12.8 Capacity of Cold Water Storage Tank

The storage capacity of cold water cistern shall be at least equal to the size of hot water storage cylinder if the cold water cistern supplies water only to the hot water heater cylinder. This capacity shall be at least twice the capacity of hot water heater cylinder if the cold water storage tank also supplies water to the cold water draw off taps.

5.12.9 Safety Devices

The temperature relief valve or pressure relief valve or a combination of temperature and pressure relief valves shall be installed for the equipment for heating or storage of hot water. The temperature relief valve shall be set at a maximum temperature of 99°C. The maximum pressure rating of water heater shall not be more than 1000 kPa. The temperature relief valve shall be placed directly above the cylinder it serves but in no case more than 75 mm away from the cylinder. The location of pressure relief valve shall be close to the equipment it serves. There shall be no valve connection in between a relief valve and the hot water cylinder it serves.

5.12.10 Wastes from Relief Valve

The outlet of pressure, temperature or such other valve shall not be directly connected to the drainage system.

5.12.11 Drain Cock

Adequate size of drain cock shall be provided with all storage tanks for their proper cleaning.

5.13 Materials, Fittings and Appliances

- (a) The materials and fittings for water supply and distribution pipe and for storage tank shall comply with the standards listed in Part 5 of this Code and those specified in this Section.
- (b) The materials chosen shall be resistant to corrosion, both inside and outside or shall be suitably protected against corrosion and free from all toxic and harmful substances. Polyethylene and un-plasticized PVC pipes shall not be installed near hot water pipes or near any other heat source.
- (c) All jointing of pipes and fittings shall be done in accordance with acceptable standard practices.

5.13.1 Water Supply Service and Distribution Pipes

Water supply service and distribution pipes shall conform to the standards listed in Tables 8.5.9 and 8.5.10. The water supply pipes and tubing used outside the building or underground shall have a minimum working pressure of 1.1 MPa at 23°C. In case of water supply exceeding 1.1 MPa pressure, the piping material shall have at least a rated working pressure equal to the highest available pressure. The hot water distribution piping shall have a minimum pressure of 550 kPa at 80°C. Different types/classes of uPVC (Un-plasticized PVC pipes) are used both for service and internal distribution pipes as described in Table 8.5.11. However, Polyvinyl chloride (PVC) plastic pipes shall not be used exposed and unprotected as riser or water distribution pipe. Polythene or un-plasticized PVC pipes shall not be installed near the heaters or hot water piping. Lead pipes may be used only for flushing and overflow purposes in a water supply system.

5.13.2 Pipe Fittings

The pipe fittings shall be in accordance with the standards listed in Table 8.5.12 and specified in Part 5.

5.13.3 Concrete, Pre-stressed or Ferro-cement Structures

The properties of the materials used for storage tank or such other structures shall conform to the material standards specified in Part 5 of the Code.

Table 8.5.9: Water Supply Service Pipe

Materials	Standards
Acrylonitrile butadiene styrene (ABS) plastic pipe	ASTM D1527, ASTM D2282
Brass pipe	ASTM B43
Cast iron water pipe	ASTM D377
Copper or Copper-alloy pipe	ASTM B42, ASTM B302
Copper or Copper-alloy tubing	ASTM B75, ASTM B88, ASTM B251, ASTM B447
Chlorinated polyvinyl chloride (CPVC) pipe	ASTM D2846, ASTM F441, ASTM F442
Galvanized steel pipe	ASTM A53
Polybutylene (PB) plastic pipe and tubing	ASTM D2662, ASTM D2666, ASTM D3309
Polyethylene (PE) plastic pipe and tubing	ASTM D2239, ASTM D2737
PVC plastic pipe	ASTM D1785, ASTM D2241, ASTM D2672

Table 8.5.10: Water Distribution Pipe

Material	Standard
Brass pipe	ASTM B43
Copper or Copper-alloy pipe	ASTM B42, ASTM B302
Copper or Copper-alloy tubing	ASTM B75, ASTM B88, ASTM B251, ASTM B477
CPVC plastic pipe and tubing	ASTM D2846, ASTM F441, ASTM F442
Galvanized steel pipe	ASTM A53
PVC plastic pipe	ASTM D1785, ASTM D2241, ASTM D2672

Table 8.5.11(a): Working Pressure Range of Different Types of uPVC (Un-plasticized PVC) Pipes

Material	Type/Class	(Bar)	(kg/cm ²)
uPVC Pipe	Class – B	6.0	6.12
uPVC Pipe	Class – C	9.0	9.19
uPVC Pipe	Class – D	12.0	12.25
uPVC Pipe	Class – E	15.0	15.30

Table 8.5.11 (b): Average Wall Thickness Class- E uPVC (Un-plasticized PVC) Pipes

Average Wall Thickness of Class- E Type uPVC Pipe			
3/8 inch (9.5 mm) Ø	½ inch (13 mm) Ø	¾ inch (19 mm) Ø	1.0 inch (25 mm) Ø
1.9 mm	2.1 mm	2.5 mm	2.7 mm

Table 8.5.12: Pipe Fittings

Material	Standard
Asbestos cement	ISO 160, ISO 881, ISO 392
Cast iron	ASME B164, ASME B16.12
Copper or copper alloy	ASME B16.15, ASME B16.18, ASME B1622, ASME B16.23, ASME B16.26, ASME B16.29, ASME B16.32
Grey iron and ductile iron	AWWA C110, ISO 2531
Malleable iron	ASME B16.3
Plastic , uPVC	ASTM D2464, ASTM D2466, ASTM D2467, ASTM D2609, ASTM F409, ASTM F437, ASTM F438, ASTM F439
Steel	ASME B16.9, ASME B16.11, ASME B16.28

5.14 General Requirement for Pipe Work

5.14.1 Public Water Mains

The following principles shall apply for the mains:

- (a) Service mains shall be of adequate size to give the required rate of flow.
- (b) The mains shall be divided into sections by making loop system and with the provisions of sluice valves and other valves so that any part of water main may be shut off for repairs without affecting major part of pipe network
- (c) To avoid dead ends, the mains shall be arranged in a grid formation or in a network.
- (d) Where dead ends are unavoidable, a hydrant shall be provided to act as a wash-out
- (e) The wash-out valve shall not discharge directly into a drain or sewer, or into a manhole or chamber directly connected to it; an effectively trapped chamber shall be interposed, into which the wash-out shall discharge.
- (f) Air valves shall be provided at all summits, and wash-out at low points between summits.
- (g) Mains need not be laid at unvarying gradients, but may follow the general contour of the ground. They shall, however, fall continuously towards the wash-out and rise towards the air valves. The gradient shall be such that there shall always be a positive pressure at every point under working conditions.
- (h) The cover for the mains shall be at least 900 mm under roadways and 750 mm in the case of footpaths. This cover shall be measured from the top of the pipe to the surface of the ground.
- (i) The mains shall be located sufficiently away from other service lines like electric and telegraph cables to ensure safety and where the mains cannot be located away from such lines, suitable protective measures shall be accorded to the mains.

5.14.2 Interconnection Pipes from Water Main

- (a) Every premises that is supplied with water by the Authority shall have its own separate communication pipe. In the case of a group or block of premises belonging to the same owner the same communication pipe may supply water to more than one premises with the prior permission of the Authority.
- (b) The communication pipe between the water main and the stop-cock at the boundary of the premises shall be laid by the Authority.

- (c) Connections up to 50 mm diameter may be made on the water main by means of screwed ferrules, provided the size of the connections does not exceed one-third the size of the water main. In all other cases, the connection shall be made by a T-branch off the water main.
- (d) As far as practicable, the communication pipe and the underground service pipe shall be laid at right angles to the main and in approximately straight lines to facilitate location for repairs. It is also recommended that the communication pipe be laid in a pipe in pipe sleeve of larger dia. Made of non-corrosive material to protect the communication pipe.
- (e) Every communication pipe shall have a stopcock and meter inserted in it. The waterway of each such fitting shall not be less than the internal sectional area of the communication pipe and the fittings shall be located within the premises at a conspicuous place accessible to Authority which shall have exclusive control over it.

5.14.3 User/Consumer Pipes

- (a) No consumer pipe shall be laid in the premises to connect the communication pipe without the approval of the Authority.
- (b) The consumer pipe within the premises shall be laid underground with a suitable cover to safeguard against damage from traffic and extremes of weather.
- (c) To control the branch pipe to each separately occupied part of a building supplied by a common service pipe, a stop tap shall be fixed to minimize the interruption of the supply during repairs. All such stop valves shall be fixed in accessible positions and properly protected. To supply water for drinking or for culinary purposes, direct taps shall be provided on the branch pipes connected directly to the consumer pipe. In the case of multi-storied buildings, down-take taps shall be supplied from overhead tanks.
- (d) Pumps shall not be allowed on the service pipe, as they cause a drop in pressure on the suction side, thereby affecting the supply to the adjoining properties. In cases where pumping is required, a properly protected storage tank of adequate capacity shall be provided to feed the pump.
- (e) No direct boosting (by booster pumps) shall be allowed from the service pipes (communication and consumer pipes).
- (f) Consumer pipes shall be so designed and constructed as to avoid air-locks. Draining taps shall be provided at the lowest points from which the piping shall rise continuously to draw-off taps.
- (g) Consumer pipes shall be so designed as to reduce the production and transmission of noise as much as possible.

- (h) Consumer pipes in roof spaces and unventilated air spaces under floors or in basements shall be protected against corrosion.
- (i) Consumer pipes shall be so located that they are not unduly exposed to accidental damage and shall be fixed in such positions as to facilitate cleaning and avoid accumulations of dirt.
- (j) All consumer pipes shall be so laid as to permit expansion and contraction or other movements.

5.14.4 Prohibited Connections

- (a) A service pipe shall not be connected into any distribution pipe; such connection may permit the backflow of water from a cistern into the service pipe, in certain circumstances, with consequent danger of contamination and depletion of storage capacity. It might also result in pipes and fittings being subjected to a pressure higher than that for which they are designed, and in flooding from overflowing cisterns.
- (b) No pipe for conveyance or in connection with water supplied by the Authority shall communicate with any other receptacle used or capable of being used for conveyance other than water supplied by the Authority.
- (c) Where storage tanks are provided, no person shall connect or be permitted to connect any service pipe with any distributing pipe.
- (d) No service or supply pipe shall be connected directly to any water-closet or a urinal. All such supplies shall be from flushing cisterns which shall be supplied from storage tank.
- (e) No service or supply pipe shall be connected directly to any hot water system or to any other apparatus used for heating other than through a feed cistern thereof.

5.15 Safe Conveyance and Distribution of Water & Prevention of Backflow

5.15.1 Basic Principles

- (a) Wholesome water supply provided for drinking and culinary purposes shall not be liable to contamination from any less satisfactory water. There shall, therefore, be no cross-connection whatsoever between the distribution system for wholesome water and any pipe or fitting containing unwholesome water, or water liable to contamination, or of uncertain quality, or water which has been used for any other purpose. The provision of reflux or non-return valves or closed and sealed stop valves shall not be construed as a permissible substitute for complete absence of cross connection.

- (b) The design of the pipe work shall be such that there is no possibility of backflow towards the source of supply from any cistern or appliance, whether by siphonage or otherwise. Reflux non-return valves shall not be relied upon to prevent such backflow.
- (c) Where a supply of less satisfactory water than wholesome water becomes inevitable as an alternative or is required to be mixed with the latter, it shall be delivered only into a cistern and by a pipe or fitting discharging into the air gap at a height above the top edge of the cistern equal to twice its nominal bore and in no case less than 150 mm. It is necessary to maintain a definite air gap in all appliances or taps used in water closets.
- (d) All pipe work shall be so designed, laid or fixed and maintained as to remain completely water-tight, thereby avoiding wastage, damage to property and the risk of contamination.
- (e) No water supply line shall be laid or fixed so as to pass into or through any sewer, scour outlet or drain or any manhole connected therewith nor through any ash pit or manure pit or any material of such nature that is likely to cause undue deterioration of the pipe, except where it is unavoidable.
- (f) Where the laying of any pipe through corrosive soil or previous material is unavoidable, the piping shall be properly protected from contact with such soil or material by being carried through an exterior cast iron tube or by some other suitable means as approved by the Authority. Any existing piping or fitting laid or fixed, which does not comply with the above requirements, shall be removed immediately by the consumer and re-laid by him in conformity with the above requirements and to the satisfaction of the Authority.
- (g) Where lines have to be laid in close proximity to electric cables or in corrosive soils, adequate precautions/protection should be taken to avoid corrosion.
- (h) Underground piping shall be laid at such a depth that it is unlikely to be damaged by frost or traffic loads and vibrations. It shall not be laid in ground liable to subsidence, but where such ground cannot be avoided, special precautions shall be taken to avoid damage to the piping. Where piping has to be laid across recently disturbed ground, the ground shall be thoroughly consolidated so as to provide a continuous and even support.
- (i) Undesigning and planning the layout of the pipe work, due attention shall be given to the maximum rate of discharge required, economy in labor and materials, protection against damage and corrosion, water hammer, protection from frost, if required, and to avoidance of airlocks, noise transmission and unsightly arrangement.

- (j) To reduce frictional losses, piping shall be as smooth as possible inside. Methods of jointing shall be such as to avoid internal roughness and projection at the joints, whether of the jointing materials or otherwise.
- (k) Change in diameter and in direction shall preferably be gradual rather than abrupt to avoid undue loss of head. No bend or curve in piping shall be made which is likely to materially diminish or alter the cross section.
- (l) No boiler for generating steam or closed boilers of any description or any machinery shall be supplied directly from a service or supply pipe. Every such boiler or machinery shall be supplied from a feed cistern.

5.15.2 Backflow Prevention

- (a) The installation shall be such that water delivered is not liable to become contaminated or that contamination of the public water supply does not occur.
- (b) The various types of piping and mechanical devices acceptable for backflow protection are:
 - (i) Barometric loop,
 - (ii) Air gap,
 - (iii) Atmosphere vacuum breaker,
 - (iv) Pressure vacuum breaker,
 - (v) Double check valve, and
 - (vi) Reduced pressure backflow device.
- (c) The installation shall not adversely affect drinking water:
 - (i) by materials in contact with the water being unsuitable for the purpose;
 - (ii) as a result of backflow of water from water fittings, or water using appliances into pipe work connected to mains or to other fittings and appliances;
 - (iii) by cross-connection between pipes conveying water supplied by the water undertaker with pipes conveying water from some other source; and
 - (iv) by stagnation, particularly at high temperatures.
- (d) No pump or similar apparatus, the purpose of which is to increase the pressure in or rate of flow from a supply pipe or any fitting or appliance connected to a supply pipe, shall be connected unless the prior written permission of the water supplier has been obtained in each instance. The use of such a pump or similar apparatus is likely to lead to pressure reduction in the upstream pipe work which, if significant, increases the risk of backflow from other fittings.
- (e) The water shall not come in contact with unsuitable materials of construction.

- (f) No pipe or fitting shall be laid in, on or through land fill, refuse, an ash pit, sewer, drain, cesspool or refuse chute, or any manhole connected with them.
- (g) No pipe susceptible to deterioration by contact with any substance shall be laid or installed in a place where such deterioration is likely to occur. No pipe that is permeable to any contaminant shall be laid or installed in any position where permeation is likely to occur.
- (h) If a liquid (other than water) is used in any type of heating primary circuit, which transfers heat to water for domestic use, the liquid shall be non-toxic and noncorrosive.
- (i) A backflow prevention device shall be arranged or connected at or as near as practicable to each point of delivery and use of water. Appliances with built-in backflow prevention shall be capable of passing the test. All backflow prevention devices shall be installed so that they are accessible for examination, repair or replacement. Such devices shall be capable of being tested periodically by the Authority to ensure that the device is functioning efficiently and no backflow is occurring at any time.

5.16 Laying of Pipes on Site

5.16.1 Excavation of Trenches and Refilling

- (a) The bottoms of the trench excavations shall be so prepared that the barrels of the pipes, when laid, are well bedded for their whole length on a firm surface and are true to line and gradient.
- (b) In the refilling of trenches, the pipes shall be surrounded with fine selected material, well rammed so as to resist subsequent movement of the pipes.
- (c) No stones shall be in contact with the pipes; when resting on rock, the pipes shall be bedded on fine selected material or (especially where there is a steep gradient) on a layer of concrete.
- (d) The width of excavation trench shall be at least 0.4 m more than the outside diameter of the pipe.
- (e) The depth of ground cover shall be at least 0.9 m under roadway or 0.75 m under garden from the top surface of the pipe to the ground surface.
- (f) The bottom of the trench shall be carefully prepared so that the pipe will be bedded well for its entire length on firm surface.

5.16.2 Laying of Pipe

- (a) The pipes shall be carefully cleared of all foreign matter before being laid.
- (b) In sloping ground, the pipe laying shall proceed in upward direction. The pipe shall be provided with anchor blocks to withstand hydraulic pressure.

5.16.3 Laying of Pipe Through Ducts, Chases, Notches or Holes

Provisions for laying pipes in ducts or chase shall be made during the time of construction. When these will be cut into existing walls, they shall be large enough with smooth finishing for fixing the pipe and to accommodate thermal expansion. Piping subject to external pressure shall not be laid in notches or holes.

5.16.4 Lagged Piping

Lagged piping shall be entirely covered with waterproof and fire insulating materials before their attachment to the walls outside the building and shall be anchored with the wall keeping a gap in between the wall and the piping.

5.16.5 Jointing of Pipes

All joints and connections shall be gas tight and water tight for the pressure required by the test in accordance with Sec 5.22.2. The joints between different piping and fittings for water supply shall conform to the standards cited against them in Table 8.5.13. The requirements for the joints not specified in the table shall be subject to the approval of the Authority.

5.16.6 Special Care for Rat Proofing

The location and installation of water meter box shall be such as not to permit the entrance of rats into the building. The openings through walls, floors or ceilings for the installation of piping shall be closed by using proper collars to prevent the entrance of rats.

Table 8.5.13: Joints between Different Piping and Fittings

Material	Standard
ABS plastic pipe and fittings	ASTM D2235, ASTM D2661, ASTM D3139, ASTM F628 ASME B1.20.1
Asbestos, cement pipe and fittings	ASTM D1869
Brass pipe and fittings	ASME B1.20.1
Cast iron pipe and fittings	ASTM C564
Copper and copper alloy pipe and fittings	ASTM B32, ASME B1.20.1
PVC plastic pipe and fittings	ASTM D2846, ASTM D3139, ASTM F493, ASME B1.20.1
Galvanized steel pipe and fittings	ASME B1.20.1
PB plastic pipe, tubing and fittings	ASTM D2657, ASTM D3140, ASTM D3309
PE plastic pipe, tubing and fittings	ASTM D2657
PVC plastic pipe and fittings	ASTM D2564, ASTM D2855, ASTM D3139, ASTM D3212, ASTM F402, ASTM F656, ASME B1.20.1

5.17 Hangers and Support

The piping system shall be installed with proper hangers and support to minimize undue strains and stresses.

All fixtures and fittings shall be provided with hangers and support to secure them properly.

5.17.1 Galvanic action

Hangers, anchors and strapping materials shall be strong and ductile and shall not promote galvanic action.

5.17.2 Hanger Spacing

Vertical and horizontal piping shall be supported in accordance with Table 8.5.14.

Table 8.5.14: Hanger Spacing

Piping Material	Max. Horizontal Spacing (m)	Max. Vertical Spacing (m)
Galvanized steel pipe	3.5	4.5
Copper pipe or copper-alloy tubing > 38 mm diameter	3.5	3.0
Copper pipe or copper-alloy tubing < 38 mm diameter	2.0	3.0
PVC pipe and tubing	1.0	1.0
Aluminum tubing	3.0	4.5
Brass pipe	3.0	3.0

5.18 Protection of Potable Water Supply

5.18.1 Cross-connection

Potable water supply system shall be protected against non-potable water sources or wastes (solid, liquid or gases). There shall be no cross-connection between potable water distribution system and non-potable water distribution or waste disposal system.

5.18.2 Submerged outlet

From potable water supply system shall be avoided. Connection of potable water to boiler feed water system, or heating or cooling system shall be made through proper air gap.

5.18.3 Cooling water

Water used for cooling or for other purposes shall not be returned to the potable water supply system.

5.18.4 Back flow

Potable water supply system shall be protected against backflow either by air gap or by back flow preventer. Reflex non-return valve shall not be used for this purpose. Details are explained in Sec 5.15.2.

5.18.5 Back flow Protections

The connection of potable water for health care plumbing fixtures shall be protected against backflow in accordance with Table 8.5.15.

5.18.6 Air Gap

The minimum air gaps for different water supply openings or outlet shall be at least 3 times the effective opening when they will be placed close to a wall. The minimum air gaps shall be at least 2 times the effective opening when they will be located away from a wall.

5.18.7 Potable water supply connection to any cistern or apparatus containing chemical(s) shall be done only with the special approval for such connection by the Authority and shall be marked by a tag.

5.18.8 All piping and fitting shall be designed, installed and maintained as to be and to remain completely air-tight and thereby avoiding waste of water, damage to property and to avoid the risk of contamination.

5.18.9 Non-potable water supply system shall have to be painted and marked by a tag.

5.18.10 Flushing valve operated water closets when installed in any building shall be supplied through a separate branch pipe with a back flow preventer at the starting point where the branch pipe is taken off from the supply pipe.

Table 8.5.15: Water Supply Protection for Hospital Fixtures

Fixtures	Protections Required
Aspirators	Vacuum breaker
Bedpan washer	Vacuum breaker
Boiling type sterilizer	Air gap
Exhaust condenser	Vacuum breaker
Flush floor drain	Vacuum breaker
House connection	Vacuum breaker
Pressure sterilizer	Vacuum breaker
Vacuum system (cleaning and fluid section)	Air gap or vacuum breaker

5.19 Health Care Water Supply

5.19.1 General Requirement

All hospitals shall have at least two service pipes from the individual water supply source or from the water main for supplying water without any interruption. For roof storage system, the hospital shall have at least two storage tanks such that each of them is capable of serving the water distribution system in absence of the other. All special fixtures shall be installed without interference to transportation and the safety of patient and staff.

5.19.2 Hot Water Supply

All hospitals shall be equipped to supply hot water as required by different fixtures and equipment.

5.19.3 Water Supply Protection

The water supply connection to all special equipment or fixtures shall be protected against backflow, flooding, fouling and contamination of water supply system in accordance with Sec 5.9.

5.20 Cleaning and Disinfecting the System

5.20.1 General

The new and repaired potable water supply system including storage tank shall be disinfected before their use. The existing water supply system shall be cleaned and disinfected depending upon the quality of water. The storage tank shall be cleaned and disinfected at least once a year.

5.20.2 Disinfection Procedure

The following procedure may be adopted to disinfect the plumbing system:

- (a) The water supply system or storage tank shall be flushed with potable water until clean water appears at the outlets.
- (b) The system or part thereof which requires disinfection shall be filled up with chlorinated water containing 50 mg/l of chlorine for 24 hours or for 3 hours with a chlorinated water of chlorine concentration of 200 mg/l.
- (c) After the period of disinfection, the system shall be flushed with potable water until the chlorine is completely removed from the water in the system.
- (d) The above procedure shall be repeated until the bacteriological examination shows presence of no water contamination within the system.

5.21 Inspection, Testing and Completion Certificate

5.21.1 Inspection

Piping and joints shall not be enclosed, concealed or covered until they have been inspected and approved by the Authority. All piping and fixtures shall be inspected for satisfactory supports and protection from damage and corrosion.

5.21.2 Testing

After installation of the entire water supply system or part thereof, it shall be tested and approved by the Authority before its use.

- (a) Testing of Water Mains: The section of the main to be tested shall be charged with water carefully by providing a 25 mm inlet with a stop cock to expel all air from the main. The main shall be allowed to stand full of water for a few days. After that the mains shall be tested to a pressure of 500 kPa or double the maximum working pressure, whichever is greater for at least 5 minutes. The system shall be able to maintain the above test pressure.
- (b) Testing of Distribution Pipes and Fixtures: The distribution system to be tested shall be slowly and carefully charged with water to expel all air from the system and to avoid all shocks and water hammer. The piping and fittings shall be absolutely water tight when all draw off taps are closed. The system shall be able to maintain the pressures and flow required under working conditions.
- (c) Testing of Hot Water System: The entire hot water system shall be tested for the maximum rated temperature and pressure of hot water storage system. The system shall be able to maintain the required test pressure. All safety devices shall be tested for their proper operation.

5.21.3 Completion Certificate

The licensed plumber shall issue completion certificate in a prescribed form (Appendix N) to the Authority on completion of the water supply system or part thereof for inspection and testing. After testing, the Authority will allow the water connection from the water main (if any) and give the final approval (Appendix N) to use the system.

5.22 Guide to Maintenance

The owner or his/her designated agent shall maintain the water supply system in a safe operating condition as specified by the Code.

5.22.1 Frequency of Cleaning

The storage tank shall be inspected regularly and shall be cleaned and disinfected periodically. Metal tanks showing the sign of corrosion shall be coated as specified in Sec 5.9.2.2.

5.22.2 Over flow Pipe

The overflow pipes of storage tank shall be inspected regularly to keep the flow free from obstruction.

5.22.3 Water Quality

A periodical examination of water quality may be made.

5.23 Individual Water Supply System**5.23.1 General**

In the absence of a public water supply, the individual potable water source shall be used to supply water in a distribution system. The following water sources may be used for individual water supply purposes: drilled well, dug well, driven well, spring, infiltration gallery.

5.23.2 Water Requirements

The capacity of source shall be sufficient to supply water as specified in Sec 5.5.

5.23.3 Quality of Water

Water from developed well or cistern shall meet the potable water quality standard requirements specified by the Department of Environment, Bangladesh.

5.23.4 Chlorination

The well or cistern shall be chlorinated after their construction or repair.

5.23.5 Location of Water Source

The minimum distance of water source and pump suction line from potential sources of contamination shall be in accordance with Table 8.5.16.

Table 8.5.16: Distance from Potential Sources of Contamination

Potential Source of Contamination	Distance (m)
Pump floor drain of cast iron, draining to ground surface	1
Sewer	3
Farm silo	8
Septic tank	8
Subsurface pit/Seepage pit	15
Subsurface disposal field	15
Barnyard	30
Pasture	30

5.23.6 Well Construction

5.23.6.1 Location of water table

The individual water supply shall not be developed from a water bearing stratum with water table at a depth less than 3 m below the ground surface.

5.23.6.2 Outside casing

The outside watertight casing shall have to be installed for each well up to a depth of at least 3 m below the ground surface and shall project at least 150 mm above the ground surface. The lower end of the casing shall be sealed in an impermeable stratum or extend into the water bearing stratum. The size of the casing shall be large enough to permit the installation of an independent drop pipe. The casing may be of concrete, tile, or galvanized or corrugated metal pipe. The annular space between the casing and the earth shall be filled with grout to a minimum depth of 3 m. For flood prone regions, top of the casing or pipe sleeve shall be at least 300 mm above the flood level.

5.23.6.3 Well cover

All potable wells shall be equipped with a watertight cover overlapping the top of the casing or pipe sleeve. For dug or bored well, the overlap and downward extension of the cover shall be at least 50 mm outside the well casing or well. The annular space between the casing or pipe sleeve and the drop pipe shall have a watertight sealing.

5.23.6.4 Drainage from well platform or pump house

The construction of well platform or pump house shall be such that this will drain away from the well by gravity.

5.23.7 Pumping Equipment

The design, installation and construction of pumps shall be such that they will not permit the entrance of any contaminating material into the well or water supply system. The pump shall be accessible for inspection, maintenance and repair.

5.24 List of Related Appendices

Appendix M	Application for Permit to Construct Water Supply and Distribution System
Appendix N	Completion Certificate (Water Supply Works)
Appendix O	Sizing of Cold Water Supply and Distribution Piping
Appendix P	Recommended Water Quality for Domestic Purposes.

PART VIII
Chapter 6
Sanitary Drainage

6.1 Purpose

The purpose of this Chapter is to set forth provisions for planning, design and installation of waste disposal systems in and out of buildings

6.2 Scope

6.2.1 This Chapter specifies the general requirements for environmental sanitation for different categories of buildings according to their occupancy classification.

6.2.2 This Chapter also covers the design, installation and maintenance of drainage systems together with all ancillary works such as manholes and inspection chambers used within the building and from the building to public sewers or to offsite waste disposal system (i.e. into septic tanks and seepage pits or subsurface drainage system).

6.2.3 The disposal of wastes from industries, nuclear plants, slaughter houses, etc. are not covered by this Code. These wastes shall be properly treated as specified by environmental quality standards of Bangladesh before their disposal into public sewers or into natural bodies of water.

6.3 Terminology

This Section provides an alphabetical list of all terms used and applicable to this Chapter of the Code. In case of any conflict or contradiction between a definition given in this Section and that in any other Chapter or Part of the Code, the meaning specified in this Chapter shall govern for interpretation of the provisions of this Chapter.

BEDDING FACTOR	The ratio of the product of design load and factor of safety to the minimum crushing strength.
BRANCH	Any part of the piping system other than a main, riser, or stack.
BRANCH INTERVAL	The length of soil or waste stack corresponding in general to a storey height, but in no case less than 2.5 m within which the horizontal branches from one floor or storey of building are connected to the stack.
BRANCH VENT	The vent connecting one or more individual vents with a vent stack or stack vent.
BUILDING DRAIN	The building (house) drain is that part of the lowest piping or open channel of a drainage system which receives the discharges from soil, waste, and other drainage systems inside the walls of the building and conveys the same to the building (house) sewer, beginning at 0.9 m outside the building wall.

BUILDING SEWER	The building (house) sewer is that part of the horizontal piping of a drainage system which extends from the end of the building drain and which receives the discharge of the building drain and conveys it to a public sewer, private sewer, individual sewage disposal system, or other point of disposal. Also known as SEWER.
CIRCUIT VENT	Venting of branch drainage pipe with which multiple fixtures are connected in battery.
DRAIN	A drain is any pipe or open channel which carries waste water or waterborne wastes in a building drainage system.
DRAINAGE SYSTEM	A drainage system (drainage piping) includes all the piping within public or private premises, which conveys sewage, rain water, or other liquid wastes to a legal point of disposal, but does not include the mains of a public sewer system or a private or public sewage treatment or disposal plant.
DRINKING FOUNTAIN	A fountain or a tap with potable water supply connection.
EXISTING WORK	The existing work is a plumbing system or any part thereof which was installed prior to the date of enforcement of this Code.
FIXTURE UNIT	A fixture unit is a quantity in terms of which the load producing effects on the plumbing system of different kinds of plumbing fixtures are expressed on some arbitrarily chosen scale.
FLUSH VALVES	A flush valve is a device installed on the fixtures for the purpose of flushing those fixtures.
FRENCH DRAIN	A shallow trench filled with coarse rubble, clinker or similar material with or without field drain pipes.
GRADE	The grade is the slope or fall of a pipe in reference to a horizontal plane. In drainage it is usually expressed as the fall in mm per m length of pipe.
HORIZONTAL BRANCH	A horizontal branch is a drain pipe extending laterally from a soil or waste stack or building drain, with or without vertical sections or branches, which receives the discharge from one or more fixture drains and conducts it to the soil or waste stack or to the building drain.

HORIZONTAL PIPE	A horizontal pipe is any pipe or fitting which is installed in a horizontal position or which makes an angle of less than 45° with the horizontal.
IMHOFF TANK	These are two-storeyed settling cum digestion tanks used for primary treatment of domestic sewage in a very anaerobic environment.
INDIVIDUAL VENT	An individual vent is a pipe installed to vent a fixture trap and which connects with the vent system above the fixture served or terminates in the open air.
INTERCEPTOR	An interceptor is a device designed and installed so as to separate and retain deleterious, hazardous, or undesirable matter from normal wastes and permit normal or liquid wastes to discharge into the disposal terminal by gravity.
INVERT	The lowest point of the internal surface of a pipe or channel at any cross-section.
KITCHEN SINK	Sink or washing facilities raised above or at the level of the floor fitted with a tap.
LEADER	A vertical drainage pipe that carries rainwater from roof or gutter drain to building storm drain or building drain or private disposal system. Also called Rainwater Down Pipe (RDP)
LIQUID WASTE	The liquid waste is the discharge from any fixture, appliance, or appurtenance in connection with a plumbing system which does not receive faecal matter.
LOAD FACTOR	The load factor is the percentage of the total connected fixture unit flow rate which is likely to occur at any point in the drainage system. It varies with the type of occupancy, the total flow unit above the point being considered, and with probability factor of simultaneous use.
LOCAL VENT STACK	A vertical piping to which connections are made from discharge side of traps and through which vapour or foul gas is removed from the fixture or device used on bedpan washer.
LOOP VENT	Also called Circuit vent. See CIRCUIT VENT.
MAIN	The main of any system of continuous piping is the principal artery of the system, to which branches may be connected.

MAIN SEWER	See Public Sewer.
MAIN VENT	The main vent is the principal artery of the venting system, to which vent branches are connected.
MANHOLE	An opening through which a man may enter or leave a drain, a sewer or other closed structure for inspection, cleaning and other maintenance operations, fitted with a cover.
MANHOLE CHAMBER	A chamber constructed on a drain or sewer so as to provide access thereto for inspection, testing or the clearance of obstruction.
NON SERVICE LATRINE	A latrine other than service latrine.
OFFSET	An offset in a line of piping is a combination of elbows or bends which brings one section of the pipe out of line but into a line parallel with the other section.
PIPE SYSTEM	The system to be adopted will depend on the type and planning of the building in which it is to be installed and will be one of the following : (a) Single Stack System (Sec 6.9.3): One pipe system without trap ventilation pipe work. (b) One Pipe System (Sec 6.9.3): The plumbing system in which the waste from sinks, bath rooms and wash basins, and soil pipe branches are all collected into one main pipe connected directly to the drainage system. Gully traps and waste pipes are completely dispensed with but all the traps of water closets, basins, etc. are completely ventilated to preserve the water seal. (c) Two Pipe System (Sec 6.9.3): A discharge pipe system comprising two independent discharge pipes, one conveying soil directly to the drain, and other conveying waste water to the drain through a trapped gully. The system may also require ventilating pipes.
PLUMBING	The plumbing includes the practice, materials, and fixtures used in the installation, maintenance, extension, and alteration of all piping, fixtures, appliances, and appurtenances in connection with any of the following: sanitary drainage or storm drainage

	facilities, the venting system and the public or private water supply systems, within or adjacent to any building, structure, or conveyance; also the practice and materials used in the installation, maintenance, extension, or alteration of the storm water, liquid waste, or sewerage, and water supply systems of any premises to their connection within any point of public disposal or other acceptable terminal.
PLUMBING FIXTURES	The plumbing fixtures are installed receptacles, devices, or appliances which are supplied with water or which receive or discharge liquids or liquid borne wastes, with or without discharge into the drainage system with which they may be directly or indirectly connected.
PLUMBING SYSTEM	The plumbing system includes the water supply and distribution pipes, plumbing fixtures and traps, soil, waste and vent pipes, building drains and building sewers, including their respective connections, devices, and appurtenances within the property lines of the premises, and water treating or water using equipment.
PUBLIC SEWER	A common sewer directly controlled by public authority. Also known as MAIN SEWER.
RELIEF VENT	A relief vent is a vent the primary function of which is to provide circulation of air between drainage and vent systems (Sec 6.9.6).
RISER	A water supply pipe that extends vertically one full storey or more to convey water to branches or fixtures.
SANITARY SEWER	A sanitary sewer is a pipe which carries sewage and excludes storm, surface, and ground water. Also known as SEWER.
SEEPAGE PIT	See SOAK PIT.
SEPTIC TANK	A septic tank is a watertight settling tank which receives the discharge of a drainage system or part thereof and is designed and constructed so as to separate solids from the liquid, digest organic matter through a period of detention, and allow the liquids to discharge into the soil outside the tank through a system of open joint or perforated piping or disposal pit (Sec 6.9.13).
SERVICE LATRINE	A latrine from which the excreta are removed by manual agency and not by water carriage.
SEWAGE	The sewage is any liquid waste containing animal or vegetable matter in suspension or solution and may include liquids containing chemicals in solution.

SEWER	See BUILDING SEWER or PUBLIC SEWER or SANITARY SEWER or STORM SEWER.
SLUDGE	A settled portion of the sewage or waste water effluent from a sedimentation tank in semi-solid condition.
SOAK PIT	A pit, dug into permeable soil lined to form a covered perforated chamber or filled with sand at the bottom and gravel or broken bricks at the top into which effluent from septic tank or storm water is led and from which these may soak away into the ground. Also known as SEEPAGE PIT or SOAK WELL.
SOAK WELL	See SOAK PIT.
SOIL PIPE	A soil pipe is any pipe which conveys the discharge of water closets, urinals, or fixtures having similar functions, with or without the discharge from other fixtures, to the building drain or building sewer.
SOIL VENT	See Stack Vent.
STACK	A stack is the vertical main of a system of soil, waste, or vent piping.
STACK VENT	A stack vent (sometimes called a waste vent or soil vent) is the extension of soil or waste stack above the highest horizontal drain connected to the stack. Also known as SOIL VENT.
STACK VENTING	Stack venting is a method of venting a fixture or fixtures through the soil or waste stack.
STERILIZER VENT	A separate pipe or stack, indirectly connected to the building drainage system at the lower terminal, which receives the vapour from non-pressure sterilizers or the exhaust from pressure sterilizers and conduct the vapour directly to the outer air.
SUBSOIL DRAIN	A subsoil drain is a drain which receives only subsurface or seepage water and conveys it to a place of disposal.
SULLAGE	The discharge from wash basins, sinks and similar appliances, which does not contain human or animal excreta.
SUMP	A sump is a tank or pit which receives sewage or liquid waste, located below the normal grade of the gravity system, and which must be emptied by mechanical means.

SUPPORTS	The supports, hangers, and anchors are devices for supporting and securing pipe and fixtures to walls, ceilings, floors, or structural members.
TRAP	A trap is a fitting or device so designed and constructed as to provide, when properly vented, a liquid seal which will prevent the back passage of air or gas without materially affecting the flow of sewage or waste water through it.
TRAP SEAL	The trap seal is the maximum vertical depth of liquid that a trap will retain, measured between the crown weir and the top of the dip of the trap.
VENT PIPE	See Vent System.
VENT STACK	A vent stack is a vertical vent pipe installed primarily for the purpose of providing circulation of air to and from any part of the drainage system.
VENT SYSTEM	A vent system is a pipe or pipes installed to provide a flow of air to or from a drainage system or to provide a circulation of air within such system to protect trap seals from siphonage and back pressure. Also known as VENT PIPE.
VERTICAL PIPE	A vertical pipe is any pipe or fitting which is installed in a vertical position or which makes an angle of not more than 45° with the vertical.
WASTE PIPE	A waste pipe is a pipe which conveys only liquid waste free of faecal matter.
YOKE VENT	A yoke vent is a vent provided between drainage and vent stacks to provide circulation of air between drainage and vent systems (Sec 6.9.6).

6.4 Drainage and Sanitation Plans

6.4.1 Requirement of Permit

Drainage and sanitation system shall not be installed until a permit for such work has been issued by the Authority for existing (only for addition or for alteration) or new building or for any other premises.

6.4.2 Application for Permit

An application for a permit for drainage and sanitation work shall be made on a prescribed form (see Appendix Q) by the licensed plumber and the owner, or by his appointed person or agent to install all or a self-contained or workable part of such work. The application shall accompany building drainage plans and adequate description of the proposed drainage and sanitation installation in a drawing (drawn to a scale not less than 1:100) with the following details:

- (a) Plan(s) of the building with typical arrangement of plumbing fixtures
- (b) Sanitary waste disposal system
- (c) Venting system in the building drainage system
- (d) Materials, sizes and gradients of all proposed piping
- (e) Position of manhole, traps, waste pipe, rainwater pipe, vent pipe, water closet, urinal, lavatory, sink or other appliances in the premises and their connection with sewerage/drainage system or with private waste disposal system; the following colours may be used to indicate sewers, waste water pipes, rainwater pipes and existing works:

Proposed sanitary sewers and sanitary waste disposal pipes : Red

Proposed sanitary sewers and sanitary waste disposal pipes : Blue

Existing network : Black

- (f) Position of refuse chute, inlet hopper and collection chamber for high rise buildings.

6.4.3 In addition to drainage plan a separate site plan of the building shall be submitted with the following particulars:

- (a) Adjoining plots and streets with their identification
- (b) Position and invert level of the public sewers (if any) and the direction of flow in it
- (c) Level of the proposed drains connecting to the sewers (if any)
- (d) Position and layout of private waste disposal system (in absence of public sewers); and
- (e) Alignment, size and gradients of all drains.

6.4.4 For high rise buildings, design calculations and specifications for various items of the work involved shall be submitted along with the drawings.

6.4.5 Permits and Approvals

The building official shall examine or cause to be examined all applications for permits and, amendments thereto within 45 days. If the application does not conform to the requirements of all pertinent laws, such application shall be rejected in writing, stating the reasons therefore. If the proposed work satisfies all the Code requirements, the Authority shall issue a nontransferable permit.

6.5 Licensing of Plumber

6.5.1 License Requirement

No individual, partnership, corporation or firm shall engage in the business of installation, repair, alteration or maintenance of plumbing, drainage and sanitation work without obtaining a license from the Authority.

6.5.2 Examination and Certification

The Authority shall establish a plumber's examination board. The board will determine the requirements for the qualification and procedures for examination of applicants for license. The Authority will issue license to such applicants who meet the qualifications therefore and successfully pass the examination conducted by the board.

6.5.3 Annulment of License

The license of a licensed plumber may be nullified by the Authority, if it is proved that a plumbing work has been completed and certified by the licensed plumber violating the provisions of this Code deliberately setting aside the approvals given in the permit or without receiving the permit from the Authority.

6.6 Drainage and Sanitation Requirement

6.6.1 General

- (a) Each family dwelling unit on premises abutting a public sewer or with a private waste disposal system shall have at least one water closet and one kitchen sink or washing facilities. It is recommended to have at least one bathroom with a bath tub or shower to meet the basic requirements of sanitation and personal hygiene and in that case bath and water closet shall be separately accommodated.
- (b) All other structures for human occupancy or use on premises abutting a sewer or with a private waste disposal system shall have adequate sanitary facilities but in no case less than one water closet and one other fixture for cleaning purposes.
- (c) There shall be one water tap and arrangement for drainage in the vicinity of each water closet in all buildings.

- (d) There shall be at least one water tap and arrangement for drainage in the vicinity of each urinal or group of urinals in all buildings.
- (e) There shall be separate facilities for each sex for public toilets and for public bathing places based on the percentage of each anticipated sex.
- (f) Where drinking water fountain is provided, it shall not be installed in toilet room.
- (g) Rooms containing water closets or urinals shall be separated by partition wall from places where food will be prepared and served.
- (h) All water closets and urinals shall be provided with flushing system.

6.6.2 Minimum Number of Fixtures

Table 8.6.1, Sections 6.6.2.1 and 6.6.2.2 provide the minimum number of fixtures required for different categories of buildings according to their occupancy classifications. The fixture requirement for the occupancy not provided in these sections shall be subject to the approval of the Authority.

6.6.2.1 The size of drainage pipe of fixtures shall be provided as shown in Table 8.6.2.

6.6.2.2 Automatic clothes washers

Waste connection: The waste from an automatic clothes washer shall discharge through an air break into a standpipe in. The trap and fixture drain for an automatic clothes washer stand pipe shall be a minimum of 2 inches (50 mm) in diameter.

6.6.2.3 Floor drains

Floor drains shall have removable strainers. The floor drain shall be constructed so that the drain is capable of being cleaned. Access shall be provided to the drain.

6.6.2.4 Physically handicapped plumbing facilities

All buildings other than residential, educational, storage and hazardous according to building occupancy classification, having public toilet facilities with required number of fixtures shall have at least one water closet for each sex (or one unisex water closet facility) and one drinking fountain accessible to and usable by physically handicapped persons. The water closet compartment for physically handicapped persons shall be in accordance with Sec 6.9.4.

6.6.2.5 Drainage and sanitation requirements for traffic terminal stations

- (a) The minimum sanitary conveniences provided at any traffic terminal station like railway station, bus station etc. shall consist of non-service type latrines one for each sex, and one non-service type urinal for males for a daily passenger volume up to 300 persons. For large stations and airports, sanitary arrangements shall be in accordance with Table 8.6.1.

- (b) There shall be adequate arrangements for satisfactory drainage of all sewage, sullage and waste water. The drainage shall be so designed as to cause no stagnation at the maximum discharge rate for which the different units are designed.
- (c) Adequate scavenging arrangements shall be provided to keep the stations or terminals clear of all refuse. Refuse containers shall be placed at convenient points.

6.6.3 Accessibility

The fixtures specified in Sec 6.6.2 for public building shall be located not more than one floor above nor more than one floor below the floor occupied by the people for whose use the fixtures are intended, unless elevator service is available, except that in buildings which are accessible to the physically handicapped, there shall be minimum facilities as specified by the Code. It is desirable that the path of travel to the facilities shall not exceed a travel distance of 150 m.

Table 8.6.1: Plumbing Fixtures Requirement

Type of Building Occupancy	Water Closets*	Wash Basins ***	Bathtubs or Shower	Urinals** (For male)	Drinking Fountains	Other Fixtures
A Residential						
A1 Single Family Dwelling	1 per dwelling or apartment	1 per dwelling or apartment	1 per dwelling or apartment	-	-	1 kitchen sink per dwelling/apartment
A3 Flats or Apartments						
A2 Two Families Dwelling	2 per dwelling	2 per dwelling	2 per dwelling	-	-	2 kitchen sink per dwelling
A4 Mess, Boarding Houses and Hostels						
For residence and residential staff	MALE			1 per 25-150 Add 1 per additional 50	1 per 75	1 kitchen sink in each kitchen
	1 per 8	1 per 8	1 per 8			
	FEMALE					
	1 per 6	1 per 6	1 per 6			
For nonresidential staff	MALE		-	0 up to 6 1 per 7-20 2 per 21-45 3 per 46-70 4 per 71-100	1 per 100	-
	1 per 1-15	1 per 1-15				
	2 per 16-35	2 per 16-35				
	3 per 36-65	3 per 36-65				
	4 per 66-100	4 per 66-100				
	FEMALE					
	1 per 1-12	1 per 1-12				
	2 per 13-25	2 per 13-25				
	3 per 26-40	3 per 26-40				
	4 per 41-57	4 per 41-57				
	5 per 58-77	5 per 58-77				
	6 per 78-100	6 per 78-100				

Type of Building Occupancy	Water Closets*	Wash Basins ***	Bathtubs or Shower	Urinals** (For male)	Drinking Fountains	Other Fixtures
Rooms wherein outsiders are received	MALE		-	1 per 50	-	-
	1 per 100 - 400 Add 1 per additional 250	1 per water closet and 1 per urinal or group of urinals				
	FEMALE					
	2 per 100- 200 Add 1 per additional 100.	1 per water closet				
A5 Hotels and Lodging Houses						
For residential public and staff	1 per 8 omitting the occupant of the room with attached water closet; Minimum of 2 if both sex are lodged.	1 per 10 omitting the wash basins installed in the room or suite.	1 per 10 omitting the occupants of the room with bath in suite.	-	1 per 100	1 kitchen sink in each kitchen
For public rooms	MALE		-	1 per 50	1 per 100	-
	1 per 100-400. Add 1 per additional 250 or part thereof.	1 per water closet and 1 per urinal or group of urinals				
	FEMALE					
	2 per 100-200. Add 1 per additional 100 or part thereof.	1 per water closet				
For nonresidential staff	MALE		-	0 up to 6 1 per 7-20 2 per 21-45 3 per 46-70 4 per 71-100	1 per 100	-
	1 per 1-15	1 per 1-15				
	2 per 16-35	2 per 16-35				
	3 per 36-65	3 per 36-65				
	4 per 66-100	4 per 66-100				
	FEMALE					
	1 per 1-12	1 per 1-12				
	2 per 13-25	2 per 13-25				
	3 per 26-40	3 per 26-40				
	4 per 41-57	4 per 41-57				
5 per 58-77	5 per 58-77					
6 per 78-100	6 per 78-100					

Type of Building Occupancy	Water Closets*	Wash Basins ***	Bathtubs or Shower	Urinals** (For male)	Drinking Fountains	Other Fixtures
B Educational Facilities						
B1 Education Facilities up to Higher Secondary Levels	BOYS		-	1 per 20	1 per 50	Service sink: 1 per floor.
	1 per 40	1 per 60 but minimum 2				
	GIRLS					
	1 per 25	1 per 40 but minimum 2				
B2 Facilities for Training and Above Higher Secondary Levels	MALE		-	1 per 20	1 per 50	Service sink: 1 per floor.
	1 per 30	1 per 40 but minimum 2				
	FEMALE					
	1 per 20	1 per 30 but minimum 2				
B3 Preschool Facilities	1 per 15 children	1 per 15 children	-	-	1 per 50 children	Service sink: 1 per floor
C Institution for Care						
C1 Institution for Care of Children	BOYS			Urinals may be provided in boys toilet rooms in lieu of water closets but for not more than ½ of the required number of water closets	1 per 50	Service sink: 1 per floor
	1 per 8	1 per 8	1 per 8			
	GIRLS					
	1 per 6	1 per 6	1 per 8			
C2 Custodial Institutions per Physically Capable Adults	1 unisex facility or 1 for each sex per 1-100	1 per 200	1 per10 Minimum 1 for use by both sexes.	-	1 per 100	Service sink: 1 per floor
C3 Custodial Institution per the Incapable Adults	2 unisex facilities or 1 unisex facility and 1 for each sex per 100-200. Over 200 one additional unisex facility or 1 for each sex for each additional 100 .	1 per 200	1 per10 Minimum 1 for use by both sexes.	-	1 per 100	Service sink: 1 per floor

Type of Building Occupancy	Water Closets*	Wash Basins ***	Bathtubs or Shower	Urinals** (For male)	Drinking Fountains	Other Fixtures
C4 Penal and Mental Institutions for Children	1 per cell	1 per cell	1 per 15 children	-	1 per 100 children	Service sink
C5 Penal and Mental Institutions for Adults	1 per cell	1 per cell	1 per 15	-	1 per 100	Service sink
D Health Care Facilities						
D1 Normal Medical Facilities (Indoor Patient Ward)	MALE			-	1 per 75 patients	Service sink: 1 per each ward. Bed pan washing sink: 1for each ward. Kitchen sink: 1 per each kitchen
	1 per 8 patient	2 up to 30 patients and add 1 fixture per additional 30 patients	1 per 8 patients			
	FEMALE					
	1 per 8 patient	2 up to 30 patients and add 1 fixture per additional 30 patients	1 per 8 patients			
D1 Normal Medical Facilities (Outdoor Patient Ward)	MALE		-	1 per 50	1 per 500	Service sink: 1 per each ward
	1 per 100	1 per 100				
	FEMALE					
D2 Emergency Medical Facilities	2 per 100	1 per 100				
E Business						
E1 Offices	MALE		-	0 up to 6 1 per 7-20 2 per 21-45 3 per 46-70 4 per 71-100 Add@ 3% for 101-200 and @ 2.5% for over 200	1 per 100	Service sink: 1 per floor.
E2 Research and Testing Laboratories	1 per 25	1 per 25				
E3 Essential Services	FEMALE					
	1 per 15	1 per 25				

Type of Building Occupancy	Water Closets*	Wash Basins ***	Bathtubs or Shower	Urinals** (For male)	Drinking Fountains	Other Fixtures
F Mercantile						
F1 Small Shops and Markets F2 Large Shops and Markets	1 per 500	1 per 750	-	Urinals may be provided in toilet room in lieu of water closets per men but per not more than ½ of the required number of water closets.	1 per 1000	Service sink:1
F3 Refuelling Station	MALE		-	0 up to 6 1 per 7-20 2 per 21-45 3 per 46-70 4 per 71-100 Add@ 3% for 101-200 and @ 2.5% for over 200	1 per 100	Service sink: 1 per floor.
	1 per 25	1 per 25				
	FEMALE					
	1 per 15	1 per 25				
G Industrial Buildings						
G1 Low hazard industries G2 Moderate hazard industries	MALE		As required by particular trades or occupations	0 up to 6 1 per 7-20 2 per 21-45 3 per 46-70 4 per 71-100 Add @ 3% for 101-200 and @ 2.5% for over 200	1 per 100	Service sink: 1 per floor
	1 per 1-15 2 per 16-35 3 per 36-65 4 per 66-100	1 per 25				
	FEMALE					
	1 per 1-12 2 per 13-25 3 per 26-40 4 per 41-57 5 per 58-77 6 per 78-100	1 per 25				
H Storage Buildings						
H1 Low fire risk storage H2 Moderate fire risk storage	1 per 100	1 per 100	Provisions per emergency shower	-	1 per 1000	Service sink:1
I Assembly						
I1 Large assembly with fixed seats I2 Small assembly with fixed seats I3 Large assembly without fixed seats I4 Small assembly without fixed seats						

Type of Building Occupancy	Water Closets*	Wash Basins ***	Bathtubs or Shower	Urinals** (For male)	Drinking Fountains	Other Fixtures
Mosque	1 per 30	-	1 per 100	-	1 per 100	Water taps with drainage arrangement: 1 per 10
Junction Stations, Intermediate Stations, Terminal Stations and Bus Terminals	MALE		-	Min 2, 4 per 1000 Add 1 per additional 1000	1 per 300	Service sink: 1 per floor
	Min 2, 4 per 1000 Add 1 per additional 1000.	Min 2, 4 per 1000 Add 1 per additional 1000				
	FEMALE					
	Min 2, 5 per 1000 Add 1 per additional 1000	Min 2, 6 per 1000 Add 1 per additional 1000				
Domestic Airport	MALE		-	1Minimum 2 per 200 4 per 400 5 per 600 6 per 800 7 per 1000	1 per 300	Service sink: 1 per floor
	2 Minimum 4 per 200 6 per 400 8 per 600 9 per 800 10 per 1000	2 Minimum 4 per 200 6 per 400 8 per 600 9 per 800 10 per 1000				
	FEMALE					
	2 Minimum 5 per 200 8 per 400 10 per 600 13 per 800 13 per 1000	2 Minimum 4 per 200 6 per 400 8 per 600 9 per 800 10 per 1000				
	MALE					
	6 per 200 12 per 600 18 per 1000	10 per 200 20 per 600 25 per 1000				
	FEMALE					
	10 per 200 20 per 600 29 per 1000	10 per 200 20 per 600 25 per 1000				
International Airport	MALE		4 shower stalls in the female or male toilet in the transit and departure lounge and also in the main concourse	8 per 200 22 per 600 22 per 1000	1 per 300	Service sink: 1 per floor
	6 per 200 12 per 600 18 per 1000	10 per 200 20 per 600 25 per 1000				
	FEMALE					
	10 per 200 20 per 600 29 per 1000	10 per 200 20 per 600 25 per 1000				

Type of Building Occupancy	Water Closets*	Wash Basins ***	Bathtubs or Shower	Urinals** (For male)	Drinking Fountains	Other Fixtures
Cinemas, Concert halls, Theatres (for public use)	MALE		-	1 per 25	1 per 500	Service sink 1
	1 per 100- 400 Add 1 per additional 250	1 per 200				
	FEMALE					
	3 per 100- 200 Add 2 per additional 100	1 per 200				
Cinemas, Concert halls, Theatres (for permanent employee use)	MALE		-	0 up to 6 1 per 7-20 2 per 21-45	1 per 500	-
	1 per 1-15 2 per 16-35	1 per 1-15 2 per 16-35				
	FEMALE					
	1 per 1-12 2 per 13-25	1 per 1-12 2 per 13-25				
Art Galleries, Libraries, Museums (for public use)	MALE		-	1 per 50	1 per 500	Service sink: 1
	1 per 200 - 400 Add 1 per additional 250	1 per 200 Over 200 add 1 per additional 250				
	FEMALE					
	1 per 100 - 200 Add 1 per additional 150	1 per 200 Add 1 per additional 150				
Art Galleries, Libraries, Museums (for permanent employee use)	MALE		-	0 up to 5 1 per 7-20 2 per 21-45	1 per 100	-
	1 per 1-15 2 per 16-35	1 per 1-15 2 per 16-35				
	FEMALE					
	1 per 1-12 2 per 14-25	1 per 1-12 2 per 14-25				
I5 Sports Facilities	MALE		1 per 50	1 per 75	1 per 300	Service sink: 1
	1 per 75	1 per 60				
	FEMALE					
	1 per 50	1 per 60				

[illegible]

Table 8.6.2: Sanitary Appliance Minimum Internal Diameter of Waste Outlet (mm)

Soil appliances		Waste appliances	
(a) Indian and European type water closets	100	(a) Drinking fountain	25
(b) Bed pan washers and slop sinks	100	(b) Wash basin	32
(c) Urinal with integral traps	75	(c) Bidets	32
(d) Stall urinals (with not more than 50- 120 mm of channel drainage)	40	(d) Domestic sinks and baths	40
(e) Lipped small urinal	40	(e) Shower bath trays	40
		(f) Domestic bath tubs	50
		(g) Hotel and canteen sinks	50
		(h) Floor traps (outlet diameter)	65

6.7 Materials and Appliances

Different sanitary appliances, materials and fittings shall conform to the requirements of the Standards as provided in Tables 8.6.3 of this Chapter and Table 8.5.9 of Chapter 5 Part 8. For other appliances, materials and fittings that are not listed in the above mentioned Tables shall be subject to the approval of the Authority.

Table 8.6.3(a): Recommended Standards for Sanitary Appliances

Appliances	Standard	Appliances	Standard
Ceramic wash basin and pedestals	BDS 1162-87	Metal sink for domestic purpose	BS 1244
Ceramic wash down water closet pans	BS 1213	Urinals (bowl type) vitreous china	BDS 1163-87 part 1, 3
Foot rest vitreous china	BDS 1163-87 part 1, 4	Wash-down water closet pans, vitreous china	BDS 1163-87 part 1, 2
Integrated squatting pans vitreous china	BDS 1163-87 part 1, 5	Water closet seat plastic	BS 1254
Metal hand rinse basin	BS 1329	Water closet flushing cisterns and pipes	BS 1125

Table 8.6.3(b): Recommended Standards for Building Drainage and Vent Pipe

Material	Standards
Acrylonitrile butadiene styrene (ABS plastic pipe)	ASTM D2661, ASTM F 628
Aluminum tubing	ASTM B429, ASTM B745M
Brass pipe	ASTM B43
Cast iron pipe	ASTM A74
Copper or Copper-alloy tubing	ASTM B75M, ASTM B88M, ASTM B251M, ASTM B306
Galvanized steel pipe	ASTM A53
Polyvinyl chloride plastic pipe	ASTM D2665, ASTM D2949, ASTM F891

Table 8.6.3(c): Recommended Standards for Building Sewer or Building Storm Sewer Pipe

Material	Standards
Acrylonitrile butadiene styrene (ABS plastic pipe)	ASTM D2261, ASTM D2751, ASTM F628, ASTM D2321
Bihuminized fibre pipe	ASTM D1861, ASTM D1862
Cast iron pipe	ASTM A74
Concrete pipe	ASTM C14M, ASTM C76M
Copper or Copper-alloy tubing	ASTM B75, ASTM B88M, ASTM B251M
Unplasticized Polyvinyl chloride (uPVC) plastic pipe	ASTM D2665, ASTM D2949, ASTM D3034, ASTM D2321, ASTM F891
Vitrified clay pipe	ASTM C4, ASTM C700

Table 8.6.3(d): Recommended Standards for Subsoil Drainage Pipe

Material	Standard
Bituminous fibre pipe	ASTM D2311
Cast iron pipe	ASTM A74
Concrete pipe	ASTM C654 M
Polyethylene (PE) plastic pipe	ASTM F405
Unplasticized Polyvinyl chloride (uPVC) plastic pipe	ASTM D2729, ASTM F891
Vitrified clay pipe	ASTM C4, ASTM C700

Table 8.6.3(e): Recommended Standards for Joints between Different Pipes and Fittings

Material	Standard
ABS plastic pipe and fittings	ASTM D2235, ASTM D2661, ASTM D3212, ASTM F628 ASME B1.20.1
Aluminium tubing	ASTM C564
Asbestos cement pipe and fittings	ASTM D1869
Brass pipe and fittings	ASME B1.20.1
Cast iron pipe and fittings	ASTM C564
Concrete pipe and fittings	ASTM C443
Copper or Copper-alloy pipe and fittings	ASTM B32, ASME B1.20.1
Copper-alloy tubing and fittings	ASTM B32
CPVC plastic pipe and fittings	ASTM F493, ASME B1.20.1
Galvanized steel pipe and fittings	ASME B1.20.1
PE plastic pipe and fittings	ASTM D2657
PVC plastic pipe and fittings	ASTM D2657, ASTM D2855, ASTM D3139, ASTM D3212, ASTM F402, ASTM F656, ASME B1.20.1
Vitrified clay pipe and fitting	ASTM C425

6.8 Hangers and Support and Pipe Jointing

6.8.1 Hangers and Support

The piping, fixtures and equipment used for plumbing, water supply and drainage system shall be provided with hangers and support in accordance with Sec 5.13 in Chapter 5.

6.8.2 Pipe Joints

The joints between different piping and fittings shall conform to the standards cited against them in Table 8.6.3. The requirements for the joints not specified in the table shall be subject to the approval of the Authority.

6.9 Design Considerations

6.9.1 Objective

For the design of drainage and sanitation system of different buildings according to building classification, the objective shall be to safeguard against fouling, deposition of solids and clogging and with adequate cleanouts and inspection chambers so arranged that the drains may be readily cleaned without the risk of health hazard.

6.9.2 General

- (a) The plumbing system shall be designed and adjusted to use the minimum quantity of water consistent with proper performance and cleaning.
- (b) Plumbing fixtures, devices and appurtenances shall be supplied with required volume of water at pressures adequate to enable these to function properly and without undue noise under normal conditions of use.

6.9.3 Different Building Drainage Systems

For the design and installation for drainage piping, one of the following building drainage systems shall be adopted: (i) single stack system; (ii) one-pipe system, and (iii) two-pipe system.

- (a) Single stack system may be used with 100 mm diameter stack for buildings up to 5-storey height. The fixtures in each floor shall be connected to a single stack for increasing the rate of discharge in the downward direction. The recommended depth of water seal trap for different fixtures shall be in accordance with Table 8.6.4. There shall be at least 200 mm vertical distance between the waste branch and the soil branch connection, while the soil pipe will be connected to stack above the waste pipe. The size of soil branch shall not be less than 100 mm. The horizontal branch distance for fixtures from stack and bend(s) at the foot of stack to avoid back pressure as well as the vertical distance between the lowest connection and the invert of drain shall be as shown in Figure 8.6.1.
- (b) Where all types of waste from the building are desired to be discharged into a common sewer or into same waste disposal system, one pipe system may be used (Figure 8.6.2).
- (c) Where the sullage from kitchen and bath will be dealt with separately and where soil waste shall be discharged into septic tank or Imhoff tank, the two pipe system shall be used (Figure 8.6.3).

Table 8.6.4: Recommended Depth of Water Seal Trap for Different Fixtures

Fixture	Water Seal (mm)
Water Closets	50
Floor Traps	50
For Waste Branch of 75 mm diameter or More	40
For Waste Branch of Less Than 75 mm diameter	75

6.9.4 Water Closet Compartment for Physically Handicapped

6.9.4.1 Provision for Wheelchair Users

The water closet compartment for wheelchair users shall have at least the dimensions and fittings as shown in Figure 8.6.4.

6.9.4.2 Provision for Ambulant Disabled People

The minimum dimension for water closet compartment and the fittings for ambulant disabled people shall be as shown in Figure 8.6.5.

6.9.5 Installation of Drainage System

6.9.5.1 All plumbing fixtures shall be made of smooth and nonabsorbent materials, free from concealed fouling surfaces and may be located in ventilated enclosures.

6.9.5.2 Whenever possible, all drainage system shall be drained to the public sewer or private waste disposal system by gravity.

Horizontal drainage piping up to 75 mm diameter shall be installed with a fall of not less than 20 mm/m and for larger than 75 mm diameter the fall shall not be less than 10 mm/m. It is a good policy to design the system for the highest possible velocity. However, consideration should be given to the fact that the high velocities in pipes with slopes greater than 20 mm per m may cause self-siphoning of trap seal.

6.9.5.3 Where conditions do not permit building drains and sewers to be laid with a fall as great as that specified, a lesser slope may be permitted provided the computed velocity in the drains will not be less than 0.6 m per second. The maximum recommended velocity will be 2.5 m per second.

6.9.5.4 Soil pipe conveying any solid or liquid filth to a drain shall be circular with a minimum dia of 100 mm.

The waste branch from bath room, wash basin or sink shall be of 32 mm to 50 mm diameter and shall be trapped immediately beneath such wash basins or sink by an efficient siphon trap with adequate means of inspection and cleaning. The minimum recommended size of waste stack is 75 mm.

6.9.5.5 The soil and waste stack shall be continued upward undiminished in size 0.6 m above the roof surface when the roof will be used only for weather protection. Where the roof will be used for any purpose other than weather protection, the soil and vent stack shall run at least 2 m above the roof surface so that there shall be least possible nuisance.

6.9.5.6 Soil and waste stack shall be firmly attached to the wall with a minimum clearance of 25 mm from wall.

All soil, waste, vent (ant siphoning) stacks shall be covered on top with cowl of same pipe material.

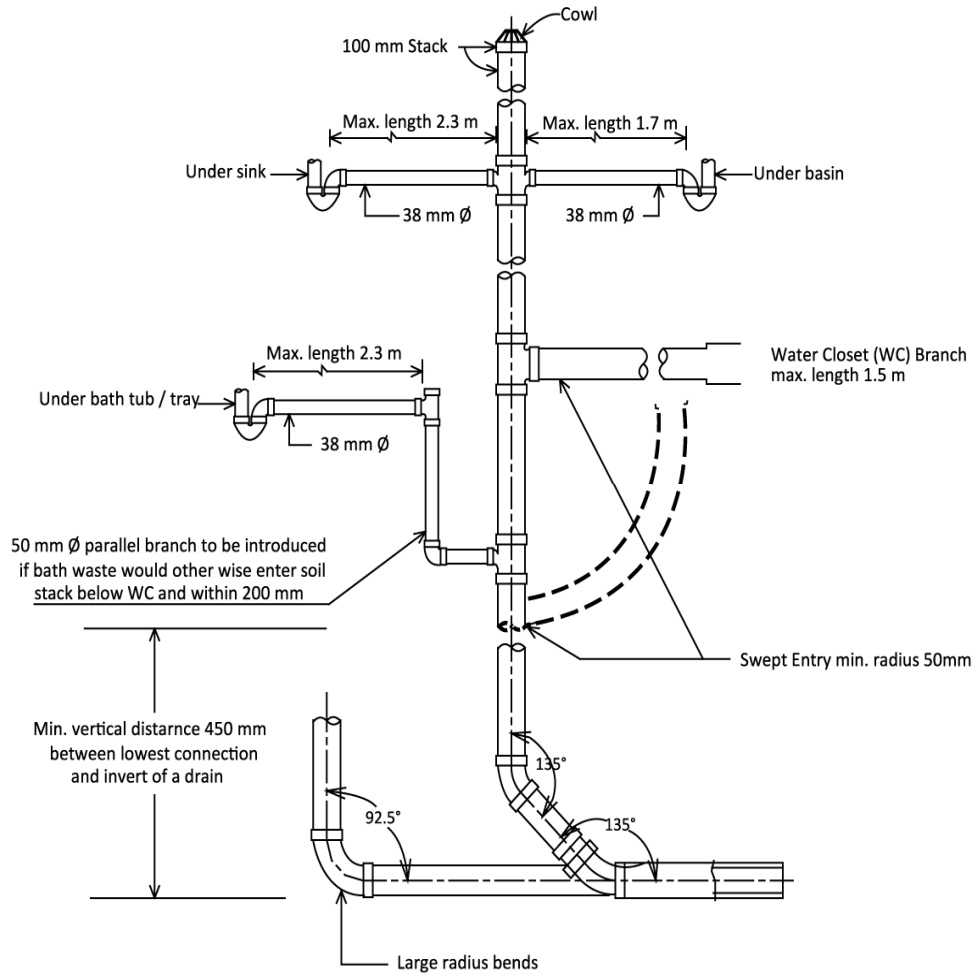


Figure 8.6.1 Single stack system

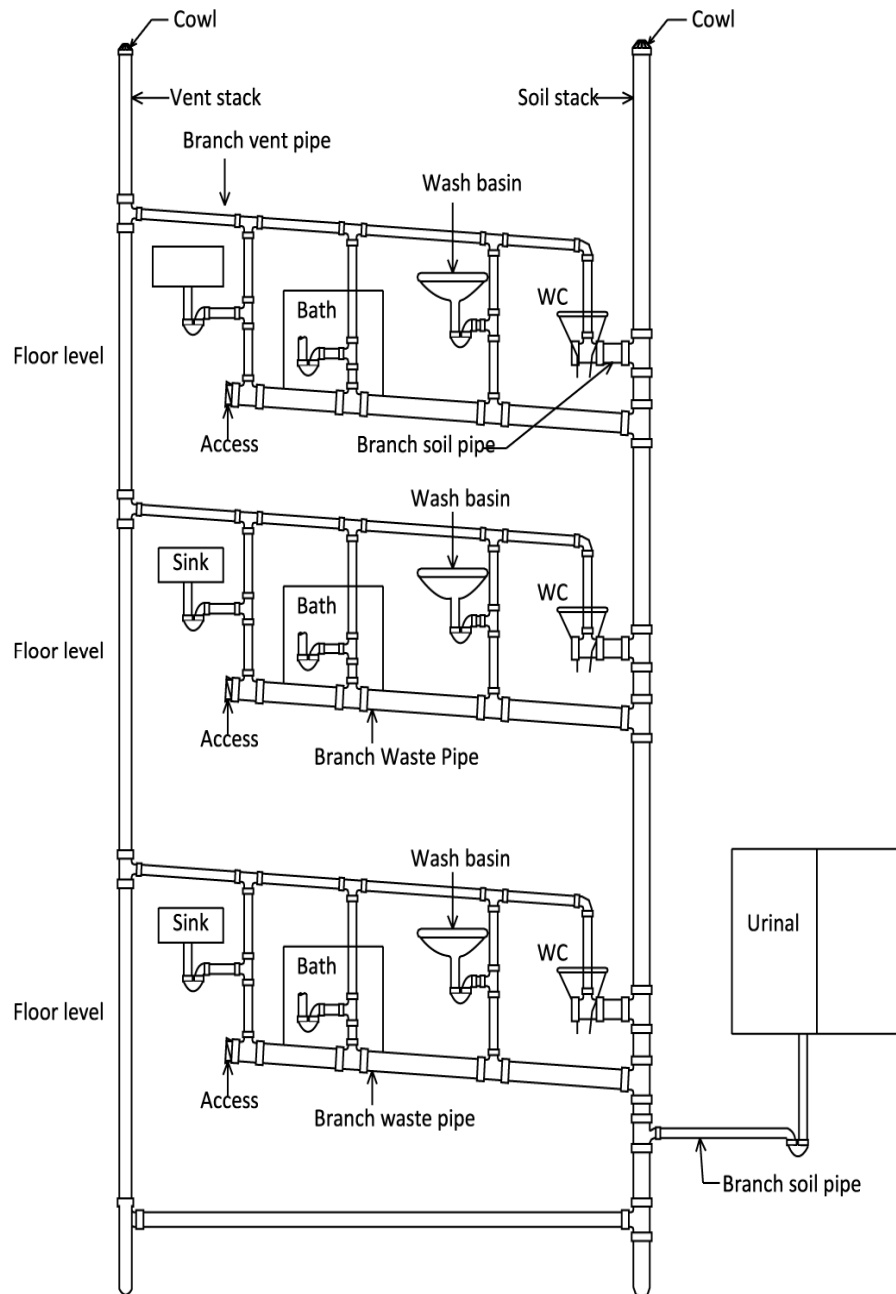


Figure 8.6.2 Diagram of one - pipe system

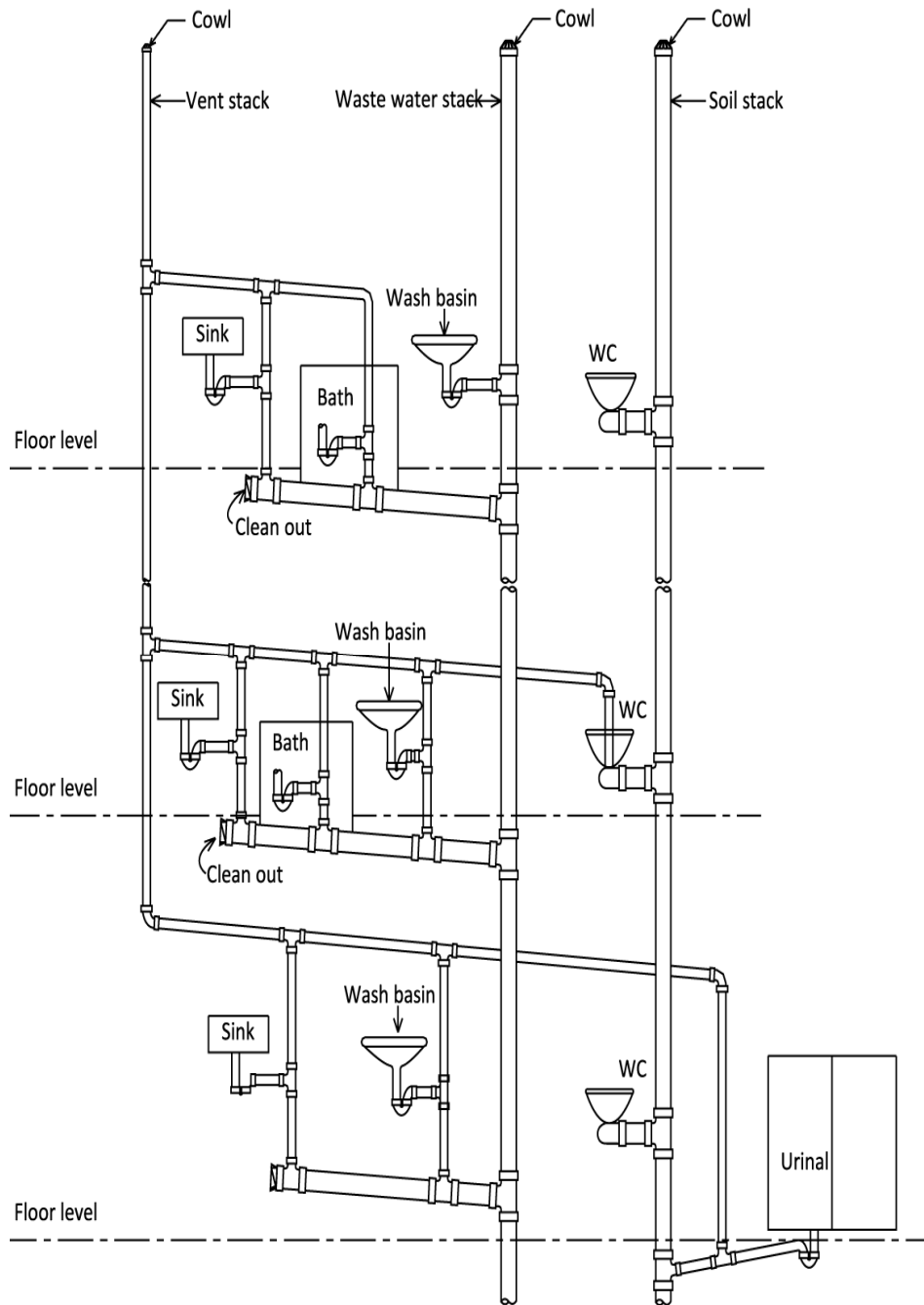


Figure 8.6.3 Diagram of two pipe system

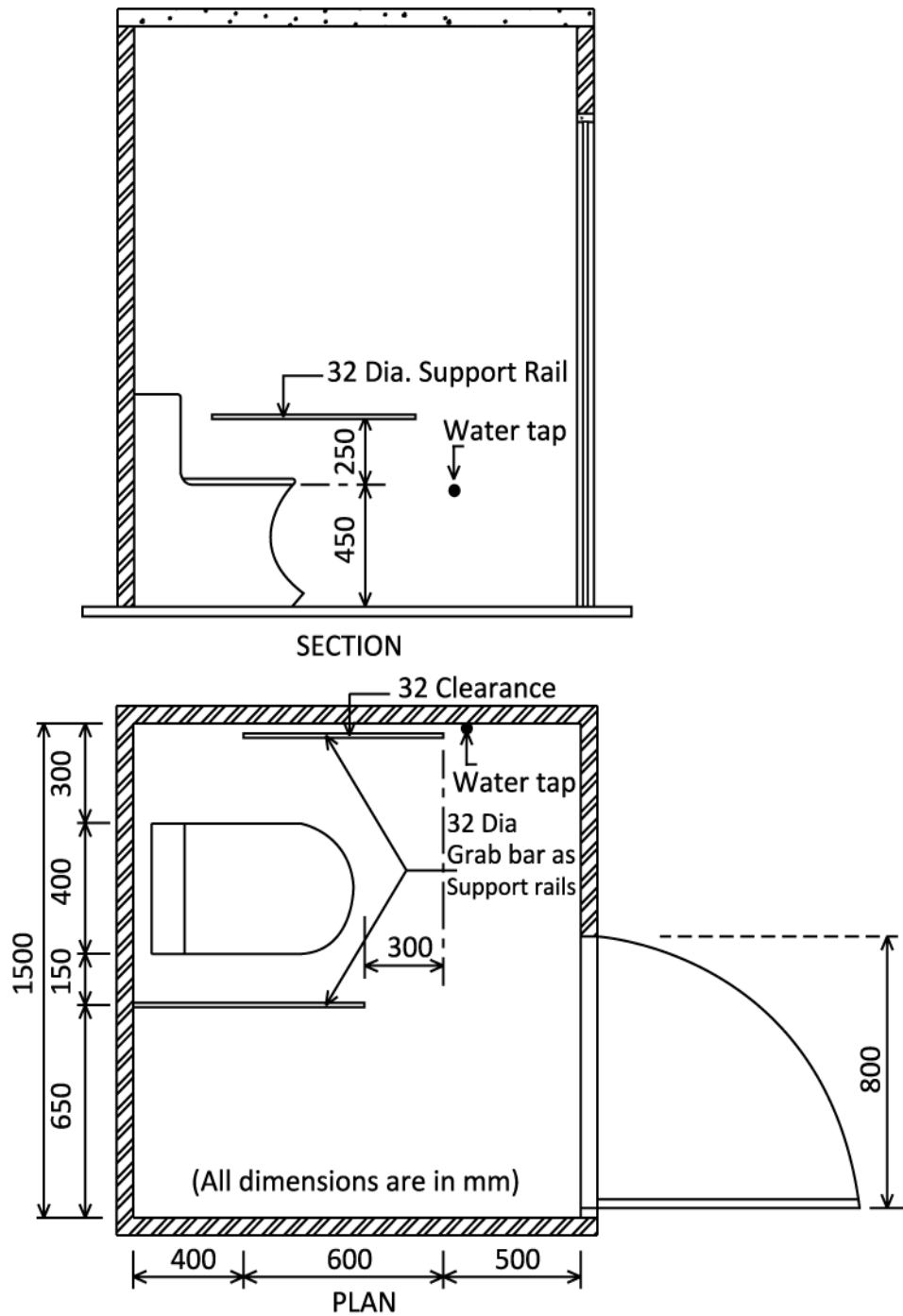


Figure 8.6.4 Water closet compartment for wheelchair user

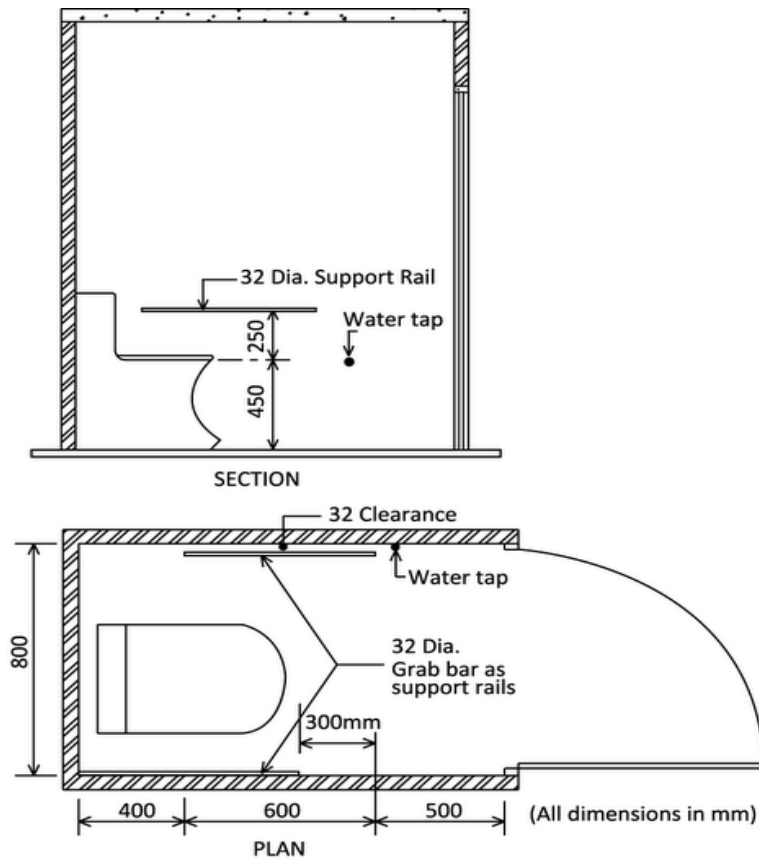


Figure 8.6.5 Water closet stall for ambulant disabled people

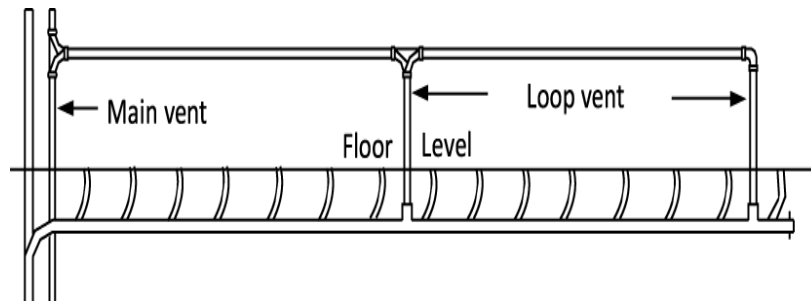


Figure 8.6.6 Circuit vent for a battery of water closets

6.9.6 Installation of Venting System

6.9.6.1 The vent stack or main vent shall be installed in conjunction with a soil or waste stack in a building. One vent stack may serve not more than two soil or waste stacks.

6.9.6.2 Ventilating pipes should be so installed that water cannot be retained in them. They should be fixed vertically. Whenever possible, horizontal runs should be avoided. Ventilating pipe shall be carried to such a height and in such a position as to afford by means of the open end of such pipe or vent shaft, a safe outlet for foul air with the least possible nuisance.

6.9.6.3 The building with building drain shall have at least one 100 mm vent stack or stack vent carried full size to outdoor air above the roof in accordance with Sec 6.9.5.7 above.

6.9.6.4 The diameter of a vent stack shall not be less than 50 mm.

6.9.6.5 The diameter of a branch vent pipe on a waste pipe shall not be less than 25 mm or two-thirds of the diameter of the branch waste pipe ventilated.

6.9.6.6 The branch vent pipe on a soil pipe shall not be less than 32 mm in diameter.

6.9.6.7 All main vents or vent stacks shall connect full size at their base to the building drain or to the soil or waste stack at or below the level of the lowest drainage connection to them. All vent stacks shall extend undiminished in size above the roof or shall be reconnected to a vent header or to the stack vent portion of the soil or waste stack, at least 150 mm above the flood level of the highest fixture connection discharging into the soil or waste stack. Where the roof is to be used for any purpose other than weather protection, the vent extension shall be in accordance with the Sec 6.9.5.7.

6.9.6.8 In case of offsetting of stacks a relief vent shall be provided at the base of upper stack just above the start of offset and at top of the lower stack portion just below the end of offset.

6.9.6.9 In high rise buildings yoke vent shall be provided at 10 storey intervals counting down from top.

6.9.6.10 In case huge number of fixtures are installed in battery to a single branch drainage pipe, circuit or loop vents shall be provided after 8 fixtures interval for 100 mm drain pipe and 24 fixtures interval for 150 mm drain pipe as shown in Figure 8.6.6

6.9.6.11 Offset in the stack vent portion of soil or waste stack, offset in vent stack and connection of vent stack at the bottom to soil or waste pipe or to the building drain shall be at an angle of at least 45° to the horizontal.

6.9.6.12 All vent and branch vent pipe shall be so graded and connected that sufficient slope is provided for condensation to drain back to soil or waste pipe by gravity.

6.9.6.13 Where fixtures, other than water closets discharge into the stack downstream of a water closet, each fixture connecting downstream shall be individually vented.

6.9.6.14 Soil and waste stacks in a building having more than 10 branch intervals shall be provided with a relief vent (Figure 8.6.7) at each tenth interval counting from the top floor.

6.9.6.15 In case the adjoining building is taller, the ventilating pipe shall be carried higher than the roof of the adjacent building, wherever it is possible.

6.9.6.16 The building drain intended for carrying waste water and sewage from a building shall be provided with at least one ventilating pipe situated as near as practicable to the building and as far away as possible from the point at which the drain empties in to the sewer or other earner.

6.9.7 Clearance of Blockages

6.9.7.1 There shall be sufficient and suitable access points at every change of alignment, gradient or diameter or at bends and junctions for clearing blockages from drains which cannot be reached by any other means.

6.9.7.2 In case of straight run of pipes, access points shall be provided at intervals of 15 meter.

Tables 8.6.5 and 8.6.6 show the maximum spacing and the recommended minimum dimensions for access fittings and chambers for the specified depth.

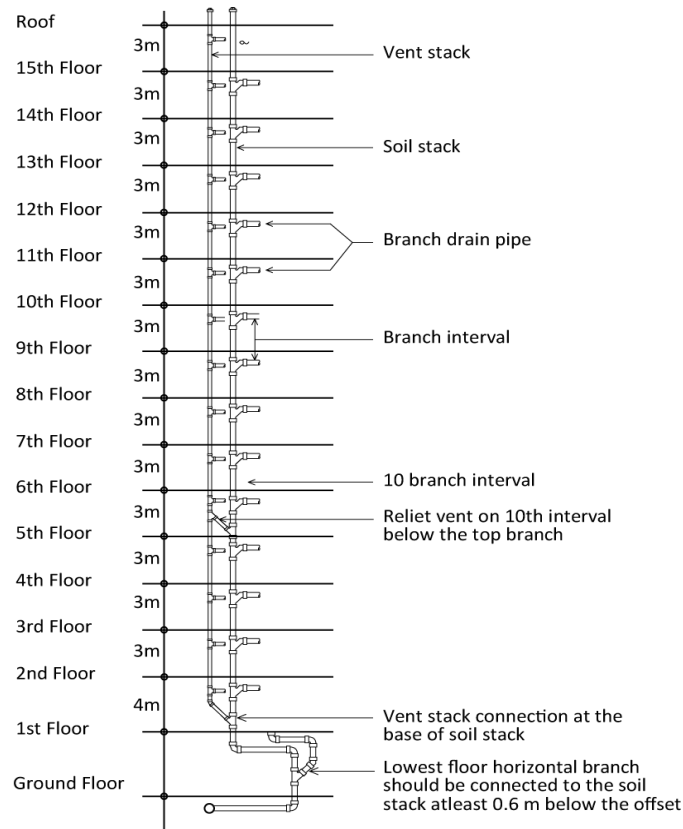


Figure 8.6.7 Relief vents for stack of more than ten branch intervals

Table 8.6.5: Maximum Spacing of Access Points

From	To Access Fitting (m)	To Junction (m)	To Inspection Chamber (m)	To Manhole (m)
Start of external drain	12	-	22	45
Rodding eye	22	22	45	45
Access fitting †	-	12	22	22
Inspection chamber	22	22	45	45
Manhole	22	-	45	90

† higher spacing may be used for larger size access fitting.

Table 8.6.6 Minimum Dimensions for Access Points

Access Points	Depth (m)	Internal Sizes		Cover Sizes	
		Length x width (mm x mm)	Diameter (mm)	Length x width (mm x mm)	Circular (mm)
Rodding eye		min. 100 mm or size of drains			
Access fitting	0.6 or less	150 x 100	150	150 x 100	150
Inspection chamber	1.0 or less	450 x 450	450*	450 x 450	450*
Manhole	1.5 or less	1200 x 750	1050	600 x 600	600
	over 1.5	1200 x 750	1200	600 x 600	600
	over 2.7	1200 x 840	1200	600 x 600	600

* 190 mm dia may be used for depth ≤ 0.6 m

6.9.7.3 Access should be one of the following four types :

- (i) Rodding eyes - capped extensions of the pipes,
- (ii) Access fittings - small chambers (or an extension of the pipes) but not with an open channel,
- (iii) Inspection chambers - chambers with working space at ground level, and
- (iv) Manholes - large chambers with working space at drain level.

6.9.7.4 Inspection chambers and manholes shall have removable non-ventilating covers of durable material and be of suitable strength. Inspection chambers and manholes in buildings shall have mechanically fixed airtight covers unless the drain itself has watertight access covers. Manholes deeper than 1 m shall have non-corrosive steps or fixed ladders. Figures 8.6.8 and 8.6.9 show the details of typical manholes at smaller depth (<1 m) and at higher depth (>1 m) respectively. Figure 8.6.9 shows the details of a drop manhole. The drop manhole is a manhole that serves as a junction and receives sewer lines at two different elevations.

6.9.7.5 Spacing of manholes

The spacing of manholes for a given pipe size should be as follows :

Pipe Diameter (mm)	Spacing of Manhole (m)
Up to 300	45
301 to 500	75
501 to 900	90

Beyond 900 Spacing shall depend upon local condition and shall be gotten approved by the Authority

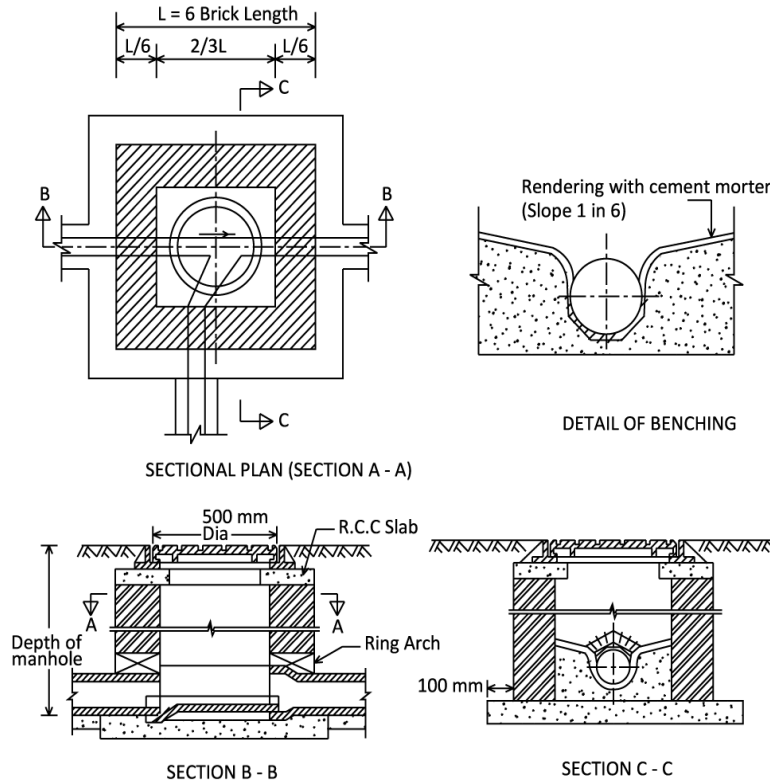


Figure 8.6.8 Detail of Manhole (Depth 1m and Below)

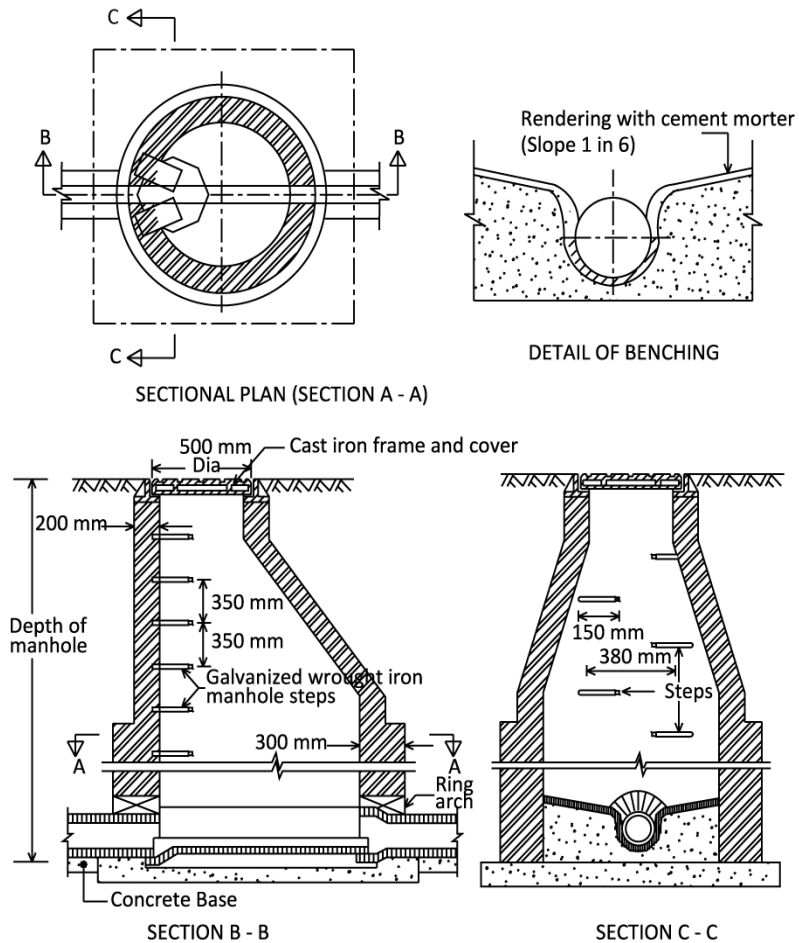


Figure 8.6.9 Detail of manhole (depth more than 1 m)

6.9.8 Protection Against Rodent

Holes through walls shall be such that they will not provide passage of rodent or other insects from room to room or from floor to floor. Materials used for embedding pipes shall be rodent proof.

6.9.9 Bedding and Backfilling

The choice of bedding and backfilling depends on the depth of the bed, and size and strength of the materials. Table 8.6.7 and Figure 8.6.10 show two types of bedding and backfilling, and minimum and maximum depth of cover for each type of bedding for rigid pipings. The bedding and backfilling for flexible pipings is shown in Figure 8.6.11. The minimum depth of bedding for flexible pipings shall be 0.3 m where there will be no extra surcharge load coming on pipe other than back filling.

The depth shall not be more than 10 m. The flexible pipe may be laid with less cover in fields and gardens. The bedding and backfilling shall be in accordance with Figure 8.6.12.

Table 8.6.7: Limits of Cover (m) for Standard Strength Rigid Pipes in any Width of Trench

Pipe Bore (mm)	Bedding Class	Fields and Gardens		Light Traffic Roads		Heavy Traffic Road	
		Min	Max	Min	Max	Min	Max
100	Type 1	0.3	7.4	0.4	7.4	0.4	7.2
	Type 2	0.3	5.8	0.5	5.8	0.5	5.5
150	Type 1	0.6	5.0	0.6	5.0	0.6	4.6
	Type 2	0.6	3.9	0.7	3.8	0.7	3.3

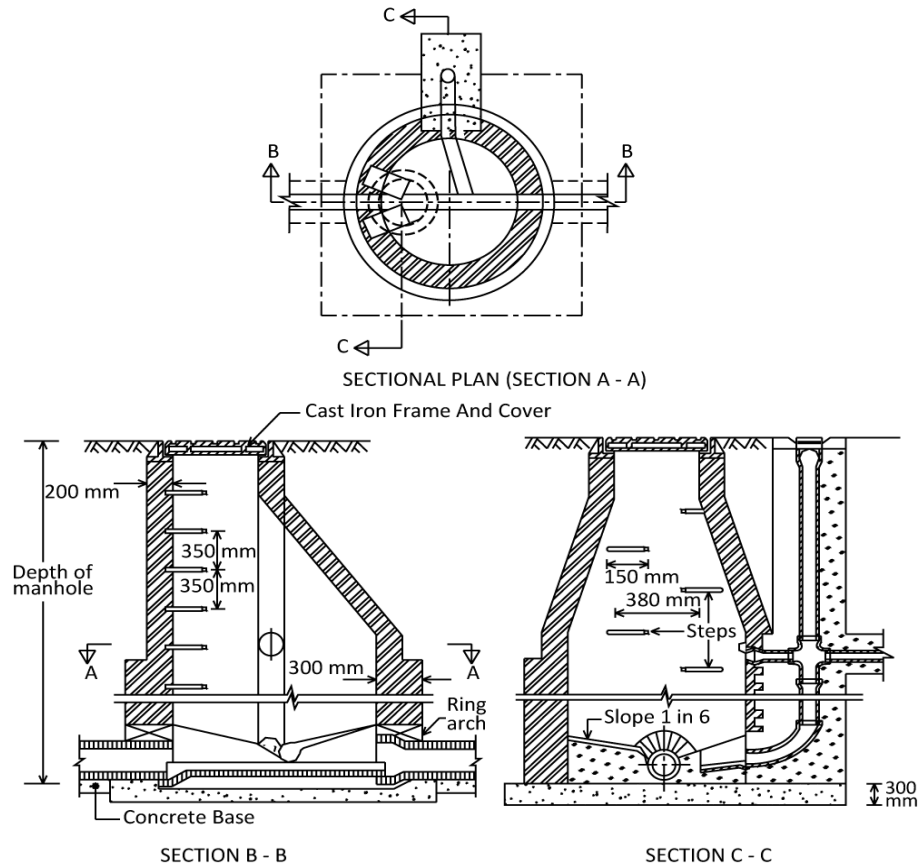


Figure 8.6.10 Drop manhole

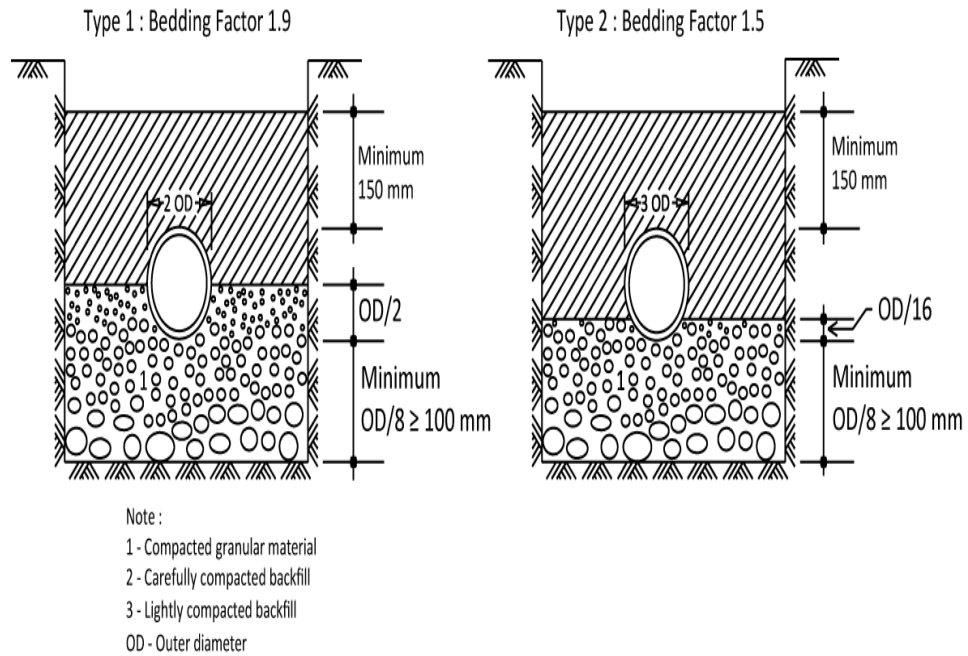


Figure 8.6.11 Bedding for rigid pipes

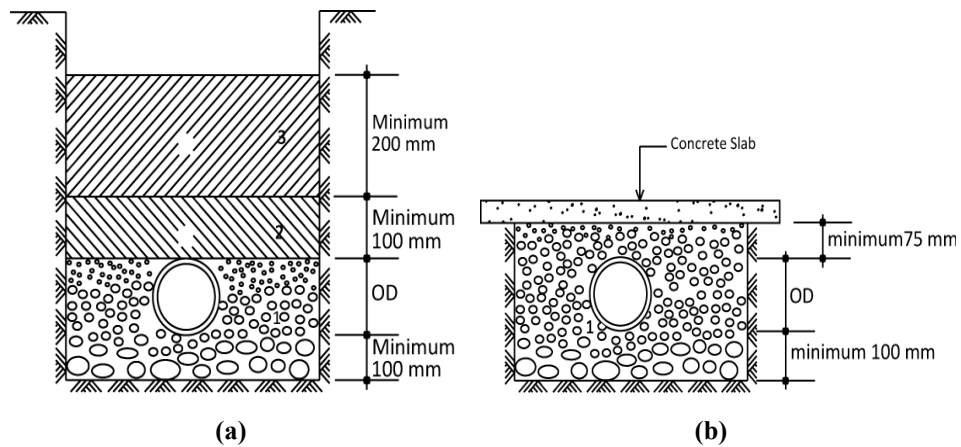


Figure 8.6.12 (a) Bedding for flexible pipes; (b) Flexible pipes bedding under concrete slab

6.9.10 Grease Traps

Oil and grease is found in wastes generated from kitchens in hotels, industrial canteens, restaurant, butcheries, some laboratories and manufacturing units having a high content of oil and greases in their final waste.

Waste exceeding temperature of 60° C should not be allowed in the grease trap. When so encountered it may be allowed to cool in a holding chamber before entering the grease trap.

Oil and greases tend to solidify as they cool within the drainage system. The solidified matter clogs the drain and the other matter in the waste stick to it due to the adhesion properties of the grease. Oil and greases are lighter than water and tend to float on the top of the waste water.

Grease traps shall be installed in building having the above types of wastes. In principle the grease laden water is allowed to retain in a grease trap which enables any solids to be settled or separated for manual disposal. The retention time allows the incoming waste to cool and allow the grease to solidify. The clear waste is then allowed to discharge into the building's drainage system.

6.9.11 Oil Interceptors

Oils and lubricants are found in wastes from vehicle service stations, workshops manufacturing units whose waste may contain high content of oils. Oils, for example, petroleum, kerosene and diesel used as fuel, cooking, lubricant oils and similar liquids are lighter than water and thus float on water in a pipe line or in a chamber when stored. Such oils have a low ignition point and are prone to catch fire if exposed to any flame or a spark and may cause explosion inside or outside the drainage system. The flames from such a fire spread rapidly if not confined or fire vented at the possible source. Lighter oils and lubricants are removed from the system by passing them through an oil interceptor/petrol gully. They are chambers in various compartments which allow the solids to settle and allow the oils to float to the top. The oil is then decanted in separate containers for disposal in an approved manner. The oil free waste collected from the bottom of the chamber is disposed in the building drainage system.

6.9.12 Septic Tank

6.9.12.1 Septic tank (Figures 8.6.13 and 8.6.14) discharging into either a subsurface disposal field or one or more seepage pits shall be required for the approval of drainage and sanitation plans for the places where public sewers are not available.

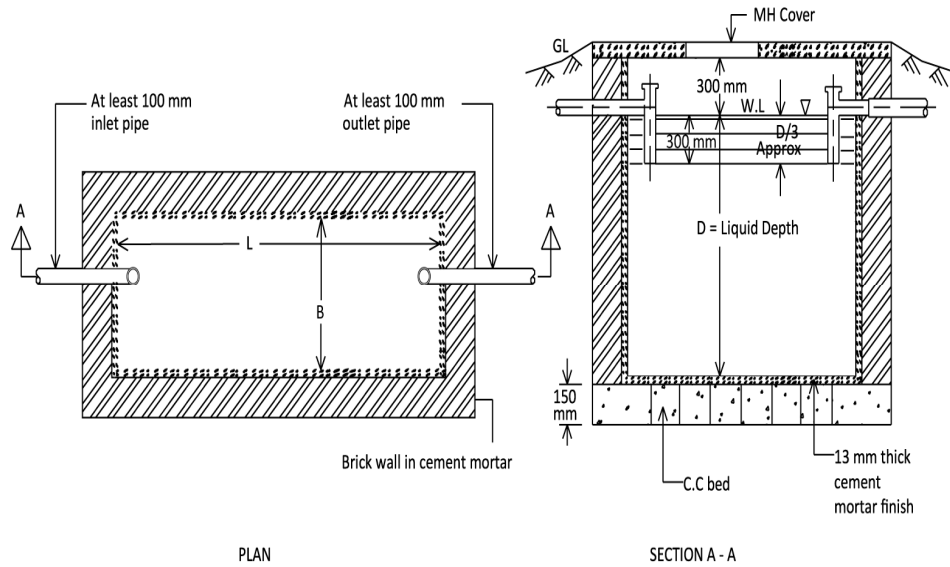


Figure 8.6.13 Typical one chamber brick septic tank

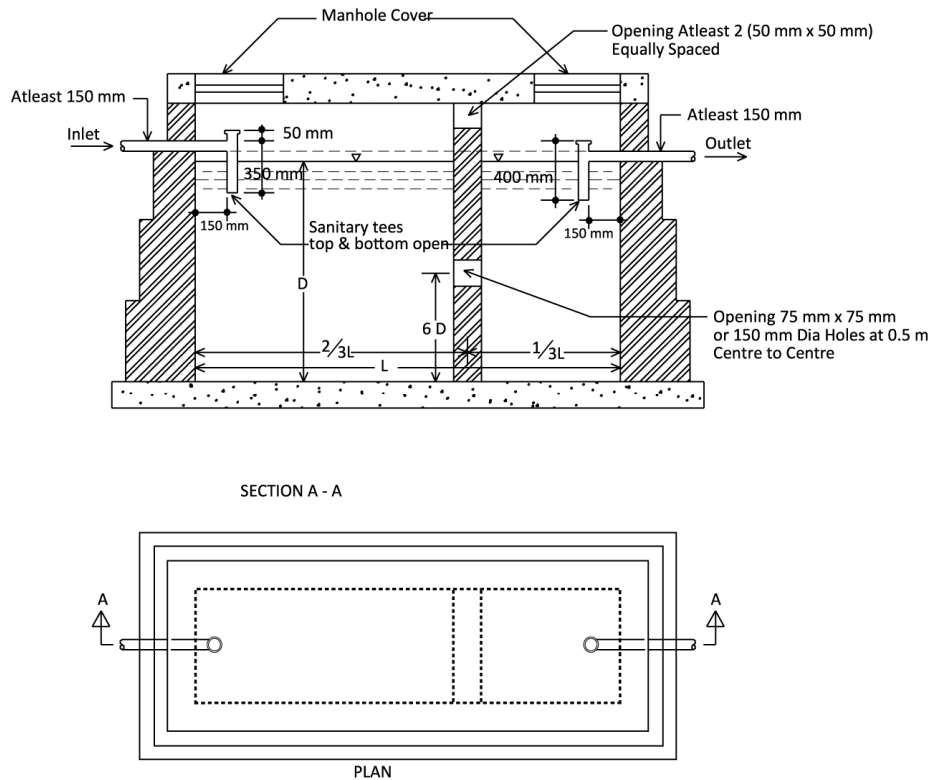


Figure 8.6.14 Typical two chamber concrete septic tank

6.9.12.2 Such disposal method shall be designed by a licensed professional in accordance with the requirement of the provisions of this Code and regulations of the concerned authorities.

6.9.12.3 The design of such system shall be on the basis of location with respect to wells or other sources of water, soil permeability, ground water elevation, area available and maximum occupancy of the building.

6.9.12.4 Sullage water shall not be discharged into the septic tank.

6.9.12.5 Effluent from septic tank(s) shall not discharge into open water courses.

6.9.12.6 The minimum distance for various components of the disposal system shall be in accordance with Table 8.6.8.

6.9.12.7 The flow into a septic tank may be calculated on the basis of plumbing fixtures discharging soil wastes simultaneously into it. The capacity of septic tank for residential buildings shall be determined according to the formula in Appendix S. For other occupancies a reduction factor shall be used as shown in Appendix S.

6.9.12.8 The septic tank shall have a minimum liquid capacity of 2000 liters, minimum width 1 m and minimum liquid depth 1 m. The minimum length of a septic tank shall be at least thrice its width. It is recommended that the maximum length of a septic tank shall be not more than 4 times its width.

6.9.12.9 The maximum size of a septic tank shall be limited to the number of users not exceeding 300 persons for residential buildings.

6.9.12.10 The volume required for digested sludge and scum may be computed on the basis of $0.04 \text{ m}^3/\text{capita}/\text{year}$. There shall be a clearance between top of the liquid level and bottom of the tank cover slab which shall be at least 300 mm.

6.9.12.11 The liquid retention time of a septic tank shall be at least 1 day.

6.9.12.12 The desludging frequency of a septic tank shall be at least 6 months interval and maximum once a year.

6.9.12.13 It is recommended to use a two chamber septic tank when the capacity of a septic tank exceeds 3000 liters. The inlet compartment of a two chamber septic tank shall have a capacity not less than two-third of its total capacity (Figure 8.6.14).

Table 8.6.8: Location of Components of Sewage Disposal System

System Component	Distance (m)				
	Building Foundation	Well	Stream	Seepage Pit	Dry Well
Septic tank	1.5	8	-	1.5	-
Disposal field	3	15	7.5	6	6
Seepage pit	4.5	15	15	6	6
Dry well	3	15	-	6	-

6.9.12.14 The septic tank shall be constructed of corrosion resistant material and be of permanent water tight construction. The manhole cover and the roof of the tank shall be designed for at least 7 kPa live load. The inlet compartment shall be provided with a manhole. Outlet compartment shall also be provided with a manhole. The design guideline of a septic tank is presented in Appendix S.

6.9.13 Imhoff tank(s)

6.9.13.1 Imhoff tank (Figure 8.6.15) discharging into either a subsurface disposal field or one or more seepage pits shall be required for the approval of drainage and sanitation plans for the places where public sewers are not available.

6.9.13.2 Imhoff tank shall be used where more than 300 peoples of residential buildings are to be served.

6.9.13.3 The settling chamber shall be designed for a detention period of 1.5 to 4.0 hours and an over flow rate of 0.95 to 1.4 m/hr. The displacement velocity should not be more than 18 m/hr. Sloping sides of settling chamber shall have inclination 1.0 horizontal to 1.2 vertical. The slot at the bottom is 0.15 to 0.25 meter measured along the slope of the hopper. The width of the side spaces should not be less than 0.45 meter. The digestion chamber should have a capacity to store about 6 to 12 months digested sludge. Capacity of the digestion chamber can be found from the following formula-

$$C = \left[V_f - \frac{2}{3}(V_f - V_d) \right] t \quad (8.6.1)$$

Where, C is the volume of the digestion tank in m^3/capita below the neutral zone which is 0.2 m below the slot. V_f is the volume of fresh sludge in $m^3/\text{capita}/\text{day}$ and V_d is the volume of the digested sludge in $m^3/\text{capita}/\text{day}$ and t is the time required for digestion. The digestion chamber should have sloping side with a ratio of 4 horizontal to 5 vertical.

6.9.13.4 The Capacity found in Sec. 6.9.12.3 is true for residential buildings. For other occupational buildings use a reduction factor as shown in Appendix S.

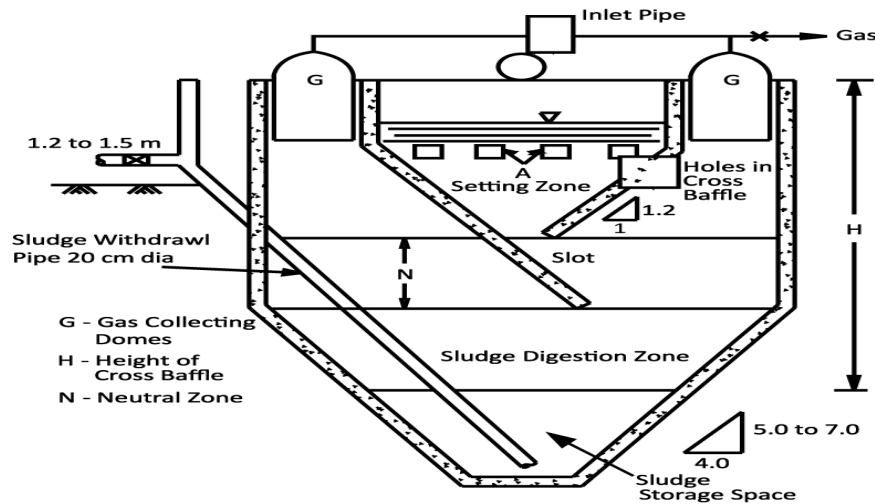


Figure 8.6.15 Cross section of an imhoff tank

6.9.14 Installation

Septic and Imhoff tanks shall be located with a horizontal distance not less than specified in Table 8.6.9 between various elements. Tanks installed in ground water shall be securely anchored. Three inch (75 mm) thick compacted bedding shall be provided for all septic and other treatment tank installations. The bedding material shall be sand, gravel, granite, lime rock or other noncorrosive materials of such size that the material passes through a 0.5 inch (12.5 mm) screen.

Table 8.6.9: Minimum horizontal separation distances for treatment tanks element distance

Elements	Distance (m)	Elements	Distance (m)
Building	1.5	Reservoir	7.5
Cistern	7.5	Spring	15
Foundation wall	1.5	Stream or watercourse	7.5
Lake, high water mark	7.5	Swimming pool	4.5
Lot line	0.6	Water service	1.5
Pond	7.5	Well	7.5

Each cleanout shall be installed so that it opens to allow cleaning in the direction of flow of the soil or waste or at right angles thereto, and except in the case of Wye branch and end-of-line cleanouts, shall be installed vertically above the flow line of the pipe.

The bath tub shall be fitted with overflow and waste pipe of nominal diameter of not less than 32 mm and 40 mm respectively.

6.9.15 Disposal Field and Seepage Pit

6.9.15.1 A distribution box shall be provided to receive the effluent from the septic tank or Imhoff tank to assure equal distribution to each individual line of disposal field. The distribution box shall be connected to the septic tank or Imhoff tank by a watertight sewer line and shall be located at the upper end of disposal field. Figure 8.6.16 shows the plans and sections of typical distribution boxes.

6.9.15.2 Soil percolation tests (at least for three holes) shall be performed at the site of a proposed individual sewage disposal system installation to determine the suitability of soil and site.

6.9.15.3 The liquid capacity (volume below inlet line) of seepage units (disposal field or seepage pit) shall be at least twice that of a septic tank or Imhoff tank. Effective absorption area of seepage unit may be computed in accordance with Table 8.6.10.

6.9.15.4 No seepage unit shall be extended into water table directly. The bottom of seepage unit shall be at least 1 meter above the highest water table.

6.9.15.5 Each disposal field shall have at least two outlet distribution lines from the distribution box. No portion of disposal field shall be installed under any pavement or any area where there will be vehicular traffic or parking

6.9.15.6 Minimum standards for disposal field construction shall be as shown in Table 8.6.11.

6.9.15.7 Seepage pit (soak pit) shall be lined with stone, brick or concrete blocks laid up dry with open joints that are backed up with at least 75 mm coarse aggregate. The joints above the inlet shall be sealed with cement mortar. A reinforced concrete cover shall be provided. For cover area more than 0.75 m² the pit shall have an access manhole. The bottom of the pit shall be filled with coarse gravel, or crushed stone/brick to a depth of 300 mm. Figure 8.6.17 provides the details of a seepage pit.

6.9.15.8 Large dry well shall be constructed in accordance with the requirements for seepage pit (Sec 6.9.12). However, for small dry wells handling limited quantities of waste water the pit may consist of a 2.0 m deep and 1.0 m diameter pipe filled with crushed bricks/stone, Figures 8.6.18 and 8.6.19.

6.9.15.9 French drains may be employed as surface water drains for drainage of unpaved surfaces.

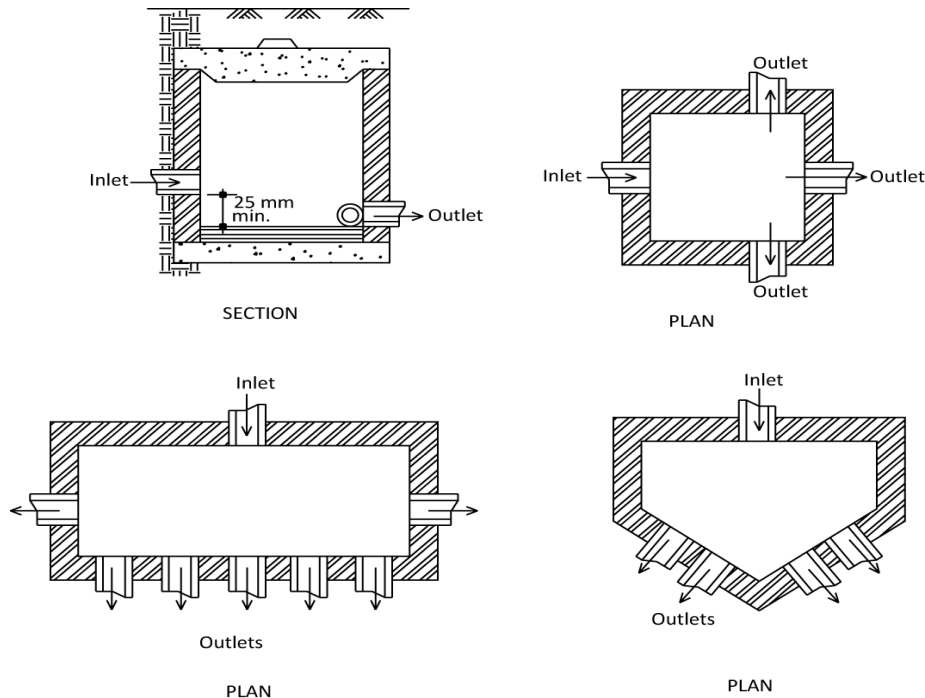


Figure 8.6.16 Distribution boxes

Table 8.6.10: Absorptive Capacity of Disposal Field and Seepage Pit

Percolation Test Rate in Minutes for Water to Fall 25 mm	Effluent Allowance Rate of Seepage Unit (l/m ² /day)	
	Disposal Field Trenches (bottom of trench)	Seepage Pit (wall area)
2 or less	128	172
5	96	128
10	68	92
30	32	44
60 (not recommended)	16	24
over 60 (not suitable)	-	-

Table 8.6.11: Design Features of Disposal Field

Design Items	Requirements
Number of lateral branches	2
Maximum length of branch	20 m
Minimum diameter of field distribution pipe	100 mm
Maximum slope of field distribution pipe	3.3 mm per m
Depth of trench	0.45 m to 9 m
Trench bottom, minimum above ground water	0.61 m
Trench bottom width	0.45m to 0.75m
Depth of coarse material	
under pipe	150 mm
over pipe	50 mm
Size of coarse material	12 mm to 63 mm

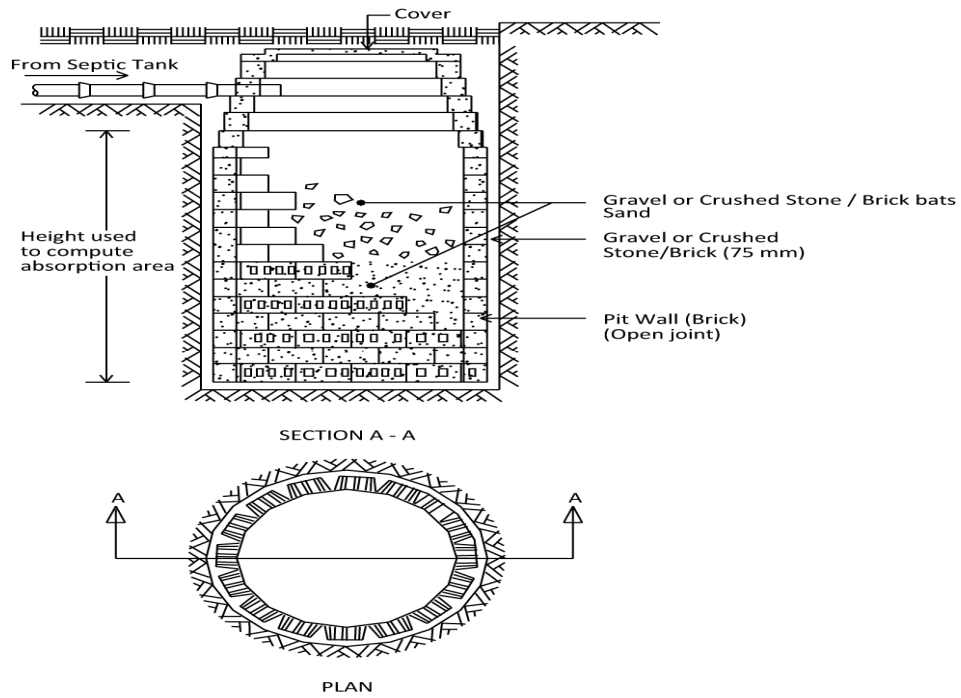


Figure 8.6.17 Typical seepage pit

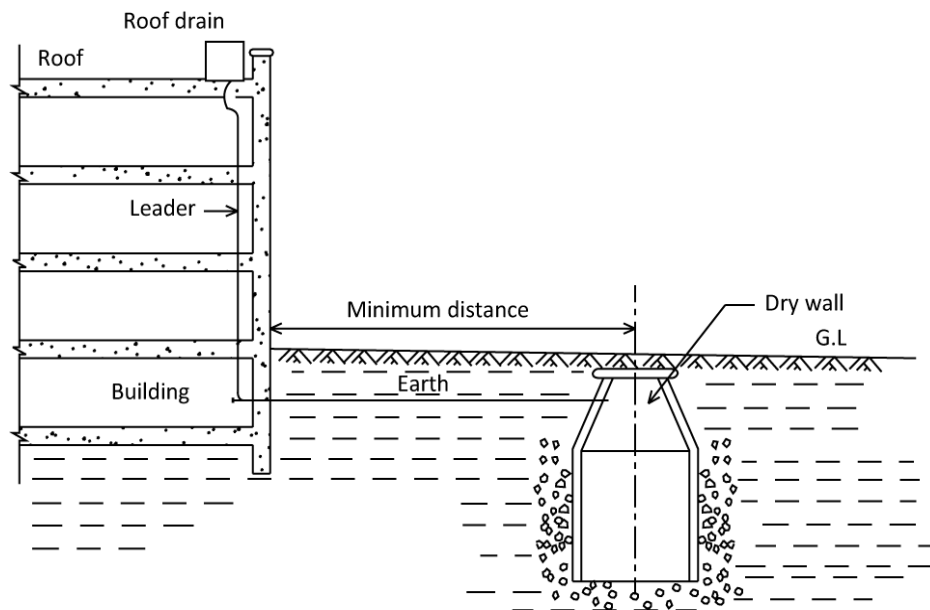


Figure 8.6.18 Typical location of a dry wall

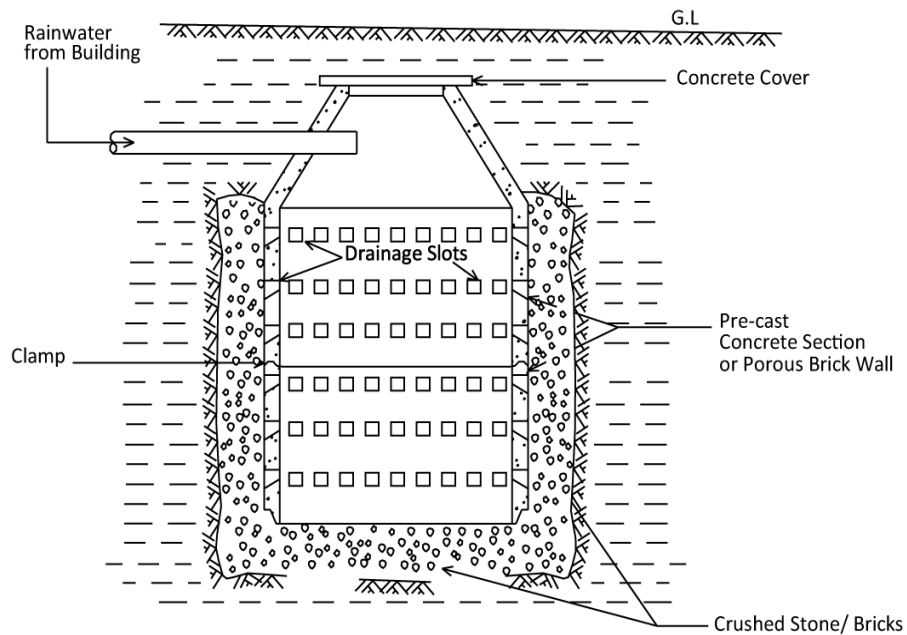


Figure 8.6.19 Installation details of a dry wall

6.10 Design of Drainage and Sanitation System

6.10.1 Estimation of Maximum Load Weight of Waste Water

To estimate the total load weight carried by a soil or waste pipe, the relative load weight for different kinds of fixtures are provided in Table 8.6.12. Table 8.6.13 provides an approximate rating of those fixtures not listed in Table 8.6.12.

Table 8.6.12: Fixture Units for Different Sanitary Appliances or Groups

Type of Fixture	Fixture Unit Value as Load Factor
One bathroom group consisting of water closet, wash basin and bath tub or shower stall :	
a) Flush Tank water closet	3
b) Flush-valve water closet	6
Bathtub*	2
Bidet	2

Type of Fixture	Fixture Unit Value as Load Factor
Combination sink and tray (drain board)	2
Drinking fountain	0.5
Floor traps†	1
Kitchen sink, domestic	2
Wash basin, ordinary‡	1
Wash basin, surgeon's	2
Shower stall, domestic	2
Shower (group) per head	3
Urinal, wall hung	4
Urinal, stall	4
Water closet, tank operated	3
Water closet, valve operated	6
* A shower head over a bath tub does not increase the fixture unit value.	
† Size of floor trap shall be determined by the area of surface water to be drained.	
‡ Wash basin with 32 mm and 40 mm trap have the same load value.	

Table 8.6.13: Fixture Unit Values for Fixtures Based on Fixture Drain or Trap Size

Fixture Drain on Trap Size	Fixture Unit Value
30 mm and smaller	1
40 mm	2
50 mm	3
65 mm	4
75 mm	5
100 mm	6

6.10.2 Gradient and Size of Pipe

6.10.2.1 The building drains and sewer shall be designed to discharge the peak simultaneous load weight flowing half-full with a minimum self-cleansing velocity of 0.75 m per second. However, flatter gradient may be used if required but the minimum velocity shall not be less than 0.6 m per second. Again, it is undesirable to employ gradients giving a velocity of flow greater than 2.5 m per second.

6.10.2.2 The maximum number of fixture units that may be connected to a given size of building sewer, building drain, horizontal branch or vertical soil or waste stack is provided as in Tables 8.6.14 and 8.6.15.

6.10.3 Size of Vent Piping

6.10.3.1 The size of vent piping shall be determined from its length and the total number of fixture units connected thereto in accordance with Table 8.6.16.

6.10.3.2 The branch vent shall be sized in accordance with Table 8.6.17.

Table 8.6.14: Maximum Number of Fixture Units that can be connected to Branches and Stacks

Diameter of Pipe (mm)	Any Horizontal Fixture Branch ^a	One Stack of 3 Storeys in Height or 3 Intervals	More than 3 Storeys in Height	
			Total for Stack	Total at One Storey or Branch Interval
30	1	2	2	1
40	3	4	8	2
50	6	10	24	6
65	12	20	42	9
75	20	30	60	16
100	160	240	500	90
125	360	540	1100	200
150	620	960	1900	350
200	1400	2200	3600	600
250	2500	3800	5600	1000
300	3900	6000	8400	1500
375	7000	b	b	b

^a Does not include branches of the building sewer. ^b Sizing load based on design criteria

Table 8.6.15: Maximum Number of Fixture Units that can be connected to Building Drains and Sewers

Diameter of Pipe (mm)	Maximum Number of Fixture Units that can be Connected to any Portion* of the Building Drain or the Building Sewer for Various Slopes			
	1/200	1/100	1/50	1/25
100	-	180	216	250
150	-	700	840	1000
200	1400	1600	1920	2300
250	2500	2900	3500	4200
300	2900	4600	5600	6700
375	7000	8300	10000	12000

* Includes branches of building sewer

6.11 Construction Relating to Conveyance of Sanitary Wastes

6.11.1 Conveyance of Sanitary Wastes

6.11.2 The layout of drainage systems shall be simple. Change of direction and gradient shall be minimized and shall be as easy as practicable.

6.11.3 The excavation, where necessary, shall be made in accordance with Table 8.6.18.

6.11.4 The depth of cover shall be in accordance with Sec 6.9.9.

6.11.5 The pipe shall be laid to even gradients and change of gradient shall be combined with an access point (Sec 6.10.2). However, access points shall be provided only if blockages could not be cleared without them.

6.11.6 The joints and connection in drainage and venting system shall be gastight and watertight for the pressures required by the test, with the exception of those portions of perforated or open joint piping which will be installed for the purpose of collecting and conveying ground or seepage water to the underground storm drains.

6.11.7 Piping in drainage and venting system shall be installed without undue strains and stresses and provision shall be made for expansion, contraction and structural settlement. Vertical piping shall be secured at sufficiently close intervals to keep the pipe in alignment and carry the weight of the piping and its content. The horizontal piping shall be supported at sufficiently close intervals to keep it in alignment and to prevent sagging.

Table 8.6.16: Size and Length of Vent Stacks and Stack Vents

Diameter of Soil or Waste Stack (mm)	Total Fixture Unit Connected to fixture	Maximum Development Length of Vent (m)* for Diameter (mm) of Vent Pipes											
		30	40	50	65	75	100	125	150	200	250	300	
30	2	9											
40	8	15	45.5										
40	10	9	30.5										
50	12	9	22.5	61									
50	20	8	15	45.5									
65	42	9			30.5	91.5							
75	10	12.5		45.5	109.5	317							
75	21	9.5		33.5	82	247							
75	53	8		28.5	70	207							
75	102	7.5		26	64	189							
100	43	10.5			26	76	298.5						
100	140	8			19.5	61	228.5						
100	320	7			16.5	52	195						
100	540	6.5			15	45.5	176.5						
125	190	8.5				25	97.5	301.5					
125	490	6.5				19	76	231.5					
125	940	5.5				16	64	204					
125	1400	4.5				15	58	180					
150	500	10					39.5	122	305				

Diameter of Soil or Waste Stack (mm)	Total Fixture Unit Connected to fixture	Maximum Development Length of Vent (m)* for Diameter (mm) of Vent Pipes										
		30	40	50	65	75	100	125	150	200	250	300
150	1100					6.5	30.5	94.5	237.5			
150	2000					6	25.5	79	201			
150	2900						23.5	73	183			
200	1800						9.5	29	73	286.5		
200	3400						7	22	58	219.5		
200	5600						6	19	48.5	186		
200	7600						5.5	17	42.5	170.5		
250	4000							9.5	23.5	94.5	292.5	
250	7200							7	18	73	225.5	
250	11000							6	15.5	61	192	
250	15000							5.5	14	55	173.5	
300	7300								9.5	36.5	116	286.5
300	13000								7	28.5	91.5	219.5
300	20000								6	24	76	186
300	26000								5.5	22	70	152.5
375	15000									12	39.5	94.5
375	25000									9.5	29	73
375	38000									8	24.5	61
375	50000									7	22.5	55

* The development length shall be measured from the vent connection to the open air

Table 8.6.17: Minimum Diameter and Maximum Length of Individual, Branch, and Circuit Vents for Horizontal Drainage Branches

Diameter of Horizontal Drainage Branch (mm)	Slope of Horizontal Drainage Branch (mm/m)	Maximum Development Length of Vent (m) for Diameter (mm) of Vent Pipe									
		30	40	50	65	75	100	125	150	200	250
30	20	NL*									
	40	NL									
40	20	NL	NL								
	40	NL	NL								
50	10	NL	NL	NL							
	20	88	NL	NL							
	40	45	115	NL							
65	10	54	137	NL							
	20	29	73	NL	NL						
	40	15	39	NL	NL						
75	10	-	58	NL	NL	NL					
	20		29.5	128	NL	NL					
	40		15	67	NL	NL					
100	10		-	58	NL	NL	NL				
	20			30	94	NL	NL				
	40			14.5	48.5	125	NL				
125	10					149	NL	NL			
	20				58	76	NL	NL			
	40				29.5	39.5	NL	NL			
150	10						NL	NL	NL		
	20				14	58	76	NL	NL		
	40					29.5	39.5	NL	NL		
200	10							NL	NL	NL	
	20					14	58	94.5	NL	NL	
	40						27.5	45.5	125	NL	NL

Diameter of Horizontal Drainage Branch (mm)	Slope of Horizontal Drainage Branch (mm/m)	Maximum Development Length of Vent (m) for Diameter (mm) of Vent Pipe									
		30	40	50	65	75	100	125	150	200	250
250	10								152	NL	NL
	20						11.5	58	73	NL	NL
	40							26	33.5	NL	NL
300	10							9.5	54.5	NL	NL
	20								24	128	NL
	40								8	61	NL

* NL means no limit; Actual value in excess of 150 m.

Table 8.6.18: Minimum Width at Bottom of Trench

Depth of Trench (m)	Width of Trench (m)
Up to 1.2	Diameter of pipe plus 0.4
Above 1.2	Diameter of pipe plus 0.45

Note: Trench top width shall not be less than 0.75 m for depths exceeding 0.9 m

6.12 Refuse Chute System

6.12.1 All buildings higher than 6-storeys shall be provided with refuse chute system for transporting and collecting refuse from different floors in a sanitary way. The refuse shall be received from the respective floor through an inlet hopper in to the chute which conveys refuse and discharges into the collection chamber. The refuse from the collection chamber shall be cleared at suitable intervals.

6.12.2 The refuse chute, inlet hopper and collection chamber shall be constructed with smooth and nonflammable materials.

6.12.3 The hopper shall be self-cleaning and shall be fitted with self-closing shutter to prevent the passage of foul gases inside the building.

6.12.4 The diameter of the chute shall not be less than 300 mm. It shall be adequately ventilated at the top. The chute shall be provided with suitable arrangements for flushing with water for the full length.

6.13 Basement Floor Drainage System

6.13.1 All buildings having basement floor below the surrounding sewer system and area more than 1000 m² shall have pumping system to drain out wastewater.

6.13.2 All buildings having basement floor below the surrounding sewer system and area more than 1000 m² shall have one sump pit for every 1000 m².

6.13.3 For more than one sump pit, pits shall be connected to a master sump pit from where pumping shall be done to drain out the waste. Minimum diameter of sump pit connection drain pipe shall be 75 mm.

6.13.4 For wastewater and sewage drainage from basement floor separate drainage system shall be provided.

6.13.5 The collection chamber shall be of suitable size and located at ground level.

6.13.6 The design and arrangement of the system shall be in accordance with established engineering practices.

6.14 Health Care Drainage System

6.14.1 General

The health care drainage system shall comply with applicable drainage and venting requirements specified in this Chapter and with this Section.

6.14.2 Special Fixtures and Equipment

The hospital shall be provided with clinical sink, bedpan washer and such other fixtures and equipment for disposal of bedpan contents and for the cleansing and disinfection of such fixtures. A clinical sink shall not be considered as a substitute for service sink.

6.14.3 Bedpan Washer and Clinical Sink

Bedpan washer and clinical sink shall be connected to soil branches and soil stacks through a water seal trap. The bedpan washer with vapour vent connection shall be provided with additional local vent stack. The minimum vent stack size for bedpan washer shall be in accordance with Table 8.6.19.

Table 8.6.19: Minimum Vent Stack Size in Bedpan Drainage

No of Bedpan Washer	Diameter (mm)
Up to 3 bedpan washers at different floors	50
4 to 6 bedpan washers	75
7 to 12 bedpan washers	100

The bottom of the bedpan local vent stack (except for one bedpan washer) shall be drained indirectly into sanitary drainage system through traps. The size of the trap and connecting pipe shall be at least the size of the vent stack. At least 6 mm diameter water supply piping shall be taken from each flush supply of each bedpan washer on the discharge side of vacuum breaker, trapped to form 75 mm or more trap seal and connected to the local vent stack on each floor.

6.14.4 Sterilizer Vent Stack

The pressure or non-pressure sterilizer shall have vent connection to the sterilizer vent stack. This vent connection shall be accessible for inspection and cleaning. The size of sterilizer vent stack shall be as follows:

6.14.4.1 Pressure sterilizers

The minimum diameter for pressure sterilizer vent stack shall be 63 mm. The stack size for combinations of pressure sterilizer exhaust shall be in accordance with Table 8.6.20.

Table 8.6.20: Pressure Sterilizer Vent Stack Size

Stack Size (mm)	Number of Connections Permitted for Different Combination Sizes (mm)			
	19	25	31	38
38	3	-	-	-
	-	2	-	-
	-	-	1	-
	2	1	-	-
50	6	-	-	-
	-	3	-	-
	-	-	2	-
	-	-	-	1
	3	2	-	-
	2	1	1	-
	1	1	-	1
75	15	-	-	-
	-	7	-	-
	-	-	5	-
	-	-	-	3
	1	5	-	1
	-	1	2	2

6.14.4.2 Pressure instrument washer sterilizers

The minimum size of sterilizer vent stack for instrument washer sterilizer shall be 50 mm for up to two sterilizers. The 75 mm stack will serve up to four sterilizers.

6.14.4.3 Non-pressure sterilizers

The minimum diameter of non-pressure sterilizer vent stack shall be 50 mm for utensil sterilizer and 38 mm for instrument sterilizer. Multiple installations shall be sized in accordance with Table 8.6.21.

6.14.4.4 Bedpan steamers

The diameter for one bedpan steamer shall be 38 mm. The stack size for combinations shall be in accordance with Table 8.6.21.

6.14.5 Vent Extension

The termination of vent stack shall be in accordance with Sections 6.9.5.6 and 6.9.6.7.

Table 8.6.21: Non-pressure Sterilizer Vent Stack or Bedpan Steamer Sizes

Stack Sizes (mm)	No of Connections Permitted for Different Connection Sizes(mm)	
	38	50
38	1	-
50	2	-
	-	1
	1	1
75	4	-
	-	2
	2	2
100	8	-
	-	4
	4	4

6.14.6 Special Fixture Drainage

The device, appurtenance and appliance required for special purposes such as refrigerators, ice boxes, cooling or refrigerating coils etc. shall be protected against backflow with adequate air gap between the equipment inlet and drainage outlet.

6.14.7 Mental Health Care Centre and Prisoners Cell

The pipes and traps used in mental health care buildings and prisoner cells shall not be exposed and all fixtures shall be securely bolted through walls.

6.15 Inspection, Testing and Completion Certificate

6.15.1 Inspection

The new drainage and sanitation system or part of the existing system shall not be covered or enclosed or put into operation until it has been inspected, tested and approved by the Authority. The Authority may examine the appliances and fittings before their installation or during the progress of the work. An installation of plumbing work, whether new or existing, which is found to be defective or unsafe shall not be allowed to continue in use unless corrections have been made to comply with the Code requirements.

6.15.2 Testing**6.15.2.1 Drainage and venting system**

The piping of drainage and venting system shall be tested first with water. The final test of completed drainage and venting system may be done by smoke test. The water and smoke tests shall be performed as described below:

- (a) **Water Test:** The water test shall be applied to the drainage and venting system either for the whole system or part (section) thereof. For the entire system, all openings in the piping except the highest opening shall be closed, and the system filled with water to the point of overflow. For the system to be tested in sections each opening shall be tightly plugged except the highest opening of the section under test and each section shall be filled with water but no section shall be tested with less than a 3 m head of water. In testing successive sections, at least the upper 3 m of the next preceding section shall be tested so that no joint or pipe in the building (except the top most 3 m of the system) shall have been submitted to a test of less than a 3 m head of water. The water shall be kept in the system or in the portion under test for at least 15 minutes before the inspection starts. The system or the part of the system under test shall be watertight at all points.
- (b) **Smoke Test:** The final test for gas and water tightness of the completed drainage and venting system may be performed by smoke test. The test is performed by filling all traps with water and then introducing smoke into the system produced by one or more smoke machines. When the smoke appears at the stack openings on the roof, they shall be closed and a pressure equivalent to 25 mm head of water shall be built and maintained for 15 minutes before inspection starts.

6.15.2.2 Building sewer

The Connection between building sewer and public sewer or individual sewage disposal system shall be closed by inserting a test plug. The building sewer shall be filled with water under a pressure of not less than 30 kPa for at least 15 minutes. The system shall be able to maintain the test pressure.

6.15.3 Completion Certificate

After the installation of drainage and sanitation system, the licensed plumber shall give a completion certificate to the authority in a prescribed form (Appendix T) for inspection and testing. After testing, the Authority will give the final approval (as presented in completion certificate form, in the Appendix T) to use the system.

6.16 Guide to Maintenance

6.16.1 The drainage and sanitation system shall be maintained in a sanitary and safe operating condition by the owner or his designated agent. All device or safeguards required by the Code shall be maintained in working order.

6.16.2 The following operations shall be carried out during periodical cleaning of a drainage and sanitation system:

- (a) The covers of inspection chambers and manholes shall be removed and the side benching and channels shall be scrubbed.
- (b) All lengths of main and branch drains shall be rodded by means of drain rods and a suitable rubber or leather plunger. After rodding, the drains shall be thoroughly flushed with clean water.
- (c) The ladders/rings in deep manholes and the manhole covers shall be painted.
- (d) All surface drains shall be cleaned.
- (e) All subsoil drains shall be examined for obstruction at the open joints.
- (f) Refuse chute system shall be cleaned.

6.17 List of Related Appendices

Appendix Q	Application for Permit to Construct Drainage and Sanitation System
Appendix R	One-hour Rainfall
Appendix S	Design Guideline of a Septic Tank
Appendix T	Completion Certificate (Drainage and Sanitation Works)

PART VIII
Chapter 7
Rainwater Management

7.1 Purpose

The purpose of this Chapter is to set forth provisions for planning, design and installation of rainwater management systems in buildings.

7.2 Scope

7.2.1 This Chapter specifies the general requirements for rain water harvesting for different categories of buildings according to their occupancy classification together with all ancillary works of ground water recharging such as perforated piping, pits and inspection chambers.

7.2.2 This Chapter also covers the design, installation and maintenance of elements for rain water drainage systems around the building and led to public storm sewers or to nearby suitable surface water system.

7.2.3 The storm water drainage on ground is not covered by this Code.

7.3 Terminology

This Section provides an alphabetical list of all terms used and applicable to this Chapter of the Code. In case of any conflict or contradiction between a definition given in this Section and that in any other chapter or part of the Code, the meaning specified in this Chapter shall govern for interpretation of the provisions of this Chapter.

BEDDING FACTOR	The ratio of the product of design load and factor of safety to the minimum crushing strength.
BRANCH	Any part of the piping system other than a main or riser.
BUILDING DRAIN	The building (house) drain is that part of the lowest piping or open channel of a drainage system which receives the discharges from soil, waste, and other drainage systems inside the walls of the building and conveys the same to the building (house) sewer, beginning at 0.9 m outside the building wall.
BUILDING SEWER	The building (house) sewer is that part of the horizontal piping of a drainage system which extends from the end of the building drain and which receives the discharge of the building drain and conveys it to a public sewer, private sewer, individual sewage disposal system, or other point of disposal. Also known as SEWER.

BUILDING STORM DRAIN	A building (house) storm drain is a building drain used for conveying rain water, surface water, ground water, subsurface water, condensate, cooling water, or other similar discharge to a building storm sewer or a combined sewer, extending to a part not less than 0.9 m outside the building wall. Also known as STORM DRAIN.
DRAIN	A drain is any pipe or open channel which carries waste water or waterborne wastes in a building drainage system.
DRAINAGE SYSTEM	A drainage system (drainage piping) includes all piping within public/private premises which conveys sewage, rain water, or other liquid wastes to a legal point of disposal, but does not include mains of public sewer system or private or public sewage treatment or disposal plant.
DRINKING FOUNTAIN	A fountain or a tap raised from the floor with potable water supply connection.
EXISTING WORK	The existing work is a plumbing system or any part thereof which was installed prior to the date of enforcement of this Code.
FIXTURE UNIT	A fixture unit is a quantity in terms of which the load producing effects on the plumbing system of different kinds of plumbing fixtures are expressed on some arbitrarily chosen scale.
FLUSH VALVES	A flush valve is a device located at the bottom of the tank for the purpose of flushing water closets and similar fixtures.
FRENCH DRAIN	A shallow trench filled with coarse rubble, clinker or similar material with or without field drain pipes.
GRADE	The grade is the slope or fall of a line of pipe in reference to a horizontal plane. In drainage it is usually expressed as the fall in mm per m length of pipe.
HORIZONTAL BRANCH	A horizontal branch is a drain pipe extending laterally from a soil or waste stack or building drain, with or without vertical sections or branches, which receives the discharge from one or more fixture drains and conducts it to the soil or waste stack or to the building (house) drain.
HORIZONTAL PIPE	A horizontal pipe is any pipe or fitting which is installed in a horizontal position or which makes an angle of less than 45 degrees with the horizontal.

INTERCEPTOR	An interceptor is a device designed and installed so as to separate and retain deleterious, hazardous, or undesirable matter from normal wastes and permit normal or liquid wastes to discharge into the disposal terminal by gravity.
INVERT	The lowest point of the internal surface of a pipe or channel at any cross-section.
LEADER	A vertical drainage pipe that carries rainwater from roof or gutter drain to building storm drain or building drain or private disposal system.
LIQUID WASTE	The liquid waste is the discharge from any fixture, appliance, or appurtenance in connection with a plumbing system which does not receive fecal matter.
LOAD FACTOR	The load factor is the percentage of the total connected fixture unit flow rate which is likely to occur at any point in the drainage system. It varies with the type of occupancy, the total flow unit above the point being considered, and with the probability factor of simultaneous use.
MAIN	The main of any system of continuous piping is the principal artery of the system, to which branches may be connected.
MAIN SEWER	See Public Sewer.
MANHOLE	An opening by which a man may enter or leave a drain, a sewer or other closed structure for inspection, cleaning and other maintenance operations, fitted with a suitable cover.
MANHOLE CHAMBER	A chamber constructed on a drain or sewer so as to provide access thereto for inspection, testing or the clearance of obstruction.
OFFSET	An offset in a line of piping is a combination of elbows or bends which brings one section of the pipe out of line but into a line parallel with the other section.
PUBLIC SEWER	A public sewer is a common sewer directly controlled by public authority. Also known as MAIN SEWER.
RISER	A water supply pipe that extends vertically one full storey or more to convey water to branches or fixtures.
SANITARY SEWER	A sanitary sewer is a pipe which carries sewage and excludes storm, surface, and ground water. Also known as SEWER.
SEEPAGE PIT	See SOAK PIT.

SEWAGE	The sewage is any liquid waste containing animal or vegetable matter in suspension or solution and may include liquids containing chemicals in solution.
SEWER	See BUILDING SEWER or PUBLIC SEWER or SANITARY SEWER or STORM SEWER.
SLUDGE	A settled portion of the sewage or waste water effluent from a sedimentation tank in semi-solid condition.
SOAK PIT	A pit, dug into permeable soil lined to form a covered perforated chamber or filled with sand at the bottom and gravel or broken bricks at the top into which effluent from a septic tank or storm water is led and from which these may soak away into the ground. Also known as SEEPAGE PIT or SOAK WELL.
SOAK WELL	See SOAK PIT.
SOIL PIPE	A soil pipe is any pipe which conveys the discharge of water closets, urinals, or fixtures having similar functions, with or without the discharge from other fixtures, to the building drain or building sewer.
STACK	A stack is the vertical main of a system of soil, waste, or vent piping.
STORM DRAIN	See Building Storm Drain.
STORM SEWER	A storm sewer is a sewer used for conveying rain water, surface water, condensate, cooling water, or similar liquid wastes, exclusive of sewage and industrial waste. Also known as SEWER.
SUBSOIL DRAIN	A subsoil drain is a drain which receives only subsurface or seepage water and conveys it to a place of disposal.
SULLAGE	The discharge from wash basins, sinks and similar appliances, which does not contain human or animal excreta.
SUMP	A sump is a tank or pit which receives sewage or liquid waste, located below the normal grade of the gravity system, and which must be emptied by mechanical means.
SUPPORTS	The supports, hangers, and anchors are devices for supporting and securing pipe and fixtures to walls, ceilings, floors, or structural members.

TRAP	A trap is a fitting or device so designed and constructed as to provide, when properly vented, a liquid seal which will prevent the back passage of air without materially affecting the flow of sewage or waste water through it.
TRAP SEAL	The trap seal is the maximum vertical depth of liquid that a trap will retain, measured between the crown weir and the top of the dip of the trap.
VERTICAL PIPE	A vertical pipe is any pipe or fitting which is installed in a vertical position or which makes an angle of not more than 45 degrees with the vertical.
WASTE PIPE	A waste pipe is a pipe which conveys only liquid waste free of fecal matter.

7.4 Rainwater Harvesting Requirements

7.4.1 General

Every building proposed for constructing on plots having extent of 300 sqm or above shall have facilities for conserving and harvesting rainwater.

7.5 Rainwater Harvesting Plans

7.5.1 Requirement of Permit

Rainwater harvesting and drainage system shall not be installed until a permit for such work has been issued by the Building Authority for existing (only for addition or for alteration) or new building or for any other premises.

7.5.2 Application for Permit

An application for a permit for rainwater harvesting and drainage work shall be made on a prescribed form (see Appendix Q) by the licensed plumber and the owner, or by his appointed person or agent to install all or a self-contained or workable part of such work. The application shall accompany building rainwater harvesting and drainage plans and adequate description of the proposed rainwater harvesting and drainage installation in a drawing (drawn to a scale not less than 1:100) with the following details:

- (a) Plan of the building and site lay out;
- (b) Rainwater harvesting;
- (c) Ground recharging system;
- (d) Storm drainage system;
- (e) Catchment areas;

- (f) Materials, sizes and gradients of all proposed piping;
- (g) The position of manhole, rainwater pipe, gutters, rainwater inlets, etc. in the premises and their connection with storm sewer system or surface waters; the following colours may be used to indicate sewers, rainwater pipes and existing works:

Proposed storm sewers and disposal pipes:	-	violet
Existing network:	-	black
- (h) The slope of catchments.

7.5.3 In addition to rainwater harvesting and drainage plan a separate site plan of the building shall be submitted with the following particulars:

- (a) Adjoining plots and streets with their identification;
- (b) The position and invert level of the storm sewers, (if any), surface drain water and the direction of flow in it
- (c) The level of the proposed drainage pipe connecting to the storm sewers (if any);
- (d) The position and layout of private storm drainage system (in absence of public storm sewers); and
- (e) The alignment, size and gradients of all harvesting and drainage piping.

7.5.4 For high rise and public buildings, design calculations and specifications for various items of the work involved shall be submitted along with the drawings.

7.5.5 Permits and Approvals

The building official shall examine or cause to be examined all applications for permits and, amendments thereto within 45 days. If the application does not conform to the requirements of all pertinent laws, such application shall be rejected in writing, stating the reasons therefore. If the proposed work satisfies all the Code requirements, the Authority shall issue a nontransferable permit.

7.6 Licensing of Plumber

7.6.1 License Requirement

No individual, partnership, corporation or firm shall engage in the business of installation, repair or alteration of rainwater harvesting and drainage work without obtaining a license from the Authority.

7.6.2 Examination and Certification

The Building Authority shall establish a plumbers' examination board. The board will determine the requirements for the qualification and procedures for examination of applicants for license. The Authority will issue license to such applicants who meet the qualifications therefore and successfully pass the examination conducted by the board.

7.6.3 Annulment of License

The license of a licensed plumber may be nullified by the Building Authority, if it is proved that a plumbing work has been completed and certified by the licensed plumber violating the provisions of this Code deliberately setting aside the approvals given in the permit or without receiving the permit from the Building Authority.

7.7 Rain-Water Harvesting

7.7.1 General

Rainwater can be conserved and used in all useful purposes related to use of water. The amount of rain water to be conserved depends upon the purpose of use, rainfall intensity at the locality and the available catchments from where rainwater shall be collected. Rainwater can also be used for artificial ground water recharging. Two major aspects of rainwater harvesting are as follows: (i) Roof top rain-water harvesting, and (ii) Artificial ground water recharge.

7.8 Roof Top Rainwater Harvesting

Water can be collected through roof gutters and rainwater down pipes. Provision shall be made to divert the first rainfall after dry spell to avoid dust, soot, and leaves etc. in the water to be collected into the water tank. The capacity of water tank should be enough for storing water required for consumption between 2 dry spells.

7.8.1 Precautions in Rainwater Harvesting

Following precautions shall be taken:

- (a) No sewage or waste water should be admitted into the system.
- (b) Wastewater from areas likely to have oil, grease or other pollutants shall not be connected to the system.
- (c) Each rainwater seepage well shall have an inlet chamber with a silt trap to prevent any silt from finding its way into the sub-soil water.
- (d) The wells should be terminated at least 5 m above the natural static sub-soil water at its highest level so that the incoming flow passes through the natural ground condition and prevents contamination hazards.
- (e) No recharge structure or a well shall be used for drawing water for any purpose.

7.8.2 Qualifying Rainwater for Harvesting.

Rainwater shall be treated adopting following methods mentioned according to the purpose of use:

- (a) For using rainwater in drinking, cooking, washing utensils bathing and ablution it shall be disinfected along with filtration.
- (b) For cloth washing, floor washing, fountain, water fall cascade etc., rainwater shall be filtered.
- (c) For using in sprinkler firefighting, air conditioning etc. sedimentation of suspended particles will be required.
- (d) For toilet flushing, gardening, cleaning artificial ground, parking lots etc. screening floating materials are needed.

7.8.3 Catchments area for Collecting Rainwater

Rainwater can be collected from following built up areas for harvesting:

- (a) Roof top surfaces
- (b) External walls and other vertical surfaces
- (c) Balconies, sunshades etc.
- (d) Metal surface of play grounds, open yards etc.

7.8.4 Determining Catchment Area

For flat surface, the catchment area is its plan area plus 50 percent of the adjoining vertical wall contributing rainwater accumulation on the concerned catchment. See Appendix U.

For sloping roof, catchment is considered to be the actual inclined roof area.

7.8.5 Storing Rainwater

Where rainwater will be used for domestic purpose, rainwater from roof or terrace may be led straight from conductor (or leader) to one or more storage tanks. Storage tanks shall be provided with ventilating covers. An arrangement shall be made in the rainwater leader to divert the first washings from the roof or terrace catchments as they will contain more undesirable materials. The open end of all pipes shall be covered with mosquito (insect) proof wire net.

7.8.6 Flushing out First Rainwater

Before storing initial rainwater, just after starting raining, shall be drained out for a period as mentioned below.

<u>Location</u>	<u>Time</u>
Dhaka metropolitan area	20 min
Sylhet	15 min
Chittagong	15 min
Other urban areas	15 min

7.8.7 Precautions for Rainwater Storage

Following precautionary measures shall be taken for rainwater storage.

- Storage tank shall be made water tight in all respect.
- Tank shall always be kept covered.
- Regular cleaning at least once a year, preferably at end of dry periods, shall be done.
- Disinfection shall be done after cleaning operation.
- Tank shall be ventilated and vent pipe shall be covered by mosquito net.
- The tank must have an overflow pipe leading to a natural water course.
- If raw rainwater and potable water is to be stored in the same storage tank separated by a separating wall, then the separating wall shall have no openings.

7.8.8 Rainwater Treatment

For portable systems, a plain galvanized roof or a metal roof with epoxy or latex paint is recommended. Composite or asphalt shingles are not advisable, to improve water quality, disinfection shall be done in the following way.

7.8.8.1 Chlorination

Chlorine must be present in a concentration of 1 ppm to achieve disinfection. Liquid chlorine, in the form of laundry bleach, usually has 6 percent available sodium hypochlorite. For disinfection purposes, 2 fluid ounces (¼ cup) must be added per 1,000 gallons of rainwater. Bleach products, however, not labeled cannot be used in water treatment. A purer form of chlorine, which comes in solid form, is calcium hypochlorite, usually with 75 percent available chlorine. At that strength, 0.85 ounces by weight in

1,000 gallons of water would result in a level of 1 ppm. Chlorine contact times are shown in Table 8.7.1. To filter out Giardia and Cryptosporidium cysts, an absolute 1-micron filter shall be used.

Table 8.7.1: Contact Time with Chlorine

Water pH	Water temperature		
	50°F or warmer	45°F	40°F or colder
	Contact time in minutes		
6.0	3	4	5
6.5	4	5	6
7.0	8	10	12
7.5	12	15	18
8.0	16	20	24

7.8.9 Determining Volume of Rainwater Storage

Determining volume of rainwater storage

$$\text{Rainwater storage volume in m}^3 = D \times N \times D_p / 1000 + \text{Floating}$$

Where, D = Rainwater demand In liter per capita per day.

N = Population number.

D_p = Number of days for which water will be stored. Consider 90 days for drinking, cooking, utensils cleansing, bathing and ablution purposes; 210 days for other purposes.

7.8.10 Sizing of Rainwater Down Piping

The size and number of vertical leaders or rainwater down pipes shall be based on the maximum projected roof area according to Table 8.7.2. Minimum two drains and vertical leaders shall be provided for any independent roof surface. The size of semi-circular gutter shall be based on maximum projected roof area according to Table 8.7.3.

7.8.11 Inlet of Leaders

Inlet of leaders on any open surfaces shall be provided with dome shaped gratings. Size of the Inlet of any vertical leader shall be one size bigger than the size of the corresponding vertical leader.

7.8.12 Design of Rainwater Distribution System

The design of rainwater distribution piping shall be in accordance with Sec 5.10 of Chapter 5.

7.9 Artificial Ground Water Recharge

7.9.1 General

Storm runoff should be used for augmentation of ground water reservoir where ground water depletion rate is more than one meter per year. The natural movement of surface water can be modified by constructing recharge structures, Figure 8.7.1, in order to attain the following objectives:

- Enhancement of sustainable yield in areas where there is over development and depletion of the aquifers.
- Conservation and storage of excess underground water in the aquifers.
- Improvement of the quality of the existing ground water through dilution.
- Maintaining the natural balance of the ground water and its usage as the rain-water is a renewable supply source providing constant, dependable and safe water supply.

Table 8.7.2: Size of Vertical Leaders*

Size of Leader ** (mm)	Maximum Projected Roof Area and Flow (m ²)	(l/min)
50	202	87
65	367	155
75	598	253
100	1287	544
125	2336	986
150	3790	1602
200	8180	3450

* This Table is based upon a maximum rainfall of 25 mm per hour for a 1-hour duration. The figure for drainage area shall be adjusted to local conditions (Appendix U).

** The equivalent diameter of square leader will be the diameter of that circle which can be inscribed within the cross-sectional area. The equivalent diameter of the rectangular leader will be the short dimension of the rectangular leader. However, the ratio of width to depth of rectangular leader shall not exceed 3:1.

Table 8.7.3: Size of Semicircular Roof Gutters*

Dia of Gutter (mm)	Maximum Projected Roof Area for Gutter of Various Slopes							
	5 mm per m		10 mm per m		20 mm per m		40 mm per m	
	m	l/min	m ²	l/min	m ²	l/min	m ²	l/min
75	61	25	87	36	123	51	174	73
100	130	55	185	77	260	110	370	155
125	227	96	320	136	455	192	645	273
150	350	148	495	210	700	296	1010	425
175	503	210	710	300	1000	425	1420	600
200	725	307	1020	430	1300	610	2040	862
250	1300	555	1850	785	2610	1110	3650	1540

* Table 8.7.4 is based upon a maximum rainfall of 25 mm per hour for 1-hour duration. The figure for drainage area shall be subject to local conditions in accordance with Appendix U.

7.9.2 Designing Recharge Pit

Volume of recharge pit shall be on the basis of maximum intensity of rainfall in a shorter period of at least 15 minute. It is about one fourth peak hourly rainfall

$$\text{Capacity of recharge tank} = A l_s f / F$$

Where,

A = Catchments area

l = Intensity of rainfall in 15 minutes

f = Runoff co-efficient and

F = Porosity of filter bed = 0.5

Large dry well can be constructed in accordance with the requirements for seepage pit (Sec 6.9.15). However, for small dry wells handling limited quantities of rainwater, the pit may consist of a 1 m long 0.45 m diameter pipe filled with crushed bricks stone.

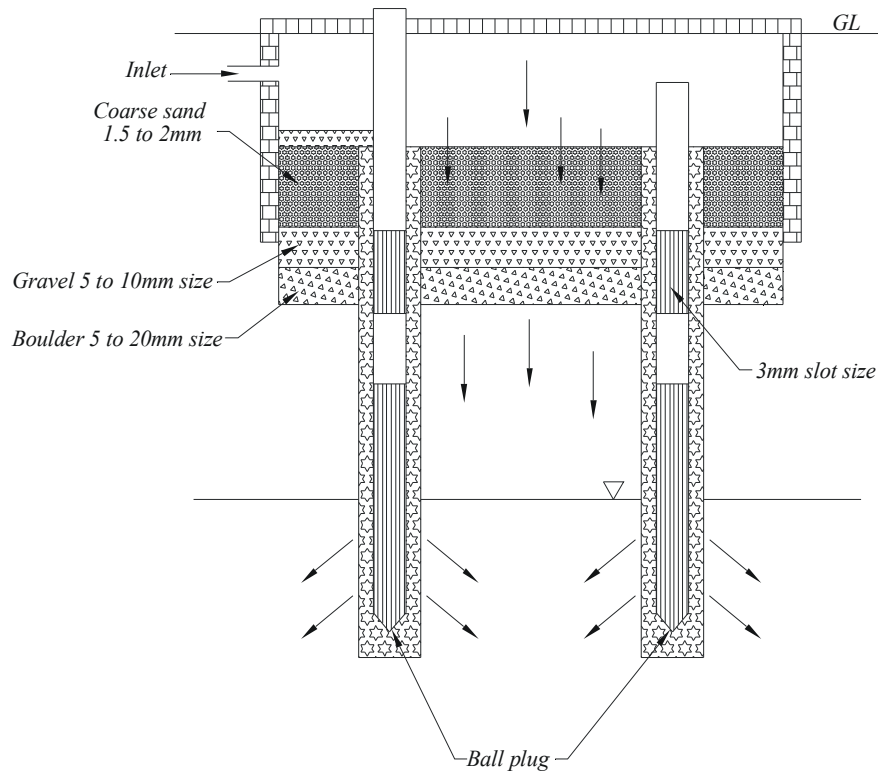


Figure 8.7.1. Artificial ground water recharge structure

7.10 Drainage and Sanitation Requirement

7.10.1 General

The object of storm water drainage is to collect and carry, the excess rain-water accumulating within the premises of the building, for suitable disposal. In rainwater drainage system there shall be safeguard against fouling, deposition of solids and clogging and with adequate inspection chambers so arranged that the drains may be readily cleaned without the risk of health hazard.

7.10.2 Design Factors

Estimation of the quantity of storm water shall be based on the following factors:

- (a) Imperviousness of the surface
- (b) Ground slope and time of concentration
- (c) Intensity of the rainfall for a design period and
- (d) Duration of the rainfall.

7.10.3 Imperviousness of the Surface

The percentage of imperviousness of the drainage area may be obtained from available data for a particular area. In the absence of such data, the following values may serve as a guide:

<u>Type of area (percent)</u>	<u>Imperviousness factor</u>
Commercial and industrial areas	70-90
Residential areas (high density)	60-75
Residential areas (low density)	35-60
Parks and underdeveloped areas	10-20

7.10.4 Time of Concentration

Time of concentration to reach the farthest point of any drainage system or the outfall under consideration should be considered between 5 min to 30 min.

7.10.5 Intensity of the Rainfall

Rainwater Intensity data for the locality of the building shall be studied to arrive at the design parameters for rainwater harvesting in accordance with Appendix R.

7.10.6 Rainwater Disposal

Rainwater from roof or from building premises shall not be discharged into septic tank. This shall be drained into storm sewer or combined sewer system where available or into private disposal methods, water course or dry well, Figures 8.6.18 and 8.6.19 of Chapter 6 Part 8.

7.10.6.1 Individual rain water traps shall be installed on the rainwater drain branch serving each leader or a single trap shall be installed in the main rainwater drain (building storm drain) just before its connection with the combined building sewer, main drain or public sewer.

7.10.6.2 No traps shall be required for rainwater drains which will be used for rainwater collection and connected to a sewer draining rainwater exclusively.

7.10.6.3 Subsurface drainage pipings for rainwater drainage shall not be less than 100 mm in diameter. The subsoil drainage system shall be protected by an accessibly located backwater valve in case the building is subject to backwater or flooding. Subsoil drains shall discharge to a trapped area drain, sump, dry well or an approved location above grade.

7.10.6.4 Rainwater pipes shall not be used as soil, waste or vent pipes.

7.10.6.5 All roof areas, except those draining to hanging gutters, shall be equipped with roof drains with strainers extending not less than 100 mm above the surface of the roof and shall have an available inlet area not less than two times the area of the leader to which the drain will be connected.

7.10.6.6 It is recommended to have more than one rainwater drainage pipe for primary roof drainage system to minimize blockage.

7.10.6.7 It is recommended to provide secondary rainwater drainage system at a suitable elevation from the roof that has been considered in the calculation of rainwater load to design the building structure. The secondary drainage system shall be a separate drainage piping up to a storm sewer or private waste (rainwater) disposal system. The size of secondary rainwater drainage piping shall not be less than the size required for primary rainwater drainage piping.

7.10.6.8 French drains may be employed as surface water drains for drainage of unpaved surfaces. Construction of French drains (if used) shall be in accordance with established engineering practices.

7.10.6.9 The design of such system shall be on the basis of location with respect to wells or other sources of water, soil permeability, ground water elevation, area available and maximum occupancy of the building.

7.11 Materials and Appliances

The piping, fixtures and equipment to be used for rainwater harvesting and drainage system shall be in accordance with Sec 6.7 in Chapter 6.

7.12 Construction of Rainwater Storage Tank

The construction of storage tanks for rainwater storage shall be in accordance with Sec 5.9 in Chapter 5 Part 8.

7.13 Installation and Construction of Rainwater Harvesting and Drainage System

7.13.1 All junctions and joints of rainwater harvesting and drainage piping shall be watertight.

7.13.2 Roof gutters shall be of suitable material of required thickness. All joints shall be watertight.

7.13.3 The depth of cover shall be in accordance with Sec 6.9.9 Chapter 6 Part 8.

7.13.4 The pipe shall be laid to even gradients and change of gradient shall be combined with an access point Sec 6.10.2 Chapter 6 Part 8. However, access points shall be provided only if blockages could not be cleared without them.

7.13.5 The joints and connection in drainage and venting system shall be gastight and watertight for the pressures required by the test, with the exception of those portions of perforated or open joint piping which will be installed for the purpose of collecting and conveying ground or seepage water to the underground storm drains.

7.13.6 Piping in rainwater drainage and harvesting system shall be installed without undue strains and stresses and provision shall be made for expansion, contraction and structural settlement. Vertical piping shall be secured at sufficiently close intervals to keep the pipe in alignment and carry the weight of the piping and its content. The horizontal piping shall be supported at sufficiently close intervals to keep it in alignment and to prevent sagging.

7.14 Hangers and Support

The piping, fixtures and equipment used for rainwater harvesting and drainage system shall be provided with sufficient hangers and support in accordance with Sec 5.17 in Chapter 5 Part 8.

7.15 Pipe Joints

The joints between different piping and fittings shall conform to the standards in accordance with Sec 5.17 in Chapter 5 Part 8.

7.16 Protection Against Rodent

Installation of pipes through any walls ceilings etc. shall be made rodent proof in accordance with Sec. 6.9.8 of Chapter 6 Part 8.

7.17 Gradient of Pipes

Lesser slope of rainwater drainage pipe shall be provided. The computed velocity in the storm water drains shall not be less than 0.6 m per second. The maximum recommended velocity shall be 2.5 m per second. For drain pipes to be used for recharging the flow velocity should be 0.6 m per second.

7.18 Inspection Chambers and Manholes

Inspection chambers and manholes shall Incorporated and constructed in accordance with 6.9.7.4 of Chapter 6 Part 8.

7.19 Bedding and Backfilling

The operation of bedding and backfilling shall be in accordance with Sec. 6.9.9 of Chapter 6 Part 8.

7.20 Design of Rainwater or Storm Water Drainage Piping

Designing of rainwater down pipe and storm water drainage pipe includes determining the number of rainwater down pipe and sizes and determining the size of drain pipes according to the fall of drain pipe.

7.21 Sizing and Finding the Number of Rainwater Drainage Piping

7.21.1 The size and number of vertical leaders shall be based on the maximum projected catchment area according to Table 8.7.2. Minimum two numbers of roof drains and vertical leaders shall be provided for any independent roof surface.

7.21.2 Sizing of Storm Water Drainage Piping

The size of building storm drain, storm sewer or any of their horizontal branches shall be based on the maximum catchment area including projected roof or paved area to be drained in accordance with Table 8.7.4. The size of semi-circular gutter shall be based on maximum projected roof area according to Table 8.7.3.

Table 8.7.4: Size of Horizontal Building Storm Drains and Building Storm Sewer *

Diameter of Drain (mm)	Maximum Catchment Area and Flow for Various Slopes					
	10 mm per m		20 mm per m		40 mm per m	
	m ²	l/min	m ²	l/min	m ²	l/min
75	299	125	422	177	599	252
100	668	288	965	406	1370	577
125	1215	515	1715	725	2430	1030
150	1950	823	2745	1157	3900	1645
200	4185	1765	5940	2500	8380	3540
250	7550	3185	10650	4500	15100	6370
300	12140	5100	17140	7236	24280	10250
375	21700	9120	30600	12900	43400	18300

* Table 8.7.3 is based upon a maximum rainfall of 25 mm per hour for 1-hour duration. The figure for drainage area shall be adjusted to local conditions in accordance with Appendix U.

7.22 Inspection, Testing and Completion Certificate

The inspection, testing and completion certification of rainwater harvesting and drainage system or part of existing system shall be in accordance with Sec 6.15 Chapter 6 Part 8.

7.23 Guide to Maintenance

The maintenance of rainwater harvesting and drainage and system shall be in accordance with Sec 6.16 Chapter 6 Part 8.

7.24 List of Related Appendices

Appendix R One-hour Rainfall

Appendix U Determining Catchments Area for a Flat Surface

PART VIII
Chapter 8
Fuel Gas Supply

8.1 General

8.1.1 Scope

8.1.1.1 This Chapter provides the requirements aimed at safeguarding life and property in nonindustrial gas piping systems for use with fuel gases such as natural gas (NG) and liquefied petroleum gas (LPG) in the vapour phase used for fuel or lighting purposes in consumers' premises.

8.1.1.2 This Chapter does not cover the safety requirements and rules for gas burning appliances. The requirements of National Fuel Gas Code - 2009 edition (NFPA-54/ANSI Z223.1) and NFPA 58-2008 edition and International Fuel Gas Code-2009 of the USA shall apply for such appliances.

8.1.1.3 The requirements of this Chapter do not apply to gas piping systems for industrial installation and applications.

8.1.1.4 This Chapter covers the aspects of design, fabrication, installation, test, operation, inspection and maintenance of gas piping systems from the point of delivery to the connections with each utilization device. The point of delivery is defined in this Chapter as the outlet of the service regulator or the service shutoff valve where there is no meter.

8.1.1.5 Piping systems covered here are limited to a maximum operating pressure of 3.45 kPa (14 inches of water column or ½ psig).

8.1.1.6 While applying the provisions of this Chapter, reference should also be made to the manufacturers' instructions, gas supply company's regulations and other applicable codes and standards listed in this Chapter or required by the authority having jurisdiction.

8.1.2 Terminology

This Section provides definitions of terms used in this Chapter of the Code. Where terms are not defined in this Chapter, they shall be defined using their ordinarily accepted meanings such as the context implies. The definitions in the Chapter are the same as used in National Fuel Gas Code-2009 edition (NFPA-54/ANSI Z223.1), International Fuel Gas

Code-2009 and Liquefied Petroleum Gas Code-2008 (NFPA-58) and National Building Code of India-2005.

APPLIANCE	Any device that utilizes gas as a fuel or raw material to produce light, heat, power, refrigeration, or air conditioning.
APPLIANCE VALVE	A device that will shut off the gas supply to burner(s)
APPROVED	Acceptable to the authority having jurisdiction
AUTHORITY HAVING JURISDICTION (AHJ)	An organization, office, or individual responsible for enforcing the requirements of a code or standard, or for approving equipment, materials, and installation, or a procedure.
BRANCH LINE	Gas piping that conveys gas from a supply line to the appliance
BURNER/COOKERS	A device for the final conveyance of gas, or a mixture of gas and air, to the combustion zone.
CONCEALED GAS PIPING	Gas piping that, when in place in a finished building, would require removal of permanent construction to gain access to the piping.
CONSUMER'S/CUSTOMER'S CONNECTION	Piping tapped on the riser to supply gas to individual customer/consumer
CYLINDER	A portable container designed, fabricated, tested and marked (or stamped) in accordance with a recognized standard/code such as ASME, or the regulations of the US Department of Transportation (DOT) used for transporting or storing LPG. The maximum size permitted inside the building is 320 kg water capacity.
DIVERSITY FACTOR	Ratio of the maximum probable demand to the maximum possible demand.
DRIP	The container placed at a low point in a system of piping to collect condensate and from which it may be removed.
EQUIVALENT	Nothing in this Code is intended to prevent the use of system, methods or device of equivalent or superior quality, strength, fire resistance, effectiveness, durability and safety over those prescribed by this Code. However, technical documentation shall be submitted to the authority having jurisdiction to demonstrate equivalency; and the proposed method, system or device shall be approved for the intended use by the same authority.

FUEL GAS	A natural gas, manufactured gas, liquefied petroleum gas or mixtures of the gases (This Chapter only recognizes natural gas and LPG as fuel gas).
GASES	Include natural gas, manufactured gas, liquefied petroleum (LP) gas in the vapor phase, liquefied petroleum gas-air mixtures and mixtures of these gases, plus gas-air mixtures within the flammable range, with the fuel gas or the flammable component of a mixture being a commercially distributed product.
GAS FITTER	An employee of the gas supplying company
GAS MANIFOLD	The conduit of an appliance that supplies gas to the individual burner(s).
LABELED	Equipment or materials to which has been attached a label, symbol or other identifying mark of an organization that is acceptable to the authority having jurisdiction and concerned with product evaluation, that maintains periodic inspection of production of labeled equipment or materials, and by whose labeling the manufacturer indicates compliance with appropriate standards or performance in a specified manner.
LIQUEFIED PETROLEUM GAS (LPG)	Liquefied petroleum gas composed predominantly of propane, propylene, butanes or butylenes or mixtures thereof that is gaseous under normal atmospheric conditions but is capable of being liquefied under moderate pressure at normal temperatures.
LISTED	Equipment, materials or services included in a list published by an organization that is acceptable to the authority having jurisdiction and concerned with evaluation of products or services, that maintains periodic inspection of production of listed equipment or materials or periodic evaluation of services, and whose listing states either that the equipment, materials, or service meets appropriate designated standards or has been tested and found suitable for a specified purpose.
METER	An instrument installed to measure the volume of gas delivered through it.
OUTLET	The point at which gas-fired appliance connects to the gas piping system.
PILOT	A small flame that is utilized to ignite the gas at the main burner or burners.

PIPE	Rigid conduit of iron, steel, copper, brass, aluminum, or plastic.
PIPING SYSTEM	All piping, valves and fittings from the outlet of the point of delivery from the supplier to the outlets of the equipment shutoff valves.
PRESSURE REGULATOR	Equipment placed in a gas line for reducing, controlling, and maintaining the pressure in that portion of the piping system downstream of the equipment.
PRESSURE TEST	An operation performed to verify the gastight integrity of gas piping following its installation or modification.
PURGE	To free a gas conduit of air or gas, or a mixture of gas and air.
QUALIFIED AGENCY	An individual, firm, corporation, or company that either in person or through a representative is engaged in and is responsible for (a) the installation, testing, or replacement of gas piping or (b) the connection, installation, testing, repair, or servicing of appliances and equipment; that is experienced in such work; that is familiar with all precautions required; and that has complied with all the requirements of the authority having jurisdiction.
RISER	A vertical pipe supplying fuel gas
SAFETY SHUTOFF DEVICE	A device that will shut off the gas supply to the controlled burner(s) in the event the source of ignition fails. This device may interrupt the flow of gas to main burner(s) only or to pilot (s) and main burner (s) under its supervision.
SERVICE METER ASSEMBLY	The piping and fittings installed by the serving gas supplier to connect the inlet side of the meter to the gas service and to connect the outlet side of the meter to the customer's house or yard piping.
SERVICE REGULATOR	A pressure regulator installed by the serving gas supplier to reduce and limit the service line gas pressure to delivery pressure.
SERVICE SHUTOFF VALVE	A valve, installed by the serving gas supplier between the service meter or source of supply and the customer piping system, to shut of the entire piping system.
SHALL	Indicates a mandatory requirement
TUBING	Semi rigid conduit of copper, steel, aluminum, CSST (corrugated stainless steels tubing) or plastic

VALVE	A device used in piping to control the gas supply to any section of a system of piping or to an appliance.
VENT	A passageway used to convey flue gases from appliances or their vent connectors to the outdoors.
WATER HEATER	An appliance for supply hot water for domestic or commercial purpose.

8.1.3 General Precautions

8.1.3.1 Turn Gas Off: All gas piping work or gas appliance installation shall be performed with the gas turned off to eliminate hazards from leakage of gas.

8.1.3.2 Notification of Interrupted Service: It shall be the responsibility of the installing agency, when if the gas supply is to be turned off, to notify all affected consumers.

8.1.3.3 Before Turning Gas Off: Before turning off the gas to premises for the purpose of installation, repair, test, inspection, replacement or maintenance of gas piping or appliances, all burners shall be turned off. When two or more consumers are served from the same supply system, precautions shall be taken to ensure that only supply to the concerned consumer is turned off.

8.1.3.4 Checking for Gas Leaks: Soap and water solution or other material approved for the purpose, shall be used in locating gas leakage. Use of matches, candles, flames or other sources of ignition shall be prohibited for this purpose.

8.1.3.5 Use of Lights: Artificial illumination used in connection with a search for gas leakage shall be restricted to battery operated flashlights (preferably of the safety type) or approved safety lamps. In searching for leaks, electric switches should not be operated. If electric lights are already turned on, they should not be turned off.

8.1.3.6 Working Alone: An individual shall not work alone in any situation where working practice desires that two or more persons are necessary to carry out the work safely.

8.1.3.7 Handling Liquid from Drips: Liquid which is removed from a drip in an existing gas piping shall be handled with proper precautions, and shall not be left on the consumers' premises.

8.1.3.8 No Smoking: When working on piping which contains or has contained gas, smoking shall be prohibited.

8.1.3.9 Handling of Flammable Liquids: Flammable liquids used by the installer shall be handled with proper precautions and shall not be left within the premises from the end of one working day to the beginning of the next.

8.1.3.10 Work Interruption: When interruptions in work occur, the system shall be left in a safe and satisfactory condition.

8.1.3.11 Certain requirements related to work on the gas supply system are listed in Appendix V.

8.1.4 Notification of Completion

8.1.4.1 When regulations so require, the completion of installation shall be notified to the gas supply company or the Authority.

8.2 Gas Piping Installation

8.2.1 Piping Plan and Approval

8.2.1.1 Plans for installation of gas piping system and gas appliances shall be prepared in accordance with requirements of the gas supply company and the Authority. Necessary approvals shall be obtained from the gas supply company and the Authority before installation of the gas piping system and the appliances. (See Appendix W).

8.2.1.2 The plan shall include proposed location of the piping, layout and sketch of the piping system, sizes of different branches, and present and future gas demands.

8.2.1.3 Approved plans shall bear the authorized seal and signatures of the gas supply company and the Authority.

8.2.2 Size of Piping to Gas Appliances

8.2.2.1 Gas piping shall be of such size and so installed to ensure adequate supply of gas to meet the maximum demand without undue pressure drop between the meter, or service regulator when there is no meter, and the appliance or appliances.

8.2.2.2 The size of gas piping depends upon the following factors:

8.2.2.3 Allowable pressure drop from meter or service regulator, when there is no meter, to appliance

8.2.2.4 Maximum gas consumption to be provided

8.2.2.5 Length of piping and number of fittings

8.2.2.6 Specific gravity of the gas

8.2.2.7 Diversity factor

8.2.2.8 The size of each gas piping system shall be determined by standard engineering methods acceptable to the gas supply company and the Authority.

8.2.2.9 Gas pipes smaller than 12 mm in diameter shall not be used.

8.2.2.10 Straight lengths of piping shall be used as far as practicable. Where there are bends in the pipeline, or as approved by the gas supply company, the inside radius of a bend shall be not less than 6 times the outside diameter of the pipe.

8.2.3 Acceptable Piping Materials

8.2.3.1 Piping material shall be one of the materials listed in Table 8.8.1 conforming to the corresponding standards, or other materials as may be approved by the gas supply company or the Authority. Cast iron pipe shall not be used.

Table 8.8.1 Fuel Gas Pipe Materials

Material	Standards
Black steel pipe*	ASTM A106
Galvanized steel pipe*	ASTM A53
Wrought steel and wrought iron pipe*	ANSI B36.10 M
Corrugated stainless steel tubing	ANSI LCI/CSA 6.26
Copper or copper-alloy tubing (Type K or L)	ASTM B88; ASTM B280
Aluminium pipe and tubing	ASTM B210; ASTM B 241
Plastic pipe and tubing	ASTM D2513
* Minimum Schedule 40	

8.2.3.2 Fittings shall be of an approved type and material for gas piping systems accepted to the gas supply company. Bushings shall not be employed.

8.2.3.3 All joints and connections shall be of an approved type and material for gas piping system acceptable to the gas supply company. Joints and connections shall be gas tight at the test pressure. (see Sec 8.2.9.3 and 8.2.9.4)

8.2.3.4 Flexible metal pipes or heavy rubber pressure tubing acceptable to the gas supply company may be used only for direct connections to burners.

8.2.4 Fabrication of Piping for Installation

8.2.4.1 Gas pipe or tubing and fittings shall be clean and free from cutting burrs and defects in structure or threading and shall be thoroughly brushed with chips and scale blown. Defects in pipe or tubing or fittings shall not be repaired when defective pipe, tubing or fittings have been identified. The defective material shall be replaced.

8.2.4.2 Pipe, tubing, fittings and valves removed from any existing installation shall not be used again until they have been thoroughly cleaned, inspected and ascertained to be equivalent to new material.

8.2.4.3 Metallic pipes with threads which are damaged or defective shall not be used.

8.2.4.4 Metallic pipes shall be threaded in accordance with approved standard acceptable to supply company.

8.2.4.5 When used in a corrosive environment, metallic pipes and fittings shall be protected with corrosion resistant coating.

8.2.4.6 For any thread joint proper sealant shall be used on the made threads only.

8.2.4.7 Joints and joining compounds if used in LPG installation shall be resistant to the action of liquid petroleum gas.

8.2.5 Installation of Gas Pipes

8.2.5.1 Installation, repair and replacement of gas piping or appliances shall be performed only by a qualified installing agency or gas fitter.

8.2.5.2 Protection of Piping: Piping shall be buried to a minimum depth of 1 m covered in a manner so as to protect the piping from physical damage. It shall be protected from physical damage when it passes through flower beds, shrub beds and other such cultivated areas.

8.2.5.3 Protection against Corrosion: Gas piping in contact with earth or other materials which will corrode the piping shall be protected against corrosion in an approved manner. When dissimilar metals are joined underground, an insulating coupling or fitting shall be used. Piping shall not be laid in contact with cinder or ash.

8.2.5.4 All the piping within the premises where it has to run on the wall shall be exposed and should not be in contact with wall to ensure that no corrosion takes place. Epoxy sealant or polyethylene conduit shall be used to ensure no contact of pipe with the wall in the situation of pipe crossing the wall. Uncoated threaded or socket welded joints shall not be used in piping in contact with soil or where internal or external crevice corrosion is known to occur.

8.2.5.5 Piping Through Foundation Wall: Underground gas piping, when installed below grade through the outer foundation or basement wall of a building, shall be either encased in a protective sleeve or protected by an approved device or method. The piping or sleeve shall be sealed at the foundation or basement wall to prevent entry of gas or water.

8.2.5.6 Piping Underground beneath Buildings: If the laying of gas piping underground beneath buildings cannot be avoided, the piping shall be encased in an approved conduit designed to withstand super imposed load. The conduit shall extend into a normally accessible portion of the building and, at the point where the conduit terminates in the building, the space between the conduit and the gas piping shall be sealed to prevent the entrance of gas from any possible leakage. The conduit shall extend at least 100 mm outside the building, be vented outdoors above finished ground level and be installed in such a way as to prevent the entrance of water and insects.

8.2.5.7 Building Structure: The building shall not be weakened by the installation of any gas piping. Existing beams or joists shall not be cut or notched.

8.2.5.8 Piping Supports: Gas piping in buildings, shall be supported with pipe hooks, metal pipe straps, bond or hangers of an approved type and material suitable for the size of piping, and of adequate strength and quality and located at specified intervals so that the piping cannot be moved accidentally from the installed position. Gas piping shall not be supported by other piping.

8.2.5.9 Piping Entrance to Buildings: When gas pipe enters a building through a wall or floor of masonry or concrete, it shall be sealed against the entrance of water, moisture or gas.

8.2.5.10 Piping in Floors: Piping in solid floors, such as concrete, shall be laid in channels in the floor suitably covered to provide access to the piping with a minimum damage to the building.

8.2.5.11 Single pipe without joints shall be used for wall crossing in any building.

8.2.5.12 Changes in direction of gas pipe shall be made by the use of approved fittings, factory bends or field bends. Field bends shall be made by employing approved procedures and equipment.

8.2.5.13 Gas piping inside any building shall not be run in or through an air duct, chimney or gas vent, ventilating duct or elevator shaft. Gas piping shall not be taken through inaccessible or concealed areas where its condition cannot be inspected and accumulation of gas due to undetected leakage may create a dangerous condition.

8.2.5.14 Provide Drips where Necessary: A drip shall be provided at any point in the line of pipe where condensate may collect. When condensation is excessive, a drip should be provided at the outlet of the meter where required by the authority or the gas supply company. This drip shall be so installed as to constitute a trap wherein an accumulation of condensate will shut off the flow of gas before it will runback into the meter. All drips installed shall be readily accessible to permit cleaning, inspection or emptying.

8.2.5.15 Cap All Outlets: Each outlet, including a valve or cock outlet, shall be firmly closed gas tight with a threaded plug or cap immediately after installation and shall be left closed until an appliance is connected thereto. Similarly, when an appliance is disconnected from an outlet and the outlet is not to be used again immediately, it shall be capped or plugged gas-tight. The outlet shall not be closed with tin caps, wooden plugs, corks or by other improvised means or objects. Use of a listed quick disconnect device is acceptable.

8.2.5.16 Prohibited Devices: No device shall be placed inside the gas pipe or fittings that will reduce the cross-sectional area or otherwise obstruct the free flow of gas.

8.2.5.17 Branch Pipe Connection: All branch pipe connections and outlets shall be taken from the top or sides of horizontal lines and not from the bottom.

8.2.5.18 Electrical Bonding and Grounding: The gas piping shall be electrically continuous throughout its length and properly earthed except in stretches where cathodic protection system is used for protection against corrosion. The piping shall not be used to ground any electrical equipment.

8.2.5.19 Distance from Electrical Wiring: The distance between the gas piping and electrical wiring system shall be at least 60 mm. They shall be securely fixed to prevent contact due to movement. The gas piping should be installed below the electrical wiring for heavier gas like LPG and for natural gas the piping should be above the electrical wiring.

8.2.5.20 Distance from Steam Piping: The gas piping and steam piping, if installed parallel, shall be at least 150 mm apart. The gas piping should preferably be installed below the steam piping.

8.2.5.21 Gas Piping to be Graded: All gas piping shall be graded not less than 1 in 750 to prevent accumulation of condensate or liquids in the line. All horizontal lines shall grade to risers, and from the risers to the meter, or service regulator when there is no meter, or to the appliance.

8.2.5.22 The gas piping shall be painted red in order to differentiate it from other piping. Where the piping is exposed to sun rays, it shall be painted silver gray.

8.2.5.23 Documentation shall be maintained for all gas supply installations. (See Appendix W).

8.2.6 Pressure Regulators

8.2.6.1 Where the pressure of gas supplied to domestic installation or other low pressure gas piping systems in buildings is in excess of 3.45 kPa, a gas pressure regulator of approved type and size shall be installed in the service pipe of each such system to prevent pressure in excess of 3.45 kPa from being introduced into such a building piping. If the building pipe is of welded construction the pressure regulator may be located upstream of the gas meter in each consumer's premises. In these cases, the gas pressure in the piping downstream of the gas pressure regulator shall not exceed 3.45 kPa.

8.2.6.2 If installed inside a building, the regulator shall comply with the following:

8.2.6.3 If any of the diaphragms of the regulator ruptures, the gas shall be directed to an outlet vent pipe made of brass or plastic in order to vent the gas out of the building. The vent pipe shall be installed about 1m above the topmost story of the building in open air. Means shall be employed to prevent water from entering the pipe and also to prevent blocking it by insects or other foreign bodies.

8.2.6.4 If the gas pressure at the outlet of the regulator falls below 50 percent of the operating gas pressure or rises above twice the operating pressure, the gas supply to the pressure regulator shall shut off.

8.2.6.5 In the event of malfunctioning of this safety device, a supplementary device shall connect the low pressure circuit to the vent pipe as soon as the exit pressure reaches 6.90 kPa.

8.2.6.6 The gas supply company shall ensure that the heating value and supply pressure of gas shall not exceed the stated values for the type of gas being supplied.

8.2.7 Service Shutoff Valves

8.2.7.1 Service shutoff valves shall be provided on all new services including replacements and shall be installed in a readily accessible location.

8.2.7.2 Service shutoff valves shall be located upstream of the meter if there is no regulator or upstream of the regulator if there is one.

8.2.7.3 All gas services installation operating at pressure greater than 3.45 kPa shall be equipped and with shutoff valve of approved type installed on the service pipe outside the building.

8.2.7.4 Underground shutoff valves shall be located in a covered durable kerb box, manhole, vault, or stand pipe which is designed to permit ready operation of the valve. The covers so provided shall be clearly marked "GAS".

8.2.7.5 Every gas outlet shall have an individual shutoff valve. The shutoff valve shall be accessible and adjacent to the appliance.

8.2.8 Existing Work

8.2.8.1 Nothing herein shall prohibit the continued use of an existing gas piping system without further inspection or test unless the Authority has reason to believe that defects which make the system dangerous to life or property exist.

8.2.9 Inspection of Services

8.2.9.1 No person shall use or permit the use of a new system or an extension of an old system of gas piping in a building or structure before the same has been inspected and tested to ensure that the system is safe and a certificate has been issued by the Authority. (See Appendix W).

8.2.9.2 Test of Piping for Tightness: Before any system of gas piping is finally put in service, it shall be carefully tested to ensure that it is gas tight and safe. Where any part of the system is to be enclosed or concealed, this test should precede the work of closing in. The test medium shall be air, nitrogen, carbon dioxide or an inert gas. OXYGEN SHALL NEVER BE USED.

8.2.9.3 Before appliances are connected, the piping systems shall be tested at a pressure of at least 159-mm mercury for a period not less than 10 minutes without showing any pressure drop. The source of pressure shall be isolated before the pressure tests are performed.

8.2.9.4 The Authority shall, within a reasonable time after being requested to do so, inspect and test the gas piping system that is ready for such inspection and test. If the system is found to comply with the requirements of inspection and test as laid down, it shall issue the certificate.

8.2.9.5 It shall be unlawful to supply gas in a building before the required certificate has been issued (see Sec 8.2.9.4 above), except that the Authority may give temporary permission for a limited time to supply and use gas before such an installation has been fully completed and the certificate issued.

8.2.10 Check of Leakage

8.2.10.1 Close All Gas Outlets: Before turning gas under pressure into any piping, all openings and outlets from which gas can escape shall be closed.

8.2.10.2 Check for Leakage: No matches, flames or other sources of ignition shall be used to check for gas leakage from meters, piping or appliance. Checking for gas leakage with soap and water solution shall be recommended (see Sec 8.1.3.4 and 8.1.3.5 also).

8.2.10.3 Checking for Leakage with Meter: Immediately after turning gas into the piping, the system shall be checked to ascertain that no gas is escaping. This may be checked by carefully watching the test dial of the meter to determine whether the gas is flowing through the meter. Under no circumstances shall a leakage test be made using a gas meter unless immediately prior to such test it has been determined that the meter is in operating condition.

8.2.10.4 Checking of Leakage not using a Meter: This may be performed by attaching to an appliance orifice, a manometer or equivalent device (gauge) so that it can be read in increments of 2.5 mm water column and momentarily turning on the gas supply and observing the gauging device for pressure drop with gas supply shutoff. No drop in pressure shall occur during a period of 3 minutes.

8.2.10.5 When Leakage is Indicated: If the meter test hand moves or a pressure drop on the gauge occurs, all appliances or outlets supplied through the systems shall be checked to ensure that they are shut off and do not leak. If they are shut off firmly there is a leak in the piping system. The gas supply shall be shut off until the necessary repairs have been made, after which the test specified in Sec 8.2.10.3 or Sec 8.2.10.4 above shall be repeated.

8.2.11 Purging

8.2.11.1 Purging All Gas Piping: After piping has been checked, all gas piping shall be fully purged. Piping shall not be purged into the combustion chamber of an appliance. A suggested method for purging the gas piping to an appliance is to disconnect the pilot piping at the outlet of the pilot valve.

8.2.11.2 Lighting Pilots: After the gas piping has been fully purged, all appliances shall be purged and the pilots lighted. The installing agency shall satisfy itself that all piping and appliances are fully purged and safe for use before leaving the premises.

8.2.12 Rules for Turning Gas On

8.2.12.1 A person, who is an employee of the gas supply company and authorized by the gas supply company, shall turn on the gas at a service shutoff valve or at any valve that controls the supply of gas to more than one consumer.

8.2.12.2 Gas shall not be turned on at any meter valve without specific permission from the gas supply company or the Authority if any of the following conditions are found:

8.2.12.3 If the gas piping, appliances or meter supply through the meter valve are known to leak or otherwise be defective;

8.2.12.4 If required inspection of the piping or appliance has not been performed;

8.2.12.5 If the gas supply company or the Authority has requested that the gas be left turned off;

8.2.12.6 If the meter valve is found shutoff for some reason not known to the gas fitter.

8.2.12.7 The gas shall not be turned on in the event of fire.

8.2.12.8 Gas shall not be turned on at any branch line valve if any of the conditions listed in Sec 8.2.12.2 above are found. Where a branch line valve is found closed, a gas fitter shall again turn the gas on at such valve only. If proper precautions to prevent leakage are taken and no other unsafe conditions are created thereby.

8.2.12.9 Gas shall not be turned on at either the meter valve or the service line unless all gas keys/ cocks or valves installed on all outlets in the piping system are closed or all outlets in the piping system are capped or plugged.

8.2.13 Rules for Shutting Off the Gas

8.2.13.1 The gas fitter shall put the gas off to any appliance, pipe or piping system and shall leave the gas turned off, until the cause for interruption of supply has been removed in any one the following cases:

8.2.13.2 If ordered to do so by the Authority;

8.2.13.3 If leakage of gas is found, which appears to be sufficient to cause fire, explosion or asphyxiation;

8.2.13.4 If an installation is found to be such as to seriously endanger life and property;

8.2.13.5 If any condition exists which threatens interruption of gas supply that may cause burner failure or otherwise lead to dangerous conditions.

8.2.13.6 Before turning off the gas at the meter, for the purpose of installation, repair, replacement, test, inspection or maintenance of piping or appliances, all burners and pilot valves on the premises supplied with gas through the meter shall be turned off and the meter test hand observed for a sufficient length of time to ascertain that there is no flow of gas through the meter. Where there is more than one meter on the premises, precautions shall be exercised to ensure that the concerned meter is turned off.

8.2.14 Provision for Meter Location

8.2.14.1 The meter location shall be such that the meter can be easily read and the connections are readily accessible for servicing. Location, space requirements, dimensions and type of installation shall be acceptable to the gas supply company and be approved by the same.

8.2.14.2 Meters shall be installed in such a way that there shall be no load transfer from the pipeline to the inlet/outlet of the meter.

8.2.14.3 Gas piping at multiple meter installations shall be clearly marked by a metal tag or other permanent means provided by the installing agency, designating the building or the part of the building being supplied.

8.3 Use of Liquefied Petroleum Gas (Lpg)

- (a) The cylinders used for the storage and transportation of liquefied petroleum gas (LPG) shall conform to the accepted standards approved by the Authority.
- (b) The handling, use, storage and transportation of liquefied petroleum gas in cylinders exceeding 500 ml of water capacity shall be done in accordance with the good practice approved by the Authority and the guidelines of the gas supply company.
- (c) The cylinders shall be marked as provided in the regulations, rules or code under which they are fabricated.

8.3.1 LPG Cylinder Installation

The following recommendations apply to installations in residential, commercial, industrial, educational and institutional premises.

8.3.1.1 Personnel engaged and responsible for the installation of cylinders, equipment and piping should understand the characteristics of LPG and be trained in good practice of handling, installing, inspection, test and maintenance of installation.

8.3.1.2 The joining compounds used in the piping system shall be resistant to the action of liquefied petroleum gas and shall be decided by the Qualified Installation Agency. Hemp and similar materials shall not be used at the joint. In any joint in which the thread provides a gas tight seal, joining compound shall be used on the male thread.

8.3.1.3 Fire extinguishers of dry power or carbon dioxide type conforming to accepted standards shall be provided in places where LPG cylinder installations are situated and shall be located near such installations. Two buckets filled with sand and two with water shall also be installed nearby. The guidelines of the LPG supply company and the Authority shall be followed in this respect. The number, type and size of the fire extinguishers shall be as follows:

For installation with LPG	Number	Type	Capacity
40 kg to 200 kg	2	Dry powder	10 kg
more than 200 kg upto 320 kg	2	Dry powder	10 kg

8.3.1.4 Liquefied petroleum gas shall not be transferred from the cylinders in which it is supplied to any other container.

8.3.2 Cylinder Location**8.3.2.1 Stationary Installations**

8.3.2.2 Stationary installation not exceeding 40 kg of LPG may be installed indoors on any floor. Recommended minimum floor area per installation is 5 m².

8.3.2.3 Stationary installations each not exceeding 40 kg of LPG may be installed indoors on any floor within the same workspace provided the minimum distance between two such installations is 3 m. Recommended minimum floor area per installation is 5 m² and the aggregate of all such installations should not exceed 200 kg.

8.3.2.4 Stationary installation not exceeding 80 kg of LPG may be installed indoors on any floor provided the floor area per installation is not less than 12 m².

8.3.2.5 Stationary installations each not exceeding 80 kg of LPG may be installed indoors on any floor and within the same workspace provided the minimum distance between two such installations is 3 m. The Recommended floor area per installation is 12 m² and the aggregate quantity of all such installations should not exceed 200 kg.

8.3.2.6 Stationary installation not exceeding 320 kg of LPG may be installed indoors in an enclosed section of a building or a room reserved exclusively for this purpose and ventilated at low level directly to the outside air.

8.3.2.7 Stationary installation above 320 kg (200 kg in case provision as in (e) above is not possible) but not exceeding 1000 kg shall be installed outdoors on the ground level only. A minimum distance of 3 m shall be maintained between such an installation and any building, public place, roadways and other surroundings. The installation shall be protected against weathering by sun, rain, etc. and from tampering by unauthorized persons. A suitable shade of approved type and material may be provided for the purpose. Adequate ventilation at ground level to the outside air shall be provided. The distance between any two such installations shall be 3 m unless separated by a solid wall of fire resistant material up to at least 1 m above the height of the manifold valve.

8.3.2.8 The position of the cylinders shall facilitate: Changing and quick removal of any cylinder in case of necessity, and access to cylinder valve connections and regulating devices

8.3.2.9 Cylinders shall be placed upright with the valve uppermost.

8.3.2.10 Cylinders shall be placed on a firm and dry base such as concrete or brick floor. For outdoor installations the base shall be elevated.

8.3.2.11 Cylinders shall not be placed close to steam pipes or any other source of heat and shall be protected from the weather and direct sun. Cylinders shall be placed at a distance of 3 m from any other source of heat which is likely to raise the temperature of cylinders above the room temperature unless separated by metal sheet or masonry partition.

8.3.2.12 When cylinders are being connected or disconnected, there shall be no open flame or any source of ignition nearby and smoking shall be prohibited.

8.3.2.13 Cylinders shall not be installed below ground level and shall be at least 1 m away from drains, culverts or entrances and openings leading to cellars and other depressions in which gas might accumulate.

8.3.2.14 Cylinders shall not be installed at a place where they are likely to cause an obstruction, to be damaged or to be exposed to conditions likely to affect their safety.

8.3.2.15 Cylinders which have safety relief valve or similar devices incorporated in them shall be so positioned that if the relief devices operates escaping gas is not hazardous.

8.3.2.16 Portable Installations

8.3.2.17 When portability of cylinders is desired, the following requirements shall be met:

8.3.2.18 The sum total capacity of the cylinders connected to each manifold shall not exceed 80 kg of LPG. The total quantity of gas thus installed in a workspace shall not exceed 200 kg.

8.3.2.19 The regulator shall be connected directly to the cylinder valve or to a manifold which shall be connected to the cylinder valve by means of rigid connections to have the regulator firmly secured.

8.3.2.20 At any time the total quantity of gas at portable installations shall not exceed the limits in proportion to the floor area specified in Sec 8-3.2.1 (a) to (f).

8.3.2.21 If cylinders are mounted on a trolley shall be stable, where necessary the cylinders shall be secured to prevent them from falling.

8.3.3 Manifolds and Pressure Regulators

8.3.3.1 If pressure regulators, manifold headers, automatic change over devices, etc. are connected to cylinders by flexible or semi-flexible connectors, they shall be rigidly secured. Copper tube pigtails and reinforced high pressure hoses are considered to be flexible or semi-flexible connectors for this application.

8.3.3.2 Pressure regulator fitted with a safety valve shall be either:

8.3.3.3 Installed in the open air or

8.3.3.4 Vented to the open air by means of a metal vent pipe connected to the safety valve discharge line.

8.3.3.5 Precautions shall be taken that safety valve outlets do not get blocked with dust or other substances.

8.3.3.6 Suitable line shutoff valves shall be provided with each appliance or burner when more than one appliance is connected to the gas supply. Both ends of the connection to portable appliances shall be firmly attached with clips. Hose shall be resistant to the action of LPG.

8.3.3.7 The manifold headers which do not have to be taken off in normal use should be brazed or welded using material conforming to approved standards and having a melting point not less than 540°C.

8.3.3.8 All materials, fittings, etc. used in cylinder manifold system shall comply with the distributing company's stipulations.

8.3.3.9 The individual component parts of manifolds, that is, piping, fittings, pigtails, etc., which are subject to cylinder pressure shall be capable of withstanding a test pressure without bursting of 2.5 N/mm² or one and half times the maximum pressure corresponding in the maximum assessed temperature of the cylinder, whichever is more.

8.3.3.10 Where cylinder installations are made up with service and reserve batteries of cylinders, suitable change-over devices or valves shall be incorporated in the manifold header to prevent undue escape of the gas when cylinders are changed.

8.3.3.11 It is recommended that joints in manifold headers which do not have to be taken in normal use should be welded or brazed using a material and which shall have melting point of at least 540°C.

8.3.3.12 All joints between manifold headers and cylinder connectors shall be readily accessible.

8.3.3.13 Pressure regulators and other devices used to control the gas shall comply with the distributing company's stipulations and accepted standards.

8.3.3.14 Care shall be taken that safety of a metal vent pipe connected to the safety valve outlets do not become choked with dust or other foreign matter.

8.3.3.15 If the regulator is fitted with a relief valve, care should be taken in positioning the regulator to avoid unnecessary hazards if the relief valve functions.

8.3.3.16 Pressure regulators and other control devices shall be adequately supported.

8.3.3.17 Instructions to Consumers: Necessary instructions dealing with the following aspects shall be supplied by the LPG supply company to each consumer in the form of a manual:

8.3.3.18 Operation of the whole system;

8.3.3.19 How to recognize and detect gas leakage;

8.3.3.20 Action to be taken in case of leakage;

8.3.3.21 Action to be taken in case of fire; and

8.3.3.22 Action to be taken in case of damage to, or failure of any part of the installation.

8.3.3.23 For detailed information regarding installation of LPG cylinders in commercial, educational and institutional premises, the LPG supply company shall be consulted.

8.4 Lpg Bulk Storage Installations

NFPA 58: Liquefied Petroleum Gas Code-2008 edition shall be followed along with approval of the gas supply company and the Authority having jurisdiction for LPG for bulk storage installations where storage tanks over 450 liters water capacity are used for domestic consumers' premises.

The maximum capacity an individual tank and group of tanks at domestic premises shall be as follows:

Maximum water capacity of an individual tank: 20,000 litre

Maximum water capacity of group of tanks: 80,000 litre

The LPG Bulk Storage Installations shall strictly adhere to the provisions laid down in NFPA 58: Liquefied Petroleum Gas Code-2008 edition or its equivalent for the followings:

Location and spacing of storage tanks

- (a) Bunding
- (b) Protection and Safety
- (c) Good house keeping
- (d) Warning Signs
- (e) Fire protection and fire extinguishers
- (f) Water supply
- (g) Sound engineering practice for design, layout and operation of the entire installations
- (h) Training of personnel on both operations and on action to be taken in an emergency
- (i) The gas supply company and the Authority having jurisdiction shall approve every item mentioned above and can add anything more that will improve safety of installation and people living around it.

8.5 Installation of Specific Appliances

8.5.1 General

8.5.1.1 Gas appliances, accessories, and equipment shall be "Approved". Listed and labeled appliances shall be installed in accordance with the manufacturer's installation instruction.

8.5.1.2 It shall be determined whether the appliance has been designed for use with the gas to which it will be connected. No attempt shall be made to convert the appliance from the gas specified on the rating plate for use with a different gas without consulting the gas supply company or the appliance manufacturer for complete instructions.

8.5.1.3 Safety shutoff devices of the complete shutoff type shall be installed on manually controlled water heaters and automatically controlled appliances, except domestic ranges.

8.5.1.4 Gas appliances shall not be installed in any location where flammable vapours are likely to be present or accumulate, unless the design, operation and installation are such as to eliminate the possibility of ignition of the flammable vapours.

8.5.1.5 Appliances shall be vented in accordance with the instructions of their manufacturers or the procedures of the gas supply company.

8.5.1.6 Gas appliances shall be firmly supported. They shall not exert undue strain on the connected piping and connections.

8.5.1.7 The installing agency shall conform to the appliance manufacturer's specific recommendations in completing an installation that will provide satisfactory performance and serviceability. The installing agency shall also leave the manufacturer's installation, operating and maintenance instructions in a readily accessible location on the premises for reference and guidance of the Authority, servicemen, and the consumer or operator.

8.5.1.8 All appliances shall be located with respect to building construction and other equipment so as to permit ready access to the appliance. Sufficient clearance shall be maintained to permit cleaning of heating surfaces, replacement of parts, adjustment, cleaning of burners and pilots and maintenance.

8.5.1.9 Connecting Appliances and Equipment: Appliances and equipment shall be connected to the building piping system by one of the following:

8.5.1.10 Rigid metallic pipe and fittings

8.5.1.11 Semi-rigid metallic tubing and metallic fittings. Aluminum-alloy tubing shall not be used in exterior locations.

8.5.1.12 Listed appliance connectors and only one connector shall be used per appliance

8.5.1.13 Semi-rigid tubing in lengths up to 2 m that are in the same room as the appliance

8.5.1.14 Listed gas hose connectors to be used as approved.

8.5.1.15 The connector or tubing shall be protected against physical and thermal damages.

8.5.1.16 Aluminum-alloy tubing and connectors shall be factory coated to protect against external corrosion where they are in contact with masonry, plaster or insulation or are subject to frequent wettings by such liquids as water (except rain water), detergents or sewage.

8.5.1.17 Any appliance connected to a piping system shall have an accessible approved manual shutoff valve with a displaceable valve member or a listed gas convenience outlet and shall be located within 2 m of the appliance it serves except as permitted by the Authority.

8.5.1.18 Appliance connectors may be connected to building piping by a listed quick disconnect device, and when installed indoors, a manual shutoff valve shall be installed upstream of the quick disconnect device.

8.5.1.19 Electrical connection between gas appliances and the building wiring shall conform to the approved electrical code.

8.5.1.20 No devices using or dependent upon electricity shall be used to control or ignite a gas supply if of such type that failure of the electricity would result in the escape of unburned gas, or in failure to reduce the supply of gas under conditions which would normally result in its reduction, unless other means are provided to prevent the creation of dangerous temperatures, pressures or the release of gas.

8.5.2 Cookers/Burners

8.5.2.1 Domestic cooking appliances shall be installed be listed and labeled as household type appliances for domestic use. These are installed in accordance with its listing and the manufacturer's instruction.

8.5.2.2 Listed cookers/burners when installed on combustible floors shall be set on their own bases or legs firmly and shall be installed in accordance with their listing and the manufacturer's instructions. In absence of clearance information the appliances shall be installed in consultation with the gas supply company. The clearances shall not interfere with the flow of combustion air, accessibility for operation and servicing.

8.5.2.3 Unlisted appliances when acceptable with the authority shall be installed with at least a 150 mm clearance at the back and sides to combustible material. Combustible floors under unlisted appliances shall be protected in an approved manner. Guidelines of the Authority shall be followed.

8.5.2.4 Appliances shall have a vertical clearance above the cooking top of not less than 750 mm to combustible material or metal cabinets.

8.5.2.5 Appliances shall be installed so that the top or oven racks are level.

8.5.3 Illuminating Appliances

8.5.3.1 Listed (labeled) illuminating appliances shall be installed in accordance with their listing and manufacturer's instructions.

8.5.3.2 Unlisted illuminating appliances may be used when acceptable to the Authority and they shall be installed in accordance with the guidelines of the Authority.

8.5.3.3 Illuminating appliances designed for wall or ceiling mounting shall be firmly attached to substantial structures in such a manner that they are not dependent on the gas piping for support.

8.5.3.4 Illuminating appliance designed for post mounting shall be firmly attached to a post which has proper strength and rigidity. Posts shall be rigidly erected.

8.5.4 Water Heaters

8.5.4.1 Water heater installation in bedrooms and bathrooms shall comply with one of the following:

8.5.4.2 Water heaters shall be installed in a closet equipped with a weather-stripped door with no openings and with a self-closing device. All combustion air shall be obtained from the outdoors through one or two permanent openings having cross-sectional area $1 \text{ in}^2/3000 \text{ Btu/hr}$ ($700 \text{ mm}^2/\text{kW}$). The minimum dimension of air opening shall not be less than 3 in (75 mm).

8.5.4.3 Water heater shall be of direct vent type

8.5.4.4 Listed (labeled) water heaters shall be installed in accordance with their listing and manufacturer's instructions. The clearances shall not be such as to interfere with combustion air, draft hood clearance and relief, and accessibility for servicing.

8.5.4.5 Unlisted water heaters shall be installed with a clearance of 300 mm on all sides and rear and they shall be installed with the approval of the authority following its guidelines.

8.5.4.6 Water heaters shall be connected in a manner to permit observation, inspection, maintenance and servicing.

8.5.4.7 Water heaters shall be fitted with limiting switches for pressure and temperature and also with temperature, pressure and vacuum relief devices as per a nationally recognized standards for such devices.

8.5.5 Stationery Gas Engine Generators

Stationary gas engine generators for generating power to meet emergency needs during power outage shall be installed in accordance with the manufacturer's installation instructions and shall meet the requirements of UL 2200-04 (Standard for Stationary Engine Generator Assemblies-2004) of USA or equivalent. The Equipment powered by internal combustion engines and turbines shall not be rigidly connected to the gas supply piping.

Installation of stationary gas engine generators in the building shall be approved by the Authority and the gas supply company; and all their regulations applicable shall be complied with for installation, testing, operation and maintenance.

8.6 Related Codes and Standards

The list includes those codes and standards that are acceptable as “good practice” and “accepted codes and standards” for complying with the requirements of this Code. The latest version of a code or standard shall be used. The list may be used by the authority as a guide to fulfill the requirements mentioned in this Code.

National Fuel Gas Code-2009 edition (NFPA 54/ANSI Z223-1).

Liquefied Petroleum Gas Code-2008 edition (NFPA 58).

International Fuel Gas Code-2009.

National Building Code of India-2005 (Part 9, Section 2, Gas supply).

ASME2004, Boiler and Pressure Vessel Code, Section; VIII; ‘Rules for the Construction of Unfired Pressure Vessels.

ASTM A53, Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc Coated Welded and Seamless, 2007.

ASTM A106, Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Services, 2006a

ASTM A539, Standard Specification for Electric Resistance-Welded Coiled Steel Tubing for Gas Fuel Oil Lines, 1999

ASTM B 43, Standard Specification for Seamless Red Brass Pipe, Standard sizes.

ASTM B88, Standard Specification for Seamless Copper Water Tube, 2003.

ASTM B210, Standard Specification for Aluminum and Aluminum-Alloy Drawn Seamless Tubes, 2004.

ASTM B210, Standard Specification for Aluminum and - Aluminum - Alloy Drawn Seamless Tubes, 2004.

ASTM B241, Standard Specification for Aluminum and Aluminum Alloy Seamless Pipe and Seamless Extruded Tube, 2002.

ASTM B280, Standard Specification for Seamless Copper Tube Air Conditioning and Refrigeration Field Service, 2008.

ASTM D2513, Standard Specification for Thermoplastic Gas Pressure Pipe, Tubing and Fittings, 2008a.

ANSI/ASME B36.10M, Welded and Seamless Wrought Steel Pipe, 2004.

ANSI LC 1/CSA 6.26, Fuel Gas Piping Systems Using Corrugated Stainless Steel Tubing (CSST), 2005.

ANSI Z21.1 Household Cooking Gas Appliances, 2005.

NFPA 37, Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines, 2006 edition.

NFPA 10, Standard for Portable Fire Extinguishers, 2007 edition.

Title 49, Code of Federal Regulations, Parts 191, 192 and 195, "Transportation of Hazardous Liquids by Pipeline".

UL 2200-04 Standard for Stationary Engine Generator Assemblies-2004.

8.7 List of Related Appendices

Appendix V	Work on the Gas Supply System
Appendix W	Documentation for Piping Installation

PART VIII
Appendix A
Maximum Demand and Diversity

Some information on the determination of the maximum demand for an electrical installation is provided in this appendix. It also includes some notes on the application of allowances for diversity. It is impossible however, to specify the appropriate allowances for diversity for every type of installation since determination of such allowances calls for special knowledge and experience. The figures shown in Table 8.A.1 are therefore, intended to act as guideline. The current demand of a final circuit is determined by summing the current demands of all points of utilization and equipment in the circuit. Typical values to be used for this summation are given in Table 8.A.2. For blocks of residential dwellings, large hotels, and industrial and large commercial premises, allowances are to be assigned by a competent Engineer.

The current demand of a circuit supplying a number of final circuits may be assessed by applying the allowances for diversity given in Table 8.A.2 to the total current demand of all the equipment supplied by that circuit. In the Table, the allowances are appraised either as percentages of the current demand or, where followed by the letters f.l., as percentages of the rated full load current of the current using equipment. After the design currents for all the circuits have been determined, enabling the conductor sizes to be chosen, it is necessary to check that the limitation on voltage drop is met.

Table 8.A.1: Allowances for Diversity

Purpose of final circuit fed from conductors or switchgear to which diversity applies	Type of Premises		
	Individual household installations, including dwellings of a block	Small shops, stores, offices and business premises	Small hotels, boarding houses, guest houses, etc.
1. Lighting	66% of total current demand	90% of total current demand	75% of total current demand
2. Cooking appliances	10 amperes + 30% f.l. of connected cooking appliances in excess of 10 amperes + 5 amperes if socket outlet is incorporated in unit.	100% f.l. of largest appliance + 80% f.l. of 2nd largest appliance + 60% f.l. of remaining appliances	100% f.l. of largest appliance + 80% f.l. of 2nd largest appliance + 60% f.l. of remaining appliances

Purpose of final circuit fed from conductors or switchgear to which diversity applies	Type of Premises		
	Individual household installations, including dwellings of a block	Small shops, stores, offices and business premises	Small hotels, boarding houses, guest houses, etc.
3. Motors (other than lift motors which are subject to special consideration)	————	100% f.l. of largest motor + 80% f.l. of 2nd largest motor + 60% f.l. of remaining motors	100% f.l. of largest motor + 50% f.l. of remaining motors.
4. Water heater (thermostatically controlled)		No diversity allowable	

Table 8.A.2: Current Demand to be Assumed for Points of Utilization and Current using Equipment

Point of Utilization or Current-using Equipment	Current Demand to be Assumed
15 A socket outlets	15 A with diversity applied
13 A socket outlets	13 A with diversity applied
5 A socket outlets	At least 0.5 A
Protected outlets other than the above mentioned socket outlets	Rated current
Lighting outlet	Current equivalent to the connected load, with a minimum of 100 W per lamp holder
House hold cooking appliance	The first 10 A of the rated current plus 30% of the remainder of the rated current plus 5 A if a socket outlet is incorporated in the control unit
All other stationary equipment / Appliances	Standard rated current or nominal current.

PART VIII
Appendix B
Useful Tables Relating to Conductor Sizes

Table 8.B.1: Number of Single-core Wire of Different Sizes for Various Sizes of Metal Conduits

Conductor Cross-sectional Area (mm ²)	Conduit Diameter					
	19 mm	25.4 mm	31.8 mm	38 mm	51 mm	63.5 mm
1.5	5	10	14	-	-	-
2.5	5	8	12	-	-	-
4.0	3	6	10	-	-	-
6.0	2	5	8	-	-	-
10.0	-	3	5	6	-	-
16.0	-	-	3	6	-	-
25.0	-	-	2	4	6	7
35.0	-	-	-	3	5	6
50.0	-	-	-	-	4	5

Table 8.B.2: Number of Single-core Wires of Different Sizes for Various Sizes of PVC Conduits

Conductor Cross-sectional Area (mm ²)	Conduit Diameter				
	19 mm	25 mm	32 mm	38 mm	51 mm
1.5	6	10	14	-	-
2.5	5	10	14	-	-
4.0	3	6	10	14	-
6.0	2	5	8	11	-
10.0	-	4	7	9	-
16.0	-	2	4	5	12
25.0	-	-	2	2	6
35.0	-	-	2	2	5
50.0	-	-	-	2	3

Table 8.B.3: Cross-sectional Area and Weight of Wire Gauges

Gauge System		Diameter	Cross-sectional Area	Weight of Copper	Weight of Aluminium
AWG	SWG	(mm)	(mm ²)	(kg/km)	(kg/km)
6/0	-	14.73	170.46	1515.4	460.4
5/0	-	13.11	134.92	1199.4	365.0
-	7/0	12.70	126.68	1126.2	342.1
-	6/0	11.79	109.09	969.8	294.6
4/0	-	11.68	107.22	953.2	289.4
-	5/0	10.97	94.56	840.7	255.4
3/0	-	10.41	85.16	757.2	229.6
-	4/0	10.16	81.70	720.7	219.6
-	3/0	9.449	70.12	623.4	189.4
2/0	-	9.271	67.51	600.1	182.1
-	2/0	8.839	61.36	545.5	165.7
0	-	8.255	53.52	475.8	144.4
-	0	8.230	53.19	472.9	143.6
-	1	7.620	45.60	405.4	123.1
1	-	7.341	42.22	376.2	114.5
-	2	7.010	38.60	343.1	104.2
2	-	6.553	33.94	299.8	90.80
-	3	6.401	32.18	286.1	86.89
-	4	5.893	27.27	242.5	73.63
3	-	5.817	26.57	236.2	72.01
4	-	5.182	21.09	187.5	57.11
-	6	4.877	18.68	166.1	50.44
5	-	4.623	16.78	149.2	45.25

Gauge System		Diameter	Cross-sectional Area	Weight of Copper	Weight of Aluminium
AWG	SWG	(mm)	(mm ²)	(kg/km)	(kg/km)
-	7	4.470	15.70	139.5	42.39
6	-	4.115	13.30	118.2	35.91
-	8	4.065	12.97	115.3	35.02
7	9	3.658	10.507	93.41	28.49
8	10	3.251	8.302	73.80	22.59
-	11	2.948	6.818	60.61	18.41
9	-	2.896	6.585	58.54	17.91
-	12	2.642	5.480	48.72	14.80
10	-	2.591	5.272	46.87	14.21
-	13	2.337	4.284	38.08	11.58
11	-	2.311	4.196	37.30	11.26
12	-	2.057	3.325	29.55	8.934
-	14	2.032	3.243	28.83	8.756
13	15	1.828	2.627	23.35	7.093
14	16	1.626	2.075	18.45	5.621
15	-	1.448	1.646	14.64	4.460
-	17	1.422	1.589	14.13	4.290
16	-	1.295	1.318	11.72	3.532
-	18	1.291	1.168	10.38	3.154
17	-	1.143	1.026	9.122	2.808
18	19	1.016	0.8107	7.207	2.223
19	20	0.9144	0.6567	5.838	1.773
20	21	0.8128	0.5189	4.613	1.401
21	-	0.7239	0.4156	3.695	1.111

Gauge System		Diameter	Cross-sectional Area	Weight of Copper	Weight of Aluminium
AWG	SWG	(mm)	(mm ²)	(kg/km)	(kg/km)
-	22	0.7112	0.3973	3.532	1.073
22	-	0.6428	0.3243	2.883	0.8756
-	23	0.6096	0.2919	2.595	0.7881
23	-	0.5733	0.2588	2.301	0.6990
-	24	0.5588	0.2453	2.181	0.6620
24	-	0.5105	0.2047	1.820	0.5527
-	25	0.5086	0.2021	1.797	0.5473

PART VIII
Appendix C

Completion Certificate Form (Electrical Works)

I/we certify that the installation detailed below has been installed by me/us and tested and that to the best of my/our knowledge and belief, it complies with the requirements of Bangladesh National Building Code and the Electricity Act of Bangladesh.

Electrical Installation at : _____

Voltage and system of supply : _____

**PARTICULARS OF
WORKS**

(a) Internal Electrical Installation

	<u>No.</u>	<u>Total load</u>	<u>Type or system of wiring</u>
(i) Light points			
(ii) Fan points			
(iii) Socket points			
2-pin 5 A			
3-pin 13 A Flat pin			
3-pin 15 A Round Pin			

(b) Others

	<u>Description</u>	<u>hp/kW</u>	<u>Type of starting</u>
(1) Motors			
(i)			
(ii)			
(iii)			
(2) Other Plants			

(c) If the work involves installations of overhead line and/or underground cable

- (1) (i) Type and description of overhead line
(ii) Total length and number of spans
(iii) Number of street lights and its description

(2) (i) Total length and size of underground cable

(ii) Number of joints

End joint

Tee joint

Straight through joint

(d) Earthing

(i) Description of earthing electrode

(ii) Number of earth electrodes

(iii) Type and short description of Brass Earthing Clamp

(iv) Number of Brass Bolt Nuts provided in the Brass Earthing Clamp for the termination of Earth Lead Cables

(v) Size of each of the main earth lead cables

(vi) Number of main earth lead cables

TEST RESULTS

(a) Insulation Resistance values:

(i) Insulation resistance of the whole system of conductors to earth _____ mega ohms

(ii) Insulation resistance between the phase conductor and neutral

Between phase R and neutral _____ mega ohms

Between phase Y and neutral _____ mega ohms

Between phase B and neutral _____ mega ohms

(iii) Insulation resistance between the phase conductors in case of polyphase supply

Between phase R and phase Y _____ mega ohms

Between phase Y and phase B _____ mega ohms

Between phase B and phase R _____ mega ohms

(b) Polarity test

Polarity of nonlinked single pole branch switches

(c) Earth continuity test

Maximum resistance between any point in the earth continuity conductor including metal conduits and main earthing lead _____ ohms

(d) Earth electrode resistance

Resistance of each earth electrode

(i) _____ ohms

(ii) _____ ohms

(iii) _____ ohms

(iv) _____ ohms

(e) Lightning protective system

Resistance of the whole of lightning protective system to earth before any bonding is effected with earth electrode and metal in/on the structure _____ ohms

Signature of Supervisor

Signature of Contractor

Name and Address _____

Name and Address _____

PART VIII
Appendix D

NC, NCB and Recommended Criteria for Sound Insulation

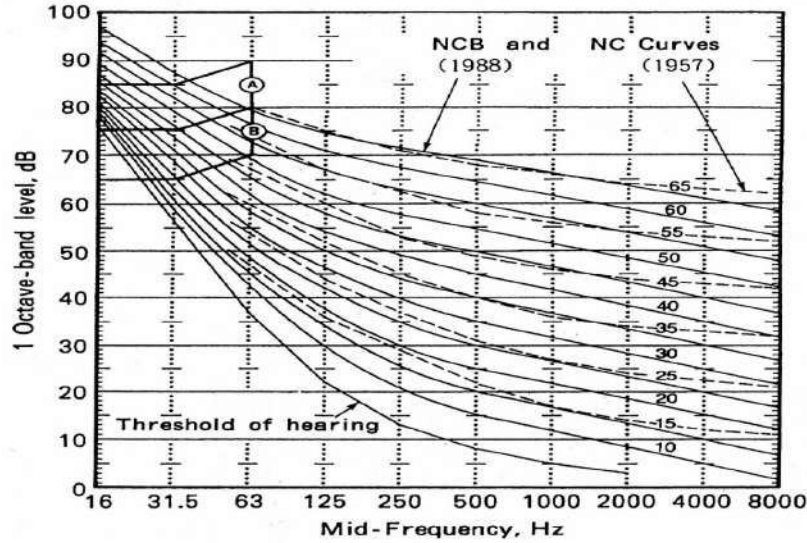


Figure 8.D.1 Noise Criteria (NC) and Balanced Noise Criteria (NCB) Curve
(Source: L. L. Beranek, 1957, 1988)

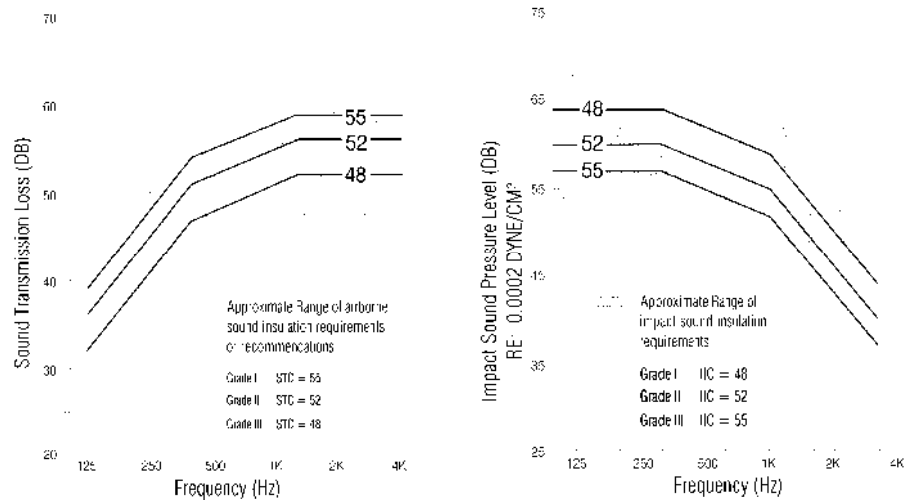


Figure 8.D.2 Recommended Criteria for Sound Insulation between Dwelling Units

(Source: Cavanaugh, W. J., Wilkes, J. A. 1999. *Architectural Acoustics Principles and Practice*, John Wiley & Sons, Inc, New York)

PART VIII**Appendix E****STC, Aural Field and Proportion of Space****Table 8.E.1: Typical STC Ratings for Building Elements**

(Source: Stein, et al. 2006. *Mechanical and Electrical Equipment for Buildings*. John Wiley & Sons, New Jersey)

Improvements in STC Ratings of Stud ^a Partitions ^b		STC Ratings of Masonry Walls	
Description	STC	Description	STC
Basic partition: single wood studs,	35	100 mm light weight hollow block	36
400 mm on centres, 13 mm gypsum board on both sides, air cavity		100 mm dense hollow block	38
Add to basic partition		150 mm lightweight hollow block	41
Double gypsum board, one side	+2	150 mm dense hollow block	43
Double gypsum board, both sides	+4	200 mm lightweight hollow block	46
Single thickness absorbent material in air cavity	+3	200 mm dense hollow block	48
Double-thickness insulation	+6	300 mm lightweight hollow block	51
Resilient channel supports for gypsum board	+5	300 mm dense hollow block	53
Staggered studs	+9	100 mm brick	41
Double studs	+13	150 mm brick	45
		200 mm brick	49
^a For application to metal stud partition, use adders as in note b, but begin with STC=40 for a 90mm basic partition.		300 mm brick	54
^b When using two improvements, add an additional +2; for three improvements, add +3.		150 solid concrete	47
^c The STC figures are conservative. Other sources list the same constructions with 1 to 5 points higher STC.		200 solid concrete	50
		250 solid concrete	53
		300 solid concrete	56
Typical STC Value for Doors		^a The STC figures are conservative. Other sources list the same constructions with 1 to 5 points higher STC.	
Door Construction	STC	^b All ratings of lightweight block assume sealing with paint. Note that this reduces absorption.	
Louvered door	15	Modifications	
Any door, 50 mm undercut	17	Add sand to core of hollow blocks	+3
38 mm hollow core door, no gasket	22	Add plaster to one side	+2
38 mm hollow core door, gaskets and drop closure	25	Add plaster to both side	+4
44 mm solid wood door, no gasket	30	Add furring strip, lath and plaster:	
44 mm solid wood door, gaskets and drop closure	35	One side	+6
Two hollow core doors, gasket all around, with sound lock	45	Two sides	+10
Two solid core doors, gasket all around, with sound lock	55	Add plaster via resilient mounting:	
Special commercial construction, with lead lining and full sealing	45-65	One side	+10
		Two sides	+15

Typical STC Ratings of Interior Partitions		Typical STC Value for Windows	
Type of Partition	STC	Window Construction	STC
Demountable partition	20-30	Operable wood sash, 3 mm glass unsealed	23
Drywall partition up to acoustical ceiling	30	Operable wood sash, 6 mm glass unsealed	25
Drywall partition extending 300 mm above acoustical ceiling tile system into ceiling plenum	35	Operable wood sash, 6 mm glass, with gasket	30
Drywall partition with cavity insulation, full height to the underside of slab above	40-45	Operable wood sash, laminated glass, unsealed	28
Two-layer drywall partition with insulation, erected full height to underside of slab above	50	Operable wood sash, double-glazed, 3mm panes, 10 mm air space, with gasket	29
		Fixed sash, double 3 mm panes, 75 mm air space, with gasket	44
		Fixed sash, double 3 mm panes, 100 mm air space, with gasket	48

Appendix E (Contd.)

Table 8.E.2: Recommended STC for Partitions for Specific Occupancies
(Source: Stein, B. et al., 2006. *Mechanical and Electrical Equipment for Buildings*.
John Wiley & Sons, New Jersey)

Type of Occupancy	Wall, Partition, or Panel Between		Sound Isolation Requirement: Background Level in Room Being Considered	
	Room Being Considered	and Adjacent Area	Quiet	Normal
Normal school buildings without extraordinary or unusual activities or requirements	Classrooms	Adjacent classrooms	STC 42	STC 40
		Corridor or public areas	STC 40	STC 38
		Kitchen and dining areas	STC 50	STC 47
		Shops	STC 50	STC 47
		Recreation areas	STC 45	STC 42
		Music rooms	STC 55	STC 50
		Mechanical equipment rooms	STC 50	STC 45
		Toilet areas	STC 45	STC 42
	Music practice rooms	Adjacent practice rooms	STC 55	STC 50
		Corridor and public areas	STC 45	STC 42
Executive areas, doctors suites; confidential privacy requirements	Office	Adjacent offices	STC 50	STC 45
		General office areas	STC 48	STC 45
		Corridor or lobby	STC 45	STC 42
		Washrooms and toilet areas	STC 50	STC 47
Normal office; normal privacy requirements; any occupancy using rooms for group meetings	Office	Adjacent offices	STC 40	STC 38
		Corridor, lobby, exterior	STC 40	STC 38
		Washrooms, kitchen, dining	STC 42	STC 40
	Conference rooms	Other conference rooms	STC 45	STC 42
		Adjacent offices	STC 45	STC 42
		Corridor or lobby	STC 42	STC 40
		Exterior of building	STC 40	STC 38
		Kitchen and dining areas	STC 45	STC 42
Large offices, drafting areas, banking floors, etc.	Large general office areas	Corridors, lobby, exterior	STC 38	STC 35
		Data-processing area	STC 40	STC 38
		Kitchen and dining areas	STC 40	STC 38
Motels and urban hotels, Hospitals and dormitories	Bedrooms	Adjacent bedrooms ^a	STC 52	STC 50
		Bathroom ^a	STC 50	STC 45
		Living rooms ^a	STC 45	STC 42
		Dining areas	STC 45	STC 42
		Corridor, lobby, or public spaces	STC 45	STC 42

^a Separate occupancy

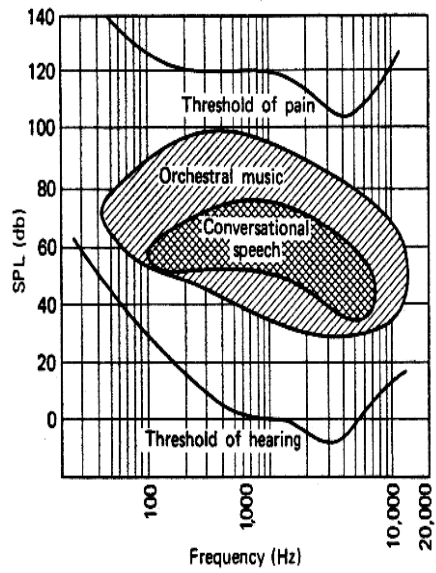


Figure 8.E.1 Speech and Music in Aural Field

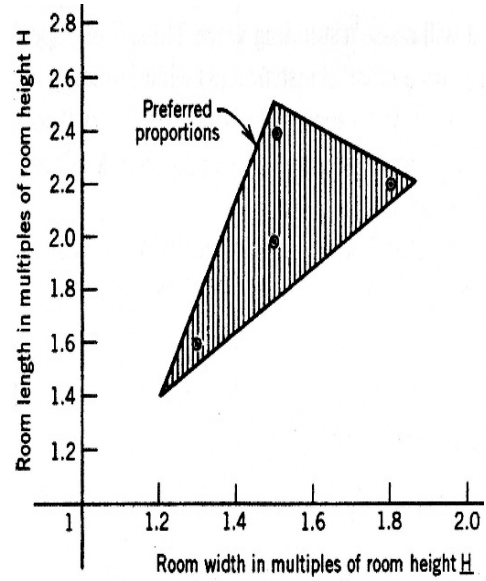


Figure 8.E.2 Recommended proportion of a space

PART VIII

Appendix F

Activity Flow Diagram: Planning, Design, Assessment and Construction in Building Acoustics

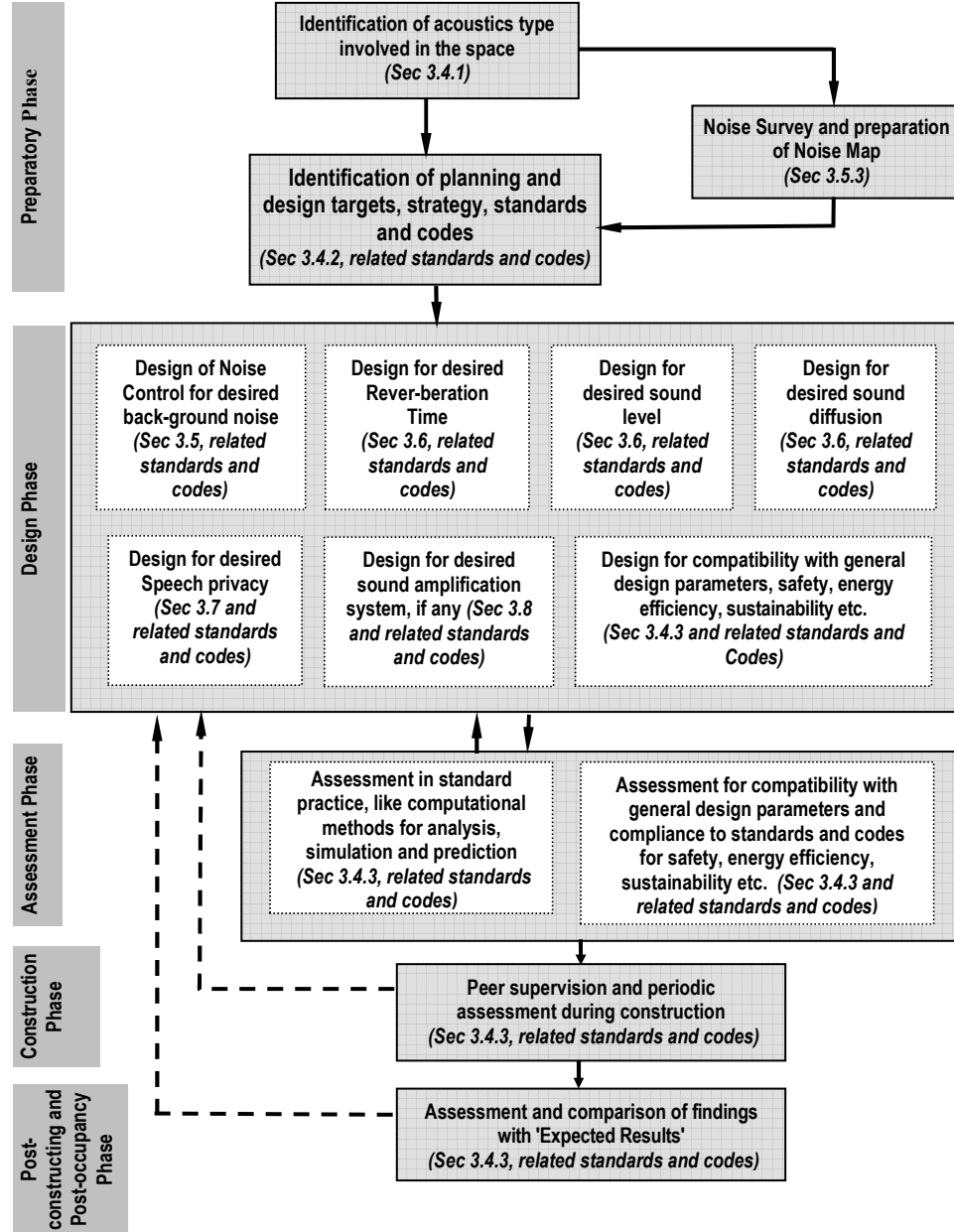


Figure 8.F.1 Activity Flow Diagram: Planning, Design, Assessment and Construction in Building Acoustics

PART VIII
Appendix G

Checklist for Acoustical Planning, Design and Post-occupancy Assessments

Form G: Checklist - Acoustical Planning, Design and Post-occupancy Assessments

Instructions:

- a) A separate form should be filled in for each space/room/building
b) Fill in with text or put tick mark, as applicable
c) After filling in and signing **Expected Result in Planning Design Phase**, submit the form to the project owner/ client
d) A photocopy of the submitted Form (as mentioned above) shall be filled in and signed for **Post- occupancy Findings**

1. Name of the space/room/building for Acoustical Planning and Design:
2. Project name :
3. Location :
4. Owner/Client :
5. Lead Consultant :
6. Acoustical Consultant :
7. Type of acoustics involved in the space: ☐ Speech ☐ Music ☐ Multipurpose
8. Noise Survey on Site and Surroundings: ☐ Conducted ☐ Not Conducted
9. Noise Map: ☐ Prepared ☐ Not prepared
10. Design Parameters of the Space/Room/ Building:

Sl. No.	Design Parameters	Expected Results in Planning and Design Phase	Post-occupancy Findings
a)	Method Adopted:	☐ Assumptions ☐ Only Manual Calculations ☐ Computational Analysis, Simulation and Prediction	☐ Instrumental Survey ☐ Opinion Survey ☐ Both of above
b)	Ambient/Background level: i) L_{10} ii) L_{eq}	i) dBA ii) dBA	i) dBA ii) dBA
c)	Reverberation Time: i) At 1 kHz ii) Average of all applicable frequencies	i) s ii) s	i) s ii) s

Sl. No.	Design Parameters	Expected Results in Planning and Design Phase	Post-occupancy Findings
d)	Sound level in audience area (Average)	 dBA	 dBA
e)	Diffusion in audience area - Difference between max. and min. sound level:	 dBA	 dBA
f)	Signal-to-Noise Ratio (SNR)	 dBA	 dBA
g)	Intelligibility:		
	i) PSA	i) %	i) %
	ii) AI	ii) %	ii) %
	iii) STI	iii) %	iii) %
h)	Reverberance: Based on average absorption coefficient $\bar{\alpha}$: <0.2 = Live, 0.2 - 0.4 = Average, >0.4 = Dead	☐ Live ☐ Average ☐ Dead	☐ Live ☐ Average ☐ Dead
i)	Speech Privacy Rating:		
J)	Overall quality (perceptual evaluation):	☐ Excellent ☐ Good ☐ Average ☐ Poor	☐ Excellent ☐ Good ☐ Average ☐ Poor
Signature of the Acoustical Consultant:		 Date:	 Date:

PART VIII
Appendix H

Noise Levels and Subjective Evaluation

Table 8.H.1: Allowable upper limit of Outdoor Noise Levels

Source: *Noise Pollution (Control) Rules 2006*, the Government of the People's Republic of Bangladesh

Sl. No.	Category of Zones	Upper Limit of Noise Level in dBA $L_{Aeq,T}$	
		Day Time (06:00 AM - 09:00 PM)	Night Time (09:00 PM - 06:00 AM)
01	Quiet Zone	50	40
02	Residential Zone	55	45
03	Mixed Use Zone	60	50
04	Commercial Zone	70	60
05	Industrial Zone	75	70

Table 8.H.2: Typical Noise Levels Generated by Aircrafts

Source: *Noise Levels for U.S. Certified and Foreign Aircraft, Advisory Circular-AC No: 36-1H*, U.S. Department of Transportation, Federal Aviation Administration, 2012

Sl. No.	Aircraft Manufacturer and Model (Arranged in an alphabetical order)	Flyover Noise Levels with Take-off Thrust (EPN dBA) ^a
01	Airbus A300	103.1
02	Airbus A380	98.1
03	Boeing 707	108.3
04	Boeing 717	92.1
05	Boeing 727	105.8
06	Boeing 737	105.3
07	Boeing 747	107.8
08	Boeing 757	100.3
09	Boeing 767	103.0
10	Boeing 777	101.3
11	Boeing 787	99.6
12	Lockheed L-1011	103.3
13	McDonnell Douglas DC-08	102.7
14	McDonnell Douglas DC-10	106.6

^a Noise is measured at 1.2 m above the ground for aircrafts flying overhead at a height of 450 m from the ground with takeoff thrust. The highest noise level among different variations in the same model is listed.

Table 8.H.3: Subjective Evaluation and Pressure Levels of Familiar Sounds

Description of Sound	Subjective Evaluation	Pressure Level (dBA)
Near jet engine	Deafening	140
Threshold of pain		130
Threshold of feeling hard rock band		120
Accelerating motorcycle at a few metre away (Note: 15 m from motorcycle equals noise at about 600 m from a 4-engine aircraft)	Very loud	110
Full orchestra, loud passage		95
Noisy urban street, noisy factory	Loud	90
School cafeteria		80
Loud speech, 1 m distant; stenographic room	Moderate	70
Near freeway auto traffic		60
Conversational speech, 1 m distant; average office	Faint	50
Attentive theatre audience, total sound. Soft radio music in apartment.		40
Faint whisper, 1 m distant; average residence without stereo playing	Very faint	30
Average whisper		20
Rustle of one program, 8 m distant		15
Rustle of leaves in wind; human breathing		10
Threshold of audibility		0

PART VIII Appendix I PSA and Liveliness

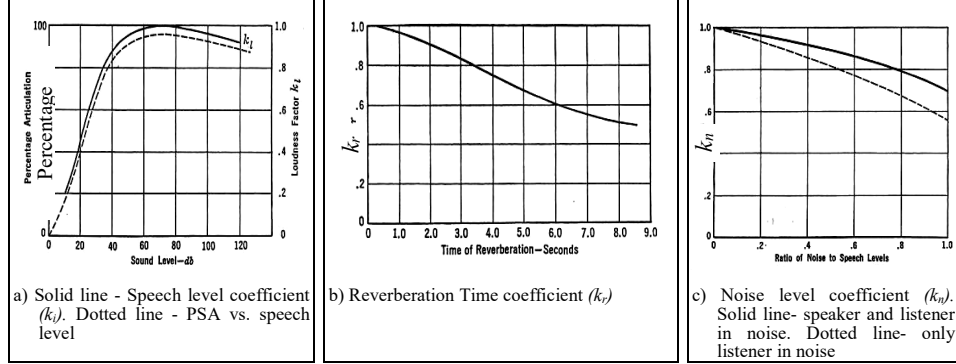


Figure 8.1.1 Coefficients for Percentage Syllable Articulation (PSA)
(Source: Knudsen, V. O. 1932. *Architectural Acoustics*, John Wiley & Sons Inc., New York)

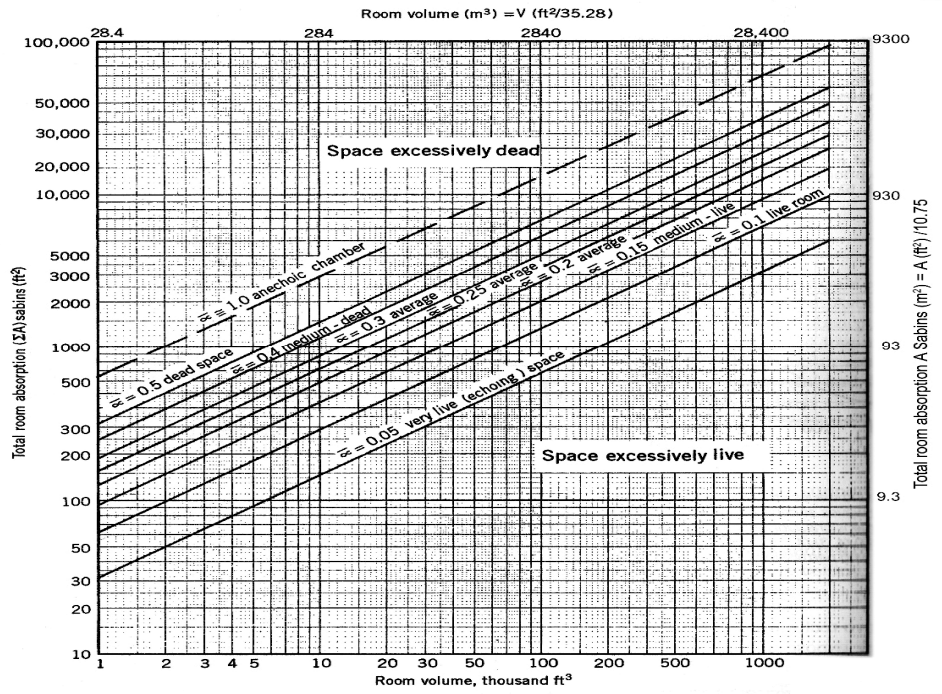


Figure 8.1.2 Liveliness of a Room as a Function of its Volume and Total Absorption

(Source: Stein, B. et al. 2006. *Mechanical and Electrical Equipment for Buildings*. John Wiley & Sons, New Jersey)

Note: The room proportions chosen as Length : Width : Height = $2H : 1.5H : H$; average of the three extreme points of the recommended room proportion triangle in Fig. 1.2. For these proportions, $S = 6.25 V^{2/3}$ and thus $A = S \alpha = 6.25 V^{2/3}$.

PART VIII
Appendix J
Speech Privacy Analysis Sheet

SPEECH PRIVACY: Analysis Sheet**Anticipated response to privacy situation**

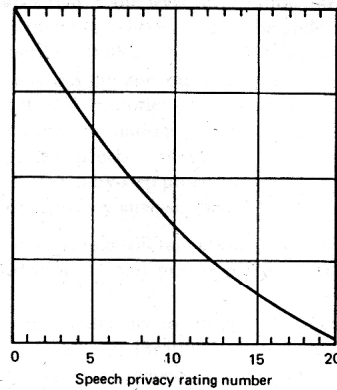
Apparent satisfaction

Mild dissatisfaction

Moderate dissatisfaction

Strong dissatisfaction

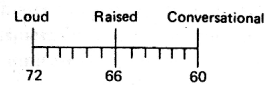
Serious dissatisfaction



NOTE: Curve shows average response of people to intruding speech based on rating number figured below.

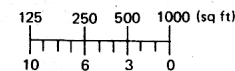
Speech rating

1. Speech effort — how people talk in source room

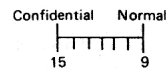


A. B. C.

2. Source room floor area (A_1) — approximates effect of source room absorption



3. Privacy allowance — degree of privacy desired

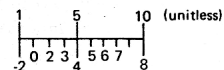


■ Speech rating total

Isolation rating

4. Sound transmission class (STC) — accounts for transmission loss of common barrier

5. Noise reduction factor (A_2/S) — approximates effect of receiving room sound absorption and common barrier size



6. Adjacent room background noise level (dBA) — masking sound available

■ Isolation rating total

Speech privacy rating number

Find speech privacy rating number by subtracting isolation rating total from speech rating total. Then use graph at top of sheet to predict degree of satisfaction.

Figure 8.J.1 Speech Privacy Analysis Sheet

(Source: Egan, M. D. 1972. *Concepts in Architectural Acoustics*, McGraw-Hill Book Company, New York)

PART VIII
Appendix K
Sound Absorption Coefficients

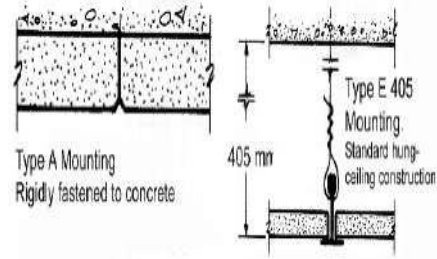
Table 8.K.1: Sound Absorption Coefficients ^a

Source: Stein, B. et al., 2006. *Mechanical and Electrical Equipment for Buildings*. John Wiley & Sons, New Jersey

	Absorption Coefficients (α)						NRC ^c
	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	
General Building Materials and Furnishings ^b							
Brick, unglazed	0.03	0.03	0.03	0.04	0.05	0.07	0.005
Brick, unglazed, painted	0.01	0.01	0.02	0.02	0.02	0.03	0.00
Carpet, heavy, on concrete	0.02	0.06	0.14	0.37	0.60	0.65	0.29
Carpet, heavy, on 1.36 kg/m ² hair felt or foam rubber	0.08	0.24	0.57	0.69	0.71	0.73	0.55
Concrete block, coarse	0.36	0.44	0.31	0.29	0.39	0.25	0.35
Concrete block, painted	0.10	0.05	0.06	0.07	0.09	0.08	0.05
Fabrics							
Light velour, 0.34 kg/m ² , hung straight, in contact with wall	0.03	0.04	0.015	0.17	0.24	0.35	0.15
Medium velour, 0.47 kg/m ² , draped to half area	0.07	0.31	0.49	0.75	0.70	0.60	0.55
Heavy velour, 0.61 kg/m ² , draped to half area	0.14	0.35	0.55	0.72	0.70	0.65	0.60
Floors							
Concrete or terrazzo	0.01	0.01	0.015	0.02	0.02	0.02	0.00
Linoleum, asphalt, rubber, or cork tile on concrete	0.02	0.03	0.03	0.03	0.03	0.02	0.05
Wood	0.15	0.11	0.10	0.07	0.06	0.07	0.10
Glass							
Large panes of heavy plate glass	0.18	0.06	0.04	0.03	0.02	0.02	0.05
Ordinary window glass	0.35	0.25	0.18	0.12	0.07	0.04	0.15
Gypsum board, 13 mm nailed to 50x100 mm stud 400 mm c/c	0.10	0.08	0.05	0.03	0.03	0.03	0.05
Marble or glazed tile	0.01	0.01	0.01	0.01	0.02	0.02	0.00
Openings							
Stage, depending on furnishings				0.25-0.75			
Deep balcony, upholstered seats				0.50-1.00			
Grillese ventilating				0.15-0.50			

General Building Materials and Furnishings ^b	Absorption Coefficients (α)							
	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	NRC ^c	
Plaster, gypsum or lime, smooth finish on tile or brick	0.013	0.015	0.02	0.03	0.04	0.05	0.05	
Plaster, gypsum or lime, on lath	0.14	0.10	0.06	0.05	0.04	0.03	0.05	
Plywood panelling, 9 mm thick	0.28	0.22	0.17	0.09	0.10	0.11	0.15	
Rough wood, as tongue-and-groove cedar	0.24	0.19	0.14	0.08	0.13	0.10	0.14	
Slightly vibrating surface (e.g., hollow core door)	0.02	0.02	0.03	0.03	0.04	0.05	0.03	
Readily vibrating surface (e.g., thin wood panelling on 400 mm studs)	0.10	0.07	0.05	0.04	0.04	0.05	0.05	
Water surface, as in swimming pool	0.008	0.008	0.013	0.015	0.020	0.025	0.00	
Absorption of Seats and Audience ^d	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	NRC ^c	
Audience, in upholstered seats, per 0.093 m ² of floor area	0.60	0.74	0.88	0.96	0.93	0.85	-	
Unoccupied cloth-upholstered seats, per 0.093 m ² of floor area	0.49	0.66	0.80	0.88	0.82	0.70	-	
Wooden pews, occupied, per 0.093 m ² of floor area	0.57	0.61	0.75	0.86	0.91	0.86	-	
Students in tablet-arm chairs, per 0.093 m ² of floor area	0.30	0.42	0.50	0.85	0.85	0.84	-	
Acoustic Absorptive Materials	Mtg ^e	125 Hz	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	NRC ^c
High-performance vinyl-faced fibreglass								
Ceiling panels								
25 mm thick	E405	0.73	0.88	0.71	0.98	0.96	0.77	0.90
38 mm thick	E405	0.79	0.98	0.83	1.03	0.98	0.80	0.95
Painted nubby glass cloth panels								
6 mm thick	E405	0.81	0.94	0.65	0.87	1.00	0.96	0.85
25 mm thick	E405	0.78	0.92	0.79	1.00	1.03	1.10	0.95
Random fissured 19 mm-thick panels	E405	0.52	0.58	0.60	0.80	0.92	0.80	0.70
Perforated metal panel with infill 1 in. thick	E405	0.70	0.86	0.74	0.88	0.95	0.86	0.85
Typical averages, mineral fibre tiles and panels								
19 mm fissured	E405	0.47	0.50	0.52	0.76	0.86	0.81	0.65
19 mm textured	E405	0.49	0.55	0.53	0.80	0.94	0.83	0.70
16 mm fissured	E405	0.28	0.33	0.66	0.73	0.74	0.75	0.60
16 mm textured	E405	0.29	0.35	0.66	0.63	0.44	0.34	0.50
16 mm perforated	E405	0.27	0.29	0.55	0.78	0.69	0.53	0.60
75 mm in thick x 400 mm square on 600 mm centres	A	0.40	0.61	1.92	2.54	2.62	2.60	

- a This table primarily useful for preliminary calculations. Complete tables of coefficients may be found in books on architectural acoustics.
- b Selected data courtesy of Owens-Corning Fibreglass.
- c NRC: the arithmetic average of the α values at 250, 500, 1000, and 2000 Hz.
- d If audience is randomly spaced, use an average of 0.47 m² sabins per person.
- e See figure at right for mounting methods Type A and Type E 405.



PART VIII
Appendix L

Particulars of Lifts, Escalators and Moving Walks

Format for Particulars of Lifts, Escalators and Moving Walks

LIFT

- a) Use (Tick as appropriate): Passenger/Goods/Services/Hospital
- b) Number of lifts required: _____
- c) Capacity per lift: Number of passengers _____ Kg _____
- d) Rated Speed: _____ m/s
- e) Travel height: _____ m
- f) Size and type of car doors: _____
- g) Number of car doors: _____
- h) Number of stops: _____
- i) Size and type of landing doors: _____
- j) Number of stops: _____
- k) Type of Drive system: _____
- l) Method of control(s): _____
- m) Size(s) and location(s) of lift well(s): _____
- n) Available Over Head Height: _____
- o) Pit Depth : _____
- p) Location and size of machine room: _____
- q) Position of counter weight: _____
- r) Size of car platform: _____
- s) Construction, design and finish of car bodywork: _____
- t) Construction, design and finish of car platform: _____
- u) Particulars of ventilation of the car: _____
- v) Particulars of control buttons in car: _____
- w) Particulars of position indicators in car: _____
- x) Particulars of call buttons in landing: _____

- y) Particulars of car position indicators in landings: _____
- z) Electric supply:
Power: _____ volts, ac/dc, _____ phase, _____ Hz/wire system _____
Lighting: _____ volts, ac/dc, _____ Hz/sec, _____
- aa) Layout of Lifts: _____
- bb) Details of Access to machine room: _____
- cc) Buffer type: _____
- dd) Additional requirements, if any: _____

ESCALATOR

- a) Number of escalators required: _____
- b) Capacity of each escalator (No. of people/hr): _____
- c) Rated speed _____ m/s
- d) Travel height _____ m, Travel length _____ m
- e) Width of escalator _____ m
- f) Construction, design and finish of balustrade: _____
- g) Details of steps: _____
- h) Materials of landing plate: _____
- i) Electric supply:
Power: _____ volts, ac/dc, _____
Phase _____ Hz/wire system _____
- j) Additional requirements, if any: _____

MOVING WALK

- a) Number of moving walks required : _____
- b) Capacity of each moving walk (No. of people/hr) : _____
- c) Rated Speed _____ m/s
- d) Inclination _____ degrees
- e) Width of moving walk _____ m
- f) Construction, design and finish of balustrade: _____
- g) Material of landing plate: _____
- h) Electric supply:
Power: _____ volts, ac/dc, _____
Phase _____ Hz/wire system _____
- i) Additional requirements, if any: _____

PART VIII
Appendix M
Application Format for Permit to Construct Water Supply and Distribution System

Application for Permit to Construct Water Supply and Distribution System

1. Name of the Applicant (Owner of holdings): _____
 Father's Name: _____
 Mother's Name: _____
2. Address & Location of Plot/Holdings:
 Ward No. _____ Road No. _____ House No. _____
 Boundary Description _____
3. (a) Description of Holdings, Buildings and Premises (Type of Building/Structure) _____
 (b) Occupancy Classification _____
 (c) No. of Floors, Room Nos. _____
4. (a) Total Population _____
 (b) Type of Connection (Residential _____ / Commercial _____ / Industrial _____)
 (c) Total Water Requirement (Calculation Sheet should be attached) _____ and
 (d) Size of Water Connection (diameter of pipe) required _____
5. Under-ground water Reservoir Capacity (where required) _____
6. Certificate of Bill Payers _____
7. This application is accompanied by all required plans, drawing of buildings, design calculations of water requirement in accordance with Section 5.3.2 and 5.3.3 of Bangladesh National Building Code.

Signature of the licensed plumber

Signature of the Owner or his/
her designated person/agent

Name and Address _____

Name and Address _____

PART VIII

Appendix N

Format of Completion Certificate (Water Supply Works)

Completion Certificate (Water Supply Works)

Certified that I/we have completed the plumbing work of water connection No _____
for the premises as detailed below. This may be inspected and connection given.

Ward No. _____ Road No. _____ House No. _____

Existing water connection No. (if any) _____

Owned by _____

Owner's address : _____

Applicant's name : _____

son of _____

Address : _____

Situated _____ Size of
main _____ on _____ Street _____

Where main is _____ Size of service pipe _____

Size of ferrule _____ No. of taps _____ No. of closets _____

Capacity of Underground Reservoir (where required) _____

No. of other fittings and appliances _____

Road cutting and repairing fee _____

Paid Tk. _____ (Receipt No. _____ dated _____ (receipt enclosed)

Dated _____ Signature of the licensed plumber _____

Name and address of the licensed plumber _____

The Authority's Report/ Certificate :

Certified that the communication and distribution pipes and all water fittings have been laid,
applied and executed in accordance with Chapter 5 of Part 8 of the Bangladesh National Building
Code, and satisfactory arrangements have been made for draining off waste water.

The water supply connection will be made on _____

Date _____

Water Supply Authority _____

PART VIII Appendix O

Sizing of Cold Water Supply and Distribution Piping

The water distribution within the building may be an upfeed or downfeed system. The design principles are the same for both the systems. The principal difference in the calculation procedure is that in the up feed system the difference in elevation between the fixtures and the water supply main exerts the pressure that supplies water into the fixtures, but in the down-feed system the difference in elevation between the fixtures and the roof storage tank provides the pressure to overcome the pipe friction. The following are the recommended pipe sizing procedures:

- O.1 The sketch of the main lines, risers and branches serving different fixtures will have to be drawn.
- O.2 Determine the number and types of fixture that will be required on the basis of the Table 8.6.1 in Chapter 6.
- O.3 The demand weight of different fixture units may be computed in terms of water supply fixture unit (wsfu) in accordance with Table 8.O.1.
- O.4 The peak demand load (or maximum probable flow) in liter per minute may be estimated with the data obtained in Sec O.3 using Figure 8.O.1 or on the basis of the number of occupants according to their occupancy classification specified in Table 8.5.1.
- O.5 The equivalent length of the main lines, risers and branches will be determined. The equivalent length of different fittings may be estimated on the basis of the data presented in the Tables 8.O.2(a), 8.O.2(b) and 8.O.2(c) or from manufacturer's specification. The total equivalent length is the sum of the equivalent lengths of all pipes and fittings.
- O.6 The pressure loss through water meter may be determined on the basis of their operating characteristics as shown in Figure 8.O.6 (Disk type water meter). The data for other types of water meter may be obtained from the manufacturer.
- O.7 The minimum pressure required at different fixtures to produce adequate flow may be estimated on the basis of the type of fixtures or minimum size of supply pipe in accordance with Table 8.5.6.
- O.8 The average pressure drop in kPa per meter of equivalent pipe length may be computed as follows:

$$F_p = (P \pm 9.807 H - f) / L \quad 8.O.1)$$

Where, F_p = Average available pressure loss (kPa) per meter of equivalent length of pipe

P = Pressure (kPa) in the water main or zero for overhead gravity storage tank.

H = Height (m) of the highest fixture above the water main or difference (m) in elevation between storage tank and the fixture under consideration.

f = Pressure loss (kPa) through water meter or such other fittings plus pressure (kPa) required to produce adequate flow through the most remote fixture in upfeed system or the fixture under consideration in down-feed system.

L = Equivalent pipe length (m)

* + sign is for down-feed system and - sign is for up-feed system

O.9 The pipe size may also be estimated from Figures 8.0.5 to 8.0.10 for different types of piping materials on the basis of the expected rate of flow determined in Sec 0.4 and the average pressure available for friction loss (F_p) in Sec O.8.

O.10 Alternative simple procedure of pipe size computation has also been explained in the Sec 5.10.3.

Table 8.O.1: Water Supply Fixture Unit (wsfu) Values for Various Plumbing Fixtures

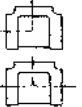
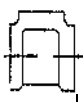
Fixture or group	Supply Control	wsfu		Total
		Cold	Hot	
Bath group	Flush tank	4.5	3	6
Bath group	Flush valve	6	3	8
Bathtub	Faucet	1.5	1.5	2
Bidet	Faucet	1.5	1.5	2
Combination	Faucet	2	2	3
Kitchen sink	Faucet	1.5	1.5	2
Laundry tray	Faucet	2	2	3
Laundry	Faucet	1.5	1.5	2
Pedestal urinal	Flush valve	10	-	10
Restaurant sink	Faucet	3	3	4
Service sink	Faucet	1.5	1.5	2
Shower head	Mixing Valve	3	3	4
Stall or wall urinal	Flush tank	3	-	3
Stall or wall urinal	Flush valve	5	-	5
Water closet	Flush tank	5	-	5
Water closet	Flush valve	10	-	10

* Fixture with both cold and hot water supplies, the weight for maximum separate demands may be considered 75% of total wsfu.

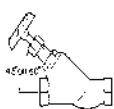
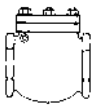
Nominal Pipe or Tube Size (mm)	Smooth Bend Elbows					
	90°	90° Long	90°	45°	45°	180°
	Std*	Rad.**	Street*	Std*	Street*	Std*
						
10	0.43	0.27	0.70	0.21	0.34	0.70
13	0.49	0.31	0.76	0.24	0.40	0.76
19	0.61	0.43	0.98	0.27	0.49	0.98
25	0.79	0.52	1.25	0.40	0.64	1.25
32	1.01	0.70	1.71	0.52	0.92	1.71
38	1.22	0.79	1.92	0.64	1.04	1.92
50	1.53	1.01	2.50	0.79	1.37	2.50
63	1.83	1.25	3.05	0.98	1.59	3.05
75	2.29	1.53	3.66	1.22	1.95	3.66
88	2.75	1.80	4.58	1.43	2.23	4.58
100	3.05	2.04	5.19	1.59	2.59	5.19
125	3.97	2.50	6.41	1.98	3.36	6.41
150	4.88	3.05	7.63	2.41	3.97	7.63
200	6.10	3.97	-	3.05	-	10.07
250	7.63	4.88	-	3.97	-	12.81
300	9.15	5.80	-	4.88	-	15.25
350	10.37	7.02	-	5.49	-	16.78
400	11.59	7.93	-	6.10	-	18.91
450	12.81	8.85	-	7.02	-	21.35
500	15.25	10.07	-	7.93	-	24.71
600	18.30	12.20	-	9.15	-	28.67

* = R/D approximately equal to 1, ** = R/D approximately equal to 1.5

Table 8.O.2(b): Fitting Losses in Equivalent Metre of Pipe - Screwed, Welded, Flanged, Flared and Brazed Connections

Nominal Pipe or Tube Size (mm)	Smooth Bend Tees				Metre Elbows			
	Flow Thru Branch	Straight-Thru Flow			90° Ell	60° Ell	45° Ell	30° Ell
		No Reduction	Reduced ¼	Reduced ½				
								
10	0.82	0.27	0.37	0.43	0.82	0.34	0.18	0.09
13	0.92	0.31	0.43	0.49	0.92	0.40	0.21	0.12
19	1.22	0.43	0.58	0.61	1.22	0.49	0.27	0.15
25	1.53	0.52	0.70	0.79	1.53	0.64	0.31	0.21
32	2.14	0.70	0.95	1.01	2.14	0.92	0.46	0.27
38	2.44	0.79	1.13	1.22	2.44	1.04	0.55	0.34
50	3.05	1.01	1.43	1.53	3.05	1.37	0.70	0.40
63	3.66	1.25	1.71	1.83	3.66	1.59	0.85	0.52
75	4.58	1.53	2.14	2.29	4.58	1.95	0.98	0.61
88	5.49	1.80	2.44	2.75	5.49	2.23	1.22	0.73
100	6.41	2.04	2.75	3.05	6.41	2.59	1.37	0.82
125	7.63	2.50	3.66	3.97	7.63	3.36	1.83	0.98
150	9.15	3.05	4.27	4.88	9.15	3.97	2.14	1.22
200	12.20	3.97	5.49	6.10	12.20	5.19	2.75	1.56
250	15.25	4.88	7.02	7.63	15.25	6.41	3.66	2.20
300	18.30	5.80	7.93	9.15	18.30	7.63	3.97	2.44
350	20.74	7.02	9.15	10.37	20.74	8.85	4.58	2.75
400	23.79	7.93	10.68	11.59	23.79	9.46	5.19	3.05
450	25.93	8.85	12.20	12.81	25.93	11.29	5.80	3.36
500	30.50	10.07	13.42	15.25	30.50	12.51	6.71	3.97
600	35.08	12.20	15.25	18.30	35.08	14.95	7.63	4.88

* = R/D approximately equal to 1, ** = R/D approximately equal to 1.5

Nominal Pipe or Tube Size (mm)	Globe	60° – Y	45° – Y	Angle*	Gate	Swing Check**	Lift Check
							
10	5.19	2.44	1.83	1.83	0.18	1.53	Globe Lift and Vertical Lift: Same as Globe Valve**
13	5.49	2.75	2.14	2.14	0.21	1.83	
19	6.71	3.36	2.75	2.75	0.27	2.44	
25	8.85	4.58	3.66	3.66	0.31	3.05	
32	11.59	6.10	4.58	4.58	0.46	4.27	
38	13.12	7.32	5.49	5.49	0.55	4.88	
50	16.78	9.15	7.32	7.32	0.70	6.10	
63	21.05	10.68	8.85	8.85	0.85	7.63	
75	25.62	13.12	10.68	10.68	0.98	9.15	
88	30.50	15.25	12.51	12.51	1.22	10.68	
100	36.60	17.69	14.34	14.34	1.37	12.20	
125	42.70	21.66	17.69	17.69	1.83	15.25	
150	51.85	26.84	21.35	21.35	2.14	18.30	Angle Lift: Same as Angle Valve
200	67.10	35.08	25.93	25.93	2.75	24.40	
250	85.40	44.23	32.03	32.03	3.66	30.50	
300	97.60	50.33	39.65	39.65	3.97	36.60	
350	109.8	56.43	41.18	41.18	4.58	41.18	
400	125.05	64.05	54.90	54.90	5.19	45.75	
450	140.3	73.20	61.00	61.00	5.80	50.33	
500	158.6	83.88	71.68	71.68	6.71	61/00	
600	186.05	97.60	80.83	80.83	7.63	73.20	

* These losses do not apply to valves with needle point type seat, ** Losses also apply to the in-line, ball type check valve.

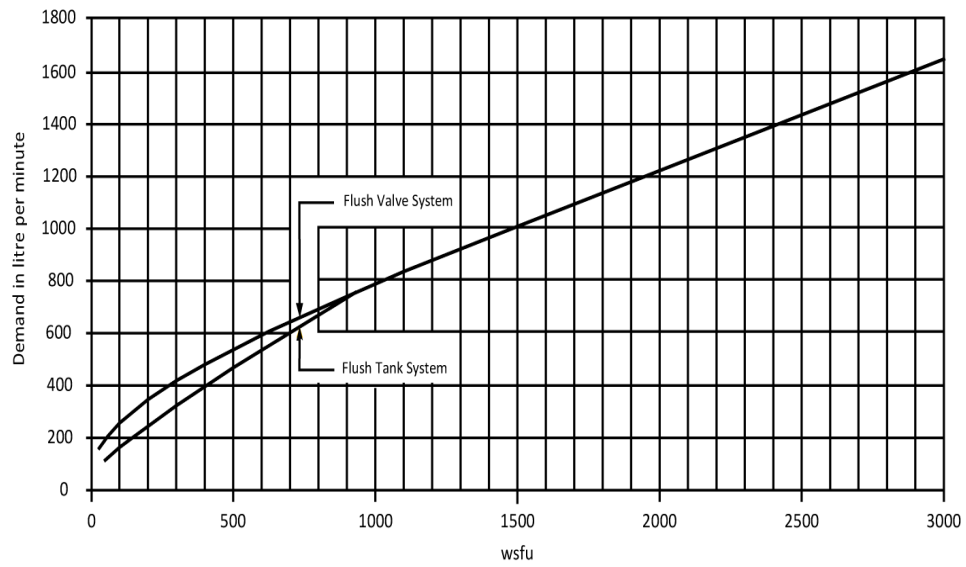


Figure 8.O.1 Water supply demand for various load in water supply fixture units (Wsfu)

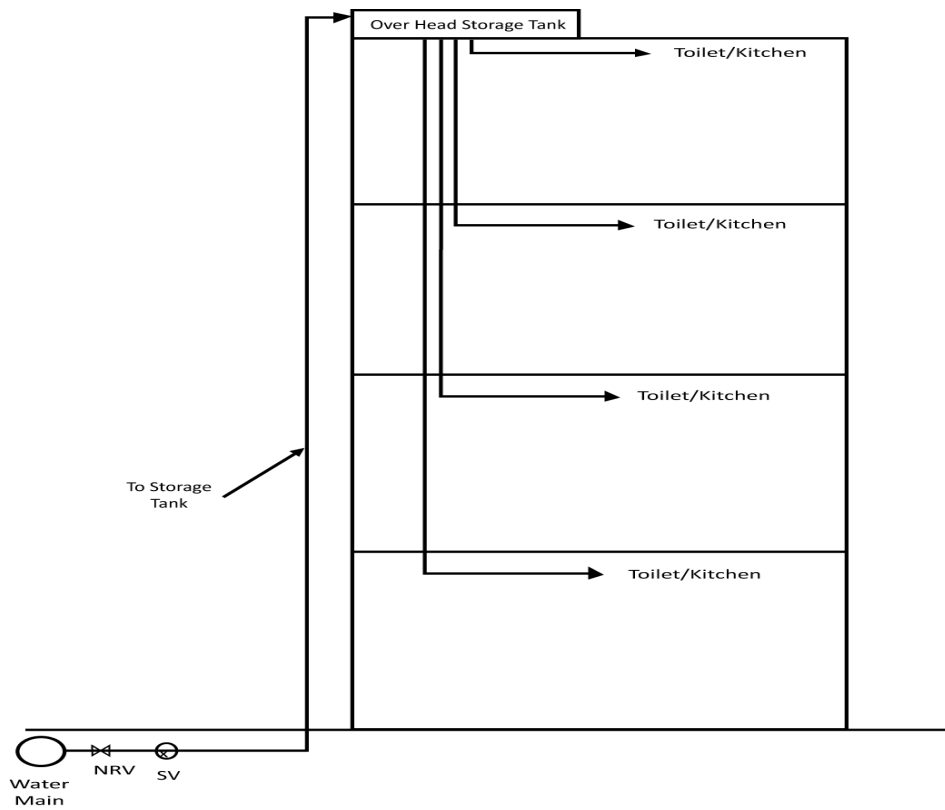


Figure 8.O.2(a) Direct connection of roof storage tank with public water main

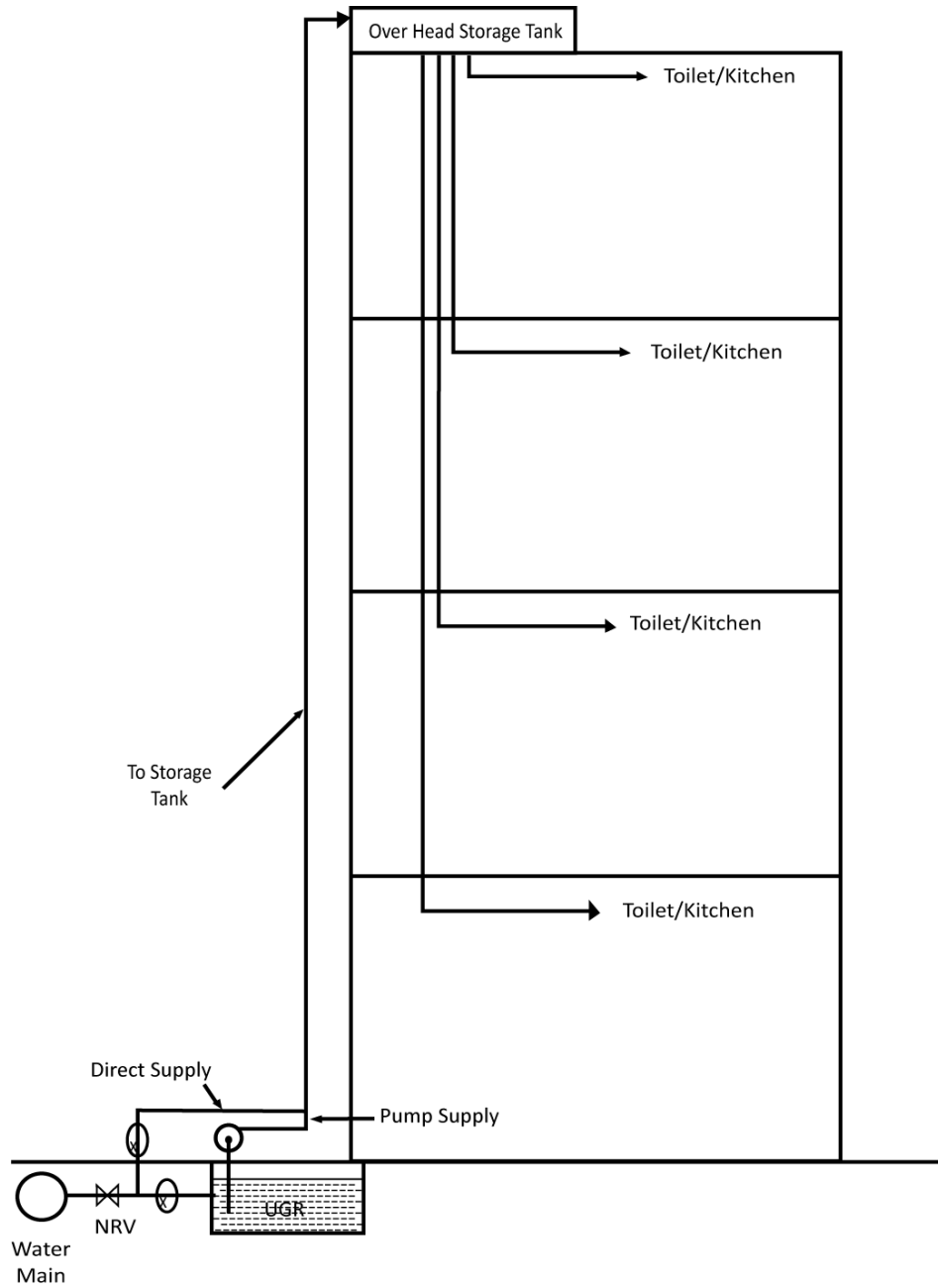


Figure 8.O.2(b) System incorporating balancing roof tank and direct water main connection

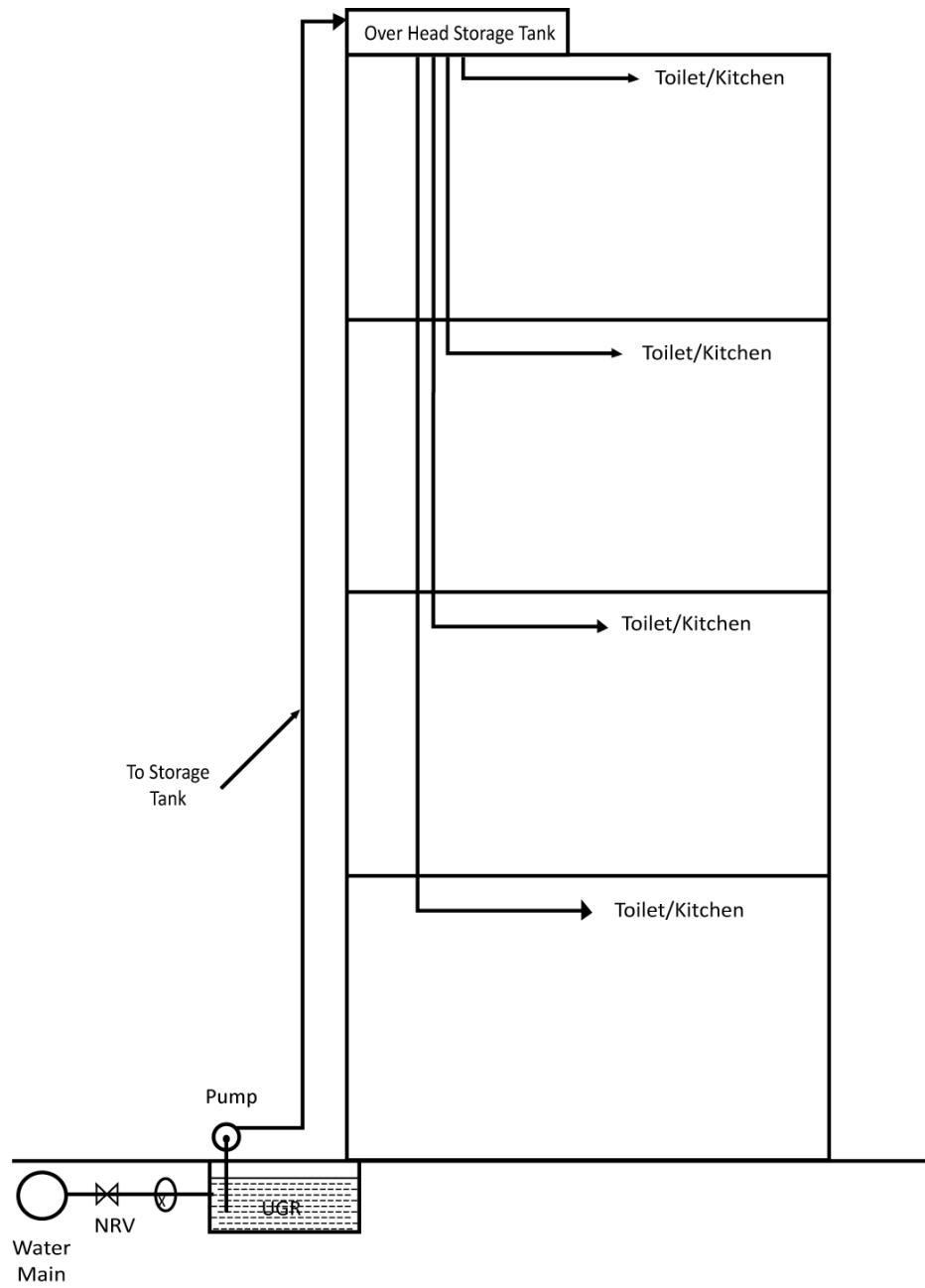


Figure 8.O.2(c) System incorporating underground tank

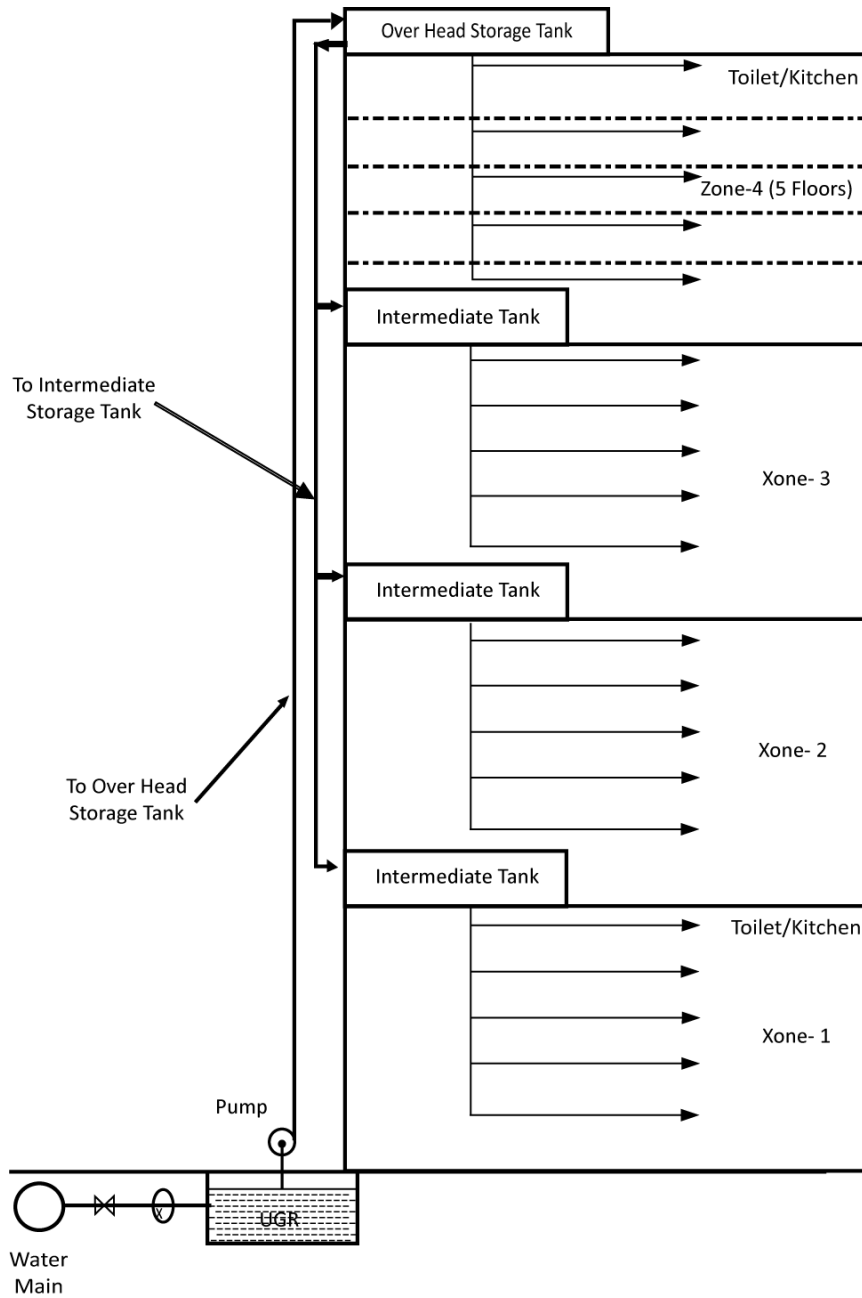


Figure 8.O.3(a) Zoning floors (5) by intermediate tanks supplied by OH storage tank

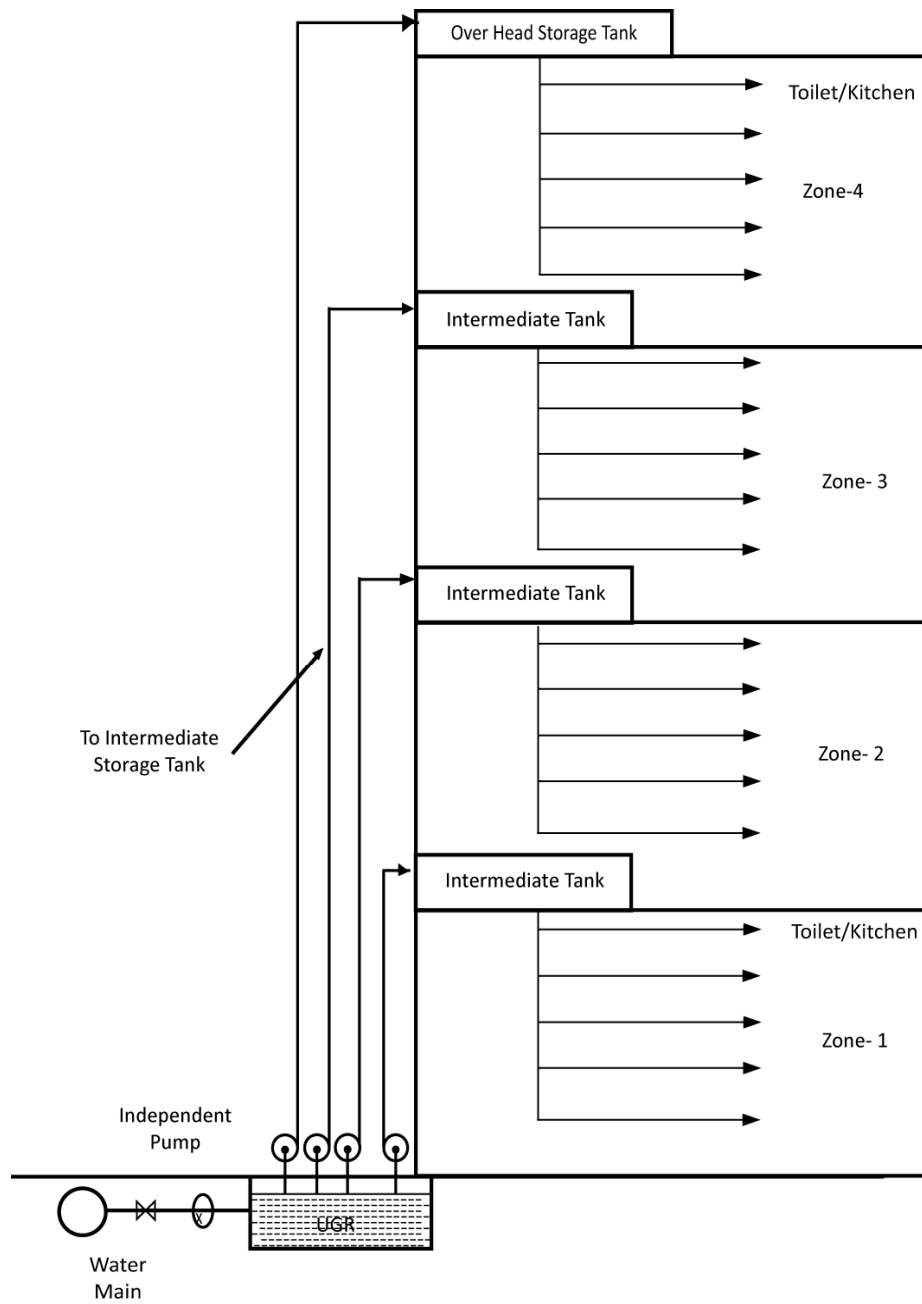


Figure 8.O.3(b) Zoning floors (5) by intermediate tanks supplied by independent pumps

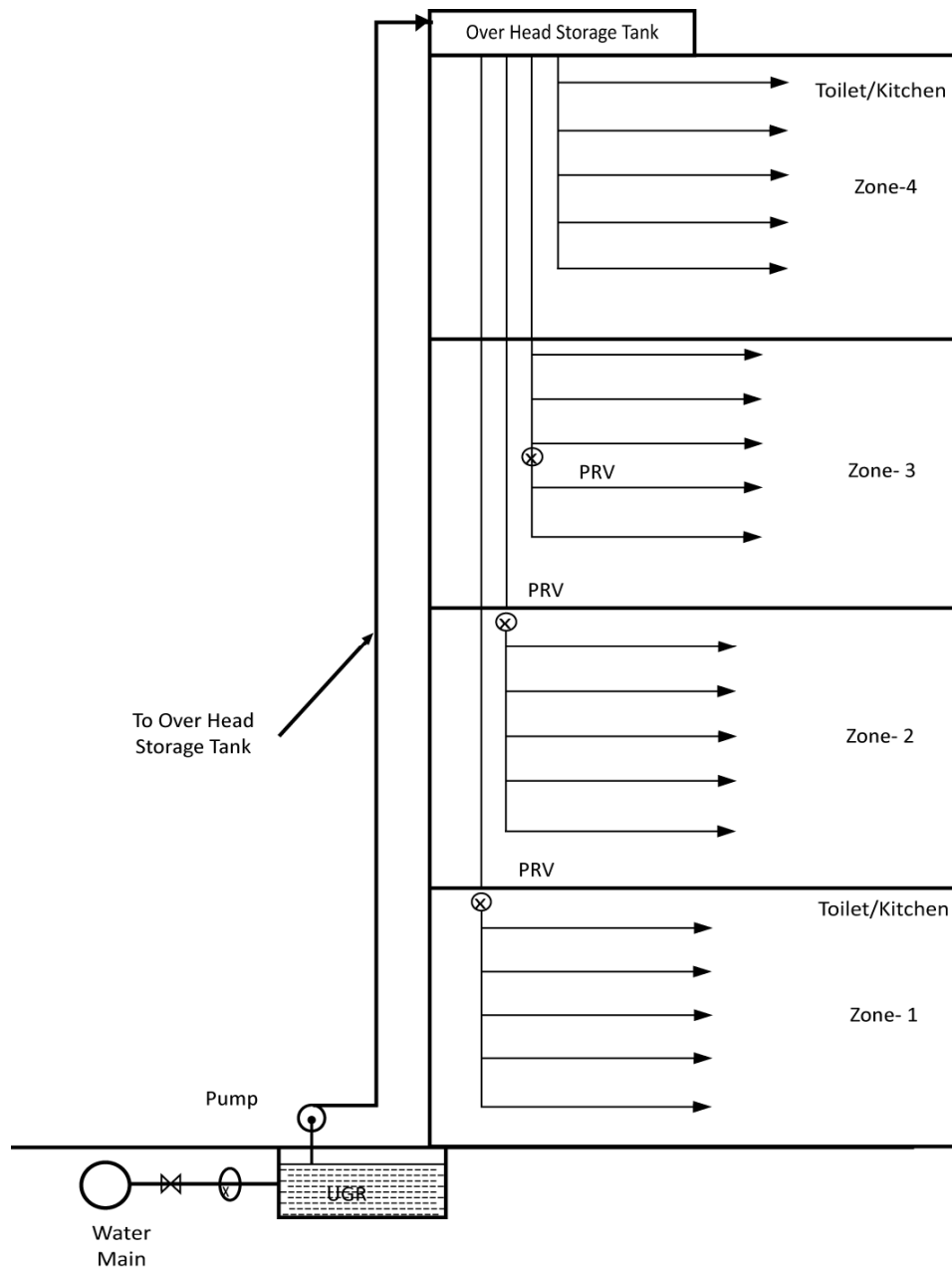


Figure 8.O.3(c) Gravity water supply system with pressure-reducing valves

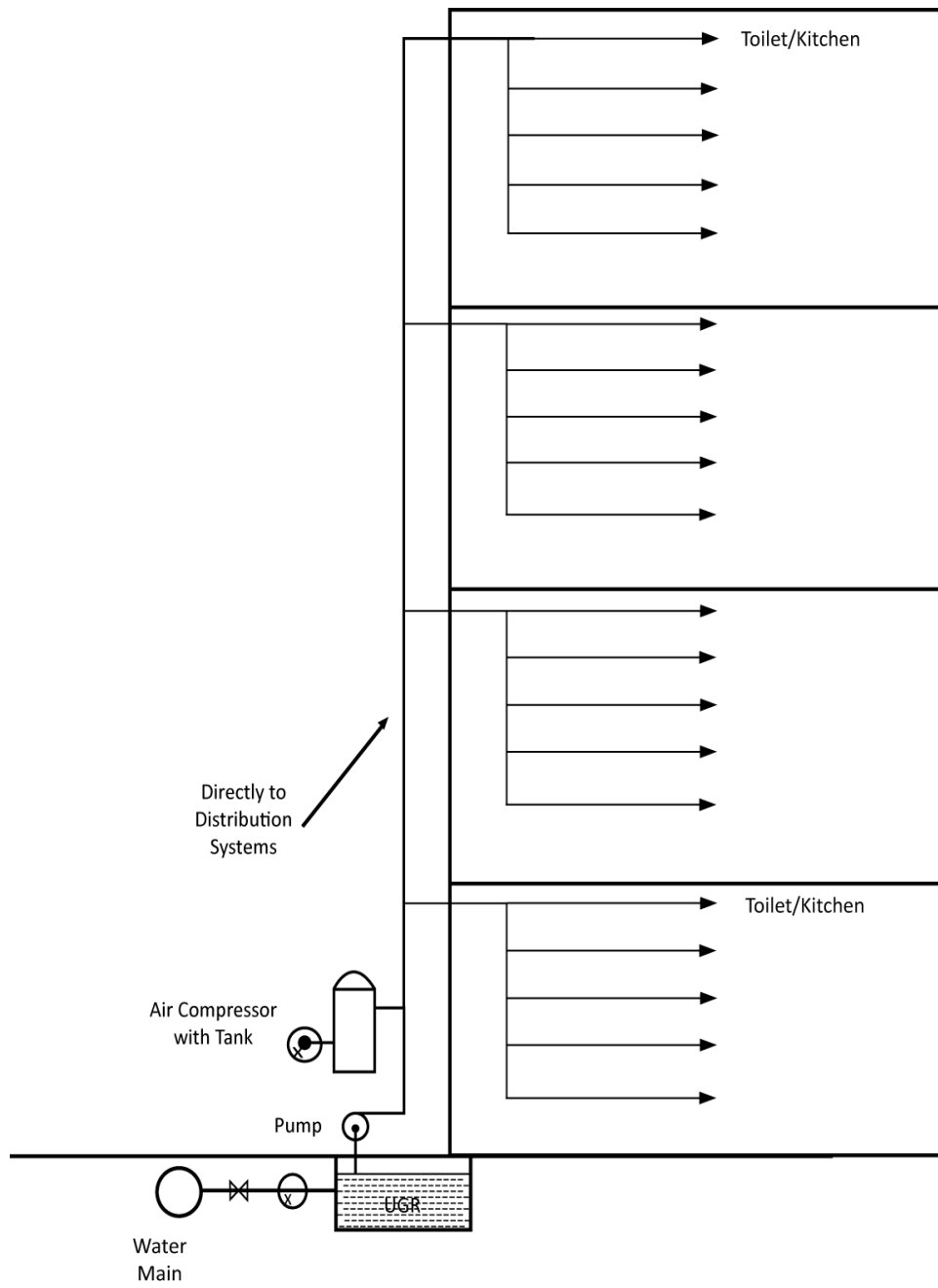
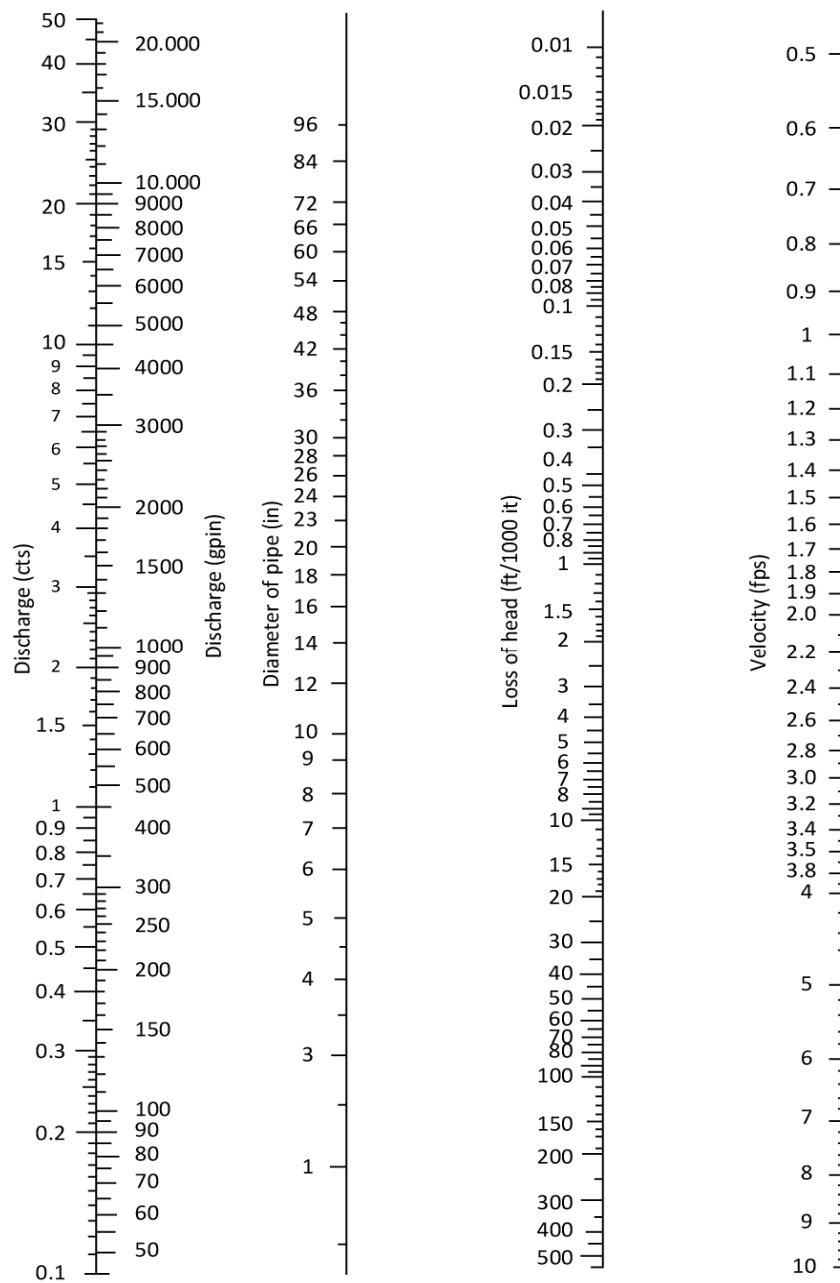


Figure 8.O.3(d) Hydro-pneumatic system of water supply in a building

Figure 8.O.4 Hazen-Williams nomograph with roughness coefficient, $C = 100$

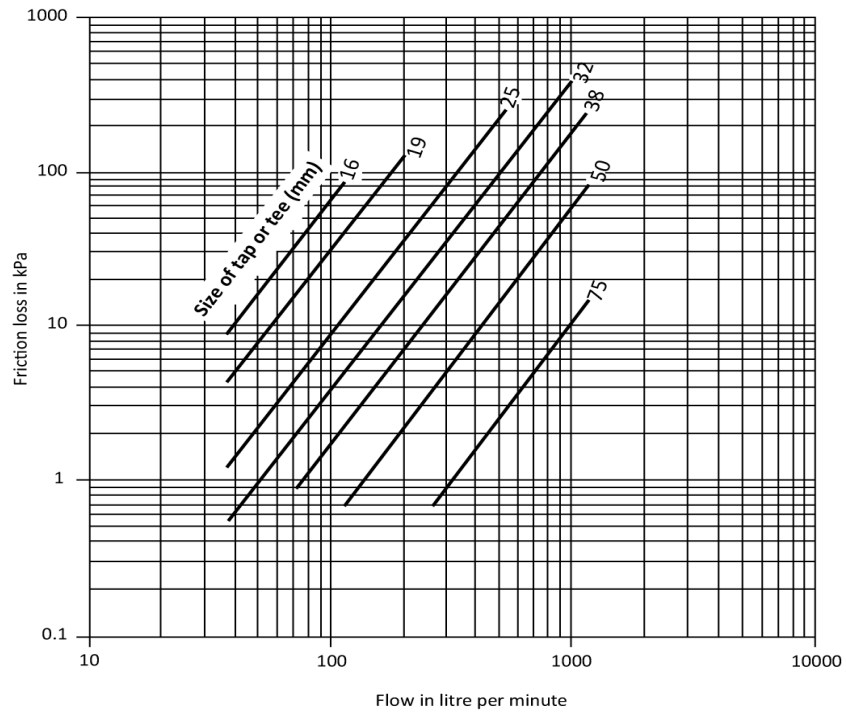


Figure 8.O.5 Friction loss through taps and tees

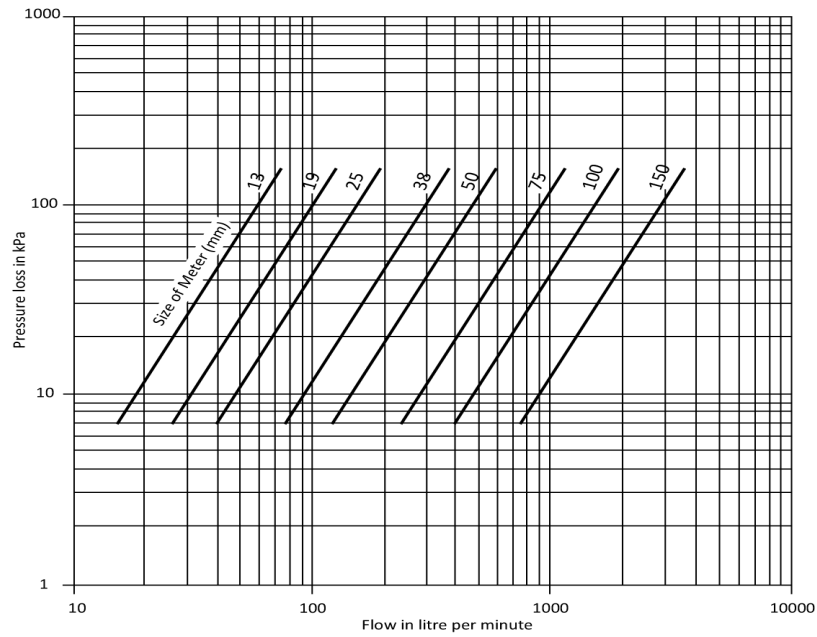


Figure 8.O.6 Loss of pressure through disc - type meter in kPa

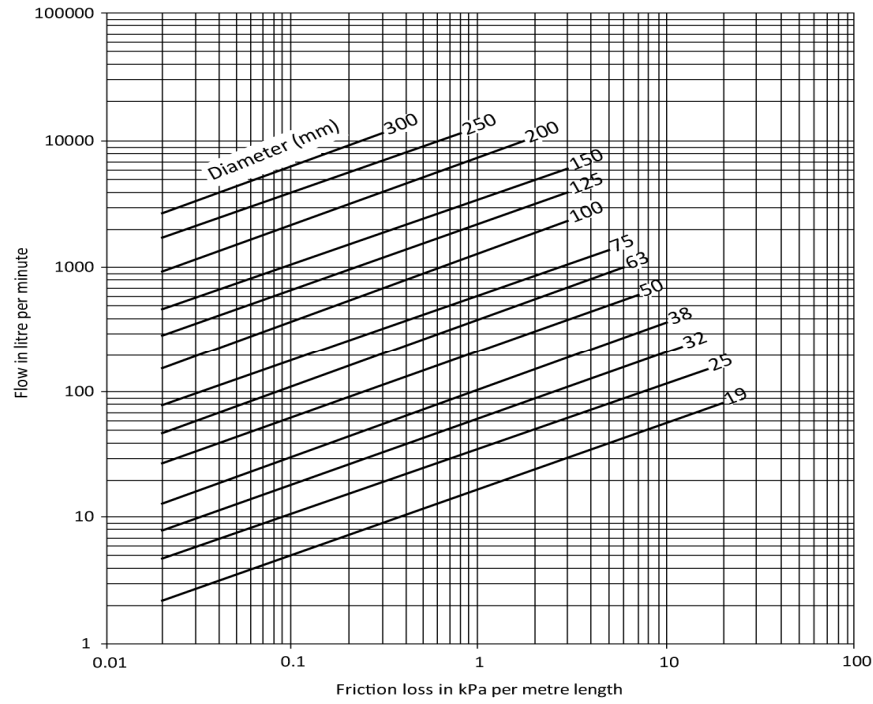


Figure 8.O.7 Friction loss in fairly rough pipe

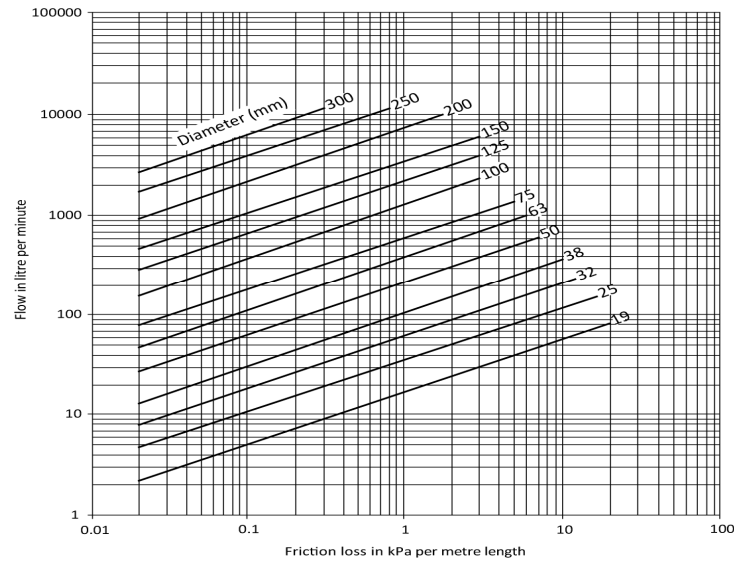


Figure 8.O.8 Friction loss in rough pipe

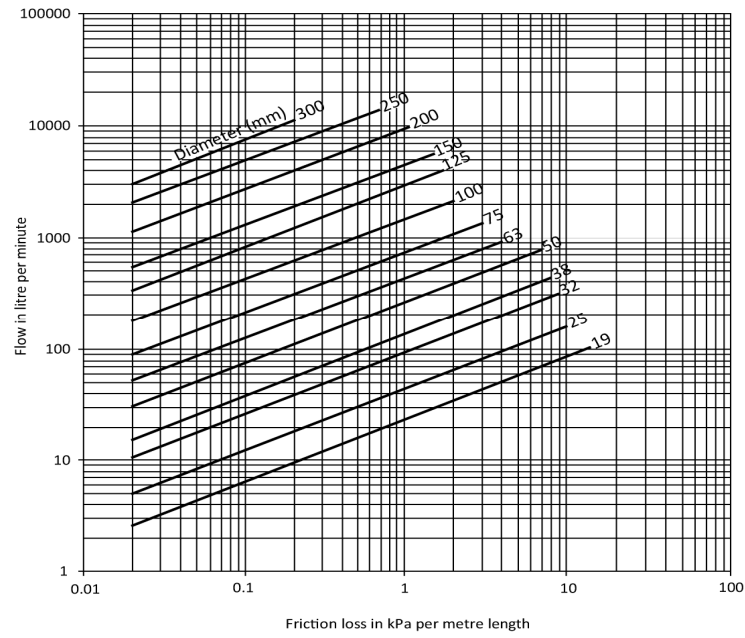


Figure 8.O.9 Friction loss in fairly smooth pipe

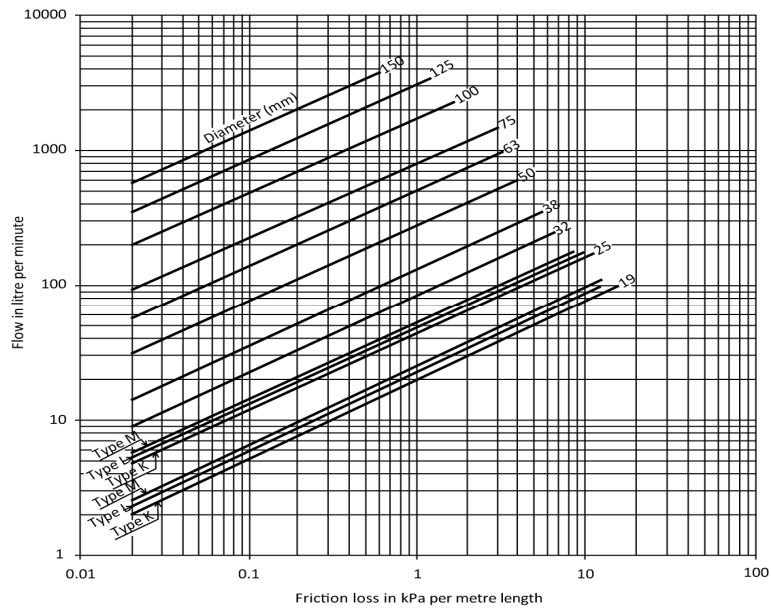


Figure 8.O.10 Friction loss in copper pipe

PART VIII
Appendix P

Recommended Water Quality for Domestic Purposes

Table 8.P.1: Water Quality Standards

Water Quality Parameters	Unit	WHO Guideline Values (2004)	Environmental Conservation Rules (ECR, 1997)
1. pH	Value	6.5 - 8.5	6.5 - 8.5
2. Turbidity	NTU	5	10
3. Color	TCU	15	15
4. Odor	TON	--	Odorless
5. Hardness (as CaCO ₃)	mg/l	500	200-500
6. Iron (Fe)	mg/l	0.3	0.3-1.0
7. Manganese, Mn	mg/l	0.4*	0.1
8. Sodium, Na	mg/l	200	200
9. Calcium, Ca	mg/l	--	75
10. Magnesium, Mg	mg/l	--	30-35
11. Potassium, K	mg/l	--	12
12. Chloride, Cl	mg/l	250	150-600
13. Nitrate, NO ₃	mg/l	50	10
14. Ammonia-Nitrogen, NH ₃ -N	mg/l	1.5	0.5
15. Sulfate, SO ₄	mg/l	250	400
16. Phosphate, PO ₄	mg/l	--	6
17. Fluoride, F	mg/l	0.6 -1.5	1
18. Mercury, Hg	mg/l	0.001	0.001
19. Arsenic, As	mg/l	0.01	0.05
20. Lead, Pb	mg/l	0.01	0.05
21. Chromium, Cr	mg/l	0.05	0.05
22. Cadmium, Cd	mg/l	0.003	0.005
23. Boron, B	mg/l	0.5	1.0
24. Aluminum, Al	mg/l	0.2	0.2
25. Nickel, Ni	mg/l	0.02	0.1
26. Copper, Cu	mg/l	2.0	1.0
27. Total Dissolved Solids, TDS	mg/l	1000	1000
28. Total Suspended Solids, SS	mg/l	--	10
29. Total Coliform, TC	CFU	0	0
30. Faecal Coliform, FC	CFU	0	0

* Health Risk Guideline Value

PART VIII
Appendix Q**Application for Permit to Construct Drainage and Sanitation System**

1. Occupancy Classification: _____
2. Number of Storeys: _____
3. Location: _____
4. This application is accompanied by all required plans, drawings (showing details of materials, sizes, gradient and location of pipes and fixtures) and other details as specified in Sections 6.4.2, 6.4.3, 6.4.4 and 7.5.2 of Part 8 of the Bangladesh National Building Code.

Signature of the Plumber_____
Signature of the Owner or his/
her appointed person

Name and Address _____

Name and Address _____

PART VIII Appendix R One-Hour Rainfall

One hour rainfall values for a 25-year return period for various locations in Bangladesh may be taken from Figure 8.R.1. The figure shows different regions of equal rainfall intensity as well as isohyets at 5 mm intervals.

Rainfall for a particular location shall be obtained as follows:

- (a) When the location lies within any region (shown shaded in the map), the value marked for that region shall be taken.
- (b) For a location lying on any isohyet in this map, the value of that isohyet shall be taken.
- (c) For a location lying outside the positions (a) and (b) above, linear interpolation shall be made between the adjacent isohyets to obtain the required rainfall value.

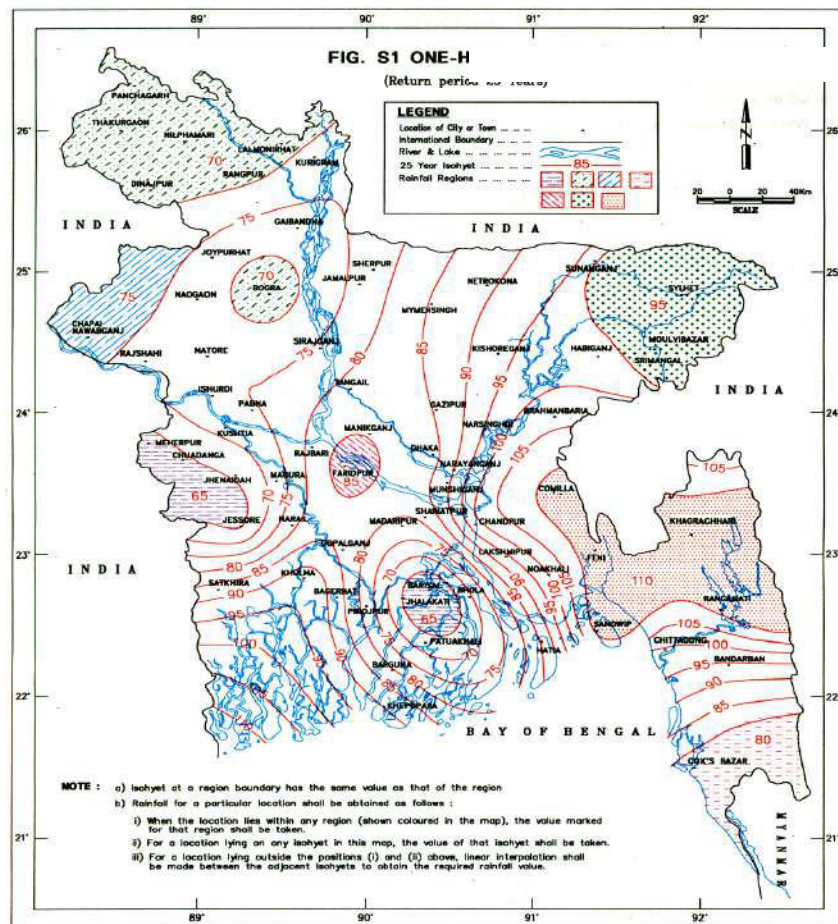


Figure 8.R.1 One-hour rainfall (in mm) map for Bangladesh

PART VIII
Appendix S
Design Guideline of a Septic Tank

The volume of a septic tank may be computed using the following equation:

$$V = PQ + PSY \quad (8.S.1)$$

Where,

V = Volume of the septic tank (litre)

P = Number of persons served

Q = Flow, litre/capita/day (*lpcd*)

The flow may be computed considering waste water flow 60% to 70% of the water consumption or on the basis of the plumbing fixtures discharging simultaneously into the septic tank, Sec 6.9.12.7. In absence of these data the waste water flow for the Occupancy groups A, C and D may be considered 120 *lpcd* for cities, 50 *lpcd* for district town and 20 *lpcd* for thanas and rural areas. For other Occupancy groups a waste water flow of 10 *lpcd* may be considered.

t = Liquid retention time, day (minimum 1 day Sec 6.9.12.11)

S = Volume required for sludge and scum (0.04 m³/capita/year, Sec 6.9.12.10)

v = Desludging frequency, year (Minimum 1 year, Sec 6.9.12.12).

If computed volume is less than 2000 litres then minimum volume of the septic tank shall be 2000 litre (Sec 6.9.12.8).

Table 8.S.1: Reduction factors for various occupancies.

Type of Development		Factor
(i) Church:	Hall-Seating capacity	0.01
	Residential Staff	1.00
(ii) School:	Primary & Secondary School-student and staff population per session	0.25
(iii) School:	Institution of Higher Learning-Total enrolment and staff	0.25
(iv) Community Centre:	Staff/worker	0.25
	Resident	1.00
	Kindergarten-student and staff population per session	0.08
	Maximum gathering	0.01

Type of Development		Factor
(v) Factory:	One shift per 24 hours	0.50
	Two/three shifts per 24 hours	1.00
(vi) Market:	Hawker /employee	0.25
(vii) Hawker/Cooked Food center:	No. of stall	1.75
(viii) Cinema:	Hall-seating capacity	0.03
	Staff	0.50
(ix) Bar & Snack Bar, Coffee- House, Restaurant:	Seating capacity	0.20
(x) Hotel:	No. of beds	0.70
	Staff/worker	0.50
(xi) Office:	No. of staff	0.50

PART VIII
Appendix T

Completion Certificate (Drainage and Sanitation Works)

Ref. No. of the Authority's permit: _____ Date: _____

This is to certify that I/We have completed the drainage and sanitation system for the building and the premises at: _____

Detailed description of the work: _____

This may be inspected, tested and approved.

Signature of the Plumber

Name and Address: _____

The Authority's Certificate

This is to certify that the above drainage and sanitation system has been completed in accordance with the provisions of the Bangladesh National Building Code. The drainage connection to the main sewer (if any) may now be made _____.

Signature of the Building Official or his

Authorized deputy

Seal

PART VIII
Appendix U

Determining Catchments Area for a Flat Surface

Figure 8.U.1 shows a building having flat roofs at different levels. In level 3 the area ABCD is the catchment area contributing to the rainwater down pipe RDP1.

In level 2 horizontal area EFGH and 50% of the vertical wall surface area AFHM, projecting above is the catchment area of rainwater down pipe RDP2.

For the rainwater down pipe RDP3 the catchment area will be the terrace at level 2, 3 and fifty percent of the adjacent contributory wall AFHM, GHIJ and MJLC.

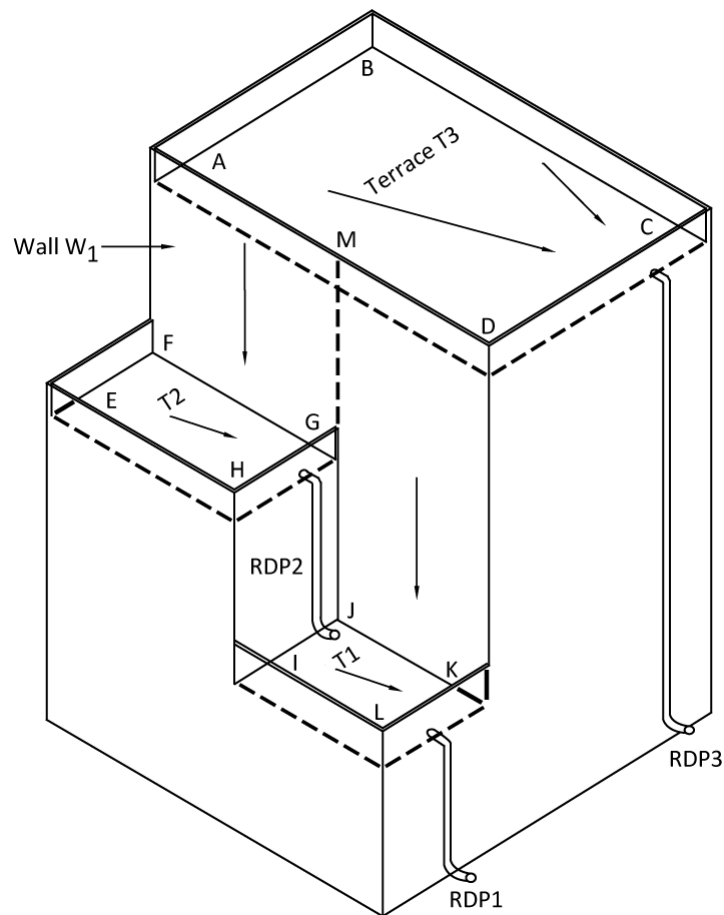


Figure U.1 Catchment area of flat roof and surface wall

PART VIII
Appendix V
Work on the Gas Supply System

This Appendix applies only to work on gas supply systems ahead of the outlet of the meter set assembly, or of the service regulator when there is no meter.

(a) Serving Gas Supplier's Main

No person, unless in the employ of or authorized by the gas supply company shall open or make connections with gas main.

(b) Serving Gas Piping

No person, unless in the employ of or authorized by the gas supply company, shall repair, alter, open or make connections to the services gas piping or do any other work on the parts of the gas supply system up to the meter set assembly or the service regulator when there is no meter.

(c) Meter or Service Regulator When a Meter is not provided

No person, unless in the employ of or authorized by the gas supply company, shall disconnect the inlet of the gas meter or service regulator when there is no meter, nor move such meter or regulator. A gas fitter may disconnect the outlet of such a meter or regulator from the house piping only when necessary. He shall make the joint at the meter or service regulator outlet when there is no meter, carefully replacing all insulating fittings or insulating parts of such fittings, and shall leave the gas turned off at the meter or regulator unless the gas supply company's rules require or allow deviation from this procedure

(d) Notify Gas Supply Company of any Repairs Needed

In case any work done by a gas fitter reveals the need for repairs or alterations on any part of the gas supply system, the gas supply company shall be notified promptly of this fact.

(e) Notify Gas Supply Company of any Leakage

If gas is leaking from any part of the gas supply system, a gas fitter or plumber not in the employ of the gas supply company may make necessary repairs and shall promptly notify the gas supply company.

PART VIII
Appendix W
Documentation for Piping Installation

The gas supply company requires the following to be fulfilled by the consumer(s) for having gas supply from its distribution piping system:

- (a) Application for gas connection in prescribed form along with the approved plan of the building where the gas is to be used. If the building is not approved by the Authority its plan is to be prepared by an approved contractor of the gas supply company.
- (b) An approved contractor of the gas supply company shall prepare the plan of the gas piping system for the building. The plan is to include plan and elevation of the proposed piping system. The plan so prepared shall be submitted to the gas supply company.
- (c) Installation of the piping system as approved by the gas supply company shall be carried out by an approved contractor of the gas supply company.
- (d) Completion report of the installation of the piping system (using the approved drawing) along with the pressure and leak tests by the approved contractor shall be submitted to the gas supply company. The pressure test is to be witnessed by the approved official of the gas supply company.
- (e) Completion report mentioned in (d) above is to include the papers related to the permission from the Authority for digging/cutting the road for taking the connection from the main supply line lying under the road, needed.
- (f) The legal owner of the building shall sign an agreement with the gas supply company using the prescribed agreement document of the gas supply company prior to having the gas supplied to his/her premises.

Chapter 1

Applicability and Implementation

1.1 General

The provisions of this Part are intended to maintain or increase the current degree of public safety as well as health and general welfare in existing buildings while permitting alteration, addition to or change of use. In addition, certain environmental measures to make the existing building stock sustainable, is also a requirement that needs to be addressed to make buildings more energy efficient and environmentally responsive. This concept is introduced in Chapter 2 of this Part.

1.2 Applicability

1.2.1 General

The provision of this Part shall apply to existing buildings that will continue to be or are proposed to be in occupancy groups A, B, C, D, E, F, G, H, I, J, K, L and M. The provisions shall not apply to buildings of historical or architectural value identified and classified by designated authorities. For buildings of historical or architectural value buildings, the provisions of Sec 1.5: Historic or Architecturally Valuable Buildings, of Part 1 Sec 1.16 Buildings and Places of Historical or Architectural Value, of Part 3 and Part 9 Chapter 3 shall be applicable.

Certain provisions to existing buildings and land-use even when additions, alterations or amendments not proposed are included in Sec 2.3 of this Part.

1.2.2 Change in Use

1.2.2.1 No change in use of any existing building shall be made without permission from the permitting authority.

1.2.2.2 No change in use of an existing building shall be allowed if the proposed use is not in conformity with land use pattern as referred in Sec 1.3 Part 3 and within the permitted occupancy classes of Sec 1.4 Part 3 as may be determined by the city or area development authorities having jurisdictions.

1.2.2.3 Where an existing building is changed to a new use group classification, the provisions for the new use group in this Code shall be used to determine compliance.

1.2.3 Part Change in Use

1.2.3.1 No change in use of any part of an existing building shall be made without permission from the permitting authority.

1.2.3.2 Where a portion of the building is changed to a new use group classification, and that portion is separated from the remainder of the building with fire separation assemblies having a fire resistance rating as required by Part 4, Fire Protection, for the separate uses, the portion changed shall be made to conform to the provisions of this Code.

1.2.3.3 Where a portion of the building is changed to a new use group classification, and that portion is not separated from the remainder of the building with fire separation assemblies having a fire resistance rating as required by Part 4, Fire, for the separate uses, the provisions of this Code which apply to each use shall apply to the entire building. Where there are conflicting provisions, those requirements which secure the greater public safety shall apply to the entire building or structure.

1.2.4 Additions

1.2.4.1 No addition to any existing buildings shall be made without permission from the permitting authority.

1.2.4.2 Additions to existing buildings shall comply with all the requirements of this Code for new constructions as set forth in Part 3 of this Code and shall comply with fire requirements set forth in Part 4 of this Code.

1.2.4.3 The combined height and area of the existing buildings and new additions shall not exceed the height and open space requirements for new buildings specified in Part 3 of this Code.

1.2.4.4 Where a separating wall that complies with Part 4, Fire Protection, is provided between the addition and the existing building, the addition shall be considered as a separate building.

1.2.5 Alterations

1.2.5.1 An existing building or portion thereof which does not comply with the requirements of this Code for new construction (Part 3) shall not be altered in such a manner that results in the building being less safe or sanitary than such building is at present.

1.2.5.2 If, in the alteration the present level of safety or sanitation is to be reduced, the portion altered shall conform to the requirements of this Code.

1.2.6 Removal

1.2.6.1 Any construction within the site which does not have approval of the appropriate authority must be removed before any new addition, alteration or change of use is carried out. All other types of existing construction and their changes shall comply with Sections 1.2.4 and 1.2.5 of this Part.

1.2.6.2 Before demolishing a building of construction Type 1 Part 4 Fire Protection the owner shall notify all utilities having service connections within the building or plot, such as water, gas, electricity, sewer and other connections.

1.2.6.3 A permit to demolish a building shall not be issued until a clearance is obtained from utilities stating that their respective service connections and appurtenant equipment, such as, meters and regulators have been removed or sealed and plugged in a safe manner.

1.3 Implementation

1.3.1 Investigation and Evaluation

For the proposed works relating to alteration, addition to and change of use, the owner of the building shall cause the existing buildings to be investigated and evaluated by competent professionals in accordance with the provisions of this Code.

1.3.2 Structural Analysis

1.3.2.1 The owner shall have a structural analysis of the existing building carried out by professional experts to determine the adequacy of all structural systems for the proposed alteration, addition or change of use.

1.3.2.2 The existing building together with the addition or alteration shall be capable of supporting the minimum load requirements specified in Part 6: Structural Design of this Code.

1.3.2.3 For Detail Engineering Assessment (DEA) of existing light manufacturing factory including readymade garment factory buildings, minimum live load of manufacturing floor shall be considered as 3 kN/m² except stacking and storage area where the minimum live load shall be 6 kN/m². However, if retrofitting is required live load for retrofitting design shall be taken according to the load requirements specified in Part 6: Structural Design of this Code.

1.3.3 Submittal

The results of the investigation and evaluation as required in Sec 1.3.1 above along with all proposed compliance alternatives shall be submitted to the Building Official for permission to implement addition, alteration or change of use.

1.3.4 Determination of Compliance

The Building Official shall examine all relevant documents, as specified by the authorities, and determine whether the existing building, with the proposed additions, alterations or change of use, complies with the provisions specified in this Code for the occupancy classification and type of construction.

Chapter 2 Evaluation and Compliance

2.1 Evaluation

The evaluation of the existing building with the proposed additions, alterations or change of use, shall take into consideration the planning requirements as well as those relating to public safety and environmental sustainability.

2.1.1 Planning Requirements

2.1.1.1 The permitting authority through building officials shall determine if any provisions of this Code are violated by the proposed change of use, addition or alteration works.

2.1.1.2 The general requirements for buildings in various occupancy classes and types of construction are specified in Part 3 and Part 8 of this Code. These include :

	Item Description	Reference
(a)	Land use classification and permitted use	Sec 1.3 Chapter 1 Part 3
(b)	Requirement of plots	Sec 1.5 Chapter 1 Part 3
(c)	Means of access	Sec 1.7 Chapter 1 Part 3
(d)	Staircase	Sec 1.14.5 Chapter 1 Part 3
(e)	Universal accessibility requirements	Chapter 3 and Appendix D, Part 3
(f)	Open spaces within a plot; minimum separation of buildings in the same plot; road front, side and rear open spaces, Permitted Construction in the Mandatory Open Space	Sec 1.8 Chapter 1 Part 3
(g)	General height and area limitations based on Road Width	Sec 1.9.2 Chapter 1 Part 3
(h)	General height and area limitations based on FAR	Sec 1.9.3 Chapter 1 Part 3
(i)	Off street (on-site) parking spaces	Sec 1.10 Chapter 1 and Appendix F, Part 3
(j)	Street encroachment	Sec 1.11 Chapter 1 Part 3

	Item Description	Reference
(k)	Community open space and amenities for various types of buildings	Sec 1.12 Chapter 1 Part 3
(l)	Ventilation requirements :	
	Natural	Sec 1.19 Chapter 1 Part 3
	Artificial	Chapter 2 Part 8
(m)	Lighting and illumination requirements	Sec 1.19 Chapter 1 Part 3
	Electric lighting	Chapter 1 Part 8
	Day lighting and Supplementary lighting system	Sec 4.5.1 Chapter 4 Part 3
	Electrical and electronic engineering services for buildings	Chapter 1 Part 8
(n)	Sanitation and drainage requirements	Sec 1.19 Chapter 1 Part 3
	Site drainage and run-off coefficient	Sec 4.3.2 Chapter 3 Part 3
	Internal water management	Sec 4.6 Chapter 4 Part 3
	Water Supply	Chapter 5 Part 8
	Sanitary Drainage	Chapter 6, Part 8
(o)	Minimum dimension of habitable and non-habitable parts of the buildings	Sec 1.14.2.2 and Sec 1.19.6, Chapter 1 Part 3

2.1.1.3 The existing building with proposed alteration or addition shall conform to the requirements of new buildings in the proposed occupancy classification. The proposed alteration or addition shall not make the building less sanitary than present.

2.1.1.4 To promote the preservation of historic buildings, open space or unique architectural/ cultural resources, the local permitting authority shall transfer development rights (TDR), as decided by the authority, to owners as compensation, in accordance with Appendix E, Part 3 of this Code.

2.1.2 Safety Requirements

2.1.2.1 Additions or alterations to an existing building or structure is not to be made if such additions or alterations cause the building or structure to be unsafe or more hazardous based on fire safety, life and structural safety and/or environmental degradation.

2.1.2.2 The fire safety requirement shall take into consideration the structural fire resistance, smoke and fire detection, fire protective signaling and fire suppression system feature of the facility and shall satisfy the requirements of various elements for Types 1, 2 and 3 fire resistive buildings.

2.1.2.3 The fire resistance of the wall used for compartmentalization of a building shall not be less than that specified in subsections of Sec 2.4, Part 3 of this Code.

2.1.2.4 Duct penetrations of this wall shall not be permitted. Ferrous or copper piping and conduit shall be allowed to penetrate or pass through the wall if the openings around such piping and conduit are sealed with impervious noncombustible materials sufficiently tight to prevent fire transfer of smoke or combustible gases from one side of the wall to the other side and are so maintained.

2.1.2.5 The fire door between compartments serving as horizontal exit shall be so installed, fitted and provided with gaskets that such fire door will provide a substantial barrier to the passage of smoke.

2.1.2.6 The floor/ceiling shall be of such construction that the fire resistive integrity between stories is maintained.

2.1.2.7 The smoke detection capability within the facility based on the location and operation of automatic fire detectors shall be evaluated with respect to the requirements of Chapters 3 and 4 of Part 4 for various Occupancies.

2.1.2.8 Where a fire protecting alarm and signaling system is provided, the capability of the system shall also be evaluated (Part 4).

2.1.2.9 The ability of the natural or mechanical venting, exhaust or pressurization system to control the movement of smoke from a fire shall be evaluated.

2.1.2.10 The shaft and exit enclosures shall satisfy the fire protection requirements as specified in Part 4 of this Code.

2.1.2.11 The configuration, characteristics and support features for means of egress in the facility including the capacity of and the number of exit routes available to the building occupants shall be evaluated and the adequacy of the means of egress routes leading to a safe area shall be examined. The length of the exit access travel path in which building occupants are confined to a single path of travel shall be evaluated. The length of exit access travel to an approved exit shall also be evaluated with respect to the exit requirements for various occupancies as detailed in Part 4 of this Code.

2.1.2.12 The efficiency and effectiveness of the elevator equipment and controls that are available to the fire department to rescue building occupants from upper floors during a fire when such equipment is installed shall be evaluated.

2.1.2.13 The presence of and reliability of means of egress emergency lighting system shall be evaluated.

2.1.2.14 The ability to suppress fire based on the installation of automatic sprinkler and stand pipe systems shall be evaluated (Chapters 4 and 5, Part 4).

2.1.2.15 The lightning protection of the building shall satisfy the requirements specified in Chapter 1 Part 8.

2.1.3 Egress Requirements

2.1.3.1 Addition to or alteration or changes of use of a building shall not be permitted if such addition or alteration or change of use causes violation of the egress requirements specified in Chapter 3 Part 4.

2.1.3.2 Permissions shall not be granted for any such work that will obstruct or block or hamper the existing means of egress of the building or any other building unless an equivalent and adequate means of egress is provided.

2.1.4 Structural Requirements

2.1.4.1 Additions to, alterations or change of use of an existing building or structure shall not be permitted if they cause an overloading or instability of the structural elements including the foundation.

2.1.4.2 On evaluation, if the structure of the existing building is found in non-conformity with the safety provisions of this code, the building will be declared as 'unsafe building' by the authority.

2.1.4.3 For all buildings identified as an 'unsafe building' the authorities shall identify measures that remove danger to the structure, life or property so that the building can be made safe for occupation.

2.1.4.4 Buildings that cannot be made structurally safe will be declared 'unfit for occupation' and demolished by the authorities.

2.1.5 Environmental Requirements

Additions to, alterations or change of use of an existing building or structure shall not be permitted if they cause environmental degradation and unsustainable development in the locality.

2.2 Compliance

2.2.1.1 When an evaluation is carried out as described above and the existing building with the proposed alteration, addition and/or change of use satisfy the requirement specified for the relevant occupancy classification, the results of the evaluation shall be accepted by the Building Official.

2.2.1.2 The owner shall fully comply with the results of such evaluation for the existing building with the proposed alteration, addition and/or change of use for the relevant occupancy classification and its recommended remedies if relevant as set forth in Sec 2.1.4 above.

2.3 Sustainability Measures for Existing Buildings

2.3.1 When there are no proposals for additions, alterations or change of use, to make existing building stock sustainable, environmentally responsive and energy efficient, the Authority may propose necessary measures after evaluating the condition of the existing buildings. When it is found that environmental degradation is taking place or such is feared by experts, the measures to be proposed may include, but not restricted to the following:

- (a) introduction of renewable energy
- (b) installation of rainwater harvesting
- (c) installation of ground water recharge systems
- (d) introduction of low energy sources

2.3.2 Any sustainability and/or energy efficiency measure suggested shall comply with relevant requirements in Chapter 4 Part 3 of this Code.

Chapter 3 Conservation

3.1 Scope

Historical places, buildings, objects and manifestation of cultural, scientific, symbolic, spiritual and religious value are important expressions of the culture and heritage, identity and religious beliefs of societies. Their role and importance, particularly in the light of the need for cultural identity and continuity in a rapidly changing world, need to be promoted.

Buildings, spaces, places and landscapes charged with historical, cultural, spiritual and religious value represent an important element of stable and humane social life and community pride. Without appropriate restoration / conservation, the architectural evolution in relation to socio cultural concept of a country's heritage remains misinterpreted, and can lead to virtual disappearance.

National legislation and international treaties and regulations aim to strike a balance between the need for development and the need to conserve the environment for the future.

3.2 Terminology and Conceptual Definitions

3.2.1 Conservation

This is the process of retention of existing buildings or groups of buildings, landscapes etc. and taking care not to alter or destroy character or detail, even though repairs or changes may be necessary. Conservation conventionally is concerned to preserve as much original fabric as possible.

3.2.2 Restoration

This is the process of carrying on alterations and repairs to a building with the intention of restoring it to its original form, often involving reinstatement of missing or badly damaged parts, so it usually includes replication. As far as possible, efforts are made to replicate the materials and construction techniques of the original in this endeavor.

While often necessary after a disaster, it is generally regarded as more drastic than conservation, which suggests retention, repair and maintenance.

3.2.3 Area Conservation/Historic Districts

Area conservation includes conservation of any building or portion thereof, that fall under historic districts and/or abutting Heritage Streets as listed by the National Committee for Heritage listing.

3.2.4 Structures of Architectural Significance

Structures of architectural significance include any building or portion thereof with architectural significance, which may not be a historic building, but is designated as of significance for its architectural qualities as designated by the standing committee for Heritage listing.

3.3 Aims of Conservation

3.3.1 Conservation, rehabilitation and culturally sensitive adaptive reuse of urban, rural and architectural heritage shall be in accordance with the sustainable use of natural and human made resources. Access to culture and the cultural dimension of development is of utmost importance, benefiting all the people who have such access.

3.3.2 In order to promote historical and cultural continuity and to encourage broad civic participation in all kinds of cultural activities, the Government at the appropriate levels, including the local authorities, should undertake the following:

- (a) Identify and document, whenever possible, the historical and cultural significance of areas, sites, landscapes, ecosystems, buildings and other objects and manifestations
- (b) Establish conservation goals relevant to the cultural and spiritual development of the society;
- (c) Promote awareness of heritage in order to highlight its value and the need for its conservation and the financial viability of rehabilitation;
- (d) Encourage and support the local heritage and cultural institutions, association and communities in their conservation and rehabilitation efforts and inculcate in children and youth an adequate sense of their heritage;
- (e) Promote adequate financial and legal support for the effective protection of cultural heritage;
- (f) Promote education and training in traditional skills in all disciplines appropriate to the conservation and promotion of heritage.

3.4 Objectives of Conservation

To integrate development with conservation and rehabilitation goals, the Government at appropriate levels, including Ministries, local authorities and municipalities, shall undertake the following:

- (a) Recognizing that historical and cultural heritage is an important asset, strive to maintain the social, cultural and economic viability of historically and culturally important sites and communities.

- (b) Preserve the inherited historical settlement and landscape forms, while protecting the integrity of the historical urban fabric and thereby guiding new construction in historical areas;
- (c) Provide adequate legal and financial support for the implementation of conservation and rehabilitation activities, in particular through adequate training of specialized human resources;
- (d) Promote incentives for such conservation and rehabilitation to public, private and non-profit developers.
- (e) Promote community based action for the conservation, rehabilitation, regeneration and maintenance of neighborhoods.
- (f) Support public and private sector and community partnership for the rehabilitation of inner cities and neighborhoods.
- (g) Ensure the incorporation of environmental concerns in conservation and rehabilitation projects;
- (h) Take measures to reduce acid rain and other types of environmental pollution that damage to buildings and other items of cultural and historical value.
- (i) Adopt human settlement planning policies, including transport and other infrastructure policies that avoid environmental degradation of historical and cultural areas.
- (j) Ensure that the accessibility concerns of people with disabilities are incorporated in conservation and rehabilitation projects.

3.5 Ethics of Conservation

The following standards of conservation ethics shall be rigorously observed in conservation works:

- (a) The condition of the building must be recorded before any intervention.
- (b) Historic evidence must not be destroyed, falsified or removed.
- (c) Any intervention must be the minimum necessary.
- (d) Any intervention must be governed by unswerving respect for the aesthetic, historical and physical integrity of cultural property.
- (e) All method and material used during treatment must be documented.

3.6 Guidelines for Conservation

Any proposed intervention should:

- (a) be reversible or repeatable, if technically possible, or
- (b) not prejudice a future intervention whenever this may become necessary;
- (c) not hinder the possibility of later access to all evidence incorporated in the objects;
- (d) allow the maximum amount of existing material to be retained;
- (e) be harmonious in color, tone, texture, form and scale, if addition is necessary. But should be less noticeable than original material, while at the same time being identifiable;
- (f) not be undertaken by conservator/restorer who are insufficiently trained or experienced.

3.7 Listing of Heritage Building or Area

3.7.1 Aim and Objectives of Listing

Since a large part of Bangladesh's cultural heritage has so far remained undocumented, preparing an inventory of heritage buildings worthy of conservation is the most important task with which to begin the process of conservation.

This inventory will become National Register of Historic and Architectural Properties. It attempts to create a systematic, accessible and retrievable record of the built heritage of this country. It will serve as resource material for developing heritage conservation policies and regulations.

The primary aim of listing is to document the fast disappearing built heritage and then present it to scholars and the general public in a user-friendly format, which aids conservation by generating public awareness. Once a property/ building is included in such a list, it becomes justifiable to undertake necessary conservation activities by formulating special regulations for its conservation or according it due protection under Town Planning Acts or Detail Area Plan. Ideally, the footprints of all listed buildings along with the guidelines referred in Sec 3.8.2 below should be included in the Master Plan documents of all cities.

3.7.2 Listing and Protection of Buildings or Area of Historic Value

All historic buildings and places identified, listed and classified so by the appropriate agency of the Government under the Act of Antiquity shall be deemed to be protected. Any change of use, repair, alteration or extension of such buildings shall be in compliance with the requirements of this Code as referred in Sec 1.18 Chapter 1 Part 3, and with those of the Department of Archaeology of the Government.

3.7.3 Listing of Projects or Work of Architectural Value

Buildings and works under the jurisdiction of and identified by the Authority as having architectural value shall be deemed protected. The owner of any such building may also apply to the Building Official for enlistment as a building with architectural value. To be so identified, a building shall have been in existence for at least 20 years from the date of its completion. To determine whether the building or work is architecturally valuable, the Authority shall appoint a standing committee comprising noted experts from the fields of Architecture, Planning, Engineering, History, Art, Literature or any other discipline which may be deemed relevant. The committee shall identify a building as architecturally valuable if, in their judgment, the building possesses distinctive architectural features, has cultural or symbolic value, has become part of the heritage, or bears testimony of some historical event. In addition to satisfying the requirements of Sec 1.18.2 Chapter 1 Part 3 any proposed repair, alteration or addition to such buildings must also have the approval of the standing committee who shall have to be satisfied that the proposed changes will not impair the aesthetic quality and architectural character of the building.

3.7.4 Publication and Web Notification

Periodically, this database of listed heritage should be made more comprehensive and the information compiled should be available in published form and online. It should also be made compatible with similar registers of other countries to facilitate international research. The list of gazetted historic buildings shall be available on the websites of all concerned departments or development authorities or in the Government website on heritage conservation.

3.7.5 Selection Criteria for Listing

Conservation must preserve and if possible enhance the message and values of cultural properties. These values help systematically to set overall priorities that will determine whether a property is worthy of listing. The values assigned to cultural property come under three major headings:

A. Cultural values	B. Use values	C. Emotional Values
• Documentary • Historic • Archaeological age and scarcity • Aesthetic • Architectural • Townscape, landscape and ecological • Technological and scientific.	• Functional • Economic • Social • Political and ethnic.	• Wonder • Identity • Continuity • Spiritual and symbolic.

After analyzing these values they should be condensed into a statement of the significance of the cultural property, whether or not the property is worthy for listing.

3.7.6 Detailed Format for all Listed Properties

Information for each building or site should be recorded as per format as described below.

- (a) Each proforma shall contain information about persons engaged in the process of listing and review. Listing shall be carried out by or under the supervision of experienced conservation architects.
- (b) At least one photograph of the property/building shall be recorded for identification purposes. All significant elements of the property shall also be photographed. All photographs should be properly catalogued.
- (c) A conceptual plan (if available, a measured drawing) should be given for each building/area to be listed.
- (d) Any additional information related to or affecting the built heritage of the city/town/region documented and its conservation should be included as appendices, for example: laws and regulations on planning and conservation, etc.

3.7.7 Grading of Historic Buildings

The primary objective of listing is to record extant architectural heritage and sites. But the outcome of this process should invariably be to grade the listed heritage into a hierarchical series. This process must be undertaken in a rigorous and transparent manner by a multi-disciplinary team of experts whose recommendations should be available for public scrutiny. The importance of this process determines the basis for subsequent conservation decisions. Such hierarchical categorization facilitates the prioritization of decisions relating to the future of architectural heritage and sites.

The listed historic buildings and sites may further be classified in descending order of importance into the following categories:

- (i) Grade I and I* – Buildings of outstanding merits, in which every effort should be made to preserve. (Buildings and sites classified as Grade I* are already listed as per Antiquity Act)
- (ii) Grade II – Buildings of special merits; in which efforts should be made to preserve selectively.
- (iii) Grade III – Buildings of some merit; in which preservation in some form would be desirable and alternative means could be considered if full preservation is not practicable.

Grade I and II should be conserved in accordance with the provisions of official and legal manuals of practice (for example, Department of Archaeology's Works Manual). Endorsed grading may be subject to review and revision. The grading system is an internal administrative mechanism carrying no legal effect.

3.7.8 Upgrading of the List

The process of listing should be constantly upgraded and the updated list should maintain the availability of fresh information of the listed building or site.

3.8 Implementation Strategy

3.8.1 Community Participation

In order to ensure community participation in the conservation process, scaled drawings of the proposed conservation shall be appended to the site during construction so that the public may become accustomed to the proposal and have the option of registering their opinion regarding the same.

If necessary the proposals shall be amended if strong public opinion is found against the proposal for conservation / restoration.

3.8.2 Protection of the Building or Site

3.8.2.1 A clear space around each monument shall be formed as an immediate environmental protection to the monument. This area should follow, as the case may be, the original line of the enclosure wall.

3.8.2.2 In rural or suburban setting no new structures/built forms of any size, shape shall be allowed to develop within the UNESCO/UNDP suggested area of half a mile radius from the epicenter of the monument under consideration.

3.8.2.3 In rural or suburban settings, a parcel of land representing approximately a circular area of a half mile radius of the farmland immediately surrounding the heritage building/monument shall be acquired to protect it from encroachment

3.8.2.4 In urban areas or metropolitan cities the radius of this protected zone shall be 250 m from the heritage building or area.

3.8.2.5 In both rural and urban areas, an immediate buffer zone shall beset by the permitting authorities, but not less than 9 m in width, adjacent to the heritage building or area, must be kept absolutely free of any structure or establishment surrounding the heritage building or its part thereof.

3.8.2.6 In general, authority having jurisdiction for planning and development, shall guide the height of any proposed buildings within the protected zone, depending upon the historic and architectural value of the building or area.

3.8.2.7 To protect the monument against human impact, movement of visitors within the structure and site shall be controlled. Climbing over the fabric of the monument shall be illegal and be prevented at all times.

3.8.2.8 Signage shall be installed within the site and building premises to guide visitors by creating circumambulatory path or circulation area around each of the heritage buildings/monuments.

3.8.3 Original Elements

3.8.3.1 Structures that impinge directly on the monument/s, identified to be additions, alterations or extensions of any form or kind on to the original monument/s, shall be removed/relocated elsewhere, in an attempt to recognize the original form/feature of the monument.

3.8.3.2 All original structures and architectural elements are to be retained and restored.

3.8.3.3 In the event that such elements have to be repaired, their features are to be retained intact.

3.8.4 Land Use, Building Height, Setback and Ground Coverage in the Listed Zone

3.8.4.1 For area conservation, to retain the urban fabric of the area, the height, setback and ground coverage for any new development within the listed zone shall comply with the height, setback and ground coverage of the heritage buildings or area to be conserved.

3.8.4.2 For all new developments in area conservation and for the conservation of any listed building or monument the guidelines of Sec 1.17 of Chapter 1 Part 3 shall be applicable.

3.8.5 Upgrading and Adaptive Reuse

3.8.5.1 Adaptive reuse shall follow logical consequence considering public requirements of the region, tourist influx and socio-cultural context of the region under influence.

3.8.5.2 The decision of including different functions shall be made only after thorough investigation by a committee working under the guideline of experts in archaeology and architecture of the region.

3.9 Incentives to Private Owners Affected by Conservation

3.9.1 Listed historic buildings, sites and areas are usually under huge pressure from competing economic activities, and hence are giving rise to sharp land values. Thus listed properties become easy target for demolition or replacement by more use-intensive and denser and taller structures, with apparently better economic return. Hence to safeguard those threaten landowner, special financial incentive should be offered to the stakeholders to preserve our national and universal outstanding values. Considering them as affected people, Transfer of Development Right's (TDR) certificate shall be introduced to protect these heritages.

3.9.2 Transfer of Development Rights (TDR) is a market-based mechanism that promotes responsible growth, while conserving culturally and environmentally sensitive sites and areas. Transfer of Development Rights (TDR) means making available certain amount of additional built up area in lieu of the area relinquished or surrendered by the owner of the land, so that he can use extra built up area either himself or transfer it to another in need of the extra built up area for an agreed value. It is designed to steer growth not to limit or stop development.

3.9.3 Local Governments may use TDR to direct development in specific zones. They may use this tool to preserve heritage areas etc. by allowing its owners to give up their rights to develop these spaces commercially in lieu of TDR. The affected land owner gets the benefit as private sector funds are used to purchase the development rights and build more in areas as per permission of the authority. The Government also gets the benefit by avoiding large public expenditures to protect historically and environmentally sensitive areas and development occurs in suitable areas, resulting in more efficient public services.

3.9.4 In implementing TDR, a Development Rights Certificate (DRC) is a certificate shall be issued by the Authority having jurisdiction to the affected owner of a property, when such property gets listed for conservation or gets affected by falling within the protected zone as mentioned in Sec 3.8.4 where his/her property (either partly or wholly) is reserved for the purpose of public utilities or conservation.

3.9.5 The DRC certificate shall state clearly the following issues-

- (a) The extra FAR (Floor Area Ratio) credit in square meter of the built-up area and the occupancy type to which the owner or lessee of the listed site/plot or affected site/plot is entitled,
- (b) The place and user zone in which the development rights are earned, and
- (c) The areas or zones in which they may be utilized.

3.9.6 To control the unplanned density, this TDR should be used only in the planned non-residential plots developed by the Government. The property owner may use DRC for himself or transfer to any other person.

Chapter 1 Scope And General

1.1 Purpose and Scope

1.1.1 The purpose of this part of the Code is to provide minimum standards to safeguard life, health, property and public welfare by regulating and controlling the design, location, construction and maintenance of signs and outdoor display structures. The requirements of the Code cover the spatial, structural and fire safety aspects of signs located on, within or outside the building. Official traffic signs erected on roads and highways are not covered by this Code.

1.1.2 The unrestrained placement of advertising signs mars the landscape with hoardings and advertising signs within the cityscape as well as in rural areas. Therefore, all signs and billboards shall be liable to take permit as per guidelines of Sec 1.4 of this Chapter and shall also comply with the guidelines of Chapter 3 Part 10 (see Sections 3.10 to 3.11).

1.1.3 The regulations of this Code are not intended, and shall not be understood to permit violation of the provisions of other ordinances, regulations or official requirements in force at or near railway stations, roads, railways, terminals or other places controlled by public agencies having jurisdiction to formulate such regulations.

1.1.4 No sign shall be erected in a manner that would confuse or obstruct the view or interfere with official signs, signals or devices installed for the purpose of guiding or controlling road, rail, marine or air traffic.

1.2 Terminology

This section provides an alphabetical list of the terms used in and applicable to this part of the Code. In case of any conflict or contradiction between a definition given in this Section and that in Part 1, the meaning specified in this part shall govern for interpretation of the provisions of this Part.

ADVERTISING SIGN	Any surface or structure with characters, letters or illustrations applied thereto and displayed in any manner whatsoever out of doors for purposes of advertising or to give information regarding or to attract the public to any place, person, public performance, article or merchandise whatsoever, and which surface or structure is attached to, forms part of or is connected with any building, or is fixed to a tree or to the ground or to any pole, screen, fence or hoarding or displayed in space.
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ALLEY	Public way having width between 2.5 m and 4 m that has been dedicated to public use.
APPROVED PLASTIC	Plastic materials conforming to the requirement of Sec 2.2.5
DIRECTION SIGN	Usually included with an arrow and used for indicating a change in route or confirmation to a correct direction.
DISPLAY SURFACE	The area used to display the advertising message in a sign structure.
IDENTIFICATION SIGN	A sign that gives specific location information identifies specific items, for example, Parking Lot B, Building No. 5, First Aid, etc.
ILLUMINATED SIGN	The display of signs, permanent or otherwise, functions of which depend upon its being illuminated by direct or indirect light, and other than an electric sign.
INFORMATIONAL SIGN	Used for overall information for general organization of a series of elements that is, campus plan, bus route, building layout, shopping mall plan, etc.
MARQUEE	A roofed structure attached to and supported by a building and projecting beyond building line or property line.
NONCOMBUSTIBLE MATERIAL	A material no part of which ignites or burns when subject to fire. Any material conforming to ASTM E136 shall be considered noncombustible.
PUBLIC PASSAGE	A public way having a width less than 2.5 m.
PUBLIC PROPERTY	The land property which has been dedicated or deeded to the public for use.
REGULATORY SIGN	Sign that gives operational requirements, restrictions or gives warnings, usually used for traffic delineation or control, for example 'stop', 'No parking', 'one Way', etc.

1.3 Classification of Signs

For the purpose of this Code, and the regulations and provisions thereof, outdoor displays shall be classified into one of the following types of sign.

BALCONY SIGN	An advertising sign attached to, hung from or posted on a balcony or verandah of a building.
COMBINATION SIGN	An outdoor sign incorporating any combination of features of other signs.
ELECTRIC SIGN	An outdoor advertising display sign, usually animated, wired for lights or luminous tubing and containing other electric fittings, excluding those illuminated by external light source. They shall meet the requirements of the provisions of this Code governing ground, roof, wall, projecting, marquee or other sign as set forth below, depending upon where the electric signs are built.
FIN SIGN	An outdoor sign affixed to the wall or exterior surface of a building with the plane of the sign perpendicular to or at an angle with the building surface projecting more than 300 mm from the surface, the property line or the building line.
GROUND SIGN	An outdoor advertising display sign painted or otherwise displayed on the ground, supported by a pole or a structure erected independently on the ground, or mounted on a vehicle or mobile structure.
MARQUEE SIGN	A projecting sign attached to or hung from a marquee canopy or covered structure projecting from and supported by a building and extending beyond the building line or property line.
PROJECTING SIGN	An outdoor sign other than a wall sign affixed to the wall or exterior surface of a building, with the exposed face of sign in a plane parallel to the wall surface projecting more than 300 mm from the building surface, building line or property line.
ROOF SIGN	An outdoor sign painted on the roof surface or erected upon or above a roof or parapet of a building.
TEMPORARY SIGN	Any sign, banner, pendant or other display constructed of cloth, canvas, fabric, cardboard, bamboo or other light material, with or without a structural frame, intended to be displayed for a limited period only. They shall include signs on gas filled balloons afloat in the air, free or anchored to the ground or a structure, signs or words traced by smoke emitted by flying aircraft or a stationary source and signs tied to a flying aircraft.
WALL SIGN	An outdoor sign directly painted on or pasted or attached to or erected against the wall or exterior surface of a building, projecting not more than 300 mm from the wall.

1.4 Application and Permit

1.4.1 Requirement of Permit

No sign shall be erected, re-erected, constructed, altered or maintained except as provided by this Code. A written permit shall be obtained from the Authority for all signs except those exempted from such permit in Sec 1.4.2.

1.4.2 Exemptions

Signs or alterations of signs described in this section and its subsections shall be exempted from the requirement of a permit. These exemptions shall not be construed as relieving the owner of the sign from the responsibility of ensuring compliance with the provisions of this Code or other regulations or laws pertaining to signs. The following works may be undertaken on existing signs without a permit:

- (a) Changing of the advertisement copy or message on a painted or printed sign only. Except for cinema or theatre marquee signs or similar signs specifically designed for replaceable copy, electric signs shall not be included in this exemption.
- (b) Cleaning, painting or repainting of a sign or sign structure not involving any structural alteration.

The following signs may be erected without a written permit :

- (a) Signs, the top of which is less than 2 m above grade.
- (b) Signs exhibited within the window of a building provided they do not affect the light and ventilation of the building.
- (c) Signs relating to the trade or business carried on within the premises on which such signs are displayed or notices of meeting or sale etc. to be held within the premises.
- (d) Signs painted or pasted on buses or other public transport vehicles provided that no part of such signs projects beyond the body of the vehicle⁶ or obstructs the visibility of the driver or passengers to endanger safe ride.

No permission shall be required for signs covered by the provisions of Sections 1.4.2.1 to 1.4.2.4.

1.4.2.1 Wall Signs: Wall signs listed below shall not require a permit:

- (a) Shop Signs: Signs erected over a display window or entrance of a shop or business establishment which announce the name of the shop and the business carried on provided such signs are less than 1 m in height.

- (b) Building Names: Wall signs erected on public or private buildings which announce the name of the establishment and the nature of occupancy.
- (c) Name Plates: Any wall sign less than 0.5 m² in area announcing the name and identity of the occupier.
- (d) Boundary Signs: Any wall sign less than 0.5 m² in area erected on boundary walls or fences surrounding the premises, or on other ornamental fences, announcing the identity of the premises.

1.4.2.2 Ground Signs Erected on or Visible from Highways: Ground signs erected on or visible from the highways bearing warning, cautionary, informative, identifying and temporary advertising messages of the following description shall not require a permit. When placed on highways these ground signs shall conform to the requirements of Sec 1.6.3.

- (a) Official warning signs, traffic directions and notices displayed on the highway by public authorities or by the requirement of a judicial order.

Example: FERRY AHEAD

- (b) Direction indicators to places of public service.

Examples: POLICE STATION PETROL PUMP

- (c) Signs bearing names of places

Example: SHERPUR TOWN

- (d) Defense warning signs.

Example: SHOOTING RANGE

- (e) Private warning signs not more than 0.2 m² in area.

Example: PRIVATE ROAD

- (f) Signs not more than 0.2 m² in area placed sufficiently clear of the carriageway showing direction to a private property.
- (g) Signs bearing identification, direction or warning messages with respect to the premises on which they are displayed, not exceeding 0.2 m² in area.

Example: MIND THE STEP

- (h) Advertising signs relating to businesses, trades or professions carried out in the buildings on which they are displayed, limited to one sign not exceeding 0.3 m² in area for each business, trade or profession.
- (i) Identification signs relating to educational, medical, social, religious, cultural, recreational or similar institutions, hotels, blocks of flats, hostels, rest houses etc. situated on the premises on which such signs are displayed, limited to one sign not exceeding 1.2 m² in area for each such institution.

- (j) Temporary advertisements relating to the sale or rent of the premises on which they are displayed, limited to one sign not exceeding 2 m² in area for each sale or rent.
- (k) Advertisements announcing sale of goods or livestock, limited to one sign not exceeding 1.2 m² in area, displayed on the land or premises where such sale is held or goods or livestock are situated.
- (l) Advertisements relating to construction works in progress on the land on which the sign is displayed not exceeding 2 m² in area.
- (m) Signs announcing noncommercial local events of a religious, cultural, recreational or educational character, limited to a total of 0.6 m² of display on any premises on which such events are to take place.

1.4.2.3 Temporary Signs: Temporary signs of the following character shall not require a permit :

- (a) Construction site signs displaying the owner or builder's identity or the project description.
- (b) Special and decorative displays used for festivities, fairs, circuses, public demonstrations or promotion of civic welfare and charitable activities, provided such displays do not involve the use of electricity.

1.4.2.4 Other Displays: Signs displayed on the concerned premises meeting the general requirements of Table 10.1.1 and conforming to the other requirements of this Code shall not require a permit.

1.4.3 Application for Permit

Application for a sign permit shall be made in writing by the erector of the sign and the owner/lessee of the premises or site where the sign is to be erected on prescribed form (Appendix A). The application shall be accompanied by the following information:

- (a) Full specification of the sign in a sketch showing the length, height and weight of the sign, and, where applicable, number and disposition of electric lights and fittings and wiring diagram.
- (b) A location plan showing the position of the sign on the site, an elevation showing the sign in relation to the facade and detail drawing of the sign.
- (c) Detail drawings of the sign structure indicating the size of all members and the foundation and, if required by the Authority, design calculations of the structure.
- (d) Any other information as may be required by the Authority.
- (e) Required fee as may be decided by the Authority from time to time.

1.4.4 Condition for Grant of Permit

The grant of permit or license to erect signs shall be subject to the following conditions:

- (a) An application for the grant of permit shall be made to the Authority in the manner prescribed in this Code.
- (b) The applicants must possess right over the property, building or premises upon which the sign is proposed to be erected either by way of ownership or by lease.
- (c) The permission shall be granted at any one time for a period not exceeding three years, on expiry of which the Authority shall have the right to have the sign removed at the erector's expense unless a renewal of the permit has been obtained.
- (d) A permit shall be renewed only upon continued satisfaction of all the conditions under which the original permit was issued.
- (e) The permission or the renewal of permission granted by the Authority shall become void, if
 - any addition or alteration is made to the sign structure except as may be directed by the Authority to make it secure;
 - any change is made in the sign or part thereof, involving a change in the type of sign;
 - any addition or alteration is made to the building which supports the sign involving disturbance of the sign or any part thereof;
 - the sign or any part thereof falls due to accident or other causes; or
 - the building or structure supporting the sign is demolished or destroyed or declared 'unsafe building' by the concerned authority.
- (f) The Authority may, in the interest of the public, suspend the license before expiry of the period in which case the licensee shall remove the sign forthwith.
- (g) The licensee shall maintain the sign and the building or structure supporting or occupied by the sign in safe, neat and sanitary condition.
- (h) The sign shall not, in the opinion of the Authority, mar the aesthetic beauty of the locality.
- (i) No sign other than that pertaining to the building concerned shall be allowed to obstruct or obscure buildings such as hospitals, public offices, educational institutions, places of worship, museums and buildings of national and historic importance.

- (j) No sign or hoarding shall be allowed to obstruct light and ventilation of buildings situated near it.
- (k) No sign shall bear any objectionable, unlawful or obscene display.
- (l) Any hoarding or sign erected on the highways shall also require the express permission of the authority or agency in charge of the highway.
- (m) In addition, all signs shall at all times conform to the requirements given in this Code.

1.4.5 Sanction or Refusal of Permit

Upon receipt of an application for permit the Authority may sanction, sanction with modification or refuse such a permit. The decision of the Authority shall be communicated within 30 days of the date of receipt of all information desired by them, failing which the permit shall be deemed to have been sanctioned. The Authority may withdraw a permit at any time in the interest of public safety and welfare.

Table 10.1.1: Signs Displayed on Concerned Premises which shall not Require a Permit (Sec1.4.2.4)

Description	Example	Number Permitted	Dimensional Limits	Maximum Size of Letters, Symbols etc.	Maximum Height Above Grade to Top of Sign	Electrification Restriction	Other Special Requirements
Identification signs of public undertakings	FIRE BRIGADE PATHOLOGY	As required	As required	As required	As required	As required	
Warnings, direction or identification of utilities	NO PARKING	No limit	Maximum area 4 m ²	800 mm (400 mm in restricted sign zone*)	6 m (5 m in restricted sign zone)	Only for medical services and to indicate danger	
Identification of company, trade, business, profession or person	ABCD & CO.	One at each entrance	Maximum area 0.3 m ² each	800 mm (400 mm in restricted sign zone*)	6 m (5 m in restricted sign zone)	Only for medical services and to indicate danger	
Name of building or premises	CENTRAL LAW ACADEMY	One on each entrance	Maximum area 1.2 m ² each	800 mm (400 mm in restricted sign zone*)	6 m (5 m in restricted sign zone)	Only for medical services and to indicate danger	

Description	Example	Number Permitted	Dimensional Limits	Maximum Size of Letters, Symbols etc.	Maximum Height Above Grade to Top of Sign	Electrification Restriction	Other Special Requirements
Temporary signs relating to sale or letting of property on which displayed	FOR SALE TO LET	No restriction but total area not to exceed 2.4 m ²	Width to depth ratio 2:1, total area of all signs not to exceed 2.4 m ²	800 mm (400 mm in restricted sign zone*)	6 m (5 m in restricted sign zone)	No electrification permitted	Shall not be displayed earlier than 1 month prior to date the sale or letting is due and shall be removed within 2 weeks after conclusion of sale or letting.
Construction site signs	SHOPPING CENTRE Consultant Contractor 	One for each frontage for each contractor	Maximum area 4 m ²	800 mm (400 mm in restricted sign zone*)	6 m (5 m in restricted sign zone)	No electrification permitted	Permitted only while the construction is in progress
Temporary signs announcing events displayed on the premises where these take place	INTERNATIONAL BOOK FAIR	No restriction but total area not to exceed 4 m ²	Maximum aggregate area 4 m ²	800 mm (400 mm in restricted sign zone*)	6 m (5 m in restricted sign zone)	Electrification shall require permission	Shall not be displayed earlier than 1 month prior to the start of event and shall be removed within 2 weeks of end of event
Identifying and advertising signs on business premises in restricted sign zone	XYZ BANK LTD. ABC TRAVEL	No restriction but total area not to exceed 10% of area of face	Maximum aggregate area not to exceed 10% of area of face	400 mm	5 m	Electrification shall require permission except for medical services and to indicate danger	

1.4.6 Application for Alteration of Sign

A fresh application for permit shall be made each time the ownership of the sign changes or any change in type or structure of the sign is proposed.

1.4.7 Existing Signs

Signs in existence on the date of promulgation of this Code, that would otherwise require a permit, shall be exempted from the requirement of permit for a period of two years after the promulgation of the Code. On expiry of this period these signs shall require a permit as for a new sign.

1.5 Unsafe and Unlawful Signs**1.5.1 Responsibility of the Owner**

It shall be the responsibility of the owner to ensure the safety and legality of the sign irrespective of any reference from the Authority.

1.5.2 Notice of the Authority

If any sign is unlawfully installed or maintained violating any of the provisions of the Code, or if any sign becomes insecure or unsafe, the owner of the sign shall, upon written notice of the Authority, make the sign conform to the provisions of the Code or remove it, within a time period specified by the Authority which shall in no case exceed three days. If the owner fails to comply within the specified time, the Authority shall remove the sign and charge the expenses to the owner.

1.5.3 Prohibited Signs

Any sign which in the opinion of the Authority, fits any of the following descriptions, shall not be permitted under any circumstance:

- (a) Signs of obscene, repulsive, vulgar, revolting or otherwise objectionable character,
- (b) Signs displaying messages prejudicial or subversive to state discipline,
- (c) Signs producing pernicious or injurious effect on a class of persons or the public,
- (d) Signs that disfigure any neighbourhood or destroy its sanctity, or
- (e) Signs those are likely to affect the sentiment of public.

1.6 Restrictions**1.6.1 Restricted Sign Zone**

1.6.1.1 The Authority may, in the interest of aesthetics, moral values or public welfare, designate an area as a restricted sign zone. Kindergartens, elementary schools, primary schools and secondary schools, Parks, playgrounds, places of historical interest, places of architectural heritage, important national monuments, animal and bird sanctuaries, nature reserves etc. may also be designated as restricted sign zones.

1.6.1.2 The erection or display of signs within the restricted sign zone shall be prohibited or restricted in any manner deemed necessary by the Authority.

1.6.1.3 The wording on any sign in the restricted sign zone shall be limited to messages for identification of, direction to and information about the owner of the premises, the nature of business carried on within the premises and location of utilities and services.

1.6.1.4 The restriction on signs in a restricted sign zone shall apply within a distance of 30 m outside the zone.

1.6.2 Prohibition of Advertisement

The erection of any advertising sign shall be prohibited on a site, when in the opinion of the Authority:

- (a) the site is unsuitable for display of advertising signs by virtue of the historic, cultural, architectural or similar characteristics of the locality, or
- (b) the display of an advertising sign is likely to pose a hazard for any form of transport.

1.6.3 Signs on Highways and Roads

Any advertising sign at or near highways or roads shall conform to the following requirements in addition to the requirements of the respective authority.

- (a) No sign shall obscure or otherwise hinder interpretation of official traffic signs.
- (b) Signs on the sides of over-bridges or flyovers across the carriageway shall not project beyond any edge of the parapet of the over-bridge or flyover.
- (c) No sign shall be erected within the right of way of the road nor within 10 m of the edge of the carriageway. The size of the sign erected at 10 m away from the edge of the carriageway shall not be more than 3 m². This limit on size shall be increased by 0.3 m² for each extra meter of setback from the edge of the carriageway.
- (d) No sign shall be erected in such manner or at such places as to obstruct or interfere with the visibility of approaching, intersecting or merging traffic.
- (e) No sign shall be erected within 100 m of a bridge, railway level crossing or road junction. For urban roads this distance may be reduced to 50 m.
- (f) No sign shall be erected in such a way as to hinder visibility at pedestrian crossings.

1.6.4 Illuminated Displays

The following illuminated advertisements that threaten traffic safety shall not be permitted:

- (a) Illuminated or electric advertisements of such brilliance or intensity as to produce glare and impair vision of the driver and the pedestrians;
- (b) Advertisements containing flashing or intermittent lights of intense brightness;
- (c) Advertisements containing lights of such colour, disposition, brightness, movement or flashes as to obscure or reduce effectiveness of official traffic lights, signals, signs or devices.

1.7 Maintenance and Inspections**1.7.1 Maintenance**

Signs and their supporting structures, together with the supports, braces, guys and anchors shall be maintained in sound condition and any deterioration shall be repaired immediately. All non-galvanized and corrosion susceptible metal components shall be painted at least once in every two years. The Authority may order removal of signs which are not maintained in proper order and such removal shall be at the owner's expense.

1.7.2 Inspection

Every sign shall be subject to inspection by the Authority from time to time as required by the Authority. The owner of the sign shall ensure access of the inspector and facilitate the inspection.

1.8 Location Restrictions

1.8.1 An outdoor display sign shall not be erected, constructed or maintained so as to obstruct any fire escape or any window or door or opening used as a means of egress or so as to prevent free passage from one part of a roof to the other.

1.8.2 A sign shall not be attached in any form, shape or manner to a fire escape, nor be placed in such manner as to interfere with openings required for lighting and ventilation.

1.9 Projection Over Public Property

1.9.1 No part of a sign or sign structure shall project into a private property not owned or leased by the owner/erector of the sign, unless explicit agreement has been entered into with the owner of that property.

1.9.2 Signs and sign structures shall project neither into public property nor into roads wider than 4 m, below a height of 2.5 m above grade, nor project more than 0.3 m when the sign is erected 2.5 m above finished grade. The signs may project 0.3 m plus 0.12 m for each 0.25 m of clearance above 2.5 m when the height is between 2.5 m and 5 m above grade. Signs may not project more than 1.5 m into a public property under any circumstance.

1.9.3 No sign or part thereof shall project into public passages. Projection of signs and sign structures over public alleys shall be limited to the values provided in Table 10.1.

Table 10.1.2: Projection of Signs over Public Alleys

Height Above Finished Grade (m)	Maximum Projection (m)
Less than 4	No projection permitted
4 to 5	0.3
over 5	0.6

1.10 CLEARANCE FROM POWER LINES

All signs and sign structures shall maintain the clearances from overhead electrical conductors as specified in Table 10.1.3

Table 10.1.3: Clearance from Power Lines

Voltage of the Line	Vertical distance (m)	Horizontal Distance (m)
Low and Medium Volt	2.5	1.25
33 KV	3.5	1.75
Over 33 KV	3.5 Plus 0.3 m for each additional 33 KV or part thereof	1.75 Plus 0.3 m for each additional 33 KV or Part thereof

1.11 Related Appendix

Appendix A Application for Permit to Erect or Alter Outdoor Signs

Chapter 2 General Requirements

2.1 Design

2.1.1 Loads

All outdoor signs and sign structures shall be designed to resist wind, gust during storm, seismic and other forces as specified in Chapter 2: Loads, Part 6 of this Code. Combination of wind and seismic loads shall not be required. Loading that produces higher stresses shall be used.

2.1.2 Design Consideration

All outdoor signs and sign structures shall be designed fulfilling the design requirements as set out in Chapter 1: General Design Requirements: Structural Design, Part 6 of this Code.

2.2 Construction

All outdoor signs and sign structures shall be constructed and erected in accordance with the requirements of Chapter 1: Constructional Responsibilities and Practices, Part 7 of this Code.

2.1.2 Use of Materials

All materials for outdoor signs and sign structures shall conform to the specification as set out in Part 5: Building Materials of this Code.

2.2.2 Use of Combustible Materials

Ground signs not higher than 6 m may be constructed of any material that meets the requirements of this Code. No combustible material other than approved plastic as defined in Sec 2.2.5, shall be used in the construction of electric signs. Roof, wall, projecting, fin, balcony, marquee and combination signs shall be constructed of noncombustible materials except as provided below:

- (a) On roofs of combustible construction, the roof sign may be constructed of combustible materials.
- (b) On roofs of any type of construction, roof signs not higher than 1.5 m and not exceeding 5 m² in area may be constructed of combustible materials.
- (c) On walls of combustible construction, wall signs not involving the use of electricity may be constructed of combustible materials.

2.2.3 Anchorage

Foundation for all unbraced signs shall be designed to resist horizontal, vertical and overturning forces. All braced ground signs shall be anchored to resist the specified wind and seismic forces in any direction. Anchors shall be designed for safe soil bearing capacity and for an effective uplift force which is 25% more than the force required to resist overturning. Anchorage of signs shall not be connected to an unbraced parapet wall unless the wall is designed for seismic load.

2.2.4 Display Surfaces

Display surfaces of outdoor signs may be made of metal, glass or approved plastic. If the surface of the sign is made of glass, the thickness and area shall be as set forth in Table 10.2.1.

Table 10.2.1: Type, Size and Thickness of Glass Panels Used in Signs

Maximum Size of Glass Panel		Minimum Thickness (mm)	Type of Glass
Any dimension (m)	Area (m ²)		
0.75	0.30	3	Plain, Plate or Wired
1.15	0.45	5	Plain, Plate or Wired
3.65	2.30	6	Plain, Plate or Wired
above 3.65	above 2.30	6	Wired Glass

Plastic of approved type may be used for wall signs in sections not exceeding 20 m² in area. Plastics sections on wall signs shall be separated 1 m laterally and 2 m vertically. Approved plastics of unlimited area may be used on any sign other than wall sign, if approved by the Authority

2.2.5 Approved Plastics

Plastic materials which burn at a rate no faster than 65 mm per minute when tested in accordance with ASTM D 635 shall be deemed approved plastic. Only approved plastic shall be used for plastic display surfaces provided for in Sec 2.2.4 above. Approved plastics may also be used for ornamental purposes, decorations, lettering, facings etc. on signs and outdoor display structures.

2.2.6 Draining Arrangements

Signs constructed on ground or at places where possibility of accumulation of water exists shall have adequate provision for proper drainage.

2.3 Use of Glass In Signs

Glass when used in outdoor signs shall be at least 3 mm thick and shall conform to the requirements of Sec 2.16.14: Glass and Glazing, of Part 5 of this Code. The area of each glass panel shall not exceed 6 m² and shall be securely fixed with the frame independently. Appropriate protection against damage by falling objects shall be provided to all glass panels by metal canopies or other approved means. Use of glass may be discouraged or avoided wherever possible for signs placed overhead. For such uses, if permitted, the authority may impose condition of using 'tempered glass' so as to produce no sharp edges or large pieces if broken accidentally.

2.4 Servicing Devices

All servicing devices (ladders, platforms, hooks, rings etc.), used for cleaning, painting, repainting of sign shall have adequate safety devices and shall be of approved type and quality.

2.5 Interference by Signs

Signs shall not be placed at such locations that would obstruct the use of fire hydrants or other firefighting appliances. Signs in bends and curves shall be placed in such a location so as not to obstruct the view of traffic at intersecting streets.

2.6 Hours of Operations

No electric sign or other type of signs using electric power source, other than those necessary in the opinion of the Authority in the interest of public amenity, health and safety, shall be operated between midnight and sunrise.

Chapter 3

Specific Requirements for Various Types of Signs

3.1 Electric Signs

3.1.1 Materials

Materials for the construction of electric signs shall be noncombustible except as provided in Sec 2.2.2

3.1.2 Location

Electric signs in colour erected at a height lower than two stories or 6 m above the pavement shall be provided with suitable screen to avoid confusion with traffic signals. No sign in red, amber or green colours shall be erected within a horizontal distance of 10 m from any traffic signal.

3.1.3 Installation

All electric signs including the electrical equipment in connection with the sign shall be installed in accordance with the provisions of Chapter 1 Part 8 of this Code.

3.1.4 Illumination

Electric signs shall not be of such intense illumination as to cause inconvenience or disturbance to residents of adjacent buildings or to the drivers on road.

3.1.5 Operational Hours

The operational hours of electric signs will be restricted (as referred in Sec 2.6 Chapter 2 of this Part)

3.2 Ground Sign

3.2.1 Material

All ground signs over 6 m in total height shall be constructed of noncombustible materials meeting the requirements of this Code, or of approved plastics as defined in Sec 2.2.5 Chapter 2 of this Part. Materials used for the construction of ground sign supporting structures may be treated timber, masonry, concrete or corrosion resistant metal.

3.2.2 Height

The height of ground signs excluding the lighting reflectors shall be limited to 9 m.

3.2.3 Design

The design and construction of ground signs shall conform to the requirements of Part 6 and Part 7 of this Code. All ground signs shall have a firm support and shall be anchored to the ground.

3.2.4 Clearance

All ground signs shall be provided with a clearance height of 0.6 m from the ground. The intervening space may be filled with open lattice work. Under no circumstance shall any ground sign obstruct or interfere with entrance or exit of a building.

3.2.5 Visual Obstruction

All ground signs must be sited in a manner so that they don't pose any visual obstruction to the traffic signs or to any emergency signs (e.g. hospital entry)

3.3 Roof Signs

3.3.1 Material

All roof signs including the frames shall be constructed of noncombustible materials except as provided in Sec 2.2.2 Chapter 2 of this Part.

3.3.2 Design

Design and construction of roof signs shall conform to the requirements of Part 6 and Part 7 of this Code. Roof signs shall be properly secured and anchored to the building and the building shall be designed to avoid overstress due to the sign.

3.3.3 Clearance

Roof signs shall not prevent free passage from one part of the roof to the other. Such passages shall be not less than 1 m wide and 1.25 m high.

3.3.4 Projection

No roof sign shall project beyond the roof in any directions.

3.4 Projecting Signs

3.4.1 Material

All projecting signs and their supporting frames shall be of noncombustible material except as provided in Sec 2.2.2 Chapter 2 of this Part

3.4.2 Design

The supporting frame of projecting signs and the building element to which it is anchored shall be designed to withstand, in addition to dead, live and wind loads calculated in accordance with Part 6 of this Code, appropriate loads due to servicing personnel and equipment.

3.4.3 Height and Clearance

A minimum of 2.5 m clearance from the road surface shall be provided for any projecting sign. The maximum height of a projecting sign shall be 15 m when erected against buildings having a height of more than 8 storey or 36 m. For buildings 5 to 8 stories high but not exceeding 36 m, the height of the projecting signs shall be limited to 12 m. The height of the sign shall be limited to 9 m when attached to a building less than 4 storey or 18 m high.

3.4.4 Projection

Projecting sign or any part of its supporting structures shall not project more than 2 m beyond the building. When such sign faces the street, it shall not project beyond the property line. Projecting sign shall not extend above the eaves of the roof of the building to which it is attached. Projection over public property or alley of projecting signs shall be limited to the values as specified in Sec 1.9 Chapter 1 of this Part.

3.4.5 Attachment

All projecting signs shall be so constructed or attached to the building that movement in any direction is prevented by rods, anchors, brackets, chains etc.

3.4.6 Visual Obstruction

All projection signs must be sited in a manner so that they don't pose any visual obstruction to the traffic signs or to any emergency signs (e.g. hospital entry)

3.5 Fin Signs

3.5.1 Material

Materials for fin sign shall conform to the requirements of Sec 2.2.2 Chapter 2 of this Part.

3.5.2 Design

Design and construction of fin signs shall conform to the requirements of Part 6 and Part 7 of the Code.

3.5.3 Clearance

Fin signs shall not obstruct windows and reduce light and ventilation to such a point as the Code or other regulations prohibit. Such signs and their frames shall not obstruct fire escape, exit and entrance of the building to which they are attached. Projection of fin signs over public property shall conform to the requirements of Sec 1.9 Chapter 1 of this Part.

3.6 Balcony Signs

3.6.1 Materials

Materials for balcony signs shall conform to the requirements of Sec 2.2.2 Chapter 2 of this Part.

3.6.2 Location

Balcony signs shall be placed above eaves of the balcony and shall not project beyond rear of the roof gutter.

3.6.3 Size

The height of a balcony signs shall be limited to 1 m. The hanging balcony sign shall not exceed 2.5 m in length and 50 mm in thickness. For hanging box type signs the maximum allowable depth shall be 200 mm.

3.6.4 Projection

Balcony signs shall not extend beyond the balcony line. Hanging balcony signs shall maintain a clearance height of 2.5 m from the finished ground level. Projection of balcony sign over public property or alley shall be limited to values as specified in Sec 1.9 Chapter 1 of this Part.

3.7 Marquee Signs

3.7.1 Materials

Materials for the construction and erection of marquee signs shall conform to the requirements of Sec 2.2.2 Chapter 2 of this Part.

3.7.2 Size

The height of a marquee sign shall be limited to 2 m. The length of such sign may be equal to the length of the marquee and no projection beyond the full length shall be allowed.

3.7.3 Clearance

A clearance height of at least 2.5 m shall be provided for marquee signs.

3.8 Combination Signs

All combination signs shall meet the general and specific requirements of all the component classes of sign.

3.9 Temporary Signs

3.9.1 Size

The area of a temporary sign shall not ordinarily exceed 10 m². Temporary signs of a larger area may be erected upon explicit approval of the Authority which shall only be granted, for signs of a duration not exceeding 7 days. Temporary signs made of rigid material shall not exceed 3 m² in area, nor shall they have a height more than 2 m. Such rigid signs shall be anchored to the ground or a building by a rigid structure.

3.9.2 Duration

Temporary signs shall be removed from the location within 60 days of first erection, unless explicit permission of the Authority for extending this period has been obtained. In no case a temporary sign shall be permitted to be maintained for more than 90 days.

3.9.3 Support

Temporary signs shall be adequately supported and secured in place. No part of the sign shall be permitted to dangle, sway or otherwise become loose or detached. In order to reduce wind resistance on signs made of fabric, adequate perforations shall be provided.

3.9.4 Location

All temporary signs shall be subject to the approval of the Authority and shall be erected in such a place so as not to obstruct any public way, foot path or entrance and exit of any building.

3.9.5 Projection

Temporary cloth sign may extend over public property. A clearance of 6 m shall be maintained when such signs are placed over a public street. Other temporary signs when placed 2.5 m above the ground may project not more than 300 mm over public property or beyond the legal setback line.

3.10 Additional Guidelines for Signs in Urban and Rural Areas

3.10.1 Erecting, maintaining and owning signs in rural areas shall be encouraged so as to create awareness, dissemination of information and to boost economic status of the rural population.

3.10.2 The tolerance criteria for the permission granted towards putting up any signs for any urban area shall be as given in Sections 3.10.2.1 to 3.10.2.4.

3.10.2.1 Small towns

The traffic hazards in small towns are few and the defacement due to excessive advertising signs has not occurred. Therefore, orderly development of signs may enliven the town environment and boost the economy. The tolerance here may be high.

The following guidelines may be followed for signage :

ADVERTISING SIGN	Electric sign, ground sign, building sign, illuminated sign, sky sign and temporary sign are permissible.
DIRECTIONAL SIGN	Electric sign, ground sign, building sign, illuminated sign and temporary sign are permissible while sky sign is not permissible.
INFORMATIONAL SIGN	Electric sign, ground sign, building sign, illuminated sign and temporary sign are permissible while sky sign is not permissible.
IDENTIFICATION SIGN	Electric sign, ground sign, building sign, illuminated sign and temporary sign are permissible while sky sign is not permissible.
REGULATORY SIGN	Electric sign, ground sign, illuminated sign and temporary sign are permissible while building sign and sky sign are not permissible.

3.10.2.2 Medium (District level) towns

The traffic hazards in medium towns are few and the defacement due to excessive advertising signs has not occurred. Proper design, erection and maintenance of the signs shall be encouraged. The following guidelines may be followed for signage :

ADVERTISING SIGN	Electric sign, ground sign, building sign, illuminated sign, sky sign and temporary sign are permissible.
DIRECTIONAL SIGN	Electric sign, ground sign, illuminated sign are permissible while building sign, sky sign and temporary sign are not permissible.
INFORMATIONAL SIGN	Electric sign, ground sign, illuminated sign, and temporary sign are permissible while building sign and sky sign are not permissible.
IDENTIFICATION SIGN	Electric sign, ground sign, building sign, illuminated sign and temporary sign are permissible while sky sign is not permissible.
REGULATORY SIGN	Electric sign, ground sign, illuminated sign and temporary sign are permissible while building sign and sky sign are not permissible.

3.10.2.3 Large (Divisional level) cities

Accident risks due to density and complexity of traffic in large cities is high. Unplanned advertisements may add more risk to this situation. Defacement of buildings, roads and urban spaces due to advertisements has to be controlled. Therefore, the permissivity and tolerance for erecting signs is very low. The following guidelines may be followed for signage:

ADVERTISING SIGN	Electric sign, ground sign, illuminated sign and sky sign are permissible while building sign and temporary sign are not permissible.
DIRECTIONAL SIGN	Ground sign, illuminated sign are permissible while electric sign, building sign, sky sign and temporary sign are not permissible.
INFORMATIONAL SIGN	Ground sign, illuminated sign, building sign and temporary sign are permissible while electric sign and sky sign are not permissible.
IDENTIFICATION SIGN	Electric sign, ground sign, building sign, illuminated sign and temporary sign are permissible while sky sign is not permissible.
REGULATORY SIGN	Ground sign, illuminated sign and temporary sign are permissible while electric sign, building sign and sky sign are not permissible.

3.10.2.4 Mega and metro cities

The traffic hazards in mega and metro cities are many and the defacement due to excessive advertising signs has marred the urban environment. The density of population is very high and the danger of greater loss of life due to disasters is self-evident. Therefore, the permissivity for erecting signs is very low and no tolerance exists for law breakers. The following guidelines may be followed for signage :

ADVERTISING SIGN	Electric sign, ground sign, illuminated sign and sky sign are permissible while building sign and temporary sign are not permissible.
DIRECTIONAL SIGN	Ground sign, illuminated sign are permissible while electric sign, building sign, sky sign and temporary sign are not permissible.

INFORMATIONAL SIGN	Ground sign, illuminated sign and temporary sign are permissible while electric sign, building sign and sky sign are not permissible.
IDENTIFICATION SIGN	Electric sign, ground sign, building sign, illuminated sign and temporary sign are permissible while sky sign is not permissible.
REGULATORY SIGN	Ground sign, illuminated sign and temporary sign are permissible while electric sign, building sign and sky sign are not permissible.

3.11 Environmental Graphics and Graphics for Universal Accessibility for Cityscape

3.11.1 The urban environment may be susceptible to confusion and chaos due to improper graphics, hoardings and advertisements. Therefore, the signage should be installed following requisite guidelines keeping the functional, safety and aesthetic aspects in view. The scale of the project should also be considered for implementing signage design. In urban design/planning projects and landscape projects on a large scale, the following criteria should be followed for signs and outdoor display structures :

- (a) The aesthetic and harmonious development of the visual environment.
- (b) Signage for the handicapped at all grade changes, entry points to buildings and public conveniences and facilities. Braille strips used should be displayed not above 1.5 m height for the benefit of the visually impaired at all important nodes, entrances and routes, Ramps for the people on wheelchair should be highlighted with the appropriate international sign of the wheelchair. These signs are needed to be lighted adequately even at night time.
- (c) Environmental graphics should be creatively designed to cater to the basic function of information, identity and way finding, with the objective of improvement of urban-scape.
- (d) Safety aspects.
- (e) Protection of trees and other vegetation from harm due to signs.

3.11.2 For People with disabilities or any kind of impairment all signs and symbols shall comply with the guidelines of Sec D.26 of Appendix D of Part 3.

Appendix A

Application for Permit to Erect or Alter Outdoor Signs

Application for Permit to Erect or Alter Outdoor Signs

1. Type of Sign : _____
2. Location of Sign : _____

3. Details of Sign
 - a) Dimension
 - i) Length _____ m
 - ii) Width _____ m
 - iii) Thickness _____ mm
 - b) Clearance and Projection
 - i) Clearance _____ m
 - ii) Projection _____ m
4. Type of Materials Used
 - i) Noncombustible
 - ii) Approved plastic
 - iii) Combination
5. Details of Electrical Installations (if any) : _____

6. Structural details of signs and supporting structures including foundations : _____

7. This application is accompanied by all required plans, drawings and other details as required by Sec 1.4.3 of this code.

Yes _____ No _____

Signature of owner of
the Building/Premise

Date _____

Name in Full _____

Address _____

Signature of Applicant

Date _____

Name in Full _____

Address _____